Reductive group schemes

GPT-6.1 Sol (OpenAI)
Codex, Ultra setting

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Course sources: https://github.com/KokunoYumeto/open-math-courses/tree/main/docs/courses/AG-RG.

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Algebra and sheaf cohomology before reductive groups

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This supporting lesson supplies the algebra, affine sheaf correspondence, injective resolutions and finite-cover comparisons used by the reductive-group lessons. Rings are commutative with identity. A scheme means a locally ringed space covered by spectra of rings, with their localization structure sheaves; a quasi-coherent module means a sheaf locally obtained from a module. These are definitions, not vanishing or descent theorems. No Noetherian or finite-generation assumption is implicit.

1. The algebra of localization

Lemma 1.0 (free modules and the tensor calculations). For every commutative ring A, modules and index sets may be arbitrary. Free modules exist and lift maps through surjections. Tensor products have their bilinear universal property, are right exact, commute with direct sums and module colimits, and have the associativity, scalar-extension and field-basis identifications used below. None of these assertions requires flatness or finite generation.

Proof. For a set I, let $A^{(I)}$ be the functions $I \to A$ of finite support, with coordinatewise operations. If e_i has coordinate 1 at i and zero elsewhere, every function has the unique finite expression $\sum_i a_i e_i$. Given arbitrary elements m_i of an A-module M, the rule $\sum_i a_i e_i \mapsto \sum_i a_i m_i$ is therefore a unique linear map. Taking I to be the underlying set of M and $m_i = i$ gives a free surjection onto M. A map from a free module lifts through a surjection by choosing a preimage of each basis image and using this formula; this is the only choice needed for the lifting assertion. Kernels are submodules and cokernels are quotient modules, so repeating free surjections onto the successive kernels gives a free resolution. More generally $\bigoplus_i M_i$ consists of finite-support tuples. Direct sums preserve exact sequences: kernels and images are computed coordinate by coordinate, and a finite-support tuple in a quotient has finitely many coordinates to lift. For finitely many indices every tuple has finite support, so the direct sum is exactly the product, including the empty zero module.

Construct $M \otimes_A N$ as the quotient of the free module on pairs (m, n) by the submodule generated by

$$(m + m', n) - (m, n) - (m', n), \quad (m, n + n') - (m, n) - (m, n'),$$

$$(am, n) - a(m, n), \qquad (m, an) - a(m, n).$$

Denote a pair's class by $m \otimes n$. Every element is a finite sum of these classes. A map out of the free module kills the displayed relations exactly when its values on pairs are additive and A-linear in both variables. It thus descends uniquely exactly for an A-bilinear map $M \times N \to P$. This proves the universal property and uniqueness of the tensor product: the maps between two such objects fix every pure tensor and are inverse. Applying maps to the two entries defines the tensor

of module maps; compositions and sums are verified on pure tensors. The maps $m \otimes n \mapsto n \otimes m$ and $a \otimes n \mapsto an$ have inverses obtained by swapping again and by $n \mapsto 1 \otimes n$, respectively. Their well-definedness is precisely the displayed relations.

Let $K \subset M$. In $T = (M \otimes_A N)/\langle k \otimes n : k \in K, n \in N\rangle$, the formula $(m + K, n) \mapsto [m \otimes n]$ is independent of the representative and is bilinear. It gives $(M/K) \otimes_A N \to T$. The quotient map $M \to M/K$ gives the inverse. Both composites fix every pure-tensor class. Consequently $K \otimes_A N \to M \otimes_A N \to (M/K) \otimes_A N \to 0$ is exact. For any right exact sequence $L \to M \to Q \to 0$, put $K = \mathrm{im}(L \to M)$. Every generator $k \otimes n$ has a preimage from $L \otimes_A N$, proving right exactness for that sequence. Swapping factors gives the same assertion in the other variable. In particular $M \otimes_A A/I \simeq M/IM$: the maps are $m \otimes \bar{a} \mapsto am \bmod IM$ and $m \bmod IM \mapsto m \otimes \bar{1}$, and multiples from I vanish by the tensor relations.

The map $M \otimes_A \bigoplus_i N_i \to \bigoplus_i (M \otimes_A N_i)$ sends $m \otimes (n_i)$ to $(m \otimes n_i)$. Its values have finite support. The inverse sends the ith pure tensor to $m \otimes \iota_i(n_i)$ and sums the finitely many components. These formulas respect addition and scalar moves and their composites fix all pure generators. Thus they are inverse; swapping proves the first-variable version. Combining with $A \otimes_A N = N$ gives $A^{(I)} \otimes_A N = N^{(I)}$. The coordinate exactness already proved makes free modules flat and proves tensor compatibility with finite products. For a small diagram of modules, its colimit is the quotient of the direct sum of all its modules by the elements $\iota_i(m) - \iota_j(f(m))$ for every transition $f : i \to j$. A linear map out of this quotient is exactly a compatible family, which proves this construction's universal property. Tensoring its right exact presentation and using the direct-sum formula produces the quotient by the elements $\iota_i(m \otimes n) - \iota_j(f(m) \otimes n)$: these generate all images of transition relations, since a general element of their submodule is a finite linear combination of those relations and pure tensors generate each tensor product. The same elements impose exactly the transition relations in the direct sum of the tensors. This is their colimit, proving that tensor commutes with such colimits without an adjoint-functor theorem.

To rebracket three factors, fix $p \in P$ and extend $(m, n) \mapsto m \otimes (n \otimes p)$ by the bilinear universal property to $M \otimes_A N$. These extensions are additive and A-linear in p, as is checked on the generating $m \otimes n$. They therefore define $(M \otimes_A N) \otimes_A P \to M \otimes_A (N \otimes_A P)$. Reversing the construction sends $m \otimes (n \otimes p)$ to $(m \otimes n) \otimes p$. The composites fix every triple pure tensor, hence are identities. For $A \to B$, make $B \otimes_A M$ a B-module by multiplying its first factor. If N is a B-module, the maps

$$N \otimes_B (B \otimes_A M) \longrightarrow N \otimes_A M, \qquad n \otimes (b \otimes m) \longmapsto bn \otimes m,$$

and $n \otimes m \mapsto n \otimes (1 \otimes m)$ are inverse. In the forward map the B-balance is $(cn) \otimes (b \otimes m) = n \otimes (cb \otimes m)$, and the inner A-balance moves an A-scalar from b to m. The inverse respects A-balance by moving that scalar first into $1 \otimes m$ and then across the outer B-tensor. These checks and additivity establish well-definedness; the composites fix pure generators. This also proves transitivity of scalar extension. An A-linear map $f : M \to N$ into a B-module corresponds to the B-linear map $B \otimes_A M \to N$, $b \otimes m \mapsto bf(m)$; restriction to $1 \otimes m$ is its inverse. This is the scalar-extension tensor adjunction. All the identifications above commute with module maps because their formulas agree on every pure generator.

If P is flat, the right-exact quotient calculation together with preservation of injections makes tensor with P exact. For a map $u : M \to N$, tensor $0 \to \ker u \to M \to \mathrm{im}\, u \to 0$ and the injection $\mathrm{im}\, u \to N$. The resulting kernel is $(\ker u) \otimes_A P$ and the resulting image is $(\mathrm{im}\, u) \otimes_A P$. For a complex, apply this to cycles and boundaries and then to $0 \to B^n \to Z^n \to H^n \to 0$. It proves $H^n(C^\bullet \otimes_A P) = H^n(C^\bullet) \otimes_A P$, including unbounded complexes; no sum over degrees is used. In particular flat tensor preserves the finite kernels and products used in affine descent.

For commutative A-algebras B, C, the rule $(b \otimes c)(b' \otimes c') = bb' \otimes cc'$ gives a ring structure on their module tensor product. Multiplication by a pure tensor preserves each defining additive and scalar relation, by distributivity and the commuting base actions, so it descends in both arguments. Checking on pure triples proves associativity; swapping factors proves commutativity, and $1 \otimes 1$ is

the unit. Two A-algebra maps $B \to D$, $C \to D$ extend uniquely by $b \otimes c \mapsto f(b)g(c)$, since this map is balanced and multiplicative. Restriction to $b \otimes 1$ and $1 \otimes c$ gives the inverse correspondence. Thus tensor is the coproduct of commutative A-algebras. In particular $B \otimes_A A[X_i]/I = B[X_i]/(IB[X_i])$: send $b \otimes \bar{f}$ to $b\bar{f}$, with coefficients mapped to B. Conversely map B and each X_i to $b \otimes 1$ and $1 \otimes \bar{X}_i$; every polynomial in I vanishes. These algebra maps are inverse on B and the variables. Variable and relation sets may be infinite, since every expression is finite.

Finally every vector space V over a field F has a basis. Order its linearly independent subsets by inclusion. A chain has an independent union because a relation involves finitely many vectors, all in one member of the chain. Zorn's lemma supplies a maximal independent subset; if it failed to span, adjoining a vector outside its span would still be independent. Its coordinate map identifies V with $F^{(I)}$. Therefore for every F-algebra B the vectors $1 \otimes e_i$ are a B-basis of $B \otimes_F V$, by the free-tensor formula. Tensoring any injection of F-vector spaces with B is injective: choose an F-basis of B and tensoring is a direct sum of copies of that injection. If V, W have respective bases $(v_i), (w_j)$, the same two direct-sum formulas identify $V \otimes_F W$ with $F^{(I \times J)}$, carrying $v_i \otimes w_j$ to its coordinate vector. This gives the finite dimension product and preservation of a finite basis under field extension; it also proves that the tensor of two nonzero vector spaces, in particular two field extensions, is nonzero. The argument includes zero modules and empty bases. □

For a multiplicative subset S of a ring A, define $S^{-1}M$ by pairs (m, s) with $(m, s) = (n, t)$ when $u(tm - sn) = 0$ for some $u \in S$. Addition and scalar multiplication have the usual common-denominator formulas. The map $S^{-1}A \otimes_A M \to S^{-1}M$, $(a/s) \otimes m \mapsto am/s$, has inverse $m/s \mapsto (1/s) \otimes m$; the defining equivalence makes both maps well defined.

Lemma 1.1. Localization is exact. Filtered colimits of modules are exact. A module is zero if all its localizations at prime ideals are zero.

Proof. A fraction m/s maps to zero precisely when some $u \in S$ kills the image of m. Replace m/s by $um/(us)$ to lift it from the kernel. An element of the localization of an injected module maps to zero only when the same multiplier kills it in that module. Surjectivity is obtained by lifting the numerator. This proves exactness. In a filtered colimit every element is represented at one stage, and an equality between finitely many representatives holds at a common later stage. Applying these two facts to kernels and surjections proves exactness of filtered colimits. If $m \neq 0$, its annihilator is a proper ideal. It lies in a maximal ideal $\mathfrak{m}$, by Zorn's lemma applied to proper ideals; then $m/1 \neq 0$ in $M_{\mathfrak{m}}$, since an element outside $\mathfrak{m}$ cannot annihilate m. This proves the last assertion. □

A module is **flat** when tensoring with it preserves injections, and consequently short exact sequences by the right-exactness calculation of Lemma 1.0. Exactness of localization proves that $S^{-1}A$ is flat. The kernel, image and quotient calculations in Lemma 1.0 show that flat tensor commutes with the cohomology of every complex and with the finite products used below.

Lemma 1.2 (Nakayama and local finite projectivity). If M is finitely generated and $M = JM$, then some $a \in J$ satisfies $(1 - a)M = 0$. If J is contained in the Jacobson radical, $M = 0$. A finitely presented flat module over a local ring is finite free; over any ring it is locally finite free.

Proof. If $M = 0$, choose $a = 0$; all assertions about this module are immediate. Otherwise choose a finite generating list $m_1, \ldots, m_n$, with $n > 0$. From $M = JM$ write $m_i = \sum_j a_{ij} m_j$, with $a_{ij} \in J$, and put $B = I - (a_{ij})$. The matrix equation is $Bm = 0$. Here is the cofactor calculation over an arbitrary commutative ring. Define the determinant by the finite signed permutation sum, and define the adjugate as the transpose of the matrix of signed minors. Grouping that sum by the chosen entry in a row or column gives its cofactor expansion. If two rows or two columns are equal, pair each permutation with the permutation interchanging those positions: the two products are equal and their signs are opposite, so their sum is zero in every characteristic. In the product of B with its adjugate, a diagonal entry is the cofactor expansion of $\det B$; an off-diagonal entry is the expansion of a determinant with two equal rows or columns, hence is zero. The same argument in the other order gives $\operatorname{adj}(B)B = B \operatorname{adj}(B) = (\det B)I$. For a matrix of size one its minors have size

zero and determinant one. Every term in the expansion of $\det(I-(a_{ij}))$ other than the constant term 1 contains at least one a_{ij}, and therefore belongs to the ideal J. Thus $\det B = 1-a$ for some $a \in J$. Multiplying $Bm = 0$ by $\operatorname{adj}(B)$ gives $(1-a)m_i = 0$ for every generator, hence $(1-a)M = 0$. If J lies in every maximal ideal, $1-a$ is a unit: a nonunit generates a proper ideal contained in a maximal ideal, and that maximal ideal would then contain both a and $1-a$, hence one. The existence of that maximal ideal is the Zorn argument of Lemma 1.1. Multiplication by the inverse of $1-a$ now gives $M = 0$. In particular this applies over every local ring with J contained in its maximal ideal, without a Noetherian hypothesis.

First record the finite-kernel calculation for an arbitrary finite free surjection onto a finitely presented module. Over any commutative ring A, let Q be finitely presented. Choose a finite presentation $A^a \to A^b \xrightarrow{\pi} Q \to 0$; its kernel $L \subset A^b$ is generated by the images of the finitely many basis vectors of A^a. Given any finite free surjection $\rho : A^r \to Q$ with kernel K, form $H = \{(u,v) \in A^r \oplus A^b : \rho(u) = \pi(v)\}$. Projection onto A^r gives $0 \to L \to H \to A^r \to 0$, and projection onto A^b gives $0 \to K \to H \to A^b \to 0$. Each projection is surjective because the other presentation map is surjective, and each splits by the free-basis lifting calculation in Lemma 1.0. Hence $H \simeq L \oplus A^r$ is finitely generated, while K is a direct summand of H and is generated by the projections of any finite generating list of H. Thus K is finite, including when one of the free ranks is zero; no Noetherian assumption is used. Over a local ring $(A, \mathfrak{m})$ lift a basis of $M/\mathfrak{m}M$ to obtain a surjection $A^r \to M$ by Nakayama. Its kernel K is finite, by finite presentation. Flatness of M makes $0 \to K \otimes A/\mathfrak{m} \to (A/\mathfrak{m})^r \to M/\mathfrak{m}M \to 0$ exact: tensor the presentation and use preservation of kernels in a free presentation, equivalently resolve the last module by free modules and use exact flat tensor. The last map is an isomorphism, so $K/\mathfrak{m}K = 0$. Nakayama gives $K = 0$. For a prime $\mathfrak{p}$ of a general ring, finite presentation localizes by applying localization to its finite generators and relations. Flatness also localizes: the tensor identifications of Lemma 1.0 identify tensoring an $A_\mathfrak{p}$-module with $M_\mathfrak{p}$ with tensoring it over A with M, so it preserves injections. The local argument therefore makes $M_\mathfrak{p}$ finite free. Choose its basis. Each basis vector is m_i/s_i with $m_i \in M$ and $s_i \notin \mathfrak{p}$. Multiplying the basis vectors by these units still gives a basis. Sending the basis of A^r to the m_i gives $f : A^r \to M$ whose localization at $\mathfrak{p}$ is an isomorphism. Its cokernel is finitely generated, and each of its finitely many generators is zero after localization at $\mathfrak{p}$. By the fraction-zero criterion of Lemma 1.1, a product $g \notin \mathfrak{p}$ annihilates all these generators. Thus f_g is surjective. Its kernel K_g is finite by the finite-kernel calculation just proved, because M_g is finitely presented. Exactness of localization gives $(K_g)_\mathfrak{p} = 0$. For each of its finitely many generators, the same fraction-zero criterion supplies an element of A_g outside $\mathfrak{p}A_g$ annihilating it; its numerator lies outside $\mathfrak{p}$ and also annihilates it. The product $h \notin \mathfrak{p}$ of these numerators kills all of K_g. Hence $f_{gh} : A_{gh}^r \to M_{gh}$ is an isomorphism, proving freeness on a principal neighbourhood. For an empty generating list use product 1, and the argument includes rank zero. This proves local finite freeness. □

Here the elementary tensor-kernel assertion can be established without derived notation: if $0 \to K \to F \to M \to 0$ with F free and M flat, and N is presented as $F_1 \to F_0 \to N \to 0$ with the F_i free, lift a proposed kernel element to $K \otimes F_0$. Its image in $F \otimes F_0$ comes from $F \otimes F_1$. Its image in $M \otimes F_1$ maps to zero in $M \otimes F_0$; flatness identifies this kernel with the image of $M \otimes \ker(F_1 \to F_0)$. Lift those elements to F, subtract them, and then lift the remaining element from $K \otimes F_1$. The proposed element was zero in $K \otimes N$.

Lemma 1.3 (the Noetherian inputs). A finitely generated algebra over a Noetherian ring is Noetherian. Submodules of finite modules over a Noetherian ring are finite.

Proof. For an ideal $I \subset A[x]$, the leading coefficients of elements of degree at most d form an ascending sequence of ideals L_d of A. This sequence stabilizes, and every L_d before stabilization is finitely generated. Choose polynomials of the corresponding degrees whose leading coefficients generate these finitely many ideals. Given any polynomial in I, subtract suitable multiples of these chosen polynomials to cancel its highest term; for degrees above stabilization multiply the stabilized generators by powers of x. Induction on degree expresses it in the ideal of the finitely

many chosen polynomials. Thus $A[x]$ is Noetherian. Iterate and take a quotient. For a submodule of A^r, projection to the last coordinate has image an ideal, and its kernel is a submodule of A^{r-1}. Induction and lifting ideal generators make the submodule finite. A finite module is a quotient of some A^r; apply the same statement to the inverse image of its submodule. □

Lemma 1.A (Tor for arbitrary modules). Let R be any commutative ring and M, N any modules. Choose a free resolution $P_\bullet \to M$ by taking a free surjection onto the module and then onto each successive kernel, and define $\operatorname{Tor}_i^R(M,N) = H_i(P_\bullet \otimes_R N)$. This definition is independent of the resolution, naturally symmetric in M, N, and has the natural long exact sequence of a short exact sequence in either variable.

Proof.

For completeness, the resolution facts used here have the following algebraic proof. Given free resolutions $P_\bullet \to M$ and $P'_\bullet \to M'$ and a map $M \to M'$, lift it to $P_0 \to P'_0$. If maps have been chosen through degree $i-1$, the image of $d(P_i)$ lies in the cycles of P'_{i-1}, which are the image of P'_i; freeness of P_i therefore supplies the next lift. Two such lifts are chain homotopic: after the homotopy has been chosen below degree i, their degree-i difference minus the already prescribed homotopy term maps to zero under the next differential, and freeness lifts it to P'_{i+1}. The same induction starts at degree zero because both lifts induce the same map on the augmentation. Tensoring preserves the homotopy identity. Thus induced homology maps are independent of the lifts, preserve compositions, and comparison maps between two resolutions of the same module are inverse on homology.

Let $Q_\bullet \to N$ be another free resolution and form the first-quadrant double complex $D_{p,q} = P_p \otimes_R Q_q$, with total differential $d_P \otimes 1 + (-1)^p 1 \otimes d_Q$. Augmentation gives maps from its total complex to $P_\bullet \otimes_R N$ and to $M \otimes_R Q_\bullet$. Each is a homology isomorphism. Indeed, filter by the degree in the unaugmented direction. In the other direction each augmented row or column is exact because its fixed factor is free. To check the resulting homology assertion directly, start with the outermost nonzero component of a cycle in the augmented kernel and use this exactness to subtract a total boundary that removes that component; proceed to the next component. A chain in any fixed total degree has only finitely many components, so this process terminates. The same process applied to boundaries proves injectivity. The comparison maps just constructed make these two augmentation isomorphisms natural. Interchanging the two resolutions, with the sign $(-1)^{pq}$ on $P_p \otimes_R Q_q$, consequently gives

$$\operatorname{Tor}_i^R(M,N) \simeq \operatorname{Tor}_i^R(N,M).$$

In particular Tor can be computed using a free resolution of either variable; a projective resolution works as well, since projectives are summands of free modules and the same lifting and exactness arguments apply.

For $0 \to N' \to N \to N'' \to 0$, tensoring in each degree with the free module P_i gives a short exact sequence of complexes. If a cycle in the quotient complex is lifted to the middle complex, its differential lies in the subcomplex and is a cycle there. Its homology class is independent of the lift and of the cycle representative: changing either changes that differential by a boundary. This defines the connecting map. The equations saying that a class maps to zero say precisely that one can subtract a boundary and lift it to the preceding complex; they prove exactness at each position. The construction commutes with maps of short exact sequences and gives

$$\cdots \longrightarrow \operatorname{Tor}_i^R(M,N') \longrightarrow \operatorname{Tor}_i^R(M,N) \longrightarrow \operatorname{Tor}_i^R(M,N'') \longrightarrow \operatorname{Tor}_{i-1}^R(M,N') \longrightarrow \cdots,$$

ending in

$$\operatorname{Tor}_1^R(M,N'') \longrightarrow M \otimes_R N' \longrightarrow M \otimes_R N \longrightarrow M \otimes_R N'' \longrightarrow 0.$$

Balance gives the corresponding sequence in the first variable. These arguments require neither Noetherianity nor finite generation. □

2. Modules on an affine scheme

Lemma 2.1 (the principal-cover calculation). If $f_1, \dots, f_r$ generate the unit ideal in A, the augmented alternating complex

$$0 \longrightarrow M \longrightarrow \prod_i M_{f_i} \longrightarrow \prod_{i<j} M_{f_i f_j} \longrightarrow \cdots$$

is exact for every A-module M.

Proof. Localize at f_a. Regard an ordered cochain as alternating, with repeated indices zero. Inserting a as its first index defines a homotopy h lowering degree, including projection to M_{f_a} in degree zero. Every extra denominator f_a is already a unit. Expanding the alternating differential gives $dh + hd = 1$. The localized augmented complex is contractible. Localization commutes with its finite products and its cohomology by Lemma 1.1. If an element of a cohomology module vanishes at every f_a, powers $f_a^{n_a}$ all kill it. These powers generate one: an ideal they generate, if proper, lies in a maximal ideal containing every f_a, contradicting the unit-ideal hypothesis. A linear combination equal to one kills the element. Thus the original cohomology is zero. □

The opens $D(f)$ form a basis of $\operatorname{Spec} A$. A collection of them covers precisely when its elements generate one: a proper generated ideal lies in a maximal ideal, and otherwise a finite linear combination already equals one. This also proves quasi-compactness and existence of finite principal refinements of arbitrary open covers.

Define $\widetilde{M}$ on this basis by $\widetilde{M}(D(f)) = M_f$. The first three terms of Lemma 2.1 give the sheaf gluing axiom on every finite principal cover. An arbitrary cover has a finite principal refinement, so these basis values extend to a sheaf, uniquely. Its stalk at $\mathfrak{p}$ is $M_{\mathfrak{p}}$. Exactness of localization therefore makes $M \mapsto \widetilde{M}$ exact.

Proposition 2.2 (affine module-sheaf correspondence). Every quasi-coherent module on $\operatorname{Spec} A$ is $\widetilde{M}$ for $M = \Gamma(\operatorname{Spec} A, \mathcal{F})$. Maps between associated sheaves correspond bijectively to module maps. In particular taking sections on the affine is exact on quasi-coherent modules.

Proof. Choose a finite principal cover $D(f_i)$ on which $\mathcal{F}$ is associated to an A_{f_i}-module M_i. Restrictions on the overlaps identify $(M_i)_{f_j}$ with $(M_j)_{f_i}$ and satisfy the cocycle equations. Put

$$M = \ker\left(\prod_i M_i \longrightarrow \prod_{i<j} (M_i)_{f_j}\right),$$

where the arrow is the difference after these identifications. The sheaf axiom identifies M with the global sections. Localize this finite kernel diagram at f_a. Its cover now has a member equal to the whole spectrum. Compatibility then says that every tuple is uniquely determined by its a-th component, and every such component supplies the other components by restriction and the overlap identifications. Thus $M_{f_a} \simeq M_a$. The associated sheaf is therefore isomorphic to $\mathcal{F}$ on this cover, compatibly on overlaps, and hence globally. A sheaf map gives a map of global sections; on each $D(f)$ its action is the localization of that map, since a fraction m/f^n must map to its image divided by f^n. Conversely localizing a module map defines a sheaf map. These operations are inverse. Exactness of associated sheaves and the equivalence prove exactness on global sections. □

Kernels, images and cokernels of maps between quasi-coherent modules are quasi-coherent: on an affine they are the corresponding module operations by Proposition 2.2, and exactness can be checked on stalks. Closure under extensions is proved after affine vanishing in Section 5. Localization also identifies pullback along $A \to B$ with $\widetilde{M \otimes_A B}$.

Proposition 2.3 (maps into an affine). For any locally ringed space X, maps $X \to \operatorname{Spec} A$ correspond bijectively to ring maps $A \to \Gamma(X, \mathcal{O}_X)$.

Proof. From a ring map ϕ send x to the inverse image of $\mathfrak{m}_x$ under $A \to \mathcal{O}_{X,x}$. This is prime. The inverse image of $D(a)$ is the open set where $\phi(a)$ is a unit in the local ring; a local inverse exists on a neighbourhood, so this set is open. On it the universal property of localization defines $A_a \to \Gamma(X_{\phi(a)}, \mathcal{O}_X)$. The maps agree on intersections and define the sheaf map. The stalk maps are local by the definition of the image prime. Conversely locality forces that image prime, and localization forces each basis sheaf map; hence the construction is unique. It recovers ϕ on global sections and is inverse to taking global sections. □

3. Injectives, resolutions and extensible sections

An object is **injective** when a map to it extends across every injection. We supply the existence assertions used to define cohomology.

Lemma 3.1. Module categories and sheaf-module categories have enough injectives. An injective sheaf restricts to an injective on every open and is flasque, meaning that every restriction of sections is surjective.

Proof. First give the integer relation calculation. For $z \in \mathbf{Z}$ and $d > 0$, there are integers q, r with $z = qd + r$ and $0 \le r < d$. For $z \ge 0$, choose the largest $q \in \{0, \dots, z\}$ satisfying $qd \le z$. If $z - qd \ge d$, then $q + 1$ also lies in that set and satisfies the inequality, a contradiction. For $z < 0$, apply this calculation to $-z = q_0 d + r_0$; use $q = -q_0, r = 0$ when $r_0 = 0$, and $q = -q_0 - 1, r = d - r_0$ otherwise. If $J \subset \mathbf{Z}$ is a nonzero subgroup, choose its least positive member d. For $z \in J$, the remainder $z - qd$ belongs to J and is smaller than d, so it is zero. Thus $J = d\mathbf{Z}$; the zero subgroup is $0\mathbf{Z}$. In particular every ideal of $\mathbf{Z}$ is principal.

Write $D = \mathbf{Q}/\mathbf{Z}$ additively. It is divisible: for a class $t + \mathbf{Z}$ and $d > 0$, the class $t/d + \mathbf{Z}$ has d-fold multiple $t + \mathbf{Z}$. Let $H \subset G$ be abelian groups and let $\chi : H \to D$ be a homomorphism. To extend over $a \in G$, set $J = \{k \in \mathbf{Z} : ka \in H\}$. This is a subgroup of $\mathbf{Z}$. If $J = 0$, choose $y = 0$. If $J = d\mathbf{Z}$ with $d > 0$, choose $y \in D$ satisfying $dy = \chi(da)$, using divisibility. Define

$$\chi'(h + ka) = \chi(h) + ky \qquad (h \in H,\ k \in \mathbf{Z}).$$

If $h + ka = h' + k'a$, then $(k - k')a = h' - h$. When $J = 0$, this forces $k = k'$ and $h = h'$. Otherwise $k - k' = qd$ and $h' - h = qda$, so $\chi(h' - h) = q\chi(da) = qdy$. The two displayed values therefore agree. The formula is additive and restricts to χ, so it defines a homomorphism on $H + \mathbf{Z}a$.

Order all extensions of χ to subgroups between H and G by inclusion and restriction. The set is nonempty, and a chain has an upper bound given by the union of its subgroups and the compatible union of its maps. Zorn's lemma supplies a maximal extension. If its domain omits an element a, the preceding calculation extends it to the strictly larger subgroup obtained by adjoining a, a contradiction. Its domain is therefore G. This proves that D is injective as an abelian group, directly from the extension definition.

For $0 \ne x \in G$, apply the integer subgroup calculation to $\{k \in \mathbf{Z} : kx = 0\}$. If this is zero, define a character on $\mathbf{Z}x$ by $kx \mapsto k/2 + \mathbf{Z}$. If it is $d\mathbf{Z}$, then $d \ge 2$ and $kx \mapsto k/d + \mathbf{Z}$ is well defined: changing k by a multiple of d changes k/d by an integer. In either case the image of x is nonzero. Injectivity extends this character to G. Thus homomorphisms to D separate all nonzero elements.

For any ring R, the module $E = \mathrm{Hom}_{\mathbf{Z}}(R, \mathbf{Q}/\mathbf{Z})$ is injective, with $(r\lambda)(s) = \lambda(rs)$: evaluation at one identifies $\mathrm{Hom}_R(N, E)$ with $\mathrm{Hom}_{\mathbf{Z}}(N, \mathbf{Q}/\mathbf{Z})$. Map N into the product of copies of E indexed by its additive characters, sending n in the character-λ component to $r \mapsto \lambda(rn)$. Separation makes the map injective. Products of injectives are injective because extension can be made component by component.

For a ringed space choose, for each point x, an injection $\mathcal{F}_x \hookrightarrow E_x$ into an injective $\mathcal{O}_{X,x}$-module. The sheaf $i_{x*}E_x$ has sections E_x on an open containing x, zero otherwise, and the induced ring action. The adjunction

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, i_{x*}E_x) = \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{G}_x, E_x)$$

and exactness of stalks make it injective. The natural map $\mathcal{F} \to \prod_x i_{x*}E_x$ is injective: projecting its stalk at x to the x-component recovers the chosen injection. Thus sheaf modules have enough injectives.

For an open immersion $j : U \hookrightarrow X$, extension by zero $j_!$ is an exact left adjoint to restriction; its stalks are unchanged on U and zero outside. Adjunction makes restriction preserve injectives. To extend a section of an injective $\mathcal{I}$ from U to V, use the injection $j_!\mathcal{O}_U \to \mathcal{O}_V$. Homomorphisms from these two modules to $\mathcal{I}|_V$ are exactly sections on U and V. Injectivity gives the desired extension. □

Repeatedly embed a sheaf and its successive cokernels into injectives. This gives an exact resolution $0 \to \mathcal{F} \to \mathcal{I}^0 \to \mathcal{I}^1 \to \cdots$. Define $H^q(U, \mathcal{F})$ as the cohomology of its section complex. A map between sheaves extends to a chain map between resolutions: after degrees below n have been treated, exactness makes the required degree-n map well defined on the image of the previous differential, and target injectivity extends it. Two extensions are homotopic by the same induction, applied to their difference. This proves independence of resolutions and naturality. For a short exact sequence, choose injectives for the two end terms and extend into their direct sum; iterating on cokernels produces a degreewise split short exact sequence of resolutions. Taking sections and lifting cocycles then gives the long exact cohomology sequence, with its usual connecting map. These constructions prove the properties of derived cohomology needed here.

Lemma 3.2 (flasque lifting and acyclicity). In $0 \to \mathcal{A} \to \mathcal{B} \to \mathcal{C} \to 0$, if $\mathcal{A}$ is flasque, every section of $\mathcal{C}$ lifts to $\mathcal{B}$ on every open. If $\mathcal{B}$ is also flasque, so is $\mathcal{C}$. A flasque sheaf has zero positive cohomology on every open.

Proof. Choose local lifts of a given section and well order their covering opens. At a successor step the new lift and the already glued lift differ on their overlap by a section of $\mathcal{A}$. Extend that difference to the new open using flasqueness, correct its lift, and glue. At a limit step the compatible previous lifts glue by the sheaf axiom. This constructs the lift on the whole open. To extend a section of $\mathcal{C}$ from a smaller open, lift it to $\mathcal{B}$, extend there by flasqueness, and project. Finally embed a flasque sheaf into an injective. Its cokernel is flasque by the assertion just proved. Iterate. Every successive short exact sequence is exact on sections by the lifting assertion, so the section complex of this injective resolution is exact in positive degrees. □

More generally a resolution by section-acyclic sheaves computes cohomology: writing its successive kernels, their long exact sequences identify the cohomology of its section complex degree by degree with the groups defined by an injective resolution. This dimension-shifting argument uses only finitely many kernels in each degree. Direct image sends flasque sheaves to flasque sheaves because its restriction maps are restrictions between inverse-image opens. Its higher direct images are the sheafification of $V \mapsto H^q(f^{-1}V, \mathcal{F})$: both are obtained from the same section complexes, and passing to a stalk commutes with kernels and cokernels by exactness of filtered colimits. Consequently flasque sheaves are also direct-image-acyclic. Applying the acyclic-resolution comparison twice proves

$$R\Gamma(Y, Rf_*\mathcal{F}) = R\Gamma(X, \mathcal{F})$$

for a sheaf in degree zero, with the actual scalar action induced by the ring map. The notation here denotes the resolution complexes up to quasi-isomorphism; the equality follows from their common section complex, not from a separate composition theorem.

4. Finite covers and the spectral-sequence calculation

For a finite open cover $U_1, \dots, U_r$, set $U_I = \bigcap_{i \in I} U_i$ for increasing index tuples I, and let $j_I : U_I \to X$. The augmented sheaf complex of finite products $\prod_{|I|=p+1} (j_I)_*(\mathcal{G}|_{U_I})$ is exact. At x choose a with $x \in U_a$ and restrict to a neighbourhood contained in U_a. Inserting a as the first index gives a homotopy, exactly as in Lemma 2.1; it works on that neighbourhood even for a boundary stalk of a direct image. This proves exactness on stalks.

Apply this complex to every term of an injective resolution. Restriction preserves injectives by Lemma 3.1; direct image under an open immersion preserves them because it is right adjoint to exact restriction. The augmented rows are finite exact rows of injectives. Their first injection splits; its cokernel is an injective direct summand, so induction splits the row. Thus taking sections leaves the rows exact. The section double complex

$$C^{p,q} = \prod_{|I|=p+1} \Gamma(U_I, \mathcal{I}^q), \qquad 0 \le p < r, \quad q \ge 0,$$

with differential $d_C + (-1)^p d_I$, has total complex quasi-isomorphic to $\Gamma(X, \mathcal{I}^\bullet)$ by the horizontal-row calculation.

For precision we give the filtered-complex calculation used in both directions. Write $K = \operatorname{Tot} C$ and $F^p K^n = \bigoplus_{a \ge p} C^{a,n-a}$. For $s \ge 1$ put

$$Z_s^{p,q} = \{ z \in F^p K^{p+q} : dz \in F^{p+s} K^{p+q+1} \}, \qquad E_s^{p,q} = Z_s^{p,q} / (Z_{s-1}^{p+1,q-1} + d Z_{s-1}^{p-s+1,q+s-2}),$$

with $Z_0^{p,q} = F^p K^{p+q}$. Both terms in the denominator lie in the numerator: their differentials lie in F^{p+s} or are zero. The differential of a representative gives a well-defined map $d_s : E_s^{p,q} \to E_s^{p+s,q-s+1}$. If its class is zero, write $dz = a + db$ with $a \in Z_{s-1}^{p+s+1,q-s}$ and $b \in Z_{s-1}^{p+1,q-1}$. Replacing z by $z - b$ raises the filtration of its differential by one. Conversely every such raised cycle lies in the kernel. Quotienting also by the image of d_s adds the boundaries with preimages in filtration $p - s$. These are precisely the numerator and denominator for E_{s+1}. Thus $E_{s+1} = H(E_s, d_s)$. At the first page this gives vertical cohomology, followed by the horizontal restriction differential.

The filtration has only r pieces. For large s its numerator consists of genuine cycles and its denominator consists of the next filtration of cycles plus all boundaries lying in F^p. Hence $E_\infty^{p,q}$ is the corresponding successive quotient of the filtration on $H^{p+q}(K)$. No infinite convergence assertion is being assumed. We have proved

$$E_1^{p,q} = \prod_{|I|=p+1} H^q(U_I, \mathcal{F}) \quad \Longrightarrow \quad H^{p+q}(X, \mathcal{F}).$$

4.1. The general Leray comparison

Theorem 4.1 (Leray). For a morphism of ringed spaces $f : X \to Y$ and a sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, there is a first-quadrant spectral sequence

$$H^p(Y, R^q f_* \mathcal{F}) \Longrightarrow H^{p+q}(X, \mathcal{F}).$$

The $\mathcal{O}_Y$-action is induced by the given morphism of structure sheaves.

Proof. We give the double resolution used in this assertion. First, if $0 \to A \to B \to C \to 0$ is a short exact sequence of module sheaves and injective resolutions of A and C have been chosen, it has a compatible injective resolution of B whose degree-p term is the direct sum of their degree-p terms. At degree zero extend the map $A \to I_A^0$ to $B \to I_A^0$ by injectivity and combine it with

$B \to C \to I_C^0$. This map $B \to I_A^0 \oplus I_C^0$ is injective: a vector killed by its second component belongs to A, where its first component is the chosen injection. Its cokernels form a short exact sequence $0 \to I_A^0/A \to (I_A^0 \oplus I_C^0)/B \to I_C^0/C \to 0$. For the middle kernel calculation, subtract a lift in B of the element of C represented by the second coordinate; the remaining representative has second coordinate zero, and its ambiguity is exactly A. Repeat the construction on these cokernels. Composing each quotient with its next injection defines the differentials, and proves exactness and compatibility in every degree. The resulting short exact sequence of resolutions is split in each degree by its direct-sum description.

Now let $K^\bullet$ be any complex of sheaves in nonnegative degrees. Put $B^q = \operatorname{im} d^{q-1}$, $Z^q = \ker d^q$ and $H^q = Z^q/B^q$. Choose injective resolutions of the B^q and H^q, taking $B^0 = 0$. Apply the preceding construction to $0 \to B^q \to Z^q \to H^q \to 0$ and then to $0 \to Z^q \to K^q \to B^{q+1} \to 0$. This gives resolutions $J^{\bullet,q}$ of K^q, with degreewise split exact sequences for both rows. Define their vertical differential by the composite

$$J^{p,q} \longrightarrow I_B^{p,q+1} \longrightarrow I_Z^{p,q+1} \longrightarrow J^{p,q+1}.$$

Here the first arrow is the second resolution's quotient map and the other arrows are its compatible inclusions. Its square is zero because the quotient in the next degree kills the included Z^{q+1}. These are maps of resolutions, so the vertical and horizontal differentials commute. At fixed p, the vertical boundaries, cycles and cohomology are respectively $I_B^{p,q}$, $I_Z^{p,q}$ and $I_H^{p,q}$. All these sheaves are injective, and their short exact sequences split. Taking sections therefore preserves this description. Every term $J^{p,q}$ is injective, and the horizontal complexes resolve K^q.

Take an injective resolution $\mathcal{I}^\bullet$ of $\mathcal{F}$ on X and apply this construction to $K^\bullet = f_*\mathcal{I}^\bullet$. Each K^q is flasque, by Section 3, hence acyclic for sections on Y. The horizontal cohomology of $\Gamma(Y, J^{\bullet,q})$ is consequently $\Gamma(Y, K^q)$ in degree zero and zero in positive degrees. The corresponding filtration of the section double complex identifies its total cohomology with that of $\Gamma(Y, K^\bullet) = \Gamma(X, \mathcal{I}^\bullet)$, namely $H^*(X, \mathcal{F})$.

For the other filtration, the split boundary and cycle sequences just constructed identify vertical cohomology with $\Gamma(Y, I_H^{p,q})$. Its horizontal cohomology is $H^p(Y, H^q(K^\bullet))$, since $I_H^{\bullet,q}$ resolves that sheaf. By the definition of higher direct image, $H^q(K^\bullet) = R^q f_*\mathcal{F}$. Use the same alternating-sign total differential and filtered-complex calculation given above. In each total degree only finitely many pairs $p, q \geq 0$ occur, so the filtration is finite and converges to the stated total cohomology. This proves the claimed spectral sequence, including its target, without presuming a composition theorem. □

5. Affine vanishing without a cited replacement proof

Theorem 5.1. For every ring A, every A-module M and every $q > 0$, $H^q(\operatorname{Spec} A, \widetilde{M}) = 0$.

Proof. Induct on q, simultaneously for all rings and modules. A positive-degree cohomology class is locally zero: represent it by a cocycle in an injective resolution; exactness of the resolution as sheaves makes this cocycle a boundary on a neighbourhood of each point. Choose a finite principal cover on which the class is zero. Apply the spectral sequence of Section 4. Its intersections are principal affines with the localized module. For $q = 1$ the kernel of restriction to the $E_1^{0,1}$ terms is the first filtration piece; the only possible term there is the positive-degree cohomology of the section complex in row zero, which is zero by Lemma 2.1. For general q, all rows $1, \ldots, q-1$ vanish by induction on those affine intersections. The row-zero positive cohomology is again zero by Lemma 2.1. A class restricting to zero on the covering members lies in filtration at least one, whose total-degree-q successive quotients are therefore all zero. It is zero. This completes the induction. □

Corollary 5.2 (extensions remain quasi-coherent). In an exact sequence of module sheaves $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$, quasi-coherence of the two end terms implies quasi-coherence of the

middle term.

Proof. Work on $X = \operatorname{Spec} A$. Theorem 5.1 and the long exact sequence give an exact sequence $0 \to M_1 \to M_2 \to M_3 \to 0$ of their section modules. The same holds on every principal $D(f)$. Localize the first sequence and compare it with the latter. Proposition 2.2 identifies the two end modules with their localizations, so an elementary kernel-and-lift argument identifies $M_{2,f}$ with $\Gamma(D(f), \mathcal{F}_2)$: lift an element through the right term and correct its difference in the left term; the same diagram detects a zero element. These identifications commute with restriction, and therefore give $\widetilde{M_2} \simeq \mathcal{F}_2$. This is local on X and proves the assertion. □

For a separated scheme over an affine base, finite intersections of affine opens are affine: the closed diagonal pulls their affine product back to that intersection. Thus the row-zero section complex computes quasi-coherent cohomology by Section 4 and Theorem 5.1. For a quasi-compact quasi-separated scheme, intersections in a finite affine cover are quasi-compact separated opens of one affine member. Keep their cohomology in the spectral sequence instead of treating them as affine.

6. Flat base change and localization of quasi-coherent cohomology

Statement. Let X be a quasi-compact quasi-separated scheme over $\operatorname{Spec} A$, let $\mathcal{F}$ be quasi-coherent, and let B be any flat A-algebra. Write $g : X_B \to X$ for the projection and $\mathcal{F}_B = g^*\mathcal{F}$. For every $q \geq 0$, the canonical map is an isomorphism

$$H^q(X, \mathcal{F}) \otimes_A B \longrightarrow H^q(X_B, \mathcal{F}_B).$$

Theorem 5.1 above proves affine vanishing: for every ring R and every R-module M, $H^q(\operatorname{Spec} R, \widetilde{M}) = 0$ for $q > 0$. Proposition 2.2 above proves the affine module-sheaf correspondence, and Sections 3-4 construct the injective and filtered-cover comparisons used in the proof. No finite generation, Noetherianity or finite presentation is assumed for A, B or $\mathcal{F}$.

The separated case and the ordered cover construction.

First suppose that X is separated over the affine base. Choose a finite affine open cover $U_1, \dots, U_r$. For a strictly increasing tuple $I = (i_0 < \cdots < i_p)$ put $U_I = U_{i_0} \cap \cdots \cap U_{i_p}$, and let $j_I : U_I \hookrightarrow X$. Empty intersections contribute zero. Every U_I is affine: the intersection of two affine opens is the inverse image of their affine product under the closed diagonal, and induction gives the assertion for finite intersections.

We construct the cohomological comparison rather than assume that a section complex computes it. For any sheaf of modules $\mathcal{G}$, form the augmented alternating sheaf complex

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{C}^0(\mathcal{G}) \longrightarrow \cdots \longrightarrow \mathcal{C}^{r-1}(\mathcal{G}) \longrightarrow 0, \qquad \mathcal{C}^p(\mathcal{G}) = \prod_{i_0 < \cdots < i_p} (j_I)_*(\mathcal{G}|_{U_I}).$$

The differential is the alternating sum of restriction maps. This complex is exact on stalks. Indeed, near a point x choose a with $x \in U_a$ and restrict to a neighbourhood $V \subset U_a$. Regard an ordered cochain as alternating in all its indices, with repeated-index components zero. The map lowering the degree is

$$(hc)_{i_0 \dots i_{p-1}} = c_{a i_0 \dots i_{p-1}}.$$

It is well defined because $V \cap U_I \cap U_a = V \cap U_I$. In degree zero it takes c_a to a section on V, the augmented term. Expanding the alternating differential cancels all terms except c, so $dh + hd = 1$, including the augmented degree. This proves exactness even when a direct-image stalk at a boundary point is nonzero.

Take an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$ in sheaves of $\mathcal{O}_X$-modules. Restriction to an open preserves injectives: its left adjoint, extension by zero, is exact, as one checks on stalks. Direct image under an open immersion also preserves injectives because it is right adjoint to exact restriction. Thus each term of the augmented sheaf complex for $\mathcal{I}^q$ is injective. Its first injection splits, and successive cokernels are injective direct summands; inductively the whole finite exact row splits. Applying global sections therefore leaves it exact.

Consequently the augmentation from $\Gamma(X, \mathcal{I}^\bullet)$ to the total complex of

$$C^{p,q} = \prod_{i_0<\cdots<i_p} \Gamma(U_I, \mathcal{I}^q|_{U_I}), \qquad 0 \le p < r, \quad q \ge 0,$$

is a quasi-isomorphism. Here the total differential is $d_{\text{cover}} + (-1)^p d_{\text{resolution}}$. Exactness of the augmented horizontal rows proves the assertion by the row filtration. Each total degree contains finitely many terms.

The column filtration $F^p \operatorname{Tot}^n = \bigoplus_{a \ge p} C^{a,n-a}$ gives the ordered-cover spectral sequence

$$E_1^{p,q} = \prod_{i_0<\cdots<i_p} H^q(U_I, \mathcal{F}|_{U_I}) \quad \Longrightarrow \quad H^{p+q}(X, \mathcal{F}),$$

with d_1 the alternating restriction differential. Explicitly, for $s \ge 1$ set

$$Z_s^{p,q} = \{z \in F^p \operatorname{Tot}^{p+q} : dz \in F^{p+s} \operatorname{Tot}^{p+q+1}\}, \qquad E_s^{p,q} = \frac{Z_s^{p,q}}{Z_{s-1}^{p+1,q-1} + dZ_{s-1}^{p-s+1,q+s-2}},$$

where $Z_0^{p,q} = F^p \operatorname{Tot}^{p+q}$. The total differential induces $d_s : E_s^{p,q} \to E_s^{p+s,q-s+1}$, and taking its cohomology gives the next page as follows. If $d_s[z] = 0$, write $dz = u + dv$ with $u \in Z_{s-1}^{p+s+1,q-s}$ and $v \in Z_{s-1}^{p+1,q-1}$. Replacing z by $z - v$ puts its differential in F^{p+s+1}, giving a representative in $Z_{s+1}^{p,q}$. Conversely every such representative is a d_s-cycle. Quotient these representatives by the old denominator and by the d_s-boundaries. The resulting denominator is exactly $Z_s^{p+1,q-1} + dZ_s^{p-s,q+s-1}$, the denominator in the formula for $E_{s+1}^{p,q}$. Thus the induced cohomology is that next page. Extend the filtration by $F^p = \operatorname{Tot}$ for $p \le 0$ and $F^p = 0$ for $p \ge r$. For sufficiently large s, the cycle condition is $dz = 0$ and the denominator accounts for the next filtration piece and all boundaries lying in F^p. Thus the stable terms are the associated graded pieces of cohomology. There are only r columns, so each abutment has a finite filtration and convergence involves no infinite-product or completion issue. This construction works for any finite open cover, including covers with non-affine intersections.

For our separated affine cover, affine vanishing makes every row with $q > 0$ on E_1 zero. The augmentation therefore canonically identifies cohomology with that of the finite section complex

$$\check{C}^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0<\cdots<i_p} \Gamma(U_I, \mathcal{F}).$$

If $U_I = \operatorname{Spec} R$ and $\mathcal{F}|_{U_I} = \widetilde{M}$, its base change has sections

$$M \otimes_R (R \otimes_A B) = M \otimes_A B.$$

These identifications commute with restriction. Tensor also commutes with the finite products, so the section complex on X_B is the section complex on X tensored with B. Flat tensor preserves kernels and images and hence cohomology. This proves the claimed isomorphism in the separated case.

Canonical comparison and nonseparated intersections.

To identify the preceding calculation with the canonical map and compare the general cover spectral sequences, choose an injective resolution $\mathcal{F}_B \to \mathcal{J}^\bullet$ on X_B. The projection g is flat, so $g^*\mathcal{I}^\bullet$ is a resolution of $\mathcal{F}_B$. Target injectivity extends its identity on $\mathcal{F}_B$ degree by degree to a chain map $g^*\mathcal{I}^\bullet \to \mathcal{J}^\bullet$. The same extension argument makes any two choices chain homotopic. Pulling back sections and multiplying by elements of B gives a map of augmented double complexes

$$C^{p,q}(X, \mathcal{I}) \otimes_A B \longrightarrow C^{p,q}(X_B, \mathcal{J}).$$

Its augmentation is the usual cohomology base-change map; its column maps are the canonical comparisons on the intersections. It is independent of the resolution choices and compatible with restriction and successive base changes. Since B is flat, the source total complex has cohomology $H^n(X, \mathcal{F}) \otimes_A B$. Flat tensor also commutes with the cycles, boundaries and finite sums defining every spectral-sequence page.

Now let X be merely quasi-compact and quasi-separated. Choose a finite affine cover as before. Each U_I is quasi-compact, by quasi-separatedness and finite intersection, and is separated over $\operatorname{Spec} A$, since it is an open subscheme of the affine U_{i_0}. It need not be affine. Apply the separated result to each U_I. It shows that every term of the comparison on E_1 is an isomorphism, for all $q \geq 0$. Therefore every subsequent page and every associated graded piece of the abutment is an isomorphism. Induction through the finite filtrations gives an isomorphism of the abutments themselves. This proves the statement for every flat A-algebra B.

Localization and higher direct images.

Taking $B = S^{-1}A$ for any multiplicative set gives, canonically,

$$S^{-1}H^q(X, \mathcal{F}) \cong H^q(X \times_A \operatorname{Spec} S^{-1}A, \mathcal{F}_{S^{-1}A}).$$

This includes $A \to A_a$ and $A \to A_{\mathfrak{p}}$.

For completeness, let $f : X \to Y$ be a quasi-compact quasi-separated morphism and let $\mathcal{F}$ be quasi-coherent. The sheaf $R^q f_*\mathcal{F}$ is the sheafification of $V \mapsto H^q(f^{-1}V, \mathcal{F})$. Indeed, use an injective resolution on X, restrict it to $f^{-1}V$, and take stalks of $f_*\mathcal{I}^\bullet$; filtered colimits of modules are exact, so taking cohomology commutes with that stalk limit. On an affine $V = \operatorname{Spec} A$, put $M = H^q(f^{-1}V, \mathcal{F})$. The localization result identifies the presheaf on every distinguished $D(a) \subset V$ with M_a, compatibly with restriction. Its sheafification is therefore $\widetilde{M}$. This proves quasi-coherence of $R^q f_*\mathcal{F}$ and supplies the canonical identification of its sections on V with M.

If $h : Y' \to Y$ is flat, cover Y' by affines $W = \operatorname{Spec} B$ mapping into affines $V = \operatorname{Spec} A$ of Y. Here B is flat over A. For the Cartesian base change $f' : X \times_Y Y' \to Y'$, the theorem identifies the modules describing both sides of

$$h^* R^q f_*\mathcal{F} \longrightarrow R^q f'_* \mathcal{F}_{Y'}.$$

Naturality makes these identifications agree on overlaps, so this is an isomorphism of sheaves. In particular, at $y \in V$ corresponding to $\mathfrak{p}$,

$$(R^q f_*\mathcal{F})_y \cong H^q(f^{-1}V \times_A \operatorname{Spec} A_{\mathfrak{p}}, \mathcal{F}_{A_{\mathfrak{p}}}).$$

This is the flat-localization identity needed for the stalk calculation in AG-RG-05, §9. □

7. Projective space by its affine charts

Define projective space by gluing the affine charts with coordinates T_j/T_i using their ratio transitions. The twists are glued rank-one modules with the corresponding degree transitions. Thus the displayed

section modules below follow from this construction; no Proj existence theorem is an additional proof input.

Projective space is separated over A, by a direct diagonal calculation. On the product of charts $U_i \times_A U_j$ put $x_l = T_l/T_i$ and $y_l = S_l/S_j$, with $x_i = 1$ and $y_j = 1$. Its diagonal has equations $x_j y_i = 1$ and $x_l = x_j y_l$ for every l. The quotient of the product-chart coordinate ring by these equations is $A[x_l\ (l \neq i), x_j^{-1}]$, the coordinate ring of $U_i \cap U_j$. If $i = j$ the equations simply identify $x_l = y_l$. Hence the diagonal is a closed immersion on every product chart. These charts cover the product, so the diagonal is a closed immersion and the asserted separatedness follows.

Lemma 7.A (projective-space properness). For every scheme S and integer $n \geq 0$, the projection $\mathbf{P}^n_S \to S$ is proper. Here proper means separated, of finite type and universally closed; no Noetherian or reducedness hypothesis is imposed.

Proof. The diagonal calculation above proves separatedness over every ring. The $n+1$ standard charts are polynomial algebras in n variables over the base, so they give finite type. We prove closedness after every base change directly.

Work over an arbitrary affine base $\operatorname{Spec} A$ and put $P = A[T_0, \ldots, T_n]$, with each variable of degree one. Every closed subset Z of $\mathbf{P}^n_A$ is $V_+(I)$ for a homogeneous ideal $I \subset P$, possibly with infinitely many generators. To see this from the chart construction, a principal open of the ith chart has an equation $g(T_j/T_i) \neq 0$. Homogenize g to a homogeneous polynomial f; this open is $D_+(T_i f)$, since the extra factor restricts it to the ith chart. These opens form a basis. Express the complement of Z as their union and let I be generated by their homogeneous equations. The corresponding quotient $B = P/I$ has exactly Z as its projective support: its ith chart is the quotient of $A[T_j/T_i]$ by the degree-zero localization of I.

Each degree module B_d is a finite A-module, since it is a quotient of the finite free module on the degree-d monomials. For any finite A-module M,

$$\{\mathfrak{p} : M \otimes_A \kappa(\mathfrak{p}) \neq 0\} = \operatorname{Supp} M = V(\operatorname{Ann}_A M).$$

Indeed $M_\mathfrak{p} = 0$ exactly when an element outside $\mathfrak{p}$ kills all of a finite generating list: clear the finitely many localization denominators. This is equivalent to $\operatorname{Ann} M \not\subset \mathfrak{p}$. The residue quotient is zero exactly when $M_\mathfrak{p} = 0$, by the determinant Nakayama proof of Lemma 1.2. Thus the displayed set is closed even when M is not finitely presented.

The fibre of Z at $\mathfrak{p}$ is the projective support of $B \otimes_A \kappa(\mathfrak{p})$. This follows on each chart by quotient and localization, which commute with scalar extension. Its fibre is empty exactly when some degree of this graded algebra is zero. In fact an empty fibre makes every standard chart empty. Its degree-zero localization ring is then zero, by the maximal-ideal argument of Lemma 1.1. Consequently a power of each T_i is zero in the graded algebra. If these powers are $T_i^{e_i}$, every monomial of degree greater than $\sum_i (e_i - 1)$ has one of them as a factor, so that degree is zero. Conversely, if degree d is zero, then each T_i^d is zero (if $d = 0$, the whole algebra is zero), making every chart empty. All larger degrees are zero too, since the algebra is generated in degree one. Right exactness of tensor identifies its degree d with $B_d \otimes_A \kappa(\mathfrak{p})$.

It follows that

$$\operatorname{image}(Z \to \operatorname{Spec} A) = \bigcap_{d \geq 0} V(\operatorname{Ann}_A B_d).$$

An arbitrary intersection of closed sets is closed. The argument works over every ring A and for every closed subset, including ideals that are not finitely generated. Apply it on each affine open after an arbitrary scheme base change. Closedness is local on the target: a subset is closed if its intersections with a covering family of opens are closed there. Hence the projection is universally closed. Together with its already proved separation and finite type this proves properness. □

Corollary 7.B (the properness consequences used below). Proper morphisms are preserved by base change and composition. Closed immersions are proper. If $X \to S$ is proper and $Y \to S$ is separated, every S-morphism $X \to Y$ is proper; when it is an immersion, it is a closed immersion.

Proof. Universally closed maps remain so under base change, since an iterated base change is a base change, and under composition, since images of closed sets are closed at both steps. Finite type remains so under base change by tensoring a finite algebra generating list, and under composition by joining finite generating lists on affine charts; the finite covers expressing quasi-compactness also pull back and compose. Separatedness remains so under base change by the diagonal formula. For $X \to Y \to S$, the diagonal $X \to X \times_S X$ factors through its closed relative diagonal $X \to X \times_Y X$ and the closed immersion $X \times_Y X \to X \times_S X$ obtained by pulling back $\Delta_{Y/S}$. This proves its preservation under composition.

On affine charts a closed immersion is a quotient map $A \to A/J$. It is of finite type with zero algebra generators, even for an infinite ideal J. Its diagonal is an isomorphism, and its images of closed sets are closed; quotients have the same properties after base change. Thus it is proper. For a morphism $g : X \to Y$ as in the statement, the graph $X \to X \times_S Y$ is closed, being the pullback of $\Delta_{Y/S}$. Projection $X \times_S Y \to Y$ is the base change of the proper map $X \to S$. Their composite g is therefore proper. If g is an immersion, it is a closed immersion on an open neighbourhood U of its image by the definition of immersion. Properness makes that image closed in Y. On U the map is closed, and on the open complement of its image it is the empty closed immersion. These opens cover Y, and their quotient ideals glue, proving that g is a closed immersion. □

The statements can be compared with the freely readable Stacks, Tag 01WC. The graded-module and graph arguments above provide the programme proofs used here.

The monomial section complex

Fix a ring A, an integer $n \geq 0$, and the standard graded ring

$$S = A[T_0, \dots, T_n], \qquad \deg T_i = 1.$$

Write $\mathbf{P}^n_A = \operatorname{Proj} S$. The twist $\mathcal{O}(d)$ is the sheaf of the graded module $S(d)$, with $S(d)_m = S_{d+m}$. On a nonempty intersection of standard charts,

$$\Gamma(D_+(T_{i_0}) \cap \cdots \cap D_+(T_{i_p}), \mathcal{O}(d)) = S[T_{i_0}^{-1}, \dots, T_{i_p}^{-1}]_d.$$

The subscript means homogeneous degree d. Inverting the product of these variables is equivalent to inverting each of them. These intersections are affine; for example, the chart indexed by i has coordinates T_j/T_i. Thus the finite-cover comparison of Sections 4-5 above identifies cohomology with the ordered Čech complex

$$C^p(d) = \bigoplus_{i_0 < \cdots < i_p} S[T_{i_0}^{-1}, \dots, T_{i_p}^{-1}]_d.$$

It stops in degree n. The differential is the alternating sum of restrictions, with increasing indices and the sign of the deleted index's position.

Each term is a free A-module on Laurent monomials. This remains true when A has torsion or zero divisors: the monomials are formal basis vectors of the polynomial localization. A weight is an integer vector $e = (e_0, \dots, e_n)$, with total $\sum e_i = d$. The monomial T^e occurs in the factor indexed by a subset I exactly when

$$N(e) = \{i : e_i < 0\} \subset I.$$

The differential preserves weights. The whole complex is therefore a direct sum of complexes $C^\bullet(e)$, each having a copy of A for the nonempty subsets I containing $N(e)$. Its differential uses only zero maps and signed identity maps between these copies.

These are direct sums rather than products over the weights: every Laurent polynomial has finite support, and only finitely many chart subsets occur in a fixed degree. Direct sums are exact, so cohomology can be computed separately at each weight.

Negative exponents determine cohomology

Lemma 7.1 (the three weight cases). For a weight e, the cohomology of $C^\bullet(e)$ is:

Negative indices	Surviving cohomology
None	One copy of AT^e in degree zero
All $n+1$ indices	One copy of AT^e in degree n
A nonempty proper subset	Zero in every degree

Proof. If every exponent is negative, only the subset $I = \{0, \dots, n\}$ is allowed. Its copy of A is the whole complex, in degree n.

If no exponent is negative, every nonempty subset is allowed. This is the ordinary simplex cochain complex, with A as its degree-zero cohomology. More explicitly, augment it by a copy of A in degree -1, whose map is the diagonal. Inserting a fixed index gives a contracting homotopy of this augmented complex. Thus the original complex has the stated cohomology.

Now suppose $N(e)$ is nonempty and proper. Choose $j \notin N(e)$. Regard an ordered cochain as an alternating function of distinct indices, with value zero on any tuple whose set does not contain $N(e)$, and zero on repeated indices. Define

$$(hc)_{i_0 \dots i_{p-1}} = c_{j i_0 \dots i_{p-1}}.$$

Insertion is followed by reordering with its sign. This map is well-defined: adjoining j does not change whether a subset contains $N(e)$. The identity $dh + hd = 1$ follows by pairing each deletion other than the inserted j with its opposite-sign counterpart; deletion of j contributes the original cochain. In degree zero the would-be augmented term is absent because $N(e) \neq \varnothing$, and the same calculation still works: the value at the singleton $\{j\}$ is zero. This is a contraction of the unaugmented complex, so all its cohomology is zero. □

For a concrete example with three variables, take $N(e) = \{0\}$. The weight complex is

$$A \longrightarrow A^2 \longrightarrow A, \qquad a \longmapsto (-a, -a), \qquad (b, c) \longmapsto b - c.$$

The two middle entries correspond to $\{0, 1\}$ and $\{0, 2\}$. The last map is surjective and its kernel is the diagonal, exactly the preceding image. With $N(e) = \{0, 1\}$, the only terms are $A \xrightarrow{1} A$ in degrees one and two. With all three exponents negative, only the last A survives. These small complexes illustrate the general cancellation without dividing by an integer.

Put $r = n + 1$. For $n \geq 1$, let

$$L_d = \bigoplus_{\substack{e_i < 0 \text{ for all } i \\ \sum e_i = d}} AT^e.$$

It is zero unless $d \leq -r$. It can also be written as the homogeneous degree-d part of

$$\frac{1}{T_0 \cdots T_n} A[T_0^{-1}, \dots, T_n^{-1}].$$

The expression is a module of reciprocal monomials, not a ring.

Theorem 7.2 (cohomology of every twist). If $n \geq 1$, then

$$H^q(\mathbf{P}^n_A, \mathcal{O}(d)) = \begin{cases} S_d, & q = 0,\ d \geq 0, \\ L_d, & q = n,\ d \leq -n-1, \\ 0, & \text{otherwise.} \end{cases}$$

All the nonzero modules are finite free over A. For $n = 0$, instead,

$$H^0(\mathbf{P}^0_A, \mathcal{O}(d)) = A \quad \text{for every integer } d, \qquad H^q = 0 \quad (q > 0).$$

Proof. For positive dimension, Lemma 7.1 and the weight direct sum identify degree-zero classes with monomials having nonnegative exponents, and top-degree classes with monomials having strictly negative exponents. All other weights are contractible. The sum condition gives the stated ranges. For fixed d, there are only finitely many vectors of nonnegative exponents summing to d, or of strictly negative exponents summing to d, proving finite freeness.

In dimension zero there is one chart, and its degree-d section module is AT_0^d for every integer d. This is A; geometrically $\mathbf{P}^0_A = \operatorname{Spec} A$, and the trivializing generator of the twist is T_0^d, including negative powers. There are no higher Čech terms. □

The separate zero-dimensional clause prevents a false negative-twist vanishing statement when degree zero is also top degree. For $n \geq 1$, all groups vanish in the entire interval $-n \leq d \leq -1$. The calculation is [Stacks, Tag 01XT], with the two cohomological degrees kept distinct when necessary.

A pairing with no factorials

The top group of $\mathcal{O}(-r)$ is free of rank one. Fix its trace by

$$\operatorname{tr}\left[\frac{1}{T_0 \cdots T_n}\right] = 1.$$

For $m \geq 0$, multiplication by global sections gives a bilinear pairing

$$S_m \times H^n(\mathbf{P}^n_A, \mathcal{O}(-m-r)) \longrightarrow H^n(\mathbf{P}^n_A, \mathcal{O}(-r)) \xrightarrow{\operatorname{tr}} A.$$

Theorem 7.3 (perfect monomial pairing). This pairing is perfect: each module is the A-linear dual of the other.

Proof. Index the monomial basis of S_m by vectors $\alpha_i \geq 0$ with $\sum \alpha_i = m$. A corresponding basis of the other group is

$$u_\alpha = T_0^{-\alpha_0 - 1} \cdots T_n^{-\alpha_n - 1}.$$

The product $T^\beta u_\alpha$ has exponent $\beta_i - \alpha_i - 1$. If $\beta = \alpha$, it is the trace generator. If the vectors differ but have the same total, at least one $\beta_i > \alpha_i$, so that product has a nonnegative exponent and represents zero in top cohomology by Lemma 7.1. The pairing matrix is therefore the identity. This proves perfection over every ring, including rings with positive characteristic or nilpotents. □

The proof also applies to $n = 0$ using the generators of its one-chart section modules. There is no appeal to integration and no normalization by a factorial. This will be the local calculation behind Serre duality later in the course.

Proposition 7.4 (base change and polynomial multiplication). The preceding identifications commute with every ring map $A \to B$, including nonflat ones. If $f \in S_s$, multiplication by f on top cohomology is the dual of

$$S_{-d-s-r} \xrightarrow{f} S_{-d-r},$$

where negative homogeneous pieces are zero and $n \geq 1$.

Proof. The section complex over B is obtained by replacing the coefficient ring in every monomial factor by B. The weight contractions use only signed identities, so they remain contractions after any coefficient change. The surviving bases also change by extension of scalars. The resulting isomorphism

$$H^q(\mathbf{P}^n_A, \mathcal{O}(d)) \otimes_A B \simeq H^q(\mathbf{P}^n_B, \mathcal{O}(d))$$

is the natural one induced by the chart section complexes. No general nonflat base-change theorem is being assumed here.

For the multiplication statement, let u be a top class and g a homogeneous polynomial of degree $-d-s-r$. Associativity of multiplication gives

$$\operatorname{tr}(g(fu)) = \operatorname{tr}((fg)u).$$

The perfect pairing identifies this equality with the claimed dual map. Products that leave the negative-exponent range are zero classes, as already proved. □

This functorial form agrees with [Stacks, Tag 01XV]. In particular, the finite free modules in the calculation retain their ranks under every specialization of the base ring.

8. Projective finiteness, ample embeddings and vanishing

Clearing denominators in a line bundle

For a line bundle L and a section $s \in H^0(X, L)$, write X_s for the open where the section generates the line. Multiplication by s gives maps $F \otimes L^a \to F \otimes L^{a+1}$.

Lemma 8.1 (extension after a power). If X is quasi-compact and quasi-separated, F is quasi-coherent, and s is as above, the natural map

$$\operatorname*{colim}_{a \geq 0} H^0(X, F \otimes L^a) \longrightarrow H^0(X_s, F), \qquad t \longmapsto t/s^a,$$

is an isomorphism. In particular every section on X_s extends after multiplying by a sufficiently high power of s.

Proof. Choose a finite affine cover U_i trivializing L. On U_i, write $s = f_i$ in that frame and $F = \widetilde{M}_i$. Then $U_i \cap X_s = D(f_i)$, and sections there are elements of $(M_i)_{f_i}$. Each is a fraction with a finite denominator, so a section on X_s has local lifts in $F \otimes L^a$ for a common exponent a.

On each overlap the difference of two lifts becomes zero after inverting s. The overlap is quasi-compact by quasi-separatedness. Cover it by finitely many affines trivializing L; on each, an element that is zero in a localization is killed by a power of its local equation. A common power kills the difference on the whole overlap. There are finitely many pairs, so one further power makes all lifts agree. They glue to a global twisted section. This proves surjectivity. If a global twisted section restricts to zero, the same localization argument on the finite cover kills it by a common power of s; it is therefore zero in the colimit. This proves injectivity. □

The same argument works with a section of L^b, using powers L^{ab}. It does not require s to be a nonzerodivisor. The associated section-module recovery is [Stacks, Tag 0AG5]; here the denominator argument also explains its compatibility with the usual maps from homogeneous elements.

Now let R be Noetherian, $S = R[T_0, \dots, T_N]$, and $P = \mathbf{P}^N_R$. The scheme P is Noetherian by Lemma 1.3 above: each standard chart is a polynomial ring over R. A coherent sheaf on a locally Noetherian scheme is a quasi-coherent sheaf with finite modules on affine opens. Kernels and cokernels of maps between such sheaves are coherent, because submodules of finite modules over Noetherian rings are finite.

Proposition 8.2 (finite generation by twists). For every coherent F on P, there is a finite surjection

$$\mathcal{O}_P(-a)^{\oplus m} \twoheadrightarrow F$$

for some $a \geq 0$. Moreover $F(d)$ is generated by finitely many global sections for every sufficiently large d.

Proof. On $D_+(T_i)$, choose finitely many module generators of F. Lemma 8.1 extends each generator, after multiplication by a power of T_i, to a global section of some $F(a_{ij})$. Choose a

common $a \geq 0$ at least as large as every a_{ij}, and multiply these sections by $T_i^{a-a_{ij}}$. On the i-th chart the multiplier is a unit in the chart's twist frame, so the resulting global sections of $F(a)$ still generate there. All charts together give a finite global generating family, which is equivalent to the displayed surjection. For $d \geq a$, tensor it with $\mathcal{O}(d)$. The source is a sum of copies of $\mathcal{O}(d-a)$, generated by its polynomial monomials; their images generate $F(d)$. This also treats $N = 0$, where the scheme is affine and every twist is trivial. □

Global generation means that the evaluation map from global sections tensored with $\mathcal{O}_P$ is surjective. It does not mean that every local generator is itself the restriction of an untwisted global section. The power of the chart coordinate is what makes local information extend.

Lemma 8.3 (an ample power gives an immersion). Let X be quasi-compact and quasi-separated and of finite type over an affine scheme $\operatorname{Spec} R$. Let L be ample. There are $b > 0$ and an immersion $i : X \to \mathbf{P}_R^N$ with $L^b \cong i^*\mathcal{O}(1)$. If X is proper over R, the immersion is closed.

Proof. We use the affine-section definition of ampleness: the affine opens X_s, for sections s of positive tensor powers of L, form a basis. Quasi-compactness supplies a finite cover by such opens. Taking powers of the finitely many sections replaces their degrees by a common positive multiple m, without changing their nonvanishing opens. Denote the resulting sections by $s_i \in H^0(X, L^m)$ and write $X_{s_i} = \operatorname{Spec} B_i$. Each B_i is a finite-type R-algebra, with the following explicit affine-chart argument. Write $W = X_{s_i} = \operatorname{Spec} B_i$. At a point of W, choose a finite-type affine chart V of X/R, then a principal open W_g containing that point and contained in V, and finally a principal open V_h containing the point and contained in W_g. As $W_g \subset V$, the restriction of h is a fraction b/g^q in $(B_i)_g$. Hence $V_h = W_g \cap V_h = (W_g)_h = W_{gb}$: it is a principal open of W, and its algebra is finite type over R by localization of the chart algebra of V. Quasi-compactness gives finitely many such opens W_{b_j}. Choose finite generators for every $(B_i)_{b_j}$, their finitely many numerators in B_i, and coefficients $e_j \in B_i$ with $\sum_j e_j b_j = 1$. Let B' be the R-subalgebra generated by these numerators, all b_j and all e_j. Then $B'_{b_j} = (B_i)_{b_j}$. For $b \in B_i$, some power $b_j^{N_j} b$ belongs to B' for each j. A sufficiently large power of $\sum_j e_j b_j = 1$ expresses 1 as a B'-linear combination of the $b_j^{N_j}$. Multiplying by b gives $b \in B'$. Thus $B_i = B'$ is finite type. Choose finitely many algebra generators a_{ij} for each B_i.

Apply Lemma 8.1 with the line bundle L^m and $F = \mathcal{O}_X$. For every a_{ij}, it gives a global section whose quotient by a power of s_i is a_{ij}. There are finitely many generators and charts, so multiplication by additional powers gives a common exponent $e > 0$ and sections $u_{ij} \in H^0(X, L^{me})$ such that $u_{ij}/s_i^e = a_{ij}$ on X_{s_i}. Set $t_i = s_i^e$. The t_i alone generate L^{me}, because their nonvanishing opens cover X. The finite collection of all t_i, u_{ij} therefore defines a morphism $i : X \to \mathbf{P}_R^N$, by the quotient convention, with $i^*\mathcal{O}(1) = L^{me}$.

On the target chart where the coordinate for t_i is nonzero, the inverse image is exactly X_{s_i}. The coordinate ratios for u_{ij} pull back to a_{ij}; hence the homomorphism from that chart's polynomial coordinate ring to B_i is surjective. The restriction of i on this chart is a closed immersion. Let V be the union of these target charts. It contains the image, and the chartwise closed immersions glue to a closed immersion $X \to V$. Thus $X \to \mathbf{P}_R^N$ is an immersion, with $b = me$.

If X is proper over R, its graph in $X \times_R \mathbf{P}_R^N$ is closed: the graph is the inverse image of the diagonal of the separated scheme $\mathbf{P}_R^N$. Projection from this product to $\mathbf{P}_R^N$ is the base change of the proper map $X \to \operatorname{Spec} R$. The graph followed by this projection shows that i is proper. Its image is therefore closed in $\mathbf{P}_R^N$. On the open V it is already a closed immersion, and on the open complement of its closed image it is the empty closed immersion. These opens cover the target, so i is a closed immersion. □

Descending induction proves Serre's theorems

Theorem 8.4 (finiteness and vanishing on projective space). For a coherent F on $\mathbf{P}^N_R$, with R Noetherian:

- $H^q(P, F) = 0$ for $q > N$.
- Every $H^q(P, F)$ is a finite R-module.
- There exists $d_0(F)$ such that $H^q(P, F(d)) = 0$ for every $q > 0$ and every $d \geq d_0(F)$.
- $F(d)$ is globally generated for every sufficiently large d.

Proof. The $N + 1$ standard charts and all their intersections are affine. Their ordered Čech complex computes quasi-coherent cohomology and has no terms above degree N. This proves the first assertion, even without coherence.

Prove finiteness and eventual vanishing simultaneously, by descending induction on q, for all coherent sheaves. The assertions hold in degrees greater than N. Choose a finite quotient by twists, with coherent kernel:

$$0 \longrightarrow K \longrightarrow E \longrightarrow F \longrightarrow 0, \qquad E = \bigoplus_j \mathcal{O}(b_j).$$

The exact cohomology segment

$$H^q(P, E) \longrightarrow H^q(P, F) \longrightarrow H^{q+1}(P, K)$$

has finite modules at both ends: the left one by Theorem 7.2, the right one by induction. Its middle term is an extension of a quotient of the left module by a submodule of the right module. The latter submodule is finite because R is Noetherian. Hence the middle term is finite. This includes $q = 0$.

For $q > 0$, twist the short exact sequence by $\mathcal{O}(d)$, which is an exact operation because this sheaf is invertible. In the corresponding segment, $H^q(E(d))$ is zero for sufficiently large d by Theorem 7.2; $H^{q+1}(K(d))$ is zero for sufficiently large d by induction. Thus $H^q(F(d)) = 0$ eventually. Only the finitely many degrees $1, \ldots, N$ can occur, so their thresholds have a maximum. Proposition 8.2 proves the last assertion. □

This is [Stacks, Tag 01YS]. The proof uses as many coherent kernels as the induction needs, rather than asserting that every coherent sheaf has a finite resolution by line bundles. Such a resolution would require further hypotheses and is unnecessary here.

For a closed immersion $i : X \hookrightarrow P$, pushforward identifies coherent sheaves on X with coherent sheaves on P annihilated by the defining ideal. Affine-locally this is just restriction of scalars along a quotient of Noetherian rings. Pushforward is exact, and

$$H^q(X, F) = H^q(P, i_*F), \qquad i_*(F \otimes i^*\mathcal{O}(d)) = (i_*F)(d).$$

The cohomology equality follows from the common affine-cover complex of Sections 4-5: restriction of the standard affine charts to the closed subscheme gives affine charts, and extension of its sheaf along the closed immersion has exactly the same section modules on their intersections. The tensor equality can be checked in a local line-bundle frame. Theorem 8.4 therefore applies to every closed projective subscheme with its hyperplane bundle.

Theorem 8.5 (proper schemes with an ample line bundle). Let $X \to \operatorname{Spec} R$ be proper, with R Noetherian, and let L be ample. For coherent F, all $H^q(X, F)$ are finite over R; the groups $H^q(X, F \otimes L^d)$, $q > 0$, vanish for all sufficiently large d; and $F \otimes L^d$ is globally generated for all sufficiently large d.

Proof. Lemma 8.3 gives a closed immersion $i : X \hookrightarrow \mathbf{P}^N_R$ and an integer $b > 0$ with $L^b = i^*\mathcal{O}(1)$. Its quasi-compactness and quasi-separatedness hypotheses hold because X is proper. Apply Theorem

8.4 to each of the finitely many coherent sheaves $i_*(F \otimes L^j)$, $0 \le j < b$. If $d = bm + j$, its twist by $\mathcal{O}(m)$ is $i_*(F \otimes L^d)$. The theorem gives vanishing and global generation for all sufficiently large m in each residue class. Taking their finitely many thresholds proves the assertion for every large d. Finiteness follows already from i_*F. □

This residue-class step is essential: an embedding given by L^b initially controls only multiples of b. The result is [Stacks, Tag 0B5T].

9. Serre's affine criterion

For a global function $f \in \Gamma(X, \mathcal{O}_X)$, let X_f be its invertibility locus. It is open; on an affine chart it is the usual distinguished open of the restricted function.

We will use two elementary facts. Every closed subset Z of a scheme has a reduced closed subscheme structure, hence a quasi-coherent radical ideal. On an affine chart $\operatorname{Spec} B$, take the radical ideal cutting out Z there. Localization of radical ideals agrees on smaller distinguished charts, so these ideals glue. Also every nonempty quasi-compact closed subset of a scheme has a closed point. For completeness, among its nonempty closed subsets, every descending chain has nonempty intersection by quasi-compactness. Zorn's lemma gives a minimal nonempty closed subset C. For each $x \in C$, its closure is C by minimality. A scheme is T_0, so two different points cannot have the same closure; thus C is a singleton. The point is closed in the original scheme when the original subset is closed.

Lemma 9.1 (an affine principal cover detects affineness). Suppose finitely many global functions f_i generate the unit ideal in $R = \Gamma(X, \mathcal{O}_X)$, and the opens X_{f_i} are affine and cover X. Then X is affine.

Proof. The finite affine cover makes X quasi-compact. Its intersections are affine: $X_{f_i} \cap X_{f_j}$ is the distinguished open defined by f_j in the affine X_{f_i}. Hence X is quasi-separated. The global ring R defines a map $X \to \operatorname{Spec} R$: on any affine chart U use the restriction ring map $R \to \Gamma(U, \mathcal{O}_X)$. On intersections these chart morphisms agree by restricting to an affine cover there, so they glue; its inverse image of $D(f)$ is exactly X_f. Since this finite principal cover has affine intersections, it is quasi-separated. Apply the degree-zero case of the complete Section 6 flat-base-change proof to this map and $R \to R_{f_i}$. Its fibre product is X_{f_i}, and it gives

$$R_{f_i} = \Gamma(X_{f_i}, \mathcal{O}_X).$$

The canonical morphism $X \to \operatorname{Spec} R$ restricts on X_{f_i} to the isomorphism with $D(f_i)$ given by this ring identification. The $D(f_i)$ cover $\operatorname{Spec} R$, since the f_i generate one. These isomorphisms agree on overlaps as restrictions of the canonical morphism, and give an isomorphism of the whole schemes. □

Theorem 9.2 (Serre's criterion). For a quasi-compact scheme X, the following are equivalent:

1. X is affine.
2. $H^p(X, \mathcal{F}) = 0$ for every quasi-coherent $\mathcal{F}$ and every $p > 0$.
3. $H^1(X, \mathcal{I}) = 0$ for every quasi-coherent ideal $\mathcal{I} \subset \mathcal{O}_X$.

No quasi-separatedness hypothesis is required in this statement.

Proof. Theorem 5.1 proves the first implication, and the second plainly implies the third. Assume the third condition.

Let x be a closed point and choose an affine neighbourhood $U = \operatorname{Spec} B$. Let $Z = X \setminus U$, and let $\mathcal{I}$ and $\mathcal{I}'$ be the reduced ideals of Z and $Z \cup \{x\}$, respectively. There is an exact sequence

$$0 \to \mathcal{I}' \to \mathcal{I} \to \mathcal{I}/\mathcal{I}' \to 0.$$

On U, the quotient is the module sheaf of $B/\mathfrak{m}_x$; away from x it is zero. It is the skyscraper with value $\kappa(x)$. This can also be checked by gluing its descriptions on U and $X \setminus \{x\}$, whose overlap has zero quotient.

Vanishing of $H^1(X, \mathcal{I}')$ lifts the section $1 \in \kappa(x)$ to a global $f \in \Gamma(X, \mathcal{I})$. The function is invertible at x and has zero residue at every point of Z. Consequently

$$x \in X_f \subset U,$$

and X_f is a distinguished affine open of U.

The union of all affine opens of this form contains every closed point. Its complement is a quasi-compact closed subset, and if nonempty would have a closed point by the fact proved above. Thus the union is all of X. Choose finitely many such $f_1, \dots, f_r$ whose invertibility loci cover X.

We must still show that these functions generate one in the ring of global sections. The map

$$\mathcal{O}_X^{\oplus r} \xrightarrow{(f_1,\dots,f_r)} \mathcal{O}_X$$

is surjective on stalks, since one f_i is invertible at each point. Its kernel $\mathcal{K}$ is quasi-coherent. Put $\mathcal{K}_i = \mathcal{K} \cap \mathcal{O}_X^{\oplus i}$, using the first i summands. Projection onto the i-th summand embeds $\mathcal{K}_i/\mathcal{K}_{i-1}$ as a quasi-coherent ideal of $\mathcal{O}_X$. These quotients therefore have zero first cohomology. Starting with $\mathcal{K}_0 = 0$, their long exact sequences inductively give $H^1(X, \mathcal{K}_i) = 0$, and in particular $H^1(X, \mathcal{K}) = 0$.

The global-section sequence is now surjective onto $\Gamma(X, \mathcal{O}_X)$. There are global a_i with $\sum_i a_i f_i = 1$. Lemma 9.1 proves affineness. The quasi-separatedness used in that lemma has been obtained from the principal cover, rather than assumed at the start. □

This is the ideal-sheaf form of [Stacks, Tag 01XF]. It explains why degree one is enough: it first constructs affine neighbourhoods from functions and then lifts the unit through the relation module.

Free source and proof record

The scheme correspondence, affine-target construction, affine vanishing and filtered-cover statements can be compared with the freely accessible Stacks Project, Tags 01IA, 01I1, 01XB, 01ES, 01EW, 012M and 012W. The proofs actually consumed above are written in this lesson. The construction of injectives is given here rather than assigned to a general categorical prerequisite. No paid work is a source or citation.

The detailed Laurent-monomial, denominator-extension and descending-induction arguments in Sections 7-9 reuse the actual independently authored CC0 programme proofs *Cohomology of projective space*, *Coherent sheaves on projective schemes: Serre's theorems*, and *Cohomology of affine schemes and Serre's criterion*. Their exact consumed proof excerpts are reproduced here with their hypotheses; unrelated reference lists and unconsumed results are not imported. The original authorship and self-check are GPT-6.1 Sol (OpenAI), Codex, Ultra. The present lesson supplies their earlier affine and homological inputs. Free comparison sources are Stacks Tags 01XT, 01YS, 01VT and 01XF.

Projective cohomology and smooth affine models

This supporting lesson proves the projective formal functions and smooth model statements needed before the reductive group lessons. It also records the precise earlier programme proofs used for completion, projective finiteness and curve duality. It follows the supporting lesson Algebra and sheaf cohomology before reductive groups, whose Sections 1–9 supply the affine, homological, projective-space and Serre calculations. Both supporting lessons precede AG-RG-01. All mathematical sources listed here are freely readable. Source citations identify the material used to write the proofs; they do not replace any proof below.

The formal functions exposition follows the freely readable Stacks Project proofs. The smooth affine component argument below specializes the Stacks model route and supplies its intermediate arguments. The text of this lesson is dedicated to the public domain under CC0 1.0; the Stacks Project itself is distributed under the GNU FDL 1.2 or later.

1. Earlier programme proofs used in this lesson

The following references are to proofs already written in programme lessons. Their numbered statements delimit exactly what is used.

- **AG-CA-03, Noetherian and Artinian rings:** Proposition 1.2 (finite modules over a Noetherian ring are Noetherian), Theorem 2.1 (Hilbert basis), Theorem 5.1 Artin–Rees, and Theorem 6.1 (Krull intersection). In particular Artin–Rees is proved by finite generation of the graded submodule of the Rees module; it is not taken from a book.
- **AG-CA-19, Completion:** Theorem 1.2 (exact inverse limits with a Mittag–Leffler kernel), Lemma 2.1 (completion preserves surjections with the induced kernel filtration), Proposition 2.2 (finite ideal powers in a completion), Theorems 3.1–3.3 (finite-module exactness, completion tensor comparison, flatness, faithful flatness and Noetherianity). These statements have full proofs there. Thus a finite module over an already complete Noetherian ring is complete, and local Noetherian completion is faithfully flat.
- **AG-CA-17, Formally smooth, unramified and étale ring maps:** Theorem 3.1 (formal smoothness is the split conormal criterion), Theorems 4.1 and 5.1 (standard smooth presentations and local standard form), and Lemma 6.2 (unique lifting of idempotents across a square-zero ideal). These are written lifting and matrix proofs. Lemma 1.1 extends idempotent lifting to a nilpotent ideal by successive square-zero steps.
- **AG-CA-18, Smooth algebras over a field and the Jacobian criterion:** Theorem 2.1 proves regularity of smooth field-algebra local rings. Its proof uses standard smooth equations and the regular parameter quotient theorem. Base change of smoothness, proved in AG-CA-17, consequently gives geometric regularity.
- **AG-CA-14, Regular local rings:** Theorem 1.1 and Proposition 1.4 prove regular parameter quotients and generation of a regular quotient's ideal by part of a regular system of parameters.

The graded and regular-sequence assertions used there are the full proof of Theorem 6.1 in **Regular sequences, depth and Cohen–Macaulay modules**. The Koszul resolution and its Hom calculation are the corresponding earlier depth lesson proofs.

- **AG-CA-07, Tor and flat modules:** Theorems 5.3 and 6.1 contain the finite-projective and flat-kernel calculations corresponding to Section 5. That section writes its universal-tail flatness and global splitting arguments directly, without taking the Tor criterion as an additional prerequisite.
- **AG-CA-05, Integral extensions:** Theorems 3.2–3.4 prove lying over and incomparability. Section 6 below contains the complete DVR argument in Lemma 6.C and the separable trace-lattice, finite normalization, function-field rational-map spreading and dense equalizer arguments in Lemma 6.4. The homogeneous-coordinate extension over every DVR is written there as well.
- **AG-MO-03, Limits of schemes and Noetherian approximation:** Theorem 1.1, Lemma 2.1 and Theorems 4.1–4.2 prove affine limits, retention of finite open covers, finite-presentation descent, eventual equality of maps and descent of closed immersions. Only the proved affine approximation Theorem 5.1 is needed here. The external-proof-only absolute approximation Theorem 5.2 is not used.
- **AG-MO-06, Quasi-finite morphisms and Chevalley's theorem:** Theorem 3.1 proves the Noetherian constructibility criterion and Theorem 4.2 proves Chevalley's theorem. The argument in Section 4 below uses these particular proved statements.
- **AG-FSE, Flat morphisms:** Lemma 5.1 and Corollary 5.2 prove generic freeness over a domain, including for a finite module over a finite type algebra. No openness-of-flatness theorem is needed in Section 4.
- **AG-CA-18, Smooth algebras over a field and the Jacobian criterion:** Theorem 6.1 and Corollary 6.2 prove smooth and étale flatness over arbitrary bases. Together with AG-CA-17 Theorem 5.1 and Lemma 7.1 these give the separating generic coordinates used below.
- **AG-RG-S01, Sections 4–6:** the finite ordered Čech comparison, affine quasi-coherent vanishing and flat base change are proved before this lesson. A finite affine cover of a separated scheme therefore computes quasi-coherent cohomology by its finite section complex. The same written complex calculation treats finite and arbitrary direct sums and flat localization.
- **AG-RG-S01, Sections 7–8:** Theorems 7.2–7.3 compute every projective-space twist and the perfect monomial pairing. Proposition 8.2 and Theorem 8.4 prove finite quotients by twists, finite coherent cohomology and eventual vanishing, including the closed-immersion passage to projective schemes. Lemmas 8.1 and 8.3 give denominator extension and the ample-power immersion; Corollary 7.B makes it closed for a proper source. Theorem 8.5 treats every sufficiently large power, as applied in Section 7 below.
- **AG-QC-14, Ext sheaves and Serre duality on projective space:** Proposition 1.1, Proposition 1.2, Proposition 2.1 and Theorem 5.1 give the local Ext calculation, vector-bundle tensor-Hom comparison, local-to-global Ext spectral sequence and ambient projective Serre duality. The duality proof uses the monomial trace, a two-stage twist presentation and explicit dimension shifting. Its reference to "proper finiteness" in Theorem 4.1 is supplied here by the full projective finiteness proof in Algebra and sheaf cohomology before reductive groups, Section 8 (AG-QC-05 Theorem 2.1), without using the general proper-finiteness provider. Section 6 below also spells out the single-row local-to-global Ext comparison being consumed, using the earlier finite filtered-complex construction; no general proper duality theorem is needed here.

These references are a finite proof dependency list, not a declaration that unnamed results are ordinary foundations. In particular the further smoothness descent and connected model results are proved in Sections 3–4.

1.1. The consumed Artin–Rees and completion arguments

The following records the actual arguments of AG-CA-03 Theorem 5.1 and AG-CA-19 Theorems 3.1–3.3 before their uses. Let A be Noetherian and $I \subset A$ an ideal.

Lemma 1.1 (Artin–Rees). If $N \subset M$ are finite A-modules, there is c such that

$$N \cap I^n M = I^{n-c}(N \cap I^c M) \quad (n \geq c).$$

Proof. The Rees algebra $B = \bigoplus I^n t^n$ is generated by the finitely many generators of I in degree one, so is Noetherian by Hilbert basis. The module $\bigoplus I^n M t^n$ is finite over it, generated by any finite list of generators of M in degree zero. Its graded submodule $\bigoplus (N \cap I^n M) t^n$ is finite. Choose homogeneous generators of degrees at most c. For $n \geq c$, the degree n component is a sum of $I^{n-d}(N \cap I^d M)$, for $d \leq c$; each is contained in $I^{n-c}(N \cap I^c M)$. The reverse inclusion follows by multiplication and the submodule property. □

The consumed Krull intersection statement also follows from this proof. For a finite module over a Noetherian local ring, with I in its maximal ideal, the submodule $N = \bigcap_n I^n M$ is finite. Lemma 1.1 gives $N = IN$ by taking $n = c + 1$, since $N \cap I^n M = N$ at every index. Nakayama gives $N = 0$. This is the actual AG-CA-03 Theorem 6.1 argument used in the earlier regular-ring proof.

Lemma 1.2 (exact finite-module completion). Writing $\widehat{M} = \varprojlim M/I^n M$, completion is exact on finite modules and

$$\widehat{M} = M \otimes_A \widehat{A}.$$

Proof. For $0 \to N \to M \to L \to 0$, the quotients give

$$0 \to N/(N \cap I^n M) \to M/I^n M \to L/I^n L \to 0.$$

The kernel system has surjective transitions. Compatible lifts in the middle system can therefore be chosen successively: given a lift at stage n, take any lift of the desired next quotient element and adjust its image's difference at stage n by lifting that difference from the kernel system. This proves surjectivity on inverse limits. Injectivity and equality with the limit kernel hold coordinatewise. By Lemma 1.1, $I^n N \subset N \cap I^n M \subset I^{n-c} N$; hence the two filtrations of N are cofinal. Their limits identify explicitly by taking a sufficiently far coordinate to define a coordinate of the other system, independent of that choice by compatibility. Thus the displayed inverse-limit sequence is $0 \to \widehat{N} \to \widehat{M} \to \widehat{L} \to 0$.

A finite module has a finite presentation, since submodules of finite modules are finite over a Noetherian ring. Complete $A^r \to A^s \to M \to 0$, using the just-proved exactness. Tensor the same presentation with $\widehat{A}$, using tensor's right exactness. Finite direct sums complete coordinatewise, so both cokernels are the cokernel of the same matrix $\widehat{A}^r \to \widehat{A}^s$. This gives the canonical tensor comparison. In particular, when A is already complete, every finite module is complete. □

Lemma 1.3 (flat, faithfully flat and Noetherian completion). The ring $\widehat{A}$ is Noetherian, is complete for $I\widehat{A}$, and is A-flat. If I is contained in the Jacobson radical, the completion is faithfully flat.

Proof. Flatness follows directly from the preceding finite-module exactness. Every module is the filtered union of its finite generated submodules. A tensor element uses finitely many such generators, and an equality to zero holds in a sufficiently large finite submodule, because tensor commutes with filtered colimits by its generators and relations. For an injection $M' \to M$, put a proposed tensor kernel element in a finite submodule $N \subset M'$, and put its vanishing equality in some finite submodule of M containing the image of N. Exactness of completion for these finite modules proves injectivity there, hence kills the proposed kernel. Thus tensor with $\widehat{A}$ preserves every injection.

For faithfulness, if $M \neq 0$, choose $0 \neq x \in M$, with cyclic submodule $Ax = A/J$. Choose a maximal ideal $\mathfrak{m} \supset J$. Since $I \subset \mathfrak{m}$, the module $A/\mathfrak{m}$ is unchanged by completion. The tensor comparison gives $\widehat{A}/\mathfrak{m}\widehat{A} = A/\mathfrak{m} \neq 0$, a quotient of $(A/J) \otimes \widehat{A}$. Flatness injects the latter into $M \otimes \widehat{A}$, so this is nonzero. This proves faithful flatness and, by applying it to a tensor kernel, detection of exactness.

Apply Lemma 1.2 to $0 \to I^a \to A \to A/I^a \to 0$. The last module is unchanged by completion, since a power of I kills it. The image of $I^a \otimes \widehat{A}$ is exactly $I^a\widehat{A}$, whence

$$\widehat{A}/I^a\widehat{A} = A/I^a.$$

Writing $J = I\widehat{A}$, its powers are $I^a\widehat{A}$; these identities and the defining inverse limit prove J-adic completeness and separatedness. Their successive kernels identify $\operatorname{gr}_J \widehat{A}$ with $\operatorname{gr}_I A$. The latter is a quotient of a polynomial ring over A/I on the finite generators of I, and is Noetherian.

For any ideal $Q \subset \widehat{A}$, its initial classes form a homogeneous ideal of this Noetherian graded ring. Choose finite homogeneous generators and lift them to $q_i \in Q$, with orders d_i. For $q \in Q$, subtract at stage n a combination $\sum_i a_{i,n} q_i$ matching its residual initial class, with $a_{i,n} \in J^{n-d_i}$ and zero when $n < d_i$. The new residual is in J^{n+1}. Each coefficient series converges, since its orders tend to infinity. Its sums a_i satisfy $q = \sum_i a_i q_i$ by separatedness. Thus every ideal is finite and $\widehat{A}$ is Noetherian. No prior assertion that Q is closed was used. □

For a local ring with maximal ideal $\mathfrak{m}$, its completion is local: an element with nonzero residue is a unit in every quotient $A/\mathfrak{m}^n$, and the compatible inverses give its inverse in the limit. Projection onto the residue field is surjective by successive lifting. The preceding quotient identity makes its kernel $\mathfrak{m}\widehat{A}$. Thus Lemma 1.3 gives the exact local completion statement used by the group lessons.

Lemma 1.4 (idempotents across a nilpotent ideal). Every idempotent of R/J, for a nilpotent ideal J, has a unique idempotent lift to R.

Proof. First suppose $J^2 = 0$. For a lift e, put $\delta = e^2 - e \in J$. Then $e' = e - (2e-1)\delta$ is idempotent: expand its square and use $\delta^2 = 0$ and $(2e-1)^2 = 1 + 4\delta$. For two idempotent lifts the difference $u \in J$ satisfies $(2e-1)u = 0$; the element $2e-1$ is a unit since its square is one, so $u = 0$. Apply this square-zero argument successively to J^a/J^{a+1}, starting with $a = 1$; each is square-zero because $2a \geq a + 1$, and nilpotence makes the list finite. □

2. Formal functions on a projective scheme

Let A be Noetherian, $I \subset A$ an ideal, $X \to \operatorname{Spec} A$ projective and F coherent. Write

$$B = \bigoplus_{n \geq 0} I^n t^n \subset A[t], \qquad M_n^q = H^q(X, I^n F), \qquad M^q = M_0^q.$$

The transition maps $M_m^q \to M_n^q$, for $m \geq n$, come from inclusion of ideal powers. Multiplication $I^d \otimes M_n^q \to M_{n+d}^q$ is compatible with these transitions: both sheaf maps are multiplication followed by the same inclusion. Applying cohomology preserves their equality.

Lemma 2.1. The graded B-module $\bigoplus_{n \geq 0} M_n^q t^n$ is finite for every q.

Proof. Generators of I, placed in degree one, generate B as an A-algebra; Hilbert basis makes B Noetherian. Let $X_B = X \times_A \operatorname{Spec} B$, with affine projection π. On $U = \operatorname{Spec} R \subset X$, put $F|_U = \widetilde{N}$. The graded module $\bigoplus I^n N$ is finite over $R \otimes_A B$: finitely many R-generators of N, placed in degree zero, generate it. Here $R \otimes_A B \to \bigoplus I^n R$ is a surjection, and need not be an isomorphism. No flatness assumption is being made.

Localization gives the same modules on overlaps. They glue to a coherent sheaf G on X_B, with $\pi_* G = \bigoplus I^n F$. Coherence holds because $R \otimes_A B$ is Noetherian and the module is finite. Base

changing a projective closed immersion shows that X_B is projective over B; AG-QC-05 Theorem 2.1 and its closed-immersion passage give finite $H^q(X_B, G)$ over B.

Use the inverse images of a finite affine cover of X. Each intersection is affine on both schemes, and its sections of G are the sections of $\bigoplus I^n F$ on the original intersection. Thus their ordered section complexes are identical. Finite products in that complex commute with direct sums, and direct sums of modules are exact. Taking cohomology therefore identifies

$$H^q(X_B, G) = \bigoplus_{n \geq 0} H^q(X, I^n F),$$

with the graded B-action preserved. This proves finiteness. □

This writes the projective case of the free proofs Stacks 02O8 and 0897. General proper coherent pushforward and its dévissage are unnecessary: both spaces just used are projective.

Put

$$J_n^q = \operatorname{im}(M_n^q \to M^q), \qquad K_n^q = \ker(M_n^q \to M^q).$$

Lemma 2.2. There are constants $a_q, b_q \geq 0$ such that

$$I^n M^q \subset J_n^q \subset I^{n-a_q} M^q \quad (n \geq a_q),$$

and $K_m^q \to K_n^q$ is zero whenever $m \geq n + b_q$.

Proof. Multiplication by $u \in I^n$ on F factors through $I^n F$, proving the first inclusion. The image and kernel of the graded map from Lemma 2.1 to the corresponding degreewise copies of M^q are graded subquotients of a finite B-module. More explicitly, multiplication and transition compatibility show that $\bigoplus J_n^q t^n$ is the graded image module and $\bigoplus K_n^q t^n$ is a graded submodule of $\bigoplus M_n^q t^n$. Both are finite, because B is Noetherian.

Choose homogeneous generators of the image module of degrees $d \leq a_q$. Its degree n is a sum of terms $I^{n-d} J_d^q \subset I^{n-a_q} M^q$, proving the other inclusion. Choose homogeneous kernel generators $z_j \in K_{d_j}^q$, with $d_j \leq b_q$. An element in degree m is a sum of $u_j z_j$, where $u_j \in I^{m-d_j}$. If $m \geq n + b_q$, express u_j as a sum of products vw, $v \in I^n$, $w \in I^{m-d_j-n}$. Transition of vwz_j to M_n^q factors as transition of z_j to M_0^q, followed by multiplication by w, then by v into M_n^q. The first map kills z_j. Every such product, and hence the transition, is zero. □

Theorem 2.3. There is a canonical isomorphism

$$H^q(X, F)^\wedge_I \xrightarrow{\sim} \varprojlim_n H^q(X, F/I^n F).$$

It is a homeomorphism for the quotient inverse-limit topologies.

Proof. Write $T_n^q = H^q(X, F/I^n F)$. The sheaf short exact sequence gives compatible exact sequences

$$0 \longrightarrow M^q/J_n^q \longrightarrow T_n^q \longrightarrow K_n^{q+1} \longrightarrow 0.$$

For $m \geq n + b_{q+1}$, the transition on the last term is zero. Hence the image of $T_m^q \to T_n^q$ is contained in the first term M^q/J_n^q. It equals that term: transition $M^q/J_m^q \to M^q/J_n^q$ is onto and these modules inject into T_m^q, T_n^q. This proves the Mittag–Leffler assertion and its stable image, with a written uniform bound.

If (u_n) is a compatible family in T_n^q, its image in K_n^{q+1} is zero at every coordinate, because that image is the transition of a sufficiently distant kernel term. Thus u_n has a unique preimage in M^q/J_n^q; uniqueness makes those preimages compatible. Consequently

$$\varprojlim_n T_n^q = \varprojlim_n M^q/J_n^q.$$

Lemma 2.2 gives $I^n M^q \subset J_n^q$ and $J_{n+a_q}^q \subset I^n M^q$. Projection and projection from the shifted index define inverse maps between the two quotient limits. A common finer index proves that their composites are the identity. Each map is continuous coordinate by coordinate, proving the topological assertion. Every construction is the canonical quotient map, so the isomorphism is the asserted one. □

This supplies the critical image, kernel and limit arguments of Stacks 02OA, 02OB and 02OC. In particular, the proof uses the kernel transition bound directly; it does not rely on the incorrect part-number reference presently printed in 02OB. No inverse limit has been interchanged with an unexplained cohomology operation.

Corollary 2.4. If A is complete local Noetherian, every compatible idempotent on the maximal-ideal thickenings of a projective X comes from a unique idempotent of $\Gamma(X, \mathcal{O}_X)$.

Proof. Projective finiteness makes $M = \Gamma(X, \mathcal{O}_X)$ finite. Lemma 1.2 gives $M = M \otimes_A \widehat{A} = \widehat{M}$. Theorem 2.3 identifies this ring with the limit of the thickened section rings. The equality $e^2 = e$ holds in this limit coordinatewise and hence holds for the corresponding actual section. Uniqueness follows from the same ring isomorphism. Idempotents at successive levels are compatible whenever they lift the same initial idempotent, by Lemma 1.4's unique nilpotent lifting. □

3. Smoothness and line bundles at a finite stage

Lemma 3.N (integer finite-stage rings). Every ideal of $\mathbf{Z}$ is principal. Hence $\mathbf{Z}$, its finite-type algebras and their localizations are Noetherian.

Proof. For $d > 0$, induction on a nonnegative integer a gives $a = qd + r$ with $0 \leq r < d$: increment the remainder at each step, replacing d by zero and incrementing the quotient when that remainder reaches d. Negating this equality, and adjusting the quotient and remainder when $r \neq 0$, gives division with such a remainder for every integer a. A nonzero ideal contains a positive integer; let d be its least positive member. The remainder on dividing any member by d belongs to the ideal and is smaller than d, so it is zero. Thus the ideal is $d\mathbf{Z}$; the zero ideal is principal too. The Noetherian module criterion in AG-CA, *Noetherian rings and modules*, Theorem 1.1, now applies to $\mathbf{Z}$. Its Theorem 2.1 proves the assertions for finite-type algebras and localizations. □

Lemma 3.M (finite character presentations). Every finitely generated abelian group is a direct sum of a finite-rank free group and finitely many finite cyclic groups. In particular a finitely generated torsion-free abelian group is free. If M is finitely generated and R is any ring, $R[M]$ is finitely presented and faithfully flat over R. On a nonempty base, finite type of a diagonalizable group conversely forces its exponent group to be finitely generated.

Proof. Choose a surjection $\mathbf{Z}^n \to M$. Its kernel is finitely generated: Lemma 3.N makes $\mathbf{Z}$ Noetherian, and the Noetherian module criterion and submodule proof in AG-CA-03, Theorem 1.1 and Proposition 1.2, apply to $\mathbf{Z}^n$. Write the chosen kernel generators as columns of a finite integer matrix C.

Integer row and column operations diagonalize C. If the matrix is nonzero, take the least positive absolute value of a nonzero entry among all matrices obtainable by invertible integer row and column operations. This least value exists among positive integers. In a matrix attaining it, move such an entry to position $(1, 1)$ and change its sign to make it positive, say d. Every other entry in its first column is divisible by d: otherwise division with remainder and subtraction of a multiple of the first row would give an equivalent matrix with an entry strictly between zero and d. Clear that column by these row operations. The same argument with column operations makes every entry in the first row divisible by d and clears that row. Those column operations leave the cleared column

below d zero. Repeat in the smaller remaining block. Induction gives a diagonal matrix with positive nonzero entries $d_1, \dots, d_s$ and a zero block. Row operations change the basis of $\mathbf{Z}^n$; column operations change the generating list of its kernel without changing that subgroup. Therefore its quotient is $\mathbf{Z}^{n-s} \oplus \bigoplus_{i=1}^{s} \mathbf{Z}/d_i$. Terms with $d_i = 1$ are zero. Every term with $d_i > 1$ has a nonzero torsion element, proving the torsion-free assertion.

For finite presentation one can use the kernel generators directly. If they are $v_1, \dots, v_a \in \mathbf{Z}^n$, then

$$R[M] = R[z_1^{\pm 1}, \dots, z_n^{\pm 1}]/(z^{v_j} - 1)_{j=1}^{a}.$$

Indeed $z^{v+w} - 1 = z^v(z^w - 1) + (z^v - 1)$ and $z^{-v} - 1 = -z^{-v}(z^v - 1)$ put the relation for every kernel vector in this ideal. Distinct cosets give distinct exponent basis elements in $R[M]$. A linear combination mapping to zero has coefficient sum zero within each coset; subtracting a representative monomial in that coset expresses it as a sum of the relations just described. Thus the displayed algebra map has exactly this kernel. Present the Laurent algebra by the $2n$ variables z_i, w_i and the n equations $z_i w_i - 1$; negative powers in the finitely many further equations are represented by the w_i. This is a finite polynomial presentation over R.

The coordinate module $R[M] = \bigoplus_{m \in M} Re^m$ is free. Tensoring an injection acts separately on each exponent summand, proving flatness. Its zero-exponent summand is R, so tensoring a nonzero module remains nonzero, proving faithfulness. Finally, over a nonzero affine base ring, a finite list of algebra generators of $R[M]$ involves only finitely many exponents. Its generated algebra lies in the subgroup algebra generated by those exponents. Independence of the exponent basis forces that subgroup to be all of M. A nonempty base has a nonzero affine chart, so the converse follows there. □

Lemma 3.F (differentials over a perfect field). Let k be perfect, let $t_1, \dots, t_d$ be algebraically independent over k, and let K be a finite extension of $F = k(t_1, \dots, t_d)$. Then $\dim_K \Omega_{K/k} = d$. No separability assumption on K/F is required.

Proof. In characteristic zero, express the finite extension as a finite tower adjoining algebraic elements. An irreducible polynomial has nonzero derivative, and the Euclidean algorithm makes its derivative nonzero at its root. Differentiating that polynomial extends a derivation uniquely across the root: if $f(X) = \sum_i a_i X^i$, its value at a root u must be $-\sum_i \delta(a_i) u^i / f'(u)$. This value also gives a derivation of the polynomial quotient, because it kills its defining relation, and extends to the field by the quotient rule. Iterating the tower proves $\Omega_{K/k} = K \otimes_F \Omega_{F/k}$. Polynomial differentials and localization make the latter module free on $dt_1, \dots, dt_d$.

Suppose the characteristic is $p > 0$. Since k is perfect, $F^p = k(t_1^p, \dots, t_d^p)$. The monomials $\prod_i t_i^{a_i}$ with $0 \le a_i < p$ form a basis of F over F^p: polynomial exponents split uniquely into their multiples of p and their remainders, and localization preserves this basis. Inverting polynomials over the smaller polynomial ring gives all of F, since $h^{-1} = h^{p-1}/h^p$. Hence $[F : F^p] = p^d$. Frobenius is an isomorphism from the extension K/F to K^p/F^p, so $[K^p : F^p] = [K : F]$. Multiplication of finite tower bases, with independence checked by grouping coefficients in the last basis, gives $[K : F^p] = [K : F]p^d = [K : K^p][K^p : F^p]$. Therefore $[K : K^p] = p^d$.

Starting with K^p, adjoin an element u whenever the current field E is smaller than K. Its pth power belongs to E. Its monic minimal polynomial divides $(X - u)^p$ over K, so is $(X - u)^r$ there. If $1 \le r < p$, its coefficient $-ru$ at degree $r - 1$ would put u in E, a contradiction. Thus each nontrivial step has degree p. Exactly d steps produce elements $u_1, \dots, u_d$ with the monomials $\prod_i u_i^{a_i}$, $0 \le a_i < p$, a basis of K/K^p. They give the algebra presentation $K = K^p[U_1, \dots, U_d]/(U_i^p - u_i^p)_{i=1}^{d}$: reduction of powers gives a spanning list of size p^d, and the displayed basis makes it independent. Every k-derivation kills K^p, while $k \subset K^p$. Consequently $\Omega_{K/k} = \Omega_{K/K^p}$. The differential presentation has zero differentials for all its defining relations, so this module is free on $du_1, \dots, du_d$. This proves the formula, including $d = 0$. □

Polynomial calculations for the smooth charts. Over a field k, division by a nonzero polynomial $g \in k[T]$ is proved by repeatedly subtracting the multiple of g that cancels the current

leading term. The degree strictly decreases, so induction gives $f = qg + r$ with $\deg r < \deg g$, or $r = 0$. In a nonzero ideal choose a monic polynomial of least degree; dividing any other member by it makes the remainder a member of smaller degree and hence zero. Thus every ideal of $k[T]$ is principal. A nonzero prime ideal has an irreducible monic generator: a factorization would contradict primality and least degree. Conversely, if g is irreducible and does not divide f, the principal ideal (g, f) has a generator dividing both, which must be a unit; a relation $ag+bf = 1$ then shows that every nonzero class modulo g is invertible. The nonzero prime ideals are consequently maximal, and the localizations at the zero prime and at a nonzero prime have dimensions zero and one respectively.

For $k[T_1, \dots, T_D]$, order monomials lexicographically. The largest monomial in a product is the product of the two largest monomials, with nonzero coefficient equal to the product of their coefficients. Hence this polynomial ring is a domain. Its degree-j monomials correspond to nonnegative integer lists $(a_1, \dots, a_D)$ with sum j. For $D > 0$, inserting $D - 1$ separators among j identical marks counts these lists as $\binom{j+D-1}{D-1}$. Summing from $j = 0$ to $n - 1$, Pascal's identity and induction give $\binom{n+D-1}{D}$. For $D = 0$, only degree zero contributes. These are the polynomial-domain and growth calculations used in the regular-parameter proof.

The cotangent map used there also follows directly from the product rule. Let $P = k[X_1, \dots, X_n]$, let $B = P_{\mathfrak{Q}}$ have maximal ideal $\mathfrak{m}$ and residue field κ, and extend the polynomial derivation by $d(a/u) = u^{-1}da - au^{-2}du$. In $\Omega_{P/k} \otimes_P \kappa$, the differential of a product of two elements of $\mathfrak{m}$ is zero. For $a \in B$ and $f \in \mathfrak{m}$, the same rule gives $d(af) = \bar{a}\, df$. Therefore $[f] \mapsto df$ defines a κ-linear map from $\mathfrak{m}/\mathfrak{m}^2$. An invertible Jacobian minor makes the corresponding differential vectors independent, so their cotangent classes are independent. This argument requires no separability of κ/k, and does not assert that the whole cotangent map is injective.

Flatness on affine charts. A ring map $R \to S$ is flat if every $S_{\mathfrak{q}}$ is flat over $R_{\mathfrak{p}}$, where $\mathfrak{p}$ is the contraction of $\mathfrak{q}$. Indeed, for an injection $U \to V$ of R-modules let K be the kernel after tensoring with S. This is an S-module. Exact localization identifies $K_{\mathfrak{q}}$ with the kernel after tensoring with $S_{\mathfrak{q}}$. That kernel is zero: $R_{\mathfrak{p}}$ is flat over R, and composition with the assumed flat map to $S_{\mathfrak{q}}$ preserves the injection. Local detection makes $K = 0$, proving flatness. Conversely, if S is flat over R, localization and composition make $S_{\mathfrak{q}}$ flat over R. Tensoring $R_{\mathfrak{p}}$-modules with $S_{\mathfrak{q}}$ over R or over $R_{\mathfrak{p}}$ gives the same module, since the inverted elements already act invertibly. Hence $S_{\mathfrak{q}}$ is flat over $R_{\mathfrak{p}}$. This proves both directions over arbitrary rings.

Lemma 3.0 (the split conormal criterion). For a polynomial presentation $C = P/J$, the algebra is formally smooth precisely when $d : J/J^2 \to \Omega_{P/B} \otimes_P C$ has a C-linear left inverse.

Proof. Formal smoothness lifts the identity of C to a section $s : C \to P/J^2$. The map $D(p) = p \bmod J^2 - s(p \bmod J)$ takes values in J/J^2, satisfies the product rule because this ideal is square-zero, and sends $j \in J$ to its class. The universal derivation for the polynomial ring therefore factors it through $\Omega_{P/B} \otimes C$, giving a left inverse of d. Conversely a left inverse gives this derivation by composition with the polynomial universal derivation. The map $p \mapsto p \bmod J^2 - D(p)$ is multiplicative by the product rule and the square-zero condition. It kills J and is the identity after reduction to C, so gives a section. In any square-zero lifting test, lift the polynomial variables arbitrarily. The resulting map from P sends J into the square-zero test ideal and kills J^2; composing its factor through P/J^2 with the section gives the required lift. Polynomial differentials are free on the variable symbols by the product rule, so no unproved splitting assertion enters this argument. □

Lemma 3.A (maps between smooth schemes). Let X and Y be smooth schemes locally of finite presentation over any scheme S, and let $h : X \to Y$ be an S-map. If the map on relative tangent spaces at a point x, with scalars extended to $\kappa(x)$, is surjective, then h is smooth near x. If that map is an isomorphism, then h is étale near x. In particular, on a smooth morphism of relative dimension d, local functions $q_1, \dots, q_d$ whose differentials form a basis at x give an étale coordinate map to $\mathbb{A}^d_S$ near x.

Proof. Work on compatible affine neighbourhoods of x, $h(x)$ and their image in S. Write their algebras as C, D, A. The map $D \to C$ is finitely presented: choose finite presentations of D and C over A, and add, to the presentation of C, the finitely many equations equating each generator of D with a polynomial representing its image in C. This gives a finite presentation over D.

The split conormal criterion makes $\Omega_{C/A}$ and $\Omega_{D/A}$ finite projective. Indeed, in a finite polynomial presentation the conormal injection has a left inverse, so its cokernel is a direct summand of the finite free module of polynomial differentials. Such a summand is flat, and the local finite-freeness proof in AG-RG-S01, Lemma 1.2, applies. Shrink the affine neighbourhoods to trivialize these modules. Dualizing the asserted tangent surjection gives an injection on the fibre of

$$u : C \otimes_D \Omega_{D/A} \longrightarrow \Omega_{C/A}.$$

A full-size minor of its matrix is nonzero at x; invert it. Projecting to the corresponding rows and multiplying by that square matrix's inverse gives a C-linear left inverse of u. In the tangent-isomorphism case its matrix is square and invertible after this shrink.

Consider an arbitrary square-zero surjection $R \to R/J$, an A-algebra map $C \to R/J$, and a prescribed D-algebra structure on R agreeing with it after reduction. Formal smoothness of C/A, as proved by Lemma 3.0, supplies an A-algebra lift $v : C \to R$. Its restriction to D differs from the prescribed map by an A-derivation $\delta : D \to J$: expanding a product leaves the two first-order terms, since $J^2 = 0$. It corresponds to a map $C \otimes_D \Omega_{D/A} \to J$, where the C-action on J is through the given reduced map. Compose it with the left inverse of u to obtain $\Omega_{C/A} \to J$, hence an A-derivation $\widetilde{\delta} : C \to J$. The map $v - \widetilde{\delta}$ is again an A-algebra map by the same square-zero product expansion. Its restriction to D is the prescribed map. Thus C/D is formally smooth. Its finite presentation makes it smooth.

If u is an isomorphism, two such D-lifts differ by a derivation $C \to J$ whose differential map vanishes on the image of u. That image is all of $\Omega_{C/A}$, so the lifts agree. The map is formally étale and finitely presented, hence étale. The coordinate assertion is this case with $Y = \mathbb{A}^d_S$: its pulled-back differentials have basis $dq_1, \ldots, dq_d$, and the given basis condition makes u an isomorphism near x. No Noetherian, reducedness, characteristic or residue-field separability hypothesis was used. □

Lemma 3.1. Let $A = \varinjlim A_i$ be a filtered ring colimit. A finitely presented algebra map $B_0 \to C_0$ over A_0 whose base change over A is smooth becomes smooth after a finite stage.

Proof. Present $C_0 = B_0[x_1, \ldots, x_r]/(f_1, \ldots, f_s)$. At stage i, let $P_i = B_i[x_1, \ldots, x_r]$, $J_i = (f_{1,i}, \ldots, f_{s,i})$ and $C_i = P_i/J_i$. Lemma 3.0, the split conormal criterion of AG-CA-17 Theorem 3.1, says that the map

$$J/J^2 \xrightarrow{d} C^r$$

over the colimit has a C-linear left inverse ρ. Each $\rho(dx_k)$ is a finite sum of classes of the f_j with coefficients in C. Lift these finitely many coefficients to one stage and define $\rho_i : C_i^r \to J_i/J_i^2$ by those images. The equations $\rho_i(df_{j,i}) = f_{j,i} \bmod J_i^2$ are finitely many equations. Their truth in the colimit is witnessed by finite sums of products of the f_j, representing membership in J^2. Lift these witnesses and enlarge the stage until the polynomial equalities hold. Then $\rho_i d$ fixes all generators of J_i/J_i^2, so is its identity. The same criterion gives formal smoothness at that stage, and the existing finite presentation gives smoothness. □

This is the actual finite witness proof in Stacks 0C0B.

Proposition 3.2. A smooth scheme of finite presentation over an affine filtered limit has a smooth finite-presentation model at a finite stage. A projective embedding and a specified line bundle can be retained simultaneously.

Proof. First descend the finite-presentation scheme, the embedding if specified, and finitely many affine source and target charts by AG-MO-03 Theorem 4.1. Retain the closed immersion by its

Theorem 4.2. Choose a finite affine cover of the original source on which the chart algebra maps are smooth. Descend these opens and the maps, and retain their covering property by Lemma 2.1 of that lesson. Apply Lemma 3.1 on the finitely many chart maps and take a common later stage. Smoothness is now true on a cover, hence globally.

For an invertible sheaf, refine the cover so that it is trivial on each chart. On finite affine overlap covers record each transition unit and its inverse. Record their inverse equations, their agreements on overlap refinements, and the cocycle equations on finite covers of triple overlaps. The sections and their finite equality witnesses descend by the section and eventual-equality parts of AG-MO-03. At a common stage these equations hold, so gluing the trivial rank-one sheaves gives an invertible sheaf there. Retain the finite chart covers, smoothness and the projective closed immersion at that same stage. Pullback recovers the original sheaf and scheme. □

This fills the finite-cover step omitted in Stacks 0C0C. It proves only the finite chart and line bundle case being used, without importing general coherent sheaf descent or a general perfection theorem.

4. Geometrically connected smooth affine models

The model needed for the normalizer argument is smooth and affine. We prove this case; the unrestricted finite-presentation connected-fibre theorem is not needed.

Lemma 4.C (Noetherian constructibility and images). In a Noetherian space, a subset is constructible exactly when it is a finite union of locally closed subsets. Equivalently, its intersection with every irreducible closed subset is either not dense or contains a nonempty open of that subset. The image of a constructible subset under a finite type morphism between Noetherian schemes is constructible.

Proof. First every Noetherian space is a finite union of irreducible closed subsets. Otherwise choose, by the descending chain condition, a minimal closed counterexample. It is reducible, hence the union of two proper closed subsets; minimality expresses each as a finite union of irreducible closed subsets, a contradiction. Removing contained members from such a finite list gives its finitely many irreducible components. Every open of a Noetherian space is quasi-compact: among finite unions of members of an open cover choose a maximal one by the ascending chain condition on opens, and maximality forces it to be the whole covered open. Thus all opens are retrocompact. A difference of two opens is locally closed, and an open intersected with a closed subset is such a difference. This proves the first description. On a Noetherian affine scheme, a locally closed subset is a finite union of pieces $V(J) \cap D(g)$: write its open part as a finite union of principal opens and its closed part as $V(J)$, with J finite by Noetherianity.

For the criterion, a dense finite union of locally closed subsets of an irreducible closed Z has one member with closure Z. That member is a nonempty open in Z, proving necessity. For sufficiency, choose a minimal closed F on which the given subset E has nonconstructible intersection. If F is reducible, its finitely many proper irreducible components have constructible intersections by minimality, giving the contradiction on their union. If $E \cap F$ is not dense, use its proper closed closure and minimality. Otherwise the assumed criterion supplies a nonempty open $W \subset F$ lying in E. The rest lies in the proper closed $F \setminus W$ and is constructible by minimality. Together with W it makes $E \cap F$ constructible, the final contradiction. In these steps a subset locally closed in a closed subset is locally closed in the ambient space, so each finite description remains valid there.

We prove the needed generic-image assertion. For domains $A \subset B$ with B of finite type over A, put $K = \operatorname{Frac} A$. The integral domain B_K is nonzero. The complete Noether-normalization proof in AG-CA-09 Corollary 3.2 supplies $K[t_1, \dots, t_d] \subset B_K$ over which B_K is finite. Choose finite algebra generators b_j of B. After inverting a common nonzero $a \in A$, all t_i belong to B_a, and the finitely many monic equations for the b_j have coefficients in $A_a[t_1, \dots, t_d]$ and hold in B_a. Their equalities hold after such a localization because an element zero in B_K is killed by a nonzero base

denominator. Algebraic independence keeps the polynomial subring injective. The monic equations express every monomial in the b_j as a combination of finitely many bounded-exponent monomials, so B_a is finite and integral over this subring, exactly as in AG-CA-05 Theorem 1.1. Its complete lying-over proof, Theorem 3.2, makes its spectrum surject onto that polynomial spectrum. The latter surjects onto $\operatorname{Spec} A_a$, since a polynomial ring over each residue field is nonzero and has a prime. Hence the image contains $D(a)$.

Now let $f : X \to Y$ be finite type between Noetherian schemes. For an irreducible closed $Z \subset Y$ with $f(X) \cap Z$ dense, the closed inverse image has finitely many irreducible components. Their image closures cover Z, so one is dominant. Give it and Z reduced structures. Choose an affine open in Z and a nonempty affine open in its inverse image on that component. This remains a dominant finite type map of integral schemes: their generic points are retained, and restricting to an affine open is quasi-compact in a Noetherian scheme. Its ring map is an inclusion of domains. The generic-image assertion supplies a nonempty open of Z in $f(X)$. The criterion just proved therefore makes the full image constructible. Finally express a constructible subset of X as a finite union of locally closed subsets. Their reduced locally closed structures map to Y by finite type morphisms: a closed immersion is finite type and the open part is quasi-compact on a Noetherian scheme. Apply the full-image result to each and take the finite union. This proves the whole assertion. □

Lemma 4.1. Over an algebraically closed field k, an affine finite type connected scheme remains connected after every field extension. A smooth affine finite type scheme has disjoint irreducible components, and each connected component is integral and remains integral after every field extension of an algebraically closed ground field.

Proof. Connectedness of $\operatorname{Spec} B$ is equivalent to absence of nontrivial idempotents. Indeed an idempotent gives the disjoint opens $D(e), D(1-e)$; conversely a disjoint open-and-closed decomposition gives a section with values one and zero, hence an idempotent by affine sections.

Suppose an extension L/k supplies a nontrivial idempotent $e \in B \otimes_k L$. Express e and $1-e$ using finite lists of k-linearly independent elements of B, with coefficients in L. Let $R \subset L$ be the finite type k-algebra generated by these coefficients, inverted nonzero coefficients ensuring that both expressions stay nonzero, and the finitely many coefficients needed for $e^2 = e$. The relation already holds over R, since $B \otimes_k R \to B \otimes_k L$ is injective. The nonzero finite type algebra R has a k-valued point by the proved Nullstellensatz in AG-CA-06. Specializing at it gives a nontrivial idempotent in B, a contradiction.

For smooth B, every local ring is a regular local domain by AG-CA-18 and AG-CA-14. Two distinct irreducible components cannot meet at a point, since their distinct minimal primes would give distinct minimal primes in that local ring. There are finitely many irreducible components by the Noetherian decomposition proved in Lemma 4.C; they are consequently both closed and open. Smoothness also makes the ring reduced. Each connected component is thus reduced and irreducible, hence integral. After extending an algebraically closed field, connectedness just proved and smoothness preserved by base change give the same conclusion. □

It follows that the number of geometric connected components of a smooth affine finite type scheme is invariant under field extensions. To check this, embed the chosen algebraic closures of the two fields into a common algebraically closed extension. Apply Lemma 4.1 to each of the finitely many open-and-closed components. Extension is faithfully flat and preserves nonemptiness, so each component remains one component.

Lemma 4.A (field and integer calculations before primitive elements). Let F be any field. Polynomial division and Bezout hold in $F[T]$. Minimal-polynomial quotients, distinct-root bounds, the derivative test for a separable algebraic generator, finite splitting fields, and finite tower degree and embedding counts have the following direct proofs. Every finite subgroup of a field's units is cyclic. Every integer ideal is principal and $\mathbf{Z}$ is Noetherian. The integer division, Bezout and prime-factorization facts needed for the group calculation are proved here as well.

Proof. First give the integer arithmetic used in the finite-group calculation. For integers z and $d > 0$, there are q, r with $z = qd + r$ and $0 \le r < d$. For $z \ge 0$, choose the largest q in the finite set $\{0, \dots, z\}$ with $qd \le z$; if the remainder were at least d, the next integer would contradict maximality. For $z < 0$, divide $-z$; negate its quotient when the remainder is zero, and otherwise use quotient $-q-1$ and remainder $d-r$. Thus integer division also covers negative integers. A nonzero subgroup $J \subset \mathbf{Z}$ contains a positive integer and has a least positive member d. The remainder on dividing any member of J by d is in J and smaller than d, so is zero. Hence $J = d\mathbf{Z}$. The zero ideal is generated by zero, so every integer ideal is principal. For an ascending chain of integer ideals, its union is an ideal generated by one integer d. If $d = 0$, every member is zero. Otherwise d lies in one member, which then contains the whole union, so the chain stabilizes. This proves that $\mathbf{Z}$ is Noetherian. For a, b not both zero, apply this to the subgroup of integer combinations $ua + vb$. Its positive generator d divides both a, b, every common divisor divides d, and $d = ua + vb$ for some integers u, v. This proves the positive gcd and Bezout identity. If $\gcd(a, b) = 1$ and $a \mid bc$, multiplication of a Bezout identity by c gives $a \mid c$. A positive prime integer is consequently a prime element. Every integer $n > 1$ factors into positive primes by induction: if it is composite, write it as a product of two smaller positive integers and factor those. Prime divisibility and integer cancellation prove uniqueness. The integer 1 has the empty factorization. The least common multiple of a nonempty finite list of positive integers is therefore the product of the largest occurring powers of each prime; it is 1 if all entries are 1. Different prime powers have gcd one. For coprime positive a, b, the divisibilities $a \mid r$ and $b \mid r$ imply $ab \mid r$: write $r = bc$ and apply the coprime divisibility assertion to obtain $a \mid c$. These are all the integer facts required below.

If $g \neq 0$ and f has degree at least that of g, subtract the leading coefficient of f divided by that of g, times the appropriate power of T times g. This strictly decreases the degree. Induction gives $f = qg + r$ with $r = 0$ or $\deg r < \deg g$. The leading coefficient of a product of two nonzero polynomials is the product of their nonzero leading coefficients, so degrees add. Consequently two such division expressions have the same remainder and quotient. For polynomials a, b not both zero, repeat division on the pair and its successive remainders. Nonnegative degrees strictly decrease, so the process ends. Its last nonzero remainder, scaled to be monic, is d. Back substitution expresses $d = ua + vb$. Each earlier remainder is divisible by d, and any common divisor of a, b divides every remainder, so d is their monic greatest common divisor. In particular coprime polynomials satisfy $ua + vb = 1$. An irreducible f is prime: if f does not divide a, Bezout for f, a, multiplied by b, shows that $f \mid ab$ forces $f \mid b$. Factor existence follows by induction on degree, and prime divisibility gives uniqueness after fixing monic factors and a scalar.

For an algebraic element α, choose a monic nonzero annihilating polynomial of least degree, denoted f. Division shows that every annihilating polynomial is a multiple of f. If f were a product of two nonconstant factors, evaluating in the extension field would make one smaller factor vanish, a contradiction. Thus f is irreducible. Every nonzero class modulo f has a polynomial inverse by Bezout, so $F[T]/(f)$ is a field. Evaluation gives $F[\alpha] = F(\alpha)$ and identifies it with this quotient. The remainders $1, T, \dots, T^{\deg f - 1}$ form a basis: they span by division, and a linear combination of smaller degree cannot be a nonzero multiple of f.

Division by $T - c$ leaves the constant $f(c)$. If c is a root, $f = (T - c)g$. Every other root is a root of g, since its difference from c is a nonzero field element. Induction proves that a nonzero polynomial of degree d has at most d distinct roots, over every extension field. A finite collection of polynomials has a finite splitting field: if one polynomial has a nonsplit nonconstant factor, factor it into irreducibles by the degree induction, adjoin a root of one such factor using the just-proved field quotient, remove the linear factor, and repeat. At most the sum of the original degrees many roots need be adjoined. The extensions at every step are finite by the displayed power basis, and adjoining these roots therefore yields a finite extension in which every original polynomial splits.

The formal derivative obeys the product rule by expanding monomials. In a splitting field, if $f = (T - c)g$, then $f'(c) = g(c)$. Hence c is a repeated root exactly when f and f' both vanish at c. For an irreducible f with $f' \neq 0$, the derivative has smaller degree, so Bezout makes f, f' coprime

and there are no repeated roots. If $f' = 0$ and f is nonconstant, every root is repeated. Thus the minimal polynomial of an algebraic element is separable precisely when its derivative is nonzero, equivalently when $f'(\alpha) \neq 0$.

The square-matrix test used with the differential presentation also follows directly over a field. Define the determinant by its finite signed permutation sum. That sum is linear in each row and each column. With two equal rows its terms cancel in pairs under their transposition, so it is zero; row interchange changes its sign. Consequently adding a multiple of another row leaves it unchanged, and scaling one row multiplies it by the scale. The same assertions hold for columns. Starting with an n by n matrix, move a nonzero entry to the first position by row and column exchanges, scale it to one, and clear the other entries in that row and column by these additions. Repeat on the remaining square block. Every operation is invertible and alters the determinant by a nonzero scalar. The result is a diagonal matrix with some entries one and the others zero. If any zero entry remains, that output coordinate cannot occur in the image, so the map is not surjective. Hence a surjective square map has only ones and has nonzero determinant. Conversely such a diagonal matrix is invertible. The determinant of a triangular matrix is the product of its diagonal entries: expand in the first column for an upper triangular matrix, or in the first row for a lower triangular matrix, and induct. For size zero the determinant is one and the map is surjective.

If $u_1, \ldots, u_r$ is an F-basis of a finite extension E and $v_1, \ldots, v_s$ is an E-basis of L, the $u_i v_j$ span L over F. A relation between them is first a relation between the v_j over E and then, coefficient by coefficient, a relation between the u_i over F. Both sets of coefficients are zero. Thus they form an F-basis and $[L : F] = [L : E][E : F]$. A finite extension has a finite generating list, for example any finite F-basis, and hence a finite simple-generator tower.

To count embeddings, fix a finite generating list $\alpha_1, \ldots, \alpha_m$ of L/F and a finite splitting field H/F of their minimal polynomials over F. Put $E_i = F(\alpha_1, \ldots, \alpha_i)$. The minimal polynomial f_i of α_i over E_{i-1} divides its minimal polynomial over F, by polynomial division. Under any F-embedding $\sigma : E_{i-1} \to H$, the polynomial $\sigma(f_i)$ therefore divides that already split F-polynomial and splits in H. An extension of σ to $E_i = E_{i-1}[T]/(f_i)$ is uniquely specified by the image of α_i, and each distinct root of $\sigma(f_i)$ defines such an extension by evaluation. The evaluation map is injective because its domain is a field. There are at most $[E_i : E_{i-1}]$ such roots, and exactly that many if f_i is separable. An element separable over F stays separable over an intermediate field because its new minimal polynomial divides the old one. Thus a separable extension L/F has exactly $[L : F]$ embeddings into H, by multiplying the tower counts. Conversely, if every step is separable, these counts also give $[L : F]$ embeddings. If some $\beta \in L$ were inseparable over F, its minimal polynomial would have fewer than $[F(\beta) : F]$ distinct roots. Enlarge H to split that polynomial and all polynomials for a generator tower above $F(\beta)$; no new embeddings of L appear, since every generator image was already a root in the original H. Restricting to $F(\beta)$ and applying the upper bound on extensions gives fewer than $[F(\beta) : F][L : F(\beta)] = [L : F]$ embeddings, a contradiction. This proves the transitivity of separability and all finite embedding counts used here. Only a finite splitting field was required.

Let G be any finite subgroup of a field's units. The order of $g \in G$ is the least positive d with $g^d = 1$, which exists by equality of two powers in the finite group. Integer division shows $g^r = 1$ exactly when d divides r. Let e be the least common multiple of the finitely many orders, computed by the maximum exponent of each prime in their factorizations; integer prime factorization and Bezout were proved above. For each prime ℓ dividing e, select an element g_ℓ of order $\ell^s m$ with ℓ not dividing m and ℓ^s the maximal occurring power. Then $h_\ell = g_\ell^m$ has order exactly ℓ^s, by the divisibility test just proved. For commuting elements x, y of coprime orders a, b, if $(xy)^r = 1$, then $x^r = y^{-r}$ belongs to both cyclic subgroups. Its ath and bth powers are one; an integer Bezout identity makes the element one. Thus both a and b divide r, and ab divides r by the same Bezout identity. Since $(xy)^{ab} = 1$, its order is ab. Inducting over the primes shows that $h = \prod h_\ell$ has order e. Every element of G is a root of $T^e - 1$, so the polynomial root bound gives $|G| \leq e$. The e different powers of h give the reverse inequality. Therefore G consists of those powers and is cyclic.

If G is trivial, the empty prime list gives $e = 1$ and $h = 1$, so this case is included. □

Lemma 4.B (finite separable trace and normalization). Let E/F be a finite separable field extension of degree n. Define $\mathrm{Tr}_{E/F}(z)$ to be the trace of multiplication by z on the F-vector space E. It is independent of basis, and the pairing $(z, w) \mapsto \mathrm{Tr}_{E/F}(zw)$ is nondegenerate. If R is a Noetherian normal domain with fraction field F, its integral closure S in E is a finite R-module; for every nonzero $a \in R$, S_a is the integral closure of R_a in E.

Proof. We use the polynomial division, finite splitting-field construction, power bases, tower-degree formula and separable embedding counts proved in Lemma 4.A. The trace calculation itself is supplied here. The free Stacks, Tag 0BIL provides further reading for the nondegenerate pairing.

Choose a finite generating list $\alpha_1, \dots, \alpha_h$ for E/F, put $E_j = F(\alpha_1, \dots, \alpha_j)$, and let f_j be the monic minimal polynomial of α_j over E_{j-1}. Each α_j is separable over F. Choose a finite extension H/F splitting their minimal polynomials $g_j \in F[T]$, by Lemma 4.A. We construct the isomorphism

$$H \otimes_F E_j \longrightarrow \prod_{\sigma : E_j \hookrightarrow H} H, \qquad c \otimes z \longmapsto (c\sigma(z))_\sigma. \tag{T1}$$

For $j = 0$, it is $H \otimes_F F = H$. At the next step the presentation $E_j = E_{j-1}[T]/(f_j)$ identifies $H \otimes_F E_j$ with $(H \otimes_F E_{j-1})[T]/(f_j)$: send T to $1 \otimes \alpha_j$; the powers of T below $\deg f_j$ and the corresponding powers of α_j are bases, so this map is bijective. Applying the previous product decomposition gives the product of $H[T]/(\sigma(f_j))$. All polynomial operations and the inverse product map are componentwise.

Division shows $f_j \mid g_j$ over E_{j-1}, since the remainder would be a polynomial of smaller degree annihilating α_j. Thus $\sigma(f_j)$ divides the already split separable polynomial g_j, and has distinct roots $r_1, \dots, r_m \in H$, with $m = \deg f_j$. Evaluation on these roots gives

$$H[T]/(\sigma(f_j)) \longrightarrow H^m, \quad [v] \longmapsto (v(r_1), \dots, v(r_m)).$$

Its inverse is the explicit interpolation map

$$(b_1, \dots, b_m) \longmapsto \left[\sum_{i=1}^{m} b_i \prod_{\ell \neq i} \frac{T - r_\ell}{r_i - r_\ell} \right]. \tag{T2}$$

Each denominator is nonzero. The displayed polynomial takes the prescribed values, proving one composite is the identity. For the other, reduce a polynomial modulo the monic polynomial to degree below m. Its difference from the interpolation polynomial has m roots, so the root bound proved in Lemma 4.A makes the difference zero. Each resulting component extends σ by sending α_j to the chosen root. Conversely every embedding restricts to a previous component and sends α_j to one of those roots. The tower-degree formula gives exactly $[E_j : F]$ components. This proves (T1), including degree-one steps and the empty tower.

For matrices over a field, direct finite sums give

$$\mathrm{tr}(UV) = \sum_{i,j} U_{ij} V_{ji} = \mathrm{tr}(VU).$$

If P is an invertible change-of-basis matrix, applying this identity to $P^{-1}M$ and P gives $\mathrm{tr}(P^{-1}MP) = \mathrm{tr}(M)$. Thus the trace of multiplication is well defined. Extending scalars leaves the matrix entries and their diagonal sum unchanged. Under (T1) for E, multiplication by z becomes the diagonal map with entries $\sigma(z)$. Hence

$$\mathrm{Tr}_{E/F}(z) = \sum_{\sigma} \sigma(z), \qquad \mathrm{Tr}_{E/F}(zw) = \sum_{\sigma} \sigma(z)\sigma(w). \tag{T3}$$

After extension to H, the pairing is the dot product on H^n. A vector with a nonzero i-th coordinate pairs nontrivially with the i-th coordinate unit vector. The pairing matrix is therefore invertible

over H; its determinant, computed in F, is nonzero, so the matrix is also invertible over F by Lemma 4.A. This proves nondegeneracy. It applies even when the characteristic divides n; no assertion that $\mathrm{Tr}(1) \neq 0$ is used.

We now prove the integral-closure statement, using the closure and transitivity proof in Integral extensions, Theorem 1.2 and the cofactor and Noetherian submodule proofs in S01, Lemmas 1.2 and 1.3. Start with any F-basis of E. If a vector v satisfies $v^s + c_1 v^{s-1} + \cdots + c_s = 0$ with $c_i \in F$, choose a nonzero common denominator $b \in R$. Then bv satisfies a monic equation with coefficients $b^i c_i \in R$. Scaling each basis vector this way gives an integral basis $v_1, \ldots, v_n$. Put

$$G = (\mathrm{Tr}_{E/F}(v_i v_j))_{i,j}, \qquad \delta = \det G \neq 0.$$

Each entry lies in R: every $\sigma(v_i v_j)$ is integral over R, their sum is integral by the earlier closure proof, and that sum lies in F by the definition of trace; normality places it in R. The same argument gives $\mathrm{Tr}_{E/F}(zv_i) \in R$ for $z \in S$. Writing $z = \sum_j a_j v_j$, with $a_j \in F$, gives

$$G(a_j)_j = (\mathrm{Tr}_{E/F}(zv_i))_i.$$

The proved identity $\mathrm{adj}(G)G = \delta I$ implies $\delta a_j \in R$ for every j. Thus S, an R-submodule by the integrality closure proof, is contained in the finite free module $\sum_j \delta^{-1} R v_j$. Noetherianity makes this submodule finite. The basis vectors are in S, so $\mathrm{Frac}\, S = E$. Transitivity also proves normality of S: an element of E integral over S is integral over R and therefore belongs to S.

Finally, elements of S_a are integral over R_a by their monic equations and multiplication by elements of R_a. If z is integral over R_a, write the coefficients of its monic equation as u_i / a^{h_i}, with $u_i \in R$. Choose N with $Ni \geq h_i$ for every coefficient. The monic equation for $a^N z$ has coefficients $a^{Ni - h_i} u_i \in R$. Hence $a^N z \in S$, so $z \in S_a$. This proves the asserted equality of closures. □

Lemma 4.2. Let A be a Noetherian domain with fraction field K. If a smooth affine finite type A-scheme has geometrically integral generic fibre, there is a nonempty principal open of $\mathrm{Spec}\, A$ on which all its fibres are geometrically integral.

Proof. Write the algebra as B. Smoothness makes B A-flat, so $B \hookrightarrow B_K$. Since the latter is a domain, B is a domain. Let $L = \mathrm{Frac}(B_K)$. An étale coordinate chart at the generic point gives algebraically independent $t_1, \ldots, t_d \in L$ with L finite separable over $K(t_1, \ldots, t_d)$: use the standard smooth chart and its zero-relative-dimension coordinate map from AG-CA-17 Theorem 5.1, applied over the field, followed by AG-CA-18 Corollary 6.2 and AG-CA-17 Lemma 7.1. A primitive element u writes

$$L = K(t_1, \ldots, t_d)(u).$$

Here is the complete primitive-element argument, using Lemma 4.A. Let L/F be finite separable of degree n. Its finite generating list $a_1, \ldots, a_r$ has exactly n distinct embeddings $\sigma_1, \ldots, \sigma_n$ into one finite splitting field H. If F is infinite, for each pair $i \neq j$ the form $\sum_t (\sigma_i(a_t) - \sigma_j(a_t)) X_t$ is a nonzero polynomial over H, since the embeddings differ on a generator. Their finite product is nonzero: the leading coefficient of a product of nonzero polynomials over a field is nonzero, iterating by the number of variables. A nonzero polynomial over H does not vanish on all tuples in the infinite subfield F. Indeed, induction on the number of variables first chooses a tuple in F where one nonzero coefficient polynomial is nonzero, and then avoids the finitely many roots in the last variable. Choose $c_t \in F$ where this product does not vanish. The element $a = \sum c_t a_t$ has n distinct images. Its minimal polynomial therefore has degree at least n by the root bound, and at most n because $F(a) \subset L$. The tower-degree equality gives $L = F(a)$. In a nontrivial extension this element cannot be zero; for a trivial extension use 1.

If F is finite, its finite basis makes L a finite set. Lemma 4.A proves that $L^\times$ is cyclic, including the trivial group case. Choose a generator u of this group. Every nonzero element of L is a power of u, and zero belongs to every subfield; hence $F(u) = L$. The generator is nonzero, and for a trivial

extension we may choose 1. Apply this with $F = K(t_1, \dots, t_d)$ and the field $L = \mathrm{Frac}(B_K)$ above to obtain the required u in the displayed simple-extension presentation.

Clear denominators in the minimal polynomial of u, removing the coefficient content, to obtain a primitive polynomial $P \in K[t_1, \dots, t_d, T]$. It is absolutely irreducible. Indeed $B_K \otimes_K \overline{K}$ is a domain, and localizing it gives the domain $L \otimes_K \overline{K}$. The algebra $K[t_1, \dots, t_d, T]/(P)$ embeds in L by its primitive minimal-polynomial presentation. Tensoring this injection with the field $\overline{K}$ keeps it injective. Its target is the just-obtained domain, so the polynomial quotient over $\overline{K}$ is a domain, proving absolute irreducibility without any assertion about contents after field extension.

Here is the needed polynomial Gauss argument. A product of primitive polynomials over a UFD stays primitive: modulo any prime coefficient factor, the two reductions are nonzero polynomials over a domain, so their product is nonzero. Starting with a field and inducting on the number of variables, factor the coefficient content and then factor the primitive part over the fraction field, whose one-variable polynomial ring has Euclidean division. Clear the denominators and normalize each factor to primitive content. The preceding product calculation shows that this gives a factorization over the coefficient ring, and uniqueness follows from the fraction-field uniqueness and the content. Thus each polynomial coefficient ring is a UFD; its primitive minimal polynomial generates the kernel of evaluation in the field extension, as used above.

Enlarge a nonzero localization of A until P has coefficients there. There is a further nonempty localization on which its specialization is absolutely irreducible in every residue field. Here is a finite proof of this assertion. A triangular variable change $x_i = y_i + y_n^{N_i}$ makes the weights of the finitely many monomials distinct, using successive powers of an integer larger than all occurring exponents. The unique largest weight gives a nonzero leading coefficient in y_n. Invert it and divide by it, making P monic of degree $r > 0$ in that variable. Let D_i bound its degrees in the other variables. A nontrivial factorization over a field has monic factors of positive y_n-degrees $e, r - e$, $1 \le e < r$, with other degrees at most D_i; degrees add in each variable over a domain. For each e, introduce the finitely many coefficients of these possible factors and impose the coefficients of $P - Q_1 Q_2$. This gives a finite type A-algebra R_e. Its generic fibre is zero by absolute irreducibility and the Nullstellensatz over $\overline{K}$. Therefore $1 = 0$ after tensoring with K, so some nonzero $a_e \in A$ makes $(R_e)_{a_e} = 0$. Inverting the finite product excludes every nontrivial factorization over every field extension of every residue field. Degree-zero factors in the monic variable are units and are excluded from this list. This is the finite coefficient proof of the free Stacks 0557 result, with only positive factor degrees used.

Put $B' = A[t_1, \dots, t_d, T]/(P)$ over the localization just obtained. Its generic fraction field and that of B are the same. The two generic affine integral schemes have a common nonempty affine principal open: represent the finite generators of each algebra as fractions in the other, invert their finitely many denominators, and invert the denominators needed for these descriptions to be inverse. AG-MO-03's map descent and eventual equality spread this isomorphism after another nonzero base localization. Thus we have $B_h \cong B'_{h'}$, with $h \neq 0$ and nonempty generic open.

Apply AG-FSE Corollary 5.2 in **Flat morphisms** to the finite B-module B/hB, and shrink so that it is A-flat. In the sequence $0 \to B \xrightarrow{h} B \to B/hB \to 0$, flatness of the last term makes multiplication by h injective after every field base change. The smooth fibre is reduced and Noetherian. A nonzerodivisor belongs to no minimal prime: if it did, localization at that minimal prime would make it zero in the field of a reduced Noetherian ring, giving an annihilator outside the prime. Therefore $D(h)$ is dense in every geometric fibre.

The finite type image theorem proved in Lemma 4.C makes the image of the common open constructible. It contains the generic point, so contains a nonempty open of the base. Shrink to a principal open in it; now the common open is nonempty in every fibre, and remains so after field extension. The B'-fibre is geometrically integral by the polynomial argument. Its nonempty open $D(h')$, identified with $D(h)$, is geometrically irreducible. Being dense in the smooth, geometrically reduced B-fibre, it forces that whole fibre to be irreducible and reduced. This proves the lemma. □

Lemma 4.3. For a smooth affine finite type morphism over a Noetherian base, the number of geometric connected components of its fibres is a constructible function.

Proof. By the Noetherian constructibility criterion proved in Lemma 4.C, it suffices to show constancy on a nonempty open of every irreducible reduced closed subscheme of the base. Restrict to such an integral affine base $\operatorname{Spec} A$, with fraction field K.

If the generic fibre is empty, its flat affine algebra injects into zero and is zero, so the component number is constantly zero. Otherwise the generic geometric fibre has finitely many open-and-closed integral components by Lemma 4.1. Their finitely many orthogonal idempotents, summing to one, descend to a finite separable extension K'/K. To justify separability, first express them over $\overline{K}$. In positive characteristic each expression uses finitely many coefficients which become separable after some common p-power. Since an idempotent equals its own p-power, raising its expression to this power puts its coefficients in K^{sep}. In characteristic zero this step is unnecessary. The finite list of algebraic separable coefficients lies in one finite separable extension. Its components are geometrically integral, as checked after extension to $\overline{K}$.

Write $K' = K[v]$ by the primitive element argument in Lemma 4.2. Clear denominators in its monic minimal polynomial and a Bezout identity with its derivative. After a nonzero localization, $A' = A[v]$ is finite free étale over A, with generic fibre the field K'. It is a domain because its monic polynomial presentation is free over A and embeds in K'. The finite generic idempotents extend after inverting finitely many elements of A'; their finite orthogonality and sum equations hold after shrinking, or directly by the generic injection coming from flatness. They split the base-changed smooth algebra into finitely many smooth factors. Each has geometrically integral generic fibre. Apply Lemma 4.2 to all factors and invert a common nonzero $a \in A'$. At every point of $D(a)$ the factors are exactly the geometric connected components.

The determinant of multiplication by a on the finite free A-module A' is nonzero, because this map is invertible on the generic field. Invert this determinant in A. The adjugate makes a invertible on the whole finite étale cover. That cover is surjective, since a nonzero finite free algebra has nonzero fibres, and those fibres have points. Field-extension invariance from Lemma 4.1 now proves the same constant component number on this nonempty base open. This verifies the criterion on every irreducible closed base. Noetherian induction, removing such a nonempty open on each of finitely many irreducible components and then applying the induction hypothesis to their proper closed complement, gives a finite locally closed stratification with constant component count. This proves constructibility as a function, and in particular of its one-component locus. □

This proves the smooth affine case needed from the free chain 055I, 055H, 055G. The hypersurface and flat quotient arguments above supply the generic constancy input explicitly; none of those tags is being substituted for a proof.

Lemma 4.4. Suppose $A = \varinjlim A_i$, A_0 is Noetherian, and $E \subset \operatorname{Spec} A_0$ is constructible. If the image of $\operatorname{Spec} A$ lies in E, the image of $\operatorname{Spec} A_i$ lies in E at some finite stage.

Proof. Lemma 4.C expresses the complement as a finite union of pieces $V(J) \cap D(g)$, with J finitely generated. Its pullback has empty spectrum, so $(A/JA)_g = 0$. This says $g^m \in JA$ for some m, with a finite expression in generators of J. Its coefficients and the finite equality descend to a stage. The same localized quotient there is zero, so that complement piece has empty pullback. Take a common stage for the finitely many pieces. □

This is a direct affine proof of the needed Stacks 05F4 step.

Theorem 4.5. A smooth affine finite-presentation scheme with geometrically connected fibres over an arbitrary affine ring has a smooth affine model with geometrically connected fibres over a finitely generated $\mathbf{Z}$-algebra. Finitely many specified maps, group laws and closed immersions can be retained.

Proof. Descend the finite presentations and finite identities to A_0, a finitely generated $\mathbf{Z}$-subalgebra of the given ring, using AG-MO-03. Retain smoothness by Section 3 and closed immersions by the

earlier proved closed-immersion descent. On this Noetherian model let E be the locus of exactly one geometric connected component. It is constructible by Lemma 4.3. Its pullback contains the image of the original base, by field-extension invariance and the original hypothesis. Apply Lemma 4.4 to enlarge the coefficient ring to a finite stage whose whole image lies in E. Pullback of the model at that stage has geometrically connected fibres and remains smooth. Previously descended finite identities and closed immersions remain true under this base change. □

This is the consumed smooth affine case of Stacks 05FI. It applies simultaneously to the smooth affine group and its smooth affine subgroup in the normalizer lemma. Projectivity of quotient fibres has not been imposed on unrelated fibres of the model.

5. A universal finite complex for a line bundle

The following is the previously written finite complex proof, retained here before its uses.

Lemma 5.1. Let A be Noetherian and C a complex of flat A-modules concentrated in $[a, b]$, with finite cohomology modules. There is a complex K of finite projective modules in $[a, b]$ and a quasi-isomorphism $K \to C$ which remains a quasi-isomorphism after tensoring with every A-module.

Proof. Construct a bounded-above complex P of finite free modules and a map $f : P \to C$, descending from degree b. With terms above j chosen, the cone has terms

$$D^i = C^i \oplus P^{i+1}, \qquad d(c, p) = (d_C c + f(p), -d_P p).$$

Its cohomology is finite: the cone long exact sequence uses finite cohomology of C and finite cohomology of the finite portion of P chosen so far. Choose finitely many cocycles (c_ℓ, p_ℓ) generating $H^j(D)$. Let P^j be free on e_ℓ, and set $f(e_\ell) = c_\ell$, $d_P(e_\ell) = -p_\ell$. The cocycle equations give $d_P^2 = 0$ and $d_C f = f d_P$. The new cone differential sends $(0, e_\ell)$ to (c_ℓ, p_ℓ), killing H^j without changing higher cohomology. Continue to the left. Each degree stabilizes, giving a quasi-isomorphism $P \to C$.

Its cone is acyclic, bounded above and termwise flat. Such a complex stays acyclic after every tensor: start at its last degree and descend through $0 \to Z^i \to D^i \to Z^{i+1} \to 0$. The quotient is flat, so the kernel is flat and the sequence remains exact under tensor. Thus $H^i(P \otimes_A M) = H^i(C \otimes_A M)$ for every M.

Put $Q = \operatorname{coker}(P^{a-1} \to P^a)$. The left tail is a free resolution of Q, and its first positive homology after tensoring with every M is $H^{a-1}(P \otimes_A M) = 0$, because $C \otimes_A M$ starts in degree a. Consequently Q is flat: tensor that free resolution with a short exact sequence $0 \to M' \to M \to M'' \to 0$. Its free terms give a short exact sequence of complexes; the homology exact sequence and the just-proved degree-one vanishing give an injection $Q \otimes M' \to Q \otimes M$. This is the definition of flatness.

The finite presentation of Q makes it locally finite free by Algebra and sheaf cohomology before reductive groups, Lemma 1.2. It is finite projective as follows. Take a surjection $A^r \to Q$ and a finite principal cover on which it splits. A local splitting extends after multiplication by a power of that chart element, since Hom from a finite presented module commutes with localization: use its finite presentation and exact localization to identify the kernel describing Hom. Increase that power until its composite with the surjection is the same power times the identity globally; equality can be checked on finitely many generators. Thus there are maps $\tau_i : Q \to A^r$ whose composites are $f_i^{N_i} \operatorname{id}_Q$. These powers generate the unit ideal, since the $D(f_i)$ cover. A linear combination of the τ_i whose coefficients combine the powers to one is a global splitting. Hence Q is a direct summand of a finite free module and is finite projective. Set $K^a = Q$, $K^i = P^i$ for $a < i \leq b$. The map to C factors through Q since $C^{a-1} = 0$. Truncation preserves cohomology, and the same acyclic flat cone argument proves the assertion after every tensor. □

Proposition 5.2. Let $X \to S$ be smooth projective of finite presentation and L invertible. If all positive cohomology of L on every geometric fibre vanishes, then $f_* L$ is finite locally free, all

positive higher direct images vanish, and these statements commute with every base change.

Proof. Work over $S = \operatorname{Spec} A$. Section 3 gives a smooth projective model $X_0 \to \operatorname{Spec} A_0$, A_0 Noetherian, and an invertible L_0 pulling back to L. A finite affine cover of X_0 has affine intersections. Its section complex C_0 computes cohomology by the earlier finite-cover and affine-vanishing proofs. Its terms are A_0-flat: an invertible module is projective over its chart algebra, and that algebra is base-flat by smoothness. Projective finiteness makes its cohomology finite. After every algebra base change the section complex is $C_0 \otimes_{A_0} B$, by affine modules and localization. Lemma 5.1 therefore gives a bounded finite projective K_0 computing cohomology after every such change.

Put $K = K_0 \otimes_{A_0} A$, and trivialize its terms near a point s. If a differential has an entry invertible at s, shrink and use row and column operations to make it an identity block. The equation $d^2 = 0$ makes the adjacent differentials zero on that block, splitting off a contractible pair $A \xrightarrow{1} A$. After finitely many cancellations the remaining differentials vanish over $\kappa(s)$. Each remaining free term tensored with $\kappa(s)$ is then the cohomology in its degree. In positive degrees it is zero by the hypothesis and faithful field extension; in negative degrees it is zero because the section complex begins in degree zero. All remaining terms other than degree zero have rank zero. The degree-zero term is finite free. Cancelled pairs remain contractible under every tensor, so the same description computes all base changes. It proves the assertions on every affine open and every affine base change. The canonical maps from sections and the section complexes commute with restriction, so glue to the claimed direct-image and base-change isomorphisms. □

Vanishing was required on the original fibres only. No vanishing on unrelated Noetherian model fibres, reduced-base assumption or flat-base-change assumption was used.

6. The smooth curve regular immersion calculation

The following retains the full independently authored CC0 proof of AG-QC-14, Sections 3–5, with its formerly general proper-finiteness input replaced by the projective proof already given in Algebra and sheaf cohomology before reductive groups. Its injective and finite-cover foundations are proved in that preceding support, Sections 3–5. We first record the vector-bundle Ext identity used in the argument. For finite locally free V, tensor-Hom adjunction gives $\mathcal{H}om(V, I) = V^\vee \otimes I$. Tensoring by V is exact, so its right adjoint $V^\vee \otimes -$ takes injectives to injectives: Hom into it is Hom from an exact functor into an injective. Applying this adjunction to an injective resolution gives

$$\operatorname{Ext}^j_P(V, G) = H^j(P, V^\vee \otimes G).$$

Hom into each injective is contravariantly exact. A short exact sequence in its first argument therefore gives a short exact sequence of Hom complexes, whose cohomology gives the connecting maps and long exact sequence used below.

6.0. Ambient projective-space trace and duality

Put $P = \mathbf{P}^n_k$ for a field k, and define

$$\omega_P = \mathcal{O}_P(-n-1).$$

Its identification with $\det \Omega_P$ is explicit. On $U_i = D_+(T_i)$, use the ordered frame $\sigma_i = (-1)^i \bigwedge_{j \neq i} d(T_j/T_i)$. On $U_0 \cap U_i$, the substitution $T_0/T_i = 1/(T_i/T_0)$, $T_j/T_i = (T_j/T_0)/(T_i/T_0)$ gives Jacobian determinant $(-1)^i (T_i/T_0)^{-n-1}$; thus $\sigma_i = (T_i/T_0)^{-n-1}\sigma_0$. These are exactly the transitions of $\mathcal{O}_P(-n-1)$, proving the identification. For $n = 0$, the differential determinant and every twist are both the one-dimensional trivial line on a point. The cohomological normalization uses the ordered standard affine cover $D_+(T_0), \ldots, D_+(T_n)$, with alternating Čech differential. For

$n \geq 1$, the projective-space calculation gives

$$H^n(P, \omega_P) = k \cdot \left[\frac{1}{T_0 \cdots T_n}\right].$$

Define the trace $\mathrm{tr} : H^n(P, \omega_P) \to k$ by sending that class to one. For $n = 0$, P is a point; the same expression is the generator of the degree -1 part of $k[T_0, T_0^{-1}]$, so this convention still identifies its one-dimensional space of sections with k.

Multiplication followed by this trace pairs

$$H^0(P, \mathcal{O}(d)) \ \times \ H^n(P, \mathcal{O}(-d-n-1)) \longrightarrow k.$$

For $n \geq 1$ and $d \geq 0$, the first basis consists of T^b, where $b_i \geq 0$ and $\sum b_i = d$. The matching top-cohomology basis consists of $T_0^{-b_0-1} \cdots T_n^{-b_n-1}$. Their products have trace one precisely for matching exponent vectors, and zero otherwise. For $d < 0$, both spaces in the trace pairing vanish. Thus the trace pairing is perfect for every d. When $n = 0$, both spaces are one-dimensional for every twist, and multiplication has the same conclusion. Negative twists on a point do not vanish.

More generally, projective-space cohomology and this coefficient computation give perfect pairings

$$H^j(P, \mathcal{O}(d)) \ \times \ H^{n-j}(P, \mathcal{O}(-d-n-1)) \longrightarrow k$$

for all j: intermediate groups vanish and the two end cases are the computed pairing and its transpose. Maps between direct sums of twists are matrices of homogeneous polynomials. Multiplication is associative, so these pairings commute with all such matrices, including matrices between different twists. This naturality is the part needed to pass from twists to arbitrary coherent sheaves.

Duality in degree zero

For any coherent F, a morphism $u : F \to \omega_P$ induces a functional on its top cohomology:

$$\theta_F^0(u) = \mathrm{tr} \circ H^n(u).$$

This construction is contravariantly natural in F.

Ambient Lemma 6.A. The trace map is an isomorphism

$$\mathrm{Hom}_P(F, \omega_P) \ \cong \ H^n(P, F)^\vee$$

for every coherent F.

Proof. Choose a surjection $P_0 \twoheadrightarrow F$ with P_0 a finite sum of twists, by the proved finite generation by twists in Algebra and sheaf cohomology before reductive groups, Proposition 8.2. Its coherent kernel K is likewise a quotient of a finite sum of twists P_1. Thus $P_1 \to P_0 \to F \to 0$ is a presentation. Hom into ω_P gives

$$0 \to \mathrm{Hom}(F, \omega_P) \to \mathrm{Hom}(P_0, \omega_P) \to \mathrm{Hom}(P_1, \omega_P).$$

The standard cover has $n+1$ affine opens with affine intersections. Consequently cohomology of quasi-coherent sheaves vanishes above n. The two short exact sequences defining this presentation therefore give

$$H^n(P, P_1) \to H^n(P, P_0) \to H^n(P, F) \to 0.$$

Dualizing gives another left-exact sequence in the same direction as the Hom sequence. All spaces are finite-dimensional, by the proved projective finiteness in Algebra and sheaf cohomology before reductive groups, Theorem 8.4. The maps θ^0 give a morphism of these sequences. The two maps for P_0, P_1 are isomorphisms by the trace pairing, and commute with the presentation map by its polynomial-matrix description. Their kernels are therefore isomorphic, which proves the assertion. □

Only two stages of a presentation were used. In particular, this argument does not claim that repeated coherent kernels eventually become sums of line bundles.

Serre duality in every degree

Ambient Theorem 6.B (Serre duality on projective space). For every coherent sheaf F and every integer i, there are natural isomorphisms

$$\operatorname{Ext}^i_P(F, \omega_P) \;\cong\; H^{n-i}(P, F)^\vee.$$

They extend the trace map, using its fixed trace.

Proof. For $i \geq 0$, consider the contravariant cohomological sequences

$$A^i(F) = \operatorname{Ext}^i_P(F, \omega_P), \qquad B^i(F) = H^{n-i}(P, F)^\vee.$$

Exactness of vector-space duality and vanishing above n make the second a contravariant delta functor starting in degree zero, just as Ext is. Choose $q > 0$ sufficiently large so that $F(q)$ is generated by its sections and all positive cohomology of $\omega_P(q)$ vanishes. There is an epimorphism

$$Q = \mathcal{O}_P(-q)^{\oplus m} \twoheadrightarrow F$$

with coherent kernel K. The vector-bundle identity above and projective Serre vanishing in Algebra and sheaf cohomology before reductive groups, Theorem 8.4 give $A^j(Q) = 0$ for every $j > 0$. The twist calculation gives $B^j(Q) = 0$ for every $j > 0$: if $0 \leq n - j < n$, the corresponding cohomology of a strictly negative twist vanishes, and if $n - j < 0$ it vanishes by convention. This also covers $n = 0$, where every positive B^j is zero.

Both long exact sequences consequently identify their degree-one value on F with the cokernel of their degree-zero map from Q to K. Ambient Lemma 6.A identifies these cokernels. For $i \geq 2$, the sequences give

$$A^i(F) \cong A^{i-1}(K), \qquad B^i(F) \cong B^{i-1}(K).$$

Induction gives the desired isomorphisms for every $i \geq 0$.

These isomorphisms are canonical. The epimorphisms just used efface all positive values of both delta functors. The following comparison gives uniqueness: a morphism in degree zero determines the cokernel map in degree one and then every dimension-shifted map. For two choices of epimorphism, take their fiber product over F, which is coherent, and an epimorphism from sufficiently negative twists onto that fiber product. The resulting maps of short exact sequences compare both choices. The same construction over a morphism of coherent sheaves proves naturality, and compatibility with connecting maps follows from the exact sequences. Thus the induction is the unique extension of the trace map.

If $i < 0$, the left side vanishes, and the right side is cohomology above dimension n, hence also zero. This proves the displayed duality for all integers. □

The following direct DVR argument supplies this local algebra before the degree calculations.

DVR Lemma 6.C. A one-dimensional Noetherian normal local domain is a discrete valuation ring.

Proof. Let $(R, \mathfrak{m})$ be such a ring and choose $0 \neq a \in \mathfrak{m}$. Every prime containing a is $\mathfrak{m}$, since any other nonzero prime would give a chain of length at least two. Thus the radical of (a) is $\mathfrak{m}$. Finite generation of $\mathfrak{m}$ gives $\mathfrak{m}^n \subset (a)$ for some n: choose a power of each generator lying in (a) and use the pigeonhole principle on monomials. Choose the least n and $b \in \mathfrak{m}^{n-1} \setminus (a)$. Then $c = b/a \notin R$ and $c\mathfrak{m} \subset R$.

If $c\mathfrak{m} \subset \mathfrak{m}$, apply the determinant trick to multiplication by c on the finite module $\mathfrak{m}$. Its resulting monic polynomial annihilates this module, hence is zero in $\operatorname{Frac}(R)$ because the domain has a nonzero element in $\mathfrak{m}$. Normality would imply $c \in R$, a contradiction. Consequently cx is a unit for some $x \in \mathfrak{m}$. It follows that $\pi = c^{-1} = x/(cx) \in \mathfrak{m}$ and $\mathfrak{m} = \pi R$.

The intersection $N = \bigcap_j \pi^j R$ is zero. In fact N is a finite ideal, since R is Noetherian, and cancellation of $\pi \neq 0$ shows $N = \pi N$: if $x = \pi y$ lies in every $\pi^{j+1}R$, then y lies in every $\pi^j R$.

Nakayama gives $N = 0$. Repeated division of a nonzero nonunit by π must therefore stop. Every nonzero element is uniquely a unit times π^j, and every nonzero ideal is generated by the element of smallest such exponent. This is the discrete valuation ring assertion. □

DVR Lemma 6.D (the regular curve case). A one-dimensional Noetherian local domain whose maximal ideal is generated by one element is a normal discrete valuation ring. In particular this applies to a one-dimensional regular local domain.

Proof. Write the maximal ideal as πR. The element π is nonzero, since the domain has dimension one and is not a field. The ideal $N = \bigcap_{j\geq 0} \pi^j R$ is finite by Noetherianity. Cancellation shows $N = \pi N$: if $x = \pi y$ lies in every $\pi^{j+1}R$, then y lies in every $\pi^j R$. Nakayama, proved in S01 Lemma 1.2, gives $N = 0$. Repeated division of a nonzero element by π must therefore terminate, giving a unit times π^r for a unique $r \geq 0$. Uniqueness follows because a positive power of π cannot be a unit. A nonzero ideal contains an element with smallest exponent; that element divides every element of the ideal and generates it. This proves the DVR description without assuming normality.

Every nonzero fraction is now uniquely a unit times π^r for an integer r. Suppose a fraction z is integral over R and has negative exponent. In a monic equation $z^m + c_1 z^{m-1} + \cdots + c_m = 0$, divide by z^m. Since $z^{-1} \in \pi R$, every term $c_i z^{-i}$ lies in the maximal ideal, forcing $1 \in \pi R$, a contradiction. Thus the exponent is nonnegative and $z \in R$; the zero fraction is already in R. This proves normality. A one-dimensional regular local domain has a single generator of its maximal ideal by the definition of regularity, so the asserted special case follows. □

Koszul Lemma 6.E. Let $f_1, \ldots, f_c$ be a regular sequence in a ring A, and put $I = (f_1, \ldots, f_c)$. Its Koszul complex is a finite free resolution of A/I, the classes of the f_i form a basis of I/I^2, and for an invertible A-module L its dual complex has cohomology only in degree c, where it is

$$(A/I) \otimes_A L \otimes_{A/I} \det(I/I^2)^*.$$

This last identification agrees under invertible changes of the regular generating list.

Proof. Define $K_r = \bigwedge^r A^c$, with basis e_i and differential

$$d(e_{i_1} \wedge \cdots \wedge e_{i_r}) = \sum_{j=1}^{r} (-1)^{j-1} f_{i_j} e_{i_1} \wedge \cdots \widehat{e_{i_j}} \cdots \wedge e_{i_r}.$$

Pairing the two possible orders of deletion gives $d^2 = 0$. On adjoining f_c, this complex is the mapping cone of multiplication by f_c on the preceding Koszul complex. The cone's short exact sequence gives its homology exact sequence directly by lifting cycles. Induction, starting with the empty list and $K_0 = A$, leaves only degree-zero homology: multiplication by f_c is injective on $A/(f_1, \ldots, f_{c-1})$ by regularity, and its cokernel is A/I. Thus K is the stated resolution.

The map $(A/I)^c \to I/I^2$ is surjective. If $\sum_i a_i f_i \in I^2$, subtract expressions $\sum_j b_{ij} f_j$ from the a_i to make their resulting linear combination of the f_i zero. Exactness at K_1 makes this relation a boundary from K_2; the displayed differential makes every coefficient of such a boundary belong to I. Hence the original a_i also belong to I, proving injectivity and the claimed basis.

The perfect exterior-product pairing identifies

$$\operatorname{Hom}_A(K_r, L) = L \otimes_A K_{c-r} \otimes_A (\bigwedge^c A^c)^*.$$

Define the dual differential by precomposition with d. For $u \in K_{r+1}$ and $v \in K_{c-r}$, the identity $0 = d(u \wedge v) = du \wedge v + (-1)^{r+1} u \wedge dv$ gives $du \wedge v = (-1)^r u \wedge dv$. Thus the dual differential corresponds to $(-1)^r d$ under the displayed pairing; multiplying the degree-r identification by $(-1)^{r(r-1)/2}$ gives the reversed Koszul differential without those signs. Its cohomology therefore vanishes except in degree c, where the canonical top-term quotient is reduction modulo I. The conormal basis just proved identifies that quotient with the displayed determinant line. An invertible matrix changing the f_i gives an isomorphism of the two Koszul complexes by the same basis change

on their exterior terms. The top exterior term changes by its determinant and the dual by its inverse. These are exactly the transition maps of $\det(I/I^2)^*$, so the identification is independent of the chosen regular list. The empty list has determinant A and gives the asserted degree-zero statement as well. □

Lemma 6.1. A closed immersion $i : C \hookrightarrow \mathbf{P}^N_k$ of a smooth pure one-dimensional projective curve is locally cut out by a regular sequence of length $c = N - 1$, and

$$0 \to I/I^2 \to i^*\Omega_{\mathbf{P}^N/k} \to \Omega_{C/k} \to 0$$

is exact.

Proof. First extend k to an algebraic closure. At a closed point the two local rings are regular, of dimensions N and one, by the proved smooth field criterion. AG-CA-14 Proposition 1.4 gives a regular sequence of c parameters generating the ideal. Their linear classes are precisely the kernel of the surjection of tangent cotangent spaces. Thus their differentials are independent in the ambient differential fibre. The complete relation calculation in Koszul Lemma 6.E makes the conormal module free on their classes. The conormal map therefore has rank c on a neighbourhood of that closed point. Its cokernel is $\Omega_{C/k}$ by the proved differential conormal sequence. Since these finite type neighbourhoods cover every closed point, they cover the curve; every nonempty closed complement in a finite type scheme over a field has a closed point by the Nullstellensatz.

Here is the descent back to k. The finite module I/I^2 becomes locally free of rank c after this faithful field extension, so it is flat before extension: any kernel of a tensor injection vanishes after the faithfully flat field extension, hence vanishes already. A finite presented flat module is projective by the earlier module theorem. Locally choose a basis and lift it to c generators of I. At each ambient local ring, $I = (f_1, \dots, f_c) + I^2$; Nakayama applied to the finite module $I/(f_1, \dots, f_c)$, with I in the maximal ideal, gives generation of I. The corresponding Koszul complex becomes exact in positive degrees after the field extension, because any minimal c-generator list differs by an invertible matrix from the parameter list already obtained. Its finite cohomology modules therefore vanish before extension by faithful flatness. Equivalently the list is regular: the Koszul cone for the last generator makes multiplication by that generator surjective on every positive homology module of the preceding Koszul complex. Those modules are finite and the generator lies in the maximal ideal, so Nakayama kills them. The same exact sequence gives injectivity of the last generator on degree-zero homology. Induction proves regularity. Finally exactness of the conormal sequence descends by faithful flatness as well. □

This is the field-curve case actually needed from the more general free Stacks 067U regular immersion theorem. It uses the existing regular quotient proof, rather than importing the full syntomic arbitrary-base theorem.

Theorem 6.2. For such a curve and an invertible L, there is a natural isomorphism

$$H^1(C, L)^\vee \cong H^0(C, \Omega_{C/k} \otimes L^{-1}).$$

Proof. Lemma 6.1 gives the local regular sequence, and the complete resolution, dual and change-of-generators calculation in Koszul Lemma 6.E gives its Koszul resolution. Dualizing into $\omega_{\mathbf{P}^N} = \det \Omega_{\mathbf{P}^N/k}$ has cohomology only in degree c, with that sheaf equal to

$$i_*\big(L^{-1} \otimes i^*\omega_{\mathbf{P}^N} \otimes \det(I/I^2)^*\big).$$

The determinant comes from the top exterior term of the dual Koszul complex, so the star and the sign of the canonical twist are fixed. Taking determinants of the conormal exact sequence identifies $i^*\omega_{\mathbf{P}^N} \otimes \det(I/I^2)^*$ with $\Omega_{C/k}$. Here is the full consumed local-to-global comparison from AG-QC-14 Proposition 2.1, with its finite filtered-complex input supplied by Algebra and sheaf cohomology before reductive groups, Section 4. Resolve $\omega_{\mathbf{P}^N}$ by injectives $I^\bullet$ and put $E^\bullet = \mathcal{H}om(i_*L, I^\bullet)$. Each

term is flasque: on $U \subset V$, extension by zero gives a stalkwise injection $j_!((i_*L)|_U) \hookrightarrow (i_*L)|_V$, and injectivity of $I^q|_V$ extends every morphism from that submodule. Thus Hom sections restrict surjectively. Form the double complex of sections of E^q on the finite standard affine intersections of projective space. Its rows have Čech cohomology only in degree zero, since the terms and all their open restrictions are flasque and the earlier finite-cover comparison applies. One filtration therefore identifies total cohomology with $H^*\Gamma(E^\bullet) = \operatorname{Ext}^*_{\mathbf{P}^N}(i_*L, \omega_{\mathbf{P}^N})$.

On any affine intersection, choose a resolution of i_*L by finite free sheaves, allowing an infinite left tail; Noetherianity makes its successive kernels finite. The first-quadrant Hom double complex with $I^\bullet$ compares its Hom into ω with $E^\bullet$. In the resolution direction injectivity makes Hom exact. In the other direction its finite free source reduces to finite sums of ω, whose positive cohomology on this affine is zero by affine vanishing. Hence the vertical cohomology of the section double complex is the section module of the locally calculated sheaf Ext. Lemma 6.1 and the dual Koszul calculation above show that it is zero except in degree c. The other filtration therefore has only that row, the finite Čech complex of $i_*(L^{-1} \otimes \Omega_{C/k})$. Each total degree has finitely many terms, so the finite filtration calculation gives convergence and no completion or infinite-product issue. In total degree c it gives

$$\operatorname{Ext}^c_{\mathbf{P}^N}(i_*L, \omega_{\mathbf{P}^N}) = H^0(C, L^{-1} \otimes \Omega_{C/k}).$$

Ambient Theorem 6.B, the full trace and dimension-shifting proof of AG-QC-14 Theorem 5.1 reproduced above, identifies the same group with $H^{N-c}(\mathbf{P}^N, i_*L)^\vee = H^1(C, L)^\vee$. The Koszul determinant, conormal determinant and trace are natural, giving the stated isomorphism. □

Lemma 6.3 (divisors, Euler characteristic and genus). Suppose now that k is algebraically closed and C is connected. Put $g = \dim_k H^1(C, \mathcal{O}_C)$. For every invertible L,

$$\chi(C, L) = \deg L + 1 - g, \qquad \deg \Omega_{C/k} = 2g - 2.$$

A line bundle of negative degree has no nonzero section.

Proof. Smooth local rings are regular domains. Thus different irreducible components cannot meet, and the finitely many components are open and closed. Connectedness makes C integral. Projective finiteness makes $H^0(C, \mathcal{O}_C)$ a finite-dimensional domain over k. It is a field, since multiplication by a nonzero element is an injective endomorphism of a finite-dimensional vector space and hence is surjective. Algebraic closedness makes this field k.

There are two affine opens covering C. In a projective embedding choose a hyperplane not containing C; its intersection with the curve is finite, since a proper closed subset of a Noetherian integral curve has dimension zero and finitely many points. For this finiteness, every Noetherian closed subset is a finite union of irreducible closed subsets: a minimal counterexample splits into two proper closed subsets and contradicts minimality. In dimension zero those irreducible subsets are single closed points. Choose another hyperplane avoiding the finitely many intersection points. The field k is infinite: for a finite field the polynomial $\prod_{a \in k}(T - a) + 1$ would have no root. Each forbidden hyperplane condition is a nonzero linear equation on its coefficients; their product is a nonzero polynomial. Induction on the number of variables, using the polynomial root bound in the last variable, gives coefficients where that product is nonzero. The complements of these hyperplanes on C are affine and cover C. Their intersection is affine by separatedness. The proved finite Čech computation consequently gives $H^q(C, L) = 0$ for $q > 1$. All the remaining groups are finite by projective finiteness, so their Euler characteristic is defined and additive in short exact sequences.

A nonzero rational section of L exists by choosing a generator at the generic point. Each closed-point local ring is a DVR: a one-dimensional regular local ring has its maximal ideal generated by a parameter, and Lemma 6.D proves directly that this domain is a normal DVR, without a prior normality assumption. Relative to a local generator, the section has an integer valuation. There are

only finitely many nonzero valuations, since a finite trivializing affine cover represents it by rational functions with finitely many zeros and poles. These valuations define a divisor $D = \sum_x n_x x$ and identify L with $\mathcal{O}_C(D)$: at x, the identification sends the local module $\pi_x^{-n_x}\mathcal{O}_{C,x}$ to the section's generated fractional module. The descriptions agree at the generic point and hence on overlaps.

For every divisor and closed point there is an exact sequence

$$0 \longrightarrow \mathcal{O}_C(D) \longrightarrow \mathcal{O}_C(D+x) \longrightarrow k_x \longrightarrow 0.$$

At x the quotient is $\pi^{-n_x-1}R/\pi^{-n_x}R \cong k$; elsewhere it is zero. The skyscraper k_x has $H^0 = k$ and no higher cohomology (its restrictions are surjective, so the previously proved flasque acyclicity applies). Additivity, successively adding or subtracting points, gives

$$\chi(C, \mathcal{O}_C(D)) = \sum_x n_x + \chi(C, \mathcal{O}_C).$$

This both proves that the integer $\deg L = \sum n_x$ is independent of the rational section and gives the first formula. Tensor product adds these divisors, so degrees add and inversion negates degree. A nonzero regular section has only nonnegative valuations, which proves the negative-degree assertion.

Theorem 6.2 with $L = \mathcal{O}_C$ gives $h^0(\Omega_{C/k}) = g$. The same theorem with $L = \Omega_{C/k}$ gives $h^1(\Omega_{C/k}) = h^0(\mathcal{O}_C) = 1$. Its Euler characteristic is therefore $g - 1$. The first formula applied to $\Omega_{C/k}$ gives $g - 1 = \deg \Omega_{C/k} + 1 - g$, as claimed. In particular $T_C = \Omega_{C/k}^{-1}$ has degree $2 - 2g$. □

Lemma 6.4 (degree under a separable curve map). Let $f : D \to C$ be a nonconstant map between smooth connected projective curves over algebraically closed k, and suppose the function-field extension is separable of degree n. Then f is finite and locally free of rank n, and $\deg f^*L = n \deg L$.

Proof. Put $F = k(C)$ and $E = k(D)$. The hypothesis gives the finite separable extension E/F of degree n. By the smooth connected curve argument in Lemma 6.3, both curves are integral and their local rings are normal. For each nonempty affine open $U = \operatorname{Spec} R \subset C$, R is a Noetherian normal domain with fraction field F. Here is the global normality check: a fraction integral over R belongs to every maximal localization by its already proved DVR normality; a proper denominator ideal for that fraction would lie in a maximal ideal, contradicting the denominator outside that ideal supplied by membership in the localization. Let S be its integral closure in E. Lemma 4.B proves that S is finite over R, normal, and has fraction field E, and that these closures agree on principal overlaps. They therefore glue to a finite normal integral curve $q : C' \to C$.

Here are the local properties needed for this construction. A finite R-algebra is Noetherian: each of its ideals is an R-submodule of a finite R-module, so is finitely generated over R, and those same generators generate it as an ideal. The integral-extension lying-over and incomparability proofs in Integral extensions, Theorems 3.2 and 3.3 show that C' has dimension one, and that each nongeneric point is closed with one-dimensional local ring. Indeed, strict prime chains contract to strict chains in the one-dimensional base, while a maximal ideal over a nonzero base maximal ideal is nonzero. A maximal ideal upstairs contracts to a maximal ideal downstairs. Its local ring is therefore a one-dimensional Noetherian normal domain, and is a DVR by Lemma 6.C. The generic local ring is E.

The given map factors through C'. For $d \in f^{-1}(U)$, the inclusion $R \to \mathcal{O}_{D,d}$ sends any monic equation of $s \in S$ to a monic equation in this normal local domain. As $s \in E$, it follows that $s \in \mathcal{O}_{D,d}$. Such a rational function is regular on a neighbourhood of d: its local fraction presentation has an invertible denominator after shrinking. Thus each s is a section on $f^{-1}(U)$. These functions define the factorization $g : D \to C'$, and the localization equality in Lemma 4.B makes the factorizations agree on overlaps. On function fields g induces the identity of E.

There is an inverse rational map $C' \dashrightarrow D$. Choose a nonempty affine open of D. Its finite

algebra generating list consists of elements of E; on an affine open of C', clear their finitely many denominators. This defines a morphism on a nonempty open with the inverse function-field map. Embed D in a projective space and represent its generic point by finitely many homogeneous coordinates in E. At a nongeneric point of C', its DVR valuation has a least value among the nonzero coordinates. Multiply all coordinates by the inverse of that parameter power. They then belong to the local ring and one is a unit. They define a map there, and their local fraction presentations spread to a neighbourhood with that coordinate invertible. Every homogeneous equation defining D still vanishes, since it vanishes in E and the neighbourhood is integral. The rational map consequently extends to a map into D at every point.

These local extensions glue, and the two composites are identities. To justify both assertions, two maps from an integral scheme to a separated scheme that agree generically agree everywhere: their equalizer is the inverse image of the closed diagonal. On each affine open of the source, every equation of this equalizer is zero in the function field and therefore zero in the domain. For gluing use that D is separated. For the composites use that both D and C' are separated; the latter is finite over the separated curve C, hence separated. This last assertion also follows on affine pieces because the diagonal of an affine morphism is induced by the surjective multiplication map $S \otimes_R S \to S$. Thus g is an isomorphism and f is finite.

The finite algebra of f is torsion-free over each local domain of C, since it embeds in E. A finite torsion-free module M over a DVR is free: lift a basis of $M/\pi M$, and Nakayama gives generators. If there were a nonzero relation, divide all its coefficients by the smallest occurring parameter power. Torsion-freeness makes the divided relation valid, while reduction modulo π gives a nonzero relation in that residue-field basis, a contradiction. At the generic point the module is the F-vector space E, so every such free rank is n. These bases hold on neighbourhoods: the cokernel of a proposed basis map is finite and vanishes at the point, hence vanishes after shrinking; its kernel is finite by Noetherianity and vanishes there as well. Clearing denominators of generators kills both on a common neighbourhood. Hence f is locally free of rank n.

Let x be a closed point of C. Its fibre algebra A has dimension n over k. Every prime quotient of A is a finite-dimensional domain over k, hence a field by the injective-multiplication argument of Lemma 4.A. Each residue field is k: every element is algebraic, and its irreducible minimal polynomial over the algebraically closed k has degree one. There are finitely many maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_r$. Indeed, distinct maximal ideals are comaximal, and any list of length t gives a surjection $A \to k^t$ by the Chinese remainder calculation, forcing $t \le n$. For completeness, if $I + J = A$, choose $u \in I, v \in J$ with $u + v = 1$. The pair of residues represented by a, b has lift $va + ub$; the kernel is $I \cap J = IJ$, because $z = z(u + v)$ for $z \in I \cap J$. Iterating proves the asserted surjection.

The nilradical $N = \bigcap_i \mathfrak{m}_i$ is nilpotent. It has finitely many generators, as A is Noetherian, and each generator is nilpotent; a sufficiently long monomial in the generators must contain a vanishing power. Choose q with $N^q = 0$. The ideals $\mathfrak{m}_i^q$ remain pairwise comaximal: expand $1 = (u + v)^{2q-1}$ with $u \in \mathfrak{m}_i, v \in \mathfrak{m}_j$, and each term belongs to one of those two powers. Their intersection equals their product, which is $(\prod_i \mathfrak{m}_i)^q = N^q = 0$. The same Chinese remainder calculation therefore decomposes A into its local factors, equivalently its localizations at the fibre points.

At a point y over x, let π_x be a parameter of $\mathcal{O}_{C,x}$. In the source DVR its image is a unit times $\pi_y^{e_y}$ for an integer $e_y > 0$: it is nonzero by the function-field injection and belongs to the maximal ideal by locality. The corresponding fibre factor is $\mathcal{O}_{D,y}/(\pi_x)$, whose filtration by powers of π_y has exactly e_y successive quotients equal to its residue field k. Its dimension is therefore e_y, and the decomposition gives $\sum_{f(y)=x} e_y = n$. Pulling back the local equation of the divisor x gives $\sum e_y y$. Lemma 6.3 represents every invertible sheaf L by a divisor. Apply this calculation to each of its finitely many coefficients, including negative coefficients, to obtain $\deg f^* L = n \deg L$. $\square$

Proposition 6.5. A smooth connected projective curve over algebraically closed k dominated by $\mathbf{P}^1_k$ has genus zero. A genus-zero smooth connected projective curve over such k is isomorphic to $\mathbf{P}^1_k$.

Proof. First suppose $f : \mathbf{P}^1 \to C$ is separable of degree n. The differential map $f^*\Omega_{C/k} \to \Omega_{\mathbf{P}^1/k}$ is generically an isomorphism. To prove the field calculation directly, a k-derivation on the smaller function field extends uniquely across a separable generator u with minimal polynomial $g(T) = \sum_i a_i T^i$: its value on u must be $-\sum_i \delta(a_i)u^i/g'(u)$. The denominator is nonzero by S02 Lemma 4.A; that value kills the polynomial relation and therefore defines a derivation of the quotient and of its fraction field. Iterating a finite generator tower proves that the absolute differential module upstairs is scalar extension of the one downstairs, by the universal differential property. Both have rank one by S02 Lemma 3.F. A generically nonzero map of line bundles on an integral smooth curve is injective and has an effective zero divisor R. Its local matrix is a nonzero DVR element, whose valuation is the coefficient of that divisor. Thus

$$-2 = \deg \Omega_{\mathbf{P}^1/k} = n(2g-2) + \deg R.$$

Here the degree on $\mathbf{P}^1$ is -2, either by the two-coordinate calculation $dz = -t^{-2}dt$, or by Lemma 6.3 and $H^1(\mathbf{P}^1, \mathcal{O}) = 0$ from the complete twist calculation in S01 Theorem 7.2. Since $n > 0$ and R is effective, $g \geq 1$ is impossible.

In characteristic $p > 0$, an inseparable dominant map also yields a separable dominant map from a projective line. Regard $k(C)$ as a subfield of $k(t)$. Choose the largest e such that $k(C) \subset k(t^{p^e})$. Such an e is bounded: for one nonconstant rational function of $k(C)$, containment in $k(t^{p^e})$ makes its reduced numerator and denominator polynomials in t^{p^e}, so p^e is bounded by their maximum degree. Indeed a coprime presentation in that variable remains coprime after substitution by its polynomial Bezout identity; uniqueness of a reduced fraction then gives the assertion. Put $z = t^{p^e}$. The kernel of d/dz on $k(z)$ is exactly $k(z^p)$. Indeed the basis $1, z, \ldots, z^{p-1}$ over $k(z^p)$ makes the derivative's kernel its constant summand; independence follows by clearing the polynomial denominators in a relation: the summands then have exponents in distinct congruence classes modulo p, so every coefficient is zero. The same reduction writes every polynomial as such a sum, and $h^{-1} = h^{p-1}/h^p$ for any nonzero polynomial h shows that localizing at polynomials preserves this basis, since k is perfect. Thus these elements are a field basis. Maximality supplies $a \in k(C)$ with derivative nonzero. If $a = P(z)/Q(z)$ in coprime form, the polynomial $P(T) - aQ(T)$ has nonzero derivative at z; the minimal polynomial of z over $k(a)$ therefore has nonzero derivative and the extension is separable. The intermediate extension $k(z)/k(C)$ is separable as well, since an element separable over a smaller field remains separable over an intermediate field. Its field embedding gives a rational map $\mathbf{P}^1_z \dashrightarrow C$, extended everywhere by the homogeneous-coordinate DVR argument above. The separable calculation applies and gives genus zero.

Conversely let $g = 0$ and fix $q \in C(k)$. For $m \geq 0$, Lemma 6.3 and duality give

$$h^0(\mathcal{O}_C(mq)) = m + 1,$$

because $\Omega_{C/k}(-mq)$ has degree $-2 - m < 0$. Choose x complementing the constants in $H^0(\mathcal{O}_C(q))$. It is a rational function with its only pole at q, of exact order one. The functions $1, x, \ldots, x^m$ have different pole orders and hence are linearly independent; their number is $m + 1$, so they form the displayed space's basis.

Every rational function h lies in $k(x)$. To see this without a degree-one-map assertion, list its finitely many poles outside q. At each such a, the function $x - x(a)$ is nonzero and has positive valuation. Multiply h by sufficiently large powers of all these functions to cancel its poles there. The product then lies in some $H^0(\mathcal{O}_C(mq))$, and therefore is a polynomial in x. Dividing by the multiplying polynomial proves the claim. Thus $k(C) = k(x)$. Inverse rational maps between C and $\mathbf{P}^1$ extend at every DVR by their projective homogeneous coordinates and agree globally by the equalizer argument. They are inverse isomorphisms. □

These proofs must be placed before AG-RG-01. In particular, for a nonempty finite boundary divisor D on a smooth completion, $\deg T_C(-D) = 2 - 2g - \deg D < 0$ when $g \geq 1$, and Lemma 6.3

gives $H^0(C, T_C(-D)) = 0$. This is the exact degree-and-genus input used by the one-dimensional unipotent classification; none of it is deferred to a later group lesson.

7. Applying the proof route in the group lessons

The normalizer construction in AG-RG-03 uses Theorem 4.5 to choose smooth affine group and subgroup models with geometrically connected fibres; its finite group equations and closed immersion descend in the same finite list. The projective homogeneous quotient proof uses Theorem 2.3 on its actual projective closure and Corollary 2.4 for the formal idempotent. Lemma 1.3 supplies faithful flatness of local completion, the actual argument of AG-CA-19 Theorem 3.2. Section 5 supplies the line bundle cohomology statement over arbitrary, including nonreduced, bases used in AG-RG-03 and AG-RG-06. Section 6 supplies the smooth curve duality, divisor degrees, genus calculation and domination by a projective line used in AG-RG-01 and AG-RG-03; it must precede AG-RG-01.

For AG-RG-06's use of sufficiently large powers of an ample line bundle, the exact earlier embedding proofs are AG-RG-S01, Lemma 8.1 and Lemma 8.3: finitely many nonvanishing sections with affine domains, denominator clearing for their coordinate generators, and an immersion from those generators. AG-RG-S01, Corollary 7.B turns that immersion into a closed immersion when the source is proper. AG-RG-S01, Theorem 8.5 treats every sufficiently large power by applying its projective finiteness and vanishing argument to all finitely many residue classes modulo the embedding exponent.

Freely readable sources and history

Mathematical source material for this lesson is the freely readable Stacks Project, especially the exact tags linked next to the proved statements. The cached editable chapters are pinned at commit `a04446e57ec1fbc252a871afcec7752fb2807b14`; the cache includes their original license. The proof differences were written on 5 October 2026. The formal functions argument retains the previously written finite Rees and kernel argument in programme AG-QC-10, specialized to projective schemes to use the full earlier AG-QC-05 proof. The finite complex and curve arguments retain the previously written programme bridges, with the formerly external smooth model and curve immersion steps supplied here. No paid book supplies an argument or a public citation in this lesson.

History

Source: *The Stacks Project*, by the Stacks Project authors, published openly under the GNU FDL 1.2 or later. Exact freely readable source tags are identified in this lesson.

Edition: *Projective cohomology and smooth affine models*, 5 October 2026, written by GPT-6.1 Sol (OpenAI), Codex, Ultra setting, for Open Mathematics Courses: formal functions, affine smooth models, universal cohomology complexes and the curve arguments, with the intermediate proof steps supplied. Self-checked by the writing AI. The text is dedicated to the public domain under CC0 1.0.

Completing a smooth affine curve and extending its group action

Original proof draft, October 2026. CC0. This lesson precedes the reductive-group lessons. Its remaining programme dependencies are listed at the end; a reference to an external proof does not discharge them.

Let k be algebraically closed. A **curve** here is an integral scheme of finite type over k of dimension one. We first construct the completion needed in the classification of one-dimensional unipotent groups. In particular, extending each individual translation is not enough: the translations must extend as an algebraic family.

1. Finite normalization for curves

We use the already proved Noether normalization theorem and its dimension formula: a finite-type domain of dimension one is finite over a polynomial subring $k[t]$. The exact earlier proof is AG-CA, *Krull dimension and Noether normalization*, Lemma 2.1, Theorem 3.1, Corollary 3.2 and Theorem 4.2, with the integral-extension proofs specified there. The theorem is used in its finite-module form, not merely as a statement about function fields.

Lemma 1.1 (a separating finite parameter). If A is a finite-type k-domain of dimension one and $K = \operatorname{Frac}(A)$, some $u \in A$ makes A finite over $k[u]$ and $K/k(u)$ finite separable.

Proof. Choose t by Noether normalization. In characteristic zero use $u = t$. Suppose the characteristic is $p > 0$. Write $n = [K : k(t)]$. Frobenius identifies the extension $K/k(t)$ with $K^p/k(t^p)$, since k is perfect. Thus

$$[K : k(t^p)] = np, \qquad [K^p : k(t^p)] = n, \qquad [K : K^p] = p.$$

The tower formula gives the last equality. Some $z \in A$ lies outside K^p, because otherwise its fraction field would lie in K^p. Hence $K = K^p(z)$ and $1, z, \dots, z^{p-1}$ is a basis over K^p. A k-derivation annihilates K^p, and arbitrary assignment of its value on z defines a derivation: use the presentation $K^p[Z]/(Z^p - z^p)$. Consequently $\Omega_{K/k}$ is one-dimensional with basis dz.

The element z satisfies a monic equation

$$z^d + c_1(t)z^{d-1} + \cdots + c_d(t) = 0, \qquad c_i(t) \in k[t].$$

Choose a p-power P so large that $\deg c_i < iP$ for every nonzero c_i, and put $u = t^P + z$. Substitution of $z = u - t^P$ gives a polynomial equation in t over $k[u]$. Its term of highest t-degree is $(-t^P)^d$, of degree Pd and coefficient $(-1)^d$; every other term has smaller degree by the stated inequalities. Dividing by that unit makes a monic equation for t. Thus $k[t, u]$ is finite over $k[u]$. A finite $k[t]$-module generating list for A also generates it over $k[t, u] \subset A$, so A is finite over $k[u]$.

We have $du = dz \neq 0$, hence $\Omega_{K/k(u)} = 0$: a $k(u)$-derivation kills du, which is a basis of $\Omega_{K/k}$, so every such derivation vanishes. Here u is transcendental, since otherwise the finite $k[u]$-module A would be finite-dimensional over k. A finite-dimensional domain is a field, because multiplication by a nonzero element is injective and hence surjective; its dimension is zero, a contradiction.

For clarity, for a finite field extension E/F the condition $\Omega_{E/F} = 0$ is equivalent to separability. A separable generator has a minimal polynomial with nonzero derivative, whose differential forces its differential to vanish; successively adjoining separable generators proves one direction. For the converse in positive characteristic, the elements of E separable over F form a subfield F_s: a finite list of such elements generates a separable extension by the complete tower-and-embedding calculation in S02 Lemma 4.A, so its sums, products and inverses are separable too. Every element of E has a p-power in F_s. Indeed its irreducible polynomial is $g(T^{p^r})$ with $g' \neq 0$, obtained by repeatedly removing zero derivatives. The polynomial g remains irreducible, since a factorization would give one of the original polynomial after substitution; thus the p^rth power of the element is separable. Consequently E/F_s is purely inseparable.

If this extension is nontrivial, choose a maximal proper intermediate field E'. Such a choice exists because the intermediate degrees are integers bounded by $[E : F_s]$. Adjoining any $\alpha \notin E'$ generates E by maximality. Its minimal polynomial is $T^{p^e} - a$, where $e \geq 1$ is least with $\alpha^{p^e} \in E'$. Indeed that polynomial is $(T - \alpha)^{p^e}$ over E, so its monic minimal factor is $(T - \alpha)^d$. If $d = p^r m < p^e$ with $p \nmid m$, its coefficient at degree $d - p^r$ is $-m\alpha^{p^r}$; that coefficient in E' would contradict the choice of e. Thus $d = p^e$. If $e > 1$, then $E'(\alpha^p)$ has degree p^{e-1} over E': a lower degree would give an annihilating polynomial for α of degree smaller than p^e after substituting T^p. It would therefore be a proper intermediate field, a contradiction. Hence $e = 1$, and $E = E'[Z]/(Z^p - a)$. Sending Z to 1 and E' to zero defines a nonzero F-derivation of this quotient, since it kills the defining polynomial. This contradicts $\Omega_{E/F} = 0$. In characteristic zero every irreducible polynomial is separable by the nonzero-derivative and Bezout calculation of S02 Lemma 4.A. The criterion and the lemma follow in all characteristics. □

Lemma 1.2 (finite integral closure). The integral closure of A in K is a finite A-module.

Proof. Lemma 1.1 gives a subring $R = k[u] \subset A$ such that A is a finite R-module and K/F, with $F = k(u)$, is finite separable. The parameter u is transcendental: otherwise algebraic closedness would give $R = k$, and a finite-dimensional domain over k would be a field, of dimension zero, contradicting $\dim A = 1$.

We check the hypotheses of the earlier S02 Lemma 4.B, whose complete proof constructs the splitting product, both interpolation inverses, the trace formula and the integral-closure bound. Polynomial division, proved in S02 Lemma 4.A, makes every nonzero ideal of $k[u]$ principal: choose a nonzero member of least degree and divide every other member by it. The remainder must be zero by minimality. The zero ideal is principal too. Thus R is Noetherian by the ideal finite-generation criterion proved in S01 Lemma 1.3.

The ring R is normal. Write a fraction in lowest terms as a/b, with $a, b \in R$, $b \neq 0$, and $ra + sb = 1$ for some $r, s \in R$, using the same polynomial Bezout proof. If it satisfies a monic equation of degree m, multiplication by b^m shows $b \mid a^m$. Expanding $(ra + sb)^m = 1$ shows that b then divides 1: the first term is divisible by b, as are all the other terms. Hence b is a unit and the fraction lies in R.

Apply S02 Lemma 4.B to this Noetherian normal domain and the finite separable extension K/F. Its integral closure B in K is a finite R-module. In particular the trace pairing and its nondegeneracy are supplied by an earlier programme proof, including the case when the characteristic divides $[K : F]$; they are not assumed from a Vandermonde assertion.

The integral closure of A in K is precisely B. Every element integral over R is integral over A, since the same monic equation has coefficients in A. Conversely, A is integral over R, and an element integral over A is integral over R by the complete transitivity proof in the earlier AG-CA *Integral extensions*, Theorem 1.2. Its finite-coefficient argument applies here: the finitely many coefficients of the monic equation lie in a finite R-algebra, adjoining the element gives a finite module over that

algebra and hence over R, and the determinant argument gives a monic equation over R. Finally an R-module generating list for B also generates it as an A-module, since $R \subset A \subset B$. This proves the assertion. $\square$

Integral closure in the fraction field commutes with localization. One inclusion follows by localizing a monic equation; sums and products of integral elements remain integral by the earlier AG-CA Theorem 1.2. Conversely, let x satisfy a monic equation of degree m over A_s, whose ith coefficient is a_i/s^{h_i}. Choose N with $Ni \geq h_i$ for every coefficient. Multiplying the equation by s^{Nm} gives the monic equation for $s^N x$ with ith coefficient $s^{Ni-h_i}a_i \in A$. Thus $s^N x$ belongs to the integral closure of A and x belongs to its localization. For a general multiplicative set, use the product of the finitely many denominators appearing in the equation.

2. Normal local curves and the smooth completion

Lemma 2.1. A one-dimensional Noetherian normal local domain is a discrete valuation ring.

Proof. Let $(R, \mathfrak{m})$ be such a ring and choose $0 \neq a \in \mathfrak{m}$. Every prime containing a is $\mathfrak{m}$, since any other nonzero prime would give a chain of length at least two. Thus the radical of (a) is $\mathfrak{m}$. Finite generation of $\mathfrak{m}$ gives $\mathfrak{m}^n \subset (a)$ for some n: choose a power of each generator lying in (a) and use the pigeonhole principle on monomials. Choose the least n and $b \in \mathfrak{m}^{n-1} \setminus (a)$. Then $c = b/a \notin R$ and $c\mathfrak{m} \subset R$.

If $c\mathfrak{m} \subset \mathfrak{m}$, apply the determinant trick to multiplication by c on the finite module $\mathfrak{m}$. Its resulting monic polynomial annihilates this module, hence is zero in $\mathrm{Frac}(R)$ because the domain has a nonzero element in $\mathfrak{m}$. Normality would imply $c \in R$, a contradiction. Consequently cx is a unit for some $x \in \mathfrak{m}$. It follows that $\pi = c^{-1} = x/(cx) \in \mathfrak{m}$ and $\mathfrak{m} = \pi R$.

The intersection $N = \bigcap_j \pi^j R$ is zero. In fact N is a finite ideal, since R is Noetherian, and cancellation of $\pi \neq 0$ shows $N = \pi N$: if $x = \pi y$ lies in every $\pi^{j+1}R$, then y lies in every $\pi^j R$. Nakayama gives $N = 0$. Repeated division of a nonzero nonunit by π must therefore stop. Every nonzero element is uniquely a unit times π^j, and every nonzero ideal is generated by the element of smallest such exponent. This is the discrete valuation ring assertion. $\square$

Lemma 2.A (normal curves are smooth over the algebraically closed field). Every normal integral finite-type curve over k is smooth over k.

Proof. Let Z be such a curve and z a closed point. Its local ring is a one-dimensional Noetherian normal local domain. Indeed, on an affine neighborhood $\operatorname{Spec} B$, the maximal prime of the closed point is nonzero: if it were zero, the domain B would be a field finite over k by the earlier Nullstellensatz, contradicting dimension one. Every chain below this nonzero prime has length at most one. The denominator calculation in the next paragraph gives normality of the localized ring. Lemma 2.1 therefore makes the local ring a DVR. Its maximal ideal is generated by one parameter, so its embedding dimension and its dimension are both one. Its residue field is k by the Nullstellensatz.

The coordinate domain B of an affine chart of a normal scheme is normal: an integral fraction belongs to every maximal localization by stalk normality. If it were absent from B, its proper denominator ideal would lie in a maximal ideal; membership in that localization would then give a denominator outside that ideal, a contradiction. Localization normality also follows directly. If a fraction y is integral over $B_{\mathfrak{p}}$, choose one $s \notin \mathfrak{p}$ clearing the finitely many coefficients of its monic equation. Multiplication of that equation by s^m, where m is its degree, makes sy integral over B: its ith coefficient is s^i times the original coefficient and belongs to B. Normality of B gives $sy \in B$, hence $y \in B_{\mathfrak{p}}$.

The earlier AG-CA *Smooth algebras over a field and the Jacobian criterion*, Proposition 3.1, proves the isomorphism

$$\mathfrak{m}_z/\mathfrak{m}_z^2 \simeq \Omega_{B/k} \otimes_B k$$

at this rational point, using the split conormal sequence of the structural field section. The right side consequently has dimension one. In the notation of its Theorem 2.1, $a_z = d_z = 1$, so its complete Jacobian proof gives a principal smooth neighborhood of z. This application needs neither an unproved geometric-regularity converse nor an identification of maps between completed local rings.

Corollary 2.2 of that earlier lesson proves that the smooth locus is open. If its closed complement in Z were nonempty, intersect it with an affine neighborhood of one of its points. A nonzero finite-type quotient algebra has a maximal ideal; the Nullstellensatz makes its residue field k. Such a rational point is closed also in Z: on any affine neighborhood of a specialization its residue map to k is surjective, hence has maximal kernel, so there is no proper specialization. The complement would thus contain a closed point, contrary to the preceding argument. It is empty, proving smoothness everywhere. □

Theorem 2.2 (smooth projective completion). Every smooth connected affine curve X/k is an open subscheme of a smooth projective curve C/k, and $D = C \setminus X$ is a nonempty finite reduced set of closed points.

Proof. Smooth local rings are regular by the earlier AG-CA Jacobian Theorem 2.1, and regular local rings are domains by AG-CA *Regular local rings*, Theorem 1.1. Thus different irreducible components of X cannot meet: at a point of intersection the local ring would have distinct minimal primes. A finite-type scheme is Noetherian, so it has finitely many components. To justify this topological step, a closed subset failing to be a finite union of irreducible closed subsets has a minimal such subset by the descending chain condition; splitting it into two proper closed subsets contradicts minimality. Disjoint components are then both closed and open. Connectedness leaves one component, and local domains give reducedness, so X is integral. Its closed-point local rings have dimension one and are regular domains; the earlier S02 DVR Lemma 6.D proves directly that they are normal DVRs. Its generic local ring is its function field and is normal as well. These are all points of an integral curve, so X is normal. Its affine coordinate domain is normal too: an integral fraction belongs to every maximal localization by the just proved local normality. If it were absent from the coordinate ring, its proper denominator ideal would lie in a maximal ideal, whereas membership in that localization supplies a denominator outside that maximal ideal, a contradiction. No normality hypothesis is used to establish this conclusion.

Choose finitely many generators of its affine coordinate algebra, obtaining a closed immersion $X \hookrightarrow \mathbb{A}^n_k$. Let Y be its reduced projective closure in $\mathbb{P}^n_k$. The closure of the irreducible X is irreducible; reducedness makes Y integral. Its dense open $Y \cap \mathbb{A}^n_k = X$ has the same function field and dimension one, by the earlier finite-type dimension theorem. Normalize its affine charts in $k(X)$. Lemma 1.2 makes their closures finite modules, and the localization argument following that lemma identifies the closures on all principal overlaps. They therefore glue to a finite birational morphism $\nu : C \to Y$. The integral-closure construction is normal: an element integral over a closure is integral over the original ring by transitivity and hence already lies in the closure. The source is integral of dimension one by the integral-extension dimension theorem, and $\nu^{-1}(X) \to X$ is an isomorphism because X is normal. Lemma 2.A makes C smooth.

The morphism ν is proper. On an affine chart its diagonal is the closed immersion induced by the surjection $B \otimes_A B \to B$, $b \otimes b' \mapsto bb'$; this proves separatedness. It is finite type. For universal closedness, every base change is still finite. On any affine chart after that base change the image of a closed set $V(J) \subset \operatorname{Spec} B$ is $V(J \cap A)$: apply the earlier lying-over theorem to the injective integral map $A/(J \cap A) \hookrightarrow B/J$. These affine assertions prove that every base change is closed. The projective-properness and composition proofs in S01 Lemma 7.A and Corollary 7.B now show that C is proper over k.

We also prove projectivity. Pull back the standard coordinate sections of $\mathcal{O}_Y(1)$ to sections of the invertible sheaf $L = \nu^*\mathcal{O}_Y(1)$. Their nonvanishing opens cover C and are affine, since ν is finite. Principal opens in each of these affine charts form a basis. S01 Lemma 8.1 extends a function on

such a chart to a section of a power of L, after multiplication by a sufficiently high power of the chart's section; if necessary one further multiplication makes it vanish off the chart. Its nonvanishing locus is exactly the chosen principal open. Thus the nonvanishing loci of sections of powers of L form a basis: L is ample in the sense used by S01 Lemma 8.3. That lemma's complete construction gives an immersion $C \to \mathbb{P}^N_k$. This immersion is proper: its graph is closed because projective space is separated, and projection from $C \times_k \mathbb{P}^N_k$ is proper by base change from $C \to \operatorname{Spec} k$. The proper-immersion proof in S01 Corollary 7.B makes the immersion closed. Therefore C is projective.

The complement $C \setminus X$ is a proper closed subset of an integral Noetherian curve. Every point in it is closed, since the generic point belongs to X; a zero-dimensional Noetherian closed subset has finitely many irreducible components, each a single point. Endow this finite set with its reduced closed-subscheme structure and call it D.

We retain the extension calculation needed for the action theorem below. A rational map from a smooth integral curve to a closed projective scheme extends at every missing closed point. In that point's DVR choose homogeneous coordinates in the function field and divide all coordinates by a common power of its parameter so that their minimum valuation is zero. All coordinates then belong to the local ring and at least one is a unit. Represent them by regular functions on a neighborhood and shrink it so that this coordinate stays invertible. They define a morphism to projective space there. The homogeneous equations of the target remain zero, because they vanish in the function field and the neighborhood is integral. Two extensions agree on overlaps: the inverse image of the target's closed diagonal contains their dense common domain, and its defining ideal is zero in a reduced integral source. This proves uniqueness and gluing.

Finally D is nonempty. A regular function f on the proper integral C defines a morphism $C \to \mathbb{A}^1_k \subset \mathbb{P}^1_k$. This morphism to projective space is proper by the same graph and base-change argument, so its image is closed. If f were nonconstant, it would be transcendental over the algebraically closed field k: an element algebraic over k satisfies a polynomial split into linear factors and hence lies in k. Thus the map on generic points is dominant and its image contains the generic point of $\mathbb{P}^1_k$. A closed image containing that point is all of $\mathbb{P}^1_k$, impossible for a map factoring through $\mathbb{A}^1_k$. Every regular function on C is consequently constant. If D were empty, $C = X$ would be affine with coordinate ring k, so $C = \operatorname{Spec} k$, contradicting dimension one. This proves every assertion. $\square$

The curve C is geometrically integral. Its global functions are k by the final argument of Theorem 2.2. Here is the precise base-change calculation required for this assertion. Choose a finite affine cover V_i of the separated C. Each intersection $V_i \cap V_j$ is affine: it is the inverse image of $V_i \times_k V_j$ under the closed diagonal. The sheaf axiom identifies global functions with the kernel of the difference-of-restrictions map from $\prod_i \Gamma(V_i, \mathcal{O})$ to $\prod_{i,j} \Gamma(V_i \cap V_j, \mathcal{O})$. For a field extension L/k, tensoring with the vector space L preserves this kernel and these finite products. The affine coordinate and localization calculation gives the base-changed coordinate rings, and hence identifies the tensored kernel with $H^0(C_L, \mathcal{O})$. Thus $H^0(C_L, \mathcal{O}) = L$. Consequently C_L is connected: a nontrivial partition into open-and-closed subsets would give a nontrivial idempotent among its global functions. Smoothness persists after base change. Regular local domains make its finitely many irreducible components disjoint and open, by the same Noetherian argument as in Theorem 2.2; connectedness leaves one, and reducedness gives integrality. Its nonempty open X_L is integral too. Products of these geometrically integral schemes are integral: on affine charts the tensor product injects into the tensor product with the function field of either integral factor, because tensoring a vector space over k preserves injections; the latter ring is a domain by geometric integrality of the other factor. This proves the density assertions used next.

3. Extending translations as a family

Theorem 3.1. Let H/k be a smooth connected affine group of finite type and dimension one. In its smooth projective completion C, multiplication extends uniquely to a morphism $H \times C \to C$ defining an action. Every point of $D = C \setminus H$ is fixed.

Proof. Each translation on H is a birational map of C. The rational-map argument in Theorem 2.2 extends it to a morphism of C, and does the same for its inverse. Their composites agree with the identity on H, hence on C. Thus each translation is an automorphism of C.

We now prove regularity in the group variable. Multiplication gives a rational map $H \times C \dashrightarrow C$ defined on $H \times H$. Over the generic point η of H it is a birational map of the smooth projective curve $C_{k(H)}$, with inverse the generic inverse translation. The same DVR argument over that field extends it everywhere on the generic fibre. Each extension chart is given by finitely many rational functions and finitely many equations, so clearing their denominators extends it to an open neighbourhood in $H \times C$. These finitely many charts agree where they overlap: their maps to the separated target agree on the dense open $H \times H$, and the source is integral. Their union contains the whole generic fibre. The largest domain of this regular rational map is therefore an open W containing that fibre.

Since C is proper, projection sends the closed complement of W to a closed subset of H. It misses the generic point. Consequently there is a nonempty open $U \subset H$ with $U \times C \subset W$. Choose $a \in U(k)$, which exists by the Nullstellensatz. Let τ_a denote the already extended fixed translation. On the neighbourhood $V = a^{-1}U$ of the identity the formula

$$\beta(g, x) = \tau_a^{-1}\big(\alpha(ag, x)\big)$$

is regular and equals multiplication on $V \times H$. For every $g_0 \in H(k)$, the formula

$$(g, x) \longmapsto \tau_{g_0}\big(\beta(g_0^{-1}g, x)\big)$$

is regular on $g_0V \times C$. These opens cover H: they contain every closed point, and a nonempty closed complement in a finite-type scheme over k has a closed point. They agree on overlaps by density and separatedness, so glue to $\alpha : H \times C \to C$. The unit and associativity identities hold on the dense open with all curve variables in H, and hence everywhere. Smoothness and connectedness make H, C and their products geometrically integral, so this density argument applies. Uniqueness follows the same way.

Every translation preserves H, thus its complement D. The orbit map $H \to C$ of a point of D has image in the finite set D on closed points. Its scheme-theoretic factorization through that reduced set follows from reducedness of H and detection by its rational points. Explicitly, a nonzero function on a reduced affine finite-type k-scheme is not nilpotent, so its localization is a nonzero finite-type algebra. A maximal ideal of that localization has residue field k by the weak Nullstellensatz, giving a rational point where the function is nonzero. Apply this on affine charts to the equations of D. Connectedness makes the orbit map constant. Thus each boundary point is fixed on all test rings. Differentiating the action at the identity gives a regular vector field on C. At a boundary point its derivation carries a local parameter into its own ideal, because the orbit of that reduced point is constant; the field therefore vanishes there. It is consequently a section of $T_C(-D)$. Its restriction to H is the translation field whose value at the identity is any chosen nonzero element of the one-dimensional $\mathrm{Lie}(H)$, so it is nonzero. □

4. How this enters the one-dimensional unipotent proof

Lemma 4.1 (characters of a unipotent group). Let H be a reduced affine group of finite type over the algebraically closed k, with a closed embedding into GL_n whose closed-point matrices are unipotent. Every group homomorphism $\chi : H \to \mathbb{G}_m$ is trivial.

Proof. In characteristic $p > 0$, choose $p^e \geq n$. For a matrix $h = I + N$ with $N^n = 0$, the binomial identity gives $h^{p^e} = I + N^{p^e} = I$. Hence $\chi(h)^{p^e} = 1$. A field of characteristic p has no nonidentity root of that equation, so $\chi(h) = 1$.

In characteristic zero fix such an h and define the polynomial matrix

$$\gamma(t) = \sum_{j=0}^{n-1} \binom{t}{j} N^j.$$

For every nonnegative integer m it equals h^m. Every defining polynomial of the closed subgroup H therefore vanishes on $\gamma(t)$ at infinitely many values of t, and vanishes identically. Its determinant is one, since N is nilpotent, so this gives a morphism $\mathbb{A}^1 \to H$. The identity $\gamma(s)\gamma(t) = \gamma(s+t)$ follows from the usual binomial formula at all pairs of nonnegative integers; a polynomial in two variables vanishing on that grid is zero, by first fixing one integer variable and then the other. Thus γ is a group homomorphism $\mathbb{G}_a \to H$, and $\gamma(1) = h$. The pullback of a character is an invertible element of $k[t]$, hence a nonzero constant. The identity at zero makes it one, so again $\chi(h) = 1$.

In either characteristic $\chi - 1$ vanishes at every closed point of H. The Nullstellensatz and reducedness make that regular function zero, proving triviality as a scheme morphism. This proof uses no triangularization or solvable-group classification. □

Theorem 4.2 (one-dimensional unipotent groups). Let H/k be a smooth connected affine group of finite type and dimension one, with a closed embedding into GL_n whose closed-point matrices are unipotent. Then H is isomorphic to $\mathbf{G}_a$ as a group scheme.

Proof. Theorems 2.2 and 3.1 give its smooth projective completion C, a nonempty finite reduced boundary D, and a nonzero section of $T_C(-D)$. The complete curve-duality and degree proofs in S02 Theorem 6.2 and Lemma 6.3 give $\deg T_C(-D) = 2 - 2g - |D|$ and the vanishing of sections of negative-degree line bundles. Thus $g \geq 1$ would contradict the nonzero section. Hence $g = 0$, and S02 Proposition 6.5 identifies C with $\mathbf{P}^1$.

For completeness, an automorphism of $\mathbb{P}^1_k$ is fractional linear. Its map on function fields sends t to a nonconstant rational function $r(t)$ generating $k(t)$. Write $r = a(t)/b(t)$ in reduced form. The polynomial $a(T) - Ub(T)$ is irreducible in $k[T, U]$: since its degree in U is one, a factorization would have a nonconstant factor depending only on T, dividing both coprime polynomials a and b. It is primitive as a polynomial over $k[U]$ too. Indeed a common nonconstant factor of its coefficients would have a root $u \in k$, forcing $a(T) = ub(T)$ and making r constant. The complete Gauss calculation in the earlier AG-CA *Integral extensions*, Proposition 2.3, therefore makes $a(T) - rb(T)$ irreducible in $k(r)[T]$. Thus $[k(t) : k(r)] = \max(\deg a, \deg b)$. Since that degree is one, $r = (at + b)/(ct + d)$ with $ad - bc \neq 0$. After putting a fixed point at infinity its stabilizer consists of $t \mapsto at + b$; after putting two individually fixed points at zero and infinity the transformations are $t \mapsto at$. Fixing a third distinct point then forces $a = 1$. The curve action is faithful, since on its open orbit H it is left translation. With three or more boundary points its action would be trivial, an impossibility. With two individually fixed boundary points it acts by scalings, and with one it acts by affine transformations.

Here these restrictions on the automorphism family are scheme-theoretic. With three boundary points every translation on a closed point of H is the identity. The family is consequently the identity: on affine charts its coordinate differences vanish on all closed points of the reduced finite-type source, so the Nullstellensatz makes them zero. With two fixed boundary points put them at zero and infinity. On their complementary $\mathbb{G}_m$ write the pullback of t as a Laurent

polynomial in t with coefficients in $k[H]$. For every closed point of H it is at; all other coefficients vanish on all closed points and hence are zero. The inverse action makes a a unit. The action law makes $a : H \to \mathbb{G}_m$ a group homomorphism. With one boundary point put it at infinity. The pullback of t on $H \times \mathbb{A}^1$ is a polynomial in t. Each closed-point automorphism has degree one, so the same coefficient argument leaves $at + b$, with a a unit. Again a is a regular group homomorphism, not merely a character on rational points.

For a unipotent H as in Lemma 4.1, every homomorphism $H \to \mathbb{G}_m$ is trivial. Thus two boundary points would make the action trivial and are impossible. With one boundary point the slope character is trivial, so the action is $t \mapsto t + b$. The action identity says $b(gh) = b(g) + b(h)$ over every test algebra. After choosing the coordinate to give the identity of H coordinate zero, its orbit map is precisely $b : H \to \mathbb{A}^1$ and is the original open embedding $H = C \setminus D = \mathbb{A}^1$. It therefore identifies H with the additive group $\mathbb{G}_a$, as schemes and with its group law. In particular the result applies to the one-dimensional quotients embedded in upper unitriangular groups in the later solvable-group proof. □

Internal proof dependencies and free comparison

The construction above consumes the complete earlier AG-CA proofs of Noether normalization, integral extensions, the Nullstellensatz, regular local rings, and the smooth Jacobian criterion. It consumes AG-RG-S01 Lemma 7.A and Corollary 7.B for projective-space properness, projections and graphs, and Sections 8.1–8.3 of that supporting lesson for denominator extension and ample immersion. Section 4 additionally consumes the earlier curve-duality and genus-zero proofs; its character-vanishing proof is Lemma 4.1 above and does not appeal to the later solvable-group lesson. These are internal proof dependencies to be included in the programme order before use.

For freely readable comparison of the completion statement, see Stacks, Tag 0H1F. Its citation is additional reading; Theorems 2.2 and 3.1 above provide the completion and family-extension arguments themselves.

Affine descent, Zariski Main and recognition of spaces

This supporting lesson puts the actual arguments before the group-theoretic applications. Every base and test scheme is arbitrary unless the statement gives an additional hypothesis. A finite algebra is finite as a module; finite presentation is never silently imposed. A quasi-finite morphism is quasi-compact and locally quasi-finite. Locally quasi-finite spaces in block E need not be quasi-compact.

The proof order is P: elementary integral algebra; A: faithfully flat and quasi-affine descent; B: polynomial normality; C: the conductor argument and affine completion; D: the étale structural lemmas and general scheme Zariski Main; E: locally quasi-finite effectivity, finite quotients and scheme recognition. Labels retain their source section number with the block letter, so Lemma C3.2 and Theorem D5.2 have unambiguous positions. The recognition theorem never supplies an input to its own scheme-descent proof.

The earlier programme inputs are the full elementary field, integral-extension, affine/relative-Spec, quasi-finite-fibre and scheme-limit proofs, together with Algebra and sheaf cohomology before reductive groups.

The text of this lesson is dedicated to the public domain under CC0 1.0. Its arguments follow the cited Stacks Project proofs; the Stacks Project itself is distributed under the GNU FDL 1.2 or later. Source citations identify freely accessible writing and comparison material; none substitutes for a programme proof.

P. Elementary integral algebra before descent and completion

This block reconstructs the consumed elementary arguments from freely readable Stacks material. It gives the full proofs. Localization, filtered colimits, Nakayama and the Noetherian polynomial argument have already been proved in Algebra and sheaf cohomology before reductive groups, Section 1.

Lemma P0.1 (prime separation and radicals). If an ideal I avoids a multiplicative set S, it is contained in a prime avoiding S. The radical of I is the intersection of primes containing I.

Proof. A chain of ideals containing I and avoiding S has its union as an upper bound still avoiding S. Choose a maximal such ideal J by Zorn. If $ab \in J$ with $a, b \notin J$, both larger ideals $J + (a)$ and $J + (b)$ meet S. Multiplying one chosen element from each gives an element of S in J, a contradiction. Thus J is prime. A power in I lies in every prime over I. Conversely, if no power of a lies in I, apply separation to $S = \{1, a, a^2, \ldots\}$ to obtain a prime over I missing a. This proves the radical statement. Primes of a quotient correspond to primes containing its kernel; primes of a localization correspond to primes avoiding its denominators, since extension and contraction are inverse by the fraction equality test. These identifications preserve inclusion. □

Lemma P0.2 (finite integral control). An element is integral over R exactly when its generated algebra is finite as an R-module. Integral elements form a subalgebra, integrality is transitive, and

an integral finite-type algebra is finite.

Proof. A monic degree-d relation bounds all powers by $1, b, \dots, b^{d-1}$. Conversely, on any finite module subalgebra containing 1, write multiplication by b on module generators as a matrix with coefficients in R. Its adjugate makes its monic characteristic polynomial annihilate every generator, hence 1, so it annihilates b in the algebra. For finitely many integral elements, the monomials whose exponents are below their relation degrees span their algebra. The preceding determinant argument makes every element of that algebra integral, proving closure under sums and products. For a monic relation over an integral intermediate ring, collect its finitely many coefficients. Their subalgebra is finite over the original ring; adjoining the element is finite over that subalgebra. Products of the two finite generating lists give finiteness over the original ring and hence integrality. This proves transitivity and the last assertion. A finite module algebra is of finite type and integral by the same argument. □

Lemma P0.3 (localization of integral closure). For any R-algebra B, its integral subalgebra C satisfies

$$S^{-1}C = \{z \in S^{-1}B : z \text{ integral over } S^{-1}R\}.$$

Proof. Divide a monic equation for c by its leading denominator power to prove the forward inclusion. Conversely, take a monic equation for $z = b/s$. Choose a common product $v \in S$ of all coefficient denominators and s, so $c = vz$ is the image of an element of B and has a monic equation $c^d + r_1 c^{d-1} + \cdots + r_d = 0$ with $r_i \in R$ in the localization. Some $h \in S$ kills its error in B. Then hc satisfies the genuine monic equation

$$(hc)^d + hr_1(hc)^{d-1} + \cdots + h^d r_d = 0$$

in B, since the left side is h^d times that error. Thus $hc \in C$ and $z = (hc)/(hv)$. Exact localization identifies $S^{-1}C$ with its image in $S^{-1}B$. The zero localization is included. □

Lemma P0.4 (prime lifting and closed images). An integral inclusion $R \subset B$ has lying over and going up. Distinct primes with equal contraction are incomparable. Integral maps of spectra are closed, also after every base change.

Proof. First, for integral domains $A \subset D$, one is a field exactly when the other is. If A is a field, each nonzero d acts injectively and hence surjectively on the finite-dimensional domain $A[d]$, so has an inverse. If D is a field, a monic relation for a^{-1}, multiplied by a^{d-1}, puts a^{-1} in A.

Localize an integral inclusion at $R \setminus \mathfrak{p}$ and choose a maximal ideal upstairs. Its contraction is maximal by the field assertion, so is the local base's unique maximal ideal. Contracting before localization gives lying over. Quotient by a prescribed prime upstairs and apply lying over to the quotient inclusion to obtain going up. A residue-field fibre algebra is integral over its field: monic equations survive quotient and localization. Every prime quotient is consequently a field. Its primes are all maximal, proving incomparability. For an arbitrary ideal $J \subset B$, apply lying over to $R/(J \cap R) \subset B/J$. Its image is $V(J \cap R)$, proving closedness. After any base change, elements from B remain integral, and each tensor expression uses finitely many of them; Lemma P0.2 proves integrality of the new algebra. The same closed-image proof therefore applies. □

Lemma P0.4a (the field calculations used below). Let F be any field. Polynomial division and Bezout hold in $F[T]$. Minimal-polynomial quotients, distinct-root bounds, the derivative test for a separable algebraic generator, finite splitting fields, and finite tower degree and embedding counts have the following direct proofs.

Proof. If $g \neq 0$ and f has degree at least that of g, subtract the leading coefficient of f divided by that of g, times the appropriate power of T times g. This strictly decreases the degree. Induction gives $f = qg + r$ with $r = 0$ or $\deg r < \deg g$. The leading coefficient of a product of two nonzero polynomials is the product of their nonzero leading coefficients, so degrees add. Consequently two

such division expressions have the same remainder and quotient. For polynomials a, b not both zero, repeat division on the pair and its successive remainders. Nonnegative degrees strictly decrease, so the process ends. Its last nonzero remainder, scaled to be monic, is d. Back substitution expresses $d = ua + vb$. Each earlier remainder is divisible by d, and any common divisor of a, b divides every remainder, so d is their monic greatest common divisor. In particular coprime polynomials satisfy $ua + vb = 1$. An irreducible f is prime: if f does not divide a, Bezout for f, a, multiplied by b, shows that $f \mid ab$ forces $f \mid b$. Factor existence follows by induction on degree, and prime divisibility gives uniqueness after fixing monic factors and a scalar.

For an algebraic element α, choose a monic nonzero annihilating polynomial of least degree, denoted f. Division shows that every annihilating polynomial is a multiple of f. If f were a product of two nonconstant factors, evaluating in the extension field would make one smaller factor vanish, a contradiction. Thus f is irreducible. Every nonzero class modulo f has a polynomial inverse by Bezout, so $F[T]/(f)$ is a field. Evaluation gives $F[\alpha] = F(\alpha)$ and identifies it with this quotient. The remainders $1, T, \ldots, T^{\deg f - 1}$ form a basis: they span by division, and a linear combination of smaller degree cannot be a nonzero multiple of f.

Division by $T - c$ leaves the constant $f(c)$. If c is a root, $f = (T - c)g$. Every other root is a root of g, since its difference from c is a nonzero field element. Induction proves that a nonzero polynomial of degree d has at most d distinct roots, over every extension field. A finite collection of polynomials has a finite splitting field: if one polynomial has a nonsplit nonconstant factor, factor it into irreducibles by the degree induction, adjoin a root of one such factor using the just-proved field quotient, remove the linear factor, and repeat. At most the sum of the original degrees many roots need be adjoined. The extensions at every step are finite by the displayed power basis, and adjoining these roots therefore yields a finite extension in which every original polynomial splits.

The formal derivative obeys the product rule by expanding monomials. In a splitting field, if $f = (T - c)g$, then $f'(c) = g(c)$. Hence c is a repeated root exactly when f and f' both vanish at c. For an irreducible f with $f' \neq 0$, the derivative has smaller degree, so Bezout makes f, f' coprime and there are no repeated roots. If $f' = 0$ and f is nonconstant, every root is repeated. Thus the minimal polynomial of an algebraic element is separable precisely when its derivative is nonzero, equivalently when $f'(\alpha) \neq 0$.

The square-matrix test used with the differential presentation also follows directly over a field. Define the determinant by its finite signed permutation sum. That sum is linear in each row and each column. With two equal rows its terms cancel in pairs under their transposition, so it is zero; row interchange changes its sign. Consequently adding a multiple of another row leaves it unchanged, and scaling one row multiplies it by the scale. The same assertions hold for columns. Starting with an n by n matrix, move a nonzero entry to the first position by row and column exchanges, scale it to one, and clear the other entries in that row and column by these additions. Repeat on the remaining square block. Every operation is invertible and alters the determinant by a nonzero scalar. The result is a diagonal matrix with some entries one and the others zero. If any zero entry remains, that output coordinate cannot occur in the image, so the map is not surjective. Hence a surjective square map has only ones and has nonzero determinant. Conversely such a diagonal matrix is invertible. The determinant of a triangular matrix is the product of its diagonal entries: expand in the first column for an upper triangular matrix, or in the first row for a lower triangular matrix, and induct. For size zero the determinant is one and the map is surjective.

If $u_1, \ldots, u_r$ is an F-basis of a finite extension E and $v_1, \ldots, v_s$ is an E-basis of L, the $u_i v_j$ span L over F. A relation between them is first a relation between the v_j over E and then, coefficient by coefficient, a relation between the u_i over F. Both sets of coefficients are zero. Thus they form an F-basis and $[L : F] = [L : E][E : F]$. A finite extension has a finite generating list, for example any finite F-basis, and hence a finite simple-generator tower.

To count embeddings, fix a finite generating list $\alpha_1, \ldots, \alpha_m$ of L/F and a finite splitting field H/F of their minimal polynomials over F. Put $E_i = F(\alpha_1, \ldots, \alpha_i)$. The minimal polynomial f_i of α_i

over E_{i-1} divides its minimal polynomial over F, by polynomial division. Under any F-embedding $\sigma : E_{i-1} \to H$, the polynomial $\sigma(f_i)$ therefore divides that already split F-polynomial and splits in H. An extension of σ to $E_i = E_{i-1}[T]/(f_i)$ is uniquely specified by the image of α_i, and each distinct root of $\sigma(f_i)$ defines such an extension by evaluation. The evaluation map is injective because its domain is a field. There are at most $[E_i : E_{i-1}]$ such roots, and exactly that many if f_i is separable. An element separable over F stays separable over an intermediate field because its new minimal polynomial divides the old one. Thus a separable extension L/F has exactly $[L : F]$ embeddings into H, by multiplying the tower counts. Conversely, if every step is separable, these counts also give $[L : F]$ embeddings. If some $\beta \in L$ were inseparable over F, its minimal polynomial would have fewer than $[F(\beta) : F]$ distinct roots. Enlarge H to split that polynomial and all polynomials for a generator tower above $F(\beta)$; no new embeddings of L appear, since every generator image was already a root in the original H. Restricting to $F(\beta)$ and applying the upper bound on extensions gives fewer than $[F(\beta) : F][L : F(\beta)] = [L : F]$ embeddings, a contradiction. This proves the transitivity of separability and all finite embedding counts used here. Only a finite splitting field was required. □

Lemma P0.5 (factorial and polynomial field rings). A factorial domain is normal. If k is a field, $k[T_1, \ldots, T_n]$ is factorial.

Proof. A fraction a/b with no common irreducible factor satisfying a monic equation has $b \mid a^d$ after clearing denominators. Thus b is a unit and the fraction belongs to the domain. For a factorial R, a product of primitive polynomials is primitive: reduction modulo an irreducible divisor of all product coefficients would contradict the domain property of $R/(p)$. Hence contents multiply. Clear denominators in a factorization over $\operatorname{Frac} R[T]$ and divide out contents. Two primitive polynomials differing by a fraction scalar differ by a unit: a reduced denominator must divide all coefficients, and then the numerator would divide the other content. The Euclidean factorization over the fraction field therefore gives existence and uniqueness of primitive factorization over $R[T]$, together with factorization of contents. This proves $R[T]$ factorial. Starting with a field and iterating proves the assertion. Euclidean factorization itself follows by dividing repeatedly into irreducibles; degrees decrease, and Bezout makes each irreducible prime, giving uniqueness. □

Lemma P0.6 (monic factors over a normal domain). Let R be normal with fraction field K. A monic factor $Q \in K[T]$ of a monic $F \in R[T]$ lies in $R[T]$. If all nonleading coefficients of F lie in I, those of Q lie in $\sqrt{I}$.

Proof. Adjoin the finitely many roots of F to a splitting field. Every root is integral over R. The coefficients of Q are elementary symmetric expressions in a sublist of those roots, counting repetitions. Lemma P0.2 makes them integral; they already lie in K, so normality puts them in R. Monic division puts the complementary factor in $R[T]$ too. Modulo each prime over I, their product is $T^{\deg F}$. Over that residue domain's fraction field its monic factors are powers of T. Thus every nonleading coefficient of Q vanishes modulo every prime over I, and Lemma P0.1 gives membership in $\sqrt{I}$. In particular the minimal polynomial of an integral element over K has coefficients in R. □

Lemma P0.7 (integral elements of an extended ideal). In an integral algebra B/R, every $y \in IB$ satisfies a monic equation whose coefficient of degree $d - i$ belongs to I^i.

Proof. Express $y = \sum_j a_j b_j$ with $a_j \in I$. The algebra generated by the finitely many integral b_j is finite over R by Lemma P0.2. Multiplication by y carries it into I times itself. On finite module generators this multiplication is therefore expressed by a matrix with entries in I. In its monic characteristic polynomial the coefficient i places below the leading term is a sum of products of i entries, hence lies in I^i. The adjugate kills the module generators, and then its unit, giving the required equation for y in B. □

Theorem P0.8 (going down over a normal domain). For an integral inclusion of domains $R \subset B$ with R normal, if $\mathfrak{p} \subset \mathfrak{p}'$ and $\mathfrak{q}'$ lies over $\mathfrak{p}'$, there is $\mathfrak{q} \subset \mathfrak{q}'$ lying over $\mathfrak{p}$.

Proof. We prove $\mathfrak{p}B_{\mathfrak{q}'} \cap R = \mathfrak{p}$. Write an element of the left side as $z = y/g$, with $y \in \mathfrak{p}B$ and $g \notin \mathfrak{q}'$. In the domain B, $zg = y$. Suppose $z \neq 0$. The minimal polynomial of g is $T^n + a_1 T^{n-1} + \cdots + a_n$

with $a_i \in R$ by Lemma P0.6. Some a_i misses $\mathfrak{p}$, since otherwise its equation puts g^n in $\mathfrak{q}'$. The minimal polynomial of zg is $T^n + za_1T^{n-1} + \cdots + z^n a_n$, by invertible scaling in K. It divides a monic polynomial for y with nonleading coefficients in $\mathfrak{p}$, supplied by Lemma P0.7. Lemma P0.6 gives $z^i a_i \in \mathfrak{p}$. The chosen a_i forces $z \in \mathfrak{p}$. The case $z = 0$ is immediate. Thus $\mathfrak{p}B_{\mathfrak{q}'}$ avoids $R \setminus \mathfrak{p}$. Prime separation, Lemma P0.1, gives a prime containing it and avoiding that set. Contracting from $B_{\mathfrak{q}'}$ gives the required prime inside $\mathfrak{q}'$. □

Lemma P0.9 (the field-algebra and zero-dimensional tests). A field finitely generated as an algebra over k is finite over k. A zero-dimensional Noetherian ring is Artinian, with finitely many local factors. In particular a finite-type zero-dimensional fibre algebra has finitely many points and finite residue fields.

Proof. In a field algebra choose a maximal algebraically independent sublist of its algebra generators. The other generators are algebraic over its rational function field. Invert one nonzero polynomial g clearing their monic-equation denominators. The field is then finite integral over $k[t_1, \ldots, t_r, 1/g]$, which must be a field by Lemma P0.4's field test. If $r > 0$, choose an irreducible polynomial in $k[t_1]$ not dividing g. Such polynomials are infinite by the product-plus-one argument and factorization by degree; only finitely many divide the finitely many coefficients of g. Its principal prime remains proper after inverting g, a contradiction. Thus $r = 0$, proving the field assertion.

A Noetherian ring has finitely many minimal primes. To prove this, choose by ACC an ideal maximal among those whose closed set is not a finite union of irreducible closed sets. That closed set is reducible, so is a union of two proper closed subsets, each a finite union by maximality, a contradiction. An irreducible closed set has prime radical: if ab vanishes on it, its cover by $V(a)$ and $V(b)$ forces one of them to contain it. Thus the finite irreducible components give the finite minimal primes. In dimension zero these are all maximal and all primes. The nilradical is generated by finitely many nilpotents, so a sufficiently high power is zero by expanding monomials. Modulo it, the Chinese remainder map gives the product of the finitely many residue fields. Each successive nilradical-power quotient is a finite module over that field product, hence has finite length. The ring therefore has finite length and is Artinian. The same Chinese remainder argument using sufficiently high powers of the maximal ideals gives its finite product of local Artinian factors. Hilbert's basis theorem in S01 makes a finite-type fibre algebra Noetherian; the two assertions just proved finish the statement. □

Lemma P0.10 (finite torsion-free lattices). Every finitely generated torsion-free abelian group is isomorphic to $\mathbb{Z}^r$ for a finite $r \geq 0$. Every subgroup of $\mathbb{Z}^n$ is free of rank at most n.

Proof. A nonzero subgroup $J \subseteq \mathbb{Z}$ has a least positive element a. Integer division of $x \in J$ by a leaves a remainder in J between 0 and $a - 1$, so that remainder is zero. Hence $J = a\mathbb{Z}$; the zero subgroup has rank zero.

Induct on n, starting with $\mathbb{Z}^0 = 0$. Project a subgroup $H \subseteq \mathbb{Z}^n$ onto its last coordinate. If its image is zero, use induction inside $\mathbb{Z}^{n-1}$. Otherwise its image is $a\mathbb{Z}$ with $a > 0$; choose $v \in H$ projecting to a. For each $h \in H$, subtract the unique multiple of v with the same last coordinate. Thus

$$H = \mathbb{Z}v \oplus H_0, \qquad H_0 = H \cap (\mathbb{Z}^{n-1} \times \{0\}).$$

The sum is direct because a nonzero multiple of v has nonzero last coordinate. Induction gives a basis of H_0, and adjoining v gives a basis of H with at most n elements.

Let now M be finitely generated and torsion-free. The localization obtained by inverting nonzero integers is $\mathbb{Q} \otimes_{\mathbb{Z}} M$: the maps $(a/b) \otimes m \mapsto am/b$ and $m/b \mapsto (1/b) \otimes m$ are inverse by the fraction and bilinear relations. The fraction equality test says that $m/1 = 0$ precisely when a nonzero integer kills m. Consequently M embeds in this rational vector space. A finite generating set spans it; discarding each vector in the span of its predecessors gives a finite basis, say of size s. In these coordinates the finitely many generators have rational entries. Multiplying all coordinates by one common positive denominator embeds M in $\mathbb{Z}^s$. The subgroup assertion just proved makes its image finite free. This also includes $M = 0$. □

Freely readable proof material: the Stacks Project, embedding finite torsion-free modules in free modules and finite torsion-free modules over a principal ideal domain, part (3). The complete integer case, including the splitting and fraction calculations used for character lattices, is proved above.

Freely readable comparison sources: the Stacks Project's finite and integral extensions, localization of integral closure, monic factors over a normal domain, minimal polynomials, integrality over an ideal, going down, finite-type field algebras, and Noetherian zero-dimensional rings. These are writing and comparison sources; the complete arguments used by blocks A–E are supplied above.

A. Faithfully flat and quasi-affine descent

A1. Faithfully flat algebra, topology and maps

Lemma A1.0 (gluing schemes along opens). Let $(X_i)_{i\in I}$ be a set-indexed family of schemes. For every pair let $X_{ij} \subset X_i$ be open, with isomorphisms $\phi_{ij} : X_{ij} \to X_{ji}$. Assume $X_{ii} = X_i$, $\phi_{ii} = 1$, $\phi_{ji} = \phi_{ij}^{-1}$, and the precise triple conditions

$$\phi_{ij}(X_{ij} \cap X_{ik}) = X_{ji} \cap X_{jk}, \qquad \phi_{jk}\phi_{ij} = \phi_{ik} \quad \text{on } X_{ij} \cap X_{ik}.$$

There is a scheme X, with open embeddings $X_i \to X$, having precisely these overlaps and identifications. Maps from X to any scheme are exactly compatible maps from the X_i. The construction is unique up to its unique compatible isomorphism. Empty overlaps give arbitrary set-indexed disjoint unions.

Proof. Form the disjoint union of the underlying spaces and identify $x \in X_{ij}$ with $\phi_{ij}(x)$. The inverse and triple conditions make this an equivalence relation; a chain of identifications between two chosen pieces reduces to their specified direct identification. No two distinct points of one piece become identified. Give the quotient the quotient topology. The saturation of an open subset of one piece is a union of its images under the open identifications, hence is open in every piece. Thus each X_i embeds openly with its original topology, and its intersection with the image of X_j is exactly X_{ij}.

For an open $W \subset X$, define $\mathcal{O}_X(W)$ to consist of tuples (s_i), with $s_i \in \mathcal{O}_{X_i}(W \cap X_i)$, whose restrictions agree under every ϕ_{ij}. Restrict tuples componentwise. A compatible family of these tuples on an open covering of W glues uniquely in each $\mathcal{O}_{X_i}$; their overlap equalities persist by the sheaf uniqueness axiom. Hence $\mathcal{O}_X$ is a sheaf of rings. On X_i the restriction is $\mathcal{O}_{X_i}$: a section on that piece determines all its restrictions to the other pieces by the specified identifications. Therefore the stalk at a point of X_i is its original local ring. The quotient is a locally ringed space covered by open subspaces which are schemes, and the affine charts in the pieces make it a scheme by the definition of a scheme. Compatible morphisms on the pieces glue their continuous maps and sheaf maps, remain local on stalks, and give the asserted unique morphism. Applying this universal property in both directions proves uniqueness of the constructed scheme. □

Lemma A1.1 (the equalizer). If $R \to A$ is faithfully flat, then for every R-module M the sequence

$$0 \longrightarrow M \longrightarrow A \otimes_R M \rightrightarrows A \otimes_R A \otimes_R M$$

is exact, with the two arrows inserting a unit in the first or second A position.

Proof. Tensor the augmented alternating unit-insertion complex with A. Regard the added factor as the coefficient ring. The new covering algebra is $A \otimes_R A$, and its multiplication map to A is a retraction of its structure map. Applying that retraction to the first covering-algebra factor is a contracting homotopy: the term inserting a unit in that position gives the identity; every other

insertion cancels the corresponding insertion after the homotopy, with opposite sign. The same formula in augmentation degree is a left inverse of the augmentation. Thus the tensored complex is exact. Flatness commutes with the kernels and cokernels computing its cohomology, and faithful flatness detects their vanishing. The original complex, and in particular the displayed beginning, is exact. Taking $M = R$ gives $R = \mathrm{Eq}(A \rightrightarrows A \otimes_R A)$. □

Lemma A1.2 (module and algebra effectivity). An A-module with descent datum for a faithfully flat map $R \to A$ is uniquely the scalar extension of its invariant R-module. The same holds for algebras and compatible morphisms, without finite generation assumptions.

Proof. Use the convention that the datum is an $(A \otimes_R A)$-linear isomorphism

$$\phi : N \otimes_R A \xrightarrow{\sim} A \otimes_R N.$$

Put $\rho(n) = \phi(n \otimes 1)$ and $M = \{n : \rho(n) = 1 \otimes n\}$. The cocycle and its diagonal restriction give

$$\rho(an) = (a \otimes 1)\rho(n), \qquad \mu\rho(n) = n, \qquad (1 \otimes \rho)\rho(n) = \sum_i a_i \otimes 1 \otimes n_i$$

when $\rho(n) = \sum_i a_i \otimes n_i$ and $\mu(a \otimes n) = an$. For diagonal normalization, the cocycle on the triple diagonal says that an invertible endomorphism is idempotent, hence is the identity. The last formula is the triple cocycle on $n \otimes 1 \otimes 1$.

Flatness identifies $A \otimes_R M$ with the kernel of $1 \otimes (\rho - \eta)$ in $A \otimes_R N$, where $\eta(n) = 1 \otimes n$. The last formula says that ρ lands in this kernel. Hence it defines $\beta : N \to A \otimes_R M$. Multiplication defines $\alpha : A \otimes_R M \to N$. The second formula gives $\alpha\beta = 1$, and the first, applied to invariant m, gives $\beta\alpha(a \otimes m) = a \otimes m$. Thus α is an isomorphism. Its compatibility with the datum is the equation $\phi((am) \otimes b) = a \otimes bm$. Compatible morphisms preserve invariants and are the scalar extensions of their restrictions. For the canonical datum on $A \otimes_R M_0$, Lemma A1.1 identifies its invariants with M_0. This proves the equivalence, including recovery and uniqueness. If N is an algebra and ϕ is multiplicative, invariants form a subalgebra containing the unit, and α is an algebra isomorphism. □

Lemma A1.3 (finite modules descend). For faithfully flat $R \to A$, finite generation, finite presentation, flatness and finite local freeness of an R-module M are equivalent to the respective property of $A \otimes_R M$.

Proof. From finitely many generators of $A \otimes_R M$ collect their finitely many M components. Their span has cokernel with zero scalar extension, so generates M. To descend finite presentation choose a finite free surjection $R^n \to M$. Its kernel becomes finitely generated after tensoring: for a finitely presented target, the kernel of any finite free surjection is finitely generated, as one sees by comparing with a finite presentation using the pullback of the two surjections and splitting its projections. Descend generation of this kernel by the first argument. To descend flatness, tensor any injection with M; its kernel vanishes after flat extension to A, because $A \otimes_R M$ is flat, and therefore vanishes. Ascent follows by tensoring presentations and injections. Finally, a finitely presented flat module is finite locally free: locally lift a basis of its residue-field fibre, use finite presentation for the kernel of the resulting free surjection, use flatness to inject that kernel's fibre, and use Nakayama to kill it. Finite presentation spreads the resulting basis to a neighbourhood. The converse is immediate from local bases. □

Lemma A1.3a (finite presentation and smoothness descend). Let $R \to A$ be faithfully flat and let B be an R-algebra. Finite presentation and smoothness of B/R are equivalent to the respective property of $(A \otimes_R B)/A$. Consequently finite presentation and smoothness of affine scheme morphisms descend under fpqc coverings, over arbitrary bases.

Proof. Suppose $B_A = A \otimes_R B$ is finitely presented. Express its finite algebra generators as finite tensor sums, and collect their B-components. Let $B_0 \subset B$ be the subalgebra they generate. The

map $A \otimes_R B_0 \to B_A$ is surjective, so $(B/B_0) \otimes_R A = 0$; faithful flatness gives $B = B_0$. Take the resulting surjection $P = R[X_1, \dots, X_n] \to B$, with kernel I. Flatness identifies its scalar-extended kernel with I_A. This is a finite ideal of P_A: independence of a finite algebra presentation from its chosen generator list is proved by polynomial substitution in S04, Lemma A1.8 and its earlier algebra provider. Write its finitely many generators as tensor sums of elements of I, and let $J \subset I$ be the ideal generated by their finitely many I-components. Then $J_A = I_A$, and faithful flatness gives $I/J = 0$. Thus B is finitely presented. Ascent follows by tensoring its finite polynomial presentation.

Now suppose B_A/A is smooth. Use the finite presentation just obtained. The conormal sequence is $I/I^2 \xrightarrow{d} B^n \to \Omega_{B/R} \to 0$. Flat scalar extension identifies $(I/I^2)_A$ with I_A/I_A^2: tensoring preserves the quotient, and the extended square ideal is generated by the products of extended ideal elements. Differential base change is the explicit derivation calculation in AG-CA-15, Theorem 2.1. The split conormal criterion in S02, Lemma 3.0 makes this sequence split exact after scalar extension. Hence d is injective before extension, by faithful detection of its kernel, and $(\Omega_{B/R})_A$ is finite projective. The ring map $B \to B_A$ is faithfully flat, so Lemma A1.3 descends finite presentation and flatness of $\Omega_{B/R}$. These properties give finite projectivity by the full simultaneous-relation splitting proof of AG-CA-07, Theorem 5.3. Thus the surjection $B^n \to \Omega_{B/R}$ splits. Its complementary projection is a left inverse of d, and S02, Lemma 3.0 proves formal smoothness of B/R. With its finite presentation this is smoothness. Conversely, flat scalar extension preserves a split conormal sequence, so the same criterion proves smoothness upstairs.

For fpqc families, work on an affine base open and choose a finite affine refinement, whose finite disjoint union gives one faithfully flat affine cover. Lemma A1.8 supplies the algebra assertions on every compatible affine pair. Apply the two algebra arguments there. Their conclusion is local on the target by that lemma and by the definition of smoothness, so holds on the original base. No Noetherian, reduced-base or residue-field separability hypothesis was used. □

Lemma A1.4 (quotient topology). A surjective flat morphism of affine schemes is a quotient map on underlying topological spaces, universally after base change.

Proof. Write $q : \operatorname{Spec} A \to \operatorname{Spec} R$ and suppose $q^{-1}(C)$ is closed. Give spectra their constructible topology, generated by the clopen sets $D(f)$ and $V(f)$. This topology is compact and Hausdorff: an ultrafilter determines the prime of f for which $V(f)$ belongs to it; closure under sums and products follows from the vanishing-set identities, primality from $V(fg) = V(f) \cup V(g)$, and properness from $V(1) = \varnothing$. This gives convergence of every ultrafilter. If a family of closed sets has the finite-intersection property, its generated proper filter extends to an ultrafilter by Zorn's lemma. A limit point lies in each closed member: otherwise its open complement is a neighbourhood belonging to the same filter, a contradiction. Taking complements proves that every open cover has a finite subcover. Distinct primes are separated by $D(f), V(f)$, so the topology is Hausdorff. Ring-induced maps are continuous for this topology. A closed subset of a compact space is compact, since adjoining its open complement to a cover gives a cover of the whole space. Pulling back open covers proves that a continuous image of a compact space is compact. A compact subset K of a Hausdorff space is closed: for a point outside K, choose disjoint neighbourhoods separating it from each point of K; finitely many of the latter neighbourhoods cover K, and the intersection of the corresponding former neighbourhoods avoids K. The closed inverse image is therefore constructibly compact; its image C, by surjectivity, is constructibly compact and closed.

It is also stable under specialization. If $\mathfrak{p} \subset \mathfrak{p}'$ with $\mathfrak{p} \in C$, choose $\mathfrak{q}'$ above $\mathfrak{p}'$. The map $R_{\mathfrak{p}'} \to A_{\mathfrak{q}'}$ is flat and local, hence faithfully flat. Indeed a proper ideal of the local source extends inside the maximal ideal of the local target; a nonzero cyclic submodule remains nonzero, and flatness preserves its injection into an arbitrary module. Its spectrum is therefore surjective, by applying this to each residue-field fibre. A prime above $\mathfrak{p}$ gives $\mathfrak{q} \subset \mathfrak{q}'$. Closedness of the inverse image puts $\mathfrak{q}'$ in it and hence $\mathfrak{p}'$ in C.

A constructibly compact specialization-stable subset is Zariski closed. If a point $\mathfrak{p}$ outside it had

every $D(f)$, $f \notin \mathfrak{p}$, meeting it, compactness would supply $\mathfrak{r} \in C$ avoiding all such f. Then $\mathfrak{r} \subset \mathfrak{p}$, contradicting specialization stability. Thus some principal neighbourhood avoids C. Applying complements proves the quotient assertion for opens. After base change, use affine opens of the new base; the same proof applies and glues. □

Lemma A1.5 (maps and refinement). Morphisms of schemes satisfy fpqc descent and are uniquely recovered from their compatible pullbacks. This includes morphisms between scheme-valued descent data after refining an fpqc covering.

Proof. First take a surjective flat affine cover $\operatorname{Spec} A \to \operatorname{Spec} R$ and a compatible map $u : \operatorname{Spec} A \to Z$. Two points above the same base point have a common point above them on the fibre product: their residue fields have nonzero tensor product over the base residue field, and a prime of that product gives the point. Compatibility therefore defines a set map from $\operatorname{Spec} R$ to Z, and Lemma A1.4 makes it continuous. Around a base point choose a principal neighbourhood mapping into an affine open of Z. The coefficient map into the corresponding localization of A has equal two pullbacks, and Lemma A1.1 factors it uniquely through the localization of R. These maps agree on overlaps, since equality can be checked after the faithfully flat extension on a common affine target chart. They glue to the unique descended map.

For an arbitrary fpqc family, each affine base open has a finite affine refinement; their finite disjoint union is a faithfully flat affine cover. The preceding construction gives a map there. On every original member it recovers the prescribed map after pulling back the refinement, and uniqueness gives recovery before that pullback. The local maps consequently agree on intersections and glue. This works after any base change, so also applies to compatible maps on the pullbacks of any source scheme.

For refinement full faithfulness, let $U_i \to V_{a(i)} \to S$ refine a cover $\{V_j \to S\}$. Given compatible maps between two data on U_i, transport them to $U_i \times_S V_j$ using the original transition isomorphisms. The transported maps agree on $U_i \times_S U_k \times_S V_j$ by the two cocycles and their compatibility. The already proved morphism descent gives a unique map over V_j. Its compatibility on the original overlaps is checked after the refinement and then follows by uniqueness. This gives full faithfulness; it does not assert effectivity of arbitrary object data. □

Lemma A1.6 (invariant opens and gluing). An open of the pullback of a scheme along an fpqc covering descends to an open if its two inverse images agree on overlaps. Invariant open subobjects of an effective scheme datum descend, even when they are not quasi-compact.

Proof. On a standard affine refinement, take the image of the invariant open under the surjective flat affine projection. The common-residue-field argument in Lemma A1.5 says that saturation is exactly being the full inverse image of this image. Lemma A1.4 says that the image is open. Give it its open subscheme structure; equality of underlying opens is equality of open subschemes. Uniqueness makes these constructions agree on original covering members and affine base overlaps. The result glues. An open-immersion map compatible with effective data therefore descends to this open immersion. Descended overlaps of a covering by invariant effective opens have unique compatible identifications by Lemma A1.5, satisfy the cocycle by faithfulness, and glue by ordinary scheme gluing. □

Lemma A1.7 (local quotient coordinates and unique arrows). Let $R \rightrightarrows U$ be an equivalence relation of fppf sheaves on every scheme test, with endpoint map a monomorphism. In particular this applies when U, R are schemes. On the big fppf site, the associated quotient sheaf F has local representatives in U, and

$$U \times_F U = R.$$

These assertions require no flatness or finite-presentation hypothesis on the projections of R.

Proof. If U, R are schemes, morphism descent in Lemma A1.5 makes them fppf sheaves. In the general case this is already a hypothesis. Construct $F(T)$ explicitly as follows. A representative is

an fppf covering $\{T_i \to T\}$, together with sections $u_i \in U(T_i)$, such that on each $T_i \times_T T_j$ the two maps are endpoints of an arrow to R. Such an arrow is unique by the endpoint monomorphism. The equivalence-relation identities therefore make the arrows obey the triple equation. Two representatives are equivalent when their coordinates are joined by arrows on the mixed overlaps, after a common fppf refinement if necessary. A locally existing arrow glues uniquely by the sheaf property of R, so the phrase "after a refinement" can be removed once the endpoints have been fixed on an overlap. Common refinements are fibre products of coverings. Reflexivity, symmetry and transitivity of the resulting identification follow respectively from the identity, inverse and composition of the arrows. Pullback of coverings and coordinates defines restriction to every test scheme.

This presheaf is a sheaf. Given compatible classes on an fppf covering $\{T_a \to T\}$, choose coordinate coverings of each T_a. Their composites form a covering of T. Compatibility of the classes means that on every mixed overlap the coordinates are locally joined by an arrow. The unique arrows glue there as just proved. Hence the combined coordinate family is a representative over T, giving existence of the glued class. If two classes restrict equally on the covering, the same mixed-overlap test and uniqueness of arrows make their representatives equivalent, giving uniqueness.

The map from the presheaf $U(T)/R(T)$ sends a global coordinate to its one-member covering. If a map from this presheaf to an fppf sheaf H is given, apply it to each coordinate u_i. The images agree on overlaps, since the arrows relate their coordinates, so they glue uniquely to a section of $H(T)$. Equivalent representatives give the same section, and the construction commutes with pullback. This is the unique extension of the original map. Thus the constructed F has exactly the universal property of the associated quotient sheaf. Its construction immediately provides local representatives in U.

Finally two maps $a, b : T \to U$ have the same class precisely when they are locally joined by an arrow. Those local arrows have the fixed endpoints a, b; uniqueness makes them agree on overlaps, and the sheaf property of R glues them to one arrow over T. Conversely an arrow gives equal classes. This identifies $U \times_F U$ with R naturally on every test, proving the displayed identity. □

Lemma A1.8 (finite algebra presentations on affine charts). Suppose a morphism $f : X \to S$ has an affine cover X_i and affine opens $S_i \subset S$ with $f(X_i) \subset S_i$, such that $\Gamma(S_i) \to \Gamma(X_i)$ is of finite type, respectively of finite presentation. For every affine open $V \subset S$ and every affine open $U \subset f^{-1}(V)$, the algebra $\Gamma(U)$ has the same property over $\Gamma(V)$. Thus either property can be tested on one compatible affine cover, holds on open restrictions and is local on the target for the Zariski topology. No quasi-compactness or separation assumption on the morphism is required.

Proof. First two affine opens in a scheme have common distinguished neighbourhoods at every point of their intersection. Choose $D_{V_1}(g)$ about the point inside the intersection, then $D_{U_1}(h)$ about it inside $D_{V_1}(g)$. By S01, Section 2, the restriction of h belongs to the localized coordinate ring of V_1, say b/g^n. Its unit locus inside $D_{V_1}(g)$ is $D_{V_1}(gb)$; by the chosen containment it is also exactly $D_{U_1}(h)$. This proves the required simultaneous distinguished neighbourhood, without assuming the intersection affine. It is the complete common-neighbourhood argument of Properties of schemes, Lemma 1.1.

We use the complete algebra patching proof of Finiteness of morphisms, Lemma 2.1: if finitely many b_j generate the unit ideal in an A-algebra B and every B_{b_j} is of finite type or finite presentation over A, then B has the same property. Its preceding Section 2 proves localization and composition by adjoining finite generators and their equations, and proves independence of the finite generating list by polynomial substitution. These algebra calculations precede the every-affine-pair assertion and do not use it.

Fix U, V as in the statement, and put $A = \Gamma(V)$. For $x \in U$, choose X_i containing x. On the base choose a neighbourhood W of $f(x)$ distinguished in both V and S_i, using the first paragraph. Write $W = D_V(a) = D_{S_i}(c)$. Then $X_i' = X_i \cap f^{-1}(W)$ is distinguished affine in X_i, with coordinate ring obtained by inverting the image of c. The localization calculation supplies the required finite

algebra property over $\Gamma(W) = \Gamma(S_i)_c$.

On the source choose H containing x, distinguished in both U and X'_i. Localizing the source algebra preserves the required property: to invert an element b, adjoin one variable Z and the relation $bZ - 1$. Thus $\Gamma(W) \to \Gamma(H)$ has the property. Also $A \to \Gamma(W) = A_a$ is finitely presented by $A[T]/(aT - 1)$. Composing the finite generator lists, and their relation lists for finite presentation, supplies the property for $A \to \Gamma(H)$.

These H form a distinguished cover of the affine scheme U. There is a finite subcover: the defining elements generate the unit ideal, since otherwise Lemma P0.1 supplies a prime outside their opens, and an expression for 1 uses only finitely many of them. Write this subcover as $D_U(b_1), \ldots, D_U(b_r)$ and $B = \Gamma(U)$. The b_j generate the unit ideal of B, and each B_{b_j} has the property over A. The preceding algebra patching lemma now proves it for B. If U is empty, $B = 0 = A/(1)$ is already finitely presented.

The assertion for every affine pair applies to restrictions of source and target opens. For a target open cover, choose compatible affine source and target covers over its members and combine the source covers. They cover X, so the first assertion applies. Conversely, the every-pair assertion applies to any chosen compatible cover. This proves all the stated locality assertions. □

Lemma A1.9 (flat monomorphisms). A flat monomorphism of schemes which is locally of finite presentation is an open immersion. In particular an étale monomorphism is an open immersion, over an arbitrary base.

Proof. Choose compatible affine neighbourhoods around a source point, with coordinate map $B \to C$ flat and finitely presented; Lemma A1.8 supplies this finite presentation on the chosen affine pair. Its monomorphism property says that multiplication $C \otimes_B C \to C$ is an isomorphism. Consequently the two maps from C into that tensor product agree. For any B-algebra E, two B-maps $C \to E$ are therefore equal: apply their tensor-product map to $c \otimes 1 = 1 \otimes c$.

Let $q \in \operatorname{Spec} C$, with contraction $p \in \operatorname{Spec} B$. The local map $B_p \to C_q$ is faithfully flat. Flatness is preserved by these localizations. For faithfulness, a nonzero B_p-module contains a nonzero cyclic submodule B_p/I. The proper ideal I lies in the maximal ideal, whose image lies in the maximal ideal of C_q, so $C_q/IC_q \neq 0$. Flatness preserves that cyclic submodule's injection, proving that tensoring cannot kill the original module. Localizing the displayed tensor-product isomorphism in both factors gives $C_q \otimes_{B_p} C_q = C_q$, with the two maps again equal. The faithfully flat equalizer proved in Lemma A1.1 now identifies B_p with C_q.

The resulting map $C \to C_q \simeq B_p$ descends to $C \to B_g$ for some $g \in B \setminus p$. Here is the finite-data argument. In a finite polynomial presentation of C/B, express the images of its finitely many generators by fractions with denominators outside p, and invert their product. Each of the finitely many defining relations evaluates to zero in B_p; an additional element outside p kills its numerator. Inverting the product of these additional elements makes every relation zero already in B_g, giving the desired algebra map. It extends to $v : C_g \to B_g$, since the image of g is a unit. Write $j : B_g \to C_g$ for the original map. Then $vj = 1$. The two B-maps $C \to C_g$, the natural map and the map through B_g, are equal by the tensor-product calculation in the first paragraph. Localizing makes $jv = 1$ too. Thus $C_g \simeq B_g$.

Every source point consequently has a target principal neighbourhood with an inverse section into the chosen affine source chart. It is an inverse for the whole morphism over that open: on any test scheme a point x and the inverse section evaluated at $f(x)$ have the same image, so monomorphy makes them equal. The local inverse sections agree on target overlaps for the same reason and glue to an inverse on the union of these opens. Every source point lies over that union. Thus the original morphism is an isomorphism onto an open subscheme. This proves the assertion without a Noetherian or reducedness assumption. An étale morphism is locally of finite presentation by its definition and is smooth of relative dimension zero; its flatness is proved in the earlier AG-CA-18, Theorem 6.1. The particular étale assertion follows. □

This result also appears in the freely readable Stacks Project, Theorem 41.14.1, Tag 025G. The complete local-ring and finite-presentation proof above supplies the result used here.

A2. The affine hull under flat base change

Lemma A2.1. If V is quasi-compact and separated over $\operatorname{Spec} R$, then, for every flat R-algebra A,

$$A \otimes_R \Gamma(V, \mathcal{O}_V) \simeq \Gamma(V_A, \mathcal{O}_{V_A}).$$

Proof. First make the closed-subscheme calculation explicit. A closed immersion into $\operatorname{Spec} D$ is defined by a quasi-coherent ideal, which is $\widetilde{J}$ for an ideal $J \subset D$ by S01, Proposition 2.2. The primes of D/J are exactly the primes containing J, with the same distinguished-open topology. On these opens, and then on stalks, its structure sheaf is the quotient of the structure sheaf of $\operatorname{Spec} D$, since $(D/J)_f = D_f/JD_f$. Thus the closed subscheme is $\operatorname{Spec}(D/J)$ and is affine. After a ring base change $D \to D'$, the maps $d' \otimes (d + J) \mapsto d'd + JD'$ and $d' + JD' \mapsto d' \otimes 1$ identify $D' \otimes_D (D/J)$ with D'/JD'. The tensor relations make these maps well defined and inverse. The affine-map correspondence in S01, Proposition 2.3, identifies fibre products of affines with spectra of tensor products: ring maps out of the tensor product are exactly the pairs agreeing on the base, on every locally ringed test space. These calculations prove the affine closed-immersion base-change assertion; restricting to affine charts proves it for arbitrary base changes. Now choose a finite affine open cover V_i. The product $V_i \times_R V_j$ is affine by the same tensor calculation. The inverse image of this product under the closed diagonal is $V_i \cap V_j$, which is therefore a closed subscheme of that affine product and hence affine. Global sections are the kernel of the differences of restrictions from $\prod_i \Gamma(V_i, \mathcal{O})$ to $\prod_{i,j} \Gamma(V_i \cap V_j, \mathcal{O})$, by the sheaf axiom. Flat tensor preserves this kernel and the finite products, as proved in S01, Section 1. On each affine chart and intersection, scalar extension replaces its ring by its tensor product with A. The resulting diagram is consequently the same sections diagram for the base-changed cover. □

Lemma A2.2. For a quasi-compact open $V \subset \operatorname{Spec} C$, the canonical map $V \to \operatorname{Spec} \Gamma(V, \mathcal{O}_V)$ is a quasi-compact open immersion.

Proof. Choose finitely many principal opens $D_C(f_i)$ contained in V and covering it, and put $B = \Gamma(V, \mathcal{O}_V)$. Lemma A2.1 with $C \to C_{f_i}$ gives $B_{f_i} = C_{f_i}$, because $V_{f_i} = D_C(f_i)$. Thus the canonical map is an isomorphism from each $D_C(f_i)$ to $D_B(f_i)$. These identifications agree on the principal intersections and glue. The image is the finite union of the $D_B(f_i)$, hence is quasi-compact. □

A3. Effective quasi-affine descent

Theorem A3.1. Quasi-affine scheme morphisms with data for an fpqc covering descend effectively to quasi-affine scheme morphisms, with all compatible morphisms descending uniquely.

Proof for an affine faithfully flat cover. Let $R \to A$ be faithfully flat and let V/A be quasi-affine with a datum. Put $B = \Gamma(V, \mathcal{O}_V)$. The two projections from $\operatorname{Spec}(A \otimes_R A)$ are flat. Lemma A2.1 therefore turns the original datum into algebra descent data on B, compatibly with multiplication, unit and the triple cocycle. Lemma A1.2 gives an R-algebra B_0 with $A \otimes_R B_0 = B$. Lemma A2.2 embeds V as a quasi-compact open of W_A, where $W = \operatorname{Spec} B_0$, and this embedding respects the datum. Its two pulled-back opens agree. Lemma A1.6 descends it to an open $U \subset W$ with $U_A = V$ and precisely the given datum. Its topology is quasi-compact, since it is the continuous image of V. Thus U is quasi-affine over R. Morphisms descend by Lemma A1.5.

Passage to arbitrary fpqc coverings. Over an affine base open choose a finite affine refinement of the covering, and combine its members into one affine scheme. Finite disjoint unions of quasi-affine schemes over an affine base are quasi-affine: take the product of their affine hull algebras and the

union of the corresponding component opens. Apply the affine construction. Its comparison with every original member descends uniquely after pulling back the refinement, by Lemma A1.5; the cocycle supplies the requisite compatibility. Construct the result on each affine base open. The uniquely descended comparisons on their intersections satisfy the cocycle and glue to a scheme $U \to S$.

The resulting morphism is globally quasi-affine, and not merely quasi-affine separately on these pieces. On an affine base open, localization and Lemma A2.1 show that $f_*\mathcal{O}_U$ is the quasi-coherent algebra associated to its section algebra. These algebras agree under restriction, so form a quasi-coherent algebra on S. The canonical map $U \to \operatorname{Spec}_S(f_*\mathcal{O}_U)$ is a quasi-compact open immersion on every affine base open, by Lemma A2.2, hence globally. This proves the asserted type and effectivity. □

Corollary A3.2. Affineness, closed immersions and open immersions of existing scheme morphisms descend under fpqc coverings. Quasi-affineness of an existing scheme morphism is fpqc-local on the target.

Proof. Affine objects descend by Lemma A1.2 on the affine refinements; their comparison isomorphisms with an existing scheme and their inverses descend by Lemma A1.5. For closed immersions descend the quotient algebra and its defining ideal using the same module construction; the surjectivity of its algebra map is detected by faithfully flat tensoring. This constructs a closed immersion with the given data, hence identifies it with the existing map. For open immersions use Lemma A1.6. Finally apply Theorem A3.1 to the canonical datum of an existing morphism and descend its comparison and inverse. □

A4. Locally discrete descent and relative characters

Lemma A4.0 (intersections and their affine hull). Let V be a disjoint union of affine schemes over $\operatorname{Spec} R$, and let $E \subset V$ be an intersection of a nonempty family of quasi-compact opens. Put $A = \Gamma(E, \mathcal{O}_V|_E)$. The canonical map of locally ringed spaces embeds E into $\operatorname{Spec} A$ as an intersection of quasi-compact opens, with identical local rings. Every open W of V contained in E maps to an open of $\operatorname{Spec} A$. If $R \to C$ is flat and E_C is the inverse image of E in V_C, then

$$\Gamma(E_C, \mathcal{O}_{V_C}|_{E_C}) = C \otimes_R A.$$

Proof. Close the family of opens under finite intersections and restrict it below one member U_0. This does not change E. A quasi-compact open meets only finitely many of the disjoint affine components of V. Their union Y is affine, with coordinate ring the finite product of their rings. Every $U_i \subset U_0$ is a quasi-compact open of Y, hence a finite union of principal opens. It is therefore clopen for the constructible topology of Y. The complete compactness proof in Lemma A1.4 makes $E = \bigcap_i U_i$ constructibly compact. Any open neighbourhood O of E contains some U_i: otherwise the constructibly closed subsets $U_i \setminus O$ of the compact U_0 would have the finite-intersection property and give a point in $E \setminus O$.

For any sheaf $\mathcal{F}$ on V one has $\Gamma(E, \mathcal{F}|_E) = \operatorname{colim}_i \Gamma(U_i, \mathcal{F})$. Here are the gluing details. A section on E is locally represented by sections on neighbourhoods in V. Choose finitely many quasi-compact neighbourhoods N_a covering E on which such representatives exist. This is possible because E is compact and these opens form a basis. On each overlap, agreement of the germs gives an open $L_{ab} \subset N_a \cap N_b$ containing $E \cap N_a \cap N_b$ on which the representatives agree. The set $D_{ab} = (N_a \cap N_b) \setminus L_{ab}$ is constructibly compact and misses E. Its Zariski closure also misses E. Indeed any point e of that closure is a specialization of a point of D_{ab}: compactness applied to $D_{ab} \cap D(f)$, for all $f \notin e$, gives a prime contained in e, exactly as in Lemma A1.4. Since E is stable under generization, such a prime would lie in E, a contradiction. Remove these finitely many closures from $\bigcup_a N_a$. The remaining open contains E, hence contains some U_i. On U_i the representatives agree and glue. Conversely, two sections with the same restriction to E agree on

a neighbourhood of E, by equality of germs, and hence on some smaller U_j. This proves both surjectivity and injectivity of the colimit formula.

Apply it to the structure sheaf and write $A_i = \Gamma(U_i, \mathcal{O}_V)$, so $A = \operatorname{colim}_i A_i$. Lemma A2.2 identifies U_i with a quasi-compact open of $\operatorname{Spec} A_i$. The affine-limit proof in AG-MO, *Limits of schemes*, Theorem 1.1, identifies $\operatorname{Spec} A$ with the inverse limit of those affine spectra. The inverse images of the opens U_i have intersection exactly E: a compatible family of points belonging to the U_i is one point of V lying in every U_i, since their transition maps are inclusions. Conversely such a point supplies its compatible prime kernels. A principal open in the limit is represented at one stage, so this correspondence is a homeomorphism onto the stated intersection. At a point of E, all the localized stage rings are canonically the same local ring of V. Localization of the colimit gives that same ring. This proves the locally ringed-space assertion; explicitly the canonical map sends a point to the kernel of $A \to \kappa(e)$ and uses these localizations on stalks.

Let W be open in V and contained in E. Fix i. Cover W in the open $U_i \subset \operatorname{Spec} A_i$ by principal opens $D(a)$ contained in W. For every smaller U_j, that principal open is still contained in U_j. Lemma A2.1 applied to localization therefore gives $(A_j)_a = (A_i)_a$. Passing to the colimit gives $A_a = (A_i)_a$. Thus each of these principal pieces maps isomorphically to $D_A(a)$, and their union makes W open in $\operatorname{Spec} A$. Finally every U_i is quasi-compact and separated over R, so Lemma A2.1 gives $\Gamma((U_i)_C) = C \otimes_R A_i$. The same intersection calculation applies in the disjoint affine union V_C. Tensor product commutes with filtered colimits by its universal property, proving the displayed formula. □

Theorem A4.1 (effective descent of locally discrete schemes). For an fpqc covering $S' \to S$, scheme descent data on a scheme which is Zariski locally over S' a disjoint union of copies of S' are effective. The descended scheme is separated and étale over S.

Proof. Work over an affine base $S = \operatorname{Spec} R$. Take a finite affine refinement of the fpqc cover, refining also the open trivializations of the locally discrete scheme. Combining those affine members gives a faithfully flat $X = \operatorname{Spec} B \to S$. The pulled-back scheme V is a disjoint union of affines, each a copy of one of the finitely many open and closed affine components of X. It is separated and étale over X. Lemma A1.5 will recover the original datum after this refinement.

The datum gives a groupoid $Q \rightrightarrows V$ with affine, flat, quasi-compact projections s, t, pulled back from the two projections of $X \times_S X$. Given a quasi-compact open $U \subset V$, its saturation $E_0 = t(s^{-1}U)$ is quasi-compact, invariant and stable under generization. Composition and inversion prove invariance; flat going down, as proved in Lemma A1.4, proves stability under generization. Such a set is an intersection of quasi-compact opens. For a point outside it, its closure misses the set; choose basic opens around its points avoiding that closure and take a finite union by quasi-compactness. Intersect these separating opens over all outside points.

Choose a quasi-compact open U' containing E_0, and let T be the saturation of $V \setminus U'$. Every point of $\overline{T}$ specializes from a point of T. To verify this locally, on an affine target the quasi-compact preimage under t has a finite affine cover. Intersect that cover with the closed set $s^{-1}(V \setminus U')$. The images of these finitely many affine closed subschemes are constructibly compact by Lemma A1.4, and so is their union. The compactness argument used in Lemma A4.0 then supplies the required generization in T. Flat going down along s lifts this generization through any arrow whose source is in $\overline{T}$; invariance of T puts the other endpoint in $\overline{T}$ as well. Thus $\overline{T}$ is invariant. Also $E_0 \cap \overline{T} = \varnothing$, since E_0 is generization-stable and misses T. Consequently

$$U \subset W = V \setminus \overline{T} \subset U'$$

is invariant and open. Its containment in U' uses $V \setminus U' \subset T$. The saturation $E = t(s^{-1}U')$ is invariant, is an intersection of quasi-compact opens by the preceding argument, and contains W.

Put $A = \Gamma(E, \mathcal{O}_V|_E)$. Invariance of E, the datum and the flat base-change formula in Lemma A4.0 give an algebra descent datum on A; restriction and base change also carry the triple equation to its cocycle. Lemma A1.2 gives $A = B \otimes_R A_0$ for an R-algebra A_0. Lemma A4.0 embeds W as an open

of Spec A. These embeddings respect the datum, since they are defined by restricting the same sections on both overlaps. Lemma A1.6 descends this invariant open to an open $W_0 \subset \operatorname{Spec} A_0$, with exactly the prescribed datum on W. These W cover V as U ranges over affine neighbourhoods. Their intersections descend as opens of their effective pieces by Lemma A1.6; Lemma A1.5 descends the unique overlap maps and their cocycle, and Lemma A1.0 glues the W_0. This constructs a scheme descending V. The same uniqueness recovers its datum on the original cover and glues the construction over affine opens of S.

Its diagonal becomes closed after the cover, so Corollary A3.2 descends separatedness. Smoothness and finite presentation on affine charts descend by Lemma A1.3a, because the pullback is locally a copy of its base. Relative differentials pull back to zero, by their base-change formula, and faithful flatness detects zero modules. Hence they vanish before pullback. The difference-of-lifts and differential criterion in AG-CA-17, Proposition 2.1 and Theorem 2.2, give uniqueness of every square-zero lift; smoothness gives existence. This is formal étaleness, and the already proved finite presentation makes the morphism étale. □

The affine-intersection and invariant-neighbourhood construction is the freely accessible Stacks Project proof: Lemma 40.15.1, Lemma 40.15.2, Lemma 40.15.3 and Lemmas 39.19.2–4. The preceding proofs include the steps needed here; these references supply credit and source access.

Theorem A4.2 (étale splitting of multiplicative type). A finite-presentation group which is fpqc locally $D(M)$, with M finitely generated, is affine, commutative and faithfully flat and is étale locally diagonalizable. No smoothness assumption is needed.

Proof. The algebra and group-law descent in Lemmas A1.2 and A1.5 gives affineness and commutativity. S02 Lemma 3.M supplies finite presentation and faithful flatness of each split character algebra. Finite presentation and flatness descend by Lemmas A1.3 and A1.3a. Faithfulness also descends: after the faithfully flat cover, tensoring a nonzero base module with the split algebra remains nonzero, since its zero-exponent summand is the base module. Thus tensoring with the descended algebra cannot have killed that module.

The abstract exponent group's isomorphism type is locally constant on the base. On splitting charts its type loci are open and closed. They agree on overlaps: a geometric fibre's characters recover its exponent group by the full group-like coefficient calculation in AG-GS, *Diagonalizable groups*, Lemma 2.1 and Theorem 2.2. Lemma A1.6 descends these invariant opens. Work on one type piece and fix its finitely generated group M, choosing isomorphisms with that same M on the splitting cover.

The fpqc sheaf $I = \underline{\mathrm{Isom}}(G, D_S(M))$ becomes the constant scheme $\mathrm{Aut}(M)$ on this cover. Indeed a Hopf map is determined by the images of finitely many character generators. The group-like coefficient calculation makes each image a locally constant exponent, and finite generation allows one common neighbourhood for all these choices. Multiplicativity gives a group map $M \to M$ there. Applying the same calculation to an inverse shows that exactly the automorphisms give isomorphisms. Conversely each locally constant automorphism defines the Hopf map on the exponent basis. This is the complete local calculation of AG-GS, Proposition 7.4. Lemma A1.5 makes the changes of trivialization into scheme descent data on I. Theorem A4.1 therefore represents I by a separated étale scheme over S. It is surjective, since its pullback to the splitting cover has the identity automorphism in every fibre. Its universal group isomorphism identifies G_I with $D_I(M)$. Thus $I \to S$ is an étale splitting cover. This applies in particular to finite nonreduced groups such as μ_p. □

B. Polynomial normality before the conductor

B1. A preliminary coefficient proof for polynomial normality

Lemma B1.1 (integral polynomial coefficients). Let $R \to B$ be any ring map. A polynomial $b(T) = \sum_{j<d} b_j T^j$ in $B[T]$ is integral over $R[T]$ if and only if every b_j is integral over R.

Proof. Integral elements form a ring, so the reverse implication follows from finite integral generators and the fact that T already belongs to the base.

Here is an interpolation argument valid in every characteristic and with nilpotents. First supply the integer arithmetic. Lemma P0.10 proves that every nonzero integer ideal has a positive generator. Applied to (a, b) this gives $\gcd(a, b) = ua + vb$ for integers u, v. If $\gcd(a, b) = 1$ and $a \mid bc$, multiplication of that identity by c shows $a \mid c$. A prime integer is therefore a prime element. Induction factors every integer $n > 1$ into primes, by splitting a composite into smaller positive factors; prime divisibility and cancellation give uniqueness. Thus $\mathbf{Z}$ is factorial. A prime divisor of the product of any finite prime list plus one is absent from that list, so prime integers are arbitrarily large. Distinct prime powers have greatest common divisor one and satisfy the same Bezout identity. Choose a prime integer $p > d$ and put

$$R_p = R[1/p], \qquad C_p = R_p[Z]/(1 + Z + \cdots + Z^{p-1}).$$

Monic division gives a unique remainder of degree below $p-1$ in this quotient: subtract the leading coefficient times the appropriate monomial multiple of the monic divisor until the degree drops. Uniqueness follows because a nonzero multiple of a monic degree-$p-1$ polynomial has degree at least $p-1$, even over a ring with zero divisors. Thus C_p is finite free over R_p with basis $1, Z, \ldots, Z^{p-2}$. The coefficient-of-one projection is an R_p-linear retraction of its unit inclusion. Tensoring that retraction with B_p splits $B_p \to B_p \otimes_{R_p} C_p$, so this map is injective without a flatness hypothesis. Zero rings are included.

In $\mathbf{Z}[Z]/(1 + \cdots + Z^{p-1})$ there is the identity

$$T^p - 1 = \prod_{i=0}^{p-1} (T - Z^i). \tag{1.1}$$

To justify it before arbitrary base change, write $\Phi(Z) = 1 + \cdots + Z^{p-1}$. After replacing Z by $V + 1$, it becomes $((V+1)^p - 1)/V$; every nonleading coefficient is divisible by p, since $p \mid \binom{p}{i}$ for $0 < i < p$ (the denominator $i!$ has no factor p), and its constant term is p, not divisible by p^2. If it had two nonconstant monic integral factors, reduction modulo p would make both powers of V, so both constant terms would be divisible by p, a contradiction. The primitive polynomial argument of Lemma P0.5, using the factoriality of $\mathbf{Z}$ proved above, converts a rational factorization to such an integral one. Thus Φ is irreducible over $\mathbf{Q}$. Put $E = \mathbf{Q}[Z]/(\Phi)$ and let z be the image of Z. This quotient is a field: for a nonzero remainder, Euclidean Bezout with the irreducible Φ supplies its inverse. The identity $(Z-1)\Phi(Z) = Z^p - 1$ gives $z^p = 1$, whereas $z \neq 1$ since $\Phi(1) = p \neq 0$ in E. If $z^n = 1$ for $0 < n < p$, the integer Bezout identity for n, p would give $z = 1$. Hence $1, z, \ldots, z^{p-1}$ are distinct. Division by $T - a$ leaves remainder $f(a)$; after dividing off one root, evaluation at each other distinct root still vanishes, because their differences are nonzero in the field. Repeating division and comparing degrees proves (1.1) in $E[T]$. Monic division by Φ shows $\mathbf{Z}[Z]/(\Phi) \to E$ injective: its unique integer remainder of degree below $p-1$ can vanish in the rational quotient only if every coefficient is zero. Therefore (1.1) holds already in that integral quotient.

Differentiate (1.1) and evaluate at $T = Z^i$. The product $\prod_{j \neq i}(Z^i - Z^j)$ equals $pZ^{i(p-1)}$. It is a unit in C_p because p is inverted and $Z^p = 1$. Each of its factors is consequently a unit. The d elements $\alpha_i = Z^i$, $0 \leq i < d$, have pairwise invertible differences.

In $B_p \otimes C_p$, evaluation of a monic integral equation for $b(T)$ at α_i makes $b(\alpha_i)$ integral over C_p. Since C_p is finite over R_p, these values are integral over R_p. Lagrange interpolation gives

$$b(T) = \sum_{i<d} b(\alpha_i) \frac{\prod_{j\neq i}(T-\alpha_j)}{\prod_{j\neq i}(\alpha_i-\alpha_j)}. \tag{1.2}$$

For completeness, prove the determinant identity used here. In independent variables $X_0, \dots, X_{d-1}$, the determinant of $(X_i^j)_{0\le i,j<d}$, viewed as a polynomial in X_{d-1} over $\mathbf{Q}(X_0, \dots, X_{d-2})$, has degree at most $d-1$ and vanishes at each X_i with $i < d-1$, because two rows then coincide. Its leading coefficient is the preceding Vandermonde determinant by expansion in the last row. Division by those distinct linear factors and induction, starting with the empty determinant 1, give $\det(X_i^j) = \prod_{i<j}(X_j - X_i)$. Both expressions are integer polynomials; their embedding in the rational-function polynomial ring is injective, so this is an integer-polynomial identity and remains true after substitution into every commutative ring. The difference of the two sides of (1.2) has degree below d and vanishes at all α_i; its coefficient vector is killed by this Vandermonde matrix. Its determinant is a unit, and the adjugate gives an inverse, so the difference is zero. Formula (1.2) makes each coefficient integral over C_p, and hence over R_p. Its monic equation holds already in B_p, by the displayed injectivity.

Repeat with a distinct prime $q > d$. If $A \subset B$ is the integral closure of the image of R, localization of integral closure, proved in Lemma P0.3, gives $b_j/1 \in A[1/p]$ and $b_j/1 \in A[1/q]$. Hence powers p^a and q^c kill the class of b_j in the R-module B/A. Bezout's identity for those two coprime integers kills the class itself. Thus $b_j \in A$, as required. The degree-zero and zero-polynomial cases are included. □

Corollary B1.2 (polynomial normality). If R is a normal domain, then $R[T]$ is a normal domain.

Proof. Put $K = \operatorname{Frac} R$. An element $z \in K(T)$ integral over $R[T]$ is integral over $K[T]$. The Euclidean ring $K[T]$ is factorial and therefore normal, by Lemma P0.5. Thus $z \in K[T]$. Lemma B1.1 applied to $R \subset K$ makes all its coefficients integral over R. Normality of R puts them in R. Therefore $z \in R[T]$. □

This is the precise preliminary input consumed by Lemma C3.2 below. The earlier proof of normality of polynomial rings over a field alone would not suffice.

C. Conductor and affine Zariski Main

C1. Extracting an integral remainder

Lemma C1.1. Let $A[X] \to B$ be a ring map and let $v \in B$ be integral over the image of $A[X]$. If $vP(X)$ belongs to that image for a monic $P \in A[X]$, then there is $Q \in A[X]$ such that $v - Q(X)$ is integral over A.

Proof. Write $vP = R$ in B for $R \in A[X]$. Divide R by the monic P to obtain $R = QP + R_0$, with $\deg R_0 < \deg P$. Replace v by $v - Q(X)$, which is still integral over $A[X]$ and now satisfies $vP = R_0$.

In B_v the element X satisfies the monic equation $P(X) - v^{-1}R_0(X) = 0$ over the image of $A[v^{-1}]$. It is therefore integral over that ring. Transitivity makes v integral over the same ring. Write the coefficients of its monic equation as finite sums $\sum_j a_{ij}v^{-j}$. Choose N large enough to make all exponents nonnegative when the equation is multiplied by v^N, and, if necessary, enlarge N to kill its localization error in B. The equation becomes

$$v^{d+N} + \sum_{i<d,j} a_{ij} v^{i+N-j} = 0$$

in B. Every exponent in the sum is nonnegative and strictly less than $d+N$. This is a monic equation over A, proving the claim. If $P=1$, the remainder is zero and the conclusion follows immediately. □

Lemma C1.2. With the same integral v, suppose $vP(X)$ belongs to the image for an arbitrary polynomial $P = a_0 + \cdots + a_d X^d$. Then $a_d^N v - Q(X)$ is integral over A for some $N \geq 0$ and $Q \in A[X]$.

Proof. Localize at a_d. The polynomial P/a_d is monic there, so Lemma C1.1 gives an integral remainder $v - Q_0(X)$, with $Q_0 \in A_{a_d}[X]$. Multiply by a power of a_d to clear the coefficients of Q_0. The resulting element $w = a_d^n v - Q(X)$ is integral after localization. In its localized monic equation, multiplying w by a sufficiently high power of a_d clears every coefficient denominator. A further power kills the error if that equation initially holds only after localization. The monic equation is then valid before localization. Thus a further multiple $a_d^m w$ is integral over A, as asserted. This argument also covers a nilpotent a_d: its localization is zero, and sufficiently high powers make the required element zero. □

C2. The conductor and all coefficients

Let $A \subset B$ be an inclusion such that A is integrally closed **in** B, and suppose B is finite over the subalgebra $A[x]$. The latter phrase does not require x to be transcendental. Define

$$J = \{u \in B : uB \subset A[x]\}.$$

This is an ideal of B, is contained in $A[x]$, and is called the conductor. For every $u \in J$, localization gives $B_u = A[x]_u$: write $b = (ub)/u$. Choose $A[x]$-module generators $b_1, \ldots, b_s$ for B.

Lemma C2.1. If $u(a_0 + a_1 x + \cdots + a_d x^d) \in J$, then $u a_d^N \in J$ for some $N \geq 0$.

Proof. For each i the element ub_i is integral over $A[x]$, since B is finite over it, and its product with the displayed polynomial lies in $A[x]$ by the conductor condition. Lemma C1.2 gives $a_d^{N_i} ub_i - Q_i(x)$ integral over A. This element belongs to B, so integral closedness puts it in $A \subset A[x]$. Hence $a_d^{N_i} ub_i \in A[x]$. With $N = \max_i N_i$, all $a_d^N ub_i$ belong to $A[x]$; module generation then gives $u a_d^N B \subset A[x]$, which is the assertion. □

Lemma C2.2. If $u(a_0 + a_1 x + \cdots + a_d x^d) \in \sqrt{J}$, then $u a_i \in \sqrt{J}$ for every i.

Proof. Choose $r > 0$ such that $u^r (a_0 + \cdots + a_d x^d)^r \in J$. Applying Lemma C2.1 to this polynomial gives $u^r a_d^{rN} \in J$ for some N. If $N = 0$, then $u \in \sqrt{J}$ and the required assertion is immediate. If $N > 0$, multiplying by a sufficiently high power of u shows that a power of $u a_d$ lies in J, so $u a_d \in \sqrt{J}$. Subtract the term $u a_d x^d$ from the original expression. The result is still in $\sqrt{J}$. Repeat with its lower-degree polynomial until every coefficient has been treated. □

An element x of a reduced algebra B is **strongly transcendental over** A if

$$u(a_0 + a_1 x + \cdots + a_d x^d) = 0 \quad \Longrightarrow \quad u a_i = 0 \text{ for every } i$$

for all $u \in B$ and all coefficients $a_i \in A$. Lemma C2.2 says that the image of x in $B/\sqrt{J}$ has this property over the embedded reduced ring $A/(A \cap \sqrt{J})$.

C3. Why a strongly transcendental finite extension has no isolated fibre point

The full pointwise equivalences are proved in the earlier programme lesson Quasi-finite morphisms and Chevalley's theorem, Theorem 1.1. For a finite-type ring map, a point is quasi-finite exactly when it is isolated in its fibre; equivalently its local fibre algebra is finite-dimensional over the base residue field. Its residue extension is then finite. The whole Proposition 2.1 of that lesson proves

preservation under arbitrary base change and composition, including composition with a closed source immersion. Its finite-dimensional fibre argument retains nilpotents and uses no Noetherian hypothesis on the base.

For the finite-extension case, let B be finite as an A-module. Its module generators also generate it as an algebra, so $A \to B$ is of finite type. For any $\mathfrak{p} \in \operatorname{Spec} A$, tensoring its finite module generating list shows that $B \otimes_A \kappa(\mathfrak{p})$ is finite-dimensional. Every localization at one of its primes is still finite-dimensional: this algebra is Artinian, since descending ideal chains are descending chains of vector subspaces; the local-factor decomposition proved in Lemma P0.9 makes its localization one of its finite-dimensional direct factors. The pointwise test therefore makes every point quasi-finite. The affine morphism is quasi-compact, by the affine-spectrum finite-cover proof in S01, so the finite extension is quasi-finite in the stated global convention. Empty fibres and the zero algebra are included.

Lemma C3.1. If reduced $A \subset B$ has a strongly transcendental $x \in B$, then for every minimal prime $\mathfrak{q}$ of B the image of x is transcendental over $A/(A \cap \mathfrak{q})$ in the domain $B/\mathfrak{q}$.

Proof. The localization $B_{\mathfrak{q}}$ is a field: it is reduced, its only prime is its maximal ideal, and its nilradical, the intersection of all primes, is zero; thus its maximal ideal is zero. If $u \sum_i a_i x^i \in \mathfrak{q}$, it vanishes there, so some $v \notin \mathfrak{q}$ makes $vu \sum_i a_i x^i = 0$ in B. Strong transcendence gives $vua_i = 0$, so $ua_i \in \mathfrak{q}$ for all i. Taking $u = 1$ proves that no nonzero polynomial over the embedded $A/(A \cap \mathfrak{q})$ can annihilate the image of x. □

Lemma C3.2. Suppose $A \subset B$ are domains, x is transcendental over A, and B is finite over $A[x]$. Then $\operatorname{Spec} B \to \operatorname{Spec} A$ is quasi-finite at no point.

Proof. First let A be normal. Corollary B1.2 above makes $A[x]$ normal. At a prime $\mathfrak{q}$ put $\mathfrak{p} = \mathfrak{q} \cap A$ and $\mathfrak{r} = \mathfrak{q} \cap A[x]$. If $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is not finite, the fibre criterion already excludes quasi-finiteness. Otherwise $\kappa(\mathfrak{r})/\kappa(\mathfrak{p})$ is finite, and $\mathfrak{r}$ strictly contains $\mathfrak{p}A[x]$: the latter prime has residue field $\kappa(\mathfrak{p})(x)$, which is transcendental. Going down for the integral domain extension $A[x] \subset B$ supplies $\mathfrak{q}' \subsetneq \mathfrak{q}$ over $\mathfrak{p}A[x]$. Both primes are in the fibre over $\mathfrak{p}$. Thus $\mathfrak{q}$ has a distinct generalization in that fibre, and cannot be isolated.

For general A, let A' be its integral closure in $\operatorname{Frac}(A)$ and let B' be the subring of $\operatorname{Frac}(B)$ generated by A' and B. The ring B' is integral over B and finite over $A'[x]$: it is a quotient of $B \otimes_A A'$, with the same finite module generating list. The element x is still transcendental over A', since A' lies in $\operatorname{Frac}(A)$. If $\mathfrak{q}$ were quasi-finite over A, its base changes to $\operatorname{Spec}(B \otimes_A A')$ would be quasi-finite over A', and so would their restrictions to $\operatorname{Spec} B'$. Lying over for $B \subset B'$ gives a prime of B' above $\mathfrak{q}$. The normal-base case contradicts quasi-finiteness at that prime. □

Corollary C3.3. If reduced $A \subset B$ contains a strongly transcendental x, and B is finite over $A[x]$, then no point is quasi-finite over A.

Proof. At a prime $\mathfrak{q}$ choose a minimal prime $\mathfrak{q}_0 \subset \mathfrak{q}$. Lemma C3.1 and Lemma C3.2 apply to $A/(A \cap \mathfrak{q}_0) \subset B/\mathfrak{q}_0$. A quasi-finite point would remain quasi-finite on this closed source subscheme, a contradiction. □

C4. Algebraic Zariski Main

Lemma C4.1 (one generator). If $B = A[b]$ and C is the integral closure of the image of A in B, every quasi-finite point $\mathfrak{q}$ admits $g \in C \setminus \mathfrak{q}$ with $C_g = B_g$.

Proof. Put $\mathfrak{p} = \mathfrak{q} \cap A$ and present $B = A[T]/I$. Some polynomial of I has a coefficient outside $\mathfrak{p}$: otherwise the fibre is $\kappa(\mathfrak{p})[T]$, whose points are not isolated with finite residue field. Thus there is an equation $a_m b^m + \cdots + a_0 = 0$ in B, with coefficients in the image of A, and some coefficient outside $\mathfrak{q}$.

Its leading multiple $a_m b$ is integral: multiply the equation by a_m^{m-1} to give a monic equation for that multiple. Hence $a_m b \in C$. If $a_m \notin \mathfrak{q}$, localizing at $g = a_m$ puts b in C_g and proves equality.

Otherwise combine the first two terms as $(a_m b + a_{m-1})b^{m-1}$ and keep the lower terms. This is a shorter equation with coefficients in C and at least one coefficient outside $\mathfrak{q}$, since $a_m b \in \mathfrak{q}$. The same argument applies: transitivity puts every new integral leading multiple in C. Repetition must find a leading coefficient outside $\mathfrak{q}$, since a degree-zero equation with its coefficient outside that prime is impossible. This coefficient gives the required g. □

Lemma C4.2 (finite over one generator). Suppose $A \subset B$ is integrally closed in B, B is finite over $A[b]$, and $\mathfrak{q}$ is quasi-finite over A. Then $A_h = B_h$ for some $h \in A \setminus \mathfrak{q}$.

Proof. Use the conductor J of $A[b] \subset B$. If $J \subset \mathfrak{q}$, Lemma C2.2 makes the quotient extension

$$A/(A \cap \sqrt{J}) \subset B/\sqrt{J}$$

strongly transcendental and finite over its single polynomial generator. Corollary C3.3 says it has no quasi-finite point. But the point from $\mathfrak{q}$ remains quasi-finite on that closed fibre subscheme, a contradiction. Choose $u \in J \setminus \mathfrak{q}$. Then $A[b]_u = B_u$. Its corresponding point is quasi-finite over A, since the local fibre agrees with the fibre of B there. Apply Lemma C4.1 to $A[b]$, whose relative integral closure is A, to obtain $a \in A \setminus \mathfrak{q}$ with $A_a = A[b]_a$. In this ring express $u = c/a^N$ with $c \in A \setminus \mathfrak{q}$. Localizing at $h = ac$ makes a and u invertible and gives $A_h = B_h$. □

Theorem C4.3 (algebraic Zariski Main). For a finite-type map $R \to B$, let $C \subset B$ be the integral closure of its image. At every quasi-finite prime $\mathfrak{q}$ there is $g \in C \setminus \mathfrak{q}$ with $C_g = B_g$.

Proof. Replace R by its image. Induct on the least n for which B is finite over $R[b_1, \dots, b_n]$. For $n = 0$, $C = B$. For $n = 1$, replace R by C and use Lemma C4.2: relative integrality is unchanged by transitivity, the same list generates B as a finite module over $C[b_1]$, and its fibre at the point is a subspace of the original fibre with the same finite residue-field condition.

For $n > 1$, take the relative integral closure D of $R[b_1, \dots, b_{n-1}]$ in B. Lemma C4.2 supplies $v \in D \setminus \mathfrak{q}$ with $D_v = B_v$. Choose finitely many numerators in D for a finite algebra generating list of B_v over R. Let $E \subset D$ be generated by those numerators, v, and $b_1, \dots, b_{n-1}$. Then $E_v = B_v$. It is finite over $R[b_1, \dots, b_{n-1}]$, because its finitely many added generators are integral over that ring. The contracted point is quasi-finite over R, by equality of the localized rings and fibres.

Induction applied to the finite-type E gives its relative integral closure F and $w \in F \setminus \mathfrak{q}$ such that $F_w = E_w$. Write $v = c/w^M$ there, with $c \in F \setminus \mathfrak{q}$. Put $g = wc$. Then $F_g = E_g = B_g$. Since $F \subset C \subset B$, also $C_g = B_g$. The finite intermediate E is essential: no finiteness of the entire integral closure D has been assumed. □

C5. Affine completion and the open quasi-finite locus

Theorem C5.1. The quasi-finite locus of a locally finite-type scheme morphism is open. A quasi-finite affine morphism $\operatorname{Spec} B \to \operatorname{Spec} R$ factors as a quasi-compact open immersion into a finite affine R-scheme.

Proof. On an affine chart Theorem C4.3 gives $C_g = B_g$ near a chosen quasi-finite point. Select finitely many integral numerators that, together with g^{-1}, generate B_g over R, and let $D \subset C$ be generated by them and g. Its generators are integral, so D is a finite R-module algebra. Since $D_g = B_g$, the open $D_B(g)$ is quasi-finite over R, being the open part of a finite map. These neighbourhoods prove openness.

If every point is quasi-finite, choose finitely many such g_i whose opens cover $\operatorname{Spec} B$. Take the finite R-subalgebra generated by all the selected numerators and all g_i. Its localizations at g_i are B_{g_i}. Thus $\operatorname{Spec} B$ identifies with the finite union of opens $D_D(g_i)$ in $\operatorname{Spec} D$, giving the stated quasi-compact open immersion. The construction allows extra points in the finite completion. □

D. Étale structure and general scheme Zariski Main

D2. The structural étale inputs, with proofs

Lemma D2.1 (maps between étale algebras). Every R-algebra map $P \to Q$ between étale R-algebras is étale. A surjective such map is a principal localization, equivalently the projection onto an open-and-closed component.

Proof. Write finite presentations of P and Q over R, and express the images of the finitely many generators of P as polynomials in the generators of Q. Over P, the presentation of Q consists of its finitely many old relations together with the equations identifying those images with the generators of P. Hence Q is finitely presented over P.

In a square-zero lifting problem over P, formal étaleness of Q/R gives a unique R-lift. Its restriction to P and the specified P-structure are two R-lifts of the same residue map. Formal unramifiedness of P/R makes them equal. The lift is therefore a P-lift and is unique. This proves étaleness using its finite-presentation and formal-lifting definition.

If $Q = P/I$, finite presentation makes I a finite ideal. One direct justification is to present the quotient over P using finitely many variables and relations, choose lifts of their images in P, and substitute those lifts into the finitely many relations; their values generate the kernel. Formal smoothness over P applied to $P/I^2 \to P/I$ gives a P-algebra section. For $a \in I$, its image in P/I^2 must then be the image of zero, so $I/I^2 = 0$.

Write finite generators a_i as $a_i = \sum_j c_{ij} a_j$ with $c_{ij} \in I$. The adjugate of $(\delta_{ij} - c_{ij})$ shows that its determinant annihilates I. That determinant is $1 - e$ for an $e \in I$. Thus $(1 - e)I = 0$, and in particular $(1 - e)e = 0$. Consequently $e^2 = e$ and $I = Pe$. The quotient P/I is P_{1-e}, proving the final assertion. □

Lemma D2.2 (one monic étale equation near a point). If $R \to S$ is étale and $\mathfrak{q} \in \operatorname{Spec} S$, a principal neighbourhood of $\mathfrak{q}$ is isomorphic over R to

$$(R[T]/(F))_h, \qquad F \text{ monic}, \quad F' \text{ invertible}. \tag{2.1}$$

Proof. Étale field fibres are finite products of finite separable fields, by the full field proof in the earlier étale lesson. Hence S/R is quasi-finite. Theorem C4.3 gives a finite R-subalgebra $D \subset S$ and $v \in D \setminus \mathfrak{q}$ with $D_v = S_v$. Let $\mathfrak{r} = \mathfrak{q} \cap D$ and $\mathfrak{p} = \mathfrak{q} \cap R$. The selected local factor of $D \otimes_R \kappa(\mathfrak{p})$ is $\kappa(\mathfrak{q})$, because localization at v is the étale fibre. The other finitely many factors are local Artinian rings.

Choose a nonzero primitive element α of $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$. Here is the complete field argument, using Lemma P0.4a. Let L/F be finite separable of degree n. Its finite generating list $a_1, \dots, a_r$ has exactly n distinct embeddings $\sigma_1, \dots, \sigma_n$ into one finite splitting field H. If F is infinite, for each pair $i \neq j$ the form $\sum_t (\sigma_i(a_t) - \sigma_j(a_t))X_t$ is a nonzero polynomial over H, since the embeddings differ on a generator. Their finite product is nonzero: the leading coefficient of a product of nonzero polynomials over a field is nonzero, iterating by the number of variables. A nonzero polynomial over H does not vanish on all tuples in the infinite subfield F. Indeed, induction on the number of variables first chooses a tuple in F where one nonzero coefficient polynomial is nonzero, and then avoids the finitely many roots in the last variable. Choose $c_t \in F$ where this product does not vanish. The element $a = \sum c_t a_t$ has n distinct images. Its minimal polynomial therefore has degree at least n by the root bound, and at most n because $F(a) \subset L$. The tower-degree equality gives $L = F(a)$. In a nontrivial extension this element cannot be zero; for a trivial extension use 1.

If F is finite, then L is finite as a set by its finite basis. Put $G = L^\times$. The following calculation proves that any finite subgroup of a field's units is cyclic. The order of $g \in G$ is the least positive d with $g^d = 1$, which exists by equality of two powers in the finite group. Integer division shows $g^r = 1$ exactly when d divides r. Let e be the least common multiple of the finitely many orders, computed

by the maximum exponent of each prime in their factorizations; integer prime factorization and Bezout were proved in Lemma B1.1. For each prime ℓ dividing e, select an element g_ℓ of order $\ell^s m$ with ℓ not dividing m and ℓ^s the maximal occurring power. Then $h_\ell = g_\ell^m$ has order exactly ℓ^s, by the divisibility test just proved. For commuting elements x, y of coprime orders a, b, if $(xy)^r = 1$, then $x^r = y^{-r}$ belongs to both cyclic subgroups. Its ath and bth powers are one; an integer Bezout identity makes the element one. Thus both a and b divide r, and ab divides r by the same Bezout identity. Since $(xy)^{ab} = 1$, its order is ab. Inducting over the primes shows that $h = \prod h_\ell$ has order e. Every element of G is a root of $T^e - 1$, so the polynomial root bound gives $|G| \leq e$. The e different powers of h give the reverse inequality. Therefore G consists of those powers and is cyclic. Its generator generates L as a field, because every nonzero element is a power of it and zero belongs to every subfield. It is nonzero. If G is trivial, take its generator to be 1; the same conclusion holds. This supplies the required α.

In the fibre of D, choose $\bar{t} = (\alpha, 0, \ldots, 0)$. Clearing a base denominator gives $t \in D$ with residue $(c\alpha, 0, \ldots, 0)$ for a nonzero $c \in \kappa(\mathfrak{p})$; scaling preserves primitivity and nonzeroness. Put $E = R[t] \subset D$. This is finite over R, since t is integral. The prime $\mathfrak{r}' = \mathfrak{r} \cap E$ has exactly one prime of D above it: all other fibre points have $t = 0$, whereas the chosen one has $t = c\alpha \neq 0$.

The finite algebra $D_{\mathfrak{r}'}$ over the local ring $E_{\mathfrak{r}'}$ is therefore local, with localization $D_{\mathfrak{r}}$. To see this, maximal ideals of a finite algebra contract to the maximal ideal, so the unique prime above $\mathfrak{r}'$ is its unique maximal ideal; localizing further at that ideal changes nothing. Its quotient by $\mathfrak{p}$ is the field $\kappa(\mathfrak{q})$. The residue field of $E_{\mathfrak{r}'}$ also equals that field, because it contains $c\alpha$ and $\kappa(\mathfrak{p})$. Nakayama applied to the finite module $D_{\mathfrak{r}'}/E_{\mathfrak{r}'}$ gives

$$E_{\mathfrak{r}'} = D_{\mathfrak{r}'} = D_{\mathfrak{r}}.$$

The modules D/E and D/vD are finite over E and vanish after localization at $\mathfrak{r}'$. A common element $u \in E \setminus \mathfrak{r}'$ kills their chosen finite generators. Thus $E_u = D_u$, v is invertible there, and $E_u = D_u = S_{uv}$ is étale over R.

Present $E = R[T]/I$ by $T \mapsto t$. Choose a monic $M \in I$ using integrality of t. The fibre ideal $I\kappa(\mathfrak{p})[T]$ is principal. Its factor belonging to the selected point is the separable minimal polynomial H_1 of $c\alpha$ with multiplicity one, because $(E \otimes \kappa(\mathfrak{p}))_{\mathfrak{r}'}$ is the field $\kappa(\mathfrak{q})$. Choose $H \in I$ whose reduction is a nonzero scalar multiple of a generator of that fibre ideal. This is possible by expressing that generator as a finite combination of images of elements of I and clearing the finitely many denominators outside $\mathfrak{p}$.

For $N \geq 2$ with $N \deg M > \deg H$, put $F = M^N + H \in I$. It is monic. Modulo $\mathfrak{p}$, the term M^N is divisible by H_1^2, whereas H is divisible by H_1 exactly once. Therefore F' is nonzero at $c\alpha$. Choose a polynomial lift of u and localize at its product with F'. We get a surjection from the standard étale algebra $(R[T]/F)_{uF'}$ onto the étale algebra $E_{uF'}$. Lemma D2.1 makes this surjection a principal localization. Its target is a principal neighbourhood of our original point in S, and its source and all principal localizations have the form (2.1). □

No Noetherian reduction was used here. Finiteness of D/E is sufficient for the localization step; no finiteness of the full relative integral closure was assumed.

Lemma D2.3 (flatness and openness). Étale morphisms are flat and open.

Proof. The algebra $R[T]/F$ for monic F is finite free over R. Localization at an element of this algebra is a filtered colimit of copies of that free R-module with multiplication transition maps. Tensoring an injection with every copy is injective, and exactness of filtered colimits proves flatness of the localization. Lemma D2.2 covers an étale algebra by such principal neighbourhoods. A kernel of a tensor map which vanishes on that principal cover is zero, so the entire algebra is flat.

For openness, let $C = R[T]/F$, of rank $d > 0$, and let $a \in C$. A prime $\mathfrak{p}$ belongs to the image of $D_C(a)$ exactly when a is not nilpotent in $C \otimes_R \kappa(\mathfrak{p})$: Lemma P0.1 supplies a prime avoiding precisely each nonnilpotent element of this fibre. Write M for the matrix of multiplication by a in

the power basis, and put

$$\chi_M(Z) = \det(ZI - M) = Z^d + c_1 Z^{d-1} + \cdots + c_d.$$

These determinant coefficients commute with passage to the residue field, since the determinant is its finite signed permutation sum. On any fibre, M^n is multiplication by a^n; applying this map to 1 shows that M is nilpotent exactly when a is nilpotent.

Here is the full matrix test over that field. If $M^N = 0$, form the finite filtration $0 \subset \ker M \subset \ker M^2 \subset \cdots \subset \ker M^N = V$. Extend a basis at each step by adjoining vectors outside its span until the next finite-dimensional subspace is spanned. Since $M(\ker M^i) \subset \ker M^{i-1}$, the matrix in this ordered basis is strictly upper triangular. Changing the basis preserves $\det(ZI - M)$: reduce its change-of-basis matrix by the elementary operations of Lemma P0.4a; the corresponding row and inverse column operations multiply determinants by reciprocal scalars. The triangular determinant calculation in that lemma therefore gives $\chi_M(Z) = Z^d$.

For the converse we prove Cayley–Hamilton over the original, arbitrary commutative ring R. Cofactor expansion gives

$$(ZI - M)\operatorname{adj}(ZI - M) = \chi_M(Z)I.$$

Every entry of the adjugate has degree at most $d-1$. Write it as $\sum_{j=0}^{d-1} B_j Z^j$. Comparing coefficients in the displayed identity gives

$$B_{d-1} = I, \qquad B_{j-1} = MB_j + c_{d-j}I \quad (1 \le j \le d-1), \qquad -MB_0 = c_d I.$$

Descending substitution yields $B_0 = M^{d-1} + c_1 M^{d-2} + \cdots + c_{d-1}I$, with $B_0 = I$ if $d = 1$. The constant relation now gives $M^d + c_1 M^{d-1} + \cdots + c_d I = 0$. Thus $\chi_M(Z) = Z^d$ implies $M^d = 0$. We have proved both directions of the fibre test, so the image of $D_C(a)$ is exactly $\bigcup_{i=1}^{d} D_R(c_i)$, an open. If $d = 0$, the monic polynomial is 1, $C = 0$, and both the image and the empty union are empty.

A principal open of a localization C_g is $D_C(ga)$ for a suitable numerator a. Its image is open by the same calculation. Principal opens form a basis, and Lemma D2.2 covers arbitrary étale morphisms by these charts. Taking unions proves openness. □

D3. Finite pieces after an elementary étale neighbourhood

An **elementary étale neighbourhood** of (S, s) is an étale morphism $T \to S$ with a chosen $t \mapsto s$ inducing $\kappa(t) = \kappa(s)$.

Lemma D3.1 (the universal coprime-factor construction). Suppose $F \in R[T]$ is monic and its reduction at $\mathfrak{p} \subset R$ factors into coprime monic polynomials $\bar{G}\bar{H}$ of degrees r, s. There is an elementary étale neighbourhood $(\operatorname{Spec} R', \mathfrak{p}')$ of $\mathfrak{p}$ on which $F = GH$ with those degrees and residues, and G, H generate the unit ideal of $R'[T]$.

Proof. If $r + s = 0$, monicity gives $F = 1$. Take $G = H = 1$ and the identity base neighbourhood with its original point; the factorization and unit-ideal assertion are immediate. Suppose henceforth $r + s > 0$. Introduce $r + s$ variables for the nonleading coefficients of monic G, H, impose the $r + s$ coefficient equations $GH = F$, and invert their Jacobian determinant Δ. At the prescribed residue point the Jacobian is the matrix of

$$\kappa(\mathfrak{p})[T]_{<r} \oplus \kappa(\mathfrak{p})[T]_{<s} \longrightarrow \kappa(\mathfrak{p})[T]_{<r+s}, \qquad (U, V) \longmapsto \bar{H}U + \bar{G}V.$$

Its kernel is zero: coprimality forces $\bar{G} \mid U$, then the degree bound gives $U = 0$ and $V = 0$. Equal dimensions make Δ nonzero at that point. The square Jacobian lifting proof in the earlier étale lesson proves that this finitely presented localized coefficient algebra R' is étale. Evaluation at the prescribed coefficients gives the point with residue field exactly $\kappa(\mathfrak{p})$.

Over R' the same coefficient matrix is invertible. Apply its inverse to the constant polynomial 1. This gives U, V with $HU + GV = 1$, proving the last assertion. Degree-zero factors are included, with the evident empty coefficient lists. □

The construction below uses this coprime-factor matrix at an actual finite neighbourhood; it requires no henselization theorem.

Lemma D3.2 (one finite open piece). Let $f : X \to S$ be locally of finite type and let x be isolated in X_s, $s = f(x)$. There is an elementary étale neighbourhood (T, t) of (S, s) and an open $V \subset X_T$ finite over T, whose fibre over t has exactly one point v mapping to x, with $\kappa(v) = \kappa(x)$. A further elementary étale base change preserves these properties.

Proof. Choose compatible affine neighbourhoods of x and s. Openness of the quasi-finite locus from Theorem C5.1 lets us choose a smaller affine principal neighbourhood Y of x on which the map is quasi-finite. Its affine completion embeds Y openly into $\operatorname{Spec} D$, with D finite over the chosen base ring R.

The finite fibre $D \otimes_R \kappa(s)$ is a product of local Artinian rings. Let e_0 be the component idempotent for the point x in this completion. Work first over $R_{\mathfrak{p}}$, where $\mathfrak{p}$ is the prime of s, and lift e_0 to $b \in D_{\mathfrak{p}}$. There is a monic $F \in R_{\mathfrak{p}}[T]$ with $F(b) = 0$ by the finite-module determinant argument. If the fibre has just this one component, take $e = 1$ and skip the next splitting paragraph.

Otherwise e_0 and $1 - e_0$ are nonzero. The identity

$$\bar{F}(e_0) = \bar{F}(0)(1 - e_0) + \bar{F}(1)e_0 = 0$$

forces $\bar{F}(0) = \bar{F}(1) = 0$, since a nonzero scalar cannot annihilate a nonzero idempotent. Factor $\bar{F} = (T - 1)^a \bar{H}$, with $a \geq 1$, $\bar{H}(1) \neq 0$ and $\bar{H}(0) = 0$. Lemma D3.1 lifts this coprime factorization after an elementary étale extension. Bezout gives an idempotent in the quotient by F, equal to one in the factor for $(T - 1)^a$ and zero in the other. Evaluation at b gives an idempotent e of $D \otimes_R R'$ whose special-fibre image is e_0: its residue polynomial takes value one at 1 and zero at 0.

All the data used so far are finite: the coefficients of F, the lift b, the equality $F(b) = 0$, and the coefficient equations and Jacobian. Clear their finitely many denominators outside $\mathfrak{p}$, and one further denominator to kill any equality error. The construction is then defined over R_a for one $a \notin \mathfrak{p}$. Its coefficient algebra, including its inverted Jacobian, is étale over R_a and therefore over R, and the chosen point has residue $\kappa(s)$. Thus it is an actual elementary étale neighbourhood of the original affine base, not only a ring over $R_{\mathfrak{p}}$.

The component $\operatorname{Spec}(e(D \otimes_R R'))$ is finite over R' and its special fibre is precisely the local Artinian factor for x. Its complement of $Y_{R'}$ is closed. Its image in $\operatorname{Spec} R'$ is closed, by the full closed-image proof for finite morphisms, and avoids the selected base point. Shrink to a principal base neighbourhood avoiding that image. The whole finite component now lies in $Y_{R'}$. Call it V. It is open in the completion and hence in $Y_{R'}$ and $X_{R'}$, finite over the shrunk base, with the required unique point and unchanged residue field.

Under a further elementary neighbourhood the selected fibre is identified with its old fibre, since the base residue field is unchanged. Finiteness and openness survive base change. This proves the final assertion. □

Proposition D3.3 (several finite pieces and the separated decomposition). For distinct isolated points $x_1, \ldots, x_n$ in X_s, a single elementary étale neighbourhood admits finite open pieces V_i with the properties of Lemma D3.2 for all i. If f is separated, after shrinking that neighbourhood there is a decomposition

$$X_T = W \amalg V_1 \amalg \cdots \amalg V_n \tag{3.1}$$

into open-and-closed subschemes, and W_t has no point mapping to any x_i.

Proof. Construct the pieces successively. At each elementary neighbourhood the fibre is the original fibre, so the next point is uniquely identified and still isolated; Lemma D3.2's final assertion preserves all earlier pieces. Finite induction gives a common neighbourhood.

For a separated map, each finite T-scheme V_i is also closed in X_T. Indeed its graph into $V_i \times_T X_T$ is closed by separatedness of X_T/T, and the projection of this graph to X_T is finite, being a closed immersion followed by the base change of the finite map $V_i \to T$. Its image, which is V_i, is closed.

The intersections $V_i \cap V_j$ are closed in the finite T-scheme V_i and have empty fibre at the chosen base point, because the original x_i are distinct. Their finitely many closed images in T can all be avoided by a further principal shrink. The pieces are then disjoint and open-and-closed. Their complementary open-and-closed scheme is W. Each original x_i has exactly one point above it in the elementary fibre and that point belongs to V_i, so none belongs to W_t. □

Proposition D3.4 (separable residue extension before the finite splitting). Without separatedness, one can instead choose an étale neighbourhood (T, t) with finite separable residue extension so that, above each x_i, all points y_{ij} have finite purely inseparable residue extension over $\kappa(t)$ and each has a finite open piece over T.

Proof. Put $k = \kappa(s)$ and $K_i = \kappa(x_i)$, finite over k by the isolated fibre criterion. We first make the finite field calculation explicit. In characteristic $p > 0$, the binomial coefficients $\binom{p}{j}$ for $0 < j < p$ are divisible by p: the identity $j!(p-j)!\binom{p}{j} = p!$ and the integer prime-divisibility calculation in the earlier Lemma 4.A of the field lesson prove this, since neither factorial on the left is divisible by p. Hence $(u+v)^p = u^p + v^p$ in every commutative ring of characteristic p, and iteration gives the same identity with p^N. Products also commute with these powers.

Choose a finite generating list α_j for K_i/k. A nonconstant irreducible minimal polynomial with zero derivative has every positive exponent divisible by p, so it is $g(T^p)$. Repeat until the resulting polynomial has nonzero derivative; its degree strictly decreases. The final expression is $g(T^{p^{a_j}})$, with g irreducible, since a factorization of g would factor the original polynomial. Thus g is the separable minimal polynomial of $\alpha_j^{p^{a_j}}$, by Lemma P0.4a. These powers generate a finite separable field F_i/k: the separable tower and embedding argument of that lemma proves this for their finite composite. With N at least every a_j, taking p^N-powers of any rational expression in the α_j puts it in F_i, by the preceding power identities. Consequently K_i/F_i is finite purely inseparable. In characteristic zero every irreducible minimal polynomial has nonzero derivative, so take $F_i = K_i$.

By the primitive-element proof in Lemma D2.2, each F_i/k has a primitive generator. Construct a finite splitting field L/k for their finitely many separable minimal polynomials using Lemma P0.4a. Every root adjoined is separable over k, and the same separable tower argument makes L/k finite separable. Lift the monic minimal polynomial of a primitive generator of L/k to coefficients over a principal neighbourhood of s. Invert its derivative and select the point evaluating that generator to its value in L. Lemma D2.2's standard monic calculation gives an étale neighbourhood with residue field L.

The fibre above x_i is $\operatorname{Spec}(K_i \otimes_k L)$. The primitive polynomial of F_i/k splits into distinct linear factors in L. Evaluation at its roots identifies its quotient algebra with the product of copies of L. Indeed division leaves a remainder of degree below their number, and the root bound in Lemma P0.4a makes a remainder vanishing at all roots zero. For each root r, the polynomial $\prod_{r' \neq r}(T - r')/(r - r')$ evaluates to one at r and zero at all other roots, proving surjectivity. Thus

$$F_i \otimes_k L \simeq \prod_{\sigma : F_i \hookrightarrow L} L, \qquad K_i \otimes_k L \simeq \prod_{\sigma : F_i \hookrightarrow L} C_{i,\sigma}, \quad C_{i,\sigma} = K_i \otimes_{F_i,\sigma} L.$$

Each $C = C_{i,\sigma}$ is nonzero and finite-dimensional over L: an F_i-basis of K_i becomes an L-basis after this tensor product. In characteristic zero it is simply L. In characteristic p, it is generated as an L-algebra by the images z_j of the α_j, and for a common N each satisfies $z_j^{p^N} = a_j \in L$.

In characteristic p, the prime calculation also allows nilpotents. Lemma P0.1 supplies a prime of the nonzero ring C. For any prime $\mathfrak{q}$, the quotient $E = C/\mathfrak{q}$ is a finite-dimensional L-domain and hence

a field: multiplication by each nonzero element is injective, therefore surjective, and its inverse is the preimage of 1. Construct one finite splitting field H/L of the finitely many polynomials $T^{p^N} - a_j$, using Lemma P0.4a. The field E embeds over L in H. Indeed, adjoin its generator images successively. At each step the new minimal polynomial divides the corresponding L-polynomial $T^{p^N} - a_j$; after applying the previously constructed embedding it still divides that split polynomial. Choose a root in H and extend by evaluation on the minimal-polynomial quotient. The quotient is a field, so this evaluation is injective. This is precisely the finite tower construction proved in Lemma P0.4a.

Every polynomial $T^{p^N} - a_j$ has exactly one root in H: it has a root by construction, and two roots u, v would satisfy $(u - v)^{p^N} = u^{p^N} - v^{p^N} = 0$, whence $u = v$ in the field H. Thus every composite $C \to E \hookrightarrow H$ sends every z_j to the same forced root and fixes L. Since the z_j generate C, all these maps coincide. Their kernels are exactly their respective primes $\mathfrak{q}$, so C has exactly one prime. Its residue field is finite purely inseparable over L: taking p^N-powers of a polynomial expression in the generator images gives an element of L. This argument makes no reducedness assumption on C.

The finitely many resulting points are isolated: base change of a finite-dimensional open fibre algebra is still finite-dimensional. Apply the multiple-piece part of Proposition D3.3 to all of them, using a further elementary neighbourhood with unchanged residue L. Its finite pieces have the stated residues, and their points exhaust all points above the selected x_i. □

D4. Relative integral closure through étale base change

Lemma D4.1 (the derivative coefficient identity). Let $F \in R[T]$ be monic of degree d, let $R \to B$ be a ring map, and let $h \in B[T]/F$ be integral over R. The unique degree-below-d representative of $F'h$ has all coefficients integral over R.

Proof. If $d = 0$, then $F = 1$, the quotient is the zero ring, and its degree-below-zero representative has no coefficients; the assertion is empty. Assume $d > 0$. Division by a monic polynomial subtracts leading terms without inverting coefficients and gives a remainder below its degree. That remainder is unique, since a nonzero multiple of a monic polynomial has at least its degree. Thus its quotient has the free power basis $1, T, \ldots, T^{n-1}$, where n is the polynomial degree, and the base ring injects via constant polynomials. Construct a splitting extension of B by adjoining one root of F, dividing by its monic linear factor, adjoining a root of the remaining monic polynomial, and continuing. At each step the quotient is therefore finite free with a basis containing 1. The composite $B \subset B'$ is an injective finite free extension and

$$F(T) = \prod_{i=1}^{d} (T - \alpha_i) \quad \text{in } B'[T].$$

Every α_i satisfies F, so is integral over R. Write $h = \sum_{j<d} h_j T^j$. In $B'[T]/F$ the identity is

$$F'h = \sum_{i=1}^{d} h(\alpha_i) \prod_{j \neq i} (T - \alpha_j). \tag{4.1}$$

Here is a proof covering repeated roots and nilpotents. First put independent variables α_i, h_j in $\mathbf{Z}[\alpha_1, \ldots, \alpha_d, h_0, \ldots, h_{d-1}]$. This ring is a domain: a product of nonzero integers has nonzero absolute value, and the leading coefficient of a product of nonzero polynomials in one variable over a domain is a product of nonzero coefficients. Induct on the number of variables. Reduce the left side by the monic F; both sides then have degree below d. Their difference evaluates to zero at every α_i, because $F'(\alpha_i) = \prod_{j \neq i} (\alpha_i - \alpha_j)$. In the fraction field of this domain, the independent variables $\alpha_1, \ldots, \alpha_d$ are distinct. A polynomial of degree below d with these d distinct roots is zero by the root bound proved in Lemma P0.4a. Thus the difference polynomial is zero in that field.

Injectivity of the domain into its fraction field makes every coefficient zero already in the universal domain. Specializing its independent variables proves (4.1) over every commutative ring, including repeated roots and nilpotents.

A monic equation for h over R evaluates to the same monic equation for each $h(\alpha_i)$ over R. Hence each is integral over R. Integral elements form a ring, so the coefficients on the right side of (4.1) are integral over R. They are the coefficients of the monic remainder on the left, which already belong to B. A monic equation for such a coefficient in B' holds in B by the displayed injection. □

Theorem D4.2 (étale base change for integral closure). Let $R \to P$ be étale, let $R \to B$ be any ring map, and let $A \subset B$ be the integral closure of the image of R. Then

$$P \otimes_R A \;\cong\; \{\text{elements of } P \otimes_R B \text{ integral over } P\}. \tag{4.2}$$

Proof. The left side injects into $P \otimes_R B$ by flatness from Lemma D2.3. Every element of the left side is integral over P, since it is built from finitely many base changes of integral elements.

It remains to prove surjectivity, which may be checked on the principal cover of $\operatorname{Spec} P$ from Lemma D2.2. Indeed integral closure commutes with localization, and a module whose localizations on that cover vanish is zero. Thus assume $P = (R[T]/F)_g$, with F monic and F' invertible there.

Put $C = R[T]/F$ and $D = B[T]/F$. If A'' is the integral closure of C in D, localization of integral closure says that the right side of (4.2) is $(A'')_g$. An element $a \in A''$ is integral over R, by transitivity, since C is finite over R. Lemma D4.1 writes the monic remainder of $F'a$ with coefficients in A. Consequently $F'a$ belongs to the image of $A[T]/F$. After localization, F' is a unit from P, so a belongs to $P \otimes_R A$. Dividing by powers of g gives the same assertion for all elements of $(A'')_g$. □

Let now $f : X \to S$ be quasi-compact and quasi-separated. On $V = \operatorname{Spec} R \subset S$, its inverse image is qcqs. The finite-cover localization proof in the earlier quasi-coherent-sheaf lesson gives

$$\Gamma(f^{-1}V, \mathcal{O}_X)_a = \Gamma(f^{-1}D(a), \mathcal{O}_X).$$

Thus $f_*\mathcal{O}_X$ is quasi-coherent. Taking the integral closure of R in these section algebras commutes with localization, by the full algebraic localization theorem. The closures therefore form a quasi-coherent integral $\mathcal{O}_S$-algebra $\mathcal{A}$. Define

$$Z = \operatorname{Spec}_S \mathcal{A}, \qquad X \xrightarrow{j} Z \xrightarrow{\nu} S. \tag{4.3}$$

The relative spectrum and its universal property can be constructed here. For a quasi-coherent algebra $\mathcal{B}$ on S, take $\operatorname{Spec} \mathcal{B}(V)$ over every affine $V = \operatorname{Spec} R$. On $D(a) \subset V$, the affine module–sheaf correspondence of AG-RG-S01, Proposition 2.2, identifies $\mathcal{B}(D(a))$ with $\mathcal{B}(V)_a$; its spectrum is precisely the principal inverse-image open in $\operatorname{Spec} \mathcal{B}(V)$. Refining intersections of base affine charts by principal opens therefore identifies these spectra canonically. Their identifications satisfy the triple equation because localization is associative, so scheme gluing constructs $\operatorname{Spec}_S \mathcal{B}$. For an S-scheme T with structure map h, an S-map $T \to \operatorname{Spec}_S \mathcal{B}$ is equivalent to an $\mathcal{O}_S$-algebra map $\mathcal{B} \to h_*\mathcal{O}_T$: over an affine base chart, the affine-target universal property of AG-RG-S01, Proposition 2.3, turns its section-algebra map into a unique map from T_V to that affine spectrum. These maps agree on principal restrictions and hence glue; conversely pullback of functions gives the algebra map. The two constructions are inverse on every test scheme. Under a base change $S_1 \to S$, compatible affine charts $R \to P$ give the spectrum of $P \otimes_R \mathcal{B}(V)$ on both sides; these identifications respect localization and glue, proving arbitrary base-change compatibility. Apply this construction to $\mathcal{A}$. Its inclusion into $f_*\mathcal{O}_X$ gives j by the just-proved universal property, and each affine chart of ν is integral by the definition of $\mathcal{A}$. This is **relative integral closure**; it retains nilpotents and is not a reduced normalization.

Corollary D4.3 (the scheme compatibility). For an étale morphism $S_1 \to S$, the base change Z_{S_1} in (4.3) is the relative integral closure of S_1 in X_{S_1}.

Proof. Work on compatible affine base charts $R \to P$. Since P is flat, the finite-cover equalizer for sections on the qcqs scheme $f^{-1}V$ stays exact after tensoring by P. Its chart and overlap terms become the corresponding section modules after base change. Hence

$$P \otimes_R \Gamma(f^{-1}V, \mathcal{O}_X) = \Gamma((f^{-1}V)_P, \mathcal{O}).$$

This is also the degree-zero case of the full flat cohomology base-change proof already in Algebra and sheaf cohomology before reductive groups, Section 6. Theorem D4.2 identifies the integral subalgebra after this base change with $P \otimes_R A$. Relative Spec commutes with base change by its earlier construction. These canonical chart identifications agree on all restrictions and glue to the asserted identification. □

D5. Descending the local isomorphism

Lemma D5.1 (isomorphisms descend along an étale cover). If $g : Y \to Z$ becomes an isomorphism after a surjective étale base change $P \to Z$, then g is an isomorphism.

Proof. The inverse of the base-changed isomorphism gives a map $\sigma_P : P \to Y$ over Z. Its two pullbacks to $P \times_Z P$ coincide: they are both inverses to the same base-changed isomorphism. We describe the descent of this compatible map using only faithfully flat affine algebra.

First, if $R \to A$ is faithfully flat, then

$$0 \longrightarrow M \longrightarrow A \otimes_R M \longrightarrow A \otimes_R A \otimes_R M, \quad a \otimes m \longmapsto 1 \otimes a \otimes m - a \otimes 1 \otimes m \tag{5.1}$$

is exact for every R-module M. After tensoring by A, multiplication of the first two A factors retracts the first map. For a tensor in the kernel of the second map, multiplication of those first two factors in its kernel equation expresses that tensor as the first-map image of its retraction. Thus the tensored sequence is exact. Flatness and faithfulness detect its kernels and quotient, proving (5.1).

Every affine open $W = \operatorname{Spec} R$ of Z has a finite affine refinement of its étale cover: affine source neighbourhoods have open images by Lemma D2.3, and quasi-compactness of W selects finitely many of them. Their disjoint union is $\operatorname{Spec} A$, flat and surjective over W, hence faithfully flat. For the last assertion, any nonzero module contains a nonzero cyclic submodule R/I; a point above a maximal ideal containing I makes IA proper, and flatness preserves the cyclic submodule's injection.

Points of this cover above the same point of W have a common point on their fibre product, because the tensor product of their two residue fields over the base residue field is nonzero. Compatibility therefore gives a well-defined set map $\sigma : W \to Y$. The cover is open and surjective, so it is a quotient map; the compatibility makes the inverse image of an open of Y invariant, and its image is the inverse image under σ. This proves continuity.

Around any point of W choose a principal neighbourhood $D(r)$ whose image under σ lies in an affine open $\operatorname{Spec} B$ of Y. The coefficient map $B \to A_r$ has equal two pullbacks to $A_r \otimes_{R_r} A_r$. Formula (5.1), with $M = R_r$, factors it uniquely through R_r. It defines the desired local scheme map. The local maps agree on overlaps by the same faithful equalizer and glue. On each affine base open the map recovers σ_P after pullback, so those descended maps also agree on overlaps of the base opens.

We obtain $\sigma : Z \to Y$ with $g\sigma = \mathrm{id}_Z$. Its composite σg becomes id_Y after the surjective étale cover pulled back to Y. Equality of scheme maps can be checked there by the same affine equalizer and open-target argument just used. Hence $\sigma g = \mathrm{id}_Y$. □

This is the earlier affine descent block in block A above, Lemmas A1.1 and A1.5, specialized with its proof to étale covers. It does not use effectivity of separated locally quasi-finite descent or any algebraic space.

Theorem D5.2 (relative Zariski Main). Let $f : X \to S$ be finite type and separated, and let (4.3) be its relative integral closure. If $U \subset X$ is its quasi-finite locus, then $U' = j(U)$ is open in Z,

$$j^{-1}(U') = U, \qquad U \xrightarrow{\sim} U'. \tag{5.2}$$

Proof. Theorem C5.1 proves that U is open. Fix $x \in U$, with image s. Proposition D3.3 gives an elementary étale neighbourhood $T \to S$ and a decomposition

$$X_T = V \amalg W, \qquad V \to T \text{ finite}, \quad (x,t) \in V.$$

The relative integral closure of T in X_T is $V \amalg W'$, where W' is the relative integral closure of T in W. On an affine base chart its section algebra is a product $D \times B$, with D finite over the base ring. The integral subalgebra is exactly $D \times C$, where C is the integral subalgebra of B. Membership implies componentwise integrality by projection. Conversely the product of two monic annihilating polynomials annihilates the pair, proving membership. Since every element of D is integral, its component is unchanged. Taking relative spectra gives the asserted disjoint union.

Corollary D4.3 identifies this closure with Z_T. Thus the clopen target component $V \subset Z_T$ has full inverse image $V \subset X_T$, mapping isomorphically onto it. The image $O \subset Z$ of that component is open by étale openness, and contains $j(x)$. Its map $V \to O$ is a surjective étale cover. The pullback of $j^{-1}(O) \to O$ along this cover is precisely the isomorphism just obtained. Lemma D5.1 makes $j^{-1}(O) \to O$ an isomorphism.

Every point of $j^{-1}(O)$ is quasi-finite over S. Indeed O is locally finite type over S by this isomorphism with an open of X, and it is open in the integral S-scheme Z. Integral fibre algebras are integral over a field, so all their prime quotients are algebraic fields and all primes are maximal, by Lemma P0.4. An affine locally finite-type chart in such a fibre is a zero-dimensional Noetherian algebra, hence Artinian and discrete by Lemma P0.9. Its residue fields are finite by Lemma P0.9. The isolated fibre criterion therefore gives quasi-finiteness.

The opens O constructed for all $x \in U$ consequently have full inverse images contained in U; they cover $j(U)$ and their inverse images cover U. Their isomorphisms are restrictions of the same map j and agree on overlaps. Taking their union proves all three assertions in (5.2). □

The scope of quasi-compactness. Let $A = k[t_1, t_2, \ldots]$, $I = (t_1, t_2, \ldots)$ and $B = A[z]/(Iz)$. Over a prime not containing I, the fibre of $\operatorname{Spec} B \to \operatorname{Spec} A$ is the single reduced point $z = 0$. Over a prime containing I, the fibre is an affine line, with no isolated point. Hence its quasi-finite locus is the copy of $D(I) = \bigcup_i D(t_i)$ at $z = 0$. This is not quasi-compact: given finitely many indices, the prime generated by those variables avoids an unselected variable and belongs to none of the selected opens. This finite-type separated affine example shows why Theorem D5.2 asserts an open immersion without adding quasi-compactness.

D6. Finite integral subalgebras and finite-stage immersions

We first reproduce the exact finite-submodule argument already proved in *Quasi-coherent sheaves on schemes*, Sections 3 and 4, so its arbitrary qcqs scope is explicit.

Lemma D6.1 (finite submodules on a qcqs scheme). Let S be qcqs, let $\mathcal{F}$ be quasi-coherent, let $Q \subset S$ be a quasi-compact open, and let $\mathcal{G} \subset \mathcal{F}|_Q$ be a finite-type quasi-coherent submodule. It extends to a finite-type quasi-coherent submodule of $\mathcal{F}$. Every $\mathcal{F}$ is the directed union of its finite-type quasi-coherent submodules.

Proof. Pushforward by the quasi-compact open immersion $i : Q \hookrightarrow S$ preserves quasi-coherence. This follows from the finite-cover localization calculation: over an affine chart of S, its intersection with Q is qcqs; finite affine covers of that intersection and its overlaps identify sections on a principal

subopen with localization. The affine module–sheaf correspondence then identifies the pushforward with the sheaf of its section module.

If $S = \operatorname{Spec} R$, the kernel

$$\mathcal{H} = \ker(\mathcal{F} \longrightarrow i_*(\mathcal{F}|_Q/\mathcal{G}))$$

is quasi-coherent, lies in $\mathcal{F}$ and restricts to $\mathcal{G}$. Write $\mathcal{H} = \widetilde{N}$. Cover Q by finitely many principal opens $D(a_j)$. Each N_{a_j} is finite because its associated sheaf there is finite type; this affine implication follows by selecting finitely many principal local generating sets and testing the quotient by their numerators. Choose numerators for finite generating sets of all N_{a_j} and let $N_0 \subset N$ be their finite span. Then $(N_0)_{a_j} = N_{a_j}$, so $\widetilde{N_0}$ is the required extension.

For general S, add finitely many affine opens to Q. At each step the intersection with the already treated quasi-compact open is quasi-compact by quasi-separatedness. Apply the affine construction on the newly added affine, and glue along equality of the two submodules inside $\mathcal{F}$. Finite induction gives the extension.

Finally any germ of $\mathcal{F}$ is represented on an affine neighbourhood. Its cyclic submodule there is finite type and extends by the assertion just proved. Thus these submodules exhaust every stalk. Their family is directed under sums, since the image of a finite direct sum remains quasi-coherent and finite type. Stalkwise exhaustion proves their directed union is $\mathcal{F}$. □

Lemma D6.2 (the finite subalgebras actually needed). An integral quasi-coherent $\mathcal{O}_S$-algebra $\mathcal{A}$ on a qcqs scheme is the directed union of its finite quasi-coherent subalgebras.

Proof. For a finite-type quasi-coherent submodule $\mathcal{N} \subset \mathcal{A}$, form the tensor algebra $\bigoplus_{n\geq 0} \mathcal{N}^{\otimes n}$, with $\mathcal{N}^{\otimes 0} = \mathcal{O}_S$ and multiplication given by concatenating tensor factors. Its quotient by the ideal generated by $u \otimes v - v \otimes u$ is $\operatorname{Sym}(\mathcal{N})$. Multiplication in the commutative algebra $\mathcal{A}$ kills these relations, so the inclusion of $\mathcal{N}$ induces a map $\operatorname{Sym}(\mathcal{N}) \to \mathcal{A}$. Let $\mathcal{A}_{\mathcal{N}}$ be its image. On affine charts, localization commutes with tensor powers, direct sums and their quotient and image maps, by the earlier module calculations. These constructions therefore give quasi-coherent sheaves, and the image is the subalgebra generated by $\mathcal{N}$ and the unit. On any affine chart, finitely many module generators of $\mathcal{N}$ generate this image as an algebra. Each is integral over the base. Their finitely many monic equations bound their powers, so the corresponding bounded monomials span $\mathcal{A}_{\mathcal{N}}$ as a module. It is therefore finite on every affine chart.

By Lemma D6.1 each germ of $\mathcal{A}$ belongs to a finite-type quasi-coherent submodule $\mathcal{N}$, and hence to the finite subalgebra $\mathcal{A}_{\mathcal{N}}$. Two finite subalgebras are contained in the finite subalgebra generated by their sum: their sum is a finite-type quasi-coherent submodule, and the same integral-generator argument applies. This family is therefore directed and exhausts every stalk. Its directed union as quasi-coherent sheaves is $\mathcal{A}$, proving the assertion. □

This proves the entire part of the finite integral approximation theorem consumed here. No approximation by finitely presented **algebras**, and no assumption of finite presentation of X/S, is used.

Lemma D6.3 (finite-stage closed immersions and immersions). Let $Z = \varprojlim_i Z_i$ be a directed limit over a scheme S, with affine transition maps, and let $Y \to Z$ be an S-map.

1. If it is a closed immersion, all Z_i are quasi-compact, and Y is locally finite type over S, then $Y \to Z_i$ is a closed immersion eventually.
2. If it is an immersion, all Z_i are quasi-separated, Y is quasi-compact and locally finite type over S, then $Y \to Z_i$ is an immersion eventually.

Proof. For the first assertion choose one stage and a finite affine cover whose members map into affine opens $\operatorname{Spec} R$ of S. Their inverse images at later stages and the limit are affine, since the

transitions and projections are affine. The inverse image $V \subset Y$ of each such chart is closed in its affine limit, hence affine and quasi-compact. It is locally finite type over $\operatorname{Spec} R$, hence its ring B is a finitely generated R-algebra, by the proved affine finite-type criterion.

Writing the coordinate rings of the chart system as C_i and its limit as $C = \varinjlim_i C_i$, the closed immersion gives a surjection $C \to B$. A finite R-algebra generating list of B has preimages represented in one common C_i. Thus $C_i \to B$ is surjective at that stage and at all later stages. Applying this to finitely many charts at once proves the eventual closed immersion. No finite relation list for B was required.

For the second assertion fix a stage i_0. Quasi-compactness of Y lets us choose a quasi-compact open $Z'_{i_0} \subset Z_{i_0}$ containing its image. Replace later stages by its inverse images. They are now qcqs, and the map still factors through their limit Z'. An immersion is closed in some open of its target. Choose a quasi-compact open $O \subset Z'$ containing Y inside that open, using finitely many affine neighbourhoods of its image. The immersion $Y \to O$ is closed.

The full finite-open-data proof in the earlier *Limits of schemes and Noetherian approximation* lesson, Lemma 2.1, gives a quasi-compact stage open $O_i \subset Z'_i$ whose inverse image is O. Its later inverse images have affine transition maps. The first part applied to $Y \to O$ gives a closed immersion $Y \to O_i$ eventually. Composing with the open immersion $O_i \hookrightarrow Z_i$ proves the assertion. □

The finite-open-data proof just used needs no Noetherian hypothesis: on a finite affine cover of a qcqs stage, a quasi-compact limit open is a finite union of principal opens with finitely many defining elements. Represent these at a common stage. Equal inverse-image opens eventually agree, since their finitely generated ideals have the same radical in the colimit and the finitely many power-membership equations hold at a common stage. This also proves that a finite stage-open covering at the limit eventually covers the stage. These are exactly the local calculations in the earlier complete proof.

D7. The finite completion over a qcqs base

Theorem D7.1 (general finite factorization). If S is qcqs and $f : X \to S$ is quasi-finite and separated, then

$$X \hookrightarrow \overline{X} \longrightarrow S$$

factors f as a quasi-compact open immersion followed by a finite morphism. The scheme S is arbitrary qcqs; X/S need not be finitely presented.

Proof. Theorem D5.2 identifies all of X with an open subscheme of $Z = \operatorname{Spec}_S \mathcal{A}$, since every point is quasi-finite. The algebra $\mathcal{A}$ is integral. By Lemma D6.2 write

$$\mathcal{A} = \varinjlim_i \mathcal{A}_i, \qquad Z_i = \operatorname{Spec}_S \mathcal{A}_i,$$

where each $\mathcal{A}_i \subset \mathcal{A}$ is a finite quasi-coherent subalgebra. Each Z_i is finite over S, hence qcqs. Their transition morphisms are affine. On affine base charts, the elementary spectrum-of-a-ring-colimit construction gives $Z = \varprojlim_i Z_i$; the chart identifications glue by relative Spec.

The source X is quasi-compact because f is quasi-compact and S is quasi-compact, and it is locally finite type by quasi-finiteness. Lemma D6.3 makes the induced map $j_i : X \to Z_i$ an immersion at some stage. We now show it is open, a step not supplied by the limit lemma alone.

By definition of immersion there is an open $O \subset Z_i$ in which X is closed. Work over an affine base chart $\operatorname{Spec} R \subset S$, and put $C = \mathcal{A}_i(\operatorname{Spec} R)$ and $B = \Gamma(X_R, \mathcal{O})$. The map $C \to B$ is injective, since $\mathcal{A}_i$ is an actual subalgebra of $\mathcal{A} \subset f_*\mathcal{O}_X$. Cover $O \cap \operatorname{Spec} C$ by principal opens $D_C(c)$ contained in it. The qcqs localization formula gives

$$\Gamma((X_R)_c, \mathcal{O}) = B_c,$$

and exact localization preserves the injection $C_c \hookrightarrow B_c$. But the closed immersion into O makes this same ring map a surjection, with kernel its defining ideal on $D_C(c)$. Hence that ideal is zero and the map is an isomorphism. These principal charts cover O over all base affines. Therefore the closed immersion $X \to O$ is an isomorphism, and j_i is an open immersion. Take $\overline{X} = Z_i$.

Finally this open immersion is quasi-compact. The scheme $\overline{X}$ is quasi-separated, and its open image is homeomorphic to the quasi-compact scheme X. Its intersection with every affine open of $\overline{X}$ is therefore quasi-compact. This is exactly quasi-compactness of the open immersion. □

Corollary D7.2 (the relative quasi-affine conclusion). Every quasi-finite separated morphism of schemes is quasi-affine, without a quasi-compactness assumption on the base.

Proof. The relative integral closure $Z \to S$ is affine, and Theorem D5.2 embeds X openly in Z. This open immersion is quasi-compact locally on S: over an affine base chart, X is quasi-compact by quasi-finiteness, and the target Z is affine, hence quasi-separated. Thus the open immersion is quasi-compact. It is a quasi-compact open immersion into an affine S-scheme, which is the relative quasi-affine assertion. If S is qcqs, Theorem D7.1 strengthens the affine target to a finite one. □

Corollary D7.3 (proper quasi-finite morphisms). A proper quasi-finite scheme morphism is finite over an arbitrary base.

Proof. The claim is local on the target, so take an affine target. Theorem D7.1 gives an open immersion $j : X \hookrightarrow Z$ with Z finite over the target S. The map j is proper: its graph into $X \times_S Z$ is closed because Z/S is separated, and the projection to Z is proper by base change of X/S. Thus its open image is also closed, by the closed-image part of properness. An open immersion with closed image identifies X with a clopen subscheme of Z, hence with a direct factor of its finite coordinate algebra. That factor is finite over the base. These affine target conclusions glue to finiteness. □

Corollary D7.4 (normal birational target). A quasi-finite separated birational morphism of integral schemes with normal target is an open immersion.

Proof. Restrict to an affine normal target $S = \operatorname{Spec} R$, and put $K = \operatorname{Frac} R$. Birationality identifies the function field of X with K. Every regular section on X injects into K: on each nonempty affine open its coordinate domain injects into the common function field, and restrictions agree there. The relative integral algebra is therefore an R-subalgebra of K. Its elements integral over R are precisely R, by normality; the pullback inclusion supplies all of R. The relative integral closure is thus S itself. Every point of X is quasi-finite, so Theorem D5.2 identifies X with an open of S. This argument applies on every affine target open, and the canonical open immersions agree on overlaps, proving the assertion. □

Free sources and the proof order

The freely accessible writing material is the Stacks Project's maps between étale algebras, Tag 00U7, monic étale neighbourhoods, Tag 00UE, finite elementary étale pieces, Tags 02LK–02LN, integral-closure derivative identity, Tag 03GD, étale integral-closure base change, Tag 03GE, polynomial coefficient argument, Tag 03GG, scheme base change, Tag 03GV, relative Zariski Main, Tag 03GW, finite integral subalgebras, Tag 0817, and finite-stage immersions, Tag 081B. Full local source texts and their GNU FDL licence accompany the draft. These references identify sources for writing and comparison; the proofs consumed above are in this lesson or the exact earlier programme texts.

The logical order of the proofs is: the elementary ring, affine-sheaf, field-algebra and integral-extension proofs; Section B1 of this lesson; Theorem C4.3; Section D2 here; the earlier complete henselian splitting proof, if used; Sections D3–D4 here; the earlier affine faithfully flat descent block;

Sections D5–D7 here; and only then separated locally quasi-finite scheme effectivity, algebraic-space recognition and normalized space factorization. In particular the latter three results never justify a step in this lesson.

E. Locally quasi-finite effectivity and space recognition

E4. Two finite scheme facts and the étale finite piece

Lemma E4.1. A finite monomorphism of schemes is a closed immersion, without a Noetherian or finite-presentation assumption.

Proof. Over an affine target write $R \to B$, with B a finite R-module. The monomorphism says that multiplication $B \otimes_R B \to B$ is an isomorphism. For every residue field k of R, the finite-dimensional k-algebra $E = B \otimes_R k$ therefore satisfies $E \otimes_k E = E$. Its vector-space dimension d satisfies $d^2 = d$, so it is either zero or one. In the latter case its unit identifies it with k. In either case $k \to E$ is surjective. The finite cokernel of $R \to B$ consequently has zero fibre at every maximal ideal. Nakayama kills its localization at each maximal ideal, and a module with all such localizations zero is zero. Hence $R \to B$ is surjective, proving the closed immersion. □

Lemma E4.2. A finite étale scheme morphism is finite locally free.

Proof. On affine charts let $R \to B$ be finite, flat and finitely presented as an algebra. Pick algebra generators b_i. Each is integral by the finite-module determinant argument. Choose monic equations $P_i(b_i) = 0$. The algebra $C = R[T_i]/(P_i(T_i))$ is finite free, with basis the bounded monomials. A finite presentation of B as an algebra says that the kernel of $C \to B$ is a finitely generated ideal: lift its finite defining relations to the polynomial algebra and add the finitely many P_i. The products of its ideal generators with the bounded monomial basis generate it as an R-module. Thus B is finitely presented as an R-module. Flatness and Lemma A1.3 make it finite locally free. □

Lemma E4.3 (finite piece without flatness). Let T and U be affine schemes and let $U \to T$ be quasi-finite. For $u \in U$, with image $p \in T$, there are an affine elementary étale neighbourhood $(T', p') \to (T, p)$ and a clopen subscheme $U' \subset U_{T'}$, finite over T', having a point above u. One may arrange that $U'_{p'} = U_p$ under $\kappa(p') = \kappa(p)$.

Proof. The fibre U_p is a finite scheme over $\kappa(p)$: its coordinate algebra is finite type of dimension zero, hence Artinian, and its residue fields are finite, by the earlier field-algebra and quasi-finite fibre proofs. List all its finitely many points as $u_1, \ldots, u_n$. Proposition D3.3 applies to the separated affine morphism $U \to T$ and all these isolated fibre points. It supplies a single elementary étale neighbourhood with disjoint clopen finite pieces $V_i \subset U_{T'}$, each containing the unique point above u_i, and complementary clopen piece W whose selected fibre has no point above any u_i. Thus $W_{p'}$ is empty. Take $U' = \coprod_i V_i$. It is finite, clopen and has the whole selected fibre. Shrink the elementary neighbourhood to an affine principal neighbourhood of its marked point; all these properties persist. Its residue field equals $\kappa(p)$ by the elementary-neighbourhood construction, giving the claimed equality of fibres and a point above the original u. No flatness or finite presentation of the finite completion is used. □

E5. Effective separated locally quasi-finite scheme descent

Theorems D5.2 and D7.1, and Corollary D7.2, have already proved the **general scheme Zariski Main theorem**: a quasi-finite separated morphism to a qcqs scheme is an open subscheme of a finite scheme over that base, and consequently is quasi-affine. Their proofs appear in block D above and use no theorem on algebraic spaces or separated locally quasi-finite scheme effectivity.

Theorem E5.1. Let $\{X_i \to S\}$ be an fppf covering. A scheme datum $(V_i/X_i, \phi_{ij})$ whose structure maps are separated and locally quasi-finite is effective as a scheme. The descended morphism

is separated and locally quasi-finite. No quasi-compactness of the local objects or the result is assumed.

Proof for a single affine cover. First take $S = \operatorname{Spec} R$ and $X = \operatorname{Spec} A$, with $X \to S$ faithfully flat and finitely presented. Let $V \to X$ be separated and locally quasi-finite, with datum $\phi : V \times_S X \to X \times_S V$. For each affine open $W^1 \subset V$ put

$$W = \mathrm{pr}_V(\phi(W^1 \times_S X)) \subset V.$$

We first prove the openness assertion for a flat locally finitely presented scheme morphism used here. On affine source and target charts it is a finitely presented algebra $R \to B$. Write $B = R[T_1, \ldots, T_m]/(F_1, \ldots, F_n)$. By the affine approximation of AG-MO-03, Theorem 5.1, choose a finitely generated $\mathbb{Z}$-subalgebra $R_0 \subset R$ containing all coefficients of the finitely many F_i. Put $B_0 = R_0[T_1, \ldots, T_m]/(F_1, \ldots, F_n)$. The presentation gives $R \otimes_{R_0} B_0 = B$, and R_0 is Noetherian. Lemma 4.C of AG-RG-S02 proves that its image is constructible. Images of scheme morphisms commute with arbitrary base change: if two points lie over the same base point, the tensor product of their residue fields over that base residue field is nonzero (a nonzero vector space stays nonzero under field extension), so its spectrum supplies a common point; this proves both inclusions of the image equality. Thus the image over R is the inverse image of that constructible set, and is constructible, since finite unions of $V(J) \cap D(a)$ pull back to the same form. Flat local going down, proved in Lemma A1.4, makes the image stable under generization. Its complement is constructibly compact and stable under specialization, and the last paragraph of Lemma A1.4 proves that this complement is Zariski closed. The image is therefore open. Apply the argument to affine charts of any source open and take their union. Flatness and local finite presentation persist after base change, so this proves universal openness. In the present situation the projection defining W is a base change of the flat finitely presented cover, hence is open; its image W is open. It is quasi-compact, being the image of the affine scheme $W^1 \times_S X$. Diagonal normalization of the datum gives $W^1 \subset W$.

The cocycle makes W invariant. One can check this on points after common residue-field extensions: membership in W means being joined by the descent relation to a point of W^1; a further relation arrow composes with that arrow. Conversely use the inverse arrow. Equivalently, on every test scheme local lifts of such arrows can be refined and composed, giving the equality of opens

$$\phi(W \times_S X) = X \times_S W.$$

Thus W carries the restricted datum. Its map to the affine scheme X is quasi-compact, separated and locally quasi-finite, hence quasi-finite and separated. Scheme Zariski Main makes W/X quasi-affine. Theorem A3.1 descends W effectively to a quasi-affine scheme U_W/S.

These invariant W cover V, because the original affine W^1 do. For two of them, $W \cap \widetilde{W}$ is an invariant open of each. Lemma A1.6 descends this open inside U_W and $U_{\widetilde{W}}$; Lemma A1.5 descends the comparison isomorphism and its inverse. The comparisons satisfy the triple equation after the cover and hence before it. Ordinary open gluing produces a scheme whose pullback is V, with its original datum. This step does not require the intersections to be affine, nor invoke Theorem E5.1 recursively.

General fppf families and arbitrary bases. On an affine open of S, affine pieces of covering members have open images, because their maps are flat and locally finitely presented. Finitely many such images cover this quasi-compact affine open. Their finite disjoint union is an affine faithfully flat finitely presented cover. Pullback preserves separatedness and local quasi-finiteness, and the disjoint union retains them. The preceding construction applies. On every original member the comparison isomorphisms descend after this refinement by Lemma A1.5; their inverses descend too. Repeat on affine base opens, compare on affine opens of intersections using the original data, and glue. This constructs the effective scheme datum over any S.

For completeness its stated properties also descend. Closedness of the diagonal follows from Corollary A3.2 after the covering. Local finite type descends by the following affine argument: if $A \otimes_R B$ is generated by finitely many elements as an A-algebra, collect their finitely many B components; the algebra they generate in B has quotient module killed by faithfully flat tensoring, so is all of B. Apply this to affine charts of the existing descended scheme. On a residue-field fibre over k, choose a field extension L/k induced by a point of the covering. The whole pulled-back fibre is locally quasi-finite over L. Let $\operatorname{Spec} B$ be an affine open in the original fibre; B is a finite-type k-algebra. If B had a strict prime chain $\mathfrak{q} \subsetneq \mathfrak{q}'$, faithful flatness of $B \to B \otimes_k L$ would give a prime above $\mathfrak{q}'$, and the flat local going-down argument of Lemma A1.4 would give a smaller prime above $\mathfrak{q}$. This would be a strict chain in the pulled-back fibre. Such a chain is impossible there: local quasi-finiteness makes every point isolated in that fibre, and a distinct generalization belongs to every neighbourhood of a point. Thus B has dimension zero. Hilbert basis makes B Noetherian, and Lemma P0.9 makes its spectrum a finite discrete set with finite residue fields over k. Each original fibre point is consequently isolated with finite residue extension. The finite-type fibre criterion stated before Lemma C3.1 proves local quasi-finiteness of the descended morphism. □

E6. The finite affine quotient used in recognition

Theorem E6.1. Let $P \rightrightarrows U = \operatorname{Spec} A$ be a scheme equivalence relation, with both projections finite locally free. Write $P = \operatorname{Spec} B$, let $s, t : A \to B$ be its endpoint maps, and set

$$C = \{a \in A : s(a) = t(a)\}.$$

Then A is faithfully flat finite locally free over C, $B = A \otimes_C A$ through the endpoint maps, and $\operatorname{Spec} C$ represents the fppf quotient U/P on all test schemes.

Proof. The empty object has the empty quotient, so suppose $U \neq \varnothing$. The identity section makes each source fibre nonempty. The rank of B over A through s is therefore positive, locally constant, and invariant under the relation. Invariance follows because composition with an arrow identifies the two source fibres, after base change to the scheme of arrows. The rank strata are finitely many clopen subsets of the affine quasi-compact U. Their idempotents have equal pullbacks under s, t (equality of the clopen subsets is equality of their idempotents), so lie in C. Decompose by them and assume the rank is $r > 0$.

For $a \in A$, multiplication by $t(a)$ on the finite locally free $s(A)$-module B has a characteristic polynomial. Here is the determinant calculation needed on its free charts, over any commutative ring. Expanding $\det(M_1 M_2)$ by the signed permutation formula groups its terms by indices $j_1, \ldots, j_r$. Each group is $\prod_i (M_1)_{ij_i}$ times the determinant of the matrix whose ith row is row j_i of M_2. Repeated indices give repeated rows, so their determinants vanish by the signed-pair cancellation proved in S01, Lemma 1.2. Otherwise the indices form a permutation τ, and reindexing the determinant sum makes that factor $\operatorname{sgn}(\tau) \det M_2$. Summing over τ gives $\det(M_1 M_2) = \det M_1 \det M_2$. Consequently, for an invertible matrix Q, the identity $\det Q \det Q^{-1} = 1$ gives $\det(TI - QMQ^{-1}) = \det(TI - M)$ over the polynomial ring. The same finite permutation formula commutes with every base change. Thus the characteristic polynomials on different free charts agree on their overlaps and define the stated polynomial globally. All its coefficients lie in C. Indeed composition gives an isomorphism, over the entire arrow scheme, between the two pulled-back source-fibre bundles, preserving target evaluations. It conjugates their multiplication operators. Characteristic polynomials commute with base change and conjugation, so the two coefficient pullbacks agree as functions, including nilpotents. Cayley–Hamilton follows on local free charts from the adjugate identity and hence globally. Apply the identity section to its equation: a satisfies a monic polynomial over C. Thus $C \subset A$ is integral and has surjective spectrum by lying over. The same conjugation proves that

$$N(a) = \det_s(\text{multiplication by } t(a) \text{ on } B) \in C.$$

The endpoint map $P \to U \times_{\mathbf{Z}} U$ is finite: factor it as the closed graph of the second endpoint, followed by the base change of the finite first projection. The graph is closed because an affine scheme is separated over $\mathbf{Z}$. This endpoint map is also a monomorphism: it factors through the equivalence-relation monomorphism into $U \times_S U$ and then the monomorphism to $U \times_{\mathbf{Z}} U$. Lemma E4.1 makes it a closed immersion. Consequently the endpoint map gives a surjection

$$\theta : A \otimes_C A \longrightarrow B, \qquad a \otimes a' \longmapsto s(a)t(a').$$

This works for a base S which is neither separated nor affine.

Invariant rings commute with flat base changes of C: tensor the kernel sequence $0 \to C \to A \xrightarrow{s-t} B$. Fix a prime $\mathfrak{p}$ of C, and extend $C_{\mathfrak{p}}$ faithfully flat to the local ring

$$D = (C_{\mathfrak{p}}[z])_{\mathfrak{p}C_{\mathfrak{p}}[z]}.$$

Polynomial extension and localization are flat; the map is local, hence faithful. Its residue field is the infinite field $\kappa(\mathfrak{p})(z)$. It is enough to prove the claimed algebra statements after these extensions at every prime. By the kernel observation the new base is still the invariant ring. We may therefore assume C is local with maximal ideal $\mathfrak{m}$ and infinite residue field k.

The ring A is semilocal. Integrality makes every maximal ideal lie over $\mathfrak{m}$. Each topological orbit of maximal ideals is finite: its targets occur in the finite source fibre over one of its objects. Inversion and composition give equivalence and transitivity; common residue-field extensions allow composition when only the underlying endpoints initially agree. Suppose there were two such orbits. By the Chinese remainder theorem choose $f \in A$ having residue zero at every ideal of the first orbit and residue one at every ideal of the second. On the finite Artinian source fibre at an object of the first orbit, $t(f)$ has zero image in every prime quotient and is therefore nilpotent, by Lemma P0.1. On the corresponding fibre of the second orbit, $t(f) - 1$ is nilpotent for the same reason. For a nilpotent endomorphism H of a finite-dimensional vector space, choose a basis respecting the successive kernels $0 \subset \ker H \subset \ker H^2 \subset \cdots$. Since H lowers this filtration, its matrix is strictly triangular. The determinant permutation formula gives $\det H = 0$ in positive dimension and $\det(1 + H) = 1$. Apply this to multiplication by $t(f)$ in the first fibre and by $t(f) - 1$ in the second. Their common rank r is positive, so $N(f)$ has residue zero in the first case and one in the second. But $N(f) \in C$ has a single residue in k, which cannot become zero in one field extension and one in another. Hence there is only one orbit, containing all maximal ideals, so there are finitely many.

The surjection θ says that B is generated over $s(A)$ by elements of $t(A)$. Choose a finite list $t(f_1), \ldots, t(f_m)$ generating the finite module B. At each of the finitely many maximal ideals of A, there is an r-tuple of linear combinations of this list giving a basis of the fibre. Its determinant is a nonzero polynomial in the combination coefficients over that residue field. It cannot vanish on all tuples from the infinite common subfield k: expand its finitely many coefficients in a basis of their span over k and retain a nonzero coordinate polynomial over k. The product of finitely many such nonzero polynomials is nonzero. A nonzero polynomial over an infinite field has a nonvanishing field-valued tuple, by induction on the number of variables, choosing nonzero coefficient values first and then avoiding the finitely many roots in the last variable. Avoid the chosen coordinate polynomials simultaneously, and lift the resulting coefficients to C. We get $x_1, \ldots, x_r \in A$ with $t(x_i)$ a basis at all maximal ideals. Nakayama makes $A^r \to B$ surjective; localizing shows it is an isomorphism between free modules of the same rank. Hence

$$B = \bigoplus_i s(A)t(x_i).$$

Write uniquely $t(a) = \sum_i s(a_i)t(x_i)$. In the composable-pair algebra $E = B \otimes_{s,A,t} B$, where an arrow in the left factor follows one in the right factor, composition satisfies

$$c^*t(a) = t(a) \otimes 1, \qquad c^*s(a) = 1 \otimes s(a).$$

Apply composition to the expansion, and also include the expansion in the left factor. The elements $t(x_i) \otimes 1$ are a basis over the right factor. Its tensor relation identifies $s(a_i) \otimes 1$ with $1 \otimes t(a_i)$. Comparing the two expansions gives $s(a_i) = t(a_i)$ for every i, so $a_i \in C$. Applying the identity section gives $a = \sum_i a_i x_i$. Applying t and the basis just constructed gives linear independence over C. Thus $A = \bigoplus_i Cx_i$ and θ is an isomorphism, since it carries the A-basis $1 \otimes x_i$ to $t(x_i)$.

Return to the original C. At each prime, faithfully flat detection over the chosen local extension shows that the kernel of θ vanishes and that A is flat over C. To verify the latter explicitly, tensor any injection with A, localize its kernel at the prime and extend to D; the computed free D-algebra makes this kernel zero, so faithfulness kills it. Vanishing at every prime kills the original kernel. Lying over gives surjectivity, hence faithful flatness. Now $A \otimes_C A = B$ is finite locally free over A. Lemma A1.3 descends this property of the module A through the faithfully flat $C \to A$. Thus A/C is finite locally free, and its relation is exactly P.

Finally, $q : \operatorname{Spec} A \to \operatorname{Spec} C$ is an fppf covering. Here is its local finite-presentation proof. On a free module chart over C_h, choose a basis $x_1, \dots, x_n$ of A_h and write $1 = \sum_i u_i x_i$ and $x_i x_j = \sum_k m_{ijk} x_k$, with coefficients in C_h. The algebra map $C_h[X_1, \dots, X_n] \to A_h$ has the finite relations $1 - \sum_i u_i X_i$ and $X_i X_j - \sum_k m_{ijk} X_k$. These relations reduce every polynomial to a linear combination of the X_i. If that combination maps to zero in A_h, independence of the chosen module basis makes all its coefficients zero. Thus the quotient by these relations is exactly A_h. The free charts prove local finite presentation, and the flatness and surjectivity already established make q an fppf covering. On every scheme test of its target, its base change is again a finite locally free surjective cover, and its double overlap is the pullback of P. Sections therefore locally lift to U, and two such sections have equal image exactly when they are related by P. This local-lifting and equality description, followed by the sheaf condition, identifies the represented target with the fppf sheaf quotient. This is an identity on all test schemes, not just on geometric points. □

E7. Descending an affine neighbourhood of a space

Lemma E7.0 (the étale diagonal). The diagonal of an étale scheme morphism is an open immersion.

Proof. Around a diagonal point use the earlier proved local standard form $E = (R[T]/(f))_g$, with f monic and f' invertible in E. In the self-overlap, write the two coordinate classes t_0, t_1. The divided difference h defined by $f(t_0) - f(t_1) = (t_0 - t_1)h(t_0, t_1)$ restricts on the diagonal to the unit $f'(t_0)$. On $D(h)$ the two equations $f(t_0) = f(t_1) = 0$ force $t_0 = t_1$. The self-overlap localized at h is therefore exactly the diagonal algebra E, giving an open neighbourhood which is the diagonal. These local charts cover the diagonal and prove it is an open immersion. □

We use algebraic spaces as fppf sheaves with scheme-representable diagonal and a representable surjective étale scheme atlas. For an open in an atlas, its image defines an open subspace: on every scheme test the pulled-back atlas is a scheme étale cover, and the image of its selected open is an open subset of the test scheme. These images commute with base change, since two points above the same point can be joined after a common residue-field extension. Thus their open subscheme structures agree on overlaps and define the open subfunctor. This description also proves the following fact without a bootstrap theorem.

Lemma E7.1 (an open atlas quotient). If $U \to X$ is a surjective étale scheme atlas of an algebraic space, and $U_0 \subset U$ is open, then $U_0/(U_0 \times_X U_0)$ is the open image of U_0 in X as an fppf sheaf.

Proof. On every scheme test of the image, its section locally lifts through U_0 after an étale, hence fppf, refinement. Two local lifts have the same image exactly when their pair factors through

$U_0 \times_X U_0$. Local lifts and this equality description identify the sheaf quotient with the open image. All arrows with given endpoints are unique because this is an equivalence relation. Consequently the equality arrows, as well as the object sections, descend on refinements. □

Lemma E7.2. Let $X \to T$ be a separated locally quasi-finite morphism of algebraic spaces, with T affine. Let $T' \to T$ be an étale map with T' affine, and let $V' \subset X \times_T T'$ be an affine open subspace. Then its image $W \subset X$ is an open subspace represented by a scheme.

Proof. Its image is open because the map to X is étale. It is quasi-compact as the image of the affine V'. There is a uniform finite bound n on the cardinalities of the geometric fibres of $T' \to T$: this is an affine quasi-finite morphism, so Theorem C5.1 embeds T' in a finite scheme over T. A finite list of module generators of that completion bounds every fibre algebra's dimension and hence its number of geometric points. We induct on such a bound; no boundedness of the fibres of $X \to T$ is required.

For $n = 0$ the image is empty. For $n = 1$, the étale map $T' \to T$ is an open immersion. To recall the precise reason, its diagonal is an open immersion by Lemma E7.0; the fibre bound makes its image all of $T' \times_T T'$, because there is at most one geometric point in each fibre. Thus the diagonal is an isomorphism and the map is a monomorphism. An étale monomorphism is an open immersion: its open image O is a scheme open, and the map to O is a surjective étale cover. Its double overlap is its diagonal, hence is the source itself. The identity map of the source is consequently compatible on that overlap. Lemma A1.5 descends it to an inverse O-map; both inverse equations hold after the cover and hence before it by the same uniqueness. Thus the map is an isomorphism onto O. In this case $X \times_T T'$ is an open subspace of X, and $W = V'$ is a scheme.

For $n > 1$, the diagonal of $T' \to T$ is both open, by étaleness, and closed, since a map of affines is separated. Thus

$$T' \times_T T' = \Delta(T') \amalg T^*$$

is a decomposition into affine clopen schemes. Each projection $T^* \to T'$ is étale and has geometric fibre bound $n - 1$, because the diagonal removes exactly one point from the corresponding full base-changed fibre.

Write $X' = X \times_T T'$ and $X^* = X \times_T T^*$. In $X' \times_X X'$ use projections p_0, p_1, with its part over T^* identified with X^*. The open

$$V^* = p_0^{-1}(V') \cap X^* = T^* \times_{\mathrm{pr}_0, T'} V'$$

is affine, since both displayed factors and their base T' are affine. It is an open in $X' \times_{T', \mathrm{pr}_1} T^*$. Apply the induction hypothesis with base T', cover $\mathrm{pr}_1 : T^* \to T'$, ambient space X', and selected affine open V^*. Its image $p_1(V^*)$ is a scheme open in X'. The diagonal part contributes V', so the saturation is

$$p_1(p_0^{-1}(V')) = V' \cup p_1(V^*).$$

It is a scheme by gluing these two scheme opens over their intersection, which is an open subscheme in each. It is also exactly $X' \times_X W = T' \times_T W$ as an open subspace. Indeed, over every test scheme a point in this pullback locally has one lift in V' and its chosen lift in X'; these two lifts give the required overlap point. Conversely every overlap point with its first endpoint in V' has image in W. This proves equality of the open subfunctors and includes infinitesimal test schemes.

Let $O \subset T$ be the open image of T'. The morphism $W \to T$ factors through O, and $T' \to O$ is a surjective étale, hence fppf, cover. Its pullback of W is the scheme just constructed, separated and locally quasi-finite over T' by base change. Its canonical scheme datum is effective by Theorem E5.1. Let W_0/O be the descended scheme. For an arbitrary scheme $Z \to O$, pull back the cover

$T' \to O$. The sections of W_0 and W are respectively the compatible sections of exactly the same pulled-back scheme with the same equality datum. Both are fppf sheaves, so their functors on Z agree naturally. Thus $W = W_0$ as a sheaf and is a scheme. This completes the induction. □

E8. Recognition and representability on every test scheme

Theorem E8.1 (scheme recognition). If $X \to T$ is a separated locally quasi-finite morphism of algebraic spaces and T is a scheme, then X is a scheme. The source need not be quasi-compact.

Proof. Restrict to affine target opens; the inverse images cover X and will glue as schemes once recognized. Over an affine T, choose a scheme atlas of X and cover it by affine opens. Each has an open image by the preceding construction. It suffices to recognize each such image, because they cover X and their scheme opens then glue. We may therefore suppose an affine scheme U maps étale surjectively to X.

The composite $U \to T$ is locally quasi-finite, by composition with the étale atlas and the definition on scheme coordinates. Since source and target are affine it is quasi-compact and separated, hence quasi-finite. The scheme relation $R = U \times_X U$ is a closed subscheme of $U \times_T U$, because X/T is separated. Thus R is affine, and its two projections are étale.

Fix $u \in U$ with image $p \in T$. Lemma E4.3 gives an affine elementary étale neighbourhood (T', p') and a clopen finite $U' \subset U_{T'}$ containing a point above u. Let

$$R' = U' \times_{X_{T'}} U'.$$

Its endpoint map is closed in $U' \times_{T'} U'$ by separatedness. Since U'/T' is finite, each projection $R' \to U'$ is finite. It is also étale, being the restriction to an open of a base-changed projection of R. Lemma E4.2 makes these projections finite locally free. In particular U' and R' are affine, and Theorem E6.1 constructs an affine quotient $V' = U'/R'$.

Lemma E7.1 identifies this quotient with the open image of U' in $X_{T'}$. It contains the image of the selected point above u. Lemma E7.2 makes its image in X a scheme open containing the image of u. Repeating for every point of U gives scheme opens covering X. Their intersections are opens in each scheme; the identifications and their cocycle are the original identifications as subfunctors of X. Ordinary scheme gluing therefore represents X. The argument applies over every affine target open and gives the asserted result over any scheme T. □

Corollary E8.2. A separated locally quasi-finite morphism $f : X \to Y$ of algebraic spaces is representable by schemes.

Proof. For every scheme $T \to Y$, the fibre product X_T is an algebraic space separated and locally quasi-finite over T, by base change. Theorem E8.1 recognizes it as a scheme. These are all scheme tests, including non-Noetherian and non-quasi-compact ones, which is the definition of representability by schemes. □

E9. The normalization factorization consumed by AG-RG-03

The scheme input needed in this section is stronger than the quasi-affineness consequence used in Section E5. It is the **representable finite-type separated normalization theorem**: for a finite-type separated scheme morphism, the quasi-finite locus maps isomorphically onto an open of the relative integral closure; formation of that integral closure commutes with étale base change. Theorem D5.2 and Corollary D4.3 above prove these two scheme inputs. The present section explains every additional space descent step, so no separate algebraic-space normalization theorem is being invoked without proof.

Theorem E9.1 (normalization factorization). Let $f : X \to Y$ be quasi-finite and separated between algebraic spaces. Let Y' be the relative integral closure of Y in X. Then $X \to Y'$ is a

quasi-compact open immersion and $Y' \to Y$ is integral. In particular f is quasi-affine. If Y is a scheme, then X is a scheme.

Proof. Corollary E8.2 makes f representable by schemes. Choose an étale scheme atlas $U \to Y$, and cover U by affine opens U_i. Each $X_i = X \times_Y U_i$ is a scheme, and $X_i \to U_i$ is quasi-finite separated. On these schemes form the relative integral closure U_i' and the scheme Zariski Main factorization. Over $U_i \times_Y U_j$ the two constructions agree: this overlap is a scheme étale over both atlas members, and the scheme integral-closure base-change theorem gives the same algebra. Its functoriality gives the triple cocycle. Consequently the integral, hence affine, U_i'/U_i have affine algebra descent data.

Here is the space construction explicitly. For every scheme test $y : T \to Y$, the atlas pullbacks $T \times_Y U_i$ form an étale scheme covering of T. Apply effective affine descent from Lemma A1.2 and its finite affine refinement argument to these pulled-back integral algebras. The result is an affine T-scheme T_y'. It is independent of the refinement and compatible with every further scheme test, by uniqueness of algebra descent. Integrality descends as well. On a faithfully flat affine refinement $R \to A$, if $b \in B$ becomes integral in $A \otimes_R B$, the injection

$$A \otimes_R R[b] \hookrightarrow A \otimes_R B$$

identifies its image with $A[b]$, a finite A-module. Lemma A1.3 descends finite generation of $R[b]$, and the finite-module determinant argument makes b integral over R. Thus every such T_y'/T is integral.

Define $Y'(T)$ to consist of pairs of a section $y \in Y(T)$ and a section of the affine scheme T_y'/T. Compatibility with base change defines pullbacks. This is an fppf sheaf: the y sections glue because Y is a sheaf, and the sections of the resulting fixed affine pullback glue by Lemma A1.5. Its pullback over each fixed $y : T \to Y$ is represented by T_y', so $Y' \to Y$ is affine and integral. The U_i' give a surjective étale scheme atlas: on a test pair (y, s) the pullback of its ith member is identified with the scheme $T \times_Y U_i$, using the base-change identification of the integral algebra and the section s. These schemes form an étale cover of T.

Its diagonal is scheme-representable. On a pair of scheme tests, first impose equality of their underlying Y sections, obtaining a scheme by the diagonal of Y. Over that scheme the two chosen sections lie in one affine integral pullback, and their equality is a closed subscheme, because an affine scheme is separated over its base. Thus the tested equality functor is a scheme. This proves the algebraic-space conditions, and the tested affine construction proves integrality. By construction this space is precisely the relative integral closure.

The canonical maps $X_i \to U_i'$ agree on overlaps by the same scheme construction, and glue as sheaf maps to $g : X \to Y'$. Each is an open immersion. To justify descent on all tests, first note that g is representable by schemes. For a scheme $Z \to Y'$, its pullback of X over Y is a scheme by Corollary E8.2. The desired $X \times_{Y'} Z$ is the inverse image of the diagonal of Y'/Y in that scheme; that diagonal is a closed immersion because Y'/Y is affine. Hence it too is a scheme. After the étale scheme cover of Z induced by the U_i', this morphism to Z is an open immersion. Its local open images agree on overlaps; Lemma A1.6 descends them to an open subscheme of Z, and their comparison isomorphisms descend by Lemma A1.5. Thus the tested morphism is an open immersion. Since this holds for every Z, g is an open immersion of spaces.

It is quasi-compact. Its graph $X \to X \times_Y Y'$ is a closed immersion, since Y'/Y is separated. The projection $X \times_Y Y' \to Y'$ is a base change of the quasi-compact f. Their composite is g, so it is quasi-compact. For every affine scheme $T \to Y$, the integral $T \times_Y Y'$ is affine, and $X \times_Y T$ is a quasi-compact open subscheme of it. This is quasi-affineness on all affine scheme tests, equivalently the quasi-affine morphism property. The final scheme assertion also follows directly from Theorem E8.1. □

For the monomorphism from the reductive quotient to its Grassmannian, Theorem E8.1 already

supplies the essential assertion that the quotient is a scheme. Quasi-affineness needs scheme Zariski Main, and the claimed specific integral-normalization factorization needs its full normalization and étale-base-change version. These three conclusions must not be collapsed into a source citation.

Freely accessible sources and exact proof use

The complete proofs appear above or in the specified earlier programme lessons. Comparison material is freely readable: the Stacks Project's conductor argument, Tags 00PT–00Q9, polynomial coefficients, Tag 03GG, étale local structure, Tags 00U7 and 00UE, finite étale-local pieces, Tag 02LN, relative integral closure and Zariski Main, Tags 03GE and 03GW, finite integral subalgebras, Tag 0817, finite-stage immersions, Tag 081B, quasi-affine descent, Tag 0247, locally quasi-finite effectivity, Tag 02W8, finite affine quotients, Tag 03BM, scheme recognition, Tag 03XX, and normalized factorization for spaces, Tag 0ABS. The editable comparison sources and copying notices are retained with the reconstruction records.

History

Source: *The Stacks Project*, by the Stacks Project authors, published openly under the GNU FDL 1.2 or later. Exact freely readable source tags are identified in this lesson.

Edition: *Affine descent, Zariski Main and recognition of spaces*, 5 October 2026, written by GPT-6.1 Sol (OpenAI), Codex, Ultra setting, for Open Mathematics Courses: faithfully flat descent, polynomial coefficients, conductor and affine completion, general scheme Zariski Main, finite quotients and space recognition, with the proofs ordered before their uses and the finite-piece and open-immersion arguments expanded. Self-checked by the writing AI. The text is dedicated to the public domain under CC0 1.0.

Artin approximation for polynomial equations and its desingularization proof

Written by GPT-6.1 Sol (OpenAI), Codex, Ultra. Self-checked by the writing AI. CC0 1.0.

A formal solution satisfies polynomial equations in a completion. Approximation asks for an exact solution on an étale neighbourhood with the same prescribed finite-order values. To keep both requirements, we encode the equations and the congruences in a single finite presentation.

The proof proceeds from smooth factorization to a pointed étale neighbourhood. Sections 1–3 give that passage and the finite coefficient system for an embedded torus. Section 4 proves the desingularization theorem that supplies the smooth factorization; Section 5 contains the algebra used in its proof.

1. The exact approximation statement

A map of Noetherian rings is **regular** if it is flat and every fibre is geometrically regular. An algebra over a field is geometrically regular if after every finite field extension its local rings are regular. A **G-ring** is a Noetherian ring whose maps from localizations to their completions are regular. The relevant implication for a local G-ring is therefore immediate from its definition: $R \to \widehat{R}$ is regular. The broader local criterion and permanence statements, including the property for local rings essentially of finite type over $\mathbb{Z}$, are supplied among the foundation arguments below.

Theorem 1.1 (Artin approximation in a pointed étale neighbourhood). Let $(R, \mathfrak{m})$ be a Noetherian local G-ring. Let

$$f_1, \dots, f_q \in R[X_1, \dots, X_n], \qquad \widehat{a} \in \widehat{R}^n, \qquad f_j(\widehat{a}) = 0.$$

For every positive integer N, there exist a finitely presented étale R-algebra R', a maximal ideal $\mathfrak{m}' \subset R'$ over $\mathfrak{m}$, and a solution $b \in (R')^n$, such that

$$\kappa(\mathfrak{m}') = \kappa(\mathfrak{m}), \qquad f_j(b) = 0, \qquad \rho(b_i) - \widehat{a}_i \in \mathfrak{m}^N \widehat{R}. \tag{1.1}$$

Here $\rho : R' \to \widehat{R}$ is the canonical lift of the specified residue-field point. This is a homomorphism; no assertion that the whole étale algebra embeds into the completion is required. The statement allows arbitrary characteristic, imperfect residue fields, nilpotent base elements, and nonregular local rings.

Lemma 1.2 (the canonical comparison map). A finitely presented étale R-algebra and a specified R-map $R' \to R/\mathfrak{m}$ have a unique compatible map $R' \to \widehat{R}$.

Proof. Starting with the specified map to $R/\mathfrak{m}$, lift successively to $R/\mathfrak{m}^r$. The kernel of $R/\mathfrak{m}^{r+1} \to R/\mathfrak{m}^r$ is $\mathfrak{m}^r/\mathfrak{m}^{r+1}$. Its square is zero because $2r \geq r+1$. Formal etaleness gives a unique lift of

the already chosen map at each step. Denote these compatible R-maps by $\phi_r : R' \to R/\mathfrak{m}^r$. By definition $\widehat{R} = \varprojlim_r R/\mathfrak{m}^r$: it is the subring of the product formed by compatible tuples, with addition and multiplication componentwise. Therefore $x \mapsto (\phi_r(x))_r$ is an R-algebra map $R' \to \widehat{R}$, inducing the specified residue map. Conversely any such map, composed with the coordinate projections $\widehat{R} \to R/\mathfrak{m}^r$, gives these same ϕ_r by induction and formal uniqueness. Its value on every x is consequently the same tuple. This proves existence and uniqueness using the inverse-limit definition alone, with no Noetherian or completion-quotient theorem needed for this lemma. Formal etaleness is part of the defining lifting property of an etale algebra in AG-CA-17; its nilpotent lifting criterion is Theorem 1.2 of that earlier lesson. □

Lemma 1.3 (a smooth residue-field point has an étale lift). If B is a smooth finitely presented R-algebra and $B \to \kappa = R/\mathfrak{m}$ is an R-map, there are a finitely presented étale R-algebra R', a point $R' \to \kappa$, and an R-map $B \to R'$ inducing the original point.

Proof. Use the local standard form proved in AG-CA, Formally smooth, unramified and étale ring maps, Theorem 5.1. At the kernel of the given point, localize once to write

$$B_g = R[X_1, \dots, X_s]/(h_1, \dots, h_c), \qquad \Delta = \det(\partial h_i/\partial X_j)_{i,j\le c} \quad \text{a unit.}$$

This presentation includes any localization variable and its equation. Choose $r_{c+1}, \dots, r_s \in R$ lifting the point's coordinates in the remaining variables. Substitute these elements in the equations, keeping the first c variables, and invert the substituted Δ and the original denominators. Call the resulting finitely presented algebra R'. Here is its entire lifting test. Given a map $R' \to D/J$ with $J^2 = 0$, lift its variable values to a vector $x \in D^c$. The inverted polynomials stay units: if $uv = 1 + j$ with $j \in J$, then u has inverse $v(1 - j)$. The equation errors form a vector $h(x) \in J^c$, and the Jacobian $H(x)$ is invertible by its unit determinant and the cofactor identity. Taylor expansion of each monomial, dropping every product of two elements of J, gives $h(x + z) = h(x) + H(x)z$ for $z \in J^c$. Thus $z = -H(x)^{-1}h(x)$ gives a solution. If y is any second lifted solution, the same identity gives $H(x)(y - x) = 0$, so $y = x$. The inverse variables are then forced by the unit equations. For $J^N = 0$, repeat this unique lifting through $D/J^{r+1} \to D/J^r$, whose kernel is square-zero for $r \ge 1$. This proves formal etaleness, including $c = 0$; finite presentation proves etaleness by the earlier programme definition. This is the square-Jacobian specialization of AG-CA-17, Theorem 4.1, with uniqueness proved here as well. The original residue coordinates define $R' \to \kappa$, because all inverted elements have nonzero residues. Its kernel is maximal with residue field κ, since $R \to \kappa$ is surjective. Substitution gives $B_g \to R'$, hence $B \to R'$, inducing exactly the original point. □

Proof of Theorem 1.1. The equality $R/\mathfrak{m}^N \simeq \widehat{R}/\mathfrak{m}^N\widehat{R}$ is proved in the earlier AG-CA lesson *Completion*, Proposition 2.2. Choose $c_i \in R$ representing $\widehat{a}_i$ modulo $\mathfrak{m}^N\widehat{R}$, and choose finitely many generators $d_1, \dots, d_M$ of $\mathfrak{m}^N$. Write

$$\widehat{a}_i = c_i + \sum_{l=1}^{M} d_l \widehat{u}_{il}.$$

Put

$$g_j(U) = f_j\left(c_1 + \sum_l d_l U_{1l}, \dots, c_n + \sum_l d_l U_{nl}\right), \qquad A = R[U_{il}]/(g_j). \tag{1.2}$$

The tuple $\widehat{u}$ gives an R-map $A \to \widehat{R}$. Since $R \to \widehat{R}$ is regular, the desingularization theorem proved by the sequence ending at Theorem D.38, and the elementary factorization criterion proved at the corresponding foundation lemma, give a factorization

$$A \longrightarrow B \longrightarrow \widehat{R} \tag{1.3}$$

with B smooth and finitely presented over R. Compose with the residue map $\widehat{R} \to \kappa$, then apply Lemma 1.3. The images $u_{il} \in R'$ solve the g_j exactly. Set $b_i = c_i + \sum_l d_l u_{il}$. Substitution in (1.2) gives $f_j(b) = 0$ exactly. Under the canonical map of Lemma 1.2,

$$\rho(b_i) - \widehat{a}_i = \sum_l d_l \big(\rho(u_{il}) - \widehat{u}_{il}\big) \in \mathfrak{m}^N \widehat{R}.$$

This proves the prescribed congruence without demanding that the auxiliary u_{il} themselves be close to $\widehat{u}_{il}$. The exact written foundations and the earlier programme proofs used in this argument are registered in §6. □

Corollary 1.4 (henselian formulation). If R is henselian, the solution can be chosen in R. In general it gives a solution in the henselization R^h, descending to the pointed étale neighbourhood just constructed.

Proof. The section criterion proved in AG-CA, Henselian local rings and henselization, Theorem 1.2 gives $R' \to R$ with the specified residue map. Its composition with $R \to \widehat{R}$ is the map ρ by Lemma 1.2, so (1.1) is preserved. For R^h, use the pointed étale colimit and its universal property proved in that lesson, Lemma 4.1 and Theorem 4.2. Compatibility with the map into the complete henselian ring follows from the same uniqueness. Here the principal monic chart consumed by that section-criterion proof is supplied by the full earlier structural étale proof, Lemma D2.2, whose integral closure input is the affine conductor theorem, Theorem C4.3. Its polynomial normality input is proved in the preliminary coefficient argument, Corollary B1.2. To spell out the actual section step, localize the selected étale algebra to $(R[T]/(F))_h$, with F monic and F' nonzero at the given residue root $\bar{a}$. The henselian simple-root property gives $a \in R$ with $F(a) = 0$ and $a \bmod \mathfrak{m} = \bar{a}$. Every denominator has nonzero residue at $\bar{a}$, hence is a unit at a, and evaluation defines the section. The resulting map $R' \to R$ induces the prescribed residue map; formal étale uniqueness along the quotients $R/\mathfrak{m}^n$ proves its composition with completion is ρ. Thus this corollary uses these written earlier proofs rather than an external conductor proof tag. □

2. Returning from a local model to the original base

Lemma 2.1 (spreading finite presentation data). Let R_0 be Noetherian and $\mathfrak{p} \subset R_0$. A pointed étale algebra, a finite tuple, and finitely many equations over $(R_0)_{\mathfrak{p}}$, as in Theorem 1.1, spread to $(R_0)_s$ for some $s \notin \mathfrak{p}$. Their solution therefore gives a pointed étale neighbourhood over R_0.

Proof. In the construction of Lemma 1.3, every polynomial coefficient, every fixed nonpivot coordinate, and every chosen inverse has a denominator outside $\mathfrak{p}$. Invert their product in R_0. The finitely many equations for the tuple and the finitely many witnesses verifying them likewise hold after multiplying by finitely many additional denominators; invert their product too. The standard étale presentation spreads with its invertible Jacobian determinant, so its spread algebra is étale by the same direct Newton calculation. Regard it as an R_0-algebra through the étale localization $R_0 \to (R_0)_s$; composing two formally étale, finitely presented maps remains étale because a square-zero lifting problem is solved uniquely in two successive steps. The chosen residue map survives and gives a point over $\mathfrak{p}$ with unchanged residue field. The defining equations remain exact, and localization at the selected point gives the original local comparison. □

This proof uses the specific standard étale algebra produced above and therefore does not invoke an external spreading theorem for an arbitrary presentation. Pulling the resulting neighbourhood back to the original scheme preserves every defining equation and the specified fibre point.

3. The finite system for an embedded torus

The approximation step in the torus lesson begins after the formal deformation and effectivity argument has produced an affine torus $\widehat{T}$ over the completed local model and a closed homomorphism $\widehat{T} \hookrightarrow G_{\widehat{R}}$. The present lesson proves the approximation input; it does not claim to supply that preceding effectivity argument.

Lemma 3.A (finite splitting over a complete local ring). Let $(A, \mathfrak{m}, \kappa)$ be a local ring with

$$A \simeq \varprojlim_{n \geq 1} A/\mathfrak{m}^n.$$

No Noetherian, reducedness, normality or perfection hypothesis is imposed. Suppose a torus T/A has an affine etale splitting chart $U \to \operatorname{Spec} A$ and a point of its closed fibre with finite separable residue field K/κ. Then a finite free, faithfully flat, etale algebra E/A splits T. We may choose E local and complete, with residue field a finite Galois extension L/κ containing K, and with an action of $\Gamma = \operatorname{Gal}(L/\kappa)$ for which

$$E \otimes_A E \xrightarrow{\sim} \prod_{\sigma \in \Gamma} E, \qquad x \otimes y \longmapsto \big(x\sigma(y)\big)_\sigma. \tag{3.2}$$

Here a torus is etale locally a product of copies of $\mathbf{G}_m$. Starting instead with the fpqc multiplicative-type definition gives this same condition by AG-GS-05, Theorem 7.7; its finite torsion-free character-group step is proved in AG-RG-S04, Lemma P0.10. Every torus therefore has such a chart and residue point: choose an affine finite-presentation part of an etale splitting cover meeting the closed fibre, and a closed point of that fibre. Its residue field is finite over κ by the finite-type field-algebra argument in AG-RG-S04, Lemma P0.9. The quotient sequence for differentials gives $\Omega_{K/\kappa} = 0$, because the chart is formally unramified. The finite-field differential calculation in AG-CA-17, Lemma 7.1 then proves separability.

Furthermore, a torus T_0/κ, with character lattice split by L, has a torus lift over A split by this same E. Its character lattice and descent action reduce compatibly modulo every $\mathfrak{m}^n$.

Proof. A finite separable extension embeds in a finite normal separable extension: adjoin all the roots of the minimal polynomials of a finite generating list. The finite splitting-field construction and separable-tower calculations of AG-RG-S02, Lemma 4.A, prove finiteness and separability. Every embedding permutes the root sets, so the resulting extension L/κ is normal. The same embedding count gives $|\Gamma| = [L : \kappa]$. We give the whole primitive-element step here. Write $n = [L : \kappa]$, choose a finite generating list $b_1, \ldots, b_s$, and list the n embeddings σ_i into a finite splitting field. If κ is infinite, every pair of distinct embeddings differs on some generator, so $\sum_j (\sigma_i(b_j) - \sigma_{i'}(b_j))X_j$ is a nonzero polynomial. Their finite product is nonzero, because a polynomial ring over a field is a domain, as follows by multiplying leading coefficients successively in each variable. It cannot vanish on every tuple in κ^s: induction on the variables first makes a nonzero coefficient polynomial nonzero and then avoids the finitely many roots in the last variable, using the root bound of Lemma 4.A. Choose such a tuple (c_j). Then $\alpha = \sum_j c_j b_j$ has n different embedding images. Its minimal polynomial has at least n roots, hence degree at least n; the tower-degree formula bounds that degree by n, so $L = \kappa(\alpha)$. If κ is finite, a finite basis makes L a finite set. The complete cyclic-subgroup calculation of Lemma 4.A makes $L^\times$ cyclic. A generator α generates L as a field because every nonzero element is its power. Degree one is included by choosing $\alpha = 1$. Let $\bar{f} \in \kappa[X]$ be the monic minimal polynomial of this α, and lift its coefficients to a monic $f \in A[X]$. Put

$$E = A[t] = A[X]/(f), \qquad d = \deg(f).$$

Successively subtracting leading multiples of the monic polynomial f reduces every polynomial to degree less than d. A nonzero multiple qf has degree $\deg(q) + d$, since its leading coefficient is the nonzero leading coefficient of q. Thus the remainder is unique, and $1, t, \ldots, t^{d-1}$ is a basis making E free of positive finite rank over A; it is faithfully flat because tensoring a module with

this free module gives a direct sum of d copies. Its quotient by $\mathfrak{m}E$ is L. If $e \in E$ has nonzero residue, multiplication by e on this basis is invertible modulo $\mathfrak{m}$. Its determinant is a unit of A, and the cofactor calculation proved in Lemma 3.B gives an inverse to multiplication by e. Thus e is a unit, E is local with maximal ideal $\mathfrak{m}E$, and coordinatewise completeness gives

$$E \simeq \varprojlim_n E/\mathfrak{m}^n E.$$

Separability makes $f'(t)$ a unit by the same argument. A root in a quotient by a square-zero ideal lifts uniquely by replacing any lift a with $a - f'(a)^{-1}f(a)$. Polynomial expansion proves existence, and $f(b) - f(a) = (b-a)(f'(a) + (b-a)H)$ proves uniqueness. Iteration gives the corresponding assertion for every nilpotent ideal. This proves formal etaleness of E/A; its finite presentation makes it etale.

The residue point of U gives an algebra map $\mathcal{O}(U) \to K \to L$. Formal etaleness of U/A lifts this map uniquely through $E/\mathfrak{m}^{n+1}E \to E/\mathfrak{m}^n E$, whose kernel is square-zero for $n \geq 1$. Taking the compatible limit gives a map $\mathcal{O}(U) \to E$, hence a section of $U_E \to \operatorname{Spec} E$. Pulling back the splitting on U therefore splits T_E.

We now construct (3.2), including its automorphisms. For every $\sigma \in \Gamma$, the residue $\sigma(\alpha) \in L$ is a simple root of $\bar{f}$. In the complete local ring E, Newton's recursion

$$a_{j+1} = a_j - f'(a_j)^{-1}f(a_j)$$

starting from any lift of this residue gives $f(a_j) \in (\mathfrak{m}E)^{2^j}$ and $a_{j+1} - a_j \in (\mathfrak{m}E)^{2^j}$. Completeness and separatedness give a root a_σ, and the difference formula above gives its uniqueness with that residue. Distinct a_σ have unit differences. If a polynomial P vanishes at all these roots, monic division by $X - a_1$ gives $P = (X - a_1)Q$, since the remainder is $P(a_1) = 0$. At each other root a_σ, the factor $a_\sigma - a_1$ is a unit, so $Q(a_\sigma) = 0$. Repeating the division shows that $\prod_\sigma (X - a_\sigma)$ divides P. If $\deg(P) < d$, monicity forces $P = 0$. The difference between f and this product has degree less than d and vanishes at every root, so

$$f(X) = \prod_{\sigma \in \Gamma} (X - a_\sigma).$$

The map $t \mapsto a_\sigma$ defines an A-algebra endomorphism g_σ of E. The two roots $g_\sigma g_\tau(t)$ and $a_{\sigma\tau}$ have the same residue, so uniqueness gives $g_\sigma g_\tau = g_{\sigma\tau}$. Also $a_1 = t$; hence these maps are automorphisms and lift the Galois action on L.

The isomorphism $E \otimes_A E = E[X]/(f)$ and evaluation at the a_σ now give (3.2). Explicitly, the polynomials

$$\ell_\sigma(X) = \prod_{\tau \neq \sigma} \frac{X - a_\tau}{a_\sigma - a_\tau}$$

give the inverse to evaluation: a tuple $(b_\sigma)_\sigma$ is represented by $\sum_\sigma b_\sigma \ell_\sigma(X)$. Thus the isomorphism uses no division by $|\Gamma|$. Its compatibility on triple products is exactly $g_\sigma g_\tau = g_{\sigma\tau}$. This is the required finite etale Galois torsor.

For the final assertion use the specified finite Galois field L splitting T_0, so that the algebra E just constructed is the same one as in the statement. Its character group M is finite free by AG-RG-S04, Lemma P0.10, and the split character calculation of AG-GS-05, Theorem 2.2. The descent of T_0 along L/κ gives its Γ-action on M: pull back a character along each descent isomorphism; the cocycle gives the group law. Theorem 2.2 identifies these maps with the corresponding lattice maps. On $E[M]$ put

$$\sigma(ae^m) = g_\sigma(a)e^{\sigma m}. \tag{3.3}$$

The group law and (3.2) make this a descent datum. Multiplication and every Hopf operation commute with it, since addition in M commutes with its action. The actual module and algebra effectivity proof in AG-RG-S04, Lemmas A1.1–A1.2, gives the invariant Hopf algebra

$$C = E[M]^\Gamma, \qquad E \otimes_A C \simeq E[M]. \tag{3.4}$$

Tensor products and Hopf identities descend by this isomorphism and faithful flatness. Finite presentation descends here by a finite-generator argument. Express the finitely many Laurent algebra generators over E as finite E-linear combinations of elements of C. Those elements generate C, since the cokernel of their generated subalgebra becomes zero over E. For a resulting finite polynomial surjection $A[Z] \to C$, its kernel becomes a finite ideal after tensoring with E, because $E[M]$ is finitely presented. To see that finiteness for this particular surjection, compare it with a finite presentation of $E[M]$: express each presentation's generators in the other, and substitute these expressions in the finitely many relations and composite-generator identities. They generate the new kernel. Write a finite list of its generators as finite sums of elements of the original kernel with coefficients in E. Their finitely many kernel components generate the original kernel by faithful flatness. Thus C is finitely presented. It defines a torus T_A, because (3.4) splits it on the finite etale faithfully flat cover.

Put $A_n = A/\mathfrak{m}^n$ and $E_n = E \otimes_A A_n$. Tensoring (3.2)–(3.4) with A_n gives the same torsor and the same character action over E_n:

$$E_n \otimes_{A_n} (C \otimes_A A_n) \simeq E_n[M]. \tag{3.5}$$

The algebra on the left descends the datum (3.3) modulo $\mathfrak{m}^n$; uniqueness in Lemma A1.2 proves that it is its effective descent. This proves compatibility for every n, even though $A \to A_n$ need not be flat. It does not assume that taking invariants commutes with an arbitrary quotient. At $n = 1$, the descent datum is precisely the given one on $L[M]$, so its descended torus is T_0.

More generally, any compatible character data split by these E_n and identified with the same lattice have no additional nilpotent transition parameters. Comparing coefficients in $\Delta(\sum c_m e^m) = (\sum c_m e^m) \otimes (\sum c_m e^m)$ gives $c_m^2 = c_m$, $c_m c_{m'} = 0$ for $m \neq m'$, and $\sum c_m = 1$. Each E_n is local, so exactly one coefficient is 1. A character is therefore one monomial; a torus automorphism is one integral lattice automorphism. A compatible automorphism reducing to a fixed one is that same one. Hence a compatible lattice descent action is the fixed action in (3.3), and (3.5) gives its actual compatible descent. This completes both assertions. □

Choose a finite presentation of the Hopf algebra of $\widehat{T}$. Lemma 3.A supplies a finite free, etale, faithfully flat splitting algebra $E/\widehat{R}$; choose the resulting splitting isomorphism

$$\mathcal{O}(\widehat{T}) \otimes_{\widehat{R}} E \simeq E[Z_1^{\pm 1}, \dots, Z_r^{\pm 1}]. \tag{3.1}$$

Lemma 3.B (perfect trace, over an arbitrary base). Let A be a commutative ring and C a commutative A-algebra free of finite rank d. If the determinant of the pairing $(c, z) \mapsto \mathrm{Tr}_{C/A}(cz)$ is a unit, then C/A is finitely presented and formally etale, hence etale. If $d > 0$, it is faithfully flat. These assertions require no Noetherian, reducedness or characteristic hypothesis.

Proof. First, the elementary matrix identities used here hold over every commutative ring. Define a determinant by the signed permutation sum and a cofactor by the corresponding smaller signed determinant. Expanding along a row or column gives the determinant. A matrix with two equal rows or columns has determinant zero: pair each permutation with the permutation interchanging those positions; their contributions have opposite signs, including in characteristic two. Consequently, if M is square, multiplying it on either side by its transposed cofactor matrix gives $(\det M)I$: a diagonal entry is the row or column expansion, and an off-diagonal entry is the expansion of a determinant with two equal rows or columns. A unit determinant therefore gives the inverse $(\det M)^{-1} \operatorname{adj}(M)$. For size zero the determinant is one. To verify the converse implication as well, expand $\det(UV)$ by multilinearity in its columns. The expansion chooses, for each column of UV, one column of U with its coefficient from V. A tuple repeating a column of U contributes zero by the cancellation just proved. Every remaining tuple is a permutation of all the columns of U, whose determinant is its permutation sign times $\det(U)$. Summing its coefficients with these signs is precisely $\det(V)$. Thus $\det(UV) = \det(U)\det(V)$, including for size zero. If $UV = I$, this equation gives $\det(U)\det(V) = 1$, so an invertible matrix has a unit determinant. For square matrices U, V,

the finite double sum also gives $\operatorname{tr}(UV) = \operatorname{tr}(VU)$. Thus conjugating a multiplication matrix by an invertible change of basis preserves its trace. Multiplication matrices, their traces and their determinants commute with every scalar extension, entry by entry.

Suppose first that $d > 0$, and choose a basis $u_1, \dots, u_d$. Write $u_i u_j = \sum_k m_{ijk} u_k$ and $1 = \sum_i a_i u_i$. There is an explicit finite algebra presentation

$$C = A[X_1, \dots, X_d]/(X_i X_j - \sum_k m_{ijk} X_k,\ 1 - \sum_i a_i X_i).$$

Indeed, repeatedly replace a product of two variables by the displayed linear combination, reducing each monomial to degree at most one. Replace its constant term by that constant times $\sum_i a_i X_i$. Every quotient element is therefore represented by $\sum_i b_i X_i$. The map to C is surjective, and if this representative maps to zero, linear independence of the u_i makes every b_i zero. This proves the presentation without an assumption on ideals of A.

The inverse pairing matrix supplies a trace-dual basis $v_1, \dots, v_d$, characterized by $\operatorname{Tr}(u_j v_i) = \delta_{ij}$. In particular $x = \sum_i u_i \operatorname{Tr}(v_i x)$ and $x = \sum_i \operatorname{Tr}(u_i x) v_i$. To check the second equality, pair both sides with each u_j and use invertibility of the pairing; the first is the ordinary basis-coordinate equality. Put

$$e = \sum_i u_i \otimes v_i \in C \otimes_A C.$$

The A-linear map

$$\Phi : C \otimes_A C \longrightarrow \operatorname{End}_A(C), \qquad \Phi(a \otimes b)(x) = a \operatorname{Tr}(bx)$$

is an isomorphism, with inverse $F \mapsto \sum_i F(u_i) \otimes v_i$. The first coordinate equality proves $\Phi\Phi^{-1}(F) = F$; the second proves $\Phi^{-1}\Phi(a \otimes b) = a \otimes b$, and pure tensors generate. Since $\Phi(e)$ is the identity, both $(c \otimes 1)e$ and $(1 \otimes c)e$ have image multiplication by c. Thus

$$(c \otimes 1)e = (1 \otimes c)e \qquad (c \in C). \tag{3.6}$$

Let $s = \sum_i u_i v_i$. The diagonal coefficient of multiplication by c in position i is $\operatorname{Tr}(v_i c u_i)$. Summing those coefficients gives $\operatorname{Tr}(c) = \operatorname{Tr}(cs)$ for every c. Invertibility of the pairing now gives

$$\sum_i u_i v_i = 1. \tag{3.7}$$

No division by d, or assertion that d is a unit, occurs.

We prove the lifting property directly. Let $J \subset D$ be a square-zero ideal of a commutative A-algebra, and let $\bar{f} : C \to D/J$ be an A-algebra map. The free-module lifting calculation in AG-RG-S01, Lemma 1.0 gives an A-linear lift $l : C \to D$. Its value $l(1)$ is $1 + j$ with $j \in J$, and has inverse $1 - j$. Multiplying the whole lift by that inverse makes $l(1) = 1$ while preserving A-linearity and its reduction. Give J its C-module action via $\bar{f}$: multiply by any lift of $\bar{f}(c)$, which is independent of that lift because $J^2 = 0$.

The defect $b(c, z) = l(cz) - l(c)l(z)$ lies in J, is A-bilinear and has $b(c, 1) = b(1, c) = 0$. Associativity in C and D gives, by expanding the four terms,

$$c\, b(z, t) - b(cz, t) + b(c, zt) - b(c, z)\, t = 0. \tag{3.8}$$

Define the A-linear correction $h(c) = \sum_i u_i b(v_i, c)$ in J. Apply the A-linear map $a \otimes z \mapsto ab(z, t)$ to (3.6); it gives $ch(t) = \sum_i u_i b(cv_i, t)$. Using (3.8) with (v_i, c, t) and then (3.7), we obtain

$$ch(t) - h(ct) + h(c)t = \sum_i u_i \big(b(v_i c, t) - b(v_i, ct) + b(v_i, c)t\big) = \sum_i u_i v_i b(c, t) = b(c, t).$$

Consequently $f = l + h$ is multiplicative: the defect of this new map is $b(c, t) + h(ct) - ch(t) - h(c)t = 0$, since $h(c)h(t) = 0$. Also $h(1) = 0$. Thus f is a unital A-algebra lift of $\bar{f}$.

For uniqueness, the difference δ of two such lifts takes values in J and satisfies $\delta(ct) = c\delta(t) + \delta(c)t$; expand their products and use $J^2 = 0$. Apply $a \otimes z \mapsto \delta(a)z$ to (3.6). The difference between its two sides is $\delta(c)\sum_i u_i v_i = \delta(c)$, by that identity and (3.7). It is zero, so every lift is unique. If $J^N = 0$, lift successively through $D/J^{r+1} \to D/J^r$, for $1 \le r < N$. Its kernel is square-zero because $2r \ge r + 1$. Existence and uniqueness at each step prove formal etaleness for every nilpotent ideal. Together with the explicit finite presentation, this is etaleness by the programme definition.

For $d = 0$, C is the zero algebra with presentation $A/(1)$. If a unital map $C \to D/J$ exists and J is nilpotent, then $D/J = 0$, so $1 \in J$ and nilpotence forces $D = 0$. There is exactly one lift. Finally, for positive d, tensoring an A-module with C gives d copies of that module; this is exact and detects whether it is zero. This proves faithful flatness. □

This is the finite-free case of the discriminant criterion in the freely accessible Stacks Project, Lemma 49.3.1, Tag 0BJF. The complete lifting proof above supplies the result used here.

Return to the splitting algebra $E/\widehat{R}$ from Lemma 3.A. Its trace determinant is a unit, as we now check rather than adding an unsupported inverse variable. By (3.2), after scalar extension from $\widehat{R}$ to E the algebra becomes $\prod_{\sigma\in\Gamma} E$. Multiplication in this product is diagonal in the coordinate idempotent basis. Its trace pairing matrix is the identity, hence invertible. The preceding trace and basis-change calculations imply that the original pairing matrix becomes invertible over E, so its determinant Δ becomes a unit there. Then $(\widehat{R}/\Delta\widehat{R}) \otimes_{\widehat{R}} E = 0$. Faithful flatness from Lemma 3.A detects zero modules, giving $\widehat{R}/\Delta\widehat{R} = 0$, which means that Δ is already a unit of $\widehat{R}$.

Fix the finite free basis of E supplied by Lemma 3.A. Its multiplication table, unit and multiplication matrices define finitely many coefficient variables; associativity, commutativity and the unit laws are finite polynomial equations. Add a variable w and the equation $w\Delta = 1$ for the trace determinant. The original completed coefficient tuple satisfies this equation by the preceding argument. After approximation it still satisfies the same equation exactly. Lemma 3.B makes the resulting finite free algebra etale and, since its unchanged rank is positive, faithfully flat.

Add coefficients for the finite presentations, Hopf maps, the isomorphism (3.1), its inverse, and the homomorphism to G. Laurent expressions can be written polynomially by adding inverse variables and relations $Z_i W_i = 1$. The maps need to be specified on finite generating sets. Relations between their images need finitely many witnesses in the finitely generated relation ideals. The two composites of the splitting isomorphism and its inverse are tested on finite generators; equality of these composites with the identity is recorded by their witnesses. The same method records comultiplication, counit, inverse, and the homomorphism identities on a finite list of generators. Coassociativity and every needed descent identity are likewise checked on finite generators, using the finite multiplication tables on the finite overlap algebras. Each expression and witness has finite support.

Retain the closed immersion in this same system. The pullback $\mathcal{O}(G_{\widehat{R}}) \to \mathcal{O}(\widehat{T})$ is surjective. Choose preimages of a finite list of algebra generators of $\mathcal{O}(\widehat{T})$, and include the equations asserting that their images are those generators. After approximation these equations still prove surjectivity; consequently the resulting homomorphism is a closed immersion.

We have obtained one finite polynomial system over the local model, solved by the original completed coefficient tuple. Apply Theorem 1.1 at order one. The defining equations hold exactly after approximation, while all the chosen coefficients retain their original residues. The preserved splitting maps identify the source Hopf algebra with a Laurent polynomial Hopf algebra on the finite étale faithfully flat cover. Thus it is a torus, with the specified special fibre and closed embedding. If a prescribed higher jet is needed, use that order instead. Lemma 2.1 spreads the data and the neighbourhood back to an open part of the finite-type model; base change returns them to the original base.

The local model is essentially of finite type over $\mathbb{Z}$. The G-ring permanence argument in §5 is therefore a consumed part of this application, including its coefficient-ring inputs. It must be

proved before the approximation step is declared complete. An excellence label or an external Stacks link does not replace this work.

Scope of the supporting algebra. The approximation theorem and the desingularization theorem concern Noetherian rings. Every base ring of the finite-type algebra diagrams in their proof is Noetherian. In the foundational complete-intersection arguments below we state and prove the exact Noetherian forms consumed there; arguments explicitly stated over arbitrary rings retain that wider scope. Polynomial presentations with arbitrarily many variables occur only in the two-term cotangent comparison, whose elementary presentation and filtered-colimit proofs are written without Noetherian assumptions. This distinction preserves arbitrary characteristic, imperfect residue fields, and the full Noetherian local G-ring statement.

4. Desingularization proof sequence

We prove approximation by passing through presentation changes, infinitesimal lifts and improvements of singular ideals. Local resolutions are constructed separately for separable and inseparable residue fields, then assembled into the smooth-colimit theorem. Smoothness throughout means the conormal condition F.6. Each displayed list of arrows specifies the maps in the relevant commuting diagram.

Singular ideals

The singular ideal records the complement of the smooth locus. Its radical formulation lets us compare smooth neighbourhoods across different finite presentations.

Definition D.1: singular ideal

Reading source: Stacks, Tag 07C5.

For any ring map $R \to A$, let $U_{A/R}$ be its smooth locus in $\operatorname{Spec} A$, with smoothness at a prime understood as in F.7. Its *singular ideal* $H_{A/R}$ is the radical ideal characterized by

$$V(H_{A/R}) = \operatorname{Spec} A \setminus U_{A/R}.$$

No finite-presentation hypothesis on A/R, or quasi-compactness hypothesis on $U_{A/R}$, is part of this definition.

Verification. The definition of smoothness at a prime writes

$$U_{A/R} = \bigcup_{\substack{g \in A \\ R \to A_g \text{ smooth}}} D(g).$$

Indeed a witnessing g gives smoothness at every prime avoiding it, and every smooth prime has such a witness. Thus $U_{A/R}$ is open. Write its closed complement as $V(J)$ and set $H_{A/R} = \sqrt{J}$. A prime contains J exactly when it contains $\sqrt{J}$, since a power belonging to a prime forces its base element into that prime. This proves existence.

Here is the radical-ideal uniqueness argument used below. For any ideal J and $x \notin \sqrt{J}$, the ideals containing J and avoiding $\{1, x, x^2, \ldots\}$ have a maximal member L, by the chain-union argument and Zorn's lemma. If $uv \in L$ while $u, v \notin L$, maximality provides a power of x in each of $L + (u)$ and $L + (v)$. Their product is again a power of x and belongs to L, a contradiction. Hence L is a prime containing J and avoiding x. It follows that

$$\sqrt{J} = \bigcap_{\mathfrak{q} \in V(J)} \mathfrak{q},$$

where the empty intersection is A. Two radical ideals with the same closed set are therefore equal. This also covers $A = 0$ and an empty singular locus. The smooth open need not have a finite subcover, as the fully proved example D.6 shows. □

Lemma D.2: find strictly standard

Reading source: AI Integrated Stacks, pinned smoothing, lines 141–181.

Fix a finite presentation

$$P = R[x_1, \dots, x_n], \qquad I = (f_1, \dots, f_m), \qquad A = P/I.$$

At a prime $\mathfrak{q}$ where A/R is smooth, one can select equally many equations and variables, indexed by sets V and U of size $c \le \min(n, m)$, and find $a, a' \in A$ such that

$$a \notin \mathfrak{q}, \qquad a = a' \det(\partial f_j / \partial x_i)_{j \in V,\ i \in U}, \qquad \widetilde{a} f_\ell \in (f_j : j \in V) + I^2 \quad (1 \le \ell \le m).$$

Here $\widetilde{a}$ is any polynomial lift of a. The last condition is equivalently the relation $a[f_\ell] \in \langle [f_j] : j \in V \rangle$ in I/I^2.

Proof. The task is to choose a nonvanishing minor and then kill a finite conormal quotient. Let $M = I/I^2$ and $K = \kappa(\mathfrak{q})$. Smoothness at the prime supplies $s \notin \mathfrak{q}$ with A_s/R smooth, by F.7. The full comparison F.11, localization F.65, and definition F.6 give

$$0 \longrightarrow M_s \xrightarrow{\mathrm{d}} A_s^n \longrightarrow \Omega_{A_s/R} \longrightarrow 0.$$

This sequence splits because its quotient is projective. Consequently the differential on $M \otimes_A K$ is injective. The finite list $[f_j]$ spans that vector space; extract a basis from this list and call its index set V. If its length is c, injectivity implies $c \le n$, while its construction implies $c \le m$.

To justify a choice of U, use the following row calculation over K. When $c > 0$, a nonzero entry in the first column supplies a pivot row. Subtract its multiples from the other rows. The lower block of the remaining $c - 1$ columns is independent: a dependence there could be completed by a multiple of the first column to give a dependence of all c columns. Induction selects $c - 1$ lower rows with nonzero determinant. The determinant on these rows and the pivot row is the pivot times this nonzero determinant. Subtracting multiples of the pivot row did not change it, so it is a minor of the original columns. For $c = 0$ use the determinant 1. Thus the corresponding Jacobian minor δ belongs to $A \setminus \mathfrak{q}$.

Now form the finite A-module

$$Q = M / \langle [f_j] : j \in V \rangle.$$

Its residue at $\mathfrak{q}$ vanishes, and F.10 gives $Q_{\mathfrak{q}} = 0$. For each generator of Q choose a localization witness outside $\mathfrak{q}$ annihilating it. Their finite product t annihilates Q and still lies outside $\mathfrak{q}$; use $t = 1$ when $Q = 0$. Put $a = t\delta$ and $a' = t$. Then $aQ = 0$ gives every stated conormal relation. Lifting its coefficients to P gives the displayed polynomial-ideal relation. A different lift of a changes its product with f_ℓ by an element of I^2, so the relation does not depend on that lift. No cancellation of a, δ or another possible zero divisor occurs. The empty equation and variable cases are included by the same construction. □

The next definition distinguishes witnesses using a single Jacobian minor from witnesses using a sum of minors. D.15 compares these two conditions with smoothness after localization.

Definition D.3: strictly standard

Reading source: Stacks, Tag 07C7.

Let $R \to A$ be finitely presented. A witness for either standard-element condition consists of

$$P = R[x_1, \dots, x_n], \qquad I = (f_1, \dots, f_m), \qquad A = P/I, \qquad 0 \le c \le \min(n, m).$$

All Jacobian entries below are reduced to A. For $a \in A$, choose any lift $\widetilde{a} \in P$ when writing a polynomial-ideal condition.

The element a is *elementary standard* if some such witness and $a' \in A$ satisfy both

$$a = a' \det \left(\frac{\partial f_j}{\partial x_i} \right)_{i,j=1}^{c} \tag{D.3e1}$$

and

$$\widetilde{a} f_{c+j} \in (f_1, \dots, f_c) + I^2 \quad (1 \le j \le m - c). \tag{D.3e2}$$

The element a is *strictly standard* if some witness and coefficients $a_U \in A$, indexed by the c-element subsets of $\{1, \dots, n\}$, satisfy both

$$a = \sum_{|U|=c} a_U \det \left(\frac{\partial f_j}{\partial x_i} \right)_{j=1,\dots,c,\ i \in U} \tag{D.3s1}$$

and

$$\widetilde{a} f_{c+j} \in (f_1, \dots, f_c) + I^2 \quad (1 \le j \le m - c). \tag{D.3s2}$$

Use increasing order on each selected variable set in the determinants. Changing the lift $\widetilde{a}$ adds an element of I; multiplying this difference by $f_{c+j} \in I$ puts it in I^2. Thus both relation conditions are independent of the lift. Equivalently they assert $a[f_{c+j}] \in \langle [f_1], \dots, [f_c] \rangle$ in the A-module I/I^2. The minor condition and this relation condition are separate requirements.

For $c = 0$ there is one minor, the empty determinant 1, and the first equation imposes no restriction on a. The second equation then reads $\widetilde{a} I \subset I^2$. For $m = c$ the second equation has an empty list of tests. These conventions include empty polynomial presentations and the zero algebra. □

Lemma D.4: parse equation strictly standard one

Reading source: Stacks, Tag 07ET.

In the presentation of D.3, the strict minor condition for a gives an A-linear map $\psi : A^n \to A^c$ for which the composite

$$A^c \xrightarrow{e_j \mapsto [f_j]} I/I^2 \xrightarrow{\mathrm{d}} A^n \xrightarrow{\psi} A^c$$

is $a 1_c$. Conversely the existence of this map gives the strict minor condition for a^c. These assertions concern the minor condition alone; the remaining-equation condition still has to be imposed independently.

Proof. Identify A^n with the free differential module on $dx_1, \dots, dx_n$. Its conormal map is well-defined: for $u, v \in I$, the product rule gives $d(uv) = u\,dv + v\,du$, which vanishes after reduction to A. The composite of its first two maps has the $n \times c$ matrix

$$M_{ij} = \frac{\partial f_j}{\partial x_i} \bmod I.$$

For a selected variable set U, let $E_U : A^n \to A^c$ be coordinate selection and put $M_U = E_U M$. The determinant in D.3s1 is $\det M_U$: transposition preserves the determinant by reversing the permutation indexing in its signed expansion.

The arbitrary-ring adjugate and determinant identities are proved in F.70. If $a = \sum_U a_U \det M_U$, they give the explicit map with matrix

$$B = \sum_U a_U \operatorname{adj}(M_U) E_U, \qquad BM = a1_c.$$

This is the required ψ. Conversely write B for any proposed ψ. The same proved multilinear expansion gives

$$a^c = \det(BM) = \sum_{|U|=c} \det(B^U) \det(M_U),$$

where B^U selects the columns indexed by U. This is the required minor expression for a^c, without cancellation or an invertibility assumption. If $c = 0$, both modules are zero, the map is unique, and $a^0 = 1$ has its expression using the empty minor. Zero rings obey the same formulas. □

Lemma D.5: elkik

Reading source: Stacks, Tag 07CA.

For a finitely presented R-algebra A, let J_e and J_s be the ideals generated by all its elementary standard and strictly standard elements, respectively. Then

$$H_{A/R} = \sqrt{J_e} = \sqrt{J_s}.$$

In particular A_a is smooth over R whenever a is strictly standard.

Proof. First fix a strictly standard witness from D.3 and put $B = A_a$ and $N = (I/I^2) \otimes_A B$. Its relation condition says that the first c equation classes generate N: divide each relation by the unit a. Write $q : B^c \to N$ for this surjection and $d : N \to B^n$ for the conormal differential. D.4 gives $\psi : B^n \to B^c$ with

$$\psi dq = a1_c.$$

If $q(v) = 0$ this identity gives $av = 0$, hence $v = 0$. Consequently q is an isomorphism. The map $\chi = q\, a^{-1} \psi : B^n \to N$ satisfies $\chi d = 1_N$. Thus

$$B^n = d(N) \oplus \ker \chi.$$

Its quotient by $d(N)$ is a finite projective module, being a direct summand of B^n, and d has zero kernel.

The actual localization still has a finite presentation over R: if $h \in P$ lifts a, it is $P[z]/(f_1, \ldots, f_m, hz - 1)$. The quotient isomorphism follows by adjoining the specified inverse of h. The full presentation comparison F.11 and localization calculation F.65 identify its first cotangent homology and differential module with the kernel and cokernel just calculated. Therefore the smoothness definition F.6 proves that $R \to B$ is smooth. If $B = 0$, these modules are zero and the same finite presentation and criterion apply.

Each singular prime contains every such a, since a prime avoiding a has the smooth neighbourhood A_a. Hence $V(H_{A/R}) \subset V(J_s)$. Setting every other minor coefficient to zero makes an elementary witness a strict witness, so $J_e \subset J_s$. If a prime is smooth, D.2 supplies an elementary witness outside it; relabel its selected variables and equations to place them first, absorbing any ordering sign in the coefficient. Such a prime cannot contain J_e. These statements give

$$V(J_e) \subset V(H_{A/R}) \subset V(J_s) \subset V(J_e).$$

The radical/prime equality proved in D.1 now proves both asserted equalities of ideals. Neither this argument nor the conormal splitting assumes that the base is Noetherian or that the smooth locus is quasi-compact. □

Example D.6: not quasi compact

Reading source: AI Integrated Stacks, pinned smoothing, lines 285–292.

A smooth locus can fail to be quasi-compact even for a finitely presented map. For any field k, take

$$R = k[x, y_1, y_2, \ldots]/(xy_i \mid i \geq 1), \qquad A = R/(x).$$

In $\operatorname{Spec} A$, its smooth locus is precisely

$$U = \bigcup_{i \geq 1} D(y_i) = \operatorname{Spec} A \setminus \{\mathfrak{m}\}, \qquad \mathfrak{m} = (y_1, y_2, \ldots).$$

No finite family of these distinguished opens covers U. In particular U is not quasi-compact.

Proof. The quotient identifies A with $k[y_1, y_2, \ldots]$. The map $R \to A$ is of finite presentation: it is the quotient by the single generator x, with no algebra variables required. The base ring R is not assumed Noetherian. Evaluation at all $y_i = 0$ gives $A/\mathfrak{m} = k$, so $\mathfrak{m}$ is maximal and is the only prime containing every y_i.

Here is the ideal calculation. Setting all y_i to zero gives a retraction $R \to k[x]$ of the natural inclusion $k[x] \to R$, so that inclusion is injective. For $I = (x) \subset R$, the relations $xy_i = 0$ give $xr = xr(x, 0, 0, \ldots)$ for every $r \in R$. Therefore

$$I = xk[x], \qquad I^2 = x^2 k[x], \qquad I/I^2 = k\,\bar{x} \simeq A/\mathfrak{m}.$$

In the last identification every y_i annihilates $\bar{x}$. For the zero-variable polynomial presentation $R \twoheadrightarrow A$, the conormal complex is $[I/I^2 \to 0]$. The presentation comparison in Lemma F.11 identifies it with the naive cotangent complex used in Definition F.6.

First suppose $y_i \notin \mathfrak{q}$. Inverting y_i in R makes $x = 0$, hence $A_{y_i} = R_{y_i}$. As an R-algebra this is

$$R[t]/(h), \qquad h = y_i t - 1.$$

The element h is a nonzerodivisor over this arbitrary base ring. Indeed, if $h \sum_{n=0}^{N} b_n t^n = 0$, its constant coefficient gives $b_0 = 0$, and its successive coefficients $y_i b_{n-1} - b_n = 0$ give every $b_n = 0$. Consequently $(h)/(h^2)$ is free on $\bar{h}$ over $R[t]/(h)$: multiplication by h gives a surjection onto it, and cancellation of the nonzerodivisor h shows that its kernel is exactly (h). An R-derivation of $R[t]$ is determined by its value on t; the product rule gives $d(t^n) = nt^{n-1}dt$. Thus $\Omega_{R[t]/R} = R[t]dt$, and this presentation has conormal complex

$$A_{y_i}\,\bar{h} \xrightarrow{y_i} A_{y_i}\,dt.$$

Its differential is an isomorphism, since y_i is a unit. Both first homology and the differential module are zero, and the zero module is finite projective. The displayed presentation is finite, so Definition F.6 proves $R \to A_{y_i}$ smooth. Definition F.7 proves smoothness at every point of $D(y_i)$.

We must also exclude the exceptional point $\mathfrak{m}$ from every smooth neighbourhood. Let $g \in A \setminus \mathfrak{m}$. Its constant coefficient $c = g(0, 0, \ldots)$ is nonzero. Regard its polynomial expression as a lift $\widetilde{g} \in R$. A presentation of A_g over R is

$$P = R[t] \twoheadrightarrow A_g, \qquad J = \ker(P \to A_g) = (x, \widetilde{g}t - 1).$$

The class of x in J/J^2 has relative differential zero, because x belongs to the base ring R. This class is nonzero. To see this, put $C = (R/(x^2))_{\widetilde{g}}$. The R-algebra map $P \to C$ sending t to $\widetilde{g}^{-1}$ maps J into the square-zero ideal $(x)C$, and maps J^2 to zero. It therefore maps the class of x in J/J^2 to $x \in C$. The assignments $x \mapsto \epsilon$ and $y_i \mapsto 0$ define

$$R/(x^2) \longrightarrow k[\epsilon]/(\epsilon^2).$$

They respect all the relations, and send $\widetilde{g}$ to the unit c. The map extends to C and sends x to the nonzero element ϵ; its nonzeroness follows directly from the degree bound for a multiple of ϵ^2. Hence $x \neq 0$ in C, and the class of x is a nonzero kernel element in the conormal complex $[J/J^2 \to A_g dt]$. By the same presentation comparison, $H_1(\mathrm{NL}_{A_g/R}) \neq 0$. Definition F.6 shows that $R \to A_g$ is not smooth. This holds for every $g \notin \mathfrak{m}$, so Definition F.7 shows that $R \to A$ is not smooth at $\mathfrak{m}$.

Every other prime omits some y_i, proving that the smooth locus is exactly the displayed open U. In the notation of Definition D.1, this also gives $H_{A/R} = \mathfrak{m}$.

Finally the open cover $U = \bigcup_{i \geq 1} D(y_i)$ has no finite subcover. Given a finite set F of indices, choose $j \notin F$. Evaluation $y_j \mapsto 1$ and $y_i \mapsto 0$ for $i \neq j$ defines a surjection $A \to k$ with prime kernel $\mathfrak{q}_j$. This point lies in $D(y_j) \subset U$ but in none of the $D(y_i)$ with $i \in F$. Thus no finite subfamily covers U, and U is not quasi-compact. □

Lemma D.7: strictly standard base change

Reading source: Stacks, Tag 07CC.

Let $R \to A$ be finitely presented and $R \to R'$ any ring map. An elementary standard element a remains elementary standard as $a \otimes 1$ in $A \otimes_R R'$. The same assertion holds for strictly standard elements. No flatness assumption on R' is needed.

Proof. Take the finite presentation and witness from D.3 and set $P' = R'[x_1, \ldots, x_n]$. The polynomial base-change map $P \otimes_R R' \to P'$ sends $x^E \otimes r'$ to $r'x^E$. Its inverse sends the coefficient of every monomial to the same tensor; the polynomial monomials are the free R-basis of P, so these maps are inverse. Tensor right exactness identifies

$$A \otimes_R R' = P'/I', \qquad I' = (f'_1, \ldots, f'_m),$$

where f'_j is obtained by changing coefficients along $R \to R'$.

For a monomial rx^E, differentiation multiplies its coefficient by the integer E_i and reduces the ith exponent by one. Changing coefficients commutes with this operation, including when that integer becomes zero in R'. Consequently every selected Jacobian entry and its determinant is the image of the old one. The elementary single-minor identity, or the strict sum-of-minors identity, maps to exactly the required new identity with coefficients $a' \otimes 1$ or $a_U \otimes 1$.

For the relation condition, choose a polynomial lift $\widetilde{a}$ and write each old relation with its finitely many coefficients in $(f_1, \ldots, f_c) + I^2$. Its image expresses $\widetilde{a}' f'_{c+j}$ in $(f'_1, \ldots, f'_c) + (I')^2$, because products of two elements of I map to products of two elements of I'. These are exactly D.3e2 or D.3s2 for the new witness. No injectivity of a tensor map was used. Empty minors, empty relation lists and a zero base-changed algebra satisfy the same identities. □

Lemma D.8: final solve

Reading source: Stacks, Tag 07EU.

Suppose $R \to A \xrightarrow{\rho} \Lambda$, with A finitely presented over R and $H_{A/R}\Lambda = \Lambda$. Then ρ factors through an R-smooth algebra:

$$A \longrightarrow B \longrightarrow \Lambda.$$

Proof. If $\Lambda = 0$, choose $B = 0$. Its presentation $R/(1)$ is finite and its cotangent homology and differential module are zero, so F.6 makes it smooth. Assume henceforth $\Lambda \neq 0$.

Let J_s be the ideal generated by strictly standard elements. D.5 gives $H_{A/R} = \sqrt{J_s}$. We first check that extending the smaller ideal also gives the unit ideal. Choose $h_1, \ldots, h_t \in H_{A/R}$ and $\lambda_i \in \Lambda$

with $\sum_i \rho(h_i)\lambda_i = 1$. For each i some $n_i \geq 1$ has $h_i^{n_i} \in J_s$. Raise that identity to

$$N = 1 + \sum_i (n_i - 1).$$

Every term in its distributive expansion uses some h_i at least n_i times: otherwise its total degree would be at most $N - 1$. Each term therefore belongs to $J_s\Lambda$. Thus $1 \in J_s\Lambda$.

Expanding the finite expressions in this ideal gives strictly standard elements $a_1, \ldots, a_r$ and $\mu_i \in \Lambda$ with $\sum_i \rho(a_i)\mu_i = 1$. Here $r > 0$, since the target is nonzero. Define

$$B = A[T_1, \ldots, T_r] / \left(\sum_i a_i T_i - 1 \right).$$

The map sending T_i to μ_i gives $B \to \Lambda$ and extends the prescribed ρ. Appending these finite variables and this one relation to a finite presentation of A proves that B/R is finitely presented.

In B, the elements a_i generate one. Their principal opens therefore cover its spectrum: a prime containing all of them would contain the displayed unit. For each i there is the explicit isomorphism

$$B_{a_i} \cong A_{a_i}[T_j : j \neq i].$$

The map from right to left includes the remaining variables; its inverse substitutes $T_i = a_i^{-1}(1 - \sum_{j \neq i} a_j T_j)$. Substitution kills the relation and both composites fix their generators. D.5 proves A_{a_i}/R smooth. Adjoining the finitely many remaining polynomial variables is smooth directly: its empty-relation presentation has zero first cotangent homology and a finite free differential module on those variables, by F.11 and AG-CA-15, Theorem 5.1. F.6 therefore gives smoothness over A_{a_i}. The entire composition proof F.29 makes B_{a_i}/R smooth. The proved finite principal-cover criterion F.67 now makes B/R smooth. Zero localizations create empty charts and are already included in those criteria. This proves the required factorization over an arbitrary base ring. □

Presentations of algebras

The presentation change in D.9 uses a symmetric algebra over A. Over a Noetherian base its finite generators also give finite presentation. These constructions are among the results originating with Elkik.

Lemma D.9: improve presentation

Reading source: Stacks, Tag 07CE.

For a finitely presented algebra A over an arbitrary ring R, there is an R-algebra map $A \to C$ of finite type with a retraction $C \to A$ such that:

1. Whenever $R \to A_a$ is a local complete intersection, $A_a \to C_a$ is smooth and C_a has a finite polynomial presentation over R with free conormal module.
2. Whenever A_a/R is smooth, $\Omega_{C_a/R}$ is free.

Proof. Fix $P = R[X_1, \ldots, X_n] \twoheadrightarrow A$ with $I = (f_1, \ldots, f_m)$ and let

$$M = I/I^2, \qquad K = \ker(A^m \longrightarrow M), \qquad C = \mathrm{Sym}_A(M).$$

The map $A^m \to M$ sends e_j to $[f_j]$. The symmetric algebra and its degree-zero augmentation are constructed in F.128; their composite on A is identity. Its universal property identifies C with the quotient of $P[Y_1, \ldots, Y_m]$ by an ideal J generated by I and the linear forms whose coefficient classes belong to K. In fact maps out of this quotient are exactly maps $A \to D$ together with linear

maps $A^m \to D$ killing K, hence linear maps $M \to D$. The resulting maps are inverse because they fix A and every Y_j. This proves the kernel description and finite type even when K is not finite.

Fix $a \in A$ and write $B = A_a$. Suppose first $R \to B$ is a local complete intersection. The arbitrary-ring presentation comparison F.94.A supplies local Koszul-regular equation lists for the inverse-variable polynomial presentation of B. Such a list gives a conormal basis: correct a relation modulo the square of its ideal by coefficients in the ideal; the resulting exact relation is a first Koszul cycle, hence a boundary, whose coefficients belong to the ideal. This proves independence as well as generation of its equation classes. F.72 identifies this locally free conormal module with $M_a \oplus B$. Its summand M_a is locally finite flat and finitely presented, since both summands are generated by projections of a finite basis. The local finite-flat proof S01, Lemma 1.2 therefore makes M_a free at each local ring; its finite presentation permits extending that basis to a principal neighbourhood. Thus M_a is locally finite free.

Here are the finiteness and global splitting steps. Choose finitely many principal charts $D(b_i)$ covering $\operatorname{Spec} B$ on which M_a is free. A finite subcover exists because the covering elements generate the unit ideal, witnessed by a finite expression for 1. On each chart the surjection $B_{b_i}^m \to (M_a)_{b_i}$ splits and its kernel $(K_a)_{b_i}$ is a finite direct summand. Lift its finite generating list to fractions of elements of K_a and collect their numerators in a finite submodule $K_0 \subset K_a$. Then $(K_a/K_0)_{b_i} = 0$ for every i. Any module element vanishing on these charts is zero: a power of each b_i kills it, and these powers generate 1, since a prime containing them would contain all the b_i. Hence $K_0 = K_a$, so M_a is finitely presented.

For each local splitting $s_i : M_{ab_i} \to B_{b_i}^m$, finite presentation identifies

$$\operatorname{Hom}_B(M_a, B^m)_{b_i} = \operatorname{Hom}_{B_{b_i}}(M_{ab_i}, B_{b_i}^m).$$

To verify this identity, present M_a by finite free modules; both Hom modules are the kernels of the same finite matrix map after exact localization. Clear denominators of s_i, then multiply by a further power of b_i to kill the error on a finite generating list. This gives $t_i : M_a \to B^m$ with $\pi t_i = b_i^{e_i} 1_{M_a}$. Choose r_i satisfying $\sum_i r_i b_i^{e_i} = 1$. The map $\sum_i r_i t_i$ splits π. Consequently

$$M_a \oplus K_a \cong B^m,$$

and both summands are finite projective. This argument also handles $B = 0$ directly by its zero modules.

The identities in F.128 give $C_a = \operatorname{Sym}_B(M_a)$, which is smooth and finitely presented over B. The inverse-variable presentation makes B/R finitely presented, and finite substitution F.28 therefore makes C_a/R finitely presented as well. The conormal calculation F.102 identifies the kernel conormal of $B[Y] \to C_a$ with $K_a \otimes_B C_a$. On principal charts where the split modules are free, this kernel is generated by a subset of polynomial coordinates. Those coordinates form a regular sequence: multiplication by each new variable is injective by polynomial coefficient comparison, even over an arbitrary ring. The regular-list cone calculation in F.118 gives their Koszul exactness in positive degrees; that direction requires no Noetherian hypothesis. Thus F.130 supplies the actual left injection in the conormal transitivity sequence.

If $h \in P$ lifts a, introduce an inverse variable X_0 and use the equation ideals $I' = (I, hX_0 - 1)$ and $J' = (J, hX_0 - 1)$. The explicit localization calculation F.72 and the conormal sequence just justified give

$$0 \longrightarrow C_a \oplus (M_a \otimes_B C_a) \longrightarrow J'/(J')^2 \longrightarrow K_a \otimes_B C_a \longrightarrow 0.$$

The last module is projective because a split finite free presentation stays split after tensoring. Hence

$$J'/(J')^2 \cong C_a \oplus ((M_a \oplus K_a) \otimes_B C_a) \cong C_a^{m+1}.$$

Together with the explicit polynomial quotient for C_a this proves (1). This conormal calculation uses only finite projectivity of M_a and K_a, so it also applies when those properties are supplied by the smooth branch below.

For (2), suppose B/R is smooth, without first assuming (1). F.6, the full presentation comparison F.11 and localization F.65 give the split exact conormal sequence for the original n variables:

$$0 \longrightarrow M_a \longrightarrow B^n \longrightarrow \Omega_{B/R} \longrightarrow 0.$$

Thus M_a is finite projective and $M_a \oplus \Omega_{B/R} \cong B^n$. F.128 gives smoothness of C_a/B; its derivation universal property also identifies $\Omega_{C_a/B} = M_a \otimes_B C_a$, since a derivation from the symmetric algebra is exactly a linear map on its degree-one generators. The full cotangent transitivity calculation F.42 now gives

$$0 \longrightarrow \Omega_{B/R} \otimes_B C_a \longrightarrow \Omega_{C_a/R} \longrightarrow M_a \otimes_B C_a \longrightarrow 0.$$

The right term is projective, so the sequence splits and its middle term is C_a^n. This proves (2). Zero algebras and empty variable or equation lists obey the same identities. □

Proposition D.10: lift smooth

Reading source: Stacks, Tag 07M8.

Let $R \twoheadrightarrow R_0 = R/I$ be a surjection of Noetherian rings. A syntomic R_0-algebra A_0 has a syntomic lift A over R, with a specified isomorphism $A/IA \cong A_0$. If A_0/R_0 is smooth, the lift can be chosen smooth. This is the auxiliary Noetherian lifting assertion used in desingularization.

Proof. If $A_0 = 0$, take $A = 0$: its presentation $R/(1)$ is finite, it is flat, its cotangent modules are zero, and its fibres are empty. Assume $A_0 \neq 0$ and syntomic. Choose a finite polynomial presentation $A_0 = R_0[X]/J_0$ and put $P_0 = J_0/J_0^2$. F.88 makes P_0 finite projective. By F.129 the map is a local complete intersection, so D.9 constructs

$$C_0 = \mathrm{Sym}_{A_0}(P_0)$$

with a free-conormal polynomial presentation and a smooth map $A_0 \to C_0$. Its finite presentation follows from F.128 and finite substitution F.28. The smooth/syntomic and composition proofs F.83 and F.32 make C_0/R_0 syntomic.

The complete equation-basis argument F.55 yields

$$C_0 = R_0[Y_1, \ldots, Y_v]/(\bar{f}_1, \ldots, \bar{f}_c),$$

with these classes a conormal basis. Flatness identifies that conormal after residue-field base change with the fibre conormal, by right exact tensor and the injection of the equation ideal into the polynomial ring. The fibres are locally complete intersections by syntomicness. F.57 therefore makes their point-dimensions $v - c$; applying this at their finitely many minimal primes gives dimension $v - c$ for every nonempty fibre.

Lift each $\bar{f}_j$ to the full polynomial ring $R[Y_1, \ldots, Y_v]$ and set $C^{\mathrm{pre}} = R[Y]/(f_1, \ldots, f_c)$. Its reduction is exactly C_0. F.66(1) gives $g \equiv 1 \pmod{IC^{\mathrm{pre}}}$ such that $C = (C^{\mathrm{pre}})_g$ is a relative global complete intersection. Thus $C/IC = C_0$, with no localization of the special fibre, and F.76 makes C/R syntomic.

Choose a complementary finite projective module Q_0 with $P_0 \oplus Q_0 = A_0^N$. Apply the complete lifting proof F.117 to the finite projective C_0-module $Q_0 \otimes_{A_0} C_0$. It gives a finitely presented étale $C \to C'$ inducing $C'/IC' \cong C_0$ and a finite projective C'-module Q reducing to that specified module. Put $D = \mathrm{Sym}_{C'}(Q)$. It is smooth over C' by F.128, and the proved composition statements make D/R syntomic: étale maps are smooth by their zero cotangent complex and finite presentation, smooth maps are syntomic, and the base rings here are Noetherian finite-type algebras. The actual symmetric-algebra identities from F.128 give

$$\begin{aligned} D/ID &\cong \mathrm{Sym}_{C_0}(Q_0 \otimes_{A_0} C_0) \\ &\cong \mathrm{Sym}_{A_0}(Q_0) \otimes_{A_0} \mathrm{Sym}_{A_0}(P_0) \\ &\cong \mathrm{Sym}_{A_0}(Q_0 \oplus P_0) \cong A_0[Z_1, \ldots, Z_N]. \end{aligned}$$

Choose $z_i \in D$ lifting these variables and set $E = D/(z_1, \dots, z_N)$. Reduction of this quotient gives the specified $E/IE = A_0$.

We verify the claimed syntomic neighbourhood of every $\mathfrak{q} \in V(IE)$ rather than assuming it. Let $\mathfrak{t}$ be its preimage in D and $\mathfrak{p} = \mathfrak{q} \cap R$. Then $I \subset \mathfrak{p}$. By F.87 choose $u \notin \mathfrak{t}$ and a relative global complete-intersection presentation

$$D_u = R[W_1, \dots, W_m]/(a_1, \dots, a_s).$$

In the fibre over $k = \kappa(\mathfrak{p})$ this is the localization of $(A_0 \otimes_{R_0} k)[Z_1, \dots, Z_N]$ at the image of u. Its ordered Z_i are a regular sequence at the prime in question: multiplication by a polynomial variable is injective over any coefficient ring, successive quotients remove one variable, and exact localization preserves each injection. Their ideal stays proper at this prime because it comes from $\mathfrak{q}$ of the quotient.

Lift each z_i to a polynomial b_i in $R[W]$. By F.77, the fibre $\bar{a}_j$ form a regular sequence in the corresponding polynomial local ring. The just proved regularity of Z_i in its quotient makes $(\bar{a}_1, \dots, \bar{a}_s, \bar{b}_1, \dots, \bar{b}_N)$ a regular sequence there. F.57 gives fibre point-dimension $m - s - N$ for

$$E_{\bar{u}} = R[W]/(a_1, \dots, a_s, b_1, \dots, b_N).$$

Apply F.66(3) to this actual $s + N$-equation presentation. It supplies a relative global complete-intersection neighbourhood of $\mathfrak{q}$, hence a syntomic one by F.76. A further localization of $E_{\bar{u}}$ is a principal localization of E: write its inverted element as $e/\bar{u}^r$ and invert $\bar{u}e$. This proves the claim at every point of $V(IE)$.

Let W be the union of these syntomic neighbourhoods and write its closed complement as $V(L)$ in $\operatorname{Spec} E$. Since $V(L) \cap V(IE) = \varnothing$, the ideal $L + IE$ is all of E; otherwise a containing maximal ideal would give a point of that intersection. Choose $h \in L$ with $h \equiv 1 \pmod{IE}$. Then $D(h) \subset W$, and $E_h/IE_h = A_0$ exactly. The finite principal-cover criterion F.62 makes $A = E_h$ syntomic over R: its spectrum has a finite subcover by the already constructed syntomic charts, using the finite expression for 1 in their covering ideal. This proves the first assertion, including $N = 0$.

Now assume A_0/R_0 smooth. For every prime $\mathfrak{q} \in V(IA)$ the fibre of A/R is the same as the fibre of A_0/R_0 over its contracted base prime. Those fibres are smooth by the following calculation, which does not assume that the residue-field map is flat. F.83 covers A_0 by standard smooth principal charts; choose finitely many by their unit-ideal cover. Tensor right exactness changes each polynomial quotient to the quotient by the coefficient-changed equations. Monomial differentiation commutes with changing coefficients, so its chosen Jacobian determinant is the image of the old unit and is still a unit. F.85 makes every fibre chart smooth. The finite expression for 1 in the covering ideal remains such an expression after base change, so these charts cover the fibre; F.67 makes the whole fibre smooth. Since A/R is finitely presented and flat, F.45 gives a smooth neighbourhood of each such $\mathfrak{q}$. Write the complement of their union as $V(L')$. Exactly as above choose $h' \in L'$ with $h' \equiv 1 \pmod{IA}$. Then $A_{h'}/IA_{h'} = A_0$, while F.67 makes $A_{h'}/R$ smooth. No hypothesis that I is nilpotent or lies in the Jacobson radical is needed for either shrinking step. □

For Noetherian R, F.87 gives relative global complete-intersection neighbourhoods of a syntomic map. The next lemma proves the required arbitrary-base extension and constructs a symmetric-algebra bundle over $\operatorname{Spec}(A)$ with a retraction whose total space is a relative global complete intersection over R.

Lemma D.11: syntomic complete intersection

Reading source: Stacks, Tag 07CG.

For a syntomic map $R \to A$ over any ring R, there is a smooth A-algebra C and a retraction $C \to A$ such that

$$C = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$$

is flat over R and every nonempty fibre has dimension $n - c$.

Proof. We first prove the arbitrary-base local complete-intersection assertion needed to apply D.9. Fix a finite presentation $A = R[X_1, \dots, X_N]/(u_1, \dots, u_s)$ and a prime $\mathfrak{q}$ of A over $\mathfrak{p}$ of R. In the polynomial local ring of its $k = \kappa(\mathfrak{p})$ fibre, syntomicness and F.58 give a complete-intersection equation ideal. Its conormal is free by F.57. Select from the u_i a list whose conormal residue classes are a basis. Nakayama and the unit determinant of its coordinate matrix make its local conormal classes a basis; the final assertion of F.57 makes that selected list generate the fibre ideal regularly.

Model the local map $R_{\mathfrak{p}} \to A_{\mathfrak{q}}$ using F.59. Explicitly, for finite-type $\mathbf{Z}$-subalgebras $R_0 \subset R$ containing all equation coefficients, put $T_0 = R_0[X]/(u_i)$ and contract the chosen primes to $\mathfrak{p}_0, \mathfrak{q}_0$. The stages are $R_{0,\mathfrak{p}_0} \to (T_0)_{\mathfrak{q}_0}$. Adding numerators and denominators of any finite set of fractions proves cofinality in the stated local-model system. The models are Noetherian, their transitions are exactly localized base changes, and their colimits are the given local rings. Since $A_{\mathfrak{q}}$ is flat over $R_{\mathfrak{p}}$ and is a finite module over itself, F.21 gives a stage where $(T_0)_{\mathfrak{q}_0}$ is flat over $R_{0,\mathfrak{p}_0}$.

We verify its closed-fibre complete-intersection property. Let $k_0 = \kappa(\mathfrak{p}_0)$. There is an embedding $k_0 \subset k$ by contraction of primes and the fraction-field construction. The map from the polynomial fibre local ring V_0 at the contracted prime to the final polynomial fibre local ring V is local and flat, being field extension followed by localization. The selected equations generate the final fibre ideal. The finite V_0-module consisting of the model ideal modulo their span becomes zero after tensoring with V, so it is zero: a finite module with zero tensor along a local flat map has zero residue-field tensor, and Nakayama kills it. Here the target residue field extends the source residue field, so that residue-field tensor is faithful on vector spaces.

The same argument applies to each positive homology module of the selected finite free Koszul complex. These modules are finite because V_0 is Noetherian. Flat tensor commutes with their kernels and images, so their tensors are zero, since the final list is regular. Nakayama again gives zero homology in the model. The list lies in the maximal ideal, and F.118 makes it regular in V_0. Thus the model has precisely the closed fibre required by the Noetherian-base theorem F.87.

Apply that theorem to obtain $g_0 \notin \mathfrak{q}_0$ with $B_0 = (T_0)_{g_0}$ a relative global complete intersection over R_0. Its scalar extension is exactly A_g, with $g \notin \mathfrak{q}$. Write $B_0 = P_0/(b_1, \dots, b_t)$ using its finite polynomial complete-intersection presentation. By F.126 its augmented Koszul complex is exact, and by F.76 its last quotient B_0 is R_0-flat. Every complex term is also R_0-flat, being finite free over the polynomial ring.

All successive kernels in this augmented complex are R_0-flat. For $0 \to Z \to F \to Q \to 0$ with F, Q flat, quotient flatness makes tensor preserve the left injection. For any inclusion of modules, the resulting map after tensoring with Z is injective by embedding its two tensor products into those for F. Thus Z is flat. Induct upwards from the flat augmentation quotient. Tensoring each short exact sequence with R now remains exact. The actual base-changed equation list for A_g is therefore Koszul-regular. This is the needed arbitrary-base assertion; it does not apply F.87 directly to a non-Noetherian base. The comparison F.94.A transports these presentations to the fixed finite polynomial presentation of A at every prime, proving that $R \to A$ is a local complete intersection.

Let $I = (u_i)$ and $M = I/I^2$. D.9 with $a = 1$ constructs $C = \mathrm{Sym}_A(M)$ with its degree-zero augmentation $C \to A$, proves M finite projective and $A \to C$ smooth, and supplies a polynomial presentation over R with free conormal. The finite-presentation proof F.128 and finite substitution F.28 make C/R finitely presented. The complete equation-basis construction F.55 gives a presentation $C = R[x_1, \dots, x_n]/J$ with $J = (f_1, \dots, f_c)$ and those classes a basis of J/J^2.

On a finite principal cover of $\mathrm{Spec}\, A$ trivializing M, the symmetric-algebra identities in F.128 make C a polynomial algebra over each chart. Those algebras are A-flat. Tensoring an injection with C and detecting its kernel on the cover proves A-flatness of C globally. The given flatness of A/R consequently makes C/R flat.

For a base residue field k, the same identities give $C_k = \mathrm{Sym}_{A_k}(M \otimes_A A_k)$. The original syntomic

hypothesis makes A_k locally a complete intersection. Trivializing the module and restricting to complete-intersection charts makes C_k a polynomial algebra over such a chart. Its regular equations remain regular after polynomial extension and localization, because both are flat; the ambient polynomial local rings are regular. F.58 therefore proves that C_k is locally a complete intersection.

Flatness of C/R makes tensoring $0 \to J \to R[x] \to C \to 0$ injective on the left. Hence $J \otimes_R k$ is the fibre equation ideal $\bar{J}$. Right exactness applied to $J^2 \to J \to J/J^2 \to 0$ identifies its conormal: the image of $J^2 \otimes_R k$ consists exactly of products of the fibre ideal generators. Thus

$$\bar{J}/\bar{J}^2 \cong (J/J^2) \otimes_R k \cong C_k^c.$$

At a minimal prime of a nonzero C_k, F.57 identifies this rank with the height of its polynomial preimage, since the quotient local ring has dimension zero. Each component therefore has dimension $n - c$ by AG-CA-09, Theorem 4.3. The field algebra has finitely many components, so its dimension is their maximum, also $n - c$. The augmentation fixes A, and is the required retraction.

If $A = 0$, take $C = 0$ with presentation $R/(1)$. It is flat, the identity $A \to C$ is smooth, and all fibres are empty. Empty variable or regular lists use the same constructions. No condition has been added to the arbitrary base ring. □

Lemma D.12: smooth standard smooth

Reading source: Stacks, Tag 07CH.

Every smooth algebra A over an arbitrary ring R is a retract of an R-algebra B that is standard smooth over R, with the map $A \to B$ itself smooth. Thus B has a finite polynomial presentation with an invertible square Jacobian minor on its first equation variables.

Proof. If $A = 0$, set $B = 0$. The presentation $R[X]/(1)$ has Jacobian entry 0, which is a unit in the zero ring, and the identity is its smooth retraction. Suppose $A \neq 0$.

The smooth branch of D.9 constructs $C = \mathrm{Sym}_A(I/I^2)$, with $A \to C$ smooth and an augmentation $C \to A$. In this branch the conormal sequence is split by F.6, so the same finite-projective conormal calculation in D.9 applies, independently of any syntomic implication over the base. Appending the inverse variable for 1 gives a finite polynomial presentation of C with free conormal. F.55 replaces it by

$$C = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$$

whose equation classes are a conormal basis. F.29 makes C/R smooth. It is nonzero, because it retracts onto A.

By F.6 and the full comparison F.11, the conormal differential in this presentation is a split injection. Write its matrix with equations as rows as $J = (\partial f_j/\partial X_i)$, of size $c \times n$. A left inverse of the transposed conormal map gives an $n \times c$ matrix T over C such that $JT = 1_c$. Reduction at any maximal ideal of C gives $c \leq n$ by vector-space dimension. Lift its entries to polynomials $\widetilde{T}_{i\ell}$.

Set $E = C[Y_1, \dots, Y_c]$ and introduce redundant coordinates Z_i by the equations

$$h_i = X_i - \sum_{\ell=1}^{c} \widetilde{T}_{i\ell}(X) Y_\ell - Z_i.$$

Eliminating Z_i gives inverse generator maps for

$$E = R[X, Y, Z]/(f_1, \dots, f_c, h_1, \dots, h_n).$$

Let G be the determinant of its square Jacobian on the selected variables $(X_1, \dots, X_n, Y_1, \dots, Y_c)$, using the displayed equation order. Modulo Y in C its matrix is

$$\begin{pmatrix} J & 0 \\ 1_n & -T \end{pmatrix}.$$

Interchange the two row blocks, which contributes $(-1)^{nc}$, then subtract J times the first block row from the second. The resulting matrix is block upper triangular with diagonal $1_n, JT = 1_c$. Its determinant is 1. Thus the image g of G in E satisfies $g \bmod (Y) = (-1)^{nc}$.

Put $B = E_g$. Evaluation at $Y = 0$ sends g to that unit and therefore extends the augmentation to $B \to C$. Its composite with $C \to B$ is identity; composing with $C \to A$ gives the required retraction of $A \to B$.

We verify both remaining smooth maps directly. The polynomial extension $C \to E = C[Y_1, \ldots, Y_c]$ has an empty-relation conormal complex: its degree-one term is zero, and its differential module is finite free on $dY_1, \ldots, dY_c$, by AG-CA-15, Theorem 5.1. Presentation comparison F.11 and the definition F.6 therefore give smoothness. The localization is the finite presentation $B = E[U]/(gU - 1)$, with inverse maps adjoining g^{-1} or evaluating U there. Its conormal is generated by the single equation class, whose differential is $g\,dU$. This coefficient is a unit in B. Hence the surjection $B \to (gU-1)/(gU-1)^2$, followed by the conormal differential to $B\,dU$, is an isomorphism. Both maps are consequently isomorphisms. Its kernel and differential cokernel are zero, so F.11 and F.6 prove smoothness of $E \to B$. Applying the entire composition proof F.29 to $A \to C \to E \to B$ proves that $A \to B$ is smooth. These calculations use no Noetherian or fibre-dimension hypothesis.

Finally adjoining the inverse of G writes

$$B = R[X, Y, U, Z]/(f_1, \ldots, f_c, h_1, \ldots, h_n, UG - 1).$$

Order the selected variables as (X, Y, U), before all the Z_i. There are $n + c + 1$ equations and selected variables. The Jacobian has the earlier square block in its upper left, zeros above the last column, and G in its last diagonal position. Its determinant is therefore G^2, a unit since $UG = 1$. Renaming the variables in this order gives the standard smooth presentation asserted. Empty blocks have determinant 1; the argument includes $c = 0$ and $n = 0$ over any base ring. □

Lemma D.13: colimit standard smooth

Reading source: Stacks, Tag 07CI.

If an R-algebra Λ is a filtered colimit of smooth R-algebras, then it is a filtered colimit of standard smooth R-algebras, over an arbitrary base ring R.

Proof. First consider a map $A \to \Lambda$ with A finitely presented over R. Choose representatives for its finite generator list at one common smooth stage. Each of its finitely many defining relations is zero at a later stage, by the equality rule for filtered colimits. A common successor therefore gives a factorization $A \to B \to \Lambda$ with B/R smooth. By D.12, B is a retract of a standard smooth R-algebra C. The composite $A \to B \to C$ followed by $C \to B \to \Lambda$ is the original map. Thus every such finite-presentation map factors through a standard smooth algebra.

Take as objects finite polynomial presentations of standard smooth R-algebras together with maps to Λ, and as arrows algebra maps respecting those maps. This is a small category: the finite coefficient lists form a set, and each map to Λ is determined by a finite tuple of elements. The presentation of R with empty variable and equation lists is an object. Two objects have a common successor by applying the factorization just proved to the map from their tensor product to Λ; that tensor product has the finite presentation with their disjoint variables and both relation lists. To equalize parallel arrows $u, v\colon B \rightrightarrows C$, choose generators $b_1, \ldots, b_s$ of B and append $u(b_i) - v(b_i)$ to the relations of C. The resulting finitely presented quotient maps to Λ; its factorization through a standard smooth object supplies an arrow out of C equalizing u, v, since equality on generators implies equality on all polynomials in them. These constructions prove that the category is filtered.

Let L be its colimit. The canonical map $L \to \Lambda$ is surjective: factor $R[T] \to \Lambda$ with $T \mapsto \lambda$ to represent any chosen λ at one of the objects. For injectivity, if $b \in B$ has zero image, the finitely presented algebra $B/(b)$ maps to Λ and factors through an object. The resulting successor kills b, so its colimit class is zero. For two representatives with equal images, first move them to a common

successor and use the same argument on their difference. Hence $L \to \Lambda$ is an isomorphism and every stage is standard smooth. Empty lists and zero rings obey the same constructions. □

Lemma D.14: standard smooth include generators

Reading source: Stacks, Tag 07EY.

Suppose A/R is standard smooth and $E = \{a_1, \dots, a_n\} \subset A$ is finite. There is a standard smooth presentation

$$A = R[x_1, \dots, x_{n+m}]/(f_1, \dots, f_c), \qquad c \geq n,$$

whose first n coordinate classes are the specified a_i in any chosen order, and whose Jacobian minor on its first c variables is a unit.

Proof. Start with $A = R[y_1, \dots, y_m]/(g_1, \dots, g_d)$ and rename its variables so that the $d \times d$ minor on $y_1, \dots, y_d$ is a unit. Choose polynomials $h_i(y)$ mapping to a_i. Adjoin variables x_i with equations $x_i - h_i(y)$. Substitution defines an isomorphism

$$R[x_1, \dots, x_n, y_1, \dots, y_m]/(x_i - h_i, g_j) \longrightarrow A.$$

Its inverse sends the old y_j classes to themselves. The two composites fix all y_j, and fix x_i because its new relation identifies it with $h_i(y)$; they are therefore identity maps.

Order the equations by the n new graph equations followed by the d old equations. With equations as rows, the minor on $(x_1, \dots, x_n, y_1, \dots, y_d)$ is

$$\begin{pmatrix} 1_n & -(\partial h_i/\partial y_j) \\ 0 & (\partial g_i/\partial y_j) \end{pmatrix}.$$

In the signed permutation formula, each of its first n columns has only its diagonal entry nonzero. The surviving terms therefore give exactly the old $d \times d$ determinant, which is a unit. Transposing yields the determinant convention with variables as rows and equations as columns. Now set $c = n + d$ and rename $y_j = x_{n+j}$, placing the unselected y variables last. Since $d \leq m$, this is the required minor on the first c variables. The first n classes are the prescribed elements. Empty blocks use determinant 1, so the proof includes an empty E or an empty old equation list. □

Lemma D.15: compare standard

Reading source: AI Integrated Stacks, pinned smoothing, lines 718–850.

For a finitely presented R-algebra A and $a \in A$, put $B = A_a$. Number the following six properties:

1. B/R is smooth.
2. B/R is smooth and $\Omega_{B/R}$ is stably free.
3. B/R is smooth and $\Omega_{B/R}$ is free.
4. B/R admits a standard smooth presentation.
5. a satisfies the strictly standard condition of D.3.
6. a satisfies the elementary standard condition of D.3.

Their comparisons are:

- **(a)** (4) ⇒ (3) ⇒ (2) ⇒ (1).
- **(b)** (6) ⇒ (5).
- **(c)** (6) ⇒ (4).
- **(d)** (5) ⇒ (2).
- **(e)** Property (2) gives an e_0 such that a^e is strictly standard for every $e \geq e_0$.

- **(f)** Every sufficiently large positive power a^e is elementary standard exactly when B/R is smooth and $\Omega_{B/R}$ has a finite basis $\mathrm{d}b_1, \ldots, \mathrm{d}b_d$ represented by elements $b_i \in A$. Having even one elementary positive power implies both these conditions.

In (f) the representatives must come from A, rather than merely from B. Property (4) by itself does not give elementary powers over an arbitrary base. The proof ends with a complete example establishing that distinction.

Proof of (a)–(e). By F.85, a standard smooth algebra is smooth and its differentials are free. A free module is stably free by adding the zero free module; forgetting this condition gives (a). For (b), take the coefficients of the other minors to be zero in D.3s1; transposing the one square minor does not change its determinant. D.3e2 and D.3s2 are the same relation condition.

For (c), use an elementary presentation $A = P/I$, where $P = R[x_1, \ldots, x_n]$, $I = (f_1, \ldots, f_m)$ and the first c equations have minor Δ. Lift a to $h \in P$. Write

$$B = A_a = P[x_0]/J, \qquad J = (x_0 h - 1, f_1, \ldots, f_m),$$

and $K = (x_0 h - 1, f_1, \ldots, f_c)$. The localization conormal calculation F.72 and D.3e2 give $J = K + J^2$: after inverting a, the classes of the remaining f_j are combinations of the first c classes. The module J/K is finite, since J has the displayed finite generating list, and satisfies $J/K = J(J/K)$. Choose generators v_i and write $v_i = \sum_j b_{ij} v_j$ with $b_{ij} \in J$. Multiplication by the adjugate of $(\delta_{ij} - b_{ij})$, whose identity is proved in F.70, shows that

$$g = \det(\delta_{ij} - b_{ij}) \in 1 + J, \qquad gJ \subset K.$$

For the zero module choose $g = 1$. Thus $J_g = K_g$. Since g maps to 1 in B, adjoining an inverse for it does not change B, and

$$B = P[x_0, z]/(x_0 h - 1, f_1, \ldots, f_c, gz - 1).$$

For these ordered equations select $(x_0, x_1, \ldots, x_c, z)$. The first $c + 1$ rows have zero last column. Their first $c + 1$ columns form the block matrix with upper left entry h, lower left zero, and lower right Jacobian minor Δ. The last diagonal entry of the whole matrix is g. Its determinant is therefore $gh\Delta$. In B, $g = 1$, $h = a$ is a unit, and D.3e1 gives $a = a'\Delta$, so $\Delta^{-1} = a'/a$. This proves standard smoothness, including the empty minor case $c = 0$.

For (d), D.3s2 makes the first c equation classes generate $M = (I/I^2)_a$. D.4 and D.3s1 give a map back from B^n whose composite with the map $B^c \to M \to B^n$ is multiplication by the unit a. Thus $B^c \to M$ is both surjective and injective, and the conormal differential is a split injection. D.5 proves that $B = A_a$ is smooth. The localized conormal sequence, by F.65 and F.72, now gives

$$M \oplus \Omega_{B/R} \cong B^n, \qquad M \cong B^c.$$

Consequently $\Omega_{B/R} \oplus B^c \cong B^n$, exactly the required stable freeness.

For (e), start with a finite presentation $A = R[X_1, \ldots, X_n]/I$ and $B = A_a$. If $B = 0$, the equality $1/1 = 0$ means $a^L = 0$ for some L. For every $e \geq L$, take $c = 0$ in any finite presentation: the empty minor is 1 and all multiplied relation classes are zero. Such powers are elementary, hence strictly standard. Assume $B \neq 0$. Smoothness, the conormal description of the naive cotangent complex, and F.65/F.72 give the split exact sequence

$$0 \to M = (I/I^2)_a \to B^n \to \Omega_{B/R} \to 0.$$

If $\Omega_{B/R} \oplus B^s \cong B^t$, then $M \oplus B^t \cong B^{n+s}$. Add t variables Y_i mapping to zero in A. The kernel of this enlarged presentation is $J = (I, Y_1, \ldots, Y_t)$, and

$$J/J^2 \cong I/I^2 \oplus A^t.$$

Indeed, evaluation at $Y = 0$ gives the first component, and the coefficients of the terms linear in Y modulo I give the second; their kernel consists precisely of products of two elements of J. The differential sends the new generators to the independent $\mathrm{d}Y_i$. Thus $(J/J^2)_a$ is finite free and its differential is a split injection into B^{n+t}.

Choose representatives $f_1, \dots, f_c \in J$ of a basis after localization: a basis element is a fraction of an element of J/J^2, and multiplying each basis element by its own denominator, a unit in B, still gives a basis. Extend them to a finite generating list $f_1, \dots, f_m$ of J. Finiteness follows from F.44, or directly from the old finite list and the new variables. Each remaining class is zero in the localization of

$$Q = (J/J^2)/\langle f_1, \dots, f_c \rangle.$$

For each of its finitely many generators some power of a kills that class. Choose a common exponent L. For a polynomial lift h of a this says

$$h^L f_j \in (f_1, \dots, f_c) + J^2 \qquad (j > c).$$

Multiplying by further powers of h gives D.3s2 for every a^e, $e \geq L$.

Let F be the $(n+t) \times c$ matrix of the differentials of the first c equations over A. The split injection after localization has a matrix left inverse P over B. Choose u and a matrix P_0 over A with $(P_0)_a = a^u P$. Every entry of $P_0 F - a^u 1_c$ becomes zero in B, so a common further power a^v kills all those finitely many entries. Hence

$$P_1 F = a^{u+v} 1_c, \qquad P_1 = a^v P_0,$$

as an exact equality over A. Increase $u + v$ to a positive integer k by multiplying P_1 by an additional power of a if necessary. No power of a has been cancelled. F.70 puts a^{ck} in the ideal of the $c \times c$ minors of F. Reduction at a prime of the nonzero ring B shows $c \leq n + t$. For every $e \geq \max(L, ck, 1)$, multiply the minor coefficients by a^{e-ck} to obtain D.3s1, together with the already established D.3s2. When $c = 0$, the minor ideal is A and the same conclusion uses the empty determinant 1. This proves (e) over arbitrary rings, including zero divisors. □

Proof of (f). Put $B = A_a$. If $B = 0$, some power a^L is zero. The zero differential module has the empty basis, and, as in (e), every a^e with $e \geq \max(L, 1)$ is elementary using $c = 0$ and the empty minor 1. Assume $B \neq 0$.

First suppose B is smooth and $\mathrm{d}b_1, \dots, \mathrm{d}b_d$ is a basis with $b_i \in A$. Choose finite algebra generators $z_1, \dots, z_n$ of A over R and the surjection

$$P = R[X_1, \dots, X_n, Y_1, \dots, Y_d] \longrightarrow A, \qquad X_i \longmapsto z_i, \quad Y_j \longmapsto b_j.$$

Its kernel J is finitely generated by F.44. The smoothness definition F.6, the presentation description of the naive cotangent complex F.11, and localization F.65 give the exact sequence

$$0 \longrightarrow M = (J/J^2)_a \xrightarrow{\mathrm{d}} B^n \oplus B^d \longrightarrow \Omega_{B/R} \longrightarrow 0.$$

Here the two summands have coordinates $\mathrm{d}X_i$ and $\mathrm{d}Y_j$. The map from the second summand to $\Omega_{B/R}$ is an isomorphism, because its images are the chosen basis. Consequently projection onto the first summand restricts to an isomorphism $M \to B^n$: for each $v \in B^n$, there is a unique $w \in B^d$ whose differential image cancels that of v, so (v, w) is in the image of M; a vector of M with first component zero has second component zero as well. Injectivity of d proves uniqueness in M.

Let $\eta_1, \dots, \eta_n$ be the resulting basis of M whose projected differentials are the standard basis of B^n. Write each η_i as a fraction of a class represented by $f_i \in J$, with denominator a^{k_i}. Thus the classes of f_i are $a^{k_i}\eta_i$, still a basis over B. Their square Jacobian on $X_1, \dots, X_n$ is diagonal after passage to B, with entries a^{k_i}. In particular the image of

$$\delta = \det(\partial f_i / \partial X_j)_{i,j=1}^n \in A$$

is a unit in B. Extend $f_1, \dots, f_n$ to a finite generating list $f_1, \dots, f_m$ of J. The finite module

$$Q = (J/J^2)/\langle f_1, \dots, f_n \rangle$$

has zero localization at a. A common power a^L kills its finite generating list. For a polynomial lift $h \in P$ of a, this means

$$h^L f_j \in (f_1, \dots, f_n) + J^2 \qquad (j > n).$$

All higher powers satisfy the same relation condition.

Since δ is a unit in B, choose $u \in A$ and $r \geq 0$ with $\delta(u/a^r) = 1$ in B. The equality in this localization supplies $v \geq 0$ such that

$$a^v(\delta u - a^r) = 0 \quad \text{in } A, \qquad a^{r+v} = \delta(a^v u).$$

Increase the exponent to a positive K by multiplying this equality by an additional power of a if necessary. This gives $a^K = \delta u_0$ in A, without cancelling a power of a possible zero divisor. Every $e \geq \max(K, L, 1)$ then satisfies both D.3e1 and D.3e2 for the presentation $A = P/J$, with $c = n$ and the selected variables X_i. This proves the forward direction, including $n = 0$, when δ is the empty determinant 1.

Conversely, suppose a^r is elementary for some $r \geq 1$. Inverting a^r is the same as inverting a, since $a^{-1} = a^{r-1}(a^r)^{-1}$. Part (c), applied to a^r, proves that B is standard smooth, hence smooth. Take its elementary presentation $A = R[x_1, \dots, x_N]/I$ with $I = (g_1, \dots, g_m)$ and selected minor on $x_1, \dots, x_c$. In B, D.3e2 makes every remaining conormal class a combination of the first c. Therefore

$$\Omega_{B/R} = B^N/\langle \mathrm{d}g_1, \dots, \mathrm{d}g_c \rangle.$$

The selected $c \times c$ Jacobian block is invertible. Subtracting a unique combination of these c differential rows eliminates the first c coordinates of any vector in B^N; a combination with those coordinates zero has all its coefficients zero. Hence the images of $\mathrm{d}x_{c+1}, \dots, \mathrm{d}x_N$ form a basis of this quotient. Their representatives are elements of A. The forward direction now proves that every sufficiently large power is elementary. This proves (f). □

Why standard smoothness alone is insufficient. Take

$$R = \mathbf{Z}, \qquad A = \mathbf{Z}[t^2, t^3] \subset \mathbf{Z}[t], \qquad a = t^2.$$

This algebra has the finite presentation

$$A \cong \mathbf{Z}[X, Y]/(Y^2 - X^3), \qquad X \longmapsto t^2, \quad Y \longmapsto t^3.$$

Indeed division by the monic polynomial $Y^2 - X^3$ writes every class uniquely as $p(X) + Yq(X)$. After substitution, the first summand has even exponents and the second has odd exponents at least 3; their disjoint monomials prove injectivity. The image is precisely the indicated subalgebra.

Inverting a gives

$$B = A_a = \mathbf{Z}[t, t^{-1}], \qquad t = Y/X, \quad t^{-1} = Y/X^2.$$

These identities follow from $Y^2 = X^3$ and give both inclusions with the Laurent ring. The presentation $B = \mathbf{Z}[T, U]/(TU - 1)$ is standard smooth: select U for its one equation, whose derivative is the unit T. Every derivation of B is uniquely determined by its value on t, with $\mathrm{d}(t^{-1}) = -t^{-2}\mathrm{d}t$. Thus $\Omega_{B/\mathbf{Z}} = B\,\mathrm{d}t$.

The units of $\mathbf{Z}[t, t^{-1}]$ are exactly $\pm t^k$. To check this, the lowest and highest nonzero exponents of a product add, since their coefficients are nonzero products in the domain $\mathbf{Z}$. For a product equal to 1, each factor must therefore have width zero and be a monomial; its integer coefficient must be 1 or -1.

Every $b \in A$ has the form $b_0 + \sum_{k\geq 2} b_k t^k$, with integer coefficients and a finite sum. Consequently

$$\mathrm{d}b = \left(\sum_{k\geq 2} k b_k t^{k-1} \right) \mathrm{d}t.$$

This coefficient cannot be a unit: if it were $\pm t^j$, necessarily $j \geq 1$, and its sole coefficient would require $(j+1)b_{j+1} = \pm 1$, impossible in $\mathbf{Z}$. A basis of the free rank-one module $B\,\mathrm{d}t$ must contain one generator; tensoring with $\mathbf{Q}(t)$ verifies that its length is one. Hence it has no basis consisting of differentials of elements of A. By the converse in (f), no positive power of a is elementary standard, despite standard smoothness of B.

There is no conflict with (e). In the original cusp presentation the two one-by-one minors are $-3X^2$ and $2Y$, and in A

$$X^3 = (-X)(-3X^2) + (-Y)(2Y).$$

The relation condition is empty because there is only one equation. Thus a^3, and every a^e with $e \geq 3$ after multiplication by a^{e-3}, is strictly standard. This exact example distinguishes the sum-of-minors condition from the single-minor condition over the same arbitrary base.

The lifting problem

D.18 lifts a smooth filtered presentation across a nilpotent ideal when the target remains flat. Its two preliminary steps use the explicit smooth presentations constructed in the preceding section.

Lemma D.16: lift once

Reading source: Stacks, Tag 07CK.

Let $R \to \Lambda$ be a ring map and $I \subset R$ an ideal with $I^2 = 0$. Suppose that $\Lambda/I\Lambda$ is a filtered colimit of smooth R/I-algebras. Every map $\varphi : A \to \Lambda$ from a finitely presented R-algebra factors as

$$A \longrightarrow B/J \longrightarrow \Lambda,$$

where B/R is smooth and $J \subset IB$ is a finitely generated ideal. Neither R nor I is required to be Noetherian or finite.

Proof. Write $\bar{R} = R/I$ and $\bar{A} = A/IA$. By D.13, the induced map $\bar{A} \to \Lambda/I\Lambda$ factors through a standard smooth $\bar{R}$-algebra C. This factorization occurs at a finite stage: take representatives for the finite generators of $\bar{A}$ and then a common successor where all its finitely many equations vanish.

There is a finite presentation $C = \bar{A}[t_1, \ldots, t_r]/(\bar{g}_1, \ldots, \bar{g}_s)$ relative to this specified map. To see the finiteness directly, choose $C = \bar{R}[t]/(k_1, \ldots, k_v)$ and generators $a_1, \ldots, a_u$ of $\bar{A}$. Represent their images by polynomials $v_i(t)$. Over $\bar{A}[t]$, the finite equations $k_j(t)$ and $a_i - v_i(t)$ give exactly C: substituting for the a_i recovers its displayed $\bar{R}$-presentation. Include these equations in the list $\bar{g}$.

Lift each $\bar{g}_i$ to $g_i \in A[t]$ and the images of the t_j in $\Lambda/I\Lambda$ to $\lambda_j \in \Lambda$. Evaluation gives a finite expression

$$\varphi(g_i)(\lambda) = \sum_{j=1}^{q_i} \epsilon_{ij}\mu_{ij}, \qquad \epsilon_{ij} \in I, \quad \mu_{ij} \in \Lambda.$$

Introduce one variable z_{ij} for each of these terms and set

$$T = A[t, z]/(g_i - \sum_j \epsilon_{ij} z_{ij}).$$

This is finitely presented over R, by combining the finite presentation of A with the displayed lists. The assignments $a \mapsto \varphi(a)$, $t_j \mapsto \lambda_j$, $z_{ij} \mapsto \mu_{ij}$ define $T \to \Lambda$, and reduction gives

$$T/IT \cong C[z].$$

Adjoining free variables to a standard smooth presentation preserves its invertible Jacobian minor: retain the equations and their pivot variables. Thus choose

$$T/IT = (R/I)[X_1, \dots, X_n]/(\bar{f}_1, \dots, \bar{f}_c)$$

with $\bar{\Delta} = \det(\partial \bar{f}_i/\partial X_j)_{1\le i,j\le c}$ invertible. Lift the f_i to $R[X]$, put $\Delta = \det(\partial f_i/\partial X_j)_{1\le i,j\le c}$, and form

$$B = R[X_1, \dots, X_n, W]/(f_1, \dots, f_c, W\Delta - 1).$$

On pivots $X_1, \dots, X_c, W$, its Jacobian has upper right block zero and lower right entry Δ; its determinant is Δ^2, a unit in B. It is therefore standard smooth, hence smooth by F.85. Its reduction is precisely T/IT, since W becomes the inverse of $\bar{\Delta}$.

We construct a map $\psi : B \to T$ inducing that isomorphism. Choose lifts $y_i \in T$ of the displayed coordinate classes in T/IT, and let $L = IT$, so $L^2 = 0$. The element $\Delta(y)$ is a unit: a lift d of its modular inverse satisfies $\Delta(y)d = 1 + \ell$ with $\ell \in L$, and $(1 + \ell)^{-1} = 1 - \ell$. Let $H = (\partial f_i/\partial X_j(y))_{1\le i,j\le c}$. Its determinant is $\Delta(y)$, so the cofactor identity proved in F.70 gives an inverse matrix. Every $f_i(y)$ lies in L. Choose

$$(u_1, \dots, u_c)^{\mathsf{t}} = -H^{-1}(f_1(y), \dots, f_c(y))^{\mathsf{t}} \in L^c$$

and set $u_j = 0$ for $j > c$. Expanding each monomial, all terms with two correction factors vanish in L^2, and hence

$$f_i(y + u) = f_i(y) + \sum_{j=1}^{c} \frac{\partial f_i}{\partial X_j}(y)u_j = 0.$$

The new determinant is still a unit because its reduction is unchanged. Set $\psi(X_i) = y_i + u_i$ and $\psi(W) = \Delta(y + u)^{-1}$. These assignments satisfy every equation of B and lift the required reduction map.

The map ψ is onto without a finiteness assumption on I. For $t \in T$, lift its reduction to $b \in B$ and write $t - \psi(b) = \sum_\nu \eta_\nu t_\nu$ with $\eta_\nu \in I$. Lift each reduction of t_ν to $b_\nu \in B$. Since $I(IT) = 0$, one obtains $t = \psi(b + \sum_\nu \eta_\nu b_\nu)$. Its kernel J lies in IB, because $B/IB \to T/IT$ is injective. To prove that J is finite, compose the finite polynomial presentation $R[X, W] \twoheadrightarrow B$ with ψ. The kernel of the resulting surjection to the finitely presented algebra T is finite by F.44. Its images in B generate J. Thus $T \cong B/J$, and the maps $A \to T \to \Lambda$ give the claimed factorization. Empty equation lists use $\Delta = 1$ and require no correction coordinates. □

Lemma D.17: lift twice

Reading source: Stacks, Tag 07CL.

Let $R \to \Lambda$ be flat, let $I^2 = 0$, and suppose that $\Lambda/I\Lambda$ is a filtered colimit of smooth R/I-algebras. If B/R is smooth, $\varphi : B \to \Lambda$ is an R-map, and $J \subset IB$ is a finitely generated ideal with $\varphi(J) = 0$, there are maps

$$B \xrightarrow{\alpha} B' \xrightarrow{\beta} \Lambda$$

with B'/R smooth, $\alpha(J) = 0$, and the exact equality $\beta\alpha = \varphi$.

Proof. We first give the square-zero lifting calculation needed below. For any smooth R-algebra B, any R-algebra E, any ideal $L \subset E$ with $L^2 = 0$, and any R-map $\gamma : B \to E/L$, there is a lift $B \to E$. Choose a finite polynomial presentation $P = R[Z_1, \dots, Z_n] \twoheadrightarrow B$, with kernel K, and lifts $e_i \in E$ of $\gamma(Z_i)$. Evaluation of a relation $f \in K$ lies in L and vanishes on K^2. It gives a B-linear map

$$\delta : K/K^2 \longrightarrow L,$$

where L is a B-module through γ; its E/L-module structure is well defined because $L^2 = 0$. Indeed evaluation of pf is evaluation of p times evaluation of f, and that multiplication on L depends only on p modulo K and E modulo L.

By the definition F.6 and presentation comparison F.11, the sequence

$$0 \longrightarrow K/K^2 \xrightarrow{d} B^n \longrightarrow \Omega_{B/R} \longrightarrow 0$$

is exact and its last module is finite projective. This sequence splits: realize that projective module as a direct summand of a finite free module, lift the free basis through the surjection, and restrict the lift to the summand. The resulting section defines a retraction $r : B^n \to K/K^2$ of d. Put $\ell = \delta r : B^n \to L$ and $c_i = \ell(dZ_i)$. Evaluate polynomials instead at $e_i - c_i$. The monomial expansion, using $L^2 = 0$, gives for each $f \in K$

$$f(e - c) = f(e) - \sum_i \frac{\partial f}{\partial Z_i}(e)c_i = \delta([f]) - \ell(d[f]) = 0.$$

In the second equality the derivative coefficients act on L through their images under γ, so no choice of their lifts changes the product. Evaluation therefore descends to $B \to E$ and lifts γ. This proves the needed lifting statement in full arbitrary-ring scope.

Now suppose first that $J = (h)$. Write $h = \sum_{i=1}^{q} \epsilon_i b_i$ with $\epsilon_i \in I$ and $b_i \in B$. Consider the row map $v : R^q \to R$, $v(a) = \sum_i \epsilon_i a_i$, and let $K_v = \ker v$, $H_v = \operatorname{im} v$. Flatness preserves the two exact sequences $0 \to K_v \to R^q \to H_v \to 0$ and $0 \to H_v \to R$. Consequently

$$\ker(v \otimes_R \Lambda) = \operatorname{im}(K_v \otimes_R \Lambda \longrightarrow \Lambda^q).$$

The vector $(\varphi(b_i))_i$ belongs to this kernel, since $\varphi(h) = 0$. An element of a tensor product is a finite sum of pure tensors. Thus there are finitely many $\lambda_j \in \Lambda$ and columns $(a_{ij})_i \in K_v$ with

$$\varphi(b_i) = \sum_{j=1}^{m} a_{ij}\lambda_j, \qquad \sum_i \epsilon_i a_{ij} = 0 \quad \text{for each } j.$$

This derives the finite relation directly from flatness, including the case of a zero vector and an empty tensor sum.

Set $C = B[T_1, \dots, T_m]/(b_i - \sum_j a_{ij}T_j)$. Its finite presentation follows by appending these variables and equations to a finite presentation of B. The displayed relations define a map $C \to \Lambda$ extending φ, with $T_j \mapsto \lambda_j$. By D.16 this map factors through E/J', where E/R is smooth and $J' \subset IE$ is finite. In particular $(J')^2 = 0$. Apply the lifting calculation above to $B \to C \to E/J'$ and obtain $\alpha : B \to E$. Let $\beta : E \to E/J' \to \Lambda$. Since the reduction of α is the original map through C, one has $\beta\alpha = \varphi$ exactly.

Choose lifts $\xi_j \in E$ of the images of T_j. The equations in C give $\alpha(b_i) = \sum_j a_{ij}\xi_j + \theta_i$ for some $\theta_i \in J'$. Therefore

$$\alpha(h) = \sum_j \Big(\sum_i \epsilon_i a_{ij}\Big)\xi_j + \sum_i \epsilon_i \theta_i = 0.$$

The first sum vanishes by the column relations, the second because $IJ' \subset I^2E = 0$. This proves the one-generator case, with $B' = E$.

For $J = (h_1, \dots, h_s)$, apply that construction successively to the image of h_i in the smooth algebra obtained at the preceding step. Each such image lies in I times that algebra, and its map to Λ is zero because the composite to Λ remains φ. Each new map preserves the previously killed generators. After finitely many steps their composite kills every generator, hence the whole ideal J, and still factors φ exactly. If $J = 0$, take $B' = B$, $\alpha = 1$, and $\beta = \varphi$. □

Proposition D.18: lift

Reading source: Stacks, Tag 07CM.

Suppose that $R \to \Lambda$ is flat, $I \subset R$ is nilpotent, and $\Lambda/I\Lambda$ is a filtered colimit of smooth R/I-algebras. Then Λ is a filtered colimit of smooth R-algebras.

Proof. Begin with $I^2 = 0$. Given any R-map $A \to \Lambda$ from a finitely presented algebra, D.16 gives maps $A \to B/J \to \Lambda$ with B/R smooth, $J \subset IB$ finite. Let $\varphi : B \to \Lambda$ be the induced map; it kills J. The flat map and the same square-zero ideal meet every hypothesis of D.17. Obtain $B \to B' \to \Lambda$ with B'/R smooth and J killed in B'. The first arrow descends to $B/J \to B'$, and its composite to Λ is the given quotient map because D.17 gives equality of the maps from B. Thus $A \to B' \to \Lambda$ is an exact factorization of the original map. Criterion F.90, whose filtered-category construction and colimit isomorphism are proved there, gives the assertion for square-zero ideals.

Induct on a positive integer n with $I^n = 0$. For $n = 1$ the ideal is zero and the assertion is the hypothesis. For $n \geq 2$, set $J = I^{n-1}$, so $J^2 = 0$ because $2n - 2 \geq n$. Over $\bar{R} = R/J$ put $\bar{\Lambda} = \Lambda/J\Lambda$ and $\bar{I} = I/J$. Then $\bar{I}^{n-1} = 0$ and $\bar{\Lambda}/\bar{I}\bar{\Lambda} = \Lambda/I\Lambda$. The map $\bar{R} \to \bar{\Lambda}$ is flat: for every $\bar{R}$-module M the maps on pure tensors give

$$M \otimes_{\bar{R}} (\Lambda/J\Lambda) \cong M \otimes_R \Lambda.$$

Here tensors involving $J\Lambda$ vanish because $JM = 0$; this verifies the inverse maps as well. An exact sequence of $\bar{R}$-modules is exact over R, so the original flatness makes the displayed tensor functor exact. The induction hypothesis therefore expresses $\bar{\Lambda}$ as a filtered colimit of smooth $\bar{R}$-algebras. Apply the square-zero case already proved to $J \subset R$ and the original flat map $R \to \Lambda$. Its quotient is exactly this smooth colimit, hence it expresses Λ as a filtered colimit of smooth R-algebras. This completes the induction without assuming that the nilpotent ideal is finitely generated. □

The lifting lemma

D.19 constructs a finitely presented algebra whose smooth locus lifts the prescribed quotient data.

Lemma D.19: lifting

Reading source: Stacks, Tag 07CP.

Let R be Noetherian, let $\pi \in R$, let Λ be an R-algebra, and assume

$$\operatorname{Ann}_R(\pi) = \operatorname{Ann}_R(\pi^2), \qquad \operatorname{Ann}_\Lambda(\pi) = \operatorname{Ann}_\Lambda(\pi^2).$$

Given a finitely presented R/π^2R-algebra $\bar{C}$ and a map $\bar{C} \to \Lambda/\pi^2\Lambda$, there is a finitely presented R-algebra D, a map $D \to \Lambda$, and commuting maps

$$\begin{array}{ccccc} R/\pi^2R & \longrightarrow & \bar{C} & \longrightarrow & \Lambda/\pi^2\Lambda \\ \downarrow & & \downarrow & & \downarrow \\ R/\pi R & \longrightarrow & D/\pi D & \longrightarrow & \Lambda/\pi\Lambda. \end{array}$$

The map $R \to D$ is smooth at every prime avoiding π. It is also smooth at each prime containing π whose inverse image in $\bar{C}$ is smooth over R/π^2R. At these latter primes, $\bar{C}/\pi\bar{C} \to D/\pi D$ is smooth as well.

Proof. Put $P = R[x_1, \dots, x_n]$ and choose lifts $f_1, \dots, f_m \in P$ of a finite relation list for

$$\bar{C} = (R/\pi^2R)[x_1, \dots, x_n]/(\bar{f}_1, \dots, \bar{f}_m).$$

Write $\bar{P} = (R/\pi^2R)[x]$ and $\bar{J} = (\bar{f}_1, \dots, \bar{f}_m)$. Thus the kernel of $P \to \bar{C}$ is $(f_1, \dots, f_m) + \pi^2 P$; π^2 need not be in the chosen list.

At a smooth prime, D.2 supplies a principal neighbourhood, selected equation indices E, and equally many variable indices U. On that neighbourhood the selected Jacobian minor δ is a unit and the selected classes generate $\bar{J}/\bar{J}^2$. They are a basis: their composite with the conormal

differential projected onto the U-coordinates is invertible, so a relation among them is zero. Shrink the polynomial neighbourhood to make $\bar{J}$ itself generated by these selected equations. Here are the details of that shrinking. The finite module

$$M = (\bar{J}/(\bar{f}_j : j \in E))_a$$

satisfies $M = \bar{J}_a M$. The determinant form of F.10 gives $b' \in \bar{J}_a$ with $(1 + b')M = 0$. Write $b' = b/a^N$, with $b \in \bar{J}$. Replacing the polynomial a by $a(a^N + b)$ inverts $1 + b'$. Its image in $\bar{C}$ is the old a^{N+1}, so this replacement preserves the principal open in $\operatorname{Spec} \bar{C}$.

These principal opens cover the smooth locus. They have a finite subcover for an algebraic reason: $\bar{C}$ is Noetherian by S01, Lemma 1.3, so the ideal generated by all their defining elements is generated by finitely many of those elements. To obtain such a list, express a finite ideal generating list as finite sums of the original elements and collect those that occur. Their principal opens have the same union. Index the resulting charts by $k = 1, \dots, r$. An empty smooth locus uses $r = 0$.

Clear the finitely many denominators in the equality of localized relation ideals on each chart and replace its a_k by a sufficiently large positive power. There are then exact identities in P

$$a_k f_\ell = \sum_{j \in E_k} h^j_{k\ell} f_j + \pi^2 g_{k\ell}. \tag{D.19a}$$

For $\ell \in E_k$, take $h^j_{k\ell} = a_k \delta_{\ell j}$ and $g_{k\ell} = 0$. The selected conormal classes remain a basis on $\bar{C}_{a_k}$, and the selected Jacobian minor δ_k remains invertible there.

In $Q = P[z_1, \dots, z_m]$, put

$$F_\ell = f_\ell - \pi z_\ell, \qquad p_{k\ell} = a_k z_\ell - \sum_{j \in E_k} h^j_{k\ell} z_j - \pi g_{k\ell},$$

and define $D = Q/(F_\ell, p_{k\ell})_{\ell,k}$. All lists are finite. Choose lifts $\lambda_i \in \Lambda$ of the prescribed images of x_i, and write $f_\ell(\lambda) = \pi^2 \mu_\ell$. Send x_i to λ_i and z_ℓ to $\pi \mu_\ell$. The F_ℓ vanish. Evaluation of (D.19a) shows that

$$\pi^2 \left(a_k(\lambda)\mu_\ell - \sum_{j \in E_k} h^j_{k\ell}(\lambda)\mu_j - g_{k\ell}(\lambda) \right) = 0.$$

The annihilator hypothesis in Λ makes its product with π zero; that product is exactly the image of $p_{k\ell}$. This defines $D \to \Lambda$. Reduction of $F_\ell = 0$ modulo π defines $\bar{C} \to D/\pi D$, and the variable assignments verify every square in the displayed diagram.

The polynomial identity

$$\pi p_{k\ell} = -a_k F_\ell + \sum_{j \in E_k} h^j_{k\ell} F_j \tag{D.19b}$$

follows by substituting (D.19a). After inverting π, all $p_{k\ell}$ follow from the F_ℓ, and $F_\ell = 0$ eliminates $z_\ell = f_\ell/\pi$. Thus $D_\pi = R_\pi[x]$, a polynomial localization smooth over R by the empty-equation and inverse-variable cases of F.85.

Fix k, and retain only its selected equations and its own compatibility equations:

$$D_k = Q/(F_j : j \in E_k,\ p_{k\ell} : \ell \notin E_k).$$

There is a surjection $D_k \to D$. Let $\mathfrak{q} \subset D$ contain π and lie over a smooth prime of $\bar{C}$; choose k with $a_k \notin \mathfrak{q}$, and let $\mathfrak{q}_k$ be its inverse image in D_k. In $D_k/\pi D_k$, after inverting a_k, each z_ℓ for $\ell \notin E_k$ is uniquely

$$z_\ell = a_k^{-1} \sum_{j \in E_k} h^j_{k\ell} z_j.$$

Substitution and its reverse on the remaining variables give the actual isomorphism

$$(D_k/\pi D_k)_{a_k} \cong (\bar{C}/\pi\bar{C})_{a_k}[z_j : j \in E_k]. \tag{D.19c}$$

Here the equality of the relation ideals modulo π^2 also gives their equality modulo π.

There are m equations in this presentation of D_k. Choose as pivot variables x_i, $i \in U_k$, followed by z_ℓ, $\ell \notin E_k$. Their Jacobian modulo π, with selected equations first, has blocks

$$\begin{pmatrix} (\partial f_j/\partial x_i)_{j\in E_k, i\in U_k} & 0 \\ * & a_k 1_{m-|E_k|} \end{pmatrix}.$$

Its determinant is $\delta_k a_k^{m-|E_k|}$, invertible at $\mathfrak{q}_k$. If Δ_k denotes the determinant before reduction, $(D_k)_{a_k\Delta_k}$ is standard smooth: adjoining an inverse of $a_k\Delta_k$ adds one equation and one pivot, whose block determinant is $(a_k\Delta_k)\Delta_k$. F.85 proves smoothness. F.83, with the Noetherian base R, proves flatness of this smooth chart. Localization is exact by S01, Lemma 1.1, so $T = (D_k)_{\mathfrak{q}_k}$ is flat over R.

We now show that every additional relation of $D_k \to D$ vanishes in T. First (D.19b), with the k-relations zero and a_k invertible, gives $F_\ell = 0$ for all ℓ. For another chart k', set

$$b^j_{kk'\ell} = a_{k'} h^j_{k\ell} - \sum_{j'\in E_{k'}} h^{j'}_{k'\ell} h^j_{kj'} \quad (j \in E_k).$$

Reduce (D.19a) modulo π^2 and express $a_k a_{k'}[\bar f_\ell]$ in the conormal basis indexed by E_k in the two possible orders. The difference is $\sum_{j\in E_k} b^j_{kk'\ell}[\bar f_j] = 0$. Basis independence therefore makes every coefficient zero in $\bar C_{a_k}$. Equivalently,

$$b^j_{kk'\ell} \in ((f_j : j \in E_k) + \pi^2 P)_{a_k}.$$

Its image in $(D_k)_{a_k}$ belongs to $\pi(D_k)_{a_k}$, since every selected f_j equals πz_j. Directly expanding the definitions, modulo π, gives

$$a_k p_{k'\ell} - a_{k'} p_{k\ell} + \sum_{j'\in E_{k'}} h^{j'}_{k'\ell} p_{kj'} = \sum_{j\in E_k} b^j_{kk'\ell} z_j.$$

Consequently $p_{k'\ell} \in \pi T$. Identity (D.19b) for k' also gives $\pi p_{k'\ell} = 0$ in T.

Flat tensor applied to the multiplication-kernel sequences identifies

$$\mathrm{Ann}_T(\pi^e) = \mathrm{im}(\mathrm{Ann}_R(\pi^e) \otimes_R T \to T) \qquad (e = 1, 2).$$

Indeed, factor multiplication through its image and tensor its short exact kernel sequence and that image's injection into R; flatness preserves both. The assumed equality over R thus gives equality of these annihilators in T. For $p = p_{k'\ell} = \pi u$, the equality $\pi p = 0$ means $\pi^2 u = 0$; it follows that $p = \pi u = 0$. This kills all additional generators.

Finally this local calculation supplies genuine neighbourhoods. The kernel of $D_k \to D$ is generated by the finite lists of additional F_ℓ and $p_{k'\ell}$. For each generator, vanishing at $\mathfrak{q}_k$ gives an element outside $\mathfrak{q}_k$ killing it. Their product s kills the entire kernel. The surjection therefore becomes an isomorphism after inverting $a_k\Delta_k s$, which still avoids $\mathfrak{q}_k$. This identifies a principal neighbourhood of $\mathfrak{q}$ with a localization of the standard smooth chart, proving the required smoothness of $R \to D$ by F.85 and F.7. Modulo π, (D.19c) identifies the same neighbourhood with a localization of a polynomial algebra over $\bar C/\pi\bar C$: denominators in the a_k-localized polynomial algebra can be cleared by a power of a_k. F.85 proves its smoothness over that base. This gives the second claimed smoothness assertion as well. Empty equation or chart lists use determinant and denominator product 1; the same constructions cover zero algebras. □

The desingularization lemma

D.20 combines a lifting witness with extra equations to enlarge the singular ideal in the target.

Lemma D.20: desingularize

Reading source: Stacks, Tag 07CR.

Let R be Noetherian, let $\pi \in R$, and let Λ be an R-algebra with $\operatorname{Ann}_\Lambda(\pi) = \operatorname{Ann}_\Lambda(\pi^2)$. Suppose A is finitely presented over R, a map $A \to \Lambda$ is given, the image of π is strictly standard in A/R, and a section

$$\rho : A/\pi^4 A \longrightarrow R/\pi^4 R$$

agrees with the prescribed map to $\Lambda/\pi^4\Lambda$. Then there is a factorization $A \to B \to \Lambda$, with B/R finitely presented, such that

$$\mathfrak{a}B \subset H_{B/R}, \qquad \mathfrak{a} = \operatorname{Ann}_R(\operatorname{Ann}_R(\pi^2)/\operatorname{Ann}_R(\pi)).$$

No annihilator equality in R is assumed.

Proof. Choose the strict witness from D.3:

$$A = R[x_1, \dots, x_n]/(f_1, \dots, f_m), \quad 0 \le c \le \min(n, m), \quad I = (f_1, \dots, f_m).$$

It gives a sum-of-minors identity for π and the relations $\pi f_\ell \in (f_1, \dots, f_c) + I^2$ for $\ell > c$. Subtract lifts in R of the finitely many values $\rho(x_i)$ from the variables. This translation is a polynomial automorphism, preserves both witness identities, and makes $\rho(x_i) = 0$. Hence $f_j(0) \in \pi^4 R$, and the actual image of x_i in Λ has the form $\pi^4\lambda_i$.

Write the linear terms of f_j as $\sum_i r_{ji}x_i$, and let $M = (r_{ji})_{j\le c, i\le n}$. Evaluate the sum-of-minors identity at ρ. There are coefficients $d_U \in R$ and $u \in 1 + \pi^3 R$ such that

$$u\pi = \sum_{|U|=c} d_U \det(M_U).$$

Apply F.70 to M^{t} and transpose its resulting identity. This supplies an $n \times c$ matrix $S = (s_{ij})$ with $MS = u\pi 1_c$.

Introduce $v_1, \dots, v_c, w_1, \dots, w_n$, set $L_i = \sum_j s_{ij}v_j + \pi w_i$, and substitute $x_i = \pi^2 L_i$. For $j \le c$, constant terms are divisible by π^4, linear terms become

$$\pi^2 \sum_i r_{ji}L_i = \pi^3 u v_j + \pi^3 \sum_i r_{ji}w_i,$$

and every monomial of degree $d \ge 2$ becomes divisible by π^{2d}. Dividing these specified powers in the coefficient expressions, rather than cancelling a ring element, constructs a polynomial G_j and an exact identity

$$f_j(\pi^2 L) = \pi^3 g_j, \qquad g_j = uv_j + \sum_i r_{ji}w_i + \pi G_j. \tag{D.20a}$$

For example, a constant term $\pi^4 a_j$ contributes a_j to G_j, and a degree-d term with coefficient b_α contributes $b_\alpha \pi^{2d-4} L^\alpha$. Thus this construction is valid even when π is a zero divisor.

Let $h_i = x_i - \pi^2 L_i$, and define

$$B = R[x, v, w]/(h_i,\ g_j : j \le c,\ f_\ell : 1 \le \ell \le m).$$

Eliminating x_i using the h_i is an actual inverse substitution isomorphism. The two required maps send the old x_i to their classes in B, and send

$$x_i \longmapsto \pi^4\lambda_i, \qquad v_j \longmapsto 0, \qquad w_i \longmapsto \pi\lambda_i$$

in Λ. The h_i and f_ℓ vanish there. Each g_j has image $t \in \pi\Lambda$ by (D.20a) modulo π, and satisfies $\pi^3 t = 0$. The annihilator equality implies $\operatorname{Ann}_\Lambda(\pi^e) = \operatorname{Ann}_\Lambda(\pi)$ for every $e \ge 1$: if $\pi^e b = 0$ for

$e \geq 2$, apply the equality to $\pi^{e-2}b$ and reduce the exponent successively. Writing $t = \pi b$ now gives $\pi^4 b = 0$, hence $t = 0$. All relations therefore vanish, and the factorization is exact.

After inverting π, the h_i uniquely eliminate w_i, and (D.20a) makes the g_j consequences of the f_j. Consequently

$$B_\pi \cong A_\pi[v_1, \ldots, v_c].$$

D.5 makes A_π smooth over R; a polynomial extension is smooth by F.85, and F.29 proves smoothness of B_π/R.

For primes containing π, first use the algebra

$$B' = R[v, w]/(g_1, \ldots, g_c).$$

The Jacobian on the pivot variables v_j is $u1_c + \pi(\partial G_j/\partial v_k)$, so its determinant Δ is 1 modulo π. Thus B'_Δ is standard smooth by the inverse-variable Jacobian calculation in F.85. It is flat over the Noetherian base R by F.83. Notice also the explicit reduction $B'/\pi B' \cong (R/\pi R)[w]$, obtained by eliminating $v_j = -\sum_i r_{ji} w_i$.

The natural surjection $B' \to B$ has kernel generated by

$$F_\ell = f_\ell(\pi^2 L) \quad (\ell > c);$$

the first c equations have already vanished by (D.20a). Let $\mathfrak{q} \subset B$ contain π and avoid some $r \in \mathfrak{a}$, and let $\mathfrak{q}'$ be its inverse image in B'. Then $\Delta \notin \mathfrak{q}'$, and $T = B'_{\mathfrak{q}'}$ is flat over R, since it is a localization of that flat chart. Every F_ℓ belongs to $\pi^2 T$: its substituted variables have this divisibility and its constant term belongs to $\pi^4 R$.

Substitute in the strict relation condition and use the first c equations already killed in B'. It gives

$$\pi F_\ell = \sum_{a,b>c} b_{\ell ab} F_a F_b.$$

For the finite T-module $N = (\pi F_\ell : \ell > c)$, divisibility of F_a by π^2 gives $N \subset \pi N$; the reverse inclusion holds for every ideal. Since π lies in the maximal ideal of T, F.10 gives $N = 0$. Thus $\pi F_\ell = 0$.

Flatness identifies each multiplication kernel and preserves their quotient sequence, giving the canonical isomorphism

$$(\operatorname{Ann}_R(\pi^2)/\operatorname{Ann}_R(\pi)) \otimes_R T \cong \operatorname{Ann}_T(\pi^2)/\operatorname{Ann}_T(\pi).$$

For the kernel identification, tensor the short exact sequence through the image of multiplication by π^e, and its injection into R, for $e = 1, 2$. Tensor their annihilator inclusion and quotient sequence next. The element r kills the left-hand module and is a unit in T, so the quotient on the right is zero. Write $F_\ell = \pi b_\ell$. From $\pi F_\ell = 0$ and the resulting annihilator equality in T, we obtain $F_\ell = 0$.

To obtain a smooth neighbourhood rather than only an isomorphism of local rings, choose for each of the finitely many F_ℓ an element of $B' \setminus \mathfrak{q}'$ killing it; these exist by the localization equality criterion. Their product s kills the whole kernel. Hence $B'_{\Delta s} \to B_{\Delta s}$ is an isomorphism, and Δs avoids $\mathfrak{q}$. This is a localization of a standard smooth algebra, so F.85 and F.7 prove smoothness of $R \to B$ at $\mathfrak{q}$.

Every prime avoiding π is smooth by the earlier description of B_π. Every other prime avoiding $\mathfrak{a}B$ has some such r and is smooth by the chart just constructed. Thus every nonsmooth prime contains $\mathfrak{a}B$. The radical/prime intersection formula proved in D.1, applied to $H_{B/R}$, gives $\mathfrak{a}B \subset H_{B/R}$. Empty lists use determinant and product 1, so the argument includes $c = 0$, $c = m$, and zero rings. □

Lemma D.21: desingularize strictly standard

Reading source: Stacks, Tag 07CT.

Let R be Noetherian, let $\pi \in R$, and let $A \to \Lambda$, $D \to \Lambda$ be R-algebra maps, with A/R and D/R finitely presented. Assume

$$\mathrm{Ann}_R(\pi) = \mathrm{Ann}_R(\pi^2), \qquad \mathrm{Ann}_\Lambda(\pi) = \mathrm{Ann}_\Lambda(\pi^2),$$

that π is strictly standard in A/R, and that a map $A/\pi^4 A \to D/\pi^4 D$ agrees with both prescribed maps to $\Lambda/\pi^4\Lambda$. There are compatible maps $A \to B$, $D \to B$, $B \to \Lambda$, with B/R finitely presented, such that

$$H_{D/R}B \subset H_{B/D}, \qquad H_{D/R}B \subset H_{B/R}.$$

Proof. Let $C = A \otimes_R D$, with map to Λ sending $a \otimes d$ to the product of their prescribed images. A finite presentation $A = R[x]/(f)$ gives $C = D[x]/(f)$, so C/D is finitely presented. The strict witness changes coefficients to D by D.7. The given map $\phi : A/\pi^4 A \to D/\pi^4 D$ gives the section

$$C/\pi^4 C \longrightarrow D/\pi^4 D, \qquad a \otimes d \longmapsto \phi(\bar{a})\bar{d}.$$

The tensor relations make this well-defined; on the $D/\pi^4 D$ factor it is the identity. Its composition to $\Lambda/\pi^4\Lambda$ is the original product map by the assumed compatibility.

The ring D is Noetherian by S01, Lemma 1.3. Apply D.20 with base D and algebra C. Its hypotheses require the annihilator equality in Λ, already assumed, and require no such equality in D. We obtain $C \to B \to \Lambda$, with B/D finitely presented, and

$$\mathfrak{a}B \subset H_{B/D}, \qquad M = \mathrm{Ann}_D(\pi^2)/\mathrm{Ann}_D(\pi), \quad \mathfrak{a} = \mathrm{Ann}_D(M).$$

The two tensor-factor maps give $A \to B$ and $D \to B$, with the desired exact composites to Λ. If $D = R[u]/(g)$ and $B = D[v]/(h)$, choose polynomial lifts $\widetilde{h}$ in $R[u, v]$. Substitution in both directions gives $B = R[u, v]/(g, \widetilde{h})$, proving finite presentation over R.

Fix a prime $\mathfrak{q} \subset D$ where $R \to D$ is smooth. A smooth principal neighbourhood is flat over the Noetherian ring R by F.83; exact localization S01, Lemma 1.1 makes $D_\mathfrak{q}$ flat as well. Tensor the multiplication-kernel sequences for π, π^2 as in D.20. The equality of the original annihilators over R gives

$$\mathrm{Ann}_{D_\mathfrak{q}}(\pi) = \mathrm{Ann}_{D_\mathfrak{q}}(\pi^2), \qquad M_\mathfrak{q} = 0.$$

The module M is finite: its two annihilator ideals are finite over the Noetherian ring D, and M is their quotient. Choose generators m_i. For each, vanishing at $\mathfrak{q}$ gives $s_i \notin \mathfrak{q}$ with $s_i m_i = 0$. Their product s kills every generator; hence $s \in \mathfrak{a} \setminus \mathfrak{q}$. For $M = 0$, use $s = 1$.

At a prime $\mathfrak{r} \subset B$ above $\mathfrak{q}$, this element remains outside $\mathfrak{r}$. Since $s \in \mathfrak{a}B \subset H_{B/D}$, D.1 shows that $D \to B$ is smooth at $\mathfrak{r}$. To verify smooth composition at the level of actual neighbourhoods, choose $t \in D \setminus \mathfrak{q}$ with D_t/R smooth. Inside a smooth D-neighbourhood of $\mathfrak{r}$, F.83 supplies a principal neighbourhood B_g that is standard smooth over D. The denominator conversion in its proof makes g an element of $B \setminus \mathfrak{r}$. Base change to D_t preserves the standard smooth presentation and its unit determinant by F.85(5), so B_{tg}/D_t is smooth. F.29 now makes B_{tg}/R smooth. Thus $R \to B$ is smooth at every such $\mathfrak{r}$.

An element of $H_{D/R}$ avoided by a prime of B is avoided by its contraction in D, so that contraction is smooth. We have just proved that the prime of B is smooth for both B/D and B/R. Consequently every nonsmooth prime for either map contains $H_{D/R}B$. Intersecting these primes and using the radical/prime formula in D.1 proves both claimed ideal inclusions. Empty singular loci and zero algebras use the same convention that the intersection of no primes is the unit ideal. □

Lemma D.22: desingularize lifting apply

Reading source: Stacks, Tag 07F0.

Let R be Noetherian, let $\pi \in R$, and let $A \to \Lambda$ be an R-algebra map with A/R finitely presented. Assume

$$\operatorname{Ann}_R(\pi) = \operatorname{Ann}_R(\pi^2), \qquad \operatorname{Ann}_\Lambda(\pi) = \operatorname{Ann}_\Lambda(\pi^2),$$

and that π is strictly standard in A/R. Given a factorization

$$A/\pi^8 A \longrightarrow \bar{C} \longrightarrow \Lambda/\pi^8\Lambda$$

with $\bar{C}$ finitely presented over $R/\pi^8 R$, there is a factorization $A \to B \to \Lambda$, with B/R finitely presented, such that B_π/R_π is smooth and

$$H_{\bar{C}/(R/\pi^8 R)}(\Lambda/\pi^8\Lambda) \subset (\sqrt{H_{B/R}\Lambda} + \pi^8\Lambda)/\pi^8\Lambda.$$

Proof. If multiplication by u on a module has $\ker u = \ker u^2$, its kernels for all positive powers agree. Indeed $u^{r+1}v = 0$ puts $u^{r-1}v$ in $\ker u^2 = \ker u$, hence $u^r v = 0$; repeating lowers the exponent to one. Thus the annihilators of π^4 and π^8 agree in both R and Λ.

Use D.19 with parameter π^4 and algebra $\bar{C}$. It constructs D/R finitely presented, a map $D \to \Lambda$, and commuting maps

$$\bar{C}/\pi^4\bar{C} \longrightarrow D/\pi^4 D \longrightarrow \Lambda/\pi^4\Lambda.$$

It proves smoothness of D/R away from π, and also at a prime over π if its contraction to $\bar{C}$ is smooth over $R/\pi^8 R$. Compose the reduced map from A with this diagram. The resulting $A/\pi^4 A \to D/\pi^4 D$ agrees with the prescribed maps to $\Lambda/\pi^4\Lambda$. Therefore D.21 constructs compatible maps $A \to B$, $D \to B$, $B \to \Lambda$, with B/R finitely presented and

$$H_{D/R}B \subset H_{B/R}. \tag{D.22a}$$

If a prime $\mathfrak{b} \subset B$ avoids π, its contraction to D is smooth and hence avoids $H_{D/R}$. Inclusion (D.22a) makes $\mathfrak{b}$ avoid $H_{B/R}$ as well, so B/R is smooth at that prime by D.1. Its smooth principal neighbourhood can be localized further at π by F.67. Viewed over R_π, it is smooth: a polynomial presentation over R in which π is invertible localizes to the same presentation over R_π. Exact localization S01, Lemma 1.1 identifies its kernel ideal and conormal quotient with their localizations. Since π is already invertible on the presented algebra, its two conormal terms and their differential are unchanged. Definition F.6 and comparison F.11 give the assertion. Every prime of B_π has such a neighbourhood, and F.67 gives global smoothness over R_π.

Set $J = H_{B/R}\Lambda$. For each prime $\mathfrak{l} \subset \Lambda$ containing J, let $\mathfrak{b}, \mathfrak{d}$ be its contractions to B, D. They contain $H_{B/R}, H_{D/R}$, respectively, so both contracted points are nonsmooth. The preceding conclusion forces $\pi \in \mathfrak{l}$. The prime $\mathfrak{d}/\pi^4 D$ now contracts along the quotient diagram to a prime $\mathfrak{c}$ of $\bar{C}$. If this were smooth, D.19 would make $\mathfrak{d}$ smooth, a contradiction. Consequently $H_{\bar{C}/(R/\pi^8 R)} \subset \mathfrak{c}$.

The contraction $\mathfrak{c}$ is also the inverse image of $\mathfrak{l}/\pi^8\Lambda$ under $\bar{C} \to \Lambda/\pi^8\Lambda$. To check this equality, reduce that map modulo π^4 and use the commuting diagram through $D/\pi^4 D$; both quotient primes are the contractions of the same $\mathfrak{l}$, which contains π. Thus the ideal generated by the images of $H_{\bar{C}/(R/\pi^8 R)}$ lies in every such $\mathfrak{l}/\pi^8\Lambda$.

For an explicit radical argument, let $K \subset \Lambda$ be the inverse image of that generated ideal. The preceding paragraph and $\pi \in \mathfrak{l}$ give $K \subset \mathfrak{l}$ for every prime containing J. The prime-intersection proof in D.1 yields $K \subset \sqrt{J}$, as well as $\pi \in \sqrt{J}$. In particular $\pi^8\Lambda \subset \sqrt{J}$; taking the image of $K \subset \sqrt{J}$ in $\Lambda/\pi^8\Lambda$ is precisely the claimed inclusion. If no prime contains J, D.1 gives $\sqrt{J} = \Lambda$, and the same argument holds. This proves the result for an arbitrary target Λ, including zero quotients. □

Warmup: reduction to a base field

The preceding lifting constructions allow D.26 to reduce the smooth-colimit theorem to a field base. We fix the global assumptions before proving that reduction.

Situation D.23: global

Reading source: AI Integrated Stacks, pinned smoothing, lines 1915–1918.

Throughout the reduction to a field, R and Λ are Noetherian and the map $R \to \Lambda$ is regular. For this fixed map, the abbreviation *PT holds* means that there is a filtered system of smooth R-algebras with colimit Λ as an R-algebra.

Lemma D.24: product

Reading source: Stacks, Tag 07F3.

For $i = 1, 2$, let $R_i \to \Lambda_i$ be a regular map of Noetherian rings as in D.23. If each Λ_i is a filtered colimit of smooth R_i-algebras, then $\Lambda_1 \times \Lambda_2$ is a filtered colimit of smooth $R_1 \times R_2$-algebras.

Proof. Put $R = R_1 \times R_2$, with orthogonal idempotents $e_1 = (1, 0)$, $e_2 = (0, 1)$. For any R-module M, the maps

$$M \longrightarrow e_1M \times e_2M, \quad m \longmapsto (e_1m, e_2m), \qquad (m_1, m_2) \longmapsto m_1 + m_2$$

are inverse, since $e_1 + e_2 = 1$ and $e_1e_2 = 0$. Tensor products split by the same rule:

$$M \otimes_R (\Lambda_1 \times \Lambda_2) \cong (e_1M \otimes_{R_1} \Lambda_1) \times (e_2M \otimes_{R_2} \Lambda_2).$$

The maps send a pure tensor to its two component tensors; summing the inclusions gives the inverse, and cross tensors vanish because the idempotent can be moved to either factor. This verifies the formula by S01, Lemma 1.0. Exact sequences split into their two component sequences, so component flatness proves flatness of the product map.

Every ideal in a finite product splits in this fashion into two ideals. Finite generator lists for those component ideals give a finite generator list for the original ideal. Hence R and $\Lambda_1 \times \Lambda_2$ are Noetherian. A prime contains exactly one of e_1, e_2: it contains at least one since their product is zero, and cannot contain both since their sum is one. It therefore comes from exactly one component, with that component's residue field. Tensoring with this residue field kills the other component and identifies the fibre with the corresponding fibre of $R_i \to \Lambda_i$. Geometric regularity of these fibres proves regularity of the product map, so it is again in Situation D.23.

We prove smoothness of each product algebra used below. Let C_i/R_i be smooth and let $C = C_1 \times C_2$. Then $C_{e_i} = C_i$, with inverse maps given by projection and inclusion of the selected component; the other component vanishes when e_i is made invertible. Also $R_{e_i} = R_i$. Indeed inverting an idempotent forces it to equal one, so this localization is the quotient $R/(1-e_i)$. The inverse-variable presentation $R[T]/(e_iT - 1)$, whose derivative e_i is a unit, proves R_i/R smooth by F.85. Thus F.29 makes C_i/R smooth. The two elements e_i generate one in C, so F.35 gives finite presentation of C/R from that of the C_{e_i}/R. At every prime one e_i is avoided; the corresponding localization is smooth over R. The full gluing result F.67 therefore proves C/R smooth.

Choose the assumed presentations

$$\Lambda_i = \operatorname*{colim}_{a \in I_i} C_{i,a}.$$

The category $I_1 \times I_2$ is filtered: it is nonempty, successors of two objects are chosen in each coordinate, and parallel arrows are equalized in each coordinate. The preceding argument makes every $C_{1,a} \times C_{2,b}$ smooth over R. Component maps induce

$$\operatorname*{colim}_{(a,b) \in I_1 \times I_2} (C_{1,a} \times C_{2,b}) \longrightarrow \Lambda_1 \times \Lambda_2.$$

Any pair on the right has representatives at a pair of stages, proving surjectivity. If a represented pair maps to zero, each component becomes zero at a later stage by the filtered equality criterion. The pair of these stages kills the represented pair, proving injectivity. The same criterion applied to differences proves equality detection, and the maps preserve sums, products and units componentwise. The displayed map is consequently an R-algebra isomorphism. Iterating gives the same conclusion for any finite nonempty product; zero components obey these identical idempotent calculations. □

Lemma D.25: delocalize base

Reading source: Stacks, Tag 07F4.

Let $R \to A \to \Lambda$ be ring maps with A/R finitely presented, and let $S \subset R$ be multiplicative. Suppose $S^{-1}A \to B' \to S^{-1}\Lambda$ is a factorization with $B'/S^{-1}R$ smooth. Then the original map factors as $A \to B \to \Lambda$, with B/R finitely presented and the image of some $s \in S$ elementary standard in B/R in the sense of D.3.

Proof. If $0 \in S$, take $B = A$ and $s = 0$. In its finite presentation choose $c = 0$: the empty minor is one with coefficient zero, and multiplying every extra equation by zero gives zero. This verifies both elementary standard conditions. Hence assume $0 \notin S$.

The retraction D.12 replaces B' by a standard smooth algebra: compose the original map into its retract inclusion, and compose its retraction with the original target map. These composites preserve the original factorization. Write $A = R[x_1, \dots, x_n]/(g_1, \dots, g_t)$, with coordinate images $\lambda_i \in \Lambda$. The graph-equation construction D.14, applied to this ordered list of images, gives

$$B' = S^{-1}R[x_1, \dots, x_{n+m}]/(f_1, \dots, f_c), \qquad n \le c \le n + m,$$

with the first n variables representing the given images and with the minor on its first c variables invertible. Repeated images are permitted: each graph equation has its own variable and the same identity Jacobian block.

For $i > n$, write the image of the old coordinate x_i in $S^{-1}\Lambda$ as λ_i/d_i, with $d_i \in S$. Introduce the new coordinate $y_i = d_i x_i$, leaving $y_i = x_i$ for $i \le n$. Express each old equation in the new coordinates by substituting $x_i = d_i^{-1} y_i$. The reverse substitution is $y_i = d_i x_i$, so these changes give inverse polynomial-algebra maps over $S^{-1}R$. The new coordinates map to $\lambda_i/1$. Differentiating the substituted equations multiplies the column for an added selected variable by d_i^{-1}; the determinant remains a unit. Clear the finite coefficient denominators in each equation by multiplying it by $u_j \in S$. This does not change the localized relation ideal and multiplies the determinant by the unit $\prod_j u_j$. Rename the new coordinates x_i and the scaled equations f_j. Now

$$P = R[x_1, \dots, x_{n+m}], \qquad f_j \in P, \qquad \Delta = \det(\partial f_j/\partial x_i)_{1 \le j,i \le c}$$

has invertible image in B'.

A polynomial representative of its inverse gives $b_0, \dots, b_c \in S^{-1}P$ with $1 = b_0\Delta + \sum_j b_j f_j$. Choose $d \in S$ and $c_j \in P$ representing all db_j. The polynomial $d - c_0\Delta - \sum_j c_j f_j$ is zero after localization; some $e \in S$ therefore annihilates it in P. With $s_0 = ed$ and $a_j = ec_j$, we have the exact identity

$$s_0 = a_0\Delta + \sum_{j=1}^{c} a_j f_j \quad \text{in } P. \tag{D.25a}$$

The annihilating multiplier is required even when P has S-torsion; no element of S has been cancelled in P.

Each original relation g_k is zero in $S^{-1}(P/(f_1, \dots, f_c))$, so choose $s_k \in S$ and $v_{kj} \in P$ with $s_k g_k = \sum_j v_{kj} f_j$. Each $f_j(\lambda)$ is zero in $S^{-1}\Lambda$, so choose $t_j \in S$ with $t_j f_j(\lambda) = 0$ in Λ. Set

$$h_j = t_j f_j, \quad B = P/(h_1, \dots, h_c, g_1, \dots, g_t), \quad s = s_0 \prod_{k=1}^{t} s_k \prod_{j=1}^{c} t_j.$$

These finite relations give finite presentation. The original coordinates define $A \to B$; evaluation of all coordinates at λ_i defines $B \to \Lambda$, since it kills both relation lists. Their composite is the given map.

Here is the full witness for elementary standardness. For each k, multiply its relation identity by $s_0 \prod_{\ell \neq k} s_\ell \prod_j t_j$. Replacing $t_j f_j$ by h_j gives

$$sg_k = \sum_{j=1}^{c} s_0 \Big(\prod_{\ell \neq k} s_\ell \Big) \Big(\prod_{q \neq j} t_q \Big) v_{kj} h_j \quad \text{in } P.$$

Thus $sg_k \in (h_1, \dots, h_c)$, which implies D.3's extra-relation condition even without its allowed I^2 term. Multiplication of (D.25a) by $\prod_j t_j$ gives

$$s_0 \prod_j t_j = a_0 \det(\partial h_j / \partial x_i)_{1 \leq j, i \leq c} + \sum_{j=1}^{c} a_j \Big(\prod_{q \neq j} t_q \Big) h_j.$$

The determinant equality follows by factoring t_j, a base scalar, from equation row j. In B, the second sum is zero. Multiplying by $\prod_k s_k$ expresses the image of s as this selected minor times the image of $a_0 \prod_k s_k$, exactly D.3's other condition. Every product with an empty index set is one. This proves the assertion also for empty equation lists and zero target algebras. □

Lemma D.26: reduce to field

Reading source: Stacks, Tag 07F5.

If PT holds for every regular map $K \to \Lambda$ of Noetherian rings with K a field, then PT holds for every regular map $R \to \Lambda$ of Noetherian rings, as in D.23.

Proof. Suppose a map fails PT. Each quotient $R/I \to \Lambda/I\Lambda$ is regular by F.123; its rings are Noetherian because ideals in a quotient lift to ideals in the original ring. The set of failing quotient ideals is nonempty and has a maximal member by the ascending-chain condition. Replace the map by that quotient. Hence every further nonzero ideal gives PT. The new R is nonzero: a unital map from the zero ring has zero target, which is already a smooth algebra over itself and a constant filtered colimit.

The nilradical N is finitely generated because R is Noetherian. If its generators satisfy $a_i^{n_i} = 0$, every monomial of degree $1 + \sum_i (n_i - 1)$ contains some $a_i^{n_i}$, so that power of N is zero. If $N \neq 0$, maximality gives PT after quotienting by N, and flatness of the original regular map allows D.18 to lift PT back to R. This is a contradiction. Thus R is reduced.

We supply the finite-prime decomposition needed for its total fraction ring. Every radical ideal of a Noetherian ring is a finite intersection of prime ideals. Otherwise take a maximal counterexample I by the ascending-chain condition. It is proper and not prime, so there are $a, b \notin I$ with $ab \in I$. Both $\sqrt{I + (a)}$ and $\sqrt{I + (b)}$ are strictly larger and hence finite intersections of primes (the unit ideal is the empty intersection). Their intersection is I: if x lies in both, then $x^m \in I + (a)$ and $x^n \in I + (b)$ for some positive m, n, whence $x^{m+n} \in I + (ab) = I$; radicalness gives $x \in I$. Combining the two finite lists contradicts the choice of I.

Apply this to the radical zero ideal of the nonzero reduced ring, and remove repeated primes and any prime containing another in the list. We obtain

$$0 = \bigcap_{i=1}^{r} \mathfrak{p}_i$$

with a nonempty, pairwise incomparable list. These are exactly the minimal primes. Indeed the product of the listed ideals is contained in their intersection, so any prime over zero contains at least one listed prime: otherwise choose an element from each $\mathfrak{p}_i$ outside it, whose product

contradicts primality. If a prime lies strictly inside $\mathfrak{p}_i$, the same observation puts a listed $\mathfrak{p}_j$ inside it, contradicting incomparability. This proves finiteness, minimality and the zero intersection directly.

For each i, choose

$$y_i \in \bigcap_{j \neq i} \mathfrak{p}_j \setminus \mathfrak{p}_i$$

by multiplying one element of $\mathfrak{p}_j \setminus \mathfrak{p}_i$ for each $j \neq i$; for $r = 1$ take $y_1 = 1$. An element outside all $\mathfrak{p}_i$ is a nonzerodivisor: test an annihilated product in the domains $R/\mathfrak{p}_i$, then use their zero intersection. Conversely $a \in \mathfrak{p}_i$ satisfies $ay_i = 0$, while $y_i \neq 0$, and is a zerodivisor. Hence the multiplicative set of nonzerodivisors is $S = R \setminus \bigcup_i \mathfrak{p}_i$.

Let $d = \sum_i y_i \in S$, $Q = S^{-1}R$, and $e_i = y_i/d$. Fractions define an injection

$$Q \longrightarrow \prod_i \operatorname{Frac}(R/\mathfrak{p}_i) :$$

the denominators are nonzero in every component, and a fraction in its kernel has numerator in their zero intersection. In component i, e_i equals one; in all other components it equals zero. Therefore $e_i^2 = e_i$, $e_i e_j = 0$ for $i \neq j$, and $\sum_i e_i = 1$. Projection by these idempotents and summation give inverse maps $Q \cong \prod_i e_i Q$.

Each $K_i = e_i Q$ is a field. A nonzero $e_i a/s$ has $a \notin \mathfrak{p}_i$. The element

$$u = e_i a + (1 - e_i) = \frac{y_i a + \sum_{j \neq i} y_j}{d}$$

is a unit of Q: its numerator avoids $\mathfrak{p}_i$ because $y_i a$ does, and avoids every other $\mathfrak{p}_j$ because only y_j survives there. Its numerator is thus in S, giving the fraction inverse. The inverse of $e_i a/s$ inside $e_i Q$ is $e_i s u^{-1}$. This proves the claimed field decomposition with actual inverse ring maps.

The localization $Q \to \Lambda_Q = S^{-1}\Lambda$ is a regular map of Noetherian rings. To verify Noetherianness of either localization, extend the contraction of an ideal: each fraction in the ideal has its numerator in that contraction, and finite generators of the contraction generate the localized ideal. For a Q-module M, the tensor maps of S01, Lemma 1.0 give

$$M \otimes_Q \Lambda_Q \cong M \otimes_R \Lambda.$$

Here elements of S are already invertible on M, so the maps on pure tensors and their inverse respect the fraction relations. Exactness of the right side gives flatness. For a prime $\mathfrak{q} \subset Q$, its contraction $\mathfrak{p} \subset R$ avoids S, and $\mathfrak{q} = S^{-1}\mathfrak{p}$: a fraction belongs to $\mathfrak{q}$ exactly when its numerator does, because its denominator is a unit. Conversely a prime disjoint from S extends to a prime, since $S^{-1}(R/\mathfrak{p})$ is a domain. The quotient $Q/\mathfrak{q}$ is this localized domain and its fraction field is $\operatorname{Frac}(R/\mathfrak{p})$, so its residue field equals $\kappa(\mathfrak{p})$. Tensor associativity from S01, Lemma 1.0 identifies its fibre with $\Lambda \otimes_R \kappa(\mathfrak{p})$, geometrically regular by hypothesis. The idempotents split $\Lambda_Q = \prod_i e_i \Lambda_Q$. Each component $K_i \to e_i \Lambda_Q$ is flat by the same component tensor calculation in D.24, and has the corresponding regular fibre. Its rings are Noetherian, since it is a quotient of the respective localization. The assumed field case and the finite-product conclusion of D.24 give PT for $Q \to \Lambda_Q$.

Take any finitely presented R-algebra $A \to \Lambda$. Localizing its finite generator and relation lists gives a finitely presented Q-algebra. Criterion F.90 factors it through a smooth Q-algebra. By D.25 there is $A \to A_1 \to \Lambda$, with A_1/R finitely presented and $\pi \in S$ elementary standard in A_1/R. It is strictly standard by D.3: put its elementary coefficient at the selected variable set and put all other minor coefficients equal to zero.

Since $R \neq 0$ and multiplication by π is injective, $\pi^8 \neq 0$. Maximality of the failing quotient therefore gives PT for $R/\pi^8 R \to \Lambda/\pi^8\Lambda$, including when this quotient is zero. Apply F.90 to obtain

$$A_1/\pi^8 A_1 \longrightarrow \bar{C} \longrightarrow \Lambda/\pi^8\Lambda$$

with $\bar{C}/(R/\pi^8 R)$ smooth. All powers of π act injectively on R, and flatness makes them injective on Λ by tensoring $0 \to R \xrightarrow{\pi^n} R$, using S01, Lemma 1.0. Thus both annihilator equalities hold for parameters π and π^4.

Apply D.19 with parameter π^4. It supplies D/R finitely presented, $D \to \Lambda$, and the compatible map

$$A_1/\pi^4 A_1 \longrightarrow \bar{C}/\pi^4 \bar{C} \longrightarrow D/\pi^4 D \longrightarrow \Lambda/\pi^4 \Lambda.$$

Every prime of D avoiding π is smooth by that lemma. Every prime containing π contracts to a smooth point of $\bar{C}$, since the whole map $\bar{C}/(R/\pi^8 R)$ is smooth, and is therefore smooth as well. The actual neighbourhood criterion F.67 makes D/R globally smooth.

Finally D.21, applied to A_1, D, π and the displayed quotient map, constructs $A_1 \to B \to \Lambda$, with B/R finitely presented and $H_{D/R} B \subset H_{B/R}$. Definition D.1 gives $H_{D/R} = D$, so $H_{B/R} = B$. Thus every prime of B is smooth, and F.67 makes B/R smooth. Composing with $A \to A_1$ gives the required smooth factorization of every finitely presented input. F.90 now gives PT for the supposedly failing map, a contradiction. Consequently no failing quotient exists and PT holds for the original map. □

Local tricks

Situation D.27: local

Reading source: Stacks, Tag 07F7.

Fix a Noetherian ring R, an R-algebra A of finite presentation, an R-algebra map $A \to \Lambda$, and a prime $\mathfrak{q}$ of Λ. The target need not be Noetherian. For any intermediate R-algebra T with a specified map to Λ, write

$$\mathfrak{h}_T = \sqrt{H_{T/R}\Lambda},$$

where $H_{T/R}$ is the singular ideal of D.1 and its extension is generated by its images in Λ.

A *resolution at* $\mathfrak{q}$ is a factorization $A \to B \to \Lambda$ through an R-algebra B of finite presentation for which

$$\mathfrak{h}_A \subset \mathfrak{h}_B, \qquad \mathfrak{h}_B \not\subset \mathfrak{q}.$$

We say that the specified local problem is *resolvable* when such a factorization exists. Both ideals in this condition belong to the same target ring Λ; finite presentation of B is over the fixed base R. The second condition means that some element of $\mathfrak{h}_B$ lies outside $\mathfrak{q}$, while the first requires preservation of the entire original singular radical.

Lemma D.28: lift solution

Reading source: Stacks, Tag 07F8.

Work in D.27. Let $r \geq 1$, and choose $\pi_1, \dots, \pi_r \in R$ whose images lie in $\mathfrak{q}$. For each i, suppose that π_i is strictly standard in A/R, and that multiplication by π_i and by π_i^2 has the same kernel on each of

$$R/(\pi_1^8, \dots, \pi_{i-1}^8)R, \qquad \Lambda/(\pi_1^8, \dots, \pi_{i-1}^8)\Lambda.$$

If the problem obtained by quotienting R, A, Λ by all π_i^8, at the image of $\mathfrak{q}$, is resolvable, then the original problem is resolvable.

Proof. First establish a base-change inclusion used in the single-element step. If $R \to R'$ is any map and $A' = A \otimes_R R'$, then

$$H_{A/R} A' \subset H_{A'/R'}. \tag{D.28a}$$

Indeed, a prime of A' whose contraction is smooth has a standard smooth principal neighbourhood after taking a smooth neighbourhood of that contraction and applying F.83. The coefficient

base-change construction F.85(5) makes its inverse image standard smooth over R'. Thus every nonsmooth prime of A' contracts to a nonsmooth prime of A, and contains the images of $H_{A/R}$. The prime-intersection verification in D.1 gives (D.28a), including an empty nonsmooth locus. No flatness was required.

Consider $r = 1$, and abbreviate $\pi = \pi_1$. Put $\bar{R} = R/\pi^8 R$, $\bar{A} = A/\pi^8 A$, $\bar{\Lambda} = \Lambda/\pi^8\Lambda$, and $\bar{\mathfrak{q}} = \mathfrak{q}/\pi^8\Lambda$. The assumed resolution provides $\bar{A} \to \bar{C} \to \bar{\Lambda}$, with $\bar{C}/\bar{R}$ finitely presented, such that

$$\sqrt{H_{\bar{A}/\bar{R}}\bar{\Lambda}} \subset \sqrt{H_{\bar{C}/\bar{R}}\bar{\Lambda}} \not\subset \bar{\mathfrak{q}}. \tag{D.28b}$$

The two kernel equalities and strict standardness are exactly the hypotheses of D.22. Let $A \to B \to \Lambda$ be its factorization, and set $\mathfrak{h}_B = \sqrt{H_{B/R}\Lambda}$. The proof of D.22 shows that every prime containing $H_{B/R}\Lambda$ contains π; D.1 consequently gives $\pi \in \mathfrak{h}_B$. Its other conclusion is

$$H_{\bar{C}/\bar{R}}\bar{\Lambda} \subset \mathfrak{h}_B/\pi^8\Lambda. \tag{D.28c}$$

The ideal on the right is radical. To check this directly, lift a class whose power belongs to it to $x \in \Lambda$. Then $x^a \in \mathfrak{h}_B + \pi^8\Lambda = \mathfrak{h}_B$, so $x \in \mathfrak{h}_B$. Taking radicals in (D.28c) is therefore legitimate.

Write $\rho : \Lambda \to \bar{\Lambda}$ for the quotient. Inclusion (D.28a) gives

$$\rho(\mathfrak{h}_A) \subset \sqrt{H_{\bar{A}/\bar{R}}\bar{\Lambda}} \subset \mathfrak{h}_B/\pi^8\Lambda :$$

an element of $\mathfrak{h}_A$ has a power in $H_{A/R}\Lambda$, whose image belongs to $H_{\bar{A}/\bar{R}}\bar{\Lambda}$. Since $\ker\rho = \pi^8\Lambda \subset \mathfrak{h}_B$, this proves $\mathfrak{h}_A \subset \mathfrak{h}_B$. If $\mathfrak{h}_B \subset \mathfrak{q}$, then (D.28c), after taking radicals, would put $\sqrt{H_{\bar{C}/\bar{R}}\bar{\Lambda}}$ inside $\bar{\mathfrak{q}}$, contradicting (D.28b). Thus $\mathfrak{h}_B \not\subset \mathfrak{q}$. D.22 supplies the required finite presentation of B/R, so this is a resolution with both radical conditions verified.

For $r > 1$, quotient first by π_1^8. The base $R' = R/\pi_1^8 R$ is Noetherian: any ascending chain of its ideals lifts to one in R. The finite polynomial presentation of A changes coefficients to give the finite presentation of $A' = A/\pi_1^8 A$ over R'. The image $\mathfrak{q}'$ is prime because $\pi_1^8\Lambda \subset \mathfrak{q}$. For the remaining elements $\pi_2, \dots, \pi_r$, each successive quotient is precisely the original quotient by $\pi_1^8, \dots, \pi_{i-1}^8$; hence its two kernel hypotheses are unchanged. Strict standardness survives by D.7. The induction hypothesis gives a resolution of this first quotient problem. Apply the single-element step just proved to π_1 to lift it to the original problem. This completes the induction, with no Noetherian assumption on Λ. □

Lemma D.29: delocalize weak

Reading source: AI Integrated Stacks, pinned smoothing, lines 2147–2227.

Use the local setting D.27, and set $\mathfrak{p} = R \cap \mathfrak{q}$. Suppose $\mathfrak{q}$ is minimal over $\mathfrak{h}_A$ and the problem over $R_{\mathfrak{p}}$ with input $A_{\mathfrak{p}}$ and target $\Lambda_{\mathfrak{q}}$ is resolvable. Then the original map from A factors through a finitely presented R-algebra C whose singular ideal satisfies

$$H_{C/R}\Lambda \not\subset \mathfrak{q}.$$

Proof. We construct the required algebra by turning a localized smooth presentation into a finite system over R. A resolution over the local target has singular radical outside its maximal ideal. Its extended singular ideal must therefore be the unit ideal: a proper ideal in that local ring lies in the maximal ideal, and so does its radical. Apply D.8 to replace that intermediate algebra by a smooth one, then D.12 to obtain a standard smooth factorization of the prescribed map from $A_{\mathfrak{p}}$.

Write $A = R[x_1, \dots, x_n]/(g_1, \dots, g_t)$ and denote the target images of x_i by λ_i. By D.14, the smooth factorization has a presentation

$$D = R_{\mathfrak{p}}[x_1, \dots, x_n, u_1, \dots, u_m]/(f_1, \dots, f_c), \qquad c \geq n,$$

containing that specific generator list, with a chosen Jacobian determinant invertible. D.14 uses a separate identity equation for each prescribed generator, so repeated images are allowed.

Multiplying each f_j by a coefficient denominator outside $\mathfrak{p}$ puts every equation in $P = R[x, u]$. These multipliers are units over $R_{\mathfrak{p}}$, so the chosen determinant δ remains invertible there. Lift its inverse and the identity proving it is an inverse to polynomial expressions over $R_{\mathfrak{p}}$. Clear their finite denominator lists and annihilate the remaining localization error by an element outside $\mathfrak{p}$. This yields $s_0 \notin \mathfrak{p}$ for which δ is invertible in $P_{s_0}/(f_j)$. Likewise, since each g_i vanishes in D, there is $s_i \notin \mathfrak{p}$ such that $s_i g_i \in (f_j)$ in P. For

$$s = s_0 \prod_{i=1}^{t} s_i, \qquad D_0 = P_s/(f_j),$$

we have a standard smooth R_s-algebra receiving A, and D_0/R is smooth by principal localization and smooth composition.

The target images of the finitely many u_ℓ can be written ν_ℓ/h with one $h \in \Lambda \setminus \mathfrak{q}$. If there are no such variables, choose $h = 1$. For integers d_j large enough to remove negative powers, form

$$F_j(x, U, v) = v^{d_j} f_j(x, U/v) \in R[x, U, v].$$

Each value $F_j(\lambda, \nu, h)$ becomes zero in $\Lambda_{\mathfrak{q}}$. Choose a localization annihilator for each value and multiply them to obtain $\lambda_0 \notin \mathfrak{q}$ with

$$\lambda_0 F_j(\lambda, \nu, h) = 0 \quad \text{in } \Lambda \quad \text{for every } j.$$

Define the finite presentation

$$C = R[w, x_1, \dots, x_n, U_1, \dots, U_m, v]/\big(g_i(x),\ wF_j(x, U, v)\big).$$

The assignments $w \mapsto \lambda_0$, $x_i \mapsto \lambda_i$, $U_\ell \mapsto \nu_\ell$, $v \mapsto h$ respect all equations and give $A \to C \to \Lambda$.

Let $b = wvs$ in C. All three factors of its image avoid $\mathfrak{q}$, so its image does too. In C_b substitute $u_\ell = U_\ell/v$. Inverting w, v, s changes each equation $wF_j = 0$ into $f_j = 0$; the g_i are already consequences of these equations over R_s. The inverse substitution $U_\ell = vu_\ell$ proves, on generators and relations, the exact isomorphism

$$C_b \cong D_0[w, w^{-1}, v, v^{-1}].$$

The right side is smooth over R by polynomial extension, localization and F.29. Thus $b \in H_{C/R}$ by D.1 and F.67. Its image outside $\mathfrak{q}$ proves the claimed noncontainment. Every annihilator used belongs to a finite list; the argument requires no Noetherian property of Λ. □

Lemma D.30: delocalize height zero

Reading source: AI Integrated Stacks, pinned smoothing, lines 2229–2274.

In the setting D.27, put $\mathfrak{p} = R \cap \mathfrak{q}$. Assume $\mathfrak{q}$ is minimal over $\mathfrak{h}_A$, that the localized problem $R_{\mathfrak{p}} \to A_{\mathfrak{p}} \to \Lambda_{\mathfrak{q}}$ is resolvable, and that $\dim \Lambda_{\mathfrak{q}} = 0$. Then a resolution exists over the original base R and target Λ.

Proof. We need an algebra that improves the singular ideal at $\mathfrak{q}$ while retaining the original singular radical. The Noetherian ring here is R. Its finite-type algebra A is Noetherian by AG-CA-03, Theorem 2.1, so write $H_{A/R} = (a_1, \dots, a_r)$.

A zero-dimensional local ring has just one prime, its maximal ideal: another prime would give a strict chain of length one. The radical/prime identity proved in D.1 therefore identifies the nilradical of $\Lambda_{\mathfrak{q}}$ with $\mathfrak{q}\Lambda_{\mathfrak{q}}$. Since each a_i maps into this prime, it has nilpotent image in the localization.

Choose one positive exponent N annihilating their finite list there. Clearing the localization zeros gives $\lambda \in \Lambda \setminus \mathfrak{q}$ such that

$$\lambda a_i^N = 0 \quad (1 \le i \le r)$$

in Λ. If the list is empty, take $N = 1, \lambda = 1$. This step uses nilpotence of individual elements, without asserting that the whole maximal ideal is nilpotent or that the local target is Artinian.

By D.29 there is a finite-presentation factorization $A \to C \to \Lambda$ with $\mathfrak{h}_C \not\subset \mathfrak{q}$. Choose $c \in H_{C/R}$ with image outside $\mathfrak{q}$; otherwise that ideal and its radical would be contained in the prime. We can present C as

$$C = A[x_1, \dots, x_n]/(f_1, \dots, f_m).$$

Indeed a finite R-generating list also generates over A, and the resulting polynomial ring over Noetherian A has a finitely generated kernel.

Introduce r variables y_i, one variable z, and rm variables t_{ij}. Set

$$B = A[x_1, \dots, x_n, y_1, \dots, y_r, z, t_{ij}]/\left(f_j - \sum_{i=1}^{r} y_i t_{ij},\ z y_i\right).$$

Send x_i to its given image, y_i to a_i^N, z to λ, and every t_{ij} to zero. The equations vanish by the chosen annihilators and the map on C, so this defines $A \to B \to \Lambda$. The finite lists make B finitely presented over R.

There are two kinds of smooth principal neighbourhoods. On the first, invert z; the equations force every $y_i = 0$, giving

$$B_z = C[z, z^{-1}, t_{ij}].$$

Since c lies in the singular ideal, C_c/R is smooth by F.67. Polynomial extension and localization then make B_{zc}/R smooth. For the second kind, invert y_ℓ. We must have $z = 0$, and each equation eliminates exactly one variable:

$$t_{\ell j} = y_\ell^{-1}\left(f_j - \sum_{i \ne \ell} y_i t_{ij}\right).$$

The assignments in this formula and their inverse inclusions identify

$$B_{y_\ell} = A[x_j, y_i, y_\ell^{-1}, t_{ij}\ (i \ne \ell)].$$

After also inverting a_ℓ, this is smooth over R because A_{a_ℓ}/R is smooth. Consequently zc and every $a_\ell y_\ell$ belong to $H_{B/R}$.

Their target images are λc and a_ℓ^{N+1} respectively. The first avoids $\mathfrak{q}$, proving $\mathfrak{h}_B \not\subset \mathfrak{q}$. The others put all a_ℓ in the radical $\mathfrak{h}_B$, hence give $\mathfrak{h}_A \subset \mathfrak{h}_B$. These are precisely both resolution requirements. □

For separable residue extensions, a suitable sequence in the base permits the local lifting construction to produce a resolution.

Lemma D.31: ogoma

Reading source: Stacks, Tag 07FC.

Let A be Noetherian, M a finite A-module, $S \subset A$ multiplicative, and $\pi \in A$. Suppose that multiplication by π and π^2 has equal kernels on $S^{-1}M$. There is one $s \in S$ such that, for every positive integer n,

$$\ker(s^n\pi : M \to M) = \ker((s^n\pi)^2 : M \to M).$$

Proof. Let

$$N = \{m \in M : \pi^2 m/1 = 0 \text{ in } S^{-1}M\}.$$

It is a submodule of the finite module M, so S01, Lemma 1.3 gives a finite generating list $u_1, \dots, u_t$. By the assumed equality of localized kernels, $\pi u_j/1 = 0$ for every j. The fraction-zero calculation of S01, Lemma 1.1 gives $s_j \in S$ with $s_j \pi u_j = 0$ in M. Put $s = \prod_j s_j$, using $s = 1$ for an empty list. Multiplication by $s\pi$ kills every generator and hence all of N.

Fix $n > 0$ and suppose $(s^n\pi)^2 m = 0$. Since s is a unit after localization, this implies $\pi^2 m/1 = 0$, so $m \in N$. Therefore $s\pi m = 0$, and multiplication by s^{n-1} gives $s^n \pi m = 0$. The reverse inclusion follows by applying $s^n\pi$ a second time to an element of its kernel. The same s works for all n. The argument also covers $0 \in S$ and $M = 0$, without cancelling s inside M. □

Lemma D.32: find sequence

Reading source: Stacks, Tag 07FD.

Let Λ be Noetherian, let $I \subset \mathfrak{q}$ with $\mathfrak{q}$ prime, and fix positive integers n, e. Suppose that $L = \Lambda_{\mathfrak{q}}$ is regular local of dimension d and that $\mathfrak{q}^n L \subset IL$. There are $\pi_1, \dots, \pi_d \in \Lambda$ satisfying

$$(\pi_1, \dots, \pi_d)L = \mathfrak{q}L, \qquad \pi_i^n \in I,$$

and, for every $1 \le i \le d$,

$$\operatorname{Ann}_{\Lambda/(\pi_1^e,\dots,\pi_{i-1}^e)\Lambda}(\pi_i) = \operatorname{Ann}_{\Lambda/(\pi_1^e,\dots,\pi_{i-1}^e)\Lambda}(\pi_i^2).$$

The prescribed exponents n, e are retained.

Proof. Set $S = \Lambda \setminus \mathfrak{q}$. Regularity says that $\mathfrak{q}L/(\mathfrak{q}L)^2$ has dimension d. Lift a basis to $\mathfrak{q}L$; finite-module Nakayama, S01, Lemma 1.2, makes these d lifts generate $\mathfrak{q}L$. Each lift is a fraction with numerator in $\mathfrak{q}$ and denominator in S. Multiplying each by its denominator replaces it by $a_i \in \mathfrak{q}$ and preserves the minimal generating property, since these denominators are units in L.

The containment $\mathfrak{q}^n L \subset IL$ gives $a_i^n/1 = 0$ in $S^{-1}(\Lambda/I)$. Exact localization and the fraction-zero criterion S01, Lemma 1.1 provide $v_i \in S$ with $v_i a_i^n \in I$. Replace a_i by $v_i a_i$. Its nth power is $v_i^{n-1}(v_i a_i^n) \in I$, and the list still minimally generates the localized maximal ideal. No change of n was made. Later multiplication of an entry by an element of S also preserves these two properties.

Now choose the entries in order. Suppose $\pi_1, \dots, \pi_{i-1}$ have been fixed, with their required annihilator equalities, and keep the remaining entries as their current a_j. The whole list is a minimal generating list of $\mathfrak{q}L$, so it is regular by the full parameter proof AG-CA-12, Theorem 6.1. The power theorem AG-CA-12, Proposition 1.3, applied with exponents e on the first $i-1$ entries and exponent one on the rest, therefore makes a_i injective on

$$L/(\pi_1^e, \dots, \pi_{i-1}^e)L.$$

Its square is injective too, so their kernels are both zero.

Take the finite Λ-module $M_i = \Lambda/(\pi_1^e, \dots, \pi_{i-1}^e)\Lambda$. S01, Lemma 1.1 identifies its localization with the displayed quotient of L. Apply D.31 to M_i, S, a_i. It supplies $w_i \in S$ for which multiplication by $w_i a_i$ and its square has the same kernel on M_i; choose the positive power one in that lemma and set $\pi_i = w_i a_i$. This is exactly the required ith annihilator equality.

The earlier equalities involve only their own entry and the entries preceding it, none of which has changed. The nth-power condition remains true because I is an ideal. In L, the unit w_i preserves the generating list and its successive regularity: the preceding power ideals stay the same, and multiplication by a unit preserves injectivity on every quotient. Thus the next step has the same hypotheses. After d steps the list satisfies all assertions. If $d = 0$, Nakayama gives $\mathfrak{q}L = 0$; the empty list already satisfies them, with both given exponents unchanged. □

Lemma D.33: resolve special

Reading source: Stacks, Tag 07FE.

In D.27, let the base be a field k, let Λ be Noetherian, and let $\mathfrak{q}$ be minimal over $\mathfrak{h}_A$. Assume that $L = \Lambda_{\mathfrak{q}}$ is regular local and that its residue field $K = \kappa(\mathfrak{q})$ is separable over k. Then the problem has a resolution.

Proof. Write $\mathfrak{m} = \mathfrak{q}L$ and $d = \dim L$. If $d = 0$, regularity says that $\mathfrak{m}/\mathfrak{m}^2 = 0$. The finite-module Nakayama lemma F.10 makes $\mathfrak{m} = 0$, so $L = K$. By F.23, this field is a filtered colimit of smooth k-algebras. Finite presentation and F.90 factor $A \to L$ through one such algebra. The resulting localized resolution returns to Λ by D.30. Henceforth $d > 0$.

Set $I = H_{A/k}\Lambda$. Minimality gives $\sqrt{IL} = \mathfrak{m}$. Choose finite generators of $\mathfrak{m}$, and, for each generator, a positive power belonging to IL. A monomial whose degree exceeds the sum of these powers minus the number of generators contains one of those powers. Consequently there is $n > 0$ with $\mathfrak{m}^n \subset IL$. The Hilbert theorem AG-CA-03, Theorem 2.1 makes A Noetherian; choose $H_{A/k} = (a_1, \dots, a_r)$.

Introduce the polynomial base $P = k[x_1, \dots, x_d]$ and the finitely presented algebra

$$B = A[x_1, \dots, x_d, z_{ij}]/(x_i^n - \sum_j a_j z_{ij}).$$

On the chart where a_j is invertible, each equation eliminates z_{ij}. This chart is a polynomial algebra over $A_{a_j}[x_1, \dots, x_d]$, and is smooth over P. On inverting x_i, the equation gives $1 = \sum_j a_j x_i^{-n} z_{ij}$; thus these a_j-charts cover B_{x_i}. Smooth gluing F.67 proves that B_{x_i}/P is smooth.

Apply D.9 over P. It supplies $B \to C$ and a retraction $C \to B$, with each C_{x_i}/P smooth and its differential module finite free. Its symmetric-algebra construction is smooth also on every a_j-chart. The algebra C is finitely presented, since it is finite type over the Noetherian ring P. By D.15(e), choose a common $c > 0$ such that every x_i^c is strictly standard in C/P, and put $e = 8c$.

Use D.32 with the fixed n, e, I. Obtain elements $\pi_i \in \Lambda$ generating $\mathfrak{m}$ in L, with

$$\pi_i^n = \sum_j a_j \lambda_{ij} \quad \text{in } \Lambda,$$

and equal kernels for π_i and π_i^2 on each quotient by the earlier π_j^e. Sending x_i to π_i and z_{ij} to λ_{ij} defines $B \to \Lambda$; the retraction defines $C \to \Lambda$.

It is enough to resolve this new problem over P. In fact the smooth a_j-charts put every a_j in $H_{C/P}$. A P-resolution through T therefore preserves $\mathfrak{h}_A$. A smooth P-neighbourhood is smooth over k, so $H_{T/P} \subset H_{T/k}$. Finite presentation over P is finite presentation over k, by adjoining the finitely many x_i to a presentation. These observations give both conditions of an original k-resolution.

For the new problem apply D.28 to $x_1^c, \dots, x_d^c$. On the source quotients multiplication by a remaining polynomial variable is injective, as equality of polynomial coefficients shows. On a target quotient, equality $\ker u = \ker u^2$ implies $\ker u^s = \ker u$ for all $s \geq 1$: if $u^{s+1}v = 0$, applying the equality to $u^{s-1}v$ gives $u^s v = 0$, and induction finishes the argument. Thus D.32 supplies the equal kernels for π_i^c and π_i^{2c}. Since $(x_i^c)^8 = x_i^e$, D.28 reduces the task to the quotient by these e-th powers.

Localize that quotient at its target prime. The base and target are

$$S = P_{(x_1,\dots,x_d)}/(x_1^e, \dots, x_d^e), \qquad L_0 = L/(\pi_1^e, \dots, \pi_d^e).$$

The contracted prime is $(x_1, \dots, x_d)$, because reduction of a polynomial in the π_i is its constant in $k \subset K$. The polynomial local base is regular by iterating AG-CA-14, Proposition 3.3 from the field. The regular local ring L has dimension d, and its maximal ideal is generated by the d elements π_i. The minimal-parameter theorem AG-CA-12, Theorem 6.1 makes this list regular. The criterion F.46 gives flatness of $P_{(x)} \to L$; quotient base change gives flatness of $S \to L_0$.

The maximal ideals of S and L_0 are nilpotent: every monomial of degree $d(e-1)+1$ contains an e-th power. The closed fibre of $S \to L_0$ is K/k. F.23 expresses it as a smooth colimit, and the nilpotent lifting theorem D.18 does the same for L_0/S. By F.90, the finitely presented algebra $C \otimes_P S$ factors through a smooth S-algebra.

To return to the unlocalized quotient, let its target prime be $\overline{\mathfrak{q}}$. Its target localization has dimension zero by the preceding nilpotence calculation. If the quotient input's singular radical escapes $\overline{\mathfrak{q}}$, its identity factorization already resolves the problem. Otherwise $\overline{\mathfrak{q}}$ is minimal over that radical, and D.30 delocalizes the smooth factorization. D.28 lifts the quotient resolution to $P \to C \to \Lambda$. The change-of-base argument above then gives the required resolution over k. □

Inseparable residue fields

For inseparable residue extensions, we first enlarge the coefficient field and control the necessary denominators. The resulting finite system then permits a local resolution.

Lemma D.34: helper

Reading source: Stacks, Tag 07FG.

Let k be a field of characteristic $p > 0$, and let $(\Lambda, \mathfrak{m}, K)$ be an Artinian local k-algebra. If $H_1(L_{K/k})$ is finite-dimensional over K, then Λ is a directed union of Artinian local k-subalgebras A, essentially of finite type over k, such that $A \to \Lambda$ is faithfully flat and $\mathfrak{m}_A\Lambda = \mathfrak{m}$.

Proof. Keep the specified map $i : k \to \Lambda$ throughout. Its reduction identifies k with a subfield of K. By AG-CA-03, Theorem 4.2, choose generators $u_1, \dots, u_d$ of $\mathfrak{m}$ and a least $N \geq 1$ with $\mathfrak{m}^N = 0$. We use the following internal prime-field lifting calculation. For a field E of characteristic p, F.119 proves the p-basis presentation

$$E = E^p[T_b : b \in \mathcal{B}]/(T_b^p - b^p : b \in \mathcal{B}).$$

In a square-zero test $f : E \to C/J$, define a map on E^p by sending a^p to the pth power of any lift of $f(a)$. The pth root in E is unique, and the value is independent of the lift because $j^p = 0$ for $j \in J$. The Frobenius identities prove addition and multiplication for this map. Choose any lifts of the images of the basis elements b. Their pth powers are exactly the prescribed values of b^p, so the displayed presentation gives an $\mathbb{F}_p$-map $E \to C$ lifting f. This proves formal smoothness for every such field, including an infinite p-basis.

It lifts maps across a nilpotent ideal by its successive powers: the kernel from the quotient by J^{a+1} to the quotient by J^a is square zero for $a \geq 1$. In particular a residue section $s : K \to \Lambda$ exists. Define $\psi_s : K[X_1, \dots, X_d] \to \Lambda$ by this section and $X_j \mapsto u_j$.

We first prove that the section can be chosen so that, for a field $k \subset F_0 \subset K$ finitely generated over k, the subring $\psi_s(F_0[X])$ contains $i(k)$. This is automatic for $N = 1$, with s the identity and $F_0 = k$. Suppose $N > 1$ and put $I = \mathfrak{m}^{N-1}$. Then $I^2 = 0$ and $\mathfrak{m}I = 0$, so scalar multiplication on I factors through K. Applying the induction assertion to Λ/I gives a section s_0 and a finitely generated field $F \supset k$ containing the prescribed image in $s_0(F)[\bar{u}]$. Lift s_0 to s. The ring

$$T = F[X_1, \dots, X_d]/(X_1, \dots, X_d)^N$$

maps onto $s_0(F)[\bar{u}]$. The kernel is contained in the ideal of positive-degree terms, because a nonzero constant from F has nonzero residue. It is therefore nilpotent. Formal smoothness of $k/\mathbb{F}_p$ lifts its specified map to a map $v : k \to T$. The difference $e = i - \psi_s v : k \to I$ is an $\mathbb{F}_p$-derivation: expand a product, use agreement modulo I, and discard the product of the two errors in I^2.

Choose the absolute differential basis $(dz_j)_{j \in J}$ supplied by F.119. Formal smoothness and F.17 give $H_1(L_{K/\mathbb{F}_p}) = 0$. Consequently F.131 identifies the kernel in

$$\Omega_{k/\mathbb{F}_p} \otimes_k K \longrightarrow \Omega_{K/\mathbb{F}_p}$$

with $H_1(L_{K/k})$. A finite basis of this kernel uses only finitely many coordinates dz_j; let J_0 contain their supports. The images of the other basis vectors are linearly independent, since a relation supported outside J_0 would also have to be supported inside it. Define a K-linear map on their span by $dz_j \mapsto e(z_j)$. Extend a basis of this span to a basis of $\Omega_{K/\mathbb{F}_p}$, assign zero on the extra vectors, and use the universal differential map to obtain a derivation $\delta : K \to I$.

The formula $s_1 = s + \delta$ gives another ring section: the derivation identity accounts for the two cross terms, and $I^2 = 0$ discards the last term. In evaluating $v(a)$, the constant coefficient is $a \in k \subset F$. Changes to positive-degree coefficients are annihilated by their monomials, using $\mathfrak{m}I = 0$. Thus $\psi_{s_1} v(a) - \psi_s v(a) = \delta(a)$. The remaining error has the form

$$e(a) - \delta(a) = \sum_{j \in J_0} a_j w_j, \qquad da = \sum_j a_j dz_j, \quad a_j \in k, \quad w_j = e(z_j) - \delta(z_j) \in I.$$

Each w_j is a linear combination of the finitely many degree-$(N-1)$ monomials u^E, with coefficients in K. Indeed these monomials generate I, and a coefficient may be replaced by its residue because $\mathfrak{m}I = 0$. Adjoin all these coefficients to F to obtain F_0. When multiplying a w_j, the scalar $i(a_j)$ can equally be represented by $s_1(a_j)$, since their difference belongs to $\mathfrak{m}$. The displayed error and $\psi_{s_1} v(a)$ therefore both lie in $\psi_{s_1}(F_0[X])$. This proves the assertion for the actual structure map. Fix this section henceforth and call it s.

The map ψ_s is surjective. Subtract a residue representative, write the remainder using the u_j, and do the same for its coefficients. After N steps the remainder lies in $\mathfrak{m}^N = 0$. This produces a polynomial expression for each element. By the Hilbert basis theorem AG-CA-03, Theorem 2.1, its kernel has finitely many generators $g_1, \ldots, g_b$. Enlarge F_0 by their finitely many coefficients to a field F_*. For each finitely generated intermediate field $F_* \subset F \subset K$, set

$$A_F = F[X_1, \ldots, X_d]/(g_1, \ldots, g_b).$$

The presentation gives $K \otimes_F A_F = \Lambda$. Choose a basis of K/F containing 1. As an A_F-module this tensor product is a direct sum of copies of A_F; its 1-coordinate makes the natural map injective, exact, and faithful on every module. Thus $A_F \subset \Lambda$ and this inclusion is faithfully flat. It contains $i(k)$ by the preceding assertion.

The ideal generated by the X_j has Nth power zero, as may be checked in the injective image. Its quotient is F, since each g_j has zero constant coefficient. The monomials of degree below N span A_F over F. Finite dimension makes it Artinian; the nilpotent ideal is its unique maximal ideal, since an element with nonzero constant term is invertible by the finite geometric series. That ideal extends to $\mathfrak{m}$. The residue field F/k is finitely generated, hence is the fraction field of the finite-type domain generated by a field-generator list. F.40 now makes A_F essentially of finite type for its actual, possibly different, k-structure map.

Two finite field-generator lists have a common successor, their joint generated field. Every polynomial expression in Λ uses finitely many coefficients of K; adjoining them to F_* places that element in some A_F. These actual subrings therefore form a directed union equal to Λ, with all the required properties. □

Lemma D.34A: Artinian denominator enlargement

Reading sources: the unit enlargement in Stacks, Tag 07FI and the finite étale construction in Tag 04GK. The auxiliary statement and the calculations below are written here.

Let k have characteristic $p > 0$, and let $A \to B$ be a flat local map of Noetherian Artinian local k-algebras with $\mathfrak{m}_A B = \mathfrak{m}_B$ and $\mathfrak{m}_B^N = 0$. Denote their residue fields by $F \subset K$. Given finitely many units $b_j \in B$, there is a local factorization $A \to A' \to B$ such that $A \to A'$ is essentially smooth, $A' \to B$ is faithfully flat, $\mathfrak{m}_{A'} B = \mathfrak{m}_B$, and a p-power of each b_j belongs to the subring

$A' \subset B$. The residue extension is a finite tower of simple transcendental and finite separable extensions. At a transcendental step the unit itself is adjoined.

Proof. The local faithful-flatness theorem AG-CA-08, Theorem 2.1 makes the given map faithfully flat, hence injective. For one unit b, write $\beta \in K^\times$ for its residue.

If β is transcendental over F, take $A' = A[T]_{\mathfrak{m}_A A[T]}$ and send T to b. Every denominator has a nonzero polynomial residue at β, hence maps to a unit of B. This defines a local, essentially smooth map with maximal ideal $\mathfrak{m}_A A'$ and residue $F(\beta)$. The ring is Noetherian by the Hilbert basis theorem and localization. Its maximal ideal is nilpotent. Each successive ideal-power quotient is a finite-dimensional vector space over its residue field, since each power is a finite module. These quotients give finite length, so A' is Artinian.

In the remaining algebraic case, take $A' = A$ if $\beta \in F$. Otherwise let f be its monic irreducible polynomial. If $f' = 0$, every positive exponent occurring in f is divisible by p, so $f(T) = f_1(T^p)$. The polynomial f_1 is irreducible, since a factorization would give one of f by substitution. Repeat this operation while the derivative vanishes. The degrees decrease, so it stops at an irreducible polynomial with nonzero derivative. Its roots include β^{q_0} for a p-power q_0. Bézout with the derivative proves this root separable. Thus $F_1 = F(\beta^{q_0})$ is finite separable over F.

Lift its monic minimal polynomial to $h \in A[T]$, and put $A' = A[T]/(h)$. Monic division makes A' finite free over A. Its reduction is the field F_1; the ideal $\mathfrak{m}_A A'$ is nilpotent and is its unique maximal ideal. It is Artinian by the same finite-length argument. Bézout for $\bar{h}, \bar{h}'$ makes h' invertible in the reduction, and a finite geometric series lifts its inverse across the nilpotent ideal. The one-equation Jacobian proof AG-CA-17, Proposition 6.1 makes A'/A étale.

Here is the actual map to B. Choose $r_0 \in B$ reducing to β^{q_0}. Its derivative value is a unit, and $h(r_0) \in \mathfrak{m}_B$. Define

$$r_{a+1} = r_a - h'(r_a)^{-1} h(r_a).$$

Polynomial expansion shows that $h(r_{a+1})$ belongs to the square of the ideal containing $h(r_a)$: the constant and linear terms cancel, and all remaining terms contain at least two copies of the correction. Consequently $h(r_a) \in \mathfrak{m}_B^{2^a}$. The residues stay fixed and the derivatives stay units. Once $2^a \geq N$, this gives an exact root and the required homomorphism $A' \to B$. Two roots with that residue are equal: their difference times the divided-difference polynomial is zero, and that polynomial has the nonzero derivative as residue and is a unit. This is the simple-root construction without an external lifting step.

In all three cases A' is A-flat and $A'/\mathfrak{m}_A A' = F'$, its residue field. Tensor a free A-resolution of F with A'; flatness makes it a free A'-resolution of F'. Tensoring further with B gives

$$\operatorname{Tor}_1^{A'}(F', B) = \operatorname{Tor}_1^{A}(F, B) = 0.$$

Apply the Noetherian local criterion AG-CA-08, Theorem 4.2 with local source A', local target B, and the finite B-module B. It proves $A' \to B$ flat. Locality makes it faithfully flat and injective, so A' is the asserted subring. Its maximal ideal still generates $\mathfrak{m}_B$.

In the two algebraic cases choose a unit $x \in A'$ with residue β^{-q_0}, taking $q_0 = 1$ if $\beta \in F$. Then $xb^{q_0} - 1 \in \mathfrak{m}_B$. For a p-power $q_1 \geq N$, the characteristic-p binomial identity yields

$$(xb^{q_0})^{q_1} = 1, \qquad b^{q_0 q_1} = x^{-q_1} \in A'.$$

In the transcendental case b itself already belongs to A'. Repeat for the finite unit list. Each earlier power remains in every later subring, and the maps remain essentially smooth by composition.

For use in D.35, retain the natural differential information. A simple transcendental residue step has

$$\Omega_{F(\beta)/k} = (\Omega_{F/k} \otimes_F F(\beta)) \oplus F(\beta)\, d\beta,$$

by polynomial differentials and the inverse rule in localization. In a finite separable step, differentiating the monic equation solves its new differential uniquely, because its derivative is invertible;

it leaves precisely the old differential space after scalar extension. Both residue extensions are formally smooth by F.49, so their relative first homology vanishes by F.17. The exact field sequence F.131 therefore gives the natural isomorphism

$$H_1(L_{F'/k}) = H_1(L_{F/k}) \otimes_F F'.$$

Composition gives these formulas for the whole finite tower. In particular its first-homology image in $H_1(L_{K/k})$ is unchanged. □

Lemma D.35: solution modulo

Reading source: Stacks, Tag 07FH. The global representatives and their denominators are constructed explicitly below.

Let k have characteristic $p > 0$, let Λ be a Noetherian geometrically regular k-algebra, and choose a prime $\mathfrak{q}$, an integer $n \geq 1$, and a finite subset $E \subset \Lambda_{\mathfrak{q}}/\mathfrak{q}^n\Lambda_{\mathfrak{q}}$. There is a map $\varphi : P = k[y_1, \ldots, y_m] \to \Lambda$, for some $m \geq 0$, such that, with $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$:

1. $P_{\mathfrak{p}} \to \Lambda_{\mathfrak{q}}$ is flat and $\mathfrak{p}\Lambda_{\mathfrak{q}} = \mathfrak{q}\Lambda_{\mathfrak{q}}$.
2. Its quotient map has a local Artinian factorization

$$P_{\mathfrak{p}}/\mathfrak{p}^n P_{\mathfrak{p}} \longrightarrow D \longrightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n\Lambda_{\mathfrak{q}},$$

   with the first arrow essentially smooth and the second faithfully flat.
3. The image of D contains E.

Proof. Write $L = \Lambda_{\mathfrak{q}}$, $\mathfrak{m} = \mathfrak{q}L$, $K = L/\mathfrak{m}$, and $B = L/\mathfrak{m}^n$. The injection in F.133 embeds $H_1(L_{K/k})$ in the finite-dimensional space $\mathfrak{m}/\mathfrak{m}^2$. By D.34, choose a local Artinian subring $A \subset B$ containing E, essentially of finite type over k, faithfully flat to B, with $\mathfrak{m}_A B = \mathfrak{m}_B$. Let $F \subset K$ be its residue field. This field is finitely generated over k: present A as a localization of a finite-type algebra, take its residue quotient, and express the resulting field as fractions of the image of the finite algebra-generator list.

The differentials of such a field-generator list span $\Omega_{F/k}$, since the product and inverse rules differentiate every rational expression. Select a basis $d\theta_1, \ldots, d\theta_t$ from this spanning list and lift its elements to A. Each lift has a representative $a_j/b_j \in L$, with $a_j \in \Lambda$ and $b_j \notin \mathfrak{q}$.

By the left injection in F.131, $H_1(L_{F/k}) \otimes_F K$ embeds in $H_1(L_{K/k})$. Denote its image in $\mathfrak{m}/\mathfrak{m}^2$ by V. If $n \geq 2$, the elements of $\mathfrak{m}_A$ span this cotangent space over K: its identification with the cotangent space of B and the equality of extended maximal ideals prove that assertion. Choose elements whose images are a basis of a complement to V. Lift them to fractions c_i/d_i, with $c_i \in \mathfrak{q}$ and $d_i \notin \mathfrak{q}$. For $n = 1$, select instead representatives $c_i \in \mathfrak{q}$ of a complement in $\mathfrak{m}/\mathfrak{m}^2$ and put $d_i = 1$. Their images in $B = K$ are zero, so are already in A.

Apply D.34A to the finitely many units represented by b_j, d_i in B. Obtain $A \subset A' \subset B$, with $A \to A'$ essentially smooth, $A' \to B$ faithfully flat, and unchanged extended maximal ideal. For appropriate p-powers q_j, r_i, all four kinds of global elements

$$a_j b_j^{q_j - 1}, \quad b_j^{q_j}, \quad c_i d_i^{r_i - 1}, \quad d_i^{r_i}$$

have images in A'. The first and third are the fractional lifts multiplied by the indicated powers; the second and fourth are supplied by the enlargement. Record in addition every original denominator which was adjoined at a transcendental residue step. It itself has its image in A'.

Let F' be the residue field of A'. The residues of the first two kinds of elements, together with these recorded transcendental denominators, have differentials spanning $\Omega_{F'/k}$. Indeed their quotient is θ_j, so the inverse rule expresses each old $d\theta_j$ in their span. D.34A adds just one new differential at

each transcendental step and none at a finite separable step. Select a differential basis from this finite list, and call its actual global representatives $\lambda_1, \dots, \lambda_s \in \Lambda$. This argument does not set $d(b_j^{q_j})$ to zero in F': the root b_j need not belong to that field. Every selected representative has its class in A'.

Define $P' = k[y_1, \dots, y_s] \to \Lambda$ by these elements, let $\mathfrak{p}'$ be the inverse image of $\mathfrak{q}$, and set $K_0 = \kappa(\mathfrak{p}') \subset F'$. Since all the basis differentials come from K_0, $\Omega_{F'/K_0} = 0$. The extension is finitely generated, because F'/k is finitely generated. The equality proved in F.98 gives

$$-\dim_{F'} H_1(L_{F'/K_0}) = \operatorname{trdeg}_{K_0} F'.$$

Its two nonnegative dimensions must both be zero. Finitely many algebraic field generators give a finite extension by multiplying their degrees; the zero first homology gives separability by F.91. Thus F'/K_0 is finite separable. The field sequence F.131, whose relative first homology and differential module here vanish, identifies

$$H_1(L_{K_0/k}) \otimes_{K_0} K = H_1(L_{F'/k}) \otimes_{F'} K = H_1(L_{F/k}) \otimes_F K.$$

The last identity is the natural one retained in D.34A, so its image in $\mathfrak{m}/\mathfrak{m}^2$ is still V.

Apply F.109 using the selected differential basis. It gives a flat map $P'_{\mathfrak{p}'} \to L$ and a regular quotient $L/\mathfrak{p}'L$. Its written conormal square identifies the image of the source cotangent space with the preceding first-homology image V. Append the global elements $\lambda_{s+i} = c_i d_i^{r_i - 1}$ to form $P = k[y_1, \dots, y_m] \to \Lambda$. Their cotangent classes are the originally chosen complementary classes multiplied by the nonzero scalars $\overline{d}_i^{r_i}$, and therefore still form a complementary basis. Their residues are zero. Hence $P_{\mathfrak{p}}$ is the localization of $P'_{\mathfrak{p}'}[y_{s+1}, \dots, y_m]$ at the prime generated by $\mathfrak{p}'$ and these variables.

Polynomial local regularity is proved in AG-CA-14, Proposition 3.3. The old regular parameters and the appended variables are a regular parameter list for this source: its maximal ideal is generated by them and its cotangent space is the old cotangent space plus one independent class for each new variable. Their images are a regular sequence in L. The old list remains regular by the flat map from $P'_{\mathfrak{p}'}$, applying flat base change to each successive multiplication injection. The new list is a minimal parameter list on the regular local quotient $L/\mathfrak{p}'L$, hence is regular by AG-CA-12, Theorem 6.1. Concatenating these lists proves the assertion. F.46 now makes $P_{\mathfrak{p}} \to L$ flat. Their cotangent classes span $\mathfrak{m}/\mathfrak{m}^2$; Nakayama F.10 gives $\mathfrak{p}L = \mathfrak{m}$. This proves (1).

All polynomial generators have their classes in A', so the map to B factors through A'. An element outside $\mathfrak{p}$ has nonzero residue and maps to a unit of this local ring. Moreover $\mathfrak{p}^n$ maps to zero in B, hence in $A' \subset B$. We obtain

$$R = P_{\mathfrak{p}}/\mathfrak{p}^n P_{\mathfrak{p}} \longrightarrow A' \longrightarrow B.$$

The ring R is Artinian: it is Noetherian local with nilpotent maximal ideal, and its finite ideal-power quotients have finite dimension over its residue field. Its composite map to B is the flat base change of (1). Since $A' \to B$ is faithfully flat, F.48 makes $R \to A'$ flat. The ideals $\mathfrak{p}A'$ and $\mathfrak{m}_{A'}$ become equal in B; faithful flatness detects equality by tensoring their quotient, so they are equal. The residue extension is the finite separable extension F'/K_0 just proved. F.40 makes A' a finite R-module. Since R is Noetherian, it is a finitely presented algebra, by taking finite module generators and a finite kernel in their polynomial presentation. Its only fibre is the finite separable residue field. The proved étale criterion F.16 therefore makes $R \to A'$ étale, in particular essentially smooth. Take $D = A'$. It retains $E \subset A$, proving (2) and (3). □

Lemma D.36: enlarge solution modulo

Reading source: Stacks, Tag 07FI.

Use the notation and a local Artinian factorization of D.35:

$$R = k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n \longrightarrow D \longrightarrow B = \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}},$$

where $R \to D$ is essentially smooth and $D \to B$ is flat. For every $\lambda \in \Lambda \setminus \mathfrak{q}$, there are a positive p-power q and a local Artinian factorization $R \to D \to D' \to B$ with $D \to D'$ essentially smooth, $D' \to B$ faithfully flat, and λ^q in the image of D'.

Proof. The polynomial-source maximal ideal extends to that of B, by D.35(1). We verify that the maximal ideal of D does also, even for a factorization other than the particular étale one constructed there. The closed fibre $D/\mathfrak{m}_R D$ is Artinian local and essentially smooth over the field $R/\mathfrak{m}_R$. The regularity theorem AG-CA-18, Theorem 2.1, followed by localization, makes it regular. An Artinian local ring has dimension zero. For a regular local ring of dimension zero its cotangent dimension is zero; its finite maximal ideal equals its square, and Nakayama makes that ideal zero. Thus this fibre is a field and $\mathfrak{m}_D = \mathfrak{m}_R D$. Extending to B gives $\mathfrak{m}_D B = \mathfrak{m}_B$.

All rings required by D.34A are Noetherian. The source R is Noetherian, and the essentially smooth algebra D is a localization of a finitely presented R-algebra, so the Hilbert basis theorem and localization make it Noetherian. The ring B is the displayed Artinian quotient of the Noetherian local ring $\Lambda_{\mathfrak{q}}$, and $\mathfrak{m}_B^n = 0$. The flat local map $D \to B$ is faithfully flat by the local criterion. The image of λ is a unit because its residue is nonzero. Apply the one-unit case of D.34A. It supplies $D \subset D' \subset B$, an essentially smooth first map, a faithfully flat last map, and the exact power containment $\lambda^q \in D'$. Composition with $R \to D$ gives the required factorization. □

Lemma D.37: resolve general

Reading source: Stacks, Tag 07FJ.

Let k have characteristic $p > 0$. In D.27, suppose that Λ is Noetherian and geometrically regular over k, and that $\mathfrak{q}$ is minimal over $\mathfrak{h}_A$. Then the problem has a resolution.

Proof. Put $L = \Lambda_{\mathfrak{q}}$, $\mathfrak{m} = \mathfrak{q}L$, $d = \dim L$, and $H = H_{A/k} = (a_1, \dots, a_r)$. These generators exist by the Hilbert theorem AG-CA-03, Theorem 2.1. The ring L is regular by geometric regularity. As in D.33, minimality and a finite maximal-ideal generating list give $N > 0$ with $\mathfrak{m}^N \subset HL$.

Over $P_0 = k[x_1, \dots, x_d]$ form

$$B = A[x_1, \dots, x_d, z_{ij}]/(x_i^{2N} - \sum_j a_j z_{ij}).$$

Eliminating z_{ij} on an a_j-chart proves smoothness there; the equation makes those charts cover B_{x_i}. By F.67, B_{x_i}/P_0 is smooth. Apply D.9 to obtain $B \to C$ with retraction $C \to B$. The smooth charts of C have finite free differentials. Since C/P_0 is finitely presented by the Hilbert theorem, D.15(e) gives one $c > 0$ for which all x_i^c are strictly standard. If $d = 0$, take $c = 1$. Fix, before making any Artinian approximation,

$$e = 8c, \qquad n = \max\{N + dc,\ d(e-1) + 1\}. \tag{D.37a}$$

Apply D.35 to the images of a finite k-algebra generating list of A in $\overline{L} = L/\mathfrak{m}^n$. It gives a polynomial algebra $P = k[y_1, \dots, y_m]$ mapping to Λ, its contracted prime $\mathfrak{p}$, and local maps

$$S_0 = P_{\mathfrak{p}}/\mathfrak{p}^n P_{\mathfrak{p}} \longrightarrow D \longrightarrow \overline{L}.$$

The first map is essentially smooth, the second is flat, and the generator images belong to D. Also $P_{\mathfrak{p}} \to L$ is flat and $\mathfrak{p}L = \mathfrak{m}$. Its closed fibre is a field, so the dimension formula AG-CA-11, Theorem 5.1 gives $\dim P_{\mathfrak{p}} = d$.

Choose $\pi_1, \dots, \pi_d \in \mathfrak{p}$ representing a regular parameter list of $P_{\mathfrak{p}}$. Polynomial local rings are regular by iteration of AG-CA-14, Proposition 3.3, starting with the field. The minimal-parameter theorem AG-CA-12, Theorem 6.1 makes their minimal generating list regular. Their images generate $\mathfrak{m}$ and form a regular parameter list of L, by the same theorem and $\dim L = d$. Let

$$R = P[t_1, \dots, t_d], \qquad \gamma_i = \pi_i t_i.$$

Throughout, localization of this polynomial ring at the coefficients means

$$R_S = (P \setminus \mathfrak{p})^{-1} R = P_{\mathfrak{p}}[t_1, \dots, t_d].$$

It is distinct from localization at a prime of R.

First arrange the source kernel conditions. By F.75, the γ_i form a regular sequence in R_S. Powers of earlier entries preserve regularity by AG-CA-12, Proposition 1.3. Inductively, on the finite R-module $R/(\gamma_1^e, \dots, \gamma_{i-1}^e)$, the localized kernels for γ_i and γ_i^2 are zero. D.31 supplies $s_i \in P \setminus \mathfrak{p}$ such that replacing π_i by $s_i \pi_i$ makes these kernels equal globally. Earlier equalities are unchanged, because their entries and quotient ideals have not changed. Such scaling is by local units, so both local parameter lists remain regular. Continue through all i, using the modified π_i from now on.

Next construct units $\delta_i \in \Lambda \setminus \mathfrak{q}$, coefficients $\lambda_{ij} \in \Lambda$, and an essentially smooth Artinian enlargement $D \to D'$ with flat local map $D' \to \overline{L}$, such that

$$(\delta_i \pi_i)^{2N} = \sum_j a_j \lambda_{ij} \quad \text{in } \Lambda, \tag{D.37b}$$

and both δ_i and λ_{ij} have images in D'. We additionally require equal kernels for $\delta_i \pi_i$ and its square modulo the earlier $(\delta_h \pi_h)^e$. A flat local map into the nonzero local ring $\overline{L}$ is faithfully flat, and hence injective, by AG-CA-08, Theorem 2.2. Thus every ring in this construction may be regarded as a subring of $\overline{L}$. In particular all relations among the chosen A-generator images already hold in D, so there is a map $A \to D$.

Suppose the construction has reached $i-1$. The element π_i^N lies in $H\overline{L}$; faithful flatness contracts this ideal to HD'. Write $\pi_i^N = \sum_j a_j d_j$ in D', and lift the d_j to $u_j \in L$. The error lies in $\mathfrak{m}^n \subset H\mathfrak{m}^{n-N}$, since $n \geq N$. Thus choose $v_j \in \mathfrak{m}^{n-N}$ with

$$\pi_i^N = \sum_j a_j (u_j + v_j) \quad \text{in } L.$$

Set $w_j = \pi_i^N (u_j + v_j)$. Then $\pi_i^{2N} = \sum_j a_j w_j$, and the residue of w_j in $\overline{L}$ is $\pi_i^N d_j$, because $\pi_i^N v_j \in \mathfrak{m}^n$.

Here is the global denominator calculation. Choose $s_0 \notin \mathfrak{q}$ so that every $s_0^{2N} w_j$ is represented by $\widetilde{\mu}_j \in \Lambda$. The error

$$r_i = (s_0 \pi_i)^{2N} - \sum_j a_j \widetilde{\mu}_j$$

vanishes in L. Hence some $v \notin \mathfrak{q}$ annihilates r_i, by the fraction-zero criterion of S01, Lemma 1.1. Set $s = s_0 v$ and $\mu_j = v^{2N} \widetilde{\mu}_j$. Now $(s\pi_i)^{2N} = \sum_j a_j \mu_j$ in Λ, and $\mu_j / s^{2N} = w_j$ in L.

Use D.36 to put s^h in a further D', for some $h > 0$. Replace s by s^h and μ_j by $s_{\text{old}}^{2N(h-1)} \mu_j$. The exact equation survives, and the new coefficient residue is $s^{2N} \pi_i^N d_j \in D'$. In $L/((\delta_1 \pi_1)^e, \dots, (\delta_{i-1} \pi_{i-1})^e)$, multiplication by $s\pi_i$ is injective, because the preceding entries are powers of a regular parameter list scaled by units. Apply D.31 on the corresponding finite Λ-module to obtain $s' \notin \mathfrak{q}$ for which every $(s')^q s \pi_i$, $q > 0$, has equal global kernels with its square. D.36 puts one such $(s')^q$ in an enlargement of D'. Take

$$\delta_i = (s')^q s, \qquad \lambda_{ij} = (s')^{2Nq} \mu_j.$$

This proves (D.37b), the residue assertions, and the required target kernel equality. Later enlargements retain all earlier assertions. The finite induction constructs the full list; when $d = 0$ this construction is empty and $D' = D$.

Map $R \to \Lambda$ by the given P-map and $t_i \mapsto \delta_i$. Give R its P_0-structure by $x_i \mapsto \gamma_i$. Equation (D.37b) defines $B \to \Lambda$ by $x_i \mapsto \delta_i \pi_i$ and $z_{ij} \mapsto \lambda_{ij}$. The retraction defines $C \to \Lambda$, and these maps induce

$$C_R = C \otimes_{P_0} R \longrightarrow \Lambda.$$

Every a_j-chart of C_R/R is smooth, so the singular radical of this new input contains $\mathfrak{h}_A$. A resolution over R returns to one over k: finite presentation composes with the polynomial base, and smooth composition gives $H_{T/R} \subset H_{T/k}$.

Let $I = (\gamma_1^e, \dots, \gamma_d^e) \subset R$. D.7 transports the strictly standard elements x_i^c to γ_i^c. The arranged source and target kernel equalities hold also for their c-th powers, by the kernel induction in D.33. D.28 therefore lifts a resolution modulo I to an R-resolution. For $d = 0$, $I = 0$ and this lifting step is unnecessary.

We construct the quotient resolution explicitly. Put $J = (\pi_1^e, \dots, \pi_d^e) \subset P$, and let $\mathfrak{r}$ be the inverse image of $\mathfrak{q}$ in R. Each $t_i \notin \mathfrak{r}$, since $\delta_i \notin \mathfrak{q}$. Consequently

$$IR_{\mathfrak{r}} = JR_{\mathfrak{r}}, \qquad IL = JL. \tag{D.37c}$$

The parameters generate $\mathfrak{p}P_{\mathfrak{p}}$ and $\mathfrak{m}$. The bound $n \geq d(e-1)+1$ in (D.37a) gives, by expansion into monomials,

$$\mathfrak{p}^n P_{\mathfrak{p}} \subset JP_{\mathfrak{p}}, \qquad \mathfrak{m}^n \subset JL. \tag{D.37d}$$

For $d = 0$, both maximal ideals are zero, so these containments hold as well.

Set $S_1 = S_0[t_1, \dots, t_d]$. Bars below denote elements in $D' \subset \overline{L}$. Each $\overline{\delta}_i$ is a unit of D': its image is a unit and the map is local. Define

$$B \longrightarrow D'[t_1, \dots, t_d], \qquad x_i \longmapsto \overline{\pi}_i t_i, \quad z_{ij} \longmapsto \overline{\lambda}_{ij} \overline{\delta}_i^{-2N} t_i^{2N}. \tag{D.37e}$$

Injectivity of $D' \to \overline{L}$ transports (D.37b) to D'. The image of the i-th defining equation in (D.37e) is therefore

$$t_i^{2N} \left(\overline{\pi}_i^{2N} - \overline{\delta}_i^{-2N} \sum_j \overline{a}_j \overline{\lambda}_{ij} \right) = 0.$$

Its restriction to P_0 is exactly the prescribed $x_i \mapsto \pi_i t_i$ in S_1, so the map extends to $B \otimes_{P_0} S_1$. Evaluation $t_i \mapsto \overline{\delta}_i$ recovers the prescribed images $\delta_i \pi_i, \lambda_{ij}$ in $\overline{L}$.

The map $S_0 \to D'$ is essentially smooth by construction and composition. Express it as a localization of a smooth finitely presented S_0-algebra. Localization is the filtered colimit of its principal localizations at finite products of the inverted elements. These stages are smooth, and adjoining the t_i gives smooth S_1-stages with colimit $D'[t_i]$. Finite presentation and F.90 factor (D.37e) through one smooth S_1-stage T, followed by the specified map to $\overline{L}$. Composing with $C \to B$ gives the same smooth factorization for $C \otimes_{P_0} S_1$.

By (D.37d) there are quotient maps $S_1 \to P_{\mathfrak{p}}[t_i]/J$ and $\overline{L} \to L/JL$. Base changing T gives a smooth factorization over $P_{\mathfrak{p}}[t_i]/J$. Localize further at $\mathfrak{r}$. Every denominator then has a unit image in the local target, so its map extends. By (D.37c) this is a smooth factorization of the localized problem modulo I.

The target L/JL has nilpotent maximal ideal: degree $d(e-1)+1$ monomials vanish. Thus its dimension is zero. If the singular radical of the unlocalized quotient input escapes the image of $\mathfrak{q}$, the identity factorization resolves that quotient. Otherwise this prime is minimal over the radical, and D.30 delocalizes the smooth factorization just obtained. D.28 then lifts it before quotienting by I. Finally the polynomial change of base returns the resolution to k. All constructions above include $d = 0$, empty parameter lists, and the specified target maps. □

The main theorem

The final argument selects a maximal unsuccessful singular radical and uses the local resolutions to enlarge it, giving a contradiction.

Theorem D.38: popescu

Reading source: Stacks, Tag 07GC.

If $R \to \Lambda$ is a regular map between Noetherian rings, then Λ, as an R-algebra, is a filtered colimit of smooth R-algebras.

Proof. The reduction D.26 reduces the assertion to a field k and a Noetherian geometrically regular k-algebra Λ. By D.8, it is enough that every finite type factorization $k \to A \to \Lambda$ can be followed by $A \to B \to \Lambda$ with B finite type and $\mathfrak{h}_B = \Lambda$. All such algebras are finitely presented by the Hilbert theorem.

Assume some factorization has no such successor. Among the singular radicals of all these unsuccessful factorizations choose a maximal one, say $\mathfrak{h}_A$. A nonempty collection of ideals in a Noetherian ring has a maximal member: otherwise successive strict enlargements would contradict the ascending chain condition. The ideal $\mathfrak{h}_A$ is proper, since if it were Λ, A itself would be a successor.

There is a prime $\mathfrak{q}$ minimal over $\mathfrak{h}_A$. Here are the existence details. Zorn's lemma first gives an ideal maximal among the proper ideals containing $\mathfrak{h}_A$: a chain union is still proper, since it cannot contain 1. Such an ideal is prime, because if a is outside it, the ideal and a generate 1, and multiplication of that relation by b shows that ab in the ideal forces b into it. Now order all primes containing $\mathfrak{h}_A$ by reverse inclusion. A chain intersection is a proper ideal containing $\mathfrak{h}_A$, and is prime: if ab belongs to the intersection while a, b do not, two members excluding them have a smaller member excluding both, contradicting its primality. Zorn's lemma therefore supplies the desired minimal prime.

Geometric regularity makes $\Lambda_{\mathfrak{q}}$ regular local. In positive characteristic D.37 resolves this problem. In characteristic zero every field extension is separable in the sense of F.5: each finitely generated subextension has a finite transcendence basis followed by a finite algebraic extension, whose irreducible polynomials have nonzero derivatives and hence distinct roots. Thus D.33 resolves it in that case.

In either case obtain $A \to C \to \Lambda$ with

$$\mathfrak{h}_A \subset \mathfrak{h}_C, \qquad \mathfrak{h}_C \not\subset \mathfrak{q}.$$

Because $\mathfrak{h}_A \subset \mathfrak{q}$, the first inclusion is strict. Maximality of the unsuccessful ideal $\mathfrak{h}_A$ means that C has a successor with unit singular radical. Composing its factorization with $A \to C$ supplies the forbidden successor of A, a contradiction. Hence every finite type input has the required successor. D.8 supplies the smooth filtered presentation, and D.26 returns the assertion to the original regular map. □

5. Foundation statements and their free-source proofs

The flatness arguments use projective resolutions, the two ways of computing Tor, and the connecting maps of a short exact sequence. These constructions are proved for arbitrary modules in Algebra and sheaf cohomology before reductive groups, Lemma 1.A (Reductive group schemes).

We begin with the two-term conormal complex and the field and dimension conventions used below. The numbered arguments then establish the presentation, flatness, regularity and completion results needed in Section 4. Section 6 collects the earlier algebra lessons used along the way.

The two-term conormal complex. A polynomial presentation $P = R[X_s] \twoheadrightarrow S$ with kernel I gives the complex $[I/I^2 \to \bigoplus_s S\, dX_s]$, in degrees -1 and 0; the map sends the class of f to df. We denote it by $\mathrm{NL}(P \to S)$. One presentation is always available: take a variable X_s for each $s \in S$ and send it to s. For maps $R \to R'$ and $S \to S'$ forming a commuting square, a map of presentations is a compatible map of their polynomial algebras lifting $S \to S'$. The comparison

and homotopy formulas below prove that the resulting two-term object does not depend on the presentation. The localization argument concerns this complex itself.

The inseparable exponent calculation. Let α be algebraic over a field F of characteristic p. Its irreducible polynomial is $f(T) = g(T^{p^s})$ for a maximal $s \geq 0$. Such s exists because f has finite positive degree. Maximality gives $g' \neq 0$. Any factorization of g would give a factorization of f, so g is irreducible, and its relatively prime derivative makes it separable. The element α^{p^s} is a root of g, hence is separable over F. This proves the field-theoretic exponent choice used in the enlargement argument.

Dimension conventions. The dimension of a ring is the supremum of the lengths of its finite chains of primes. For a prime $\mathfrak{p}$, its height is $\dim R_{\mathfrak{p}}$. In a Noetherian local ring of dimension d, a system of parameters consists of d elements whose ideal has radical the maximal ideal. Existence is proved in AG-CA-11, Theorem 2.1. Minimal primes and irreducible components are treated in AG-CA-01, Theorem 4.1, Lemma 4.2 and Theorem 4.3. Polynomial-ring dimension and the finite-type domain height formula are AG-CA-11, Theorem 6.1 and AG-CA-09, Theorem 4.3, respectively.

Definition F.1: lci

Reading sources: Stacks, Tag 00SL; AI Integrated Stacks, pinned Algebra, lines 37874–37881.

For an arbitrary ring map $R \to S$, the word *syntomic* means that the following three requirements hold together: S is a finitely presented R-algebra, it is flat as an R-module, and, for every $\mathfrak{p} \in \operatorname{Spec} R$, the fibre algebra

$$S(\mathfrak{p}) = S \otimes_R \kappa(\mathfrak{p})$$

is a local complete intersection over $\kappa(\mathfrak{p})$ in the sense of F.2. The phrase *flat local complete intersection over* R has this same meaning.

Definition F.2: lci field

Reading sources: Stacks, Tag 00S9; AI Integrated Stacks, pinned Algebra, lines 37249–37264.

Let S be a finite-type algebra over a field k. To call S a *global complete intersection over* k is to require a finite presentation

$$S = k[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$$

for which its Krull dimension is exactly $n - c$. A *local complete intersection over* k means that one can cover $\operatorname{Spec} S$ by principal opens $D(g_i)$, with each algebra S_{g_i} satisfying this global condition. The zero algebra is included among the global complete intersections by convention; it therefore causes no exception when an open or a fibre is empty.

Definition F.3: lci local ring

Reading sources: Stacks, Tag 00SD; AI Integrated Stacks, pinned Algebra, lines 37446–37458.

Suppose S is a local k-algebra essentially of finite type, where k is a field. The condition *complete intersection over* k asks for a regular local k-algebra R, itself essentially of finite type, together with a regular sequence

$$f_1, \ldots, f_c \in \mathfrak{m}_R$$

and a k-algebra isomorphism $R/(f_1, \ldots, f_c) \cong S$. Thus the presenting ring is local and regular, and the equations lie in its maximal ideal. All these requirements belong to the definition.

Definition F.4: relative global complete intersection

Reading sources: Stacks, Tag 00SP; AI Integrated Stacks, pinned Algebra, lines 37943–37951.

For a ring map $R \to S$, a *relative global complete-intersection presentation* consists of

$$S = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$$

such that, for each $\mathfrak{p} \in \operatorname{Spec} R$ with nonempty fibre, the ring $S \otimes_R \kappa(\mathfrak{p})$ has dimension $n - c$. The map is a *relative global complete intersection* if it has at least one such presentation. When a particular displayed presentation is called a relative global complete intersection, the fibre condition refers to its displayed values of n and c. Empty fibres impose no dimension condition.

Definition F.5: separable field extension

Reading sources: Stacks, Tag 030O; AI Integrated Stacks, pinned Algebra, lines 10135–10146.

Consider a field extension K/k. It is *separably generated* when there is a transcendence basis $(x_i)_{i \in I}$ such that

$$K/k(x_i \mid i \in I)$$

is algebraic and separable. The index set may be infinite. The condition *separable over* k used here is tested on every intermediate field K' finitely generated over k: each extension K'/k must be separably generated in the preceding sense. These are the two field-extension conditions used in the separable and inseparable arguments.

Definition F.6: smooth

Reading sources: Stacks, Tag 00T2; AI Integrated Stacks, pinned Algebra, lines 38526–38533.

Smoothness of $R \to S$ requires finite presentation together with the following condition on the naive cotangent complex:

$$H_1(\mathrm{NL}_{S/R}) = 0, \qquad \Omega_{S/R} \text{ is a finite projective } S\text{-module.}$$

Equivalently, that two-term complex is quasi-isomorphic to a finite projective module concentrated in degree zero. To verify this equivalence directly, map the two-term complex to its degree-zero cokernel. This map induces an isomorphism on degree-zero homology; it is a quasi-isomorphism exactly when the degree-minus-one kernel, denoted H_1, vanishes. Its cokernel is $\Omega_{S/R}$. Presentation independence is proved in F.11.

Definition F.7: smooth at prime

Reading sources: Stacks, Tag 00TB; AI Integrated Stacks, pinned Algebra, lines 38953–38960.

Given $R \to S$ and $\mathfrak{q} \in \operatorname{Spec} S$, *smoothness at* $\mathfrak{q}$ means that some principal neighbourhood of this point has a smooth coordinate algebra over R. Explicitly, there must be

$$g \in S \setminus \mathfrak{q} \quad \text{with} \quad R \to S_g \text{ smooth.}$$

This definition applies to any ring map; finite presentation is required of the witnessing localization through F.6.

Definition F.8: unramified

Reading sources: Stacks, Tag 00UT; AI Integrated Stacks, pinned Algebra, lines 42897–42912.

For $R \to S$, both unramified conditions require $\Omega_{S/R} = 0$. Their finiteness requirements differ: *unramified* requires finite type, whereas *G-unramified* requires finite presentation.

At a prime $\mathfrak{q}$ of S, each condition means that it holds on some principal neighbourhood $D(g)$ of $\mathfrak{q}$. Thus *unramified at* $\mathfrak{q}$ requires $g \notin \mathfrak{q}$ with S_g/R of finite type and $\Omega_{S_g/R} = 0$; *G-unramified at* $\mathfrak{q}$ requires such a g with S_g/R finitely presented and the same differential-module vanishing.

Lemma F.9: CM over regular flat

Reading source: AI Integrated Stacks, pinned Algebra, lines 34036–34074.

Let $R \to S$ be local, with both rings Noetherian and local. Suppose R is regular, S is Cohen–Macaulay, and

$$\dim S = \dim R + \dim(S/\mathfrak{m}_R S).$$

Then S is flat as an R-module.

Proof. We reduce the dimension of the regular source by one while preserving the dimension equality. Write $d = \dim R$. If $d = 0$, regularity says $\mathfrak{m}_R/\mathfrak{m}_R^2 = 0$. Nakayama, F.10, applies to the finite ideal $\mathfrak{m}_R$ and gives $\mathfrak{m}_R = 0$. Thus R is a field, and a basis makes every R-module free, hence flat.

For $d > 0$, consider the finitely many associated primes of S. The Cohen–Macaulay component theorem AG-CA-12, Theorem 4.1, says that they are its minimal primes $\mathfrak{q}_i$ and that $\dim(S/\mathfrak{q}_i) = \dim S$. None contains $\mathfrak{m}_R S$. Indeed that containment would give a surjection $S/\mathfrak{m}_R S \to S/\mathfrak{q}_i$, forcing the dimension of the latter to be at most that of the former, whereas the hypothesis gives

$$\dim(S/\mathfrak{m}_R S) = \dim S - d < \dim S.$$

Consequently every contraction $\mathfrak{p}_i = R \cap \mathfrak{q}_i$ is strictly smaller than $\mathfrak{m}_R$.

The prime-avoidance calculation F.80 now chooses

$$x \in \mathfrak{m}_R \setminus (\mathfrak{m}_R^2 \cup \textstyle\bigcup_i \mathfrak{p}_i).$$

To check its hypotheses, $\mathfrak{m}_R$ is not contained in any $\mathfrak{p}_i$, and is not contained in $\mathfrak{m}_R^2$, since otherwise Nakayama would again give $d = 0$. There is just one nonprime ideal in this avoidance list.

The element x avoids all associated primes of S, so it is a nonzerodivisor on S. The complete regular-sequence and Cohen–Macaulay arguments of AG-CA-12, Sections 2 and 4, make S/xS Cohen–Macaulay with dimension $\dim S - 1$. AG-CA-14, Theorem 1.1 and Corollary 1.2, make x a nonzerodivisor of R and R/xR regular of dimension $d - 1$. The quotient map is still local, and its closed fibre is $S/\mathfrak{m}_R S$. Hence it satisfies

$$\dim(S/xS) = \dim(R/xR) + \dim(S/\mathfrak{m}_R S).$$

Induction gives flatness of S/xS over R/xR.

The two-term free resolution of R/xR has differential multiplication by x. Its tensor with S therefore has zero first homology, because x is a nonzerodivisor on S. Thus $\operatorname{Tor}_1^R(R/xR, S) = 0$. Apply the entire proper-ideal local criterion AG-CA-08, Corollary 4.4, with ideal (x) and module S, which is finite over the stipulated Noetherian local overring S. The quotient is flat and its required Tor vanishes. This proves flatness over R and completes the induction. □

Lemma F.10: NAK

Reading sources: Stacks, Tag 00DV; AI Integrated Stacks, pinned Algebra, lines 3732–3813.

For a ring R, an ideal I, and an R-module M, the following twelve forms of Nakayama's lemma hold. Write $\mathrm{rad}(R)$ for the Jacobson radical.

1. If M is finite and $M = IM$, some $f \in 1 + I$ annihilates M.
2. Under the conditions of (1), $I \subset \mathrm{rad}(R)$ implies $M = 0$.
3. If $N, N' \subset M$, $M = N + IN'$ and N' is finite, some $f \in 1 + I$ satisfies $fM \subset N$ and $M_f = N_f$.
4. Under the conditions of (3), $I \subset \mathrm{rad}(R)$ implies $M = N$.
5. For a module map $N \to M$ with M finite, surjectivity of $N/IN \to M/IM$ implies surjectivity of $N_f \to M_f$ for some $f \in 1 + I$.
6. Under the conditions of (5), $I \subset \mathrm{rad}(R)$ makes the original map $N \to M$ surjective.
7. If M is finite and the images of $x_1, \ldots, x_n$ generate M/IM, these same elements generate M_f over R_f for some $f \in 1 + I$.
8. Under the conditions of (7), $I \subset \mathrm{rad}(R)$ implies that $x_1, \ldots, x_n$ generate M.
9. If I is nilpotent, $M = IM$ implies $M = 0$, with no finiteness requirement on M.
10. If I is nilpotent and $M = N + IN'$ for submodules N, N', then $M = N$, with no finiteness requirement on N'.
11. If I is nilpotent and $N/IN \to M/IM$ is surjective, then $N \to M$ is surjective, without module finiteness assumptions.
12. If I is nilpotent and a possibly infinite family (x_α) generates M/IM, it generates M.

Proof. The determinant assertion F.10 / nakayama, namely (1), supplies all finite-module cases. Choose a generating column $y = (y_1, \ldots, y_r)^{\mathsf{t}}$ of M. The equality $M = IM$ expresses each y_i as an I-linear combination of the y_j, so for a matrix Z with entries in I we have $(1_r - Z)y = 0$. By the proved adjugate identity F.70,

$$\mathrm{adj}(1_r - Z)(1_r - Z) = \det(1_r - Z)1_r.$$

Consequently $f = \det(1_r - Z)$ kills every generator. Reduction modulo I gives $f \equiv 1$, as required. An empty generating list uses $f = 1$.

If $I \subset \mathrm{rad}(R)$, F.34 proves that every element of $1 + I$ is a unit. Multiplying $fM = 0$ by its inverse proves (2).

For (3) and (4), put $Q = M/N$. Since the image of N' generates Q and N' is finite, Q is finite. The hypothesis $M = N + IN'$ gives $Q = IQ$. Apply (1) to obtain $fQ = 0$, equivalently $fM \subset N$, and apply (2) for the equality $Q = 0$ when the radical condition holds. Once $fM \subset N$, any fraction m/f^a in M_f equals $(fm)/f^{a+1}$ and comes from N_f. The inclusion $N_f \subset M_f$ follows from exact localization. Thus the stated equality of localized modules is proved as well.

For a map $u : N \to M$, let $Q = \operatorname{coker} u$. Right exactness identifies its quotient Q/IQ with the cokernel of $N/IN \to M/IM$. The assumed surjectivity therefore gives $Q = IQ$. If M is finite, so is Q. Assertion (1) makes $Q_f = 0$, and exact localization makes u_f surjective, proving (5). Assertion (2) gives $Q = 0$ in the radical case, proving (6). Apply these conclusions to $R^n \to M$, $(r_i) \mapsto \sum r_i x_i$, to get (7) and (8).

Finally suppose $I^a = 0$. Repeated substitution into $M = IM$ yields $M = I^a M = 0$, proving (9) for an arbitrary module. In (10), $Q = M/N$ satisfies $Q = IQ$, whether or not the image of N' is finite, so (9) gives $Q = 0$. For (11), take the same cokernel Q of the given map and use its zero quotient modulo I. For (12), quotient by the submodule generated by all x_α; its quotient modulo I is zero, so (9) kills it. This proves all twelve assertions, including empty generating lists and arbitrary nilpotent ideals. □

Lemma F.11: NL homotopy

Reading sources: Stacks, Tag 00S1; AI Integrated Stacks, pinned Algebra, lines 36700–36815.

Let $u : R \to R'$ and $\phi : S \to S'$ be compatible with the structure maps $R \to S$ and $R' \to S'$. Choose polynomial presentations $\alpha : P \twoheadrightarrow S$ and $\alpha' : P' \twoheadrightarrow S'$, allowing arbitrary sets of variables. Then:

1. There is a map $\psi : P \to P'$ whose restriction to R is u and for which $\alpha'\psi = \phi\alpha$.
2. Any two such maps induce chain-homotopic maps $\mathrm{NL}(\alpha) \to \mathrm{NL}(\alpha')$.
3. Composing presentation maps composes their induced complex maps.
4. If u and ϕ are isomorphisms, every induced map is a homotopy equivalence and a quasi-isomorphism.

In particular the comparison with the canonical presentation identifies $\mathrm{NL}(\alpha)$ with $\mathrm{NL}_{S/R}$ up to homotopy, and these identifications respect the stated functorialities up to homotopy.

Proof. Write $P = R[x_a \mid a \in E]$, $I = \ker \alpha$ and $I' = \ker \alpha'$. For each a, choose a polynomial in P' lifting $\phi(\alpha(x_a))$. Sending the variables to these choices and the coefficients through u gives ψ by the polynomial universal property. Each polynomial uses only finitely many variables, so this construction also works when E is infinite. It proves (1).

The two terms of the complex are

$$C_1 = I/I^2, \qquad C_0 = \bigoplus_{a \in E} S\, dx_a, \qquad d : C_1 \to C_0, \quad [f] \mapsto df.$$

The map is well-defined because $d(fg) = f\,dg + g\,df$ vanishes modulo I when $f, g \in I$. Its target counterpart is $C_1' = I'/(I')^2 \to C_0' = \Omega_{P'/R'} \otimes_{P'} S'$. Regard these target modules as S-modules through ϕ. The induced maps are

$$\psi_1([f]) = [\psi(f)], \qquad \psi_0(dx_a) = d\psi(x_a).$$

The map on I/I^2 is well-defined since $\psi(I) \subset I'$. On a monomial the product rule gives $\psi_0(df) = d\psi(f)$, since differentiation over R' kills the transformed coefficient from R. Addition gives this identity for every polynomial. Thus these are complex maps.

For another choice χ, both evaluations of any $p \in P$ have the same residue in S'. Set

$$D(p) = \psi(p) - \chi(p) \pmod{(I')^2} \in I'/(I')^2.$$

Both choices give the same action of P on this module: their difference lies in I', which kills $I'/(I')^2$. That action factors through S, because the image of I is in I'. The exact product identity

$$\psi(pq) - \chi(pq) = \psi(p)\big(\psi(q) - \chi(q)\big) + \big(\psi(p) - \chi(p)\big)\chi(q)$$

therefore becomes the derivation rule for D. It is additive and vanishes on R. Define the S-linear map

$$h : C_0 \longrightarrow C_1', \qquad h(dx_a) = D(x_a).$$

Expanding a monomial by the derivation rule shows $h(dp) = D(p)$ for every polynomial p, including constant and empty-variable cases.

Both homotopy identities can now be checked without a diagram omission. For $f \in I$,

$$h(d[f]) = D(f) = \psi_1([f]) - \chi_1([f]).$$

For each free differential generator,

$$d'h(dx_a) = d\big(\psi(x_a) - \chi(x_a)\big) = \psi_0(dx_a) - \chi_0(dx_a).$$

The first identity holds on all relation classes and the second on a free basis, so

$$\psi_1 - \chi_1 = hd, \qquad \psi_0 - \chi_0 = d'h.$$

These are exactly a chain homotopy, proving (2).

For a subsequent presentation map $\eta : P' \to P''$, the relation-class map sends $[f]$ successively to $[\psi(f)]$ and $[\eta\psi(f)]$. The differential map sends dx_a to $d\psi(x_a)$ and then to $d(\eta\psi(x_a))$, by the same monomial product-rule calculation. Thus the composite equals the map of the composite presentation, proving (3).

If the two structural square maps are isomorphisms, apply (1) to their inverse square to get a presentation map in the reverse direction. Both composites lie over identity squares. By (2), their complex maps are homotopic to the respective identity maps. This is the required homotopy equivalence. A homotopy changes a degree-one map by hd and a degree-zero map by $d'h$; the former is zero on the kernel and the latter has zero class in the cokernel. Thus homotopic maps act equally on both homology modules. The two reverse maps consequently induce inverse isomorphisms on kernel and cokernel, proving the quasi-isomorphism assertion. Taking identity structural maps and the canonical polynomial presentation proves the last assertion, with composition compatibility already established. □

Lemma F.12: Noetherian field extension

Reading source: AI Integrated Stacks, pinned Algebra, lines 6331–6347.

If A is a Noetherian algebra over a field k, then extending scalars from k to a finitely generated field extension K leaves it Noetherian.

Proof. Select a finite field-generating list $\xi_1, \dots, \xi_r$ for K/k, and put $B = k[\xi_1, \dots, \xi_r] \subset K$. The field of fractions of B is K, since it contains both k and all chosen field generators. Therefore

$$K = (B \setminus \{0\})^{-1} B.$$

A polynomial surjection $k[T_1, \dots, T_r] \to B$ gives a surjection $A[T_1, \dots, T_r] \to A \otimes_k B$. The Hilbert-basis and quotient arguments in AG-CA-03, Theorem 2.1, make the latter algebra Noetherian.

Localizing it at the images of the nonzero elements of B gives $A \otimes_k K$: the maps in both directions send a pure tensor with a fraction to that fraction of the corresponding pure tensor. They are inverse by the tensor and localization universal properties. AG-CA-03, Theorem 2.1, also proves that a localization of a Noetherian ring is Noetherian, giving the result. For a finite extension the same conclusion follows from its finite k-basis, which makes the tensor algebra a finite free A-module; its ideals are then finite A-submodules. The argument requires finite field generation, rather than finite degree. □

Lemma F.13: another variant local criterion flatness

Reading source: AI Integrated Stacks, pinned Algebra, lines 24942–24990.

Consider a commutative square of local homomorphisms between Noetherian local rings

$$\begin{array}{ccc} R & \longrightarrow & R' \\ \downarrow & & \downarrow \\ S & \longrightarrow & S'. \end{array}$$

Let $I \subsetneq R$ and let M be a finite S-module. Put $I' = IR'$ and $M' = M \otimes_S S'$. Assume S' is a localization of $B = S \otimes_R R'$, the module M/IM is flat over R/I, and the natural map

$$\operatorname{Tor}_1^R(M, R/I) \longrightarrow \operatorname{Tor}_1^{R'}(M', R'/I')$$

is zero. Then M' is flat over R'.

Proof. We prove the needed Tor surjection and then apply the proper-ideal criterion. All free-resolution comparisons, naturality and balance used here are the complete arbitrary-module calculus of AG-RG-S01, Lemma 1.A.

Choose an R-free presentation $0 \to K \to F \to M \to 0$, with no finiteness requirement on F. Set $T = \operatorname{Tor}_1^R(M, R/I)$, and let U be the image of K/IK in F/IF. The free-presentation kernel formula gives exact sequences

$$0 \to T \to K/IK \to U \to 0, \qquad 0 \to U \to F/IF \to M/IM \to 0.$$

Since the last quotient is flat, the proved flat-syzygy criterion AG-CA-07, Theorem 6.1, makes U flat over R/I. Tensoring the first sequence with an R/I-module N is consequently left exact. Tensoring the second is left exact because its last quotient is flat. Their kernel comparison gives the natural identity

$$\operatorname{Tor}_1^R(M, N) = T \otimes_{R/I} N.$$

This uses only the kernel formula for a free presentation; it does not assume that K is flat.

Take $N = R'/I'$. In $F_0 = F \otimes_R R'$ let K_0 be the image of $K \otimes_R R'$. Then F_0 is R'-free and $F_0/K_0 = M_0 = M \otimes_R R'$. The surjection $K \otimes_R R' \to K_0$ remains surjective after tensoring with R'/I'. A cycle in

$$\ker\bigl(K_0 \otimes_{R'} R'/I' \to F_0 \otimes_{R'} R'/I'\bigr)$$

therefore has a lift in $K \otimes_R R'/I'$. That lift maps to zero in the same free-module target, so is already a source cycle. Thus

$$\operatorname{Tor}_1^R(M, R'/I') \twoheadrightarrow \operatorname{Tor}_1^{R'}(M_0, R'/I').$$

The preceding natural identity identifies its source with $T \otimes_R R'$.

Write $S' = V^{-1}B$, so $M' = V^{-1}M_0$. We justify localization of the last Tor group with its natural B-action. Choose an R'-free resolution of R'/I'. Balance identifies Tor with the homology after tensoring this resolution with M_0; multiplication by B acts on each tensor term. Exact localization of B-modules commutes with kernels and images, and tensoring the same resolution with $V^{-1}M_0$ is the localization of each term. Hence

$$\operatorname{Tor}_1^{R'}(M', R'/I') = V^{-1} \operatorname{Tor}_1^{R'}(M_0, R'/I').$$

The preceding surjection is B-linear: multiplication by S acts by the functorial module maps on M, while R' acts by scalar extension; the comparison calculus in Lemma 1.A makes these actions compatible. Localizing that surjection consequently gives

$$V^{-1}(T \otimes_R R') \twoheadrightarrow \operatorname{Tor}_1^{R'}(M', R'/I').$$

The hypothesis says the original natural map from T to this target is zero. Its R'-linear extension is zero, and its extension across V is still zero, because the target is a module over $V^{-1}B$. The displayed surjection is therefore the zero map, forcing its target to vanish.

Finally $M'/I'M'$ is obtained from the flat R/I-module M/IM by scalar extension to R'/I' and localization at V. Scalar extension preserves flatness. For the localization, an injection of R'/I'-modules remains injective after tensoring with the unlocalized flat module, and remains injective after exact localization of the resulting B-modules. This proves flatness of the localized quotient. The ideal I' is proper because $R \to R'$ is local and $I \subset \mathfrak{m}_R$. The module M' is finite over the stipulated Noetherian local ring S'. AG-CA-08, Corollary 4.4, applies with this quotient flatness and the Tor vanishing just proved, and gives R'-flatness. No flatness assumption on $R \to R'$ was added. □

Lemma F.14: base change relative global complete intersection

Reading sources: Stacks, Tag 00SS; AI Integrated Stacks, pinned Algebra, lines 38076–38106.

For a relative global complete-intersection presentation

$$S = R[x_1, \dots, x_n]/(f_1, \dots, f_c),$$

all three of the following presentations have the same property:

1. For any $R \to R'$, the coefficientwise base change $S \otimes_R R' = R'[x_1, \dots, x_n]/(f_1, \dots, f_c)$.
2. If $g \in S$ is represented by $h \in R[x_1, \dots, x_n]$, the principal localization
$$S_g = R[x_1, \dots, x_n, Z]/(f_1, \dots, f_c, hZ - 1).$$
3. If $f \in R$ already has a unit image in S, its presentation over R_f obtained by viewing the same equations in $R_f[x_1, \dots, x_n]$.

Proof. The fibre condition of F.4 is a dimension condition. We first prove the field-extension calculation it requires. Let A be a finite-type k-algebra and K/k any field extension. Its nilradical is nilpotent and has finitely many minimal primes, by AG-CA-03, Proposition 2.2. Nilpotence persists after tensoring with K and does not affect dimension, since every prime contains every nilpotent element.

For the reduced quotient, let its minimal primes be $\mathfrak{q}_1, \dots, \mathfrak{q}_r$. Their product is zero, being contained in their zero intersection. After field extension the product of their extended ideals is still zero. Every prime contains one of these extended ideals: otherwise choose one element outside the prime from each ideal and multiply, contradicting primality. Thus the extended spectrum is the finite union of the closed spectra of

$$(A/\mathfrak{q}_i) \otimes_k K.$$

Every prime chain belongs to one of these closed sets, by choosing an ideal contained in its smallest prime. Dimension is therefore the maximum of their dimensions.

For each domain $A/\mathfrak{q}_i$, Noether normalization AG-CA-09, Corollary 3.2, supplies an injective finite polynomial subalgebra $k[t_1, \dots, t_{d_i}]$. Exact field tensor preserves its injection and finite module generators. The corresponding K-algebra is thus integral and injective over $K[t_1, \dots, t_{d_i}]$. The entire integral-dimension theorem AG-CA-05, Theorem 5.1, gives dimension d_i. Taking the maximum proves $\dim(A \otimes_k K) = \dim A$. The zero algebra has empty spectrum and causes no nonempty-fibre case.

For (1), a fibre at a prime of R' is the scalar extension of the original fibre at its contraction, by associativity of tensor products and the induced residue-field inclusion. A nonempty new fibre therefore has the original dimension $n - c$ by the calculation above. The displayed coefficientwise presentation proves (1).

For (2), the field case of F.77 proves that each original nonempty fibre is equidimensional of dimension $n - c$. Indeed its minimal-prime height calculation applies to its polynomial presentation over the fibre field, without any Noetherian assumption on R. A principal open in a surviving irreducible component has the same dimension as that component: its coordinate domain has unchanged fraction field, and AG-CA-09, Theorem 4.2, computes dimension by its transcendence degree. The localized nonempty fibre therefore still has dimension $n - c$. Adjoining Z and $hZ - 1$ adds one variable and one equation, preserving the difference $n - c$. Evaluation at $Z = g^{-1}$ and adjoining that inverse give both directions of the claimed algebra isomorphism.

For (3), multiplication gives $S \otimes_R R_f \cong S$, because the image of f already has an inverse. Its inverse sends s to $s \otimes 1$; the forced inverse of f verifies the other composite on fractions. Apply (1) to this base localization. All three arguments apply over arbitrary base rings and to empty fibres. □

Lemma F.15: change base NL

Reading sources: Stacks, Tag 00S4; AI Integrated Stacks, pinned Algebra, lines 36963–36981.

Let $\alpha : P \twoheadrightarrow S$ be a polynomial presentation over R, with an arbitrary variable set, and let $R \to R'$ be flat. Put $P' = P \otimes_R R'$, $S' = S \otimes_R R'$, and let $\alpha' : P' \twoheadrightarrow S'$ be the resulting presentation. There is an isomorphism of two-term complexes

$$\mathrm{NL}(\alpha) \otimes_R R' = \mathrm{NL}(\alpha) \otimes_S S' \cong \mathrm{NL}(\alpha').$$

Consequently the canonical comparison

$$\mathrm{NL}_{S/R} \otimes_S S' \longrightarrow \mathrm{NL}_{S'/R'}$$

is a homotopy equivalence.

Proof. Write $I = \ker \alpha$. Tensoring $0 \to I \to P \to S \to 0$ with the flat R' identifies $I' = \ker \alpha'$ with $I \otimes_R R'$. The square $(I')^2$ is the image of $I^2 \otimes_R R'$: a product of two elementary tensors from I' is the image of the product of their I-entries, and every product generator of I^2 is obtained this way. Tensoring the quotient therefore gives

$$I'/(I')^2 \cong (I/I^2) \otimes_R R'.$$

For any S-module E, the pure-tensor maps

$$E \otimes_S S' \longrightarrow E \otimes_R R', \quad e \otimes (s \otimes r') \longmapsto se \otimes r', \qquad e \otimes r' \longmapsto e \otimes (1 \otimes r')$$

are inverse. They give the other tensor description of the conormal term.

The polynomial differential term is free on the same dx_a over S and over S'. Differentiating a polynomial commutes with changing its coefficients, since every monomial derivative multiplies by the same integer exponent and lowers the same variable degree. Thus both differentials agree on every relation class, proving the complex isomorphism. Each polynomial has finite support, so no finite-variable assumption enters.

Apply F.11 to compare each presentation with the canonical one. Tensoring its explicit homotopy identities with S' preserves those identities. The complex isomorphism just proved, between these comparison homotopy equivalences, therefore identifies the displayed canonical map up to homotopy and supplies its inverse up to homotopy. This proves the final assertion without assuming that S is flat over R. □

Lemma F.16: characterize etale

Reading source: AI Integrated Stacks, pinned Algebra, lines 40942–40975.

Let $\mathfrak{q} \in \operatorname{Spec} S$ lie over $\mathfrak{p} \in \operatorname{Spec} R$. The map $R \to S$ is étale at $\mathfrak{q}$ if the following four hypotheses hold: it is finitely presented; $R_{\mathfrak{p}} \to S_{\mathfrak{q}}$ is flat; the ideal $\mathfrak{p}S_{\mathfrak{q}}$ is the maximal ideal of $S_{\mathfrak{q}}$; and $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is finite separable.

Proof. Write $k = \kappa(\mathfrak{p})$ and $L = \kappa(\mathfrak{q})$. The stipulated maximal-ideal equality identifies the local fibre ring with the field L. Since L/k is finite, the complete isolated-fibre argument F.56 supplies a principal neighbourhood of $\mathfrak{q}$ having no other point in this fibre. Its finite presentation is preserved by F.35. On this neighbourhood the fibre algebra $B = S \otimes_R k$ has exactly one prime. It is local: every nonunit lies in a maximal ideal, and there is only one such ideal. Therefore its localization at that prime is B itself. This localization is the given local fibre L, so $B = L$.

We prove directly the smoothness and differential vanishing of this finite separable field fibre. A finite k-basis generates L as a field; adjoining its entries gives a finite tower $k = E_0 \subset E_1 \subset \cdots \subset E_t = L$, with $E_i = E_{i-1}(\alpha_i)$. Let $p_i(T)$ be the monic minimal polynomial of α_i over E_{i-1}. The complete

monic-division and Bezout arguments AG-RG-S02, Lemma 4.A give $E_i = E_{i-1}[T]/(p_i)$. Each p_i divides the separable minimal polynomial of α_i over k, now regarded over E_{i-1}, so $p_i'(\alpha_i) \neq 0$.

Choose polynomial representatives in the preceding variables for all coefficients of p_i, obtaining $P_i(T_1, \ldots, T_i)$. Successive monic division gives

$$L = k[T_1, \ldots, T_t]/(P_1, \ldots, P_t).$$

The Jacobian over L is triangular with diagonal $p_i'(\alpha_i)$, so is invertible. The equation classes give a surjection $L^t \to I/I^2$, where $I = (P_1, \ldots, P_t)$, and its composite with the conormal differential to L^t is this invertible Jacobian. The surjection must also be injective, and the conormal differential is an isomorphism. Hence its kernel and cokernel vanish. Definition F.6 and presentation comparison give smoothness of L/k and $\Omega_{L/k} = 0$. For $L = k$, use the empty tower and the two zero modules.

The local flatness hypothesis and the finite presentation now meet every condition of F.45, giving a smooth principal neighbourhood of $\mathfrak{q}$. The complete arbitrary-base chart result F.83 gives a further neighbourhood with a standard smooth presentation

$$S = R[X_1, \ldots, X_n]/(f_1, \ldots, f_c).$$

Every inverted element has nonzero residue in L, so the fibre remains exactly L. By F.85(2), the differential module is free on $dX_{c+1}, \ldots, dX_n$. The full differential scalar-extension calculation AG-CA-15, Theorem 2.1, identifies its fibre with $\Omega_{L/k} = 0$. Thus $L^{n-c} = 0$, forcing $n = c$.

We check the lifting definition of étaleness on this final chart. Given a map to D/J with $J^2 = 0$, lift its variable tuple to $a \in D^n$. The error vector $f(a)$ lies in J^n. The square Jacobian $A(a)$ is invertible: its determinant is a unit modulo J, a lift of an inverse gives a product $1 + j$ with inverse $1 - j$, and the adjugate identity F.70 gives the matrix inverse. Take

$$b = a - A(a)^{-1} f(a).$$

Expanding each monomial and dropping terms with two entries from J gives $f(b) = f(a) + A(a)(b - a) = 0$. Thus b defines a lift. A second lifting root has the form $b + \epsilon$ for $\epsilon \in J^n$. The same expansion gives $A(b)\epsilon = 0$, and $A(b)$ is invertible by its unchanged reduction modulo J, so $\epsilon = 0$. Every square-zero test therefore has exactly one lift. With finite presentation this is étaleness by the programme definition. The chart contains $\mathfrak{q}$, proving the claim for the original map. Empty variable lists use the unique empty tuple and determinant 1; no Noetherian or perfect-field hypothesis was introduced. □

Lemma F.17: characterize formally smooth field extension

Reading source: AI Integrated Stacks, pinned Algebra, lines 45959–45969.

For any field extension K/k, the vanishing of $H_1(\mathrm{NL}_{K/k})$ is equivalent to formal smoothness of K as a k-algebra.

Proof. Choose a polynomial surjection $P = k[X_a \mid a \in E] \to K$, allowing an infinite set E, with kernel I. By F.11, the homology in question is the kernel of

$$d : I/I^2 \longrightarrow \Omega_{P/k} \otimes_P K.$$

Its cokernel is $\Omega_{K/k}$, by the conormal calculation. When this kernel is zero, the sequence with these three modules is short exact. A vector-space basis of $\Omega_{K/k}$ can be lifted to the middle module. Extending those lifts K-linearly splits the surjection, and hence the sequence.

The entire arbitrary-variable criterion AG-CA-17, Theorem 3.1, proves that this split conormal sequence is equivalent to formal smoothness. Its hypotheses apply to the chosen polynomial presentation over k with target K and impose no finite generation. Conversely, formal smoothness

gives a split exact conormal sequence by that same complete theorem, so its left map has zero kernel. F.11 identifies that kernel with the presentation-independent first cotangent homology, also denoted $H_1(L_{K/k})$ in the reading source. This proves both directions for arbitrary field extensions. □

Lemma F.18: characterize smooth kbar

Reading sources: Stacks, Tag 00TS; AI Integrated Stacks, pinned Algebra, lines 39962–40064.

Let S be a finite-type algebra over an algebraically closed field k, and let $\mathfrak{m}$ be a maximal ideal. Write

$$d = \dim S_{\mathfrak{m}}, \qquad e = \dim_{\kappa(\mathfrak{m})}\bigl(\Omega_{S/k} \otimes_S \kappa(\mathfrak{m})\bigr).$$

The following four conditions are equivalent: $S_{\mathfrak{m}}$ is regular local; $e \le d$; $e = d$; and some $g \in S \setminus \mathfrak{m}$ has S_g/k smooth.

Proof. The complete rational-point result AG-CA-18, Proposition 3.1, gives $\kappa(\mathfrak{m}) = k$ and

$$\Omega_{S/k} \otimes_S k \cong \mathfrak{m}/\mathfrak{m}^2.$$

The embedding-dimension theorem of AG-CA-14, Section 1, says $d \le \dim_k \mathfrak{m}/\mathfrak{m}^2$, with equality exactly for regularity. Thus the first three conditions are equivalent. The same complete AG-CA-18 proposition proves that smoothness at this closed rational point implies regularity, giving the implication from the fourth to the first.

For the converse we construct a smooth chart and identify its entire relation ideal. Suppose $S_{\mathfrak{m}}$ is regular. Choose a finite polynomial presentation $S = k[x_1, \dots, x_n]/I$, shifting each generator by its residue so that the preimage of $\mathfrak{m}$ is $\mathfrak{n} = (x_1, \dots, x_n)$. The exact conormal sequence at this point is

$$I/\mathfrak{n}I \longrightarrow \mathfrak{n}/\mathfrak{n}^2 \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \longrightarrow 0.$$

Since the middle dimension is n and the last is d, select $c = n - d$ elements $f_1, \dots, f_c \in I$ whose linear classes form a basis of the kernel. Put $J = (f_1, \dots, f_c)$ and $T = k[x]/J$, with maximal ideal $\mathfrak{m}' = \mathfrak{n}/J$. The corresponding conormal sequence shows that the cotangent space of $T_{\mathfrak{m}'}$ has dimension d. This local ring surjects onto $S_{\mathfrak{m}}$, so its dimension is at least d; its embedding dimension bounds it above by d. It is therefore regular of dimension d.

By AG-CA-14, Theorem 1.1, $T_{\mathfrak{m}'}$ is a domain. The surjection to $S_{\mathfrak{m}}$ has zero kernel. To prove this, a nonzero kernel would put every lifted quotient prime strictly above (0). A prime chain of length d in the quotient exists because its dimension is the finite integer d. Prepending (0) to the lifted chain would give length $d + 1$ in $T_{\mathfrak{m}'}$, a contradiction. Thus the local surjection is an isomorphism.

The independent linear coefficient rows of the f_i have a nonzero c-variable Jacobian minor Δ at this point. Existence of such a minor follows from the explicit pivot induction in D.2; for $c = 0$ take $\Delta = 1$. Reorder the pivot variables first. The presentation

$$T_\Delta = k[x_1, \dots, x_n, u]/(f_1, \dots, f_c, u\Delta - 1)$$

has a selected Jacobian determinant Δ^2: the last column is zero in the first c rows, its last entry is Δ, and the first square block has determinant Δ. This determinant is a unit in the quotient. The complete principal-localization standard-smooth calculation F.85 therefore proves T_Δ/k smooth.

It remains to pass from T to S. The kernel of $T \to S$ is finite since T is Noetherian, and becomes zero at $\mathfrak{m}'$ by the local isomorphism. For each of its finite generators take a localization annihilator outside $\mathfrak{m}'$. Their product a still avoids $\mathfrak{m}'$ and kills the entire kernel. Exact localization of the surjection then gives $T_a \cong S_{\bar{a}}$, hence

$$T_{a\Delta} \cong S_{\overline{a\Delta}}.$$

The left side is a principal localization of the smooth algebra T_Δ, and F.85(4) supplies its smooth presentation. On the right $g = \overline{a\Delta}$ avoids $\mathfrak{m}$. This gives the fourth condition, with the exact kernel

eliminated on the whole neighbourhood. Empty equation lists obey the same construction. The algebraically closed ground-field assumption is retained throughout. □

Lemma F.19: charpoly

Reading source: Stacks, Tag 00DX.

For a square matrix M of size n over any commutative ring R, let $P(T) = \det(T1_n - M)$. Then $P(M) = 0$ as an endomorphism of R^n.

Proof. Define determinants by the signed permutation formula. Grouping its terms according to the entry in a fixed row gives the cofactor expansion. If that row is replaced by another row, the same expansion is zero: permutations pair with the transposition of the two equal rows, and the paired terms have opposite signs. The column calculation is identical. Thus, for every square matrix C, the transpose of its signed cofactor matrix, denoted $\operatorname{adj}(C)$, satisfies

$$C \operatorname{adj}(C) = \operatorname{adj}(C)C = (\det C)1_n.$$

This argument uses addition and multiplication only, so it applies over $R[T]$ as well.

Assume $n > 0$, write

$$\operatorname{adj}(T1_n - M) = \sum_{j=0}^{n-1} C_j T^j, \qquad P(T) = \sum_{j=0}^{n} p_j T^j, \quad p_n = 1.$$

The degree bound follows because each cofactor is a determinant of size $n-1$. Comparing coefficients in the first adjugate identity gives

$$-MC_0 = p_0 1_n, \qquad C_{j-1} - MC_j = p_j 1_n \ (1 \le j < n), \qquad C_{n-1} = 1_n.$$

Multiply the equation indexed by j on the left by M^j, the constant equation by 1_n, and the last equation by M^n. Adding them cancels every $M^{j+1}C_j$ against its next occurrence. The result is $\sum_{j=0}^{n} p_j M^j = 0$. No substitution into a polynomial with noncommuting matrix coefficients, and no division, is involved. If $n = 0$, the empty determinant gives $P = 1$; its value is the identity of the zero module, which is the zero endomorphism. □

Lemma F.20: codimension

Reading source: AI Integrated Stacks, pinned Algebra, lines 29578–29596.

Let $S' \twoheadrightarrow S$ be a surjection of finite-type algebras over a field k. For a prime $\mathfrak{p} \subset S$, let $\mathfrak{p}' \subset S'$ be its inverse image, and let x, x' denote the corresponding points. Then

$$\dim_{x'} \operatorname{Spec} S' - \dim_x \operatorname{Spec} S = \operatorname{ht}_{S'} \mathfrak{p}' - \operatorname{ht}_S \mathfrak{p}.$$

Proof. The induced map $S'/\mathfrak{p}' \to S/\mathfrak{p}$ is an isomorphism: surjectivity descends, and the definition of $\mathfrak{p}'$ makes its kernel zero. Its fraction-field map therefore identifies $\kappa(\mathfrak{p}')$ with $\kappa(\mathfrak{p})$ as extensions of k. Call their common transcendence degree e.

The complete point-dimension argument AG-CA-09, Theorem 6.1, applies separately to the two finite-type algebras and gives

$$\dim_{x'} \operatorname{Spec} S' = \dim S'_{\mathfrak{p}'} + e, \qquad \dim_x \operatorname{Spec} S = \dim S_{\mathfrak{p}} + e.$$

Here point dimension is the minimum of the dimensions of open neighbourhoods. Localization identifies chains of primes in $S'_{\mathfrak{p}'}$ with chains in S' ending below $\mathfrak{p}'$; taking their supremum gives $\dim S'_{\mathfrak{p}'} = \operatorname{ht}_{S'} \mathfrak{p}'$. The same chain correspondence gives $\dim S_{\mathfrak{p}} = \operatorname{ht}_S \mathfrak{p}$. All dimensions are finite by that earlier finite-type theorem. Subtract the two displayed equalities; the identical terms e cancel and give the assertion. No equidimensionality assumption is needed. □

Lemma F.21: colimit eventually flat

Reading source: AI Integrated Stacks, pinned Algebra, lines 34105–34145.

Consider a directed system of local maps $R_\lambda \to S_\lambda$ between Noetherian local rings, with local transition maps, and let $R = \operatorname{colim}_\lambda R_\lambda$ and $S = \operatorname{colim}_\lambda S_\lambda$. Suppose that for $\mu \geq \lambda$, S_μ is the specified localization of $S_\lambda \otimes_{R_\lambda} R_\mu$. Let M_λ be finite S_λ-modules with $M_\mu = M_\lambda \otimes_{S_\lambda} S_\mu$, and put $M = \operatorname{colim}_\lambda M_\lambda$. These are exactly the kinds of models supplied by F.59 and F.60. If M is R-flat, then M_μ is R_μ-flat at some stage.

Proof. Fix λ, write $\mathfrak{m} = \mathfrak{m}_\lambda$, and form

$$T = \operatorname{Tor}_1^{R_\lambda}(M_\lambda, R_\lambda/\mathfrak{m}) = \ker\bigl(\mathfrak{m} \otimes_{R_\lambda} M_\lambda \longrightarrow M_\lambda\bigr).$$

The free-resolution and ideal-kernel calculation is AG-RG-S01, Lemma 1.A. The ideal $\mathfrak{m}$ is finite over R_λ, so its tensor with the finite S_λ-module M_λ is finite over S_λ. Since S_λ is Noetherian, its kernel T is finite as well. Choose generators $\xi_1, \dots, \xi_a$ over S_λ.

For $\nu \geq \lambda$, put $I_\nu = \mathfrak{m}R_\nu$. The transition of each ξ_i lies in

$$\ker(I_\nu \otimes_{R_\nu} M_\nu \longrightarrow M_\nu) = \operatorname{Tor}_1^{R_\nu}(M_\nu, R_\nu/I_\nu).$$

The colimit of the ideals I_ν is $\mathfrak{m}R$: a finite sum of images of generators has a representative at a common stage, and equality of two representatives holds at a later stage. The same finite sums and the finitely many tensor relations show

$$\operatorname*{colim}_{\nu \geq \lambda}(I_\nu \otimes_{R_\nu} M_\nu) \cong (\mathfrak{m}R) \otimes_R M.$$

Multiplication from the last module into M is injective, because tensoring the inclusion $\mathfrak{m}R \subset R$ with the flat module M preserves injectivity. Each ξ_i consequently has zero image in this colimit. The definition of a directed colimit says that its representative is already zero at some later stage. Choose a common successor μ for these finitely many witnesses. The entire natural map

$$T \longrightarrow \operatorname{Tor}_1^{R_\mu}(M_\mu, R_\mu/\mathfrak{m}R_\mu)$$

is now zero, since it is compatible with the scalar map $S_\lambda \to S_\mu$ and kills every chosen generator.

Apply the full local-square criterion F.13 to $(R_\lambda, S_\lambda, R_\mu, S_\mu)$ and $I = \mathfrak{m}$. This ideal is proper and its extension is proper because $R_\lambda \to R_\mu$ is local. The module $M_\lambda/\mathfrak{m}M_\lambda$ is flat over the field $R_\lambda/\mathfrak{m}$: any vector-space basis expresses it as a free module, without a finite-dimensional assumption. The localized tensor description and finite S_λ-module hypothesis are exactly those stipulated above. The zero Tor map just proved supplies the last hypothesis of F.13, giving R_μ-flatness of M_μ. □

Lemma F.22: colimit formally etale

Reading source: AI Integrated Stacks, pinned Algebra, lines 42732–42747.

Let (S_i) be a filtered system of formally étale R-algebras. Then $\operatorname{colim}_i S_i$ is formally étale over R.

Proof. Set $S = \operatorname{colim}_i S_i$. Test the formal lifting property on an R-algebra A, an ideal $J \subset A$ with $J^2 = 0$, and an R-map $u : S \to A/J$. The restriction $u_i : S_i \to A/J$ has exactly one lifting $v_i : S_i \to A$, by the assumed formal étaleness.

If $\phi_{ij} : S_i \to S_j$ is a transition map, then $v_j\phi_{ij}$ and v_i both lift u_i. Uniqueness on S_i forces $v_j\phi_{ij} = v_i$. The colimit universal property therefore gives a map $v : S \to A$ whose restrictions are all v_i. Its reduction agrees with u on every stage and hence on S. If v' is another lifting, its restriction to each S_i equals v_i, so the same universal property gives $v' = v$.

For a general nilpotent ideal J with $J^N = 0$, apply the square-zero test successively to $A/J^{n+1} \to A/J^n$ for $1 \leq n < N$. Its kernel J^n/J^{n+1} has square zero because $2n \geq n + 1$. Each lifting step

exists uniquely; induction reaches A and also proves uniqueness there. Thus either formulation of the formal étale test holds. No finite-presentation condition on any stage or on the colimit enters the proof. □

Lemma F.23: colimit syntomic

Reading source: AI Integrated Stacks, pinned Algebra, lines 46090–46123.

Every extension of fields K/k is a filtered union of k-subalgebras that are global complete intersections. If K/k is separable, it is a filtered union of smooth k-subalgebras.

Proof. We first place an arbitrary finite set $E \subset K$ in one of the asserted complete intersections. The subfield $L = k(E)$ has a finite transcendence basis $t_1, \dots, t_d$ and is finite over $F = k(t_1, \dots, t_d)$. Choose a finite tower

$$F = L_0 \subset L_1 \subset \cdots \subset L_r = L, \qquad L_i = L_{i-1}(y_i).$$

Write e_i for the degree of the monic minimal polynomial of y_i over L_{i-1}. Successive monic division, proved in AG-RG-S02, Lemma 4.A, makes the products $\prod_{j<i} y_j^{a_j}$ with $0 \le a_j < e_j$ an F-basis of L_{i-1}. Expand each coefficient of the minimal polynomial in this basis. Its basis coefficients belong to F; finitely many rational-function denominators occur.

Choose one nonzero $h \in k[t_1, \dots, t_d]$ clearing these denominators, and put $B = k[t_1, \dots, t_d, 1/h]$. The expansions give monic polynomials

$$P_i \in B[Y_1, \dots, Y_i], \qquad C = B[Y_1, \dots, Y_r]/(P_1, \dots, P_r).$$

At each successive quotient monic division gives a free module on $1, Y_i, \dots, Y_i^{e_i-1}$. Hence C is finite free over B on the corresponding product monomials. Multiplication by a monic polynomial in a polynomial ring is injective even over a ring with zero divisors: in a nonzero product, the leading coefficient of the other factor survives unchanged as the leading coefficient. Applying this fact at each stage proves that the displayed equation list is a regular sequence.

After tensoring with F, these same quotients are the chosen minimal-polynomial tower, so $C \otimes_B F = L$. Freeness over the domain B makes the map $C \to C \otimes_B F$ injective: a nonzero element of B cannot kill a nonzero coordinate in a free basis. We may thus regard C as a subalgebra of L. Expand the finitely many elements of E in its product basis over F, and enlarge h to clear their coefficients as well. The construction with this enlarged B then contains E.

The coefficient ring B has dimension d by the finite-type-domain dimension theorem AG-CA-09, Theorem 4.2; its fraction field is F. The finite injective integral map $B \to C$ preserves dimension by AG-CA-05, Theorem 5.1. Encoding $1/h$ by a variable Z gives

$$C \cong k[t_1, \dots, t_d, Y_1, \dots, Y_r, Z]/(P_1, \dots, P_r, hZ - 1),$$

where the coefficients of the P_i are polynomial representatives modulo $hZ - 1$. This presentation has $d + r + 1$ variables, $r + 1$ equations, and dimension d, so it satisfies the global complete-intersection definition F.4 over k.

If the extension is separable, F.5 makes the finitely generated subfield L separably generated. The entire forward construction in F.64 supplies a smooth finite-type domain $D \subset L$ with fraction field L. For every element of E choose a numerator and a nonzero denominator in D. Localizing at their finite product gives a smooth algebra containing E, by the principal-localization calculation F.85.

Finally, in either case two constructed subalgebras have finite sets of algebra generators over k. Apply the corresponding construction to the union of those generator sets. The resulting subalgebra contains both, proving directedness under inclusion. Applying the construction to a singleton includes each element of K. The union, and thus its filtered colimit, is exactly K. Empty towers and transcendence bases use the same argument with their empty products equal to one. □

Lemma F.24: colimits NL

Reading source: AI Integrated Stacks, pinned Algebra, lines 36983–36999.

For a filtered system of ring maps $R_\lambda \to S_\lambda$, put $R = \operatorname{colim} R_\lambda$ and $S = \operatorname{colim} S_\lambda$. The canonical two-term conormal complexes satisfy

$$\mathrm{NL}_{S/R} \cong \operatorname*{colim}_\lambda \mathrm{NL}_{S_\lambda/R_\lambda}.$$

Proof. Use the canonical presentation $P_\lambda = R_\lambda[X_s \mid s \in S_\lambda] \twoheadrightarrow S_\lambda$, with kernel I_λ. A transition sends coefficients along the base map and sends X_s to the variable named by the image of s. Thus distinct variables may become equal; injective transition maps are not assumed. Let $P = R[X_s \mid s \in S]$ and $I = \ker(P \to S)$.

Every element of P uses finitely many coefficients, variables and monomials. They all have representatives at one common stage, giving surjectivity of $\operatorname{colim} P_\lambda \to P$. If a representative polynomial is zero in P, the finitely many identifications among its variable labels and the finitely many coefficient equalities hold at a common later stage. The representative is zero there. This proves injectivity and identifies the two polynomial rings.

A representative polynomial belongs to I precisely when its evaluation becomes zero in S. That one finite equality holds at a later stage, so the polynomial can then be represented in I_μ. If an element of I_λ becomes zero in $I \subset P$, it is zero at a later polynomial stage, and hence at a later ideal stage. Therefore $I = \operatorname{colim} I_\lambda$. An element of I^2 is a finite sum of products of two elements of I. Choose a common stage for all those factors and their kernel equalities; their products lie in I_μ^2. Vanishing is again detected in a later polynomial ring. This proves $I^2 = \operatorname{colim} I_\lambda^2$.

Filtered colimits of the relevant modules preserve short exact sequences. Indeed, if a left representative maps to zero in the middle colimit, it is zero in the middle at a later stage, where injectivity makes it zero on the left. If a middle representative maps to zero on the right, at a later stage exactness expresses it as an image from the left. A right representative has a middle preimage at its own stage. These arguments also work for modules over the varying rings: the actions of the colimit rings are defined using common representatives. Applying them to $I_\lambda^2 \subset I_\lambda$ gives

$$I/I^2 \cong \operatorname*{colim}_\lambda I_\lambda/I_\lambda^2.$$

The degree-zero term is $\bigoplus_{s\in S} S\,dX_s$. Its vectors have finite support; a common-stage argument for their coefficients and labels, with later-stage detection of zero, identifies it with the colimit of $\bigoplus_{s\in S_\lambda} S_\lambda\,dX_s$. Formal differentiation of a polynomial is a finite coefficient calculation and respects every transition, including variable identifications. Consequently the two module identifications intertwine the maps $[f] \mapsto df$. They give the asserted isomorphism of entire two-term complexes. □

Lemma F.25: complete local Noetherian domain finite over regular

Reading source: AI Integrated Stacks, pinned Algebra, lines 46654–46712.

Every complete Noetherian local domain A contains a regular complete local subring A_0 such that A is finite over A_0 and the residue fields agree. The subring can be chosen as a power-series ring over a field or over a Cohen ring.

Proof. Write $\mathfrak{m}$ for the maximal ideal and $D = \dim A$. The complete coefficient-ring construction is AG-CA-19S, Theorem 5.1. Its arbitrary-residue-field step uses F.119, including the full degree-p argument: $T^p - a^p = (T-a)^p$, and the coefficient $-da$ of a proper monic factor, with $1 \le d < p$, forces the claimed degree. Thus this construction applies without a perfect-residue-field assumption.

In equal characteristic it embeds a coefficient field $k \subset A$. Choose a system of D parameters $x_1, \dots, x_D$ by AG-CA-11, Theorem 2.1. In mixed characteristic the coefficient ring is a Cohen DVR

C with uniformizer p and residue field k. It embeds in A: a nonzero kernel ideal in this DVR would contain a power of p, whereas the domain A has characteristic zero. The element p is consequently a nonzerodivisor in A, and AG-CA-11, Theorem 3.2, gives $\dim A/pA = D - 1$. Choose parameters $x_1, \dots, x_{D-1}$ in that quotient and lift them to A.

In the respective cases put

$$B = k[[X_1, \dots, X_D]] \quad \text{or} \quad B = C[[X_1, \dots, X_{D-1}]],$$

and let $\mathfrak{n}$ be its maximal ideal. The ideal I generated by the proposed images of $\mathfrak{n}$ is $(x_1, \dots, x_D)$ or $(p, x_1, \dots, x_{D-1})$. The parameter property gives $\sqrt{I} = \mathfrak{m}$. AG-CA-03, Proposition 2.2, supplies an integer b with $\mathfrak{m}^b \subset I$, while $I \subset \mathfrak{m}$. Hence

$$\mathfrak{m}^{bn} \subset I^n \subset \mathfrak{m}^n \qquad (n \geq 1).$$

The two filtrations are cofinal. Completeness and separatedness of A for its maximal-ideal filtration therefore imply the same properties for its I-adic filtration. Evaluation $X_i \mapsto x_i$ defines a ring map $B \to A$: the tail of a power series of degree at least n evaluates into I^n, and finite truncation followed by a separated limit respects addition and multiplication. This is the evaluation calculation of AG-CA-19S, Theorem 6.1.

The local quotient A/I has nilpotent maximal ideal and residue field k. Each maximal-ideal layer is finite-dimensional over k, since $\mathfrak{m}$ is finite and its powers are generated by finitely many products. The finite-layer proof in AG-CA-03, Theorem 4.2, therefore makes A/I finite-dimensional. Choose elements $a_1, \dots, a_s \in A$ lifting a k-basis of this quotient.

These elements generate A over B. To verify this directly, approximate any $a \in A$ modulo I by $\sum_j c_{j1} a_j$ with coefficients lifted from k to B. If coefficients have been chosen through order n, the error lies in I^n. Express it as a finite sum of degree-n products of the generators of I times elements of A. Replace each latter element modulo I by its residue-basis expansion. Lifting those coefficients to B gives corrections in $\mathfrak{n}^n$ and leaves an error in I^{n+1}. Induction constructs compatible coefficients c_{jn} modulo $\mathfrak{n}^n$. The complete ring B supplies limits c_j; continuity of evaluation and I-adic separatedness give $a = \sum_j c_j a_j$. Thus A is a finite B-module.

The ring B is the maximal-ideal completion of the polynomial local ring over k or C: their quotients by each maximal-ideal power have the same finite polynomial truncations, including the coefficient reductions modulo powers of p in the second case. AG-CA-14, Proposition 3.3, proves regularity of that polynomial local ring, and AG-CA-11, Theorem 5.1, gives its dimension D. Completion preserves this dimension and regularity by AG-CA-19, Theorem 4.1. Hence B is a regular local domain of dimension D.

Let J be the evaluation kernel. If it contains a nonzero element f, the one-equation theorem AG-CA-11, Theorem 3.2, gives $\dim B/(f) = D - 1$, and the further quotient B/J has dimension at most $D - 1$. But B/J embeds in the finite integral extension A, so AG-CA-05, Theorem 5.1, gives $\dim B/J = \dim A = D$, a contradiction. Therefore $J = 0$. Take A_0 to be the resulting embedded copy of B. Its residue map is the coefficient identification with k, and the finiteness has already been proved. If $D = 0$, the Noetherian local domain is its own residue field, so $A_0 = A$ gives the empty-variable case directly. □

Lemma F.26: completion at quasi finite prime

Reading source: AI Integrated Stacks, pinned Algebra, lines 32412–32464.

Let R be Noetherian, let $R \to S$ be finite type, and let $\mathfrak{q} \subset S$ be quasi-finite over $\mathfrak{p} = \mathfrak{q} \cap R$. There is an algebra D with

$$\widehat{R_{\mathfrak{p}}} \otimes_R S \cong \widehat{S_{\mathfrak{q}}} \times D,$$

whose first projection is the natural map to the completed local ring.

Proof. The full affine conductor theorem Zariski Main, Theorem C4.3 produces an integral subalgebra $C \subset S$ and $g \in C \setminus \mathfrak{q}$ such that $C_g = S_g$. Choose a finite set of R-algebra generators of S_g. Each is a fraction with numerator in C and denominator a power of g. Let $S' \subset C$ be generated over R by those numerators and by g. All these generators are integral. Repeated monic reduction, as in AG-CA-05, Section 1, shows that their bounded product powers generate S' as an R-module. Thus S' is finite over R, and the chosen generators give $S'_g = S_g$.

Put $\mathfrak{q}' = \mathfrak{q} \cap S'$. Localizing the last equality at the corresponding primes gives $S'_{\mathfrak{q}'} = S_{\mathfrak{q}}$, including their maximal ideals, so their maximal-adic completions are naturally identified. By F.27,

$$T' = \widehat{R_{\mathfrak{p}}} \otimes_R S' \cong A \times D', \qquad A = \widehat{S_{\mathfrak{q}}}.$$

Let $e \in T'$ be the idempotent $(1, 0)$. Its image in $T = \widehat{R_{\mathfrak{p}}} \otimes_R S$ gives an explicit product map

$$T \longrightarrow eT \times (1-e)T, \qquad t \longmapsto (et, (1-e)t),$$

with inverse addition. The structural map $A = eT' \to eT$ is unital. The image of g in A is a unit because $g \notin \mathfrak{q}'$, and remains a unit in eT.

Localizing $T' \to T$ at g gives an isomorphism, by $S'_g = S_g$ and the tensor/localization universal properties. Its e-factor is

$$A = A_g \xrightarrow{\sim} (eT)_g = eT.$$

This identifies the first factor with the required completion, and the second is $D = (1-e)T$.

It remains to determine the projection on all of T. On $\widehat{R_{\mathfrak{p}}}$ and S' it agrees with the completed-local structural maps by the construction in F.27. Given $s \in S$, the equality $S'_g = S_g$ writes $s/1 = b/g^n$ with $b \in S'$. Equality of these fractions means that some g^a kills $g^n s - b$ in S. Therefore $g^{n+a}s = g^a b \in S'$. Both the factor projection and the natural map $S \to \widehat{S_{\mathfrak{q}}}$ have the same value on this element and on the unit g. Multiplying by the inverse of g^{n+a} makes their values on s equal. The values on all pure tensors are now equal, proving the assertion about the first projection. □

Lemma F.27: completion finite extension

Reading source: AI Integrated Stacks, pinned Algebra, lines 24144–24184.

For a finite map of Noetherian rings $R \to S$ and a prime $\mathfrak{p} \subset R$, let $\mathfrak{q}_1, \ldots, \mathfrak{q}_r$ be all primes of S above $\mathfrak{p}$. Then

$$\widehat{R_{\mathfrak{p}}} \otimes_R S \cong \prod_{i=1}^{r} \widehat{S_{\mathfrak{q}_i}},$$

where each local ring is completed at its own maximal ideal.

Proof. Replace the base by $R_{\mathfrak{p}}$ and the finite algebra by $S_{\mathfrak{p}}$; tensor associativity gives the same left side, and localizing at each prime gives the same right side. Write $\mathfrak{m}$ for the resulting base maximal ideal. The integral lying-over and maximal-ideal results AG-CA-05, Theorems 3.3–3.4, show that the maximal ideals of this finite algebra are precisely the finitely many $\mathfrak{q}_i$, and all contract to $\mathfrak{m}$.

For $n \geq 1$, the ring $R/\mathfrak{m}^n$ has a finite filtration whose successive quotients are finite-dimensional over the residue field: each $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ is generated by finitely many products of generators of $\mathfrak{m}$. It is therefore Artinian. The finite module $S/\mathfrak{m}^n S$ has finite length over this ring, by the finite-module argument AG-CA-03, Theorem 3.3, and consequently is itself Artinian. Its primes are the images of the $\mathfrak{q}_i$: every prime over $\mathfrak{m}^n$ contracts to $\mathfrak{m}$ and is maximal by integrality.

The entire Artinian product theorem AG-CA-03, Theorem 4.2, now gives the canonical localization decomposition

$$S/\mathfrak{m}^n S \cong \prod_{i=1}^{r} S_{\mathfrak{q}_i}/\mathfrak{m}^n S_{\mathfrak{q}_i}.$$

These maps commute with reduction from order $n+1$ to order n, because every projection is the corresponding localization map. Inverse limits of a finite product are computed componentwise: a compatible tuple sequence is exactly a tuple of compatible sequences. Hence the $\mathfrak{m}S$-adic completion of S is

$$\varprojlim_n S/\mathfrak{m}^n S \cong \prod_i \varprojlim_n S_{\mathfrak{q}_i}/\mathfrak{m}^n S_{\mathfrak{q}_i}.$$

In the ith local ring the radical of $\mathfrak{m}S_{\mathfrak{q}_i}$ is its maximal ideal $\mathfrak{q}_i S_{\mathfrak{q}_i}$: a prime containing the former corresponds to a prime above $\mathfrak{m}$, and localization leaves only $\mathfrak{q}_i$. AG-CA-03, Proposition 2.2, supplies b_i such that

$$(\mathfrak{q}_i S_{\mathfrak{q}_i})^{b_i n} \subset \mathfrak{m}^n S_{\mathfrak{q}_i} \subset (\mathfrak{q}_i S_{\mathfrak{q}_i})^n.$$

These cofinal filtrations have canonically identical completions. The right-hand limits are thus $\widehat{S_{\mathfrak{q}_i}}$. Finally AG-CA-19, Theorem 3.1, identifies the left-hand limit with $\widehat{R} \otimes_R S$, because S is a finite R-module. Every identification respects the maps from R and S, giving the asserted natural product. If there are no primes above $\mathfrak{p}$, then $S_{\mathfrak{p}} = 0$, and both sides are the zero ring, with the empty product interpreted as that ring. □

Lemma F.28: compose finite type

Reading source: AI Integrated Stacks, pinned Algebra, lines 547–571.

Finite-type maps compose, and finitely presented maps compose. Given $R \to S' \to S$, if S/R is finite type then S/S' is finite type. If S/R is finitely presented and S'/R is finite type, then S/S' is finitely presented.

Proof. Choose generators $y_1, \dots, y_a$ for S' over R and $x_1, \dots, x_b$ for S over S'. Each coefficient of a polynomial expression in the x_j is itself a polynomial in the y_i with coefficients in R. Substitution shows that the combined finite list generates S over R, proving the first claim.

For the second claim take finite presentations

$$S' = R[Y_1, \dots, Y_a]/(u_1, \dots, u_c), \qquad S = S'[X_1, \dots, X_b]/(v_1, \dots, v_d).$$

Lift the finitely many coefficients of the v_j through $R[Y] \twoheadrightarrow S'$ to obtain $\widetilde{v}_j \in R[Y, X]$. Then

$$R[Y, X]/(u_1, \dots, u_c, \widetilde{v}_1, \dots, \widetilde{v}_d) \cong S.$$

Indeed quotienting first by the u_i gives $S'[X]$, where the remaining equations are exactly the v_j. This is a finite presentation of the composite.

If S is generated by a finite list over R, every element is already an R-coefficient polynomial in that list, hence also an S'-coefficient polynomial. This proves the third claim without a finiteness assumption on S'.

For the fourth claim write

$$S = R[X_1, \dots, X_b]/(f_1, \dots, f_d), \qquad S' = R[Y_1, \dots, Y_a]/J.$$

The ideal J need not be finite. For each i choose $h_i(X)$ representing the image of $\bar{Y}_i$ in S. Form the finitely presented S'-algebra

$$Q = S'[X_1, \dots, X_b]/(f_1, \dots, f_d, \bar{Y}_1 - h_1(X), \dots, \bar{Y}_a - h_a(X)).$$

Evaluation at the specified elements of S defines $Q \to S$. Conversely the images of the X_j in Q satisfy every f_j, giving an R-map $S \to Q$. Their composite on S fixes its R-algebra generators. The other composite fixes each X_j, and it fixes every coefficient from S' because it fixes $\bar{Y}_i = h_i(X)$ and S' is generated by those elements over R. Both composites are identities, so $Q \cong S$ as S'-algebras. This proves finite presentation over S' without making any extra assumption on J. □

Lemma F.29: compose smooth

Reading source: AI Integrated Stacks, pinned Algebra, lines 39039–39053.

If $R \to S$ and $S \to T$ are smooth ring maps, then $R \to T$ is smooth.

Proof. First translate the cotangent condition into the earlier lifting criterion. For a polynomial presentation of S/R, F.11 identifies its conormal complex with the canonical one. F.6 gives zero kernel and a projective differential cokernel, so the conormal sequence is split exact. AG-CA-17, Theorem 3.1 then proves that S is formally smooth over R. The same argument proves that T is formally smooth over S.

In a lifting problem $T \to C/J$ over R, where J is nilpotent, first lift its restriction $S \to C/J$ to an R-map $S \to C$. This supplies an S-algebra structure on C whose reduction is exactly the given one. Formal smoothness of T/S now lifts the original map to an S-map $T \to C$. In particular it is an R-map. Thus T/R is formally smooth, by an explicit composition of the two lifts.

Finite presentation is equally explicit. Write $S = R[x_1, \dots, x_n]/(f_1, \dots, f_r)$ and $T = S[y_1, \dots, y_m]/(g_1, \dots, g_s)$, and lift each coefficient in the g_j to a polynomial in the x_i. The resulting polynomials $\widetilde{g}_j$ give $T = R[x_i, y_l]/(f_i, \widetilde{g}_j)$: the two maps fixing the generators are inverse by the quotient universal property. Applying the same entire conormal criterion AG-CA-17, Theorem 3.1 to this finite presentation gives zero first homology and finite projective differentials. F.11 transports them to the canonical complex, so F.6 proves that $R \to T$ is smooth. The lifting and presentation arguments include zero algebras and empty lists. □

Lemma F.30: compose standard smooth

Reading source: AI Integrated Stacks, pinned Algebra, lines 38815–38878.

A composite of standard smooth ring maps is standard smooth.

Proof. Take standard smooth presentations

$$S = R[X_1, \dots, X_n]/(f_1, \dots, f_c), \qquad T = S[Y_1, \dots, Y_m]/(g_1, \dots, g_d).$$

Reorder the variables in each list so that the first c and first d variables respectively give the chosen square Jacobian minors. Denote their determinants by Δ_f and Δ_g; they are units in S and T. Choose polynomial representatives $\widetilde{g}_j \in R[X, Y]$ for the coefficients of every g_j. Quotienting first by (f) and then by the remaining equations gives

$$T \cong R[X_1, \dots, X_n, Y_1, \dots, Y_m]/(f_1, \dots, f_c, \widetilde{g}_1, \dots, \widetilde{g}_d).$$

Use the pivot variables $X_1, \dots, X_c, Y_1, \dots, Y_d$ and write Jacobian rows by equations. In T its selected matrix has the form

$$\begin{pmatrix} A & 0 \\ B & C \end{pmatrix}, \qquad A = (\partial f_i/\partial X_j), \quad C = (\partial g_i/\partial Y_j).$$

The upper-right entries are zero because the f_i contain no Y variables. Reduction of the derivatives of $\widetilde{g}_i$ with respect to Y_j equals the derivatives of g_i: changing polynomial coefficients commutes with the monomial differentiation rule.

In the permutation expansion of this determinant, a nonzero term must assign the first c rows to the first c columns; otherwise one of those rows uses a zero upper-right entry. The remaining rows then use the remaining columns. Factoring the two independent permutation sums gives $\det A \det C = \Delta_f \Delta_g$. The image of the first unit and the second given unit are units in T, so their product is a unit. The displayed finite presentation is therefore standard smooth.

If a presentation includes a principal localization at h, replace it by one more variable Z and equation $hZ - 1$. With Z added last to the pivot list, the old equation rows have zero Z entry and

the new last diagonal entry is h. The selected determinant is the old determinant times h, which is invertible in the localized quotient. This calculation encodes all the stated localized presentations before applying the preceding composition argument. Empty equation lists use determinant one and the same permutation calculation. □

Lemma F.31: composition essentially of finite type

Reading source: AI Integrated Stacks, pinned Algebra, lines 13129–13138.

Composites of essentially finite-type ring maps are essentially finite type. Composites of essentially finitely presented ring maps are essentially finitely presented.

Proof. Let $R \to S \to T$ be the maps. Write $S = U^{-1}A$, where A is a finite-type R-algebra, and write $T = V^{-1}(S[Y_1, \dots, Y_m]/J)$. Set $P = A[Y_1, \dots, Y_m]$ and let K be the kernel of $P \to S[Y]/J$. Its quotient $B = P/K$ is finite type over R, even if J and K have infinitely many generators.

The map $U^{-1}P \to S[Y]/J$ has kernel exactly $U^{-1}K$. Indeed every element of $U^{-1}P$ is f/u; its image is zero precisely when the image of f is zero, because the image of u is a unit. This identifies $U^{-1}B$ with $S[Y]/J$. For each $v \in V$, choose a fraction b_v/u_v representing it in $U^{-1}B$. Let $W \subset B$ be generated multiplicatively by the images of U and all the b_v. Since the u_v are already inverted, inverting b_v/u_v is equivalent to inverting b_v. The localization universal property in both directions gives $T \cong W^{-1}B$, proving the finite-type assertion. No single denominator for all relations has been assumed.

For finite presentation, choose A finitely presented over R and $J = (j_1, \dots, j_d)$ in $U^{-1}P$. Each j_i has the form f_i/u_i with $f_i \in P$ and $u_i \in U$. The images of $f_1, \dots, f_d$ generate the same ideal after localizing at U, since all their denominators are units there. Therefore

$$S[Y]/J \cong U^{-1}(A[Y]/(f_1, \dots, f_d)).$$

The algebra in parentheses is finitely presented over R: combine the finite presentation of A with these finitely many polynomial equations, as in F.28. As before, incorporate the numerator representatives of V into the final multiplicative set. Thus T is a localization of a finitely presented R-algebra. The two arguments also cover empty variable lists and the zero algebra. □

Lemma F.32: composition syntomic

Reading source: AI Integrated Stacks, pinned Algebra, lines 38404–38453.

For arbitrary ring maps $R \to S \to T$:

1. If both maps are syntomic, their composite is syntomic.
2. If both maps admit relative global complete-intersection presentations, their composite admits such a presentation.

Proof. Begin with (2). Choose presentations

$$S = R[X_1, \dots, X_n]/(f_1, \dots, f_c), \qquad T = S[Y_1, \dots, Y_m]/(h_1, \dots, h_d).$$

Lifting each coefficient of h_j to $R[X]$ gives polynomials $\widetilde{h}_j$ and the exact presentation

$$T \cong R[X, Y]/(f_1, \dots, f_c, \widetilde{h}_1, \dots, \widetilde{h}_d).$$

To check F.4, fix $\mathfrak{p} \subset R$ and let $k = \kappa(\mathfrak{p})$, $A = S \otimes_R k$ and $B = T \otimes_R k$. Empty B imposes no dimension condition. If $B \neq 0$, then $A \neq 0$, and $\dim A = n - c$. Every nonempty fibre of $A \to B$ has dimension $m - d$: it is a field extension of a corresponding fibre of $S \to T$, so the coefficient-change calculation F.14 applies.

We prove both dimension inequalities for B. Given a minimal prime η of B, let $\mathfrak{a} = \eta \cap A$, $F = \kappa(\mathfrak{a})$ and $L = \kappa(\eta)$. The finite-type domain $A/\mathfrak{a}$ has dimension $\operatorname{trdeg}_k F \leq \dim A = n - c$, by AG-CA-09, Theorem 4.2. The prime induced by η in $B \otimes_A F$ has residue field L. That fibre has dimension $m - d$, so applying the same domain dimension theorem to its prime quotient gives $\operatorname{trdeg}_F L \leq m - d$. Transcendence degrees add here. To check this, choose a finite transcendence basis U of F/k and one V of L/F. The elements of V are independent over F, hence over $k(U)$; together with the independence of U this proves independence of $U \cup V$ over k. Algebraicity of $F/k(U)$ and $L/F(V)$, and transitivity of algebraicity, show that $L/k(U, V)$ is algebraic. Thus

$$\dim(B/\eta) = \operatorname{trdeg}_k L = \operatorname{trdeg}_k F + \operatorname{trdeg}_F L \leq n - c + m - d.$$

The Noetherian finite-type algebra B has finitely many minimal primes, and its dimension is the maximum of their quotient dimensions. This proves the upper bound.

For the lower bound, the polynomial presentation of B has $n + m$ variables and $c + d$ equations. A prime minimal over its equation ideal has height at most $c+d$, by the full height theorem AG-CA-11, Theorem 3.1. The polynomial domain dimension formula AG-CA-09, Theorem 4.3, makes its quotient dimension at least $n + m - c - d$. Such a prime exists because $B \neq 0$. Consequently $\dim B \geq n + m - c - d$. Equality follows, proving (2) over arbitrary R.

For (1), first suppose that R is Noetherian. Finite presentation of both syntomic maps makes S and T Noetherian by the Hilbert basis theorem. At $\mathfrak{q} \subset T$, with contraction $\mathfrak{r} \subset S$, the entire neighbourhood criterion F.87 supplies $g \in S \setminus \mathfrak{r}$ and $h \in T \setminus \mathfrak{q}$ such that $R \to S_g$ and $S \to T_h$ have relative global complete-intersection presentations. By F.14, $S_g \to T_{gh}$ has one too. Assertion (2) gives such a presentation for $R \to T_{gh}$. These opens cover $\operatorname{Spec} T$. Its quasi-compactness AG-CA-01, Proposition 2.3 gives a finite subcover. Each chart is syntomic by F.87, and the complete gluing proof F.62 gives syntomicity of T/R.

Now let R be arbitrary. Finite presentation follows from F.28, and flatness follows by composing the two exact tensor functors, as in AG-CA-07, Proposition 3.2. For $\mathfrak{p} \subset R$ use the same k, A, B as above. The map $k \to A$ is syntomic: finite presentation and flatness survive base change, and its only field fibre is the given local complete intersection. The map $A \to B$ is also syntomic. Its finite presentation and flatness are the base changes of those of $S \to T$. At a prime $\mathfrak{a} \subset A$ over $\mathfrak{r} \subset S$, its fibre is

$$B \otimes_A \kappa(\mathfrak{a}) \cong (T \otimes_S \kappa(\mathfrak{r})) \otimes_{\kappa(\mathfrak{r})} \kappa(\mathfrak{a}).$$

Choose the principal global complete-intersection cover of the original field fibre, as in F.2. Scalar extension carries that cover to a cover, and F.14 carries each of its presentations to a global complete-intersection presentation. Thus the displayed fibre is a local complete intersection. Since the base k is Noetherian, the case already proved applies to $k \to A \to B$. It makes B/k syntomic, in particular a local complete intersection. Every field fibre of T/R has now been checked. Together with flatness and finite presentation, the syntomic definition F.1 proves (1) for arbitrary bases. □

Lemma F.33: conormal module localize

Reading source: AI Integrated Stacks, pinned Algebra, lines 37203–37225.

Suppose that $R[X_1, \ldots, X_n] \twoheadrightarrow S$ and $R[Y_1, \ldots, Y_m] \twoheadrightarrow S_g$ have kernels I and J. There is an isomorphism of S_g-modules

$$(I/I^2)_g \oplus S_g^{\oplus m} \cong J/J^2 \oplus S_g^{\oplus n}.$$

Proof. Put $C = S_g$, $P = R[X]$ and $Q = R[Y]$. Choose $f \in P$ mapping to g. Exact localization gives $P_f \twoheadrightarrow C$ with kernel I_f. Lift the image of each Y_j to $u_j \in P_f$, and lift the image of each X_i to $v_i \in Q$.

The simultaneous evaluation $W = P[Y] = Q[X] \to C$ has kernel $L = (J, X_i - v_i(Y))$: quotienting by the graph equations eliminates X and leaves exactly Q/J. Translation by $v_i(Y)$ makes $W =$

$Q[Z_1, \ldots, Z_n]$ and $L = (J, Z_1, \ldots, Z_n)$. In this polynomial ring

$$L^2 = (J^2,\ JZ_1, \ldots, JZ_n,\ Z_iZ_j \mid 1 \le i, j \le n).$$

Therefore reading the constant term modulo J^2 and each linear coefficient modulo J gives an isomorphism $L/L^2 \to J/J^2 \oplus C^n$. Its inverse sends a relation class to its constant representative and each coordinate to the class of the corresponding Z_i. Changes of representatives lie in the displayed square, so both maps are well defined and inverse.

The image of f is a unit in C, and L/L^2 is a C-module. Multiplication by f on it is consequently invertible. Localizing thus gives the same module, namely

$$L/L^2 \cong L_f/L_f^2.$$

On $W_f = P_f[Y]$, the same evaluation has kernel $(I_f, Y_j - u_j)$: eliminating Y leaves $P_f/I_f = C$. Translate Y_j by u_j. The identical constant-and-linear calculation now gives

$$L_f/L_f^2 \cong I_f/I_f^2 \oplus C^m.$$

The ideal-product fraction formulas identify I_f/I_f^2 with $(I/I^2)_g$. Combining these isomorphisms gives the assertion. No free summand was cancelled, and no flatness of C over R was used. Empty variable lists and $C = 0$ satisfy the same formulas. □

Lemma F.34: contained in radical

Reading source: Stacks, Tag 0AME.

For an ideal I of a commutative ring R, the inclusion $I \subset \mathrm{Jac}(R)$ is equivalent to invertibility of every $1 + i$, $i \in I$. Under either condition, every element invertible modulo I is already invertible in R. Here the Jacobson radical is the intersection of all maximal ideals.

Proof. We first justify the maximal-ideal test for a unit. A nonunit a generates a proper ideal. The partially ordered set of proper ideals containing (a) has an upper bound for every chain, its union: that union cannot contain 1, since then a member would contain 1. Zorn's lemma supplies a maximal proper ideal containing a. Conversely no unit belongs to a proper ideal. Therefore an element is a unit exactly when it belongs to no maximal ideal.

Suppose I is contained in every maximal ideal. If a maximal ideal contained $1 + i$, it would contain its difference with i, namely 1. This is impossible, so the unit test makes every $1 + i$ invertible. Conversely, if a maximal ideal $\mathfrak{m}$ fails to contain I, maximality gives $I + \mathfrak{m} = R$. Write $1 = i + b$ with $i \in I$ and $b \in \mathfrak{m}$. The element $b = 1 - i$ has the stipulated form, yet is not a unit. This contradiction proves the inclusion.

If $a \bmod I$ is a unit, lift one inverse to $b \in R$. Then $ab = 1 + i$ for some $i \in I$. The element $b(ab)^{-1}$ is an actual inverse of a, proving the last assertion. In the zero ring the empty intersection is the whole ring and its unique element is its identity and a unit; the conclusions hold there too. □

Lemma F.35: cover upstairs

Reading source: AI Integrated Stacks, pinned Algebra, lines 4346–4405.

Let $g_1, \ldots, g_r \in S$ generate the unit ideal. The R-algebra S is finite type, respectively finitely presented, if and only if every S_{g_i} has that property.

Proof. We will use a finite-cover calculation for arbitrary modules. In a ring A, suppose $\sum_i c_i t_i = 1$ and $N_{t_i} = 0$ for all i. For $n \in N$, choose positive e_i with $t_i^{e_i} n = 0$. In the expansion of $(\sum_i c_i t_i)^b$ with $b = 1 + \sum_i (e_i - 1)$, every monomial has an exponent at least e_i in one of the t_i. Multiplication by n kills every monomial. Since the expanded element is one, $n = 0$. Thus $N = 0$ without a finiteness hypothesis.

Choose h_i with $\sum_i h_i g_i = 1$. If every S_{g_i} is finite type, choose a finite R-algebra generator list for each, written as fractions $a_{ij}/g_i^{b_{ij}}$. Let $S_0 \subset S$ be generated over R by all a_{ij}, g_i, h_i. The map $(S_0)_{g_i} \to S_{g_i}$ is surjective by the chosen generators. It is injective: a numerator from S_0 whose fraction vanishes in S_{g_i} is killed by a power of g_i in S, hence also in the subring S_0. Therefore $(S/S_0)_{g_i} = 0$ as a module over S_0. The g_i generate one in S_0, so the cover calculation gives $S = S_0$. This proves finite type.

If every chart is finitely presented, the finite-type case supplies $S = P/J$ for a finite polynomial ring $P = R[X_1, \dots, X_n]$. Choose lifts $u_i, v_i \in P$ of g_i, h_i. The kernel of

$$P[Z] \longrightarrow S_{g_i}, \qquad Z \longmapsto g_i^{-1},$$

is $JP[Z] + (u_i Z - 1)$, since its quotient is exactly $(P/J)[Z]/(g_i Z - 1)$. The entire presentation-independence proof F.44 makes this kernel finitely generated. Express each of its finite generators as a finite sum of multiples of elements of J and of $u_i Z - 1$. Collect the finitely many coefficients $a_{ij} \in J$ used in those sums. They, together with $u_i Z - 1$, generate the same kernel: they belong to it and generate each chosen generator.

Set

$$J_0 = (\sum_i v_i u_i - 1,\ a_{ij} \text{ for all } i, j) \subset J, \qquad A = P/J_0.$$

The images of u_i generate one in A. The preceding kernel equalities give isomorphisms $A_{u_i} \cong S_{g_i}$. Indeed adjoining Z with $u_i Z - 1$ identifies both quotients with the same localization, and every element of J_0 also belongs to the original kernel. The kernel J/J_0 of $A \twoheadrightarrow S$ therefore vanishes on this cover. The arbitrary-module calculation gives $J = J_0$, so the original finite polynomial presentation has a finite relation ideal.

Conversely, localization adds only g_i^{-1} to a finite generating list. For a finite presentation $S = P/(f_1, \dots, f_d)$ and a lift u_i of g_i it gives the exact finite presentation $S_{g_i} = P[Z]/(f_1, \dots, f_d, u_i Z - 1)$. If the cover is empty, its unit-ideal condition forces $S = 0$; the presentation $R/(1)$ handles both assertions directly. Empty generator lists cause no exception. □

Lemma F.36: criterion flatness fibre

Reading source: AI Integrated Stacks, pinned Algebra, lines 34374–34496.

Let $R \to S \to S'$ be local maps of local rings and write $\mathfrak{m} = \mathfrak{m}_R$. Suppose that both S/R and S'/R are essentially finitely presented, that $M \neq 0$ is a finitely presented S'-module, that M is R-flat, and that $M/\mathfrak{m}M$ is flat over $S/\mathfrak{m}S$. Then S is R-flat and M is S-flat. The original rings need not be Noetherian.

Proof. We construct compatible Noetherian local models, then apply the Noetherian fibre criterion twice through the two flatness hypotheses. The full finite-expression construction F.59 writes $R = \operatorname{colim} R_\lambda$, where the R_λ are local rings essentially of finite type over $\mathbf{Z}$, with maximal ideals $\mathfrak{p}_\lambda$ and local transitions. Their maximal ideals have colimit $\mathfrak{m}$.

An essentially finitely presented algebra that is local is a localization at one prime of a finitely presented algebra: all elements outside the inverse image of its maximal ideal are units, so the two localization universal properties identify the rings. Choose such finite presentations for S and S'. To model their intervening map, choose images in S' of the finite generators of S, represented by polynomials with finitely many denominators. Put those denominators among the inverted elements, and impose the finite graph equations equating the chosen source generators with their images. Eliminating the graph variables recovers the presentation of S' over R after localization. The source relations vanish there; any vanishing of their finitely many representatives is witnessed by finitely many additional denominators. Include these witnesses too. All the coefficients and finite equality witnesses can be placed at one stage of F.59.

For later stages we consequently have presentations

$$S_\lambda = (R_\lambda[X]/(f_\lambda))_{\mathfrak{q}_\lambda}, \qquad S'_\lambda = (R_\lambda[X,Y]/(f_\lambda, g_\lambda))_{\mathfrak{q}'_\lambda},$$

where the primes are contracted from the actual local target rings. The two maps are local. The coefficient lists are fixed and finite, so the Hilbert basis theorem makes all these model rings Noetherian. For $\mu \geq \lambda$, tensoring the displayed polynomials and relations gives exactly

$$S_\lambda \otimes_{R_\lambda} R_\mu \longrightarrow S_\mu, \qquad S'_\lambda \otimes_{S_\lambda} S_\mu \longrightarrow S'_\mu$$

as localizations. The same description holds with the final rings R, S, S'.

Write a finite module presentation $(S')^a \xrightarrow{A} (S')^b \to M \to 0$. By F.60, its finitely many matrix entries occur at a common stage. Define $M_\lambda = \operatorname{coker}((S'_\lambda)^a \xrightarrow{A_\lambda} (S'_\lambda)^b)$. Right exactness of tensor gives

$$M_\mu \cong M_\lambda \otimes_{S'_\lambda} S'_\mu, \qquad M \cong M_\lambda \otimes_{S'_\lambda} S'.$$

In particular every M_λ is nonzero, because a zero one would make the final tensor zero. It is finite over the stipulated Noetherian local overring S'_λ.

The R-flatness of M and the localized tensor descriptions allow F.21 to be applied to $R_\lambda \to S'_\lambda$ and M_λ. It gives a stage where M_λ is R_λ-flat. At every later stage this persists: scalar extension preserves the exact tensor functor, and localization of an algebra-module preserves its injection tests. The final module and the later ones have precisely these two forms.

Next put

$$A_\lambda = S_\lambda/\mathfrak{p}_\lambda S_\lambda, \qquad B_\lambda = S'_\lambda/\mathfrak{p}_\lambda S'_\lambda, \qquad N_\lambda = M_\lambda/\mathfrak{p}_\lambda M_\lambda.$$

These are local Noetherian algebra models with finite target modules. The local transitions and right exactness give their colimits $S/\mathfrak{m}S$, $S'/\mathfrak{m}S'$ and $M/\mathfrak{m}M$. Reduction of the same polynomial presentations along the residue-field maps $R_\lambda/\mathfrak{p}_\lambda \to R_\mu/\mathfrak{p}_\mu$ shows that

$$B_\lambda \otimes_{A_\lambda} A_\mu \longrightarrow B_\mu$$

is a localization, and that $N_\mu = N_\lambda \otimes_{B_\lambda} B_\mu$. Thus this quotient system has every model property required by F.21. The assumed flatness of its limit over $S/\mathfrak{m}S$ gives a later stage where N_λ is A_λ-flat. Passing to a common successor retains the first flatness assertion as well.

At that stage the local maps $R_\lambda \to S_\lambda \to S'_\lambda$ and the nonzero finite module M_λ satisfy the full criterion F.37. It gives R_λ-flatness of S_λ and S_λ-flatness of M_λ. Finally S is a localization of $S_\lambda \otimes_{R_\lambda} R$, so it is R-flat. The module M is a localization of $M_\lambda \otimes_{S_\lambda} S$, so it is S-flat. These conclusions use the exact scalar-extension and localization functors and impose no new hypothesis on the original rings. □

Lemma F.37: criterion flatness fibre Noetherian

Reading source: AI Integrated Stacks, pinned Algebra, lines 24999–25045.

Let $R \to S \to T$ be local maps of Noetherian local rings, let $\mathfrak{m} = \mathfrak{m}_R$, and let $M \neq 0$ be finite over T. If M is R-flat and $M/\mathfrak{m}M$ is flat over $S/\mathfrak{m}S$, then M is faithfully flat over S and S is flat over R.

Proof. Set $I = \mathfrak{m}S$. Locality gives $I \subset \mathfrak{m}_S$, so it is proper. The map

$$u : \mathfrak{m} \otimes_R M \longrightarrow I \otimes_S M, \qquad x \otimes z \longmapsto x \otimes z$$

is surjective: write an element of I as a finite sum of products sx and move each s to the module factor. Its composite with multiplication $I \otimes_S M \to M$ is injective by R-flatness. Therefore u is

injective as well as surjective, and multiplication from $I \otimes_S M$ is injective. The ideal-kernel Tor calculation of AG-RG-S01, Lemma 1.A, gives $\operatorname{Tor}_1^S(S/I, M) = 0$.

Apply the full proper-ideal criterion AG-CA-08, Corollary 4.4, to the source S, local overring T, finite T-module M and ideal I. Its flat quotient condition is exactly the assumed flatness of M/IM over S/I, and the Tor condition has just been proved. Thus M is S-flat. Since M is finite and nonzero over the local ring T, Nakayama F.10 gives $M/\mathfrak{m}_T M \neq 0$. The quotient $M/\mathfrak{m}_S M$ surjects onto it, by locality of $S \to T$, so this closed fibre is nonzero too. The complete flat-module criterion AG-CA-08, Theorem 1.1, gives faithful flatness over S.

Let $L = \operatorname{Tor}_1^R(R/\mathfrak{m}, S)$. The ideal sequence over R and its Tor-kernel calculation give the exact sequence of S-modules

$$0 \longrightarrow L \longrightarrow \mathfrak{m} \otimes_R S \longrightarrow I \longrightarrow 0.$$

Tensor with the flat S-module M. Tensor associativity identifies its middle term with $\mathfrak{m} \otimes_R M$, and its final term is $I \otimes_S M$. Their map is the isomorphism u already proved, so $L \otimes_S M = 0$. Faithfulness forces $L = 0$. AG-CA-08, Theorem 4.2, applied to the local map $R \to S$ and the module S, finite over its own Noetherian local overring, gives R-flatness of S. Both conclusions are established with the stated target-module finiteness, without assuming M finite over R or S. □

Lemma F.38: descent regular

Reading source: AI Integrated Stacks, pinned Algebra, lines 48353–48371.

If $R \to S$ is faithfully flat and S is a regular ring, then R is regular. In particular the source is Noetherian even if this is not separately assumed.

Proof. First prove that R is Noetherian. For an ideal $I \subset R$, its extension IS is finite because S is Noetherian. Express a finite generating list for IS using finitely many elements $a_1, \ldots, a_r$ of I, and let $I_0 = (a_1, \ldots, a_r)$ in R. Then $I_0 S = IS$. Flatness identifies $I \otimes_R S$ with IS and $I_0 \otimes_R S$ with $I_0 S$, by tensoring their inclusions into R. Right exactness consequently gives $(I/I_0) \otimes_R S = 0$. Faithfulness implies $I = I_0$. Every ideal is finite, proving Noetherianity.

Fix $\mathfrak{p} \subset R$. Faithful flatness supplies a prime $\mathfrak{q} \subset S$ above it, by AG-CA-08, Theorem 2.1. Put $A = R_{\mathfrak{p}}$, $B = S_{\mathfrak{q}}$ and $k = \kappa(\mathfrak{p})$. The induced map $A \to B$ is flat and local, hence faithfully flat by that same theorem. The ring B is regular local by the hypothesis.

Take a free resolution $F_\bullet \to k$ over the Noetherian local ring A. Tensoring with the flat algebra B gives a free resolution of $k \otimes_A B$ over B. The complex computing its self-Tor is

$$(F_\bullet \otimes_A B) \otimes_B (k \otimes_A B) \cong (F_\bullet \otimes_A k) \otimes_A B.$$

Exact tensor with B commutes with kernels and images, so the entire arbitrary-module resolution calculus AG-RG-S01, Lemma 1.A, gives

$$\operatorname{Tor}_i^A(k, k) \otimes_A B \cong \operatorname{Tor}_i^B(k \otimes_A B, k \otimes_A B).$$

The regular-local global-dimension theorem AG-CA-14, Theorem 2.2, makes the right-hand groups zero for sufficiently large i. Faithfulness makes the corresponding source groups zero.

The full minimal-resolution construction AG-CA-13, Theorem 2.2, gives a resolution of k by finite free A-modules whose differentials have entries in $\mathfrak{m}_A$. Tensoring it with k makes every differential zero. Thus its degree-i free term modulo $\mathfrak{m}_A$ is exactly $\operatorname{Tor}_i^A(k, k)$. Eventual vanishing makes those finite free terms zero, and the resolution truncates. Therefore k has finite projective dimension. The reverse implication in AG-CA-14, Theorem 2.2, proves that A is regular local. This holds at every source prime, proving regularity of R. □

Lemma F.39: dimension fibres bounded open upstairs

Reading source: AI Integrated Stacks, pinned Algebra, lines 32674–32690.

For a finite-type ring map $R \to S$ and $\mathfrak{q} \subset S$ over $\mathfrak{p} \subset R$, put

$$n = \dim_{\bar{\mathfrak{q}}} \operatorname{Spec}(S \otimes_R \kappa(\mathfrak{p})),$$

the dimension in a neighbourhood of its point in the fibre. Some principal open neighbourhood of $\mathfrak{q}$ has fibre point-dimension at most n at every point. The base ring is arbitrary.

Proof. The fibre is a finite-type algebra over $k = \kappa(\mathfrak{p})$ and hence Noetherian. By the definition of point dimension and the complete field point-dimension theorem AG-CA-09, Theorem 6.1, choose a principal neighbourhood of $\bar{\mathfrak{q}}$ in that fibre whose algebra has dimension exactly n. Its defining element is a fraction b/a with $b \in S$ and $a \in R \setminus \mathfrak{p}$. Since a is a fibre unit, the same fibre open is defined by b; also $b \notin \mathfrak{q}$. Replace S temporarily by S_b.

Apply the full Noether-normalization proof AG-CA-09, Lemma 2.1 and Corollary 3.2, to this fibre. It supplies n algebraically independent parameters and a finite map from their polynomial algebra to the fibre. These parameters lift to elements $t_1, \dots, t_n \in S_b$. Indeed that proof repeatedly replaces a generator by it minus an integer power of the last generator, so its chosen parameters are polynomials with integer coefficients in the original fibre generators. Dividing a relation by its nonzero leading coefficient proves integrality and does not change those parameter expressions.

The finite-type map $R[T_1, \dots, T_n] \to S_b$, $T_i \mapsto t_i$, is quasi-finite at the chosen point. Its fibre there is a further fibre of the displayed finite $k[T]$-algebra, and is therefore finite-dimensional over the contracted polynomial prime's residue field. The entire open-locus proof Zariski Main, Theorem C5.1, using its algebraic Theorem C4.3 over arbitrary bases, gives a principal open around that point on which this polynomial map is quasi-finite everywhere. A defining fraction in S_b has numerator $c \in S$; the resulting neighbourhood in the original spectrum is $D(bc)$.

For any $\mathfrak{p}' \subset R$, let $U = S_{bc} \otimes_R \kappa(\mathfrak{p}')$. The map $\kappa(\mathfrak{p}')[T] \to U$ is quasi-finite. This can be checked directly on its fibres: the polynomial base ring is a quotient and localization of $R[T]$ along $\mathfrak{p}'$, so a point and its residue field are those of the corresponding prime of $R[T]$, and its fibre is the fibre of the quasi-finite polynomial map just constructed.

At a minimal prime η of a nonzero U, its residue field is finite over the residue field of its contracted polynomial prime, by the full isolated-fibre criterion F.56. The latter field has transcendence degree at most n over $\kappa(\mathfrak{p}')$, since it is generated as a field by the images of n polynomial variables. Finite algebraic extension preserves this transcendence degree. AG-CA-09, Theorem 4.2, therefore gives $\dim(U/\eta) \le n$. Taking the maximum over its finitely many minimal primes gives $\dim U \le n$. Every neighbourhood dimension at a point of this fibre is at most its total dimension, proving the required bound on $D(bc)$. Empty fibres have no point to check. The earlier conductor proof includes the polynomial-normality argument Zariski Main over an arbitrary base, Corollary B1.2; no Noetherian assumption on R or S has entered the argument. □

Lemma F.40: essentially of finite type into artinian local

Reading source: Stacks, Tag 07DT.

Let $R \to S$ be any ring map, with S Artinian local, maximal ideal $\mathfrak{m}$, and residue field $K = S/\mathfrak{m}$. Each of the following properties of S/R is equivalent to the same property of K/R: finite, of finite type, and essentially of finite type.

Proof. The Artinian length theorem AG-CA-03, Theorem 4.2 gives a finite composition series of the S-module S. Every simple factor is K: a nonzero element of a simple module generates it, and its annihilator is the unique maximal ideal. If K is finite over R, each factor is therefore a finite R-module. Finite generation passes through an extension directly: lift finitely many generators of

the quotient and adjoin finitely many generators of the submodule. Induction on the composition series makes S finite over R. The converse follows by taking the images of module generators in K. This proves the finite assertion for an arbitrary source ring.

For finite type, a quotient keeps the images of any finite list of algebra generators. Conversely, lift finite R-algebra generators of K to $s_1, \dots, s_a \in S$. The polynomial map $P = R[X_1, \dots, X_a] \to S$, $X_i \mapsto s_i$, is surjective on residues. Thus K is cyclic as a P-module. The finite assertion just proved makes S a finite P-module. Choose its module generators $u_1, \dots, u_b$. Every element is a sum of polynomials in the s_i times the u_j, so s_i, u_j together generate S as an R-algebra.

Finally, a quotient of a localization of a finite-type algebra is the localization of its quotient, by the fraction formula. Hence essential finite type passes from S to K. For the converse write $K = T^{-1}C$, with C a finite-type R-algebra, and let C_0 be the image of C in K. The domain C_0 has fraction field K: every element of the displayed localization is a fraction of its elements, and the field K contains the inverse of every nonzero element of C_0.

Lift a finite list of R-algebra generators of C_0 to S, and let $A \subset S$ be the R-subalgebra they generate. Put $\mathfrak{p} = A \cap \mathfrak{m}$. Then $A/\mathfrak{p} = C_0$, so $\kappa(\mathfrak{p}) = K$. Every element of $A \setminus \mathfrak{p}$ maps to a unit in S; thus the inclusion extends to $A_{\mathfrak{p}} \to S$. Applying the finite assertion to this map makes S a finite $A_{\mathfrak{p}}$-module. Choose generators $v_1, \dots, v_c$ and set $B = A[v_1, \dots, v_c] \subset S$. This is a finite-type R-algebra. Localizing B at $A \setminus \mathfrak{p}$ maps injectively into S, because those denominators are units there, and surjectively because the v_j generate over $A_{\mathfrak{p}}$. Therefore S is that localization of B. This proves the last equivalence without a finiteness assumption on R. □

Lemma F.41: etale standard smooth

Reading source: AI Integrated Stacks, pinned Algebra, lines 40737–40760.

Every étale ring map $R \to S$ has a standard smooth presentation with equally many variables and equations:

$$S \cong R[X_1, \dots, X_n]/(f_1, \dots, f_n), \qquad \det(\partial f_i/\partial X_j) \in S^\times.$$

Proof. If $S = 0$, take $S = R[X]/(1)$. Its Jacobian determinant is zero, which is the unit element in the zero ring, so it has the required property.

Suppose $S \neq 0$ and choose a finite polynomial presentation $P = R[X_1, \dots, X_n] \twoheadrightarrow S$ with kernel I. The complete formal/conormal comparison AG-CA-17, Theorem 3.1 and Section 4, says that the conormal differential of an étale presentation is an isomorphism:

$$I/I^2 \xrightarrow{\sim} \bigoplus_{i=1}^{n} S\, dX_i, \qquad [f] \longmapsto \sum_i (\partial f/\partial X_i) dX_i.$$

Specifically, formal smoothness gives a split injection and formal unramifiedness gives zero cokernel; étaleness supplies both. Thus I/I^2 is finite free.

The entire global construction F.55 replaces this presentation by $S = R[Y_1, \dots, Y_m]/J$ with $J = (g_1, \dots, g_c)$ and the classes of the g_j a basis of J/J^2. Apply the same étale conormal comparison to this new presentation. In its actual equation and differential bases it is the Jacobian map $S^c \xrightarrow{H} S^m$. It is an isomorphism. Choose any maximal ideal of the nonzero ring S and reduce this inverse pair modulo it. The resulting vector-space isomorphism forces $c = m$.

The matrix H is now square and has an inverse over S. The determinant product identity gives $\det(H)\det(H^{-1}) = 1$, so its determinant is a unit. Relabeling these m variables and m equations produces the asserted presentation of the whole algebra. It proves global standard smoothness, with no Noetherian-base assumption or restriction to a neighbourhood. □

Lemma F.42: exact sequence NL

Reading source: AI Integrated Stacks, pinned Algebra, lines 36838–36913.

Let $A \to B \to C$ be ring maps. Choose surjections $P = A[x_s \mid s \in \mathcal{S}] \to B$ and $Q = B[y_t \mid t \in \mathcal{T}] \to C$, with kernels I, J. Put $E = A[x_s, y_t]$ and $K = \ker(E \to C)$. Variable sets may be infinite. Define

$$N_A = (I/I^2) \otimes_B C, \quad N_B = K/K^2, \quad N_C = J/J^2,$$
$$E_A = \bigoplus_s C\, dx_s, \quad E_B = E_A \oplus \bigoplus_t C\, dy_t, \quad E_C = \bigoplus_t C\, dy_t.$$

Their conormal differentials give the commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & E_A & \xrightarrow{i} & E_B & \xrightarrow{p} & E_C & \longrightarrow & 0 \\ & & \uparrow d_A & & \uparrow d_B & & \uparrow d_C & & \\ & & N_A & \xrightarrow{a} & N_B & \xrightarrow{b} & N_C & \longrightarrow & 0. \end{array}$$

No injectivity of a is asserted. With the first cotangent homology $H_1(L_{-/-})$ computed by these presentation complexes, the resulting canonical exact sequence is

$$H_1(\mathrm{NL}_{B/A} \otimes_B C) \longrightarrow H_1(L_{C/A}) \longrightarrow H_1(L_{C/B})$$
$$\xrightarrow{\delta} C \otimes_B \Omega_{B/A} \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0.$$

It extends the differential sequence AG-CA-15, Theorem 3.1. If both $\mathrm{Tor}_1^B(\Omega_{B/A}, C)$ and $\mathrm{Tor}_2^B(\Omega_{B/A}, C)$ vanish, there is moreover a natural identification

$$H_1(\mathrm{NL}_{B/A} \otimes_B C) \cong H_1(L_{B/A}) \otimes_B C.$$

Proof. The map i includes the dx_s coordinates, and p kills those coordinates and retains the dy_t. Their row is split exact. Define

$$a([f] \otimes c) = [\widetilde{c} f] \in K/K^2, \qquad b([h]) = [h \bmod IE] \in J/J^2,$$

where $\widetilde{c} \in E$ lifts c. Changing this lift adds an element of K times $f \in I \subset K$, hence an element of K^2. Changing f by I^2, or using a different polynomial lift of a coefficient in B, also changes the result by K^2. Multiplication of coefficient representatives proves the tensor-balancing relation, so a is well defined. The map $E \to Q$ has kernel IE, maps K onto J, and maps K^2 onto J^2. Therefore the preimage of J^2 in K is $K^2 + IE$: subtract a finite sum of products of lifts from K to leave an element of IE. The image of a is exactly $(IE + K^2)/K^2$. This proves exactness at N_B and N_C.

Differentiation gives $d_B(\widetilde{c} f) = c\, df$, since f evaluates to zero in C. Hence $d_B a = i d_A$. Projecting a polynomial derivative to its dy_t part gives precisely its derivative in Q relative to B, so $p d_B = d_C b$. These identities prove commutativity of the entire diagram.

We now give the connecting map and check every exactness assertion. For $z \in \ker d_C$, choose $y \in N_B$ with $b(y) = z$. Then $p(d_B y) = d_C z = 0$, so the split exact upper row gives a unique $v \in E_A$ with $i(v) = d_B y$. Set $\delta(z) = [v] \in E_A / \operatorname{im} d_A$. If the lift y changes, its difference is $a(x)$ for some $x \in N_A$; the corresponding v changes by $d_A x$. Thus this class is well defined.

A cycle $y \in \ker d_B$ mapping to zero under b is $a(x)$, and $0 = d_B a(x) = i(d_A x)$ forces $d_A x = 0$, since i is injective. This proves exactness at $H_1(L_{C/A})$. At $\ker d_C$, the class $\delta(z)$ is zero precisely when the above v is $d_A x$ for some x; then $y - a(x)$ is a cycle lifting z. Conversely a cycle lift has $v = 0$. This proves exactness at $H_1(L_{C/B})$.

At $E_A / \operatorname{im} d_A$, a class $[v]$ maps to zero in $E_B / \operatorname{im} d_B$ precisely when $i(v) = d_B y$ for some y. Its image $b(y)$ is then a cycle in N_C, and the construction gives $\delta(b(y)) = [v]$. At the next term, if $p(w) = d_C z$, choose y with $b(y) = z$. Then $p(w - d_B y) = 0$, so $w - d_B y = i(v)$ for a $v \in E_A$. Thus

$[w]$ comes from $[v]$. Conversely every image of i is killed by p. Finally p is surjective, so its map on the cokernels is surjective as well. These calculations prove the complete displayed sequence.

The entire presentation comparison F.11 identifies the kernels and cokernels with the stated cotangent homology and differential modules. The sequence is canonical under those identifications. Indeed comparison maps are obtained by lifting each source polynomial variable to the new presentation. Make the lifts for B first, then lift the target variables for C and their coefficients; this gives comparison maps on P, Q, E respecting I, J, K. Their relation-class and derivative maps commute with a, b, i, p. Transporting a chosen lift y transports its derivative and the unique v, so the connecting classes commute too. F.11 proves that alternative choices are homotopic and that the reverse comparisons induce inverse homology maps. Hence the induced sequence and its maps are independent of the chosen presentations.

For the final assertion write $\mathrm{NL}_{B/A} = [F_1 \xrightarrow{d} F_0]$, with F_0 free, put $N = \operatorname{im} d$, $H = \ker d$ and $\Omega = \operatorname{coker} d$. The short exact sequence $0 \to N \to F_0 \to \Omega \to 0$ and the first Tor hypothesis give

$$0 \longrightarrow N \otimes_B C \longrightarrow F_0 \otimes_B C \longrightarrow \Omega \otimes_B C \longrightarrow 0.$$

The long Tor sequence and freeness of F_0 identify $\operatorname{Tor}_1^B(N, C)$ with $\operatorname{Tor}_2^B(\Omega, C)$, using the full resolution calculus of AG-RG-S01, Lemma 1.A. The second hypothesis therefore makes tensoring $0 \to H \to F_1 \to N \to 0$ give

$$0 \longrightarrow H \otimes_B C \longrightarrow F_1 \otimes_B C \longrightarrow N \otimes_B C \longrightarrow 0.$$

Since $N \otimes_B C$ injects into $F_0 \otimes_B C$, these two sequences identify $H \otimes_B C$ with the kernel of the tensor differential. This is the claimed natural first-homology identification. No flatness of F_1 and no finite-variable assumption has been added. □

Lemma F.43: extension

Reading source: Stacks, Tag 0519.

Let R be any ring and let $0 \to M_1 \to M_2 \to M_3 \to 0$ be exact. If the two end modules are finite, so is the middle module; a quotient of a finite module is finite. If both end modules are finitely presented, so is the middle module. A quotient of a finitely presented module by a finite submodule is finitely presented. Finally, if M_2 is finite and M_3 is finitely presented, then M_1 is finite.

Proof. Identify M_1 with its image in M_2. To generate the middle module, take generators of M_1 and lifts of generators of M_3. Subtracting a combination of the latter from an arbitrary element leaves an element of M_1. Images of any generating list generate a quotient. These observations prove the two assertions about finite modules.

We first establish the presentation calculation needed in the other assertions. Suppose $q : G \twoheadrightarrow N$ is a finite free presentation with finitely generated kernel L, and $p : F \twoheadrightarrow N$ is any surjection from a finite free module. Choose maps $u : F \to G$ and $v : G \to F$ by lifting their finite bases, so that $qu = p$ and $pv = q$. For every $z \in \ker p$,

$$z = v(uz) + (1 - vu)z, \qquad uz \in L.$$

Both $v(L)$ and $(1 - vu)F$ lie in $\ker p$, since $pv = q$ and $pvu = qu = p$. Thus that kernel is generated by the images under v of a finite generating list of L, together with the finitely many columns of $1 - vu$. In particular the kernel of every finite free surjection onto a finitely presented module is finite. The choices of lifts here use only the defining property of a free basis.

If M_2 is finite, choose a finite free surjection $F \to M_2$. When M_3 is finitely presented, the kernel of the composite $F \to M_3$ is finite by this calculation. Its image in M_2 is exactly M_1: an element mapping to zero in M_3 has any chosen lift in that kernel. Hence M_1 is finite.

Next write a finitely presented M_2 as F/K, with F finite free and K finite. If M_1 is generated by $a_1, \dots, a_h$, choose lifts $\widetilde{a}_i \in F$. The kernel of $F \to M_3$ is

$$K + R\widetilde{a}_1 + \cdots + R\widetilde{a}_h.$$

Indeed an element in that kernel maps to a combination of the a_i in M_2; subtracting the corresponding lifted combination puts it in K. This is a finite kernel and gives the asserted finite presentation of the quotient.

Finally choose finite free presentations $p_i : F_i \twoheadrightarrow M_i$, with finite kernels K_i, for $i = 1, 3$. Lift the basis of F_3 to obtain $\ell : F_3 \to M_2$, and use

$$F_1 \oplus F_3 \longrightarrow M_2, \qquad (a, b) \longmapsto p_1(a) + \ell(b).$$

It is onto by the first generator argument. For each generator b_j of K_3, the element $\ell(b_j)$ lies in M_1; choose $a_j \in F_1$ with $p_1(a_j) = -\ell(b_j)$. Its kernel is generated by $(K_1, 0)$ and the finitely many pairs (a_j, b_j). To see this, the second coordinate of any kernel element lies in K_3. Subtract a combination of these pairs to make that coordinate zero; the first coordinate then lies in K_1. We have constructed a finite generating list of relations, so M_2 is finitely presented. Empty generating lists and zero modules satisfy the same formulas. No Noetherian assumption is used. □

Lemma F.44: finite presentation independent

Reading source: Stacks, Tag 00R2.

If an R-algebra S is finitely presented, the kernel of every surjection $\alpha : R[X_1, \dots, X_n] \twoheadrightarrow S$ is a finitely generated ideal.

Proof. Fix one presentation

$$S = R[Y_1, \dots, Y_m]/(f_1, \dots, f_h).$$

Choose $g_i(Y)$ representing $\alpha(X_i)$, and, using the surjectivity of α, choose $h_j(X)$ representing the class of Y_j. In the polynomial ring $P = R[X_1, \dots, X_n]$, form the finite ideal

$$J = \big(f_1(h(X)), \dots, f_h(h(X)),\ X_i - g_i(h(X))\ (1 \le i \le n)\big).$$

Each generator maps to zero under α, by the choices of representatives, so $J \subset \ker \alpha$.

For the reverse inclusion, take $F(X) \in \ker \alpha$. The polynomial $F(g(Y))$ represents zero in the fixed presentation; hence there are polynomials $c_l(Y)$ with

$$F(g(Y)) = \sum_{l=1}^{h} c_l(Y) f_l(Y).$$

Substitute $Y = h(X)$. This puts $F(g(h(X)))$ in J. Also

$$F(X) - F(g(h(X))) \in \big(X_i - g_i(h(X)) : 1 \le i \le n\big).$$

Here is the polynomial identity behind this last inclusion, valid over an arbitrary ring. Replace the variables one at a time and telescope the differences. At a single variable, every monomial difference is divisible by its coordinate difference, because $U^a - V^a = (U - V)\sum_{b=0}^{a-1} U^{a-1-b}V^b$ for $a > 0$; the difference is zero for $a = 0$. Adding the monomials proves the inclusion without division by any coefficient. Thus $F \in J$, proving $\ker \alpha = J$. The argument includes either empty variable list, an empty relation list and the zero algebra. □

Lemma F.45: flat fibre smooth

Reading source: AI Integrated Stacks, pinned Algebra, lines 39115–39149.

Let $R \to S$ be a ring map and let $\mathfrak{q} \subset S$ lie over $\mathfrak{p} \subset R$. Suppose that some $s \in S \setminus \mathfrak{q}$ makes S_s finitely presented over R, that $R_{\mathfrak{p}} \to S_{\mathfrak{q}}$ is flat, and that $S \otimes_R \kappa(\mathfrak{p})$ is smooth at the point of this fibre induced by $\mathfrak{q}$. Then $R \to S$ is smooth at $\mathfrak{q}$, over an arbitrary base ring.

Proof. Replace S by S_s and keep the same notation for the prime. Choose a finite presentation $S = P/I$, where $P = R[X_1, \dots, X_n]$ and $I = (a_1, \dots, a_m)$. Let $\mathfrak{r}$ be the inverse image of $\mathfrak{q}$ in P, and put

$$A = R_{\mathfrak{p}}, \quad k = \kappa(\mathfrak{p}), \quad Q = P_{\mathfrak{r}}, \quad B = S_{\mathfrak{q}}.$$

Thus $0 \to I_{\mathfrak{r}} \to Q \to B \to 0$ is exact over A, and B is A-flat.

The full arbitrary-module Tor construction AG-RG-S01, Lemmas 1.0 and 1.A gives exactness after tensoring this sequence with k. Indeed use a free resolution of k to compute $\operatorname{Tor}_1^A(k, B)$; tensoring with flat B preserves the kernels and images of that exact resolution, so its positive homology is zero. The natural long exact Tor sequence therefore gives

$$0 \longrightarrow I_{\mathfrak{r}}/\mathfrak{p}I_{\mathfrak{r}} \longrightarrow Q/\mathfrak{p}Q \longrightarrow B/\mathfrak{p}B \longrightarrow 0.$$

Set $\bar{Q} = Q/\mathfrak{p}Q$, $\bar{B} = B/\mathfrak{p}B$ and $J = \ker(\bar{Q} \to \bar{B})$. The injection identifies J with $I_{\mathfrak{r}}/\mathfrak{p}I_{\mathfrak{r}}$. Quotient and localization commute by their fraction maps, giving $\bar{Q} = k[X]_{\bar{\mathfrak{r}}}$ and $\bar{B} = \bar{Q}/J$. These are local rings, J lies in the maximal ideal of $\bar{Q}$, and J is generated by the finitely many images of the a_j.

The smooth fibre has a smooth principal neighbourhood. Definition F.6, presentation comparison F.11, and the full conormal criterion AG-CA-17, Theorem 3.1 make that neighbourhood formally smooth over k: its injective conormal map has projective cokernel, so the sequence splits. The localization lifting proof in AG-CA-17, Theorem 1.2, gives formal smoothness of its local ring $\bar{B}$. This argument does not assert that the local ring is finitely presented.

Apply formal smoothness to the square-zero surjection $\bar{Q}/J^2 \to \bar{B}$, lifting the identity of $\bar{B}$ to a k-algebra section σ. Then

$$D(q) = q \bmod J^2 - \sigma(q \bmod J) \in J/J^2$$

is a k-derivation. Expanding $q = \sigma(\bar{q}) + D(q)$ and $q' = \sigma(\bar{q}') + D(q')$, the product of the two correction terms is zero, giving the Leibniz rule. For $j \in J$, $D(j) = [j]$. The universal differential construction thus supplies a left inverse of

$$J/J^2 \longrightarrow \Omega_{\bar{Q}/k} \otimes_{\bar{Q}} \bar{B} = \bar{B}^n.$$

The differential basis and its localization are the complete AG-CA-15, Theorems 5.1 and 2.2. In coordinates this conormal map sends $[j]$ to its partial derivatives. Consequently J/J^2 is a finite projective direct summand of $\bar{B}^n$.

Choose from $a_1, \dots, a_m$ elements $f_1, \dots, f_c$ whose reduced conormal classes form a basis over the residue field $\kappa(\mathfrak{q})$. The finite Nakayama argument F.10 makes the map $\bar{B}^c \to J/J^2$ surjective. Projectivity splits it. Its kernel is a finite direct summand of $\bar{B}^c$, and its residue quotient is zero by the chosen residue basis; a second application of F.10 makes this kernel zero. Thus the chosen classes are a free basis.

Let H be their c-by-n equation-by-variable Jacobian over $\bar{B}$. The conormal map has matrix H^{T} and a left inverse, so its reduction modulo the maximal ideal is injective; hence $c \le n$. The full maximal-minor calculation F.70, applied to this n-by-c matrix and its left inverse with scalar 1, makes its maximal minors generate one in $\bar{B}$. Some minor is therefore a unit in this local ring. Transposing the selected minor preserves its determinant. Let $\Delta \in P$ be the corresponding c-variable Jacobian determinant. Its residue in $\kappa(\mathfrak{q})$ is nonzero, so $\Delta \notin \mathfrak{r}$. If $c = 0$, use the empty basis and $\Delta = 1$.

The chosen f_i generate the whole ideal after localizing at $\mathfrak{r}$. First $J = (\bar{f}_1, \ldots, \bar{f}_c) + J^2$. Its finite quotient by the displayed generators is equal to J times that quotient. Since J lies in the local maximal ideal, F.10 gives $J = (\bar{f}_1, \ldots, \bar{f}_c)$. Now the finite Q-module

$$M = I_{\mathfrak{r}}/(f_1, \ldots, f_c)Q$$

has quotient $M/\mathfrak{p}M = 0$, by the exact tensor identification of J. The ideal $\mathfrak{p}Q$ lies in the maximal ideal of Q, so F.10 gives $M = 0$. This is where the flat local quotient is used; neither ring has needed a chain condition.

For each original generator a_j of I, write its image in Q as a finite combination of the f_i. Clear all coefficient denominators, and also the multipliers outside $\mathfrak{r}$ witnessing the resulting equalities in the localization. Their finite product $v \notin \mathfrak{r}$ gives

$$I_v = (f_1, \ldots, f_c)P_v, \qquad S_{\bar{v}\bar{\Delta}} \cong R[X]_{v\Delta}/(f_1, \ldots, f_c).$$

This uses the fraction equality criterion, including its annihilator factor, so no possible zero divisor has been cancelled.

Adjoin Z with equation $v\Delta Z - 1$. The selected Jacobian block for the c pivot variables and Z has zero Z entries in the first c rows; its determinant is $\Delta(v\Delta) = v\Delta^2$, a unit in the quotient. The entire arbitrary-base standard-smooth proof F.85 applies to this finite presentation and proves its smoothness. In terms of lifting, it inverts this square block, corrects the lifted pivot coordinates by the negative error vector, and the monomial expansion kills all terms containing two corrections in a square-zero ideal. Unit lifts then extend evaluation through the inverse-variable relation, exactly as proved in F.85.

Finally express the elements inverted in this neighbourhood of S_s as fractions with powers of s as denominators. Inverting s and their finitely many numerators gives one principal neighbourhood in the original $\operatorname{Spec} S$ containing $\mathfrak{q}$. Its smoothness proves the conclusion, including empty pivot lists. □

Lemma F.46: flat over regular

Reading source: Stacks, Tag 07DY.

Let $R \to S$ be a homomorphism of Noetherian local rings. Suppose R is regular and one regular system of parameters $x_1, \ldots, x_d$ maps to an S-regular sequence. Then S is flat over R. A regular sequence here has proper generated ideal.

Proof. The x_i generate the maximal ideal of R. Their images belong to the maximal ideal of S, because the ideal they generate is proper in that local ring. Thus the given map is local, even though this was not a separate assumption.

For $0 \le i \le d$, put

$$R_i = R/(x_1, \ldots, x_i), \qquad S_i = S/(x_1, \ldots, x_i)S.$$

These are nonzero Noetherian local rings and their induced maps are local. The written regular-parameter theorem AG-CA-12, Theorem 6.1 makes the chosen list regular in R. Therefore, for every $i < d$, multiplication by x_{i+1} is injective on R_i; the assumed regularity in S gives the same injection on S_i.

At $i = d$, the source R_d is the residue field of R, so S_d is flat over it. Indeed a vector-space basis identifies tensoring with that space with a direct sum of copies of a module, which preserves injections.

We descend from $i + 1$ to i. Since x_{i+1} is a nonzerodivisor of R_i,

$$0 \longrightarrow R_i \xrightarrow{x_{i+1}} R_i \longrightarrow R_{i+1} \longrightarrow 0$$

is a free resolution. Tensoring it with S_i computes

$$\operatorname{Tor}_1^{R_i}(R_{i+1}, S_i) = \ker(x_{i+1} : S_i \to S_i) = 0.$$

The quotient $S_i/x_{i+1}S_i = S_{i+1}$ is flat over R_{i+1} by the descending induction. Apply the complete ideal local criterion AG-CA-08, Corollary 4.4 and equation (6) to $R_i \to S_i$, the proper ideal $(x_{i+1}) \subset R_i$, and the module S_i. This module is finite over the target ring S_i, being generated by 1; it need not be finite over the source. The Tor vanishing and quotient flatness prove that S_i is R_i-flat. The induction reaches $S_0 = S$ over $R_0 = R$. If $d = 0$, the initial field argument already proves the conclusion. □

Lemma F.47: flat over regular with regular fibre

Reading source: Stacks, Tag 031E.

Let $(R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a flat local map of Noetherian local rings. If R and the closed fibre $F = S/\mathfrak{m}S$ are regular, then S is regular.

Proof. Put $d = \dim R$ and $e = \dim F$. Regularity supplies generators $x_1, \dots, x_d$ of $\mathfrak{m}$ and generators $\overline{y}_1, \dots, \overline{y}_e$ of the maximal ideal $\mathfrak{n}/\mathfrak{m}S$ of F. Choose lifts $y_j \in \mathfrak{n}$. Every $z \in \mathfrak{n}$ reduces to a combination of the $\overline{y}_j$; lifting the coefficients and subtracting that combination leaves an element of $\mathfrak{m}S = (x_1, \dots, x_d)S$. Consequently

$$\mathfrak{n} = (x_1, \dots, x_d, y_1, \dots, y_e)S.$$

Its cotangent space $\mathfrak{n}/\mathfrak{n}^2$ is therefore spanned by at most $d + e$ vectors over the residue field of S.

The flat local dimension formula, with precisely the stated Noetherian and flatness hypotheses, is proved in AG-CA-11, Theorem 5.1; it gives $\dim S = d + e$. Its Proposition 4.2 proves that the dimension of a Noetherian local ring is at most the dimension of its cotangent space: lifts of a cotangent basis generate the maximal ideal by finite Nakayama, and the parameter/height theorem bounds dimension by that generator count. Combining these inequalities gives

$$d + e = \dim S \le \dim_{\kappa(\mathfrak{n})} \mathfrak{n}/\mathfrak{n}^2 \le d + e.$$

Equality of dimension and embedding dimension is the definition of regularity, so S is regular. The empty source or fibre parameter lists when $d = 0$ or $e = 0$ obey the same calculation. No finite-type assumption on the ring map is needed. □

Lemma F.48: flatness descends more general

Reading source: Stacks, Tag 0584.

Let $S \to S'$ be a flat map of R-algebras, let M be any S-module, and put $M' = S' \otimes_S M$. If M is flat over R, so is M'. If $S \to S'$ is faithfully flat, the converse holds as well.

Proof. Test an arbitrary injection $f : N \hookrightarrow N_1$ of R-modules. Write

$$Z = \ker\bigl(N \otimes_R M \xrightarrow{f \otimes 1} N_1 \otimes_R M\bigr).$$

Both tensor products, their map and Z are S-modules, using the action on M. Flatness of S' over S preserves this kernel. The balancing relations give inverse isomorphisms

$$S' \otimes_S (N \otimes_R M) \;\longleftrightarrow\; N \otimes_R M', \qquad s' \otimes (n \otimes m) \;\longleftrightarrow\; n \otimes (s' \otimes m),$$

and the same formula for N_1. These maps commute with f, so

$$S' \otimes_S Z \cong \ker\bigl(N \otimes_R M' \longrightarrow N_1 \otimes_R M'\bigr).$$

If M is R-flat, $Z = 0$, and the displayed kernel vanishes for every f. If M' is R-flat, the right side is zero; faithful flatness then detects $Z = 0$. In either direction the injection characterization of flatness, proved for arbitrary modules in AG-CA-07, Theorem 2.1, gives the claimed flatness. No module in this calculation is required to be finite. □

Lemma F.49: formally smooth extensions easy

Reading source: AI Integrated Stacks, pinned Algebra, lines 45971–46012.

For a field extension K/k, each of the following conditions implies formal smoothness of K over k: it is purely transcendental; it is separable algebraic; or it is separable in the sense of F.5. Infinite generator sets and infinite algebraic extensions are allowed.

Proof. In the first case write $K = k(x_j \mid j \in \mathcal{J})$. Given a k-algebra D, a square-zero ideal I and a k-map $K \to D/I$, choose lifts $a_j \in D$ of the variable images. Polynomial evaluation at these lifts is well defined because each polynomial involves finitely many variables. Every nonzero polynomial in the x_j is a unit in K, so its evaluated image is a unit modulo I. Units lift: if $bc = 1 + r$ with $r \in I$, then $bc(1 - r) = 1$ since $r^2 = 0$. Thus all the nonzero polynomials have unit evaluations in D. The polynomial map extends uniquely, for these chosen variable lifts, through the fraction-field localization to a k-map $K \to D$. Its reduction is the given map, proving the lifting property.

For a separable algebraic extension, a finite subset of K generates a finite separable subextension: successive degrees of its finitely many algebraic generators are finite and multiply along the tower. These subfields are directed under inclusion, since the union of two finite generator lists is finite, and their union is K. The full separable-generator tower proof AG-CA-17, Proposition 6.3 gives a unique square-zero lift on every such finite field. The derivatives of its separable minimal polynomials are units, so its successive one-equation lifting steps apply. Consequently every finite subextension is formally étale. The entire compatible-lifts proof F.22 makes their colimit formally étale, and hence formally smooth.

In the third case use the directed family of finitely generated subfields K_i/k. By F.5 each has a separating transcendence basis and is a finite separable extension of the resulting rational-function field. The first two cases give formal smoothness of both maps in that tower; the full composition lifting proof AG-CA-17, Theorem 1.2 gives formal smoothness of K_i/k. The arbitrary-field criterion F.17 therefore gives $H_1(L_{K_i/k}) = 0$.

Now F.24 identifies the conormal complex of K/k with the filtered colimit of these complexes. Its representative proof of exactness of filtered module colimits also identifies their homology: a cycle is represented at a stage after its finite differential equality holds, and a boundary equality is witnessed at a common later stage. Thus $H_1(L_{K/k}) = \operatorname{colim}_i H_1(L_{K_i/k}) = 0$. Applying F.17 again proves formal smoothness of K/k. This uses the field criterion, rather than assuming a general permanence theorem for filtered colimits of formally smooth algebras. □

Lemma F.50: geometrically regular

Reading source: AI Integrated Stacks, pinned Algebra, lines 48753–48798.

Let A be a Noetherian k-algebra, where k is a field. Regularity of $A \otimes_k K$ for every finitely generated field extension K/k is equivalent to regularity of $A \otimes_k k'$ for every finite purely inseparable field extension k'/k.

Proof. The forward implication includes the finite purely inseparable extensions among the finitely generated ones. For the converse, fix a finitely generated K/k. The full separabilization construction F.69 gives finite purely inseparable extensions k'/k and K'/K, inside a commuting field square, with K'/k' separably generated. This last extension is finitely generated: a finite field-generating list for K/k and finitely many generators for K'/K still generate K' after enlarging the coefficient field to k'.

The entire forward smooth-domain construction F.64 supplies a smooth finite-type domain $D \subset K'$ over k' with fraction field K'. Its explicit separating-generator equations, denominator clearing and derivative localization have been proved there. They also give the smooth field model recognized by AG-CA-18, Theorem 3.2.

By assumption $A' = A \otimes_k k'$ is regular. It is Noetherian by F.12. Form $C = A' \otimes_{k'} D$. The coefficient-change polynomial presentation makes it finitely presented over A', and formal smoothness survives base change by AG-CA-17, Theorem 1.2. Hence C/A' is smooth. It is also Noetherian, being finite type over the Noetherian ring A'.

The complete smooth-flatness and regular-fibre results AG-CA-18, Theorems 6.1 and 2.1, make this map flat with regular residue-field fibres. At a prime of C, the localized source is regular, and its closed fibre is the local ring of one of those regular fibres. The full ascent calculation F.47 therefore makes the target local ring regular. This proves regularity of C.

Localizing at all nonzero elements of D gives

$$C \otimes_D K' = A' \otimes_{k'} K' \cong A \otimes_k K'.$$

Every local ring of this localization is a local ring of the original regular ring, so it is regular. Finally choose a basis of the finite field extension K'/K. As a module over $B = A \otimes_k K$, the algebra $A \otimes_k K'$ is the direct sum of $[K' : K] \geq 1$ copies of B, by tensoring this basis. Tensoring an arbitrary B-module with that algebra therefore takes this positive finite number of copies; it preserves exactness and detects zero modules. Thus the map $A \otimes_k K \to A \otimes_k K'$ is faithfully flat. The full arbitrary-source regularity descent proof F.38 makes $A \otimes_k K$ regular. This holds for every chosen K, proving the equivalence without a perfect-field assumption. □

Lemma F.51: geometrically regular descent

Reading source: AI Integrated Stacks, pinned Algebra, lines 48815–48834.

Let $A \to B$ be a faithfully flat map of k-algebras. If B is geometrically regular over the field k, then A is geometrically regular over k. No separate Noetherianity hypothesis on A is needed.

Proof. The trivial extension k/k is among the field tests for geometric regularity, so B is regular. The full regularity descent proof F.38 makes A regular as well, deriving its Noetherianity by its finite-ideal argument.

For every finite purely inseparable extension k'/k, set $A' = A \otimes_k k'$ and $B' = B \otimes_k k'$. Both are Noetherian by F.12. Tensor associativity identifies B' with $A' \otimes_A B$, and for an arbitrary A'-module M it gives the inverse pure-tensor maps

$$M \otimes_{A'} (A' \otimes_A B) \cong M \otimes_A B, \qquad m \otimes (a' \otimes b) \longmapsto a'm \otimes b, \quad m \otimes b \longmapsto m \otimes (1 \otimes b).$$

The balancing relations verify these maps and their inverse composites. Restriction from A'-modules to A-modules preserves exact sequences and detects the zero module. Since B is faithfully flat over A, the displayed tensor identification proves that B' is faithfully flat over A'.

Geometric regularity makes B' regular. Applying F.38 gives regularity of A' for every such k'. The complete finite purely inseparable criterion F.50 now applies to the Noetherian algebra A and proves its geometric regularity over k. □

Lemma F.52: geometrically regular over subfields

Reading source: AI Integrated Stacks, pinned Algebra, lines 48851–48866.

Let A be a Noetherian algebra geometrically regular over the field k.

1. For any field extension E/k such that $C = A \otimes_k E$ is Noetherian, C is regular.
2. Under this same Noetherian tensor hypothesis, C is geometrically regular over E.
3. If k is a directed union of subfields k_i and A is geometrically regular over every k_i, then A is geometrically regular over k.

Proof. For (1), write E as the union of its intermediate fields E_i generated by finite subsets over k. They are directed, since the union of two finite generator sets is finite. Put $C_i = A \otimes_k E_i$. An E_i-basis of E containing 1 gives

$$C = C_i \otimes_{E_i} E \cong \bigoplus_{\beta \in \mathcal{B}_i} C_i.$$

Here the structural inclusion $C_i \to C$ is the coordinate belonging to 1. To obtain such a basis, take a maximal independent set containing 1; if it did not span, a vector outside its span could be adjoined, contradicting maximality. The tensor/direct-sum calculation is AG-RG-S01, Lemma 1.0. It proves both injectivity of this inclusion and flatness of C over C_i, even when the basis is infinite. Every element of C uses finitely many field coefficients and so lies in one C_i.

Suppose that C is Noetherian and fix $\mathfrak{q} \subset C$. In the Noetherian local ring $T = C_{\mathfrak{q}}$ choose a finite generating list $x_1, \dots, x_r$ for its maximal ideal $\mathfrak{m}$. Write $x_j = a_j/s_j$, with $a_j \in \mathfrak{q}$ and $s_j \notin \mathfrak{q}$. Put all their finitely many field coefficients into a common E_i, so that every a_j, s_j lies in C_i. Let $\mathfrak{q}_i = \mathfrak{q} \cap C_i$ and $B_i = (C_i)_{\mathfrak{q}_i}$. The fractions x_j now belong to B_i, and their numerators lie in its maximal ideal.

The map $B_i \to T$ is local and flat: it is obtained by localizing the free scalar-extension map above, and exact localization preserves its injection tests. Its extended maximal ideal is contained in $\mathfrak{m}$ and contains every x_j. Hence

$$\mathfrak{q}_i T = \mathfrak{m}, \qquad T \otimes_{B_i} \kappa(\mathfrak{q}_i) = T/\mathfrak{m}.$$

The source B_i is Noetherian by F.12, and is regular because E_i/k is finitely generated and A is geometrically regular. The displayed closed fibre is a field, thus regular. The entire flat-local ascent proof F.47 makes T regular. An empty generating list gives the same argument when $\mathfrak{m} = 0$. This proves regularity at every prime of C; if $C = 0$, there are no primes to test.

For (2), let F/E be a finite purely inseparable extension. F.12 makes $C \otimes_E F$ Noetherian. Tensor associativity identifies it with $A \otimes_k F$, so (1), applied to the field extension F/k, makes it regular. The complete finite purely inseparable criterion F.50 proves geometric regularity of C over E.

For (3), we construct an actual finite purely inseparable field model. Let k'/k be finite purely inseparable. If it is trivial, use $k'_i = k_i$ at any index. Otherwise the characteristic is $p > 0$. Choose a k-basis $b_1 = 1, b_2, \dots, b_d$ of k'. A common exponent e satisfies $b_j^{p^e} \in k$ for every j, since the extension is purely inseparable and the list is finite. Write its finitely many multiplication constants

$$b_j b_\ell = \sum_h c_{j\ell h} b_h, \qquad c_{j\ell h} \in k.$$

Choose k_i containing all $c_{j\ell h}$ and all $b_j^{p^e}$. This is possible because the union is directed and the list is finite.

The subspace $V = \bigoplus_j k_i b_j \subset k'$ contains 1 and is closed under multiplication by those identities. It is a finite-dimensional domain over k_i. Multiplication by any nonzero $v \in V$ is an injective endomorphism of that finite-dimensional vector space, hence surjective; a preimage of 1 gives its inverse. Therefore V is a field, denoted k'_i. Moreover every $v = \sum_j a_j b_j$ in it has

$$v^{p^e} = \sum_j a_j^{p^e} b_j^{p^e} \in k_i.$$

Thus k'_i/k_i is finite purely inseparable. This checks the exponent condition for every element of the field, beyond just its chosen generators.

The multiplication map $k \otimes_{k_i} k'_i \to k'$, $a \otimes v \mapsto av$, is an algebra isomorphism: its basis $1 \otimes b_j$ maps to the given k-basis b_j, proving both bijectivity and compatibility with the prescribed products. Consequently

$$A \otimes_k k' \cong A \otimes_{k_i} k'_i.$$

The right side is regular by the hypothesis over k_i. F.50 proves geometric regularity over k. In characteristic zero all finite purely inseparable extensions are trivial, so the same conclusion follows from the trivial case. All three assertions are established. □

Lemma F.53: grothendieck regular sequence

Reading source: AI Integrated Stacks, pinned Algebra, lines 24513–24526.

Let $R \to P$ be a flat local map of Noetherian local rings, and let $\mathfrak{m} = \mathfrak{m}_R$. If the reductions of $f_1, \dots, f_c \in P$ form a regular sequence in $P/\mathfrak{m}P$, then the list is regular in P, and every $P/(f_1, \dots, f_i)$ is flat over R.

Proof. Put $P_i = P/(f_1, \dots, f_i)$, including $P_0 = P$. The regular-sequence hypothesis includes a nonzero final fibre quotient. Consequently each $\bar{f}_i$ belongs to the maximal ideal of the local closed fibre: a unit there would make the final quotient zero. Its lift f_i belongs to the maximal ideal of P as well. Every P_i is thus a nonzero Noetherian local quotient, and $R \to P_i$ remains local. Its closed fibre is

$$P_i/\mathfrak{m}P_i \cong (P/\mathfrak{m}P)/(\bar{f}_1, \dots, \bar{f}_i).$$

Induct on i, keeping R-flatness of P_i. The initial assertion is the assumed flatness of P. On P_i consider multiplication by f_{i+1}. Both its source and target are finite over this actual Noetherian local ring, and its target is R-flat by induction. The displayed fibre description and the original regular-list hypothesis make its reduction injective. The full lifting-injectivity proof AG-CA-08, Lemma 5.1 gives injectivity on P_i and R-flatness of its cokernel P_{i+1}. This proves the induction.

Every multiplication in the list has now been proved injective on its preceding quotient, and the final quotient is nonzero, so the list is regular. The same induction has proved flatness of each successive quotient. If $c = 0$, the list is empty and the only quotient is the original flat ring P, giving both assertions directly. □

Lemma F.54: henselian cat finite etale

Reading source: Stacks, Tag 04GK.

For a henselian local ring $(R, \mathfrak{m}, k)$, reduction $S \mapsto S/\mathfrak{m}S$ is an equivalence from finite étale R-algebras to finite products of finite separable extensions of k. Ring maps are unital, and the zero algebra is the empty product.

Proof. The written idempotent and finite-algebra theorem AG-CA-20, Theorem 2.2 decomposes a finite R-algebra into finitely many local factors. A factor of an étale algebra is étale, since it is localization at its component idempotent. For each such factor S_i, reduction is an étale k-algebra and is local: every maximal ideal of a finite algebra contracts to $\mathfrak{m}$, and S_i has only one. The field classification AG-CA-17, Theorem 7.2 makes this reduction a finite separable field. In particular $\mathfrak{m}S_i$ is its maximal ideal. Every S_i is henselian by AG-CA-20, Proposition 3.1.

Construct a lift of a finite separable field extension L/k as follows. Choose finitely many field generators and use their successive fields

$$k = L_0 \subset L_1 \subset \cdots \subset L_a = L, \qquad L_i = L_{i-1}(\alpha_i).$$

Each minimal polynomial is monic separable. Starting with $T_0 = R$, lift its coefficients to a monic $h_i \in T_{i-1}[X]$ and define $T_i = T_{i-1}[X]/(h_i)$. Its bounded powers of X are a free T_{i-1}-basis, by monic division. Reduction modulo $\mathfrak{m}$ is L_i. Every maximal ideal contracts to $\mathfrak{m}$ by integrality, so the field reduction makes T_i local with maximal ideal $\mathfrak{m}T_i$. The derivative h_i' has nonzero residue in that field and is consequently a unit. The one-equation Jacobian proof AG-CA-17, Theorem 4.1 and Section 6 makes T_i/T_{i-1} étale. Composition makes T_a/R finite étale, with free rank $[L : k]$ and reduction L. Finite products of these lifts provide every desired residue algebra.

We next show explicitly that every local finite étale R-algebra S, with residue L, is isomorphic to this constructed T_a. Given the residue identification, lift a map $T_{i-1} \to S$ through the next step by choosing the root of h_i reducing to α_i. The simple-root Hensel statement, AG-CA-20, Theorem 1.2,

gives exactly one such root in the henselian local ring S, and hence exactly one map $T_i \to S$ at each step. This constructs $\phi : T_a \to S$ inducing the chosen isomorphism on residues.

By the complete arbitrary-base proof in S04, Lemma E4.2, a finite étale algebra is finite locally free. Its proof first obtains a finite module presentation from monic algebra equations, then applies Lemma A1.3 to a finitely presented flat module. Since R is local, S is finite free, of rank $\dim_k(S/\mathfrak{m}S) = [L : k]$. Both modules in ϕ therefore have this same finite rank. In chosen bases its matrix reduces to an invertible matrix over k, so its determinant lies outside $\mathfrak{m}$ and is a unit of R. The adjugate matrix divided by this unit is its inverse. Thus ϕ is an isomorphism of modules, and its bijective algebra map is an algebra isomorphism.

Now let S_1, S_2 be local finite étale algebras and let $L_1 \to L_2$ be a k-field map between their residues. Replace S_1 by its isomorphic tower T_a. Starting with $R \to S_2$, lift each successive generator to the unique simple root whose residue is its prescribed image in L_2. The same Hensel calculation supplies a unique R-algebra map $S_1 \to S_2$ inducing the field map. Conversely, any algebra map has those generator values, so it is this unique lift. The map on residue fields is defined because $\mathfrak{m}S_i$ is the maximal ideal.

For products, a map to a local target sends each source component idempotent to 0 or 1; orthogonality and their sum 1 force exactly one component to be chosen. The residue map chooses precisely that component, and the local result lifts its field map uniquely. Doing this independently for each target factor proves a bijection on all Hom sets. The empty-product cases follow from unital maps directly: there is one map to the zero ring and none from it to a nonzero ring. We have proved full faithfulness and essential surjectivity, hence the asserted equivalence, over every henselian local source. □

Lemma F.55: huber

Reading source: Stacks, Tag 07CF.

Suppose a finitely presented R-algebra S has a finite polynomial presentation $S = P/I$, where $P = R[X_1, \dots, X_n]$, and I/I^2 is free over S. Then S has another finite polynomial presentation whose defining equations themselves form a basis of its conormal module.

Proof. F.44 makes I finitely generated. Thus I/I^2 has a finite basis: when $S \neq 0$, the supports of a finite generating list in any free basis have finite union, which must contain the entire basis; when $S = 0$, choose its empty conormal basis directly. Lift this basis to $f_1, \dots, f_c \in I$ and put $K = (f_1, \dots, f_c)$. The basis property gives $I = K + I^2$. The finite P-module $U = I/K$ consequently satisfies $U = IU$.

Choose generators $u_1, \dots, u_t$ of U and coefficients $b_{ij} \in I$ with $u_i = \sum_j b_{ij} u_j$. The adjugate identity proved in F.70 applied to $1_t - (b_{ij})$ shows that

$$g = \det(1_t - (b_{ij})), \qquad gU = 0, \qquad g \in 1 + I.$$

For $U = 0$ take $g = 1$. Hence $gI \subset K$ and $I_g = K_g$. The image of g in S is exactly 1, so $P_g/I_g = S$. It follows, with both inverse maps given by adjoining or evaluating g^{-1}, that

$$S = P[T]/(f_1, \dots, f_c, gT - 1).$$

Let L be the displayed equation ideal. After localization at g, the classes of f_i form the given basis of I_g/I_g^2. The explicit inverse-variable conormal calculation F.72, applied to P_g/I_g and this presentation, identifies

$$L/L^2 \cong S^c \oplus S.$$

The first basis vectors are $[f_i]$ and the last is $[gT - 1]$: the last equation has derivative g, a unit, in its new variable, and the remaining classes are read by evaluation at $T = g^{-1}$. These are exactly the claimed equation classes. No Noetherian hypothesis or cancellation of a free summand is used.

Empty bases use the same determinant convention; if $S = 0$, all the displayed modules and bases are interpreted over the zero ring. □

Lemma F.56: isolated point fibre

Reading source: Stacks, Tag 00PK.

Let $R \to S$ be of finite type and let $\mathfrak{q} \in \operatorname{Spec} S$ lie over $\mathfrak{p}$. Put $k = \kappa(\mathfrak{p})$, $T = S \otimes_R k$, and let $x \in \operatorname{Spec} T$ be the corresponding point. The following conditions are equivalent: x is isolated; $T_x = S_{\mathfrak{q}}/\mathfrak{p}S_{\mathfrak{q}}$ is finite-dimensional over k; some $g \in S \setminus \mathfrak{q}$ has x as the only point of its fibre neighbourhood; the fibre point-dimension at x is zero; x is closed and $\dim T_x = 0$; and $\kappa(x)/k$ is finite and $\dim T_x = 0$.

Proof. All the geometry can be checked in the finite-type field algebra T. Such an algebra and its principal localizations are Noetherian by AG-CA-03, Theorem 2.1. We will use the following consequence of its Theorem 4.2. A zero-dimensional finite-type k-algebra B is Artinian, with finitely many local factors and nilpotent radical J. Every residue field is finite over k by AG-CA-06, Theorem 2.1. Each J^i/J^{i+1} is finite over the product B/J of those fields: B is Noetherian, so the ideals involved are finite. Its coordinate summands are consequently finite-dimensional over the residue fields, hence over k. The finite filtration by powers of J proves that B is finite-dimensional over k. The factor idempotents give a principal singleton neighbourhood of each of its points.

If x is isolated, choose $h \in T$ with $D(h) = \{x\}$. The algebra T_h has just one prime, so has dimension zero and is finite-dimensional by the preceding calculation. Every element outside its unique maximal ideal is a unit. Its localization at that ideal is therefore itself, and $T_x = T_h$ is finite-dimensional.

Conversely, a finite-dimensional T_x is Artinian: chains of its ideals are chains of subspaces of a finite-dimensional k-space. Each of its prime quotients is a finite-dimensional domain, hence a field, because multiplication by a nonzero element is an injective, and thus surjective, linear map. Thus $\dim T_x = 0$, and its residue field is finite over k. For a prime of a finite-type k-algebra, finiteness of the residue extension implies closedness: T/x embeds into the finite-dimensional field $\kappa(x)$, so is a finite-dimensional domain and the same multiplication argument makes it a field. The converse follows from the cited weak Nullstellensatz.

Suppose now that x is closed and $\dim T_x = 0$. The prime corresponding to x is both maximal and minimal. There are finitely many other minimal primes, by AG-CA-03, Proposition 2.2. From each choose an element outside x and take their product h; use $h = 1$ if there are none. A prime in $D(h)$ contains none of those other minimal primes. Since every prime contains a minimal prime, it must contain x and therefore equal x. Hence $D(h) = \{x\}$ and x is isolated. This proves the equivalence of isolation, finite local fibre algebra, and the two residue-field/local-dimension conditions.

An isolated point has point-dimension zero, using its singleton open neighbourhood. Conversely, zero point-dimension supplies an open neighbourhood of dimension zero: the infimum defining that dimension is an attained nonnegative integer. Take a principal neighbourhood inside it. Its algebra has dimension zero, so the Artinian factor calculation above supplies a smaller principal singleton neighbourhood of x. Thus zero point-dimension implies isolation as well.

Finally, every $h \in T = S_{R\setminus\mathfrak{p}}/\mathfrak{p}S_{R\setminus\mathfrak{p}}$ is represented by g/s with $g \in S$ and $s \in R \setminus \mathfrak{p}$. The image of s is a fibre unit, so $D(h) = D(\bar{g})$, and $h \notin x$ means $g \notin \mathfrak{q}$. Conversely the fibre of $D(g)$ is exactly $D(\bar{g})$. This proves the neighbourhood condition over the original ring, with no Noetherian assumption on R. □

Lemma F.57: lci

Reading source: AI Integrated Stacks, pinned Algebra, lines 37341–37441.

Let k be a field, S a finite-type k-algebra, and $\mathfrak{q}$ a prime of S. Fix any polynomial presentation $S = P/I$, with $P = k[X_1, \dots, X_n]$, and let $\mathfrak{Q} \subset P$ lie over $\mathfrak{q}$. Put

$$c = \operatorname{ht} \mathfrak{Q} - \operatorname{ht} \mathfrak{q} = n - \dim_{\mathfrak{q}} \operatorname{Spec} S.$$

The last equality is the point-codimension formula F.20. The following are equivalent.

1. Some S_g, with $g \notin \mathfrak{q}$, is a global complete intersection over k.
2. $I_{\mathfrak{Q}}$ has a generating list of length c.
3. $(I/I^2)_{\mathfrak{q}}$ has a generating list of length c.
4. $(I/I^2)_{\mathfrak{q}}$ is free of rank c over $S_{\mathfrak{q}}$.
5. $I_{\mathfrak{Q}}$ is generated by a regular sequence in $P_{\mathfrak{Q}}$.

Whenever these hold, every list of c elements spanning $I_{\mathfrak{Q}}/\mathfrak{Q}I_{\mathfrak{Q}}$ is a regular sequence in $P_{\mathfrak{Q}}$.

Proof. Write $V = P_{\mathfrak{Q}}$ and $J = I_{\mathfrak{Q}}$. The whole polynomial-local regularity proof AG-CA-14, Proposition 3.3, starting with k, makes V regular. The whole AG-CA-12, Theorem 6.1 makes it Cohen–Macaulay. Its dimension is $\operatorname{ht} \mathfrak{Q}$, whereas $V/J = S_{\mathfrak{q}}$ has dimension $\operatorname{ht} \mathfrak{q}$. In particular J is proper and finite, and the dimension drop is c.

Generators of J give generators of $J/J^2 = (I/I^2)_{\mathfrak{q}}$. Conversely lift c generators of this quotient to $f_1, \dots, f_c \in J$. Their quotient module $N = J/(f_1, \dots, f_c)V$ satisfies $N = JN$. It is finite, and J lies in the maximal ideal of V, so the determinant proof F.10 gives $N = 0$. Thus (2) and (3) are equivalent. The entire expected-dimension criterion AG-CA-12, Theorem 4.2, now says that any such generating list is regular: it lies in the maximal ideal and lowers the dimension of the Cohen–Macaulay ring by exactly its length. A list spanning $J/\mathfrak{Q}J$ generates J by the same Nakayama argument, proving the final assertion too.

We spell out the conormal basis for a regular generating list. The regular-list cone proof F.118 makes its first Koszul homology zero. If $\sum_i a_i f_i \in J^2$, write this sum as $\sum_i b_i f_i$ with every $b_i \in J$. The vector $(a_i - b_i)$ is a first Koszul cycle, hence a boundary. Each coefficient of a degree-two Koszul boundary belongs to J, because its differential has coefficients among the f_i. Thus every $a_i \in J$. This proves independence, as well as generation, of the classes of the f_i in J/J^2. Consequently (3) implies (4); (4) plainly implies (3). The same criterion proves (2) implies (5). If (5) uses a regular list of length e, successive regular quotients of a Cohen–Macaulay local ring lower dimension by one, by the complete depth and dimension proofs in AG-CA-12, Sections 2 and 4. Hence

$$\operatorname{ht} \mathfrak{q} = \dim(V/J) = \dim V - e = \operatorname{ht} \mathfrak{Q} - e,$$

so $e = c$, proving (5) implies (2).

Suppose (2) holds. Choose generators $f_i = a_i/s_i$ with $a_i \in I$ and $s_i \notin \mathfrak{Q}$. Multiplication by these units shows that the a_i also generate J and form a regular list there. Since I is finite over the Noetherian polynomial ring, each generator of $I/(a_1, \dots, a_c)$ is killed by some element outside $\mathfrak{Q}$. Their product $h' \notin \mathfrak{Q}$ gives $I_{h'} = (a_1, \dots, a_c)_{h'}$.

Choose a principal neighbourhood $D(h'')$ of $\mathfrak{q}$ on which the dimension is its point dimension. Here is the finite-component construction from F.20: for every minimal prime $\mathfrak{p}$ of S not contained in $\mathfrak{q}$, choose an element in $\mathfrak{p} \setminus \mathfrak{q}$ and multiply these finitely many elements. The product kills exactly the components not passing through the point; any surviving component has its original dimension, since a nonempty principal open in a finite-type domain over a field has the same transcendence degree and dimension. Thus the maximum surviving component dimension is the point dimension, by the whole AG-CA-09, Theorem 6.1, calculation. Lift this product to $h'' \in P \setminus \mathfrak{Q}$ and put $h = h'h''$. Its image g is outside $\mathfrak{q}$, and

$$S_g \cong k[X_1, \dots, X_n, Z]/(a_1, \dots, a_c, hZ - 1), \qquad \dim S_g = n - c.$$

This is the required global complete-intersection presentation: there are $n+1$ variables, $c+1$ equations and the prescribed dimension difference. This proves (1).

Conversely suppose (1), and choose a global complete-intersection presentation $S_g = k[Y_1,\dots,Y_m]/H$, with $H = (b_1,\dots,b_t)$ and $\dim S_g = m-t$. Every minimal prime $\mathfrak{a}$ over H has height at most t by the entire AG-CA-11, Theorem 3.1. Its component has dimension $m - \operatorname{ht}\mathfrak{a}$ by the polynomial height formula in AG-CA-09, Theorem 4.3. Hence each component has dimension at least $m-t$, while their maximum is the given dimension $m-t$. All components therefore have dimension exactly $m-t$. The point formula of AG-CA-09, Theorem 6.1, gives this same dimension at $\mathfrak{q}S_g$.

At the polynomial prime $\mathfrak{Q}''$ over that point, F.20 consequently gives

$$t = \operatorname{ht}\mathfrak{Q}'' - \operatorname{ht}\mathfrak{q}.$$

The actual ambient polynomial local ring is regular and Cohen–Macaulay as above. The expected-dimension criterion makes its given $b_1,\dots,b_t$ regular. The conormal calculation already proved makes $(H/H^2)_{\mathfrak{q}}$ free of rank t. The entire graph-presentation calculation F.33, localized at this point, supplies

$$S_{\mathfrak{q}}^{t+n} \cong (I/I^2)_{\mathfrak{q}} \oplus S_{\mathfrak{q}}^{m}.$$

Thus the conormal module is a finite direct summand of a finite free module. Tensor injections restricted to this summand prove flatness; the finite complementary summand gives finite presentation. The full local finite-flat proof AG-RG-S01, Lemma 1.2 makes it free. Reducing the displayed isomorphism to the residue field computes its rank as

$$t + n - m = n - \dim_{\mathfrak{q}} \operatorname{Spec} S = c.$$

This proves (4). Empty equation lists are included in the conormal and dimension calculations, so every implication and the last assertion also hold when $c = 0$. □

Lemma F.58: lci at prime

Reading source: Stacks, Tag 00SG.

For a finite-type k-algebra S and $\mathfrak{q} \in \operatorname{Spec} S$, the following are equivalent: $S_{\mathfrak{q}}$ is a complete intersection in the sense of F.3; it has a global complete-intersection principal neighbourhood; and the kernel of any finite polynomial presentation of S is generated by a regular sequence in the polynomial local ring above $\mathfrak{q}$. Equivalently, there is a principal neighbourhood that is locally a complete intersection over k.

Proof. Write $C = S_{\mathfrak{q}}$ and $P = k[X_1,\dots,X_n]_{\mathfrak{q}'} \twoheadrightarrow C$. The field k is a regular ring: its only prime localization is itself, with dimension and maximal-ideal generating number both zero. Iterating the entire polynomial-regularity proof AG-CA-14, Proposition 3.3 makes $k[X_1,\dots,X_n]$ a regular ring. Its localization at the displayed prime is therefore regular local, by the definition of a regular ring. A regular generating list for its kernel therefore gives the ambient ring required by F.3. Conversely suppose F.3 gives $C = Q/(a_1,\dots,a_e)$, where Q is regular local and essentially of finite type over the same field k, and this list is regular in its maximal ideal. We compare this local ambient with P explicitly.

Present Q as a quotient of a polynomial local ring $V = k[Y_1,\dots,Y_m]_{\mathfrak{n}}$. Both V and Q are regular local. AG-CA-14, Proposition 1.4, makes the kernel of $V \to Q$ generated by part of a regular parameter system. Append lifts of the a_i. The resulting list generates $\ker(V \to C)$ and is regular: its first successive quotients are those by the parameter list, and its remaining multiplication injections are precisely those of the given regular list in Q.

Let $U = k[X,Y]_{\mathfrak{r}}$, where $\mathfrak{r}$ is the preimage of the maximal ideal of C under the map using both sets of coordinates, and put $L = \ker(U \to C)$. Lift the images of X_i to V and denote them by $u_i(Y)$;

these are fractions with denominators outside $\mathfrak{n}$. Those denominators are units in U. The quotient by $X_i - u_i(Y)$ is V, including its indicated localization: evaluating an element outside $\mathfrak{r}$ gives an element outside $\mathfrak{n}$, hence a unit, and every fraction defining V already occurs in U. Translation of polynomial variables shows that the graph equations form a regular sequence. They remain regular after localization by exactness, and their quotient is nonzero. Appending the preceding regular list in V consequently gives a regular generating list for L.

In the other direction lift the images of Y_j to fractions $v_j(X)$ in P. The identical graph calculation gives

$$U/(Y_j - v_j(X)) \cong P, \qquad L = (\ker(P \to C), Y_j - v_j(X))U.$$

The constant and linear coefficient calculation of F.33 works in this localized polynomial ring as well: first invert the finitely many denominators of the v_j, translate Y_j by v_j, and then localize at $\mathfrak{r}$. Modulo L^2 its constant terms are read modulo $\ker(P \to C)^2$, its graph-linear coefficients modulo $\ker(P \to C)$, and terms of graph-degree at least two vanish. The resulting mutually inverse maps give

$$L/L^2 \cong \ker(P \to C)/\ker(P \to C)^2 \oplus C^m.$$

The regular list for L makes its conormal finite free by F.74. Its rank is $\dim U - \dim C$ by AG-CA-12, Corollary 6.2. Hence the first summand in the display is finite projective, and is finite free over the local ring C by S01, Lemma 1.2. Reduction to its residue field computes its rank. The polynomial height formula AG-CA-09, Theorem 4.3 gives $\dim U = \dim P + m$, since the two polynomial primes have the same residue field $\kappa(\mathfrak{q})$. The rank is thus $\dim P - \dim C$. F.57 now supplies a regular generating list for $\ker(P \to C)$ and a global complete-intersection principal neighbourhood of $\mathfrak{q}$. This proves the assertion for every chosen polynomial presentation, including the local ambients in F.3.

Conversely a global complete-intersection neighbourhood has the regular local equation lists proved in F.57, so gives F.3 at $\mathfrak{q}$. It is also locally a complete intersection: apply F.57 at each of its points. A locally complete-intersection principal neighbourhood contains a further global complete-intersection principal neighbourhood of the specified point by the defining local condition. If the further inverted element is b/g^r in S_g, that neighbourhood is S_{gb}, with $gb \notin \mathfrak{q}$. This proves the last equivalence. All graph and rank calculations allow empty coordinate or equation lists. □

Lemma F.59: limit essentially finite presentation

Reading source: AI Integrated Stacks, pinned Algebra, lines 33722–33770.

Let $R \to S$ be a local map of local rings, with S essentially finitely presented over R. There are a directed set and local maps $R_\lambda \to S_\lambda$ such that their colimit is $R \to S$, each R_λ is essentially finite type over $\mathbf{Z}$, each S_λ is essentially finite type over R_λ, and, for $\lambda \leq \mu$,

$$S_\lambda \otimes_{R_\lambda} R_\mu \longrightarrow S_\mu$$

is localization at a prime. All model rings are Noetherian, although R and S need not be.

Proof. Write $S = (R[X_1, \ldots, X_n]/(f_1, \ldots, f_c))_{\mathfrak{q}}$. Such a prime localization represents the given local essentially finitely presented algebra: in any localization presentation, every element outside the inverse image of the local maximal ideal is a unit, so the two localization universal properties give the prime localization. Locality of $R \to S$ makes $\mathfrak{q} \cap R = \mathfrak{m}_R$.

Let Λ be the finite subsets of R containing all coefficients of the f_j, ordered by inclusion. It is directed under finite union. Let $A_\lambda \subset R$ be the subring generated by such a subset over the image of $\mathbf{Z}$, put $\mathfrak{p}_\lambda = A_\lambda \cap \mathfrak{m}_R$, and set $R_\lambda = (A_\lambda)_{\mathfrak{p}_\lambda}$. Elements outside $\mathfrak{p}_\lambda$ are units in R, so this localization embeds in R: a fraction becoming zero has zero numerator, since its denominator is a unit there and A_λ is a subring. The maps $R_\lambda \to R_\mu$ are therefore inclusions and are local, with maximal ideals contracted from $\mathfrak{m}_R$. Every element of R occurs in some finite subset, proving $\operatorname{colim} R_\lambda = R$.

Define

$$C_\lambda = R_\lambda[X_1, \dots, X_n]/(f_1, \dots, f_c),$$

where the fixed coefficient list lies in R_λ. Let $\mathfrak{q}_\lambda$ be the inverse image of the maximal ideal of S under $C_\lambda \to S$, and set $S_\lambda = (C_\lambda)_{\mathfrak{q}_\lambda}$. These are local rings, and $R_\lambda \to S_\lambda$ is local because the maximal-ideal contractions agree. We verify the integer base and the permanence steps explicitly. A nonzero ideal of $\mathbf{Z}$ has a least positive member d. Repeated subtraction of d writes any nonnegative member as $qd+r$ with $0 \le r < d$; the remainder is in the ideal and minimality forces $r = 0$. Negative members are handled by changing their sign. Thus every integer ideal is principal, including the zero ideal, so $\mathbf{Z}$ is Noetherian.

Each A_λ is a quotient of a finite-variable polynomial ring over $\mathbf{Z}$, and the Hilbert basis theorem AG-CA-03, Theorem 2.1, makes that polynomial ring Noetherian. An ideal of a quotient is the image of its inverse-image ideal, so is finitely generated. An ideal of a localization equals the localization of its contraction: if a/s is in the ideal, then $a/1$ is too, and the converse follows by multiplying by $1/s$. Its contracted ideal has finite generators, whose localized images generate the original ideal. These calculations prove quotient and localization permanence and make $R_\lambda, C_\lambda, S_\lambda$ Noetherian. Their finite polynomial lists and prime localizations give the two essential finite-type assertions; composition of such maps was fully proved in F.31.

Polynomial scalar extension gives $C_\lambda \otimes_{R_\lambda} R_\mu \cong C_\mu$. The tensor with S_λ initially inverts the images of $C_\lambda \setminus \mathfrak{q}_\lambda$. Those images avoid $\mathfrak{q}_\mu$, by their definition as contractions from the same target. Thus $\mathfrak{q}_\mu$ defines a prime of that tensor localization. Localizing at it inverts the remaining elements outside $\mathfrak{q}_\mu$ and gives exactly S_μ. The fraction universal property proves both inverse maps, establishing the stipulated transition formula. The identical argument with the final base R gives S as a prime localization of $S_\lambda \otimes_{R_\lambda} R$.

For the colimit of the target rings, every fraction in S uses finitely many coefficients in its numerator and denominator, so it has a representative in some S_λ. If two representatives become equal in S, their cross difference is killed in $R[X]/(f)$ by a denominator outside $\mathfrak{q}$. Lift that denominator to a polynomial and express the killed difference as a finite sum $\sum_j h_j f_j$ in $R[X]$. These polynomials and their equality use finitely many coefficients. Put them into one common later R_μ. The denominator still avoids $\mathfrak{q}_\mu$, and the same ideal-membership equality holds there, so the representatives are already equal in S_μ. This proves injectivity as well as surjectivity of the colimit map. The structural maps from the base agree throughout, giving the entire claimed colimit of local ring maps. □

Lemma F.60: limit module essentially finite presentation

Reading source: AI Integrated Stacks, pinned Algebra, lines 33804–33859.

For a local essentially finitely presented map $R \to S$ and a finitely presented S-module M, the models in F.59 can be taken at sufficiently large stages with finitely presented S_λ-modules M_λ such that

$$M_\lambda \otimes_{S_\lambda} S_\mu \cong M_\mu \quad (\lambda \le \mu), \qquad \operatorname*{colim}_\lambda M_\lambda \cong M.$$

In particular each module is finite over its model algebra, and all base/prime-localization assertions of F.59 remain valid.

Proof. Choose a finite presentation

$$S^a \xrightarrow{H} S^b \longrightarrow M \longrightarrow 0.$$

The finitely many entries of H are elements of the colimit S. The representative proof in F.59 puts them in a common S_{λ_0}. Restrict to stages $\lambda \ge \lambda_0$ and let H_λ be their images there. Define

$$M_\lambda = \operatorname{coker}\bigl(S_\lambda^a \xrightarrow{H_\lambda} S_\lambda^b\bigr).$$

This is a finite presentation over S_λ. Right exactness of tensor and the fixed free bases identify the tensor of this presentation with the presentation given by H_μ, proving the asserted transition isomorphism. The same calculation gives $M_\lambda \otimes_{S_\lambda} S \cong M$.

We verify the colimit explicitly. Any element of M is represented by a vector in S^b, whose finitely many coordinates occur at a common model stage, giving surjectivity from the module colimit. If $x_\lambda \in S^b_\lambda$ represents a class that is zero in M, its image is Hy for some finite vector $y \in S^a$. Choose a later stage representing every coordinate of y. The vector equality $x = Hy$ is a finite list of ring equalities, so F.59 detects all of them at one common successor. There $x_\mu = H_\mu y_\mu$, proving that the original class is zero in M_μ. This proves injectivity of the colimit map. The representative and transition calculations respect addition and the ring actions, so the isomorphism is an isomorphism of modules over $\operatorname{colim} S_\lambda = S$.

No ring models have been changed except for removing an initial segment of the directed set, which leaves all colimits unchanged: every earlier representative has an image at a common successor with λ_0. Hence their local maps, essential finite-type properties and prime tensor-localizations are exactly those proved in F.59. The arguments include $a = 0$, $b = 0$ and the zero module. □

Lemma F.61: local dimension zero henselian

Reading source: AI Integrated Stacks, pinned Algebra, lines 44097–44121.

Every local ring $(R, \mathfrak{m})$ of dimension zero is henselian. No Noetherian or nilpotent-maximal-ideal hypothesis is required.

Proof. The only prime of R is $\mathfrak{m}$: any other prime is properly below the maximal ideal and would give a chain of length one. Every $u \in \mathfrak{m}$ is nilpotent. Indeed, if no power of u were zero, R_u would be nonzero. A maximal proper ideal of this ring exists by the chain-union and Zorn argument, and its inverse image would be a prime of R avoiding u. It cannot be the unique prime $\mathfrak{m}$, a contradiction.

Use the simple-root definition of henselianity in AG-CA-20, Section 1. Let $f \in R[T]$ be monic, and let $\bar{a}$ be a root of its reduction with $\bar{f}'(\bar{a}) \neq 0$. Choose any lift $a_0 \in R$. The element $f'(a_0)$ is a unit because its residue is nonzero, and $\epsilon_0 = f(a_0) \in \mathfrak{m}$ has some zero power $\epsilon_0^N = 0$.

Inductively set

$$a_{j+1} = a_j - f'(a_j)^{-1} f(a_j).$$

Every a_j has residue $\bar{a}$, so each inverse in this formula exists. Expanding the polynomial by the binomial formula gives, for all a, h,

$$f(a+h) = f(a) + f'(a)h + h^2 Q(a,h)$$

for a polynomial Q over R. Substitution of the correction therefore gives

$$f(a_{j+1}) \in (f(a_j)^2) \subset (\epsilon_0^{2^{j+1}}).$$

The correction remains in $\mathfrak{m}$, as needed for the residue induction. Choose j with $2^j \geq N$; then $f(a_j) = 0$. This supplies the required lifted root in finitely many steps, using only the nilpotence of the initial error.

For completeness it is unique with this residue. If a, b are two such roots, polynomial subtraction factors as

$$0 = f(b) - f(a) = (b-a)\big(f'(a) + (b-a)H(a,b)\big).$$

The second factor has the nonzero residue $\bar{f}'(\bar{a})$ and is a unit, so $b = a$. All simple residue roots have thus been lifted, proving henselianity. □

Lemma F.62: local syntomic

Reading source: AI Integrated Stacks, pinned Algebra, lines 37927–37941.

Let $R \to S$ be any ring map. If finitely many principal opens $D(g_i)$ cover $\operatorname{Spec} S$ and every $R \to S_{g_i}$ is syntomic, then $R \to S$ is syntomic.

Proof. The assumed cover means the g_i generate the unit ideal. The whole finite-equation gluing proof F.35 gives finite presentation of S/R from the chart finite presentations.

To prove flatness, take an arbitrary injection $M' \to M$ of R-modules and let K be the kernel of $M' \otimes_R S \to M \otimes_R S$. Exactness of localization identifies K_{g_i} with the kernel after tensoring with S_{g_i}; it is zero by chart flatness. If $x \in K$, some power $g_i^{e_i}$ kills x for each i. These powers still generate the unit ideal: otherwise a maximal ideal containing all of them would contain all the g_i. A linear combination equal to one then kills x. Thus $K = 0$, proving flatness without a finiteness condition on these modules.

For every $\mathfrak{p} \subset R$, put $C = S \otimes_R \kappa(\mathfrak{p})$. The images of the g_i still generate the unit ideal, and

$$C_{g_i} \cong S_{g_i} \otimes_R \kappa(\mathfrak{p})$$

by the fraction universal property. Each is a local complete intersection over the field, by syntomicness of that chart. For a point of C, choose a chart containing it and then a global complete-intersection principal neighbourhood there. If that second denominator is a/g_i^e, its principal open in the chart is $D(g_i) \cap D(a) = D(g_i a)$ in C. Its coordinate algebra is exactly the same iterated localization. Thus these neighbourhoods give the required complete-intersection cover of the entire fibre, as in F.2. Finite presentation, flatness and all fibre conditions are now proved. If the cover is empty, its spectrum is empty and $S = 0$; this algebra is finitely presented as $R/(1)$, flat, and has empty fibres, so the conclusion holds as well. □

Lemma F.63: localization colimit

Reading source: AI Integrated Stacks, pinned Algebra, lines 1260–1278.

Let $U \subset R$ be multiplicative and M an R-module. Then $U^{-1}M$ is a filtered colimit of copies of M with transition maps given by multiplication by denominators. It is also the filtered colimit of the principal localizations M_s for $s \in U$. In particular $U^{-1}R$ is the corresponding filtered colimit of the rings R_s.

Proof. We give a directed indexing set with specified transitions. Let Λ be the finitely supported functions $\nu : U \to \mathbf{N}$, ordered coordinatewise, and put $s_\nu = \prod_{u \in U} u^{\nu(u)}$. For $\nu \leq \mu$, write

$$s_\mu = s_\nu v_{\nu\mu}, \qquad v_{\nu\mu} = \prod_{u \in U} u^{\mu(u)-\nu(u)}.$$

The maximum of two functions is a common successor. The displayed factor is specified even if the ring has zero divisors, and the factors multiply compatibly along three stages. Put one copy $M^{(\nu)} = M$ at each stage, with transition $m \mapsto v_{\nu\mu} m$.

The maps $M^{(\nu)} \to U^{-1}M$, $m \mapsto m/s_\nu$, commute with the transitions and are jointly surjective, since every denominator occurs as a one-element product. If $m/s_\nu = n/s_\mu$, the fraction criterion gives $u \in U$ with

$$u(s_\mu m - s_\nu n) = 0.$$

At the successor $\rho = \nu + \mu + \delta_u$, both representatives have become equal, their difference being the left side of this identity. Here δ_u has value one at u and zero elsewhere. Conversely equality at any successor gives equality of the fractions. This proves bijectivity. To check addition, move two representatives to a common successor; addition there becomes the usual common-denominator

addition. Scalar multiplication is preserved by every map, so the bijection is an R-module isomorphism.

For the principal-localization assertion use the same directed set, now with modules M_{s_ν}. If $\nu \le \mu$, s_ν is a unit after inverting $s_\mu = s_\nu v_{\nu\mu}$: its inverse is $v_{\nu\mu}/s_\mu$. This gives the specified localization map. Each fraction in $U^{-1}M$ is represented at a principal stage. If m/s_ν^a and n/s_μ^b become equal in the full localization, some $u \in U$ kills their cross difference

$$u(s_\mu^b m - s_\nu^a n) = 0.$$

At the principal successor $s_\rho = s_\nu s_\mu u$, all three factors are units. The identity therefore makes the two fractions equal there. This proves both directions of the colimit isomorphism, including all module operations.

Taking $M = R$, these transition maps are ring maps. The same fraction calculation preserves multiplication and one, proving the ring colimit. These diagrams represent the usual directed family of principal localizations: each product is itself in U, and any finite family of its elements is inverted at their product. The argument includes $0 \in U$, in which case every representative vanishes at a successor containing that denominator. □

Lemma F.64: localization smooth separable

Reading source: AI Integrated Stacks, pinned Algebra, lines 46075–46088.

A finitely generated field extension K/k is separably generated if and only if it is the fraction field of a smooth finite-type k-domain.

Proof. For the forward direction choose a finite separating transcendence basis $T_1, \dots, T_d$, set $F = k(T_1, \dots, T_d)$, and write the remaining finite separable extension as a finite tower

$$F = K_0 \subset K_1 \subset \cdots \subset K_r = K, \qquad K_i = K_{i-1}(y_i).$$

A finite field-generating list gives such a tower; each step is separable. Let its monic minimal polynomial have degree e_i. Successive monic division makes the products $y_1^{a_1} \cdots y_{i-1}^{a_{i-1}}$, with $0 \le a_j < e_j$, a basis of K_{i-1} over F. Express every coefficient of the next minimal polynomial in this basis. There are only finitely many coefficients and basis vectors in the entire tower. One nonzero polynomial $h \in k[T_1, \dots, T_d]$ clears every rational-function denominator.

Put $R_0 = k[T_1, \dots, T_d, 1/h]$. The coefficient expressions give successive monic equations F_i in $R_0[Y_1, \dots, Y_i]$. Define

$$C_i = R_0[Y_1, \dots, Y_i]/(F_1, \dots, F_i).$$

Monic polynomial division, applied successively, makes the same standard monomials an R_0-basis of C_i. Tensoring with F gives exactly K_i. Since R_0 is a domain and these modules are free, their maps to their tensors with F are injective. Thus each C_i is an actual subdomain of K_i.

The derivative $\delta_i = (\partial F_i/\partial Y_i)(y_1, \dots, y_i)$ is nonzero in K_i, by separability of its minimal polynomial. Form the successive rings

$$D_0 = R_0, \qquad D_i = (D_{i-1}[Y_i]/(F_i))_{\delta_i}.$$

Here the coefficient expressions are mapped from the preceding C_{i-1}, and all previously inverted derivatives remain inverted. Inductively D_i is the corresponding localization of C_i, so is a domain in K_i. Each map $D_{i-1} \to D_i$ is a standard étale one-equation map, with monic equation and inverted derivative. The entire AG-CA-17, Proposition 6.1 proves its unique square-zero lifts. The polynomial, localization and composition proofs in its Theorem 1.2 make D_r formally smooth over k. All equations, variables and inverses are finite in number, so it is finitely presented, hence smooth. It contains every T_j and y_i, and lies in K, so its fraction field is exactly K. Empty towers have no derivatives to invert and give the same construction.

Conversely let C be a smooth finite-type k-domain with fraction field K. Choose a nonzero principal standard smooth chart D by the entire AG-CA-17, Theorem 5.1. Its fraction field remains K. Reorder the variables of this finite polynomial presentation so the invertible square Jacobian minor uses the pivot variables, and call the remaining variables $U_1, \dots, U_t$. With $B_0 = k[U_1, \dots, U_t]$, the identical chart is a finitely presented algebra over B_0 with only the pivot variables and an invertible full square Jacobian. The explicit square-zero equation correction of AG-CA-17, Theorem 4.1, gives a unique lift: the square invertible matrix solves the error vector uniquely. Thus $B_0 \to D$ is étale. The whole arbitrary-base flatness proof AG-CA-18, Corollary 6.2 makes it flat.

This structural map from B_0 is injective. For $0 \neq b \in B_0$, multiplication by b is injective on the polynomial domain. Flat tensor with D preserves this injection. If b had zero structural image, that map on the nonzero ring D would be zero, a contradiction. Consequently the U_j are algebraically independent in K.

Localize this base at its nonzero elements, obtaining $F_0 = k(U_1, \dots, U_t)$. The domain $D \otimes_{B_0} F_0$ is nonzero and étale over F_0, by the full base-change proof of AG-CA-17, Theorem 1.2. Its entire Theorem 7.2 classifies it as a finite product of finite separable extensions of F_0. A nonzero domain has no nontrivial orthogonal idempotents, so this product has exactly one factor. It is therefore a finite separable field. It contains D and lies in K; every nonzero element of D is invertible in this field. Hence the field is precisely K. Thus the independent U_j form a separating transcendence basis, proving the converse at every characteristic, without using F.91. □

Lemma F.65: localize NL

Reading source: AI Integrated Stacks, pinned Algebra, lines 37118–37133.

For any ring map $A \to B$ and multiplicative subset $U \subset B$, the canonical map

$$\mathrm{NL}_{B/A} \otimes_B U^{-1}B \longrightarrow \mathrm{NL}_{U^{-1}B/A}$$

is a quasi-isomorphism.

Proof. Use the directed principal-localization diagram constructed in F.63, with stages B_{s_ν} and colimit $U^{-1}B$. At every stage the whole calculation F.72 gives a homotopy equivalence

$$\mathrm{NL}_{B/A} \otimes_B B_{s_\nu} \longrightarrow \mathrm{NL}_{B_{s_\nu}/A}\,.$$

These are the canonical maps induced by $B \to B_{s_\nu}$; they are compatible with later localizations. Presentation comparisons and their explicit homotopies are proved in F.11. No compatible choice of the individual contracting homotopies is needed for the induced homology maps.

Tensor commutes with this colimit term by term: a tensor is a finite sum of pure tensors, and each coefficient has a common-stage representative; its balancing relations and equalities hold at a later stage. The right side has colimit $\mathrm{NL}_{U^{-1}B/A}$ by the complete canonical-presentation proof F.24.

Finally filtered colimits commute with homology here. A cycle in either colimit has a stage representative; its differential is zero at some later stage, giving a cycle there. If its homology class vanishes, it is the differential of a representative vector in the colimit. Put the two representatives and their equality at one common successor to obtain a boundary there. This proves both directions of the homology-colimit isomorphism. The stage maps induce isomorphisms in degrees one and zero, so their colimit does too. All other homology groups vanish since these complexes have only two terms. This proves the claimed quasi-isomorphism, without a Noetherian or finite-generation hypothesis on U. The zero localization is included in the same diagrams. □

Lemma F.66: localize relative complete intersection

Reading source: Stacks, Tag 00ST.

Let R be Noetherian and put $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$. In each situation below there are $g \in S$ and a polynomial lift h of g for which

$$S_g = R[X_1, \dots, X_n, T]/(f_1, \dots, f_c, hT - 1)$$

is a relative global complete intersection as defined in F.4.

1. If every nonempty fibre over $V(I)$, for an ideal $I \subset R$, has dimension $n - c$, one may require $g \equiv 1 \pmod{IS}$.
2. If the fibre over $\mathfrak{p} \in \operatorname{Spec} R$ has dimension $n - c$, one may require the image of g in that fibre to be a unit.
3. If a point $\mathfrak{q}$ of the fibre over $\mathfrak{p}$ has fibre point-dimension $n - c$, one may require $g \notin \mathfrak{q}$.

Proof. Set $d = n - c$. The neighbourhood bound proved in F.39 gives, around every point whose fibre point-dimension is at most d, a principal neighbourhood on which every fibre point-dimension is at most d. The union of these neighbourhoods is an open W with that same bound. Each of its nonempty fibres has dimension at most d: a finite-type field fibre is Noetherian, and its component dimensions are its point-dimensions at the corresponding generic points, as in F.20. Open subsets retain this bound.

In case (1) choose W containing $V(IS)$; each point of those fibres has point-dimension at most their assumed dimension. Write the closed complement of W as $V(J)$ in $\operatorname{Spec} S$. The equality $V(J + IS) = \varnothing$ implies $J + IS = S$: a proper ideal would be contained in a maximal, hence prime, ideal. Write $1 = g + b$ with $g \in J$ and $b \in IS$. Then $g \equiv 1 \pmod{IS}$, and $D(g) \subset W$.

In case (2) take W containing the entire indicated fibre, and again write its complement as $V(J)$. If $k = \kappa(\mathfrak{p})$, the image ideal $J(S \otimes_R k)$ is the unit ideal. Choose a finite expression $1 = \sum_j \beta_j \bar{g}_j$, where $g_j \in J$. Each β_j is represented in $S_{R\setminus\mathfrak{p}}/\mathfrak{p}S_{R\setminus\mathfrak{p}}$. Clear their finitely many denominators to obtain $b_j \in S$ and $s \in R \setminus \mathfrak{p}$ with $\beta_j = \bar{b}_j/\bar{s}$. Thus $g = \sum_j b_j g_j \in J$ has fibre image $\bar{s}$, a unit. Again $D(g) \subset W$.

In case (3), one of the principal neighbourhoods supplied directly by F.39 is $D(g)$ with $g \notin \mathfrak{q}$ and the required upper bound. In all three cases select any polynomial lift h of g. Adjoining the inverse of h gives the presentation in the statement, by the explicit substitution $T \mapsto g^{-1}$ and its inverse. Every nonempty fibre of this presentation has dimension at least

$$(n + 1) - (c + 1) = d.$$

Indeed, in its polynomial ring over the residue field a minimal prime over $c + 1$ equations has height at most $c + 1$, by AG-CA-11, Theorem 3.1; AG-CA-09, Theorem 4.3 gives that component dimension as $n + 1$ minus its height. The upper bound on $D(g)$ makes the dimension exactly d. This proves the relative global complete-intersection assertion, without presupposing flatness; its flatness is subsequently proved in F.77. Empty fibres impose no dimension test, and a zero localization is included. □

Lemma F.67: locally smooth

Reading source: Stacks, Tag 00TC.

For any ring map $R \to S$, smoothness is equivalent to the existence, around every prime of S, of a principal localization that is smooth over R.

Proof. Use the finite-presentation and cotangent-complex definition F.6. If S/R is smooth, then S_g is finitely presented by adjoining an inverse variable. The exact calculation F.72, together with the presentation comparison F.11, identifies its conormal complex with the localized complex plus a contractible summand. Its first homology is zero and its differential module is the localization of

a finite projective module. Localization keeps the splitting of a finite free module, so that module is still finite projective. Thus S_g/R is smooth.

For the converse, the elements defining the given neighbourhoods generate the unit ideal: otherwise a maximal ideal containing them would be an uncovered prime. A finite expression for 1 gives a cover by $D(g_1), \dots, D(g_r)$. The complete gluing proof F.35 makes S/R finitely presented. Fix $S = R[X_1, \dots, X_n]/I$ with I finite. For this presentation set

$$K = \ker(I/I^2 \xrightarrow{d} S^n), \qquad M = \operatorname{coker}(I/I^2 \xrightarrow{d} S^n) = \Omega_{S/R}.$$

The derivatives of a finite list generating I generate the image of d, so M is finitely presented. Exact localization and F.72/F.11 identify K_{g_i} with the zero first homology of each smooth chart. The power-annihilator module calculation in F.35 therefore gives $K = 0$.

It remains to produce a global projective splitting for M. Write $\pi : S^n \twoheadrightarrow M$. On each chart M_{g_i} is finite projective. A section s_i of π_{g_i} exists: write that projective module as a summand of a finite free module, lift the free basis through π_{g_i}, and restrict the resulting lift to the summand.

For a finitely presented module, localization of $\operatorname{Hom}_S(M, S^n)$ is $\operatorname{Hom}_{S_{g_i}}(M_{g_i}, S^n_{g_i})$. Indeed a finite presentation of M identifies both modules with the kernel of the same matrix between finite powers of S^n, and localization is exact. Clear the denominator of s_i to obtain $t_i : M \to S^n$ whose localized composite is $g_i^{a_i} 1_M$. The error $\pi t_i - g_i^{a_i} 1_M$ vanishes on a finite generating list after localization. A further common power of g_i kills those finitely many errors. After multiplying t_i by that power, we have an actual equality

$$\pi t_i = g_i^{e_i} 1_M.$$

Increase e_i to a positive integer if needed. These powers generate 1: expand a finite unit expression in the g_i to exponent $1 + \sum_i (e_i - 1)$; every monomial contains one of the required powers. Choose c_i with $\sum_i c_i g_i^{e_i} = 1$. Then $t = \sum_i c_i t_i$ satisfies $\pi t = 1_M$. Thus $S^n = t(M) \oplus \ker \pi$, with both summands generated by projections of its finite basis. This proves finite projectivity of M directly.

We have proved finite presentation, zero first cotangent homology and finite projective differentials, exactly F.6. If $S = 0$, its finite presentation is $R/(1)$ and both cotangent modules are zero, so the conclusion also holds for the empty spectrum. No Noetherian hypothesis is needed. □

Lemma F.68: make etale map prescribed residue field

Reading source: AI Integrated Stacks, pinned Algebra, lines 41287–41319.

Let $\mathfrak{p}$ be a prime of a ring R, and let $L/\kappa(\mathfrak{p})$ be finite separable. There are a finitely presented étale R-algebra R' and a prime $\mathfrak{p}'$ over $\mathfrak{p}$ with an isomorphism $\kappa(\mathfrak{p}') \cong L$ respecting the given residue-field extension.

Proof. Choose a finite field-generating list $\alpha_1, \dots, \alpha_r$ for $L/\kappa(\mathfrak{p})$, for example a vector-space basis. Put $E_0 = \kappa(\mathfrak{p})$ and $E_i = E_{i-1}(\alpha_i)$. Each step is finite separable: the minimal polynomial over E_{i-1} divides a separable polynomial over E_0 and therefore has no repeated root.

Inductively construct a finitely presented étale R-algebra A_i with a prime $\mathfrak{q}_i$ and a compatible identification $\kappa(\mathfrak{q}_i) = E_i$. Start with $A_0 = R$ and $\mathfrak{q}_0 = \mathfrak{p}$. Suppose the preceding stage is constructed, and write $A = A_{i-1}$, $\mathfrak{q} = \mathfrak{q}_{i-1}$. Let

$$\bar{F}_i(T) = T^{e_i} + \sum_{j<e_i} \bar{a}_j T^j \in E_{i-1}[T]$$

be the monic minimal polynomial of α_i. The residue map $A_{\mathfrak{q}} \to E_{i-1}$ is surjective, so lift each $\bar{a}_j$ as a fraction a_j/s_j with $s_j \notin \mathfrak{q}$. The product $h = \prod_j s_j$ is outside $\mathfrak{q}$, and all these lifts belong to the single principal localization A_h. They give a monic polynomial $F_i \in A_h[T]$. Set

$$A_i = (A_h[T]/(F_i))_{F_i'}.$$

Its defining equation has an invertible derivative. The full one-equation square-zero correction AG-CA-17, Proposition 6.1 makes this map étale. Principal localization is formally étale by its Theorem 1.2 and finitely presented by its inverse-variable equation. Finite composition, with the equation substitution proof F.28, therefore makes A_i/R finitely presented and étale.

Evaluation at $T = \alpha_i$ defines a ring map $A_i \to E_i$: its derivative has nonzero value by separability, so its inverse can be evaluated. Let $\mathfrak{q}_i$ be the kernel. It is prime, and its contraction to A is exactly $\mathfrak{q}$, since the restriction is the given residue map and h has nonzero residue.

To compute its residue field, let D_i be the image of A_i in E_i. It is a domain, and $A_i/\mathfrak{q}_i = D_i$. The fraction field of the image of A_h is E_{i-1}: this image is $(A/\mathfrak{q})_h$, and localization does not change its fraction field $\kappa(\mathfrak{q})$. Thus $\operatorname{Frac} D_i$ contains E_{i-1} as well as α_i, and so contains $E_i = E_{i-1}(\alpha_i)$. It lies inside E_i, giving equality. Localization at the kernel identifies

$$\kappa(\mathfrak{q}_i) = \operatorname{Frac}(A_i/\mathfrak{q}_i) = \operatorname{Frac} D_i = E_i.$$

This uses the fraction field of the image; it does not require the evaluation map itself to be onto the field.

Continue through the finite tower. Taking $R' = A_r$ and $\mathfrak{p}' = \mathfrak{q}_r$ gives the required extension and prime, with all residue identifications commuting with the original map from R. For the trivial extension the empty tower gives $R' = R$ and $\mathfrak{p}' = \mathfrak{p}$. □

Lemma F.69: make separably generated

Reading source: AI Integrated Stacks, pinned Algebra, lines 10182–10253.

For any finitely generated field extension K/k, there is a commutative square of inclusions

$$\begin{array}{ccc} K & \longrightarrow & K' \\ \uparrow & & \uparrow \\ k & \longrightarrow & k' \end{array}$$

in which k'/k and K'/K are finite purely inseparable and K'/k' is separably generated.

Proof. Choose a finite field-generating list and a maximal algebraically independent sublist $T_1, \dots, T_d$. Every remaining generator is algebraic over $F = k(T_1, \dots, T_d)$: a polynomial dependence using that generator has a nonzero coefficient after substituting the independent T_j. A finite algebraic generator tower is finite. At each step, division by the monic minimal polynomial gives a power basis; multiplication of these bases gives a basis of the whole tower, since grouping a relation by the powers of the last generator proves independence inductively. Thus K/F is finite. In characteristic zero its irreducible polynomials have nonzero derivatives, so every tower step is separable. The entire separable-tower proof AG-CA-17, Lemma 7.1 gives the result with $k' = k$ and $K' = K$.

Assume now characteristic $p > 0$. We first prove the finite-extension facts used in the construction. For a finite extension E/F, the elements separable over F form a subfield E_s. Indeed if x, y are separable, the minimal polynomial of y over $F(x)$ divides its separable polynomial over F. The two separable steps and the same complete separable-tower proof make every element of $F(x, y)$ separable over F. This proves closure under sums, products and inverses. As a subspace of the finite-dimensional F-space E, this field is finite over F.

If $x \in E$ has irreducible minimal polynomial f, repeatedly remove zero derivatives to write

$$f(X) = g(X^{p^e}), \qquad g' \neq 0.$$

This process terminates as degrees decrease; g is irreducible because a factorization would substitute to one of f. Its derivative has smaller degree, so the Euclidean algorithm gives $\gcd(g, g') = 1$. Hence x^{p^e} is separable over F and lies in E_s. A common exponent for a finite field-generating list

makes every p^eth power from E belong to E_s: apply Frobenius to its rational expressions. Thus E/E_s is purely inseparable.

We will use the degree calculation without an external p-basis result. If $\beta^p \in H$ but $\beta \notin H$, its minimal polynomial over H divides $X^p - \beta^p = (X - \beta)^p$ in a field containing β. A monic proper nonconstant factor of degree $r < p$ has coefficient $-r\beta$ at degree $r - 1$. Since r is nonzero in the field, a factor over H would force $\beta \in H$, a contradiction. The minimal polynomial therefore has degree p. By adjoining the powers of each purely inseparable generator in reverse order, starting from a power in the base, each nontrivial step has this degree. Multiplication of the power bases shows that every finite purely inseparable degree is a power of p.

Apply these facts to K/F, and call its separable subfield K_s. Prove the desired assertion by strong induction on $m = [K : K_s]$, for all finitely generated extensions with their chosen transcendence bases. If $m = 1$, the chosen T_j already separate K/k. If $m > 1$, choose $x \in K \setminus K_s$ and the least $e \geq 1$ with $x^{p^e} \in K_s$. Put

$$\beta = x^{p^{e-1}}, \qquad \alpha = \beta^p \in K_s.$$

Then $\beta \notin K_s$. The degree calculation gives $[K_s(\beta) : K_s] = p$ and $[K : K_s(\beta)] = m/p$.

Let

$$P(X) = X^n + \sum_{i<n} a_i X^i \in F[X]$$

be the separable minimal polynomial of α. Write all a_i as rational functions in T, and list the finitely many coefficients of k occurring in their numerators and denominators. Adjoin their pth roots compatibly over K, one at a time. Such a root exists in a field extension: take the quotient by an irreducible factor of $X^p - c$. The quotient is a field by the Euclidean algorithm. A root is unique in any common extension, since $(u - v)^p = u^p - v^p$. All old fields therefore remain embedded in these successive root fields. Let k_1 be the field generated by these roots over k. It is finite purely inseparable, and $k_1^p \subset k$, since taking the pth power of any rational expression puts all its coefficients and generators in k.

In the same compatible extension adjoin U_j with $U_j^p = T_j$. They are algebraically independent over k_1: a nonzero polynomial relation would, after taking its pth power, be a nonzero relation among the T_j over k. Its coefficients remain nonzero under Frobenius. Put

$$F_1 = k_1(U_1, \ldots, U_d), \qquad L = KF_1.$$

This gives the actual commuting intermediate field square. The extension L/K is finite and purely inseparable; it adjoins finitely many roots with pth powers in K. The pth power of any rational expression in them lies in K.

Take coefficientwise pth roots of the chosen numerator and denominator for each a_i, and substitute the U_j. This gives $b_i \in F_1$ with $b_i^p = a_i$. The denominator is nonzero because its pth power is the original nonzero denominator. Define

$$Q(X) = X^n + \sum_{i<n} b_i X^i, \qquad Q(X)^p = P(X^p).$$

Then $Q(\beta)^p = P(\alpha) = 0$, so $Q(\beta) = 0$. The derivative calculation, using $i^p = i$ in the prime field, gives

$$Q'(\beta)^p = P'(\alpha) \neq 0.$$

Write $Q = hq$ with h the minimal polynomial of β over F_1. Evaluation of $Q' = h'q + hq'$ at β implies $h'(\beta)q(\beta) \neq 0$. Thus h is separable, and β is separable over F_1.

Every separable polynomial remains square-free after a field extension: its Euclidean identity with its derivative remains an identity equal to one after extending coefficients. The minimal polynomial of a generator after base change divides that polynomial and is therefore separable too. A finite generating tower for K_s/F consequently gives a separable tower for K_sF_1/F_1. The

minimal polynomial of the now separable β over this larger field divides its separable polynomial over F_1, so this final step is separable as well. The separable-tower proof gives

$$H = K_s(\beta)F_1 \subset L_s,$$

where L_s denotes the separable subfield of the finite extension L/F_1.

To bound its remaining inseparable degree, take a basis of K over $K_s(\beta)$ of size m/p and span it over H inside L. This span contains K and H, contains one, and is closed under multiplication, since the basis-product coefficients belong to $K_s(\beta) \subset H$. It is a finite-dimensional domain over H, hence a field: nonzero multiplication is injective, therefore surjective, and supplies an inverse. It contains F_1 and K, so equals L. Consequently

$$[L : L_s] \leq [L : H] \leq m/p < m.$$

Also L/F_1 is finite, by the same spanning argument starting with a basis of K/F. The U_j are a transcendence basis for the finitely generated extension L/k_1. Strong induction thus applies to this precise new data, giving finite purely inseparable k_2/k_1 and L'/L with L'/k_2 separably generated.

Finally finite purely inseparable extensions compose: for each element of L', a p-power lies in L and a further p-power lies in K. The same argument applies to k_2/k. Finiteness follows by multiplying their finite tower degrees. Taking $K' = L'$ and $k' = k_2$ gives the asserted square. Empty transcendence bases, trivial adjunctions and the base case are all included. □

Lemma F.70: matrix left inverse

Reading source: Stacks, Tag 07DQ.

Let A be an $n \times m$ matrix over a commutative ring R, where $0 \leq m \leq n$, and let J be the ideal of its maximal minors. Every $f \in J$ has an $m \times n$ matrix B with $BA = f1_m$. Conversely, existence of such a B implies $f^m \in J$.

Proof. For an increasing m-element subset I of $\{1, \ldots, n\}$, let E_I select those coordinates from R^n, and put $A_I = E_I A$. Its determinant is the corresponding minor. By the complete cofactor calculation in F.19,

$$\operatorname{adj}(A_I)E_I A = (\det A_I)1_m.$$

Thus, from an expression $f = \sum_I c_I \det A_I$, the finite sum

$$B = \sum_I c_I \operatorname{adj}(A_I)E_I$$

constructs the required matrix.

For the converse, here is the determinant product calculation. Expand each column of BA as the linear combination of columns of B whose coefficients are entries of that column of A. Multilinearity follows directly from the determinant permutation formula. A term choosing the same column twice is zero by the equal-column cancellation. For any fixed set I of distinct chosen columns, order those indices increasingly. Reordering them in all possible ways contributes their permutation signs; the signed sum of their coefficient products is exactly $\det A_I$. The remaining column determinant is $\det B^I$, where B^I uses the columns indexed by I. Consequently

$$\det(BA) = \sum_{|I|=m} (\det B^I)(\det A_I) \in J.$$

When $BA = f1_m$, its determinant is f^m. This proves the converse over arbitrary rings, including rings with nilpotents. If $m = 0$, the sole minor is the empty determinant 1, so $J = R$. The unique map to and from the zero module supplies the first assertion, and $f^0 = 1 \in J$ supplies the second. □

Lemma F.71: presentation sym exterior

Reading source: Stacks, Tag 00DO. The quotient maps and both kernels are proved below.

For an exact sequence $M_2 \xrightarrow{u} M_1 \xrightarrow{v} M \to 0$ of modules over any commutative ring and every integer $n \geq 1$, there are exact sequences

$$M_2 \otimes_R \operatorname{Sym}^{n-1} M_1 \longrightarrow \operatorname{Sym}^n M_1 \longrightarrow \operatorname{Sym}^n M \longrightarrow 0,$$

$$M_2 \otimes_R \bigwedge^{n-1} M_1 \longrightarrow \bigwedge^{n} M_1 \longrightarrow \bigwedge^{n} M \longrightarrow 0.$$

The first maps multiply, respectively wedge, with $u(x)$; the second maps apply v to every factor. In degree zero both second maps are the identity of R, so both kernels are zero.

Proof. Put $N = \operatorname{im} u$, so exactness identifies $M = M_1/N$. Construct the tensor algebra $T(M_1) = \bigoplus_{d \geq 0} M_1^{\otimes d}$ with multiplication by concatenation. Balancing in each tensor factor makes that product well defined. Its symmetric quotient imposes $xy = yx$ for degree-one generators; its exterior quotient imposes $x^2 = 0$ for every degree-one generator. In the latter quotient, expanding $(x + y)^2 = 0$ gives $xy = -yx$ over every ring, including characteristic two.

In either algebra let L be the two-sided homogeneous ideal generated by the degree-one image of N. The factorwise map to the corresponding algebra of M_1/N is onto: every pure tensor in the target lifts, and pure tensors generate each degree. It kills L. Its inverse on the quotient is explicit. Send a class $\bar{x} \in M_1/N$ to the class of any lift x in the source algebra modulo L. Another lift differs by an element of N, hence gives the same class. This map is linear, so its products extend to the tensor algebra by the tensor balancing relations. Those products obey commutation, respectively square-zero, relations in the indicated quotient; it therefore descends to the symmetric, respectively exterior, algebra. The two composites fix every degree-one generator and the scalar ring, hence every product and sum of generators. They are inverse graded algebra maps.

It remains to identify the degree-n part of L. It is spanned by products having one selected factor in N, since those generate the ideal and all factors have nonnegative degree. In the symmetric case move that factor to the front by commutation. In the exterior case move it to the front using $xy = -yx$, retaining the resulting sign. Every remaining product has degree $n - 1$. Thus that part is exactly the image of $N \otimes_R \operatorname{Sym}^{n-1} M_1$, respectively $N \otimes_R \bigwedge^{n-1} M_1$, under multiplication. Each selected element of N lifts through u, so these are also the images of the stated maps from M_2. The inverse quotient maps just constructed prove that these images are exactly the kernels. The ideal has no degree-zero part, proving the separate degree-zero assertion. No module is assumed finite or free. □

Lemma F.72: principal localization NL

Reading source: AI Integrated Stacks, pinned Algebra, lines 37031–37116.

Let $A \to B$ be a ring map, $g \in B$, and $P \twoheadrightarrow B$ a polynomial presentation over A with kernel I. Choose $f \in P$ mapping to g. The map $P[X] \to B_g$ extending the presentation and sending X to g^{-1} has kernel $J = (I, fX - 1)$, and:

1. $J/J^2 \cong (I/I^2)_g \oplus B_g[fX - 1]$.
2. $\Omega_{P[X]/A} \otimes_{P[X]} B_g \cong (\Omega_{P/A} \otimes_P B_g) \oplus B_g\, dX$.
3. The two-term conormal complex is isomorphic to

$$\mathrm{NL}(P \to B) \otimes_B B_g \oplus (B_g \xrightarrow{g} B_g).$$

Consequently the canonical map $\mathrm{NL}_{B/A} \otimes_B B_g \to \mathrm{NL}_{B_g/A}$ is a homotopy equivalence.

Proof. The inverse-variable universal property gives

$$P[X]/(I, fX-1) = B[X]/(gX-1) \cong B_g,$$

including a nilpotent g, when the last ring is zero. Put $q = gX - 1 \in B[X]$. Multiplication by q is injective: if a polynomial has lowest nonzero coefficient at degree j, the product's coefficient at that degree is its negative. Thus the map

$$B[X]/(q) \longrightarrow (q)/(q^2), \quad [a] \longmapsto [aq]$$

is an isomorphism. It is surjective by the definition of the principal ideal, and $aq = bq^2$ implies $a = bq$ by the proved multiplication injection. Therefore $(q)/(q^2)$ is free of rank one over B_g, with the indicated generator, also when both modules are zero.

Reduction modulo I gives an exact sequence

$$(I/I^2) \otimes_B B_g \longrightarrow J/J^2 \longrightarrow (q)/(q^2) \longrightarrow 0.$$

For exactness in the middle, if an element of J becomes a multiple of q^2, lift its coefficient to $P[X]$ and subtract that coefficient times $(fX-1)^2$. The remainder lies in $IP[X]$. Its class modulo J^2 is a finite sum of classes of elements of I, with coefficients evaluated in B_g. This is exactly the image of the left map.

That map is injective and has a retraction. Define

$$\pi : P[X] \longrightarrow (P/I^2)_f, \qquad X \longmapsto f^{-1}.$$

It maps J into the square-zero ideal I_f/I_f^2 and maps J^2 to zero. The target ideal is naturally $(I/I^2) \otimes_B B_g$: the action factors through B_g, and localization commutes with the quotient and square by its exact fraction identities. Hence π induces a B_g-linear map from J/J^2 to this module. On the left map its composite is the identity, proving injectivity. It kills $[fX-1]$, whose image generates the right quotient. Lifting a right-quotient coefficient by $a[fX-1]$, for $a \in B_g$, gives the asserted direct sum and proves (1).

For (2), derivations out of $P[X]$ are uniquely a derivation out of P together with the chosen value of X. This proves the differential direct sum by the universal property, and tensoring yields the displayed formula.

Write $E = \Omega_{P/A} \otimes_P B_g$. On the first summand the conormal differential is the original $[a] \mapsto da$. On the new generator it is

$$d(fX-1) = g^{-1}df + g\,dX = g(dX + g^{-2}df).$$

Changing the new differential generator from dX to $w = dX + g^{-2}df$ is invertible, with inverse $dX = w - g^{-2}df$. It leaves E fixed. The two direct sums now identify the entire differential with the old conormal differential plus multiplication by g, proving (3).

The new summand is contractible: multiplication by g^{-1} from degree zero to degree one has both composites with its differential equal to the identity. Inclusion of the old summand and projection onto it therefore have composite identity there, and their other composite differs from identity by this explicit homotopy. They are homotopy inverses. The inclusion is precisely the map induced by $P \to P[X]$ and $B \to B_g$. The full presentation comparison F.11 identifies it with the stated canonical map. The argument uses neither a nonnilpotence assumption on g nor a proper regular-sequence condition. □

Lemma F.73: product local

Reading source: AI Integrated Stacks, pinned Algebra, lines 13035–13062.

If a ring R has finitely many maximal ideals and locally nilpotent Jacobson radical, every prime is maximal and the natural map to the product of the corresponding localizations is an isomorphism.

Proof. Write the distinct maximal ideals as $\mathfrak{m}_1, \dots, \mathfrak{m}_r$, and put $J = \bigcap_i \mathfrak{m}_i$. Each element of J is nilpotent. Any prime contains J, hence contains $\mathfrak{m}_1 \cdots \mathfrak{m}_r \subset J$. If it contained none of the factors, choosing one element outside the prime from each factor would give a product outside the prime, a contradiction. It therefore equals one of the maximal ideals.

The maximal ideals are pairwise comaximal. For two comaximal ideals A, B, choose $a \in A, b \in B$ with $a + b = 1$; the element $ub + va$ lifts prescribed residues u modulo A and v modulo B. Also $A \cap B = AB$, since $x = x(a + b) \in AB$ when $x \in A \cap B$. These identities inductively give the Chinese remainder isomorphism

$$R/J \cong \prod_{i=1}^{r} R/\mathfrak{m}_i.$$

For the induction, the intersection of preceding factors is comaximal with the next one: multiply elements in the preceding maximal ideals whose residues are one modulo the next ideal. Their product lies in the intersection and still has residue one.

We give the nil-ideal idempotent lift explicitly. If $e_0^2 - e_0 = \epsilon$ is nilpotent, then $u = 2e_0 - 1$ is a unit, since $u^2 = 1 + 4\epsilon$ has inverse given by a finite geometric series. Set $e_1 = e_0 - u^{-1}\epsilon$. Direct expansion gives

$$e_1^2 - e_1 = u^{-2}\epsilon^2.$$

The correction lies in the nil ideal. Repeat; the error squares at each step, so a zero power of the initial error makes the process terminate at an idempotent with the prescribed residue. Uniqueness across a nil ideal follows as well: if e, f are two lifts, the idempotents $e(1 - f)$ and $f(1 - e)$ lie in that ideal. A nilpotent idempotent is zero, so $e = ef = f$.

Lift the coordinate idempotents of R/J to $e_i \in R$. Their products for distinct indices are idempotents in J, hence zero. Their sum is then idempotent, so $1 - \sum_i e_i$ is an idempotent in J and is zero. Thus

$$R \longrightarrow \prod_i e_i R, \quad r \longmapsto (e_i r)_i$$

has inverse $(r_i)_i \mapsto \sum_i r_i$. Each factor, whose identity is e_i, has exactly one prime, corresponding to $\mathfrak{m}_i$, so it is local.

The factor is precisely $R_{\mathfrak{m}_i}$. For $s \notin \mathfrak{m}_i$, $e_i s$ avoids its unique maximal ideal and is a unit in $e_i R$; this gives a map $R_{\mathfrak{m}_i} \to e_i R$. Conversely e_i is an idempotent outside $\mathfrak{m}_i$, hence becomes one in that localization, whereas the other e_j become zero. The map $e_i R \to R_{\mathfrak{m}_i}$ is therefore unital, and the two fraction maps are inverse. This identifies the displayed product with the natural localization map. If there are no maximal ideals, the ring is zero by the maximal-proper-ideal Zorn argument, and the empty product is the zero ring. □

Lemma F.74: regular quasi regular

Reading source: AI Integrated Stacks, pinned Algebra, lines 17680–17766.

Let A be any ring, M any A-module, and $f_1, \dots, f_c$ an M-regular sequence. Put $J = (f_1, \dots, f_c)$. Then the natural graded map

$$(M/JM)[T_1, \dots, T_c] \longrightarrow \bigoplus_{n \geq 0} J^n M/J^{n+1}M, \qquad \bar{m}T^a \longmapsto [mf^a]$$

is an isomorphism. Thus a regular sequence is quasi-regular, both for rings and for arbitrary modules. In particular, for a regular sequence in A, J/J^2 is free over A/J on the classes of the f_i.

Proof. Every degree is generated by products of the f_i, so the map is surjective. Degree zero is the identity. We prove injectivity in every positive degree by induction on c, without finiteness or Noetherian assumptions on M.

For $c = 1$, a relation $mf_1^n \in f_1^{n+1}M$ reads $f_1^n(m - f_1 v) = 0$ for some v. Multiplication by f_1 is injective on M, hence so is its nth power; thus $m \in f_1 M$. For $c = 0$, only degree zero is present.

Suppose now $c > 1$ and the assertion holds for the first $c-1$ entries; write $J' = (f_1, \dots, f_{c-1})$ and $f' = (f_1, \dots, f_{c-1})$. A degree-$n$ relation modulo $J^{n+1}M$ can be made an exact zero relation by subtracting, from each coefficient, an element of JM: every element of $J^{n+1}M$ is a sum of degree-n products with coefficients in JM. It suffices therefore to handle

$$\sum_{e=0}^{\ell} \sum_{|a|=n-e} m_{a,e}(f')^a f_c^e = 0 \quad (0 \le \ell \le n).$$

Induct on its largest exponent ℓ. If $\ell = 0$, the induction on c applied in degree n gives all coefficients in $J'M \subset JM$. For $\ell \ge 1$, every lower-exponent term belongs to ${J'}^{n-\ell+1}M$. Consequently

$$\sum_{|a|=n-\ell} (f_c^\ell m_{a,\ell})(f')^a \in {J'}^{n-\ell+1}M.$$

The already proved degree-$n-\ell$ assertion for the shorter list gives $f_c^\ell m_{a,\ell} \in J'M$; in degree zero this is simply membership in $J'M$. Since f_c is injective on $M/J'M$, its powers are injective there, and hence $m_{a,\ell} \in J'M$.

Write each such coefficient as $\sum_{j<c} f_j b_{a,j}$. Replace its top term by

$$m_{a,\ell}(f')^a f_c^\ell = \sum_{j<c} (f_c b_{a,j})(f')^{a+\delta_j} f_c^{\ell-1}.$$

The new relation has largest exponent at most $\ell - 1$, and the only changed coefficients differ from the old ones by elements of $f_c M \subset JM$. The induction on ℓ puts all new coefficients in JM. It therefore puts all old lower coefficients in JM, and the old top coefficients were already in $J'M$. This proves the homogeneous injectivity for every n and completes both inductions.

Taking $M = A$ proves the ring assertion. Its degree-one isomorphism is exactly $(A/J)^c \to J/J^2$, with the stated basis. No permutation of the regular list, separatedness, or finite-module hypothesis has been used. □

Lemma F.75: regular sequence in polynomial ring

Reading source: AI Integrated Stacks, pinned Algebra, lines 17576–17624.

Let $f_1, \dots, f_r \in R$ generate a proper ideal in an arbitrary ring. These conditions are equivalent:

1. Every permutation of the list is regular.
2. Every subsequence in its original order is regular.
3. $f_1 X_1, \dots, f_r X_r$ is regular in $R[X_1, \dots, X_r]$.

Proof. A subsequence can be placed first in a permutation, so (1) implies (2). To prove the converse we first check the interchange of two elements. If a, b is regular and b is also a nonzerodivisor, then b, a is regular. Indeed, if $ax = by$, injectivity of b modulo (a) gives $y = az$. Injectivity of a then gives $x = bz$, proving injectivity of a modulo (b). The final ideal is the same proper ideal.

Induct on r to deduce (1) from (2). Any proper sublist inherits (2), so by induction all its permutations are regular. Given a permutation whose last entry is f_s, the preceding entries are therefore regular, and it remains to check that f_s is injective modulo all the others. If $s = r$, this is the last injection of the original list. If $s < r$, put

$$V = R/(f_1, \dots, \widehat{f_s}, \dots, f_{r-1}).$$

Permuting the first $r-1$ entries shows that f_s is a nonzerodivisor on V. The original-order subsequence omitting f_s shows that f_r is one too. The pair f_s, f_r is regular in V, since its second injection is the final injection of the original full list. The two-element interchange just proved makes f_r, f_s regular in V, giving the required last injection. All quotient ideals are proper because they are contained in the original proper ideal.

For the polynomial condition, the coefficient of each monomial is independent of the coefficients of the others. After imposing the first i equations, this gives the exact R-module decomposition

$$R[X]/(f_1X_1, \dots, f_iX_i) \cong \bigoplus_{e \in \mathbf{N}^r} (R/J_e)\, X^e, \qquad J_e = (f_j \mid j \le i,\ e_j > 0).$$

To verify it, a multiple of f_jX_j contributes to the coefficient of X^e precisely when $e_j > 0$, and then contributes a multiple of f_j. Conversely any such coefficient is obtained by multiplying f_jX_j by that monomial with its jth exponent reduced by one. These two calculations identify the entire ideal coefficient by coefficient.

Multiplication by $f_{i+1}X_{i+1}$ shifts each exponent to $e + \delta_{i+1}$ and multiplies its coefficient by f_{i+1}. The coefficient ideal J_e is unchanged by this shift, since only the first i indices occur in it. The multiplication is injective precisely when f_{i+1} is injective on every R/J_e.

Under (2), the ordered list with indices $\{j \le i : e_j > 0\}$ followed by $i+1$ is a regular subsequence, giving every such injection. This proves (3). Conversely, for any subset $U \subset \{1, \dots, i\}$ choose $e_j = 1$ for $j \in U$ and zero otherwise. The injection supplied by (3) says that f_{i+1} is injective on $R/(f_j : j \in U)$. Apply it successively along any ordered subsequence to prove all of (2). The polynomial equation ideal is proper since evaluation at all $X_j = 0$ factors its quotient onto the nonzero ring R. For the empty list all three conditions hold directly. □

Lemma F.76: relative global complete intersection

Reading source: AI Integrated Stacks, pinned Algebra, lines 38280–38292.

A relative global complete intersection over any commutative base ring is syntomic.

Proof. Write its presentation as $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$, with every nonempty residue-field fibre of dimension $n - c$, as in the relative complete-intersection definition. The finite generator and equation lists give finite presentation. The entire arbitrary-base proof F.77 gives flatness of S over R.

Each nonempty fibre has the displayed n polynomial variables, c equations and dimension $n - c$, so is a global complete intersection over its field. It is consequently locally a complete intersection: the whole five-condition proof F.57 applies at every fibre prime, or its global presentation itself is the required neighbourhood. Empty fibres have no point to check. These are the finite-presentation, flatness and fibre conditions for syntomicness. If $S = 0$, finite presentation is $R/(1)$, flatness is immediate, and all fibres are empty, giving the same conclusion. □

Lemma F.77: relative global complete intersection conormal

Reading source: AI Integrated Stacks, pinned Algebra, lines 38204–38278.

Let R be any ring and

$$S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$$

a relative global complete intersection: every nonempty residue-field fibre has dimension $n - c$. For a prime $\mathfrak{q} \subset S$, let $\mathfrak{Q} \subset R[X]$ be its inverse image. Then:

1. $f_1, \dots, f_c$ is regular in $R[X]_{\mathfrak{Q}}$.
2. Every $R[X]_{\mathfrak{Q}}/(f_1, \dots, f_i)$, including $i = 0$, is flat over R.

3. The global S-module $(f)/(f)^2$ is free on the classes of the f_i.

In particular S is flat over R. These assertions include bases that are not Noetherian.

Proof. First take a Noetherian base. Fix $\mathfrak{Q}$ and put $\mathfrak{p} = \mathfrak{Q} \cap R$, $k = \kappa(\mathfrak{p})$. Its closed polynomial fibre is a localization of $k[X]$ at the corresponding prime. In the nonempty quotient fibre, every minimal equation prime has height at most c, by the whole AG-CA-11, Theorem 3.1. The polynomial height formula AG-CA-09, Theorem 4.3, makes each component dimension at least $n - c$. Their maximum is the assumed dimension $n - c$, so all have this dimension. The point formula of AG-CA-09, Theorem 6.1, gives dimension $n - c$ at the chosen quotient point. Thus F.20 gives local dimension drop exactly c for this polynomial presentation. The full criterion F.57 makes its actual c-element equation list regular in the ambient fibre local ring.

The map $R_{\mathfrak{p}} \to R[X]_{\mathfrak{Q}}$ is flat and local: a polynomial ring is free, and localization preserves exact tensor injection tests. Both rings are Noetherian. Apply the complete successive-quotient lifting proof F.53. It proves both regularity in the polynomial local ring and flatness of every local successive quotient over $R_{\mathfrak{p}}$, hence over R.

For the general base we construct a finite Noetherian coefficient model. We first justify the field-dimension fact needed in this descent. For a nonzero finite-type k-algebra C, the entire AG-CA-09, Corollary 3.2 gives an inclusion $k[T_1, \dots, T_d] \subset C$ with C finite over it, even when C is not reduced or a domain. For any field extension E/k, a basis makes its tensor functor free and faithfully flat. Thus the inclusion remains an inclusion of $E[T]$ into $C \otimes_k E$, and finiteness remains true. The entire AG-CA-05, Theorem 5.1 and polynomial dimension AG-CA-09, Theorem 4.2, give

$$\dim C = d = \dim(C \otimes_k E).$$

The zero algebra stays zero. This proves precisely the field-extension dimension preservation used below.

Let $R_a \subset R$ be generated over the image of $\mathbf{Z}$ by all coefficients of the equations, and put $S_a = R_a[X]/(f)$. These are finite-type Noetherian models, and $S = R \otimes_{R_a} S_a$. At any actual $\mathfrak{q} \subset S$, let $\mathfrak{p}, \mathfrak{q}_a, \mathfrak{p}_a$ be its contractions. The residue fields give an inclusion $\kappa(\mathfrak{p}_a) \subset \kappa(\mathfrak{p})$, and the two corresponding fibres are related by that scalar tensor. The latter is nonempty, since the prime $\mathfrak{q}$ lies over $\mathfrak{p}$. Its dimension is $n - c$. The proved field-dimension fact therefore gives that dimension for the model fibre as well.

The whole arbitrary-base neighbourhood proof F.39 supplies $g_{\mathfrak{q}} \in S_a \setminus \mathfrak{q}_a$ whose localized model fibres all have dimension at most $n - c$. Indeed its bound is the dimension at the chosen fibre point, which is at most its total dimension $n - c$; a bound at every point bounds the total dimension by testing the finitely many component generic points. The resulting opens pull back to a cover of $\operatorname{Spec} S$. Choose finitely many, defined by $g_1, \dots, g_m$, by affine quasi-compactness: their images generate the unit ideal, because otherwise a maximal ideal would give an uncovered point.

Choose polynomial representatives $G_j \in R_a[X]$. A unit-ideal relation in S has a finite polynomial witness

$$1 = \sum_j H_j G_j + \sum_i V_i f_i \quad \text{in } R[X].$$

Adjoin to R_a all coefficients of the finitely many H_j, V_i, obtaining a finite-type integer subalgebra $R_0 \subset R$. Set $S_0 = R_0[X]/(f)$. The same identity now holds in $R_0[X]$, so the images of the G_j cover $\operatorname{Spec} S_0$.

For every base prime of R_0, its residue field extends that of its contraction in R_a. Each chart fibre is therefore the corresponding old chart fibre after a field extension. It retains dimension at most $n - c$ by the proved dimension fact. A finite open cover has total dimension the maximum of the chart dimensions here: in a finite-type field algebra each component's generic point belongs to some chart, and that nonempty principal component open retains its dimension. Thus every nonempty fibre of S_0/R_0 has dimension at most $n - c$. The height bound for an ideal generated by c elements

in a polynomial ring gives the reverse inequality at any minimal equation prime. Every nonempty model fibre consequently has dimension exactly $n - c$. This constructs the required Noetherian relative global complete-intersection model, with the identical equation list and

$$S = R \otimes_{R_0} S_0.$$

If $S = 0$, a finite identity $1 = \sum_i V_i f_i$ already gives a model with $S_0 = 0$. The local claims are then vacuous and both conormal modules are zero; hence we may continue with a chosen actual prime.

Contract $\mathfrak{Q}$ to $\mathfrak{Q}_0 \subset R_0[X]$ and $\mathfrak{p}_0 \subset R_0$. Put

$$A_0 = (R_0)_{\mathfrak{p}_0}, \qquad B_0 = (R_0[X])_{\mathfrak{Q}_0}, \qquad B_i = B_0/(f_1, \dots, f_i).$$

The Noetherian case gives regularity and A_0-flatness of each B_i, so for every $i < c$ there is an exact sequence

$$0 \longrightarrow B_i \xrightarrow{f_{i+1}} B_i \longrightarrow B_{i+1} \longrightarrow 0.$$

Tensor it with $R_{\mathfrak{p}}$ over A_0. It remains left exact because its quotient B_{i+1} is A_0-flat: the full Tor long exact sequence AG-RG-S01, Lemma 1.A gives zero for the possible first kernel. This is the exact reason regularity survives this possibly nonflat scalar change. Base change also makes each $B_i \otimes_{A_0} R_{\mathfrak{p}}$ flat over $R_{\mathfrak{p}}$, by the tensor associativity formula for injection tests.

The further prime localization of $B_0 \otimes_{A_0} R_{\mathfrak{p}}$ at the prime induced by $\mathfrak{Q}$ is exactly $R[X]_{\mathfrak{Q}}$. To check this, the tensor first inverts the images of $R_0[X] \setminus \mathfrak{Q}_0$, all of which avoid the actual prime. The further localization inverts the remaining polynomials outside $\mathfrak{Q}$. Every fraction in the full prime localization is then represented, and the two localization universal properties give inverse maps. Taking the same localizations of the quotient tensors gives exactly $R[X]_{\mathfrak{Q}}/(f_1, \dots, f_i)$. Exact localization preserves each displayed multiplication injection and its quotient. Its last quotient is $S_{\mathfrak{q}} \neq 0$, so the resulting list is regular. Localization preserves flatness of these target modules, and $R_{\mathfrak{p}}$ is flat over R. This proves (1) and (2) over the original arbitrary base.

Finally F.74 gives the conormal basis at every prime of S. The global map $S^c \to (f)/(f)^2$ is surjective by generation. Localization commutes with this ideal quotient; its kernel vanishes at every prime by the local bases. A nonzero module element would have a proper annihilator contained in a maximal ideal and would stay nonzero there, so the kernel is zero. This proves (3). For any injection of R-modules the tensor kernel with S likewise vanishes after localization at every $\mathfrak{q}$, by (2) for its final quotient. The same module detection proves that kernel zero. Thus S is flat over R, as asserted. Empty equation lists are included at every step. □

Lemma F.78: relative global complete intersection smooth

Reading source: AI Integrated Stacks, pinned Algebra, lines 39071–39113.

For a relative global complete-intersection presentation $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$ over any ring R, smoothness at a prime $\mathfrak{q} \subset S$ is equivalent to some $c \times c$ Jacobian minor having image outside $\mathfrak{q}$.

Proof. The whole unrestricted conormal proof F.77 identifies the conormal module with S^c, on the actual equation classes. In the polynomial differential basis the conormal map is

$$J : S^c \longrightarrow S^n, \qquad e_j \longmapsto \sum_i \frac{\partial f_j}{\partial X_i}\, dX_i.$$

Thus its maximal minors are precisely the stated polynomials.

If a selected minor Δ has image $g \notin \mathfrak{q}$, then S_g has the inverse-variable presentation

$$S_g = R[X_1, \dots, X_n, Z]/(f_1, \dots, f_c, \Delta Z - 1).$$

Use the selected c variable columns together with Z. Its square Jacobian block has the original selected block at the top, a zero top-right column, and Δ at the bottom right. Its determinant is Δ^2, a unit in this algebra. The entire arbitrary-base standard smooth lifting proof AG-CA-17, Theorem 4.1 therefore makes this finitely presented algebra smooth. This proves the sufficient direction with an actual principal neighbourhood.

Conversely suppose some S_h, with $h \notin \mathfrak{q}$, is smooth. Its polynomial inverse-variable presentation has a split conormal map by the complete split-sequence proof AG-CA-17, Theorem 3.1. The entire F.72 calculation identifies its conormal complex with the original localized complex

$$S_h^c \xrightarrow{J_h} S_h^n$$

plus a contractible rank-one complex, whose differential is multiplication by the unit h. In these explicitly changed bases, restrict a left inverse of the full differential to the original degree-zero summand, and project its values to the original degree-one summand. The composite is a left inverse of J_h. Reduction to $\kappa(\mathfrak{q})$ preserves this identity. Hence the resulting n-by-c matrix has rank c, and the full minor calculation F.70 supplies a nonzero c-minor over that field. Its original image is outside $\mathfrak{q}$, proving necessity.

For $c = 0$ the empty minor is one and the presentation is a polynomial algebra, smooth by arbitrary variable lifting. Thus this case obeys the same criterion. No Noetherian condition on the base or the target is used. □

Lemma F.79: section smooth

Reading source: AI Integrated Stacks, pinned Algebra, lines 39859–39914.

Let $R \to S$ be smooth and let $\sigma : S \to R$ be its section, with $I = \ker \sigma$. Then I/I^2 is finite projective over R. If it is free of rank d, the I-adic completion of S is isomorphic as an R-algebra to $R[[T_1, \dots, T_d]]$.

Proof. Define

$$D : S \longrightarrow I/I^2, \qquad D(s) = [s - \sigma(s)].$$

Give the target its S-module action through σ. Expanding a product and discarding the product of two elements of I proves the derivation identity. It is an R-derivation. The universal differential map therefore induces $\Omega_{S/R} \otimes_{S,\sigma} R \to I/I^2$. Its inverse sends $[i]$ to $di \otimes 1$. This is well-defined since $d(ij)$ vanishes after tensoring for $i, j \in I$; on every ds its composite is $d(s - \sigma(s)) = ds$, and on every ideal class its other composite is the identity. Thus the two maps are inverse.

The entire AG-CA-17, Theorem 3.1 gives finite projectivity of the differential module of this finitely presented formally smooth algebra. Tensor of its finite free direct-summand description with R retains that description, proving the first assertion. The section ideal is finite as well: if $x_1, \dots, x_m$ are finite algebra generators, the polynomial difference identity makes the $x_j - \sigma(x_j)$ generate its kernel.

Choose $t_1, \dots, t_d \in I$ representing a free conormal basis, and put $P = R[[T_1, \dots, T_d]]$, $J = (T_1, \dots, T_d)$. Continuous evaluation defines

$$b : P \longrightarrow \widehat{S} = \varprojlim_n S/I^n, \qquad T_i \longmapsto t_i.$$

This map is surjective. To see it, every element of I is an R-linear combination of the t_i modulo I^2. Multiplying these expressions shows that the degree-n monomials in the t_i generate I^n/I^{n+1} over R for every n. Start with the constant residue of a chosen element of $\widehat{S}$, which is in $S/I = R$. At step n, the error in its coordinate modulo I^{n+1} lies in I^n/I^{n+1}, so subtract a homogeneous degree-n polynomial in the t_i with coefficients in R. The resulting sum of homogeneous corrections is a formal series in P, with image the chosen compatible sequence. This proves surjectivity without a Noetherian completion theorem.

The square-zero section decomposition $S/I^2 = R \oplus I/I^2$ defines an R-map $u_2 : S \to P/J^2$, sending the chosen t_i to T_i. Formal smoothness lifts it successively along

$$P/J^{n+1} \longrightarrow P/J^n \quad (n \geq 2),$$

whose kernel J^n/J^{n+1} has square zero. Choose each lift to extend the preceding one. Their inverse limit is an R-map $u : S \to P$ taking I into J. Therefore the maps modulo J^n factor through S/I^n and give a continuous extension $a : \widehat{S} \to P$.

The composite $v = ab$ is an R-endomorphism of P with

$$v(T_i) = T_i + \text{terms of degree at least two.}$$

We prove that such a substitution is invertible. Having chosen inverse variable series through degree $n-1$, compute the degree-n error after substitution. Adding its negative as the next homogeneous term removes it, since the identity linear part changes that degree by exactly the added term; all nonlinear changes have higher degree. This recursively constructs a right inverse. The substitution is injective: any nonzero series has a lowest nonzero homogeneous part, and the identity linear part leaves that part unchanged. If w is the right inverse, then $v(wv) = v$; injectivity gives $wv = 1$, proving that it is also a left inverse. The coefficient calculations hold over arbitrary R, including zero divisors.

Replace a by $v^{-1}a$. Now $ab = 1$; hence b is injective, and it was already proved surjective. It is the desired R-algebra isomorphism. All maps are continuous for the displayed adic topologies. When $d = 0$, the homogeneous correction argument gives $\widehat{S} = R$ directly, and the same assertion holds. □

Lemma F.80: silly

Reading source: Stacks, Tag 00DS.

Let $J, I_1, \ldots, I_r$ be ideals of any commutative ring. Suppose $J \not\subset I_i$ for every i, and at most two of the I_i are not prime. There is an element of J outside all the I_i.

Proof. Remove repetitions and every listed ideal contained in another listed ideal. Avoiding the retained larger ideals also avoids the removed smaller ones. The remaining ideals are pairwise incomparable, and still at most two are not prime. We prove the assertion for this remaining list by induction on its length.

For the empty list choose $0 \in J$; there are no avoidance conditions. For one ideal the hypothesis supplies an element outside it. For two ideals choose $a \in J \setminus I_1$ and $b \in J \setminus I_2$. If either avoids both, it suffices. Otherwise $a \in I_2$, $b \in I_1$. Their sum belongs to neither: membership in I_1 would force $a = (a+b) - b \in I_1$, and membership in I_2 would similarly force $b \in I_2$.

For a list of length at least three, place a prime ideal last, calling it $\mathfrak{p} = I_r$. Induction gives $a \in J$ outside all earlier ideals. If $a \notin \mathfrak{p}$, it already suffices. Otherwise choose $b_0 \in J \setminus \mathfrak{p}$ and, for each $i < r$, choose $b_i \in I_i \setminus \mathfrak{p}$; incomparability supplies the latter choices. The product $b = b_0 b_1 \cdots b_{r-1}$ belongs to J and every earlier I_i. It lies outside $\mathfrak{p}$, because iterating the defining implication for a prime ideal shows that a product of elements outside it stays outside. Then $a + b \in J$ still avoids every earlier ideal by subtraction of b, and avoids $\mathfrak{p}$ by subtraction of a. This completes the induction without a Noetherian hypothesis. □

Lemma F.81: size extension pth roots

Reading source: AI Integrated Stacks, pinned Algebra, lines 45787–45812.

Let k have characteristic $p > 0$. If $da_1, \ldots, da_n$ are linearly independent in $\Omega_{k/\mathbf{F}_p}$, then

$$[k(a_1^{1/p}, \ldots, a_n^{1/p}) : k] = p^n.$$

Proof. Roots can be adjoined successively by quotienting by an irreducible factor of $X^p - a_i$. Such a factor exists by descent on degree, and its quotient is a field by the Euclidean algorithm. A pth root is unique in any field containing it: equal pth powers give $(x-y)^p = 0$, hence $x = y$. Thus these compatible adjunctions give the asserted root field.

We also prove the degree at a nontrivial step. If $\beta^p = a \in H$ and $\beta \notin H$, its minimal polynomial divides $X^p - a = (X - \beta)^p$ in a field containing β. A monic proper nonconstant factor has degree r with $1 \le r < p$ and coefficient $-r\beta$ at degree $r-1$. If it were over H, the nonzero scalar r would put β in H, a contradiction. Thus the minimal polynomial has degree p, and monic division gives the basis $1, \beta, \dots, \beta^{p-1}$.

Induct on the number of roots already adjoined. The zero-root extension is k, of degree one. Suppose the first $i-1$ roots have degree p^{i-1} and the standard monomials in them form a k-basis. If a_i were a pth power in that field, a root could be written in that basis. Raising it to the pth power would give

$$a_i = \sum_{0 \le e_j < p} \lambda_e^p \prod_{j<i} a_j^{e_j}, \qquad \lambda_e \in k.$$

Every pth power in k has zero differential, directly from the derivation product rule. Differentiating this finite equality therefore gives

$$da_i = \sum_e \sum_{\substack{j<i \\ e_j>0}} e_j \lambda_e^p \, a_j^{e_j-1} \left(\prod_{\substack{\ell<i \\ \ell \ne j}} a_\ell^{e_\ell} \right) da_j.$$

This is a linear dependence on the earlier differentials, contrary to the hypothesis. For $i = 1$ the right side is empty, giving the same contradiction. Hence the next root is not in the preceding field; the degree-p calculation applies. Its power basis multiplied by the old monomial basis gives the new monomial basis, proving both independence and generation. Degrees multiply along these finite vector-space towers, so the final degree is p^n. This also covers $n = 0$, and uses no converse criterion for zero absolute differentials. □

Lemma F.82: smooth over field

Reading source: AI Integrated Stacks, pinned Algebra, lines 38622–38676.

Every smooth finite-type algebra over any field is locally a complete intersection.

Proof. Let S/k be smooth and choose a prime $\mathfrak{q}$ of S. The full principal-chart proof AG-CA-17, Theorem 5.1 supplies a standard smooth neighbourhood $D = S_h$ containing it. Write its polynomial presentation as

$$D = k[Y_1, \dots, Y_N]/(F_1, \dots, F_r),$$

with an invertible r-column Jacobian minor. Let $\mathfrak{Q}$ be the polynomial prime over the chosen point and put $V = k[Y]_{\mathfrak{Q}}$, $\mathfrak{m} = \mathfrak{Q}V$.

The classes of the F_i in $\mathfrak{m}/\mathfrak{m}^2$ are independent. Indeed the differential of a product of two elements of $\mathfrak{m}$ is zero after tensoring with its residue field. A relation among these classes would consequently give the same residue-field relation among the vectors dF_i; derivatives of its lifted coefficients multiply F_i and vanish in that residue field. The chosen nonzero Jacobian minor makes these vectors independent, proving the assertion. This does not assume that the entire cotangent-to-differential map is injective.

The entire polynomial-local regularity proof AG-CA-14, Proposition 3.3 makes V regular. Extend the independent classes to a basis of its finite-dimensional cotangent space and lift the added classes to $\mathfrak{m}$. Nakayama makes this list a minimal generating list for $\mathfrak{m}$. The whole AG-CA-12, Theorem 6.1 proves that every such list is regular in this order. Its initial list $F_1, \dots, F_r$ is therefore regular and generates the defining ideal in V.

Condition (5) of the entire criterion F.57 now gives a principal global complete-intersection neighbourhood of this point in D. Its denominator is some a/h^e with $a \in S$, so the same neighbourhood in $\operatorname{Spec} S$ is $D(ha)$, with the identical iterated localization as coordinate algebra. Every point thus has a global complete-intersection principal neighbourhood. This is precisely local complete-intersectionness over k. The argument treats arbitrary primes and fields, including imperfect fields. For $S = 0$ its empty spectrum already has the required empty cover. □

Lemma F.83: smooth syntomic

Reading source: Stacks, Tag 00TA.

A smooth map $R \to S$ admits a cover of $\operatorname{Spec} S$ by principal opens $D(g)$ for which S_g is standard smooth over R. When R is Noetherian, the map is also syntomic.

Proof. The case $S = 0$ has an empty cover. The zero algebra has the finite presentation $R/(1)$, is flat as a zero module, and has no nonempty fibres, so it also satisfies the asserted syntomic condition. Hence consider a nonzero S.

Fix a finite presentation $P = R[x_1, \dots, x_n] \twoheadrightarrow S$ with kernel $I = (f_1, \dots, f_m)$, and put $M = I/I^2$. The split conormal theorem AG-CA-17, Theorem 3.1 puts M inside S^n as a direct summand. Its complement is finite, because projections of the coordinate vectors generate it. The splitting therefore presents M finitely; it also proves flatness by restricting tensor injections for the free module. S01, Lemma 1.2 makes M free over each local ring.

For a subset E of the equation indices, let U_E be the points where the classes indexed by E are a basis of M. These loci are open and cover the spectrum. Here are the denominator details. At a prime $\mathfrak{q}$, select a basis of the residue vector space from the spanning equation classes, with r selected members. Their map $\varphi : S^r \to M$ is surjective over $S_{\mathfrak{q}}$ by Nakayama. In a local free basis its matrix is square and invertible modulo the maximal ideal, hence invertible by the determinant and adjugate identities. The finite cokernel vanishes at $\mathfrak{q}$; multiplying denominators which kill its generators gives $a \notin \mathfrak{q}$ with φ_a surjective. Since M_a is projective, its kernel is a finite summand of S_a^r. It vanishes at $\mathfrak{q}$ too. Multiplying numerators of denominators which kill its generators gives $b \notin \mathfrak{q}$ with φ_{ab} an isomorphism. Thus $D(ab)$ lies in this basis locus. The same reasoning applies at every point of any fixed U_E, proving openness. Selecting a basis from a spanning vector-space list proves covering. For rank zero the selected list is empty and empty denominator products are 1.

Choose principal opens inside these loci. On one such open $D(g)$, adjoining a variable representing g^{-1} gives a finite polynomial presentation whose conormal module, by F.72, is $M_g \oplus S_g$. It is free. F.55 then supplies a polynomial presentation of S_g whose kernel is generated by a list which is itself a conormal basis. Denote this algebra temporarily by T, and write its presentation as

$$R[y_1, \dots, y_N] \twoheadrightarrow T, \qquad K = (a_1, \dots, a_c), \qquad K/K^2 = T^c.$$

The conormal differential has a left inverse. In these bases let its matrix be $A : T^c \to T^N$, and let $B : T^N \to T^c$ satisfy $BA = 1$. At a residue field this forces $c \le N$. The determinant expansion F.70(2) shows that its c-row minors Δ_E generate 1. Each selected square map is invertible precisely on $D(\Delta_E)$, by the adjugate formula, so these principal opens cover $\operatorname{Spec} T$. The empty minor for $c = 0$ is 1.

On a selected minor open, order its c pivot variables first and let $h \in R[y]$ represent the inverted element $u = \Delta_E \in T$. Present T_u by $a_1, \dots, a_c, hz - 1$. Its Jacobian on $y_1, \dots, y_c, z$ has diagonal blocks the selected Jacobian and u, and upper right block zero. Its determinant is $u\Delta_E$, a unit. This is a standard smooth presentation. Finally a principal open in S_g is principal in S: write its defining fraction as s/g^k, so its open corresponds to $D(gs)$. The charts just constructed are consequently the required principal cover of the original spectrum.

If R is Noetherian, F.85 makes each chart a relative global complete intersection. F.76 makes it syntomic, and F.62 glues that condition on the target. This proves the last assertion. □

Lemma F.84: snake

Reading source: Stacks, Tag 07JW.

Suppose the following diagram of abelian groups commutes and both rows are exact:

$$\begin{array}{ccccccccc} & & X & \xrightarrow{a} & Y & \xrightarrow{b} & Z & \longrightarrow & 0 \\ & & \alpha\downarrow & & \beta\downarrow & & \gamma\downarrow & & \\ 0 & \longrightarrow & U & \xrightarrow{i} & V & \xrightarrow{j} & W. & & \end{array}$$

There is a canonical exact sequence

$$\ker\alpha \xrightarrow{a} \ker\beta \xrightarrow{b} \ker\gamma \xrightarrow{\partial} \operatorname{coker}\alpha \xrightarrow{i} \operatorname{coker}\beta \xrightarrow{j} \operatorname{coker}\gamma.$$

Its first map is injective if a is injective, and its last map is surjective if j is surjective.

Proof. For $z \in \ker\gamma$, choose y with $b(y) = z$. Commutativity gives $j\beta(y) = 0$, so exactness supplies u with $i(u) = \beta(y)$. The injection i makes this u unique for the chosen y. Define $\partial z = [u]$ modulo $\alpha(X)$. Replacing y by another lift changes it by $a(x)$; commutativity and injectivity of i then change u by $\alpha(x)$. Thus the class is independent of the lift. Adding lifts proves additivity. The other arrows are the restrictions or quotient maps induced by a, b, i, j, which are well defined by the commutative squares.

For exactness at $\ker\beta$, take y there with $b(y) = 0$. Write $y = a(x)$. Since $i\alpha(x) = \beta a(x) = 0$ and i is injective, $x \in \ker\alpha$. Conversely the image of such an x lies in that kernel and has zero b-image.

At $\ker\gamma$, the condition $\partial z = 0$ says that the chosen u is $\alpha(x)$ for some x. Subtract $a(x)$ from the chosen y. This still maps to z, and its β-image is zero. Conversely a lift already in $\ker\beta$ allows $u = 0$, so has zero connecting class.

At $\operatorname{coker}\alpha$, a class $[u]$ dies in $\operatorname{coker}\beta$ exactly when $i(u) = \beta(y)$ for some y. Then $z = b(y)$ satisfies $\gamma(z) = j\beta(y) = ji(u) = 0$, and the defining lift gives $\partial z = [u]$. Every connecting class has zero image for this same reason.

At $\operatorname{coker}\beta$, a class $[v]$ dies in $\operatorname{coker}\gamma$ exactly when $j(v) = \gamma(z)$ for some z. Lift z to $y \in Y$. The element $v - \beta(y)$ has zero j-image, so is $i(u)$ by bottom-row exactness. Hence $[v]$ comes from $[u]$. Conversely $ji = 0$ kills every such image. These checks prove exactness at every intermediate term. Restriction of an injective a remains injective. If j is onto, lift any representative in W to V to obtain surjectivity on the last cokernel. □

Lemma F.85: standard smooth

Reading source: Stacks, Tag 00T7.

Let $P = R[x_1, \ldots, x_n]$, $I = (f_1, \ldots, f_c)$, and $S = P/I$ be standard smooth, with the first c variables chosen as pivots. Then:

1. $R \to S$ is smooth.
2. $\Omega_{S/R}$ has the free basis $dx_{c+1}, \ldots, dx_n$.
3. I/I^2 has the free basis $[f_1], \ldots, [f_c]$.
4. Every principal localization S_g is standard smooth over R.
5. For every base map $R \to R'$, the algebra $R' \otimes_R S$ is standard smooth over R'.
6. If $r \in R$ becomes a unit in S, then S is standard smooth over R_r.
7. This presentation makes S a relative global complete intersection over R.

Proof. Let $C = (\partial f_i/\partial x_j)_{1\le i,j\le c}$; its determinant is a unit in S. The map $S^c \to I/I^2$ taking the standard vectors to the equation classes is onto. Its composite with the conormal differential and

projection onto the first c differential coordinates is an invertible matrix (the transpose of C under column-vector conventions). Thus this surjection is injective as well. The equation classes are a basis, and the conormal differential is split injective.

Write its image in S^n as the columns of $\binom{C^{\mathsf{t}}}{D}$. An arbitrary vector (v, w) differs from a unique conormal vector by

$$(v, w) - \begin{pmatrix} C^{\mathsf{t}} \\ D \end{pmatrix} (C^{\mathsf{t}})^{-1} v = (0, w - D(C^{\mathsf{t}})^{-1} v).$$

The conormal image meets the last coordinate summand only in zero. Its cokernel is therefore exactly that summand, with the asserted differential basis. Finite presentation and this split conormal sequence give smoothness by the complete criterion AG-CA-17, Theorem 3.1 and Section 4. This proves (1)–(3), including $c = 0$.

To invert g, choose a polynomial h representing it and use the equations $f_1, \dots, f_c, hz - 1$. On pivot variables $x_1, \dots, x_c, z$, the Jacobian has blocks

$$\begin{pmatrix} C & 0 \\ z(\partial h/\partial x_1, \dots, \partial h/\partial x_c) & h \end{pmatrix}.$$

Its determinant $h \det C$ is a unit in S_g. This proves (4), even when the localization is zero. Under a base map, applying the map to every coefficient presents the tensor product; the same operation takes the determinant to its image, which remains invertible. This proves (5). For (6), the original equations over R_r present $S_r = S$, since the image of r is already invertible; their determinant is unchanged.

For (7), it remains to compute every nonempty fibre's dimension. By (5) such a fibre is a nonzero standard smooth algebra $B = k[x_1, \dots, x_n]/(\bar{f}_1, \dots, \bar{f}_c)$ over its residue field k. It is locally a complete intersection by F.82. At any prime $\mathfrak{q}$ of B, let $\mathfrak{q}'$ be its polynomial preimage. The conormal module is free of rank c by (3); reducing to the residue field determines that rank uniquely. F.57 identifies it with $\operatorname{height}(\mathfrak{q}') - \operatorname{height}(\mathfrak{q})$. For a minimal prime $\mathfrak{q}$ this gives $\operatorname{height}(\mathfrak{q}') = c$. The polynomial height formula AG-CA-09, Theorem 4.3 therefore gives dimension $n - c$ for every component.

The algebra B is Noetherian by S01, Lemma 1.3, so F.136 and F.137 give a finite component cover. Dimension equals the largest component dimension: a finite chain of irreducible closed subsets lies in a component containing its largest member, and every chain inside a component is also a chain in the whole space. Thus $\dim B = n - c$. This argument works over imperfect fields and for an empty equation list. Empty fibres require no dimension assertion, completing (7). □

Lemma F.86: sum two terms

Reading source: Stacks, Tag 00S3.

Let $A_1 \xrightarrow{d_A} A_0$ and $B_1 \xrightarrow{d_B} B_0$ be two-term complexes of modules over a ring. If they are homotopy equivalent, then

$$A_1 \oplus B_0 \cong B_1 \oplus A_0.$$

Proof. Choose chain maps $\varphi : A_\bullet \to B_\bullet$ and $\psi : B_\bullet \to A_\bullet$ and homotopies $h : A_0 \to A_1$, $h' : B_0 \to B_1$ with

$$1 - \psi_1\varphi_1 = h d_A, \qquad 1 - \psi_0\varphi_0 = d_A h,$$
$$1 - \varphi_1\psi_1 = h' d_B, \qquad 1 - \varphi_0\psi_0 = d_B h'.$$

Define $U : B_1 \oplus A_0 \to A_1 \oplus B_0$ and V in the reverse direction by

$$U = \begin{pmatrix} \psi_1 & h \\ -d_B & \varphi_0 \end{pmatrix}, \qquad V = \begin{pmatrix} \varphi_1 & -h' \\ d_A & \psi_0 \end{pmatrix}.$$

Multiplying the blocks and using the four homotopy identities and the two chain-map identities gives

$$UV = \begin{pmatrix} 1 & h\psi_0 - \psi_1 h' \\ 0 & 1 \end{pmatrix}, \qquad VU = \begin{pmatrix} 1 & \varphi_1 h - h'\varphi_0 \\ 0 & 1 \end{pmatrix}.$$

Each off-diagonal matrix squares to zero, so negating that off-diagonal block gives the inverse product. Consequently $L = (VU)^{-1}V$ is a left inverse of U, and $T = V(UV)^{-1}$ is a right inverse. The equality $L = L(UT) = (LU)T = T$ makes them the same inverse. Hence U is an isomorphism, proving the claim without any finite-generation or freeness hypothesis. □

Lemma F.87: syntomic

Reading source: Stacks, Tag 00SY.

Let R be Noetherian, and let $\mathfrak{q} \in \operatorname{Spec} S$ lie over $\mathfrak{p} \in \operatorname{Spec} R$. The following neighbourhood conditions are equivalent:

1. Some $g \notin \mathfrak{q}$ makes $R \to S_g$ syntomic.
2. Some $g \notin \mathfrak{q}$ makes S_g a relative global complete intersection over R.
3. Some $g \notin \mathfrak{q}$ makes S_g finitely presented over R, while $R_{\mathfrak{p}} \to S_{\mathfrak{q}}$ is flat and $S_{\mathfrak{q}}/\mathfrak{p}S_{\mathfrak{q}}$ is a complete-intersection local ring over $k = \kappa(\mathfrak{p})$ in the sense of F.3.

Proof. A syntomic neighbourhood is finitely presented and flat and has locally complete-intersection fibres; F.58 gives (3) from (1). F.76 gives (1) from (2). We prove the remaining implication with all rings localized explicitly.

Under (3), first replace S by the indicated finitely presented neighbourhood, retaining the prime. Write $P = R[x_1, \dots, x_n] \twoheadrightarrow S$ with kernel I and polynomial preimage $\mathfrak{q}'$. In the fibre $\bar{P} = k[x_1, \dots, x_n]$, write $\bar{I}$ for the image ideal and $\bar{\mathfrak{q}}'$ for the induced prime. By F.58 and F.57, the ideal $\bar{I}_{\bar{\mathfrak{q}}'}$ has a regular generating list of length c, its minimal number of generators. Choose $f_1, \dots, f_c \in I$ whose images give a basis of $\bar{I}_{\bar{\mathfrak{q}}'}/\bar{\mathfrak{q}}'\bar{I}_{\bar{\mathfrak{q}}'}$. This is possible by selecting a basis from the images of a finite generating list of I. Nakayama gives generation in the fibre local ring, and the last assertion of F.57 makes this very list regular there.

Set $T = P/(f_1, \dots, f_c)$ and $J = \ker(T \to S)$. The polynomial ring is Noetherian by S01, Lemma 1.3, so J is finite. Localize the short exact sequence at $\mathfrak{q}'$ as P-modules:

$$0 \longrightarrow J_{\mathfrak{q}'} \longrightarrow T_{\mathfrak{q}'} \longrightarrow S_{\mathfrak{q}} \longrightarrow 0.$$

These are $R_{\mathfrak{p}}$-modules. Flatness of the last module makes tensoring with k injective on the first arrow, by the tensor-kernel calculation S01, Lemma 1.2 and Lemma 1.A. The last arrow becomes an isomorphism after tensoring, because the chosen fibre equations generate the localized fibre ideal. Hence

$$J_{\mathfrak{q}'}/\mathfrak{p}J_{\mathfrak{q}'} = 0.$$

Over the local polynomial ring $P_{\mathfrak{q}'}$, the ideal $\mathfrak{p}P_{\mathfrak{q}'}$ is contained in its maximal ideal. Applying F.10 to the finite module $J_{\mathfrak{q}'}$ gives zero. Clear annihilating denominators for a finite generating list of J: some $u \in P \setminus \mathfrak{q}'$ gives $J_u = 0$, and thus $T_u = S_u$.

The fibre of T at the induced prime has the same local quotient by the chosen regular list. F.57 gives its point dimension $n - c$. Applying F.66(3) to this c-equation presentation produces a relative global complete-intersection principal neighbourhood of that point. Apply this to T_u, using its presentation with the additional equation $uz - 1$; the fibre point dimension remains $n - c$, and the variable and equation counts both increase by one. The resulting neighbourhood is therefore a principal localization of S_u with a relative global complete-intersection presentation. Writing its inverted element as a fraction in S_u makes the composite neighbourhood S_g for some $g \notin \mathfrak{q}$. This is (2). Empty regular lists use the same argument with $c = 0$. □

Lemma F.88: syntomic presentation ideal mod squares

Reading source: Stacks, Tag 07BT.

Let R be Noetherian, let $S = R[x_1, \dots, x_n]/I$ with I finitely generated, and suppose S_g is syntomic over R. Then $(I/I^2)_g$ is finite projective over S_g.

Proof. Put $A = S_g$ and $M = (I/I^2)_g$. If $A = 0$, then $M = 0$ and the assertion follows. Otherwise A is Noetherian by S01, Lemma 1.3 and localization, so the finite module M is finitely presented.

At every prime of A, F.87 supplies a principal relative global complete-intersection neighbourhood. F.77 makes the conormal of its displayed equations finite free. The presentation comparison F.33 identifies M there, up to finite free summands, with that conormal. Therefore M is finite projective there: the comparison realizes it as a summand of a finite free module. Choose finitely many of these neighbourhoods $D(a_i)$ covering $\operatorname{Spec} A$. The finite choice follows from the principal-open cover criterion: the ideal of all covering elements is not contained in any prime, hence contains 1, which is a finite sum.

Here is the global splitting argument. Choose a finite free surjection $\pi : A^N \to M$. Its localization on $D(a_i)$ has a right inverse s_i, because M_{a_i} is projective. A finite presentation $A^r \to A^s \to M \to 0$ realizes $\operatorname{Hom}_A(M, A^N)$ as the kernel of a map between finite matrix modules. Localization is exact and commutes with these finite modules, so

$$\operatorname{Hom}_A(M, A^N)_{a_i} = \operatorname{Hom}_{A_{a_i}}(M_{a_i}, A_{a_i}^N).$$

Consequently some global $t_i : M \to A^N$ represents $a_i^{k_i} s_i$. The map $\pi t_i - a_i^{k_i} 1_M$ becomes zero after inverting a_i. Multiplying by another power of a_i kills its values on all members of a finite generating list of M. Replacing t_i by that multiple gives

$$\pi t_i = a_i^{e_i} 1_M.$$

The powers $a_i^{e_i}$ still generate 1: if a proper ideal contained them, a containing prime would contain all the corresponding a_i, contradicting the cover. Choose b_i with $\sum_i b_i a_i^{e_i} = 1$. Then $\sum_i b_i t_i$ is a global right inverse of π. Thus M is a direct summand of A^N, which proves finite projectivity. All denominator and zero-module cases are included. □

Lemma F.89: total ring fractions no embedded points

Reading source: Stacks, Tag 02LX.

Suppose a ring R has finitely many minimal primes $\mathfrak{q}_1, \dots, \mathfrak{q}_t$, and its zerodivisors are exactly their union. Its total ring of fractions has the canonical decomposition

$$Q(R) \cong \prod_{i=1}^{t} R_{\mathfrak{q}_i}.$$

Proof. If $t = 0$, the assumed zerodivisor set is empty. A nonzero ring would make 0 a zerodivisor, since it annihilates $1 \neq 0$, so $R = 0$. Both its localization and the empty ring product are zero. Now assume $t > 0$. Let D be the multiplicative set of nonzerodivisors and put $A = D^{-1}R$. Every element of D avoids every $\mathfrak{q}_i$, so each map $R \to R_{\mathfrak{q}_i}$ extends to A. The prime correspondence for localization identifies the primes of A with primes of R disjoint from D. Such a prime is contained in $\bigcup_i \mathfrak{q}_i$. By F.80 it is contained in one $\mathfrak{q}_i$, and minimality forces equality. Thus the primes of A are precisely the extensions $\mathfrak{m}_i = D^{-1}\mathfrak{q}_i$. They are distinct and incomparable, hence all maximal.

We construct the product decomposition, including possible nilpotents. For each i and $j \neq i$, choose an element in $\mathfrak{m}_j \setminus \mathfrak{m}_i$, and multiply these choices to obtain a_i. Then a_i avoids $\mathfrak{m}_i$ and belongs to every other prime. For $t = 1$ use the empty product $a_1 = 1$. Each product $a_i a_j$ for $i \neq j$ belongs to

every prime, and hence is nilpotent by AG-CA-01, Theorem 1.2. Choose a common positive integer N with $a_i^N a_j^N = 0$ for every pair. The elements a_i^N generate the unit ideal, since no prime contains all of them. Write $1 = \sum_i b_i a_i^N$ and set $e_i = b_i a_i^N$. The e_i are orthogonal and sum to 1; multiplying this sum by e_i gives $e_i^2 = e_i$. Modulo $\mathfrak{m}_i$, only the ith summand survives, so e_i is 1 there and is zero at every other prime.

The maps $a \mapsto (e_i a)_i$ and $(u_i)_i \mapsto \sum_i u_i$ are inverse ring maps between A and $\prod_i e_i A$. Each factor has only the prime induced by $\mathfrak{m}_i$, and is therefore local: a nonunit lies in a maximal ideal, which must be that prime. The projection $A \to e_i A$ consequently inverts every element outside $\mathfrak{m}_i$. Conversely, in $A_{\mathfrak{m}_i}$ the idempotent e_i is a unit and hence equals 1, so the complementary factor vanishes. These universal properties give $e_i A \cong A_{\mathfrak{m}_i} \cong R_{\mathfrak{q}_i}$. The isomorphism is the canonical map stated above. □

Lemma F.90: when colimit

Reading source: Stacks, Tag 07C3.

For an arbitrary ring map $R \to \Lambda$, the following are equivalent: Λ is a filtered colimit of smooth R-algebras; every R-algebra map from a finitely presented algebra A to Λ factors through a smooth R-algebra.

Proof. First suppose such a filtered presentation of Λ is given. Write A with finitely many polynomial generators and finitely many defining equations. The images of its generators have representatives at a common stage. Each defining equation becomes zero at a later stage, by the equality criterion in a filtered colimit. A common successor of those finitely many stages therefore satisfies every equation. The generator images define the required map to that smooth stage.

For the converse, form the category whose objects are smooth R-algebras B with a map $B \to \Lambda$, and whose arrows commute with those maps. Smooth algebras are finitely presented. This category can be taken small: finite polynomial presentations form a set, and a map from each one to Λ is specified by a finite tuple of elements of Λ; use these presentations as representatives. The object $R \to \Lambda$ makes it nonempty.

Given two objects B_1, B_2, the tensor product $B_1 \otimes_R B_2$ is finitely presented: combine their finite generator lists and equations. Factor its induced map to Λ through a smooth algebra. The two structure maps into that algebra give a common successor in the category. For parallel arrows $u, v \colon B \rightrightarrows C$, choose finite algebra generators $b_1, \ldots, b_r$ of B. The quotient

$$C/(u(b_1) - v(b_1), \ldots, u(b_r) - v(b_r))$$

is finitely presented by adding these finitely many equations to a presentation of C. Its map to Λ factors through a smooth algebra by the hypothesis. The resulting arrow from C equalizes u and v, because equality on the generators gives equality on every polynomial in them. These two constructions prove filteredness.

Let L be this category's colimit and $L \to \Lambda$ its induced map. It is surjective: for any $\lambda \in \Lambda$, apply the factoring hypothesis to $R[T] \to \Lambda$ with $T \mapsto \lambda$; the intermediate object supplies a representative of λ in L. For injectivity, a representative $b \in B$ with zero image in Λ gives a map $B/(b) \to \Lambda$. This quotient is finitely presented, so its factorization through a smooth object yields a successor at which $b = 0$. Hence its colimit class is zero. Two representatives can be moved to a common stage and this argument applied to their difference, proving equality detection as well. Thus $L \to \Lambda$ is an isomorphism of R-algebras, establishing the filtered presentation without any Noetherian hypothesis. □

Proposition F.91: characterize separable field extensions

Reading source: AI Integrated Stacks, pinned Algebra, lines 46037–46069.

For every field extension K/k, the following are equivalent: separability in F.5; geometric reducedness over k; algebraic formal smoothness over k; $H_1(L_{K/k}) = 0$; and injectivity of

$$\iota : K \otimes_k \Omega_{k/P} \longrightarrow \Omega_{K/P},$$

where P is the prime field. In characteristic $p > 0$ they are also equivalent to the condition that an absolute p-basis of k remains p-independent over K^p. In characteristic zero all the conditions hold. For finitely generated extensions they agree with the existence of a separating transcendence basis.

Proof. We give the comparisons in an order that does not assume the last finitely generated assertion.

First, every field is formally smooth over its prime field. In characteristic zero, choose a transcendence basis over $\mathbf{Q}$; the remaining algebraic extension is separable, because a nonconstant irreducible polynomial has nonzero derivative of smaller degree and hence is relatively prime to it. The two entire lifting constructions in F.49, together with the composition lifting proof AG-CA-17, Theorem 1.2, give formal smoothness, for arbitrary bases and algebraic extensions.

In characteristic p, take an absolute p-basis B of a field H by the whole F.119. Given a prime-field map $\psi : H \to D/J$, $J^2 = 0$, define a lift on H^p by

$$\theta(a^p) = \widetilde{\psi(a)}^{\,p}.$$

The root a is unique in H, and changing its chosen lift by $j \in J$ does not change this power, since $j^p = 0$. Frobenius of sums and products proves that θ is a unital ring map. Choose arbitrary lifts of the $\psi(b)$ for $b \in B$. Their pth powers are $\theta(b^p)$; the exact presentation

$$H = H^p[T_b : b \in B]/(T_b^p - b^p)$$

proved in F.119 therefore extends them to a lift on H. This establishes prime-field formal smoothness in both characteristics.

Suppose ι is injective. In a square-zero lifting problem for K/k, obtain a P-lift $u : K \to D$ by the preceding argument. Write $v : k \to D$ for the prescribed structural map. The difference $u|_k - v$ takes values in J and is a P-derivation for the common action modulo J, by expanding a product and using $J^2 = 0$. The given map $K \to D/J$ makes J a K-vector space. Its derivation corresponds to a linear map from $K \otimes_k \Omega_{k/P}$ to J; extend it along the injection ι to $\Omega_{K/P}$ by extending a vector-space basis. The resulting derivation $\delta : K \to J$ gives the corrected lift $u - \delta$. Indeed

$$(u(x) - \delta(x))(u(y) - \delta(y)) = u(xy) - u(x)\delta(y) - u(y)\delta(x) = u(xy) - \delta(xy).$$

It preserves one and agrees with v on k, proving formal smoothness.

Conversely, any P-derivation $d : k \to V$, for a K-vector space V, makes $K \oplus V$ a k-algebra by $a \mapsto (a, d(a))$; its multiplication is $(x, v)(y, w) = (xy, xw + yv)$. Formal smoothness lifts the identity through the square-zero quotient to K. The second component of that lift is a derivation extending d. Thus every linear functional on the domain of ι extends to its codomain. If a nonzero vector lay in its kernel, extend that vector to a basis and take a functional equal to one on it; such a functional could not extend. Hence ι is injective. The whole F.17 proves precisely the equivalence with the displayed first cotangent homology condition, in arbitrary field scope.

In characteristic p, F.119 identifies the differentials of an absolute p-basis B of k with a basis of $\Omega_{k/P}$. Applied over P to K, its other equivalence says their images are independent exactly when the standard monomials in B are independent over K^p. This proves the equivalence with ι's injectivity.

We next compare this condition with reducedness. A finite root algebra

$$K \otimes_k k(b^{1/p} : b \in B_0) = K[T_b : b \in B_0]/(T_b^p - b), \qquad B_0 \subset B \text{ finite},$$

is free on the standard monomials. The corresponding root-field monomials are independent over K exactly when the monomials in B_0 are independent over K^p: raise a proposed relation to its pth power, and take roots of its coefficients for the converse. If independent, the algebra maps isomorphically to the root field. If dependent, that map has a nonzero kernel. Every element of the root algebra has pth power in K; a kernel element therefore has pth power zero in the algebra itself. It is a nonzero nilpotent. The full algebra $K \otimes_k k^{1/p}$ is the union of these algebras with injective transitions, since a full p-basis expansion expresses every root using finitely many basis roots. Reducedness of this union is equivalent to reducedness of every finite stage. Thus geometric reducedness implies the p-basis condition.

For the converse we must test all field extensions. The independence persists through all root heights. More explicitly, a relation

$$\sum_a c_a^{p^r} B^a = 0, \qquad 0 \le a_b < p^r,$$

can be grouped by $a = pu + v$, $0 \le v_b < p$. The coefficient of B^v is

$$\left(\sum_u c_{pu+v}^{p^{r-1}} B^u \right)^p.$$

Independence over K^p makes each coefficient zero. Frobenius is injective in a field; induction on r then kills every c_a. Iterating the unique finite p-basis expansions in k shows that the monomials with exponents below p^r are a basis of k over k^{p^r}. Taking roots gives the standard presentation and basis of every finite subset root extension of height r. The preceding relation calculation makes all its base changes to K fields; their directed union $K \otimes_k k^{1/p^r}$ is a field too.

Every finite purely inseparable extension k'/k embeds over k in some k^{1/p^r}: choose finitely many generators and a common root height. Tensoring its inclusion with the vector space K is injective. Hence $K \otimes_k k'$ is a finite-dimensional domain over K, and is a field by the multiplication-inverse argument.

Now take a finitely generated field extension E/k. In characteristic p, the entire F.69 supplies finite purely inseparable k'/k, an extension E'/E containing k', and a separating transcendence basis for E'/k'. Put $H = K \otimes_k k'$, a field as just proved. In characteristic zero use $k' = k$, $H = K$ and $E' = E$: a maximal algebraically independent subset of finite field generators leaves a finite algebraic, necessarily separable, extension.

In either case, for a separating basis T, the ring

$$C = H \otimes_{k'} k'(T) = \left(k'[T] \setminus \{0\}\right)^{-1} H[T]$$

is a domain. The algebra $C \otimes_{k'(T)} E'$ is finite free over C, by a vector-space basis of the finite separable field extension $E'/k'(T)$. It therefore injects into its localization over $\mathrm{Frac}(C)$. That generic algebra is reduced: express the finite separable extension by a finite tower of simple generators. At each step a monic minimal polynomial is relatively prime to its derivative, and its Bézout identity remains valid after any field extension. Its irreducible factors are consequently distinct. The two-factor Bézout Chinese remainder formula and its finite induction, proved in F.73, identify its quotient with a product of fields. Repeat on every field factor at the next step of the tower. The final tensor algebra is a finite product of fields, hence reduced. Its subalgebra $C \otimes E' = H \otimes_{k'} E' = K \otimes_k E'$ is reduced as well.

The inclusion $E \subset E'$ remains injective after tensoring over k with K, since field-vector-space inclusions split by extending bases. Thus $K \otimes_k E$ is reduced. An arbitrary field extension of k is the directed union of its finitely generated subfields, with injective tensor transitions. A nilpotent and its finite power relation occur at one stage, so the union is reduced. This proves geometric reducedness from the p-basis condition; it also proves it for all extensions in characteristic zero.

It remains to connect formal smoothness with the definition of separability. For a finitely generated formally smooth field L/k, choose a finite-type domain $B = k[a_1, \dots, a_n] \subset L$ with fraction field L, and a polynomial presentation $Q = k[X_1, \dots, X_n] \to B$ with prime kernel I. Hilbert's basis theorem AG-CA-03, Theorem 2.1, makes I finite. The entire localization comparison F.65, together with F.17 for L/k, makes

$$(I/I^2) \otimes_B L \longrightarrow L^n$$

injective: exact field-of-fractions localization identifies this map's kernel with the localized first cotangent homology. Choose $f_1, \dots, f_c \in I$ whose conormal classes form a basis of this finite-dimensional space. Their differentials are independent, giving a c-column Jacobian determinant $\Delta \notin I$.

The local ring Q_I has maximal ideal IQ_I and residue field L. Its cotangent space is exactly the conormal space just used. The whole F.10 makes the f_i generate IQ_I. The finite module $I/(f_1, \dots, f_c)$ then vanishes there. Kill its finite generator list by a product $h_0 \notin I$ of annihilating denominators. For $h = h_0\Delta$ and its image $g \in B$ this gives

$$B_g = k[X_1, \dots, X_n, Z]/(f_1, \dots, f_c, hZ - 1).$$

The square Jacobian block using the selected columns and Z has determinant $h\Delta$, a unit. The full AG-CA-17, Theorem 4.1, makes this a standard smooth domain. Its fraction field is L, and the entire domain calculation F.64 gives a separating transcendence basis. This includes $c = 0$, with empty minor one. Conversely such a basis gives a purely transcendental extension followed by a finite separable one; F.49 and the full formal composition proof make L/k formally smooth.

If K/k is separable by F.5, the whole F.49(3) gives formal smoothness, hence the conditions already compared. Conversely, in characteristic p the p-basis condition descends to every finitely generated intermediate L: coefficients in L^p are also in K^p. The differential and lifting comparisons make L/k formally smooth, and the preceding finite construction makes it separably generated. This is exactly F.5's definition for K/k. In characteristic zero the finite transcendence-basis argument above gives that definition for every intermediate field. Finally F.49 gives formal smoothness there, hence differential injectivity and the homology condition. All comparisons and the finite-generation assertion are now proved. □

Definition F.92: G ring

Reading source: Stacks, Tag 07GH.

The term *G-ring* imposes two requirements on R: it is Noetherian, and at every prime $\mathfrak{p}$ its localization-to-completion map is regular. The map in this second requirement is

$$R_{\mathfrak{p}} \longrightarrow \widehat{R_{\mathfrak{p}}}, \qquad \widehat{R_{\mathfrak{p}}} = \varprojlim_{n \geq 1} R_{\mathfrak{p}}/(\mathfrak{p}R_{\mathfrak{p}})^n.$$

Thus the completion used in the definition is taken at the maximal ideal of the local ring, separately for each prime.

Definition F.93: excellent

Reading source: Stacks, Tag 07QT.

For a Noetherian ring R, *quasi-excellence* means that R satisfies the G-ring condition of F.92 and the J-2 condition. Here J-2 requires the regular locus

$$\{\mathfrak{q} \in \operatorname{Spec} B : B_{\mathfrak{q}} \text{ is regular}\}$$

to be open for each finite-type R-algebra B; this is the terminology of Stacks, Tag 07P7.

An *excellent* ring has these properties and is universally catenary. To specify that last condition, a ring is catenary when, between each fixed pair of comparable primes, the lengths of finite strict chains have a common finite bound and the chains into which no further prime can be inserted all have equal length. A Noetherian ring is universally catenary when every finite-type algebra over it has this property. These two definitions are Stacks, Tag 00NI and Tag 00NL.

Definition F.94: local complete intersection

Reading source: Stacks, Tag 07D0.

For a map $A \to B$ of finite type, choose a surjection

$$P = A[T_1, \dots, T_n] \twoheadrightarrow B, \qquad I = \ker(P \to B).$$

The map is a *local complete intersection* when I is a Koszul-regular ideal, with that ideal condition understood exactly as in F.95. Checking one such presentation suffices: the full arbitrary-ring proof F.94.A below shows that every other finite polynomial presentation then has a Koszul-regular kernel. Neither this definition nor that comparison assumes A Noetherian.

Lemma F.94.A: arbitrary-ring presentation independence

Reading source: AI Integrated Stacks, pinned More Algebra, lines 8869–8942; Stacks, Tag 07CZ.

Let A be an arbitrary ring and suppose an algebra C has finite polynomial presentations

$$P = A[T_1, \dots, T_n] \twoheadrightarrow C = P/I, \qquad Q = A[X_1, \dots, X_m] \twoheadrightarrow C = Q/J.$$

If I is a Koszul-regular ideal as defined in F.95, then J is also Koszul-regular. Moreover C is finitely presented. This proves the presentation-independence assertion in F.94 without Noetherian hypotheses.

Proof. We first supply the elementary complex calculations used to compare the two presentations. For a list $f = (f_1, \dots, f_c)$ over a ring V, take a free module with basis $e_1, \dots, e_c$ and form

$$K_j(f; V) = \bigwedge^j V^c, \qquad d(e_{i_1} \wedge \cdots \wedge e_{i_j}) = \sum_{a=1}^{j} (-1)^{a-1} f_{i_a} e_{i_1} \wedge \cdots \widehat{e_{i_a}} \cdots \wedge e_{i_j}.$$

Deleting a pair of indices in the two orders gives opposite signs, so $d^2 = 0$. Its degree-zero homology is $V/(f)$. Appending u identifies the new degree-j term with $K_j(f) \oplus K_{j-1}(f)$, with differential

$$d(a, b) = (d_f a + ub, -d_f b);$$

the sign identification multiplies the wedge ending in the added basis vector by $(-1)^{j-1}$.

If $K(f)$ has zero positive homology and u acts injectively on $V/(f)$, this cone has zero positive homology too. Indeed a degree-j cycle satisfies $d_f b = 0$ and $d_f a = -ub$. For $j > 1$, write $b = d_f c$ using the old acyclicity. For $j = 1$, the last equation makes $u[b] = 0$ in $H_0(K(f))$; injectivity likewise gives $b = d_f c$. Subtract the boundary of $(0, -c)$ to obtain $(a + uc, 0)$, an old positive-degree cycle. Its old bounding chain makes this remaining vector a boundary in the cone. Degree zero is the additional quotient by u. This proves the appending assertion, including empty initial lists.

A second calculation will detect conormal ranks. If $H_1(K(f)) = 0$, the classes of the f_i give a basis of $(f)/(f)^2$ over $V/(f)$. They generate by definition. Given $\sum_i b_i f_i \in (f)^2$, write the right side as $\sum_{i,j} a_{ij} f_i f_j$ and replace b_i by $b_i - \sum_j a_{ij} f_j$. Their scalar product is now exactly zero, so the resulting vector is a degree-one cycle. A degree-two Koszul boundary has every coordinate in (f). Vanishing of H_1 therefore puts all the corrected coefficients, and hence every b_i, in (f). This proves independence.

We next prove that the kernels are finite. The principal charts in the Koszul-regularity condition cover $V(I)$. If their denominators are $a_\lambda \in P$, then $I + (a_\lambda) = P$: a proper ideal is contained in a maximal ideal, which would give an uncovered prime containing I. Thus finitely many denominators and an element $b \in I$ satisfy

$$1 = b + \sum_{\ell=1}^{s} c_\ell a_\ell.$$

Choose numerators in I of every finite local generator list for I_{a_ℓ}, and let I_0 be generated by those numerators and b. At a prime where some a_ℓ is invertible, I_0 and I have the same localization. At any other prime, the identity makes b a unit, so both localize to the unit ideal. Therefore (I/I_0) vanishes at every prime, and is zero. Here is local detection for this possibly infinite module: a nonzero element has a proper annihilator contained in a maximal ideal; if it became zero there, an element outside that ideal would annihilate it, a contradiction. Hence $I = I_0$ is finite. This makes C finitely presented, and the entire F.44 then makes J finite. If $C = 0$, both kernels are unit ideals and their required prime tests are empty. We now assume $C \neq 0$ and fix $\mathfrak{q} \in \operatorname{Spec} C$.

Choose $p_j(T) \in P$ representing each image of X_j and $q_i(X) \in Q$ representing each image of T_i. In $W = A[T, X]$ the kernel L of the common map to C has two exact descriptions:

$$L = (I, u_1, \dots, u_m) = (J, s_1, \dots, s_n), \qquad u_j = X_j - p_j(T), \quad s_i = T_i - q_i(X).$$

Quotienting by the added equations eliminates those variables and recovers the specified maps to C, which proves both descriptions. Let $\mathfrak{r} \subset W$ be the inverse image of $\mathfrak{q}$, and $\mathfrak{p}_P, \mathfrak{p}_Q$ its contractions. Translation is a polynomial-ring isomorphism $W = Q[s_1, \dots, s_n]$. Constant and linear coefficients in these variables give an exact decomposition

$$L/L^2 = (J/J^2) \oplus C^n.$$

The constant part is taken modulo J^2, the linear coefficients modulo J, and all terms with at least two s factors disappear. Sending these data to their constant-plus-linear representatives and reading them back gives inverse maps, so the decomposition is valid over arbitrary rings.

Pick $a \in P \setminus \mathfrak{p}_P$ and a Koszul-regular list f generating I_a. Polynomial extension is free and localization is flat, by the entire tensor/localization proofs AG-RG-S01, Lemmas 1.0–1.1; exact flat tensor preserves each cycle and boundary quotient of this complex. Hence f remains Koszul-regular over W_a. In the quotient by f, the u_j are successive polynomial variables after translation. Multiplication by each is injective by comparison of coefficients, and remains so after localization. Applying the cone calculation one entry at a time proves that (f, u) is Koszul-regular over W_a. Its conormal classes consequently form a basis of L/L^2 at $\mathfrak{r}$.

The decomposition above now makes $(J/J^2)_{\mathfrak{q}}$ a finite direct summand of a finite free $C_{\mathfrak{q}}$-module. Such a summand is flat by restricting the free tensor injection test. It is finitely presented: the projection onto its complementary summand sends a finite free basis to finitely many generators of the kernel of the projection onto the original summand. The whole AG-RG-S01, Lemma 1.2, therefore makes it finite free over this local ring.

From a finite generating list of J choose $g_1, \dots, g_d$ whose conormal residue classes form a basis. Relative to a basis of the finite free conormal module, their square coefficient matrix is invertible modulo the maximal ideal, hence has unit determinant and is invertible by the cofactor formula. They thus form a basis over $C_{\mathfrak{q}}$. In $Q_{\mathfrak{p}_Q}$ this says $J = (g) + J^2$. The finite module $J/(g)$ equals J times itself; J lies in the maximal ideal. The entire Nakayama proof in AG-RG-S01, Lemma 1.2, gives $J_{\mathfrak{p}_Q} = (g)$. Since J is finite, kill the finitely many quotient generators by denominators outside $\mathfrak{p}_Q$. Their product $v_0 \notin \mathfrak{p}_Q$ gives $J_{v_0} = (g)$ on an actual principal chart.

In $W_{\mathfrak{r}}$ both (f, u) and (g, s) generate $L_{\mathfrak{r}}$ and their conormal classes are bases. Their lengths agree by reduction to the residue field. Expressing the second list in the first gives a square matrix M whose reduction modulo $\mathfrak{r}$ is invertible: $L^2 \subset \mathfrak{r}L$, so these are precisely the two bases of $L/\mathfrak{r}L$. The

determinant is therefore a unit. The exterior powers of the map on free generator modules give an isomorphism of the Koszul complexes. Indeed that map preserves their degree-one scalar-product functional, so the alternating deletion formula checks commutation with d on every wedge; the exterior powers of the inverse matrix give the inverse complex map. Permuting (g, s) to (s, g) is the same calculation with a permutation matrix.

All of this can be realized on a principal neighbourhood. Include a and v_0 in a denominator product. Include representatives outside $\mathfrak{r}$ of every denominator in the finite matrix and both finite generator comparisons, and the numerator of its determinant. For each asserted equality in the stalk, also include a denominator outside $\mathfrak{r}$ that kills its cross-difference. Their product $h \notin \mathfrak{r}$ gives the actual generator equalities over W_h and an invertible comparison matrix there. This clearing uses annihilating denominators and never cancels a zero divisor. The complex on (s, g) over W_h is consequently isomorphic to the acyclic-positive complex on (f, u).

Evaluate h by $T_i = q_i(X)$, and call the resulting element of Q v. Its image in C agrees with that of h, so $v \notin \mathfrak{p}_Q$. Elimination and localization give

$$W_h/(s) = Q_v, \qquad J_v = (g).$$

The second equality holds because v_0 was a factor of h, and hence of v.

Finally $K(s; W_h)$ is a free resolution of Q_v. Before localization this follows by appending the successive polynomial variables s_i with the cone calculation above; exact localization preserves it. Form its double complex with the finite free complex $K(g; W_h)$. The total complex is $K(s, g; W_h)$, with differential $d_s + (-1)^i d_g$ at s-degree i. Augmentation maps it to $K(g; Q_v)$ and induces an isomorphism on homology. To verify this without a spectral sequence, augment each vertical column by its quotient in degree -1. Every such column is exact because the fixed g term is free. For a total cycle, its component of greatest g degree is a vertical cycle: no component of greater g degree contributes to its differential. Lift it in that exact column and subtract the resulting total boundary. This removes that component and changes only smaller g degrees. The procedure terminates, because there are only finitely many g degrees. It proves the augmented total complex acyclic, equivalently both surjectivity and injectivity of the augmentation on homology by lifting cycles and boundaries.

The source total complex has zero positive homology. Thus $K(g; Q_v)$ does too, and its entries generate J_v. We have produced the required Koszul-regular principal neighbourhood of the specified prime of C. Repeating for every prime proves the claim for J. The equations, complexes and finite eliminations include all empty variable and generator lists. □

Definition F.95: regular ideal

Reading sources: AI Integrated Stacks, pinned More Algebra, lines 8710–8731; Stacks, Tag 061P.

Let R be any ring. For a finite list $f = (f_1, \dots, f_c)$ put $J = (f_1, \dots, f_c)$.

1. The list is **regular** when $J \neq R$ and multiplication by f_i is injective on $R/(f_1, \dots, f_{i-1})$ for every i.
2. It is **Koszul-regular** when every positive-degree homology module of the exterior-power Koszul complex $K(f; R)$ is zero.
3. It is **H_1-regular** when $H_1(K(f; R)) = 0$.
4. It is **quasi-regular** when the graded R/J-algebra map

$$(R/J)[Z_1, \dots, Z_c] \longrightarrow \mathrm{gr}_J(R) = \bigoplus_{d \geq 0} J^d/J^{d+1}, \quad Z_1^{e_1} \cdots Z_c^{e_c} \longmapsto f_1^{e_1} \cdots f_c^{e_c} \bmod J^{1+\sum_i e_i}$$

is an isomorphism.

For an ideal $I \subset R$, each of these four conditions defines the corresponding **regular**, **Koszul-regular**, H_1**-regular**, or **quasi-regular ideal** by the following local requirement: for every prime $\mathfrak{p}$ containing I, there are $g \in R \setminus \mathfrak{p}$ and a finite list in R_g generating I_g and satisfying that list condition. The denominator and list may depend on the prime. This specifies all four ideal definitions over arbitrary rings.

Verification of the formulas. The complex in (2) and (3) has term $\bigwedge^d R^c$ and differential

$$d(e_{i_1} \wedge \cdots \wedge e_{i_d}) = \sum_{a=1}^{d} (-1)^{a-1} f_{i_a} e_{i_1} \wedge \cdots \widehat{e_{i_a}} \cdots \wedge e_{i_d}.$$

Each pair of deleted indices contributes twice to d^2, with opposite signs, so this is a complex in every characteristic. Its degree-zero homology is R/J, since the degree-one differential has image J. The whole complex calculations in F.94.A also apply to the empty list.

The map in (4) is well defined on coefficients modulo J: changing a coefficient by an element of J changes a degree-d value by an element of J^{d+1}. Multiplication of representatives respects the displayed graded products, so it is a graded algebra map. It is surjective in every degree because J^d is generated by the degree-d products of the f_i. If it is an isomorphism, degree one gives precisely

$$(R/J)^c \xrightarrow{\sim} J/J^2, \qquad (a_i) \longmapsto \sum_i a_i f_i \bmod J^2.$$

Thus quasi-regularity includes the conormal basis used later.

For the empty list $J = 0$, the complex consists only of R in degree zero and the graded map is the identity of R in degree zero. It supplies all four local tests for the zero ideal at its primes. For $I = R$, no prime contains it and the ideal conditions are vacuous. These conventions handle both the zero ring and empty charts without a Noetherian hypothesis. □

Lemma F.96: G ring goes up quasi finite

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12823–12870.

Let $R \to R'$ be finite type between Noetherian rings. Let $\mathfrak{q}' \subset \mathfrak{p}'$ be primes of R' contracting to $\mathfrak{q} \subset \mathfrak{p}$ in R, and assume the map is quasi-finite at $\mathfrak{p}'$.

1. If $\widehat{R_{\mathfrak{p}}} \otimes_R \kappa(\mathfrak{q})$ is geometrically regular over $\kappa(\mathfrak{q})$, then $\widehat{R'_{\mathfrak{p}'}} \otimes_{R'} \kappa(\mathfrak{q}')$ is geometrically regular over $\kappa(\mathfrak{q}')$.
2. If all formal fibres of $R_{\mathfrak{p}}$ are geometrically regular, the same holds for $R'_{\mathfrak{p}'}$.
3. A quasi-finite algebra over a Noetherian G-ring is a G-ring.

Proof. Put $A = \widehat{R_{\mathfrak{p}}}$, $B = \widehat{R'_{\mathfrak{p}'}}$, $k = \kappa(\mathfrak{q})$ and $E = \kappa(\mathfrak{q}')$. The injection $R/\mathfrak{q} \to R'/\mathfrak{q}'$ gives a field map $k \to E$: nonzero elements remain nonzero, so invert in the target fraction field. The specified completion-factor construction F.26 supplies an isomorphism of R'-algebras

$$T = A \otimes_R R' \cong B \times D,$$

whose projection onto B is the natural completed-local map. No finiteness assertion about the unneeded second factor is required.

The ring A is Noetherian by AG-CA-19, Theorem 3.3, and T is a finite-type A-algebra, hence Noetherian by the earlier Hilbert basis theorem. Tensor associativity identifies

$$T \otimes_{R'} E \cong (A \otimes_R k) \otimes_k E.$$

The left ring is Noetherian: writing $U = R' \setminus \mathfrak{q}'$, it equals $U^{-1}(T/\mathfrak{q}'T)$. The hypothesis makes $A \otimes_R k$ geometrically regular, so the field-extension assertion F.52, with this target Noetherianness checked, makes the displayed ring geometrically regular over E.

The product splitting gives

$$T \otimes_{R'} E \cong (B \otimes_{R'} E) \times (D \otimes_{R'} E).$$

For any finite purely inseparable extension E'/E, the tensor of this product with E' is regular. Its first factor is localization at the idempotent $(1, 0)$: inverting that idempotent kills exactly the second factor. Its prime local rings are therefore prime local rings of the regular product, so it is regular. That first factor is also Noetherian, as a quotient of the Noetherian product. The same quotient argument makes $B \otimes_{R'} E$ Noetherian. The finite purely inseparable criterion F.50 proves (1), including zero factors, which have no prime local rings.

For (2), every prime of $R'_{\mathfrak{p}'}$ comes from a $\mathfrak{q}' \subset \mathfrak{p}'$ and contracts to a $\mathfrak{q} \subset \mathfrak{p}$. Its fibre is exactly the ring in (1), by localization and tensor associativity. Thus (1) proves all the required geometric-regularity assertions. The completion map is flat by AG-CA-19, Theorem 3.2, so these fibres make it regular. For (3), quasi-finiteness applies at every $\mathfrak{p}'$, and the G-ring hypothesis gives the assumption of (2) at its contracted $\mathfrak{p}$. Definition F.92 then gives a regular completion map at every prime of R', exactly the G-ring condition. □

Lemma F.97: another helper G ring

Reading source: AI Integrated Stacks, pinned More Algebra, lines 13103–13159.

Let R be a complete Noetherian local domain with fraction field K. For every prime $\mathfrak{q} \subset R[X]$, the ring $\widehat{R[X]_{\mathfrak{q}}} \otimes_R K$ is regular. Moreover, if $\mathfrak{q}$ lies over the maximal ideal of R and $0 \neq \mathfrak{r} \subset \mathfrak{q}$ contracts to zero in R, then

$$\widehat{R[X]_{\mathfrak{q}}} \otimes_{R[X]} \kappa(\mathfrak{r})$$

is geometrically regular over $\kappa(\mathfrak{r})$. The second assertion includes the maximal-ideal, positive-characteristic case of the reading source.

Proof. Choose the finite regular complete local subring $A_0 \subset R$ from F.25, and put $F = \mathrm{Frac}(A_0)$. A finite-dimensional domain over F is a field: multiplication by a nonzero element is an injective endomorphism of a finite-dimensional vector space, hence surjective and has an inverse element. Therefore $R \otimes_{A_0} F$ is a field containing R and equals K. In particular K/F is finite.

Contract $\mathfrak{q}$ to $\mathfrak{q}_0 \subset A_0[X]$. Applying F.27 to the finite map $A_0[X] \to R[X]$ gives

$$\widehat{A_0[X]_{\mathfrak{q}_0}} \otimes_{A_0[X]} R[X] \cong \prod_i \widehat{R[X]_{\mathfrak{q}_i}},$$

where the finitely many $\mathfrak{q}_i$ are all primes above $\mathfrak{q}_0$, including $\mathfrak{q}$. Tensoring over R with K identifies the left side with

$$(\widehat{A_0[X]_{\mathfrak{q}_0}} \otimes_{A_0} F) \otimes_F K.$$

The ring in parentheses is geometrically regular by the first assertion of F.110. Its tensor with K is Noetherian, since K/F is finite, and is regular by F.52. On the right it is the product of the rings $\widehat{R[X]_{\mathfrak{q}_i}} \otimes_R K$. Each factor is localization of that product at its factor idempotent, so its local rings are regular and it is Noetherian. This proves the first assertion for every prime, without a formal-smoothness criterion for regularity.

For the second assertion set $E = \kappa(\mathfrak{r})$. The prime $\mathfrak{r}$ remains nonzero in $K[X]$: choose a nonzero polynomial in it, and localization is injective because $R[X]$ is a domain. Its extension is a proper prime, by the localization correspondence, and hence is generated by an irreducible polynomial of positive degree in the Euclidean ring $K[X]$. Thus $E = K[X]/\mathfrak{r}K[X]$ is finite over K. The equality with the residue field follows because this quotient is already a field containing the image of $R[X]/\mathfrak{r}$ and all its fractions.

Fix a finite purely inseparable extension L/E. It is finite over K. Choose finitely many algebraic field generators of L/K and multiply each by a nonzero element of R so that its monic equation has

coefficients in R: for a degree-d equation, the coefficient of degree j is multiplied by the $(d-j)$th power of that scalar. The resulting integral generators define a finite domain $B \subset L$, $R \subset B$, with fraction field L. The ring B is complete local, as follows directly from F.27 applied to the complete ring R at its maximal ideal. Its left side is B, so

$$B \cong \prod_j \widehat{B_{\mathfrak{n}_j}}.$$

Integral maximal-ideal correspondence makes the $\mathfrak{n}_j$ exactly its maximal ideals. Each factor is nonzero, since its residue-field quotient is nonzero. A domain has only the idempotents zero and one: $e(1-e)=0$ proves this. Hence there is exactly one factor. The resulting unique maximal ideal makes $B = B_{\mathfrak{n}}$, and the natural isomorphism identifies it with its maximal-adic completion. Finiteness over Noetherian R also makes B Noetherian.

Let $a \in L$ be the image of X through $R[X] \to E \to L$, and let $\mathfrak{r}' = \ker(B[X] \to L)$ for the map evaluating X at a. The quotient image is a domain containing B, so its fraction field contains $\mathrm{Frac}(B) = L$ and equals L. Thus $\kappa(\mathfrak{r}') = L$, and $\mathfrak{r}'$ contracts to $\mathfrak{r}$ in $R[X]$. Its extension to $L[X]$ is exactly $(X-a)$.

Put $H = \widehat{R[X]_{\mathfrak{q}}}$. The finite map $R[X] \to B[X]$ and F.27 identify the tested extension with

$$\begin{aligned}(H \otimes_{R[X]} E) \otimes_E L &= H \otimes_{R[X]} L \\ &\cong \prod_i (\widehat{B[X]_{\mathfrak{q}_i}} \otimes_{B[X]} L),\end{aligned}$$

where $\mathfrak{q}_i$ run over the primes above $\mathfrak{q}$. In each factor tensor first with $L = \mathrm{Frac}(B)$ over B and then impose the evaluation relation in $L[X]$. Its ring is precisely

$$(\widehat{B[X]_{\mathfrak{q}_i}} \otimes_B L)/(X-a).$$

The parent ring is Noetherian regular by the already proved first assertion, applied to the complete local domain B and the prime $\mathfrak{q}_i$; no maximality assumption on this prime is needed. Polynomial differentiation $\partial/\partial X$ kills B. The three successive extensions through prime localization, completion and localization by $B \setminus \{0\}$ are furnished by F.104. Its fraction formula shows that it kills L as well, and it sends $X-a$ to one. The full hypersurface argument F.122 therefore makes this quotient regular. A zero quotient is included. A finite product of regular Noetherian rings is regular: each prime contains all but one factor idempotent and its local ring is the corresponding factor local ring. The tested fibre is consequently regular for every L/E. It is Noetherian as a localized quotient of H before the finite field extension. Criterion F.50 proves the claimed geometric regularity. □

Lemma F.98: cartier equality

Reading source: Stacks, Tag 07E1.

For a finitely generated field extension K/k, the spaces of differentials and first conormal homology are finite-dimensional, and

$$\dim_K \Omega_{K/k} - \dim_K H_1(L_{K/k}) = \mathrm{trdeg}_k K.$$

Proof. Put $d = \mathrm{trdeg}_k K$. Apply the explicit monic-tower construction in F.23 to a finite field-generating list for K/k. It gives a domain $C \subset K$ containing that list and having fraction field K. In that construction there are d transcendence variables, r algebraic tower variables and one inverse variable. Consequently its polynomial presentation has

$$N = d + r + 1 \quad \text{variables}, \qquad c = r + 1 \quad \text{equations}.$$

The first equation is $hZ-1$ for the chosen nonzero denominator polynomial h. The remaining equations, after imposing this first equation, are the successive monic minimal polynomials in the tower. Multiplication by $hZ-1$ is injective: the lowest nonzero coefficient of a polynomial in its product with $hZ-1$ is the negative of that coefficient. Each subsequent monic equation is injective on its polynomial ring by its highest-coefficient calculation. Thus this ordered list is regular, exactly as in the construction of F.23.

Write $C = k[X_1, \dots, X_N]/I$. F.74 identifies I/I^2 with C^c on the equation classes. Polynomial differentials identify the other term with C^N on dX_i. Presentation comparison F.11 and localization F.65, applied at all nonzero elements of the domain C, therefore identify the first conormal homology and the differential module of K/k with the kernel and cokernel of a linear map

$$D : K^c \longrightarrow K^N.$$

Localization is exact by S01, Lemma 1.1, so it preserves the kernel and cokernel used in this comparison.

If ρ is the rank of D, choosing bases for its image and extending them to bases gives

$$\dim_K \ker D = c - \rho, \qquad \dim_K \operatorname{coker} D = N - \rho.$$

Their difference is $N - c = d$, and both numbers are finite. This proves the asserted equality without a separability hypothesis. Equivalently the parameter count is $\dim C = d = \operatorname{trdeg}_k K$, as proved for finite-type domains in AG-CA-09, Theorem 4.2. Empty transcendence or tower lists obey the same count, including $K = k$. □

Lemma F.99: check G ring easy

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12803–12821.

A Noetherian ring R is a G-ring exactly when, for every pair of primes $\mathfrak{q} \subset \mathfrak{p}$, the algebra

$$\widehat{(R/\mathfrak{q})_{\mathfrak{p}}} \otimes_{R/\mathfrak{q}} \kappa(\mathfrak{q})$$

is geometrically regular over $\kappa(\mathfrak{q})$. The completion is at the maximal ideal of the displayed local quotient.

Proof. Fix $\mathfrak{p}$ and put $A = R_{\mathfrak{p}}$, $\mathfrak{m} = \mathfrak{p}A$. Primes of A correspond exactly to primes $\mathfrak{q} \subset \mathfrak{p}$ of R, by extension and contraction in localization. Their residue fields are the same $\kappa(\mathfrak{q}) = \operatorname{Frac}(R/\mathfrak{q})$: tensoring with that field already inverts every denominator outside $\mathfrak{p}$.

Since A is Noetherian, both $\mathfrak{q}A$ and $A/\mathfrak{q}A$ are finite A-modules. The whole completion exactness and tensor theorem AG-CA-19, Theorem 3.1 applied to

$$0 \longrightarrow \mathfrak{q}A \longrightarrow A \longrightarrow A/\mathfrak{q}A \longrightarrow 0$$

gives the natural ring identification

$$\widehat{A}/\mathfrak{q}\widehat{A} \;\cong\; \widehat{A/\mathfrak{q}A} = \widehat{(R/\mathfrak{q})_{\mathfrak{p}}}.$$

The completion topology on the quotient is the correct maximal-ideal topology: its degree-r quotient is $A/(\mathfrak{m}^r + \mathfrak{q}A)$. The kernel image is $\mathfrak{q}\widehat{A}$, since the tensor map from the finite ideal sends its generators and coefficients to their products in $\widehat{A}$.

Because $\mathfrak{q}$ annihilates $\kappa(\mathfrak{q})$, the defining balanced tensor relations now give

$$\widehat{A} \otimes_A \kappa(\mathfrak{q}) \cong \widehat{A} \otimes_R \kappa(\mathfrak{q}) \cong (\widehat{A}/\mathfrak{q}\widehat{A}) \otimes_{R/\mathfrak{q}} \kappa(\mathfrak{q}).$$

Explicitly the last map sends $\widehat{a} \otimes \lambda$ to $(\widehat{a} \bmod \mathfrak{q}\widehat{A}) \otimes \lambda$. Its inverse chooses a representative; changing the representative adds products by elements of $\mathfrak{q}$, whose tensors vanish. The first identity holds

because the inverted denominators act invertibly in both factors. These maps preserve the specified residue-field algebra structures.

The entire AG-CA-19, Theorem 3.2, makes $A \to \widehat{A}$ flat. Thus it is regular precisely when all its residue-field fibres are geometrically regular, by the regular-map definition. The identities above identify those fibres with exactly the algebras in the assertion. By F.92, being a G-ring means that this regularity holds at every $\mathfrak{p}$. Applying the equivalence at each prime proves both directions. If $R = 0$, there are no primes and the two conditions are vacuous. □

Lemma F.99A: completion of a flat local map

Reading sources: AG-CA-19, Proposition 2.2 and Theorems 3.2–3.3; AG-CA-08, Theorems 4.2 and 2.1.

If $(A, \mathfrak{m}) \to (B, \mathfrak{n})$ is a flat local map of Noetherian local rings, its induced map on maximal-ideal completions $\widehat{A} \to \widehat{B}$ is faithfully flat. The residue fields may differ.

Proof. Since $\mathfrak{m}^r$ maps into $\mathfrak{n}^r$, the finite-quotient maps $A/\mathfrak{m}^r \to B/\mathfrak{n}^r$ commute with their transitions. Their inverse limit defines the asserted map.

The entire completion proofs AG-CA-19, Proposition 2.2 and Theorem 3.3, make the completions Noetherian and complete, with

$$\widehat{A}/\mathfrak{m}\widehat{A} = A/\mathfrak{m} = \kappa, \qquad \widehat{B}/\mathfrak{n}\widehat{B} = B/\mathfrak{n}.$$

These are local rings with the indicated extended maximal ideals. Explicitly, an element whose residue is nonzero has an inverse modulo the extended ideal. Its product with a chosen inverse is $1 - j$; the series $\sum_{r \geq 0} j^r$ converges in the complete ring and inverts this product. Thus every element outside the extended maximal ideal is a unit. The completion map sends $\mathfrak{m}\widehat{A}$ into $\mathfrak{n}\widehat{B}$, so it is local.

Choose a free A-resolution $F_\bullet \to \kappa$ by the whole construction AG-RG-S01, Lemma 1.A. The full AG-CA-19, Theorem 3.2, proves $A \to \widehat{A}$ flat, so $F_\bullet \otimes_A \widehat{A}$ is an exact free $\widehat{A}$-resolution of $\kappa \otimes_A \widehat{A} = \kappa$. The same theorem makes $\widehat{B}$ flat over B, and the given flatness of B/A then makes it flat over A by tensor associativity. Consequently tensoring the original resolution with $\widehat{B}$ is exact in positive degrees, and

$$\operatorname{Tor}_1^{\widehat{A}}(\kappa, \widehat{B}) = H_1\big((F_\bullet \otimes_A \widehat{A}) \otimes_{\widehat{A}} \widehat{B}\big) = H_1(F_\bullet \otimes_A \widehat{B}) = 0.$$

Apply the complete local criterion AG-CA-08, Theorem 4.2, to the actual local map of Noetherian rings $\widehat{A} \to \widehat{B}$ and the module $\widehat{B}$. This module is finite over the stipulated target $\widehat{B}$, which is the required finiteness condition; no finiteness over $\widehat{A}$ is being assumed. The vanishing just proved gives flatness.

Its closed fibre is nonzero: $\mathfrak{m}\widehat{B} \subset \mathfrak{n}\widehat{B}$, so $\widehat{B}/\mathfrak{m}\widehat{B}$ surjects onto the nonzero field $B/\mathfrak{n}$. The entire local faithful-flatness proof AG-CA-08, Theorem 2.1, therefore makes the map faithfully flat. This proves the full assertion without identifying the two residue fields. □

Lemma F.100: check G ring maximal ideals

Reading source: AI Integrated Stacks, pinned More Algebra, lines 13001–13037.

A Noetherian ring R is a G-ring if and only if $R_\mathfrak{m}$ has geometrically regular formal fibres for every maximal ideal $\mathfrak{m}$.

Proof. If R is a G-ring, definition F.92 gives the stated property at every maximal ideal. Conversely assume those properties, and fix any prime $\mathfrak{p} \subset R$. Choose a maximal ideal $\mathfrak{m} \supset \mathfrak{p}$. Put $C = \widehat{R_\mathfrak{m}}$. The local completion map $R_\mathfrak{m} \to C$ is faithfully flat by AG-CA-19, Theorem 3.2. Hence its fibre at

$\mathfrak{p}R_{\mathfrak{m}}$ is nonzero. Choose a prime of that fibre and take its preimage $\mathfrak{p}' \subset C$; the residue-field map is injective, so its contraction is exactly $\mathfrak{p}R_{\mathfrak{m}}$. This is the prime lifting supplied by faithful flatness.

Set $B = C_{\mathfrak{p}'}$ and $\widehat{B} = \widehat{C_{\mathfrak{p}'}}$. The map $R_{\mathfrak{p}} \to B$ is local and flat: it is induced by the localized completion map, and its closed prime contraction is $\mathfrak{p}R_{\mathfrak{p}}$. A local map sends every power of the source maximal ideal into the corresponding power of the target maximal ideal. Taking inverse limits therefore constructs the map $\widehat{R_{\mathfrak{p}}} \to \widehat{B}$. Naturality of reduction makes both squares in the actual diagram commute:

$$\begin{array}{ccccc} C & \longrightarrow & B & \longrightarrow & \widehat{B} \\ \uparrow & & \uparrow & & \uparrow \\ R_{\mathfrak{m}} & \longrightarrow & R_{\mathfrak{p}} & \longrightarrow & \widehat{R_{\mathfrak{p}}}. \end{array}$$

All rings here are Noetherian. By assumption and completion flatness, the first vertical map $R_{\mathfrak{m}} \to C$ is regular. The map $C \to B$ is regular: localization is flat; at a prime of C its fibre is its residue field if the multiplicative set avoids that prime, and the zero ring otherwise. Every extension of that residue field leaves a field, hence a regular ring. Also C is complete Noetherian local, so F.132 makes it a G-ring and gives regularity of $B \to \widehat{B}$. Applying F.124 twice proves $R_{\mathfrak{m}} \to \widehat{B}$ regular.

This composite factors through $R_{\mathfrak{p}}$. It makes $\widehat{B}$ flat over $R_{\mathfrak{p}}$: tensoring any injection of $R_{\mathfrak{p}}$-modules with $\widehat{B}$ is the same as tensoring the underlying $R_{\mathfrak{m}}$-modules, by the localization universal property. Its fibres at primes of $R_{\mathfrak{p}}$ are the corresponding fibres over $R_{\mathfrak{m}}$, since their residue fields are identical. Thus $R_{\mathfrak{p}} \to \widehat{B}$ is regular as well.

The proved flat local map $R_{\mathfrak{p}} \to B$ satisfies precisely the hypotheses of F.99A, which makes the induced map $\widehat{R_{\mathfrak{p}}} \to \widehat{B}$ faithfully flat. The factorization in the right square and F.125 therefore descend regularity of $R_{\mathfrak{p}} \to \widehat{B}$ to regularity of $R_{\mathfrak{p}} \to \widehat{R_{\mathfrak{p}}}$. As $\mathfrak{p}$ was arbitrary, F.92 says R is a G-ring. For the zero ring both sets of tests are empty, so the equivalence still holds. □

Lemma F.101: conormal sequence H1 regular ideal

Reading source: Stacks, Tag 07CX.

For ideals $I \subset J$ of an arbitrary ring A, suppose the ideal J/I of A/I is H_1-regular. Then

$$I \cap J^2 = IJ.$$

Proof. The inclusion $IJ \subset I \cap J^2$ follows from $I \subset J$. We prove the other inclusion at each prime of A. Exact localization commutes with the intersection, the product and the quotient in this assertion; the fraction proof and the detection of a zero module are S01, Lemma 1.1. These facts will turn the local equalities into the global one.

At a prime outside $V(J)$, the ideal J becomes the unit ideal, and both sides become I. At a prime containing J, localize A there. The defining neighbourhood in F.95 supplies a finite list $g_1, \dots, g_s \in J$ whose images generate J/I and have zero first Koszul homology over A/I. Indeed we can lift the list from its quotient neighbourhood and localize further at the prime; exact localization preserves cycles, boundaries and their homology. In this local ring $J = I + (g_1, \dots, g_s)$.

Take $x \in I \cap J^2$. Expanding a finite expression for x as a sum of products of members of J gives

$$x = u + \sum_{i=1}^{s} g_i b_i, \qquad u \in IJ, \quad b_i \in J.$$

Terms having a factor in I were absorbed in u; all other terms have two factors from the displayed list. Reducing modulo I makes $(\bar{b}_i)$ a degree-one Koszul cycle because its scalar product with $(\bar{g}_i)$ is zero. The zero first homology says this cycle is a boundary. With the convention

$$\partial(e_i \wedge e_j) = \bar{g}_i e_j - \bar{g}_j e_i,$$

it is a finite sum of these displayed vectors with coefficients in A/I.

Lift those coefficients to A and subtract the corresponding vectors $g_i e_j - g_j e_i$ from (b_i). Their scalar products with (g_i) are exactly zero in A, so this subtraction leaves $\sum_i g_i b_i$ unchanged. Its new coefficients reduce to zero modulo I, hence belong to I. The sum is therefore in IJ, as is u, proving $x \in IJ$. An empty list gives the same calculation with sum zero. No finite generation of I and no Noetherian condition were used. Applying local detection to $(I \cap J^2)/IJ$ completes the proof. □

Lemma F.102: cotangent complex symmetric algebra

Reading source: Stacks, Tag 07EV.

Let A be any ring and let

$$0 \longrightarrow K \longrightarrow A^m \xrightarrow{q} M \longrightarrow 0$$

be exact, where M is finite projective. Set $C = \mathrm{Sym}_A(M)$ and let $\alpha : A[y_1, \dots, y_m] \twoheadrightarrow C$ be the presentation induced by q, with kernel J. The natural map

$$K \otimes_A C \longrightarrow J/J^2, \qquad (k_1, \dots, k_m) \otimes c \longmapsto c \sum_i k_i y_i \bmod J^2$$

is an isomorphism. With this identification the conormal complex and its cokernel are

$$\mathrm{NL}(\alpha) = [K \otimes_A C \longrightarrow C^m], \qquad \Omega_{C/A} = M \otimes_A C,$$

where the differential is obtained from $K \hookrightarrow A^m$ by tensoring with C.

Proof. Choose a section $s : M \to A^m$ using projectivity. The map $(k, m) \mapsto k + s(m)$ identifies $K \oplus M$ with A^m: its inverse sends a to $(a - sq(a), q(a))$. We compute the presentation using this decomposition, keeping track of its specified natural map.

For a module V, the symmetric algebra is the tensor algebra modulo the relations that interchange two adjacent factors. A linear map $V \to D$ into a commutative A-algebra extends on a pure tensor by multiplying its images; the interchange relations vanish, and these products determine the extension uniquely. This proves the symmetric-algebra universal property from the tensor construction S01, Lemma 1.0. It also gives its grading by tensor degree, with degree zero A and degree one V.

That property and the algebra tensor universal property give inverse maps

$$\mathrm{Sym}_A(K \oplus M) \;\cong\; \mathrm{Sym}_A(K) \otimes_A C.$$

In one direction send k to $k \otimes 1$ and m to $1 \otimes m$. In the other direction use the inclusions of K and M into $K \oplus M$ on the two tensor factors. Both composites fix the algebra generators, hence are identities. Likewise the symmetric algebra of A^m is the polynomial ring on its coordinate basis. Under these identifications α is the augmentation on the K factor and the identity on C.

Grade the tensor product by K degree. The direct-sum tensor formula of S01, Lemma 1.0 gives

$$J = \bigoplus_{d \geq 1} \mathrm{Sym}^d_A(K) \otimes_A C, \qquad J^2 = \bigoplus_{d \geq 2} \mathrm{Sym}^d_A(K) \otimes_A C.$$

For the second equality, a product of positive degrees has degree at least two. Conversely each degree-d term is generated by products $k_1 \cdots k_d \otimes c$; when $d \geq 2$ separate its first factor from the rest to express it as a product of two elements of J. Quotienting thus leaves precisely $K \otimes_A C$. Its generator $k \otimes c$ corresponds in the original coordinates to $c \sum_i k_i y_i$, proving exactly the natural isomorphism in the statement, independently of the auxiliary section.

Relative differentiation sends this linear polynomial to $\sum_i k_i \, dy_i$. The polynomial differentials are free on these m symbols, so the displayed complex has exactly the claimed differential. Tensoring

the split sequence with C remains split exact, and its cokernel is $M \otimes_A C$. This proves the assertion about $\Omega_{C/A}$ as well, including zero modules and the empty coordinate basis.

The projectivity hypothesis prevents additional conormal relations. For example let $A = k[\varepsilon]/(\varepsilon^2)$, let $M = k^2$ via $\varepsilon M = 0$, and let $q : A^2 \to M$ be reduction. Then $K = \varepsilon A^2$, $C = A[y_1, y_2]/(\varepsilon y_1, \varepsilon y_2)$, and $J = (\varepsilon y_1, \varepsilon y_2)$ has square zero. The source $K \otimes_A C$ is $k[y_1, y_2]^2$, whereas the vector $(y_2, -y_1)$ maps to $\varepsilon y_1 y_2 - \varepsilon y_2 y_1 = 0$ in J/J^2 and is nonzero in that source. Thus the same conormal identity need not hold after dropping the hypothesis. □

Lemma F.103: degree p extension regular

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12453–12467.

Let R be a Noetherian regular ring and let $D : R \to R$ be a derivation such that $D(f)$ is a unit. For every integer-coefficient polynomial $P(Z)$, the ring $R[Z]/(P(Z) - f)$ is regular. In particular this holds for $P(Z) = Z^n$, $n \geq 1$, including degree-p adjunctions in characteristic p.

Proof. The whole polynomial-regularity argument AG-CA-14, Proposition 3.3 makes $V = R[Z]$ regular; Hilbert's basis theorem also makes it Noetherian. Extend the derivation by the explicit formula

$$D_V\left(\sum_i a_i Z^i\right) = \sum_i D(a_i) Z^i.$$

Coefficient convolution and the product rule in R prove the product rule for D_V, and additivity is coefficientwise. The identity $D(1) = 2D(1)$ gives $D(1) = 0$; additivity and additive inverses then give $D(n) = 0$ for every integer scalar n. Hence $D_V(P) = 0$ and

$$D_V(P - f) = -D(f).$$

This is a unit already in V, and remains a unit in its quotient. The entire hypersurface result F.122, with its localized-derivation and parameter-quotient proofs, applies to V and $P - f$. It makes the quotient regular, for every specified P. Constant polynomials, units and the zero quotient obey the same calculation. □

Lemma F.104: derivation extends

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12363–12405.

Let $D : R \to R$ be a derivation of an arbitrary ring.

1. It extends uniquely to every localization $U^{-1}R$.
2. For every ideal I, it has a canonical continuous extension to $\widehat{R} = \varprojlim_n R/I^n$, with its inverse-limit topology. This extension is unique among continuous extensions.
3. If $R \subset R'$ is finite type, $g \in R$ is a nonzerodivisor in R', and the inclusions identify $R_g = R'_g$, then some $g^N D$ extends to a derivation $R' \to R'$.

Proof. For (1), the map $r \mapsto r + \epsilon D(r)$ into $(U^{-1}R)[\epsilon]/(\epsilon^2)$ is a ring map. Each $s \in U$ has image with inverse

$$s^{-1} - \epsilon s^{-2} D(s).$$

Localization's universal property therefore extends the map. Its constant component is the structural map on the localized ring, so its second component is a derivation, with formula

$$D_U(r/s) = D(r)/s - rD(s)/s^2.$$

This constructs a well-defined derivation even with zero divisors. Any extension must have that formula by differentiating $s\, s^{-1} = 1$, proving uniqueness.

For (2), a product of $n+1$ ideal elements differentiates into I^n; finite sums and coefficient derivatives give

$$D(I^{n+1}) \subset I^n \qquad (n \geq 1).$$

For a compatible family $a = (a_n) \in \widehat{R}$, choose $r \in R$ representing a_{n+1} and define its derivative's nth coordinate as $D(r) \bmod I^n$. Replacing r changes its derivative by an element of I^n, so this is well defined. Comparing representatives of a_{n+2} and a_{n+1} gives compatibility of these derivative coordinates. They therefore define $\widehat{D}(a) \in \widehat{R}$.

Represent a, b modulo I^{n+1} by r, s. Their sum and product are represented by $r+s, rs$. The identities in R, reduced modulo I^n, give

$$\widehat{D}(a+b)_n = \widehat{D}(a)_n + \widehat{D}(b)_n, \qquad \widehat{D}(ab)_n = a_n \widehat{D}(b)_n + b_n \widehat{D}(a)_n.$$

Thus the inverse-limit map is a derivation and extends D through the canonical map $R \to \widehat{R}$.

Write $K_n = \ker(\widehat{R} \to R/I^n)$. The formula gives $\widehat{D}(K_{n+1}) \subset K_n$, proving continuity. Canonical images of R are dense: any coordinate modulo I^m has a representative in R, matching all lower coordinates too. If another continuous extension were given, their difference would be a continuous additive map vanishing on this dense image. For a prescribed output coordinate choose a sufficiently deep input kernel mapped into its kernel by continuity, and choose an R-representative matching that input coordinate. The difference has zero prescribed output coordinate. This holds for every coordinate, and the inverse limit is separated, so the difference is zero. The construction uses the inverse-limit kernels, without assuming an infinitely generated ideal's extended powers agree with those kernels.

For (3), R' embeds in R'_g: if $x/1 = 0$, some $g^j x = 0$, and injectivity of multiplication by g repeatedly gives $x = 0$. Choose algebra generators $x_1, \ldots, x_s$ of R'/R. The localized equality writes each as a fraction from R with denominator a power of g; the embedding just proved makes those cleared equalities hold in R'. Thus a common $n \geq 0$ satisfies $g^n x_i \in R$ for every i.

Inside the common localization apply (1). The product rule, including $D(g^n) = ng^{n-1}D(g)$ when $n > 0$, gives

$$g^{n+1} D_U(x_i) = gD(g^n x_i) - n(g^n x_i)D(g) \in R.$$

For $n = 0$ the second term is zero and the same identity holds directly. Multiplying a derivation by the scalar g^{n+1} still gives a derivation. It sends R into R and each generator into R, so repeated product and sum rules send every polynomial expression in the generators into R'. Since this is the restriction of an actual derivation on the localization, all relations in R' are respected. Restriction therefore gives the required extension with $N = n+1$. Empty generator lists and the zero ring cause no exception. □

Lemma F.105: find D

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12469–12520.

Let p be prime, let B be a domain of characteristic p, and let $f \in B$ be outside $\mathrm{Frac}(B)^p$. If B is finite type over a complete Noetherian local ring, there is a derivation $D : B \to B$ with $D(f) \neq 0$.

Proof. Let $R_1 \to B$ be such a finite-type map. Its kernel is prime. Exact completion on finite modules AG-CA-19, Theorem 3.1 identifies the quotient $R_1/\ker(R_1 \to B)$ with its maximal-adic completion: tensor the finite quotient with $\widehat{R_1} = R_1$. Thus its image is again a complete Noetherian local domain, of characteristic p. By F.25 it is finite over an embedded power-series ring $R = k[[X_1, \ldots, X_n]]$. A finite module generating list for the image ring, together with the finite algebra generating list for B over that image, makes B finite type over R. In particular $R \subset B$ is injective.

Put $K_0 = \mathrm{Frac}(R)$. Noether normalization AG-CA-09, Corollary 3.2 supplies algebraically independent parameters in $B \otimes_R K_0$ over which that algebra is finite. Multiply each parameter by a

nonzero scalar of R to put it in B; independence and the parameter algebra over K_0 are unchanged. Write the resulting parameters as $y_1, \dots, y_d$, and put $A = R[y_1, \dots, y_d]$. Choose finite A-algebra generators b_i of B. Each satisfies a monic equation over $K_0[y]$. There is a common nonzero $g \in R$ for which every coefficient belongs to A_g.

Write such an equation as $b_i^{e_i} + \sum_{j<e_i} c_{ij} b_i^j = 0$. Choose N such that $g^{N(e_i-j)} c_{ij} \in A$ for every i, j. Multiplying by g^{Ne_i} gives a monic equation over A for $g^N b_i$. Consequently

$$B' = A[g^N b_1, \dots, g^N b_t] \subset B$$

is finite over A: reduce powers of each generator using its monic equation, obtaining a finite spanning list of bounded monomials. Moreover $B'_g = B_g$, since all b_i belong to B'_g. A sufficiently large $r \geq 0$ therefore puts $f' = g^{pr} f$ in B'. The fraction fields of B' and B agree; call this field L. Since $g^{pr} = (g^r)^p$, membership of f' in L^p would imply membership of f in L^p. Thus $f' \notin L^p$. Finiteness over A makes L finite over $K = \mathrm{Frac}(A)$: the localization $B' \otimes_A K$ is a finite-dimensional domain, hence a field, and contains all fractions of B'.

The complete power-series calculation F.120 supplies a nonempty downward-directed family of coefficient subrings

$$A_J = k_J[[X_1^p, \dots, X_n^p]][y_1^p, \dots, y_d^p] \subset A,$$

with A finite free over A_J and fraction fields K_J containing K^p, whose intersection is K^p. Here J runs over finite subsets of an absolute p-basis of k, and k_J is generated over k^p by the basis elements outside J; no perfection hypothesis is imposed. The finite-extension assertion of F.112 gives $\bigcap_J L^p K_J = L^p$. Since $f' \notin L^p$, choose J with $f' \notin L^p K_J$.

The relative p-basis theorem F.119 extends $\{f'\}$ to a p-basis of L/K_J. Indeed its first degree is p: a proper monic factor of $T^p - (f')^p = (T - f')^p$ has coefficient $-ef'$, $1 \leq e < p$, and would put f' in $L^p K_J$. The coordinate derivation from the full presentation in F.119 consequently gives $\delta : L \to L$ killing K_J with $\delta(f') = 1$.

The ring B' is finite over A_J, by the two finite module spanning lists. Choose A_J-module generators $v_1, \dots, v_s$. Write each $\delta(v_i)$ as a fraction of B' and let $h \in B' \setminus \{0\}$ be the product of their nonzero denominators. Then $D' = h\delta$ sends every v_i into B' and kills A_J. Differentiating an arbitrary expression $\sum_i a_i v_i$, $a_i \in A_J$, proves $D'(B') \subset B'$. Its value on f' is $h \neq 0$.

Finally B is finite type over B', g is a nonzerodivisor in both domains, and $B'_g = B_g$. The third assertion of F.104 therefore gives $s \geq 0$ for which $g^s D'$ restricts to a derivation $D : B \to B$. In the common fraction field, the fraction formula and $D'(g^{pr}) = prg^{pr-1} D'(g) = 0$ yield

$$D(f) = g^s D'(g^{-pr} f') = g^{s-pr} h \neq 0.$$

Its membership in B follows from the restriction just proved; the displayed expression proves its nonvanishing. Empty parameter lists or a coefficient field as base require no changes. □

Lemma F.106: formally smooth

Reading source: Stacks, Tag 07EC.

Let $\varphi : R \to S$ be continuous. If the underlying algebra map is formally smooth, it is formally smooth when R has any linear topology and S has a pre-adic topology. Moreover, if the topologies on R and S are given by the powers of ideals $\mathfrak{m}$ and $\mathfrak{n}$, formal smoothness is unchanged when the topology on R is replaced by the discrete topology.

Proof. We specify the topological test. Give a ring D and its quotient D/J the discrete topology, where $J^2 = 0$. Fix continuous maps $a : R \to D$ and $b : S \to D/J$ such that $b\varphi$ is the reduction of a. A solution is a continuous R-algebra map $c : S \to D$ reducing to b. Topological formal smoothness requires a solution for every such test; algebraic formal smoothness uses the same tests without continuity conditions. The latter definition is AG-CA-17, Section 1.

For a pre-adic topology on S, choose $\mathfrak{n}$ whose powers form a neighbourhood basis at zero. A ring map from S to a discrete ring is continuous precisely when its kernel contains some $\mathfrak{n}^r$: continuity at zero gives this condition, and translation gives continuity at every point. Algebraic formal smoothness supplies a lift c in the test. Continuity of b gives $b(\mathfrak{n}^r) = 0$ for some $r \geq 1$. Thus $c(\mathfrak{n}^r) \subset J$, and multiplication of two such ideal powers gives

$$c(\mathfrak{n}^{2r}) \subset J^2 = 0.$$

The lift is therefore continuous. The linear topology on R needs no further restriction beyond continuity of the stipulated maps.

Now assume both topologies are given by the stated ideal powers. If formal smoothness holds with discrete R, any test with continuous a is among its tests, so the same property holds with the original topology on R. In the other direction, start with a test having discrete R; the map a is initially allowed to be arbitrary. Continuity of φ gives an integer $s \geq 1$ with $\varphi(\mathfrak{m}^s) \subset \mathfrak{n}$. For the integer r supplied by b as above, compatibility implies

$$a(\mathfrak{m}^{sr}) \subset J, \qquad a(\mathfrak{m}^{2sr}) = 0.$$

Hence a was continuous for the $\mathfrak{m}$-power topology as well. Formal smoothness for that topology now gives the desired continuous lift. This proves both directions, with no finite-generation or Noetherian assumption. Neither completeness nor separatedness of S was required. □

Lemma F.107: formally smooth completion

Reading source: Stacks, Tag 07ED.

Let $R \to S$ be a continuous map of Noetherian rings with topologies defined by finitely generated ideals $\mathfrak{m} \subset R$ and $\mathfrak{n} \subset S$. Write $\widehat{R}$ and $\widehat{S}$ for the corresponding completions. The following three properties are equivalent: $R \to S$ is formally smooth for the $\mathfrak{n}$-power topology; $R \to \widehat{S}$ is formally smooth for the $\mathfrak{n}\widehat{S}$-power topology; and $\widehat{R} \to \widehat{S}$ is formally smooth for that target topology.

Proof. The continuous map induces the last completion map by applying it to compatible finite-order coordinates. AG-CA-19, Proposition 2.2 proves the exact quotient identities

$$\widehat{R}/\mathfrak{m}^q\widehat{R} = R/\mathfrak{m}^q, \qquad \widehat{S}/\mathfrak{n}^q\widehat{S} = S/\mathfrak{n}^q \quad (q \geq 1).$$

These identities use the canonical maps and commute with all transition maps. That proposition also proves completeness for the extended-ideal topologies.

A continuous homomorphism $u : S \to D$ to a discrete ring kills some $\mathfrak{n}^q$. Its factorization through $S/\mathfrak{n}^q$ and the second displayed identity define a homomorphism $\widehat{u} : \widehat{S} \to D$ killing $\mathfrak{n}^q\widehat{S}$. It extends u. It is the only continuous extension: any other extension kills a possibly different ideal power; pass to an exponent at least as large as both, where its map is determined by the same surjective finite quotient. Restriction and extension are inverse, and they commute with maps between discrete target rings. The identical argument applies to R and $\widehat{R}$.

Use the tests defined in F.106, with $D \to D/J$ square-zero. First compare $R \to S$ with $R \to \widehat{S}$. Keep the structural map from R fixed. The just-proved correspondence identifies every continuous map to D/J, every continuous lift to D, and their compatibility with R. Thus one test has a solution exactly when its counterpart does.

Next compare $R \to \widehat{S}$ with $\widehat{R} \to \widehat{S}$. With the original topology on the base, extension and restriction identify the structural maps to both discrete test rings. Compatibility after restriction implies compatibility after extension: the two continuous maps from $\widehat{R}$ agree by uniqueness of extension. The test diagrams and their dotted lifts therefore correspond in both directions. Finally F.106 identifies each of these topological-base tests with the tests using discrete base. In particular a structural map allowed in a discrete-base test is automatically continuous when it is compatible with the continuous target map modulo J; that is precisely the ideal-power calculation in F.106. This proves the three equivalences in the stipulated Noetherian scope. □

Lemma F.108: gamma commutative diagram

Reading source: Stacks, Tag 07E3.

Consider a commutative square of fields with arrows $k \to k'$, $k \to K$, $k' \to K'$ and $K \to K'$. Assume k'/k and K'/K are finitely generated. For the natural maps

$$\alpha : \Omega_{K/k} \otimes_K K' \longrightarrow \Omega_{K'/k'}, \qquad \beta : H_1(L_{K/k}) \otimes_K K' \longrightarrow H_1(L_{K'/k'}),$$

all four kernels and cokernels are finite-dimensional, and

$$\dim \ker \alpha - \dim \operatorname{coker} \alpha - \dim \ker \beta + \dim \operatorname{coker} \beta = \operatorname{trdeg}_k k' - \operatorname{trdeg}_K K'.$$

Proof. All vector spaces below are over K', and all omitted scalar extensions are to K'. Apply the full field-transitivity proof F.131 to $k \subset k' \subset K'$ and $k \subset K \subset K'$. Let

$$V = H_1(L_{K'/k}), \qquad U = \Omega_{K'/k}.$$

For the first row put $P_1 = H_1(L_{k'/k}) \otimes_{k'} K'$, $W_1 = H_1(L_{K'/k'})$, $Q_1 = \Omega_{k'/k} \otimes_{k'} K'$ and $Z_1 = \Omega_{K'/k'}$. For the second put $P_2 = H_1(L_{K/k}) \otimes_K K'$, $W_2 = H_1(L_{K'/K})$, $Q_2 = \Omega_{K/k} \otimes_K K'$ and $Z_2 = \Omega_{K'/K}$. The two exact rows are

$$0 \longrightarrow P_i \longrightarrow V \longrightarrow W_i \longrightarrow Q_i \longrightarrow U \longrightarrow Z_i \longrightarrow 0 \qquad (i = 1, 2).$$

Their first injections identify P_1, P_2 with subspaces of V. Naturality of the representative and derivative maps in F.42 identifies β with $P_2 \to V \to W_1$ and α with $Q_2 \to U \to Z_1$.

We calculate only finite defects, so no subtraction of infinite dimensions is involved. Set

$$T = P_1 \cap P_2, \quad Y = V/(P_1 + P_2), \quad A_i = \ker(Q_i \to U), \quad B_i = \operatorname{im}(Q_i \to U).$$

Write $i_0 = \dim T$, $r = \dim(P_1/T)$, $y = \dim Y$, $a_i = \dim A_i$, $j = \dim(B_1 \cap B_2)$, $s = \dim(B_1/(B_1 \cap B_2))$ and $z = \dim(U/(B_1 + B_2))$. These are finite. Indeed F.98 makes P_1, Q_1, W_2, Z_2 finite. The space Y is a quotient of V/P_2, which embeds in W_2; A_2 is a quotient of W_2; and the space measured by z is a quotient of Z_2. All other spaces on the list are subspaces or quotients of P_1 or Q_1.

The kernel of β is T. Exactness of the first row gives an injection $V/P_1 \to W_1$ with quotient A_1. Dividing its image by the image of P_2 therefore yields

$$0 \longrightarrow Y \longrightarrow \operatorname{coker} \beta \longrightarrow A_1 \longrightarrow 0.$$

For α, its kernel consists of the elements of Q_2 whose image in U lies in B_1. Mapping them into U gives a surjection onto $B_1 \cap B_2$ with kernel A_2. Its cokernel is $U/(B_1 + B_2)$. Hence

$$\dim \ker \beta = i_0, \quad \dim \operatorname{coker} \beta = y + a_1, \qquad \dim \ker \alpha = a_2 + j, \quad \dim \operatorname{coker} \alpha = z.$$

The second row identifies V/P_2 with the kernel of $W_2 \to Q_2$. Inside V/P_2, the image of P_1 is P_1/T and has quotient Y. Similarly U/B_2 contains the image $B_1/(B_1 \cap B_2)$ with quotient $U/(B_1 + B_2)$. The resulting finite exact sequences give

$$\dim P_1 = i_0 + r, \quad \dim W_2 = r + y + a_2, \qquad \dim Q_1 = a_1 + j + s, \quad \dim Z_2 = s + z.$$

Subtracting these finite numbers shows that the expression in the statement equals

$$(a_2 + j) - z - i_0 + (y + a_1) = \dim Q_1 - \dim P_1 - \dim Z_2 + \dim W_2.$$

F.98 applied to k'/k and K'/K gives the required transcendence-degree difference. This also proves finiteness of every defect without requiring K/k or K'/k' to be finitely generated. □

Lemma F.109: geometrically regular over field

Reading source: AI Integrated Stacks, pinned More Algebra, lines 9460–9501.

Let k have characteristic $p > 0$, and let $(A, \mathfrak{m}, K)$ be a geometrically regular Noetherian local k-algebra. Let $k \subset F \subset K$ be finitely generated over k. Suppose a map $\varphi : k[y_1, \dots, y_m] \to A$ has residue images $\overline{y}_i \in F$, with $d\overline{y}_1, \dots, d\overline{y}_m$ a basis of $\Omega_{F/k}$. Put $\mathfrak{p} = \varphi^{-1}(\mathfrak{m})$. Then

$$k[y_1, \dots, y_m]_{\mathfrak{p}} \longrightarrow A$$

is flat, and $A/\mathfrak{p}A$ is regular.

Proof. Write $R_0 = k[y_1, \dots, y_m]_{\mathfrak{p}}$, $\mathfrak{m}_0 = \mathfrak{p}R_0$, and $K_0 = R_0/\mathfrak{m}_0$. The residue map embeds K_0 into F, and

$$K_0 = k(\overline{y}_1, \dots, \overline{y}_m).$$

Indeed it is the fraction field of the domain image of $k[y]$ in K; localizing at the kernel inverts every nonzero image. The polynomial and fraction differential formulas in AG-CA-15, Sections 1–2 and Theorem 5.1 make $\Omega_{R_0/k}$ free on dy_i, and make their images span $\Omega_{K_0/k}$. They are independent in that last space: a K_0-linear relation maps to an F-linear relation between the assumed basis of $\Omega_{F/k}$, so every coefficient vanishes in F, and hence in K_0. Thus

$$\Omega_{R_0/k} \otimes_{R_0} K_0 \xrightarrow{\sim} \Omega_{K_0/k}.$$

The ring R_0 is regular local by iterating AG-CA-14, Proposition 3.3 from the field and using the definition of a regular ring at its prime localizations. For a finite purely inseparable k'/k, its tensor is the localization of $k'[y]$ at the images of $k[y] \setminus \mathfrak{p}$, and is Noetherian regular. Criterion F.50 therefore makes R_0 geometrically regular over k. The geometric-regularity characterization F.133 gives injectivity of the connecting map

$$\gamma_0 : H_1(L_{K_0/k}) \longrightarrow \mathfrak{m}_0/\mathfrak{m}_0^2.$$

The full quotient sequence F.42 gives

$$H_1(L_{K_0/k}) \longrightarrow \mathfrak{m}_0/\mathfrak{m}_0^2 \longrightarrow \Omega_{R_0/k} \otimes_{R_0} K_0 \longrightarrow \Omega_{K_0/k} \longrightarrow 0.$$

The last nonzero arrow is the displayed isomorphism, so exactness also makes γ_0 surjective. Hence it is an isomorphism.

All arrows in the following square are the connecting and scalar-extension maps of the same quotient presentations:

$$\begin{array}{ccc} H_1(L_{K_0/k}) \otimes_{K_0} K & \xrightarrow{\gamma_0 \otimes 1} & (\mathfrak{m}_0/\mathfrak{m}_0^2) \otimes_{K_0} K \\ \downarrow & & \downarrow \\ H_1(L_{K/k}) & \xrightarrow{\gamma} & \mathfrak{m}/\mathfrak{m}^2. \end{array}$$

It commutes: a lifted polynomial relation has the same differentiated representative whether mapped first to A or first to its residue field, as in F.42. The left arrow is injective by the first injection of the field-tower sequence F.131 for $k \subset K_0 \subset K$. The bottom arrow is injective by F.133 applied to the assumed geometrically regular A. The top arrow is an isomorphism, since tensor over a field is exact. If a vector in its upper-right space maps to zero below, take its preimage in the upper-left space. Commutativity and the two injections force that preimage, and then the vector, to be zero. This proves injectivity of the right arrow.

Choose a regular system of parameters $x_1, \dots, x_d$ of R_0. Their classes are a K_0-basis of $\mathfrak{m}_0/\mathfrak{m}_0^2$; after scalar extension they remain a K-basis. The right injection says their images have independent classes in $\mathfrak{m}/\mathfrak{m}^2$. The ring A is regular, by F.133. Extend those classes to a basis of this finite-dimensional K-space and lift the additional basis vectors to $\mathfrak{m}$. Nakayama's lemma F.10 makes the resulting list a minimal generating list for $\mathfrak{m}$. The complete parameter theorem AG-CA-12,

Theorem 6.1 makes it a regular sequence. In particular its prefix, the images of $x_1, \dots, x_d$, is regular in A, with proper generated ideal.

The map $R_0 \to A$ is local by the definition of $\mathfrak{p}$. Therefore F.46 applies to these images and proves flatness. The full successive parameter-quotient result AG-CA-14, Theorem 1.1 and Corollary 1.2 makes $A/(x_1, \dots, x_d)A$ regular local. Since the parameters generate $\mathfrak{m}_0$, their image ideal is exactly $\mathfrak{p}A$. This is the asserted regular quotient. If $d = 0$, R_0 is a field, its flatness assertion is the vector-space argument, and the quotient is A itself; the proof also covers $m = 0$. □

Lemma F.110: helper G ring

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12915–12963.

Let A_0 be a regular complete Noetherian local domain, put $P = A_0[Y_1, \dots, Y_m]$, and let $\mathfrak{q} \subset P$ be any prime. Both

$$\widehat{P_{\mathfrak{q}}} \otimes_{A_0} \operatorname{Frac}(A_0) \quad \text{and} \quad \widehat{P_{\mathfrak{q}}} \otimes_P \operatorname{Frac}(P)$$

are geometrically regular over their displayed fields. The second assertion contains the power-series helper; the first is the additional form used in F.97.

Proof. Polynomial regularity and localization are AG-CA-14, Proposition 3.3 and Theorem 3.2. Thus $P_{\mathfrak{q}}$ is regular local. Its completion $H = \widehat{P_{\mathfrak{q}}}$ is Noetherian regular local by AG-CA-19, Theorem 4.1. Each displayed ring is a localization of H, so is Noetherian and regular. If the indicated field has characteristic zero, its only finite purely inseparable extension is itself, and F.50 proves geometric regularity.

Suppose the characteristic is $p > 0$. In this case $A_0 \cong k[[X_1, \dots, X_n]]$, with no assumption that k is perfect. To check this specialization, AG-CA-19S, Theorem 6.1 evaluates the variables at a minimal list of generators of the maximal ideal of A_0, giving a surjection from $k[[X_1, \dots, X_n]]$. Regularity makes $n = \dim A_0$. The source is regular local of dimension n, by the polynomial completion calculation and AG-CA-19, Theorem 4.1. If the kernel contained a nonzero element, the one-equation dimension theorem AG-CA-11, Theorem 3.2 would make the quotient dimension at most $n - 1$, a contradiction. The evaluation kernel is therefore zero.

Treat the two assertions together by taking $Q = A_0$, $F = \operatorname{Frac}(A_0)$ in the first case, and $Q = P$, $F = \operatorname{Frac}(P)$ in the second. We prove that $H \otimes_Q L$ is regular for every finite purely inseparable extension L/F, by induction on its degree. The base case is the localization already proved regular. Such an extension has a tower of degree-p steps: take finite field generators, with successive p-powers in F, and adjoin their roots one at a time; at a nontrivial step the proper-factor argument in F.119 gives degree p. Thus at the final step write $F \subset M \subset L = M(z)$, $z^p = f \in M \setminus M^p$, and assume $E = H \otimes_Q M$ regular.

There is a finite domain B, $Q \subset B \subset M$, with fraction field M and, after rescaling z, with $f \in B$. Here are the denominator details. Choose finitely many algebraic generators u_i of M/F. For a monic equation $u_i^{d_i} + \sum_{j<d_i} c_{ij} u_i^j = 0$, multiplication by a suitable nonzero $a_i \in Q$ makes every $a_i^{d_i - j} c_{ij}$ lie in Q. Hence $a_i u_i$ is integral over Q. The algebra generated by these elements is finite by bounded monic reductions, and its fraction field is M. Write $f = b/d$ in that algebra. A monic integral equation for $d \neq 0$ can be shortened, by cancelling powers of d in this domain, until its constant coefficient $a \in Q$ is nonzero. The equation expresses a/d in B. Replace z by az. Its new pth power is $a^p f = ba^{p-1}(a/d) \in B$, and remains outside M^p, since a^p is a nonzero pth power. Relabel this generator and its power as z, f. The monic irreducible equation makes $C = B[Z]/(Z^p - f)$ a finite domain with fraction field L; injectivity in L follows from the degree-p minimal polynomial.

The base Q is a normal domain. In the first case this is regular local normality; in the second it is polynomial regularity followed by AG-CA-15, Corollary 4.5. Every element of B or C has some p-power in F, since its fraction field is finite purely inseparable over F. That power is integral over

Q, because the element is integral. Normality therefore puts the power in Q.

Set $T = B[Y_1, \dots, Y_m]$ in the first case, and $T = B$ in the second. Then $P \subset T$ is finite, and every element of T has a sufficiently high p-power in P. For a polynomial in the first case, take a common exponent for its finitely many coefficients and apply Frobenius to the sum. Lying over gives a prime $\mathfrak{r}$ above $\mathfrak{q}$, and it is unique: for $t^{p^e} \in P$, membership of t in a prime above $\mathfrak{q}$ is equivalent to membership of t^{p^e} in $\mathfrak{q}$. The full finite-completion product theorem F.27 thus has one factor and gives

$$H \otimes_P T \cong \widehat{T_{\mathfrak{r}}}.$$

Tensoring with $M = \mathrm{Frac}(B)$ over B, and using $P \otimes_{A_0} B = T$ in the first case, gives the actual identity

$$E = H \otimes_Q M \cong \widehat{T_{\mathfrak{r}}} \otimes_B M.$$

In particular E is a localization of a Noetherian completed local ring. These are the maximal-ideal completions specified in F.27, not a replacement by an unrelated topology.

The domain B is finite type over the complete local ring A_0, in either case. Since $f \notin M^p$, F.105 produces a derivation $D : B \to B$ with $D(f) \neq 0$. Extend it coefficientwise to T in the first case, assigning zero to the Y_i; coefficient convolution proves the product rule. In the second case it is already a derivation of T. By F.104 it extends successively to $T_{\mathfrak{r}}$, its maximal-adic completion, and the localization by $B \setminus \{0\}$. The resulting derivation of E sends f to the original nonzero element $D(f) \in M$. This is a unit in every nonzero M-algebra E; if $E = 0$, the next test ring is also zero and already regular.

Finally tensor associativity and $L = M[Z]/(Z^p - f)$ give

$$H \otimes_Q L \cong E \otimes_M L \cong E[Z]/(Z^p - f).$$

The polynomial quotient identity follows by reducing each polynomial to its degree-less-than-p remainder, or directly from its finite monic presentation. F.103 applies: E is Noetherian regular and its derivation has unit value on f. It proves this next ring regular. The induction covers every finite purely inseparable extension. Criterion F.50 now gives both assertions. Empty lists of X_i or Y_i are included throughout. □

Lemma F.111: helper integral

Reading source: Stacks, Tag 09XG.

Let $A \to B$ be an integral ring map, $I \subset A$ an ideal, and $y \in B$ have idempotent image in B/IB. There are an integer $d \geq 1$ and a monic polynomial $f(T) \in A[T]$ such that

$$f(y) = 0, \qquad f(T) \bmod I = T^d(T-1)^d.$$

Proof. Write $z = y^2 - y = \sum_{i=1}^{h} a_i b_i$ with $a_i \in I$ and $b_i \in B$. This is a finite sum by the definition of the extended ideal. The subalgebra $C = A[y, b_1, \dots, b_h] \subset B$ is a finite A-module: each of its finitely many generators satisfies a monic equation over A, and successive substitution reduces every monomial to one whose exponent in each generator is below that equation's degree. Those finitely many bounded monomials generate C. This argument uses the image of A; injectivity of $A \to B$ is unnecessary.

Choose a finite module generating list $c_1, \dots, c_d$ for C, including 1, so $d \geq 1$. Since $zC \subset IC$, express each zc_i as $\sum_j m_{ij} c_j$ with all $m_{ij} \in I$. For the column $c = (c_i)$ these equations say $(z1_d - M)c = 0$, where $M = (m_{ij})$. Multiplying by the adjugate, whose identity was proved completely in F.19, gives

$$g(z)c_i = 0 \quad \text{for all } i, \qquad g(U) = \det(U1_d - M) \in A[U].$$

The included generator 1 makes $g(z) = 0$ in C, hence in B. The determinant expansion makes g monic of degree d; reducing the entries of M modulo I gives $g(U) \bmod I = U^d$. Set $f(T) = g(T^2 - T)$.

Its leading coefficient is 1, it annihilates y, and its reduction is $(T^2-T)^d = T^d(T-1)^d$. The same construction permits an empty sum for z, the zero target algebra, and $I = A$. $\square$

Lemma F.112: intersection subfields

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12067–12128.

Let K have characteristic $p > 0$. Let (F_λ) be a nonempty downward-directed family of subfields of K containing K^p, with intersection K^p. Then

$$\bigcap_\lambda \ker(\Omega_{K/\mathbf{F}_p} \to \Omega_{K/F_\lambda}) = 0.$$

For every finite field extension L/K, the composita in L also satisfy

$$\bigcap_\lambda L^p F_\lambda = L^p.$$

Proof. Take an absolute p-basis of K by the entire F.119. If a nonzero absolute differential ω vanished in every indicated relative module, write it in that differential basis using a finite nonempty subset $b_1, \dots, b_c$ and coefficients not all zero. At every λ those relative differentials would be dependent. F.119's full equivalence makes their p-monomials dependent over $F_\lambda K^p = F_\lambda$.

Index those finitely many monomials by e, $0 \le e_i < p$, and form the actual K-subspace

$$V = \{(a_e) \in K^{p^c} : \sum_e a_e b^e = 0\}.$$

Its intersection with $F_\lambda^{p^c}$ is nonzero at every index. The entire newly proved F.113 gives a nonzero intersection with $(K^p)^{p^c}$, contradicting independence of the absolute p-basis monomials. Hence no such ω exists, proving the first assertion.

We prove the second assertion through finite simple steps. Here is the reduction for an arbitrary finite extension. Choose finitely many field generators α_i. For each monic irreducible polynomial over K, remove a common factor p from its positive exponents repeatedly until writing it as $G_i(T^{p^{e_i}})$ with $G_i' \neq 0$. This terminates because its degree decreases. The polynomial G_i is irreducible, since a factorization would give one of the original polynomial, and its nonzero derivative is relatively prime to it. Thus $\beta_i = \alpha_i^{p^{e_i}}$ is separable over K.

The field $M = K(\beta_i)$ is finite separable: at each successive adjunction its minimal polynomial divides the original squarefree polynomial over the larger coefficient field, and remains separable. The full separable-tower transitivity proof AG-CA-17, Lemma 7.1 proves that this tower is separable over K. From M, adjoin each successive pth root needed to recover the α_i. Each step has degree one or p: if the root is absent, the proper-factor coefficient calculation in F.119 proves its degree-p polynomial irreducible. This gives a tower of separable simple steps followed by degree-p purely inseparable steps.

If the desired intersection identity has been proved up to an intermediate field M, the fields $E_\lambda = M^p F_\lambda \subset M$ are downward directed, contain M^p, and have intersection M^p. For the next field N, one has $N^p E_\lambda = N^p F_\lambda$, since $M^p \subset N^p$. Thus the same statement for that next simple step proves it for the composite. It remains to prove the two types of simple steps.

For a separable simple extension $L = K(\alpha)$, put $\theta = \alpha^p$. The minimal polynomial over $K(\alpha^p)$ divides both the original squarefree polynomial and $T^p - \alpha^p = (T-\alpha)^p$. It must therefore have degree one, so $K(\theta) = L$ and $\theta \in L^p$. Write $d = [L : K]$. Frobenius transports the K-basis $1, \alpha, \dots, \alpha^{d-1}$ to the K^p-basis $1, \theta, \dots, \theta^{d-1}$ of L^p: raising expansions to the pth power proves spanning, and unique roots of coefficients prove independence. The same powers are a K-basis of L, because $K(\theta) = L$.

Consequently they are independent over every $F_\lambda \subset K$. Their monic relation over K^p gives spanning over F_λ, so the algebra $F_\lambda[\theta]$ is a finite-dimensional domain and therefore a field, by multiplication and inversion. It is exactly $L^p F_\lambda$, with the displayed basis. An element in every compositum has a unique expansion in the K-basis of L; its coefficients must belong to every F_λ, hence to K^p. It therefore lies in L^p, proving the identity for this step.

For a nontrivial degree-p purely inseparable step, write $L = K(\theta)$ with $\theta^p = t \in K \setminus K^p$. Frobenius of its power basis gives

$$L^p = K^p(t), \qquad 1, t, \ldots, t^{p-1} \text{ a } K^p\text{-basis.}$$

Choose λ_0 with $t \notin F_{\lambda_0}$, possible from the intersection assumption. Restrict to members contained in F_{λ_0}. This family is cofinal under refinement: downward directedness refines any given member together with λ_0. Its intersection is still K^p, by the same refinement argument in F.113, and taking the intersection of the composita over it does not change the full intersection.

For every such member, $t^p \in K^p \subset F_\lambda$ while $t \notin F_\lambda$. The proper-factor calculation gives degree p, so $L^p F_\lambda = F_\lambda(t)$ has basis $1, t, \ldots, t^{p-1}$. Use the unique expansion over the fixed coarse field F_{λ_0}; every refined-field expansion must have those same coefficients. They belong to all refined fields and hence to K^p. The element is therefore in $K^p(t) = L^p$. The reverse inclusion is automatic. This cofinal argument includes indices where t was initially already in the field, without presuming a degree-p basis there.

Both step types are proved. Applying the tower reduction establishes the identity for every finite extension, including the trivial one. Both assertions and all directed-family hypotheses have now been checked. □

Lemma F.113: intersection subfields subspace

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12027–12065.

Let (F_λ) be a nonempty downward-directed family of subfields of a field K, and put $F = \bigcap_\lambda F_\lambda$. For every K-linear subspace $V \subset K^n$,

$$V \cap F^n \neq 0 \quad \Longleftrightarrow \quad V \cap F_\lambda^n \neq 0 \text{ for every } \lambda.$$

Proof. A nonzero vector in the intersection over F belongs to every larger field's intersection. For the other implication induct on n. At $n = 0$ both nonzero-intersection assertions are false, since the family is nonempty. At $n = 1$, a K-subspace of K is zero or all of K; the hypothesis excludes zero, and the vector one lies in F.

For $n > 1$, let V_0 be the subspace of vectors of V with first coordinate zero, viewed in K^{n-1}. If $V_0 \cap F^{n-1}$ is nonzero, adjoining the zero first coordinate finishes the proof. Otherwise the induction hypothesis gives an index λ_0 with $V_0 \cap F_{\lambda_0}^{n-1} = 0$. Choose a nonzero vector of $V \cap F_{\lambda_0}^n$. Its first coordinate cannot vanish; divide by it to get a vector v with first coordinate one.

For any other vector w in this intersection, subtract $w_1 v$. The difference is in $V_0 \cap F_{\lambda_0}^{n-1}$ and hence zero. Thus its normalized vector is uniquely v. In particular, for every member $F_\mu \subset F_{\lambda_0}$, the hypothesized nonzero intersection contains a vector whose first coordinate is nonzero, and normalization gives this same v. All coordinates of v consequently belong to the intersection of these refined fields.

That intersection is F. One inclusion follows because F lies in every member. For the other, given any index λ, downward directedness supplies a member contained in both F_{λ_0} and F_λ; membership in every refinement then implies membership in F_λ. Thus $v \in F^n$, and it is nonzero because its first coordinate is one. This proves the induction and the full equivalence. □

Lemma F.114: lci local

Reading source: Stacks, Tag 07D1.

Let R be Noetherian and let S be a finitely presented R-algebra. A kernel of a finite polynomial presentation of S is a regular ideal if and only if this property holds for polynomial presentations on a finite principal cover of $\operatorname{Spec} S$. The condition does not depend on the chosen presentations.

Proof. The arbitrary-ring comparison F.94.A proves presentation independence for Koszul-regular kernels. The finite polynomial rings and their localizations are Noetherian by AG-CA-03, Theorem 2.1; its Theorem 1.1 and Proposition 1.2 make their finite-module submodules finite. In these rings, F.118 converts a Koszul-regular generating list in a local maximal ideal into a regular list. To obtain the neighbourhood condition in F.95, we will explicitly spread the successive injections, rather than infer global regularity merely from Koszul acyclicity.

Fix $P = R[X_1, \dots, X_n] \twoheadrightarrow S$ with kernel I, and a prime $\mathfrak{q} \supset I$ of P. Suppose a chart S_g containing its image has a regular presentation. Choose $u \in P$ lifting g, so $u \notin \mathfrak{q}$. A presentation of that chart is

$$Q = P[Z] \twoheadrightarrow S_g, \qquad H = (I, t), \quad t = uZ - 1.$$

The prime $\mathfrak{q}'$ of Q given by $\mathfrak{q}$ and $Z = 1/u$ satisfies

$$Q_{\mathfrak{q}'}/(t) = P_{\mathfrak{q}}.$$

Indeed quotienting first gives P_u, and then localization at the prime over $\mathfrak{q}$ gives $P_{\mathfrak{q}}$. Presentation comparison and F.118 give a regular generating list for $H_{\mathfrak{q}'}$. Its conormal classes are a basis by F.74. Meanwhile F.72 gives, at this point,

$$(H/H^2)_{\mathfrak{q}'} \cong (I/I^2)_{\mathfrak{q}} \oplus S_{\mathfrak{q}/I}\, t.$$

The first summand is a finite direct summand of a finite free module. It is flat and finitely presented, hence free over the local quotient by S01, Lemma 1.2. Choose $f_1, \dots, f_c \in I$ whose classes are a basis there; denominators may be cleared since their values outside $\mathfrak{q}$ are units. The list $t, f_1, \dots, f_c$ is a conormal basis. Nakayama applied to $H_{\mathfrak{q}'}/(t, f_1, \dots, f_c)$ makes it a generator list for $H_{\mathfrak{q}'}$.

Compare it with the regular generator list just obtained. Both have the same length, equal to the conormal rank. Express the new list in the old one. Modulo the maximal ideal the coefficient matrix takes one basis of the conormal fibre to the other, so its determinant is a unit in $Q_{\mathfrak{q}'}$. Its exterior powers identify the Koszul complexes: the differential on a wedge deletes one entry with its alternating sign, and this formula commutes with the change of basis. Thus $t, f_1, \dots, f_c$ has zero first homology. F.118 makes this very order regular, with t first. Quotienting by t proves that $f_1, \dots, f_c$ is a regular list generating $I_{\mathfrak{q}}$.

Here is the required principal neighbourhood. The module $I/(f_1, \dots, f_c)$ is finite and vanishes at $\mathfrak{q}$. For each i, the kernel of multiplication by f_i on $P/(f_1, \dots, f_{i-1})$ is also finite, because P is Noetherian; it vanishes at $\mathfrak{q}$ by the just-proved regularity. Exact localization and the finite-generator fraction-zero argument S01, Lemma 1.1 give one element $h \notin \mathfrak{q}$ annihilating all these finitely many modules. In P_h the list therefore generates I_h and every successive multiplication is injective. Its final quotient is nonzero because the prime $\mathfrak{q}P_h$ still contains I_h. The list is regular on this actual neighbourhood, as required by F.95.

The argument applies at every point lying in one of the stipulated charts, hence at every prime above I. Conversely, localization preserves the successive injections of a regular list. For a chart where a given neighbourhood also inverts u, present the localization by adjoining Z and put $uZ - 1$ first. Multiplication by this polynomial is injective by its lowest-coefficient calculation, and its quotient is P_u; the remaining localized list is regular wherever the quotient is nonzero. Charts with zero quotient have no primes to check. Thus the kernel condition restricts to every principal chart. The local neighbourhoods cover the target, and a finite subcover exists by the

finite unit-ideal calculation AG-CA-01, Proposition 2.3. Applying the same comparison to a second initial presentation proves the claimed independence. If $S = 0$, all these prime conditions are vacuous, including the empty-cover case. □

Lemma F.115: lift factorization monic

Reading source: Stacks, Tag 0ALH.

Let $I \subset A$ be any ideal. Suppose the reduction of a monic $f \in A[T]$ has a factorization $\bar{f} = \bar{G}\bar{H}$ into coprime monic polynomials over A/I. There is a finitely presented étale map $A \to A'$ such that its canonical map $A/I \to A'/IA'$ is an isomorphism and $f = G'H'$ in $A'[T]$, with monic factors reducing to the prescribed ones.

Proof. If $I = A$, take $A' = 0$, the principal localization at zero; its closed fibre and the prescribed factors are zero, so all requirements hold. Assume $A/I \neq 0$. Put $d = \deg \bar{G}$ and $e = \deg \bar{H}$. Introduce $d + e$ variables for the lower coefficients of monic polynomials G and H of those degrees. The $d + e$ lower coefficients of $GH - f$ are equations in these variables. Write C_0 for their quotient algebra and Δ for the determinant of their square Jacobian matrix.

Over $D = A/I$, the derivative of multiplication at the given factors is

$$D[T]_{<d} \oplus D[T]_{<e} \longrightarrow D[T]_{<d+e}, \qquad (v, w) \longmapsto \bar{H}v + \bar{G}w.$$

Here $D[T]_{<r}$ is the free module on $1, T, \ldots, T^{r-1}$. Monic division works over this arbitrary ring: for a monic polynomial p of degree r, subtract the current leading coefficient times $T^k p$ until the remainder has degree below r. For a nonzero polynomial q, the product qp has degree $\deg q + r$, with the same nonzero leading coefficient as q. Hence no nonzero multiple of p has degree below r, and the quotient and remainder are unique. For $r = 0$, the monic polynomial is 1 and the remainder is zero. Choose a Bezout identity $a\bar{G} + b\bar{H} = 1$. For a polynomial q in the right module, let v be the remainder of bq on monic division by $\bar{G}$. Then $q - \bar{H}v$ is divisible by $\bar{G}$; its quotient w has degree below e. This constructs a preimage of q. If the image of (v, w) is zero, reduction modulo $\bar{G}$ and invertibility of $\bar{H}$ there give $v = 0$, by its degree bound, and monic multiplication then gives $w = 0$. Thus this is an isomorphism even when D has zero divisors. Degree-zero factors are covered by empty polynomial modules.

Consequently Δ has a unit value at the prescribed coefficient tuple. Set $C = (C_0)_\Delta$. The square Jacobian calculation AG-CA-17, Theorem 4.1 gives formal smoothness, and there are no remaining differential coordinates. Theorem 2.2 of that lesson therefore gives formal unramifiedness. The finite equations and principal localization give finite presentation, so C/A is étale. The coefficient tuple defines a D-algebra retraction $\sigma : E = C/IC \to D$ of its structural map.

We select exactly this section in the closed fibre. Let $L = \ker \sigma$. Finite algebra generators x_i of E/D give finite ideal generators $x_i - \sigma(x_i)$ for L: replace each variable by its constant value in a polynomial and telescope the difference one variable at a time. The map

$$\delta : E \longrightarrow L/L^2, \qquad x \longmapsto x - \sigma(x) \pmod{L^2}$$

is a D-derivation, with E acting on L/L^2 through σ. To check products, expand $x = \sigma(x) + \ell$ and $y = \sigma(y) + m$ and discard $\ell m \in L^2$. Since E/D is étale by base change, its differentials vanish by the same Theorem 2.2. The universal differential property makes $\delta = 0$. Its restriction to L is the quotient map, so $L = L^2$.

Choose finite ideal generators ℓ_i and write $\ell_i = \sum_j m_{ij}\ell_j$ with $m_{ij} \in L$. The adjugate identity of S01, Lemma 1.2 shows that $a = \det(1 - M)$ annihilates L, while $a \equiv 1 \pmod{L}$. Put $p = 1 - a \in L$. Then $p\ell = \ell$ for every $\ell \in L$, so $p^2 = p$ and $L = pE$. Inverting $1 - p$ kills precisely L and induces $E_{1-p} \cong E/L = D$, compatibly with its structural map. Lift $1 - p$ to $h \in C$ and put $A' = C_h$. Its closed fibre is this very copy of D, and principal localization preserves finite presentation and étaleness by AG-CA-17, Theorem 1.2 and Section 4. The universal polynomials in $A'[T]$ supply G', H'. If both degrees are zero, one can directly take $A' = A$ and both factors equal to one. □

Lemma F.116: lift idempotent upstairs

Reading source: Stacks, Tag 07M4.

Suppose $A \to B$ is integral and $I \subset A$ is an ideal. Every idempotent $\bar{e} \in B/IB$ lifts to an idempotent of $B \otimes_A A'$ after a finitely presented étale map $A \to A'$ for which $A/I \to A'/IA'$ is an isomorphism.

Proof. Choose $y \in B$ reducing to $\bar{e}$. The determinant proof F.111 gives an integer $d \geq 1$ and a monic $f \in A[T]$ with

$$f(y) = 0, \qquad \bar{f} = T^d(T-1)^d.$$

The two factors generate the unit ideal: in the expansion of $(T + (1-T))^{2d-1} = 1$, every term contains either T^d or $(1-T)^d$. Apply F.115. Over its étale algebra A_1 we have $f = GH$ with reductions T^d and $(T-1)^d$. Put $B_1 = B \otimes_A A_1$, write y_1 for the image of y, and set $b_1 = G(y_1)$, $b_2 = H(y_1)$. Then

$$b_1 b_2 = 0, \qquad \bar{b}_1 = \bar{e}, \quad \bar{b}_2 = (-1)^d(1-\bar{e}).$$

Thus b_1, b_2 generate the unit ideal modulo IB_1.

The map $A_1 \to B_1$ remains integral. Indeed each tensor element uses finitely many elements of B; their monic equations generate a finite module algebra after base change, and its elements are integral by the determinant test AG-CA-05, Theorems 1.1–1.2. Let K be the inverse image in A_1 of the ideal $(b_1, b_2)B_1$. The closed-image and lying-over proof in Theorem 3.4 of that lesson gives

$$\operatorname{image} V_{B_1}(b_1, b_2) = V_{A_1}(K).$$

No prime in this image can contain IA_1, since its prime upstairs would contain IB_1 as well as b_1, b_2. Hence $V(K + IA_1)$ is empty and $K + IA_1 = A_1$, by the prime-separation proof AG-CA-01, Lemma 1.1. Choose $a \in K$ with $a \equiv 1 \pmod{IA_1}$ and set $A' = (A_1)_a$.

In $B' = B_1 \otimes_{A_1} A'$, the image of a is a unit belonging to (b_1, b_2). Choose $u, v \in B'$ with $ub_1 + vb_2 = 1$ and put $e' = ub_1$. The equations give

$$e'(1-e') = uvb_1b_2 = 0,$$

so e' is idempotent. Reducing $ub_1 + vb_2 = 1$ and multiplying by $\bar{e}$ gives $\bar{u}\bar{e} = \bar{e}$, proving that e' reduces to the specified idempotent. Finally a reduces to one, so the last localization leaves the quotient $A_1/IA_1 = A/I$ unchanged. Its composition with $A \to A_1$ is finitely presented étale by the proved stability in AG-CA-17, Theorem 1.2 and Section 4. This supplies every stated property, including the zero closed fibre. □

Lemma F.117: lift projective module

Reading source: Stacks, Tag 07M5.

For an ideal I of any ring A and a finite projective A/I-module $\bar{P}$, there is a finitely presented étale map $A \to A'$ inducing $A/I \cong A'/IA'$, and a finite projective A'-module P' with $P'/IP' \cong \bar{P}$ under that identification.

Proof. Realize $\bar{P}$ as a direct summand of $(A/I)^n$ and let $\bar{p}$ be its projector. Such a realization follows from projectivity by splitting a finite free surjection, as in S01, Lemma 1.0. Lift the finitely many entries of $\bar{p}$ to a matrix φ over A. Its characteristic polynomial $f(T) = \det(T1 - \varphi)$ is monic. The full coefficient proof of Cayley–Hamilton F.19 gives an algebra map

$$B = A[T]/(f) \longrightarrow \operatorname{End}_A(A^n), \qquad x \longmapsto \varphi,$$

where x is the class of T. For $n > 0$, subtracting the leading coefficient times the appropriate power of T times the monic polynomial f reduces any polynomial to degree below n. Conversely, for

nonzero q, the leading coefficient of qf equals that of q and its degree is $\deg q + n$; thus no nonzero multiple of f has degree below n. These two calculations show that $1, x, \dots, x^{n-1}$ are a free A-basis of B, even when A has zero divisors. The proved finite-algebra determinant criterion AG-CA-05, Theorem 1.2 now makes $A \to B$ integral. The case $n = 0$ is handled directly below.

In each residue field of A/I, an idempotent linear operator decomposes the vector space into its image and kernel: $v = \bar{p}v + (1 - \bar{p})v$, and applying $\bar{p}$ proves the intersection zero. Choose bases of the two summands. Its characteristic polynomial there is $T^{n-r}(T-1)^r$ for the dimension r of its image. Consequently at every prime of B/IB the image of x is zero or one: pass to the field of that prime and use this factorization. The nilradical calculation AG-CA-01, Theorem 1.2 then gives an integer $N \geq 1$ such that

$$x^N(1-x)^N = 0 \quad \text{in } E = B/IB.$$

This uses a single nilpotent element, with no Noetherian condition.

Set $a = x^N$, $b = (1-x)^N$ in E. They generate the unit ideal: expand $(x + (1-x))^{2N-1}$, whose every term is divisible by a or b. Also $ab = 0$. Choose u, v with $ua + vb = 1$. Then $(a+b)(u^2a + v^2b) = u^2a^2 + v^2b^2 = (ua+vb)^2 = 1$. Thus $a + b$ is a unit, and

$$\epsilon = \frac{a}{a+b}, \qquad \epsilon(1-\epsilon) = \frac{ab}{(a+b)^2} = 0$$

is an idempotent. Under the reduction of the displayed endomorphism map, a goes to $\bar{p}^N = \bar{p}$, b to $(1-\bar{p})^N = 1 - \bar{p}$, and their sum to the identity. Hence ϵ maps to exactly $\bar{p}$.

Apply F.116 to the integral algebra B/A and this idempotent of B/IB. It supplies the required étale A' and an idempotent of $B \otimes_A A'$ lifting ϵ. Its image in $\mathrm{End}_{A'}((A')^n)$ is a projector p' reducing to $\bar{p}$. Put $P' = \operatorname{im} p'$. The decomposition

$$(A')^n = \operatorname{im} p' \oplus \operatorname{im}(1 - p')$$

follows by writing each vector as the sum of its two projections; applying p' makes their intersection zero. The projections of the finite coordinate basis generate each summand, so P' is finite projective. Reduction preserves this split decomposition and identifies its first summand with $\operatorname{im} \bar{p} = \bar{P}$. If $n = 0$, take $A' = A$ and $P' = 0$ directly. This also handles $A/I = 0$ by choosing its zero module with $n = 0$. □

Lemma F.118: noetherian finite all equivalent

Reading source: Stacks, Tag 09CC, together with the regular-sequence and Koszul arguments cited there. The proof below is independently written for finite modules.

For a nonzero finite module over a Noetherian local ring and a list in its maximal ideal, regularity, vanishing of all positive Koszul homology, vanishing of first Koszul homology, and the polynomial associated-graded condition are equivalent.

Proof. Write $(A, \mathfrak{m})$ for the ring, M for the module, and $J = (f_1, \dots, f_r)$. The fourth assertion means that the natural graded map

$$\theta : (M/JM)[T_1, \dots, T_r] \longrightarrow \bigoplus_{n \geq 0} J^n M / J^{n+1} M, \qquad \bar{m} T^a \longmapsto [m f^a],$$

is an isomorphism. It is always surjective. If $r = 0$, the Koszul complex has only M in degree zero, $J = 0$, and θ is the identity in degree zero. The empty list is regular because $M \neq 0$. Thus all four assertions hold in that case; assume $r > 0$.

Use the exterior complex on the symbols $e_1, \dots, e_r$, with coefficients in M, differential $d(e_i) = f_i$, and the signed product rule. Separating wedges which contain e_r identifies it with the cone of

multiplication by f_r on the complex for the first $r-1$ entries. Its homology sequence contains

$$H_1(K_{r-1}) \xrightarrow{f_r} H_1(K_{r-1}) \longrightarrow H_1(K_r) \longrightarrow H_0(K_{r-1}) \xrightarrow{f_r} H_0(K_{r-1}).$$

For the two consequences needed here, exactness can also be checked directly. A degree-one cycle of K_{r-1} becomes a boundary in K_r when $H_1(K_r) = 0$. Writing that boundary as the sum of a wedge without e_r and a wedge containing e_r shows that its class in $H_1(K_{r-1})$ is f_r times a class represented by a cycle. This gives surjectivity of the first arrow. If $f_r m \in (f_1, \dots, f_{r-1})M$, choose coefficients for this equality to obtain a degree-one cycle in K_r with last coefficient m. A degree-two boundary has last coefficient in $(f_1, \dots, f_{r-1})M$, so $[m] = 0$ in $H_0(K_{r-1})$. This gives injectivity of the last arrow.

If $H_1(K_r) = 0$, the preceding H_1 is a finite A-module, since its complex consists of finite modules over a Noetherian ring. Nakayama, with $f_r \in \mathfrak{m}$, makes it zero by the surjectivity just proved. Induction therefore gives all successive multiplication injections, including the final one displayed above. Their last quotient is nonzero by Nakayama applied to M/JM. This proves regularity. Conversely, assume the list regular and the preceding complex acyclic in positive degrees. A cycle in K_r of degree k has a component v multiplying e_r, and v is a cycle of degree $k-1$ in K_{r-1}. For $k \geq 2$, preceding acyclicity makes v a boundary. For $k = 1$, the cycle identity says $f_r[v] = 0$ in $H_0(K_{r-1})$; regularity again makes v a boundary. Subtract the differential of a suitable wedge containing e_r to remove v. What remains is a degree-k cycle in K_{r-1} and is a boundary by preceding acyclicity. This proves acyclicity in every positive degree by induction. In particular it implies vanishing of H_1.

We next prove that a regular list makes θ injective. In degree zero this is the defining quotient M/JM. In degree one, suppose $\sum_i f_i m_i \in J^2M$. Express this element as $\sum_i f_i u_i$ with $u_i \in JM$, and subtract. The vector $(m_i - u_i)$ is a degree-one Koszul cycle. By the preceding acyclicity it is a boundary, whose coefficients all belong to JM, since the degree-two differential has coefficients f_i. Consequently every $m_i \in JM$. This proves directly the module version of the conormal calculation in F.74.

Here is a coefficient calculation for any degree $n \geq 1$. In a relation

$$\sum_{|a|=n} m_a f^a \in J^{n+1}M,$$

collect every term involving one of $f_1, \dots, f_{r-1}$ into their generated submodule. The only remaining term on the left is $m_{(0,\dots,0,n)} f_r^n$; the only remaining type on the right is $u f_r^{n+1}$. Thus

$$(m_{(0,\dots,0,n)} - u f_r) f_r^n + \sum_{i<r} f_i v_i = 0$$

for suitable $u, v_i \in M$. The list $f_1, \dots, f_{r-1}, f_r^n$ is regular by AG-CA-12, Proposition 1.3, whose power argument applies to modules. Its first Koszul homology is zero by the cone argument. Hence the last coefficient of this relation belongs to $(f_1, \dots, f_{r-1})M$, so $m_{(0,\dots,0,n)} \in JM$.

To recover the coefficients of all monomials, put $B = A[U_1, \dots, U_r, U_r^{-1}]$, $N = M \otimes_A B$, and

$$g_i = f_i - (U_i/U_r) f_r \quad (i < r), \qquad g_r = f_r/U_r.$$

The ring B is free and faithfully flat over A, with its Laurent monomials as a basis. Tensoring preserves Koszul acyclicity. The displayed change of generators is invertible, so its exterior powers identify the two Koszul complexes and give $H_1(K(g; N)) = 0$.

Fix a prime $\mathfrak{p}$ of B. If $N_{\mathfrak{p}} = 0$, every term of every Koszul complex with these coefficients is zero. If $N_{\mathfrak{p}} \neq 0$ and $JB \subset \mathfrak{p}$, the local criterion already proved makes $g_1, \dots, g_r$ regular on the nonzero finite $B_{\mathfrak{p}}$-module $N_{\mathfrak{p}}$. Its power list $g_1, \dots, g_{r-1}, g_r^n$ is regular and has zero H_1. Finally, if $JB \not\subset \mathfrak{p}$, at least one g_i is a unit in $B_{\mathfrak{p}}$. The power list still contains a unit, so choose a_i with

$\sum_{i<r} a_i g_i + a_r g_r^n = 1$. Wedge multiplication by $\sum_i a_i e_i$ satisfies $dh + hd = 1$ by the signed product rule, and contracts its Koszul complex. These three cases show that the power list has zero first homology at every prime. Localization commutes with the complex and its homology; a module whose localizations at all primes vanish is zero. The global H_1 therefore vanishes as well.

The preceding pure-last-coefficient calculation now applies to g on N. Substituting $f_i = g_i + U_i g_r$ for $i < r$ and $f_r = U_r g_r$ makes the coefficient of g_r^n equal to $\sum_{|a|=n} m_a U^a$. Thus this element belongs to JN. Since $N/JN = (M/JM) \otimes_A B$ is the direct sum of its Laurent monomial copies of M/JM, each coefficient m_a belongs to JM. This proves injectivity of θ in every degree.

For the converse, suppose θ is an isomorphism. Krull intersection, AG-CA-03, Theorem 6.1, gives $\bigcap_{n\geq 0} J^n M = 0$. Every nonzero $m \in M$ therefore has a finite order v, characterized by $m \in J^v M \setminus J^{v+1} M$. Its initial class is nonzero; multiplication by T_1 is injective on $(M/JM)[T_1, \ldots, T_r]$, so $f_1 m$ is nonzero and has order $v+1$. In particular f_1 acts injectively on M. The same calculation gives

$$f_1 M \cap J^n M = f_1 J^{n-1} M \qquad (n \geq 1) :$$

if $f_1 m \in J^n M$ and $m \neq 0$, its order forces $m \in J^{n-1} M$; the reverse inclusion follows from $f_1 \in J$.

Set $M' = M/f_1 M$ and let J' be the ideal induced by J. In degree $n \geq 1$, the last equality identifies the kernel of $J^n M/J^{n+1} M \to J'^n M'/J'^{n+1} M'$ with multiplication by T_1 from degree $n-1$. Indeed an element in that kernel is $y + f_1 m$, with $y \in J^{n+1} M$; then $f_1 m \in J^n M$, so $m \in J^{n-1} M$. Degree zero is unchanged. It follows that

$$\operatorname{gr}_{J'} M' \simeq (M/JM)[T_2, \ldots, T_r].$$

Repeat the order and quotient argument for the remaining entries. Nakayama ensures that each finite intermediate module and the final quotient are nonzero. This proves regularity and hence all four equivalences. □

Lemma F.119: p basis

Reading source: AI Integrated Stacks, pinned More Algebra, lines 11972–12025.

Let L/F be a field extension of characteristic $p > 0$ and put $E = FL^p$. A subset $C \subset L$ is *p-independent over F* when its monomials, with finite support and exponents between 0 and $p-1$, are linearly independent over E. It is a *p-basis* when those monomials are a basis of L over E. Independence holds exactly when the differentials dc are linearly independent over L in $\Omega_{L/F}$. Every such subset extends to a p-basis. Moreover C is a p-basis exactly when its differentials form an L-basis of $\Omega_{L/F}$.

Proof. All derivations over F annihilate E: they kill F, the product rule kills pth powers, and the inverse rule kills the field generated by these elements. We first construct the coordinate system for a maximal p-independent set.

Order the p-independent supersets of the prescribed set by inclusion. A chain has an independent union, since any proposed relation involves finitely many elements contained together in one member of the chain. Zorn's lemma gives a maximal set B. For each finite $B_0 \subset B$, the algebra $E[B_0]$ is spanned by its standard monomials, because $b^p \in E$. It is a finite-dimensional domain and therefore a field: multiplication by any nonzero element is an injective linear endomorphism, hence surjective, and its inverse image of one supplies an inverse. The union $E[B]$ is consequently a field as well.

Suppose $x \in L \setminus E[B]$. Its minimal polynomial over that field divides $T^p - x^p$. In $L[T]$ this polynomial is $(T-x)^p$, and unique Euclidean factorization shows that any monic divisor is $(T-x)^r$. If $0 < r < p$ and this divisor were defined over $E[B]$, its next coefficient $-rx$ would put x there, a contradiction. Thus the minimal polynomial has degree p. The power basis of this degree-p extension, together with independence of the old monomials, proves that $B \cup \{x\}$ remains p-independent. Maximality rules this out, so $E[B] = L$.

There is now an exact presentation

$$L = E[T_b : b \in B]/(T_b^p - b^p : b \in B).$$

Surjectivity follows from $E[B] = L$. For injectivity, reduce a polynomial, which involves only finitely many variables, by the displayed monic relations. Its reduced standard-monomial expression can vanish only when every coefficient is zero, by p-independence. This proves the presentation for infinite sets too.

Differentiate polynomials in a single variable T_b, killing E and all other variables. Each defining relation has derivative zero, so this operation descends through the exact presentation to an F-derivation $\partial_b : L \to L$. It sends b to one and every other member of B to zero. These derivations prove that the db are independent. The finite monomial expansion of each element, differentiated by the product rule, proves they span $\Omega_{L/F}$. This establishes the differential-basis assertion for B, and shows that every p-independent subset has independent differentials.

Conversely, let $dc_1, \ldots, dc_r$ be independent. Each coordinate functional on their span extends to all of $\Omega_{L/F}$. For completeness, order the extensions to larger subspaces by extension; chain unions give compatible upper bounds. A maximal extension must have the whole space as domain, since otherwise one can adjoin a vector outside its domain and assign it value zero. This is another application of Zorn's lemma. The universal differential property turns these extended functionals into F-derivations D_i with $D_i(c_j) = \delta_{ij}$, all killing E.

If these elements had a nonzero standard-monomial relation over E, choose its polynomial P with least total degree. This degree is positive. Some partial derivative of P is nonzero: a nonconstant monomial has a positive exponent strictly below p, and distinct monomials containing that variable have distinct differentiated monomials. Applying the corresponding D_i gives the smaller-degree relation $(\partial P/\partial T_i)(c) = 0$, a contradiction. Thus the finite set is p-independent. Since every relation on any subset is finite, this proves the equivalence in general.

Finally, if the dc form a basis, the equivalence makes C p-independent. Extend it to B by the construction above. An additional $b \in B \setminus C$ would have a differential independent of all the dc, contradicting their spanning property. Hence $B = C$ and C is a p-basis. The converse has already been proved by the presentation. Empty bases and arbitrary infinite transcendence degree are included. □

Lemma F.120: power series ring subfields

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12130–12171.

Let k have characteristic $p > 0$ and let B be a p-basis of k over $\mathbf{F}_p$. Set

$$A = k[[X_1, \ldots, X_n]][Y_1, \ldots, Y_m], \qquad K = \mathrm{Frac}(A).$$

For each finite $J \subset B$, define

$$k_J = k^p(B \setminus J), \quad A_J = k_J[[X_1^p, \ldots, X_n^p]][Y_1^p, \ldots, Y_m^p], \quad K_J = \mathrm{Frac}(A_J).$$

Then A is finite free over A_J. The subfields K_J contain K^p, are downward directed, and have intersection K^p.

Proof. The whole F.119 gives the standard monomial basis of k over k^p. In particular the subalgebra generated by any subset of B is a field: each element lies in a finite-dimensional domain generated by finitely many of those elements, and multiplication by a nonzero element gives its inverse there. Thus $k_J = k^p[B \setminus J]$.

The $p^{|J|}$ monomials in J form a basis of k over k_J. They span by grouping each finite standard-monomial expansion according to its exponents in J. For independence, expand any finitely many

coefficients from k_J in the monomials of $B \setminus J$. A relation then becomes a relation between distinct monomials in the full p-basis, so all coefficients vanish. The same uniqueness shows

$$\bigcap_{J \subset B,\ J \text{ finite}} k_J = k^p :$$

if an element's unique finite expansion has a monomial involving $b \in B$, it cannot belong to $k_{\{b\}}$.

The algebra A is a domain. For two nonzero power series, the product of their first nonzero total-degree homogeneous polynomials is nonzero: polynomial domains follow by taking a leading monomial. This product is the first nonzero homogeneous part of their product. Adjoining the polynomial variables preserves the domain property by the same leading-monomial test.

A basis of A as an A_J-module is the finite list

$$\left\{ \prod_{b \in J} b^{e_b} \prod_i X_i^{r_i} \prod_j Y_j^{s_j} : 0 \le e_b, r_i, s_j < p \right\}.$$

Indeed expand each coefficient in the finite k_J-basis just proved, and group the X and Y exponents by their residues modulo p. The grouped coefficients are power series in X^p and polynomials in Y^p over k_J. There are finitely many Y terms initially, so these expressions remain polynomials. Coefficient comparison and uniqueness of the field basis prove linear independence. The rank is $p^{|J|+n+m}$, including empty variable lists.

The characteristic-p identity for a series is computed in each finite degree, where only finitely many terms contribute. It gives $a^p \in A_J$ for every $a \in A$, hence $K^p \subset K_J$. Also $k_{J \cup J'} \subset k_J \cap k_{J'}$, so $K_{J \cup J'} \subset K_J \cap K_{J'}$. This proves downward directedness.

It remains to identify the intersection; fractions cannot be compared by their coefficients directly. Write a candidate as $u = f/g^p$ with $f, g \in A$ and $g \neq 0$, by multiplying an arbitrary numerator and denominator by the $(p-1)$st power of its denominator. Fix finite J and suppose $u \in K_J$. Writing it as a/h with $a, h \in A_J$, $h \neq 0$, gives

$$u = (ah^{p-1})/h^p.$$

Rename this numerator a and put $b = h$. Cross multiplication yields $b^p f = ag^p \in A_J$.

Any A_J-derivation $D : A \to A$ consequently gives $b^p D(f) = 0$, because the derivatives of pth powers vanish. The domain property makes $D(f) = 0$. The formal partial derivatives in every X_i and Y_j are such derivations. Their product rules hold coefficient by coefficient, since each convolution coefficient is a finite sum. Distinct monomials containing a given variable have distinct derivative monomials; their scalar exponents are nonzero precisely when not divisible by p. Therefore only exponents divisible by p can occur in f.

For $b \in J$, the coordinate derivation of k/k_J differentiates its unique J-monomial expansion. It kills k_J and exists by the exact presentation in F.119, or by restricting its b-coordinate derivation. Its extension to A applies it to each coefficient and kills X, Y; the finite convolution formula again proves the product rule. The common kernel of these derivations on k is exactly k_J: differentiation of the finite J-basis expansion kills precisely the coefficient terms with a positive exponent in an excluded variable. Thus every coefficient of f belongs to k_J.

If u belongs to every K_J, all these conclusions hold for every finite J. Its coefficients then lie in k^p by the field intersection above, and all its variable exponents are multiples of p. Choose the unique pth root of each coefficient and divide each exponent by p; this defines an element $v \in A$ with $v^p = f$. The polynomial variables still have bounded degree. It follows that $u = (v/g)^p \in K^p$. Together with the first inclusion this proves the claimed intersection, even with an infinite coefficient p-basis. $\square$

Lemma F.121: quasi regular ideal finite projective

Reading source: AI Integrated Stacks, pinned More Algebra, lines 8769–8778.

For an arbitrary ring R and quasi-regular ideal I, the module I/I^2 is finite projective over R/I.

Proof. Put $S = R/I$ and $M = I/I^2$. If $S = 0$, then $I = R$ and $M = 0$, so the assertion holds. Otherwise the explicit quasi-regularity definition F.95 gives principal neighbourhoods of all primes of S on which the ideal is generated by a finite quasi-regular list. The degree-one graded isomorphism identifies its conormal module with the free module on that list. Exact localization gives the same identification for M_g over S_g.

The chart denominators generate S: if their ideal were proper, a maximal ideal containing it would lie outside every chart. A finite list $g_1, \dots, g_s$ therefore already generates the unit ideal and supplies finite free localizations of M. We record the detection argument needed without any finiteness assumption on a module N. If $N_{g_i} = 0$ for every i, each $x \in N$ satisfies $g_i^{e_i} x = 0$ for positive integers e_i. These powers generate the unit ideal too, since a maximal ideal containing all of them would contain all the g_i. Multiply an expression $1 = \sum_i a_i g_i^{e_i}$ by x to obtain $x = 0$. Hence $N = 0$.

For every local basis, represent its finitely many elements by fractions from M and take their numerators. Multiplication by the corresponding powers of the unit g_i still gives a basis locally. The finite submodule generated by all these numerators has quotient zero on every chart; the detection argument makes that quotient zero globally. Thus M is finite. Choose a finite free surjection $S^t \to M$ with kernel H.

Localizing on a chart gives a surjection onto a finite free module, so it splits by lifting a basis. The complementary projection of the t standard vectors then generates H_{g_i}. Exact localization identifies this kernel with the localization of H; clear the finitely many denominators of its displayed generators and collect their numerators from H. They generate a finite submodule H_0 with $(H/H_0)_{g_i} = 0$ everywhere. Detection again gives $H = H_0$, so M is finitely presented.

Every stalk of M is free, because a chart contains that prime. The entire AG-CA-07, Theorem 3.3 gives flatness over S. The full proof of AG-CA-07, Theorem 5.3 then splits a finite presentation of this flat module, making it a direct summand of a finite free module. This is finite projectivity. The argument supplies the missing global finite presentation and applies without Noetherianity or finite generation of I in R. □

Lemma F.122: quotient regular

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12407–12424.

Let R be a Noetherian regular ring. If a derivation $D : R \to R$ satisfies that $D(f)$ is a unit in $R/(f)$, then $R/(f)$ is regular.

Proof. Fix a prime $\mathfrak{p}$ containing f and set $V = R_{\mathfrak{p}}$, $\mathfrak{m} = \mathfrak{p}V$. We first extend the derivation without a finiteness or continuity assumption. The map

$$r \longmapsto r + \epsilon D(r), \qquad \epsilon^2 = 0,$$

is a ring homomorphism to $V[\epsilon]/(\epsilon^2)$ by the product rule. For $s \notin \mathfrak{p}$ its image has the explicit inverse $s^{-1} - \epsilon s^{-2} D(s)$. The universal property of localization extends this map to V. Its constant component is the identity, so its ϵ component is a derivation $D_V : V \to V$ extending D, with

$$D_V(a/s) = D(a)/s - aD(s)/s^2.$$

This construction proves well-definedness and the product rule for the fraction formula.

The hypothesis gives $uD(f) + vf = 1$ for some $u, v \in R$. At $\mathfrak{p}$, this makes $D_V(f)$ a unit modulo $\mathfrak{m}$, and in particular outside $\mathfrak{m}$. If f belonged to $\mathfrak{m}^2$, express it as a finite sum $\sum_j a_j b_j$ with $a_j, b_j \in \mathfrak{m}$.

The product rule would give

$$D_V(f) = \sum_j (a_j D_V(b_j) + b_j D_V(a_j)) \in \mathfrak{m},$$

a contradiction. Thus $f \in \mathfrak{m} \setminus \mathfrak{m}^2$.

Regularity of R makes V regular local. The entire AG-CA-14, Corollary 1.2 proves V/fV regular: extend the nonzero class of f to a cotangent basis, lift it, and apply the proved successive parameter-quotient theorem. These quotients are exactly the local rings of $R/(f)$ at its corresponding primes. The quotient is Noetherian, so this proves regularity. If no prime contains f, the quotient is zero and has no local rings to test. □

Lemma F.123: regular base change

Reading source: AI Integrated Stacks, pinned More Algebra, lines 11022–11047.

For a regular map $R \to \Lambda$ of Noetherian rings and any finite-type R-algebra R', the map

$$R' \longrightarrow T = \Lambda \otimes_R R'$$

is regular.

Proof. Both R' and T are Noetherian: a finite algebra-generating list for R' makes it a quotient of a polynomial ring over R, and makes T a quotient of the corresponding polynomial ring over Λ. The entire Hilbert basis theorem AG-CA-03, Theorem 2.1, applies to each. The full tensor proof AG-CA-07, Proposition 3.2 proves flatness of T over R' by base change.

Let $\mathfrak{p}'$ be any prime of R', with contraction $\mathfrak{p}$ to R. Put $k = \kappa(\mathfrak{p})$ and $E = \kappa(\mathfrak{p}')$. The field inclusion $k \hookrightarrow E$ is induced by the domain inclusion $R/\mathfrak{p} \hookrightarrow R'/\mathfrak{p}'$. If $u_1, \dots, u_n$ are algebra generators, localizing this domain inclusion at $R \setminus \mathfrak{p}$ gives

$$(R \setminus \mathfrak{p})^{-1}(R'/\mathfrak{p}') = k[\bar{u}_1, \dots, \bar{u}_n].$$

Its fraction field is still E, since this intermediate localization does not change the fraction field of the domain. Thus $E = k(\bar{u}_1, \dots, \bar{u}_n)$ is a finitely generated field extension of k. This step requires neither algebraicity nor separability.

Set $F = \Lambda \otimes_R k$. The original regular map gives a Noetherian geometrically regular k-algebra F. Tensor associativity, with the indicated residue-field maps, gives the actual new fibre

$$T \otimes_{R'} E \cong \Lambda \otimes_R E \cong F \otimes_k E.$$

The entire F.12 makes it Noetherian from the finite field-generating list. Assertion (2) of the complete F.52 makes it geometrically regular over E; its required Noetherian tensor hypothesis has just been checked. Every new fibre satisfies the criterion, and the map is flat, so it is regular. Empty spectra require no fibre test. □

Lemma F.124: regular composition

Reading source: AI Integrated Stacks, pinned More Algebra, lines 11049–11065.

If $A \to B$ and $B \to C$ are regular maps of Noetherian rings, then $A \to C$ is regular.

Proof. The complete flat-composition calculation AG-CA-07, Proposition 3.2, makes the composite flat. Fix $\mathfrak{a} \in \operatorname{Spec} A$ and write $k_0 = \kappa(\mathfrak{a})$. Its two relevant fibres are Noetherian:

$$B \otimes_A k_0 = (A \setminus \mathfrak{a})^{-1}(B/\mathfrak{a}B), \qquad C \otimes_A k_0 = (A \setminus \mathfrak{a})^{-1}(C/\mathfrak{a}C).$$

They are localizations of quotients of the stipulated Noetherian rings.

Test a finite purely inseparable field extension K/k_0, including the trivial one. Put

$$D = B \otimes_A K, \qquad E = C \otimes_A K.$$

The full F.12 makes both Noetherian. Regularity of $A \to B$ makes D regular. Moreover $E \cong C \otimes_B D$, and the full flat base-change proof makes $D \to E$ flat.

Choose $\mathfrak{r} \in \operatorname{Spec} E$ and contract it to $\mathfrak{s} \subset D$ and then $\mathfrak{b} \subset B$. The contraction of $\mathfrak{b}$ to A is $\mathfrak{a}$: its elements vanish in K, while every element of $A \setminus \mathfrak{a}$ is inverted there. Consequently k_0 embeds in $\kappa(\mathfrak{b})$. The residue-field change relevant at this point is finite, as we now check directly.

There is a natural finite-dimensional algebra

$$Q = D \otimes_B \kappa(\mathfrak{b}) \cong K \otimes_{k_0} \kappa(\mathfrak{b}).$$

A finite k_0-basis of K gives its finite basis over $\kappa(\mathfrak{b})$. Since $\mathfrak{s}$ contracts to $\mathfrak{b}$, quotienting by $\mathfrak{b}D$ and inverting $B \setminus \mathfrak{b}$ carries it to a prime $\bar{\mathfrak{s}}$ of Q. Its quotient is a finite-dimensional domain over $\kappa(\mathfrak{b})$, hence a field: multiplication by a nonzero element is an injective finite-dimensional linear map and supplies an inverse. This quotient is also the localization of $D/\mathfrak{s}$ at the nonzero images of $B \setminus \mathfrak{b}$. A field containing that domain and obtained by this localization is its full fraction field. Hence

$$Q/\bar{\mathfrak{s}} = \kappa(\mathfrak{s}), \qquad [\kappa(\mathfrak{s}) : \kappa(\mathfrak{b})] < \infty.$$

Tensor associativity identifies the fibre of $D \to E$ at $\mathfrak{s}$ with

$$E \otimes_D \kappa(\mathfrak{s}) \cong (C \otimes_B \kappa(\mathfrak{b})) \otimes_{\kappa(\mathfrak{b})} \kappa(\mathfrak{s}).$$

The B-fibre in parentheses is Noetherian and geometrically regular. The finite residue-field extension just proved and F.12 make the displayed tensor Noetherian; the whole F.52 makes it regular. The closed fibre of the actual flat local map $D_{\mathfrak{s}} \to E_{\mathfrak{r}}$ is a localization of this ring and is regular. Its source is regular local, and both local rings are Noetherian. The entire F.47 therefore makes $E_{\mathfrak{r}}$ regular.

This holds at every target prime; if there are none, the assertion is vacuous. Thus E is regular for every finite purely inseparable K/k_0. The complete criterion F.50 now proves that the original composite fibre is geometrically regular. Applying this at every $\mathfrak{a}$ proves regularity of the composite, with all fibre Noetherianity conditions verified. □

Lemma F.125: regular permanence

Reading source: AI Integrated Stacks, pinned More Algebra, lines 11116–11129.

For maps $A \to B \to C$ of Noetherian rings, if $A \to C$ is regular and $B \to C$ is faithfully flat, then $A \to B$ is regular.

Proof. Take any inclusion of A-modules $U \hookrightarrow V$ and let

$$H = \ker(U \otimes_A B \longrightarrow V \otimes_A B).$$

Flatness of C over B identifies $H \otimes_B C$ with the kernel of this map after tensoring with C. Tensor associativity identifies that map with $U \otimes_A C \to V \otimes_A C$, which is injective because $A \to C$ is flat. Hence $H \otimes_B C = 0$; faithfulness gives $H = 0$. Tensoring every inclusion with B is therefore injective, proving flatness of $A \to B$.

For a prime $\mathfrak{a} \subset A$, put $k = \kappa(\mathfrak{a})$ and

$$F = B \otimes_A k, \qquad G = C \otimes_A k.$$

These are Noetherian localizations of $B/\mathfrak{a}B$ and $C/\mathfrak{a}C$ at $A \setminus \mathfrak{a}$. Their structural map is faithfully flat. Indeed $G \cong F \otimes_B C$, and for every F-module N the natural balancing maps give

$$N \otimes_F G \cong N \otimes_B C.$$

Restriction to B preserves exact sequences and detects a zero module. Since the right tensor functor is exact and detects zero, the left one has both properties too. This proves faithful flatness on the actual fibre rings, without a finiteness assumption on the residue-field maps.

Regularity of $A \to C$ makes G geometrically regular over k. The entire geometric descent proof F.51 therefore makes F geometrically regular. This checks every fibre of the flat map $A \to B$, so that map is regular. If a target fibre is zero, faithfulness forces its source fibre to be zero as well; the same argument and empty local-ring tests apply. □

Lemma F.126: relative global complete intersection koszul

Reading source: Stacks, Tag 07D2. The following proof gives the Noetherian form used in this lesson.

Let R be Noetherian, $P = R[X_1, \dots, X_n]$, and $S = P/(f_1, \dots, f_c)$. Suppose each nonempty fibre of $R \to S$ has dimension $n - c$, as in Definition F.4. The Koszul complex $K_P(f_1, \dots, f_c)$ has zero homology in every positive degree.

Proof. Let $I = (f_1, \dots, f_c)$ and fix a prime $\mathfrak{p}$ of P. If $I \subset \mathfrak{p}$, the complete relative conormal argument F.77 makes this ordered list regular in $P_{\mathfrak{p}}$. The cone induction in F.118 then makes its localized Koszul complex acyclic in positive degrees. That implication only needs the successive multiplication injections; it imposes no assertion of global ordered regularity outside $V(I)$.

If $I \not\subset \mathfrak{p}$, write $\sum_i a_i f_i = 1$ in $P_{\mathfrak{p}}$. On the exterior complex with basis e_i and differential $d(e_i) = f_i$, set $h(v) = (\sum_i a_i e_i) \wedge v$. The signed product rule gives

$$dh(v) + hd(v) = \Big(\sum_i a_i f_i\Big) v = v.$$

Thus every localized cycle is a boundary in this case too. Localization is exact and therefore commutes with kernels, images and homology, by AG-RG-S01, Lemma 1.1. Each positive homology module now has zero localization at every prime. The local-detection proof in that same lemma makes it zero. When $c = 0$, there are no positive complex terms. When $S = 0$, every prime is in the contracting-homotopy case. Both cases satisfy the assertion. □

Lemma F.127: relative regular immersion algebra

Reading source: Stacks, Tag 0CEP. All tensor steps are specified below.

Take ring maps $A \to B$ and $A \to A'$, a finite list $f_1, \dots, f_r \in B$, and $J = (f_1, \dots, f_r)$. Assume B/J is flat over A. Put $B' = B \otimes_A A'$, let f_i' be the images, and put $J' = (f_1', \dots, f_r')$. If the original list is quasi-regular, so is the new list. If its first Koszul homology vanishes, the new first Koszul homology vanishes. No flatness of B is assumed.

Proof. We use the exact quotient-flatness statement AG-CA-07, Theorem 6.1: a short exact sequence with flat quotient remains short exact under any tensor functor, and an extension of two flat modules is flat.

In the quasi-regular case Definition F.95 identifies

$$J^n/J^{n+1} \simeq (B/J)^{\oplus\{a \in \mathbf{N}^r : |a| = n\}}.$$

These modules are A-flat. Starting with B/J, the sequences

$$0 \to J^n/J^{n+1} \to B/J^{n+1} \to B/J^n \to 0$$

therefore prove by induction that B/J^n is flat for every $n \geq 1$. Apply quotient-flatness to $0 \to J^n \to B \to B/J^n \to 0$. Its tensor identifies $J^n \otimes_A A'$ with its image in B'. That image is exactly $(J')^n$: the finite products of the f_i generate J^n, and their images generate $(J')^n$. Right exactness applied to $J^{n+1} \to J^n \to J^n/J^{n+1} \to 0$ now gives

$$(J')^n/(J')^{n+1} \simeq (J^n/J^{n+1}) \otimes_A A'.$$

In degree zero it gives $B'/J' = (B/J) \otimes_A A'$. The identifications commute with multiplication, since every one is induced by the images of the same products of generators. They carry each T_i to f_i' and prove the required polynomial graded-ring identity.

For the first-homology assertion, put $Z = \ker(B^r \xrightarrow{(f_i)} J)$. Quotient-flatness first gives an injection $J \otimes_A A' \hookrightarrow B'$, whose image is J'. Right exactness of

$$Z \longrightarrow B^r \longrightarrow J \longrightarrow 0$$

then makes $Z \otimes_A A'$ surject onto $Z' = \ker((B')^r \xrightarrow{(f_i')} B')$. The original Koszul boundaries are the image of $\bigwedge^2 B^r \to Z$. If $H_1 = 0$, this map is surjective. Tensoring preserves its surjectivity, and the natural identification $(\bigwedge^2 B^r) \otimes_A A' = \bigwedge^2 (B')^r$ carries its alternating boundary formula to the new formula. Consequently every element of Z' is a new boundary, as required. For an empty list $J = 0$ and all first complex terms vanish; both conclusions still hold. Unit ideals and zero rings cause no change to these exact-sequence arguments. □

Lemma F.128: symmetric algebra smooth

Reading source: Stacks, Tag 07M6. The universal-property and lifting argument is written here for arbitrary modules and rings.

For an A-module M, its symmetric algebra $C = \mathrm{Sym}_A(M)$ is smooth over A precisely when M is finite projective. The base-change, direct-sum and free-module identities used in D.10 hold without finiteness hypotheses.

Proof. Start with $\bigoplus_{d \geq 0} M^{\otimes_A d}$, where degree zero is A. Concatenation of tensors defines its product: the balancing relations defining a tensor product are preserved by concatenation, and associativity holds on pure tensors, which generate. Quotient by the homogeneous relations $m \otimes n - n \otimes m$. Interchanging adjacent factors makes the quotient commutative. This constructs C, with $C_0 = A$, $C_1 = M$, and all its pieces generated by products from degree one. Projection $\epsilon : C \to A$ is an algebra map. If $J = \ker \epsilon$, the homogeneous degree calculation gives $J/J^2 = M$.

For a commutative A-algebra D, an A-linear map $u : M \to D$ extends by the formula

$$m_1 \otimes \cdots \otimes m_d \longmapsto u(m_1) \cdots u(m_d).$$

It respects balancing and the displayed commutation relations, hence defines an algebra map $C \to D$. Every algebra map has this restriction and is determined by it. These constructions are inverse, proving the universal property.

That property gives actual mutually inverse maps

$$\begin{aligned} B \otimes_A \mathrm{Sym}_A(M) &\simeq \mathrm{Sym}_B(B \otimes_A M), \\ \mathrm{Sym}_A(M \oplus N) &\simeq \mathrm{Sym}_A(M) \otimes_A \mathrm{Sym}_A(N), \\ \mathrm{Sym}_A(A^n) &\simeq A[X_1, \ldots, X_n]. \end{aligned}$$

In the first pair of maps send $b \otimes m$ to $b \otimes m$, and send m back to $1 \otimes m$; the tensor-algebra coproduct calculation in AG-RG-S01, Lemma 1.0 combines the latter map with the base map from B. For the second, send (m, n) to $m \otimes 1 + 1 \otimes n$; the inverse comes from the two inclusions into $M \oplus N$ and the same coproduct formula. For the last, send basis vectors to variables and variables

back to basis vectors. Each composite fixes the base ring and every degree-one generator, so is identity. The maps preserve degree. These verifications prove the identities, including reduction modulo a base ideal, used later in the lesson.

Suppose first C is smooth. It is finitely presented, hence has algebra generators $z_1, \dots, z_s$. The differences $y_i = z_i - \epsilon(z_i)$ generate J: for a polynomial P, successive subtraction in each variable expresses $P(z) - P(\epsilon(z))$ as a sum of multiples of those differences. For an element of J the second term is zero. Thus $M = J/J^2$ is finite. Choose a finite free surjection $\pi : F \to M$.

For any module V, form the square-zero algebra $A \oplus V$, with multiplication $(a, v)(b, w) = (ab, aw+bv)$. The map $A \oplus F \to A \oplus M$ induced by π is surjective with square-zero kernel. The linear map $m \mapsto (0, m)$ extends to an algebra map $C \to A \oplus M$. The formal-smoothness comparison proved in AG-CA-17, Theorem 3.1 and Section 4 gives a lift to $A \oplus F$. Its first coordinate on m is zero, because reduction leaves that coordinate unchanged. Write its second coordinate as $s(m)$. Restriction of the algebra map makes s linear and $\pi s = 1_M$. Hence

$$F = s(M) \oplus \ker \pi,$$

as the decomposition $f = s(\pi f) + (f - s(\pi f))$ and application of π verify. This is exactly finite projectivity of M.

Conversely write $F = A^n = M \oplus K$ with maps $i : M \to F$ and $\pi : F \to M$ satisfying $\pi i = 1_M$. The projections of a basis of F generate K. By the universal property, C is the quotient of $A[X_1, \dots, X_n]$ by the linear forms corresponding to K: maps from that quotient are precisely linear maps $F \to D$ killing K, and hence precisely linear maps $M \to D$. The generator maps give the inverse algebra identifications. Since K has the finite generating list just exhibited, this is a finite presentation of C.

Now take a square-zero surjection $D \to D/H$ of A-algebras and a map $C \to D/H$. Its restriction is a linear $u : M \to D/H$. Lift the finitely many values of $u\pi$ on the free basis of F to elements of D. The resulting linear map $v : F \to D$ reduces to $u\pi$, so vi lifts u. Its algebra extension lifts the original map from C. Thus C is formally smooth; its finite presentation and the same proved formal/conormal comparison make it smooth. The constructions include $M = 0$, $n = 0$, and the zero ring. □

Lemma F.129: syntomic lci

Reading source: Stacks, Tag 07D3. We use the Noetherian-base version and specify the syntomic conditions directly.

Let R be Noetherian. A map $R \to S$ is syntomic if and only if it is flat and a local complete intersection homomorphism in Definition F.94. Here syntomic means flat, finitely presented, with local complete intersection fibres as in Definition F.2.

Proof. If $R \to S$ is syntomic, the complete principal-neighbourhood construction F.87 covers $\operatorname{Spec} S$ by charts that have relative global complete intersection presentations. On each chart F.126 makes the presenting equations Koszul-regular. Thus each chart is a local complete intersection homomorphism. A finite collection of the principal charts suffices, and F.114 proves their local complete intersection property descends to $R \to S$. Flatness was one of the syntomic conditions.

Conversely suppose $R \to S$ is flat and a local complete intersection. The finite-kernel and arbitrary-presentation proof F.94.A makes it finitely presented. Choose $S = P/I$ with $P = R[X_1, \dots, X_n]$. For a field-valued base map $R \to k$, put $P' = k[X_1, \dots, X_n]$ and $S' = S \otimes_R k = P'/I'$, with I' the image of I.

Fix a prime $\mathfrak{q}'$ of S' and its inverse image $\mathfrak{p}'$ in P'. Contract to a prime $\mathfrak{p}$ of P. Definition F.95 and the presentation comparison give $g \notin \mathfrak{p}$ such that I_g has a finite Koszul-regular generating list $f_1, \dots, f_r$. Its first Koszul homology is zero. The quotient $P_g/I_g = S_g$ is R-flat, since localization is

flat and flat maps compose. Apply F.127 to $R \to P_g$ and $R \to k$. It proves first-homology vanishing for the image list in P'_g; this list generates I'_g by tensor right exactness. The image of g is outside $\mathfrak{p}'$ by contraction. Localize at $\mathfrak{p}'$. This is a Noetherian polynomial local ring, the list lies in its maximal ideal, and F.118 makes it regular there. The full field complete-intersection neighbourhood argument F.57 therefore gives a global complete intersection chart of S' around $\mathfrak{q}'$. Since every prime of every fibre has such a chart, the fibres meet Definition F.2. Together with flatness and finite presentation this proves syntomicity. If a fibre or S is zero there are no primes requiring charts; the same conditions and argument remain valid. □

Lemma F.130: transitive lci at end

Reading source: Stacks, Tag 07D4. The two rows and their maps are written explicitly.

Let $A \to B \to C$ be maps, with $B \to C$ a local complete intersection homomorphism. Choose $P = A[X_s : s \in \Sigma] \twoheadrightarrow B$ with kernel I, allowing an arbitrary set Σ. Choose $B[Y_1, \dots, Y_m] \twoheadrightarrow C$ with kernel J. Put $Q = P[Y_1, \dots, Y_m]$ and let K be the kernel of $Q \to C$. There are natural short exact rows of C-modules

$$0 \to (I/I^2) \otimes_B C \to K/K^2 \to J/J^2 \to 0,$$
$$0 \to \Omega_{P/A} \otimes_P C \to \Omega_{Q/A} \otimes_Q C \to \Omega_{B[Y]/B} \otimes_{B[Y]} C \to 0.$$

The conormal differential maps from the first row to the second commute. In particular the cotangent transitivity sequence is exact with zero at the left:

$$0 \to H_1(\mathrm{NL}_{B/A} \otimes_B C) \to H_1(L_{C/A}) \to H_1(L_{C/B})$$
$$\to \Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to \Omega_{C/B} \to 0.$$

Proof. The kernel of $Q \to B[Y]$ is IQ, by comparing polynomial coefficients, and $J = K/IQ$. The arbitrary-ring presentation comparison F.94.A makes J a Koszul-regular ideal in this chosen finite polynomial presentation. In particular it is H_1-regular. Apply the intersection identity F.101 to $Q \to Q/IQ$ and the ideal K. It gives

$$IQ \cap K^2 = (IQ)K.$$

Thus the kernel of $K/K^2 \to J/J^2$ is $IQ/(IQ)K$, embedded in K/K^2. On the other hand

$$IQ/(IQ)K \simeq (I/I^2) \otimes_B C.$$

To check this identity directly, polynomial coefficients identify $IQ/I^2Q = (I/I^2) \otimes_B B[Y]$. Quotienting by the products with $J = K/IQ$ gives the right-hand tensor with $C = B[Y]/J$, and gives $IQ/(IQ)K$ on the left; $I^2Q \subset (IQ)K$ already. The maps send a generator class of I to its class in K/K^2. This proves exactness of the first row, including its left injection and right surjection.

The second row is the split sequence of free differential modules on the X symbols, on both the X and Y symbols, and on the Y symbols. Inclusion and projection give the maps, and inclusion of the Y symbols is a splitting. These statements remain true for an infinite set Σ, since each polynomial and differential involves finitely many symbols. The vertical maps send a relation f to df after reduction in C. Including a relation from I, or reducing one from K modulo IQ, commutes with this differentiation; hence the square on each pair of maps commutes.

Apply the complete snake argument F.84 to these actual short exact rows. Their kernel and cokernel sequence begins with zero and ends with zero. The polynomial-presentation homotopy comparison F.11 identifies the middle and right kernels with the indicated first cotangent homologies and their cokernels with the corresponding differential modules. For the left vertical map, right exactness identifies its source and target with the terms of $\mathrm{NL}_{B/A} \otimes_B C$. Its kernel is therefore $H_1(\mathrm{NL}_{B/A} \otimes_B C)$, and its cokernel is $\Omega_{B/A} \otimes_B C$. No flatness of $B \to C$ is needed or inferred in

this identification. The representative description of the snake connecting map agrees with the cotangent map in F.42: lift a relation, differentiate and take the class in the left cokernel. Thus the displayed sequence has the specified canonical maps. Empty variable lists and the zero target obey the same quotient and complex formulas. □

Lemma F.131: transitivity gamma

Reading source: Stacks, Tag 07E2.

For any tower of fields $K \subset L \subset M$, the natural sequence

$$\begin{aligned} 0 \longrightarrow H_1(L_{L/K}) \otimes_L M \longrightarrow H_1(L_{M/K}) \longrightarrow H_1(L_{M/L}) \\ \longrightarrow \Omega_{L/K} \otimes_L M \longrightarrow \Omega_{M/K} \longrightarrow \Omega_{M/L} \longrightarrow 0. \end{aligned}$$

is exact, including its first injection. No finite-generation assumption is imposed on the fields.

Proof. First replace M by a global complete-intersection L-algebra $C = L[Y_1, \dots, Y_r]/J$, with J generated by a regular list. Choose a polynomial presentation $P = K[X_s] \twoheadrightarrow L$ with kernel I, allowing an arbitrary set of variables. Let $Q = P[Y_1, \dots, Y_r]$ and let $H = \ker(Q \to C)$; the image of H in $Q/IQ = L[Y]$ is J.

The regular-list Koszul argument in F.118 proves vanishing of its first homology; that direction uses only the successive multiplication injections. Thus F.101, applied to $IQ \subset H$ in Q, gives $IQ \cap H^2 = IH$. Polynomial coefficient expansion identifies

$$IQ/IH = (I/I^2) \otimes_L C.$$

Indeed first reduce I modulo I^2, extend polynomial coefficients to $L[Y]$, and then impose the equations of J; these are precisely the relations obtained by dividing IQ by IH. Lifting representatives of J to H gives the short exact conormal row

$$0 \longrightarrow (I/I^2) \otimes_L C \longrightarrow H/H^2 \longrightarrow J/J^2 \longrightarrow 0.$$

The first injection is exactly the intersection equality just proved. Differentiation, with coefficients evaluated in C, maps this row to the split exact row

$$0 \longrightarrow \Omega_{P/K} \otimes_P C \longrightarrow \Omega_{Q/K} \otimes_Q C \longrightarrow \Omega_{L[Y]/L} \otimes_{L[Y]} C \longrightarrow 0.$$

Its maps include the free dX_s coordinates and project to the free dY_i coordinates. Differentiating any lifted polynomial relation verifies commutativity. The complete snake proof F.84 now gives the asserted six-term kernel/cokernel sequence for C, with zero at both ends. Presentation comparison F.11 identifies its kernels and cokernels with the named first conormal homologies and differential modules. For the left terms, tensoring over the field L is exact, so they are $H_1(L_{L/K}) \otimes_L C$ and $\Omega_{L/K} \otimes_L C$.

Finally F.23 writes M as a directed union of global complete-intersection L-subalgebras of this kind. The relation and differential comparison F.24 identifies their cotangent complexes over both K and L with those of the union. The sequence maps agree at each inclusion because they were defined by the same polynomial representatives and their derivatives. Filtered colimits are exact: a representative killed in a colimit is killed at some common later stage, and finite sums and equalities can all be placed at one such stage, as proved in F.24. The tensor generators and their balanced relations likewise give $\varinjlim (C \otimes_L V) = M \otimes_L V$ for each L-vector space V. Taking the colimit of the exact sequences therefore proves the full sequence for M, including its first zero. Empty equation lists and infinitely generated intermediate fields obey the same argument. □

Proposition F.132: Noetherian complete G ring

Reading source: AI Integrated Stacks, pinned More Algebra, lines 12965–12999.

Every complete Noetherian local ring is a G-ring.

Proof. We first prove the following generic-fibre assertion for every complete Noetherian local domain B, with fraction field K: for each prime $\mathfrak{p} \subset B$,

$$\widehat{B_{\mathfrak{p}}} \otimes_B K$$

is geometrically regular over K. Choose the embedded regular complete local subring $A_0 \subset B$ of F.25. It can be chosen as a power-series ring over a field or a Cohen ring, and B is finite over it. Write $F = \mathrm{Frac}(A_0)$ and $\mathfrak{p}_0 = \mathfrak{p} \cap A_0$.

The localization $(A_0)_{\mathfrak{p}_0}$ is regular local by AG-CA-14, Theorem 3.2, and its completion is Noetherian regular local by AG-CA-19, Theorem 4.1. Its localization

$$G = \widehat{(A_0)_{\mathfrak{p}_0}} \otimes_{A_0} F$$

is therefore Noetherian regular. If F has characteristic zero, there are no nontrivial finite purely inseparable extensions of F, and F.50 makes G geometrically regular. If its characteristic is positive, the chosen subring is $k[[X_1, \ldots, X_n]]$; the Cohen-ring alternative has characteristic-zero fraction field. The case $m = 0$ of F.110 makes G geometrically regular in this positive-characteristic case as well.

The finite map $A_0 \to B$ is quasi-finite at $\mathfrak{p}$. To see the fibre condition, its algebra over any residue field is finite-dimensional. Descending chains of ideals stabilize because their vector-space dimensions decrease, so it is Artinian. The full product theorem AG-CA-03, Theorem 4.2 makes its prime points a finite discrete set. Every prime quotient is a finite-dimensional domain, hence a field, giving finite residue extensions. Equivalently the finite morphism satisfies the quasi-finite fibre criterion used in F.26. Apply assertion (1) of F.96 to the pairs $(0) \subset \mathfrak{p}$ in B and $(0) \subset \mathfrak{p}_0$ in A_0. Both rings are domains and the subring is injective, so these are the exact contractions. Its assumption is the geometric regularity of G just proved. Its conclusion is the desired generic-fibre assertion for B. This uses the single specified fibre assertion of F.96, and does not presuppose that A_0 is a G-ring.

Now let A be an arbitrary complete Noetherian local ring. For each prime $\mathfrak{q} \subset A$, the domain $B = A/\mathfrak{q}$ is complete Noetherian local. Indeed AG-CA-19, Theorem 3.1 identifies completion of this finite quotient with $\widehat{A} \otimes_A (A/\mathfrak{q}) = A/\mathfrak{q}$, with its induced maximal-adic topology. For every prime $\mathfrak{p}$ containing $\mathfrak{q}$, put $\overline{\mathfrak{p}} = \mathfrak{p}/\mathfrak{q}$. The generic assertion applied to this domain gives geometric regularity of

$$\widehat{B_{\overline{\mathfrak{p}}}} \otimes_B \mathrm{Frac}(B).$$

These are all pairs required by the exact quotient criterion F.99. That criterion identifies the displayed ring with the $\mathfrak{q}$-fibre of $A_{\mathfrak{p}} \to \widehat{A_{\mathfrak{p}}}$, and proves that geometric regularity for every such pair makes A a G-ring. Thus no other prime fibres are omitted. If A is the zero ring, there are no primes and the definition is vacuous. □

Proposition F.133: characterization geometrically regular

Reading source: Stacks, Tag 07E5. The proof below includes both finite-defect dimension arguments.

Let k have characteristic $p > 0$, and let $(A, \mathfrak{m}, K)$ be a Noetherian local k-algebra. The following conditions are equivalent:

1. A is geometrically regular over k.
2. $A \otimes_k k'$ is regular whenever $k \subset k' \subset k^{1/p}$ and $[k' : k] < \infty$.

3. A is regular and $H_1(L_{K/k}) \to \mathfrak{m}/\mathfrak{m}^2$ is injective.
4. A is regular and $\Omega_{k/\mathbf{F}_p} \otimes_k K \to \Omega_{A/\mathbf{F}_p} \otimes_A K$ is injective.

Proof. Write $E = \mathfrak{m}/\mathfrak{m}^2$, $H = H_1(L_{K/k})$, $U = \Omega_{k/\mathbf{F}_p} \otimes_k K$ and $V = \Omega_{A/\mathbf{F}_p} \otimes_A K$. The quotient presentation of K over A has no relative polynomial variables, so its naive complex is $E \to 0$. Every field is formally smooth over its prime field by AG-CA-19S, Lemma 1.1, and F.17 consequently gives $H_1(L_{K/\mathbf{F}_p}) = 0$. The complete quotient transitivity proof F.42 and the field version F.131 give

$$0 \to E \to V \to \Omega_{K/\mathbf{F}_p} \to 0,$$
$$0 \to H \to U \to \Omega_{K/\mathbf{F}_p} \to \Omega_{K/k} \to 0.$$

Naturality maps the second row into the first, with identity on $\Omega_{K/\mathbf{F}_p}$. If $u \in U$ maps to zero in V, it maps to zero in $\Omega_{K/\mathbf{F}_p}$ and hence comes from a unique $h \in H$. The first row embeds E in V, so this h maps to zero in E. Conversely an h killed in E maps to a vector of U killed in V. Thus the two kernels are canonically isomorphic, proving (3) equivalent to (4).

Assume (3), and take any finite purely inseparable k'/k. Set $A' = A \otimes_k k'$. A basis of k' containing 1 makes A' finite free over A and the structural map injective. It is integral by the determinant proof AG-CA-05, Theorem 1.2. For some e, $(k')^{p^e} \subset k$; applying Frobenius to a finite tensor sum shows $x^{p^e} \in A$ for every $x \in A'$. For a prime $\mathfrak{q}$ over $\mathfrak{p}$, the equivalence $x \in \mathfrak{q} \iff x^{p^e} \in \mathfrak{p}$ determines $\mathfrak{q}$ uniquely. Lying over supplies it. Integral maximal-ideal correspondence, AG-CA-05, Theorem 3.3, makes A' local with maximal ideal $\mathfrak{m}'$ over $\mathfrak{m}$. It is Noetherian, since it is finite over A. Write $K' = A'/\mathfrak{m}'$ and $E' = \mathfrak{m}'/\mathfrak{m}'^2$. The surjection $K \otimes_k k' \to K'$ proves $[K' : K] < \infty$. The integral inclusion preserves dimension by AG-CA-05, Theorem 5.1.

Put $H' = H_1(L_{K'/k'})$. Relative quotient transitivity gives rows

$$0 \to H \otimes_K K' \to E \otimes_K K' \to V_0 \to W \to 0,$$
$$H' \to E' \to V_0 \to W' \to 0,$$

where $W = \Omega_{K/k} \otimes_K K'$, $W' = \Omega_{K'/k'}$, and differential base change AG-CA-15, Theorem 2.1 identifies the two middle differential terms V_0. Let $\alpha : W \to W'$ and $\beta : H \otimes_K K' \to H'$ be the natural maps. Surjectivity from the common V_0 makes α surjective. Both extensions k'/k and K'/K are finite, so F.108 applies with transcendence degrees zero and proves

$$\dim \ker \alpha - \dim \ker \beta + \dim \operatorname{coker} \beta = 0.$$

All these defects are finite. Also H embeds in the finite space E, and finite cokernel of β therefore makes H' finite.

Let L and L' be the images of the two conormal terms in V_0. Then $L \subset L'$ and $L'/L = \ker \alpha$. The first row gives $\dim E = \dim H + \dim L$; the second gives $\dim E' \le \dim H' + \dim L'$. Consequently

$$\begin{aligned} \dim E' - \dim E &\le \dim H' - \dim H + \dim \ker \alpha \\ &= \dim \operatorname{coker} \beta - \dim \ker \beta + \dim \ker \alpha = 0. \end{aligned}$$

Here only finite spaces are subtracted. Since A is regular and $\dim A' = \dim A$, this says $\dim E' \le \dim A'$. Conversely a basis of E' lifts to generators of $\mathfrak{m}'$ by F.10, and AG-CA-11, Theorem 3.1 and Proposition 4.2 give $\dim A' \le \dim E'$. Equality makes A' regular. This holds for every finite purely inseparable extension, so F.50 proves (1).

It remains to prove (2) implies (4). Taking $k' = k$ first makes A regular. Choose $a_1, \dots, a_n \in k$ whose absolute differentials are independent. The full root-degree proof F.81 identifies

$$k' = k(a_1^{1/p}, \dots, a_n^{1/p}) = k[T_1, \dots, T_n]/(T_i^p - a_i), \qquad [k' : k] = p^n.$$

The preceding prime and dimension arguments apply to $A' = A \otimes_k k'$; by (2) it is regular. Its monic equations are a regular list, since successive monic quotient division makes each next monic

polynomial injective by its leading coefficient. Their conormal classes are a free basis by F.74. All relative derivatives are zero. Thus $H_1(L_{A'/A})$ and $\Omega_{A'/A}$ are free of rank n. The representative connecting-map construction in F.42 sends the ith homology basis vector to $da_i \otimes 1$, up to an immaterial common sign, and yields

$$A'^n \to \Omega_{A/\mathbf{F}_p} \otimes_A A' \to \Omega_{A'/\mathbf{F}_p} \to A'^n \to 0.$$

This row stays exact after tensoring with K'. Indeed the last surjection splits because its target is free. Its kernel therefore embeds after tensoring, while tensor right exactness preserves the preceding cokernel presentation. Combining these two statements gives a map

$$\delta : V \otimes_K K' \longrightarrow V' = \Omega_{A'/\mathbf{F}_p} \otimes_{A'} K'$$

whose kernel is spanned by the n vectors $da_i \otimes 1$ and whose cokernel is K'^n.

Use the two absolute short exact rows

$$0 \to E \otimes_K K' \to V \otimes_K K' \to \Omega_{K/\mathbf{F}_p} \otimes_K K' \to 0,$$
$$0 \to E' \to V' \to \Omega_{K'/\mathbf{F}_p} \to 0.$$

The left vertical map has finite equally dimensional domain and target: both dimensions equal $\dim A = \dim A'$. Its kernel and cokernel thus have equal dimensions. For the right vertical map, F.131 identifies the kernel with $H_1(L_{K'/K})$ and the cokernel with $\Omega_{K'/K}$. Their finite dimensions agree by F.98, applied to the proved finite extension K'/K. The full snake sequence F.84 now gives

$$\dim \ker \delta - \dim \operatorname{coker} \delta = 0.$$

Every defect in this computation is finite, although the absolute differential spaces need not be finite. Since the cokernel has dimension n, the n vectors spanning the kernel are independent. The p-basis proof in AG-CA-19S, Lemma 1.1, and its differential calculation F.81 give a basis of $\Omega_{k/\mathbf{F}_p}$ consisting of such da_i. Every prospective relation uses a finite subset. Applying the preceding calculation to that subset proves the full injection in (4). For $n = 0$ the argument is empty. Finally (1) implies (2) directly from geometric regularity. These implications prove the equivalence. □

Proposition F.134: finite type over G ring

Reading source: Stacks, Tag 07PV. The completed-local descent and each individual fibre reduction are explicit below.

If R is a G-ring and S is an essentially finite-type R-algebra, then S is a G-ring.

Proof. G-rings are Noetherian by Definition F.92; finite-type algebras and their localizations are Noetherian by AG-CA-03, Theorem 2.1 and Proposition 1.2. A localization has the same prime local rings as the original ring at the surviving primes. Thus it suffices to handle finite-type algebras. Such an algebra is a quotient of a finite-variable polynomial ring. Quotient maps are quasi-finite, and F.96 proves G-ring ascent for them. Induction on the number of variables reduces the problem to $R[X]$.

By F.100, fix a maximal ideal $\mathfrak{q}$ of $R[X]$ and prove its local completion map regular. Let $\mathfrak{p} = \mathfrak{q} \cap R$. Replacing R by $R_{\mathfrak{p}}$ leaves $R[X]_{\mathfrak{q}}$ unchanged, and this new base is a local G-ring. Write its maximal ideal as $\mathfrak{m}$, its completion as $\widehat{R}$, and put

$$A = R[X]_{\mathfrak{q}}, \qquad B = \widehat{R}[X]_{\mathfrak{q}'}.$$

There is exactly one prime $\mathfrak{q}'$ above $\mathfrak{q}$: the quotient identity $\widehat{R}/\mathfrak{m}\widehat{R} = R/\mathfrak{m}$ gives $\widehat{R}[X]/\mathfrak{q}\widehat{R}[X] = R[X]/\mathfrak{q}$, a field. This uses AG-CA-19, Proposition 2.2 and the fact $\mathfrak{m} \subset \mathfrak{q}$. The map $A \to B$ is local and flat. It is also regular, since it is a localization of the base change of the regular G-ring map $R \to \widehat{R}$; these operations preserve regularity by F.123.

If $B \to \widehat{B}$ is regular, F.124 makes $A \to \widehat{B}$ regular. The completed map $\widehat{A} \to \widehat{B}$ is faithfully flat by the full proof F.99A for this actual flat local $A \to B$. Therefore F.125 makes $A \to \widehat{A}$ regular. This reduces us to a complete Noetherian local base, with $\mathfrak{q}$ above its maximal ideal; it does not assume unrelated completions are equal.

For this complete base, fix $\mathfrak{r} \subset \mathfrak{q}$ in $R[X]$. Its completed fibre is

$$\widehat{R[X]_{\mathfrak{q}}} \otimes_{R[X]} \kappa(\mathfrak{r}).$$

Let $\mathfrak{a} = \mathfrak{r} \cap R$. Finite-module completion commutes with quotient by $\mathfrak{a}$, by AG-CA-19, Theorem 3.1. Since $\mathfrak{a}$ vanishes in the displayed residue field, replacing R by $R/\mathfrak{a}$ gives exactly the same fibre. The quotient is still complete and Noetherian, by that theorem and Theorem 3.3. Thus assume R is a complete local domain and $\mathfrak{r} \cap R = 0$.

Choose the finite regular complete local subring $R_0 \subset R$ from F.25. The map $R_0[X] \to R[X]$ is finite, hence quasi-finite at the contracted primes $\mathfrak{r}_0 \subset \mathfrak{q}_0$. The prime $\mathfrak{q}_0$ is maximal by integral maximal-ideal correspondence; it lies above the maximal ideal of R_0, since R is finite local over the local subring with the same residue field. Also $\mathfrak{r}_0 \cap R_0 = 0$. Assertion (1) of F.96 transfers geometric regularity of this individual completed fibre from $R_0[X]$ to $R[X]$. Its tensor and completion-product proof requires only that fibre, without any prior G-ring assertion for $R_0[X]$. We may therefore assume the base domain is regular and complete.

Let $K = \operatorname{Frac}(R)$. If $\mathfrak{r} = 0$, the fibre is $\widehat{R[X]_{\mathfrak{q}}} \otimes_{R[X]} \operatorname{Frac}(R[X])$, geometrically regular by the second full assertion of F.110. If $\mathfrak{r} \neq 0$, its extension to the principal ideal domain $K[X]$ is a nonzero prime, hence generated by an irreducible polynomial. Its residue field is finite over K. The second complete assertion of F.97 proves geometric regularity of this fibre: for each finite purely inseparable test field it constructs a finite complete local domain D inside that field and identifies the tested fibre with factors of

$$(\widehat{D[X]_{\mathfrak{q}_i}} \otimes_D \operatorname{Frac}(D))/(X - a).$$

Its first assertion makes the parenthesized algebra regular. The extended $\partial/\partial X$ sends $X - a$ to 1, and the proved hypersurface argument then makes the quotient regular. This quotient is used as a quotient; no localization assertion about it is required. The two prime cases exhaust the fibres. Completion is flat by AG-CA-19, Theorem 3.2, so the geometrically regular fibres make its map regular. This proves polynomial permanence and all the reductions to S. Zero target rings have no local rings to check. □

Proposition F.135: ubiquity G ring

Reading source: Stacks, Tag 07PX. We prove precisely the finite-type-over-$\mathbf{Z}$ assertion used in this programme.

Every ring essentially of finite type over $\mathbf{Z}$ is a G-ring.

Proof. The prime ideals of $\mathbf{Z}$ are (0) and (ℓ) for prime integers ℓ. At (0) its local ring is $\mathbf{Q}$. Completion at the zero maximal ideal is identity, so this local completion map is regular.

At (ℓ) put $A = \mathbf{Z}_{(\ell)}$. Factoring an integer into its power of ℓ and its remaining factor shows every nonzero element of A is $\ell^n u$ with u a unit. Every nonzero ideal is generated by the element of smallest such order. Hence A is Noetherian local with maximal ideal (ℓ) and only the primes (0) and (ℓ). These are also the DVR computations in AG-CA-15, Section 1.

Write $\widehat{A}$ for its maximal-ideal completion. The exact quotient identities and local-ring proof AG-CA-19, Proposition 2.2, Proposition 2.3 and Theorems 3.2–3.3 make it Noetherian, complete and separated, with maximal ideal $\ell\widehat{A}$ and residue field $\mathbf{F}_\ell$. They also make $A \to \widehat{A}$ flat, so multiplication by ℓ remains injective. If $0 \neq x \in \widehat{A}$, separatedness gives a finite largest n with $x \in \ell^n \widehat{A}$. Thus $x = \ell^n u$, and u cannot belong to $\ell\widehat{A}$, or maximality would fail. The local-ring unit

criterion makes u invertible. For two nonzero elements their product is a unit times a power of ℓ, which is nonzero by its multiplication injection. This proves directly that $\widehat{A}$ is a domain. The same order argument makes its nonzero ideals powers of ℓ and makes $\widehat{A}[1/\ell]$ its fraction field.

There are just two fibres of the flat map $A \to \widehat{A}$. The closed fibre is $\widehat{A}/\ell\widehat{A} = \mathbf{F}_\ell$ over itself; the generic fibre is the field $\widehat{A}[1/\ell]$ over $\mathbf{Q}$. A field of characteristic zero is geometrically regular over any characteristic-zero subfield by F.50: the only finite purely inseparable test extension is the identity, and the tested ring is a field, hence regular. The closed fibre satisfies the same criterion over itself, since a finite purely inseparable extension tensored with its base field is again that extension field. Thus both fibres are geometrically regular. Definition F.92 now makes $\mathbf{Z}$ a G-ring. Apply the fully proved finite-type permanence F.134 and its localization step to obtain the assertion. The zero algebra is included by its empty spectrum. □

Lemma F.136: Noetherian

Reading source: Stacks, Tag 0052.

Let X be a Noetherian topological space. Every subspace is Noetherian, there are finitely many irreducible components, and each component contains a nonempty subset open in X.

Proof. Recall that Noetherianity means stabilization of descending sequences of closed subsets. For a subspace T, take a descending sequence C_j closed in T and choose closed $D_j \subset X$ with $C_j = T \cap D_j$. Put $E_j = D_1 \cap \cdots \cap D_j$. Because the C_j descend, $T \cap E_j = C_j$. The descending sequence E_j stabilizes in X, so the sequence C_j stabilizes in T. This proves the assertion for every subspace, including the empty one.

We first show that every closed subset of X is a finite union of irreducible closed subsets. The empty subset is the union of an empty family. Otherwise, if there were closed subsets without such an expression, the descending-chain condition would supply a minimal one C: lacking a minimal member would allow a strictly descending sequence of counterexamples. This C is nonempty and cannot itself be irreducible. Thus $C = C_1 \cup C_2$, where both C_i are proper subsets closed in C, and hence closed in X. Minimality supplies a finite irreducible closed cover of each C_i, and their combined covers supply one of C, a contradiction.

Apply the result to X. From the finite cover remove any member contained in the union of the others; this operation terminates and still leaves a cover. By F.138, its members are exactly the irreducible components. There are consequently only finitely many of them.

Fix a component C and list the others as $C_1, \ldots, C_s$. They are closed by F.137. The open set $U = X \setminus \bigcup_{i=1}^{s} C_i$ lies in C, since the components cover X. It is nonempty: otherwise C would be the finite union of the closed subsets $C \cap C_i$. Binary irreducibility, applied repeatedly to a finite union, would force $C \subset C_i$ for some i, contradicting the distinctness and maximality of these components. If there are no other components, $U = X = C$, which is nonempty by the definition of an irreducible component. □

Lemma F.137: irreducible

Reading source: Stacks, Tag 004W.

In an arbitrary topological space, the closure of every irreducible subset is irreducible. Every irreducible subset is contained in an irreducible component; components are closed and together cover the space.

Proof. We use the nonempty convention for irreducibility. A nonempty space is irreducible exactly when any two nonempty open subsets intersect: taking complements converts this into the assertion that two proper closed subsets cannot cover it.

Let $Y \subset X$ be irreducible. An open subset of X which meets $\overline{Y}$ meets Y, by the definition of

closure. If two relatively open subsets of $\overline{Y}$ are nonempty, represent them by ambient opens. Their intersections with Y are nonempty and relatively open in Y, so they intersect. The original two subsets therefore intersect as well. This proves irreducibility of $\overline{Y}$.

Consider the set of all irreducible subsets of X containing Y, ordered by inclusion. It is nonempty because it contains Y. For a nonempty chain, its union V is nonempty. Two nonempty relatively open subsets of V meet some members of the chain. One of those members contains the other, so that member meets both opens. Its irreducibility supplies a point in their intersection. Thus V is irreducible and is an upper bound in the set. The empty chain has Y as an upper bound. Zorn's lemma now gives a maximal member C. Any irreducible subset containing C also contains Y, so this is a maximal irreducible subset of X, namely a component.

The first part shows that $\overline{C}$ is irreducible; maximality forces $C = \overline{C}$, so every component is closed. Each singleton is irreducible, since its only nonempty open subset is itself. Applying the existence argument to every singleton proves that the components cover X. For $X = \varnothing$ both the component family and its union are empty. □

Lemma F.138: pick irreducible components

Reading source: Stacks, Tag 0G2Y.

Suppose $X = C_1 \cup \cdots \cup C_n$, where every C_i is irreducible and closed in X, and no C_i lies in the union of the remaining members. Then this family is exactly the family of irreducible components of X.

Proof. Let D be any irreducible component. The sets $D \cap C_i$ form a finite closed cover of D. Irreducibility implies that one member of a finite closed cover is the whole space: for two members this is its definition, and grouping the last member against the preceding union proves the assertion by induction. Consequently $D \subset C_i$ for some i. Because C_i is itself irreducible, maximality of D gives $D = C_i$.

Conversely fix C_i. The general existence argument F.137 puts it in a component D. The preceding paragraph identifies D with some C_j. If $j \neq i$, the containment $C_i \subset C_j$ would put C_i in the union of all the other members, contrary to the hypothesis. Thus $j = i$, and C_i is a component. These two arguments prove equality of the families. When $n = 0$, the covering assumption says X is empty, and it has no nonempty irreducible components, giving the same conclusion. □

6. Earlier algebra and geometry

The following lessons supply the algebra used above. The henselian section argument uses Affine descent, Zariski Main and recognition of spaces, Lemma D2.2, Theorem C4.3 and Corollary B1.2. The completion and G-ring arguments are F.99A, F.110 and F.132–F.135. For the torus application, finite-type and finite-presentation descent on affine pairs is Lemma A1.8 of the same supporting lesson. The fixed-character splitting scheme becomes a disjoint union on its splitting cover, which is the case of AG-GS-05, Lemma 7.6 used here.

Programme proof locators

The statement numbers specify the earlier results used in this lesson.

Consumed source label	Earlier programme proof
definition regular	AG-CA-14, Section 1, definition
definition regular local	AG-CA-14, Section 1, definition
definition standard smooth	AG-CA-17, Section 4, equations (6) and (7)

Consumed source label	Earlier programme proof
lemma Artin Rees	AG-CA-03, Theorem 5.1
lemma Noetherian irreducible components	AG-CA-03, Proposition 2.2
lemma Noetherian topology	AG-CA-03, Proposition 2.2
lemma Zariski topology	AG-CA-01, Theorem 1.2 and Proposition 2.1
lemma artinian finite length	AG-CA-03, Theorem 4.2
lemma characterize henselian	AG-CA-20, Theorem 1.2 and Theorem 2.2
lemma characterize projective	AG-CA-06S, Section 1
lemma cokernel flat	AG-CA-08, Lemma 5.1
lemma compose formally smooth	AG-CA-17, Theorem 1.2
lemma differentials base change	AG-CA-15, Theorem 2.1
lemma differentials finitely presented	AG-CA-15, Theorem 5.1, finite polynomial presentation
lemma dimension base fibre equals total	AG-CA-11, Theorem 5.1
lemma dimension base fibre total	AG-CA-11, Theorem 5.1
lemma dimension prime polynomial ring	AG-CA-09, Theorem 4.3
lemma disjoint implies product	AG-CA-01, Theorem 5.2 and Corollary 5.3
lemma etale at prime	AG-CA-17, Theorem 7.2 and Corollary 7.3
lemma exact sequence differentials	AG-CA-15, Theorem 3.1
lemma faithfully flat universally injective	AG-CA-08, Theorem 2.2
lemma finite finite fibres	AG-CA-05, Theorem 3.4
lemma finite projective	AG-CA-07, Theorem 5.3
lemma finite transitive	AG-CA-05, Theorem 1.2
lemma flat	AG-CA-07, Theorem 2.1
lemma flat base change	AG-CA-07, Proposition 3.2
lemma flat eq	AG-CA-07, Theorem 5.1
lemma flat going down	AG-CA-07, Theorem 6.3
lemma flat tor zero	AG-CA-07, Theorem 2.1
lemma formally etale etale	AG-CA-17, Theorem 2.2 and Proposition 6.1
lemma integral no inclusion	AG-CA-05, Theorem 3.3
lemma intersect powers ideal module zero	AG-CA-03, Theorem 6.1
lemma lci CM	AG-CA-12, Corollary 6.2
lemma maximal chain CM	AG-CA-12, Lemma 5.4 and Theorem 5.5
lemma quasi compact	AG-CA-01, Proposition 2.3
lemma rank omega	AG-CA-18, Proposition 3.1
lemma reformulate CM	AG-CA-12, Theorem 4.2 and Corollary 4.3
lemma regular domain	AG-CA-12, Theorem 6.1
lemma regular goes up	AG-CA-14, Proposition 3.3
lemma regular ring CM	AG-CA-12, Theorem 6.1
lemma regular sequence powers	AG-CA-12, Proposition 1.3
lemma spec localization	AG-CA-02, Theorem 3.1
lemma variant local criterion flatness	AG-CA-08, Corollary 4.4 and equation (6)
proposition CM module	AG-CA-12, Theorem 4.2 and Corollary 4.3
proposition characterize formally smooth	AG-CA-17, Theorem 3.1
proposition smooth formally smooth	AG-CA-17, Theorem 3.1 and the split conormal definition of smoothness in this draft
lemma completion regular	AG-CA-19, Theorem 4.1
complete-local simple-root lifting	AG-CA-19, Theorem 5.1
finite residue fields in finite-type fibres	AG-CA-06, Theorem 2.1
minimal resolutions and Tor detection	AG-CA-13, Theorem 2.2
regular Noetherian rings are normal	AG-CA-15, Corollary 4.5
arbitrary fields lift over their prime field	AG-CA-19S, Lemma 1.1

Consumed source label	Earlier programme proof
complete local coefficient rings	AG-CA-19S, Theorem 5.1
complete local power-series presentations	AG-CA-19S, Theorem 6.1

Further reading

- The Stacks Project Authors, *Smoothing Ring Maps*, *Algebra*, and *More on Algebra*. Individual arguments above give their exact tag or source locator.
- AI Integrated Stacks, the consulted edition, including its smoothing and algebra chapters.

Lie proofs for the characteristic-zero group construction

This lesson proves the Lie-theoretic results used by the subsequent rational group construction: the trace criterion and complete reducibility, the rational Serre algebra, finite integrable highest-weight modules, and rational rank-one integration. The algebraic-group closure and integral construction have their distinct proofs in the subsequent group lesson. All field hypotheses are stated where used.

Written and self-checked by GPT-6.1 Sol (OpenAI), Codex, Ultra setting, October 2026. Original exposition is dedicated under CC0 1.0. This reconstruction draft remains under programme review.

The comparison sources are Milne, *Lie Algebras, Algebraic Groups, and Lie Groups*, version 2.00, 5 May 2013, Chapter I, and Etingof, *Lie Groups and Lie Algebras I*, full MIT OpenCourseWare notes, Sections 13–18 and 21–25. The proofs here use algebraic operations and give the required steps. These links identify freely accessible comparisons; they do not supply omitted proofs.

The full ordered-word and relation-ideal proof is supplied in Section 1.A before any use of enveloping or free Lie algebras. It proves termination, contextual ambiguity resolution, the lower-length Jacobi calculation, and both inclusions identifying the normalization kernel with the relation ideal.

Throughout, all Lie algebras and modules in Sections 2–4 are finite-dimensional unless explicitly stated otherwise. Sections 5–7 permit the indicated infinite-dimensional auxiliary algebras or cyclic modules until finiteness is proved. The characteristic-zero restriction is essential for the trace criteria, complete reducibility, and the finite-type construction.

1. Finite linear operations and scalar extension

The finite matrix reductions used here have an explicit algorithm. Choose a nonzero entry as a pivot, move it to the first available row and column, divide its row by that nonzero scalar, and subtract multiples to clear its column. Apply the same procedure to the remaining rows and columns. Each step has an inverse (reverse a swap, multiply back, or subtract the opposite multiple), and each pivot decreases both remaining sizes, so the procedure terminates. Back substitution expresses each nonpivot column in the pivot columns; the pivot columns are independent by their successive leading entries. Thus an independent list in k^m has length at most m. Applied to an augmented system, the algorithm either produces a contradictory row $0 = c \neq 0$, or obtains a solution by choosing free variables and back substituting. This proves the matrix facts about independence and consistency used below.

A finite independent list in a finite-dimensional vector space extends to a basis: append a vector outside its span whenever the span is not the whole space. Its length strictly increases and is bounded by the length m of any given finite spanning list. Indeed choose lifts of its vectors under the surjection $k^m \to V$ given by that list; a relation among those lifts would give a relation among the independent vectors, so the lifts are independent and the matrix bound applies. Thus the process stops. This also constructs a complement to every subspace. A basis of a kernel followed

by such a complement proves rank–nullity: the images of the added vectors form a basis of the image. Every linear functional on a subspace extends by prescribing zero on a complement.

For matrices, direct summation gives

$$\operatorname{tr}(AB) = \sum_{i,j} A_{ij}B_{ji} = \operatorname{tr}(BA), \qquad \operatorname{tr}([A,B]C) = \operatorname{tr}(A[B,C]). \tag{1.1}$$

The trace of an operator preserving a subspace is the sum of its trace on that subspace and on the quotient: choose a basis starting in the subspace and read the two diagonal blocks. A nilpotent operator has trace zero. Indeed, the increasing kernels of its powers give a basis in which it is strictly triangular. These calculations also show that traces and products commute with field extension.

Here are the determinant identities used in the matrix examples, over every commutative ring. Define $\det M = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_i m_{i,\sigma(i)}$, where the sign is the parity of the number of inversions. An adjacent interchange changes that parity by one; sorting by adjacent interchanges proves the sign rule for any permutation. The sum is multilinear in the columns. If two columns coincide, pair each permutation with its transposition on those columns: they are distinct and their two terms have opposite signs and the same product. Thus the determinant is zero, even when two is not invertible. Interchanging two columns changes the sign by the same bijection. Expanding arbitrary columns in the coordinate basis now determines this alternating multilinear expression uniquely from its value one at the identity matrix. Expand the columns of MN as combinations of the columns of M. Repeated-column terms vanish; each remaining term uses each column once and equals the sign of that permutation times $\det M$. Summing their coefficients is precisely $\det N$, so $\det(MN) = \det M \det N$.

Grouping the permutation sum by the chosen entry in a row gives its cofactor expansion. For the transpose cofactor matrix $\operatorname{adj} M$, the diagonal entries of $M \operatorname{adj} M$ are those expansions of $\det M$. Each off-diagonal entry is the determinant with one row repeated and is zero by the same paired-permutation argument. Expanding by columns gives the other product as well. Hence $M \operatorname{adj} M = \operatorname{adj} M M = (\det M)I$. A unit determinant gives the inverse $(\det M)^{-1} \operatorname{adj} M$, and an invertible matrix has unit determinant by multiplicativity. The determinant of a block triangular matrix is the product of the two diagonal-block determinants: in a nonzero permutation term every row of the block with zero off-diagonal entries must use a column of its own block, forcing the other rows to use their own block too. This also proves the characteristic-polynomial product for an invariant subspace and quotient.

For an n-by-n matrix M, write $p(t) = \det(tI - M) = \sum_{j=0}^{n} c_j t^j$, with $c_n = 1$. Write $\operatorname{adj}(tI - M) = \sum_{j=0}^{n-1} B_j t^j$. The preceding identity gives $B_{n-1} = I$, $B_{j-1} - MB_j = c_j I$ for $1 \le j < n$, and $-MB_0 = c_0 I$. Downward induction makes every B_j a polynomial in M, so it commutes with M. Multiplying these coefficient equations by the corresponding powers of M and summing cancels all B_j terms, leaving $p(M) = 0$. This is the entire Cayley–Hamilton calculation. Over a field the pivot algorithm makes a matrix singular exactly when its determinant is zero: row reduction has unit determinant multipliers, an invertible echelon matrix has nonzero diagonal product, and a dependent column list has zero determinant by multilinearity. Thus a root of $p(t)$ supplies an eigenvector, by the kernel part of that same algorithm.

Every symmetric polynomial in $x_1, \dots, x_n$ with coefficients in any commutative ring is a polynomial in the elementary symmetric polynomials $e_1, \dots, e_n$. Treat its finitely many homogeneous degrees separately and order each degree's monomials lexicographically. A largest monomial with nonzero coefficient has exponents $a_1 \ge \cdots \ge a_n$: if $a_i < a_j$ for $i < j$, interchanging the two variables would give a larger monomial with the same nonzero coefficient. The product $e_1^{a_1-a_2} e_2^{a_2-a_3} \cdots e_n^{a_n}$ has that largest monomial with coefficient one. Subtract its multiple by the given leading coefficient. The remainder is symmetric, of the same degree, with strictly smaller largest monomial. There are only finitely many monomials of that degree, so repetition terminates at zero. In particular

$\prod_{i<j}(x_i - x_j)^2$ is an integer polynomial in the elementary symmetric functions: permuting variables merely permutes the unordered pairs, and their squared differences remove the possible sign. Its value on the roots of a monic polynomial is consequently a polynomial in that polynomial's coefficients over every base.

Let K/k be a field extension. If (v_i) is a basis, then $(1 \otimes v_i)$ is a K-basis of $K \otimes_k V$: tensor multilinearity gives spanning, and the tensor extensions of the coefficient functionals give independence. It follows that scalar extension is exact and detects a zero vector space. In particular, it detects whether a subspace is nonzero or proper. For a finite matrix over k, Gaussian elimination uses only nonzero scalars already in k. Its rank and nullity are unchanged by extension. Thus kernels, simultaneous kernels of a finite family of maps, and consistency of finite systems of linear equations commute with field extension. For example, an inconsistent equation $0 = 1$ produced by row reduction over k cannot become consistent over K.

There is an algebraically closed extension of any field. Here is the needed construction, using the usual choice principle to well order sets. For a field F, well order its nonconstant monic polynomials. In that order, adjoin a root of each polynomial not already having one: choose an irreducible factor and pass to the field quotient $F'[t]/(q)$; polynomial division and Bézout show that a quotient by an irreducible polynomial is a field. At limit stages take unions of the nested fields. This gives an algebraic extension F^+ in which every nonconstant polynomial originally over F has a root. Starting with $F_0 = F$, repeat to obtain $F_{n+1} = F_n^+$. The union $\Omega = \bigcup_n F_n$ is algebraic over F, since each element belongs to a finite tower of algebraic extensions. A polynomial over Ω has finitely many coefficients, all in some F_n, so has a root in F_{n+1}. Factoring out that root and inducting on degree shows that every polynomial splits over Ω. This proves the existence needed below; no analytic assertion about $\mathbb{C}$ is required. Here are the field facts in this construction. Polynomial division subtracts the leading multiple and strictly lowers degree; successive remainders terminate and back substitution expresses the last nonzero remainder as a polynomial combination of the two inputs. Thus coprime inputs have a Bézout identity. A nonconstant polynomial has an irreducible divisor by choosing a nonconstant divisor of least positive degree. For an irreducible q, division shows that its quotient has basis $1, t, \ldots, t^{\deg q - 1}$, and Bézout gives an inverse to each nonzero residue; it is a field. A field generated by finitely many algebraic elements is finite-dimensional: adjoining an element satisfying a monic polynomial of degree d is spanned by its first d powers, and products of the successive spanning lists span a tower. Conversely every element of a finite-dimensional extension is algebraic, because its powers are linearly dependent. These statements also prove transitivity of algebraicity: the finitely many coefficients in a polynomial for an element generate a finite-dimensional field over the original field, and adjoining that element gives another finite-dimensional tower. At a limit stage the union is a field because any finite list of its elements lies in a common earlier field; its elements remain algebraic. Transfinite induction therefore verifies algebraicity at every stage, as well as at the final union.

An algebraically closed field is infinite, and so is a separably closed field. For if such a field had q elements, multiplication by a nonzero a would permute its nonzero elements. Cancelling their nonzero product gives $a^{q-1} = 1$, and hence every element satisfies $a^q = a$. The polynomial $t^q - t - 1$ has no root there. Its derivative is -1 (some positive multiple of 1 is zero in a finite field; the least such integer p is prime, since a proper factorization would give a product of two nonzero elements equal to zero. Its prime subfield is $\mathbf{F}_p$. The finite basis algorithm gives a basis of the field over that subfield; counting its uniquely specified coefficient tuples gives $q = p^d$ for the basis size d). Every irreducible factor is therefore separable and gives a nontrivial finite separable extension by the preceding irreducible-quotient proof. This contradicts either closure property. Over an infinite field, a nonzero polynomial does not vanish on all tuples: in one variable division by a linear factor and degree induction bound its distinct roots by its degree; in several variables first choose values making a nonzero coefficient in the last variable nonzero by induction, then choose the last value outside its finite root set. Thus every nonempty principal open in affine space has a rational point, as does every nonempty open because it contains a nonempty principal open.

We also need the polynomial Jordan decomposition over an algebraically closed field. If X acts on a finite-dimensional space, linear dependence among its powers supplies an annihilating polynomial. Its monic polynomial of least degree factors as $m(t) = \prod_\lambda (t-\lambda)^{d_\lambda}$. The factors are pairwise coprime. Euclidean division gives Bézout identities, which combine residues two at a time to prove the Chinese remainder construction for these factors. The resulting polynomials e_λ give orthogonal projections $e_\lambda(X)$ with sum one and images $V_\lambda = \ker(X-\lambda)^{d_\lambda}$. Prescribe $p(t) \equiv \lambda$ modulo each factor, and also $p(0) = 0$. If zero is an eigenvalue the last condition is automatic; otherwise add the coprime modulus t. For clarity, the annihilating polynomial exists because the m^2 matrix units are a basis of the endomorphisms of an m-dimensional space, so the first m^2+1 powers are dependent. The Euclidean calculation in the preceding paragraph constructs the projections explicitly. For a factor $q_\lambda = (t-\lambda)^{d_\lambda}$ set $r_\lambda = m/q_\lambda$ and choose $a_\lambda r_\lambda + b_\lambda q_\lambda = 1$. The residue $e_\lambda = a_\lambda r_\lambda$ is one modulo q_λ and zero modulo every other factor. A polynomial divisible by each of finitely many pairwise coprime factors is divisible by their product: Bézout proves the two-factor assertion, then induction proves it in general. It follows that $\sum e_\lambda = 1$, $e_\lambda^2 = e_\lambda$ and $e_\lambda e_\mu = 0$ modulo m. The image of $e_\lambda(X)$ is exactly $\ker q_\lambda(X)$, since $q_\lambda e_\lambda$ is divisible by m and $1 - e_\lambda$ is divisible by q_λ. This proves the direct generalized-eigenspace decomposition being used. Then

$$X = S + N, \qquad S = p(X), \quad N = X - p(X) \tag{1.2}$$

has commuting diagonalizable and nilpotent parts. On V_λ, $S = \lambda$ and $N = X - \lambda$. Uniqueness follows because any commuting decomposition preserves each V_λ: on an eigenspace of its semisimple part, the sum with its nilpotent part has that same eigenvalue, which must therefore be λ. Both parts are polynomials with zero constant term. On $\operatorname{End}(V)$, $\operatorname{ad} S$ is diagonalizable with eigenvalues $\lambda - \mu$ on $\operatorname{Hom}(V_\mu, V_\lambda)$. If $N^d = 0$, the binomial expansion of the commuting left and right multiplications proves $(\operatorname{ad} N)^{2d-1} = 0$. In that uniqueness argument, the commuting semisimple part preserves V_λ and decomposes it into its own eigenspaces by the polynomial projections just constructed. On one of them with eigenvalue ν, $X = \nu + N'$ with N' nilpotent, so its only possible eigenvalue is ν. Since it lies in V_λ, that value is λ. Thus the semisimple part is exactly the displayed scalar λ on all of V_λ, and the nilpotent part is the displayed difference. Consequently

$$(\operatorname{ad} X)_s = \operatorname{ad} S, \qquad (\operatorname{ad} X)_n = \operatorname{ad} N. \tag{1.3}$$

This identity concerns the Jordan parts. It does not assert that $\operatorname{ad}(p(X))$ is an arbitrary polynomial in $\operatorname{ad} X$.

1.A. Ordered words and their relation ideal

For a Lie algebra $\mathfrak{l}$ over a field, choose a totally ordered basis (x_i). The tensor algebra is the vector space with the finite basis words, including the empty word, with concatenation as multiplication. Let I be the two-sided ideal generated by $xy - yx - [x,y]$ for $x, y \in \mathfrak{l}$. By bilinearity it suffices to use basis pairs. Define $U(\mathfrak{l}) = T(\mathfrak{l})/I$.

Theorem 1.A (ordered words). The classes of $x_{i_1} \cdots x_{i_m}$ with $i_1 \le \cdots \le i_m$, including the empty word, are a basis of $U(\mathfrak{l})$. In particular $\mathfrak{l} \to U(\mathfrak{l})$ is injective. This holds in arbitrary characteristic and does not require $\mathfrak{l}$ to be finite-dimensional.

Proof. At an adjacent inversion replace $x_j x_i$, $j > i$, by $x_i x_j + [x_j, x_i]$, expanding the bracket in the given basis. Order a word first by its length and then by its number of inversions. The swapped term keeps its length and loses one inversion; every bracket term has smaller length. Each replacement has finitely many terms. Induction on this pair proves termination of every fully expanded reduction. One may make the well-founded bound an integer: give a length-n word the rank $\sum_{j=0}^{n-1} (j(j-1)/2 + 1)$ plus its inversion count. Both kinds of replacement strictly lower this rank. Every replacement has finitely many terms and there are finitely many first choices in a word.

Induction on this rank bounds every fully expanded reduction, including in an infinite ordered basis.

We prove by the same induction that its final linear combination of ordered words is independent of the choices. Two replacements on disjoint adjacent pairs commute: expanding both in either order gives the identical terms, all strictly smaller, whose subsequent normalizations agree by induction. An overlapping pair can only occur in zyx with $z > y > x$. Performing the swaps of the three original letters in the two orders yields respectively

$$xyz + [y, x]z + y[z, x] + [z, y]x,$$

and

$$xyz + x[z, y] + [z, x]y + z[y, x].$$

Subtract these. For any two basis elements, normalizing $ab - ba$ gives $[a, b]$: use the single replacement in the inverted order, its negative in the opposite order, and alternation for equal indices. Bilinearity gives the same identity when either entry is a bracket. Consequently the difference above normalizes to

$$[[y, x], z] + [y, [z, x]] + [[z, y], x] = 0$$

by Jacobi. The same calculation works inside any left and right word context. Every term involving a bracket has smaller total length than the original contextual triple, so its normalization is already unique by induction. The two three-swap paths end at the same larger ordered-letter term with the displayed smaller corrections. Thus their further normalizations coincide. These are all possible first-step conflicts. After comparing the two first steps, apply induction to all their smaller terms; this proves unique normalization at the original word. Linearity extends the normalization N to the tensor algebra.

Every contextual generator of I has normal form zero: for an inverted basis pair this is exactly the permitted first replacement followed by uniqueness, and the reverse or equal pair follows by alternation. Hence $I \subset \ker N$. Conversely each replacement changes an expression by an element of I, so finite reduction gives $a - N(a) \in I$ for every a. If $N(a) = 0$, then $a \in I$. Thus $I = \ker N$. Normalization fixes each ordered word, proving that their quotient classes span and are independent. Their length-one members are the original basis, proving injectivity. $\square$

This proves the entire relation-ideal assertion, including contextual relations; resolving the bare triple alone would not suffice. The associated graded algebra for the length filtration is the symmetric algebra: the ordered-word basis identifies both degree-m pieces, and the leading relation replaces a swap without its lower-length bracket. A Lie subalgebra's enveloping algebra embeds in the full one by extending its ordered basis and comparing these word bases. The symmetric algebra's monomial assertion here can itself be checked by sorting words without a bracket term: a permutation has one sorted word, and the coefficient map to that word kills every contextual commuting relation. Each swap changes the expression by such a relation, so its normalization kernel is exactly their ideal. This proves its monomial basis and the asserted graded identification.

A free Lie algebra on V is formed from bracket expressions modulo bilinearity, alternation and Jacobi; evaluation gives its universal property. The algebra maps from its enveloping algebra to an associative algebra are therefore in bijection with linear maps from V to that algebra, as are maps from $T(V)$. The two universal maps are inverse on generators, hence inverse on all words. Thus its enveloping algebra is $T(V)$, and Theorem 1.A injects the free Lie algebra into these tensor words. This is the precise separation assertion used in Section 6.

Freely accessible comparison material: Etingof's MIT OpenCourseWare notes listed at the beginning of this supporting lesson, Section 13. The proof used here is the complete normalization and ideal

calculation above.

2. Engel, Lie, and the trace criterion

For a Lie algebra $\mathfrak{l}$, let $D^0\mathfrak{l} = \mathfrak{l}$ and $D^{j+1}\mathfrak{l} = [D^j\mathfrak{l}, D^j\mathfrak{l}]$. It is solvable when one term is zero. Its lower central series is $C^1\mathfrak{l} = \mathfrak{l}$, $C^{j+1}\mathfrak{l} = [\mathfrak{l}, C^j\mathfrak{l}]$; it is nilpotent when this series terminates. Jacobi makes all these terms ideals. The series of a subalgebra lies in the ambient series, and a quotient map takes the series onto the quotient series. If an ideal I and $\mathfrak{l}/I$ are solvable, some $D^a\mathfrak{l}$ lies in I, and then $D^{a+b}\mathfrak{l} \subseteq D^b I = 0$ for some b. This proves the solvable-extension assertion. Nilpotent algebras are solvable, because $D^j\mathfrak{l} \subseteq C^{j+1}\mathfrak{l}$. Bilinearity identifies either series after field extension with the tensor extension of the original series, so solvability and nilpotence are preserved and reflected by field extension.

Proposition 2.1 (Engel). Over any field, a finite-dimensional operator Lie algebra $\mathfrak{l} \subseteq \operatorname{End}(V)$ consisting of nilpotent operators, with $V \neq 0$, has a common nonzero zero vector. It is strictly triangular in a suitable basis.

Proof. If $X^d = 0$, then $\operatorname{ad} X = L_X - R_X$ on $\operatorname{End}(V)$ satisfies $(\operatorname{ad} X)^{2d-1} = 0$: in the binomial expansion every term contains at least d left or right factors. No division is used.

Induct on $\dim \mathfrak{l}$. For any proper subalgebra $\mathfrak{a}$, its bracket action on $\mathfrak{l}/\mathfrak{a}$ is defined and consists of nilpotent operators by the preceding calculation. Its image has dimension less than $\dim \mathfrak{l}$. Induction gives a nonzero annihilated class, so the normalizer of $\mathfrak{a}$ strictly contains $\mathfrak{a}$. A maximal proper subalgebra is therefore an ideal. Its quotient has dimension one, because a higher-dimensional quotient has a proper one-dimensional subalgebra whose inverse image contradicts maximality.

Write $\mathfrak{l} = \mathfrak{a} + kZ$ for this ideal. Induction makes $W = \{v : \mathfrak{a}v = 0\}$ nonzero. Idealness gives $a(Zw) = Z(aw) + [a, Z]w = 0$, so Z preserves W. Its nilpotent restriction has a nonzero kernel, giving the common zero vector. The zero algebra begins the induction. Repeat the conclusion on successive quotient spaces of V; the resulting flag has $\mathfrak{l}V_i \subseteq V_{i-1}$. An adapted basis is strictly triangular. □

It follows that if every $\operatorname{ad} x$ on $\mathfrak{l}$ is nilpotent, then $\mathfrak{l}$ is nilpotent: strict triangularity makes every product of $\dim \mathfrak{l}$ adjoint operators zero, and their values span the corresponding lower central term. The converse follows directly from that series.

Proposition 2.2 (Lie). Over an algebraically closed field of characteristic zero, every finite-dimensional representation of a solvable algebra has an invariant flag with one-dimensional successive quotients.

Proof. Induct on the dimension of the algebra to produce a common eigenvector. A nonzero solvable algebra has proper derived ideal; choose a codimension-one ideal $\mathfrak{a}$ containing it and write $\mathfrak{l} = \mathfrak{a} + kx$. Induction gives $v \neq 0$ with $av = \lambda(a)v$. The cyclic space $W = \langle v, xv, x^2v, \ldots \rangle$ has basis $v, \ldots, x^{m-1}v$, stopping at the first dependence, and is x-stable. Simultaneous induction on j, for all $a \in \mathfrak{a}$, gives

$$ax^j v - \lambda(a)x^j v \in \langle v, \ldots, x^{j-1}v \rangle. \tag{2.1}$$

For the step, use $ax^j v = x(ax^{j-1}v) + [a, x]x^{j-1}v$ and apply the preceding step also to $[a, x] \in \mathfrak{a}$. Thus W is $\mathfrak{a}$-stable and the trace of $a|_W$ is $m\lambda(a)$. Since both a and x preserve W, equation (1.1) gives $m\lambda([x, a]) = 0$. Characteristic zero implies $\lambda([x, a]) = 0$. Hence the full common eigenspace E_λ of $\mathfrak{a}$ is x-stable. The restriction of x to this nonzero space has an eigenvector by Section 1: in a nonzero generalized eigenspace, the last nonzero iterate of its nilpotent part is an eigenvector. This vector is common to $\mathfrak{a}$ and x, giving the induction. Passing repeatedly to quotients constructs the asserted flag. □

Proposition 2.3 (Cartan trace criterion). For a finite-dimensional $\mathfrak{l} \subseteq \operatorname{End}(V)$ in characteristic

zero,

$$\mathfrak{l} \text{ is solvable} \iff \operatorname{tr}(XY) = 0 \quad (X \in [\mathfrak{l},\mathfrak{l}],\ Y \in \mathfrak{l}). \tag{2.2}$$

Proof. Extend to an algebraic closure. Lie's theorem proves necessity: the first factor is strictly triangular, the second triangular, so their product has zero diagonal. This also descends the equality to the original field.

For sufficiency, fix $X \in [\mathfrak{l},\mathfrak{l}]$ and list its eigenvalues $\lambda_1, \dots, \lambda_d$ with multiplicities. Let $E \subseteq k$ be their rational span. For a rational linear functional $b : E \to \mathbb{Q}$, define B to be multiplication by $b(\lambda)$ on the generalized λ-space. On $\operatorname{Hom}(V_\mu, V_\lambda)$, $\operatorname{ad} B$ has eigenvalue $b(\lambda - \mu)$, whereas $\operatorname{ad} S$ has eigenvalue $\lambda - \mu$. Interpolate on the finite set of these differences, including zero, to obtain q with $q(0) = 0$ and $\operatorname{ad} B = q(\operatorname{ad} S)$. By (1.3) and the polynomial Jordan construction, $\operatorname{ad} S = p_0(\operatorname{ad} X)$ with $p_0(0) = 0$. Thus $\operatorname{ad} B$ is a zero-constant polynomial in $\operatorname{ad} X$. The finite interpolation used here is explicit: for distinct differences d, the polynomial $\sum_d b(d) \prod_{d' \neq d} (t - d')/(d - d')$ has precisely those prescribed values. Zero is one of the differences, so its constant term is $b(0) = 0$. No assertion that $\operatorname{ad}(X^2)$ preserves the Lie algebra enters this calculation. Since X is in the derived ideal, this proves

$$[B, \mathfrak{l}] \subseteq [\mathfrak{l},\mathfrak{l}]. \tag{2.3}$$

Write $X = \sum_j [Y_j, Z_j]$. The trace identity and (2.3) give

$$\sum_i \lambda_i b(\lambda_i) = \operatorname{tr}(XB) = \sum_j \operatorname{tr}(Y_j [Z_j, B]) = 0. \tag{2.4}$$

The nilpotent restriction of $X - \lambda$ contributes zero trace on each generalized eigenspace, proving the first equality. Apply b to the equality in E: it gives $\sum_i b(\lambda_i)^2 = 0$ in $\mathbb{Q}$. A sum of rational squares can be zero only if every summand is zero. Every rational functional therefore vanishes on every eigenvalue, and coordinate functionals in a basis of E imply $E = 0$. Thus every X in the derived algebra is nilpotent. Engel makes that represented derived algebra nilpotent, hence solvable; the quotient is abelian, so the solvable-extension argument proves the assertion. Field extension preserves and reflects it. For the rational-square step, clear a common positive denominator: the equality becomes a sum of nonnegative integer squares equal to zero, which forces each integer to be zero. A nonzero vector in the finite rational space E is detected by a coordinate functional after extending it to a basis, by Section 1. This proves the last vanishing inference. □

The Hom-block interpolation is necessary. In general, preservation of a subspace by $\operatorname{ad} X$ says nothing of this kind about $\operatorname{ad}(X^2)$. For instance, $X = \operatorname{diag}(-1, 0, 1)$ and $A = E_{12} + E_{23}$, $B = E_{21} + E_{32}$ span a Lie algebra with $[X, A] = -A$, $[X, B] = B$, $[A, B] = -X$, whereas $[X^2, A] = E_{12} - E_{23}$ is outside it. This excludes the incorrect polynomial inference appearing in the freely available LAG proof on page 39.

3. Killing forms, ideals, and inner derivations

For a representation ρ put $B_\rho(x, y) = \operatorname{tr}(\rho(x)\rho(y))$; its adjoint instance is the Killing form κ. Equation (1.1) proves symmetry and invariance:

$$B_\rho([x, y], z) = B_\rho(x, [y, z]). \tag{3.1}$$

The orthogonal complement of an ideal for any invariant form is an ideal: if $z \perp I$, then $B([x, z], i) = -B(z, [x, i]) = 0$. For x, y in an ideal I, the quotient action of $\operatorname{ad} x$ is zero. The block trace formula

therefore gives $\kappa_{\mathfrak{g}}|_{I\times I} = \kappa_I$. Applying Proposition 2.3 to the adjoint image, whose kernel is the abelian centre, proves the intrinsic solvability criterion

$$\mathfrak{g} \text{ solvable} \quad \Longleftrightarrow \quad \kappa([\mathfrak{g},\mathfrak{g}],\mathfrak{g}) = 0. \tag{3.2}$$

Call $\mathfrak{g}$ semisimple if it has no nonzero solvable ideal. This is equivalent to having no nonzero abelian ideal: the last nonzero derived term of a solvable ideal is abelian and remains an ideal of $\mathfrak{g}$, by Jacobi. The radical exists without an additional theorem. The sum of two solvable ideals is solvable by the extension argument; the sum of all of them is already a finite sum, by choosing a basis of their total sum and collecting finitely many ideals expressing each basis vector.

Proposition 3.1. In characteristic zero, $\mathfrak{g}$ is semisimple if and only if κ is nondegenerate. Every faithful finite-dimensional representation of a semisimple algebra also has nondegenerate trace form.

Proof. The radical R of the Killing form is an ideal; its restricted Killing form is zero. Equation (3.2) makes R solvable, so it is zero for a semisimple algebra. Conversely, for an abelian ideal I, $\operatorname{ad} i \operatorname{ad} y$ maps the algebra into I and kills I. Its trace is zero. Nondegeneracy forces every i to be zero.

For a faithful representation, the radical R of its trace form is an ideal. On its represented subalgebra the trace pairing vanishes identically, so Proposition 2.3 makes R solvable. Semisimplicity forces $R = 0$. $\square$

The Killing matrix has the same entries after extension of the field; a matrix is invertible if and only if its kernel is zero, which is unchanged by extension. Hence semisimplicity is preserved and reflected under every field extension in characteristic zero.

Proposition 3.2. A semisimple algebra is a direct sum of commuting nonabelian simple ideals. Every ideal is a sum of a subset of these factors. In particular its ideals and quotients are semisimple, its centre is zero, and $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$.

Proof. For an ideal I, set $J = I \cap I^{\perp}$. It is an ideal, and for $u, v \in J$ and $z \in \mathfrak{g}$, $\kappa([u,v],z) = \kappa(u,[v,z]) = 0$. Nondegeneracy makes J abelian, so $J = 0$. Identifying $\mathfrak{g}$ with its dual by κ and extending functionals on I gives $\dim I^{\perp} = \dim \mathfrak{g} - \dim I$. Thus $\mathfrak{g} = I \oplus I^{\perp}$, and the two ideals commute because their bracket lies in their intersection.

Choose a minimal nonzero ideal. Any ideal of it is an ideal of the whole algebra, since its complement commutes with it; hence it is simple, and it is not abelian. Its complement is again semisimple, since an abelian ideal there would be an abelian ideal of the whole algebra. Dimension induction produces the claimed factors I_j. If an ideal J contains a vector with nonzero I_j-component v_j, some $a \in I_j$ has $[a, v_j] \neq 0$, since the centre of I_j is zero. Then $[a,v] \in J \cap I_j$ is nonzero, so simplicity gives $I_j \subseteq J$. This identifies every ideal as the sum of the factors it contains. Finally each simple factor equals its own derived algebra: its derived algebra is a nonzero ideal. All stated consequences follow. $\square$

Proposition 3.3. Every derivation D of a semisimple characteristic-zero algebra is inner, with a unique inducing element.

Proof. Nondegeneracy gives a unique z with $\kappa(z,x) = \operatorname{tr}(D \operatorname{ad} x)$. Set $E = D - \operatorname{ad} z$, so $\operatorname{tr}(E \operatorname{ad} x) = 0$ for all x. The derivation identity gives $[E, \operatorname{ad} x] = \operatorname{ad}(Ex)$. Therefore

$$0 = \operatorname{tr}(E \operatorname{ad}[x,y]) = \operatorname{tr}([E, \operatorname{ad} x] \operatorname{ad} y) = \kappa(Ex, y). \tag{3.3}$$

It follows that $E = 0$. Uniqueness uses the zero centre. $\square$

4. Complete reducibility with its field scope

Proposition 4.1 (Casimir operator). Suppose B is a nondegenerate invariant symmetric form on a finite-dimensional Lie algebra. If (x_i) and (y_i) are dual bases, $C_B = \sum_i x_i y_i \in U(\mathfrak{g})$ is independent of the basis and is central. For a faithful representation of a semisimple characteristic-zero algebra, use its trace form for B. The resulting operator C on that representation has trace $\dim \mathfrak{g}$.

Proof. The tensor $\sum_i x_i \otimes y_i$ corresponds to the identity endomorphism under $u \otimes v \mapsto (w \mapsto B(v,w)u)$, proving independence. Write $[a, x_i] = \sum_j c_{ji} x_j$. Invariance gives $[a, y_i] = -\sum_j c_{ij} y_j$. The adjoint action on the tensor is consequently zero: the two double sums cancel on interchanging their indices. Multiplication into $U(\mathfrak{g})$ is compatible with the commutator derivation rule, so $[a, C_B] = 0$ for every generator a and hence for every element of $U(\mathfrak{g})$. Finally $\operatorname{tr} C = \sum_i B(x_i, y_i) = \dim \mathfrak{g}$. □

Over an algebraically closed field, an endomorphism of an irreducible nonzero module commuting with its action is scalar: it has an eigenvalue, and the nonzero kernel after subtracting that eigenvalue is a submodule and therefore the whole module. A semisimple algebra acts trivially on every one-dimensional module, since it equals its derived algebra and scalars commute.

Theorem 4.2 (Weyl). Every finite-dimensional module over a semisimple Lie algebra in characteristic zero is a sum of irreducibles. Every submodule has an invariant complement, over the given ground field.

Proof over an algebraically closed field. First suppose $W \subset V$ is irreducible and $\dim(V/W) = 1$. The quotient action is trivial, so $\mathfrak{g}V \subseteq W$. If the action on V is zero, any linear complement works. Otherwise replace $\mathfrak{g}$ by its nonzero image $\mathfrak{l} \subseteq \operatorname{End}(V)$, which is semisimple by Proposition 3.2. Its trace form is nondegenerate. Its Casimir operator maps V into W and acts by a scalar a on W. Its trace is both $a \dim W$ and $\dim \mathfrak{l}$, a nonzero scalar in characteristic zero. Thus $a \neq 0$. The kernel is an invariant line disjoint from W, and is a complement.

For arbitrary W of codimension one, induct on $\dim V$. If $W = 0$ or is irreducible, the conclusion is settled. Otherwise choose $0 \neq U \subsetneq W$ invariant. By induction, W/U has a complementary invariant line in V/U. Its inverse image T satisfies $T \cap W = U$ and T/U has dimension one. Since $\dim T < \dim V$, induction complements U in T by an invariant line L. Then $L \cap W = 0$ and $\dim L = 1$, giving the desired complement to W.

For a general nonzero $W \subset V$, put the commutator action on $\operatorname{Hom}_k(V, W)$ and define

$$A = \{P : P|_W = a \operatorname{id}_W \text{ for some } a \in k\}, \qquad B = \{P : P|_W = 0\}. \tag{4.1}$$

Restriction to W gives a surjection $A \to k$, because id_W extends linearly to V. Its kernel is B. For $x \in \mathfrak{g}$ and $P \in A$, $(xP)|_W = 0$, so both spaces are invariant and A/B is trivial. The codimension-one conclusion supplies $A = B \oplus L$ with L an invariant line. Its action is trivial. Normalize $0 \neq P \in L$ to have $P|_W = \operatorname{id}_W$. This P commutes with the action, and $V = W \oplus \ker P$. The cases $W = 0$ and $W = V$ are immediate.

Over an arbitrary characteristic-zero field k, extend to an algebraic closure. The extended algebra is semisimple by Proposition 3.1, so an equivariant projection $V_\Omega \to W_\Omega$ exists. The equations specifying restriction to W as the identity and commutation with a basis of $\mathfrak{g}$ form a finite linear system with coefficients in k. Section 1 shows that consistency over Ω implies consistency over k. The resulting projection has invariant kernel and gives the complement over k. Finally choose an irreducible submodule of minimum positive dimension and split it off; induction on dimension gives a finite direct sum of irreducibles. □

No assertion about modular complete reducibility is used. In particular, the adjoint module being a sum of irreducibles does not characterize semisimple Lie algebras: a nonzero abelian algebra has a trivial adjoint action. The draft also does not import the positive-characteristic claim in LAG, Aside 5.22, from the single trace computation.

5. The finite root facts needed for the construction

Let Φ be a finite reduced crystallographic root system in a positive definite real space E: it spans E, is stable under $s_\alpha(x) = x - 2(x,\alpha)\alpha/(\alpha,\alpha)$, has integral root-coroot pairings, and has only $\pm\alpha$ on each root line. We give the Euclidean inputs directly, without assuming a semisimple Lie algebra exists. For $E = 0$ the root set is empty and every assertion has its empty interpretation.

For nonproportional roots α, β, the product of their Cartan integers is $4(\alpha,\beta)^2/((\alpha,\alpha)(\beta,\beta))$, an integer strictly smaller than 4. The strict bound follows by expanding the positive squared norm of $\beta - (\alpha,\beta)\alpha/(\alpha,\alpha)$. If the inner product is positive, those integers are positive with product $1, 2$ or 3, so one is 1. One reflection therefore gives $\alpha - \beta$ or its negative as a root. Negation is also a reflection image, so $\alpha - \beta$ is a root. Replacing β by $-\beta$ treats a negative inner product.

Choose $t \in E$ with $(t,\alpha) \neq 0$ for every root. Such a vector exists: the product of the finitely many nonzero linear coordinate polynomials is a nonzero polynomial, since a leading monomial of a product has a nonzero product coefficient. A nonzero polynomial over an infinite field does not vanish everywhere, by induction on the number of variables using the finite-root bound in one variable. Set $\Phi^+ = \{\alpha : (t,\alpha) > 0\}$ and let Δ be its roots not expressible as sums of two positive roots. Induction on the finite ordered set of positive values (t,α) expresses every positive root as a sum of members of Δ, since the two summands of a decomposition have smaller positive values. Distinct members of Δ have nonpositive inner product: otherwise their difference is a root, whose positive sign in either direction decomposes one of them. A real linear relation between them can be split into equal positive combinations with disjoint supports. The inner product between those combinations is nonpositive, but it is the squared norm of their common vector. That vector must be zero. A nonempty positive combination has positive pairing with t, so both supports are empty. Thus Δ is independent and spans E. Every root has unique integer coordinates in this base, with all coordinates nonnegative or all nonpositive.

The coroots $\alpha^\vee = 2\alpha/(\alpha,\alpha)$ themselves form a reduced root system. Orthogonality gives $s_\alpha(\beta^\vee) = (s_\alpha\beta)^\vee$, and their Cartan integers are the transposed original integers; spanning and reducedness are immediate from their root lines. Their positive set is on the same rays as Φ^+. The cone generated by it is therefore exactly the cone generated by the $\alpha_i^\vee$, because each positive root is a nonnegative combination of the α_i and every coroot is a positive multiple of its root. For any linearly independent base, the extreme rays of its nonnegative cone are precisely its basis rays: a vector with two positive coordinates splits into two nonproportional positive vectors, whereas a basis ray cannot so split by uniqueness of coordinates. Applying the base construction just proved to the dual system identifies its simple roots with these extreme rays. Reducedness leaves exactly $\alpha_i^\vee$ on each such positive ray. Hence the simple coroots are a base, and every coroot has an integral same-sign expansion in them.

Finally s_i permutes $\Phi^+ \setminus \{\alpha_i\}$ and negates α_i. A different positive root has a positive coefficient outside i, since reducedness excludes a larger multiple of α_i. Reflection changes only its i-coefficient. Its image is a root with an unchanged positive coefficient, so the same-sign property makes it positive; involutivity proves the permutation assertion. These root statements have just been proved directly, before use in the Serre construction.

The Cartan matrix in this draft is $a_{ji} = \langle \alpha_j, \alpha_i^\vee \rangle$, so its columns index coroots. The simple coroots form a base of the dual root system and are a basis of the space. The pairing matrix between two bases of a space with a nondegenerate form is invertible. Consequently the α_j, defined by these values on a formal basis (h_i), are independent rational linear forms. In coordinates the pairing matrix is a product of the two invertible basis-change matrices and the invertible Gram matrix. The Gram matrix is invertible because a vector in its kernel would have squared norm zero; positive definiteness forces that vector to vanish. Gaussian elimination from Section 1 then supplies the inverse, also over $\mathbb{Q}$ when its entries are rational.

Every positive root is moved to a simple root by a product of simple reflections. For a nonsimple

$\beta = \sum b_i \alpha_i$ with $b_i \geq 0$, positive definiteness gives some i with $(\beta, \alpha_i) > 0$. That b_i is positive, because otherwise the nonpositive cross inner products would force $(\beta, \alpha_i) \leq 0$. The integer $\langle \beta, \alpha_i^\vee \rangle$ is positive. Reflection changes only its i-coefficient; some coefficient outside i is positive unless β is simple, so the same-sign root property makes $s_i\beta$ positive with smaller height. Iteration terminates at a simple root. Negative roots are handled by negation and the reflection of a simple root. Since $s_{w\alpha} = w s_\alpha w^{-1}$, this also proves generation of the full Weyl group by the simple reflections. The group is finite because it permutes the finite spanning root set faithfully.

For completeness, there is an element w_0 taking all positive roots to negative roots, and it satisfies $w_0^2 = 1$. Define $N(w)$ to be the number of positive roots made negative. If $w\beta < 0$ for a positive root and $w = s_{i_1} \cdots s_{i_m}$, read the suffix images of β from right to left. At a sign change the immediately preceding positive root must be the simple root of that step, since s_i negates only α_i. If u is that suffix, $u\beta = \alpha_{i_j}$ and $us_\beta = s_{i_j}u$. Hence ws_β is the given word with its jth letter removed. For a shortest word this implies $\ell(ws_\beta) < \ell(w)$. Conversely if this inequality holds but $w\beta > 0$, apply the proved implication to ws_β and β to obtain the reverse strict inequality, a contradiction. For a simple reflection, the length changes by at most one. If $w\alpha_i$ is negative, the proved implication makes the change a strict decrease. If it is positive, $(ws_i)\alpha_i$ is negative, and the same implication applied to ws_i makes the change a strict increase. Thus the change is exactly minus or plus one according to that sign. The simple sign rule gives the identical change for $N(ws_i)$. Prefixes of a shortest word are shortest, proving $N(w) = \ell(w)$ by induction on such a word.

Choose an element of maximum length. Multiplying it by any s_i shortens it, so it takes all simple roots to negative roots. It therefore takes every positive root to a negative root by their positive expansions. If two elements do this, their quotient preserves all positive roots; its N is zero and its length is zero, so the two are equal. The square of this unique element preserves positives, giving $w_0^2 = 1$. Also $w_0 Q^+ = -Q^+$, where $Q^+ = \sum_i \mathbb{Z}_{\geq 0}\alpha_i$.

These deductions are the only Weyl-group inputs to Sections 6–7. No classification of complex Lie algebras, conjugacy of Cartan subalgebras, or Lie-group existence theorem is used.

6. A rational Serre algebra with the prescribed roots

Set $H = \bigoplus_i \mathbb{Q}h_i$, and define $\alpha_j(h_i) = a_{ji}$. Let $\widetilde{\mathfrak{g}}$ be the Lie algebra generated by e_i, f_i, h_i with

$$[h_i, h_j] = 0, \quad [h_i, e_j] = a_{ji}e_j, \quad [h_i, f_j] = -a_{ji}f_j, \quad [e_i, f_j] = \delta_{ij}h_i. \tag{6.1}$$

A free Lie algebra can be defined by formal bracket expressions modulo bilinearity, alternation and Jacobi; evaluating expressions proves its universal property. By Theorem 1.A it embeds in its enveloping algebra. The universal property identifies the enveloping algebra of the free Lie algebra on a space V with $T(V)$, since both have algebra maps to an associative algebra in bijection with linear maps V to that algebra. Thus free Lie expressions embed in tensor words. This supplies the separation used next.

Lemma 6.1. The auxiliary algebra has a direct decomposition

$$\widetilde{\mathfrak{g}} = \widetilde{\mathfrak{n}}^- \oplus H \oplus \widetilde{\mathfrak{n}}^+, \tag{6.2}$$

where the two side algebras are free on their respective f_i and e_i. It is graded by $Q = \bigoplus_i \mathbb{Z}\alpha_i$, with these generators of degrees $-\alpha_i, \alpha_i$ and h_i of degree zero. There are no mixed-sign degrees, the degree-zero part is H, and each nonzero degree is finite-dimensional.

Proof. Jacobi rewrites a word on the f_i as a combination of words of the form $[f_j, w]$. Induction on its length, using $[e_i, [f_j, w]] = [[e_i, f_j], w] + [f_j, [e_i, w]]$, puts its bracket with e_i in H for length one and in the negative side for larger length. The positive version follows in the same way. Induction on the length of a positive word, with $[[e_i, u], w] = [e_i, [u, w]] - [u, [e_i, w]]$, now puts every mixed

bracket in the sum in (6.2). The sum contains the generators and is a subalgebra, so it spans the quotient.

To prove the promised embeddings rather than assume them, act on $M = T(\mathbb{Q}\{F_1, \dots, F_r\}) \otimes \mathbb{Q}[z_1, \dots, z_r]$. For a word w of positive letter degree β, put

$$\begin{aligned} \mathcal{F}_j(w \otimes P) &= F_j w \otimes P, \\ \mathcal{H}_i(w \otimes P) &= w \otimes (z_i - \beta(h_i))P, \\ \mathcal{E}_i(1 \otimes P) &= 0, \\ \mathcal{E}_i(F_j w \otimes P) &= F_j \mathcal{E}_i(w \otimes P) + \delta_{ij} w \otimes (z_i - \beta(h_i))P. \end{aligned} \tag{6.3}$$

The Cartan operators commute. Prepending F_j changes its degree by α_j and gives the $[\mathcal{H}_i, \mathcal{F}_j]$ relation. Every output term of $\mathcal{E}_j$ deletes one F_j, giving $[\mathcal{H}_i, \mathcal{E}_j] = a_{ji}\mathcal{E}_j$. The last recursion is precisely $[\mathcal{E}_i, \mathcal{F}_j] = \delta_{ij}\mathcal{H}_i$. Thus all defining relations hold. On $1 \otimes 1$, the h_i give the independent polynomials z_i. Negative Lie words act by left multiplication by their free tensor commutators, and evaluating at $1 \otimes 1$ proves their independence. The involution $e_i \leftrightarrow f_i$, $h_i \mapsto -h_i$ of (6.1) gives the positive embedding. The homogeneous grading separates the three summands. A fixed side degree has a fixed finite number of letters, with only finitely many orderings and bracket patterns, so its dimension is finite. $\square$

For $i \neq j$, impose the additional relations

$$S_{ij}^{+} = (\operatorname{ad} e_i)^{1-a_{ji}} e_j = 0, \qquad S_{ij}^{-} = (\operatorname{ad} f_i)^{1-a_{ji}} f_j = 0. \tag{6.4}$$

Let I^+ be the ideal generated by S_{ij}^{+} inside the positive free algebra, and I^- its negative analogue. The ideal they generate in the whole auxiliary algebra is exactly $I^- \oplus I^+$. To verify this, first note that each S_{ij}^{+} is killed by every $\operatorname{ad} f_k$. For k distinct from i, j, all brackets with its letters vanish. For $k = i$, on a vector with $[f_i, v] = 0$, $[h_i, v] = \lambda v$, commutator induction gives

$$[f_i, (\operatorname{ad} e_i)^m v] = -m(\lambda + m - 1)(\operatorname{ad} e_i)^{m-1} v. \tag{6.5}$$

Indeed the next step contributes $-(\lambda + 2m)$ from $[f_i, e_i] = -h_i$ and the previous coefficient, whose sum is $-(m+1)(\lambda + m)$. Substituting $v = e_j$, $\lambda = a_{ji}$, $m = 1 - a_{ji}$ gives zero. For $k = j$, commute $\operatorname{ad} f_j$ past the e_i adjoints and obtain $-(\operatorname{ad} e_i)^m h_j$. If $a_{ji} < 0$, then $m \geq 2$ and two adjoints kill h_j. If $a_{ji} = 0$, the symmetric zero pattern gives $a_{ij} = 0$, so the first adjoint already kills it.

Now I^+ is spanned by repeated adjoints of positive generators on these homogeneous elements, and is H-stable. To justify this spanning description, take the span of all iterated adjoints of the generators on the specified relations. It is stable under each generator's adjoint by definition. Jacobi expresses the adjoint of a bracket as a commutator of adjoints, so it is stable under every positive Lie word. It is consequently an ideal containing those relations, and every ideal containing them contains this span. Induct on that number of adjoints and use $[f_k, [e_l, u]] = [[f_k, e_l], u] + [e_l, [f_k, u]]$. The first term is zero or a Cartan action on I^+, and the second uses induction. Thus I^+ is an ideal of the entire auxiliary algebra. The involution proves the same for I^-. Their sum is direct and contains exactly the required generating relations, proving the claim. The Serre quotient consequently has

$$\mathfrak{g}_{\mathbb{Q}} = \mathfrak{n}^- \oplus H \oplus \mathfrak{n}^+. \tag{6.6}$$

In particular the h_i survive independently, and $e_i, f_i \neq 0$ since their bracket is h_i. Their degree spaces are one-dimensional. A side word using only one generator has length one or is zero, so there are no degrees $m\alpha_i$ with $|m| > 1$.

The adjoints of e_i and f_i are locally nilpotent. On each generator this follows from (6.4) or from the three rank-one brackets; on f_i, for example, successive e_i adjoints are $h_i, -2e_i, 0$. If a

derivation kills u in p steps and v in q steps, its $(p+q-1)$st iterate on $[u,v]$ is zero by the binomial derivation formula. Induction on bracket words proves local nilpotence everywhere. Thus the finite-on-each-vector exponential automorphism

$$T_i = \exp(\operatorname{ad} e_i)\exp(-\operatorname{ad} f_i)\exp(\operatorname{ad} e_i) \tag{6.7}$$

is defined even before dimension is known. The binomial derivation formula proves bracket preservation, and $\exp(-D)$ is inverse to $\exp(D)$ by multiplication of the finite series on each vector. The rank-one brackets directly give $T_i h_i = -h_i$, $T_i e_i = -f_i$, $T_i f_i = -e_i$. The part of H in $\ker \alpha_i$ commutes with both root generators, so $T_i H' = H' - \alpha_i(H')h_i$. Transporting the weight equation gives $T_i \mathfrak{g}_\beta = \mathfrak{g}_{s_i\beta}$. No Coxeter relations among the T_i are required.

Theorem 6.2. The algebra (6.1), (6.4) is finite-dimensional, has exactly the roots Φ with one-dimensional root spaces, and is semisimple over $\mathbb{Q}$ and every characteristic-zero extension field. Its Cartan span is H.

Proof. Consider a nonzero positive degree $\beta = \sum_i b_i\alpha_i$ with $b_i \geq 0$. If only one coefficient is nonzero, the preceding single-generator observation makes it a simple root. Otherwise positive definiteness gives i with $b_i > 0$ and $\langle \beta, \alpha_i^\vee \rangle > 0$. The reflection automorphism makes $s_i\beta$ another nonzero degree. It has an unchanged positive coefficient outside i, so the absence of mixed-sign degrees makes it positive. Its height is strictly lower. Induction shows it, and hence β, is a root. The negative argument follows by the involution. Conversely each root is moved to a simple root by Section 5, and the reflection automorphisms show that its degree space exists and has dimension one. Thus the total dimension is $r + |\Phi|$.

For a connected simple-root diagram, let J be a nonzero ideal. It is stable under the commuting diagonalizable adjoints of H, so polynomial interpolation separates any finite list of weight components and places each component in J. A nonzero Cartan component h detects some simple root because the α_i form a basis of H^*; then $[h, e_i] \neq 0$ puts a root vector in J. Otherwise J already has a root vector. An ideal is preserved by every finite exponential of an inner derivation, including its inverse. Apply the T_i to move that root to a simple root, giving $e_i \in J$. Then $h_i = [e_i, f_i] \in J$ and $f_i = -[h_i, f_i]/2 \in J$. If j is adjacent to i, its nonzero Cartan integer gives e_j from $[h_i, e_j]$ and then h_j, f_j in the same way. Connectivity yields all generators. The algebra is therefore nonabelian simple.

For several diagram components, generators from distinct components commute: (6.4) with exponent one gives this for positive-positive and negative-negative pairs, while (6.1) gives all mixed and Cartan pairs. The quotient is their direct sum, by (6.6) and the grading, and each component is simple. It is semisimple. Every argument is valid over any characteristic-zero field with the same rational coefficients. Alternatively, the Killing criterion reflects the rational conclusion from an algebraic closure. The centralizer of H is H by the weight decomposition. If an element normalizes H, its nonzero root components would give nonzero root components in a bracket with a suitable h, so there are none; thus its normalizer is also H. This gives the Cartan claim directly. □

Tensor and bracket-word presentations commute with field extension: all relations and all their finite bracket consequences span a rational subspace, whose extension is the subspace of the extended relations. The tensor basis argument in Section 1 proves exactness of the quotient operation. This verifies the precise rational form and every characteristic-zero base change; it does not assert the same finite-type theorem in prime characteristic.

7. The finite integrable highest-weight presentation

Fix $\lambda \in H^*$ with $n_i = \lambda(h_i) \in \mathbb{Z}_{\geq 0}$. The fundamental case is $\lambda = \omega_j$, where $\omega_j(h_i) = \delta_{ij}$. The simple coroots form a base of the dual root system, so their integral dual lattice is $P = \bigoplus_i \mathbb{Z}\omega_i$. The root lattice lies in P by the Cartan integers.

Order a root basis of $\mathfrak{n}^-$ first, the h_i next, and a root basis of $\mathfrak{n}^+$ last. Theorem 1.A gives a vector-space multiplication isomorphism

$$U(\mathfrak{n}^-) \otimes U(H) \otimes U(\mathfrak{n}^+) \longrightarrow U(\mathfrak{g}). \tag{7.1}$$

It also embeds each subalgebra's enveloping algebra: extend its ordered basis to one of the full algebra and compare the two PBW word sets. With $\mathfrak{b} = H \oplus \mathfrak{n}^+$ and its character λ killing $\mathfrak{n}^+$, let $M(\lambda) = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} k_\lambda$. Equivalently it is the quotient by the left ideal generated by the e_i and $h_i - n_i$. The tensor generator obeys those relations; conversely they make every $\mathfrak{b}$ word act on the cyclic generator by its character, giving the inverse tensor map. The multiplication in (7.1) respects the right $U(\mathfrak{b})$-action, so it identifies $M(\lambda)$ with $U(\mathfrak{n}^-)$ as a vector space. In particular its top vector v is nonzero, its top space is one-dimensional, and its weight spaces have weights $\lambda - Q^+$.

Each of these weight spaces is finite-dimensional. A monomial with multiplicities m_α in the negative root basis has degree $-\sum_{\alpha>0} m_\alpha \alpha$. Fixing $\gamma \in Q^+$ bounds $\sum m_\alpha$ by the height of γ, since every positive root has positive integral height. Only finitely many tuples are possible. For a vector with finitely many distinct weight components, choose $h \in H$ separating their values; the excluded equalities are finitely many proper linear hyperplanes over an infinite field. Lagrange polynomials in h project onto the separate components. Every H-stable subspace and its quotient are consequently weight modules with the corresponding intersections and quotient spaces. The separating element exists by the polynomial argument in Section 5 applied to the finite product of the distinct-weight differences. For the distinct values $c_1, \dots, c_s$ on these components, the polynomial $\prod_{j \neq i}(h - c_j)/(c_i - c_j)$ is the projection onto the ith component. This proves the stated intersections and quotient-weight decomposition for each vector, even in the still infinite module.

The rank-one commutator calculation, valid on this cyclic vector without a finite-dimensional rank-one classification, is

$$h_i f_i^m v = (n_i - 2m) f_i^m v, \qquad e_i f_i^m v = m(n_i - m + 1) f_i^{m-1} v. \tag{7.2}$$

For the second formula, commute e_i past one more f_i and use $[e_i, f_i] = h_i$ and the first formula; this adds $n_i - 2m$ to the previous coefficient $m(n_i - m + 1)$, producing $(m+1)(n_i - m)$. The first formula follows by the same induction from $[h_i, f_i] = -2f_i$.

It follows that $z_i = f_i^{n_i+1} v$ is killed by every e_j: use (7.2) for $j = i$ and $[e_j, f_i] = 0$ otherwise. Thus it is a highest vector of weight $\lambda - (n_i + 1)\alpha_i$. The same induced-module construction gives a map from the Verma module of that weight to its generated submodule, so all those weights lie below it. In particular that submodule has no top λ-component. Hence

$$T(\lambda) = M(\lambda) / \sum_i U(\mathfrak{g}) z_i \tag{7.3}$$

is nonzero, cyclic, and has a one-dimensional top weight space. This proves properness of the defining sum before any finite-dimensional complete reducibility is invoked.

Each $\operatorname{ad} e_i$ and $\operatorname{ad} f_i$ is nilpotent on the finite-dimensional $\mathfrak{g}$ by Section 6. On $U(\mathfrak{g})$ the same derivation is locally nilpotent: the iterated Leibniz rule on a fixed finite word distributes more derivatives than the sum of the bounds on its letters. Finite sums of words retain a bound. For any Lie element x, multiplication and commutators give

$$x^N(uv) = \sum_{a=0}^{N} \binom{N}{a} (\operatorname{ad} x)^a(u) x^{N-a} v. \tag{7.4}$$

This follows by induction using $xu = [x, u] + ux$, with Pascal's identity combining the two sums. In (7.3), $f_i^{n_i+1} v = 0$ and $e_i v = 0$. If $(\operatorname{ad} f_i)^a u = 0$ for $a > R$, choose $N > R + n_i$; every summand in

(7.4) vanishes. For e_i, choose $N > R$. Thus both simple root operators act locally nilpotently on $T(\lambda)$.

On this possibly infinite module, the operator $S_i = \exp(e_i)\exp(-f_i)\exp(e_i)$ is well defined and invertible, since each exponential terminates on each vector. The conjugation formula for an operator A with terminating iterated commutators is $\exp(x)A\exp(-x) = \sum_m (\operatorname{ad} x)^m(A)/m!$. It follows by the binomial expansion; each expression evaluated on a fixed vector involves only finitely many terms, because x is locally nilpotent on that vector and on the finitely many commutator images. Indeed, on a fixed vector the rightmost $\exp(-x)$ uses finitely many powers x^b. Local nilpotence also bounds the leftmost powers on the finitely many vectors Ax^bv. Expanding this finite double sum and the identity $(\operatorname{ad} x)^m A = \sum_{b=0}^{m}(-1)^b\binom{m}{b}x^{m-b}Ax^b$ gives exactly the displayed conjugation formula. The latter identity follows by induction from $[x, A] = xA - Ax$ and Pascal's identity. The rank-one brackets give

$$\exp(e_i)h_i\exp(-e_i) = h_i - 2e_i, \quad \exp(-f_i)h_i\exp(f_i) = h_i - 2f_i, \quad \exp(-f_i)e_i\exp(f_i) = e_i + h_i - f_i. \tag{7.5}$$

Applying the three conjugations gives $S_ih_iS_i^{-1} = -h_i$. The part of H in $\ker\alpha_i$ is fixed, so $S_ihS_i^{-1} = h - \alpha_i(h)h_i = s_ih$. Therefore S_i takes the μ-space isomorphically to the $s_i\mu$-space. The weights and their multiplicities are Weyl-invariant before finiteness is known.

Every occurring μ satisfies $\lambda - \mu \in Q^+$. Weyl invariance also gives $\lambda - w_0\mu \in Q^+$, and applying w_0 gives $\mu - w_0\lambda \in Q^+$. Write $\lambda - w_0\lambda = \sum_i b_i\alpha_i$ and $\lambda - \mu = \sum_i d_i\alpha_i$. Since λ occurs and Weyl translates occur, the b_i are nonnegative integers; the two bounds give $0 \le d_i \le b_i$. There are finitely many integer tuples in this box. Every weight space is a quotient of a finite Verma weight space, so $T(\lambda)$ is finite-dimensional.

It is irreducible. If N is a proper submodule, weight projection gives $N_\lambda = 0$, since a nonzero vector there would be the cyclic generator. Theorem 4.2 gives an invariant complement S. The projections preserve weights, so the projection of v to N is zero and $v \in S$. Cyclicity forces $S = T(\lambda)$ and $N = 0$. We have proved the finite irreducible presentation

$$L(\lambda) = U(\mathfrak{g})\Big/\Big(\sum_i U(\mathfrak{g})e_i + \sum_i U(\mathfrak{g})(h_i - n_i) + \sum_i U(\mathfrak{g})f_i^{n_i+1}\Big). \tag{7.6}$$

This is a quotient by a left ideal; none of these module relations is asserted to be a two-sided algebra relation.

All constructions are rational. Tensor powers, bracket relations, associative relations and left ideals extend exactly under a field extension, by Section 1. Construct (7.6) over $\mathbb{Q}$ and extend to an algebraic closure. The preceding proof shows that extension is finite-dimensional. Selecting rational vectors whose extended images span it, exactness shows that their rational span is already the whole rational module. Thus the rational module is finite-dimensional. The displayed PBW weights are rational linear forms, so the rational weight spaces are simultaneous kernels of finite matrices and extend to the indicated weight spaces. Over every characteristic-zero extension field the top remains a one-dimensional nonzero space, the vector remains cyclic, and the algebra remains semisimple. The same complement argument proves irreducibility there. Consequently this particular rational module is absolutely irreducible. No assertion of absolute irreducibility for arbitrary rational modules is needed.

8. The precise consequences consumed by AG-RG-05

Take $V = \bigoplus_j L_{\mathbb{Q}}(\omega_j)$. These summands are pairwise nonisomorphic, including after any characteristic-zero extension. Indeed all weights of $L(\lambda)$ lie in $\lambda - Q^+$ and its top occurs; an

isomorphism for λ, μ would imply both $\lambda - \mu \in Q^+$ and $\mu - \lambda \in Q^+$, giving $\lambda = \mu$. The distinct dual basis elements ω_j cannot be equal.

The action of $\mathfrak{g}_{\mathbb{Q}}$ on V is faithful. Its kernel is an ideal, hence a sum of the simple diagram factors. A factor containing the index j acts nontrivially on $L(\omega_j)$: h_j acts by 1 on its top. Thus no factor is in the kernel. The same argument works after field extension.

Each root vector acts nilpotently on every summand. Its bracket with H shows that it sends V_μ into $V_{\mu+\alpha}$. The finite weight set contains no arbitrarily long sequence $\mu + m\alpha$ for a nonzero root in characteristic zero. A sufficiently high power is therefore zero on every weight space, hence on the module. This proves the polynomial exponentials for all root vectors, with no appeal to an unproved abstract Jordan-compatibility theorem.

The split torus with character lattice P acts by the weights. Its action is faithful as a group scheme: its kernel is cut out by all characters occurring in V being one, and among them are the basis characters ω_j. Over every test algebra those equations force every coordinate of the split torus to be one. Here is the closed-immersion proof. Choose the highest vector of each fundamental summand among a rational weight basis of V. For the Laurent coordinate t_j of the basis character ω_j, the corresponding diagonal matrix entry restricts to t_j. The corresponding entry of the inverse matrix restricts to t_j^{-1}; inverse matrix entries are regular functions on $\mathrm{GL}(V)$ by the adjugate formula and inversion of its determinant. These restricted functions generate $\mathbb{Q}[P] = \mathbb{Q}[t_1^{\pm 1}, \dots, t_r^{\pm 1}]$. Thus the coordinate-ring map $\mathbb{Q}[\mathrm{GL}(V)] \to \mathbb{Q}[P]$ is surjective, proving that the torus representation is a closed immersion. The determinant and adjugate formulas used in this assertion follow over every ring from the permutation sum for the determinant. In multiplying a matrix by its cofactor transpose, a diagonal entry is the cofactor expansion of the determinant, while an off-diagonal entry is the determinant with a repeated row or column. That sum is zero by pairing the terms exchanged by those repeated rows or columns. Thus both products are the determinant times the identity; after inverting it, the cofactor entries give the regular inverse matrix.

For each irreducible summand the sum of its weights with multiplicities is zero. The functional $h \mapsto \operatorname{tr}(h|_{L(\omega_j)})$ is zero on $\mathfrak{g}$ because $\mathfrak{g}$ is perfect and traces of commutators vanish. On H it is exactly the sum of weights. Independence of the h_i forces that sum to be zero as an element of P. Thus the torus determinant on each summand is identically one; this proof is a lattice computation. A root exponential has determinant one: choose the filtration by kernels of powers of its nilpotent generator, making its exponential upper triangular with every diagonal entry one. Consequently the generated algebraic group lies in the product of these special linear groups.

Over an algebraic closure, Schur's scalar lemma and the pairwise nonisomorphism imply $\mathrm{End}_{\mathfrak{g}}(V) = \bigoplus_j k \operatorname{id}_{V_j}$. The off-diagonal blocks are zero because a nonzero homomorphism between simple modules has zero kernel and full image and would be an isomorphism. If an operator A normalizes the faithful image of $\mathfrak{g}$, $[A, \rho(x)]$ gives a derivation of $\mathfrak{g}$. Proposition 3.3 supplies z with this derivation $\operatorname{ad} z$, so $A - \rho(z)$ commutes with $\mathfrak{g}$. This proves exactly the normalizer decomposition used in the Lie-algebra calculation of the generated group. If A also belongs to the Lie algebra of the product of the special linear groups, its block traces are zero. The block traces of $\rho(z)$ are zero by perfectness, so each remaining scalar c_j satisfies $(\dim V_j)c_j = 0$. Characteristic zero gives $c_j = 0$. The derivation assertion is the actual identity $[A, [X, Y]] = [[A, X], Y] + [X, [A, Y]]$, obtained by expanding both sides as products. Faithfulness transports it to the represented Lie algebra. The trace description of the special-linear Lie algebra is also direct: over dual numbers, the permutation formula gives $\det(I + \varepsilon A) = 1 + \varepsilon \operatorname{tr} A$, since every nonidentity permutation term has at least two off-diagonal factors and $\varepsilon^2 = 0$. This proves the block-trace condition used here.

Finally, the rational rank-one integration used in AG-RG-05, Lemma 6.1, needs only Theorem 4.2 and the following calculation. The trace-zero 2×2 algebra with basis e, f, h and brackets $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$ is simple: an ideal is stable under the diagonal operator $\operatorname{ad} h$ with eigenvalues $2, -2, 0$; polynomial projections isolate one of its basis lines, and bracketing any of those lines with the other generators yields all three. In an irreducible finite-dimensional

module over an algebraically closed characteristic-zero field, choose an h-eigenvector. Successive applications of e have distinct eigenvalues until they become zero; take the last nonzero vector v, with $ev = 0$ and $hv = \lambda v$. Similarly $f^j v$ eventually becomes zero. If $f^n v \neq 0$, $f^{n+1} v = 0$, the same commutator calculation as (7.2) gives $(n+1)(\lambda - n) f^n v = 0$, so $\lambda = n$. The vectors $v, fv, \ldots, f^n v$ have distinct eigenvalues and span a nonzero invariant submodule; irreducibility makes it the whole module.

The polynomial SL_2 action on $W_n = \mathrm{Sym}^n(k^2)$ realizes this string: on variables u, v its operators are $e = u\partial_v$, $f = v\partial_u$, $h = u\partial_u - v\partial_v$. Projection to individual h-weights followed by raising and lowering proves irreducibility. Theorem 4.2 decomposes every finite-dimensional rank-one module into these strings. The polynomial action itself is constructed on every test ring as follows. For $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $ad - bc = 1$, send u to $au + cv$ and v to $bu + dv$, and substitute in each homogeneous monomial of degree n. Multilinearity gives the action matrix with integral polynomial entries. Composing two substitutions is substitution by the matrix product, and substitution by the inverse matrix is inverse; hence these formulas are a group action on every commutative ring. Differentiating the upper and lower elementary matrices over dual numbers gives exactly the three displayed operators. The vectors $f^j u^n = n!/(n-j)!\, u^{n-j} v^j$, for $0 \leq j \leq n$, are a basis in characteristic zero. The upper and lower elementary substitutions are the finite exponential operators, by the binomial formula, and the diagonal substitution has weights $n - 2j$. For a rational module M, set $K_n = \ker e \cap \ker(h - n)$ and map $W_n \otimes K_n \to M$ by $f^j u^n \otimes w \mapsto f^j w$. Only finitely many K_n are nonzero, since the spectrum of h is finite. The map is equivariant by the string formulas, and becomes an isomorphism over an algebraic closure by the decomposition. Section 1 reflects that isomorphism to $\mathbb{Q}$. Transport the polynomial SL_2 action on the direct sum through it. These are polynomial matrix identities over $\mathbb{Q}$, so they define the rank-one group morphism on every commutative $\mathbb{Q}$-algebra, including nonreduced test algebras. Its diagonal and unipotent actions are the weight action and the finite exponentials just computed.

Flat quotient bootstrap over an arbitrary base

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This lesson proves the general flat quotient theorem used in lesson three, Section 5, and lesson five, Section 8. The base schemes, covering schemes and test schemes may be non-Noetherian and non-quasi-compact. Finite presentation is imposed only where stated. Freeness and equivalence relations mean uniqueness of arrows on every scheme test, including tests with nilpotents.

The proof proceeds through pointed constant fields, fibre dimensions and Cohen–Macaulay loci, regular cuts, finite-presentation descent, transverse slices, finite open subgroupoids and their division. Descent and Zariski Main supplies the complete preceding scheme proofs: blocks A and E for descent and recognition, block B for polynomial normality, block C for affine Zariski Main and block D for its general scheme form.

1. The exact earlier algebra that will be used

The following earlier programme proofs provide the algebra used below. Their hypotheses and precise statements are part of each application; links identify those proof bodies, rather than replacing an unwritten argument.

- Krull dimension and Noether normalization, Theorem 3.1 and Corollary 3.2, give normalization by polynomial changes with integer coefficients. Theorems 4.2–4.3 give dimension and height for finite-type field domains, and Theorem 6.1 gives

$$\dim_x \operatorname{Spec} B = \dim B_{\mathfrak{q}} + \operatorname{trdeg}_k \kappa(\mathfrak{q}) \tag{1}$$

  for an arbitrary finite-type k-algebra. Its proof treats all minimal components through the point.
- Regular local rings, Proposition 3.3, proves directly by polynomial induction that prime localizations of a polynomial ring over a field are regular. Regular sequences, depth and Cohen–Macaulay modules, Theorems 4.1–4.2, Corollary 4.3 and Theorems 5.1 and 6.1, prove absence of embedded associated primes, regularity of parameters, regular dimension/depth drops, localization of Cohen–Macaulay modules and Cohen–Macaulayness of regular local rings. Associated primes and primary decomposition, Theorems 1.2 and 2.2 and Solution 8.5, prove the zero-divisor, finiteness and finite prime-avoidance statements used to choose parameters.
- Discrete valuation rings, normal rings and Serre's criterion, Theorems 1.2 and 3.3, prove the discrete valuation and intersection-in-height-one tests for a Noetherian normal domain.
- Faithful flatness and the local criterion for flatness, Theorem 4.2 and Lemma 5.1, prove the Noetherian residue-Tor criterion and lifting of an injective fibre map. Their proof uses the actual Artin–Rees/Krull intersection proof, also written in Cohomology and models, Section 1.

- Flatness criteria, dimension and the flat locus, Theorem 3.1, proves the flat local dimension formula; Theorem 4.1 proves miracle flatness; Lemmas 6.1–6.2 and Theorems 6.3–6.4 prove the finite-obstruction approximation, arbitrary-base fibre criterion and arbitrary-base openness of flatness. The restriction to finite presentations in the last three assertions is essential.
- Algebra and sheaf cohomology, Theorem 8.4, proves finiteness of the cohomology of a coherent sheaf on projective space over a Noetherian ring. We need its degree-zero case for a coherent sheaf supported on a projective variety.
- Descent and Zariski Main, block D, Sections D2–D3 and D5–D7, proves the standard étale local form, elementary finite pieces, the open quasi-finite locus, compatibility of relative integral closure with étale base change, and finite completion of a quasi-finite separated morphism over a qcqs base.
- Descent and Zariski Main, block A and Sections E4–E5, proves faithfully flat algebra and morphism descent, invariant-open descent, quasi-affine effectivity and effectivity of separated locally quasi-finite scheme data. Its Sections E6–E8 prove the finite flat affine quotient and scheme recognition. These statements precede the present quotient bootstrap.
- Algebraic spaces, Lemma 2.1, Theorem 2.2 and Proposition 3.2, prove local representatives, unique equality arrows, the étale scheme quotient and open gluing. Its previously external scheme-effectivity input is now the complete preceding scheme lesson; its target-local étaleness input is Lemma 5.3 below, which is independent of quotient construction. The étale bootstrap, Proposition 1.1, Lemma 1.2 and Theorem 2.1, proves the corresponding representability-by-spaces permanence and étale bootstrap directly from these results and scheme recognition. The scheme gluing and local quotient-coordinate constructions used in that proof are written in Descent and Zariski Main, Lemma A1.0 and Lemma A1.7, including arbitrary set-indexed covers and uniqueness of equality arrows on tests with nilpotents.

Each use below specifies which of these actual proved statements applies. This list grants no exemption to an unspecified algebraic or geometric prerequisite. In the preceding local flatness and finite-obstruction proofs, the independence, balance and exact-sequence properties of Tor are supplied by Lemma 1.A below. In particular this binding applies to the residue-Tor and regular-cut arguments used in Sections 3–4 and to the Tor-obstruction transport used in Section 5; it applies over arbitrary rings and to arbitrary modules, without an additional finiteness assumption.

Lemma 1.A (the Tor calculus used in the local criteria). Let R be any commutative ring and M, N any modules. Choose a free resolution $P_\bullet \to M$ by taking a free surjection onto the module and then onto each successive kernel, and define $\operatorname{Tor}_i^R(M,N) = H_i(P_\bullet \otimes_R N)$. This definition is independent of the resolution, naturally symmetric in M, N, and has the natural long exact sequence of a short exact sequence in either variable.

Proof.

For completeness, the resolution facts used here have the following algebraic proof. Given free resolutions $P_\bullet \to M$ and $P'_\bullet \to M'$ and a map $M \to M'$, lift it to $P_0 \to P'_0$. If maps have been chosen through degree $i-1$, the image of $d(P_i)$ lies in the cycles of P'_{i-1}, which are the image of P'_i; freeness of P_i therefore supplies the next lift. Two such lifts are chain homotopic: after the homotopy has been chosen below degree i, their degree-i difference minus the already prescribed homotopy term maps to zero under the next differential, and freeness lifts it to P'_{i+1}. The same induction starts at degree zero because both lifts induce the same map on the augmentation. Tensoring preserves the homotopy identity. Thus induced homology maps are independent of the lifts, preserve compositions, and comparison maps between two resolutions of the same module are inverse on homology.

Let $Q_\bullet \to N$ be another free resolution and form the first-quadrant double complex $D_{p,q} = P_p \otimes_R Q_q$, with total differential $d_P \otimes 1 + (-1)^p 1 \otimes d_Q$. Augmentation gives maps from its total complex to $P_\bullet \otimes_R N$ and to $M \otimes_R Q_\bullet$. Each is a homology isomorphism. Indeed, filter by the degree in the unaugmented direction. In the other direction each augmented row or column is exact because its

fixed factor is free. To check the resulting homology assertion directly, start with the outermost nonzero component of a cycle in the augmented kernel and use this exactness to subtract a total boundary that removes that component; proceed to the next component. A chain in any fixed total degree has only finitely many components, so this process terminates. The same process applied to boundaries proves injectivity. The comparison maps just constructed make these two augmentation isomorphisms natural. Interchanging the two resolutions, with the sign $(-1)^{pq}$ on $P_p \otimes_R Q_q$, consequently gives

$$\operatorname{Tor}_i^R(M,N) \simeq \operatorname{Tor}_i^R(N,M).$$

In particular Tor can be computed using a free resolution of either variable; a projective resolution works as well, since projectives are summands of free modules and the same lifting and exactness arguments apply.

For $0 \to N' \to N \to N'' \to 0$, tensoring in each degree with the free module P_i gives a short exact sequence of complexes. If a cycle in the quotient complex is lifted to the middle complex, its differential lies in the subcomplex and is a cycle there. Its homology class is independent of the lift and of the cycle representative: changing either changes that differential by a boundary. This defines the connecting map. The equations saying that a class maps to zero say precisely that one can subtract a boundary and lift it to the preceding complex; they prove exactness at each position. The construction commutes with maps of short exact sequences and gives

$$\cdots \longrightarrow \operatorname{Tor}_i^R(M,N') \longrightarrow \operatorname{Tor}_i^R(M,N) \longrightarrow \operatorname{Tor}_i^R(M,N'') \longrightarrow \operatorname{Tor}_{i-1}^R(M,N') \longrightarrow \cdots,$$

ending in

$$\operatorname{Tor}_1^R(M,N'') \longrightarrow M \otimes_R N' \longrightarrow M \otimes_R N \longrightarrow M \otimes_R N'' \longrightarrow 0.$$

Balance gives the corresponding sequence in the first variable. These arguments require neither Noetherianity nor finite generation. □

2. Units and the field relation at an identity

We only need the following special case of uniqueness of a constant field: an affine integral scheme of finite type for two field structures, with a rational point that retracts both structures. Proving that case avoids importing the stronger assertion for every reduced connected scheme.

Lemma 2.1 (finite normalization over a perfect field). If k is perfect and B is a finite-type k-domain, its integral closure in its fraction field is finite over B.

Proof. Choose a finite injective polynomial normalization $P = k[x_1,\dots,x_d] \subset B$, using the normalization proof named in Section 1, and put $K = \operatorname{Frac} P$, $L = \operatorname{Frac} B$. The ring $(P \setminus \{0\})^{-1}B$ is a finite-dimensional K-domain, hence a field: multiplication by a nonzero element is injective and therefore surjective, so its image contains 1. It is consequently L, and L/K is finite. We use the proved integrality, transitivity and polynomial normality statements of Integral extensions, Theorem 1.2 and Proposition 2.3, and the Noetherian finite-submodule statement of S01, Lemma 1.3. The field calculation needed here is given next in full; Stacks, Tag 0BIL is a freely accessible reading source for the trace-pairing statement.

First consider any finite extension A/K generated by separable elements $\alpha_1,\dots,\alpha_h$. Write $A_j = K(\alpha_1,\dots,\alpha_j)$, $A_0 = K$, and let f_j be the monic minimal polynomial of α_j over A_{j-1}. Polynomial division identifies

$$A_j = A_{j-1}[T]/(f_j), \qquad 1, \alpha_j, \dots, \alpha_j^{\deg f_j - 1}$$

as a basis: the span is a finite-dimensional domain and thus a field by the multiplication argument above, and any linear dependence of these powers would contradict minimality. Let $g_j \in K[T]$ be the separable minimal polynomial of α_j over K. Then $f_j \mid g_j$ in $A_{j-1}[T]$, by division and the same independence.

There is a finite extension E/K splitting all the finitely many g_j. Indeed, for a polynomial not yet split, factor out its already found linear factors, choose an irreducible factor of the remainder and adjoin a root by taking the quotient of the polynomial ring by that factor. This quotient is a field: Euclidean Bezout supplies an inverse modulo the factor for every nonzero residue class. Its degree is finite. Factor out the newly found linear factor and repeat; the degree of the remaining factor decreases, so only finitely many adjunctions are needed. Finite products of bases show that the resulting extension is finite. A separable polynomial has distinct roots in this splitting field: Bezout's identity for it and its derivative excludes any common root, while a repeated linear factor would give such a root.

We now construct, by induction on j, an explicit isomorphism

$$E \otimes_K A_j \;\simeq\; \prod_{\sigma : A_j \hookrightarrow E} E, \qquad e \otimes a \longmapsto (e\sigma(a))_\sigma. \tag{N1}$$

At $j = 0$ this is $E \otimes_K K = E$. At a later step, the polynomial presentation gives

$$E \otimes_K A_j \simeq (E \otimes_K A_{j-1})[T]/(f_j).$$

The map sends T to $1 \otimes \alpha_j$; both sides have the displayed monic-division basis over $E \otimes_K A_{j-1}$, which proves bijectivity. Under the preceding product isomorphism this becomes the product of $E[T]/(\sigma(f_j))$. Coefficients and polynomial operations are componentwise, so this product identification has the componentwise inverse. Each $\sigma(f_j)$ divides g_j, hence splits into distinct roots $r_1, \ldots, r_m$ in E. Evaluation gives an isomorphism

$$E[T]/(\sigma(f_j)) \longrightarrow E^m, \quad [u] \longmapsto (u(r_1), \ldots, u(r_m)).$$

Its inverse sends $(b_1, \ldots, b_m)$ to

$$\left[\sum_{i=1}^{m} b_i \prod_{\ell \neq i} \frac{T - r_\ell}{r_i - r_\ell} \right]. \tag{N2}$$

All denominators are nonzero. These polynomials have the prescribed values; if a polynomial of degree below m has all m roots, successive division by $T - r_i$ makes it zero. Division by the monic polynomial therefore proves both inverse identities. Each component extends σ by sending α_j to the chosen root, and every embedding arises in this way. There are $\deg f_j$ extensions of each preceding embedding. Thus the product in (N1) has exactly $[A_j : K]$ factors, since multiplying the successive power bases also proves the degree formula. Empty towers and degree-one steps are included.

This calculation also proves that every element of A is separable over K, a fact we will use in the inseparable case. For $c \in A$, let $m_c \in K[T]$ be its minimal polynomial. The powers below its degree remain linearly independent after tensoring with E, by extending them to a K-basis and using the resulting E-basis. In (N1), form the polynomial which is the product of $T - b$ over the *distinct* coordinate values $b = \sigma(c)$. It annihilates $1 \otimes c$. Divide it by m_c in $E[T]$; the remainder still annihilates that element, so the independence just proved makes the remainder zero. Hence m_c divides a product of distinct linear factors and is separable.

For a finite extension, define $\mathrm{Tr}_{A/K}(c)$ as the trace of the K-linear multiplication map by c. This is independent of basis: the finite entry sums give $\mathrm{tr}(UV) = \mathrm{tr}(VU)$, so conjugating a matrix by an invertible change of basis preserves its trace. Extending scalars preserves the entries and their sum. Under (N1), multiplication by c is the diagonal map with entries $\sigma(c)$. Consequently

$$\mathrm{Tr}_{A/K}(c) = \sum_\sigma \sigma(c), \qquad \mathrm{Tr}_{A/K}(cd) = \sum_\sigma \sigma(c)\sigma(d). \tag{N3}$$

After extension to E, the trace pairing is therefore the coordinate dot product on $E^{[A:K]}$. It is nondegenerate: a nonzero coordinate pairs nontrivially with its coordinate unit vector. In any K-basis its matrix thus has nonzero determinant, since it becomes an invertible matrix over E.

This argument also works when the characteristic divides $[A : K]$; it does not require a nonzero value of the trace on 1.

Suppose now that L/K is separable. Choose a K-basis $v_1, \dots, v_n$ of L integral over P. To obtain it from any basis vector with monic equation $v^s + a_1 v^{s-1} + \cdots + a_s = 0$ over K, take a common nonzero denominator $b \in P$ of the coefficients. Then bv satisfies a monic equation with coefficients $b^i a_i \in P$. Scaling by such nonzero elements preserves a basis. Formula (N3) applies by choosing a finite generating list for L/K, and shows that

$$G = \left(\mathrm{Tr}_{L/K}(v_i v_j)\right)_{i,j=1}^{n}, \qquad \delta = \det G$$

has nonzero determinant. Its entries belong to P: every embedded $v_i v_j$ is integral over P, their sum is integral by the earlier integrality proof, and that sum is in K; normality of P places it in P. The same reasoning gives $\mathrm{Tr}_{L/K}(z v_i) \in P$ for every $z \in L$ integral over P. Writing $z = \sum_j a_j v_j$, the trace equations are

$$G(a_j)_j = \left(\mathrm{Tr}_{L/K}(z v_i)\right)_i.$$

The complete cofactor identity of S01, Lemma 1.2 gives $\mathrm{adj}(G)G = \delta I$. Its entries are in P, so $\delta a_j \in P$ for every j. The integral closure is a P-submodule of $\sum_j \delta^{-1} P v_j$, which is a finite P-module. The Noetherian finite-submodule result makes this closure finite over P.

In characteristic zero every irreducible polynomial is separable: its derivative is nonzero and has smaller degree, so irreducibility and Bezout give coprimality. This covers that case. In characteristic $p > 0$, let L_0 consist of the elements of L separable over K. It is a subfield. Indeed, for two such elements, the calculation for a field generated by separable elements proves that every element of their generated field is separable, including their sum, difference, product and any defined inverse. The finite-dimensional subfield L_0 has a finite K-basis, so is generated by finitely many of its separable elements and is a finite separable extension by the same calculation.

For any $u \in L$, repeatedly extract powers of T^p from its irreducible polynomial while the derivative is zero. The degree decreases, giving a polynomial $g(T^{p^a})$ with $g' \neq 0$. The polynomial g is irreducible, since a factorization of it would give a factorization after substitution. It is therefore separable, and $u^{p^a} \in L_0$. Choose finitely many field generators for L/K and a common exponent $q = p^a$ at least as large as all their exponents. The identity $(x + y)^p = x^p + y^p$, obtained from the binomial coefficients divisible by p, and its iterates show that the q-th power of every rational expression in these generators lies in L_0. Thus $L^q \subset L_0$.

Let Q be the integral closure of P in L_0, finite by the separable case, and choose P-module generators $c_1, \dots, c_t$. If $z \in L$ is integral over P, then $z^q \in Q$, so $z^q = \sum_j b_j c_j$ with $b_j \in P$. In a field extension of L, adjoin q-th roots of the finitely many x_i, c_j by the irreducible-quotient construction above. Roots are unique wherever they exist: $u^q - v^q = (u - v)^q$, and a field has no nonzero nilpotents. Perfectness of k gives the unique q-th root in k of every coefficient. Taking roots term by term therefore gives

$$P^{1/q} = k[x_1^{1/q}, \dots, x_d^{1/q}], \qquad z = \sum_j b_j^{1/q} c_j^{1/q} \in \sum_j P^{1/q} c_j^{1/q}. \tag{N4}$$

The first equality follows by taking roots of polynomial coefficients and monomials, and conversely raising each such polynomial to the q-th power. Grouping exponents modulo q shows that the monomials $\prod_i (x_i^{1/q})^{e_i}$, $0 \le e_i < q$, span $P^{1/q}$ over P. Hence the last module in (N4) is finite over P. Its submodule of elements of L integral over P is finite by Noetherianity. This includes $d = 0$, $q = 1$ and the one-dimensional separable extension.

Finally, finiteness makes B integral over P. An element of L integral over B is integral over P by transitivity; an element integral over P satisfies the same monic equation over B. The two closures in L coincide. Its finite P-module generating list also generates it over B, proving the statement. □

Lemma 2.2 (units over an algebraically closed field). For a finite-type integral k-scheme X, with k algebraically closed, $\Gamma(X, \mathcal{O}_X)^*/k^*$ is a finitely generated abelian group.

Proof. Restriction to a nonempty affine open is injective, also after quotienting by k^*; it suffices to treat that affine open. The projective closure used here has the following explicit construction. Write the affine domain as $B = k[b_1, \dots, b_n]$. The map $k[T_0, \dots, T_n] \to B[z]$, $T_0 \mapsto z$, $T_i \mapsto b_i z$, has a homogeneous prime kernel H: its target is a domain and the map preserves degree. Its homogeneous quotient defines a closed subscheme $\bar{X}$ of $\mathbb{P}^n_k$, by the projective affine-chart construction in AG-RG-S01, Section 7. Its $T_0 \neq 0$ chart has degree-zero coordinate ring B, since the ratios T_i/T_0 map to the chosen generators b_i; conversely every dehomogenized relation is exactly a relation in B. All nonempty projective charts of this homogeneous domain are domains and have the same generic point, the homogeneous zero prime. Thus $\bar{X}$ is integral, and the displayed affine chart is dense. This proves the asserted integral projective closure rather than assuming it. Normalize $\bar{X}$ in its function field. On each affine chart Lemma 2.1 gives a finite normal algebra. Integral closure commutes with localization by the earlier integral-extension lesson, so these charts glue to a finite birational morphism $\nu : \bar{X}^\nu \to \bar{X}$, with $\bar{X}^\nu$ integral and normal.

The sheaf $\nu_* \mathcal{O}_{\bar{X}^\nu}$ is coherent: on each Noetherian affine chart it is its finite normalization module. Its extension by the projective closed immersion is coherent, so projective finiteness from Section 1 gives a finite-dimensional k-algebra $\Gamma(\bar{X}^\nu, \mathcal{O})$. It is a domain; a finite-dimensional domain over a field is a field, since nonzero multiplication is an injective, hence surjective, linear map. Algebraic closedness makes this algebra equal to k.

Let $X^\nu = \nu^{-1}(X)$. Its complement has finitely many irreducible components, hence finitely many codimension-one generic points $z_1, \dots, z_r$. The height-one local rings of the normal scheme are DVRs by Section 1. A unit f on X^ν defines the vector

$$(\operatorname{ord}_{z_1} f, \dots, \operatorname{ord}_{z_r} f) \in \mathbf{Z}^r.$$

If this vector is zero, both f and f^{-1}, regarded as rational functions, are regular at every codimension-one point of $\bar{X}^\nu$: at the remaining points this follows from their regularity on X^ν. The intersection-in-height-one theorem on every normal affine chart makes both functions global. Therefore $f \in k^*$. We obtain an injection of the unit quotient into $\mathbf{Z}^r$.

A subgroup of $\mathbf{Z}^r$ is finitely generated: induct on r, project to the last coordinate, use its least positive generator when its image is nonzero, lift that generator, and apply the induction to the kernel. Units on the original X inject into units on X^ν, with the same constants, so their quotient is finitely generated too. □

Lemma 2.3 (units in a pointed domain). If B is a finite-type k-domain and $e : B \to k$ retracts its field structure, then B^*/k^* is finitely generated.

Proof. An algebraic closure can be constructed as follows. For a field K, introduce one variable Z_f for each nonconstant $f \in K[T]$, and impose all equations $f(Z_f) = 0$. Their ideal is proper: any proposed expression of 1 uses finitely many equations, whose roots can be adjoined one at a time by irreducible polynomial quotients; evaluation in the resulting field rules out that expression. A maximal ideal containing this proper ideal exists by Zorn, and its quotient is a field K_1 containing K, with a root of every such polynomial. It is algebraic over K: each generating variable satisfies its monic rescaled polynomial, and each finite set of those variables generates a finite-dimensional domain, hence a field by the nonzero-multiplication argument. Repeat this construction for $K_1, K_2, \dots$ and take their union. Every polynomial over the union has all coefficients at a finite stage and hence a root at the next stage. The union is algebraic over K and algebraically closed. Apply this to $K = k$ and call the union $\bar{k}$. The finite-type $\bar{k}$-algebra $B_{\bar{k}}$ is Noetherian. Choose an irreducible component of its reduction containing the point given by $e \otimes \bar{k}$, and write it as the integral affine scheme Y. Every minimal prime of $B_{\bar{k}}$ contracts to zero in B. Indeed its local map over the contraction is flat and local; going down would put a strictly smaller prime below it if that contraction were nonzero. Thus $B \to \Gamma(Y, \mathcal{O}_Y)$ is injective.

If a unit of B becomes a constant $\lambda \in \bar{k}^*$ on Y, evaluate at the selected $\bar{k}$-point to get $\lambda = e(b) \in k^*$. Injectivity then makes b that constant already. Thus B^*/k^* injects into $\Gamma(Y, \mathcal{O}_Y)^*/\bar{k}^*$. Lemma 2.2 and the subgroup argument prove the claim. □

Lemma 2.4 (two retracted field structures). Suppose that D is a domain, $s, t : K \to D$ make it finite type over a field K, and $e : D \to K$ satisfies $es = et = \mathrm{id}_K$. Then $s = t$.

Proof. Write $K_s = s(K)$, $K_t = t(K)$, and $K_0 = K_s \cap K_t$. Lemma 2.3 for the two rational-point structures shows that D^*/K_s^* and D^*/K_t^* are finitely generated. Every subgroup of a finitely generated abelian group is finitely generated: choose a surjection from $\mathbb{Z}^m$ onto that group; the inverse image of the subgroup is a subgroup of $\mathbb{Z}^m$, finitely generated by the induction in Lemma 2.2, and its generator images generate the given subgroup. The natural injection

$$K_s^*/K_0^* \hookrightarrow D^*/K_t^*$$

makes its left side finitely generated. Lift generators α_i. Every nonzero element of K_s is a constant in K_0^* times a Laurent monomial in the α_i. Hence

$$K_0[\alpha_1, \ldots, \alpha_n, (\alpha_1 \cdots \alpha_n)^{-1}] = K_s.$$

The fully proved Zariski lemma in the earlier Nullstellensatz lesson makes K_s/K_0 finite. Interchanging the structures makes K_t/K_0 finite as well.

The elements of D algebraic over K_s form a field: the inverse of a nonzero algebraic element is a polynomial in it, obtained from its equation with nonzero constant term. The map e injects that field into K and is already surjective on K_s. Consequently this field is exactly K_s. Since K_t is algebraic over $K_0 \subset K_s$, it lies in K_s. The reverse containment follows symmetrically. Finally e has inverse both s and t on the common field. Thus $s = t$. □

Corollary 2.5 (field equivalence relation). If $R \rightrightarrows \operatorname{Spec} K$ is a scheme equivalence relation with locally finite-type projections, then its projections are locally quasi-finite.

Proof. At the identity choose an affine neighbourhood $\operatorname{Spec} B$ finite type over K for both maps, and let $\mathfrak{q}$ be the identity prime. For a minimal prime $\mathfrak{p} \subset \mathfrak{q}$, put $D = B/\mathfrak{p}$. The identity evaluation factors through D and retracts both field structures, so Lemma 2.4 makes those structures equal. The map $\operatorname{Spec} D \to R$ therefore lies in the stabilizer of the one object. That stabilizer is the identity scheme: uniqueness of arrows on every test scheme is the monomorphism condition. Thus this component is a point. Every component through the identity has dimension zero, giving $\dim_e R = 0$.

For an arbitrary arrow over a residue field L, extension to L followed by composition with its inverse identifies the selected source-fibre point with the identity in the other source fibre. Dimension at a point is preserved by field extension, as proved in Lemma 3.1 below. It is therefore zero. For a locally finite-type map, a fibre point of dimension zero is isolated and has finite residue extension, by the finite-type field dimension and Zariski lemmas. This is precisely local quasi-finiteness. Inversion supplies the other projection. □

3. Dimension and the Cohen–Macaulay locus

Lemma 3.1 (field extension preserves point dimension). For a finite-type k-algebra B, an extension k'/k and a point x' above x, one has $\dim_{x'} \operatorname{Spec}(B \otimes_k k') = \dim_x \operatorname{Spec} B$.

Proof. Present $B = k[X_1, \ldots, X_n]/I$. At the two corresponding primes, the vertical maps on the localized polynomial rings and on the localized quotient rings are flat local maps. Their closed fibre rings are identical: both are the localization, at the selected point, of $k' \otimes_k \kappa(x)$. The flat local dimension formula from Section 1 therefore gives

$$\dim P_x - \dim B_x = \dim P'_{x'} - \dim B'_{x'}.$$

The polynomial height formula and (1) identify each side with $n - \dim_x \operatorname{Spec} B$, respectively $n - \dim_{x'} \operatorname{Spec} B'$. This proves the equality. No equality of the two local-ring dimensions is asserted. □

Lemma 3.2 (quasi-finite coordinates near a fibre point). Let $A \to B$ be finite type. If the fibre has point dimension d at $\mathfrak{q}$, then, after a principal localization not removing that point, there is a quasi-finite map $A[T_1, \dots, T_d] \to B$.

Proof. Put $\mathfrak{p} = A \cap \mathfrak{q}$. In the finite-type $\kappa(\mathfrak{p})$-fibre, remove the finitely many components not through $\mathfrak{q}$, and choose a principal neighbourhood with dimension d. The same principal element lifts to B. Polynomial normalization from Section 1 can be chosen by the integer-coefficient triangular substitutions in its proof; hence its d parameters are polynomials with integer coefficients in a fixed finite list of fibre algebra generators. Lift these polynomials to B. The fibre algebra is finite over the parameter algebra, so B is quasi-finite over $A[T_1, \dots, T_d]$ at the selected point. Its quasi-finite locus is open by the affine Zariski Main theorem: that theorem gives there a principal localization of a finite subalgebra. Shrink once more inside this open. □

Lemma 3.3 (the polynomial coordinate test). Suppose B is finite type over a field k, and $k[T_1, \dots, T_d] \to B$ is quasi-finite. At a prime $\mathfrak{q}$, the local map from the polynomial localization is flat if and only if $B_{\mathfrak{q}}$ is Cohen–Macaulay and $\dim_{\mathfrak{q}} \operatorname{Spec} B = d$.

Proof. Write $\mathfrak{r}$ for the polynomial prime. Quasi-finiteness makes the local fibre zero-dimensional and the residue extension finite. Thus the two residue fields have the same transcendence degree over k. Formula (1) and the polynomial height formula show that the point-dimension condition is equivalent to

$$\dim B_{\mathfrak{q}} = \dim k[T]_{\mathfrak{r}}. \tag{2}$$

If the local map is flat, its dimension formula gives (2). A regular parameter system in the polynomial local ring remains regular in $B_{\mathfrak{q}}$ by flatness. Its quotient is the zero-dimensional local fibre, so these are parameters there; the parameter criterion gives Cohen–Macaulayness. Conversely (2), Cohen–Macaulayness and the zero-dimensional fibre satisfy the fully proved miracle flatness theorem from Section 1. □

Theorem 3.4 (the good locus). For a flat locally finitely presented scheme morphism $X \to S$, the locus of points whose local fibre ring is Cohen–Macaulay is open and dense in every fibre, and it commutes with every base change. For a locally finitely presented morphism not initially flat, its Cohen–Macaulay locus is the corresponding open within its open flat locus. The locally quasi-finite locus is open and commutes with every base change.

Proof of openness. Work on an affine chart $A \to B$ of finite presentation, and let the selected Cohen–Macaulay fibre point have point dimension d. Lemma 3.2 gives quasi-finite coordinates $P = A[T_1, \dots, T_d] \to B$ near it. Lemma 3.3 makes the map on the local fibre flat. The arbitrary-base fibre criterion in Section 1, applied to $A \to P \to B$ and the finitely presented B-module B, gives flatness over P at the selected point; both P and B are essentially of finite presentation over A. Arbitrary-base openness of flatness gives a neighbourhood on which $P \to B$ is flat. Lemma 3.3, applied to every residue-field fibre, then gives Cohen–Macaulay fibre rings on that neighbourhood. This proves openness without replacing A by a Noetherian base.

Proof of density. A fibre is locally finite type over a field and locally Noetherian. At each generic point its local ring is zero-dimensional, so its depth and dimension are both zero; it is Cohen–Macaulay. The open just proved therefore contains all fibre generic points and is dense.

Proof of base change. It suffices to treat finite-type field algebras under a field extension. Near a chosen prime of B/k, first remove components of larger dimension not through it, and take a finite injective polynomial normalization $k[T_1, \dots, T_d] \to B$ with $d = \dim_x \operatorname{Spec} B$. It remains finite and injective after field extension. Lemma 3.1 preserves d. By Lemma 3.3, Cohen–Macaulayness at either point is exactly flatness over its polynomial coordinate local ring. Flat base change preserves this flatness. It also reflects it at each point above: the two vertical local maps are faithfully flat, and have identical fibre rings; for any injection of modules over the old polynomial local ring, its kernel after tensoring with the old B-stalk is killed by the new faithfully flat B-stalk, because the new stalk is flat over the new polynomial ring and that ring is flat over the old one. Faithfulness kills the kernel. This proves equivalence of Cohen–Macaulayness at the two points and the stated

arbitrary scheme base change assertion.

For a map not initially flat, openness of flatness was proved in the earlier finite-obstruction lesson. On that open apply the preceding result. Under a flat base change its flat locus is both preserved and reflected by the same faithfully flat local kernel test, and its fibre Cohen–Macaulay locus has the compatibility just proved.

The quasi-finite assertion follows locally from affine Zariski Main and Lemma 3.1: a finite-type fibre has an isolated point with finite residue field exactly when its point dimension is zero. Thus the zero-dimensional point locus is open and is preserved and reflected by arbitrary base change. □

4. Regular cuts with a non-Noetherian base

Lemma 4.1 (regular cut). Let $A \to B$ be a flat local map, with B essentially of finite presentation over A, and let $f \in B$ be regular in $B/\mathfrak{m}_A B$. Then multiplication by f on B is injective and B/fB is flat over A.

Proof. Use the Noetherian local models constructed explicitly in the earlier finite-obstruction Lemma 6.2: $A = \varinjlim A_i$, $B = \varinjlim B_i$, each A_i, B_i Noetherian local, with transition B_j a localization of $B_i \otimes_{A_i} A_j$. Include f in one model. That lemma makes B_i/A_i flat eventually. It remains to ensure fibre regularity eventually, rather than assuming it.

Write $k_i = A_i/\mathfrak{m}_i$, $k = A/\mathfrak{m}$. For a fixed stage the ring $\bar{B}_i \otimes_{k_i} k$, with $\bar{B}_i = B_i/\mathfrak{m}_i B_i$, is essentially finite type over k and Noetherian. The map from it to the final fibre $B/\mathfrak{m}B$ is localization at the images of all later denominators. Its kernel is a finitely generated ideal. Each of its generators vanishes after finitely many later denominators have been inverted. Their coefficients and those denominators occur at a common later stage j. Consequently $\bar{B}_j \otimes_{k_j} k \to B/\mathfrak{m}B$ is injective: any element in its kernel lifts after one of the already inverted denominators to the original finite kernel, which has been killed. This is the finite-kernel argument; no uniform control of infinitely many denominators is assumed.

Regularity of f in the final fibre therefore implies regularity in $\bar{B}_j \otimes_{k_j} k$, and faithful field extension reflects it to $\bar{B}_j$. The Noetherian lifting-injectivity lemma from Section 1, applied to $B_j \xrightarrow{f} B_j$, makes f injective there and B_j/fB_j flat over A_j. The same holds at later stages by base change and localization. Filtered colimits preserve these injections and preserve flatness after extending each flat model to A; they give the asserted conclusions over A. □

For a regular sequence in a Cohen–Macaulay local fibre, apply this lemma successively. The earlier parameter theorem makes each quotient Cohen–Macaulay, with its dimension reduced by one. Thus every successive cut stays flat over the original base, however non-Noetherian that base is.

5. The source descent needed after dividing a finite relation

Lemma 5.1 (flat finite presentations descend to a stage). Let $A = \varinjlim A_i$. A flat finitely presented A-algebra has a flat finitely presented model over some A_i. If it is faithfully flat, the model may be made faithfully flat.

Proof. A finite presentation descends by putting its finitely many coefficients at one stage. To justify eventual flatness, first view A as the union of its finite-type $\mathbf{Z}$-subalgebras and model the presentation over these Noetherian rings. At every prime of the final algebra the earlier local finite-obstruction Lemma 6.2 supplies a Noetherian model flat at the contracted prime. Its Noetherian flat-locus theorem supplies a principal neighbourhood flat over that model. The images of these principal neighbourhoods cover the affine final algebra. Choose finitely many; put their equations at a common stage. Their covering condition is an identity $1 = \sum_j b_j g_j^{n_j}$, which holds at a later stage because it has finitely many coefficients. At that stage the finitely many flat localizations

cover the algebra, so the model is flat globally.

This gives a flat model over a finite-type $\mathbf{Z}$-subalgebra $A_0 \subset A$. It is itself finitely presented over $\mathbf{Z}$. The map $A_0 \to A$, the identification with the initially chosen model and its inverse all factor through one sufficiently late A_i: choose the finitely many generator images and verify the finitely many defining equations there. Base-changing the flat A_0-model to that A_i gives the desired model over the given system. This step covers systems whose transition maps are not injective.

For faithfulness, the image of a flat finitely presented morphism is open. Let its closed complement in a chosen model be $V(J)$. Images commute with base change: existence of a fibre point is existence of a prime in a nonzero residue-field algebra, a condition unchanged by faithfully flat field extension. If the limit model is surjective, $JA = A$. A finite expression $1 = \sum a_\ell j_\ell$ occurs at a later stage, making the closed complement empty there. □

Lemma 5.2 (local finite presentation is fppf-local on the source). If $Z \to Y$ is a surjective flat locally finitely presented scheme morphism and $Z \to T$ is locally finitely presented, then $Y \to T$ is locally finitely presented.

Proof. Here is the algebra of the exact earlier Descending properties, Proposition 3.2, with its previously external approximation input replaced by Lemma 5.1. Reduce by finite affine refinements to $R \to B \to C$, where C/R is finitely presented and C/B is faithfully flat finitely presented. Then $D = C \otimes_B C$ is finitely presented over R.

For a map $B \to L = \varinjlim L_i$, the algebra $C_L = L \otimes_B C$ is faithfully flat finitely presented over L. Lemma 5.1 models it faithfully flat over some L_i, as C_i. Put $D_i = C_i \otimes_{L_i} C_i$. Finite presentation of C/R and D/R descends their coefficient maps to C_i, D_i and the two commuting squares to a common later stage. The composite $B \to C \to C_i$ has equal two images in D_i, so the faithfully flat equalizer makes it factor through L_i.

Every R-algebra is a filtered colimit of finitely presented R-algebras: use finitely many generators and finitely many relations at each stage. Apply the factorization just proved to its identity when this algebra is B. Then B is a retract of a finitely presented R-algebra P. If $P = R[X_1, \dots, X_n]/(r_1, \dots, r_m)$ and the retract idempotent sends X_i to $h_i(X)$, its image is $P/(X_i - h_i(X))$. Applying the idempotent gives the inverse to this quotient identification. Thus B is finitely presented.

For schemes, over an affine open of Y lying above an affine open of T, openness of the covering and quasi-compactness select finitely many affine source opens covering it. Their finite disjoint union yields the algebra situation. This proves the local assertion. Flatness is also local on this source cover: at local rings $R \to B \to C$, the last map is faithfully flat, and every kernel of an R-module injection tensored with B vanishes after tensoring with C, hence vanishes. □

Target descent of flatness and local finite presentation has the following full affine proof. Let $R \to A$ be faithfully flat. If $A \otimes_R B$ is flat over A, tensor an arbitrary injection of R-modules with B and then with A; its kernel vanishes after the latter tensor product, so faithfulness kills that kernel. Thus B is R-flat. If $A \otimes_R B$ is finitely presented over A, take finitely many of its algebra generators and collect their finitely many B-components. The subalgebra $B_0 \subset B$ they generate has $A \otimes_R (B/B_0) = 0$, hence $B_0 = B$. Choose the resulting finite presentation map $R[T_1, \dots, T_n] \twoheadrightarrow B$, with kernel J. Flat tensoring identifies $A \otimes_R J$ with the kernel of $A[T] \to A \otimes_R B$, a finitely generated ideal by the finite-presentation calculation of AG-MO-02. Write its finitely many ideal generators as finite sums of elements of J with coefficients in A. Let $J_0 \subset J$ be the ideal generated by all those finitely many J-components. Then $A \otimes_R (J/J_0) = 0$, so $J = J_0$; this proves finite presentation over R. On an affine target chart, a finite affine refinement of an fppf cover gives just such an $R \to A$, using the universal openness proved in S04, Section E5. Apply the algebra argument on each affine source chart and glue. This proves both target-descent assertions for existing scheme maps over arbitrary bases. Open immersions descend by the invariant-open and morphism proofs in S04, Lemmas A1.5–A1.6.

Lemma 5.3A (the arbitrary-base étale criterion). A locally finitely presented scheme morphism is étale if and only if it is flat and has zero relative differentials.

Proof. We first prove the field calculation needed in the converse. If a finite-type k-algebra E has $\Omega_{E/k} = 0$, every maximal residue field L is finite over k by the full Zariski lemma, AG-CA-06, Theorem 1.3. The quotient differential sequence gives $\Omega_{L/k} = 0$. The field calculation in Lemma 2.1 constructs the subfield L_0 of separable elements and shows that L/L_0 is purely inseparable. If this extension were nontrivial, AG-CA-15, Proposition 6.3, would make $\Omega_{L/L_0} \neq 0$, contradicting its being a quotient of $\Omega_{L/k} = 0$. Hence L/k is separable.

Put $D = E_{\mathfrak{n}}$, $\mathfrak{m} = \mathfrak{n}D$ and $V = \mathfrak{m}/\mathfrak{m}^2$. A field section $L \to D/\mathfrak{m}^2$ can be constructed without a primitive-element assumption. Choose a finite separable-generator tower for L/k, as in Lemma 2.1. At each step lift its generator to $D/\mathfrak{m}^2$, lift the minimal polynomial's coefficients using the preceding field section, and replace the lift a by $a - f(a)/f'(a)$. The denominator is a unit, and the square-zero Taylor identity makes the corrected value a root. Evaluation from the irreducible-polynomial field quotient extends the section at that step. Thus $D/\mathfrak{m}^2 = L \oplus V$ as a square-zero k-algebra. Derivations into any L-vector space vanish on the separable field L and are exactly linear maps from V. The differential quotient sequence, tensored with L, consequently gives $\Omega_{D/k} \otimes_D L = V$: differentials of $\mathfrak{m}^2$ vanish there. Therefore $V = 0$. Hilbert basis makes D Noetherian, so Nakayama gives $\mathfrak{m} = 0$. All maximal localizations of E are fields. Every prime lies in a maximal ideal, so E is zero-dimensional. The full Artinian decomposition of S04, Lemma P0.9, makes it a finite product of these finite separable fields. This proves the field calculation, including possible nilpotents rather than assuming reducedness.

Now work with a finitely presented algebra $B = A[X_1, \dots, X_n]/I$ at a prime $\mathfrak{q}$, with contraction $\mathfrak{p}$, where $B_{\mathfrak{q}}$ is $A_{\mathfrak{p}}$-flat and $\Omega_{B/A,\mathfrak{q}} = 0$. The complete conormal sequence in AG-CA-15, Theorems 3.3 and 5.1, makes $I/I^2 \to B^n$ surjective after tensoring with $\kappa(\mathfrak{q})$. Choose n relations $F_i \in I$ whose Jacobian determinant Δ is nonzero there. The square algebra $C = (A[X]/(F_i))_\Delta$ is étale directly by its definition. For a map into a square-zero quotient, choose arbitrary lifts of the coordinate images. Their equation errors lie in the square-zero ideal. The Jacobian is invertible: its determinant lifts a unit, a unit lifts across a square-zero ideal by $(1 + v)^{-1} = 1 - v$, and the adjugate identity gives the inverse matrix. Subtract the inverse-Jacobian vector of errors from the lifted coordinates. The square-zero Taylor expansion makes all equations zero. Any two lifts differ by a vector in that ideal whose product with the Jacobian is zero, so the inverse matrix makes them equal. Localization imposes no extra obstruction because its invertible element also lifts to a unit. Thus the finite presentation has unique square-zero lifts and is formally étale. Its conormal differential matrix is invertible, so its differentials vanish. It is flat by the full arbitrary-base smooth-algebra proof in AG-CA-18, Theorem 6.1. There is a surjection $C \to B_\Delta$ with finitely generated kernel, since I is finitely generated. Localize at the prime $\mathfrak{r}$ above $\mathfrak{q}$, obtaining kernel K in $C_{\mathfrak{r}} \to B_{\mathfrak{q}}$. Both closed fibre local rings are fields by the field calculation just proved; for B, first shrink its fibre chart to where its finite differential module vanishes. Their quotient map is an isomorphism. Flatness of $B_{\mathfrak{q}}$ makes reduction of the kernel sequence modulo $\mathfrak{p}A_{\mathfrak{p}}$ injective on $K/\mathfrak{p}K$. This module is zero. The finite $C_{\mathfrak{r}}$-module K is killed by Nakayama. Its finitely many generators vanish after one further principal localization avoiding $\mathfrak{r}$, making the surjection an isomorphism there. Thus B is étale at $\mathfrak{q}$. For $n = 0$ the empty determinant is 1, and the same argument applies. The forward implication is AG-CA-18, Corollary 6.2, with its complete flatness proof, and the differential definition. Affine charts prove the scheme assertion. □

Lemma 5.3 (étaleness descends on the target). An existing scheme morphism whose pullback by an fppf covering is étale is étale. Surjectivity is also detected by this covering.

Proof. Descend flatness and local finite presentation by the full target-descent argument above. On affine target and source charts, differential base change identifies the pulled-back module with $A' \otimes_A \Omega_{B/A}$, by AG-CA-15, Theorem 2.2. It is zero after a faithfully flat affine refinement, hence was zero before it. Lemma 5.3A gives étaleness. Surjectivity is detected because any point of the target has a point above it on the covering and, if the pullback is surjective, a further source point

above that one; its image is the original target point. □

This supplies the target-local input in the earlier étale quotient proof directly, without using its broader smoothness-descent assertion.

6. Transverse slicing of the relation

Let $R \rightrightarrows U$ be a scheme equivalence relation with flat locally finitely presented projections s, t. For $g : V \to U$, write

$$Z = R \times_{t,U,g} V, \quad h = s\,\mathrm{pr}_R, \qquad R_V = V \times_{g,U,t} R \times_{s,U,g} V.$$

The following is the complete geometric step underlying the earlier bootstrap Lemmas 3.3–3.4.

Lemma 6.1 (good opens descend to objects). For g locally finitely presented, the Cohen–Macaulay locus and the locally quasi-finite locus of h are inverse images of opens of V.

Proof. On $Z \times_V Z$, two arrows $r_1 : x_1 \to g(v)$, $r_2 : x_2 \to g(v)$ give $r_2^{-1} r_1 : x_1 \to x_2$. The two resulting squares with h, using s and t, are Cartesian: for example $r_1 = r_2 a$ recovers the missing arrow uniquely. Their horizontal base changes are flat and locally finitely presented. Theorem 3.4 and the faithfully flat local test for the flat locus show that the two inverse images of either good open agree. The identity gives a section $\sigma : V \to Z$. If O is the good open, set $W = \sigma^{-1} O$. Comparing each arrow with the identity at its target on this self-overlap proves $O = Z \times_V W$, with scheme equality because these are open subschemes. □

Theorem 6.2 (slicing). There is $g : V \to U$ such that $R_V \rightrightarrows V$ has flat locally finitely presented locally quasi-finite projections and $V/R_V = U/R$ as fppf sheaves.

Proof. First use Lemma 6.1 with $V = U$. It gives $U_0 \subset U$ whose inverse image under t is the Cohen–Macaulay locus of s. This open meets every source fibre by Theorem 3.4 and the identity section. Hence $t^{-1}U_0 \to U$ is flat locally finitely presented and surjective. Restricting to U_0 preserves the quotient: every object is locally joined to an object of U_0 on this covering, and equality of two such objects is exactly their restricted relation. Inversion shows both restricted projections Cohen–Macaulay. Replace the relation by this restriction.

Fix u closed in an affine open $\operatorname{Spec} A \subset U$, with maximal ideal $\mathfrak{m}$. Around its identity choose affine arrow coordinates $\operatorname{Spec} B$ with both endpoints in that open, and identity prime $\mathfrak{q}$. The local source fibre

$$D = B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}}$$

is a Noetherian Cohen–Macaulay local ring of dimension d. Restricting both endpoints to u gives $D/t(\mathfrak{m})D$; Corollary 2.5 makes its dimension zero.

Choose $f_1, \dots, f_d \in \mathfrak{m}$ whose target images are parameters in D. They can be selected from this particular ideal: at each positive-dimensional Cohen–Macaulay quotient its finitely many associated primes are its minimal primes; none contains the whole target ideal since the quotient by that ideal has dimension zero. Finite prime avoidance in A chooses an f_i outside their inverse images. Its target image is regular, and the quotient remains Cohen–Macaulay of dimension one less. Repeat; for $d = 0$ choose the empty list.

Set $V_u = \operatorname{Spec} A/(f_1, \dots, f_d)$. The map $h_u : R \times_{t,U} V_u \to U$ is locally finitely presented. At the identity its fibre is $D/(t(f_1), \dots, t(f_d))$. Lemma 4.1 applied successively makes it flat there with Cohen–Macaulay zero-dimensional fibre. Thus the identity belongs to both good opens of h_u. Lemma 6.1 shrinks an affine neighbourhood of u in V_u so that the entire h_u is flat locally finitely presented and locally quasi-finite. Base change and inversion give these properties to both restricted projections.

Take the disjoint union of these slices over all points closed in some affine open. On a mixed pair of components its source projection is a base change of one of the h_u; the same three properties hold.

The image of $h = \coprod h_u$ is open and contains all these finite-type points. If its closed complement were nonempty, intersect it with an affine open meeting it. The resulting nonempty closed subset of an affine scheme contains a point closed in that affine open, contradicting the construction. Hence h is surjective.

It is therefore an fppf cover. Every quotient object locally joins a slice object, and unique equality arrows between slice objects belong to R_V. Sheafification gives $V/R_V = U/R$ on every test scheme. The index is a set of points of U; disjoint union is allowed within the fixed universe. □

7. The finite part and invariant affine coordinates

Lemma 7.1 (finite open subgroupoid). Suppose U is affine and $R \rightrightarrows U$ is an equivalence relation with flat locally finitely presented locally quasi-finite projections. At every $u \in U$ there is an affine étale $V \to U$, with image containing u, and an open subgroupoid $P \subset R_V$ having finite locally free projections.

Proof. The endpoint map to $U \times_{\mathbf{Z}} U$ is a separated monomorphism. Both projections from this affine product are separated, so s, t are separated.

Consider the functor Y/U of clopen finite parts $Z \subset R \times_{s,U} T$ containing the identity. Such parts are flat and locally finitely presented as opens of the given map, and finite. Here is the module argument, also present in Descent and Zariski Main, the proof of Lemma E4.2. On an affine base a finite algebra C is integral by the determinant argument. For finite algebra generators c_i, choose monic equations $P_i(c_i) = 0$. The algebra $F = A[T_i]/(P_i(T_i))$ is finite free on the bounded monomials and surjects onto C. Finite presentation of C as an algebra makes this surjection's ideal kernel finitely generated; multiplying its ideal generators by the bounded monomial basis generates the kernel as an A-module. Thus C is a finitely presented module. With flatness, the complete finite-local-freeness proof in Lemma A1.3 of that lesson applies. Their inclusions and identity containment descend on fppf covers, by affine algebra, invariant-open and morphism descent.

Here is the local lifting which gives étale charts, using the now integrated scheme proof directly. A prescribed finite clopen part of R_u has finitely many selected points. Choose a quasi-compact open $W \subset R$ containing them. Its map to the affine U is quasi-finite and separated. The complete elementary finite-pieces construction in Descent and Zariski Main, Proposition D3.3, gives a single elementary étale neighbourhood and disjoint finite open pieces for all selected points. The special fibre of each such piece is the whole Artinian component at its selected point: an open of the zero-dimensional fibre containing that point and no other point is exactly that component as a scheme. Their finite disjoint union therefore has exactly the prescribed special fibre. It is open in R and finite over the neighbourhood. It is also closed in R: the graph of its map into the separated ambient R is closed, and its projection to R is finite. Pulling this clopen part back along the identity gives a clopen subset of the neighbourhood containing its marked point. Restrict to an affine principal neighbourhood there, so that the lifted part contains every identity. It is now a point of the stated functor, flat and locally finitely presented, hence finite locally free.

No henselization or finite-presentation assumption on a finite completion is used in this construction. Proposition D3.3 already constructs its elementary neighbourhood by finitely many polynomial factorization and idempotent equations. The current integrated Lemma E4.3 uses the same construction for all points of a fibre.

For two finite clopen parts on an arbitrary scheme T, their differences are clopen in finite locally free schemes, hence finite locally free. Equality is represented by the clopen subset on which the ranks of both differences are zero. This equality description commutes with every base change. In particular, parts having equal special fibre agree on a neighbourhood of that base point.

The charts cover also parts defined over field extensions. A clopen finite part of a zero-dimensional locally finite-type fibre involves only finitely many of its Artinian components. After a separable

closure these components are supported at single points; further field extension cannot split them: their reduced residue extensions are purely inseparable, and their radicals are nilpotent. Idempotents after a purely inseparable extension are unchanged, since that extension is radicial and induces a bijection of the underlying finite spectra. Each of the finitely many defining idempotents over the separable closure descends to a finite separable stage by its finite coefficients and equation. Thus any selected part after an arbitrary field extension, after a common field extension for comparison, comes from a finite separable residue stage. The complete elementary étale residue-neighbourhood construction realizes that stage; the lifting just proved supplies its chart. At any point of a general test T, compare its given finite part with the chart part after this étale residue extension. Their equality locus is open and contains the point, proving that the charts cover the functor on all tests.

The relation between these scheme charts is clopen in their overlaps, hence is a scheme étale equivalence relation. The complete earlier étale quotient theorem constructs Y as an algebraic space, with $Y \to U$ étale. The same equality locus is closed, so this map is separated. Theorem E8.1 and Corollary E8.2 of Descent and Zariski Main therefore make Y a scheme, separated and étale over U. Its universal finite part is a scheme.

Its groupoid has objects (x, Z). For an arrow $r : x \to y$ in Z, define its target as

$$(y, Zr^{-1}), \qquad Zr^{-1} = \{ar^{-1} : a \in Z\}. \tag{3}$$

Translation identifies source fibres, so this is again a clopen finite part containing the identity. It gives a morphism to Y by the representing property. Inversion sends (x, Z, r) to (y, Zr^{-1}, r^{-1}); applying it twice is the identity. The source is the universal finite locally free map, hence the target is finite locally free too. If $r' \in Zr^{-1}$, then $r'r \in Z$ and $Z(r'r)^{-1} = Zr^{-1}(r')^{-1}$, proving closure under composition. All laws follow from those of R.

Call this groupoid $P \rightrightarrows Y$. Its inclusion in R_Y is open: over the universal finite open of $R \times_{s,U} Y$, target rule (3) is a section of the étale projection $R_Y \to R \times_{s,U} Y$, and an étale section is open by the proved étale diagonal criterion. It is an equivalence relation because its endpoints factor through those of R_Y.

The Artinian component of the identity in R_u gives a $\kappa(u)$-point $y \in Y$. Its P-orbit is finite. Choose a quasi-compact open of Y containing that orbit; general scheme Zariski Main makes this open quasi-affine over the affine U. A finite set in a quasi-affine scheme has an affine neighbourhood: embed it as an open in an affine scheme, and use finite prime avoidance to find an element of the boundary ideal avoiding every selected prime. Its principal open lies inside the given open and contains the selected set. Let $A \subset Y$ be such an affine neighbourhood.

Remove from Y the finite closed source image of arrows whose target lies outside A. The remainder D is an invariant open, contained in A by the identity, and contains the entire orbit. Choose $f \in \Gamma(A, \mathcal{O})$ with that orbit in $D_A(f) \subset D$, by the same finite prime avoidance. On D, take the norm of the target pullback of f along the finite locally free source of P_D. This gives a function N. The conjugation/translation proof of norm invariance in the complete finite affine quotient proof applies on local free sheaf charts as well: the two multiplication endomorphisms are conjugate on the entire arrow scheme. Thus N is invariant, including nilpotents.

It is invertible at the selected orbit, since f is invertible at every point of each finite fibre there. Also $D_D(N) \subset D_A(f)$: invertible determinant makes the target multiplication an invertible endomorphism of the finite locally free algebra, hence the target element a unit, and the identity makes f a unit. Since $D_A(f) \subset D$,

$$D_D(N) = D_{D_A(f)}(N)$$

is affine. Set $V = D_D(N)$. It is invariant, so restricting P preserves finite locally free projections. It contains y, maps étale to U, and is affine. This proves the lemma. □

8. Division and the flat quotient theorem

The earlier bootstrap Lemma 3.1 gives the following restriction fact with its complete test-scheme proof. If $g : V \to U$ is open or étale, then $V/R_V \to U/R$ is a representable open immersion onto the saturation of V. For a general g, it is an isomorphism if $R \times_{t,U} V \to U$ is an fppf cover. The proof constructs the saturated open after local object lifts on a test scheme, descends its invariant open, and identifies its quotient sections by local arrow lifts. These are exactly invariant-open descent and uniqueness of equality arrows; the statement does not require the quotient already algebraic.

Lemma 8.1 (division). Let U be affine, $R \rightrightarrows U$ a scheme groupoid with separated locally quasi-finite endpoint map, and $P \subset R$ a finite locally free equivalence subgroupoid on all objects. There is a scheme groupoid $\bar{R} \rightrightarrows \bar{U} = U/P$ whose restriction along $q : U \to \bar{U}$ is R, and $U/R = \bar{U}/\bar{R}$.

Proof. The fully proved finite affine quotient theorem in Descent and Zariski Main, Theorem E6.1, gives q finite locally free surjective and $P = U \times_{\bar{U}} U$. The map $U \times_S U \to \bar{U} \times_S \bar{U}$ is a finite locally free cover. Give R its endpoint-transport descent datum: for $r : x \to y$ and $p : a \to y$, $p' : b \to x$ in P, send r to $p^{-1} r p' : b \to a$. Its inverse uses p and $(p')^{-1}$; associativity verifies its cocycle. Theorem E5.1 of that same lesson descends R to a scheme $\bar{R}$ over $\bar{U} \times_S \bar{U}$.

Composition descends explicitly. Locally lift two composable descended arrows to $r : x \to y$, $v : z \to w$. Their middle objects have the same q-image. The unique P-arrow $a : y \to z$, from $P = U \times_{\bar{U}} U$, gives var. Changing the middle lifts cancels the two middle transport arrows; changing outer lifts performs the specified outer transport. Thus the composite is invariant and descends by morphism descent. Identities and inverses descend in the same manner, and all laws can be checked on the covering. If R is an equivalence relation, the monomorphism of its endpoint map is faithfully flat local and descends; hence $\bar{R}$ is also an equivalence relation.

Every descended object and arrow locally lifts through q and its relation cover. Equality of two quotient sections is the local existence of an arrow. These lift and equality descriptions give $U/R = \bar{U}/\bar{R}$ as sheaves on all tests. When R is an equivalence relation, its equality arrow is unique. □

Theorem 8.2 (general flat quotient). Let $R \rightrightarrows U$ be a groupoid in algebraic spaces over a scheme S, with flat locally finitely presented source and target, and with endpoint monomorphism giving an equivalence relation. Then the fppf quotient $Q = U/R$ is an algebraic space, $U \to Q$ is flat locally finitely presented and surjective, and $R = U \times_Q U$. Equivalently, an fppf sheaf admitting a representable-by-spaces flat locally finitely presented surjection from an algebraic space is algebraic.

Proof. Replace U by an étale scheme atlas. Local object and arrow lifts preserve the quotient and produce the restricted algebraic-space relation. Its endpoint map to the product of schemes is a locally finite-type monomorphism. To check local finite type, take affine charts of the product and of an étale scheme atlas of the arrow space. On their rings $A \to B \to C$, the composite is finite type because an arrow projection is locally finitely presented. Its finite A-algebra generators also generate C as a B-algebra. Thus the endpoint map is locally finite type. A monomorphism is separated. Its locally finite-type fibres are locally quasi-finite by this explicit atlas calculation. Over a field K, let $F \to \operatorname{Spec} K$ be a nonempty fibre and choose an étale scheme atlas $V \to F$. Monomorphy on all tests gives $V \times_F V = V \times_K V$, whose first projection is étale. Pull back along any selected $\operatorname{Spec} L \to V$, with L its residue field; it follows that $V_L \to \operatorname{Spec} L$ is étale. Its points have point dimension zero by the monic étale charts in S04, Lemma D2.2. Every point of V has a point above it after the faithful field extension $K \to L$, and Lemma 3.1 reflects that zero point dimension. On finite-type affine charts the fibre criterion therefore makes V/K locally quasi-finite. This proves local quasi-finiteness on the scheme atlas, including nonreduced tests, hence for F/K and for the endpoint map. Empty fibres require no check. Theorem E8.1 and Corollary E8.2 of Descent and Zariski Main make this restricted arrow space a scheme.

Theorem 6.2 now preserves the quotient while making its projections also locally quasi-finite. Restrict to affine open object schemes. Their quotients are representable open subfunctors by the

restriction fact, and they cover. It suffices to make each such quotient algebraic, since the earlier algebraic-space open-gluing proof constructs its diagonal and atlas on these opens.

For an affine object scheme, Lemma 7.1 gives affine étale coordinates through every point with an open finite locally free subgroupoid P. Their saturated quotient opens cover as well. On one coordinate scheme apply Lemma 8.1. We obtain an affine $\bar{U}$, a scheme relation $\bar{R}$, and the same quotient.

The descended projections are flat locally finitely presented. Indeed

$$R \longrightarrow \bar{R} \times_{\bar{U}} U \longrightarrow U$$

has first map a base change of q, finite locally free and surjective, and its composite is the original projection. Lemma 5.2 and its flat kernel proof descend the two properties along the source cover; their target descent along $q : U \to \bar{U}$ descends them to $\bar{R} \to \bar{U}$.

The unit $\bar{U} \to \bar{R}$ is open. Its inverse image along the finite locally free surjective $R \to \bar{R}$ is precisely $P \subset R$. Open-immersion descent therefore applies.

This open unit makes the projections étale. Take any point x of a source fibre over an algebraically closed field $\bar{k}$, and extend that field to $L = \kappa(x)$. Its chosen L-valued arrow, composed with its inverse, identifies the selected point of the base-changed source fibre with an identity in another source fibre. The identity is an open $\operatorname{Spec} L$, since the unit is open. Its point dimension is zero; Lemma 3.1 therefore gives point dimension zero at the original x. The finite-type field dimension and Zariski lemmas make $\kappa(x)$ finite over $\bar{k}$, hence equal to $\bar{k}$. Now the arrow is already $\bar{k}$-valued; translation in the original geometric relation identifies its local ring with the identity local ring $\bar{k}$. Thus every local ring in every geometric source fibre is $\bar{k}$. Such a locally finite-type fibre is a disjoint union of these reduced points, and its differential sheaf is zero. Base change of differentials, finite generation for a locally finitely presented map, and local Nakayama give $\Omega_{\bar{R}/\bar{U}} = 0$. The complete earlier étale criterion says that a flat locally finitely presented map with this unramified condition is étale. Inversion gives the other projection. The earlier étale quotient theorem now makes $\bar{U}/\bar{R}$ an algebraic space. Open gluing completes the construction of Q.

For any two object sections with the same image in Q, sheafification supplies local arrows. The endpoint monomorphism makes those arrows unique, so their local restrictions agree and glue as sections of R. Hence $R = U \times_Q U$ on every test scheme.

For a scheme $T \to Q$, choose an fppf scheme cover $T_i \to T$ on which its quotient section lifts to U, also using a scheme atlas of U when necessary. Over T_i, its pullback of $U \to Q$ is a base change of a projection of R. Target descent on scheme atlases proves flatness and local finite presentation; local lifting proves surjectivity. These are the claimed properties of the algebraic-space map.

Finally a representable-by-spaces flat locally finitely presented surjection $X \to F$ is an epimorphism of fppf sheaves. On a scheme test of F, choose a scheme atlas of its surjective flat locally finitely presented pullback to X; affine opens supply an fppf covering and local lifts. Its self-fibre product is an algebraic-space equivalence relation with the specified projections. The preceding construction gives its quotient; local representatives and equality arrows identify that quotient with F. Conversely the quotient already has the asserted cover. □

Corollary 8.3 (free flat actions). A flat locally finitely presented group algebraic space G/B acting freely on an algebraic space X/B has an algebraic-space quotient $Q = X/G$. The map $X \to Q$ is flat locally finitely presented surjective and is an fppf G-torsor:

$$G \times_B X = X \times_Q X.$$

Proof. The action source is the base change of G/B. The isomorphism $(g, x) \mapsto (g, gx)$ identifies target with another such source, so it has the same properties. Freeness is exactly the endpoint monomorphism on all tests. Apply Theorem 8.2. Local object sections trivialize the displayed torsor identity after a covering by affine opens in a scheme atlas of X. □

The construction is compatible with every base change. Indeed local quotient representatives and their unique equality arrows give the same sheaf quotient of the base-changed relation; the identity $R = U \times_Q U$ and the covering property provide these descriptions on all tests. This assertion includes nonflat changes of base.

Comparison material actually read: the Stacks Project's constant-field result, 04MK, units, 04L5, point dimension, 00P4, polynomial coordinate test, 00RE, Cohen–Macaulay locus, 00RH, regular cuts, 046Y, slicing, 0489, finite-part representability, 04RI, finite-part groupoid, 04RU, and flat bootstrap, 04S6. These citations acknowledge free material used to write and check the proofs; they replace none of those proofs.

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History

Source: *The Stacks Project*, by the Stacks Project authors, freely published under the GNU FDL 1.2 or later. The editable free source revision used is a04446e57ec1fbc252a871afcec7752fb2807b14. The source tags are identified above.

Edition: *Flat quotient bootstrap over an arbitrary base*, 5 October 2026, written by GPT-6.1 Sol (OpenAI), Codex, Ultra setting, for Open Mathematics Courses. The consumed constant-field input is proved in its pointed-domain form; dimensions, Cohen–Macaulay loci, regular cuts, finite-presentation descent, slicing, finite-part charts, division and the general flat quotient are written before their uses. The text is dedicated to the public domain under CC0 1.0.

Supporting proofs for the consumed group-scheme foundations

This lesson supplies the additional arguments consumed from the earlier group-scheme lessons, before their use in the following reductive-group lessons. The statements retain their field, characteristic, finiteness and base hypotheses. Definitions of a group scheme, a character, an fpqc form and a quasi-coherent representation are definitions, rather than theorem obligations.

The earlier algebra proofs used below are the current programme lessons, with their actual locators specified at the point of use. Those locators refer to written proofs. External free sources at the end explain the origin of the arguments; none replaces a proof below or the stated earlier proof. The separate general flat quotient proof is Flat quotient bootstrap, Theorem 8.2 and Corollary 8.3; the further field scheme-quotient argument remains in the earlier quotient lesson.

G.0. The component structure used in the field-group proof

Lemma G.0.A (the algebraic overfields used here). Every field has an algebraic closure. Any two extensions of a field embed, over that field, in a common algebraically closed overfield.

Proof. For a field F, put one indeterminate X_f for every monic nonconstant polynomial $f \in F[T]$, and form the ideal generated by all $f(X_f)$. This ideal is proper. Indeed, a putative expression for one uses only finitely many of these relations and indeterminates. For those finitely many polynomials, adjoining a root of an irreducible factor successively gives a field in which they all have roots; the irreducible-factor quotient is a field by the division and Bezout proof of AG-RG-S02, Lemma 4.A. Evaluate the specified indeterminates at these roots and any remaining finitely many indeterminates at zero. The purported expression becomes $1 = 0$, a contradiction. Choose a maximal ideal containing the relation ideal: a union of a chain of proper ideals is proper, so Zorn applies. Its quotient is a field (each nonzero class generates the unit ideal) and contains F, since a proper ideal cannot contain a nonzero element of a field. Each indeterminate class is algebraic over F; every element of this quotient uses finitely many classes and is algebraic by the integral-element proof of AG-RG-S04, Lemma P0.2. Thus we have an algebraic extension in which every polynomial of F has a root.

Repeat this construction with the resulting field, and then countably many times, always identifying the original field with its injected image. The union is algebraic over F, by transitivity of integrality in Lemma P0.2. Each polynomial in the union has finitely many coefficients, all at one stage, and has a root at the next stage. Dividing by its leading coefficient reduces to the monic case. Repeated division by linear factors therefore splits every nonconstant polynomial. The union is an algebraic closure.

For extensions K, L of k, the tensor ring $K \otimes_k L$ is nonzero: choose vector-space bases containing one, whose pairs give a tensor basis with nonzero $1 \otimes 1$. Lemma P0.1 supplies a prime of this ring. Its quotient domain, then its fraction field, contains both K and L, since a nonzero unital map from a field is injective. Applying the preceding construction gives a common algebraically closed

overfield. □

Lemma G.0.1 (canonical flat closed structure). Let X be any scheme and let $Y \subset |X|$ be closed and stable under generalization. There is exactly one closed subscheme $X_Y \subset X$ with underlying space Y whose inclusion is flat. Every scheme morphism $f : T \to X$ with $f(|T|) \subset Y$ factors uniquely through X_Y. This structure commutes with base change, when the underlying subset is replaced by its inverse image. In particular it applies to every connected component of any scheme and retains its ambient local rings, including their nilpotents.

Proof. Begin with $X = \operatorname{Spec} A$ and define

$$I_Y = \ker\left(A \longrightarrow \prod_{\mathfrak{q}\in Y} A_{\mathfrak{q}}\right).$$

If Y is empty, this means $I_Y = A$. For every $\mathfrak{q} \in Y$, $(I_Y)_{\mathfrak{q}} = 0$. We show that $(I_Y)_{\mathfrak{p}} = A_{\mathfrak{p}}$ for $\mathfrak{p} \notin Y$. Write $Y = V(K)$, and choose $f \in K \setminus \mathfrak{p}$. Every prime of $A_{\mathfrak{q}}$, for $\mathfrak{q} \in Y$, comes from a prime contained in $\mathfrak{q}$, hence still in Y by generalization stability. Thus $f/1$ lies in every prime of $A_{\mathfrak{q}}$. The prime-intersection proof in AG-RG-S04, Lemma P0.1 makes it nilpotent. Consequently there are $n_{\mathfrak{q}} > 0$ and $s_{\mathfrak{q}} \notin \mathfrak{q}$ with $s_{\mathfrak{q}} f^{n_{\mathfrak{q}}} = 0$ in A.

The affine scheme is quasi-compact by the same prime-separation argument: if principal opens $D(a_i)$ cover its spectrum, no prime contains the ideal generated by all a_i; Lemma P0.1 makes that ideal the unit ideal. An expression for one uses finitely many a_i, and those opens already cover. An arbitrary open cover refines to principal opens. A closed subset is quasi-compact by adding its open complement to any cover. Thus the opens $D(s_{\mathfrak{q}})$ covering Y have a finite subcover. Let N be the largest of its exponents. At every $\mathfrak{r} \in Y$, some chosen $s_{\mathfrak{q}}$ is a unit in $A_{\mathfrak{r}}$, so $f^N/1 = 0$ there. Hence $f^N \in I_Y$, while $f^N \notin \mathfrak{p}$. This proves the assertion outside Y. It also proves $V(I_Y) = Y$.

Each localization of A/I_Y, as an A-module, is therefore $A_{\mathfrak{p}}$ if $\mathfrak{p} \in Y$ and zero otherwise. It is flat at every prime, hence flat over A by the earlier localization criterion in AG-CA, *Tor and flat modules*, Theorem 3.3. That criterion follows by localizing every tensor-injection kernel and using the zero-module localization test. Thus $\operatorname{Spec}(A/I_Y)$ is the required flat closed subscheme.

For uniqueness, suppose A/J is flat over A and $V(J) = Y$. For $\mathfrak{p} \in Y$, the quotient map $A_{\mathfrak{p}} \to A_{\mathfrak{p}}/J_{\mathfrak{p}}$ is a flat local map of nonzero local rings, hence faithfully flat by AG-CA, *Faithful flatness and the local criterion for flatness*, Theorem 2.1. Flatness tensors the injection $J_{\mathfrak{p}} \hookrightarrow A_{\mathfrak{p}}$ to an injection into $A_{\mathfrak{p}}/J_{\mathfrak{p}}$, whose image is zero. Faithful flatness makes $J_{\mathfrak{p}} = 0$. For $\mathfrak{p} \notin Y$, $J_{\mathfrak{p}} = A_{\mathfrak{p}}$. Thus J and I_Y have the same localizations at all primes and are equal, by the earlier zero-module localization test applied to their sums and quotients.

If $f : T \to \operatorname{Spec} A$ has image in Y, any $a \in I_Y$ is zero in $A_{f(t)}$ at each $t \in T$. Its image is therefore zero in $\mathcal{O}_{T,t}$. A section whose every stalk is zero is zero, so f kills I_Y and factors through the quotient. Uniqueness follows from the closed immersion being a monomorphism.

On an affine open of a general X, use this construction for its intersection with Y. Restrictions to further affine opens remain flat closed subschemes with that same underlying subset, so affine uniqueness identifies the restrictions. Their quotient ideals glue and give X_Y. The same local uniqueness proves global uniqueness and the factorization assertion. Under any base change, the pulled-back closed subscheme is flat and has the inverse-image underlying subset; that subset is closed and stable under generalization. Uniqueness proves the asserted compatibility.

A connected component is closed: the closure of a connected set is connected, since any separation of the closure by two nonempty relatively open sets would meet the dense original set on both sides and separate it. Maximality thus makes the closure the component itself. It is stable under generalization: if y is in the component and x generalizes y, every nonempty open in the closure of $\{x\}$ contains x, so that closure is irreducible and connected and contains y. Its union with the component is connected: a separation of the union would put both intersecting connected sets on the side containing their common point. Maximality therefore puts x in the component. The

construction thus applies. The localized quotient at a point of the component is exactly the ambient local ring, as already computed. □

Lemma G.0.2 (the algebraic-extension connectedness step). Let X be a connected scheme over a field k with a k-rational point. Then X_L is connected for every algebraic field extension L/k, including infinite extensions and imperfect fields. No finiteness or reducedness assumption on X is needed.

Proof. The projection $p : X_L \to X$ is integral and closed. On an affine chart its algebra is $A \otimes_k L$, the filtered union of the finite free A-algebras $A \otimes_k E$, where E/k ranges over finite subextensions. Every element is therefore integral over A, by AG-RG-S04, Lemma P0.2. Lemma P0.4 proves closedness, also for the quotients defining closed subsets.

Here is the needed openness calculation over an arbitrary affine chart, without a finite-presentation or reducedness assumption on A. First let E/k be finite of degree $d \geq 1$, and put $B = A \otimes_k E$. Choose its free basis $e_1, \ldots, e_d$. For $b \in B$, write $be_j = \sum_i m_{ij} e_i$, and set

$$P_b(T) = \det(TI - (m_{ij})) = T^d + c_{d-1}T^{d-1} + \cdots + c_0 \in A[T].$$

The row of basis elements annihilates $bI - (m_{ij})$. Multiplying by its adjugate, using the cofactor identity proved in AG-RG-S01, Lemma 1.2, shows that $P_b(b)e_j = 0$ for every j, hence $P_b(b) = 0$, since the basis spans one.

We also record the determinant multiplication calculation used in the next sentence. Expanding $\det(PQ)$ by permutations and then expanding each entry of PQ, group terms by the chosen intermediate indices. If two indices coincide, the inner alternating sum is a determinant with two equal rows of Q, and is zero by the paired-permutation cancellation of Lemma 1.2. If the indices form a permutation τ, that inner sum is $\operatorname{sgn}(\tau) \det Q$. Summing the remaining coefficients gives exactly $(\det P)(\det Q)$. Thus this identity holds over every commutative ring.

At a prime $\mathfrak{p} \subset A$, the fibre of $D(b)$ is empty precisely when the image $\bar{b}$ in $\kappa(\mathfrak{p}) \otimes_k E$ is nilpotent, by the prime-intersection statement of Lemma P0.1. If $\bar{b}$ is nilpotent, its multiplication matrix $\overline{M}$ is nilpotent. The finite geometric sum $\sum_{j<N} z^j \overline{M}^j$ inverts $I - z\overline{M}$ when $\overline{M}^N = 0$. Determinant multiplication makes $\det(I - z\overline{M})$ a unit in the polynomial ring over $\kappa(\mathfrak{p})$. Degrees add for nonzero polynomials over a field, so every such unit is constant; its value at zero is one. Therefore all c_i vanish in $\kappa(\mathfrak{p})$. Conversely, if all these coefficients vanish there, the already proved equation $P_b(\bar{b}) = 0$ gives $\bar{b}^d = 0$. Consequently

$$p_E(D(b)) = \bigcup_{0 \leq i < d} D(c_i),$$

which is open. Principal opens form a basis, proving openness for every finite E/k, including inseparable extensions.

For general algebraic L/k, the finitely many coefficients of any $b \in A \otimes_k L$ lie in one finite subextension E. Its principal open is the pullback of the corresponding principal open over E. The projection from the L-level to the E-level is surjective: field extension is faithfully flat by the vector-space basis calculation, and the prime-fibre argument is proved in AG-CA, *Faithful flatness and the local criterion for flatness*, Theorem 2.2. Its image in $\operatorname{Spec} A$ therefore equals the open finite-level image just calculated. This proves openness of p.

If $X_L = U \amalg V$ were a decomposition into nonempty clopen subsets, both images $p(U)$ and $p(V)$ would be nonempty clopen subsets of the connected X, hence both would equal X. The fibre at the rational point is $\operatorname{Spec} L$, which has one point. It cannot meet both disjoint U and V. This contradiction proves the assertion. □

Lemma G.0.3 (the consumed field-group identity component). Let G/k be a group scheme locally of finite type over any field. Its connected components are open, closed and irreducible. The component G^0 through the identity, with its open subscheme structure, is a normal, geometrically

irreducible, quasi-compact subgroup; its inclusion is a flat closed immersion and its formation commutes with every field extension. No reducedness or characteristic assumption is imposed.

Proof. Begin over an algebraically closed field. Every affine chart is Noetherian by AG-RG-S01, Lemma 1.3. Irreducible components therefore have reduced closed structures locally of finite type. The product of two such irreducible schemes is irreducible: cover each by finite-type integral affine opens and apply Lemma G.4.1 below to each product chart. Any two charts meet, since nonempty opens of each irreducible factor meet, and the product of these intersections is nonempty (each intersection has a rational point by the earlier Nullstellensatz). The union of irreducible open charts with pairwise nonempty intersections is irreducible: any two nonempty opens of the union meet by first meeting the charts and then their overlap.

If Z_1, Z_2 are two irreducible components through the identity, the closure of the multiplication image of $Z_1 \times Z_2$ is irreducible and contains both, using the identity in the other factor. Maximality of the components forces $Z_1 = Z_2$. Translation proves uniqueness of the component through every rational point. On a Noetherian affine chart there are finitely many irreducible components, by AG-CA, *Noetherian and Artinian rings*, Proposition 2.2. The closure of a chart component is a global component: any larger irreducible closed subset intersects that chart in an irreducible closed subset containing the chart component, hence equal to it; this nonempty open is dense in the larger subset, so the closures coincide. Conversely a global component meeting the chart intersects it in a chart component, by the same maximality and density argument. If two distinct global components met at a point, their intersections with such a chart would be distinct closed chart components with nonempty intersection. That intersection has a rational point by the Nullstellensatz, contradicting uniqueness there. Thus all components are pairwise disjoint. Their finite disjoint intersections with every Noetherian chart are open and closed there, so the global components are open and closed. They are exactly the connected components.

Return to the original field and let C be its connected identity component with the canonical flat closed structure of Lemma G.0.1. The identity factors through this structure. For an algebraic closure $\overline{k}$, Lemma G.0.2 makes $C_{\overline{k}}$ connected. Its underlying space is the inverse image of $|C|$. The preceding algebraically closed argument puts it inside the irreducible component Z through the geometric identity. Conversely the connected image of Z in G contains the identity and is contained in $|C|$, so $Z \subset |C_{\overline{k}}|$. These subsets are equal. In particular $C_{\overline{k}}$ is irreducible and open in $G_{\overline{k}}$. The projection is open by the calculation in Lemma G.0.2 and surjective by faithful flatness, so its image $|C|$ is open in G. It is already closed. Its open subscheme structure is a flat closed subscheme, and uniqueness in Lemma G.0.1 identifies that structure with C, retaining every ambient nilpotent.

We spell out why irreducibility persists under arbitrary field extensions, rather than only algebraic ones. If A is a finite-type domain over the algebraically closed $\overline{k}$ and $F/\overline{k}$ is any field extension, write any two nonzero tensors in $A \otimes_{\overline{k}} F$ with independent coefficients from F. For each tensor a nonzero coefficient from A defines a nonempty principal open where its evaluation is nonzero in F. These opens meet because A is a domain, and their intersection has a $\overline{k}$-point by the Nullstellensatz. Evaluating there gives two nonzero elements of the field F, whose product is nonzero. Thus the tensor algebra is a domain; this is the coefficient proof of Lemma G.4.1 with the second factor allowed to be any field. In a Noetherian affine chart of $C_{\overline{k}}$, its nilradical has finitely many nilpotent generators; a power with exponent larger than the sum of their nilpotence exponents is zero. Its extension remains nilpotent. Quotienting by it therefore leaves the underlying space unchanged after field extension, and the domain argument proves irreducibility of each extended chart. The original nonempty chart intersections remain nonempty by faithful flatness. Hence C_F is irreducible for every $F/\overline{k}$.

For an arbitrary K/k, Lemma G.0.A gives a common algebraically closed overfield Ω of K and $\overline{k}$. Then C_Ω is irreducible by the preceding argument, and its surjective projection to C_K makes C_K irreducible. This proves geometric irreducibility. The same affine product proof over $\overline{k}$, with nilpotent chart thickenings retained, makes $C \times C$ geometrically irreducible. Its multiplication image is connected and contains the identity, hence lies in $|C|$. Inversion has the same property.

Lemma G.0.1 factors both entire morphisms through C; their group identities follow from its inclusion being a monomorphism.

For any K/k, let D be the connected identity component of G_K with its canonical flat closed structure. The connected C_K contains the identity and is contained in $|D|$. Conversely the connected image of D in G contains the identity, so lies in $|C|$. Lemma G.0.1 factors its projection through C, hence puts $|D|$ inside $|C_K|$. Equality of underlying subsets and flat closed uniqueness give $D = C_K$, as schemes. Conjugation by every $g \in G(K)$ preserves that identity component. Every point of $G \times C$ lifts to a pair over its residue field. Thus conjugation has set-theoretic image in $|C|$, and Lemma G.0.1 factors its entire morphism through C. This proves normality on all test schemes, including nonreduced ones.

Every other connected component D of G is open, closed and irreducible as well. Choose a finite-type affine chart meeting its closed subset and give that subset its reduced closed structure. A maximal ideal supplies a point $x \in D$ with finite residue field K/k, by the earlier weak Nullstellensatz. The rational lift of x to G_K translates C_K to an open, closed, irreducible subset. Its image U in G is open and closed by the finite-extension projection calculation of Lemma G.0.2 and the integral closed-image proof. It is connected, irreducible and contains x, so $U \subset D$. A nonempty clopen subset of the connected D is all of D. Hence $D = U$, proving these assertions without losing its original scheme structure.

Finally work over $\overline{k}$ and choose a nonempty affine open $U \subset C_{\overline{k}}$. For every rational point g, irreducibility makes $gU \cap U$ nonempty, and the Nullstellensatz supplies a rational point in that intersection. Thus $g = u_1 u_2^{-1}$ for $u_1, u_2 \in U(\overline{k})$. The map $C_{\overline{k}} \times U \to C_{\overline{k}}$, $(a, b) \mapsto ab^{-1}$, is open: the shear $(a, b) \mapsto (ab^{-1}, b)$ is an automorphism of $C_{\overline{k}} \times U$, and projection is open on every finite-type affine target chart V, since $V \times U \to V$ is flat and of finite type and Lemma G.4.F below proves openness in exactly that setting. Restriction to the open $U \times U$ is still open. Its image contains every rational point. Its closed complement, if nonempty, has a rational point on a finite-type affine chart by the Nullstellensatz, a contradiction. The affine $U \times U$ is quasi-compact and surjects onto $C_{\overline{k}}$, so that component is quasi-compact. The surjective projection to C makes C quasi-compact as well. All assertions are proved. □

Lemmas G.0.1 and G.0.3 supply the precise identity-component inputs used in the following reductive-group lessons. Their geometric connectedness and product connectedness follow from the proved geometric irreducibility, rather than an additional arbitrary-scheme theorem. Lemma G.0.2 independently supplies the algebraic-extension connectedness step.

For free comparison, the corresponding canonical identity-component statement is Stacks, Proposition 39.7.11, Tag 0B7R, and the geometric irreducibility input is Tag 047M. The complete arguments used here are the proofs above and the exact earlier programme proofs they specify.

Lemma G.0.T (finite separable splitting of the consumed field groups). Let G be a finite-type group scheme over a field k. If it becomes diagonalizable after any field extension, then it is diagonalizable over a separable closure and over some finite separable extension of k. In particular a k-torus splits over a finite separable extension, including over imperfect fields.

Proof. Choose an algebraic closure using Lemma G.0.A. The elements separable over k form a subfield k_s. Here are the needed field details. Finitely many such elements generate a finite separable extension: adjoining them one at a time uses minimal polynomials which divide their original distinct-root polynomials, so every step is separable; the tower embedding-count proof of AG-RG-S02, Lemma 4.A gives separability over k. Every element of that generated field is separable by the same embedding-count proof. Thus sums, products and inverses stay in this set. If an element of the algebraic closure is separable over k_s, take its minimal polynomial and a Bezout identity between that polynomial and its derivative. Their finitely many coefficients lie in one finite separable extension E/k. The same Bezout identity makes the polynomial separable over E; its root is separable over E, hence over k by the tower proof, and lies in k_s. Therefore k_s is separably closed. A separable irreducible polynomial over k_s has a root in the fixed algebraic

closure of k. The preceding argument puts that root in k_s, so this polynomial has degree one; the same conclusion holds in every algebraically closed overfield of k_s. Finite collections of elements of k_s lie in a finite separable extension of k.

Let K/k diagonalize G. Its affine base change descends to an affine G by the full affine-effectivity and comparison proof in AG-RG-S04, Corollary A3.2; the field cover is faithfully flat by its vector-space basis. Write $A = \mathcal{O}(G)$. Lemma G.0.A gives an algebraically closed common overfield Ω of K and k_s, with

$$A \otimes_k \Omega \simeq \Omega[M]$$

for an abelian group M. We will descend each basis character using a finite system of equations, before descending a finite choice of coordinates.

Every finite subset of A lies in a finite-dimensional subcoalgebra. To see this, use the complete comodule proof of AG-GS, *Group schemes, actions and Hopf algebras*, Lemma 5.1 on the regular comodule, obtaining a finite-dimensional subcomodule W containing that subset. For a basis w_j, write $\Delta(w_j) = \sum_i w_i \otimes a_{ij}$. Coassociativity and the counit give

$$\Delta(a_{ij}) = \sum_\ell a_{i\ell} \otimes a_{\ell j}, \qquad w_j = \sum_i \epsilon(w_i) a_{ij}.$$

The span V of all a_{ij} is therefore a finite-dimensional subcoalgebra containing W. These V cover A.

For one such V, coordinates of $c \in V \otimes_k R$ satisfying

$$\Delta(c) = c \otimes c, \qquad \epsilon(c) = 1$$

define an affine finite-type parameter scheme C_V over k. There are finitely many quadratic equations from the first equality and a linear equation from the second, using bases of V and $V \otimes V$. This represents the group-like elements in V on every algebra R, and the equations commute with scalar extension.

In $\Omega[M]$, the subcoalgebra V_Ω is the span of a finite set F of basis monomials. Indeed, if $v = \sum_m d_m e^m$ lies in it, applying the coefficient functional for e^m in the second factor of $\Delta(v)$ puts $d_m e^m$ in V_Ω. Each nonzero coefficient therefore puts that monomial in V_Ω. Finite dimensionality makes the resulting set finite. Over an arbitrary Ω-algebra, comparing the coefficients of $\Delta(c) = c \otimes c$ gives

$$c = \sum_{m \in F} x_m e^m, \quad x_m^2 = x_m, \quad x_m x_n = 0 \ (m \neq n), \quad \sum_{m \in F} x_m = 1.$$

Their coordinate algebra is precisely Ω^F: the standard orthogonal idempotents map to the x_m; every monomial in the x_m is either zero or one of these idempotents, so they span the algebra; evaluation at the standard basis idempotents gives the inverse. If F is empty both algebras are zero. Thus

$$(C_V)_\Omega = \coprod_{m \in F} \operatorname{Spec} \Omega$$

on every test algebra, not just on field-valued points.

Put $E_V = \mathcal{O}(C_V)$. It is finite dimensional over k, since any independent family remains independent after extension to Ω, and its extension is Ω^F. It is reduced, since it injects into that reduced algebra. The Artinian decomposition and Chinese remainder proofs of AG-CA, *Noetherian and Artinian rings*, Lemma 4.1 and Theorem 4.2 make it a product of finite field extensions. Each factor is separable: an element with inseparable irreducible polynomial would give, after extension to the algebraically closed Ω, a nonzero nilpotent in its polynomial quotient. That quotient injects into the extended field factor by exact field tensor, contradicting reducedness of $E_V \otimes_k \Omega$. The repeated-root criterion is proved in AG-RG-S02, Lemma 4.A.

After extension to the separably closed k_s, each of these finite separable factors becomes a product of copies of k_s. Explicitly, take a finite list of generators for a factor and adjoin them in a tower.

At every step its separable minimal polynomial, in each existing k_s-factor, splits into distinct linear factors; the Chinese remainder calculation gives one k_s for each root. Iterating proves the assertion without a primitive-element assumption. The scalar extension from this product to Ω gives a bijection of its points. Consequently each group-like element of A_Ω lying in V_Ω descends uniquely to A_{k_s}. Every monomial lies in one such V_Ω, since its coefficient expression uses finitely many elements of A.

The group N of group-like elements of A_{k_s} therefore maps bijectively, compatibly with multiplication and inverse, to M. The Hopf map $k_s[N] \to A_{k_s}$ becomes the basis isomorphism over Ω. Exact faithful field tensor detects both its kernel and cokernel, hence this map is an isomorphism already over k_s.

Finally this isomorphism descends to a finite separable extension. The group M, hence N, is finitely generated: a finite list of algebra generators of $\Omega[M]$ has finite monomial support, and polynomial expressions in that list have support in the subgroup generated by that finite union. Since every e^m occurs, that subgroup is all of M. Thus both A and $k[N]$ are finite-type algebras and have finite polynomial presentations by the Hilbert basis proof of AG-RG-S01, Lemma 1.3. Express the isomorphism and inverse on their finitely many algebra generators. Their images use only finitely many coefficients of k_s; put these in one finite separable L/k. The finite relation lists vanish after extension to k_s, hence already over L by injectivity of field tensor, so the two maps are defined there. Their compositions, counits, antipodes and comultiplications agree on the generators after extension to k_s, hence already over L, using the same injectivity on the algebras and their tensor squares. They are inverse Hopf isomorphisms over L, giving $G_L = D_L(N)$. For a torus the diagonal group has $N \simeq \mathbf{Z}^r$, so this is exactly $\mathbf{G}_{m,L}^r$. □

G.1. Dimension at a point after field extension

We use the earlier proofs in AG-CA, *Krull dimension and Noether normalization*, Corollary 3.2, Theorems 4.2 and 6.1: normalization over a polynomial ring, dimension as transcendence degree for a finite-type domain, and dimension at a point as the largest dimension of a component through it. We also use *Associated primes and primary decomposition*, Theorem 3.2 and Theorem 1.2, for minimal primes being associated and the associated-prime description of zero divisors. For an integral inclusion, going up lifts every finite prime chain and incomparability makes the contraction of every strict chain strict, so the two dimensions agree. These two assertions have their complete earlier proofs in Descent and Zariski Main, Lemma P0.4; its normal going-down proof P0.8 supplies the further input used by the earlier normalization dimension argument.

Lemma G.1.1 (flat going down). If $A \to B$ is flat, $\mathfrak{q} \subset B$ contracts to $\mathfrak{p} \subset A$, and $\mathfrak{p}_0 \subset \mathfrak{p}$, there is $\mathfrak{q}_0 \subset \mathfrak{q}$ contracting to $\mathfrak{p}_0$.

Proof. The local map $A_\mathfrak{p} \to B_\mathfrak{q}$ is flat and faithfully flat. The latter assertion is proved in AG-CA, *Faithful flatness and the local criterion for flatness*, Theorem 2.1: a proper ideal of the source lies in its maximal ideal and its extension lies in the target maximal ideal, so the residue-field test for faithful flatness applies. Consequently

$$B_\mathfrak{q} \otimes_{A_\mathfrak{p}} \kappa(\mathfrak{p}_0)$$

is a nonzero ring. A maximal ideal of it gives a prime of $B_\mathfrak{q}$ over $\mathfrak{p}_0 A_\mathfrak{p}$, by the fibre prime correspondence of AG-CA, *Localization, local properties and support*, Theorem 3.2. Contracting this prime to B proves the assertion. This is also the complete earlier proof in *Tor and flat modules*, Theorem 6.3. □

Lemma G.1.2. Let X be locally of finite type over a field k, let K/k be any field extension, and let $y \in X_K$ lie over $x \in X$. Then

$$\dim_y X_K = \dim_x X.$$

Proof. Work on an affine finite-type neighbourhood, with algebra A. First suppose A is a domain of dimension d. Choose a normalization $P = k[t_1, \dots, t_d] \subset A$, finite over P. Put $F = \operatorname{Frac} P$. The inclusion

$$A \hookrightarrow A \otimes_P F \simeq F^r$$

is an injection of P-modules for some finite r. Tensoring over k preserves injection and embeds A_K into

$$(F \otimes_k K)^r.$$

Here $F \otimes_k K$ is a localization of the domain $P_K = K[t_1, \dots, t_d]$. Thus A_K is torsion-free over P_K, as well as finite over it. A minimal prime $\mathfrak{q}$ of the Noetherian ring A_K is associated. It cannot contain a nonzero element of P_K, since such an element acts injectively on A_K. Therefore $P_K \to A_K/\mathfrak{q}$ is an integral inclusion and every component of $\operatorname{Spec} A_K$ has dimension d.

For general A, flat going down shows that every minimal prime of A_K contracts to a minimal prime of A. Indeed, a smaller prime downstairs could be lifted below it, contrary to minimality. If $\mathfrak{q}$ is a minimal prime upstairs contracting to $\mathfrak{p}$, it is minimal in the quotient $(A/\mathfrak{p})_K$, so the preceding domain argument gives component dimension $\dim(A/\mathfrak{p})$.

Conversely, for every minimal $\mathfrak{p} \subset \mathfrak{p}_x$, Lemma G.1.1 lifts $\mathfrak{p}$ below the prime of y; choosing a minimal prime below that lift produces an upstairs component through y contracting to $\mathfrak{p}$. Thus the component dimensions through y are exactly those through x. The point-dimension formula of AG-CA, *Krull dimension and Noether normalization*, Theorem 6.1 takes their maximum on each side and proves the equality. □

This supplies the field-extension step in AG-GS-02, Lemma 3.1 and Theorem 3.2, without identifying point dimension with local-ring dimension at a nonclosed point.

G.2. Smoothness descends under field extension

Lemma G.2.1. For a locally finite-type k-scheme X and a field extension K/k, if X_K/K is smooth, then X/k is smooth.

Proof. It suffices to treat an affine chart $S = k[X_1, \dots, X_n]/I$. The polynomial ring is Noetherian by the earlier Hilbert basis theorem, so I has a finite generating list and S is finitely presented. Its conormal sequence is

$$I/I^2 \longrightarrow S^n \longrightarrow \Omega_{S/k} \longrightarrow 0.$$

Formation of every term and map commutes with the flat scalar extension $k \to K$, by the polynomial presentation and conormal calculation in AG-CA, *Kähler differentials*, Theorems 3.3 and 5.1. Upstairs smoothness makes the conormal arrow injective and $\Omega_{S_K/K}$ finite projective: this is proved by the lifting argument in AG-CA, *Formally smooth, unramified and étale ring maps*, Theorem 3.1. Smoothness is local on the affine chart, so these properties hold on the whole upstairs module by localization and the earlier local criterion for finite projectivity.

Faithful scalar extension detects the kernel of the conormal arrow. Finite projectivity of $\Omega_{S/k}$ descends by AG-CA, *Faithful flatness and the local criterion for flatness*, Corollary 3.2. The conormal sequence downstairs is consequently split exact. The converse lifting construction in the same Theorem 3.1 then makes S formally smooth: for a square-zero test, the error on I/I^2 extends to the free polynomial differential module using this splitting, and subtracting that derivation corrects all defining equations. Finite presentation now makes S smooth. Apply this to each affine chart. □

No perfectness hypothesis occurs in this descent argument.

G.3. The local regularity step in Cartier's theorem

The local assertions needed here have complete earlier programme proofs: Nakayama in AG-CA, *Localization, local properties and support*, Theorem 4.2; Krull intersection in *Noetherian and Artinian rings*, Theorem 6.1; the one-equation dimension theorem in *Dimension theory of Noetherian local rings*, Theorem 3.2; the dimension bound by a maximal-ideal generating list in that lesson, Theorem 2.1; and rational closed-point smoothness in *Smooth algebras over a field and the Jacobian criterion*, Proposition 3.1 and Theorem 2.1.

Lemma G.3.1 (lifting regularity). If $(R, \mathfrak{m})$ is Noetherian local, $f \in \mathfrak{m}$ is a nonzerodivisor, and R/fR is regular local, then R is regular local.

Proof. Put $D = \dim R$. AG-CA, *Dimension theory of Noetherian local rings*, Theorem 3.2 gives $\dim(R/fR) = D - 1$, since f is a nonzerodivisor. Regularity of the quotient supplies $D - 1$ generators of its maximal ideal. Lift them to R. Every element of $\mathfrak{m}$, modulo their generated ideal, belongs to fR; thus the lifts together with f generate $\mathfrak{m}$. The embedding dimension of R is at most D. Conversely, the local dimension theorem, AG-CA Theorem 2.1, bounds $\dim R$ by the number of generators of any ideal of definition, in particular any minimal generating list of $\mathfrak{m}$. Hence embedding dimension equals dimension, which is precisely regularity. □

Lemma G.3.2. Suppose R is a Noetherian local algebra over a characteristic-zero field, $f \in \mathfrak{m}$, and a derivation $\delta : R \to R$ satisfies $\delta(f) = 1$. Then f is a nonzerodivisor.

Proof. If $fa = 0$, the iterated product rule gives

$$f\delta^n(a) + n\delta^{n-1}(a) = 0 \qquad (n \geq 1).$$

Since $n!$ is invertible,

$$a = (-1)^n \frac{f^n \delta^n(a)}{n!} \in f^n R.$$

Krull intersection gives $a = 0$. □

Lemma G.3.3. Let k be algebraically closed of characteristic zero, let $\mathfrak{m}$ be a closed point of a finite-type k-algebra, and let R be its local ring. If $\Omega_{R/k}$ is free, then R is regular.

Proof. Its residue field is k, by the earlier Nullstellensatz. The rational-point cotangent calculation identifies

$$\mathfrak{m}/\mathfrak{m}^2 \simeq \Omega_{R/k} \otimes_R k.$$

Let the free rank be r, and induct on r. It is finite: if the original finite-type algebra is generated by $a_1, \ldots, a_n$, the universal product rule makes its differentials generated by $da_1, \ldots, da_n$, and the differential localization formula, AG-CA Theorem 2.2, gives the same finiteness for R. If $r = 0$, Nakayama, AG-RG-S01 Lemma 1.2, gives $\mathfrak{m} = 0$, so R is a field and is regular of dimension zero. If $r > 0$, choose $f \in \mathfrak{m}$ with df nonzero modulo $\mathfrak{m}$. One coordinate of df in a free basis is a unit. Reorder it to the first position and replace the first basis vector by df. The change-of-basis matrix has determinant that unit, so the adjugate identity makes it invertible. The functional taking value 1 on df and zero on the remaining basis vectors corresponds, by the universal differential property, to a derivation with $\delta(f) = 1$. Lemma G.3.2 makes f a nonzerodivisor.

The quotient conormal sequence gives

$$\Omega_{(R/fR)/k} = (\Omega_{R/k} \otimes_R R/fR)/(R/fR)\, df,$$

which is free of rank $r - 1$. This is the full conormal sequence of AG-CA, *Kähler differentials*, Theorem 3.3: the image of $(f)/(f^2)$ is exactly the span of df, by the product rule. The quotient remains a rational closed-point local ring of a finite-type k-algebra. Indeed write $R = A_{\mathfrak{m}_A}$ and $f = a/s$, with $s \notin \mathfrak{m}_A$. Then $R/fR = (A/(a))_{\mathfrak{m}_A/(a)}$, and its residue field is still k. The induction hypothesis therefore makes R/fR regular, and Lemma G.3.1 makes R regular. □

Theorem G.3.4 (Cartier, with the consumed scope). A group scheme locally of finite type over a characteristic-zero field is smooth.

Proof. Extend the field to its algebraic closure. The cotangent sheaf of any group scheme is

$$\Omega_{G/k} \simeq \mathcal{O}_G \otimes_k e^*\Omega_{G/k}.$$

Here is the complete identification, valid without affineness or smoothness. Write $p_1, p_2 : G \times_k G \to G$ for the projections and $s(g) = (g, e)$, $\Delta(g) = (g, g)$ for the two sections of p_1. The automorphism $a(g, x) = (g, gx)$, with inverse $(g, y) \mapsto (g, g^{-1}y)$, is over the first factor and satisfies $a \circ s = \Delta$. The differential base-change identity, AG-CA, *Kähler differentials*, Theorem 2.1, gives $\Omega_{(G\times_k G)/G} \simeq p_2^*\Omega_{G/k}$. Since a is an automorphism over G, its relative differential is an isomorphism $a^*\Omega_{(G\times_k G)/G} \simeq \Omega_{(G\times_k G)/G}$. Pulling this isomorphism back along s identifies $\Delta^* p_2^*\Omega_{G/k} = \Omega_{G/k}$ with $s^* p_2^*\Omega_{G/k} = \mathcal{O}_G \otimes_k e^*\Omega_{G/k}$, proving the display. These calculations on affine charts glue because the universal differential and pullback maps agree on intersections. In the present locally finite-type scope, a finite-type chart at the identity makes $e^*\Omega$ finite-dimensional, so the cotangent sheaf is free.

Every closed point of each finite-type affine chart is rational. Lemma G.3.3 makes its local ring regular; the earlier rational-point Jacobian criterion makes that point smooth. The nonsmooth locus is closed on the chart by the earlier open-locus proof. If nonempty it has a rational closed point by the Nullstellensatz, a contradiction. Hence the group is smooth after algebraic closure. Lemma G.2.1 descends smoothness to the original field. This proof permits infinitely many components and makes no global affineness or quasi-compactness assumption. □

Lemma G.3.5 (universal schematic density). Let $p : X \to S$ be smooth, with geometrically irreducible nonempty fibres, and let $U \subset X$ be an open whose intersection with every fibre is dense. After every base change $S' \to S$, the restriction $\mathcal{O}_{X_{S'}} \to (j_{S'})_*\mathcal{O}_{U_{S'}}$ is injective, where $j_{S'} : U_{S'} \hookrightarrow X_{S'}$. Consequently two maps from $X_{S'}$ to a separated S'-scheme which agree on $U_{S'}$ agree everywhere.

Proof. Schematic density is local on the source and the base. Choose a smooth affine chart $V \to \operatorname{Spec} R$ of finite presentation. Its image in the base is open. Over each principal base open contained in that image, replace R, V by their restrictions; then every fibre of V is nonempty. It is geometrically integral: after a geometric field extension it is a nonempty open of the smooth irreducible fibre of X, hence reduced and irreducible. An empty chart contributes no condition.

Write $V = \operatorname{Spec} A$. The principal opens $D(f) \subset U \cap V$ cover $U \cap V$. Their base images are open because their morphisms to the base are smooth. They cover $\operatorname{Spec} R$: the dense U_s meets each nonempty V_s. Quasi-compactness of this affine base therefore selects finitely many $f_1, \dots, f_m$ whose base images cover it. No quasi-compactness of U has been assumed.

Use AG-RG-S02 Theorem 4.5 to choose a smooth affine model $V_0 = \operatorname{Spec} A_0$ over a finitely generated $\mathbf{Z}$-algebra R_0, with geometrically connected nonempty fibres. These fibres are geometrically integral by the smooth component argument in its Lemma 4.1. Enlarge the coefficient ring to retain the finitely many f_i, using the finite-presentation map and equality proof. All these enlargements are base changes of that model. In the Noetherian model the base images W_i of $D(f_i)$ are open and constructible; their union E pulls back to the whole final base. AG-RG-S02 Lemma 4.4 therefore gives a finite coefficient stage at which their pullbacks cover the base. We retain this finite stage and its specified functions.

On each W_i every fibre of V_0 is integral and f_i is nonzero in it, because $D(f_i)$ has a point in that fibre. At a local ring of V_0 where f_i is not a unit, the local map from the corresponding Noetherian base local ring is flat and f_i is regular in its closed fibre. The written slicing proof in *Faithful flatness and the local criterion for flatness*, Theorem 5.2, shows that multiplication by f_i is injective and that its cokernel is flat over that base local ring. Where f_i is a unit the same conclusions are immediate, with zero cokernel. Thus, locally over W_i, the sequence

$$0 \longrightarrow \mathcal{O}_{V_0} \xrightarrow{f_i} \mathcal{O}_{V_0} \longrightarrow \mathcal{O}_{V_0}/f_i\mathcal{O}_{V_0} \longrightarrow 0$$

remains exact after every base change: the last module is base-flat, so its first Tor vanishes. This assertion is made on affine base charts and their affine inverse images, and then glues.

It follows that, after base change first to R and then to any R-algebra, restriction to $D(f_i)$ is injective on every open of the inverse image of W_i. Indeed an element killed by localization is killed by a power of f_i, and injectivity of multiplication by f_i and of all its localizations makes that element zero. A function vanishing on U vanishes on every such $D(f_i)$. The W_i cover the base, so it vanishes on V. The argument applies after every further base change and on all source charts, proving the sheaf injection. Finally the equalizer of two maps to a separated target is closed; its defining ideal restricts to zero on $U_{S'}$, so the injection makes that ideal zero everywhere. □

G.4. The reduced products used in the closed-orbit argument

Lemma G.4.1. Over an algebraically closed field, the product of two reduced finite-type schemes is reduced, and the product of two integral finite-type schemes is integral.

Proof. On affine charts let the algebras be A, B. The earlier Nullstellensatz says that an element of A vanishing at every k-point is zero when A is reduced. For a tensor $u = \sum_i a_i \otimes b_i$, choose the b_i linearly independent over k. If all evaluations of u at k-points of $\operatorname{Spec} A$ are zero in B, coefficient independence makes each a_i vanish at every such point, so $u = 0$. If u is nilpotent and B is reduced, each of those evaluations is zero. This proves reducedness.

If A, B are domains and u, v are nonzero tensors, the locus in $\operatorname{Spec} A$ where the evaluation of u is nonzero contains a nonempty open: choose one nonzero coefficient in an independent expression. The analogous open for v meets it, since $\operatorname{Spec} A$ is irreducible. Their intersection has a k-point. At this point both evaluations are nonzero in the domain B, so their product is nonzero. Thus $uv \neq 0$. Finite affine covers give the stated scheme assertions. □

Lemma G.4.A (finite-type images). The image of a morphism between finite-type schemes over an algebraically closed field is constructible. Every nonempty constructible subset of such a scheme contains a k-point.

Proof. Reduction preserves the underlying spaces and images, so first reduce the source and target. They are Noetherian by the Hilbert basis proof. They have finitely many irreducible components: a minimal closed subset which is not a finite union of irreducible closed subsets would split into two proper closed subsets, contradicting minimality.

Here is the dense-open calculation. Suppose an integral source dominates an integral target. Choose nonempty affine opens at their generic points, with coordinate domains $A \subset B$, and put $F = \operatorname{Frac} A$. The generic algebra B_F is a finite-type domain over F. Noether normalization supplies algebraically independent $t_1, \dots, t_r \in B_F$ such that B_F is finite over $F[t_1, \dots, t_r]$. Clear denominators in these elements and in the finitely many monic equations for algebra generators of B. After inverting one nonzero $a \in A$, we have an inclusion $A_a[t_1, \dots, t_r] \subset B_a$ which is finite and integral, by the complete finite-integral calculation. Lying over makes its spectrum map surjective. Projection from polynomial affine space is surjective too: a prime $\mathfrak{p}$ of A_a lifts to $\mathfrak{p}A_a[t_1, \dots, t_r]$, whose quotient is a polynomial domain over $A_a/\mathfrak{p}$. Thus the original image contains the nonempty target open $D(a)$.

Use Noetherian induction on closed target subsets, with the assertion quantified over every finite-type source mapping to that subset. If the closure of an image is proper, the induction hypothesis applies to it. If the target is reducible, apply induction to the inverse images of its finitely many proper irreducible components and take their finite union. Otherwise the target is integral and the image is dense. One of the finitely many source components dominates it, so the preceding calculation gives a nonempty open U in the image. The remaining image is the image of the source's inverse image of the proper closed subset $Y \setminus U$, and is constructible there by induction. Its union with U is constructible in Y. This proves the assertion. Affine charts used in the dense-open calculation suffice because every nonempty open of an integral scheme contains its generic point.

A dense constructible subset contains a dense open. Write it as a finite union of locally closed pieces. On each irreducible component, one piece has dense closure, since a finite union of proper closed subsets cannot contain that component. Its locally closed description then contains an open neighbourhood of the component's generic point. Remove the other components and take the union of these neighbourhoods over the finitely many components; this is a dense open of the whole scheme contained in the subset.

A nonempty constructible subset has a nonempty locally closed piece, hence a nonempty affine open in its reduced locally closed scheme. This is a finite-type k-scheme. A maximal ideal in its nonzero coordinate algebra has residue field k by the weak Nullstellensatz. The resulting point is a k-point of the original scheme and is closed there, again by that theorem. □

Lemma G.4.B (generic flatness by monic division). If A is a Noetherian domain and B a finite-type A-algebra, then B_a is a free A_a-module for some nonzero $a \in A$. The basis may be infinite or empty.

Proof. Write $B = A[X_1, \dots, X_n]/J$, where J is finite by the Hilbert basis theorem, and put $F = \operatorname{Frac} A$. Order monomials first by total degree and then lexicographically. Each bounded total degree has only finitely many monomials, so there is no infinite decreasing sequence; multiplication preserves the order. The ideal of leading monomials of J_F has a finite generating subset of those monomials, since $F[X_1, \dots, X_n]$ is Noetherian. Choose polynomials $f_i \in J_F$ realizing them and normalize their leading coefficients to one.

Division by these monic polynomials terminates by the chosen order. The remainder of an element of J_F is zero: a nonzero remainder would still belong to J_F, while its leading monomial would be divisible by one of the chosen generators, contradicting the remainder condition. Hence the monomials divisible by none of those generators form an F-basis of B_F: division spans them and the same leading-monomial argument proves independence.

Invert a single nonzero $a \in A$ which clears the coefficients of the f_i, their expressions in the original generators of J, and the expressions of those original generators in the ideal of the f_i. The latter expressions are supplied by the terminating division just proved. There are only finitely many coefficients and expressions. Thus $J_a = (f_i)$, and monic division over A_a makes the same monomials span B_a. They are independent because a relation becomes a relation over F, and $A_a \hookrightarrow F$ is injective. They are therefore an A_a-basis. This includes $J_F = (1)$, where the basis is empty and $B_a = 0$. A free module is flat by the elementary tensor calculation in S01 Lemma 1.0. □

Theorem G.4.2 (the closed-orbit argument). Let a smooth finite-type group G over an algebraically closed field k act on a nonempty reduced separated finite-type k-scheme X. The reduced orbits are locally closed, and X contains a closed orbit.

Proof. Choose $x \in X(k)$. Let O be the topological image of $G \to X$, $g \mapsto gx$, and let Y be its reduced closure. Lemma G.4.A gives constructibility, including its complete dense-open and Noetherian-induction arguments.

Consequently O is constructible and contains a dense open U of Y. The union $V = \bigcup_{g \in G(k)} gU$ is open in Y and contained in O. It contains all rational points of O: a nonempty orbit-map fibre is finite type and has a rational point, so $G(k)$ is transitive on those points. The constructible subset $O \setminus V$, if nonempty, has a rational point; hence it is empty. Thus O is open in Y, proving local closedness.

Translations preserve Y. The action restricts schematically to Y and to O. Indeed Lemma G.4.1 makes $G \times Y$ reduced. Each defining equation of Y pulls back to a function vanishing at every rational point, so the Nullstellensatz makes it zero. The inverse image of the boundary $Y \setminus O$ under the restricted action on $G \times O$ is a closed subset with no rational point and is therefore empty.

For completeness, the components of smooth G are open and disjoint: its local rings are regular domains by the earlier smoothness and regular-local proofs, so two distinct irreducible components cannot meet. There are finitely many components. Its identity component G^0 is therefore irreducible.

Lemma G.4.1 makes $G^0 \times G^0$ irreducible; multiplication and inversion preserve the component containing the identity, so G^0 is a subgroup. Every component contains a rational point and is a translate of G^0.

The closures of the images of those finitely many translated components under the orbit map are translates of one irreducible closed set, and have a common dimension d. They cover Y; after repetitions are removed they are precisely its irreducible components. The boundary meets each component in a proper closed subset, so every boundary component has dimension less than d. This strict inequality is the earlier finite-type domain dimension theorem: a nonzero prime in a finite-type domain lowers the transcendence degree and hence the dimension.

Choose an orbit of minimum dimension; this is possible among the nonnegative integers since $X(k) \neq \varnothing$. If its boundary were nonempty, that boundary would contain a rational point, whose orbit lies in it by invariance and has smaller dimension. This contradicts minimality. The chosen orbit is closed. The same argument inside any orbit closure or nonempty invariant closed reduced subscheme gives a closed orbit there. □

Theorem G.4.C (smooth connected group orbit quotients). Let a smooth connected affine finite-type group G over algebraically closed k act on a separated finite-type k-scheme X, and let $x \in X(k)$. Its reduced locally closed orbit O is smooth, and the orbit map $q : G \to O$ is faithfully flat and of finite presentation. If K is the full schematic stabilizer of x, then q is a K-torsor and O represents the fppf quotient sheaf G/K. In particular this full quotient maps to X by a locally closed immersion; no reduction of K is made.

Proof. Smoothness and connectedness make G integral: smooth local rings are regular domains, so its finitely many irreducible components are disjoint, and connectedness leaves one. The action restricts to X_{red}, since $G \times X_{\mathrm{red}}$ is reduced by Lemma G.4.1. Theorem G.4.2 gives the locally closed orbit, with its reduced structure and the restricted action. The orbit map factors through it: equations of its reduced closure pull back to zero on the reduced G by rational-point detection, and the inverse image of its boundary has no k-point and is empty. The closure and O are integral because they are dominated by integral G.

We prove flatness without assuming a homogeneous quotient exists. Take a nonempty affine open $U = \operatorname{Spec} A$ of O and a finite affine cover of $q^{-1}(U)$. Each nonempty chart contains the generic point of G, and its coordinate map from the domain A is injective because q is dominant. Its algebra is finite type over A. Apply Lemma G.4.B to each of the finitely many charts and invert the product of its nonzero chart elements. This gives a nonempty open $U_0 \subset U$ over which q is flat. Choose $u \in U_0(k)$. The orbit map is surjective onto its topological orbit; each nonempty closed-point fibre is finite type and has a k-point by the weak Nullstellensatz. Hence $G(k)$ acts transitively on $O(k)$. The translates gU_0 cover O: their open union contains every closed point and its closed complement must be empty. Equivariance identifies the orbit maps over these opens by translation on both source and target. Flatness consequently holds everywhere. The map is surjective and of finite presentation: both schemes are finite type over k, and on Noetherian affine charts a finite-type algebra has a finite presentation by the Hilbert basis theorem. Thus q is fppf.

The stabilizer K is the closed fibre over x. For every test scheme, $gx = hx$ is equivalent to $g^{-1}h \in K$. Multiplication and inversion therefore give inverse scheme maps

$$G \times K \longrightarrow G \times_O G, \quad (g, n) \longmapsto (g, gn), \qquad (g, h) \longmapsto (g, g^{-1}h).$$

After the fppf cover obtained by pulling back q, every section of O has a lift in G, and any two lifts differ by a unique section of K. This proves the torsor assertion and identifies O with the fppf quotient sheaf. Its immersion in X is the original reduced locally closed orbit immersion, now with the full quotient interpretation.

Finally O is smooth. At its generic field, Noether normalization and S02 Lemma 3.F give differential dimension equal to transcendence degree. AG-CA-18 Theorem 2.1 and Corollary 2.2 give a nonempty open smooth locus. It contains a k-point, and the transitive translations just proved give a smooth

neighbourhood of every k-point. The remaining closed nonsmooth locus has no closed point and is empty. □

Lemma G.4.D (orbit dimension, including nonreduced stabilizers). In Theorem G.4.C,

$$\dim O = \dim G - \dim K.$$

Proof. First, dimension of a finite-type algebra A over a field F is unchanged by extending F to any field E. For a domain, Noether normalization makes A finite integral over an injected $F[T_1, \dots, T_d]$. Tensoring with E preserves that injection and finiteness, so the integral-extension dimension theorem gives dimension d after extension as well. For general A, its finitely many minimal primes give its irreducible components. Their product ideal is contained in the nilradical; that nilradical is nilpotent, because its finitely many nilpotent generators give a uniform monomial bound. Thus, after field extension, the closed subschemes $(A/\mathfrak{p})_E$ still cover the spectrum. Dimension is the maximum of the dimensions of a finite closed cover, as proved in the earlier dimension lesson. Apply the domain calculation to each component to obtain the assertion.

Put $F = k(O)$ and $L = k(G)$. Each nonempty affine chart of the generic fibre of q is an integral finite-type F-algebra with fraction field L. Noether normalization gives its dimension $\operatorname{trdeg}_F L$. This is therefore the dimension of the generic fibre. After extending F to an algebraic closure, that nonempty finite-type fibre has a point. Translation by the point identifies it, on every test algebra, with K after the same extension, by the torsor identity in Theorem G.4.C. The field-extension calculation above gives $\operatorname{trdeg}_F L = \dim K$.

Finite transcendence bases in the tower $k \subset F \subset L$ add: a basis for F/k followed by one for L/F is algebraically independent over k, and L is algebraic over the field they generate, by transitivity of algebraicity. Consequently $\operatorname{trdeg}_k L = \operatorname{trdeg}_k F + \operatorname{trdeg}_F L$. The finite-type domain dimension formula for G and O now gives the claimed equality. □

Lemma G.4.E (generic recognition for affine monomorphisms). Let $f : Y \to X$ be an affine finite-type monomorphism between affine finite-type schemes over the algebraically closed field k. Let $W \subset X$ be its schematic image, defined by the kernel of the coordinate-ring map. There is an open $V \subset W$ containing all generic points of W for which $f^{-1}(V) \to V$ is an isomorphism. The schemes may be nonreduced.

Proof. A nonempty finite-type monomorphism $D \to \operatorname{Spec} F$, for any field F, is an isomorphism. Choose a closed point of an affine chart of D; the field-algebra lemma makes its residue field L/F finite. After extending to L, the monomorphism has a section. A monomorphism with a section is an isomorphism: its section followed by its projection equals the identity because both maps have the same composite with the monomorphism. The inverse descends along the faithfully flat field extension, by the morphism descent of S04 Lemma A1.5. Thus $D = \operatorname{Spec} F$, with its full scheme structure.

A finite monomorphism is a closed immersion. On an affine target write it as $C \to B$, with B finite as a C-module. At every prime of C, the fibre is empty or is the residue field itself by the preceding argument. Hence the cokernel of $C \to B$ has zero tensor product with every residue field. It is a finite module, so localization and the determinant Nakayama argument make it zero. The ring map is surjective.

Now write $W = \operatorname{Spec} C$ and $Y = \operatorname{Spec} B$. The map $C \hookrightarrow B$ is injective and of finite type. Its image on spectra is dense: an element of C whose principal open misses the image becomes nilpotent in B, and hence in C. Lemma G.4.A makes that image constructible, so it contains every generic point of W. Fix the minimal prime $\mathfrak{p}$ of one such point. The Noetherian local ring $R = C_{\mathfrak{p}}$ has just one prime; its maximal ideal $\mathfrak{m}$ is its nilradical. This ideal is finitely generated by nilpotent elements, so $\mathfrak{m}^N = 0$ for some N. The preceding field calculation gives

$$(B \otimes_C R)/\mathfrak{m}(B \otimes_C R) = \kappa(\mathfrak{p}).$$

Take finitely many algebra generators b_j of $B \otimes_C R$. Lift their residues to $r_j \in R$. Then $(b_j - r_j)^N = 0$, so each generator satisfies a monic equation over R. The integral-generator calculation makes this algebra a finite R-module. The coefficients and the finitely many equalities in these monic equations involve only finitely many denominators outside $\mathfrak{p}$. Their product gives $a \in C \setminus \mathfrak{p}$ such that B_a is finite over C_a. The restricted map remains a monomorphism, so $C_a \to B_a$ is surjective by the finite calculation. Localization preserves its original injection; it is therefore an isomorphism. The union of these principal neighbourhoods for the finitely many minimal primes is the required V. Their inverses agree on overlaps because they invert the same morphism. □

Lemma G.4.F (openness of flat finite-type maps). A flat morphism between finite-type schemes over the algebraically closed field k is open. In particular this applies to smooth morphisms, since their flatness was proved in AG-CA-18 Theorem 6.1.

Proof. For a point y of the source over x, the local map $R = \mathcal{O}_{X,x} \to S = \mathcal{O}_{Y,y}$ is flat and local. It is faithfully flat. Indeed, for any nonzero R-module M, choose a nonzero element and its cyclic submodule R/I, with I proper and hence contained in the maximal ideal. Its tensor product is $S/IS \neq 0$, since IS is contained in the maximal ideal of the local S. Flatness embeds this tensor product in $S \otimes_R M$, so the latter is nonzero. For every prime $\mathfrak{p}$ of R, it follows that $S \otimes_R \kappa(\mathfrak{p})$ is a nonzero ring. A prime of that ring gives a prime of S contracting to $\mathfrak{p}$. Interpreting local spectra, every generalization of x therefore lifts to a generalization of y.

For an open $U \subset Y$, all these generalizations of $y \in U$ stay in U. Thus its image is stable under generalization. It is constructible by Lemma G.4.A: U is quasi-compact in the Noetherian Y, and its restriction is still of finite type. A constructible subset stable under generalization is open. To check this last assertion, its complement is constructible and stable under specialization. In each irreducible component of the closure of that complement, its dense constructible part contains the generic point, by Lemma G.4.A. Stability under specialization then includes every point of that component. The complement therefore equals its closure and is closed. This proves openness. □

For arbitrary finite-type G, without smoothness or reducedness, retain AG-GS-03, Lemma 7.12 and Theorem 7.13 with their original scheme structures. Their generic monomorphism argument is compatible with nilpotents: at a minimal prime the target ring is Artinian; a finite-type algebra with the same residue field becomes finite because its lifted generators differ from scalars by nilpotents; a finite monomorphism is a closed immersion by the residue-field test and Nakayama. Schematic dominance then makes it an isomorphism on a generic open. Translating that open proves the orbit immersion once the scheme quotient G/H exists. The full quotient and schematic-orbit arguments retain their separate exact earlier proofs; the smooth connected group result above keeps its stabilizer's full scheme structure.

G.5. The semilinear descent used for character modules

Lemma G.5.1 (independence of automorphisms). Distinct automorphisms $\sigma_1, \dots, \sigma_s$ of a field K are linearly independent over K as functions $K \to K$.

Proof. In a nonzero relation with the fewest terms, normalize the first coefficient to 1. A relation with one term is impossible by evaluation at 1. Choose b with $\sigma_1(b) \neq \sigma_j(b)$ for another participating automorphism. Evaluate the relation at bx and subtract $\sigma_1(b)$ times the relation at x. The first term disappears and the j-th term remains. This is a shorter nonzero relation, a contradiction. □

Lemma G.5.2 (finite Galois descent). Let K/k be finite Galois with group Δ, and let V be any K-vector space with a semilinear Δ-action. Then

$$K \otimes_k V^{\Delta} \longrightarrow V, \qquad a \otimes v \longmapsto av$$

is an isomorphism. Equivariant maps descend uniquely. Tensor products, algebras, Hopf operations and their identities descend through this isomorphism.

Proof. Lemma G.5.1 shows that the vectors $(\sigma(b))_{\sigma\in\Delta}$, $b \in K$, span K^Δ as a K-vector space: otherwise a nonzero linear functional annihilating their span would give a relation among the automorphisms. Consequently there are finite lists $a_i, b_i \in K$ such that

$$\sum_i a_i\sigma(b_i) = \begin{cases} 1 & \sigma = 1, \\ 0 & \sigma \neq 1. \end{cases}$$

For $v \in V$ set $w_i = \sum_{\sigma\in\Delta} \sigma(b_i v)$. These vectors are invariant, and

$$\sum_i a_i w_i = v.$$

This proves surjectivity, with no finite-dimensionality hypothesis on V.

Vectors in V^Δ independent over k remain independent over K. Otherwise choose a shortest relation among them and normalize its first coefficient to 1. Applying any σ and subtracting gives a shorter relation, so all its coefficients are fixed by Δ, hence belong to k. This contradicts the original independence. A k-basis of V^Δ therefore proves injectivity.

An equivariant map preserves invariant vectors, and the displayed isomorphisms recover it from its restriction. They also identify the scalar extension of $V^\Delta \otimes_k W^\Delta$ with $V \otimes_K W$, compatibly with descent. An equivariant unit, multiplication, comultiplication, counit or antipode thus descends as a map on invariant spaces. The algebra and Hopf identities can be checked after the injective scalar extension to K.

If the K-algebra V is finitely generated, write a finite generating list as finite linear combinations of invariant vectors. Let B_0 be the k-algebra generated by those finitely many invariant vectors in $B = V^\Delta$. Then $K \otimes B_0 = V = K \otimes B$, so faithful scalar extension detects $B/B_0 = 0$. Thus $B = B_0$ is finite type and is finitely presented by the earlier Hilbert basis theorem. □

The usual field facts in this argument can be checked without an unstated descent theorem. In a finite normal separable extension, count embeddings by adjoining generators one at a time: each embedding of the preceding field extends by choosing one of the distinct roots of the next minimal polynomial. Normality puts all these embeddings in the field itself, giving $|\Delta| = [K : k]$. If $F = K^\Delta$, Lemma G.5.1 shows $|\Delta| \leq [K : F]$: apply its spanning argument to an F-basis of K. The degree tower then gives $F = k$. A finite separable extension embeds in a finite normal separable extension by adjoining the finitely many roots of the minimal polynomials of its generators; every embedding permutes those root sets. These arguments justify the fixed-field and normal-closure facts used here.

Corollary G.5.3. For a finitely generated abelian group M with a continuous action of $\mathrm{Gal}(k_s/k)$, choose a finite Galois splitting quotient $\Delta = \mathrm{Gal}(K/k)$. The semilinear action

$$\sigma(ae^m) = \sigma(a)e^{\sigma m}$$

on $K[M]$ descends all its Hopf operations to

$$B = K[M]^\Delta, \qquad K \otimes_k B = K[M].$$

The descended group has geometric character group M, with the prescribed action. Equivariant character homomorphisms descend uniquely in the reverse direction.

Proof. Intersect the open stabilizers of a finite generating list of M. This fixes all of M and lies in the action kernel. By the definition of the Krull topology, this neighbourhood contains the subgroup fixing some finite separable subextension of k_s/k. Adjoin all the finitely many conjugate roots of its generators. The normal-closure calculation of Lemma G.5.2, with the finite separability proof of S02 Lemma 4.A, gives a finite normal separable extension K/k inside k_s, whose fixing subgroup still acts trivially on M.

We also prove that restriction onto $\Delta = \mathrm{Gal}(K/k)$ is surjective. For $\sigma \in \Delta$, order the extensions of its k-embedding to intermediate subfields of k_s by extension. The union along a chain is again an embedding, so Zorn gives a maximal one, $\tau : E \to k_s$. If $E \neq k_s$, choose $b \in k_s \setminus E$. Its minimal polynomial over E divides its separable minimal polynomial over k. The polynomial obtained by applying τ to its coefficients still divides that same split separable k-polynomial, so has a root in k_s. Evaluation extends τ to $E(b)$, a contradiction. Thus its domain is all of k_s. It is onto: every element of k_s lies in a finite normal separable extension L/k by the same normal-closure construction, and a k-embedding sends L into L because its generator images are roots of their split k-polynomials. The injective endomorphism of this finite-dimensional k-space is surjective by S06 Section 1. Hence its image contains each such L and all of k_s. This gives an automorphism extending σ.

The action therefore factors through the full finite quotient Δ: two lifts have quotient fixing K and hence act identically on M. Apply Lemma G.5.2 to $K[M]$ and its displayed operations. The character calculation is the coefficient proof already written in AG-GS-05, Lemma 2.1 and Theorem 2.2: a group-like element over a connected field is one monomial, so a Hopf map is precisely its homomorphism on exponents. Transport of that monomial is exactly $e^m \mapsto e^{\sigma m}$. This proves the assertion and supplies the descent step of AG-GS-05, Theorem 5.2. □

AG-GS-05, Theorem 5.0 has its full separable-splitting argument: finite-dimensional subcoalgebras cover the Hopf algebra; the scheme of group-like elements in one is finite étale because its scalar extension is a finite disjoint union of points; over k_s each such point descends; the resulting monomials form the entire Hopf basis after faithful extension. Its finite splitting-field step is Lemma 5.1, where finitely many expressions for an algebra isomorphism and its inverse descend to a finite separable field and then a finite Galois field. These arguments do not assume perfectness of k. They remain required earlier written proofs, together with the affine descent argument of AG-GS-04.

Free provenance and rights

The source comparisons for point dimension and field descent are the freely accessible official Stacks Project, 02FX, 02FY and 0CDR. The proof arguments actually needed in this draft have been supplied above, with the named earlier programme algebra proofs.

For the character classification the verified free author version is J. S. Milne, *Algebraic Groups*, version 2.00, 20 December 2015, author PDF, Chapter 14, especially Theorem 14.17. Its numbering is specific to that free version. The later corrected author edition is also freely accessible: 5 October 2021 text, published in 2022. That edition uses Chapter 12 for character classification. Edition-specific locators must be matched to the cited free PDF; both free versions remain valid comparison sources.

History

Sources: the Stacks Project authors and the named earlier programme authors, for the algebra and descent results this lesson builds on; GPT-6.1 Sol (OpenAI), supporting arguments and explicit prerequisite reconciliation, 5 October 2026. The text of this lesson is dedicated to the public domain under CC0 1.0; the Stacks Project itself is distributed under the GNU FDL 1.2 or later.

Lesson 1

Tori, maximal tori and their conjugacy

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A diagonal matrix is easy to understand because its eigenspaces separate the action into one-dimensional pieces. A torus is the group-scheme version of this diagonal action. Over a field that is not algebraically closed, or over a scheme, the eigenspaces may appear only after changing the base. We therefore keep track of which conclusions hold on the original base and which hold locally for the étale topology.

We assume affine group schemes, smooth morphisms, faithfully flat descent, and the character description of groups of multiplicative type. Basic references are Conrad's *Reductive group schemes*, SGA 3, and Milne's *Algebraic Groups*. The first two treat arbitrary bases; the third provides complementary field arguments.

Diagonal actions and their descent

For a scheme S, a **torus** is a group scheme T that is étale locally isomorphic to $\mathbf{G}_{m,S}^r$, with r locally constant. Its character sheaf is

$$X^*(T) = \underline{\mathrm{Hom}}_{S\text{-groups}}(T, \mathbf{G}_m).$$

It is a locally constant sheaf of finite free abelian groups on the étale site. Its dual is the cocharacter sheaf $X_*(T)$. The pairing is fixed by

$$\chi \circ \lambda(t) = t^{\langle \chi, \lambda \rangle}.$$

For $T = \mathbf{G}_m^r$, the characters $e_i(t_1, \dots, t_r) = t_i$ and coordinate cocharacters form dual bases. A representation of a split torus on a finite locally free module M is a grading

$$M = \bigoplus_{\chi \in \mathbb{Z}^r} M_\chi,$$

with finitely many nonzero summands locally on S. Indeed, write its comodule map as $m \mapsto \sum_\chi m_\chi \otimes z^\chi$. Coassociativity implies that each m_χ has weight χ, and the counit gives $m = \sum m_\chi$. Linear independence of the Laurent monomials makes the sum direct. Each summand is a direct summand of M, hence finite locally free.

Lemma D.1 (maps of split tori on every base). Let X, Y be finite free abelian groups and $D_S(X) = \mathrm{Spec}_S \mathcal{O}_S[X]$. Homomorphisms $D_S(X) \to D_S(Y)$ correspond to locally constant

functions on S with values in $\mathrm{Hom}_{\mathbf{Z}}(Y, X)$. Automorphisms of $D_S(X)$ correspond to locally constant functions with values in $\mathrm{Aut}_{\mathbf{Z}}(X)$. These assertions commute with every base change, including nonreduced and non-quasi-compact schemes.

Proof. On an affine chart $\operatorname{Spec} R$, write the image of a group-algebra basis element under a Hopf map as $a = \sum_x c_x e^x$, a finite sum. Comparison of coefficients in $\Delta(a) = a \otimes a$ and the counit identity gives $c_x^2 = c_x$, $c_x c_z = 0$ for $x \neq z$, and $\sum_x c_x = 1$. For an idempotent c, the open $D(c)$ is also the closed set $V(1-c)$: a prime contains either c or $1-c$ because their product is zero, and cannot contain both. The finitely many $D(c_x)$ therefore partition the chart into open-and-closed parts. On the part $D(c_x)$, that coefficient is one and every other coefficient is zero, so $a = e^x$ there. Apply this to a finite basis of Y and refine their finite partitions together. On each resulting part, multiplicativity and inverses extend the chosen images to one homomorphism $Y \to X$. Linear independence of the exponent basis makes it unique. These locally constant exponent maps agree on chart overlaps for the same reason. Conversely a locally constant homomorphism defines the Hopf map $e^y \mapsto e^{f(y)}$ on each part where its value is fixed. The maps glue by S04 Lemma A1.5, giving the required group morphism. For an automorphism, apply the same calculation to its inverse; the two integer maps are inverse, and conversely inverse integer maps give inverse Hopf maps. Every coefficient identity persists under an arbitrary ring map. On a non-quasi-compact scheme the argument is chartwise and the image of the locally constant function need not be finite. This proves all the assertions. □

Example. Multiplication on $\mathbb{C}$ gives an embedding

$$\mathrm{Res}_{\mathbb{C}/\mathbb{R}}\, \mathbf{G}_m \longrightarrow \mathrm{GL}_{2,\mathbb{R}}, \qquad a + bi \longmapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

The determinant is $a^2 + b^2$, so the displayed functor is represented by $\operatorname{Spec} \mathbb{R}[a, b, (a^2+b^2)^{-1}]$, with matrix multiplication giving its group law and a closed inclusion into GL_2 by the two linear entry equations. After base change to $\mathbb{C}$, put $u = a + bi$ and $v = a - bi$. The inverse formulas $a = (u+v)/2$, $b = (u-v)/(2i)$ identify this coordinate algebra with $\mathbb{C}[u^{\pm 1}, v^{\pm 1}]$, since $a^2 + b^2 = uv$. This proves the torus assertion on every test algebra. Complex conjugation interchanges u, v, and hence the two character generators, by Lemma D.1. A split real torus has trivial conjugation action on its character lattice: its coordinate Laurent monomials are already real. A real isomorphism would intertwine these actions after complex base change, whereas the transposition action is nontrivial. Thus this torus is not split.

The anti-equivalence between tori and locally constant free character sheaves follows by descent from the split case: a change of basis in $\mathbb{Z}^r$ gives the corresponding monomial automorphism of $\mathbf{G}_m^r$, and these constructions carry inverse descent data to inverse descent data. The alternative definition using an fpqc splitting cover gives the same tori; we prove this after Lemma 3.1. One cannot replace the character sheaf by a single lattice with trivial action on an arbitrary base.

1. What maximality means

Lemma 1.M (the matrix groups used in the examples). For $n \geq 1$, GL_n and SL_n over every base are smooth affine reductive group schemes. Their diagonal tori are split and geometrically maximal. All assertions follow from the matrix functors themselves, in every residue characteristic.

Proof. The determinant identities, inverses and polynomial matrix calculations have their complete finite proofs in S06 Section 1. The determinant open in affine n^2-space is GL_n, hence smooth affine of finite presentation over $\mathbf{Z}$. Its geometric coordinate ring is a polynomial domain localized at its nonzero determinant, so it is geometrically irreducible by the prime-spectrum proof of AG-CA-08. The equation $\det M = 1$ defines SL_n. The cofactors of such a matrix are the entries of its inverse transpose, and therefore generate the unit ideal. On the open where a chosen cofactor is a unit,

solve for its corresponding entry in the determinant equation, which is linear in that entry and has this cofactor as coefficient. This is an open chart in affine (n^2-1)-space; the equation/inverse maps are explicit and commute with any base change. Thus SL_n is smooth affine of finite presentation over every base as well.

We prove its geometric connectedness on an algebraically closed field. In an invertible matrix, choose a nonzero pivot. Determinant-one row and column swaps are supplied in a two-coordinate block by

$$\begin{pmatrix}1&1\\0&1\end{pmatrix}\begin{pmatrix}1&0\\-1&1\end{pmatrix}\begin{pmatrix}1&1\\0&1\end{pmatrix}=\begin{pmatrix}0&1\\-1&0\end{pmatrix}.$$

Move the pivot into the first position, and use elementary additions to clear its column and row. Induct on the remaining block; its pivots are nonzero because it remains invertible. This leaves a determinant-one diagonal. Write $x(u)=\left(\begin{smallmatrix}1&u\\0&1\end{smallmatrix}\right)$ and $y(v)=\left(\begin{smallmatrix}1&0\\v&1\end{smallmatrix}\right)$. Direct two-by-two multiplication over any ring gives

$$y(-d^{-1})x(d-1)y(1)x(-(d-1)d^{-1})=\mathrm{diag}(d,d^{-1})\quad (d \text{ a unit}).$$

For a diagonal $(c_1,\dots,c_n)$ with product one, multiply these adjacent-coordinate pairs with $d_i=c_1\cdots c_i$, $1\le i<n$; the successive diagonal entries telescope to exactly $c_1,\dots,c_n$. Thus every determinant-one field matrix is an elementary word. Each fixed-length word is the image of an affine space of its additive parameters, which is connected and has identity at all parameters zero. Every such matrix consequently lies in the identity component. A nonempty complement would have a rational closed point by the complete Nullstellensatz, a contradiction. This proves connectedness; for $n=1$ the determinant-one group is the identity directly.

The diagonal tori have Laurent coordinate algebras, with characters $\mathbf{Z}^n$ for GL_n and $\mathbf{Z}^n/\mathbf{Z}(1,\dots,1)$ for SL_n, by GS05 Theorem 2.2. Conjugation on the tangent matrices has full characters e_i-e_j on E_{ij}, and zero-weight diagonal part, trace zero in the determinant-one case. These are characters of the torus scheme; they remain distinct in characteristic two. Let N be a smooth connected normal unipotent subgroup of either geometric group. Its intersection with the torus is diagonalizable by GS05 Theorem 4.2 and trivial by the entire unitriangular exclusion of Lemma 2.A below. A tangent point in both closed subgroups satisfies the sum of their defining ideals, so $\mathrm{Lie}\,N$ has zero intersection with the diagonal Lie space. Its full character grading is supplied by GS05 Lemma 2.1. If nonzero, it therefore contains some E_{ij}. Normality makes it a Lie ideal: conjugate dual-number tangent points over $k[\epsilon,\eta]/(\epsilon^2,\eta^2)$; the mixed coefficient of their commutator is $[X,Y]$, and the normal-subgroup equations force it to be in $\mathrm{Lie}\,N$. Taking $X=E_{ij}$ and $Y=E_{ji}$ gives $E_{ii}-E_{jj}$ in that Lie algebra, a nonzero diagonal vector even in characteristic two. This contradicts the intersection calculation. Hence $\mathrm{Lie}\,N=0$. The complete smooth zero-dimensional field calculation of S07 Lemma 5.3A gives an étale group with finitely many geometric points; connectedness and its identity make it the identity scheme. Thus both matrix groups have reductive geometric fibres.

A torus containing the diagonal torus is commutative, so its tangent matrices have only the displayed zero weight. Its rank is at most the diagonal rank and at least that rank by inclusion. The closed inclusion of equal-rank tori gives, by GS05 Theorem 4.2, a surjection of finite free character groups of equal rank. Its kernel is a free subgroup of rank zero by the full integer subgroup proof of Lemma 4.S below, and hence is zero. Thus the containing torus is the diagonal torus. These computations prove geometric maximality and all reductive assertions over any base by the stated geometric-fibre definition. □

A closed subtorus $T\subset G$ is **maximal** if $T_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$ at every geometric point of S. We always use this fibrewise definition. A **reductive group scheme** is a smooth affine group scheme of finite presentation whose geometric fibres are connected reductive algebraic groups. A connected smooth affine group over an algebraically closed field is reductive when it has no nontrivial smooth connected normal unipotent subgroup.

This definition of maximality is preserved by every base change. Here is the field-extension detail for the affine finite-presentation groups used here. Let $k\subset K$ be algebraically closed fields and suppose

a larger torus Q_K contains a maximal T_K. Choose split coordinates on $T = \mathbf{G}_m^t$ and $Q_K = \mathbf{G}_m^r$, with $r > t$. Lemma D.1 describes the inclusion of these tori by one fixed surjection $\mathbb{Z}^r \to \mathbb{Z}^t$. Write $k[G] = k[z_1, \dots, z_m]/(f_1, \dots, f_a)$. The closed group inclusion $Q_K \to G_K$ sends the z_i to finitely many Laurent polynomials; surjectivity of its coordinate map expresses all r coordinates and their inverses as polynomials in these images. Collect the finitely many coefficients in these expressions in a finitely generated k-subalgebra $D \subset K$. Enlarge D to include the coefficients in the finite identities expressing the $f_j = 0$, the coproduct/counit/inverse group-map identities on the z_i, and equality of the composite $T_D \to Q_D \to G_D$ with the given inclusion. These identities hold coefficient by coefficient because $D \hookrightarrow K$ is injective. The inverse-coordinate expressions consequently make $Q_D \to G_D$ a closed group inclusion, and it contains T_D through the same surjective integer map. The nonzero finite-type algebra D has a k-rational closed point by the complete Nullstellensatz. Specializing at that point gives a closed r-dimensional torus containing T in G, contradicting maximality. This proves persistence without assuming that an arbitrary specialization preserves a closed immersion.

Example 1.1. Diagonal matrices. In $\mathrm{GL}_{n,S}$, the diagonal torus $D = \mathbf{G}_m^n$ is maximal. For a matrix $A = (a_{ij})$ over an S-algebra R, commuting with the universal diagonal matrix says

$$a_{ij}(z_i - z_j) = 0 \quad \text{in } R[z_1^{\pm 1}, \dots, z_n^{\pm 1}].$$

Comparison of distinct Laurent monomials forces $a_{ij} = 0$ for $i \neq j$. Thus $C_{\mathrm{GL}_n}(D) = D$ as group schemes over every base. Any torus containing D centralizes it and is therefore equal to it.

Example 1.2. Two real tori. In $\mathrm{SL}_{2,\mathbb{R}}$, compare

$$D = \{\mathrm{diag}(t, t^{-1})\}, \qquad K = \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} : a^2 + b^2 = 1 \right\}.$$

Over $\mathbb{C}$, the second group has characters $a + bi$ and its inverse, so both groups become the same type of one-dimensional torus. An eigenbasis gives a conjugation in $\mathrm{GL}_2(\mathbb{C})$; rescaling one eigenvector makes its determinant one, giving a conjugation in $\mathrm{SL}_2(\mathbb{C})$. They are maximal because their complex fibres are maximal. They are not conjugate over $\mathbb{R}$: the character lattice of D has trivial conjugation action, while that of K has the sign action. A real conjugation would be an isomorphism of real tori and would preserve these actions.

2. Moving tori over an algebraically closed field

We first establish the classical tools used in the conjugacy argument. The proofs work in every characteristic. Orbit representability and the closed-orbit lemma are the exact supporting results of the supporting lesson Group schemes over a field; finite free-action quotients are provided by Quotients and torsors.

A line detecting a subgroup. Every closed subgroup H of an affine algebraic group K is the stabilizer of a line in a finite-dimensional representation. Let $A = k[K]$ and let I define H. For the right regular action write the coproducts of a finite list f_α as

$$\Delta(f_\alpha) = \sum_j u_{\alpha j} \otimes v_j$$

with the second factors v_j linearly independent. Put $W = \mathrm{span}_k\{u_{\alpha j}\}$. The counit puts every f_α in W. Coassociativity and linear functionals selecting the v_j give $\Delta(u_{\alpha j}) \in W \otimes A$, so W is a finite-dimensional right regular subrepresentation. This is also the complete local-finiteness calculation in AG-GS-01, Lemma 5.1.

The Hilbert basis proof makes I finitely generated. Choose W containing these generators and put $V = I \cap W$. For every k-algebra R, exact tensoring of vector spaces gives $V_R = I_R \cap W_R$. Right translation by an element of $H(R)$ preserves I_R and W_R, hence V_R. Conversely, an element stabilizing V_R preserves its generated ideal I_R. Evaluating the translated functions at the identity shows $f(g) = 0$ for every $f \in I$, so that element belongs to $H(R)$. Stabilizing V_R is equivalent to stabilizing its wedge line in $\bigwedge^{\dim V} W_R$: choose a basis of W starting with a basis of V. Preservation of that line makes the wedge coefficient in the first block a unit; wedging each transformed vector with its generator makes every component outside V_R zero. Applying the inverse transformation gives equality with V_R. For $\dim V = 0$ the wedge line is k and the generated ideal is zero, so both stabilizers are all of K. This proves the equality of subgroup schemes, including nonreduced test rings.

Finally take W containing algebra generators of A. Write its coaction as $\Delta(w_j) = \sum_i w_i \otimes a_{ij}$. Applying the counit in the first factor gives $w_j = \sum_i \epsilon(w_i) a_{ij}$. The matrix coefficients therefore generate A as an algebra. The antipode gives the inverse matrix, so the corresponding map to $\mathrm{GL}(W)$ is defined, and its coordinate-ring map is surjective. It is a closed immersion, proving linearity. □

Lemma 2.A (all representations of a unitriangular subgroup). Over any field k, let N be a closed subgroup scheme of the upper unitriangular group over k. Every finite-dimensional representation of N has a filtration whose nonzero successive quotients are one-dimensional trivial representations. Consequently N has no nontrivial character, and a diagonalizable subgroup scheme of N is trivial.

Proof. For the defining representation, the coordinate flag has trivial quotients. Tensor products inherit this property by ordering tensor basis vectors lexicographically. Direct sums inherit it by concatenating flags. A subrepresentation inherits the intersections with a flag; a quotient inherits its images, deleting zero quotients. These statements use ordinary vector-space intersections and remain statements about comodules, hence about every test algebra.

The coordinate algebra $k[N]$ is generated by the upper matrix entries. Its right regular coaction sends a row by

$$\Delta(x_{ij}) = \sum_l x_{il} \otimes x_{lj}.$$

The span of a row is therefore a quotient of the defining representation. Multiplication is a comodule map. The finite-dimensional span of polynomials of degree at most d in the entries is consequently a quotient of a finite direct sum of their tensor powers of orders at most d, and has the same trivial-quotient filtration.

Let W be an arbitrary finite-dimensional comodule. Its coaction embeds it in $W_{\mathrm{triv}} \otimes k[N]$: the counit is a left inverse to the underlying linear map, and coassociativity makes it a comodule map when N acts only on the second factor of its target. Finitely many matrix coefficients of this map occur, and all belong to one of the preceding finite-degree spaces. Thus W embeds into a finite direct sum of such spaces and inherits the required filtration.

An invariant line in W has trivial action, by intersection with that filtration. Applied to the group-like function of a character in $k[N]$, this proves character triviality as a coordinate-algebra identity, including nonreduced N. If a diagonalizable subgroup D embeds in N, each representation of D splits into weight spaces by AG-GS-05, Theorem 3.1. Its faithful defining representation has only trivial constituents by the filtration just proved, so it is trivial. Faithfulness gives $D = 1$. □

A subgroup of multiplicative type cannot have a faithful unitriangular representation: its representations split into character spaces after the splitting cover, whereas the same filtration has only trivial quotients. Its action is therefore the identity. Consequently a subgroup of multiplicative type and a unitriangular subgroup of a linear group have trivial scheme-theoretic intersection. This conclusion includes finite nonreduced groups of multiplicative type.

Lemma 2.H (the connected reduced derived subgroup). For a smooth connected affine

B/k, with k algebraically closed, the reduced closure of the subgroup generated by commutators is a smooth connected normal subgroup D. If B is solvable of derived length $l > 0$, then D has derived length at most $l - 1$.

Proof. The regular-local-domain and component argument in Lemma 2.G makes B integral. Let C_r be the image of the product of r commutators, a morphism $B^{2r} \to B$, and let J_r be its reduced closure. The product is integral by S08 Lemma G.4.1, so J_r is integral. Inserting identity commutators makes the J_r nested. Their dimensions are bounded by $\dim B$; choose R attaining their largest dimension. For every $r \geq R$ the inclusion $J_R \subset J_r$ is equality, because a proper closed subset of an integral finite-type variety has smaller dimension, by the normalization dimension proof. Put $D = J_R$.

The maps $B^{2R} \to D$ and their products are schematically dominant: the reduced image closure gives injective coordinate maps, and tensoring injections over a field preserves injectivity. Multiplication of two products of R commutators gives a product of $2R$ commutators, whose image is still in D. Thus multiplication factors through D. Inversion reverses the factors and exchanges the two inputs of each commutator, so also factors. Conjugation preserves these commutator products; the same schematic-dominance argument, now after tensoring with $k[B]$, proves that D is normal as a group scheme. Its identity is the empty product. It is integral, hence connected, and the generic field-rank, Jacobian and translation calculation in Lemma 2.G proves it smooth.

Products of commutators of $B(k)$ are dense in D: a function vanishing on their image pulls back to zero on the dense rational points of B^{2R} and is therefore zero on D. If B has derived length l, its abstract commutator subgroup satisfies the derived word of length $l - 1$. The same word vanishes on products of these dense sets. Tensoring their injective coordinate maps detects this identity on D, including all test rings. Thus the derived length drops as asserted. □

Triangularization. Every finite-dimensional representation of a smooth connected solvable group B has an invariant full flag.

Proof. Induct on the derived length, with the trivial group as base case. Lemma 2.H gives a smooth connected normal derived subgroup D of smaller length. Its representation has an invariant flag by induction, hence a nonzero common character eigenspace. The nonzero common character spaces form a finite, linearly independent collection. Here is the independence calculation: distinct group-like functions are linearly independent, since applying a linear functional to one factor of a minimal comultiplication relation and subtracting a scalar multiple eliminates one term while retaining another. Distinct group-like functions cannot be scalar multiples, because each has counit one. Applying this independence to the coaction of a relation among character vectors makes the character spaces a direct sum; there are at most $\dim V$ nonzero ones.

Normality makes $B(k)$ permute these spaces. The conditions for a fixed permutation are closed matrix conditions on B. Their finitely many closed loci cover every rational point, hence all of the reduced B by the Nullstellensatz. The integral B lies in one of them. That permutation is the identity because the locus contains the identity matrix and the distinct character spaces are disjoint. Its matrix equations vanish identically, so each space is stable under the group scheme B on every test ring.

Choose one, V_χ, of dimension d. The determinant of every commutator of B on this invariant space is one. On D the determinant is χ^d; density of commutator products therefore makes $\chi^d = 1$ as a scheme identity. Thus χ maps D into the finite scheme μ_d. Reducedness kills its nilpotents, and connectedness makes its image in the finite reduced set a single point. That point is one, so D acts trivially on V_χ. The matrices of B on this space consequently commute, since their commutators factor through D.

Commuting matrices over an algebraically closed field have a common eigenvector. If all are scalar, any line works. Otherwise an eigenspace of a nonscalar matrix is a nonzero proper space preserved by every commuting matrix, and induction on its dimension gives the eigenvector. This line is $B(k)$-stable; its matrix equations vanish on the reduced B, so it is a group-scheme subrepresentation.

Repeat on its quotient, using induction on representation dimension, and lift the resulting flag. This proves the full invariant flag in every characteristic. □

Lemma 2.M (the Borel fixed-point argument). A smooth connected solvable affine group B over algebraically closed k acting on a nonempty complete variety has a fixed k-point, fixed as a scheme morphism.

Proof. S08 Theorem G.4.2 supplies a closed reduced orbit O. It is complete because it is closed in the given complete variety. S08 Theorem G.4.C proves directly that its orbit map is an fppf torsor under its full schematic stabilizer H, and that O represents B/H. The detecting-line construction gives a finite-dimensional B-representation V and a line kv with schematic stabilizer H, including its possible nilpotents. Here $\mathbf{P}_{\text{lines}}(V) = \mathbf{P}(V^\vee)$ denotes lines in V, or equivalently rank-one quotients of its dual. The map $b \mapsto b[kv]$ is constant on H-cosets over every test scheme. The torsor identity and S04 Lemma A1.5 therefore descend it to an equivariant morphism $O \to \mathbf{P}_{\text{lines}}(V)$.

Its image is closed: the graph in $O \times \mathbf{P}_{\text{lines}}(V)$ is closed by separatedness, and projection is proper by base change of the proper $O \to \operatorname{Spec} k$. Give that image its reduced structure Z. Every k-point of Z lifts to a k-point of O, and then to one of B: both fibres are nonempty finite-type schemes, so the weak Nullstellensatz supplies the lifts. Thus $Z(k)$ is the single $B(k)$-orbit of $[kv]$. The action preserves Z schematically by reduced-product and rational-point detection, as in S08 Theorem G.4.2.

Triangularize V by the proof above, with B-stable full flag $V_0 = 0 \subset V_1 \subset \cdots \subset V_n = V$. Choose the least d for which $Z \cap \mathbf{P}_{\text{lines}}(V_d)$ is nonempty. Its intersection with $\mathbf{P}_{\text{lines}}(V_{d-1})$ is empty, so $Y = Z \cap \mathbf{P}_{\text{lines}}(V_d)$ is a closed projective subscheme entirely contained in the affine chart $\mathbf{P}_{\text{lines}}(V_d) \setminus \mathbf{P}_{\text{lines}}(V_{d-1})$. It is closed in that chart, hence affine as well. S01 Theorem 8.4 and its closed-immersion cohomology calculation make $H^0(Y, \mathcal{O}_Y)$ finite-dimensional over k. Since Y is affine, its coordinate algebra is that finite-dimensional algebra, so Y has finitely many points. Indeed each prime quotient is a finite-dimensional domain, hence a field by injective multiplication; the Noetherian zero-dimensional spectrum is a finite union of its irreducible components, each a single point. Each residue field is k by algebraic closedness.

The finite subscheme Y is B-stable. Choose $y \in Y(k)$. Its orbit map from connected B has connected image in this finite discrete space and contains y, so its image is the point y. Since B is reduced, that map kills nilpotent functions and factors through the reduced point $\operatorname{Spec} k$. Hence y is fixed on every test scheme. Choose $b \in B(k)$ with $b[kv] = y$, using the lifts above. The full schematic stabilizer of y is bHb^{-1}. Fixedness gives $B \subset bHb^{-1}$, and the reverse inclusion is automatic. Thus $H = B$, and the quotient $O = B/H$ is a point by its proved fppf quotient description. Its point in the original variety is the required fixed point. □

Solvable structure. We prove the splitting $B = U \rtimes T$ first. The conjugacy of maximal tori is then the separate consequence of Lemmas 2.B and 2.C below.

Lemma 2.J (the diagonal image and its filtration). In a faithful upper triangular representation of a smooth connected affine B/k, the diagonal map is an fppf quotient $B \to D$ onto a torus, with unitriangular kernel U. The upper-entry filtration induces a finite B-normal filtration $U = U_0 \supset \cdots \supset U_m = 1$. Each fppf quotient U_i/U_{i+1} is a closed subgroup scheme of $\mathbf{G}_a$, with linear D-action, and $[U, U_i] \subset U_{i+1}$.

Proof. The coordinate algebra of B is a domain by the argument in Lemma 2.G. The diagonal image algebra is generated by the diagonal characters and their inverses. Distinct group-like elements are independent by the preceding calculation, so this algebra is $k[M]$, where M is the finitely generated group of their distinct images. It has no torsion: an element of order $n > 1$ would give two nonzero elements $e^a - 1$ and $1 + e^a + \cdots + e^{(n-1)a}$ with product zero. Thus M is free by S02 Lemma 3.M, and the image D is a torus. Its schematically dominant image map is faithfully flat and of finite presentation by AG-GS-04, Lemma 11.1c. The kernel is B intersected with the unitriangular group, and its kernel identity identifies D with the fppf quotient.

Order the upper entries by increasing distance $j - i$, refining equal distances arbitrarily, and let F_i be the unitriangular subgroup whose first i entries vanish. The multiplication formula for an entry of distance r has cross terms of smaller distances, all zero on F_i at the stage where that entry is removed. Hence the next entry gives $F_i \to \mathbf{G}_a$ with kernel F_{i+1}. Conjugation by a unitriangular matrix changes it only by smaller-distance terms, so $[\mathrm{UT}, F_i] \subset F_{i+1}$. Diagonal conjugation scales it by the character d_a/d_b for that entry (a, b). These are polynomial identities, using the finite inverse series for a strictly upper triangular matrix, over every test ring. Thus each $U_i = U \cap F_i$ is closed and normal in B and has the stated commutator property.

The quotients $Q_i = B/U_i$ exist as smooth connected affine groups by Lemma 2.G, with $Q_0 = D$ and $Q_m = B$. The induced maps $Q_{i+1} \to Q_i$ descend by S04 Lemma A1.5. They are surjective and flat: $B \to Q_i$ is faithfully flat, and flatness descends through the faithfully flat $B \to Q_{i+1}$ by Lemma A1.3. Finite presentation follows from their finite-type affine coordinate algebras over the field and the Hilbert basis proof. Thus they are fppf. Their kernels K_i have the fppf cover $U_i \to K_i$, the base change of $B \to Q_{i+1}$, with kernel U_{i+1}. The next-entry map descends through this cover to $K_i \to \mathbf{G}_a$. Its kernel is trivial, since zero entry means membership in U_{i+1} after a local lift. It is a monomorphism and hence a closed immersion by Lemma 2.I. Therefore $K_i = U_i/U_{i+1}$ is the required closed additive subgroup. The universal conjugation formula on its entry gives the linear D-action; its descent is unique by Lemma A1.5. □

Lemma 2.E (additive subgroup quotients). A proper closed subgroup scheme $N \subset \mathbf{G}_a$ over k is finite and is the kernel of a monic additive polynomial P. Its fppf quotient is $\mathbf{G}_a$ via P, and this map is finite faithfully flat. A linear torus action preserving N induces a linear action on this quotient.

Proof. Its nonzero ideal in $k[X]$ has one monic generator P of positive degree d, by the polynomial-division calculation in S02, Section 3. The identity belongs to N, so $P(0) = 0$. The subgroup identity makes $P(X+Y)$ vanish modulo $(P(X), P(Y))$. The polynomial $P(X+Y) - P(X) - P(Y)$ has degree less than d in each variable. Its vanishing in that quotient, whose basis is $X^i Y^j$ for $i, j < d$, makes it zero. Thus P is additive and $N = \ker P$ as schemes.

The ring $k[X]$ is free over $k[P(X)]$ on $1, X, \ldots, X^{d-1}$. The monic relation expressing X^d in terms of $P(X)$ and lower powers spans all higher powers by induction. Independence follows because the leading degrees of nonzero terms $a_i(P(X))X^i$, for $0 \le i < d$, are distinct modulo d and cannot cancel. Hence $P : \mathbf{G}_a \to \mathbf{G}_a$ is finite faithfully flat. For two points over the same quotient coordinate, additivity says $P(x - y) = 0$; subtraction and addition give inverse scheme maps between $\mathbf{G}_a \times N$ and $\mathbf{G}_a \times_{P,\mathbf{G}_a,P} \mathbf{G}_a$. Local lifts through this faithfully flat map therefore identify its target with the fppf quotient.

A preserving torus action descends through this quotient: its maps and identities descend uniquely by S04 Lemma A1.5. At every k-point of the torus it is an automorphism of $k[X]$ fixing zero. Degrees multiply under composition of nonconstant polynomials, so an automorphism and its inverse both have degree one. In the universal action the coefficients of higher powers vanish at every k-point of the reduced torus, and are zero by the Nullstellensatz. The constant term is zero, the slope is a unit by the inverse action, and the action identity makes the slope a character. This proves linearity on every test scheme. □

An extension of an affine torus D by a vector line has a regular scheme section. Here is a proof for its fppf vector-line torsor over $\operatorname{Spec} A = D$. Choose a faithfully flat affine cover trivializing the torsor, with coordinate ring B. On the double overlap the two line coordinates differ by translation by $c \in B \otimes_A B$. The torsor relation makes c satisfy the additive cocycle identity on the triple overlap. Consequently the matrices $\begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}$ give descent data on B^2. The affine module-descent proof in AG-GS-04, Section "Faithfully flat affine descent", or AG-RG-S04, Section A, descends them to an A-module E and an exact sequence $0 \to A \to E \to A \to 0$: exactness is checked after the faithfully flat cover. Choose a preimage of $1 \in A$ under the surjection $E \to A$. Its local coordinate vectors are

(a,1), and the displayed transition matrices give exactly the torsor translation law. They therefore define a descended section of that torsor. This proof supplies the fppf assertion directly and needs no unproved comparison of fppf and Zariski cohomology. The multiplication defect is a regular degree-two cocycle. Exactness of torus invariants makes the regular group cohomology in positive degrees vanish, so a correction makes the section a homomorphism. The same vanishing in degree one makes two sections conjugate by the vector line. The regular cochain contraction is proved in AG-GS-05, Proposition 7.14, and the homomorphism-lifting and conjugacy calculation is AG-GS-05, Theorem 7.15.

Lemma 2.F (finite additive extensions of a torus). Let D be a torus over k, let a character $\lambda : D \to \mathbf{G}_m$ preserve a finite subgroup $N \subset \mathbf{G}_a$ by scalar multiplication, and let

$$1 \longrightarrow N \longrightarrow E \longrightarrow D \longrightarrow 1$$

be an extension of affine algebraic groups, with $E \to D$ an fppf N-torsor and conjugation on N given by λ. This extension has a group section.

Proof. Form $\mathbf{G}_a \rtimes E$ with multiplication

$$(a, e)(b, f) = (a + \lambda(\bar{e})b, ef),$$

where $\bar{e}$ is the image in D. Its subgroup $K = \{(-n, n) : n \in N\}$ is normal. Conjugation by the vector line fixes it, and conjugation by e sends $(-n, n)$ to $(-\lambda(\bar{e})n, ene^{-1})$, which belongs to K by the specified conjugation action. This verifies normality as a scheme identity.

Construct its quotient P by affine descent, without assuming a finite-action quotient theorem. Over the faithfully flat affine cover $E \to D$ it is an affine line. If two lifts of d differ by $e \mapsto en$, change its coordinate by $a \mapsto a - \lambda(d)n$. These translations and their inverses satisfy the triple cocycle because the differences of three lifts add in N. S04 Lemma A1.2 descends this affine line to $P \to D$. It represents the sheaf of classes of (a, e) under

$$(a, e) \sim (a - \lambda(\bar{e})n, en).$$

The preceding normality calculation makes multiplication and inversion well defined on these classes. S04 Lemma A1.5 descends them and their identities. Thus P is a group extension of D by the vector line, with action λ. The map $j : E \to P$, $e \mapsto [(0, e)]$, is a closed immersion: after the cover $E \to D$ it is the inclusion of a coset of N in the affine line; the scalar $\lambda(d)$ is a unit. Closed immersions descend by S04 Corollary A3.2. In particular its kernel is zero and its vector-line intersection is exactly N.

The preceding vector-line extension proof, equivalently the complete regular-cochain calculation in AG-GS-05, Proposition 7.14 and Theorem 7.15, splits P. Identify it with $\mathbf{G}_a \rtimes D$ using that section. Lemma 2.E identifies $\mathbf{G}_a/N$ with a vector line; write P_N for its additive quotient polynomial and ρ for the induced character. In each fibre $j(E)$ is a coset of N. Its quotient coordinate therefore descends through the fppf cover $E \to D$ to a regular map $c : D \to \mathbf{G}_a/N$. The subgroup multiplication gives the scheme identity

$$c(de) = c(d) + \rho(d)c(e).$$

The degree-one coefficient-of-identity contraction displayed below gives $c(d) = v - \rho(d)v$ for $v \in (\mathbf{G}_a/N)(k)$. The finite surjective map P_N has a k-point a over v, by the Nullstellensatz. Conjugate $j(E)$ by $(-a, 1)$ on the left and $(a, 1)$ on the right. Its quotient coordinate becomes

$$c(d) - P_N(a) + \rho(d)P_N(a) = 0.$$

Consequently this conjugate is $N \rtimes D$ as a scheme, not only on its k-points: equality with the inverse image of the zero quotient coordinate is checked after the N-torsor cover, where both are

the same coset subscheme. Its standard section of D transports back through j to a group section in E. □

These two lemmas supply the finite additive-kernel step in the splitting argument. The case $N = \mathbf{G}_a$ is the vector-line case already proved above.

Lemma 2.G (affine normal unitriangular quotients). Let B be smooth, connected and affine over the algebraically closed field k, and let $N \subset B$ be a closed normal subgroup admitting a closed unitriangular embedding. The fppf quotient B/N is a smooth connected affine algebraic group, and its quotient map is faithfully flat and of finite presentation. The subgroup N may be nonreduced or have positive dimension.

Proof. Choose a detecting line kv for N by the complete construction above. Its character on N is trivial by Lemma 2.A, so N fixes v over every test algebra. Let E be the span of the vectors gv for $g \in B(k)$. Normality of N makes each of these vectors N-fixed: for every test ring, conjugating an N-point by this fixed g gives another N-point. The space E is preserved by $B(k)$. It is preserved by the group scheme B too: the matrix coefficients for its map to the quotient vector space vanish on every k-point of the reduced B, hence vanish by the Nullstellensatz. This gives a representation $B \to \mathrm{GL}(E)$ in which N acts trivially. Conversely an element in its scheme-theoretic kernel fixes v and its detecting line, so belongs to N. Its kernel is therefore exactly N on all test rings.

The coordinate ring $A = k[B]$ is an integral domain. Indeed smooth local rings are regular by AG-CA-18, Theorem 2.1, and regular local rings are domains by the complete regular-parameter proof. A finite-type scheme has finitely many irreducible components; two meeting components would give two minimal primes in the local ring at an intersection point. Thus these components are disjoint and open and closed. Connectedness leaves one, and smoothness also gives reducedness.

Let $C \subset A$ be the image of $k[\mathrm{GL}(E)]$ under the representation. It is a finitely generated domain. Matrix comultiplication, inversion and the counit preserve C, and their Hopf identities restrict, since tensoring inclusions over the field is injective. Consequently $Q = \operatorname{Spec} C$ is an affine algebraic group and $q : B \to Q$ is schematically dominant with kernel N. The complete monic-division and translation proof in AG-GS-04, Lemma 11.1c, applies to this integral target and proves that q is faithfully flat and of finite presentation. It assumes no smoothness of the kernel and no pre-existing quotient.

For clarity, smoothness of Q has its own earlier proof inputs. Let $r = \operatorname{trdeg}_k \operatorname{Frac} C$. Noether normalization makes this field finite over $k(t_1, \dots, t_r)$, and S02 Lemma 3.F makes its differential dimension r. At the generic point of Q the local ring is that field, so the dimension and differential numbers in AG-CA-18, Theorem 2.1 are equal. Thus Q is smooth on a nonempty open, by that theorem and Corollary 2.2. Choose a k-point in the open. Its translates show smoothness at every k-point of Q; a nonempty closed complement would have a k-point by the Nullstellensatz. Hence Q is smooth everywhere. It is connected because C is a domain.

The kernel identity gives inverse scheme maps between $B \times N$ and $B \times_Q B$, by $(b, n) \mapsto (b, bn)$ and $(b, b') \mapsto (b, b^{-1}b')$. Every section of Q lifts fppf locally through the faithfully flat finite-presentation map q; any two lifts differ by a unique section of N. These statements identify Q with the fppf quotient sheaf, proving the assertion. □

Lemma 2.I (monomorphisms into the additive line). Every monomorphism $X \to \mathbf{G}_a$ of affine algebraic groups over an algebraically closed field is a closed immersion, including nonreduced X.

Proof. Write its coordinate map as $k[T] \to A$. If it is injective, AG-GS-04 Lemma 11.1c makes the morphism faithfully flat. Monomorphy identifies $A \otimes_{k[T]} A$ with A by multiplication, so the two maps from A to that tensor product agree. The faithfully flat equalizer in S04 Lemma A1.1 then gives $k[T] = A$, and the morphism is an isomorphism.

Otherwise its nonzero kernel cuts out a finite subscheme of the line, by polynomial division. Injectivity on rational points makes $X(k)$ finite. Every positive-dimensional reduced irreducible

component of X would have infinitely many rational points: Noether normalization gives a finite surjection to a positive-dimensional affine space, and lying over and the Nullstellensatz give a rational lift of every rational base point. The algebraically closed field is infinite, since for a finite field the polynomial $\prod_{a\in k}(T-a)+1$ would have no root. Thus all components of X have dimension zero. Normalization makes their coordinate domains finite over k, so A_{red}, which injects into the product of these finitely many domains, is finite-dimensional. The Noetherian nilradical of A is nilpotent: finitely many nilpotent generators give a uniform vanishing power by expanding products. A monic polynomial for the image of each algebra generator in A_{red}, raised to that power, annihilates the generator in A. The finite-integral calculation of S04 Lemma P0.2 therefore makes A finite over k.

In particular A is a finite $k[T]$-module. At every prime of $k[T]$ its finite fibre algebra F satisfies $F\otimes_\kappa F = F$, by monomorphy. Its dimension satisfies $r^2 = r$, so the fibre is either zero or the scalar field. The finite module $A/(k[T]\cdot 1)$ consequently has zero fibre at every prime. Local Nakayama and the localization zero test make it zero. The coordinate map is surjective, proving a closed immersion. □

Theorem 2.K (split solvable structure). Every smooth connected solvable affine group B over an algebraically closed field is $U \rtimes T$, where T is a torus and U is smooth, connected and unitriangular. Moreover U is its largest smooth connected normal unipotent subgroup.

Proof. The detecting-line construction supplies a faithful representation, and the triangularization proof supplies an invariant flag. Lemma 2.J then gives an fppf extension of a torus D by a unitriangular kernel U and a finite normal central filtration with closed additive quotients and linear D-actions. We prove splitting by induction on the length of such a supplied filtration for any smooth connected affine extension with these properties. Length zero is already D.

Let N be its last nontrivial term. It is central in U, normal in B, and a closed subgroup of the additive line. Lemma 2.G constructs the smooth connected affine quotient B/N, with its fppf map and kernel identity. Its induced map to D is fppf: surjectivity is inherited from $B \to D$, flatness descends through $B \to B/N$ by S04 Lemma A1.3, and finite presentation follows from the affine finite-type algebras over the field. The kernel U/N is affine. It is unitriangular: a faithful representation of it exists by the linearity proof; pulled back through the fppf cover $U \to U/N$, it has the trivial-quotient flag of Lemma 2.A. Its flag and trivial constituent identities are detected by the injective faithfully flat coordinate map, so that faithful representation is unitriangular already on U/N.

The other terms descend as closed subgroups along $B \to B/N$, because their defining ideals are invariant under right N-translation; apply S04 Lemma A1.2. Their quotient maps and conjugation identities descend by Lemma A1.5. Cancelling the common subgroup N preserves their fppf additive quotients: local lifts identify $(U_i/N)/(U_{i+1}/N)$ with U_i/U_{i+1}, with the same linear action and centrality. This is a shorter filtration. The induction hypothesis gives a torus section $D \to B/N$. Its inverse image in B is an affine fppf extension of D by N, whose conjugation is the specified linear action, since N is central in U. If $N = \mathbf{G}_a$, the vector-line section and regular-cochain calculation above splits it. If N is proper, Lemmas 2.E and 2.F split it, including finite nonreduced N. The resulting section of D lies in B.

For this section s, the maps

$$U \times D \longrightarrow B, \quad (u,d) \longmapsto us(d), \qquad b \longmapsto \big(bs(q(b))^{-1}, q(b)\big)$$

are inverse scheme maps and give the semidirect product law. The section is a closed immersion, since its coordinate map is surjective using $q \circ s = 1$. Its image T is the torus D.

Here smoothness of U also has a proof. Write $D = \operatorname{Spec} k[t_1^{\pm 1}, \dots, t_r^{\pm 1}]$. At any rational point $(u,1)$ of $U \times D \simeq B$, the local ring is regular. The classes $t_i - 1$ in its cotangent space are independent: evaluation at u gives a retraction to the torus local ring and its cotangent space. Complete them to a minimal parameter list. The regular-parameter theorem makes their quotient, the local ring

of U at u, regular. The rational-point criterion of AG-CA-18, Proposition 3.1, gives smoothness there. Its open smooth locus has no nonempty closed complement without a rational point, by the Nullstellensatz, so U is smooth everywhere. A disconnection of U would disconnect $U \times D \simeq B$, since D is nonempty, so U is connected. It was already a closed unitriangular subgroup by Lemma 2.J.

Finally every unipotent subgroup has trivial characters by Lemma 2.A, so its map to the torus B/U is trivial on every test ring. In particular each smooth connected normal unipotent subgroup is contained in U. This proves the largest-subgroup assertion and finishes the theorem. □

Lemma 2.L (reduced identity components). For a finite-type affine group K over algebraically closed k, its reduction is a smooth subgroup. Its component containing the identity, with its reduced structure, is a smooth connected closed subgroup K^0_{red}. Every morphism from a connected reduced k-scheme to K whose image contains the identity factors through this subgroup. Each component of K_{red} is a translate of it by a k-point.

Proof. The nilradical of $k[K]$ is a Hopf ideal. Indeed, the product of two reduced finite-type k-schemes is reduced by S08 Lemma G.4.1. Pullback of a nilpotent function under multiplication is therefore zero on $K_{\mathrm{red}} \times K_{\mathrm{red}}$; inversion and the identity kill it too. These factorizations give the subgroup structure on the reduction.

Here is its smoothness proof, including the generic input. A Noetherian scheme has finitely many irreducible components: a minimal closed subset failing this assertion would split into two proper closed subsets and give a contradiction. Remove the other components from one component, giving a nonempty integral open. On an affine part of that open, Noether normalization and its dimension formula make the generic field finite over $k(t_1, \dots, t_d)$, with d the component dimension. S02 Lemma 3.F makes its differential dimension d. The field criterion in AG-CA-18 Theorem 2.1 and its open smooth locus in Corollary 2.2 therefore give a nonempty open on which the reduction is smooth over k. The weak Nullstellensatz supplies a k-point a there. For any $g \in K_{\mathrm{red}}(k)$, translation by ga^{-1} takes that smooth neighbourhood to a smooth neighbourhood of g. The nonsmooth closed complement consequently has no closed point, and is empty by the same Nullstellensatz. Thus the reduction is smooth everywhere.

Its local rings are regular domains by the same smooth criterion and the regular-parameter proof. Distinct irreducible components cannot meet: at a common point their distinct minimal primes would remain distinct in the local ring, contradicting its being a domain. The finitely many components are therefore disjoint, open and closed, and each is integral. The identity component is one of these. Its product with itself is integral by S08 Lemma G.4.1, so multiplication has connected image containing the identity and stays in that component. Inversion does likewise. This proves the subgroup assertion.

A map from a reduced scheme kills all nilpotent functions, so factors through K_{red}. Its connected image is contained in one of the disjoint open and closed components; if it contains the identity it is contained in the identity component. This gives the asserted factorization as schemes, since that component is open and closed. Finally every component contains a k-point a by the Nullstellensatz. Translation by a^{-1} sends it to the identity component, proving the translate assertion. □

Lemma 2.B (the smooth central filtration). Let B be a smooth connected solvable affine group with the splitting $B = U \rtimes T$ just constructed, where U is smooth connected and unitriangular. There is a B-stable filtration

$$U = U_0 \supset U_1 \supset \cdots \supset U_a = 1$$

by smooth connected normal subgroups such that every successive quotient is $\mathbf{G}_a$, each T-action on it is multiplication by a character, and

$$[U, U_i] \subset U_{i+1}.$$

Proof. Use the splitting $B = U \rtimes T$ just proved. In the faithful triangular representation choose a T-weight basis compatible with the full B-stable flag: exactness of the T-weight projections lets

one choose a weight basis in each flag quotient and lift it inside that flag term. In this basis T is diagonal and B remains upper triangular. In its unitriangular group, order the entries by increasing distance $j - i$ from the diagonal and refine each distance arbitrarily. Let F_i be the subgroup on which the first i entries vanish. The multiplication formula for an entry of distance r has cross terms of smaller positive distances. On F_i, all distances smaller than the distance currently removed have vanished. The next entry therefore gives a homomorphism $F_i \to \mathbf{G}_a$ with kernel F_{i+1}. Conjugation by a unitriangular matrix changes an entry only by terms involving smaller distances; hence each F_i is normal and $[\mathrm{UT}, F_i] \subset F_{i+1}$. Diagonal conjugation scales the next entry by a character. Thus the entire upper triangular group preserves the filtration.

Put $V_i = (U \cap F_i)^0_{\mathrm{red}}$. Lemma 2.L makes these smooth connected closed subgroups. The connected reduced subgroup V_{i+1} maps into $U \cap F_i$ and contains the identity, so the factorization in that lemma gives $V_{i+1} \subset V_i$. They are B-stable: conjugation preserves $U \cap F_i$, and the morphism from the smooth connected $B \times V_i$ has image in its reduced identity component by the same factorization. Similarly, the commutator morphism $U \times V_i \to U \cap F_{i+1}$ has smooth connected reduced source and contains the identity in its image, so factors through V_{i+1}. This proves the schematic central-filtration assertion. We have $V_0 = U$ and the final $V_i = 1$.

Lemma 2.G, applied to the smooth connected affine group V_i and its normal unitriangular subgroup V_{i+1}, constructs the smooth connected affine quotient $Q = V_i/V_{i+1}$ and its faithfully flat finite-presentation torsor map q. The next entry $e : V_i \to \mathbf{G}_a$ vanishes on V_{i+1} and is constant on its cosets over every test algebra. The torsor identity and the faithfully flat equalizer in S04 Lemma A1.1 therefore descend it to a group map $\bar{e} : Q \to \mathbf{G}_a$.

Let $N_i = \ker e = V_i \cap F_{i+1}$. Its reduced identity component equals V_{i+1}. One inclusion follows by mapping it into $U \cap F_{i+1}$ and using Lemma 2.L; the other follows from the already proved $V_{i+1} \subset V_i$ and its inclusion in F_{i+1}. The same lemma makes every component of $(N_i)_{\mathrm{red}}$ a translate of V_{i+1} by a k-point. Put $L = \ker \bar{e}$. Base change of q gives the faithfully flat surjection $N_i \to L$. Each of the finitely many components of $(N_i)_{\mathrm{red}}$ maps to one k-point of L, since q is constant on the corresponding coset. Consequently the underlying space of L is a finite set of closed k-points.

This implies that L is a finite scheme, retaining its nilpotents. Explicitly, let $A = k[L]$. The prime-intersection calculation in AG-CA-01 Theorem 1.2 embeds A_{red} into the finite product of the residue fields of those points, each equal to k. Thus A_{red} is finite-dimensional. The Noetherian nilradical J has finitely many nilpotent generators, so $J^m = 0$ for some m by expanding monomials. Each J^r/J^{r+1} is a finite module over A/J, hence finite-dimensional over k. The filtration of A by these powers makes A finite-dimensional too.

Here is the dimension bound without a general torsor-dimension theorem. Lemma 2.G makes $k[Q]$ a domain. The coordinate map $k[T] \to k[Q]$ of $\bar{e}$ therefore has either zero kernel or a prime kernel $(T - a)$, since k is algebraically closed and polynomial division gives the prime ideals of $k[T]$. In the latter case the identity makes $a = 0$, so $Q = L$ is finite. In the zero-kernel case the complete Hopf-subalgebra proof in AG-GS-04 Lemma 11.1c makes $\bar{e}$ faithfully flat. Its generic fibre over $k(T)$ becomes, after algebraic closure of that field, a translate of L after choosing a point: faithful flatness gives a nonempty finite-type fibre, the weak Nullstellensatz supplies the point, and multiplication by its inverse identifies the fibre with the kernel on every test algebra. That fibre's algebra is therefore finite-dimensional after the field extension, and already finite-dimensional before it: linearly independent vectors remain independent under field extension. The domain $k[Q] \otimes_{k[T]} k(T)$ is thus a field finite over $k(T)$, and is the fraction field of $k[Q]$. Noether normalization and its dimension formula give $\dim Q = 1$. In the finite case its coordinate domain is a finite field extension of algebraically closed k, so $Q = 1$ and $V_i = V_{i+1}$ by the kernel identity of q.

Otherwise the quotient is a one-dimensional smooth connected unipotent group. A faithful representation of the affine quotient, pulled back to V_i, has a trivial-quotient filtration by Lemma 2.A. The matrix coefficients expressing preservation of the subspaces and identity on their quotients vanish in $k[V_i]$, and injectivity of $k[Q] \to k[V_i]$ makes them vanish already in $k[Q]$. Thus its faithful

image is unitriangular. The complete curve, genus and automorphism-family proof in earlier S03 Theorem 4.2 now identifies Q with $\mathbf{G}_a$ as a group scheme.

The T-action is a regular action by automorphisms of this affine line fixing zero. On each closed point of T the automorphism is a linear polynomial. In the universal action its coefficients of higher powers therefore vanish on all closed points of the reduced torus, and are zero. Its slope is a unit and the action identity makes that slope a character of T. Removing the repeated V_i gives the required filtration. This argument retains finite and infinitesimal kernels of the original entry maps; it does not mistake those maps for isomorphisms. □

Here are the degree-one and degree-two calculations explicitly. For $D = D_k(M)$, let $\int : k[M] \to k$ extract the coefficient of the identity character. It is translation invariant on every scalar extension because $\Delta(e^m) = e^m \otimes e^m$. If V is a vector representation and c is a regular cocycle

$$c(ab) = c(a) + a\,c(b),$$

put $v = \int_b c(b)$. Integration of this equality in b gives

$$v = c(a) + av, \qquad c(a) = v - av.$$

This is an equality of regular maps, also when D is finite nonreduced. It divides by no group order. For a regular two-cocycle c, the identity

$$ac(b,d) - c(ab,d) + c(a,bd) - c(a,b) = 0$$

and integration in d give $c(a,b) = ah(b) - h(ab) + h(a)$, with $h(a) = \int_d c(a,d)$. These are the two calculations used in the splitting argument.

Lemma 2.C (unipotent cocycles on all test schemes). Work over an algebraically closed field k. Let a diagonalizable D act on a smooth connected affine group U of finite type, with a given finite D-stable filtration by smooth connected normal subgroups, as specified in Lemma 2.B. Its successive fppf quotient sheaves are assumed to be represented vector lines with linear D-actions. Every regular crossed homomorphism $D \to U$ is principal. More generally, over any k-algebra R it is principal after an fppf extension of R.

Proof. Project the cocycle to U/U_1, a vector line. The displayed degree-one identity, or the complete split regular-cochain contraction in AG-GS Proposition 7.14, makes its projection principal with parameter v; the coefficient-of-the-identity integral works over every R, including one with nilpotents. Because the quotient in the hypothesis is the represented fppf quotient sheaf, v lifts to $u \in U$ fppf locally. Its fibre is a torsor under U_1 on all test schemes: two lifts differ uniquely by an element of that kernel, and translation identifies a locally chosen lift with every other lift. Over k this nonempty fibre is of finite type, so the earlier Nullstellensatz supplies a k-point and no extension is required. Replace the cocycle by

$$c'(d) = u^{-1}c(d)\,d(u).$$

Its projection to U/U_1 is now the identity, because the additive quotient cocycle is $v - d(v)$ and the projection of u is v. It takes values in the full kernel U_1. The cocycle identity survives this correction on the universal pair of D_R-points:

$$c'(ab) = u^{-1}c(a)\,a(c(b))\,ab(u) = c'(a)\,a(c'(b)).$$

Repeat through the finite filtration. Over k the same finite-type fibre argument supplies every correcting lift without extending the field. Over R compose the finitely many fppf refinements. If the corrected cocycle is $c'(d) = w\,d(w)^{-1}$, then the original one is

$$c(d) = uw\,d(uw)^{-1}.$$

Thus the correcting parameters multiply in their actual order, and induction makes the original cocycle principal. Every equation is a regular comodule identity over the universal D_R, so detects nonreduced test rings. Applying the affine assertion on an affine cover of an arbitrary test scheme gives the stated fppf local result there as well. □

Lemma 2.N (tori in a split solvable group). In the proved decomposition $B = U \rtimes T$, every torus of B is conjugate by an element of $U(k)$ to a subtorus of T. Every maximal torus is consequently conjugate to T. If B is commutative, the decomposition is $U \times T$.

Proof. Let $M \subset B$ be a torus and put $A = M \cap U$ with its full scheme structure. A torus over algebraically closed k is split. The algebra $k[A]$ is a quotient of the group algebra $k[X^*(M)]$, so is generated by group-like elements. The group of their distinct images is some abelian group Γ. The independence calculation in S02 Lemma 3.N makes its distinct group-like elements linearly independent. Hence $k[A] = k[\Gamma]$ and A is diagonalizable, including any possible finite or infinitesimal part. It embeds in the unitriangular U, so Lemma 2.A gives $A = 1$ on all test schemes.

Projection $q : M \to T$ has trivial schematic kernel. Let $\Lambda \subset X^*(M)$ be the image of its character map $X^*(T) \to X^*(M)$, calculated by Lemma D.1. The coordinate algebra of its kernel is the quotient of $k[X^*(M)]$ by $e^{\lambda} - 1$ for $\lambda \in \Lambda$. The map taking a monomial to its coset identifies this quotient with $k[X^*(M)/\Lambda]$: cosets supply its basis, and the displayed relations identify precisely the monomials in one coset. Triviality of the kernel therefore gives $X^*(M)/\Lambda = 0$, because a nontrivial group algebra has more than the identity basis element. Thus the coordinate map of q is surjective, so q is a closed immersion onto a subtorus $T' \subset T$.

Identify M with T' through q. Its inclusion in $U \rtimes T'$ is a regular section $d \mapsto c(d)d$. The multiplication law gives $c(de) = c(d)\, d(c(e))$ on every test algebra. Lemma 2.B supplies the smooth central filtration of U, with linear actions after restriction from T to T'. Lemma 2.C therefore gives $c(d) = u\, d(u)^{-1}$ for some $u \in U(k)$. Conjugation by u^{-1} sends M to the standard T'. If M is maximal, then $T' = T$ since otherwise this conjugate is properly contained in the torus T. This proves conjugacy. In a commutative B conjugation of U by T is trivial, so its semidirect product law is the direct product law. □

Lemma 2.1. If G is smooth connected affine over an algebraically closed field k, its maximal smooth connected solvable subgroups are conjugate, and G/B is complete for each such subgroup B.

Proof. We first construct the flag scheme and its equivariant projective embedding from earlier proved results. Let V be a finite-dimensional G-representation of dimension n. We use the quotient convention for Grassmannians and projective spaces. In the product

$$P = \prod_{i=1}^{n-1} \mathrm{Gr}_{n-i}(V)$$

the universal quotient $V \otimes \mathcal{O}_P \twoheadrightarrow Q_i$ has locally free kernel F_i of rank i. These universal bundles, their arbitrary-base-change property and the Grassmannian charts are proved in AG-HP, *Grassmannians*, Theorem 1.1. Impose the vanishing of each composite

$$F_i \longrightarrow V \otimes \mathcal{O}_P \longrightarrow Q_{i+1}, \qquad 1 \le i < n-1.$$

In local bundle frames this means that all entries of a finite matrix vanish. Changing either frame multiplies by invertible matrices, so the generated ideals agree on overlaps. They define a closed subscheme $\mathrm{Fl}(V) \subset P$.

For every test scheme S, these equations say exactly that $F_i \subset F_{i+1}$. The resulting surjection $Q_i \twoheadrightarrow Q_{i+1}$ splits locally, since its target is locally free. Its kernel F_{i+1}/F_i is consequently locally free of rank one. With $F_0 = 0$ and $F_n = V \otimes \mathcal{O}_S$, a point of this incidence scheme is therefore a full flag of locally direct-summand subbundles. Conversely such a flag gives the indicated locally free quotients and satisfies every equation. Thus the construction represents full flags over arbitrary

test schemes, including nonreduced schemes. For $n = 0$ or $n = 1$, the empty product and the flag scheme are the point $\operatorname{Spec} k$.

AG-HP, *Grassmannians*, Theorem 2.1 proves the closed Plücker embedding

$$\mathrm{Gr}_{n-i}(V) \hookrightarrow \mathbf{P}(\textstyle\bigwedge^{n-i} V)$$

by its complete minor-chart calculation. Products of these closed immersions are closed immersions: on affine charts a quotient map stays surjective after tensoring, and a composite of surjective maps is surjective. Factor the product map into base changes of its individual closed immersions. Iterating the Segre embedding proved in AG-MO, *Very ample invertible sheaves, Segre and Veronese embeddings*, Theorem 3.1, gives a closed immersion

$$\mathrm{Fl}(V) \hookrightarrow \mathbf{P}(W), \qquad W = \bigotimes_{i=1}^{n-1} \textstyle\bigwedge^{n-i} V.$$

For an empty product put $W = k$. The incidence equations, exterior quotients and tensor product of quotient lines are all natural under automorphisms of V. Hence the flag scheme is G-stable and this embedding is equivariant for the linear representation on W. In particular its restriction to any torus is a linear projective action. This proves both projectivity and the precise equivariance required later in the rank-one lesson. S01 Lemma 7.A and Corollary 7.B prove that this projective flag scheme is proper.

Choose a smooth connected solvable subgroup B of largest dimension. Such a subgroup exists: the trivial group is one candidate and every candidate has integer dimension at most $\dim G$. The detecting-line construction above provides a representation and a line whose schematic stabilizer is B. Add a faithful representation as a direct summand and retain that line in the first summand, obtaining a faithful V. Apply the triangularization proved above to the quotient by this line and lift the resulting flag. We obtain a B-stable full flag x beginning with the detecting line. Its schematic stabilizer G_x is exactly B: preserving the flag implies preserving that line, whereas every B-point over every test algebra preserves the full flag.

For any full flag $y \in \mathrm{Fl}(V)(k)$, its stabilizer G_y lies in an upper triangular group, since V is faithful. Lemma 2.L proves that its reduced identity subgroup $(G_y)^0_{\mathrm{red}}$ is smooth, and that every component of its reduction is a translate of that subgroup. It is connected; the polynomial upper-entry commutator identities of Lemma 2.J make the upper triangular group solvable, and hence make this subgroup solvable. Its dimension is therefore at most $\dim B$. Reduction does not change dimension, and translations identify the finitely many components of an algebraic group; consequently

$$\dim G_y = \dim (G_y)^0_{\mathrm{red}} \leq \dim B.$$

The complete generic-fibre and field-extension calculation in S08 Lemma G.4.D gives

$$\dim(G \cdot y) = \dim G - \dim G_y \geq \dim G - \dim B = \dim(G \cdot x).$$

Thus $G \cdot x$ has minimum dimension among all these orbits; this calculation includes nonreduced stabilizers.

We apply the reduced orbit argument to $\mathrm{Fl}(V)_{\mathrm{red}}$. The action restricts to this reduction: S08 Lemma G.4.1 makes $G \times \mathrm{Fl}(V)_{\mathrm{red}}$ reduced, so nilpotent defining functions pull back to zero. The reduced orbit $O = G \cdot x$ is locally closed. Its closure is irreducible, because smooth connected G is irreducible, as proved in S08, Section G.4. The boundary argument of S08 Theorem G.4.2 shows that any nonempty boundary is a closed invariant subset of strictly smaller dimension. Such a boundary has a k-point, by the earlier Nullstellensatz. Its orbit lies in the boundary and has smaller dimension than O, contradicting the preceding inequality. Therefore O has closed underlying image in $\mathrm{Fl}(V)$.

We also retain the full scheme structure. S08 Theorem G.4.C constructs the full quotient directly by the monic-division flatness and torsor calculation and gives the orbit immersion

$$G/B \longrightarrow \mathrm{Fl}(V)$$

and the fppf B-torsor $G \to G/B$. The quotient is reduced: on an affine chart its functions inject under faithfully flat pullback into functions on a reduced open subscheme of the smooth G, so it has no nonzero nilpotents. An immersion from a reduced scheme with closed underlying image identifies that scheme with the reduced closed subscheme on its image. Indeed, on an open neighborhood where the immersion is closed its defining ideal is radical; it agrees there with the radical ideal defining the reduced closed image, and outside the image the ideal is the unit ideal. These descriptions glue. Thus this orbit immersion is a closed immersion. Composing with the embedding above realizes G/B as a closed subvariety of $\mathbf{P}(W)$, with its linear G-action. It is projective and hence complete.

Now let B' be any maximal smooth connected solvable subgroup. Lemma 2.M applies to its action on the nonempty complete variety G/B, giving a B'-fixed k-point. The fppf torsor fibre above this point is nonempty and of finite type over the algebraically closed field, so the Nullstellensatz supplies a lift $g \in G(k)$. The schematic stabilizer of gB is gBg^{-1}. Fixedness consequently gives $g^{-1}B'g \subset B$ as subgroup schemes. Maximality of B' forces equality, because gBg^{-1} is smooth, connected and solvable. Conjugation identifies G/B' with G/B, transferring projectivity and completeness. This proves the assertion. For additional reading, compare the freely accessible Milne author draft, v2.00, Theorem 4.19; the complete argument used here is the one just given. □

Lemma 2.D (the no-torus criterion). A smooth connected affine group H over an algebraically closed field is unipotent if it has no nontrivial torus. Consequently any such H which is not unipotent has a positive-dimensional maximal torus.

Proof. Choose a Borel B. The solvable structure just proved gives $B = U \rtimes T$; the hypothesis makes $T{=}1$, so B is unitriangular. Detect B as the stabilizer of a line kv in a finite-dimensional H-representation. Its line character is trivial by Lemma 2.A, so B fixes v as a vector, and its schematic vector stabilizer is still B. The map $h \mapsto hv$ descends through the full fppf quotient $H \to H/B$ to a morphism $H/B \to V$. Lemma 2.1 realizes H/B as a closed subvariety of the projective flag variety. It is integral: its underlying space is the surjective image of the irreducible smooth connected H, and its charts are reduced because their functions inject under the faithfully flat pullback into functions on reduced open subschemes of H. AG-RG-S01, Theorem 8.4 and its closed-immersion cohomology calculation make $H^0(H/B, \mathcal{O})$ a finite-dimensional k-domain. A finite-dimensional domain is a field, and the algebraically closed k makes that field k. Thus every coordinate of the morphism to V is constant. Its value is v at the identity coset, so H fixes v. The vector-stabilizer identity forces $H = B = U$, proving the assertion. For the final sentence, the contrapositive gives a nontrivial torus. Choose a torus containing it with largest dimension, bounded by the dimension of H. A proper inclusion of tori strictly increases dimension, by the domain dimension theorem for their reduced integral schemes. Thus this torus is maximal and has positive dimension. □

Theorem 2.2. Any two maximal tori in a smooth connected affine group over an algebraically closed field are conjugate.

Proof. A torus is smooth, connected and solvable. Among the smooth connected solvable closed subgroups containing it, choose one of largest dimension; the torus itself is a candidate. A proper inclusion of such groups strictly increases dimension: their underlying reduced schemes are integral, and a proper closed subset of an integral finite-type scheme has smaller dimension by the earlier domain dimension theorem. Thus the chosen subgroup is maximal. For maximal tori T_1, T_2, choose B_1, B_2 containing them. Lemma 2.1 gives $gB_2g^{-1} = B_1$. Both T_1 and gT_2g^{-1} are maximal tori of B_1. Lemma 2.N conjugates them within this split solvable group, giving $u \in B_1(k)$ with $ugT_2g^{-1}u^{-1} = T_1$. □

No conclusion here asserts conjugacy under $G(k)$ when k is not algebraically closed. Example 1.2 is already a counterexample.

3. Deforming an embedding rather than an eigenvalue

We need a mechanism that transports a torus from one fibre to nearby fibres. It uses the entire torus action, which retains information even when differentiating its characters loses information in positive characteristic.

For tori $T_1, T_2 \subset G$, define

$$\mathrm{Transp}_G(T_1, T_2)(R) = \{g \in G(R) : g(T_1)_R g^{-1} = (T_2)_R\}.$$

The normalizer $N_G(T)$ is the transporter from T to itself. The centralizer $C_G(T)$ asks that the induced automorphism of T be the identity. Both conditions concern subgroup schemes after arbitrary base change, rather than just the rational points of a fibre.

Lemma 3.1. Torus deformation. Let G be smooth affine of finite presentation over S.

1. The normalizer and centralizer of a subtorus are closed subgroup schemes of finite presentation. Transporters between tori of the same rank are schemes of finite presentation.
2. These normalizers, centralizers, and transporters are smooth over S.
3. A torus embedded in a fibre G_s extends, after an étale neighbourhood with unchanged residue field, to a torus embedded in G.

Proof. We explain the equations, the deformation obstruction, and then the passage from formal to étale neighbourhoods.

After splitting a torus, its finite subgroups $T[n]$ are schematically dense, universally on the base. A Laurent polynomial has finite support; take n larger than all differences between exponent coordinates. Its distinct monomials remain distinct modulo n, so a nonzero Laurent polynomial remains nonzero on $T[n]$. This works over any ring. The closed conditions on every $T[n]$ therefore detect both inclusion and equality of maps on T.

Here is a finite presentation of the conditions directly over the original base. Work on an affine cover $\operatorname{Spec} R$ splitting the specified tori $T_i = D_R(X_i)$, and put $C = \mathcal{O}(G)$. The ideal J_i of each closed torus inclusion is finitely generated: the source and target are finitely presented over R, so the morphism is finitely presented by the affine-pair presentation calculation in AG-MO-02, Theorem 3.1. For the universal conjugation $G \times T_1 \to G$, the pullback of each chosen generator of J_2 is a finite Laurent sum in $C[X_1]$. Set each of its finitely many coefficients in C equal to zero. The quotient by these coefficients represents $gT_1g^{-1} \subset T_2$ on every R-algebra, because the exponent basis of its group algebra stays free after every base change. Impose also the coefficient conditions from $g^{-1}T_2g \subset T_1$. The two inclusions are precisely equality of the subgroup schemes, so together they represent the equality transporter as a closed subscheme of finite presentation. Taking $T_1 = T_2$ gives the normalizer. For the centralizer, choose finitely many algebra generators f_j of C and compare their pullbacks along $(g, t) \mapsto gtg^{-1}$ and $(g, t) \mapsto t$. Each difference again has finitely many Laurent coefficients; their vanishing is exactly equality of the two maps on every test algebra. These equations represent the centralizer as a closed subscheme of finite presentation. On overlaps the schemes represent the same conditions. S04 Lemmas A1.2–A1.3 descend their compatible ideal modules, their multiplication stability and their finite generation, giving the closed subgroups and transporter inside the original G. The coefficient tests commute with arbitrary base change, including to rings with nilpotents. Thus no finite generation or base-change assertion about an infinite intersection is needed.

Now let $R \to R/I$ have $I^2 = 0$. Two liftings of a torus homomorphism $f_0 : T_0 \to G_0$ differ by a regular crossed homomorphism with values in

$$M = \mathrm{Lie}(G_0) \otimes_{R/I} I,$$

with the action through $\mathrm{Ad} \circ f_0$. More explicitly, lifting a scheme map first gives a multiplication defect $c(t, u)$, and associativity is exactly the degree-two cocycle equation. Correcting a lift by a function $b(t)$ changes c by its coboundary. Once the defect vanishes, the difference between two homomorphisms is a degree-one cocycle. Conjugating by an element infinitesimally equal to one changes that cocycle by a degree-one coboundary.

For a split torus, let $\int$ take the coefficient of the trivial character in its group algebra. On an exponent basis, $(\int \otimes 1)\Delta(f) = (\int f)1$, so this integral is invariant under translation after every scalar extension. For a regular cochain c of degree $j \geq 1$, put $(hc)(g_1, \ldots, g_{j-1}) = \int_a a^{-1} c(a, g_1, \ldots, g_{j-1})$. In $dhc + hdc$, the internal multiplication faces and the last faces cancel in pairs. The first face of hdc gives c; its next face, with the substitution $b = ag_1$, cancels the first face of dhc by translation invariance. Hence $dh + hd = 1$ in positive degrees, as an identity of regular sections, over any ring. A closed degree-two defect is therefore a coboundary and can be removed. A difference cocycle b between homomorphism lifts is dw; conjugating the first lift by the kernel element corresponding to $-w$ changes it by $(-w) - g(-w) = dw(g)$ and gives the second lift. The square-zero kernel and its additive law are proved in AG-GS-03, Propositions 1.4 and 1.6. The full cochain calculation and homomorphism correction are AG-GS-05, Proposition 7.14 and Theorem 7.15.

We now check smoothness on a cover splitting both specified tori, so write $T_i = D_R(X_i)$. Let g_0 be an equality-transporter point over R/I, with $I^2 = 0$. Smooth lifting over an affine test scheme, including the gluing of local lifts, is proved in AG-FSE, *Infinitesimal lifting and invariance under thickenings*, Theorem 4.1; it gives a lift $g \in G(R)$. Conjugation by g_0 induces an isomorphism $\psi_0 : T_{1,R/I} \to T_{2,R/I}$. Every prime of R contains I, so $\mathrm{Spec}(R/I) \to \mathrm{Spec}\, R$ is a homeomorphism. Lemma D.1 identifies ψ_0 with a locally constant invertible integer matrix on this space; the same matrix defines its unique lift $\psi : T_1 \to T_2$. Thus $\mathrm{Int}(g) \circ i_1$ and $i_2 \circ \psi$, where $i_i : T_i \hookrightarrow G$, are two homomorphism lifts of exactly the same map with the same source T_1. The preceding degree-one correction gives $u \in G(R)$ reducing to one such that $\mathrm{Int}(u) \circ \mathrm{Int}(g) \circ i_1 = i_2 \circ \psi$. Consequently ug is an equality-transporter point lifting g_0.

Taking $T_1 = T_2$ proves the normalizer lifting assertion. If g_0 centralizes T, its induced map ψ_0 is the identity, and the unique lift supplied by Lemma D.1 is $\psi = 1_T$. The corrected element ug therefore centralizes the entire torus. This proves formal smoothness of all three representing schemes on the splitting cover. Their finite presentation was proved by the coefficient equations above, so the same infinitesimal criterion gives smoothness there. S04 Lemma A1.3a descends smoothness to the original base.

Finally, in the dual-number kernel, conjugation by a torus element acts by its adjoint action on $\mathrm{Lie}(G)$, by the derivation calculation of AG-GS-03, Proposition 1.4. The centralizer equations require this action to fix the tangent vector on every test algebra. They therefore give, as a sheaf on the base,

$$\mathrm{Lie}(C_G(T)) = \mathrm{Lie}(G)^T.$$

Work on an affine base $\mathrm{Spec}\, R$ at the chosen prime p. Descend the finitely presented smooth G to a finitely generated $\mathbf{Z}$-subalgebra R_0 using AG-MO-03, Theorems 4.1–4.2, and retain smoothness by the finite conormal splitting witnesses of AG-RG-S02, Lemma 3.1 and Proposition 3.2. Include the specified fibre torus in the same finite data before taking a local model. Its Hopf presentation, finite separable splitting algebra, splitting maps and inverses, descent maps, embedded coordinate pullback and finitely many surjectivity witnesses use only finitely many coefficients in $\kappa(p) = \mathrm{Frac}(R/p)$. Express those coefficients as fractions of elements of R modulo p and include all their numerators

and denominators in R_0. With $p_0 = p \cap R_0$, the residue field $\kappa(p_0)$ embeds in $\kappa(p)$. Enlarge R_0 until it contains these finite presentations, expressions and relation witnesses. Their identities then hold over $\kappa(p_0)$, because this field embeds in $\kappa(p)$; their chosen nonzero determinants remain nonzero. The resulting source is a torus split by the retained finite étale algebra and is embedded in the fibre of the smooth stage. Now take the Noetherian local ring $(R_0)_{p_0}$, essentially of finite type over $\mathbf{Z}$, and its completion. The argument below lifts that exact stage fibre torus. After the final pointed étale approximation and pullback to R, its residue data recover the originally specified torus and embedding, with unchanged residue field at p.

Over its complete local ring A, lift the fibre torus as follows. AG-RG-S05, Lemma 3.A constructs, from a finite Galois residue splitting field L/κ, a finite free etale faithfully flat local algebra E/A with lifted Galois automorphisms and $E \otimes_A E \simeq \prod_\Gamma E$. Descend $E[M]$ with its fixed semilinear character-lattice action. That lemma proves that this is an actual finitely presented torus T_A, and that every reduction $(T_A)_{A/\mathfrak{m}^n}$ is the descent of the same character data on $(E/\mathfrak{m}^n E)[M]$. Thus the source is already defined over A. Apply AG-GS-05, Theorem 7.15 to this fixed source and the smooth target to lift the specified fibre homomorphism successively modulo $\mathfrak{m}^n$, retaining each previous lifting. Its existence is supplied by the regular cochain contraction of AG-GS-05, Proposition 7.14. We now prove effectivity of the compatible homomorphisms.

Let A be Noetherian and complete for an ideal J, and first let the source be $D_A(M)$. Its completed coordinate module consists of families $(a_m)_{m\in M}$ tending J-adically to zero, while its actual coordinate module consists of families of finite support. Compatible homomorphisms modulo J^n give a map from $A[G]$ to the former. For $f \in A[G]$, write its coproduct as a finite sum $\sum_{i=1}^{r} f_i \otimes g_i$. Compatibility with the coproduct says that the coefficient matrix of its completed image has entries

$$\delta_{m,n} a_m = \sum_{i=1}^{r} b_{i,m} c_{i,n}, \tag{3.1}$$

where the right side consists of the coefficient families of the images of the f_i and g_i. Regard these columns as elements of A^M. Every $a_n e_n$ belongs to the finite module spanned by the r families $(b_{i,m})_m$. The submodule generated by all $a_n e_n$ is therefore finitely generated, since A is Noetherian. It is the direct sum of the ideals Aa_n in their distinct coordinate positions. Finite generation forces all but finitely many of these ideals to be zero. Thus the image of f has finite support. This holds for every f, so the completed map takes values in $A[M]$ itself. Its algebra and Hopf identities hold because $A[M]$ and $A[M \times M]$ inject into their completions. This proves existence and uniqueness of the actual homomorphism. For the torus source just constructed, use the particular finite free cover E/A of AG-RG-S05, Lemma 3.A. Both E and $E \otimes_A E \simeq \prod_\Gamma E$ are complete for the extended ideal, by their finite free bases; they are Noetherian because they are finite over the Noetherian ring A. Apply the split argument over E. Its two pullbacks to $E \otimes_A E$ have the same reductions at every order, since the original formal maps respected the character descent data in (3.5) of that lemma. Uniqueness of the split effectivity argument makes those pullbacks equal as actual Hopf maps. AG-RG-S04, Lemmas A1.1–A1.2 therefore descend the actual homomorphism to A, and faithful flatness verifies all of its Hopf identities and the prescribed reductions. This proves the effectivity statement needed here.

Lemma 3.E (a torus lift remains a closed embedding). Let $(A, \mathfrak{m}, \kappa)$ be any local ring, let T/A be a torus, and let G/A be an affine group scheme of finite presentation. If a homomorphism $f : T \to G$ is a closed immersion on the closed fibre, then f is a closed immersion. No Noetherian, completeness or smoothness hypothesis on the local base or target is needed.

Proof. For every positive integer n, the group scheme $T[n]$ is finite locally free. On a splitting cover its coordinate algebra is $B[M/nM]$, with the finite exponent basis of M/nM, where M is the finite free character lattice. Its rank is $n^{\operatorname{rank} M}$. The finite locally free module and finite affine algebra descend by S04 Lemmas A1.2–A1.3; thus this is the full torsion scheme, including when n is not invertible. Put $K_n = \ker f \cap T[n]$, with its scheme-theoretic structure. It is a closed subscheme

of this finite affine scheme, so its coordinate algebra C_n is a finite A-module. Its augmentation $C_n \to A$ has the unit map as a section. The augmentation ideal J_n is consequently an A-direct summand of C_n and is finite: project a finite generating list of C_n to it. The closed-fibre immersion gives $(K_n)_\kappa = 1$ as schemes. Tensoring the direct sum $C_n = A \oplus J_n$ gives $J_n/\mathfrak{m}J_n = 0$. Nakayama, S01 Lemma 1.2, makes $J_n = 0$, so $K_n = 1$ over A, for every n.

Now take any algebraically closed residue overfield at any point of $\operatorname{Spec} A$. The full geometric kernel is a closed subgroup of its split torus, hence is $D(Q)$ for a finitely generated quotient Q of M, by Lemma 4.S. Its intersection with $T[n]$ is its own n-torsion, namely $D(Q/nQ)$, by the same group-algebra kernel calculation; it is trivial by the preceding paragraph. If $Q \neq 0$, it has a maximal proper subgroup. Indeed, in a chain of proper subgroups the union remains proper: if it contained all of a finite generating list, one chain member would contain that list and would be all of Q. Zorn therefore supplies a maximal proper subgroup L. The nonzero simple abelian group Q/L is cyclic, since any nonzero element generates a nonzero subgroup; it cannot be infinite cyclic, which has the proper nonzero subgroup $2\mathbf{Z}$. A nontrivial finite cyclic simple group has prime order, say ℓ, since a composite order supplies a proper nonzero cyclic subgroup. Therefore $Q/\ell Q$ surjects onto $Q/L \neq 0$. Its group algebra has more than one exponent-basis vector, so $D(Q/\ell Q) \neq 1$, a contradiction. Thus $Q = 0$: every full geometric kernel is trivial. This uses the subgroup's finite character quotient, not ambient torsion density.

Let $K = \ker f = \operatorname{Spec} C$, and let $J = \ker(C \to A)$ be its augmentation ideal. The finite-type affine algebra C, being a quotient of the torus algebra, has generators $c_1, \dots, c_r$ over A. Then $J = (c_i - \epsilon(c_i))_i$: subtracting the evaluation of a polynomial from that polynomial, one variable at a time, expresses the difference in these generators. Thus J is finite as a C-module. The exact sequence $0 \to J \to C \to A \to 0$ is split as A-modules. Every geometric fibre of K is the identity, and faithful field extension detects the kernel and cokernel of the map from the residue field to its coordinate algebra. Hence every residue fibre is already the identity and $J \otimes_A \kappa(p) = 0$ for every prime $p \subset A$. At a prime $q \subset C$, put $p = q \cap A$. Localizing this equality gives $J_q/pJ_q = 0$; the ideal pC_q is contained in the maximal ideal of C_q. Nakayama applied to the finite C_q-module J_q gives $J_q = 0$. A module zero at all primes is zero: if an element were nonzero, its annihilator would be proper, contained in a maximal ideal by S01 Lemma 1.1, and the element would survive that localization by the fraction-zero criterion. Therefore $J = 0$, and $K = 1$ as a group scheme. For any test algebra, two T-points with the same image differ by a kernel point; they are equal. Thus f is a monomorphism on all test schemes.

It remains to prove that this monomorphism is a closed immersion over the whole base. Work on an affine faithfully flat splitting cover, write $T_B = D_B(M)$, and put $C_G = \mathcal{O}(G_B)$. Right translation by T_B, through f, grades this algebra as $C_G = \bigoplus_{m \in M} (C_G)_m$, by the complete weight decomposition of AG-GS, *Diagonalizable groups*, Theorem 3.1. Pullback maps $(C_G)_m$ into Bz^m. Its coefficients form an ideal $I_m \subset B$. The weight projections commute with every base change, since they are the coefficient projections in a direct exponent-basis decomposition. On any residue field the full kernel is trivial. Lemma 4.S, applied after splitting its torus, proves that this field homomorphism is a closed immersion; it uses the faithful linearity proof of Section 2, the generating weight lattice and the inverse determinant, with no smoothness or reducedness assumption on the target. Its coordinate pullback is therefore surjective on each residue fibre. Projecting to weight m shows that I_m is contained in no prime of B. A proper ideal has a maximal ideal containing it, by S01 Lemma 1.1. Consequently $I_m = B$ for every m. Each z^m has a preimage, so $C_G \to B[M]$ is surjective. Finally faithful flatness reflects zero cokernels, by CA08 Theorem 1.1, and descends this surjectivity to the original affine map. It follows that f is a closed immersion. □

Apply Lemma 3.E to the actual homomorphism over the complete local ring constructed above. Its closed-fibre map is the specified torus embedding, so its coordinate pullback is surjective. This supplies the actual closed embedding, and therefore the finite surjectivity witnesses, for the approximation step that follows.

To make the approximation step precise, use **Artin approximation for polynomial equations**

in the following form. If $(R, \mathfrak{m})$ is a Noetherian local G-ring, a solution $\widehat{a} \in \widehat{R}^n$ of finitely many polynomial equations over R can, for every $N \geq 1$, be approximated modulo $\mathfrak{m}^N \widehat{R}$ by a solution on an étale neighbourhood with the same residue field. Equivalently the approximating solution belongs to the henselization R^h and descends to a pointed étale neighbourhood. Comparison with the formal solution uses the canonical map from the chosen local étale algebra to $R^h \to \widehat{R}$; no embedding of an entire, possibly disconnected, étale algebra is required.

A **G-ring** is a Noetherian ring whose local completion maps are regular, meaning flat with geometrically regular fibres. In particular the map $R \to \widehat{R}$ above is regular.

Theorem 1.1 of the preceding Artin approximation lesson proves this statement by its written desingularization sequence D.1–D.38, followed by the standard smooth residue-point lift in Lemma 1.3. The finite-order congruence is imposed in that same finite polynomial system. Its G-ring proof for every local ring essentially of finite type over $\mathbb{Z}$ is F.132–F.135, including the completed-local faithful-flatness argument F.99A. These proofs retain arbitrary characteristic, imperfect residue fields, nilpotent elements and nonregular local G-rings.

Here is the finite system to which the theorem is applied. Choose finite presentations for the completed torus and a finite étale algebra splitting it. Since the completed base is local, that algebra is finite free; choose a basis and its multiplication table. Include the finitely many coefficients of the splitting isomorphism to a Laurent polynomial algebra, its inverse, the descent maps on the double overlap, and the homomorphism into G. Each chosen algebra expression is a finite polynomial or Laurent polynomial. The algebra, Hopf, inverse-map and descent identities are tested on finite generating sets; equality in a finitely presented algebra is recorded using finitely many witnesses in its relation ideal. Invertibility conditions are encoded by adjoining inverse variables. In particular, invertibility of the trace-pairing determinant keeps the chosen finite free splitting algebra étale and faithfully flat.

The closed embedding must be kept in the same system. Its coordinate pullback is surjective by the preceding argument. Choose preimages of a finite set of generators of the completed torus algebra and include the equations stating that their images are those generators. After approximation, these equations still give a surjective pullback. The splitting maps and their inverses still identify the source with a split torus on the finite étale cover, and the preserved descent identities give a torus on the approximating base. Thus approximation retains both the torus structure and the closed embedding, rather than approximating only an arbitrary scheme map.

Apply the theorem at order one to this entire coefficient tuple. The solution then agrees with the specified fibre torus, its embedding and splitting data on the residue field; the equations themselves hold exactly on the étale neighbourhood. Spread its finite presentation and all chosen data from the local model to an open part of the Noetherian model, then pull back to the original base. This gives the required embedded torus after an étale neighbourhood, without a Noetherian assumption on S. □

The positive-characteristic proof behind approximation

The full Néron–Popescu argument in Artin approximation (Reductive group schemes), Section 4, includes inseparable residue fields. The following calculation gives the positive-characteristic exponent and approximation bounds used in that construction. The Stacks Project authors' smoothing chapter supplies further reading.

Let k be a field of characteristic $p > 0$, and write $k \to A \to \Lambda$, where A is of finite type over k and Λ is Noetherian and geometrically regular. Let $H = H_{A/k}$ be the radical ideal defining the non-smooth locus, and choose a prime $\mathfrak{q}$ minimal over $\sqrt{H\Lambda}$. Put $\Gamma = \Lambda_{\mathfrak{q}}$ and $d = \dim \Gamma$. The construction chooses $N \geq 1$ with $\mathfrak{q}^N \Gamma \subset H\Gamma$, then improves an auxiliary presentation and chooses an exponent $c \geq 1$. Its lifting step requires $e = 8c$. At the order-selection step, take

$$n = \max\{N + dc,\ d(e-1)+1\}, \qquad e = 8c.$$

This replaces the printed $n = N + dc$. The earlier choices of N, the auxiliary algebras and c do not depend on n. Define the finite set of coefficient images in $\Gamma/\mathfrak{q}^n\Gamma$ with this order, then apply the Artinian approximation lemma, D.35 with its global-selection proof D.34A, which permits every positive n. All subsequent maps, Artinian subalgebras and parameters are chosen for this same quotient.

That lemma supplies a local polynomial ring $P = k[y_1, \dots, y_m]_{\mathfrak{p}}$ and a flat map $P \to \Gamma$ for which $\mathfrak{p}\Gamma = \mathfrak{q}\Gamma$. The chosen parameters $\pi_1, \dots, \pi_d$ generate $\mathfrak{p}P$ and their images generate $\mathfrak{q}\Gamma$. Put $J = (\pi_1^e, \dots, \pi_d^e) \subset P$. A degree-$n$ monomial in d parameters contains some π_i^e: otherwise every exponent is at most $e-1$ and the total degree is at most $d(e-1) < n$. Therefore

$$\mathfrak{p}^n P \subset J, \qquad \mathfrak{q}^n\Gamma \subset J\Gamma.$$

This proves the inclusions used to descend the smooth factorization from the quotient of order n to the quotient by J. The unit rescalings of the parameters preserve these inclusions. In the auxiliary polynomial ring containing the t_i, the selected target point sends t_i to a unit δ_i. Hence the ideals generated by $(\pi_i t_i)^e$ and by π_i^e agree after localization at that point. Reducing a smooth algebra modulo J is the required base change. The other lifting conditions involve e, strict standardness and the annihilator equalities; these are unchanged by the larger order.

The coefficient calculation also survives that choice. At the denominator-clearing step, write $H = (a_1, \dots, a_t)$ and $\pi_i^N = \sum_j a_j d_j$ in the Artinian intermediate algebra D'. Lift d_j to $b_j \in \Gamma$. The error in this equality lies in $\mathfrak{q}^n\Gamma \subset \mathfrak{q}^{n-N}H\Gamma$, so it is $\sum_j a_j b_j'$ with $b_j' \in \mathfrak{q}^{n-N}\Gamma$. Multiplication by π_i^N kills each b_j' modulo $\mathfrak{q}^n\Gamma$. Choose a denominator unit $s \in \Lambda \setminus \mathfrak{q}$; replace it by a suitable power and enlarge D' so that its residue also belongs to D', as explained below. Clearing denominators then gives coefficients $\mu_{ij} \in \Lambda$ whose residues satisfy

$$\mu_{ij} \bmod \mathfrak{q}^n\Gamma = s^{2N}\pi_i^N d_j \in D'.$$

The printed expression at line 2967 omits π_i^N. This factor belongs to D' through the polynomial coordinate map, so the needed coefficient membership holds with the corrected expression. The subsequent unit and annihilator adjustments preserve the exact defining relations.

One further detail concerns enlarging the Artinian intermediate algebra. Let $\overline{\Gamma}$ be its Artinian target with maximal ideal $\mathfrak{n}$ satisfying $\mathfrak{n}^n = 0$, and let $\overline{\lambda}$ be a unit whose residue α belongs to the residue field of D. Choose a unit $x \in D$ lifting α^{-1} and put $u = x\overline{\lambda} - 1 \in \mathfrak{n}$. For a power $Q = p^r \geq n$, Frobenius gives

$$(x\overline{\lambda})^Q = (1+u)^Q = 1 + u^Q = 1, \qquad \overline{\lambda}^Q = x^{-Q} \in D.$$

Thus the exponent must be a sufficiently large power of p; divisibility by p alone does not give this identity. If α is algebraic over the residue field F of D, first take a p-power Q_0 making α^{Q_0} separable over F. Lift that finite separable residue extension to the finite étale Artinian algebra D_1 and its map to $\overline{\Gamma}$. The same calculation with another sufficiently large p-power Q_1 gives $\overline{\lambda}^{Q_0Q_1} \in D_1$. The transcendental case uses the localized polynomial extension in the linked proof. These steps require no perfection assumption on k or on its residue extensions.

If $d = 0$, the localized regular target Γ is a field and the parameter lists are empty. Skip the sequence-lifting lemma, which is stated for a positive-length sequence, and use the Artinian smooth factorization and the height-zero delocalization step directly. The separable and inseparable cases therefore both cover dimension zero as well. These details are included in the preceding written

desingularization proof; its Theorem D.38 and Theorem 1.1 supply the arbitrary-characteristic approximation input used above.

Why fpqc tori split étale locally. Suppose T/S is fpqc locally $\mathbf{G}_m^r$. Affine descent makes it affine, smooth and of finite presentation, with geometric fibres tori; its relative dimension is locally constant. A fibre torus splits over a finite separable extension, by the separable-splitting proof in the earlier character lesson, with its full semilinear Hopf descent supplied in Supporting group-scheme proofs, G.5.1–G.5.3. After making that residue extension, apply Lemma 3.1(3) with target $G = T$ and source the entire fibre torus. We obtain an embedded étale torus $T' \subset T$ near the point. Shrink so that both groups have the same relative dimension. On every geometric fibre the closed subtorus T' now has the dimension of the connected torus T, so is the entire fibre. The inclusion between these smooth groups has invertible differential. AG-RG-S02, Lemma 3.A, makes it an étale monomorphism, hence an open immersion. Fibrewise surjectivity makes it an isomorphism. Finally split T' by its étale character cover. This proves the asserted equivalence of definitions, using the deformation lemma for a general smooth affine target, rather than presupposing the splitting assertion.

Subgroups obtained by taking a limit

Over a ring R, let $\lambda : \mathbf{G}_m \to H$ be a cocharacter of a smooth affine group. Define

$$\begin{aligned} P_H(\lambda)(A) &= \{h : \lambda(t)h\lambda(t)^{-1} \text{ extends to } t = 0\},\\ U_H(\lambda)(A) &= \{h : \text{that extension has value } 1\},\\ L_H(\lambda) &= C_H(\lambda(\mathbf{G}_m)). \end{aligned}$$

The extension is a morphism from $\mathbf{A}^1_A$, not a numerical limit. All definitions commute with base change.

Lemma 3.3. These are smooth closed subgroups of finite presentation. The limit map identifies

$$P_H(\lambda) = L_H(\lambda) \ltimes U_H(\lambda),$$

and multiplication gives an open immersion

$$U_H(-\lambda) \times P_H(\lambda) \longrightarrow H.$$

If $\mathfrak{h} = \bigoplus \mathfrak{h}_n$ is the cocharacter-weight decomposition, their Lie algebras are respectively $\bigoplus_{n\geq 0} \mathfrak{h}_n$, $\bigoplus_{n>0} \mathfrak{h}_n$ and $\mathfrak{h}_0$. The fibres of $U_H(\lambda)$ are connected unipotent.

Proof. Since H is smooth and affine over R, the earlier S04 Lemma A1.8 makes its coordinate algebra finitely presented over R. Give $R[H]$ its conjugation grading, with $f(t.h) = t^n f(h)$ for a homogeneous function of degree n. Existence of the limit is equivalent to vanishing of all negative-degree functions at h, so these functions generate the ideal of $P_H(\lambda)$. Killing all nonzero-degree functions gives $L_H(\lambda)$; adding to the ideal of $P_H(\lambda)$ every degree-zero $f - f(1)$ gives $U_H(\lambda)$. These ideals are finitely generated over any R. Indeed replace a finite algebra generating list of $R[H]$ by all the finitely many homogeneous components appearing in it. A negative-weight monomial contains a negative-weight generator, so those generators generate the first ideal. The nonzero-weight generators generate the second. Modulo the first ideal, a weight-zero monomial involves only weight-zero generators: a product of remaining positive-weight generators cannot have weight zero. Thus the first ideal together with $a - a(1)$ for the finitely many weight-zero generators a generates the third ideal. Adding these finite lists to a finite presentation gives finite presentation of all three subgroups. The coefficient projectors for the grading commute with every scalar extension, so these equations also prove arbitrary base-change compatibility.

Evaluation at zero is a homomorphism $P \to L$ and is the identity on L. The mutually inverse maps $(l, u) \mapsto lu$ and $p \mapsto (\lim t.p, (\lim t.p)^{-1}p)$ prove the semidirect product assertion.

Here is a direct smoothness check over every base ring. The fixed subgroup $L = H^{\mathbf{G}_m}$ is smooth by AG-GS-05, Theorem 3.4. To prove smoothness of P, test a square-zero surjection $C \to \bar{C} = C/J$. A point of $P(\bar{C})$ gives the equivariant extension $F_0 : \mathbf{A}^1_{\bar{C}} \to H_{\bar{C}}$ of its conjugation orbit. Smoothness of the affine H lifts it to a scheme map $F : \mathbf{A}^1_C \to H_C$. On these maps let the universal $a \in \mathbf{G}_m$ act by

$$(aF)(t) = \lambda(a)F(a^{-1}t)\lambda(a)^{-1}.$$

Since F_0 is fixed, the ratio $(aF)F^{-1}$ is a regular one-cocycle in the vector module

$$V = \mathrm{Lie}(H_{\bar{C}}) \otimes_{\bar{C}} J[t], \qquad (av)(t) = \mathrm{Ad}(\lambda(a))v(a^{-1}t).$$

This square-zero kernel identification is made on the universal free algebras $C[t, a, a^{-1}]$ and $C[t, a, a^{-1}, b, b^{-1}]$, so the displayed tensor ideal is exact there. The ratio identity $c(ab) = c(a) + ac(b)$ follows from composition of this action; multiplication in the square-zero kernel is addition. AG-GS-05, Proposition 7.14 supplies $w \in V$ with $c(a) = w - aw$. Left multiplication of F by its kernel section $w(t)$ therefore gives a fixed map, since $aw + c(a) = w$. A fixed map is equivariant and, on $\mathbf{G}_{m,C}$, is the conjugation orbit of its value at 1. Its extension to $\mathbf{A}^1_C$ thus makes that value a lift in $P(C)$.

For U, start with $F_0(0) = 1$. Correct any scheme lift first by the inverse of its value at 0, a square-zero kernel element, so that $F(0) = 1$. Its equivariance defect now belongs to the stable submodule $\mathrm{Lie}(H_{\bar{C}}) \otimes_{\bar{C}} J[t]t$. Apply the same degree-one calculation to this submodule. The correcting $w(t)$ vanishes at zero, so the resulting equivariant lift still has $F(0) = 1$ and gives a lift in $U(C)$. We have proved formal smoothness of P and U. Their already proved finite presentation and the infinitesimal criterion of AG-FSE, Infinitesimal lifting and invariance under thickenings, Theorem 4.1, give smoothness everywhere. Finally an identity tangent vector $X = \sum_n X_n$ has orbit $\sum_n t^n X_n$. It extends exactly when $X_n = 0$ for $n < 0$, and has value zero at the origin exactly when also $X_0 = 0$. Fixedness means only weight zero remains. These are the three asserted Lie algebras.

The intersection $U_H(-\lambda) \cap P_H(\lambda)$ is trivial: both positive and negative degrees must vanish, and the limit is the identity. The multiplication map is a monomorphism. Its differential is an isomorphism at the identity; at (u, p), translations reduce it to the decomposition of $\mathfrak{h}$ by $\mathfrak{u}_- \oplus \mathfrak{p}$, followed by $\mathrm{Ad}(p^{-1})$, which induces an automorphism modulo $\mathfrak{p}$. AG-RG-S02, Lemma 3.A, therefore makes it étale everywhere; the complete flat-monomorphism proof in the earlier S04 Lemma A1.9 makes it an open immersion.

Every point of U is connected to the identity by its extended orbit map $\mathbf{A}^1 \to U$, so the geometric fibres are connected. Over a field, embed $H \rtimes \mathbf{G}_m$ faithfully in a general linear group and choose a weight basis for the acting torus. The condition that the limit be the identity makes U strictly triangular relative to that weight ordering. It is therefore unipotent. □

Lemma 3.D (contracted one-dimensional groups). Let U be a smooth affine group of finite presentation over a scheme S. Suppose an action of $\mathbf{G}_m$ by group automorphisms extends to $\mathbf{A}^1 \times U \to U$, with zero sending every point to the identity. Suppose $\mathrm{Lie}(U/S)$ is an invertible module on which the action has a locally constant positive integer weight d. There is a unique $\mathbf{G}_m$-equivariant isomorphism

$$\mathrm{Exp}_U : \mathbf{V}(\mathrm{Lie}(U/S)) \longrightarrow U$$

whose differential is the identity. It is an isomorphism of groups and commutes with every base change. Any further group action commuting with this $\mathbf{G}_m$-action also commutes with Exp_U, using its differential action on the source. The integer d need not be invertible on S.

Proof. Work on an affine open $S = \mathrm{Spec}\, R$ where $d > 0$ is constant and the Lie line has a frame v. Put $A = R[U]$. We use the point-pullback convention for weights: a homogeneous function a

of weight j satisfies $a(t.u) = t^j a(u)$ over the universal Laurent polynomial algebra. The complete weight decomposition in AG-GS, *Diagonalizable groups and groups of multiplicative type*, Theorem 3.1 gives $A = \bigoplus_{j\in\mathbf{Z}} A_j$. The extension to $\mathbf{A}^1$ forces $A_j = 0$ for $j < 0$: the coefficient at^j must belong to $A[t] \subset A[t, t^{-1}]$. Evaluation at zero is the identity section, so a weight-zero element equals its identity value. The unit $R \to A$ and the identity evaluation are inverse on that summand. Thus

$$A_0 = R, \qquad I = \ker(A \to R) = \bigoplus_{j>0} A_j.$$

The relative tangent-functor calculation in AG-GS, *Lie algebras and smoothness*, Proposition 1.4 identifies $I/I^2 = e^*\Omega_{U/S}$ with the dual of the Lie line. Explicitly, an identity derivation kills R and I^2, and is precisely an R-map $I/I^2 \to R$. Smoothness makes this cotangent module finite projective, so its double dual is itself. In our point-pullback convention its sole weight is d: pulling a cotangent functional back along multiplication by t^d multiplies that functional by t^d. Choose a homogeneous lift $f \in A_d$ of the functional taking v to one. This lift exists because the surjection $I \to I/I^2$ is surjective in each weight, by the same weight-decomposition theorem.

Every homogeneous element of A is a polynomial in f. Here is the induction, including possible nilpotents in R. For $a \in A_j$ with $j > 0$, its class in I/I^2 is zero if $j \neq d$, and is $r\bar{f}$ for some $r \in R$ if $j = d$. Subtract rf in the latter case. The remainder lies in $(I^2)_j$. Write it as a finite sum of products of homogeneous elements of positive degrees adding to j, by decomposing both factors of each product. Each factor has degree strictly smaller than j, and induction expresses it as a polynomial in f. The degree-zero case is $A_0 = R$. This proves $A = R[f]$. In particular all A_j with $0 < j < d$ vanish. Hence $(I^2)_d = 0$, and the chosen lift f is unique. The map

$$q : U \longrightarrow \mathbf{A}^1_R, \qquad q^*(x) = f,$$

is a closed immersion, is equivariant for weight d on the target, and has invertible differential at the identity.

We prove that it is an isomorphism over the original ring. The sheaf $\Omega_{U/R}$ is an invertible sheaf. One can see its rank everywhere directly: on every geometric fibre, translation by a point identifies its tangent space with the one-dimensional tangent space at the identity. Smoothness makes $\Omega_{U/R}$ finite locally free, as proved in Proposition 1.4 and S02 Lemma 3.A, so these fibre ranks give rank one at every point. The noninvertibility locus of the differential $q^*\Omega_{\mathbf{A}^1/R} \to \Omega_{U/R}$ is closed: in local frames it is the zero locus of its single coefficient. It is $\mathbf{G}_m$-stable, because q is equivariant and multiplication by an invertible t^d is an automorphism on both tangent lines. This locus misses the identity section.

If it contained a geometric point u with algebraically closed residue field K, its preimage under the extended orbit $\mathbf{A}^1_K \to U$, $t \mapsto t.u$, would be a closed subset containing $\mathbf{G}_{m,K}$. That open subset is dense, since $K[t]$ is a domain. The preimage would contain zero, whose image is the identity, a contradiction. Thus the differential is invertible everywhere. S02 Lemma 3.A makes q étale everywhere; its proof includes the finite-presentation calculation for a map between smooth schemes of finite presentation. The earlier S04 Lemma A1.9 makes this monomorphism an open immersion. Its image is also closed, because q was already a closed immersion. On every geometric fibre it is an open and closed subset of the connected $\mathbf{A}^1_K$ containing zero, hence is the whole fibre. Every target point belongs to a geometric fibre, so the open image is the whole target. Consequently q is an isomorphism of schemes, including over nonreduced bases.

Transport the group law of U to this affine line. It is a polynomial $m(x, y) \in R[x, y]$, and equivariance gives, in $R[t, t^{-1}, x, y]$,

$$m(t^d x, t^d y) = t^d m(x, y).$$

The Laurent monomials are an R-basis. A coefficient of $x^i y^j$ can be nonzero only if $d(i+j) = d$, hence only if $i + j = 1$. This is equality of integer exponents; it uses no division by d in R. Thus $m = ax + by$, and the two identity laws give $a = b = 1$. The inverse of q, using the frame v on the source Lie line, is the required group isomorphism with identity differential.

A weight-d equivariant morphism between two such affine lines is also linear: comparing the coefficient of x^i gives $di = d$, so only $i = 1$ occurs. Its differential determines it, even when the coefficients have nilpotents. Composing two candidate isomorphisms proves uniqueness. Therefore the local constructions agree after changes of frame and glue to $\mathbf{V}(\mathrm{Lie}(U/S))$. Relative differentials and the finite-projective dual commute with arbitrary base change by Proposition 1.4, and the uniquely characterized isomorphism after any base change is the base change of this one.

Finally let another group scheme act on U by automorphisms commuting with $\mathbf{G}_m$. On its universal test algebra, applying that automorphism after Exp_U and applying its differential before Exp_U give equivariant isomorphisms with the same differential. Compose with the inverse of that differential on the Lie line; uniqueness makes the two maps equal. This is an equality of scheme morphisms over every test algebra, proving the final assertion. □

4. A torus over the ground field

The finite-field and infinite-field arguments have quite different mechanisms. The first adjusts descent by Lang's theorem. The second finds a semisimple infinitesimal direction and uses dimension induction. The infinitesimal argument is essential over imperfect fields.

Lemma 4.L (Lang's equation). Let $G/\mathbf{F}_q$ be a smooth connected algebraic group, of finite type, and let $F : G \to G$ be its q-power Frobenius. For every algebraically closed extension $K/\mathbf{F}_q$, the map $h \mapsto h^{-1}F(h)$ is surjective on $G(K)$. Affineness is not required.

Proof. S08 Lemma G.0.3 makes G_K connected. Frobenius kills every square-zero infinitesimal point at the identity. Here is the full coordinate check, including for a nonaffine group. Choose an affine $\mathbf{F}_q$-open neighbourhood U of the identity. A point δ reducing to the identity across a square-zero ideal $I \subset R'$ factors through U_K, since a nilpotent closed immersion leaves the underlying topological space unchanged. A coordinate $a \in \mathbf{F}_q[U]$ vanishing at the identity has value in I. Frobenius pulls it back to a^q, whose value is zero. These coordinates, together with constants, show $F(\delta) = 1$. The restriction of Frobenius to this open is legitimate: over $\mathbf{F}_q$ absolute Frobenius has the identity underlying topological map.

For $x \in G(K)$, consider the twisted orbit map

$$\theta_x : G_K \longrightarrow G_K, \qquad g \longmapsto gxF(g)^{-1}.$$

It is formally étale. Given a square-zero extension $R' \twoheadrightarrow R$, a point $g \in G(R)$, and a prescribed lift $y \in G(R')$ of $\theta_x(g)$, smoothness of G_K gives a lift g_0, locally on $\operatorname{Spec} R'$ if necessary. Put

$$\delta = g_0^{-1} y F(g_0) x^{-1}.$$

It reduces to the identity, so $F(\delta) = 1$, and direct multiplication gives $\theta_x(g_0\delta) = y$. Any other lift is $g_0\epsilon$ with ϵ reducing to the identity; its equation and $F(\epsilon) = 1$ force $\epsilon = \delta$. This uniqueness glues the local lifts. Thus existence and uniqueness hold on every square-zero test, including all nilpotent directions.

The map is of finite presentation. On affine source and target charts their algebras are finite type over the Noetherian field K, by S01 Lemmas 1.0 and 1.3; the relative presentation kernel is finitely generated. The finite affine covers supplied by finite type give the global assertion. Formal étaleness implies formal smoothness, so the definition and lifting criterion of AG-CA, *Formally smooth, unramified and étale maps*, Section 3, make θ_x smooth. Its flatness is the complete arbitrary-base

proof of AG-CA, *Smooth algebras over a field and the Jacobian criterion*, Theorem 6.1. S08 Lemma G.4.F consequently makes its image O_x open.

These open images are either disjoint or equal. If $O_x \cap O_z$ is nonempty, it has a K-rational point by the weak Nullstellensatz on an affine chart. Each of its nonempty finite-type orbit-map fibres has a K-rational point by the same argument. Hence x and z belong to the same twisted point orbit. The identity $g \cdot (a \cdot x) = (ga) \cdot x$ then identifies the two entire orbit-map images. The union of the O_x, for $x \in G(K)$, covers G_K: a nonempty closed complement would have a rational point, which belongs to its own orbit image. Connectedness makes this disjoint open partition consist of one member. In particular $O_1 = G_K$. Every rational target point has a rational lift in its finite-type fibre. Thus $g \mapsto gF(g)^{-1}$ is surjective on $G(K)$. Replacing g by h^{-1} gives the stated equation. □

Lemma 4.F (a fixed basis for an invertible semilinear operation). Let K be algebraically closed of characteristic $p > 0$, and let V be finite-dimensional with an invertible Frobenius-semilinear operation Φ. Then V has a basis fixed by Φ.

Proof. The assertion is empty in dimension zero. Otherwise, in a chosen basis write $\Phi(v) = Av^{(p)}$, with $A \in \mathrm{GL}_d(K)$. The group $\mathrm{GL}_d /\mathbf{F}_p$ is smooth and geometrically connected: its coordinate ring is the polynomial ring in matrix entries localized at the determinant, a domain over every field extension; entries lift across a square-zero surjection and the determinant remains a unit because an element invertible modulo a nilpotent ideal is invertible, using the finite geometric-sum inverse. This proves its finite-presentation and formal smoothness directly. Apply Lemma 4.L to choose h with $h^{-1}F(h) = A$, and put $C = h^{-1}$. Then

$$C\big(C^{(p)}\big)^{-1} = A, \qquad AC^{(p)} = C.$$

The columns of the invertible C are a basis and are fixed by Φ, as required. □

Lemma 4.H (PBW, Frobenius kernels and height-one groups). Ordinary PBW holds for every finite-dimensional Lie algebra over any field. In characteristic $p > 0$, the restricted enveloping algebra $u(L)$ of every finite-dimensional restricted Lie algebra has the ordered basis $x_1^{r_1} \cdots x_d^{r_d}$, $0 \le r_i < p$, and dimension p^d. It is a cocommutative Hopf algebra with primitive generators. Its finite dual defines a height-one group H_L, and $L \mapsto H_L$ is inverse to the Lie functor on finite height-one group schemes over the field, with their full scheme structures. In particular, for a smooth affine algebraic group G,

$$k[G[F]] \simeq k[t_1, \dots, t_n]/(t_1^p, \dots, t_n^p), \qquad \mathrm{Dist}(G[F]) = u(\mathrm{Lie}\, G),$$

where $n = \dim_k \mathrm{Lie}\, G$ and the first isomorphism concerns algebras, with its Hopf maps supplied by G. Restricted subalgebras $L \subset \mathrm{Lie}\, G$ correspond to closed subgroup schemes of $G[F]$. Their centralizers are closed and have Lie algebra $\{Y : [Y, L] = 0\}$. All these constructions commute with field extension. If L is commutative and its p-operation is geometrically invertible, then H_L is of multiplicative type and is geometrically $\mu_p^{\dim L}$. No perfect-field hypothesis is imposed.

Proof. We first prove ordinary PBW, including the confluence step. In the tensor algebra on an ordered basis, replace $x_j x_i$, $j > i$, by $x_i x_j + [x_j, x_i]$. On words, order first by length and then by inversion count. A swapped word has one fewer inversion, and every bracket term has shorter length, so all reductions terminate. Disjoint reductions commute as linear substitutions. The only overlapping pair occurs in $x_k x_j x_i$, $k > j > i$. Reducing its left or right pair and then the remaining inverted pairs gives expressions whose difference is

$$[[x_j, x_i], x_k] + [x_j, [x_k, x_i]] + [[x_k, x_j], x_i] = 0$$

by Jacobi. Here a commutator between two linear combinations of basis vectors reduces to their given Lie bracket by the defining two-letter rules. These terms have smaller length than the original triple, even inside a larger word.

For completeness, well-founded induction now proves uniqueness of the final linear combination of ordered words. Compare two first reductions. In the disjoint case apply the other substitution

to obtain the same expression; all resulting words have smaller order. In the overlap case their difference has the zero normal form just computed, using the induction for shorter words. Subsequent reductions are unique by the same induction. Extend this normal-form operation linearly. Every defining relation in any word context has normal form zero, so its entire two-sided ideal does as well. Reduction changes a word only by that ideal, while an ordered word is its own normal form. Hence the ordered monomials both span the quotient and are independent. This proves ordinary PBW in every characteristic.

We specify the restricted convention and verify the identity needed for all vectors, rather than only a basis. In characteristic p, a restricted operation satisfies $(aX)^{[p]} = a^p X^{[p]}$, $\operatorname{ad}(X^{[p]}) = (\operatorname{ad} X)^p$, and

$$(X+Y)^{[p]} = X^{[p]} + Y^{[p]} + \sum_{i=1}^{p-1} s_i(X,Y),$$

where $is_i(X,Y)$ is the coefficient of t^{i-1} in $(\operatorname{ad}(tX+Y))^{p-1}(X)$. These are defining axioms; the integers $1,\dots,p-1$ are invertible in the field. In any associative algebra the same s_i, using its commutator bracket, satisfy

$$(X+Y)^p = X^p + Y^p + \sum_{i=1}^{p-1} s_i(X,Y).$$

Indeed set $Z = tX + Y$. Differentiate the product Z^p formally in the central variable t, obtaining $\sum_{j=0}^{p-1} Z^j X Z^{p-1-j}$. Expanding the iterated commutator gives this same sum, since $\binom{p-1}{j} = (-1)^j$ in characteristic p. To see the coefficient identity, multiply $\sum_{j=0}^{p-1}(-t)^j$ by $1+t$ and compare with $(1+t)^p = 1+t^p$; its intermediate binomial coefficients vanish because their denominators are prime to p. The coefficient of t^{i-1} is consequently i times the coefficient of t^i in Z^p. The pure terms have zero derivative, and this determines every mixed coefficient. Evaluating at $t=1$ proves the identity.

In the ordinary enveloping algebra put $z_i = x_i^p - x_i^{[p]}$. The binomial expansion of an iterated commutator gives $[x_i^p, y] = (\operatorname{ad} x_i)^p(y) = [x_i^{[p]}, y]$, so z_i is central. The products $z_1^{a_1}\cdots z_d^{a_d} x_1^{r_1}\cdots x_d^{r_d}$, with $0 \le r_i < p$, have distinct leading PBW monomials, whose exponents are $pa_i + r_i$. They are independent. They span by replacing each basis exponent $\ge p$ by a z_i term plus $x_i^{[p]}$, which has lower degree, and inducting on degree. Thus the ordinary algebra is free over the polynomial algebra in its central z_i, on those truncated monomials. Quotienting by the z_i retains precisely that basis. The preceding associative identity and the restricted axioms make the defect $X^p - X^{[p]}$ additive and Frobenius-semilinear in this quotient. It vanishes on each basis vector, hence on every X. The quotient is therefore exactly $u(L)$, not a larger algebra imposing too few relations. The coproduct $\Delta X = X \otimes 1 + 1 \otimes X$, counit zero and antipode $SX = -X$ preserve the bracket relations. Each z_i is primitive, because its two tensor terms commute before taking the p-th power. Its ideal is consequently a Hopf ideal. The quotient is cocommutative and has the asserted basis and dimension.

We now prove its relation with the first Frobenius kernel from coordinates. Put $A = k[G]$ and $\mathfrak{m} = \ker \epsilon$. Relative Frobenius pulls a twisted augmentation element back to its p-th power, so

$$C = k[G[F]] = A/(a^p : a \in \mathfrak{m}).$$

The ideal $\mathfrak{m}$ is finitely generated by S01 Lemmas 1.0 and 1.3. Its images in C generate a nilpotent ideal: if there are s generators, every word of length $s(p-1)+1$ contains one with exponent p. The only underlying point of $\operatorname{Spec} C$ is the identity. Iteratively replacing coefficients by their identity values shows that C is generated by those nilpotent images over k, so it is finite-dimensional.

Choose $t_1,\dots,t_n \in \mathfrak{m}$ dual to a basis of the tangent space. The complete smooth-coordinate proof of S02 Lemma 3.A makes their map $f : U \to \mathbb{A}^n_k$ étale on an open neighbourhood U of the identity. An étale map has unique lifts across any nilpotent ideal: apply its proved square-zero uniqueness and existence successively to I^j/I^{j+1}, whose square is zero. The lifted points glue by uniqueness.

For any k-algebra R and any list $b_i \in R$ with $b_i^p = 0$, the ideal $I = (b_i)$ is nilpotent by the same word bound. The identity over R/I has a unique lift $g \in U(R)$ with coordinates b_i. The twisted étale coordinate map $f^{(p)}$ sends $F_{G/k}(g)$ to zero; it reduces to the twisted identity. Uniqueness for that étale map makes $F_{G/k}(g)$ the identity, so $g \in G[F](R)$.

Conversely any $G[F](R)$-point factors through U and has p-nilpotent coordinates. Its finitely many augmentation-generator values generate a nilpotent ideal J, and it reduces to the identity modulo J. The point constructed from its coordinates also reduces to the identity modulo J. Étale uniqueness, with their common coordinate map, makes the two points equal. This gives mutually inverse natural maps on every algebra R, including nonreduced ones. Hence $C \simeq k[t_1, \dots, t_n]/(t_i^p)$, and $\dim_k C = p^n$.

We describe invariant derivatives and their pairing, without a distribution-algebra classification assumption. A tangent vector is an identity derivation $v : A \to k$, so $v(ab) = v(a)\epsilon(b) + \epsilon(a)v(b)$. Its invariant derivation is $D_v = (\mathrm{id} \otimes v)\Delta$. The Hopf identities prove the product rule and invariance, and evaluation at the identity recovers v. Invariance conversely determines the field from that value, by translation. Commutators of these derivations are invariant. Their p-th powers are invariant derivations as well: iteration of the product rule gives the binomial formula, whose intermediate coefficients vanish in characteristic p. Embed G faithfully in $\mathrm{GL}(V)$ by Section 2. On its matrix-coordinate entries, $D_X(z) = zX$, and therefore $D_X^p(z) = zX^p$. This proves that $\mathrm{Lie}\, G$ is closed under matrix p-th powers and that its restricted operation is $X^{[p]} = X^p$. The same calculation gives its bracket as the matrix commutator and the restricted axioms just verified.

Each D_X preserves the Frobenius-kernel ideal, since $D_X(a^p) = 0$. The finite dual $C^\vee$ is a Hopf algebra; its convolution is $(vw)(a) = (v \otimes w)(\Delta a)$. Directly from the coaction formula, this is $\epsilon D_v D_w(a)$ for tangent vectors, with the same formula for longer products. These primitive elements satisfy the bracket and restricted relations, giving a Hopf map $u(\mathrm{Lie}\, G) \to C^\vee$. Choose the tangent basis dual to the t_i. Then $D_{x_i} t_j = \delta_{ij}$ modulo the augmentation ideal. On its associated graded algebra D_{x_i} lowers degree by one and acts as $\partial/\partial t_i$. Consequently a distribution monomial of total degree r vanishes on a coordinate monomial of degree greater than r, and its pairing at degree r is

$$\langle x_1^{a_1} \cdots x_n^{a_n}, t_1^{b_1} \cdots t_n^{b_n} \rangle = \begin{cases} \prod_i a_i! & a_i = b_i \text{ for all } i, \\ 0 & \text{otherwise,} \end{cases} \qquad \sum_i a_i = \sum_i b_i = r.$$

All factorials are nonzero when $a_i < p$. In any relation, evaluate on monomials of the largest participating degree to kill all its coefficients of that degree, then descend on degree. Thus the p^n truncated distribution monomials are independent. They form a basis of $C^\vee$, proving the Hopf isomorphism with $u(\mathrm{Lie}\, G)$.

For a restricted subalgebra $L \subset \mathrm{Lie}\, G$, extend an ordered basis of L to one of $\mathrm{Lie}\, G$. The PBW basis makes $u(L) \to u(\mathrm{Lie}\, G)$ injective. Finite duality gives a surjective map of commutative Hopf coordinate algebras $C \to u(L)^\vee$, hence a full closed subgroup of $G[F]$. Its order is $p^{\dim L}$.

We prove that every subgroup is obtained this way, including the needed order assertion. Let $H \subset G[F]$ be closed and put $d = \dim_k \mathfrak{m}_H/\mathfrak{m}_H^2$. Lifts $a_1, \dots, a_d$ of a cotangent basis generate $k[H]$: first $\mathfrak{m}_H = (a_i) + \mathfrak{m}_H^2$, and iteration to its nilpotence exponent gives ideal generation; expanding coefficient identity values then gives algebra generation. Their p-th powers are zero. Thus

$$P = k[T_1, \dots, T_d]/(T_i^p) \twoheadrightarrow k[H]$$

has kernel $J \subset (T_i)^2$. It has no nonzero kernel. Otherwise choose the smallest augmentation degree $r \geq 2$ of a nonzero kernel element, and let f_r be its first nonzero homogeneous component. Lift the coproduct of each a_i to $T_i \otimes 1 + 1 \otimes T_i + u_i$, with u_i in the tensor product of the two augmentation ideals. This defines an algebra map out of P: every augmentation element in $P \otimes P$ has p-th power zero. The kernel element maps into $J \otimes P + P \otimes J$: this is the tensor quotient kernel, as choosing a vector-space complement to J in P shows. Its terms of bidegree $(s, r-s)$, $0 < s < r$, must vanish, since every element of J has augmentation degree at least r. The first homogeneous component is

therefore primitive for the standard additive coproduct on P. Its bidegree $(1, r-1)$ coefficients are the partial derivatives of f_r. A nonzero positive-degree polynomial whose individual exponents are all below p has a nonzero partial derivative: differentiate in an index with positive exponent in any nonzero monomial; that exponent is a unit, and distinct monomials give distinct derivatives in that index. This contradiction proves $J = 0$ and $|H| = p^d$.

Its tangent derivations preserve its defining Hopf ideal I, because $\Delta I \subset I \otimes A + A \otimes I$ and the identity derivations kill I. Thus $\operatorname{Lie} H$ is restricted in $\operatorname{Lie} G$. The preceding distributions of its tangent vectors lie in $k[H]^\vee \subset C^\vee$. Their generated algebra $u(\operatorname{Lie} H)$ has dimension p^d, exactly that of $k[H]^\vee$, so it equals that dual. Conversely this argument applied to the subgroup built from $u(L)$ gives its Lie dimension $\dim L$; the original L consists of primitive distributions in that Lie algebra, so it is the whole Lie algebra. This proves the subgroup correspondence.

For an abstract restricted L, left multiplication on $u(L)$ gives a faithful restricted embedding $L \to \operatorname{End}_k u(L)$: it preserves the bracket and p-operation, and evaluating at 1 detects a nonzero basis-linear element by PBW. Apply the subgroup construction to this subalgebra of $\operatorname{Lie} \mathrm{GL}(u(L))$. Its coordinate algebra is exactly $u(L)^\vee$, so it is height one and has Lie algebra L. Every finite height-one group embeds in a general linear group by Section 2; Frobenius naturality puts it in that group's first kernel, and the preceding converse identifies its distributions with $u(\operatorname{Lie} H)$. A restricted linear map extends uniquely to a Hopf map of enveloping algebras because those algebras are generated by their primitive Lie elements. Finite duality gives the corresponding group homomorphism. Conversely a group homomorphism gives this same map on primitive distributions, which generate the whole algebra. This proves the equivalence, including uniqueness of every morphism.

The schematic centralizer of a finite $H_L \subset G$ is closed: expand the conjugation identities for a finite generating list of $k[G]$ in a basis of the finite algebra $k[H_L]$. Their finitely many coefficient equations in $k[G]$ express exactly that each test point fixes every H_L-point. For a tangent Y, conjugation by $1 + \epsilon Y$ acts on the distribution generator X as $X + \epsilon[Y, X]$, by matrix multiplication. It fixes the full subgroup exactly when it fixes all generators of $u(L)$, equivalently $[Y, L] = 0$. The finite-dual basis formulas remain valid over the dual-number algebra, so this tests the entire subgroup scheme. PBW bases, coordinate equations, tangent derivations and finite duals all tensor exactly with field extensions; this proves the asserted compatibility.

Finally suppose L is commutative with geometrically invertible p-operation. Over an algebraic closure Lemma 4.F gives a fixed basis, so $u(L)$ is the tensor product of the primitive Hopf algebras $\bar{k}[x]/(x^p - x)$. Evaluation at the p distinct elements of $\mathbf{F}_p$, using the Chinese remainder proof, identifies each with functions on the additive group $\mathbf{F}_p$; its coproduct is pullback of addition. The dual has basis e_a with $e_a e_b = e_{a+b}$ and $\Delta(e_a) = e_a \otimes e_a$. It is $\bar{k}[\mathbf{Z}/p\mathbf{Z}]$, the coordinate algebra of μ_p. Thus H_L is geometrically $\mu_p^{\dim L}$. S08 Lemma G.0.T gives its finite-separable splitting and multiplicative type over the original field. □

Lemma 4.J (multiplicative Jordan components). Let G be an affine algebraic group over a field k, and choose a faithful finite-dimensional linear representation. Over an algebraic closure, every $g \in G(\bar{k})$ has unique commuting factors $g = g_s g_u$, where g_s is semisimple and g_u is unipotent. Both factors belong to $G(\bar{k})$. Every homomorphism of affine algebraic groups preserves the factors. If k is perfect and $g \in G(k)$, both factors belong to $G(k)$. These conclusions do not require G to be smooth or connected.

Proof. Work first over $K = \bar{k}$, and let the representation space have dimension n. Dimension zero gives the trivial group, so assume $n > 0$. Cayley–Hamilton, S06 Section 1, and the polynomial Bézout/Chinese remainder calculation, CA03 Lemma 4.1, decompose the space as the sum of the generalized eigenspaces of g. On the summand for a nonzero eigenvalue λ, put $g_s = \lambda I$ and $g_u = \lambda^{-1} g$. Thus g_s is diagonalizable, $g_u - I$ is nilpotent, and the factors commute and multiply to g. This also proves uniqueness: if commuting semisimple S and unipotent U multiply to g, every eigenspace of S is preserved by U, and on it g has only the corresponding eigenvalue of S.

Therefore S acts by exactly λI on the generalized λ-space of g, and $U = S^{-1}g$.

We use the following elementary sequence calculation, and give its proof. Over any field, the sequences α^m, for distinct nonzero α and integers $m \geq 0$, are linearly independent. In characteristic zero the same is true of exponential polynomial sequences $\alpha^m P_\alpha(m)$: if their finite sum vanishes, each polynomial P_α is zero. Let E shift a sequence one place. On $\alpha^m P(m)$, the operator $E - \beta$ gives $\alpha^m(\alpha P(m+1) - \beta P(m))$. When $\beta = \alpha$, it lowers a positive polynomial degree by one, with leading coefficient multiplied by $\alpha \deg P$, and kills a constant. When $\beta \neq \alpha$, it preserves the degree and multiplies the leading coefficient by $\alpha - \beta$. To isolate one nonzero term, apply the product of $(E - \beta)^{\deg P_\beta + 1}$ over the other terms. It kills them and leaves $\alpha^m Q(m)$ with Q nonzero. Over a characteristic-zero field, a nonzero polynomial cannot vanish at all nonnegative integers, by the root bound in S02 Lemma 4.A. This is a contradiction. For constant polynomials the same operator proof works in every characteristic, since it leaves a nonzero constant and uses no integer denominators.

Let $\lambda_1, \ldots, \lambda_r$ be the distinct eigenvalues, and write $D = (\mathbf{G}_m)^r$, acting by $z_i I$ on the corresponding generalized eigenspace. For $a \in \mathbf{Z}^r$, put $\lambda^a = \prod_i \lambda_i^{a_i}$, and let $N = \{a : \lambda^a = 1\}$. The quotient $K[\mathbf{Z}^r] \to K[\mathbf{Z}^r/N]$ defines a closed diagonalizable subgroup $D_\lambda \subset D$, as in Lemma 4.S. A finite Laurent sum $\sum_a c_a z^a$ belongs to its defining ideal exactly when the sum of its coefficients in each coset of N is zero. This follows directly from the coset basis of the group algebra. By the constant-sequence calculation, these are exactly the Laurent sums vanishing at every $(\lambda_1^m, \ldots, \lambda_r^m)$.

Suppose the characteristic is $p > 0$, and choose $p^q \geq n$. Then $g_u^{p^q} = I$, by the binomial theorem for $I + (g_u - I)$. Hence $g^{p^q m} = g_s^{p^q m} \in G(K)$. A defining regular function f of $G \subset \mathrm{GL}_n$, restricted to D, is a finite Laurent sum. Its evaluations on these points vanish. The map $\alpha \mapsto \alpha^{p^q}$ is injective on K: $(\alpha - \beta)^{p^q} = 0$ implies $\alpha = \beta$. The constant-sequence calculation therefore makes the coefficient sum in every λ^a-class zero. Thus f restricts to zero on the full group scheme D_λ, and $D_\lambda \subset G_K$. In particular $g_s \in G(K)$, and $g_u = g_s^{-1} g \in G(K)$. This argument makes no isomorphism assertion about the Frobenius power map on a torus.

Now suppose the characteristic is zero. Put $T = g_u - I$, and define the nilpotent matrix $L = \log(I + T) = \sum_{j=1}^{n-1} (-1)^{j+1} T^j / j$. For a nilpotent matrix, $\exp(tL) = \sum_{j=0}^{n-1} t^j L^j / j!$ is a polynomial matrix. These finite logarithm and exponential formulas are inverse: in formal rational-coefficient power series, differentiating $\log(1 + x)$ gives $1/(1 + x)$, so $\exp(\log(1 + x))$ has derivative equal to itself divided by $1 + x$ and constant term one. Comparing coefficients gives $1 + x$. Substituting the nilpotent T proves $\exp L = g_u$. Multiplying the exponential series and using the binomial coefficients also gives $\exp(tL)\exp(sL) = \exp((t + s)L)$, hence $g_u^m = \exp(mL)$.

The matrix L preserves the generalized eigenspaces and commutes with D. On each summand a basis adapted to the kernels of its successive powers makes it strictly upper triangular: extend bases through that kernel filtration, and note that L lowers the filtration. Consequently $\exp(tL)$ is upper triangular with diagonal one, and has determinant one. The substitution $(z, t) \mapsto D(z)\exp(tL)$ therefore pulls every regular function on GL_n, including the inverse determinant, back to a finite sum $F(z, t) = \sum_a z^a P_a(t)$ in $K[z_1^{\pm 1}, \ldots, z_r^{\pm 1}, t]$.

For a defining function f of G, evaluation on g^m gives $\sum_a (\lambda^a)^m P_a(m) = 0$ for all $m \geq 0$. Group by the distinct values $\alpha = \lambda^a$. The exponential-polynomial calculation shows $\sum_{\lambda^a = \alpha} P_a(t) = 0$ as a polynomial for each α. Setting $t = 0$ shows f vanishes on the full D_λ. Setting every $z_i = 1$ shows $f(\exp(tL)) = 0$ as a polynomial. Thus $D_\lambda \subset G_K$, and the whole polynomial one-parameter homomorphism $t \mapsto \exp(tL)$ factors through G_K. In particular both Jordan factors belong to $G(K)$.

For functoriality, embed the target of a homomorphism $\phi : G \to H$ faithfully in a general linear group by Section 2. The restriction to D_λ has the complete weight decomposition of AG-GS, *Diagonalizable groups*, Theorem 3.1. Therefore $\phi(g_s)$ is semisimple. In characteristic p, $\phi(g_u)^{p^q} = I$, so $(\phi(g_u) - I)^{p^q} = 0$, and its image is unipotent. In characteristic zero, write the polynomial matrix of the homomorphism $\mathbf{G}_a \to H \to \mathrm{GL}$, obtained above, as $B(t)$. It satisfies $B(0) = I$

and $B(t+s) = B(t)B(s)$. Differentiate in s at zero to obtain $B'(t) = B(t)Y$, with $Y = B'(0)$. Coefficient comparison gives $B_j = Y^j/j!$. Since B is polynomial of some degree d, the coefficient of t^d in this differential equation gives $Y^{d+1} = 0$. Thus $B(1)$ is unipotent. In either characteristic the two images commute, their product is $\phi(g)$, and uniqueness makes them its Jordan factors. Applying this to two faithful representations and their direct sum also proves independence of the chosen representation.

Finally let k be perfect and $g \in G(k)$. The geometric factors have algebraic matrix coefficients. Perfectness makes their finite field of definition separable. Adjoin all roots of the finitely many separable minimal polynomials to obtain a finite normal separable extension K_0/k containing those coefficients. Its k-embeddings all map it to itself; the separable embedding count in S02 Lemma 4.A and the finite automorphism-field calculation in S08 Lemma G.5.2 therefore give $K_0^{\mathrm{Gal}(K_0/k)} = k$. Every such automorphism fixes g and, by uniqueness, fixes its two geometric factors. Their coefficients are consequently in k. Their defining equations already vanish after faithful field extension, so they belong to $G(k)$. □

Lemma 4.K (additive Jordan components and their functoriality). Let G be a smooth affine algebraic group over a field k, with a faithful finite-dimensional linear representation. Over an algebraic closure every $X \in \mathrm{Lie}\, G$ has unique commuting matrix components $X = X_s + X_n$, with X_s semisimple and X_n nilpotent. Both components belong to $\mathrm{Lie}\, G$. For every homomorphism from G to an affine algebraic group, its differential carries them to the additive Jordan components of its image, in any faithful representation of the target. Over a perfect field a rational X has rational components. In particular these assertions apply to the adjoint representation.

Proof. Work first over $K = \bar{k}$. Cayley–Hamilton and the same polynomial primary projectors used in Lemma 4.J decompose the representation space into the generalized eigenspaces of X. On its λ-summand define $X_s = \lambda I$ and $X_n = X - \lambda I$. These components commute, and X_n is nilpotent. They are unique: if a commuting semisimple matrix S and nilpotent matrix N sum to X, the eigenspaces of S are preserved by N, and on the eigenvalue-λ space the sum has only eigenvalue λ. Its generalized eigenspace is exactly that space, so S is the specified scalar on every generalized eigenspace of X. Dimension zero gives two zero matrices and is included.

In characteristic $p > 0$, choose p^q at least the matrix dimension. Since X_s and X_n commute, the characteristic-p binomial formula gives $X^{p^q} = X_s^{p^q}$. Let $V \subset K$ be the $\mathbf{F}_p$-span of the eigenvalues λ^{p^q}. The assignment $\lambda^{p^q} \mapsto \lambda$ respects every $\mathbf{F}_p$-linear relation: raising a proposed relation among the λ's to the p^q-power gives the corresponding relation in V, and Frobenius on the field is injective. It therefore defines an $\mathbf{F}_p$-linear map $h : V \to K$.

If $V = 0$, all eigenvalues are zero, so $X_s = 0$ and $X_n = X$. Otherwise choose an $\mathbf{F}_p$-basis $v_1, \dots, v_d$ of V. The square matrix $(v_i^{p^j})_{1 \le i \le d,\, 0 \le j < d}$ is invertible over K. Indeed, a vector in its kernel gives a polynomial $P(T) = \sum_{j<d} a_j T^{p^j}$ vanishing on that basis. This polynomial is additive and $\mathbf{F}_p$-linear, by the characteristic-p binomial formula, so it vanishes on all p^d elements of V. If it were nonzero its degree would be at most $p^{d-1} < p^d$, contradicting the polynomial root bound in S02 Lemma 4.A. Thus the kernel is zero; the finite-dimensional linear algebra and adjugate calculation in S01 Lemmas 1.0 and 1.2 give invertibility. Solve $\sum_{j<d} a_j v_i^{p^j} = h(v_i)$. The resulting linearized polynomial agrees with h on all of V, and hence

$$X_s = P(X^{p^q}) = \sum_{j<d} a_j X^{p^{q+j}}.$$

Lemma 4.H proves closure of $\mathrm{Lie}\, G$ under all these restricted matrix powers. It follows that X_s, and then $X_n = X - X_s$, belong to $\mathrm{Lie}\, G$.

We give the full characteristic-zero argument using only formal power series, with no analytic exponential. For a matrix X, the series $E(t) = \sum_{j \ge 0} t^j X^j/j!$ is an invertible matrix over $K[[t]]$, since its constant term is I; coefficient multiplication gives inverse $E(-t)$. It satisfies $E'(t) = E(t)X$. Let $A = K[\mathrm{GL}_n]$, and let I_G be the Hopf ideal defining G. A tangent vector X defines the invariant

derivation D_X with $D_X(z) = zX$ on the matrix of coordinate functions. The identity derivation kills I_G, and the Hopf-ideal inclusion $\Delta I_G \subset I_G \otimes A + A \otimes I_G$ shows $D_X(I_G) \subset I_G$. This is the invariant-derivation calculation already proved in Lemma 4.H, and it uses no smoothness at this step.

For every $f \in A$, the chain rule for the substitution $z \mapsto E(t)$ gives

$$\frac{d}{dt} f(E(t)) = (D_X f)(E(t)), \qquad f(E(t)) = \sum_{j \geq 0} \frac{t^j}{j!} \epsilon(D_X^j f).$$

To check the chain rule on all of A, check it first on the matrix entries, using $E' = EX$, and extend by sums and products. For the inverted determinant, differentiate the identity $b\, b^{-1} = 1$ to get $(b^{-1})' = -b^{-2} b'$; this gives exactly the localized derivation formula. Repeated differentiation and evaluation at $t = 0$ then give the coefficients in the second identity; characteristic zero makes every factorial invertible. If $f \in I_G$, every $D_X^j f$ still belongs to I_G, and its identity value is zero. Therefore $E(t)$ belongs to $G(K[[t]])$.

Write $\lambda_1, \ldots, \lambda_r$ for the distinct eigenvalues of X, including zero if it occurs, and let $D = (\mathbf{G}_m)^r$ act by scalar z_i on the corresponding generalized eigenspace. Put

$$N = \{a \in \mathbf{Z}^r \mid \textstyle\sum_i a_i \lambda_i = 0\}, \qquad D_0 = D_K(\mathbf{Z}^r / N) \subset D.$$

Lemma 4.S gives its full defining ideal and coset basis. The matrices X_n and D commute. Since X_n is nilpotent, $\exp(tX_n)$ is polynomial, is upper triangular with diagonal one in a kernel-filtration basis, and has determinant one, by the calculation in Lemma 4.J. Consequently substitution $(z, t) \mapsto D(z) \exp(tX_n)$ sends any regular f on GL_n to a finite Laurent polynomial $F(z, t) = \sum_a z^a P_a(t)$. The formal identity $E(t) = \exp(tX_s) \exp(tX_n)$, obtained by multiplying commuting series and the binomial formula, gives for $f \in I_G$

$$\sum_a e^{(\sum_i a_i \lambda_i) t} P_a(t) = 0 \quad \text{in } K[[t]].$$

The formal exponential polynomials with distinct exponents are independent. Here is the exact proof. On $e^{\alpha t} P(t)$, the operator $d/dt - \beta$ gives $e^{\alpha t}(P' + (\alpha - \beta)P)$. For $\beta = \alpha$ it lowers a positive degree by one and kills a constant; for $\beta \neq \alpha$ it preserves the degree and multiplies the leading coefficient by $\alpha - \beta$. Apply the product of $(d/dt - \beta)^{\deg P_\beta + 1}$ over all exponents but one. If that term were nonzero, the resulting identity would be $e^{\alpha t} Q(t) = 0$, with Q a nonzero polynomial. Since the formal exponential has constant term one, it is a unit, so this is impossible. This proves independence and makes every grouped polynomial $\sum_{\sum a_i \lambda_i = \alpha} P_a(t)$ zero.

Setting $t = 0$ in these coefficient-sum identities shows that f vanishes on the full D_0, since its cosets are exactly the equal-exponent classes. Setting all $z_i = 1$ shows $f(\exp(tX_n)) = 0$ as a polynomial. Thus $D_0 \subset G_K$, and $\exp(tX_n) : \mathbf{G}_a \to G_K$ is a polynomial group homomorphism. The element $z_i = 1 + \varepsilon \lambda_i$ in the dual numbers satisfies every relation $z^a = 1$, $a \in N$: products and inverses give $z^a = 1 + \varepsilon \sum_i a_i \lambda_i$. It is therefore a D_0-tangent vector whose matrix is X_s. Differentiating the polynomial one-parameter subgroup gives X_n. Both components consequently belong to $\operatorname{Lie} G$.

We verify differential functoriality, including the positive-characteristic restricted operation. For a homomorphism $\phi : G \to H$, the invariant derivations satisfy $D_X \phi^* = \phi^* D_{d\phi(X)}$, by the coproduct identity for ϕ^* and evaluation of the tangent vector. Taking p-th powers of this identity and evaluating at the identity proves $d\phi(X^{[p]}) = d\phi(X)^{[p]}$. In a closed matrix group the same invariant-derivation calculation as in Lemma 4.H identifies this operation with the matrix p-th power; no smoothness of H is required. Differentials preserve brackets too: expand the commutator of $I + \varepsilon X$ and $I + \eta Y$ over $K[\varepsilon, \eta]/(\varepsilon^2, \eta^2)$; its mixed coefficient is $[X, Y]$, and a group homomorphism preserves this commutator. Thus the images of our commuting components commute.

In characteristic p, the image of X_n is nilpotent, since $X_n^{p^q} = 0$ for large q and the differential preserves these powers. Let L be the span of $X_s, X_s^p, X_s^{p^2}, \ldots$ in $\operatorname{Lie} G$. It is a finite-dimensional

commutative restricted subalgebra. In a diagonalization its p-operation is injective, since a diagonal matrix whose p-th power is zero is zero. Over the perfect field K, an injective semilinear endomorphism of a finite-dimensional space is bijective: its images of a basis are independent by taking p-th roots of any linear relation, and hence form a basis. Lemma 4.H supplies a multiplicative-type height-one subgroup with Lie algebra L. Restrict ϕ to this subgroup. The full diagonalizable weight decomposition of AG-GS, *Diagonalizable groups*, Theorem 3.1 shows that its differential acts diagonally on every target representation. In particular $d\phi(X_s)$ is semisimple. This includes $L = 0$.

In characteristic zero, restrict ϕ to the full diagonalizable subgroup D_0 just constructed. Its weight decomposition again makes $d\phi(X_s)$ semisimple. The polynomial homomorphism $\phi(\exp(tX_n))$, in a faithful target representation, has the form $B(t)$ with $B(0) = I$ and $B(t + s) = B(t)B(s)$. The coefficient calculation in Lemma 4.J gives $B(t) = \sum_j t^j Y^j/j!$, with $Y = B'(0) = d\phi(X_n)$ nilpotent, since $B(t)$ is polynomial. In either characteristic the images are a commuting semisimple and nilpotent pair with sum $d\phi(X)$. The uniqueness proved at the start identifies them as its Jordan components. Applying this to two representations and their direct sum gives independence of faithful representation.

For perfect k and rational X, the finite-normal-separable-extension argument at the end of Lemma 4.J applies verbatim to the algebraic entries of these unique matrix components: all field automorphisms fixing X fix both components, so their coefficients lie in k. The linear equations defining $\operatorname{Lie} G$ descend through faithful field extension, hence both are rational Lie vectors. Finally conjugation on $\operatorname{Lie} G$ is an algebraic linear representation: differentiate the regular conjugation morphism at the identity in its second variable, or use the matrix formula $Y \mapsto gYg^{-1}$ in a faithful embedding. Therefore the functoriality just proved applies to the adjoint representation and gives $\operatorname{ad}(X_s)$ and $\operatorname{ad}(X_n)$ as the semisimple and nilpotent components of $\operatorname{ad}(X)$. □

Lemma 4.S (field character subgroups, quotients and closed embeddings). Let k be any field and M an abelian group. Every closed subgroup of $D_k(M)$ is uniquely $D_k(M/N)$, with its natural embedding, for a subgroup $N \subset M$. If M is finitely generated, its fppf quotient by that subgroup is $D_k(N)$; the quotient map is faithfully flat of finite presentation with that full schematic kernel. In particular the quotient of a split torus by any closed subgroup is a split torus. A homomorphism from a split torus to an affine algebraic k-group with trivial schematic kernel is a closed immersion. No characteristic or reducedness hypothesis is imposed on the closed subgroup or the target.

Proof. First, distinct group-like elements in a coalgebra over a field are linearly independent. In a shortest nonzero relation $\sum_{i=1}^{s} a_i g_i = 0$, all coefficients are nonzero. A one-term relation is impossible because $\epsilon(g_i) = 1$. The first $s - 1$ elements are independent by minimality. Apply comultiplication and replace $a_s g_s$ in the first factor by $-\sum_{i<s} a_i g_i$. This gives

$$\sum_{i<s} a_i g_i \otimes (g_i - g_s) = 0.$$

Coefficient functionals for those independent first factors give $g_i = g_s$, a contradiction.

For a closed subgroup $H \subset D_k(M)$, the coordinate algebra is a Hopf quotient $k[M] \twoheadrightarrow C$. The images g_m of the basis elements e^m are nonzero unit group-like elements, with $g_m g_{m'} = g_{m+m'}$. Their equality relation is $g_m = g_{m'}$ precisely when $m - m' \in N$, where $N = \{m : g_m = 1\}$ is a subgroup. Their distinct images span C, by surjectivity, and are independent by the preceding paragraph. Thus $C = k[M/N]$, with the group-algebra Hopf operations and the specified quotient map. This proves the assertion, uniqueness and its full scheme structure, including finite nonreduced subgroups.

For any subgroup $N \subset M$, restriction of characters defines $D_k(M) \to D_k(N)$ through $k[N] \hookrightarrow k[M]$. Its kernel has coordinate algebra

$$k[M]/(e^n - 1 : n \in N) = k[M/N].$$

Indeed each difference of basis elements in one coset is $e^{m'}(e^{m-m'}-1)$; every linear combination with zero coefficient sum in each coset is a sum of these differences. The resulting coset basis proves the equality of algebras. Choose a representative r_c of each coset. The decomposition

$$k[M] = \bigoplus_{c \in M/N} k[N] e^{r_c}$$

makes it a free $k[N]$-module on a nonempty basis. Exact tensor with this free module preserves injections, and its direct-sum formula detects a nonzero module. The map is therefore faithfully flat, also after every base change.

We give the finite-generation argument needed for finite presentation and for the torus assertion. Every subgroup $L \subset \mathbf{Z}^r$ is finite free. Induct on r. The image of its first-coordinate projection is either zero or $d\mathbf{Z}$, where $d > 0$ is its least positive element; division with remainder proves that description. In the zero case apply induction in $\mathbf{Z}^{r-1}$. Otherwise choose $z \in L$ with first coordinate d. Subtracting the appropriate multiple of z puts every element in the projection kernel, and the intersection of $\mathbf{Z}z$ with that kernel is zero. Induction gives the finite free direct sum. For a subgroup of any finitely generated abelian M, lift it to a subgroup of a finite free group surjecting onto M; the images of the just-constructed finite basis generate it. Thus N is finitely generated when M is. The group algebras $k[N]$ and $k[M]$ are generated by the finitely many basis characters of group generators and their inverses. The Hilbert basis proof of S01 Lemma 1.3 makes $k[N]$ Noetherian, and a finite polynomial presentation of $k[M]$ over it has finitely generated kernel. The quotient map is consequently of finite presentation.

It is an fppf cover and its full torsor relation is

$$D_k(M) \times D_k(M/N) \simeq D_k(M) \times_{D_k(N)} D_k(M), \qquad (g,h) \longmapsto (g, gh),$$

with inverse $(g, g') \mapsto (g, g^{-1}g')$. These maps are inverse on every test scheme by the kernel calculation. Local lifts along this cover and that relation identify the fppf quotient with $D_k(N)$. For $M = \mathbf{Z}^r$, the proved finite freeness of N makes it a split torus, as asserted.

Finally let $T = D_k(L)$, with L finite free, map to an affine algebraic group K with trivial schematic kernel. The full linearity construction of Section 2 embeds K in $\mathrm{GL}(V)$. The weight decomposition of AG-GS, *Diagonalizable groups*, Theorem 3.1 gives a basis of V in which the T-action is diagonal with characters $\lambda_1, \dots, \lambda_a \in L$. Its full kernel is $D_k(L/\Lambda)$, where Λ is the subgroup generated by these characters, by the preceding kernel calculation. Triviality forces $\Lambda = L$: a nonzero quotient group has at least two independent basis elements in its group algebra, so its diagonalizable scheme is nontrivial even when its geometric point set has only one point. The matrix coefficients supply e^{λ_i}, and the inverse determinant supplies $e^{-\sum_i \lambda_i}$. For each i,

$$e^{-\lambda_i} = e^{-\sum_j \lambda_j} \prod_{j \neq i} e^{\lambda_j}.$$

They generate all of $k[L]$. Hence $k[\mathrm{GL}(V)] \to k[T]$ is surjective and $T \to \mathrm{GL}(V)$ is a closed immersion. It factors through the closed K, so $k[K] \to k[T]$ is also surjective. This proves the final assertion on full schemes. □

Lemma 4.1. If a smooth connected affine k-group has a nontrivial k-torus, and geometrically maximal tori exist for groups of smaller dimension, then it has a geometrically maximal k-torus.

Proof. Let $M \subset G$ be a nontrivial k-torus and put $C = C_G(M)^0$. Lemma 3.1 makes the centralizer smooth. The identity-component and scalar-extension calculation of AG-RG-S08 Lemma G.0.3 makes C a smooth geometrically connected affine group containing M; the torus M is central in C. We construct the affine quotient before applying induction.

The detecting-line construction in Section 2 works over k: its finite coproduct, ideal-generation and exterior-power calculations use no algebraic closedness. It gives a finite-dimensional C-representation

V and a line kv whose full schematic stabilizer is M. Let $W \subset V$ be the subrepresentation generated by v. Explicitly, write the coaction of v as $\sum_i w_i \otimes a_i$ with linearly independent $a_i \in k[C]$; the counit puts v in the span of the w_i, and coassociativity puts the coaction of each w_i in that span tensored with $k[C]$. Thus their span is the required W. The action of M on the line is a character $\chi : M \to \mathbf{G}_m$. Centrality says, on the universal $C \times M$ point, that $m(cv) = c(mv) = \chi(m)cv$. Comparing the independent coaction coefficients proves that M acts by this same scalar on all of W, on every test algebra.

Let $\rho : C \to \mathrm{GL}(\mathrm{End}_k W)$ act by conjugation through the representation on W. Its kernel is exactly M. Indeed a matrix which commutes with every matrix unit is scalar over any k-algebra: the diagonal units kill its off-diagonal entries, and the other matrix units make all diagonal entries equal. An invertible scalar matrix has a unit scalar. Such a matrix fixes the line kv, so its preimage in C belongs to the schematic stabilizer M. Conversely the scalar action just proved makes every M-point lie in the kernel. This avoids any assumption that a projective linear image is already the desired quotient.

Let H be the schematic image of ρ. It is an affine finite-type group: its coordinate algebra is the finitely generated Hopf subalgebra

$$B = \mathrm{im}(k[\mathrm{GL}(\mathrm{End}\, W)] \longrightarrow k[C]),$$

and the kernel of the coordinate map is a Hopf ideal. To check the full Hopf structure, write the ambient coordinate algebra as P and its kernel as J. The identity $\ker(P \otimes_k P \to B \otimes_k B) = J \otimes_k P + P \otimes_k J$ follows by taking vector-space complements to J. The injection $B \otimes_k B \hookrightarrow k[C] \otimes_k k[C]$ and the group homomorphism identity make comultiplication preserve that kernel; the identity and inverse maps give the counit and antipode identities. Thus all Hopf operations descend. For every field extension L/k, the injection $B \hookrightarrow k[C]$ remains injective after tensoring. Since $C_{\bar{k}}$ is smooth connected and therefore integral by Lemma 2.L, $H_{\bar{k}}$ is integral as well. The reduced-target image proof in AG-GS, *Quotients and torsors*, Lemma 11.1c applies over $\bar{k}$ and makes $C_{\bar{k}} \to H_{\bar{k}}$ faithfully flat of finite presentation. Its full kernel is $M_{\bar{k}}$. The integral algebraic group $H_{\bar{k}}$ is smooth by the reduction argument of Lemma 2.L. Faithfully flat descent, S04 Lemmas A1.3 and A1.3a, consequently makes $C \to H$ faithfully flat of finite presentation and H smooth over k. Its geometric fibre is connected, because it is the surjective image of $C_{\bar{k}}$.

For every test scheme, two C-points have the same image in H exactly when their quotient lies in M. Multiplication and quotient give inverse maps

$$C \times M \xrightarrow{\sim} C \times_H C, \qquad (c, m) \longmapsto (c, cm), \qquad (c, d) \longmapsto (c, c^{-1}d).$$

The faithfully flat map $C \to H$ therefore is an M-torsor and identifies H with the fppf quotient C/M. Applying the full orbit-dimension calculation of S08 Lemma G.4.D over $\bar{k}$ gives

$$\dim H = \dim C - \dim M < \dim G.$$

Dimension is unchanged by field extension by that same lemma. The induction hypothesis now supplies a geometrically maximal k-torus $Q \subset H$.

Put $E = C \times_H Q$, an affine closed subgroup of C. Its map to Q is the base change of the M-torsor. Every geometric fibre is a nonempty finite-type M-torsor over an algebraically closed field. The weak Nullstellensatz gives a rational point; translating by that point identifies the full fibre scheme with the split torus M over that field. The map is flat of finite presentation; the fibre criterion of AG-FSE, *Smooth morphisms*, Theorem 3.1 makes $E \to Q$ smooth. Hence E is smooth over k. It is geometrically connected. To check this over $\bar{k}$, its smooth components are disjoint open and closed by Lemma 2.L. Their images in $Q_{\bar{k}}$ are open by S08 Lemma G.4.F. A connected fibre cannot meet two such components, so those images are disjoint; surjectivity makes them cover the connected $Q_{\bar{k}}$. There is only one component.

The quotient Q is commutative and the kernel M is central, so the commutator of E takes values in M, and the next commutator is trivial. Thus $E_{\bar{k}}$ is a smooth connected solvable group. Theorem 2.K writes it as its smooth connected unitriangular radical semidirect a torus. Every character of that radical is trivial by Lemma 2.A, so its map to $Q_{\bar{k}}$ is trivial and it lies in $M_{\bar{k}}$. Lemma 2.A also makes its intersection with this diagonalizable group trivial, with full scheme structure. The radical is therefore the identity, and $E_{\bar{k}}$ is a torus. The complete field splitting proof in AG-RG-S08, Lemma G.0.T gives a finite separable extension over which E is the split torus with its free geometric character group. Thus E is a k-torus.

Finally suppose a $\bar{k}$-torus $A \subset G_{\bar{k}}$ contains $E_{\bar{k}}$. It contains $M_{\bar{k}}$, hence centralizes it; its connected image lies in the identity component $C_{\bar{k}}$, with full scheme factorization from S08 Lemma G.0.1. Lemma 4.S represents the fppf quotient $A/M_{\bar{k}}$ as a split torus: its character group is the kernel of the surjection $X^*(A) \to X^*(M_{\bar{k}})$, a finite free subgroup of $X^*(A)$. The full torsor relation for $C \to H$ makes the map $A \to H_{\bar{k}}$ constant on the M-fibres, so S04 Lemma A1.5 descends it to this quotient.

Its induced map $A/M_{\bar{k}} \to H_{\bar{k}}$ has trivial schematic kernel. On any test scheme, lift a kernel point fppf-locally to A. Its image in H is the identity, so that lift lies in the full kernel M of $C \to H$; its quotient point is therefore the identity. Faithful descent of this equality proves triviality before the cover, including nilpotent test schemes. The closed-embedding assertion of Lemma 4.S now makes its image a closed torus in $H_{\bar{k}}$. It contains $Q_{\bar{k}}$, because $E \subset A$ and $E \to Q$ is fppf-surjective. Geometric maximality of Q makes that image $Q_{\bar{k}}$. Consequently the map of A factors through $C_{\bar{k}} \times_{H_{\bar{k}}} Q_{\bar{k}} = E_{\bar{k}}$. Since the opposite inclusion was assumed, $A = E_{\bar{k}}$ as closed subgroup schemes. Thus E is geometrically maximal in G, proving the lemma. □

Lemma 4.Q (the central affine quotients used in the induction). Let G be a smooth geometrically connected affine group over a field k, and let $D \subset G$ be any closed central subgroup. Its fppf quotient is represented by a smooth geometrically connected affine k-group H. The map $G \to H$ is faithfully flat of finite presentation with full kernel D, and is a D-torsor. If D is finite, this map is finite locally free of rank $\dim_k k[D]$.

Proof. Smoothness makes G locally of finite type, and S08 Lemma G.0.3 makes its connected identity component quasi-compact; thus G is of finite type. The complete detecting-line construction of Section 2 gives a finite-dimensional representation V and a nonzero vector v whose line has full schematic stabilizer D. Let W be the cyclic subrepresentation generated by v. Concretely write its coaction as $\sum_i w_i \otimes a_i$ with independent $a_i \in k[G]$. The counit puts v in their span; coassociativity and coefficient functionals make that span a subrepresentation. The character χ of D on kv acts by the same scalar on W: on the universal $G \times D$ point, centrality gives $d(gv) = g(dv) = \chi(d)gv$, and independence of the a_i compares every coefficient. This is an equality on every test algebra, including nilpotent ones.

Let $\rho : G \to \mathrm{GL}(\mathrm{End}_k W)$ be conjugation through the representation on W. Its kernel is D. A matrix commuting with all matrix units is scalar over any algebra: the diagonal units force every off-diagonal entry to vanish and the other units make all diagonal entries equal. Its invertibility makes that scalar a unit. Such an element fixes kv in V, hence belongs to D; conversely the scalar action of D just proved makes it lie in the kernel.

Put $B = \mathrm{im}(k[\mathrm{GL}(\mathrm{End}\, W)] \to k[G])$ and $H = \mathrm{Spec}\, B$. This is a finite-type Hopf algebra and defines an affine group. To check the Hopf ideal explicitly, the injection $B \hookrightarrow k[G]$ remains injective on tensor squares over the field. Write $P = k[\mathrm{GL}(\mathrm{End}\, W)]$ and J for the kernel. Exact field tensor gives $\ker(P \otimes P \to B \otimes B) = J \otimes P + P \otimes J$. Comultiplication of $u \in J$ maps to zero in $k[G] \otimes k[G]$, hence in its injected $B \otimes B$; therefore $\Delta(u) \in J \otimes P + P \otimes J$. Compatibility of the counit and antipode gives $\epsilon(J) = 0$ and $S(J) \subset J$. The group operations therefore descend to B. Exact field tensor also makes $B_{\bar{k}}$ the image algebra after extension. Since $G_{\bar{k}}$ is smooth connected, its algebra is a domain by Lemma 2.L; hence $B_{\bar{k}}$ is a domain.

The complete integral-target image argument of AG-GS, *Quotients and torsors*, Lemma 11.1c

applies to $G_{\bar{k}} \to H_{\bar{k}}$, whose coordinate map is injective. It is faithfully flat of finite presentation. S04 Lemma A1.3 descends flatness, and Lemma A1.3a descends finite presentation along $B \to B \otimes_k \bar{k}$. Faithfulness also descends: if tensoring a B-module with $k[G]$ gives zero, extending to $\bar{k}$ and using the faithful map upstairs makes its extended module zero, and faithful field tensor then makes the original module zero. Thus $G \to H$ is faithfully flat of finite presentation. The integral group $H_{\bar{k}}$ is smooth by Lemma 2.L and connected as the surjective image of $G_{\bar{k}}$. The full field smoothness-descent proof of S08 Lemma G.2.1 makes H smooth over k; the same geometric image argument proves geometric connectedness.

The kernel calculation gives the full torsor relation

$$G \times D \xrightarrow{\sim} G \times_H G, \qquad (g,d) \longmapsto (g,gd), \qquad (g,h) \longmapsto (g,g^{-1}h).$$

Both are scheme morphisms and are inverse on every test scheme, because equality of the two images is exactly membership of their quotient in D. The fppf cover $G \to H$, with this relation, identifies its sheaf quotient by D with H: every H-point lifts fppf-locally, and two lifts differ uniquely by D.

If D is finite, pull back the quotient map along its own faithfully flat cover $G \to H$. Its coordinate module becomes $k[G] \otimes_k k[D]$, free of rank $\dim_k k[D]$ over $k[G]$. The finite generation, finite presentation and flatness descent in S04 Lemma A1.3 makes the original B-module $k[G]$ finite locally free. The rank can be read after the same cover, and is that constant dimension. Thus the quotient map is finite locally free of precisely that rank. □

Lemma 4.A (a commutative finite extension of multiplicative-type groups). A finite commutative k-group E in an fppf exact sequence

$$1 \longrightarrow N \longrightarrow E \longrightarrow Q \longrightarrow 1$$

with N and Q of multiplicative type is of multiplicative type.

Proof. Work first over an algebraic closure. Both end groups are diagonalizable, and their invariant functors are exact by the complete weight decomposition in AG-GS, *Diagonalizable groups*, Theorem 3.1. For a finite-dimensional E-representation W, the space W^N is E-stable, since N is normal. Its action factors through Q on all test schemes. Indeed the kernel identity gives the full torsor relation $E \times N \simeq E \times_Q E$, with inverse $(a,b) \mapsto (a, a^{-1}b)$. The fppf-surjective quotient becomes N times its base after an fppf lift. The finite Q has a finite discrete spectrum by Artinian decomposition. Choose finitely many affine pieces of such lifts covering those points and combine them into one affine faithfully flat cover. The trivial projection is flat; flat descent from AG-CA, *Faithful flatness and the local criterion for flatness*, Corollary 3.2 makes the quotient flat, and its surjectivity makes it faithfully flat. Each matrix coefficient of the action on W^N has equal pullbacks along that relation, because N acts trivially. The faithfully flat equalizer proved in AG-RG-S04, Lemma A1.0 descends these coefficients to Q; its algebra and group identities descend by the same faithful comparison. Hence

$$W^E = (W^N)^Q,$$

and E-invariants are exact.

For a subrepresentation $V \subset W$, the map

$$\mathrm{Hom}(W/V, W) \longrightarrow \mathrm{Hom}(W/V, W/V)$$

is a surjective equivariant linear map: lift the images of a vector-space basis to prove surjectivity, and use the representation $(g \cdot f)(v) = gf(g^{-1}v)$ on each Hom space for equivariance. Exactness of invariants lifts its invariant identity to an equivariant section. A nonzero finite-dimensional representation has a simple subrepresentation by choosing a nonzero subrepresentation of smallest dimension. The section splits it off, and induction on dimension decomposes every finite-dimensional representation into simples.

Put $C = k[E]$ and $H = C^\vee$. Choose a basis c_j of C with dual basis h_j of H. A coaction λ defines $hw = (1 \otimes h)\lambda(w)$, and an H-action defines $\lambda(w) = \sum_j (h_j w) \otimes c_j$. These formulas are inverse by the dual-basis identity. Pairing both tensor factors with that basis identifies coassociativity with associativity of the action, and the counit identity with its unit identity. Thus C-comodules and H-modules are exactly the same representations. Since E is commutative, its comultiplication on C is cocommutative, so the finite-dimensional algebra H is commutative. Its regular module is semisimple by the preceding paragraph. Decompose it into simple ideals $H = \bigoplus_i L_i$, and project one to $e_i \in L_i$. For $x \in L_i$, the products xe_j lie in $L_i \cap L_j$, hence vanish for $j \neq i$, while $x = \sum_j xe_j = xe_i$. Thus e_i is the identity of L_i, the e_i are orthogonal idempotents and each commutative L_i has no proper nonzero ideal. Every nonzero element is invertible, so each L_i is a field. It is a finite extension of the algebraically closed base, and therefore equals that base. Hence $H = k^M$ for a finite set M.

The Hopf maps of H give a group structure on this finite set: a map between finite products of the field sends a standard idempotent to a sum of standard idempotents, and hence specifies a map between their indexing sets. The Hopf identities give associativity, identity and inverse. Commutativity of C makes the dual comultiplication cocommutative, so M is abelian. Dualizing its idempotent basis gives the group-algebra Hopf algebra $C = k[M]$: basis products encode the multiplication of M, and each basis element has comultiplication $e^m \otimes e^m$ and counit one. Thus E is diagonalizable over the algebraic closure. The full finite-separable splitting proof of AG-RG-S08, Lemma G.0.T now proves the assertion over the original k, including imperfect fields. □

Lemma 4.B (composite kernels stay central). Suppose G is smooth geometrically connected, N is a finite connected central multiplicative-type subgroup, and Q is a finite connected central multiplicative-type subgroup of G/N. Its inverse image E in G is finite connected central and of multiplicative type.

Proof. The kernel identity gives $E \times N \simeq E \times_Q E$, and fppf-local lifts identify $E \to Q$ with the projection from N times the base. The finite Q has a finite discrete spectrum by Artinian decomposition, so finitely many affine pieces of these lifts give a single affine faithfully flat cover. Effective affine descent in S04 Lemma A1.2, with comparison morphisms from Lemma A1.5, represents this torsor by an affine scheme. Its coordinate module becomes free of rank $\dim_k k[N]$ after that cover. S04 Lemma A1.3 descends finite local freeness, and the local lifts give surjectivity. Thus $E \to Q$ is finite faithfully flat and E is finite over k. After algebraic closure, Lemma G.0.2 of S08 makes the finite connected N and Q connected; each has one underlying point and has its identity as a rational point. The single geometric fibre of $E \to Q$ is a nonempty finite-type N-torsor. A rational point supplied by the weak Nullstellensatz identifies its full scheme with N. Thus E is geometrically connected, hence connected.

Since Q is commutative and N is central, the commutators in E take values in N. Replacing either lift by its product with an element of N leaves that commutator unchanged. The fppf comparison proof in S04 Lemma A1.5 therefore descends the commutator to $b : Q \times Q \to N$. The identities $[aa', c] = [a, c][a', c]$ and $[a, cc'] = [a, c][a, c']$, when the commutators are central, make b a homomorphism in each variable. They follow by inserting $a'^{-1}a^{-1}$ and the other inverse products into the commutator formula and commuting only the central commutators.

Over the algebraic closure write $Q = D(M_Q)$, $N = D(M_N)$, and put $R = k[Q]$. As a family of homomorphisms in the second variable, b is a Hopf map $R[M_N] \to R[M_Q]$. The coefficient calculation in S08 Lemma G.0.T makes the image of each character a sum of basis characters with orthogonal idempotent coefficients summing to one. A connected ring spectrum has only idempotents zero and one: $D(c)$ and $D(1-c)$ are disjoint clopen subsets covering it, and a nonzero idempotent is not nilpotent, so has a nonempty principal open by S04 Lemma P0.1. Consequently the image is one basis character, independent of the first variable even along nilpotent test directions. Evaluation at that variable's identity gives the trivial homomorphism. Thus every character maps to one and b is trivial as a scheme map. Faithful scalar extension proves that E is commutative over the original field. Lemma 4.A makes E of multiplicative type.

Conjugation preserves E, as the preimage of the normal Q. Over the algebraic closure write $E = D(M_E)$. On any nonempty affine open of G, conjugation is a family of group homomorphisms of this diagonalizable group. That open is irreducible by S08 Lemma G.0.3, hence connected. The same coefficient argument makes every character image a fixed basis character throughout the open, including its nilpotents. Any two such opens meet and their families agree there, so their basis characters agree; for example evaluate at a rational point of the nonempty intersection. The open containing the identity gives the identity homomorphism. Thus conjugation is the identity family on all of G. Faithful field descent proves centrality schematically over k.

In the affine case used in Theorem 4.2, the full central-quotient proof of Lemma 4.Q applies to the now-central E. It represents G/E as a smooth geometrically connected affine group and makes $G \to G/E$ finite locally free of rank $\dim_k k[E]$. This additional affineness assertion uses the affine hypothesis; the central-kernel statement above retains its stated generality. □

Lemma 4.C (descending the torus in a central affine group). Let Z be an affine commutative algebraic k-group. The torus factor of $(Z^0_{\bar{k}})_{\mathrm{red}}$ is defined over k. In particular, a nontrivial such torus descends even if k is imperfect.

Proof. Let n range over the positive integers invertible in k, and let $I_n \subset A = k[Z]$ define $Z[n]$. Its coordinate algebra is finite and is a product of finite separable field extensions. Here is the full calculation. The differential of multiplication at the identity is addition: its restrictions to either tangent summand are the identity, so its linear map is their sum. Iterating gives $d[n] = n$, whose kernel is zero. Thus $Z[n]$ has zero tangent space at the identity and, by translation, at every rational point over an algebraic closure. On each of its Noetherian local rings at such a point, $\mathfrak{m}/\mathfrak{m}^2 = 0$; Nakayama gives $\mathfrak{m} = 0$. Every prime lies under a maximal ideal, and the local field there has no smaller prime, so every prime is maximal. The zero-dimensional Noetherian and Artinian decomposition proofs of AG-CA, *Noetherian and Artinian rings*, Theorems 3.3 and 4.2 make the geometric coordinate algebra a finite product of copies of $\bar{k}$, using the weak Nullstellensatz for their residue fields. The finite-algebra argument in S08 Lemma G.0.T now applies: independence of vectors proves finite dimensionality before extension; injectivity proves reducedness; and an inseparable element in a field factor would give a nonzero nilpotent after extension. Thus all its finite field factors are separable. In particular every $Z[n]$ is geometrically reduced.

Define the ideal and closed subscheme directly over the original field by

$$I = \bigcap_{\substack{n \in \mathbf{Z}_{>0} \\ n \cdot 1_k \neq 0}} I_n, \qquad Y = \operatorname{Spec}(A/I).$$

Here the indexing condition means that the integer n is invertible in k. This intersection commutes with every field extension L/k. Write an element of $A \otimes_k L$ as $\sum_i f_i \otimes a_i$, with the finitely many a_i linearly independent over k. Its image in $(A/I_n) \otimes_k L$ is zero precisely when each f_i lies in I_n, by applying linear functionals selecting these coefficients. Membership in every extended ideal is therefore precisely membership in $I \otimes_k L$. Consequently $(A/I) \otimes_k L$ embeds in the product of the reduced algebras $(A/I_n) \otimes_k L$. It is reduced. Thus Y is geometrically reduced, and the closure construction commutes with all field extensions without a perfectness assumption.

Over $\bar{k}$, put $H = \bigcup_n Z[n](\bar{k})$. These finite schemes are reduced, so their rational-point evaluations detect their functions; the preceding intersection equality says exactly that $Y_{\bar{k}}$ is the reduced schematic closure of H. The set H is a subgroup, since Z is commutative and the product of elements killed by n and m is killed by nm. It is schematically dense in $Y_{\bar{k}}$, and $H \times H$ is schematically dense in its product. For the latter assertion write a tensor as $\sum_i b_i \otimes c_i$ with the c_i independent. If its values on $H \times H$ vanish, evaluation at each first-coordinate point gives an element vanishing on all second-coordinate points, hence zero. Independence makes each b_i vanish on H, hence zero. Multiplication, inverse and identity therefore restrict to $Y_{\bar{k}}$: the equations of Y pull back to zero on those dense sets. The same equations hold before extension by faithful field tensor. Thus Y is a k-subgroup of Z, with its full scheme structure.

Over $\bar{k}$, put $R = (Z^0_{\bar{k}})_{\mathrm{red}}$. Reduction is a subgroup here: products of reduced finite-type schemes are reduced by S08 Lemma G.4.1, so the group operations factor through it. This connected commutative group is smooth by Lemma 2.L. The complete solvable structure theorem 2.K gives $R = T_Z \times U_Z$, since commutativity makes the semidirect action trivial. The unitriangular factor has no nontrivial torsion of order invertible in the field. In characteristic p, a matrix $1 + N$, with N nilpotent, has p-power order by $(1 + N)^{p^r} = 1 + N^{p^r}$. In characteristic zero, if $(1 + N)^n = 1$, its binomial expansion is $N(nI + NB) = 0$ for a polynomial B in N; the second factor is invertible by a finite geometric sum, so $N = 0$. Hence $H \cap R$ consists exactly of the invertible-order torsion points of T_Z.

These points are schematically dense in T_Z. For a nonzero Laurent polynomial choose an invertible integer n larger than every absolute coordinate difference between its finitely many exponent vectors. Those vectors are distinct modulo $n\mathbf{Z}^r$, so their monomials stay independent in the coordinate algebra of $T_Z[n]$. Its restriction is nonzero, and that finite reduced scheme has a rational point where it is nonzero. The identity component of $Z_{\bar{k}}$ is open and closed by S08 Lemma G.0.3, so closure in that component is closure of the intersection with it. Its rational points factor through R. It follows that the intersection of $Y_{\bar{k}}$ with that component has exactly the underlying space of T_Z. Both are reduced closed schemes there, hence equal as schemes; equivalently both defining ideals are the radical ideal of that same closed subset, by S04 Lemma P0.1. This intersection is connected, so $Y^0_{\bar{k}} = T_Z$.

Finally put $W = Y^0$, with the canonical component structure of S08 Lemma G.0.3. It is an open and closed k-subgroup, and that lemma's full field-base-change assertion gives $W_{\bar{k}} = T_Z$ as schemes. It is affine and of finite type, being closed in the affine finite-type Y. The complete finite-separable splitting proof of S08 Lemma G.0.T makes W a k-torus with that geometric fibre. This is the required descent. □

Lemma 4.O (one order bound for a torus-free reduced centre). Let Z be a connected commutative affine algebraic group over an algebraically closed field K, retaining its possible nilpotents. If its reduction R has no nontrivial torus, there is one integer B bounding the order of every finite multiplicative-type closed subgroup $E \subset Z$.

Proof. Put $A = K[Z]$ and let J be its nilradical. S01 Lemmas 1.0 and 1.3 make A Noetherian, so J is generated by finitely many nilpotent elements u_i. If $u_i^{n_i} = 0$, every monomial of degree $1 + \sum_i (n_i - 1)$ contains one of these zero powers. Consequently choose $N \geq 1$ with $J^N = 0$. Let d generators generate the augmentation maximal ideal $\mathfrak{m}$ of A.

The reduction is a connected commutative subgroup: S08 Lemma G.4.1 makes $R \times R$ reduced, so the multiplication and inverse maps kill the nilradical and factor through R. Its underlying space is the connected space of Z. Lemma 2.L makes this reduced group smooth. The full solvable structure of Theorem 2.K and the hypothesis make R unitriangular. Over K, the subgroup E is diagonalizable by the definition of multiplicative type and S08 Lemma G.0.T. Its full intersection $E \cap R$ is diagonalizable by Lemma 4.S. Lemma 2.A says that a diagonalizable subgroup of a unitriangular group is the trivial scheme, including when that subgroup is finite and nonreduced.

Write $C = K[E]$. The intersection just computed has coordinate ring $C/JC = K$, and the identity map shows that JC is the augmentation maximal ideal $\mathfrak{n}$ of C. It is generated by the images of the d generators of $\mathfrak{m}$, since $A \twoheadrightarrow C$. Moreover $\mathfrak{n}^N = 0$. For $0 \leq j < N$, every class in $\mathfrak{n}^j/\mathfrak{n}^{j+1}$ is a K-linear combination of the at most d^j words of length j in those generators: a coefficient in C can be reduced modulo $\mathfrak{n}$, and its remainder raises the degree. For $j = 0$ the empty word gives the one-dimensional quotient. The finite filtration therefore gives the uniform bound

$$|E| = \dim_K C \;\leq\; B := \sum_{j=0}^{N-1} d^j.$$

Take $d^0 = 1$, including when $d = 0$. Both d and N depend only on Z, proving the assertion. □

Lemma 4.G (finite-field Borel and maximal-torus descent). A smooth connected affine group over $\mathbf{F}_q$ has a Borel subgroup and a geometrically maximal torus defined over $\mathbf{F}_q$.

Proof. Over $\overline{\mathbf{F}}_q$, choose a torus T of largest dimension, starting with the trivial torus. Dimensions are bounded by $\dim G$, and a proper inclusion of tori strictly increases dimension by the earlier domain dimension theorem; thus T is maximal. Choose a smooth connected solvable subgroup B of largest dimension containing T, exactly as in the proof of Theorem 2.2; it is a Borel. Borel–torus pairs are conjugate: Lemma 2.1 first conjugates their Borels, and Lemma 2.N conjugates the two maximal tori within the common Borel.

Let $\sigma(a) = a^q$ on the algebraic closure, and let $\sigma(B,T)$ denote the full pair obtained by this coefficient transport. Since G is defined over $\mathbf{F}_q$, it is again such a pair. Choose $g \in G(\overline{\mathbf{F}}_q)$ with $\sigma(B,T) = g(B,T)g^{-1}$. On rational points this coefficient transport is the Frobenius F. Lemma 4.L gives h with $h^{-1}F(h) = g^{-1}$. Consequently $F(h)g = h$, and the full transported pair $(B',T') = h(B,T)h^{-1}$ satisfies $\sigma(B',T') = (B',T')$, as closed subgroup schemes.

We give the finite descent details. The ambient coordinate algebra is finitely presented, and the two ideals have finite generating lists by S01 Lemmas 1.0 and 1.3. Their finitely many coefficients lie in a finite extension $L/\mathbf{F}_q$, since every element of the algebraic closure is algebraic. Include the finitely many coefficients of their Hopf identities and finite algebra presentations as well, so they define closed subgroup schemes over L. If $[L : \mathbf{F}_q] = n$, then $|L| = q^n$. Multiplication by any nonzero $a \in L$ permutes the nonzero elements. Comparing their nonzero products gives $a^{q^n-1} = 1$. Therefore all elements are the q^n distinct roots of $X^{q^n} - X$; its derivative is -1. All minimal polynomials over $\mathbf{F}_q$ divide this split separable polynomial, so L is normal and separable. Its Frobenius has order exactly n: its n-th power is the identity, whereas for $0 < r < n$, being the identity would give q^n roots to $X^{q^r} - X$, whose degree is q^r. The embedding count and fixed-field proof following S08 Lemma G.5.2 show that its Galois group has order n; it is consequently generated by this Frobenius.

Both ideals over L are Frobenius-stable. Indeed their scalar extensions to the algebraic closure have that stability, and extension of field scalars is injective and detects equality of subspaces. The whole finite Galois group therefore preserves them. Apply the complete invariant-space proof of S08 Lemma G.5.2 to each ideal and to $L \otimes_{\mathbf{F}_q} \mathbf{F}_q[G]$. It identifies each ideal with the scalar extension of its invariant ideal in $\mathbf{F}_q[G]$. Multiplication and all Hopf maps descend by that lemma; their ideal identities are detected after faithful field extension. This gives closed subgroup schemes B_0, T_0 over $\mathbf{F}_q$ with the prescribed geometric fibres. S08 Lemma G.2.1 descends smoothness, and Lemma G.0.T makes T_0 a torus. Its geometric maximality and the Borel property of B_0 are exactly the properties of those fibres. This proves the lemma. □

Lemma 4.R (rational semisimple directions in positive characteristic). Let k be infinite of characteristic $p > 0$, and let G/k be a smooth affine algebraic group. If $G_{\bar{k}}$ has a nontrivial torus, then $\operatorname{Lie} G(k)$ contains a nonzero matrix-semisimple vector in any faithful linear representation.

Proof. Choose a closed $\mathbf{G}_m$ in that split geometric torus. In a weight basis of the faithful representation its integer weights generate $\mathbf{Z}$, by the full trivial-kernel calculation of Lemma 4.S. Thus at least one weight is not divisible by p. The Lie generator is a nonzero diagonal matrix, so its characteristic polynomial is not a pure power of its variable. At least one coefficient of the characteristic polynomial of the universal matrix in $\operatorname{Lie} G$ is therefore a nonzero polynomial over k.

A nonzero polynomial over an infinite field has a rational point where it is nonzero. Induct on the number of variables: retain a nonzero coefficient polynomial in the last variable, choose values of the preceding variables by induction, and then choose the last value outside the finitely many roots of the resulting nonzero one-variable polynomial. The degree root bound follows by the repeated linear-factor division in S02 Lemma 4.A. Choose the resulting $Y \in \operatorname{Lie} G(k)$. Its characteristic polynomial is not T^n, so Y is not nilpotent.

We give the exact matrix-power argument. Over $\bar{k}$, the characteristic polynomial splits. Cayley–Hamilton, S06 Section 1, and the polynomial Bézout/Chinese remainder proofs decompose the

space into the generalized eigenspaces: the pairwise coprime powers $(T-\lambda)^n$ give polynomial orthogonal projectors summing to one. On a projector's image $Y = \lambda I + N_\lambda$, with $N_\lambda^n = 0$. For $p^r \geq n$, the characteristic-p binomial identity gives

$$Y^{p^r} = \lambda^{p^r} I$$

on that entire summand. Thus Y^{p^r} is semisimple and nonzero, because some eigenvalue is nonzero. The matrix Y^{p^r} is rational over k simply because Y is, and Lemma 4.H puts every iterated p-power in $\operatorname{Lie} G$. This is the required vector; no descent of the semisimple component itself is used. □

Lemma 4.P (a torus in a finite infinitesimal central extension). Suppose P is an affine algebraic k-group and $q : P \to Q$ is a faithfully flat map of finite presentation to a nontrivial k-torus. Suppose its full kernel E is finite connected central and of multiplicative type. Then P contains a nontrivial k-torus. Smoothness of P and perfection of k are not assumed.

Proof. The full kernel relation gives $P \times E \simeq P \times_Q P$. Pulling q back along its own fppf cover therefore gives the finite free coordinate module $k[P] \otimes_k k[E]$, of rank $\dim_k k[E]$. S04 Lemma A1.3 descends this property, so q is finite locally free. The commutator of P takes values in E, since Q is commutative. Changing either input by a central E-point leaves it unchanged. S04 Lemma A1.5, applied to the full torsor relation in both inputs, descends it to $Q \times Q \to E$.

Over $\bar{k}$, Lemmas G.0.2 and G.0.T of S08 make E a connected finite diagonalizable scheme with one underlying point, its identity, and reduced algebra $\bar{k}$. The split $Q \times Q$ is reduced. Its map to E kills the nilradical and hence factors through that identity. The commutator is therefore zero after scalar extension, and faithful field extension detects this equality of affine coordinate maps before extension. Thus P is commutative.

Every geometric fibre of the finite surjection q is an E-torsor. The weak Nullstellensatz supplies a rational point on that nonempty finite fibre, and translation identifies its full scheme with E. Its underlying space has one point. This remains true after every base change. A finite map is universally closed by the integral closed-map proof in S04 Lemma P0.4; hence q is a universal homeomorphism. In particular $P_{\bar{k}}$ is connected and its connected reduction R still maps onto the underlying space of $Q_{\bar{k}}$.

The reduction R is a subgroup, because products of reduced affine groups over the algebraically closed field are reduced by S08 Lemma G.4.1. Lemma 2.L makes R smooth, and the full commutative solvable structure of Theorem 2.K writes it as a torus times a smooth connected unitriangular group U. Its map $U \to Q_{\bar{k}}$ is trivial by Lemma 2.A, so U lies in the finite full kernel $E_{\bar{k}}$. Therefore U is finite. Its coordinate algebra is a finite-dimensional domain by Lemma 2.L; it is a field, and algebraic closedness makes that field $\bar{k}$. Thus $U = 1$ as a scheme. It follows that R is a torus, and it is nontrivial since its underlying image is the nontrivial $Q_{\bar{k}}$. Finally apply Lemma 4.C to the commutative affine group P: it descends precisely this torus factor of its geometric reduced identity group to k. The descended torus is nontrivial and is a closed subgroup of P. □

Theorem 4.2 (Grothendieck). Every smooth connected affine group over a field k has a geometrically maximal torus defined over k.

Proof. Induct on the finite dimension of G, for all fields at once. S08 Lemma G.0.3 shows that a connected field-group is geometrically irreducible and that its identity component commutes with field extension. A smooth connected affine group of dimension zero is trivial: over an algebraic closure, Lemma 2.L makes its coordinate ring a finite-type domain; the dimension/transcendence-degree theorem, AG-CA, *Noether normalization and dimension*, Theorem 4.2, gives transcendence degree zero. Its generators are algebraic over the algebraically closed field, so the ring is a finite field extension and equals that field. Faithful field extension descends the trivial group. Its trivial torus is maximal. For the induction step all smaller-dimensional smooth connected affine groups, over any field, have the asserted torus.

If $G_{\bar{k}}$ has a central maximal torus, it is unique, since all maximal tori are conjugate. Put $Z = Z(G)$, with its scheme-theoretic structure. The torus factor of $(Z^0_{\bar{k}})_{\mathrm{red}}$ is that maximal torus: it contains

every central torus, and maximality leaves no larger torus. Lemma 4.C descends this factor to a k-torus. Its geometric fibre is the chosen maximal torus. This proves the theorem in the central-maximal-torus case, including the trivial-torus case; the same lemma will later descend a merely nontrivial central torus.

Suppose k is infinite and $G_{\bar{k}}$ has a noncentral torus. We will find a nontrivial k-torus and then use Lemma 4.1, whose smaller-dimension hypothesis is supplied by this induction. Choose a faithful representation by Section 2 and regard $\mathfrak{g} = \mathrm{Lie}(G)$ inside its matrix algebra. Lemma 4.K proves the entire additive Jordan assertions used here: geometric Lie membership, perfect-field rationality and functoriality for all algebraic-group homomorphisms, in particular the adjoint representation.

First assume that $\mathfrak{g}(k)$ contains a semisimple element X with $\mathrm{ad}(X) \neq 0$. In characteristic zero its full stabilizer $C_G(X)$, defined by the matrix equations $\mathrm{Ad}(g)X = X$, is a finite-type closed subgroup and is smooth by S08 Theorem G.3.4. Its Lie algebra is $\ker \mathrm{ad}(X)$: substitute $g = I + \varepsilon Y$ in the conjugation equation to obtain $X + \varepsilon[Y, X] = X$. Thus the stabilizer is proper. Its smoothness makes its defining ideal radical; over $\bar{k}$, the integral coordinate ring of G and the strict-specialization/dimension proofs in AG-CA, *Noether normalization and dimension*, Lemma 4.1 and Theorem 4.2, make its dimension strictly smaller than $\dim G$. Its identity component is smooth geometrically connected by S08 Lemma G.0.3, and contains the tangent vector X, since that component is open at the identity. After algebraic closure it is not unipotent: Lemma 2.A makes the Lie action in a faithful representation strictly upper triangular for a unipotent group, which excludes a nonzero semisimple X. Dimension induction supplies a geometrically maximal k-torus in this smaller-dimensional centralizer component. Lemma 2.D makes its geometric fibre positive-dimensional, so it is nontrivial.

In characteristic $p > 0$, a version of the same argument avoids smoothness questions for a Lie-element stabilizer. Put

$$\mathfrak{h} = \mathrm{span}_k\{X, X^{[p]}, X^{[p^2]}, \ldots\}.$$

This is a commutative restricted Lie subalgebra: in a diagonalization of X, its powers and their linear combinations remain diagonal, and the p-operation preserves their span. Over $\bar{k}$ this operation has zero kernel, since a diagonal matrix with zero p-th power is zero. Its image is a vector subspace because every scalar has a p-th root in $\bar{k}$; in finite dimension injectivity therefore gives an invertible semilinear operation. The full subgroup correspondence and multiplicative-type assertion of Lemma 4.H give a closed $H \subset G[F]$ with $H_{\bar{k}} \simeq \mu_p^d$, $d = \dim \mathfrak{h} > 0$.

The centralizer $C_G(H)$ is smooth by the complete fixed-subgroup theorem in AG-GS, *Diagonalizable groups*, Theorem 3.4, whose coefficient and invariant-cohomology proof includes nonreduced multiplicative-type acting groups. Its Lie algebra is the centralizer of $\mathfrak{h}$, by Lemma 4.H, so $\mathrm{ad}(X) \neq 0$ makes it proper. Its identity component contains the connected H, with the full factorization of S08 Lemmas G.0.1 and G.0.3. After algebraic closure that component cannot be unitriangular: Lemma 2.A excludes its nontrivial diagonalizable closed subgroup $H_{\bar{k}}$. Dimension induction therefore supplies a geometrically maximal torus in this smaller-dimensional smooth geometrically connected centralizer, and the no-torus criterion of Lemma 2.D makes it nontrivial.

It remains to find a usable semisimple direction or to remove the central directions obstructing this method. Over $\bar{k}$, split the given noncentral torus into its coordinate copies of $\mathbf{G}_m$. At least one is noncentral, since if each coordinate subgroup centralized G, their product would do so on every test algebra. Its adjoint action on $\mathfrak{g}_{\bar{k}}$ has a nonzero integer weight. Indeed, if all weights were zero, the weight decomposition of AG-GS, *Diagonalizable groups*, Theorem 3.1 and the centralizer Lie formula proved in Lemma 3.1 would make its smooth centralizer have the full Lie algebra and dimension of G. The integral-group dimension argument just used, followed by radicality of the smooth centralizer ideal, would make that entire centralizer equal to G, contradicting noncentrality. In characteristic zero a nonzero integer weight has nonzero differential, so the Lie generator has a nonzero adjoint eigenvalue. Therefore at least one nonconstant coefficient of the characteristic

polynomial of $\operatorname{ad}(Y)$ is a nonzero polynomial on the k-vector space $\mathfrak{g}$. A nonzero polynomial over an infinite field has a rational point where it is nonzero: induct on the number of variables, specialize a nonzero coefficient polynomial in the preceding variables, and then choose the last variable outside the finitely many roots, using S02 Lemma 4.A. Choose such a rational Y. Its adjoint matrix is not nilpotent. The characteristic-zero field is perfect, so Lemma 4.K makes Y_s rational and identifies $\operatorname{ad}(Y_s)$ as the semisimple part of $\operatorname{ad}(Y)$; it is nonzero. This supplies the case already treated.

In characteristic p, the Lie generator of a noncentral torus can be central: the adjoint weights may have zero differential. If a semisimple rational vector has nonzero adjoint action, the preceding case applies. Otherwise all semisimple rational vectors are central in $\mathfrak{g}$. Lemma 4.R nevertheless supplies a nonzero such vector by a rational p-power of a nonnilpotent matrix. That proof establishes rationality directly, including over imperfect fields.

Choose a nonzero semisimple rational X. It is central in $\mathfrak{g}$, and so are its restricted powers, since the matrix commutator formula gives $\operatorname{ad}(X^{[p]}) = (\operatorname{ad} X)^p$. Their span $\mathfrak{h}$ is commutative with geometrically injective, hence invertible, semilinear operation. Lemma 4.H gives its nontrivial height-one multiplicative-type subgroup M. Its full centralizer is smooth by AG-GS, *Diagonalizable groups*, Theorem 3.4, and Lemma 4.H gives its whole Lie algebra as $\mathfrak{g}$. Over $\bar{k}$, its dimension is consequently $\dim G$. The smooth connected $G_{\bar{k}}$ is integral by Lemma 2.L; the earlier domain dimension theorem excludes a proper closed subset of that dimension. The centralizer's defining ideal is radical, since it is smooth, and vanishes on the entire underlying space, so it is zero in the reduced coordinate ring of $G_{\bar{k}}$. Faithful field extension descends this equality. Thus $C_G(M) = G$ as schemes, and M is central. No descent of the span of all semisimple vectors is needed.

Pass to G/M and repeat if all its rational semisimple vectors are central. Lemma 4.B makes every composite kernel E_i finite connected central and of multiplicative type and the quotient smooth geometrically connected affine. The geometric torus needed to continue persists. Indeed its intersection with a finite E_i is a finite diagonalizable subgroup by Lemma 4.S; the quotient torus is represented there and has the same positive rank, since its character kernel has finite index in a nonzero free lattice. Its induced map into $(G/E_i)_{\bar{k}}$ has trivial full kernel by fppf lifting, so Lemma 4.S makes it a closed torus. If a geometric maximal torus of the new group is central, the first case gives a nontrivial k-torus there, because its geometric group still contains the positive-rank torus just constructed; otherwise Lemma 4.R again supplies a nonzero rational semisimple direction. Whenever a quotient torus is found, Lemma 4.P recovers a nontrivial torus in G, and Lemma 4.1 completes the induction. If no quotient torus is found, each next height-one kernel is nontrivial and its order increases as follows.

If the iteration continues indefinitely, its composite kernels E_i are finite connected central and of multiplicative type by Lemma 4.B. Their orders strictly increase: the exact finite locally free rank assertion in that lemma gives $|E_{i+1}| = |E_i|\,|H_i|$, and the new nontrivial height-one kernel has order $|H_i| = p^{\dim \mathfrak{h}_i} > 1$. After algebraic closure these connected kernels lie in $Z = Z(G_{\bar{k}})^0$, with full factorization through that identity component from S08 Lemmas G.0.1 and G.0.3. If its connected reduction R had no nontrivial torus, the single bound of Lemma 4.O would bound every $|E_i|$, a contradiction. Thus R has a nontrivial torus; Theorem 2.K gives its torus factor, and Lemma 4.C descends that factor to a nontrivial k-torus. Lemma 4.1 finishes this branch of the dimension induction.

If the iteration stops at a nontrivial k-torus $Q \subset G/E_i$, its inverse image P has the full finite connected central multiplicative-type kernel E_i and is its fppf torsor over Q. Lemma 4.P supplies a nontrivial k-torus in $P \subset G$, with the full commutator descent, universal-homeomorphism and geometric-reduction arguments proved there. Lemma 4.1 makes it geometrically maximal in G. This finishes the infinite-field induction.

Finally let $k = \mathbf{F}_q$. Lemma 4.G gives a Borel and a geometrically maximal torus over k, with the Lang equation, full finite-field Galois calculation and descent of the defining Hopf ideals already proved. This closes the induction in every characteristic and over every field. □

The infinitesimal step is needed because a noncentral group torus may have a central Lie algebra. In SL_2 in characteristic two, the Lie algebra of the diagonal torus consists of scalar matrices. The torus itself still acts nontrivially on the upper and lower root groups.

5. The finite symmetry left after centralizing

For a maximal torus in a reductive group scheme, define the fppf quotient

$$W_G(T) = N_G(T)/T.$$

The quotient is a sheaf of groups: the definition of the normalizer makes T normal in it, so multiplying local coset representatives gives a well-defined local coset, compatible with their overlaps. A representative n acts on T by conjugation. Replacing it by nt for $t \in T$ gives the same automorphism, since T is commutative. Thus the action descends through the fppf sheaf quotient, and composition/inverse on characters and cocharacters gives its actions on both sheaves. Representability and finiteness are not assumed in this definition.

The general finite étale Weyl-scheme theorem is taught with its complete proof in Root data, Weyl chambers and the Bruhat decomposition, Theorem 4.2, after the centralizer and reflection arguments it consumes. The matrix examples below have their own direct proofs here.

Example 5.2. Let A normalize the diagonal torus of GL_n over a test algebra R, including its nilpotents. Its induced torus automorphism is, by Lemma D.1, locally a fixed invertible integer map f on the character lattice. On each part where f is fixed, the intertwining equation for the standard representation is

$$(z^{e_i} - z^{f(e_j)})a_{ij} = 0 \quad \text{in } R[z_1^{\pm 1}, \dots, z_n^{\pm 1}].$$

If the two characters differ, their distinct coefficients force $a_{ij} = 0$. If some $f(e_j)$ were not among the e_i, the whole jth column would vanish, contradicting invertibility. The distinct $f(e_j)$ thus permute the n distinct e_i. Only the corresponding one entry per column survives; the determinant is a signed product of these entries and is a unit, so each entry is a unit. Thus A is a permutation matrix times an invertible diagonal matrix on this part. Conversely multiplication by such matrices induces the stated character permutation and normalizes the torus on every test algebra. The local permutation is unique, and therefore glues as a section of the constant finite scheme $\mathfrak{S}_n$. This proves the entire functor identity

$$N_{\mathrm{GL}_n}(D) = D \rtimes (\mathfrak{S}_n)_S, \qquad W_{\mathrm{GL}_n}(D) = (\mathfrak{S}_n)_S.$$

The permutation is locally constant; over a disconnected algebra it need not be a single global permutation.

6. Exercises

Exercise 6.1 (first steps). Over an arbitrary scheme, compute the centralizer and normalizer of the diagonal torus of GL_3. Explain how the answer changes when the test algebra is disconnected.

Exercise 6.2 (descent). Show that the two real tori in Example 1.2 are conjugate over $\mathbb{C}$ but not over $\mathbb{R}$. Show also that the multiplication torus $\mathrm{Res}_{\mathbb{C}/\mathbb{R}}\,\mathbf{G}_m$ in $\mathrm{GL}_{2,\mathbb{R}}$ is maximal and nonsplit.

Exercise 6.3 (finite fields). For a smooth connected affine group over $\mathbb{F}_q$, construct a Borel–torus pair over $\mathbb{F}_q$ from one over $\overline{\mathbb{F}}_q$. The only cohomological input permitted is Lang's theorem.

Exercise 6.4 (a family). Over $\mathbb{Z}$, let D be the diagonal torus in SL_2. Show that its nontrivial Weyl element is represented by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ even in characteristic two, and explain why the quotient is étale there.

Solutions

Solution 6.1

The universal Laurent-polynomial computation in Example 1.1 makes the centralizer diagonal. A normalizer induces a permutation of the three rank-one character summands in the standard representation. After a decomposition by idempotents, the permutation is fixed on each piece. The normalizer is therefore $D \rtimes (\mathfrak{S}_3)_S$, with quotient the constant finite étale group $\mathfrak{S}_3$. Disconnected test algebras allow different permutations on different open and closed pieces, exactly as sections of a constant étale scheme do.

Solution 6.2

The eigenvectors $(1, -i)$ and $(1, i)$ diagonalize the rotation matrix, with eigenvalues $a + bi$ and $a - bi$. On the norm-one subgroup these are inverse, so the eigenbasis carries it to the diagonal torus of SL_2. Rescale a column to obtain a determinant-one change of basis. Conjugation on its character lattice is $m \mapsto -m$, whereas the diagonal real torus has trivial action, which rules out a real conjugation. For the multiplication torus, the two eigenvalues vary independently after complex base change; it becomes the full diagonal torus of GL_2. It is therefore maximal. Its two character generators are exchanged by conjugation, so it is nonsplit over $\mathbb{R}$.

Solution 6.3

Lemma 2.1 conjugates the two Borels, and Lemma 2.N conjugates their maximal tori inside the common Borel. Hence choose g with $F(B,T) = g(B,T)g^{-1}$. Lemma 4.L gives $h^{-1}F(h) = g^{-1}$. Direct substitution then makes both hBh^{-1} and hTh^{-1} Frobenius-stable. For the descent, their finitely generated defining ideals have coefficients in some finite extension $L/\mathbb{F}_q$, because every coefficient in the algebraic closure is algebraic. Frobenius-stability after faithful field extension detects stability of those ideals over L, so the whole cyclic Galois group preserves them. S08 Lemma G.5.2 supplies the full invariant-space proof identifying each ideal with the scalar extension of its invariant ideal; it also descends the Hopf identities. Smoothness descends by S08 Lemma G.2.1, and Lemma G.0.T identifies the descended geometric torus as a torus. Its maximality and the Borel property are those of the unchanged geometric fibres. This is precisely the explicit descent construction of Lemma 4.G. Connectedness of G is a hypothesis of the proved Lang equation.

Solution 6.4

Conjugation by the displayed matrix sends $\mathrm{diag}(t, t^{-1})$ to $\mathrm{diag}(t^{-1}, t)$. A normalizer permutes the two universal weight lines, giving two components, each a coset of D. Its quotient is the constant scheme with two points over $\mathbb{Z}$, not μ_2. In characteristic two μ_2 is nonreduced, whereas a constant two-point scheme is still the disjoint union of two copies of the field and remains étale. The representative squares to $-I \in D$; it need not itself have order two before taking the quotient.

Exact supporting prerequisites and proof order

The classical triangularization, fixed-point, stabilizer, solvable structure and conjugacy arguments are in Section 2. Section 3 proves formal multiplicative-type homomorphism effectivity and the closed-immersion criterion. Section 4 proves Lang's theorem and supplies the Jordan and restricted-Lie arguments used in the field existence theorem. Their historical sources remain credited below.

The supporting geometric proofs are Group schemes over a field, for orbits and the closed-orbit lemma; Quotients and torsors, for finite free-action quotients; Faithfully flat descent, for affine schemes, representations and character descent; Diagonalizable groups and groups of multiplicative type, for the field character classification; Limits of schemes and Noetherian approximation, for finite-presentation spreading; the complete Artin approximation proof and application, Sections 1–5, used in Section 3; and *Dualizing sheaves and Serre duality for projective schemes*, for duality of curves. The prerequisite guide records their full statements and proof locations. The tori and reductive-group theorems assigned to this lesson are proved here.

The general cocharacter construction is proved here. The relative maximal-torus application is proved in the second lesson after the centralizer equality, and the finite Weyl-scheme theorem is proved in the fourth lesson after its reflection calculation. Their obligations are retained at those proof locations.

The course prerequisite guide records the exact supporting statements, their full proof routes, and the hypotheses needed in their applications.

References

- Brian Conrad, *Reductive group schemes*, freely accessible author draft: §§2.1–2.3 and 3.2, Appendix A and §§B.2–B.3.
- M. Demazure and A. Grothendieck, with the SGA 3 contributors, *Schémas en groupes*, re-edited by P. Gille and P. Polo: Exposés III, VII A, IX–X, XII and XIV.
- Milne's freely accessible *Algebraic Groups*, version 2.00 (2015). This exact free author edition provides comparison material; its citations do not replace the programme proofs.

Lesson 2

Regular elements and centralizers

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A maximal torus describes a reductive group by diagonal symmetries. A regular semisimple element describes the same torus by one element: the torus is the identity component of its centralizer. This observation is useful in families, where the centralizer of a section is easier to specify than a choice of eigenvectors.

We use the torus deformation and conjugacy arguments of Tori, maximal tori and their conjugacy. The classical geometric input concerning Borel subgroups is used below to prove the centralizer theorem, rather than importing that theorem from a reference. All centralizers are scheme-theoretic. Over a field, a geometric assertion means an assertion after algebraic closure.

1. The geometry behind the centralizer theorem

We first work over an algebraically closed field k. A Borel subgroup of a smooth connected affine group H is a maximal smooth connected solvable subgroup. Write $B = U \rtimes T$ for a Borel subgroup and one of its maximal tori. The solvable-group construction in the preceding lesson identifies U with the unipotent radical of B.

We shall need several consequences of that construction. Every smooth connected solvable subgroup lies in a Borel: among those containing it choose one of largest dimension, bounded by $\dim H$. A proper inclusion would strictly increase dimension, since both groups are reduced connected and hence integral by RG01 Lemma 2.L, and the earlier domain dimension theorem excludes a proper closed subset of the same dimension. Thus the chosen subgroup is maximal. RG01 Lemma 2.1 proves that H/B is projective and that all Borel subgroups are conjugate. The scheme-theoretic normalizer theorem is proved below, after the centralizer and Borel-intersection arguments needed for its induction.

Lemma 1.A (unipotent radicals). A smooth connected affine H/k has a unique largest smooth connected normal unipotent subgroup $R_u(H)$. It is preserved by every automorphism. Every smooth connected normal unipotent subgroup lies in every Borel. If K is a smooth connected normal solvable subgroup of H, then $R_u(K)$ is normal in H.

Proof. We first prove the precise affine image assertion used here. If $f : E \to H$ is a homomorphism with E smooth connected affine of finite type, then $k[E]$ is a domain: Lemma 1.0 below gives irreducibility, and smooth local rings are regular and therefore domains by AG-CA, *Smooth algebras over a field and the Jacobian criterion*, Theorem 2.1, and the regular-parameter theorem. Let C be the image of $k[H] \to k[E]$. This is a finitely generated domain. Comultiplication, inversion and counit restrict to it, because they commute with f and tensoring vector-space inclusions over k is injective. Their identities therefore restrict as well. Thus $J = \operatorname{Spec} C$ is a closed subgroup of H,

and $E \to J$ is schematically dominant. The complete monic-division and translation argument of AG-GS, *Quotients and torsors*, Lemma 11.1c makes it faithfully flat and of finite presentation, in precisely its integral-target case.

The group J is smooth and connected. Put $r = \operatorname{trdeg}_k \operatorname{Frac} C$. Noether normalization, proved in AG-CA, *Krull dimension and Noether normalization*, Corollary 3.2, makes this field finite over a rational function field with r parameters. S02 Lemma 3.F gives its differential dimension r, since k is perfect. At the generic point the local dimension is zero and the residue transcendence degree is r, so the earlier Jacobian Theorem 2.1 makes that point smooth. Its Corollary 2.2 gives a nonempty smooth open. This open contains a k-point by the weak Nullstellensatz. Translation by the rational points of J makes every rational point smooth; a nonempty closed complement of the smooth locus would again have a rational point. Hence J is smooth everywhere. Its coordinate domain makes it irreducible and connected. This proves the entire finite-type affine image assertion needed here, without consuming an image theorem for arbitrary group schemes.

For smooth connected normal unipotent N_1, N_2, the semidirect product $N_1 \rtimes N_2$ has underlying scheme $N_1 \times N_2$. It is smooth and integral by Lemma 1.0, the regular-local-domain calculation above and S08 Lemma G.4.1, so is connected. Apply the preceding image proof to multiplication into H; denote its closed image by N_1N_2. It contains both factors. Normality holds schematically: conjugation by any H-point preserves both factors, hence their multiplication image; pulling back equations along the faithfully flat cover $H \times (N_1 \rtimes N_2) \to H \times (N_1N_2)$ detects their vanishing. Thus their product is a smooth connected normal closed subgroup.

This product is unipotent. In any nonzero finite-dimensional representation W, AG-RG-01, Lemma 2.A makes $W^{N_1} \neq 0$. Normality makes this subspace N_2-stable, so AG-RG-01, Lemma 2.A supplies a nonzero vector fixed by both groups, and hence by their product: the latter equality follows faithfully flatly from the surjective multiplication map. Applying the same argument to successive quotients gives a trivial-quotient full flag. A faithful representation of the affine image is therefore unitriangular.

Choose a smooth connected normal unipotent subgroup of maximal dimension, possible since the dimensions are bounded by $\dim H$. Its product with any other such subgroup is again of this type. Maximal dimension forces that product to equal it: a proper integral closed subgroup of an integral finite-type group has strictly smaller dimension, by the strict-specialization and dimension proofs in AG-CA, *Krull dimension and Noether normalization*, Lemma 4.1 and Theorem 4.2. This proves greatestness and uniqueness. Automorphisms preserve its defining property, so preserve the subgroup.

For a Borel B and such a normal N, the product NB is similarly the smooth connected closed image of $N \rtimes B$. It is solvable directly on all test rings. Projection to the solvable B makes every sufficiently iterated commutator word in $N \rtimes B$ take values in N. The unitriangular entry-distance filtration, with its polynomial commutator identities proved in RG01 Lemma 2.J, makes finitely many further derived commutators equal to the identity. Every tuple of points in the image lifts fppf locally through the multiplication map; hence the same commutator identities hold for NB. Thus maximality of B gives $NB = B$, and N lies in B.

If K is normal in H, every conjugation by $H(k)$ gives an automorphism of K and preserves its unique $R_u(K)$. The whole conjugation morphism $H \times R_u(K) \to H$ factors through that closed subgroup: the source is reduced by AG-RG-S08, Lemma G.4.1, and its defining equations vanish at all closed points. To justify the required detection, a nonzero element a of a reduced finite-type k-algebra has a nonzero localization at a, since it is not nilpotent. That localization is still finite type and has a maximal ideal, whose residue field is k by the weak Nullstellensatz. It gives a rational point where a is invertible. Therefore an element vanishing at every rational point is zero. Apply this calculation to the affine source of conjugation. This proves scheme normality in H. In the solvable K, the decomposition constructed in AG-RG-01, Theorem 2.K, $K = U \rtimes T$, identifies its radical with U: every smooth connected unipotent subgroup, including one not assumed normal

in K, projects trivially to T by AG-RG-01, Lemma 2.A and is contained in U. □

Lemma 1.0 (components of an affine group over an algebraically closed field). Let J be an affine group scheme of finite type over an algebraically closed field k. Its irreducible components are pairwise disjoint and open and closed. In particular a connected such group is irreducible. No smoothness or reducedness hypothesis is needed for this topological assertion.

Proof. The Hilbert basis theorem, proved in AG-CA, *Noetherian and Artinian rings*, Theorem 2.1, makes $k[J]$ Noetherian. Its spectrum has finitely many irreducible components: if a closed subset failed to be a finite union of irreducible closed subsets, choose a minimal such subset using the descending chain condition on closed sets. It is reducible, hence the union of two proper closed subsets, both finite unions by minimality, a contradiction.

If C_1, C_2 are components containing the identity, give them their reduced closed structures. They are integral finite-type schemes, so their product is integral by the complete coefficient-and-Nullstellensatz proof in the earlier S08 Lemma G.4.1. The closure of its multiplication image in J is irreducible and contains both C_1 and C_2, by multiplication with the identity in either factor. It is contained in a component, and maximality of C_1 and C_2 forces them to be that same component. Thus exactly one component contains the identity.

If two different components met, their closed intersection would contain a k-point by the earlier Nullstellensatz, AG-CA Theorem 2.1 of *The Nullstellensatz and Jacobson rings*. Translation by the inverse of that point is a scheme automorphism of J. It would carry the two distinct components to two distinct components containing the identity, contradicting the preceding paragraph. Hence all components are disjoint. There are only finitely many, so each has closed complement and is open as well as closed. Connectedness leaves exactly one component, proving irreducibility. This is the finite-type algebraically closed case of the earlier AG-GS, *Group schemes over a field*, Section 1 and Theorem 2.2; the proof just given supplies precisely the case consumed here. □

Lemma 1.1. Let H be a smooth connected affine group of finite type over an algebraically closed field k, and let $A \subset H$ be a torus. Its schematic centralizer $C_H(A)$ is smooth and connected.

Proof. The complete fixed-subgroup proof in AG-GS, *Diagonalizable groups and groups of multiplicative type*, Theorem 3.4 supplies smoothness. We give explicitly the cocharacter reduction needed for connectedness. Over the algebraically closed k, the torus A is split. Indeed its étale splitting cover, given by that lesson's Theorem 7.7, has a nonempty finite-type affine chart; the earlier Nullstellensatz supplies a k-point, and pulling the splitting back along that point splits A. Write $A = \mathbf{G}_m^r$ and its character lattice as $\mathbf{Z}^r$.

Choose finitely many algebra generators of $k[H]$ and replace them by all their finitely many homogeneous components for conjugation by A. The complete weight decomposition of AG-GS Theorem 3.1 shows that the resulting homogeneous list still generates. Let $\Xi \subset \mathbf{Z}^r$ be its finite set of nonzero weights. If $r > 0$, associate to $\chi = (a_1, \dots, a_r) \in \Xi$ the nonzero polynomial

$$p_\chi(Z) = a_1 + a_2 Z + \cdots + a_r Z^{r-1} \in \mathbf{Q}[Z].$$

Each has only finitely many roots: division by $Z - c$ removes a root, and induction on the degree bounds the number of distinct roots by that degree. This is also the earlier explicit polynomial calculation in S02 Lemma 4.A. Choose a positive integer M outside the finite union of their root sets, and put $\lambda(t) = (t, t^M, \dots, t^{M^{r-1}})$. Its pairing with every member of Ξ is nonzero. For an empty weight list choose $M = 1$, and for $r = 0$ use the unique cocharacter of the trivial torus.

The centralizers $C_H(A)$ and $C_H(\lambda)$ are equal as schemes, including on nonreduced test algebras. To check this, use the universal conjugation action on each homogeneous algebra generator f of weight χ. A test point fixed by A must satisfy $f(h)(e^\chi - 1) = 0$ in its group algebra. The distinct basis monomials force $f(h) = 0$ when $\chi \neq 0$, and impose no condition when $\chi = 0$. The corresponding condition for λ is $f(h)(t^{\langle \chi, \lambda \rangle} - 1) = 0$ in the Laurent polynomial algebra. Our choice makes its exponent nonzero precisely for the nonzero weights on this generating list, so coefficient comparison

imposes exactly the same equations. Checking the algebra generators checks the full action on every function. Thus the two represented fixed subgroups have the same defining ideal over k and the same points on every test algebra.

Use the general cocharacter construction proved in the preceding lesson, Lemma 3.3. Its proof is a grading and smoothness argument for arbitrary smooth affine groups and uses no reductive centralizer theorem. Multiplication gives an open immersion

$$U_H(-\lambda) \times C_H(\lambda) \times U_H(\lambda) \longrightarrow H.$$

The domain contains the triple of identity elements. Lemma 1.0 makes the connected finite-type H irreducible, so the nonempty open image, and hence the domain, is irreducible. Its projection onto $C_H(\lambda)$ is surjective: insertion of the identity in the two unipotent factors gives a section. The centralizer is therefore irreducible and connected. Together with the fixed-subgroup smoothness proof and the schematic equality already established, this proves the lemma. □

The cocharacter lemma and the classical fixed-point arguments are independent of the root classification and of the reductive centralizer theorem proved below.

Lemma 1.B (fixed unipotent groups are smooth connected). Let a diagonalizable D act on U as in AG-RG-01, Lemma 2.C. Then U^D is smooth connected. Moreover the sequence

$$1 \longrightarrow U_1^D \longrightarrow U^D \longrightarrow (U/U_1)^D \longrightarrow 1$$

is exact as fppf sheaves.

Proof. Fixed subschemes are affine finitely presented and smooth by the coordinate and infinitesimal proofs in AG-GS-05, Proposition 3.2 and Theorem 3.4, including a nonreduced D. Let x be a D-fixed quotient section on any test algebra. Choose a lift u fppf locally. The defect

$$c(d) = u^{-1}d(u)$$

lies in U_1 and is a regular crossed homomorphism. Write it fppf locally as $c(d) = vd(v)^{-1}$ by AG-RG-01, Lemma 2.C. Then u v is fixed, since $d(uv) = uc(d)d(v) = uv$. This proves surjectivity of the last map of the displayed sequence, with the kernel equality evident on every test scheme. Differences of two fixed lifts belong uniquely to U_1^D; consequently its action identifies $U_1^D \times U^D$ with $U^D \times_{(U/U_1)^D} U^D$ on every test scheme. A local fixed lift translates this torsor into the product with U_1^D.

For a vector line of character χ, its D-fixed subgroup is the whole line if $\chi = 0$, and is zero otherwise: comparison of the two distinct basis monomials e^χ and 1 forces the vector coordinate to vanish, including over nonreduced tests. We induct on the given filtration length, simultaneously after every algebraically closed extension of k; the length-zero group is the identity. The fixed kernel in the displayed sequence is smooth and geometrically connected by induction, and the quotient Q is either $\mathbf{G}_a$ or 1. On a geometric fibre the torsor has a point by the earlier Nullstellensatz, and translation identifies it with that connected kernel. Hence all geometric fibres of $q : U^D \to Q$ are connected.

Here is an explicit descent proof that the source is connected. Both source and target are affine. Write $A = k[Q]$, $B = k[U^D]$. On each affine fppf trivializing chart $\operatorname{Spec} A' \to \operatorname{Spec} A$, the algebra $B \otimes_A A'$ is flat over A': it is the base change of the coordinate algebra of the smooth k-group U_1^D, and AG-CA, *Smooth algebras over a field and the Jacobian criterion*, Theorem 6.1 proves smooth flatness, with the flat base-change proof in the flat-module lesson, Proposition 3.2. For any prime $\mathfrak{p}$ of A, choose such a chart and a prime $\mathfrak{p}'$ above it. The flat local map $A_\mathfrak{p} \to A'_{\mathfrak{p}'}$ is faithfully flat by AG-CA, *Faithful flatness and the local criterion for flatness*, Theorem 2.1. For an injection of $A_\mathfrak{p}$-modules, its tensor map with $B_\mathfrak{p}$ becomes injective after this faithfully flat extension, because the extended algebra is flat. Flatness of the extension and faithful detection make the

original tensor map injective. Thus $B_{\mathfrak{p}}$ is flat, and local flatness detection in the flat-module lesson, Theorem 3.3, proves B flat over A. The local torsor sections give a point above every prime; hence $A \to B$ is faithfully flat by the same faithful-flatness Theorem 2.1.

A partition of the source into open and closed pieces is an idempotent $e \in B$: glue the function equal to one on one piece and zero on the other. Conversely an idempotent gives these pieces, since in every local ring one of e and $1 - e$ is a unit and their product is zero.

Let $e_1, e_2 \in B \otimes_A B$ be the pullbacks of e from the two factors. On each geometric fibre of q, connectedness makes that idempotent constant, so e_1 and e_2 have equal values at every geometric point of the fibre product. Their difference therefore belongs to every prime ideal and is nilpotent. Here is the required nilradical argument for any ring C. If $z \in C$ is not nilpotent, the powers $1, z, z^2, \dots$ avoid zero. An ideal maximal among those avoiding these powers exists by the union-of-chains argument. It is prime: if ab lies in it but neither a nor b does, both larger ideals obtained by adjoining a and b meet the powers; multiplying the two resulting equations puts a power of z in the original ideal, a contradiction. This prime avoids z. Conversely a nilpotent belongs to every prime. Apply this to $C = B \otimes_A B$. Two idempotents with nilpotent difference are equal: $e_1(1 - e_2)$ and $e_2(1 - e_1)$ are both idempotent and nilpotent, hence zero; thus $e_1 = e_1 e_2 = e_2$.

The faithfully flat equalizer proved in the earlier S04 Lemma A1.1 now descends e to A. The map $A \to B$ is injective, so the descended element is also idempotent. But A is $k[x]$ or k, both domains, and has only the idempotents zero and one. Hence e was trivial. This proves connectedness, and the same argument after every algebraically closed extension proves geometric connectedness. The torsor and kernel equalities were identities on all test schemes, so the entire lemma retains the stated schematic exactness. □

For $A \subset T \subset B = U \rtimes T$, these proofs give

$$C_B(A) = U^A \rtimes T,$$

so this centralizer is smooth connected and solvable.

Lemma 1.2. A unipotent group acting on an affine variety has closed orbits, with their full scheme structures. No smoothness or connectedness assumption on the group or the schematic stabilizer is needed.

Proof. We first prove closedness for a smooth connected unitriangular group L. Give the closure Y of one point orbit its reduced structure. It is integral. S08 Theorem G.4.2 makes the orbit open in Y, and makes both Y and its reduced boundary D invariant under L, using reducedness of the products and the strong Nullstellensatz. If $D \neq \varnothing$, choose a nonzero function in its ideal. AG-GS-01 Lemma 5.1 puts that function in a finite-dimensional L-stable subspace of the ideal: in the coefficient construction all coefficient functions stay in the ideal, since its coaction maps into $k[L] \otimes I_D$. RG01 Lemma 2.A gives a nonzero invariant function in this subspace. Its values are constant on the dense point orbit and hence on Y. It vanishes on the nonempty D, so the constant is zero, a contradiction. Thus the orbit is closed. S08 Theorem G.4.C represents the full quotient by its possibly nonreduced stabilizer as this reduced locally closed orbit. Since its support is closed, it is a closed immersion in the affine target and is affine.

Now embed the given unipotent group N as a closed subgroup of the smooth connected group $L = U_m$ of upper unitriangular matrices. Its coordinate ring is the polynomial ring in the entries above the diagonal, and the matrix formulas give its group operations. Fix x in the affine variety and retain the entire schematic stabilizer $K = N_x \subset N \subset L$. The RG01 detecting-line construction supplies an L-representation and a line kv with full stabilizer K. RG01 Lemma 2.A makes the character of K on this line trivial, including its infinitesimal points. The vector stabilizer is consequently K as well. The first paragraph and S08 Theorem G.4.C now construct an affine closed vector orbit P, with an fppf K-torsor

$$q : L \longrightarrow P, \qquad L \times K \simeq L \times_P L.$$

All these identities hold on arbitrary test algebras.

The closed subscheme $N \subset L$ descends along this torsor. Indeed, in the displayed relation the two projections are $(l,k) \mapsto l$ and $(l,k) \mapsto lk$. The inverse images of N coincide as schemes: right multiplication by any section of $K \subset N$ preserves N, and its inverse does too. This is the descent datum for its defining ideal. The module, ideal and algebra effectivity proofs of S04 Lemmas A1.1, A1.2 and A1.5 give a closed subscheme $Z \subset P$ with

$$q^{-1}(Z) = N, \qquad N \longrightarrow Z \text{ an fppf } K\text{-torsor.}$$

For clarity, ideal descent applies on the affine cover $L \to P$: the compatible ideal descends as a submodule of $\mathcal{O}_P$, multiplication preserves it after faithful pullback and hence before pullback, and its quotient algebra pulls back to $\mathcal{O}_N$. Thus Z retains any nilpotents of N/K; no reduction has been taken here. The torsor identity is the restriction of that for L.

The action map $N \to X$, $n \mapsto nx$, descends to $i : Z \to X$, since its two pullbacks on $N \times K$ agree. This is a finite-type morphism between affine schemes and is a monomorphism. To check the latter on every test scheme, lift two sections of Z fppf locally to $n, n' \in N$. Equality of their images says exactly $n^{-1}n' \in K$, so the two quotient sections agree. Morphism descent then gives their equality before the cover. Left translation by N likewise descends to an action on Z, and i is equivariant.

Let $W \subset X$ be the schematic image of i. It is invariant under the entire N-action. In affine coordinates the ideal of the schematic image is a kernel, and tensoring with the k-vector space $k[N]$ preserves that kernel. Thus the schematic image of $1_N \times i$ is $N \times W$. The shear automorphisms $(n,z) \mapsto (n,nz)$ on $N \times Z$ and $N \times X$ commute with this map; the latter automorphism and its inverse consequently preserve $N \times W$. This proves invariance with its original scheme structure.

S08 Lemma G.4.E gives an open $V \subset W$ over which i is an isomorphism. Choose $z_0 \in i^{-1}(V)(k)$. Every point of $Z(k)$ lifts to $N(k)$: a fibre of its fppf torsor is nonempty and of finite type over the algebraically closed field, so has a rational point. Hence $N(k)$ acts transitively on $Z(k)$. Translating V by these elements gives opens $nV \subset W$ whose inverse images are isomorphic to them and cover all closed points of Z. They cover Z, since a nonempty closed complement in a finite-type scheme has a closed rational point. The inverses agree on overlaps. Therefore Z is isomorphic to the open union $O = \bigcup_{n \in N(k)} nV \subset W$.

Finally this open has closed support. RG01 Lemma 2.L gives the smooth connected group $J = (N_{\mathrm{red}})^0$, finitely many component translates, and $N(k) = N_{\mathrm{red}}(k)$. The first paragraph makes every J-point-orbit closed. The $N(k)$-orbit of x is the finite union of translates of such an orbit, so it is the set of rational points of a closed reduced subscheme $D \subset X$. The locally closed image O has precisely these rational points. Its support equals $|D|$: the difference in either direction, if nonempty, is constructible and has a rational point by S08 Lemma G.4.A. The schematic image W has this same support, because the image of i is dense in it. Thus the open $O \subset W$ contains every point of W, so $O = W$ as an open subscheme. It follows that $i : Z \to X$ is a closed immersion. This constructs the full possibly nonreduced orbit and proves the assertion. □

Put $X = H/B$, initially as a homogeneous space with stabilizer B. Choose a maximal torus T of H contained in B, and put $C = C_H(T)$. Such a torus exists: a maximal torus is contained in a Borel by the dimension argument of RG01 Theorem 2.2, and Borel conjugacy in RG01 Lemma 2.1 moves that Borel to B. We justify both finiteness of X^T and the assertion that C fixes it, instead of assuming either one.

Lemma 1.C (a central maximal torus). If a maximal torus T of a smooth connected affine group C is central, then

$$C = T \times U_C$$

with U_C smooth connected unipotent.

Proof. We construct the affine quotient by T, without presupposing solvability of C. Write $T = D_k(M)$, with M a finite free character group. Right translation grades $A = k[C]$ by M; put

$B = A_0$. Choose finitely many homogeneous generators a_i of A, of weights m_i. The weight-zero monomials in these generators generate B. They have a finite generating list. Here is the needed integer argument: in $\mathbf{N}^r$ every infinite sequence has an infinite subsequence nondecreasing in every coordinate, by successive extraction in the finitely many coordinates. Thus an antichain is finite. The nonzero vectors v with $\sum v_i m_i = 0$ have finitely many minimal elements; subtracting a minimal element from a comparable vector of the same weight leaves weight zero and decreases the coordinate sum. Induction expresses every such vector as a sum of minimal ones. The corresponding finitely many monomials generate B. Hilbert's basis theorem makes B finitely presented over k.

Centrality makes B a Hopf subalgebra: for $b \in B$, the function $b(c_1 c_2)$ is invariant under right translation by T in either variable. The two exact weight projectors put its comultiplication in $B \otimes B$. Inversion and the counit restrict for the same reason. Their identities restrict along the injective degree-summand inclusions. We obtain an affine group $Q = \operatorname{Spec} B$ and a homomorphism $q : C \to Q$. The projector $A \to B$ is B-linear and retracts its inclusion. Consequently $A \otimes_B \kappa(y) \neq 0$ for every $y \in Q$, so q is surjective.

The closed embedding $T \subset C$ gives a graded surjection $A \to k[M]$. For a basis m_j of M, lift e^{m_j}, e^{-m_j} to a_j, b_j of those weights and put $f = \prod_j a_j b_j \in B$. Its value at the identity is one. All these elements are units in A_f. Their products supply a unit u_m of each weight m, and

$$A_f = \bigoplus_{m \in M} B_f u_m.$$

This is a faithfully flat B_f-module. The torsor algebra map sends $a \otimes c_m$ to $a c_m \otimes e^m$, and its inverse sends $a \otimes e^m$ to $a u_m^{-1} \otimes u_m$. Thus q is a T-torsor on $D(f)$. Every $Q(k)$-point lifts to $C(k)$, by the nonzero finite-type fibre and the weak Nullstellensatz. Translates of $D(f)$ by these image points cover all closed points of Q, and hence Q. Translation in C gives the torsor calculation on each such open. Finite presentation follows from the finite-type Noetherian coordinate rings. It follows that q is an fppf T-torsor globally, representing C/T on all test schemes.

The smooth connected C is integral, and its subring B is an integral finite-type k-algebra. The perfect-field differential-rank calculation of S02 Lemma 3.F and the Jacobian criterion give a nonempty smooth open of Q; translating it covers all closed points, so Q is smooth. Surjectivity makes Q connected. Also q is smooth: its geometric fibres are translates of the smooth T, and it is flat of finite presentation, so the proved geometric-fibre criterion applies.

Suppose $S \subset Q$ is a nontrivial torus and put $E = q^{-1}(S)$. Then $E \to S$ is a smooth surjective T-torsor. The group E is smooth and connected. Indeed its smooth components are disjoint, open and closed; its geometric fibres are translates of the connected T. Its ordinary fibres are connected as well, because field-extension projections are surjective. By S08 Lemma G.4.F the images of these open components are open; connectedness of the fibres makes the images of distinct components disjoint. They cover the connected S, so there is just one component. Its commutator maps land in the central T, hence its next derived commutator is trivial. Thus E is solvable, and RG01 Theorem 2.K writes it as $V \rtimes M_E$. The homomorphism $V \to S$ is trivial because a unitriangular group has no character, by RG01 Lemma 2.A. Hence $V \subset T$; a diagonalizable subgroup of a unitriangular group is trivial by that same lemma. Thus $V = 1$ and E is a torus. It properly contains T, since its quotient S is nontrivial, contradicting maximality. We conclude that Q has no nontrivial torus.

RG01 Lemma 2.D now makes Q unipotent. A sufficiently long derived commutator in C lands in T, since its image in the unitriangular Q is trivial; the next derived commutator is one. Thus C is solvable. RG01 Theorem 2.K gives $C = U_C \rtimes M_C$, with U_C smooth connected unipotent. By RG01 Lemma 2.N the maximal torus T is conjugate to M_C. Centrality makes every conjugate of T equal to T, so $M_C = T$. The action of this central torus on U_C is trivial, giving the asserted direct product. $\square$

Apply Lemma 1.C to $C = C_H(T)$, which is smooth and connected by Lemma 1.1. The torus T is central and remains maximal in this subgroup.

Torus-fixed cosets. The scheme X^T is a finite disjoint union of k-points, and C fixes it pointwise.

Proof. RG01 Lemma 2.1 constructs the projective homogeneous orbit $X = H/B$, and S08 Theorem G.4.C gives its full fppf B-torsor and its smooth scheme structure. The complete equivariant embedding of RG01 Lemma 2.1 realizes X as a closed subvariety of a projective space with a linear H-action. Choose a weight basis for the restricted T-representation, using the weight decomposition proved in RG01. Every standard projective chart given by a nonzero weight coordinate is T-stable. Its intersection with X is affine and smooth; these intersections cover X.

On each of these affine charts, AG-GS, *Diagonalizable groups*, Proposition 3.2 constructs the fixed scheme by the ideal of nonzero-weight elements and proves that it is closed, of finite presentation and compatible with every base change. Its Theorem 3.4, with the universal coefficient calculation of Lemma 3.3, proves smoothness and identifies its tangent functor with the invariant tangent functor of the chart. The fixed schemes agree on chart overlaps because they represent the same universal fixed-point condition. They glue to the closed smooth scheme $X^T \subset X$. Thus X^T is projective and complete. This uses the earlier full infinitesimal proof directly and includes nonreduced test schemes.

At a fixed point $x = gB$, put $B' = gBg^{-1}$. The equality of its schematic stabilizer with B' shows that $T \subset B'$. The quotient map $H \to X$ is smooth: on the fppf torsor trivializations it is projection from the smooth group B, and smoothness descends. Its differential is therefore surjective, with kernel the tangent space of the fibre B'. It gives the exact T-representation sequence

$$0 \longrightarrow \mathfrak{b}' \longrightarrow \mathfrak{h} \longrightarrow T_x X \longrightarrow 0.$$

The exact weight projections of RG01 give

$$T_x(X^T) = (T_x X)^T = \mathfrak{h}^T/(\mathfrak{b}')^T = \operatorname{Lie} C/\operatorname{Lie} C_{B'}(T).$$

The last identities are the fixed-subgroup tangent calculation of that same AG-GS Theorem 3.4, applied to conjugation on H and B'. The differential of the orbit map $C \to X^T$ is consequently surjective at the identity. Lemma 1.1 already proves that C is smooth and connected. S02 Lemma 3.A now makes this orbit map smooth near the identity, and translation makes it smooth everywhere. Its image is open by S08 Lemma G.4.F. S08 Theorem G.4.C supplies the full schematic orbit, its fppf stabilizer torsor and its immersion. Flatness of the smooth orbit map descends through the fppf map from C to its homogeneous orbit, so that immersion is flat. Locally it is a quotient $A \to A/I$. Tensoring its ideal sequence with the flat A/I gives $I/I^2 = 0$; at a point of the orbit the ideal is finite and lies in the maximal ideal, so the determinant Nakayama proof of S01 Lemma 1.2 gives $I = 0$ locally. Thus the orbit immersion is an open immersion, and the full scheme orbit is precisely this open subscheme.

These disjoint open orbits cover X^T. Indeed they contain every k-point, and a nonempty closed complement would have a k-point by the Nullstellensatz. Quasi-compactness gives a finite subcover, hence finitely many orbits. Each is also closed, since its complement is the union of the other open orbits. Thus every orbit is smooth and complete. Since $C = T \times U_C$ and T fixes the entire fixed scheme, each is also the full schematic U_C-orbit of its chosen point.

Such a U_C-orbit is affine, with the following explicit earlier proof route. Let $K \subset U_C$ be its schematic stabilizer. The RG01 detecting-line construction gives a representation and a line kv whose schematic stabilizer in U_C is K. The subgroup K is unitriangular, so RG01 Lemma 2.A makes its line character trivial, including for nonreduced K. Its vector stabilizer is therefore exactly K. S08 Theorem G.4.C identifies U_C/K, retaining the entire K, with the vector orbit in affine space. Lemma 1.2 makes that orbit closed there. The original orbit and this vector orbit represent the same fppf quotient by K, so they are isomorphic. This proves affineness without replacing the stabilizer by its rational points.

An orbit just obtained is integral: it is smooth, and its underlying space is the surjective image of the irreducible smooth connected U_C. As a closed projective subscheme, its algebra of global

functions is finite-dimensional over k, by S01 Theorem 8.4 and its closed-immersion cohomology calculation. This algebra is a domain. Multiplication by any nonzero element is injective and hence surjective on that finite-dimensional vector space; thus it is a field, and algebraic closedness makes it k. The orbit is affine, so it is $\operatorname{Spec} k$. We conclude that X^T is a finite disjoint union of such reduced points. The action of C on each of these point orbits is trivial as a scheme morphism. In particular, at every gB fixed by T, its full schematic stabilizer contains C; equivalently $C \subset B'$ for every Borel B' containing T. □

The normalizer acts transitively on X^T. If gB is fixed, conjugate $g^{-1}Tg$ to T within B to obtain a representative in $N_H(T)$. Its stabilizer is $N_B(T) = C_B(T) = C$, since conjugation in the solvable decomposition induces the identity on $B/R_u(B)$. Thus X^T is the set $N_H(T)/C_H(T)$, with a simply transitive action of that quotient. A cocharacter separating the finitely many weights in an equivariant projective embedding has exactly the same fixed points as T.

Lemma 1.D (Borel intersections). For a torus $A \subset B$, the full subgroup $K = C_H(A) \cap B = C_B(A)$ is a Borel subgroup of $C = C_H(A)$. Its full quotient C/K is a closed projective smooth orbit in $(H/B)^A$.

Proof. Lemma 1.1 makes C smooth and connected, and Lemma 1.B applied to the solvable B makes K smooth and connected. The equivariant projective embedding and smooth quotient $X = H/B$ are those of RG01 Lemma 2.1 and S08 Theorem G.4.C. Choose a weight basis for A in the embedding representation. The projective weight-coordinate charts are A-stable affine opens. AG-GS-05 Proposition 3.2 and Theorem 3.4 construct their closed smooth fixed schemes with invariant tangent spaces; they agree on overlaps by their universal fixed-point condition. They therefore give a closed smooth projective scheme $X^A \subset X$, with its full fixed-point condition on arbitrary test schemes.

For any $x = gB \in X^A(k)$, put $B' = gBg^{-1}$. Its full stabilizer contains A. The map $H \to X$ is smooth: it is a flat finitely presented B-torsor whose geometric fibres are translates of the smooth B, so the geometric-fibre criterion applies. Its tangent sequence at x is exact. Exact weight projections therefore give

$$T_x(X^A) = (\mathfrak{h}/\mathfrak{b}')^A = \operatorname{Lie} C / \operatorname{Lie} C_{B'}(A).$$

These centralizer tangent identities are the fixed-subgroup calculation of AG-GS-05 Theorem 3.4. The differential of $C \to X^A$ is surjective at the identity. S02 Lemma 3.A makes the orbit map smooth near the identity, and translation makes it smooth throughout C. Its image is open by S08 Lemma G.4.F. S08 Theorem G.4.C gives its full orbit immersion and its fppf stabilizer torsor. That immersion is flat, since its composite with the faithfully flat torsor is smooth and flatness descends. In a local closed-immersion presentation $R \to R/I$, flatness gives $I/I^2 = 0$. At a point of the orbit the finite ideal I lies in the maximal ideal, so Nakayama gives $I = 0$ locally. Hence the full orbit is an open subscheme of X^A.

The disjoint open C-orbits cover all rational points of X^A. Their complement, if nonempty, would be a closed finite-type subscheme with a rational point, so they cover X^A. Quasi-compactness gives finitely many such orbits; each is also closed, because its complement is the union of the others. In particular the orbit O of eB is closed and projective, smooth and integral, and its full stabilizer is exactly K. S08 Theorem G.4.C identifies O with the full quotient C/K, without passing to pointwise stabilizers.

The smooth connected solvable K is contained in a Borel D of C: choose a containing smooth connected solvable subgroup of largest dimension, and use the strict dimension drop for proper closed integral subgroups, as in RG01 Lemma 2.1. Its action on the complete nonempty O has a fixed point by RG01 Lemma 2.M. At cK this says $D \subset cKc^{-1}$ schematically. Thus $\dim D \le \dim K$. Since $K \subset D$ and both are smooth connected integral groups, a proper inclusion would give a strict dimension increase. They are equal. Consequently K is a Borel of C. Finally RG01 Lemma 2.1 gives conjugacy of Borels in C, so all its Borels are conjugates of this intersection. □

Theorem 1.E (Borel normalizers). Every Borel subgroup of a smooth connected affine group

H over an algebraically closed field satisfies

$$N_H(B) = B$$

as group schemes.

Proof. We first construct the full normalizer. In the conjugation representation on $k[H]$, AG-GS-01 Lemma 5.1 gives a finite-dimensional stable space W containing a finite generating list for the ideal I_B. Put $J = I_B \cap W$. Its stabilizer in H is a closed finite-type subgroup: in a basis of W extending one of J, preservation of J and of J by the inverse matrix is a finite list of equations in the representation matrix and its inverse. This stabilizer represents $N_H(B)$. Indeed, J generates I_B, so stabilizing J preserves that ideal in both directions. Conversely a normalizing section preserves I_B and W, hence their intersection. These statements remain true over every k-algebra R: exact scalar extension identifies $J_R = I_{B,R} \cap W_R$, and the chosen ideal generators still generate. Thus this construction includes arbitrary nonreduced test schemes. Write N for the resulting subgroup.

Choose $T \subset B$ maximal in H, as in the preceding fixed-coset argument. The torus-fixed-coset proof gives $C_H(T) \subset B$, and hence $\mathfrak{h}^T \subset \mathfrak{b}$. We compute the tangent space of N directly. An infinitesimal element with tangent vector $X \in \mathfrak{h}$ normalizes B precisely when, for its universal section b, infinitesimal conjugation stays in B. Translating the resulting tangent at b to the identity gives

$$X - \mathrm{Ad}(b)X \in \mathfrak{b} \otimes_k k[B].$$

To check this criterion schematically, substitute the infinitesimal conjugation into every generator of I_B: its constant coefficient is zero on B, and its first-order coefficient is the pairing with this translated tangent vector. Vanishing of all such coefficients is exactly membership in the tangent space of B; translation identifies that space with $\mathfrak{b}$ on all test algebras. The inverse infinitesimal conjugation gives the negative coefficients, so the criterion gives preservation in both directions. Consequently

$$\mathrm{Lie}\, N/\mathfrak{b} = (\mathfrak{h}/\mathfrak{b})^B \subset (\mathfrak{h}/\mathfrak{b})^T = 0,$$

where the last equality uses exact weight projections and $\mathfrak{h}^T \subset \mathfrak{b}$. Thus the tangent dimension of N at the identity is $\dim B$. Its local dimension is at least $\dim B$, since it contains the smooth B, and at most its tangent dimension, by the local embedding-dimension bound. Equality makes this local ring regular; the perfect-field Jacobian criterion makes N smooth at the identity. Translation proves smoothness at every rational point, and hence everywhere. It will therefore suffice to show $N(k) = B(k)$, since equations vanishing at all points of the reduced N vanish by the strong Nullstellensatz.

We also need the following consequence of projectivity: every rational element of H lies in a Borel. Choose $t \in T(k)$ avoiding the equations $\chi(t) = 1$ for every nonzero adjoint weight χ. Such a point exists, because these are finitely many proper closed subsets of the integral torus; the weak Nullstellensatz supplies a rational point of their open complement. For $T = 1$ the same calculation has no nonzero weights. The differential of

$$H \times C_H(T) \longrightarrow H, \qquad (g, c) \longmapsto gcg^{-1},$$

at $(1, t)$, with tangent spaces translated to the identity, is $(1 - \mathrm{Ad}\, t)X + Y$. It is surjective: the first summand covers every nonzero weight and $\mathrm{Lie}\, C_H(T) = \mathfrak{h}^T$ covers weight zero. S02 Lemma 3.A and S08 Lemma G.4.F give a nonempty open image, dense in the integral H. The image lies among conjugates of B, since $C_H(T) \subset B$.

The incidence of pairs (h, x) with $hx = x$ is closed in $H \times H/B$, by the separated diagonal. Its projection to H has closed image by the projectivity of RG01 Lemma 2.1 and S01 Lemma 7.A and Corollary 7.B. This image contains the preceding dense open, so is all of H. At any rational h, its nonempty finite-type fibre has a rational fixed coset; lifting that coset through the full B-torsor gives $h \in gBg^{-1}$ for some $g \in H(k)$. In particular, if B is normal in H, then $B(k) = H(k)$ and reducedness gives $B = H$ schematically.

We now induct on $\dim H$; the zero-dimensional smooth connected group is trivial. Let $n \in N(k)$. RG01 Lemma 2.N conjugates nTn^{-1} back to T within B. Multiplying n by that element of B preserves both its membership in $N(k)$ and the question whether it belongs to $B(k)$. We may assume that n normalizes T. If $T = 1$, RG01 Lemma 2.D makes H unipotent, hence solvable; its Borel is then H itself. Otherwise consider the torus homomorphism

$$f : T \longrightarrow T, \qquad t \longmapsto ntn^{-1}t^{-1}.$$

If f is not surjective, its character matrix has rank smaller than $\dim T$. The integer basis calculation of S02 Lemma 3.M expresses its kernel as a diagonalizable group with a free character summand of positive rank. The corresponding split factor supplies a nontrivial torus $A \subset \ker f$. Thus $n \in C_H(A)$, and it normalizes $K = C_H(A) \cap B$, a Borel of this smooth connected centralizer by Lemma 1.D. If the centralizer is proper in H, its dimension is strictly smaller; induction gives $n \in K \subset B$.

If instead $C_H(A) = H$, the torus A is central. The quotient construction in Lemma 1.C uses only centrality through its construction of the smooth connected affine $Q = H/A$ and its full smooth A-torsor q. S08 Lemma G.4.D gives $\dim Q = \dim H - \dim A < \dim H$. The closed subgroup $B \subset H$ descends along q to a closed subgroup $\overline{B} \subset Q$: right multiplication by $A \subset B$ preserves its defining ideal, so the ideal effectivity argument of S04 Lemmas A1.1 and A1.2 gives

$$q^{-1}(\overline{B}) = B, \qquad B \longrightarrow \overline{B} \text{ an fppf } A\text{-torsor.}$$

Its coordinate ring injects into the reduced $k[B]$, so $\overline{B}$ is reduced and connected. RG01 Lemma 2.L makes it smooth; derived commutator identities descend through the torsor, so it is solvable. The central A acts trivially on H/B, since $ahB = haB = hB$. Morphism descent gives an action of Q there with full stabilizer $\overline{B}$, and the same local representatives identify its full quotient with the projective H/B. Any containing Borel of Q has a fixed point on this quotient by RG01 Lemma 2.M, and the stabilizer inclusion and strict dimension argument of Lemma 1.D force it to equal $\overline{B}$. Thus $\overline{B}$ is a Borel. Induction in Q gives $q(n) \in \overline{B}$, so the exact inverse-image identity gives $n \in B$.

It remains to handle surjective f. The RG01 detecting-line construction gives an H-representation and a line kv with full stabilizer N; write $\chi : N \to \mathbf{G}_m$ for its character. This character kills commutators, so $\chi(f(t)) = 1$ on every test algebra. Surjectivity of the torus homomorphism makes its character pullback injective, hence $\chi|_T = 1$. RG01 Lemma 2.A also makes χ trivial on the unipotent radical of B. The split-solvable decomposition therefore makes B fix v as a vector. The map $H \to V$, $h \mapsto hv$, descends along its full B-torsor to $H/B \to V$. The global-function algebra of this integral projective quotient is finite-dimensional by S01 Theorem 8.4; being a domain over algebraically closed k, it is k. All coordinate functions of the descended map are constant, with value v at eB. Thus H fixes v as a scheme morphism, and the detecting-line stabilizer gives $H = N$. Now B is normal in H, so the incidence argument above gives $H = B$. This proves $n \in B$ in the final case and completes the induction. The smooth normalizer and the strong Nullstellensatz turn equality of rational points into the required full scheme equality. □

We may consequently identify H/B with the variety of Borel subgroups. The preceding simply transitive action on its torus-fixed points is also the action on the Borels containing T.

Lemma 1.3. Let H be smooth connected affine over the algebraically closed field k, and let $T \subset H$ be a maximal torus. If $\mathcal{B}^T$ is the collection of Borels containing T, put

$$J = \bigcap_{B' \in \mathcal{B}^T} R_u(B'), \qquad I = (J_{\mathrm{red}})^0.$$

Then $I \subset R_u(H)$, as a schematic inclusion.

Proof. Write $X = H/B$, where B contains T. The full normalizer theorem identifies its rational points with the Borel subgroups, and the torus-fixed-coset proof makes X^T a finite disjoint union of reduced points. Thus $\mathcal{B}^T$ is finite and nonempty, and J is the full finite intersection of closed

subgroup schemes. RG01 Lemma 2.L makes I smooth connected. It is unipotent, since it embeds in the unitriangular $R_u(B)$. For each $n \in N_H(T)(k)$, conjugation permutes the Borels in $\mathcal{B}^T$ and their unipotent radicals, which are characteristic by Lemma 1.A. It preserves the full J and hence I. This invariance under actual rational conjugating elements is sufficient for the chart transport below.

RG01 Lemma 2.1 embeds X equivariantly as a closed subscheme of $\mathbf{P}_{\text{lines}}(V) = \mathbf{P}(V^\vee)$. Replace V by the span of its point lines. This span is H-stable on all test algebras: the matrix entries into its quotient vanish on $H(k)$, and hence on the reduced H by the strong Nullstellensatz. The embedding factors through this span by the same argument on reduced X. Thus X spans V.

Choose a cocharacter λ separating the distinct T-characters in V. Here is an explicit choice when $T = \mathbf{G}_m^r$, $r > 0$. Write each difference of two characters as $(a_1, \ldots, a_r)$. The nonzero integer polynomial $\sum_i a_i M^{i-1}$ has finitely many integer roots by S02 Lemma 4.A. Choose one positive integer M outside these finitely many root sets and put $\lambda(u) = (u, u^M, \ldots, u^{M^{r-1}})$. For $r = 0$ use the unique cocharacter. On weight-coordinate projective charts, the ideals of the T-fixed and λ-fixed schemes are generated by the same coordinates: a difference character is nonzero precisely when its pairing with λ is nonzero. The universal weight-ideal calculation of AG-GS-05 Proposition 3.2 therefore gives $X^\lambda = X^T$, including arbitrary test algebras.

Let V_0 be the weight space of lowest λ-weight among those in V. The open $U \subset X$ where projection to V_0 is nonzero is nonempty, since X spans V. For a point in this open, multiply its weight coordinates by u^{w-w_0}. All exponents are nonnegative, so this defines the limit at $u = 0$, in $\mathbf{P}_{\text{lines}}(V_0)$. The limit is in X: on each projective chart its equations vanish in the Laurent polynomial ring and hence in the polynomial ring. Thus projection defines a morphism

$$U \longrightarrow X \cap \mathbf{P}_{\text{lines}}(V_0) \subset X^T.$$

The integral U has irreducible image in this finite reduced scheme, so that image is one point $x = [v]$. Every nonzero projected point line is consequently kv; projection of the point lines spanning V spans V_0, so $V_0 = kv$. Choose the coordinate covector ℓ with $\ell(v) = 1$ and zero on the other weight spaces. The chart

$$X_x = X \cap D(\ell)$$

is affine, contains x, is T-stable, and λ contracts every one of its rational points to x.

The group $N_H(T)(k)$ is transitive on $X^T(k)$. Indeed a fixed coset gB has $g^{-1}Tg \subset B$; this is a maximal torus of B. RG01 Lemma 2.N supplies $b \in B(k)$ with $b^{-1}g^{-1}Tgb = T$, so $gb \in N_H(T)(k)$ represents that same coset. Apply this to two fixed cosets to obtain an element carrying one to the other. Transport X_x to an affine T-stable chart X_y containing each $y \in X^T(k)$. These charts cover X. For a rational point z, the reduced closure of its T-orbit is projective and T-stable by S08 Theorem G.4.2. RG01 Lemma 2.M gives a fixed point y in this closure. If $z \notin X_y$, its entire orbit and closure would be contained in the closed T-stable complement, contradicting $y \in X_y$. Thus all rational points lie in these finitely many charts; the Nullstellensatz rules out a nonempty closed complement.

We prove their invariance under I. Every $H(k)$-orbit in $\mathbf{P}_{\text{lines}}(V^\vee)$ meets the affine chart of covectors nonzero on v. Otherwise a nonzero covector would vanish on every translate of v; these translates span V, since H is transitive on X. On this dual chart λ^{-1} contracts to $[\ell]$, by the same coordinate calculation. S08 Theorem G.4.2 gives a closed H-orbit in this projective space. It meets the dual chart, so its closedness includes $[\ell]$. It is therefore exactly the orbit $H[\ell]$. S08 Theorem G.4.C gives this closed projective orbit with its full stabilizer P and its fppf torsor $H \to H[\ell]$.

The action of B on this nonempty projective orbit has a fixed point by RG01 Lemma 2.M. Its full stabilizer supplies an H-Borel $B_1 \subset P$. Both B_1 and the torus T, which fixes $[\ell]$, lie in the smooth connected $(P_{\text{red}})^0$, by RG01 Lemma 2.L. A maximal torus T_1 of H in B_1, and T, are maximal there too. RG01 Theorem 2.2 conjugates T_1 to T by an element of this reduced identity component. It carries B_1 to a Borel $B_2 \subset P$ containing T. By its definition $I \subset R_u(B_2) \subset P$ schematically.

It therefore preserves the covector line $k\ell$, and hence the chart $D(\ell)$, on every test algebra. This proves invariance of X_x. The previously proved invariance of I under each transporting normalizer element gives invariance of every X_y. Neither smoothness nor connectedness of P has been assumed.

For $z \in X(k)$, the reduced projective closure of Iz is I-stable, by S08 Theorem G.4.2. The smooth connected solvable I has a fixed point z_0 there, by RG01 Lemma 2.M. Choose an invariant affine chart X_y containing z_0. If the orbit met its invariant closed complement, that complement would contain the whole point orbit and its closure, contradicting $z_0 \in X_y$. Hence the orbit lies in X_y. Lemma 1.2 makes its full orbit closed in this chart, so it contains z_0; transitivity makes it a point. S08 Theorem G.4.C gives its reduced scheme structure, so the full I-action fixes z. This holds for every rational z; the reducedness of $I \times X$ and the strong Nullstellensatz give equality of the action with the projection on the whole scheme.

For the final normality argument, form the schematic intersection

$$K = \bigcap_{g \in H(k)} gBg^{-1}.$$

It is a closed finite-type subgroup: the sum of these conjugate ideals in the Noetherian $k[H]$ is generated by finitely many elements, each from a finite subsum. The resulting finite intersection represents the same condition on every test algebra. The stabilizer of gB is the entire gBg^{-1}, so the preceding scheme action gives $I \subset K$. Put $K_0 = (K_{\mathrm{red}})^0$. It is smooth connected, contains I, and is solvable because it is contained in B. Every rational conjugation by H preserves K_0; the reducedness of $H \times K_0$ and the strong Nullstellensatz make K_0 normal in H schematically.

RG01 Theorem 2.K gives $K_0 = U \rtimes D$, with $U = R_u(K_0)$ smooth connected unipotent and D a torus. Every character of I is trivial by RG01 Lemma 2.A, so its map to the torus quotient D is trivial and $I \subset U$. The uniqueness of the unipotent radical makes U invariant under every rational conjugation by H; reducedness of $H \times U$ and the same equation argument make it normal schematically. Lemma 1.A then gives $U \subset R_u(H)$, proving $I \subset R_u(H)$. $\square$

This is the geometric mechanism in Chevalley's description of the unipotent radical. It gives exactly the part needed for centralizers without presupposing roots.

2. Centralizers and weight zero

Theorem 2.1. If G is connected reductive over an algebraically closed field and T is maximal, then $C_G(T) = T$. More generally, the centralizer of any torus in G is connected reductive.

Proof. For a maximal T, Lemmas 1.1 and 1.C give the full decomposition

$$C = C_G(T) = T \times U_C,$$

where U_C is smooth connected unipotent. The torus-fixed-coset proof gives $C \subset B'$ for every Borel B' containing T. Its split-solvable quotient $B'/R_u(B')$ is a torus by RG01 Theorem 2.K. The homomorphism from U_C to this quotient is trivial: every character pulled back to a unitriangular group is trivial by RG01 Lemma 2.A, and the torus characters generate its coordinate algebra. Thus $U_C \subset R_u(B')$ on all test algebras. It lies in their full intersection J, and its reduced connected identity-containing source factors through $I = (J_{\mathrm{red}})^0$, by RG01 Lemma 2.L. Lemma 1.3 gives $U_C \subset R_u(G)$. Reductivity makes this radical trivial, so $C_G(T) = T$ as group schemes.

Now let $A \subset G$ be any torus. Choose a maximal torus $T \supset A$: among containing tori choose one of largest dimension, bounded by $\dim G$; every proper closed inclusion of integral tori increases dimension, so it is maximal. Put $C = C_G(A)$, smooth connected by Lemma 1.1, and put $R = R_u(C)$, whose existence and normality are Lemma 1.A. For every Borel B' of G containing T, Lemma 1.D makes $K' = C \cap B'$ a Borel of C, with projective full quotient C/K'. The action of the smooth connected solvable R on this quotient has a fixed point by RG01 Lemma 2.M. Lift that point

through its fppf stabilizer torsor to $c \in C(k)$; its full stabilizer gives $R \subset cK'c^{-1}$. Since R is normal in C, this is equivalent to $R \subset K' \subset B'$. Its map to $B'/R_u(B')$ is again trivial by the character argument, so $R \subset R_u(B')$. Thus R factors through the same I attached to G, T, and Lemma 1.3 gives

$$R \subset R_u(G) = 1.$$

The smooth connected affine C therefore has trivial unipotent radical, which is precisely reductivity over this field. This proves both assertions with their full scheme structures. □

The Borel-intersection step here is the complete projective fixed-orbit argument of Lemma 1.D; no additional closure or rigidity argument is needed.

Theorem 2.2. If $G \to S$ is reductive and T is a maximal S-torus, then $C_G(T) = T$. For any subtorus A, the group $C_G(A)$ is reductive over S. These centralizers commute with arbitrary base change.

Proof. A torus splits étale locally by its definition. On a split affine chart, conjugation by A grades the affine coordinate algebra of G. AG-GS-05 Proposition 3.2 represents the fixed group by the ideal generated by all nonzero-weight elements, proves finite presentation, and proves compatibility with every base change, including nonreduced bases. Its Theorem 3.4 proves smoothness of this fixed group when G is smooth. The fixed-point functor is the centralizer functor on every test scheme. Thus the constructions agree canonically on overlaps; the ideal and algebra effectivity in S04 Lemmas A1.1 and A1.2 gives the global affine closed subgroup $C_G(A)$. Finite presentation and flatness descend by S04 Lemmas A1.3 and A1.3a; its geometric fibres are smooth by the split calculation, so the geometric-fibre criterion of AG-FSE-05 Theorem 3.1 proves global smoothness. The same canonical fixed condition and the local base-change calculation prove arbitrary base-change compatibility globally.

Each geometric fibre is the connected reductive centralizer in Theorem 2.1. Consequently this smooth affine group of finite presentation is reductive by the definition of a reductive group scheme. In particular this proves the assertion for arbitrary A before treating maximal T.

For maximal T, its closed inclusion $i : T \to C_G(T)$ is an isomorphism on every geometric fibre by Theorem 2.1 and the just-proved base-change compatibility. Both groups are smooth of finite presentation over S. Their relative tangent map is an isomorphism at every point of T, since it becomes one over a geometric residue field; the exact tangent and differential scalar-extension calculation identifies that map with the one on the fibre. S02 Lemma 3.A therefore makes i étale everywhere. Its finite presentation follows also from the affine-pair presentation calculation of AG-MO-02 Theorem 3.1. Its étale smoothness makes it flat by AG-CA-18 Theorem 6.1.

Here is the scheme equality over the original base, without a fibrewise-isomorphism shortcut. On an affine chart of $C_G(T)$, the closed inclusion has algebra $D \to D/I$, with I finitely generated by finite presentation. Flatness of D/I makes the tensor of $I \hookrightarrow D$ injective; that tensor map is zero, so

$$I \otimes_D D/I = I/I^2 = 0.$$

Every target point is in the image of i: extend its residue field to an algebraic closure and use the fibre isomorphism. Thus, at every prime $\mathfrak{p}$ of D, the finite ideal $I_{\mathfrak{p}}$ is contained in the maximal ideal and satisfies $I_{\mathfrak{p}} = I_{\mathfrak{p}}^2 \subset \mathfrak{p}D_{\mathfrak{p}}I_{\mathfrak{p}}$. The determinant Nakayama proof of S01 Lemma 1.2 gives $I_{\mathfrak{p}} = 0$. Localization detects the zero module, so $I = 0$. All these affine charts therefore give $T = C_G(T)$, with the original base and every nilpotent retained. □

Theorem 2.3. A reductive group scheme admits maximal tori étale locally. Any two maximal tori are conjugate étale locally.

Proof. Fix $s \in S$. AG-RG-01 Theorem 4.2 gives a geometrically maximal torus $M \subset G_s$ over $\kappa(s)$ itself. AG-RG-01 Lemma 3.1 extends that specified closed embedding to a torus $T \subset G_U$ on a pointed étale neighbourhood $(U, u) \to (S, s)$, with $\kappa(u) = \kappa(s)$ and fibre embedding equal to $M \subset G_s$. No separate residue extension or assertion about an unrepresented embedding functor is

needed.

Theorem 2.2, applied to this subtorus before it is known to be maximal on U, makes $C = C_{G_U}(T)$ a smooth affine reductive group of finite presentation and identifies every geometric fibre with the centralizer of the fibre torus. Write $r = \dim M$. The torus rank is locally constant, so restrict to an open neighbourhood of u on which it is r. The relative Lie module

$$\mathfrak{c} = e_C^* \mathcal{T}_{C/U}$$

is finite locally free: the standard smooth-coordinate calculation in AG-RG-S02 Lemma 3.1 makes the relative differential bundle locally free, and dualizing and pulling back by the identity gives this module. Its scalar extension is the Lie algebra of the fibre centralizer, by the differential base-change calculation of AG-CA-15 Theorem 2.1. At a geometric point above u, Theorem 2.1 gives $C_{G_{\bar{u}}}(T_{\bar{u}}) = T_{\bar{u}}$, so $\mathfrak{c}$ has rank r there. A local free basis of $\mathfrak{c}$ consequently gives an open neighbourhood of u on which its rank is r. Restrict to that neighbourhood.

For every geometric point v of this smaller U, the smooth connected group C_v has dimension r, since its dimension equals its tangent dimension at the identity. The closed subgroup T_v is a torus of the same dimension. A smooth connected finite-type group over an algebraically closed field is integral by the component calculation in AG-RG-01 Lemma 2.L. A proper closed subset of an integral finite-type scheme has smaller dimension, by AG-CA-09 Theorem 4.2. Thus T_v has the whole underlying space of C_v. Its defining ideal is radical because T_v is reduced, and its radical is zero because C_v is reduced and the underlying space is the whole space. Therefore $T_v = C_v$ as schemes. Any torus containing T_v centralizes it and hence lies in $C_v = T_v$; it equals T_v. Thus every fibre torus is maximal, which makes T a maximal U-torus. These open restrictions and étale neighbourhoods exist around every s, proving local existence over the original, possibly nonreduced base.

Now let $T_1, T_2 \subset G$ be maximal S-tori. On each geometric fibre they are conjugate by AG-RG-01 Theorem 2.2. The full equality transporter

$$P = \mathrm{Transp}_G(T_1, T_2)$$

is represented by a scheme of finite presentation and is smooth by AG-RG-01 Lemma 3.1: its Laurent-coefficient equations test equality of the subgroup schemes, including on algebras with nilpotents. Each geometric fibre has a conjugating point, so $P \to S$ is surjective.

Apply AG-FSE, *Smooth morphisms*, Theorem 5.2 to this smooth surjection. Its proof first chooses a point of the smooth fibre with finite separable residue extension, realizes that extension by a pointed étale neighbourhood, and then pulls an étale coordinate chart back along a section of affine space. The resulting étale neighbourhood $V \to S$ has an S-map $V \to P$. The represented transporter condition says exactly that its image $g \in G(V)$ satisfies $gT_{1,V}g^{-1} = T_{2,V}$ as subgroup schemes. This proves étale-local conjugacy. □

The statement is local around every point, and the conjugating element belongs to the group over the chosen neighbourhood. It does not assert a global section of the transporter.

If T is split, write

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha.$$

Here Φ is the finite collection of nonzero weights on a neighbourhood where their ranks are constant. Theorem 2.2 says that the zero-weight summand is exactly $\mathfrak{t}$. In the next lesson we prove that the nonzero summands have rank one and construct their root groups. The arguments about regularity below need the weight decomposition and weight zero; they do not need that rank-one conclusion.

3. Three uses of the word regular

Let G be connected reductive over an algebraically closed field, with rank $r = \dim T$. A group element is **dimension-regular** if its centralizer has the minimum dimension r. A **regular semisimple** element is a semisimple element g such that $C_G(g)^0$ is a maximal torus. A **strongly regular semisimple** element additionally has $C_G(g)$ itself equal to that torus. This last condition is stronger.

There is also the convention used in SGA 3, Exposé XIII: a group element is regular if the generalized eigenspace for eigenvalue 1 in its adjoint action has the minimum dimension, the nilpotent rank. For reductive groups this minimum is r, and this condition selects exactly the regular semisimple elements. We verify this identification below. Thus dimension-regular unipotent elements are not regular in that convention.

For $t \in T$, define

$$T_{\mathrm{rs}} = \{t : \alpha(t) \neq 1 \text{ for every } \alpha \in \Phi\}.$$

The inequalities mean invertibility of the indicated functions over a general base. Since every α is a nontrivial character, its equation $\alpha = 1$ is a proper closed subset of a torus over an algebraically closed field. A finite union of such subsets cannot cover the irreducible torus. Hence T_{rs} is open and dense, and is nonempty in every geometric fibre in the relative case.

For a semisimple t, its centralizer is smooth: the closure of the subgroup generated by t is of multiplicative type, and invariants for such a group are exact. Its Lie algebra is the fixed space of $\mathrm{Ad}(t)$. Consequently

$$\mathrm{Lie}(C_G(t)) = \mathfrak{t} \oplus \bigoplus_{\alpha(t)=1} \mathfrak{g}_\alpha.$$

The identity centralizer equals T precisely on T_{rs}. Indeed when no moving weight is fixed, the smooth identity centralizer has the same tangent dimension as T. Its closed torus inclusion has an isomorphic differential, so S02 Lemma 3.A and the étale-monomorphism proof of S04 make it open. It is also closed; connectedness makes it the whole identity component. If a moving weight is fixed, its nonzero summand gives strictly larger tangent dimension, and smoothness makes the identity centralizer strictly larger than T.

Theorem 3.1. The regular semisimple locus G_{rs} is a dense open subset. Its elements lie in a unique maximal torus.

Proof. First construct an open set of conjugates of torus elements. Consider the conjugation map

$$q : G \times T_{\mathrm{rs}} \longrightarrow G, \qquad (g, t) \longmapsto gtg^{-1}.$$

After translations identify tangent spaces, its differential at $(1, t)$ is

$$(X, Y) \longmapsto (1 - \mathrm{Ad}(t))X + Y.$$

It is surjective: the second term covers $\mathfrak{t}$, while the first is invertible on every nonzero weight summand. AG-RG-S02, Lemma 3.A, therefore makes q smooth, so its image is nonempty and open. We justify density precisely. The affine finite-type G is Noetherian by the Hilbert basis proof and has finitely many irreducible components. If two met, their nonempty closed intersection would have a rational closed point by the Nullstellensatz. Smoothness makes its local ring regular by the field Jacobian proof; AG-CA-14 Theorem 1.1 makes that local ring a domain, contradicting two component primes there. Thus the components are disjoint and each is open and closed.

Connectedness leaves one component. Hence G is irreducible and this nonempty open image is dense.

Here is a useful consequence of that density. Choose a Borel B containing T. In $G \times G/B$ the incidence scheme

$$I = \{(g, z) : gz = z\}$$

is closed, since the flag variety is separated. Its projection to G is proper: AG-RG-01, Lemma 2.1 and AG-RG-S01, Lemma 7.A and Corollary 7.B make G/B proper, and Corollary 7.B preserves properness under this base change and restriction to the closed incidence. Its image contains the dense open just constructed, and is consequently all of G. Over our algebraically closed field, a nonempty fibre has a rational point. Thus **every element of G belongs to a Borel subgroup**.

Let b be a semisimple element of $B = U \rtimes T$. In a faithful representation diagonalize b and take the reduced closure D of its integral powers. It is a closed subgroup of a diagonal torus, hence diagonalizable by the explicit Hopf-quotient proof in AG-GS-05, Theorem 4.2. Since it is reduced, its finite character torsion has order invertible in k, by the equation calculation of AG-GS-05, Theorem 4.4.

The intersection $D \cap U$ is trivial by AG-RG-01, Lemma 2.A. Thus $D \to T$ is a monomorphism and a closed immersion by a direct character calculation. Write $D = D(M)$ and $T = D(\mathbf{Z}^r)$. The kernel has character group $M/\operatorname{im}(\mathbf{Z}^r \to M)$, by AG-GS-05 Theorem 4.1. A diagonalizable group is trivial only when its character group is zero, by its monomial basis. Thus the character map is surjective, and so is the group-algebra map defining $D \to T$. Let A denote that image. In $U \rtimes A$, the subgroup D is the graph of a regular crossed homomorphism $A \to U$. AG-RG-01, Lemma 2.B supplies a T-stable filtration of U, hence an A-stable one, and AG-RG-01, Lemma 2.C makes this cocycle principal. Conjugation by its correcting $u \in U(k)$ places D and b in T.

The incidence proof just given puts every group element in a Borel. Its multiplicative Jordan factors belong to that same closed group, by the Jordan proof in AG-RG-01, Section 4. Applying the preceding paragraph to the semisimple factor proves that every semisimple group element belongs to a maximal torus. Hence the conjugation map $G \times T_{\mathrm{rs}} \to G$, whose differential was computed above, has image exactly the regular semisimple locus.

Any maximal torus containing g belongs to $C_G(g)^0$, because a torus is connected. That identity centralizer is already a maximal torus. This proves uniqueness. □

Minimum centralizer dimension is indeed r. On the dense open just constructed it is r. Here is the complete local dimension bound needed for the converse. The universal centralizer $H \to G$ is an affine scheme of finite presentation: in the finite coordinate generators of G, impose equality of the two products with the universal element. Its identity is a section. Suppose at a rational point g_0 its fibre has dimension $d < r$. The full finite Noether-normalization proof in AG-CA-09 Theorem 4.2 gives d functions on that fibre making its coordinate algebra finite over a polynomial algebra. Lift these functions to the universal centralizer algebra and subtract their restrictions to its identity section; subtracting constants on the fibre preserves finiteness. They define $f : H \to \mathbf{A}^d_G$ with the identity sent to zero. Its fibre at g_0 is finite, so f is locally quasi-finite there. S07 Theorem 3.4 proves that its locally quasi-finite locus is open. Pulling that open back along the identity section gives an open neighbourhood V of g_0 in G.

For every rational $g \in V$, the fibre morphism is quasi-finite near its identity. The full affine Zariski Main factorization of S04 Theorem D7.1 writes that neighbourhood as an open in a finite scheme over affine d-space. The integral dimension calculation in AG-CA-09 therefore bounds its local dimension at the rational identity by d. For any finite-type group over this algebraically closed field, that local dimension is its whole dimension: choose a rational closed point on each irreducible component by the Nullstellensatz and translate it to the identity. Translation carries the entire component to one through the identity, without changing its dimension. The exact closed-point

component/local-dimension calculation of AG-CA-09 Corollary 4.4 then puts each component dimension below the local identity dimension; the reverse inequality is immediate. No smoothness of the fibre group is used. Thus $\dim H_g \le d$ throughout $V(k)$. The open V meets the dense regular semisimple open and has a rational point in their intersection, again by the Nullstellensatz. At that point its fibre dimension is r, a contradiction. This proves the minimum bound without importing an unwritten semicontinuity theorem.

For the SGA convention, write the adjoint characteristic polynomial as

$$P_g(x) = \det(x - \mathrm{Ad}(g)) = (x-1)^r Q_g(x).$$

This divisibility holds for all g, since it holds on the dense open locus already proved, and the coefficients are regular functions. The SGA regularity condition is $Q_g(1) \neq 0$. Jordan decomposition $g = g_s g_u$ preserves the eigenvalues of the adjoint action, so this condition first implies that $g_s \in G_{\mathrm{rs}}$.

Choose a Borel containing g, by the incidence argument. Its Jordan factors belong to this Borel. Conjugate within it to arrange $g_s = t \in T_{\mathrm{rs}}$. Its commuting unipotent factor g_u belongs to U, because a unipotent element in the torus quotient is one.

Let D be the reduced diagonalizable closure of the powers of t. Then $C_U(t) = U^D$: regular conjugation identities holding on all powers hold on their schematic closure, and the reverse containment is immediate. Lemma 1.B makes this group smooth connected. Its Lie algebra is the t-fixed part of $\mathrm{Lie}\, U$, by AG-GS-05, Theorem 3.4. Since $U \cap T = 1$, $\mathrm{Lie}\, U \cap \mathrm{Lie}\, T = 0$. Regular semisimplicity gives $(\mathrm{Lie}\, G)^t = \mathrm{Lie}\, T$; consequently $\mathrm{Lie}(U^D) = 0$. A smooth connected zero-dimensional algebraic group over k is the identity. Thus $g_u = 1$.

Conversely a regular semisimple element has exactly r eigenvalue-one directions by the weight calculation. This proves the claimed equivalence in all characteristics. □

The minimum generalized zero-eigenspace dimension for $\mathrm{ad}(X)$ similarly defines SGA regularity in a Lie algebra. It is distinct from minimum kernel dimension. Its openness has a complete polynomial proof: write $\det(x - \mathrm{ad}(X)) = \sum_j c_j(X) x^j$ on the affine space $\mathfrak{g}$, and let d be the least index with coefficient polynomial not identically zero. Earlier coefficients vanish identically, and the open $c_d \neq 0$ is nonempty by the polynomial-avoidance proof of S06 Section 1. The generalized-eigenspace decomposition there, followed by the block determinant calculation, identifies the zero-eigenvalue multiplicity with the dimension of the generalized zero space. Thus its minimum is exactly d, attained precisely on that open. A semisimple $X \in \mathfrak{t}$ has Lie centralizer $\mathfrak{t}$ precisely when every occurring nonzero weight differential $d\alpha(X)$ is nonzero, by the weight decomposition. Each nonzero differential is a nonzero linear polynomial; their finite product is nonzero in the polynomial domain, so avoidance supplies such a vector whenever no differential vanishes identically. In characteristic zero every nonzero integer character has a nonzero differential: in a torus basis at least one of its integer exponents is nonzero and remains nonzero in the field. In positive characteristic that exponent can vanish, and group regularity still works.

Lemma 4.P (the projective matrix group used below). Over every base on which two is invertible, PGL_2 is the smooth affine reductive group of invertible two-by-two matrices modulo locally invertible scalar multiples. Its diagonal torus is split and geometrically maximal. The scalar quotient and all centralizer computations below are sheaf statements over arbitrary test algebras.

Proof. We construct the quotient first, over $\mathbf{Z}$. Let $B = \mathbf{Z}[a,b,c,d]$, $f = ad - bc$, with each variable of degree one, and let $A = (B[f^{-1}])_0$ be its degree-zero subring. It is generated by the ten degree-two monomials divided by f: a monomial of degree $2m$ factors into m pairs. For each variable $x_i \in \{a,b,c,d\}$ put $g_i = x_i^2/f$. The four opens $D(g_i)$ cover $\mathrm{Spec}\, A$. Indeed, a prime containing all g_i would contain ad/f and bc/f, whose squares are $g_a g_d$ and $g_b g_c$; their difference is one, a contradiction. On $D(g_i)$ one has

$$x_j/x_i = (x_i x_j/f)/g_i, \qquad f/x_i^2 = g_i^{-1}.$$

Conversely these ratios express every generator of A and g_i^{-1}. Thus this localized ring is exactly the three-variable polynomial chart, with $x_i = 1$, localized at its normalized determinant. The equalities are identities inside $B[f^{-1}, x_i^{-1}]$, and remain the displayed mutually inverse presentations after every base change. Gluing these four charts by the ratio transitions of S01 Section 7 gives exactly the determinant-nonzero open in projective three-space. This proves its affineness, finite presentation and smoothness by explicit affine charts, without importing a general Proj affineness assertion.

A point of this projective open is locally a tuple of four entries, at least one a unit, whose determinant is a unit, and different local tuples differ by a unit scalar. Thus the map $\mathrm{GL}_2 \to \operatorname{Spec} A$ is a Zariski locally trivial $\mathbf{G}_m$-torsor: normalize that chosen entry to one, and retain its scalar as the other coordinate. In particular it is fppf surjective on sheaves. Matrix multiplication and the cofactor inverse are compatible with these scalar transitions. They therefore glue to a group law and inverse on the projective open, with scalar kernel, giving precisely PGL_2. No assertion that every projective class has a global matrix lift is used. On a geometric field, its space is the surjective image of the irreducible GL_2 of RG01 Lemma 1.M, and is consequently irreducible and connected.

The diagonal locus $b = c = 0$ is a closed torus with unit ratio a/d, hence is $\mathbf{G}_m$ on every test algebra. The torsor charts compute the tangent algebra as $\mathfrak{gl}_2/kI$, since multiplying a lift $I + \epsilon X$ by $1 + \epsilon q$ changes X by qI. Conjugation by the ratio torus has nonzero full weights $1, -1$ on E_{12}, E_{21} and a one-dimensional zero-weight diagonal quotient. If N is a smooth connected normal unipotent subgroup over an algebraically closed field, its intersection with this torus is diagonalizable and trivial, by GS05 Theorem 4.2 and RG01 Lemma 2.A. Thus its tangent space misses the diagonal quotient. The full weight decomposition forces any nonzero tangent space to contain E_{12} or E_{21}. Normality gives the mixed dual-number Lie-ideal identity proved in RG01 Lemma 1.M. The bracket of these two vectors is $E_{11} - E_{22}$ modulo kI, which is nonzero when two is invertible; it is diagonal, a contradiction. Hence $\operatorname{Lie} N = 0$, and the complete smooth zero-dimensional calculation of S07 Lemma 5.3A makes connected N the identity. This proves reductivity in the stated characteristics. A containing torus has tangent space in the same one-dimensional zero weight, so has rank one. Its closed inclusion of the displayed torus is an isomorphism by the free-character argument of RG01 Lemma 1.M. This proves geometric maximality, and therefore the stated assertions over every base with two invertible. □

4. Matrix centralizers and the missing component

For a diagonal matrix $d = \operatorname{diag}(a_1, \dots, a_n)$ over a field,

$$(Ad - dA)_{ij} = (a_j - a_i)a_{ij}.$$

These equations hold on every algebra over the field: a nonzero difference of field eigenvalues stays a unit. With distinct eigenvalues they force every off-diagonal entry to vanish, and invertibility makes every remaining diagonal entry a unit. The centralizer scheme is therefore precisely the diagonal torus, which is maximal by RG01 Lemma 1.M; the element is strongly regular semisimple. With multiplicities $m_1, \dots, m_a$, the same equations leave exactly the square diagonal blocks, each invertible, so the entire centralizer scheme is $\prod_i \mathrm{GL}_{m_i}$. Its dimension is $\sum_i m_i^2$: each factor is an open in affine m_i^2-space by the earlier matrix proof. This exceeds n when some multiplicity exceeds one.

In SL_2, a noncentral diagonal element has centralizer the diagonal torus. For

$$u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

commutation says $c = 0$ and $d = a$ for a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, and the determinant condition says $a^2 = 1$. Scheme-theoretically

$$C_{\mathrm{SL}_2}(u) = \left\{ \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} : a^2 = 1 \right\} \simeq \mu_2 \times \mathbf{G}_a,$$

via $(a, x) \mapsto a \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$. Its dimension is one; u is dimension-regular and is not semisimple. In characteristic two this centralizer is non-smooth. This is why tangent dimension alone cannot define dimension-regularity in every characteristic.

For PGL_2 over a field of characteristic different from two, take the class t of $\mathrm{diag}(-1, 1)$. Its root character has value -1, so t is regular semisimple. On any test algebra R, lift a projective matrix class fppf locally to an invertible matrix A. It centralizes the class precisely when $AtA^{-1} = ct$ for a unit scalar c. Taking determinants gives $c^2 = 1$. Since two is invertible, $e = (1 + c)/2$ is an idempotent, and $R = eR \times (1 - e)R$; on those factors $c = 1$ and $c = -1$ respectively. For $c = 1$, the entry equations and invertibility of two make A diagonal; for $c = -1$ they make it anti-diagonal. The remaining two entries are units by the determinant condition. Conversely all such matrices centralize the class. Taking the scalar quotient and descending these unique locally constant factors proves the entire group-scheme identity, not merely equality on field points. The standard swap matrix has square one in the projective quotient and inverts the diagonal torus, so

$$C_{\mathrm{PGL}_2}(t) = T \rtimes (\mathbb{Z}/2\mathbb{Z}).$$

The identity component is the unique torus containing t, while the full centralizer has a second component. Regular semisimple must not silently be strengthened to strongly regular semisimple.

In characteristic two, $\mathfrak{t} \subset \mathfrak{sl}_2$ consists of scalar matrices. Every $\mathrm{ad}(X)$ for $X \in \mathfrak{t}$ is zero. Nevertheless a diagonal group element with $t^2 \neq 1$ acts on the upper root space by $t^2 \neq 1$. Differentiation has lost a nontrivial character.

5. Regular sections in a family

Let $G \to S$ be reductive. Define G_{rs} fibrewise using geometric regular semisimplicity. This is an open subscheme, compatible with every base change. To prove this without making assumptions about S, choose a maximal torus étale locally by Theorem 2.3 and split it étale locally. The same differential calculation makes

$$G \times_S T_{\mathrm{rs}} \longrightarrow G$$

smooth by AG-RG-S02, Lemma 3.A. Its image is open and, on every geometric fibre, precisely the required locus. These opens agree on overlaps and descend. Their fibres are dense and nonempty.

Theorem 5.1. A geometrically regular semisimple section $s : S \to G$ of a reductive group lies in a unique maximal S-torus. This construction commutes with arbitrary base change.

Proof. The existence and conjugacy of maximal tori étale locally have already been proved in this lesson, Theorem 2.3. On such a neighbourhood choose and split a maximal torus T. Write the adjoint weight decomposition

$$\mathrm{Lie}\, G = \mathrm{Lie}\, T \oplus \bigoplus_{\chi \neq 0} M_\chi.$$

The zero-weight equality is Theorem 2.2. Here the nonzero characters are just Lie weights; their root-group construction is unnecessary. Put $T_{\mathrm{rs}} = \{t : 1 - \chi(t) \text{ is invertible for every occurring } \chi \neq 0\}$. The differential of

$$q : G \times T_{\mathrm{rs}} \longrightarrow G, \qquad (g,t) \longmapsto gtg^{-1}$$

at $(1,t)$, after translations, is $(Y,Z) \mapsto (1 - \operatorname{Ad} t)Y + Z$. It is surjective on the displayed weight decomposition. AG-RG-S02, Lemma 3.A, makes q smooth. Its image on each geometric fibre is exactly the regular semisimple locus proved in Section 3. Thus its pullback over s is smooth and surjective. A smooth surjection has sections étale locally; such a section gives $s = gtg^{-1}$ with $t \in T_{\mathrm{rs}}$, and the torus gTg^{-1} contains s.

We prove uniqueness over all test schemes, including nilpotent ones. It suffices to prove that, when $s \in T_{\mathrm{rs}}(S)$, the closed subgroup T is also open in the scheme centralizer $C = C_G(s)$. This is a local assertion on S, so write $S = \operatorname{Spec} A$. First assume that A is Noetherian. Set $B = A[G]$, let I be its augmentation ideal, let K be the ideal of T, and let $J \subset K$ be the equalizer ideal defining C.

After shrinking S, choose a homogeneous basis of I/I^2. Such a choice is possible because the torus weight summands are finite locally free, and exactness of weight projections lifts their basis elements to homogeneous elements $q_i \in I$. Smoothness of G/A identifies its formal completion along the identity with

$$\widehat{B}_I \simeq A[[q_1, \ldots, q_N]].$$

For completeness, the coordinate assertion follows by taking the associated graded along the smooth identity section: $\operatorname{gr}_I B = \operatorname{Sym}_A(I/I^2)$. The chosen lifts give an isomorphism at each successive graded piece, hence by induction modulo each power of I, and then on inverse limits. Locally this associated-graded assertion is the regular-parameter calculation for a section of a smooth morphism: the chosen-coordinate conclusion of AG-RG-S02, Lemma 3.A, makes these parameters an étale chart after a local shrink. Subtracting their values on the section sends it to zero, and its ideal is generated by those parameters.

The cotangent map to T is an isomorphism on weight zero and is zero on every nonzero weight. Indeed conjugation on T is trivial and the zero-weight Lie space of G is exactly $\operatorname{Lie} T$. Consequently the completed ideal $\widehat{K}$ is generated by the q_i of nonzero weight. One can check this directly: these elements vanish on T; the remaining weight-zero elements are formal coordinates on the smooth torus, so the quotient by the displayed generators is exactly its formal coordinate ring.

If q_i has weight $\chi_i \neq 0$, conjugation by s gives the exact equality $s \cdot q_i = \chi_i(s) q_i$. Thus $(\chi_i(s) - 1)q_i \in J$, and the coefficient is a unit. Every moving parameter belongs to $\widehat{J}$. Since $J \subset K$, we have $\widehat{J} = \widehat{K}$. In particular,

$$K \subset J + I^n \qquad \text{for every } n \geq 1.$$

Here is the ideal-separation step. At a prime $\mathfrak{p} \supset I$, work in the Noetherian local ring $R = B_{\mathfrak{p}}/J_{\mathfrak{p}}$, with ideal L the image of I. It is contained in the maximal ideal. The preceding inclusions put the image of K inside $\bigcap_n L^n R = 0$. To see the last equality without importing a separatedness theorem, set $N = \bigcap_n L^n R$. The Rees ring $R[Lt]$ is Noetherian, since L is finitely generated and this ring is a quotient of a polynomial ring. The submodule $\bigoplus_n (L^n R \cap N) t^n$ of the finite Rees module $\bigoplus_n L^n R, t^n$ has finitely many homogeneous generators. If their degrees are at most c, then

$$L^n R \cap N = L^{n-c}(L^c R \cap N) \quad (n \geq c).$$

Both intersections on the left and at degree c equal N, so taking $n = c + 1$ gives $N = LN$. The ideal N is finitely generated. Nakayama's lemma gives $N = 0$: in generators, the equality writes

the identity minus a matrix with entries in the maximal ideal times their column as zero; its determinant is a unit and annihilates every generator. Thus $K_{\mathfrak{p}} = J_{\mathfrak{p}}$.

The finite B-module K/J consequently vanishes at every point of the identity section. Its support is closed. The complementary open in C is an open W, contains the entire identity section, and is contained scheme-theoretically in T. Translation now supplies openness along all of T: the action map $T \times C \to C$ is smooth, being a projection after the isomorphism $(t, c) \mapsto (t, tc)$. Its image on $T \times W$ is therefore open. This image is contained in T and contains T, because W contains the identity section. Hence T is open in C, as claimed.

For a general ring A, descend this finite collection of data to a finitely generated $\mathbf{Z}$-algebra A_0: the finitely presented smooth group G, the closed torus embedding, its conjugation action, the section s, the finite Lie weight decomposition, the equality of its zero part with $\operatorname{Lie} T$, and the inverses of the finitely many $1 - \chi(s)$. AG-MO-03, Theorem 5.1, Lemma 3.1 and Theorems 4.1–4.2 descend these presentations, maps and finite-module isomorphisms. The finite splitting-witness proof in AG-RG-S02, Lemma 3.1 and Proposition 3.2 retains smoothness at a sufficiently large stage. Enlarging that stage makes the finite identities valid there. No spreading of reductivity is needed: the proof just given uses smoothness and these explicit weight conditions. Thus it proves that T_0 is open and closed in $C_{G_0}(s_0)$. Equalizers commute with base change, and openness and closedness persist under it. We obtain the same conclusion over A.

Let T' be any maximal torus containing s. It maps to C. The inverse image of the open and closed T is open and closed in T' and contains the identity section. Each geometric fibre of a torus is connected, so this inverse image has every point of every geometric fibre and is all of T'. Thus $T' \subset T$. The inclusion of two smooth maximal tori is an isomorphism on geometric fibres. Its differential is therefore an isomorphism. AG-RG-S02, Lemma 3.A, makes the inclusion an étale monomorphism and hence an open immersion. Fibrewise surjectivity makes it an isomorphism. This proves uniqueness on every base change.

The local tori obtained above consequently agree on overlaps as subgroup schemes, and their unique descent identifications satisfy the cocycle condition. Affine faithfully flat descent produces the desired global maximal torus. The same uniqueness shows that pulling it back gives the torus associated with the pulled-back section. □

The full centralizer can have additional components supported on special loci. Work over $A = k[a, a^{-1}, (a-1)^{-1}]$, with two invertible, and take the projective section $\operatorname{diag}(a, 1)$. On any test algebra a lift centralizes it up to a scalar c, and determinants again give $c^2 = 1$. The same idempotent decomposition separates $c = 1$ and $c = -1$. In the first piece, the equations and the unit $a - 1$ force diagonal matrices. In the second, the two diagonal entries vanish because $2a$ and 2 are units; the off-diagonal entries are units, and their equations force $a + 1 = 0$. Conversely these conditions suffice. Thus the centralizer is the disjoint union of its diagonal torus over all of $\operatorname{Spec} A$ and an anti-diagonal torus over $\operatorname{Spec} A/(a+1)$. The latter is not flat: tensoring the injection of the principal ideal $(a + 1) \hookrightarrow A$ with its nonzero coordinate algebra gives the zero map on a nonzero free rank-one module, violating AG-CA-07 Theorem 2.1. The diagonal torus remains open and closed and is the unique maximal torus containing the section.

Theorem 5.2. Let G/S be reductive. Let X be a section of $\operatorname{Lie} G$ such that every geometric fibre is conjugate into the Lie algebra of a maximal torus and every occurring nonzero Lie-weight differential there is nonzero on X. There is a unique maximal S-torus whose Lie algebra contains X, and the construction commutes with every base change.

Proof. Existence is the smooth conjugation map with the same differential calculation as Theorem 5.1: after splitting a local maximal torus, the differential of $(g, X) \mapsto \operatorname{Ad}(g)X$ is $(Y, Z) \mapsto [Y, X] + Z$. Its moving weights have invertible coefficients, and its zero part is the torus Lie algebra by Theorem 2.2. AG-RG-S02, Lemma 3.A, therefore makes it smooth; its fibrewise image contains the section, so a smooth surjective pullback supplies étale local lifts.

For uniqueness work on one such chart, with $X \in \operatorname{Lie} T$. Let C be its scheme stabilizer in G.

It is affine finitely presented: in a finite local frame of $\operatorname{Lie} G$, the finitely many coordinates of $\operatorname{Ad}(g)X - X$ are defining equations. Their formation commutes with all base changes. Let A be the local affine base, first Noetherian, let $B = A[G]$, let I be its identity ideal, let K define T, and let $J \subset K$ define C.

Choose homogeneous identity parameters of the conjugation action of T. The completed-coordinate proof in Theorem 5.1 gives

$$\widehat{B}_I = A[[z_1, \dots, z_r, m_1, \dots, m_d]], \qquad \widehat{K} = (m_1, \dots, m_d),$$

where the z_i have weight zero and the m_j have nonzero weights. The linearization of the equations defining J is $Y \mapsto [Y, X]$. It is zero on the z-part and is invertible on the m-part by the hypotheses. Select d coordinate equations f_i whose d by d matrix on these moving directions is invertible.

Each f_i vanishes on T, so its completed expansion lies in (m). Factor it as

$$f_i = \sum_j a_{ij}(z, m) m_j.$$

The constant matrix $(a_{ij}(0, 0))$ is the invertible moving differential. Its determinant has unit constant term, hence is a unit in $A[[z, m]]$. The adjugate matrix therefore expresses every m_j as a completed linear combination of the f_i. It follows that

$$\widehat{J} = \widehat{K}.$$

This is the complete implicit-system argument: it does not infer ideal equality merely from equality of tangent spaces.

Consequently $K \subset J + I^n$ for every n. At any prime containing I, pass to the Noetherian local quotient $B_{\mathfrak{p}}/J_{\mathfrak{p}}$. The Rees-module Artin–Rees and Nakayama argument written in Theorem 5.1 makes the intersection of powers of the image of I zero. Thus $K_{\mathfrak{p}} = J_{\mathfrak{p}}$. The finite module K/J has closed support avoiding the identity section, so its complementary open in C lies in T and contains that section. Translation by the smooth T makes T open along all of itself. It is already closed.

For arbitrary A, descend G, T, X, the finite weight decomposition, the zero-weight isomorphism, and the inverses of the moving differentials to a finitely generated $\mathbf{Z}$-algebra. The affine limit and presentation proofs are AG-MO-03, Theorem 5.1, Lemma 3.1, and Theorems 4.1–4.2. The finite module presentations, their maps and inverse maps involve finite lists of coefficients and identities, so the same generator-and-relation argument retains them at a common stage. Smoothness of the stage is supplied by the full finite splitting-witness proof in AG-RG-S02, Lemma 3.1 and Proposition 3.2. The proof just given uses only smoothness and those finite identities, so applies there without requiring a reductive stage. Its open-and-closed conclusion pulls back to A.

Any torus T' with $X \in \operatorname{Lie} T'$ centralizes X as a section of its Lie algebra: conjugation on the Lie algebra of a commutative group is trivial. Hence it maps to C. The inverse image of the open-and-closed T in T' contains its identity section and is all of each connected geometric torus fibre, so is all of T'. Thus $T' \subset T$. If T' is maximal, its inclusion is a fibrewise isomorphism of smooth tori; the differential is an isomorphism, so AG-RG-S02, Lemma 3.A, makes it an étale monomorphism and hence an open immersion. Surjectivity on every fibre makes it an isomorphism. These statements persist on every test scheme. The local tori consequently agree as subgroup schemes on overlaps; the affine descent proof in AG-RG-S04 Section A glues them, and uniqueness proves arbitrary base-change compatibility. □

In characteristic zero the fibre condition holds for every semisimple Lie element with centralizer dimension equal to the rank. We prove each step over an algebraically closed geometric field. RG01 Lemma 4.K gives the entire polynomial Jordan decomposition and invariant-derivation construction: the formal exponential of a matrix Lie vector lies in $G(k[[t]])$, and the distinct-exponent independence proof gives its exact Laurent relations. For a matrix-semisimple X, write

its diagonal eigenvalues λ_i. The full closed exponential image is

$$D = D_k(\mathbf{Z}^n/N), \qquad N = \{(a_i) : \textstyle\sum_i a_i\lambda_i = 0\}.$$

Indeed substitution sends a Laurent monomial to $e^{(\sum a_i\lambda_i)t}$; RG01 Lemma 4.K makes a polynomial vanish precisely when the coefficient sum in each equal-exponent class is zero. The full coset-basis ideal proof of RG01 Lemma 4.S then gives exactly the displayed group algebra and closed subgroup of G, including every scheme equation. Its character group embeds into the additive field as the group generated by the finitely many λ_i. Choose a rational basis of their finite-dimensional $\mathbf{Q}$-span, using the finite basis algorithm, and clear the finitely many coordinate denominators. This identifies that group with a subgroup of a finite $\mathbf{Z}^r$. The explicit integer subgroup induction in RG01 Lemma 4.S makes it finite free. Hence D really is a torus. Its tangent point $z_i = 1 + \epsilon\lambda_i$ satisfies every relation, so $X \in \operatorname{Lie} D$.

Choose a torus containing D of largest possible dimension; dimensions are bounded by $\dim G$, so this nonempty finite set of integer dimensions has a largest member. A strictly larger containing torus would have larger dimension: a closed equal-rank inclusion corresponds to a surjection of free character lattices of equal rank, whose free kernel is zero by the integer subgroup proof. Thus the chosen torus is maximal and its Lie algebra contains X. The full group stabilizer $C_G(X)$ is smooth by the characteristic-zero Cartier theorem proved in S08 Theorem G.3.4, and its tangent space is $\ker \operatorname{ad}(X)$ by the dual-number conjugation formula. Its dimension is therefore this kernel dimension. On the chosen torus decomposition, dimension equal to the rank says exactly that every moving weight differential is nonzero. This is the fibre condition of Theorem 5.2.

If semisimplicity is instead specified by diagonalizability of $\operatorname{ad}(X)$, it gives the same conclusion. RG01 Lemma 4.K makes its nilpotent part $\operatorname{ad}(X_n)$, so this is zero. S08 Theorem G.3.4 makes $C_G(X_n)$ smooth with Lie algebra the whole $\mathfrak{g}$, hence with dimension $\dim G$. The proper-closed-subset dimension proof of AG-CA-09 and irreducibility proved in Theorem 3.1 force $C_G(X_n) = G$. In particular X_n commutes with the maximal torus. Theorem 2.2 puts it in $\operatorname{Lie} T$, whose matrices are diagonal in a torus weight basis. A diagonal nilpotent matrix is zero, so $X_n = 0$. General SGA-regular Lie sections in small characteristic need not meet the fibre condition; the characteristic-two example above explains that restriction.

For a general smooth connected affine group over a field, a **Cartan subgroup** is $C_G(T)$ for a geometrically maximal torus. The complete field centralizer proof of Lemma 1.1 gives smoothness and geometric connectedness. Over an algebraic closure T is central and still maximal in this centralizer, so the entire central-maximal-torus proof of Lemma 1.C gives $C_G(T) = T \times U_C$ with U_C smooth connected unipotent. Here nilpotence is explicit. Embed U_C in upper unitriangular m-by-m matrices by RG01 Lemma 2.A. Let I_h be the matrices whose entries vanish unless column minus row is at least h. Matrix multiplication gives $I_h I_j \subset I_{h+j}$, and $I_m = 0$. The inverse of $1 + X$ is the finite geometric series, since $X^m = 0$. Modulo I_{h+j} the products of an I_h and an I_j entry vanish in either order, so $[1 + I_h, 1 + I_j] \subset 1 + I_{h+j}$ on every test algebra. Intersecting these subgroups with U_C makes its lower central series vanish by step m. The central product with T is therefore nilpotent as well. These nested commutator identities descend through the faithful field extension, so the original Cartan group is nilpotent. Conjugacy of maximal tori, proved in RG01 Theorem 2.2, conjugates their entire centralizers. In a reductive group Theorem 2.1 makes that centralizer the torus itself. Thus Cartan subgroups are precisely the maximal tori in the reductive case. We make no additional general-base Cartan representability assertion here.

6. Exercises

Exercise 6.1 (first steps). Compute the centralizer of a diagonal matrix with distinct eigenvalues in GL_n. If its eigenvalues have multiplicities $m_1, \ldots, m_a$, compute the dimension of its centralizer.

Exercise 6.2 (centralizers). In characteristic different from two, prove $C_{\mathrm{SL}_2}(u) = \mu_2 \times U$ for the

upper unipotent element displayed above. Determine its component group. Explain what changes in characteristic two.

Exercise 6.3 (universal equations). Over an arbitrary base scheme, prove that the centralizer of the diagonal torus in GL_n is that torus. Why is checking only the base field's rational points insufficient?

Exercise 6.4 (geometry). Show that the regular semisimple locus of GL_n is exactly the nonvanishing locus of the discriminant of its characteristic polynomial, and is dense over every algebraically closed field.

Solutions

Solution 6.1

The equations $(a_j - a_i)a_{ij} = 0$ allow matrix entries only between equal-eigenvalue subspaces. The centralizer is $\prod_i \mathrm{GL}_{m_i}$ and has dimension $\sum_i m_i^2$. For distinct eigenvalues all $m_i = 1$, giving the diagonal torus and dimension n.

Solution 6.2

Multiplying out gives $c = 0$, $a = d$ and $a^2 = 1$. The map $(a, x) \mapsto \begin{pmatrix} a & ax \\ 0 & a \end{pmatrix}$ is an isomorphism of group schemes. In characteristic different from two, μ_2 is étale with two points, so the component group is the constant group of order two. In characteristic two μ_2 is connected and nonreduced, and the centralizer has one geometric component and a two-dimensional tangent space despite having dimension one.

Solution 6.3

Use the universal diagonal matrix with independent Laurent coordinates z_i. The equation $a_{ij}(z_i - z_j) = 0$ forces $a_{ij} = 0$ for $i \neq j$ by independence of Laurent monomials over every ring. The diagonal entries are units since the matrix is invertible. Over a finite field the rational points of a torus may not separate all its characters, and over nonreduced algebras rational-point equations may miss infinitesimal conditions; the universal computation avoids both problems.

Solution 6.4

If the roots are a_i, the discriminant is $\prod_{i<j}(a_i - a_j)^2$, a symmetric polynomial in them. The complete lexicographic elementary-symmetric reduction of S06 Section 1 proves that it is a polynomial in the characteristic coefficients, including in positive characteristic. It is nonzero precisely when all eigenvalues are distinct. A matrix with this property is diagonalizable, and Exercise 6.1 gives its torus centralizer. Conversely a semisimple matrix with repeated eigenvalues has larger centralizer, so is not regular semisimple. The polynomial does not vanish identically: choose distinct nonzero elements in the infinite algebraically closed field and take their diagonal matrix. Its nonvanishing locus in the irreducible variety GL_n is therefore open dense. No assertion that this locus has a rational point over every finite field is needed.

The course prerequisite guide records the exact supporting statements, their full proof routes, and the hypotheses needed in their applications.

References

- Milne's freely accessible *Algebraic Groups*, version 2.00 (2015). This exact free author edition provides comparison material; its citations do not replace the programme proofs.
- Brian Conrad, *Reductive group schemes*, 2014, §§1.1–1.2, 2.2 and 3.2. These references receive mathematical credit; the centralizer proof above is written out independently.
- A. Grothendieck and the SGA 3 contributors, *Schémas en groupes*, Exposé XIII, Theorem 3.1 and Corollary 3.2, for the relative regular-section theorem and its convention; Exposé XIV for the applications to maximal tori.

Lesson 3

Roots and reductive groups of rank one

Written by GPT-6.1 Sol (OpenAI) in Codex, Ultra setting, October 2026. Self-checked by the writing AI, GPT-6.1 Sol (OpenAI). Original text: public domain (CC0). The normalization lemma in Section 5 follows the Stacks Project's proof, cited at the end.

The diagonal entries of a matrix act on its off-diagonal entries through ratios. Those ratios are the roots of a general linear group. For an arbitrary reductive group, roots are still characters of a maximal torus, but their one-dimensional spaces need not be globally trivial over the base. This lesson constructs the corresponding groups without using a power-series exponential, and shows why every root carries a copy of the rank-one geometry of SL_2.

We use torus conjugacy, torus deformation, and the centralizer theorem from the preceding two lessons. A reductive S-group is smooth affine of finite presentation with connected reductive geometric fibres. A semisimple S-group has semisimple geometric fibres. Statements about line bundles and subgroup schemes are made over arbitrary schemes, including nonreduced ones.

1. The centre and the two ranks

Let T be a split maximal torus of G, and, on an open set where the ranks are constant, write

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha .$$

For now Φ means the nonzero weights; we will prove that they form a reduced root system. Put

$$Z = \bigcap_{\alpha \in \Phi} \ker(\alpha : T \to \mathbf{G}_m).$$

Lemma 1.A (the torus weight kernel is central). Let G/R be smooth affine of finite presentation with connected geometric fibres, let $T = D_R(X) \subset G$ be a split torus, and let Φ be the finite set of nonzero T-weights occurring in $\mathrm{Lie}(G)$. Put $L = \mathbf{Z}\Phi \subset X$. Then $D_R(X/L) \subset T$ is central in G, is of finite presentation, and commutes with arbitrary base change as a specified subgroup of T.

Proof. The ideal of the displayed intersection of character kernels in $R[X]$ is generated by the finitely many elements $e^\alpha - 1$. Its quotient is $R[X/L]$: these relations identify exactly the exponent basis elements in the same coset of L, and the distinct cosets remain an R-basis. The Hopf operations preserve the relations. This gives the indicated diagonalizable subgroup Z, including any torsion in X/L; no smoothness of Z is assumed. Finite presentation follows by adjoining these

finitely many equations to the Laurent presentation of T. This description also proves arbitrary base-change compatibility.

For conjugation by Z, put $H = G^Z = C_G(Z)$. AG-GS-05, Proposition 3.2 and Theorem 3.4, prove that this is a closed smooth subgroup of finite presentation and that $\mathrm{Lie}(H) = \mathrm{Lie}(G)^Z$, even for a nonreduced diagonalizable Z. Every nonzero T-weight restricts trivially to Z by construction, and the zero-weight part does too. Thus this Lie inclusion is an isomorphism. Translation by points of H, after each residue-field extension, makes the relative tangent map of $H \hookrightarrow G$ an isomorphism at every point. S02 Lemma 3.A consequently makes this closed immersion étale.

Here is the scheme argument turning that conclusion into equality. A closed immersion of finite presentation which is étale is open. Indeed, locally write it as $\mathrm{Spec}(A/J) \hookrightarrow \mathrm{Spec}\, A$ with J finitely generated. Formal smoothness applied to $A/J^2 \to A/J$ lifts the identity of A/J as an A-algebra map. Compatibility with A forces every element of J to vanish in A/J^2, so $J = J^2$. At any prime containing J, Nakayama's lemma gives $J_{\mathfrak{p}} = 0$; finite generation makes J zero on a neighbourhood of that prime. Hence the closed image is also open. Apply this to $H \hookrightarrow G$. On every geometric fibre its image is open and closed and contains the identity. Connectedness of that fibre makes its image the whole fibre. Its open complement in G therefore has no point, so $H = G$ as schemes. This says exactly that Z is central on every test algebra. □

For a maximal T, the displayed intersection Z is precisely the scheme-theoretic centre. Lemma 1.A proves its centrality. Conversely, on every test algebra a central point belongs to $C_G(T) = T$ by the preceding lesson, Theorem 2.2. Its action on each $\mathfrak{g}_\alpha$ is multiplication by α. On the present rank-constant open this summand is locally free of positive rank; choose a local basis to see that a scalar acting as the identity is one. Thus $\alpha = 1$ for each $\alpha \in \Phi$, and the point lies in Z. This proves the equality of functors, including nilpotent test algebras. On splitting-cover overlaps these subgroup schemes represent the same centrality condition, so their ideals and maps descend by S04 Lemmas A1.2 and A1.5. The resulting equality and the coefficient presentations commute with every base change.

The centre need not be smooth: $Z(\mathrm{SL}_n) = \mu_n$, including when the characteristic divides n. Its largest subtorus is called the **radical**, denoted $\mathrm{Rad}(G)$. For a group of multiplicative type with finitely generated character sheaf M, it is $D(M/M_{\mathrm{tors}}) \subset D(M)$. Indeed the quotient is a finite free lattice by S04 Lemma P0.10, and the group-algebra quotient gives this closed torus inclusion. Any homomorphism from a torus into $D(M)$ induces a character map from M into its torsion-free lattice, so kills M_{tors} and factors through this inclusion, by the anti-equivalence proved in AG-GS-05 Theorem 2.2. These character maps agree on splitting-cover overlaps. Thus the construction descends, its rank is locally constant, and it commutes with every base change. The **rank** of G is the dimension of a maximal torus; its **semisimple rank** is

$$\mathrm{rank}(G) - \dim \mathrm{Rad}(G).$$

Over an algebraically closed field, every smooth connected solvable normal subgroup P of G is a central torus. RG01 Theorem 2.K writes $P = U \rtimes D$ and makes U characteristic in P. Rational conjugations by G preserve U; the reducedness of $G \times U$ and the Nullstellensatz make the whole conjugation morphism preserve it. Thus U is normal in G and is trivial by reductivity. Hence P is a torus. The integer-matrix description of torus automorphisms in RG01 Lemma D.1 makes the conjugation map from connected G locally constant, with identity value at the identity; it is the identity everywhere. Therefore P is central and belongs to the radical constructed above. Conversely that radical is smooth connected solvable and normal. Its quotient Q is reductive by the independent central-quotient construction in Theorem 1.1 below. To prove semisimplicity, let $N \subset Q$ be smooth connected solvable and normal and put $P = q^{-1}(N)$. The torsor $P \to N$ has the split radical torus as kernel. The diagonalizable-torsor algebra calculation in AG-GS-05 Theorem 4.7 and Corollary 4.8 gives local trivializations: trivialize the finitely many degree modules for a basis of its character lattice, whose homogeneous units identify the torsor algebra with a Laurent

algebra. Thus P is smooth and this map is open with connected geometric fibres. The finitely many disjoint open components of smooth P have disjoint open images: a connected fibre lies in only one component. These images partition connected N, so P is connected. It is solvable on all test algebras: a sufficiently long derived commutator word maps to one in N and hence into the central radical, and the next commutator is one. It is normal as the inverse image of N. The preceding argument gives $P \subset \mathrm{Rad}(G)$, while the reverse inclusion holds by construction; the quotient torsor therefore gives $N = 1$. This proves that Q is semisimple. The same inverse-image argument shows that a torus in Q lifts to a torus in G. Its inverse image is smooth connected and solvable by the preceding torsor and commutator arguments. Write it $U \rtimes D$ using RG01 Theorem 2.K. Every character of unitriangular U is trivial by RG01 Lemma 2.A, so its map into the quotient torus is trivial and U lies in the radical torus. Its inclusion into that torus is itself trivial by the same character calculation, giving $U = 1$. The inverse image is therefore a torus. Conversely the image of a torus in G is a torus, by the character-lattice quotient calculation of AG-GS-05 Theorems 4.1–4.2. Hence maximal tori correspond under the radical quotient, and their dimensions decrease by the dimension of its torus kernel. In particular the rank of Q is the stated semisimple rank of G.

The derived group will be constructed for arbitrary bases after the pinned existence theorem in Pinnings and the classification of split reductive groups. It is semisimple and the multiplication map

$$\mathrm{Rad}(G) \times G_{\mathrm{der}} \longrightarrow G$$

is a central isogeny. Its construction uses the root groups established here. We do not assume a relative derived group in constructing these groups. Central quotients are constructed by graded invariant algebras in Theorem 1.1 below, without using coroots. In rank one we will construct the simply connected cover explicitly below.

Central quotients before the rank-one argument

The rank-one argument below needs the affine quotient even when the centre is not smooth. The invariant-algebra construction is independent of coroots and root groups. We first supply the field-differential calculation used to check its geometric fibres.

Lemma 1.B (differentials over a perfect field). Let k be perfect, let $t_1, \dots, t_d$ be algebraically independent over k, and let K be a finite extension of $F = k(t_1, \dots, t_d)$. Then $\dim_K \Omega_{K/k} = d$. No separability assumption on K/F is required.

Proof. This is S02 Lemma 3.F. Its complete field-differential calculation proves the assertion in both characteristics, including finite inseparable extensions. □

Lemma 1.C (reduced groups over an algebraically closed field). A reduced affine group scheme E of finite type over an algebraically closed field k is smooth. Its finitely many irreducible components are open and closed translates of its identity component. That component is a smooth connected normal subgroup.

Proof. First prove uniqueness of the component through the identity, without presupposing smoothness. If Z_1, Z_2 are two such components with their reduced structures, S08 Lemma G.4.1 makes $Z_1 \times Z_2$ irreducible. The closure of its multiplication image is contained in a component of E and contains both Z_1 and Z_2, because the identity belongs to both. Maximality of components forces them to be equal. If two components met elsewhere, their nonempty closed intersection would have a k-point by the Nullstellensatz. Translation of that point to the identity would give two components there. Thus components are disjoint. They are finite in number by Noetherianity, so each is open as well as closed. Each contains a k-point, whose translation identifies it with the identity component. In particular they have one dimension d. The identity component is a subgroup: its product with itself is irreducible, and its multiplication image contains the identity;

inversion also preserves the component. Both maps factor through this open and closed component.

The restriction of E to each open component has its reduced closed structure. Indeed, at any point of that component all other minimal primes disappear under localization. The sole remaining minimal prime is the nilradical, which is zero because E is reduced. Thus the component's defining prime localizes to zero there. AG-GS-03, *Lie algebras and smoothness*, Proposition 2.7, gives $\Omega_{E/k} \simeq \mathcal{O}_E \otimes_k \omega_E$, of rank $r = \dim_k \omega_E$. At a component's generic point the local ring is a field K, since it is reduced with only one prime. Noether normalization and Theorem 4.2 of AG-CA, *Krull dimension and Noether normalization*, make K finite over a rational function field on d variables. Localization of differentials and Lemma 1.B give $r = \dim_K \Omega_{K/k} = d$. At the identity, Corollary 4.4 of the dimension lesson gives local dimension d. AG-CA-18, Proposition 3.1, identifies the identity cotangent space with ω_E, so its embedding dimension is also d and its local ring is regular. The same proposition makes E smooth at that rational point. Translations give smoothness at every k-point. The smooth locus is open by AG-CA-18, Corollary 2.2; its closed complement, if nonempty, would have a k-point. Thus E/k is smooth everywhere.

Finally conjugation by every k-point preserves the identity component. The source of the conjugation map from E times that component is reduced by S08 Lemma G.4.1. The equations of the component therefore pull back to functions vanishing at every k-point, and hence to zero by the Nullstellensatz. This proves normality of the entire group-scheme morphism. □

Theorem 1.1 (affine central quotient). Let G be a reductive group over a scheme S, and let $H \subset Z(G)$ be a closed subgroup of multiplicative type of finite presentation. The fppf quotient G/H is an affine reductive S-group. Formation of the quotient commutes with arbitrary base change.

Proof. Work étale locally on $S = \operatorname{Spec} R$, with $H = D_R(M)$. Such splitting neighbourhoods are proved in AG-GS-05, Theorem 7.7. On a nonempty affine chart M is finitely generated: a finite algebra generating set for $R[M]$ contains only finitely many exponents; its polynomial expressions contain only the subgroup generated by those exponents, and independence of the exponent basis forces that subgroup to be M. Right translation gives the grading $A = \Gamma(G, \mathcal{O}_G) = \bigoplus_{m \in M} A_m$. Put $B = A_0$ and $Q = \operatorname{Spec} B$. These degree projectors prove exactness of invariants and arbitrary base-change compatibility. Since smooth algebras are flat over arbitrary rings by AG-CA-18, Theorem 6.1, the direct summand B is flat over R too.

Here is finite presentation directly over R. Choose finitely many homogeneous algebra generators of A, of weights $m_1, \dots, m_n$, and a graded surjection $P = R[z_1, \dots, z_n] \to A$. For $d \in M$ let $E_d = \{v \in \mathbf{N}^n : \sum_i v_i m_i = d\}$. Every such subset has finitely many minimal elements under coordinatewise order. Indeed, every infinite sequence in $\mathbf{N}$ has an infinite nondecreasing subsequence: either a value repeats infinitely, or the tails are unbounded and one chooses successive larger values. Extract such a subsequence successively in each of the finitely many coordinates. An infinite antichain is impossible. A minimal element below a prescribed vector exists in the finite box below it. The minimal nonzero vectors in E_0 generate that monoid by repeated subtraction, which decreases the coordinate sum. For any d, its finitely many minimal vectors generate P_d as a P_0-module, because subtraction from a comparable vector of the same weight leaves a vector of weight zero.

The monoid algebra $P_0 = R[E_0]$ is finitely presented over R. Choose the just-proved finite monoid generators. They give a surjection from a finite polynomial ring onto $\mathbf{Z}[E_0]$; its kernel ideal is finitely generated because the polynomial ring over $\mathbf{Z}$ is Noetherian, by S02 Lemma 3.N. Tensor this presentation with R. Tensor product preserves its cokernel, and the monomial basis identifies $R \otimes_{\mathbf{Z}} \mathbf{Z}[E_0]$ with $R[E_0]$, giving the required finite presentation.

The kernel I of $P \to A$ is finitely generated: compare this generating set with a fixed finite presentation of A by adjoining both finite sets of variables and the equations for their mutual expressions, then eliminate the old variables. This is the affine presentation calculation of AG-MO-02, Theorem 3.1. Since I is graded, replace its finite generators by all their homogeneous

components, obtaining homogeneous generators r_j of weights d_j. Taking weight zero in an expression in these generators gives $I_0 = \sum_j P_{-d_j} r_j$. Each coefficient module is finite over P_0 by the preceding minimal-vector calculation. Thus I_0 is a finite ideal of P_0, and $B = P_0/I_0$ is finitely presented over R. This also supplies the whole invariant-algebra finiteness argument of AG-GS-05, Theorem 3.5 in the present application.

Centrality first makes Q a group, directly on its algebra. For $b \in B$, the function $b(g_1 g_2)$ is invariant under right translation by H in either variable: for the first variable move the central element past g_2, and for the second use the original invariance. The double grading of $A \otimes_R A$ therefore puts $\Delta(b)$ in $B \otimes_R B$. Likewise $(gh)^{-1} = g^{-1}h^{-1}$ proves that inversion preserves B, and the counit restricts to B. The Hopf identities restrict because all the relevant inclusions are split degree-summand inclusions, even after tensoring. Thus Q is an affine group and $q : G \to Q$ a homomorphism. It is finitely presented by the affine-pair presentation calculation, since A and B are finitely presented over R. It is surjective: the degree-zero projector is a B-linear retraction of $B \hookrightarrow A$, so $A \otimes_B \kappa(y)$ is nonzero for every prime y of B, and prime existence gives a point above y.

The closed embedding $H \hookrightarrow G$ gives a graded surjection $A \to R[M]$. For each of finitely many generators m_i of M, choose $a_i \in A_{m_i}$ and $b_i \in A_{-m_i}$ restricting to e^{m_i} and e^{-m_i}. Put $f = \prod_i a_i b_i \in B$. Its restriction to H, and hence its value at the identity, is one. In A_f all these factors are units. Products of their powers provide a unit u_m in every weight m. Multiplication by u_m identifies $(A_f)_m$ with the free rank-one B_f-module $B_f u_m$. The torsor algebra map

$$A_f \otimes_{B_f} A_f \longrightarrow A_f \otimes_R R[M], \qquad a \otimes c_m \longmapsto a c_m \otimes e^m$$

is an isomorphism: its inverse on each exponent basis element sends $a \otimes e^m$ to $a u_m^{-1} \otimes u_m$. The forward map is an algebra map and the displayed inverse makes it bijective, so its inverse is an algebra map as well. Also $A_f = \bigoplus_m B_f u_m$ is faithfully flat over B_f. Hence q is an H-torsor over the principal open $U = D(f) \subset Q$, which contains the identity.

To cover the entire quotient over the original ring, form the principal open $W = \{(g, y) \in G \times_R Q : q(g)^{-1} y \in U\}$. Its projection to Q is smooth of finite presentation, being a principal open in the base change of G/R. It is surjective. For a geometric point $y : \operatorname{Spec} \Omega \to Q$, the same module retraction makes $A \otimes_B \Omega$ nonzero. This is a finite-type Ω-algebra, so the proved weak Nullstellensatz supplies $g \in G(\Omega)$ above y. Then $q(g)^{-1} y = 1 \in U$, giving a point of W above y. Thus $W \to Q$ is an fppf cover, over arbitrary nonreduced bases as well.

On W, its first coordinate is an actual section g of G_W. Left translation $x \mapsto g^{-1} x$ identifies the pullback of q with the pullback of the already constructed torsor over U, through the map $(g, y) \mapsto q(g)^{-1} y$. This translation commutes with the right H-action. S04 Lemma A1.3 descends flatness through the affine faithfully flat cover $W \to Q$, and finite presentation of q was already proved. The torsor isomorphism $G \times H \to G \times_Q G$ and its inverse descend by S04 Lemma A1.5. Consequently q is an fppf H-torsor globally. Every section of Q lifts fppf locally through q, and two lifts differ by a unique section of H by this isomorphism. This identifies Q with the fppf quotient sheaf G/H.

On overlaps of the splitting cover the quotient sheaf is the same, so these affine representing groups are canonically isomorphic, with the cocycle identity forced by their quotient maps. The affine effectivity proof in S04 Appendix A1 descends their algebras, and Lemma A1.5 descends the group laws and torsor isomorphisms. Lemmas A1.3 and A1.3a descend flatness and finite presentation. The degree-zero projector on each splitting chart commutes with every scalar extension, so this descended quotient and its torsor identity commute with arbitrary base change.

We check geometric smoothness directly. Over an algebraically closed field k, smoothness of G makes A reduced: AG-CA-18, Theorem 2.1, makes its local rings regular, and the regular-local-domain proof makes them reduced. The inclusion $B \hookrightarrow A$ therefore makes B reduced. Lemma 1.C proves Q/k smooth. It is connected because $G \to Q$ is surjective.

Let $U \subset Q$ be any smooth connected normal unipotent subgroup, and put $P = q^{-1}(U)$. This is a closed normal subgroup of finite type, and $P \to U$ is the pullback of the surjective H-torsor. Over k, the product of reduced finite-type schemes is reduced, by S08 Lemma G.4.1. Multiplication and inversion therefore factor through P_{red}: nilpotent functions pull back to zero on their reduced sources. Thus P_{red} is a group. Lemma 1.C makes it smooth with open and closed irreducible components. Put $K = (P_{\mathrm{red}})^0$. In particular K is smooth connected and closed in G.

Here normality is a statement about morphisms, including their infinitesimal points. For each $g \in G(k)$, conjugation preserves P, its reduction and its identity component. Every defining equation of $K \subset G$ consequently pulls back under conjugation $G \times K \to G$ to a function vanishing at all k-points. The source is reduced by S08 Lemma G.4.1, and the Nullstellensatz makes that function zero. Hence the whole conjugation morphism factors through K, proving $K \triangleleft G$ schematically. Also K is solvable. Indeed $U(k)$ is unitriangular and hence nilpotent: commuting matrices with vanishing entries below distances a and b gives vanishing entries below distance $a + b$, as follows by multiplying their strictly upper triangular parts. A sufficiently long derived commutator word in $P(k)$ therefore lands in $H(k)$; the next derived commutator is one because H is commutative. The corresponding word map on a power of the reduced K is identically one by the same Nullstellensatz argument. This proves solvability as an algebraic group.

The preceding lesson, Theorem 2.K, writes $K = V \rtimes T$ with V smooth connected unipotent and T a torus and proves that V is the largest smooth connected normal unipotent subgroup. Thus every conjugation automorphism of K preserves V. Reducedness of $G \times V$ and the preceding equation argument then give $V \triangleleft G$. Reductivity of G forces $V = 1$. Consequently K is a torus.

There are only finitely many components of P_{red}. Each contains a k-point and is a translate $p_i K$: translation carries the identity component to the component containing p_i. Let J be the topological closure of the image of $K \to U$. Surjectivity of $P \to U$, and equality of the underlying spaces of P and P_{red}, imply $U = \bigcup_i q(p_i)J$. The smooth connected U is reduced and hence irreducible by Lemma 1.C. One of these finitely many closed translates is all of U, hence $J = U$. But the homomorphism $K \to U$ is trivial. To see this directly, pull back a faithful unitriangular representation of U. Its coordinate flag has trivial successive quotients. Exact torus weight projections split every term of this flag, so the whole representation of K is trivial. Faithfulness of the representation of U then makes $K \to U$ the identity-valued morphism. Thus $J = \{e\}$ and U is the trivial smooth group. This proves that every geometric fibre of Q is reductive. Flatness, finite presentation and these smooth geometric fibres prove smoothness of Q/S by AG-FSE-05, Theorem 3.1.

□

2. Why rank one has only two moving directions

We first prove the geometric rank-one statement, rather than assuming a root classification. Work over an algebraically closed field k.

Lemma 2.1. If a torus acts linearly on an irreducible projective variety X, it has at least $\dim X + 1$ fixed points.

Proof. Put $d = \dim X$ and let T be the acting torus. Since k is algebraically closed, an étale splitting cover of T has a k-point: take a nonempty finite-type affine chart and apply the earlier Nullstellensatz. Pulling the splitting back along this point gives $T \simeq \mathbf{G}_m^r$. The weight decomposition proved in the preceding lesson, *Diagonal actions and their descent*, applies to the finite-dimensional representation defining the linear action. Choose homogeneous coordinates in which the action on points has the form

$$t \cdot [c_0 : \cdots : c_N] = [\chi_0(t)c_0 : \cdots : \chi_N(t)c_N], \qquad \chi_j \in X^*(T) = \mathbf{Z}^r.$$

Delete any coordinates vanishing identically on X, thereby restricting to an invariant projective linear subspace.

Here is an explicit cocharacter separating all distinct characters in this finite list. If $r > 0$, write $\chi = (a_1, \dots, a_r)$. For distinct characters χ, ψ, the integer polynomial

$$\sum_{i=1}^{r} (a_i - b_i) M^{i-1}$$

is nonzero. The polynomial root bound proved in S02 Lemma 4.A, applied over $\mathbf{Q}$, makes its set of integer roots finite. Choose a positive integer M outside the union of these finitely many sets and put

$$\lambda(u) = (u, u^M, \dots, u^{M^{r-1}}) : \mathbf{G}_m \longrightarrow T.$$

The integers $w_j = \langle \chi_j, \lambda \rangle$ are equal only for equal characters. For $r = 0$, the unique cocharacter has this property since there is only the trivial character. Reorder the coordinates so that w_0 is minimal. By the deletion step, c_0 does not vanish identically on X. The coordinate hyperplane $H = V(c_0)$ is T-invariant.

We induct on d. For $d = 0$, the reduced irreducible finite-type X has one point. An affine neighborhood is a zero-dimensional domain, hence a field; the Nullstellensatz makes it k. Thus $X = \operatorname{Spec} k$, and its point is fixed.

Suppose $d > 0$. The section $X \cap H$ is nonempty. Otherwise X would be closed in the affine complement of H and hence affine. S01 Theorem 8.4 and its closed-immersion cohomology calculation prove that $H^0(X, \mathcal{O}_X)$ is finite-dimensional over k. It is a domain because X is integral. Multiplication by a nonzero element is an injective endomorphism of a finite-dimensional vector space and therefore surjective, so this algebra is a field. Algebraic closedness makes it k. Affineness would then give $X = \operatorname{Spec} k$, contradicting $d > 0$.

Every irreducible component of $X \cap H$, with its reduced structure, has dimension $d - 1$. To check the dimension rather than assume it, take a standard projective affine chart meeting the component and intersect it with X. Its algebra A is a finite-type domain of dimension d, by the function-field dimension formula in AG-CA *Krull dimension and Noether normalization*, Theorem 4.2. In a local frame the hyperplane section is defined by a nonzero element $f \in A$; its component corresponds to a prime $\mathfrak{p}$ minimal over (f). This prime is nonzero. The principal ideal theorem in AG-CA *Dimension theory of Noetherian local rings*, Theorem 3.1, gives $\operatorname{ht} \mathfrak{p} \leq 1$, while $(0) \subsetneq \mathfrak{p}$ gives the reverse inequality. The height formula, Theorem 4.3 of the former lesson, yields

$$\dim(A/\mathfrak{p}) = \dim A - \operatorname{ht} \mathfrak{p} = d - 1.$$

Passing between nonempty affine opens of this integral component preserves its function field and hence its dimension by that same Theorem 4.2.

Choose one such reduced component Y. It is T-stable. Indeed, $T \times Y$ is integral: on affine charts of Y its algebra is the domain $A[t_1^{\pm 1}, \dots, t_r^{\pm 1}]$, and these charts have nonempty intersections. The closure of its image under the action is therefore irreducible, lies in $X \cap H$, and contains Y through the identity of T. Maximality of the irreducible component forces this closure to be Y. The action factors through its reduced structure, since pullbacks of its defining ideal vanish at every point of the reduced source and therefore vanish. The restriction is a linear action on the projective variety Y. Induction supplies at least d distinct T-fixed points in $Y \subset H$.

There is one more fixed point outside H. Choose $a \in (X \setminus H)(k)$, using the Nullstellensatz on this nonempty affine chart, and normalize its coordinates so that $c_0(a) = 1$. Its λ-orbit on the same chart has coordinates

$$c_j(\lambda(u)a) / c_0(\lambda(u)a) = u^{w_j - w_0} c_j(a), \qquad j > 0.$$

All exponents are nonnegative integers, so these polynomials define a morphism $\mathbf{A}^1_k \to \mathbf{P}^N_k \setminus H$. It lands in X: the equations of the closed affine subscheme $X \setminus H$ vanish after substitution in $k[u, u^{-1}]$, hence in its subring $k[u]$. At $u = 0$, every coordinate of weight greater than w_0 is zero;

the remaining coordinates have character χ_0 by the separating property. Every element of T, over every k-algebra, scales these remaining coordinates by the same unit $\chi_0(t)$. Thus the resulting projective point is T-fixed. Its zeroth coordinate is still one, so it lies outside H and differs from all d points obtained inside H. This proves the bound $d+1$. □

The curve duality used in rank one

Lemma (curve duality). If C is a smooth projective pure one-dimensional curve over a field k and L is invertible, then naturally

$$H^1(C,L)^* \simeq H^0(C,\Omega_{C/k}\otimes L^{-1}).$$

Proof. Embed $i : C \hookrightarrow P = \mathbf{P}^N_k$, and put $c = N-1$. The regular-immersion calculation in Projective cohomology and smooth affine models, Lemma 6.1, proves over the original field, including imperfect fields, that I/I^2 is locally free of rank c, that the ideal is locally generated by a regular sequence, and that

$$0 \longrightarrow I/I^2 \longrightarrow i^*\Omega_{P/k} \longrightarrow \Omega_{C/k} \longrightarrow 0$$

is exact. Its proof first calculates at geometric closed points, then descends the conormal module and the exact Koszul complex by faithful field extension.

Dualizing that Koszul resolution, with a local frame of L, gives

$$\mathcal{E}xt^q_P(i_*L,\omega_P) = 0 \quad (q \neq c), \qquad \mathcal{E}xt^c_P(i_*L,\omega_P) = i_*(L^{-1}\otimes i^*\omega_P \otimes \det(I/I^2)^*).$$

Here $\omega_P = \det \Omega_{P/k} = \mathcal{O}_P(-N-1)$. Taking determinants of the conormal sequence identifies the sheaf in degree c with $i_*(L^{-1}\otimes\Omega_{C/k})$. The finite-cover Hom double complex written in S02 Theorem 6.2 has only this nonzero sheaf-Ext row, and therefore gives

$$\operatorname{Ext}^c_P(i_*L,\omega_P) = H^0(C, L^{-1}\otimes\Omega_{C/k}).$$

S02 Ambient Theorem 6.B proves projective-space duality by the monomial trace, a two-stage twist presentation and dimension shifting; it identifies the same Ext group with $H^{N-c}(P,i_*L)^* = H^1(C,L)^*$. Its trace sends the class of $T_0^{-1}\cdots T_N^{-1}$ to one. All the Koszul, determinant, finite-cover and trace comparisons are natural. This proves the asserted isomorphism. □

Lemma 2.2 (the curve argument). Over an algebraically closed field, a smooth projective curve dominated by $\mathbf{P}^1$ is isomorphic to $\mathbf{P}^1$.

Proof. Write C for the dominated curve. The dense image of the irreducible $\mathbf{P}^1_k$ is irreducible. Smoothness gives reducedness, so C is integral and connected. If domination is initially given by a rational map, the homogeneous-coordinate DVR extension in the earlier S03 Theorem 2.2 makes it a morphism. Thus the hypotheses of S02 Proposition 6.5 hold: its complete separable differential calculation and inseparable-subfield reduction prove $g = \dim_k H^1(C,\mathcal{O}_C) = 0$ in every characteristic. Its proof is earlier in the programme; no unproved external form of Lüroth's theorem is used.

We supply explicitly the concluding rational-function calculation from that proposition. Choose a closed point $q \in C(k)$, which exists by the earlier Nullstellensatz. S02 Lemma 6.3 proves $H^0(C,\mathcal{O}_C) = k$, divisor Euler characteristics, the canonical degree $\deg\Omega_{C/k} = 2g-2 = -2$, and vanishing for negative-degree line bundles. Its earlier Theorem 6.2 gives duality. For every integer $m \geq 0$, these results yield

$$H^1(C,\mathcal{O}_C(mq))^* = H^0(C,\Omega_{C/k}(-mq)) = 0, \qquad h^0(C,\mathcal{O}_C(mq)) = m+1.$$

The first vanishing uses degree $-2-m<0$; the second is the proved Euler characteristic formula with $g=0$.

Take x in $H^0(C, \mathcal{O}_C(q))$ independent of the constants. In the divisor description of this sheaf, it is a rational function regular away from q with pole order at most one at q. It has a pole: otherwise it would be a global regular function and belong to k. Its pole order is therefore exactly one. The functions $1, x, \ldots, x^m$ belong to $H^0(C, \mathcal{O}_C(mq))$ and have distinct pole orders. A nonzero linear combination has the pole order of its highest nonzero power, since all coefficients lie in k and that term cannot be canceled by terms of smaller order. They are independent, and their number $m+1$ is the dimension just computed. They form a basis.

Let $h \in k(C)$ be any nonzero rational function. The earlier divisor proof shows that it has finitely many poles. List its poles $a_1, \ldots, a_r$ outside q. The function x is regular at each of them, and $x(a_i) \in k$. Each nonzero function $x - x(a_i)$ has positive valuation at a_i and no pole outside q. Choose integers $N_i \geq 0$ large enough that

$$P(x)h, \qquad P(T) = \prod_{i=1}^{r} (T - x(a_i))^{N_i},$$

has no pole outside q. Multiplying cannot introduce a pole at another such point. Its remaining pole order at q is finite, so the product is a section of $\mathcal{O}_C(mq)$ for some $m \geq 0$. By the basis calculation, it equals a polynomial $Q(x)$. The polynomial P is nonzero and x is transcendental over the algebraically closed k, so $P(x) \neq 0$. Consequently $h = Q(x)/P(x) \in k(x)$. The empty pole list gives $P = 1$, and the zero function is already in $k(x)$. Hence $k(C) = k(x)$.

This equality gives inverse rational maps between C and $\mathbf{P}^1_k$: on a nonempty affine chart, send its finitely many coordinate generators to their rational functions in the other function field and clear their denominators; their equations remain satisfied. Both targets are projective and both sources are smooth curves. The earlier S03 Theorem 2.2 extends these maps at every DVR by homogeneous coordinates. Their composites are the identity on the common function field and hence on a dense open. The closed-diagonal equalizer argument for the reduced integral sources makes the composites the identity everywhere. The resulting morphisms are inverse isomorphisms. This argument needs no degree formula for an unchecked inseparable map. □

Proposition 2.3. A semisimple group of rank one has exactly two Borel subgroups containing a maximal torus, its Borel quotient is $\mathbf{P}^1$, and the action on that quotient has image PGL_2 and kernel its centre.

Proof. Fix a maximal torus $T \subset B$ and put $X = G/B$. RG01 Lemma 2.1 constructs it as a closed orbit with a linear equivariant projective embedding and the full fppf B-torsor $G \to X$. The complete orbit proof S08 Theorem G.4.C makes this actual quotient smooth. It is integral, being reduced and the surjective image of the irreducible smooth connected G. RG02's complete torus-fixed-coset proof makes X^T a finite disjoint union of reduced k-points. Its following transitivity argument identifies this set with $N_G(T)/C_G(T)$. RG02 Theorem 2.1 gives $C_G(T) = T$, so this quotient acts faithfully on the rank-one character lattice $\mathbf{Z}$, whose automorphisms are precisely multiplication by 1 or -1. Hence X^T has at most two points. It has at least one, the coset B. If $\dim X = 0$, integrality and the Nullstellensatz make $X = \operatorname{Spec} k$, so the torsor stabilizer forces $G = B$, contradicting nontrivial semisimplicity. Lemma 2.1 now gives $|X^T| \geq \dim X + 1 \geq 2$. Therefore there are exactly two fixed points and $\dim X = 1$. The Borel-normalizer theorem already proved in RG02 identifies these with exactly two Borels containing T.

Let K be the schematic kernel of the action on X. It is the intersection of the conjugates of B. This retains the full kernel on every test algebra: in the equivariant projective embedding the fixed-point equations are the coordinate two-by-two minors. If they vanish on every constant k-point of the reduced X, each of their finitely many linearly independent scalar coefficients vanishes on every such point, hence vanishes identically by the Nullstellensatz. Thus these equations vanish after arbitrary scalar extension as well. Its reduced identity component is a smooth connected normal solvable subgroup, hence is trivial by semisimplicity. Normality here holds schematically: conjugation by the smooth connected G preserves that component, and its reduced source factors through its reduced structure, as in S08 Lemma G.4.1. Thus K is zero-dimensional. It is finite:

its affine algebra is Noetherian, has finitely many minimal primes, and in dimension zero they are maximal, with residue fields k. The nilradical is generated by finitely many nilpotent elements and hence is nilpotent, by the bound on monomial exponents. The reduced algebra is a finite product of copies of k; the successive nilradical quotients are finite modules over it. This gives a finite-dimensional algebra for K. A positive-dimensional Borel consequently has a nonconstant orbit map to X. Otherwise all its point orbits would be points, and density of rational points in the reduced $B \times X$ would make the action trivial schematically, contradicting $B \subset K$.

We supply the rational coordinates required for restricting such an orbit map to a line. For a smooth connected solvable group $B = U \rtimes T$, RG01 Lemma 2.B gives

$$U = U_0 \supset U_1 \supset \cdots \supset U_a = 1, \qquad U_i/U_{i+1} \simeq \mathbf{G}_a.$$

The quotients U/U_i are smooth connected and affine by the complete earlier RG01 Lemma 2.G: each U_i is a closed normal unitriangular subgroup of the smooth connected U. The resulting quotient maps are fppf torsors. The map

$$U/U_{i+1} \longrightarrow U/U_i$$

is explicitly a $U_i/U_{i+1} \simeq \mathbf{G}_a$-torsor. Pull it back along the fppf cover $U \to U/U_i$. If two local representatives have the same U_i-coset, their ratio belongs to U_i; its U_{i+1}-coset is unchanged on replacing either representative by its U_{i+1}-equivalent lift. Multiplication and division therefore give inverse maps between this pullback and $(U_i/U_{i+1}) \times U$. This is a local trivialization with the translation action, and its equality relation gives the asserted torsor on every test scheme. All these subgroups are normal, so either coset convention gives these quotient groups. Every such torsor over an affine algebra has a section by the explicit vector-line module-descent calculation in RG01's solvable-structure proof: the translation cocycle descends $0 \to A \to E \to A \to 0$, and a lift of 1 gives the section. This calculation works over any affine A. Translation by a section identifies the torsor with its base times $\mathbf{A}^1$. Starting with $U/U_0 = 1$ and repeating gives $U \simeq \mathbf{A}^a$ as varieties. The torus is split over the algebraically closed field, by the splitting argument in Lemma 2.1. Thus

$$B \simeq \mathbf{A}^a \times \mathbf{G}_m^r, \qquad k(B) = k(z_1, \ldots, z_d), \quad d = a + r.$$

These are scheme coordinates; no assertion about abstract group coordinates is needed.

Choose a nonconstant orbit map $b \mapsto bx$ and a nonconstant $h \in k(X)$. The map is dominant because its image has positive dimension in the integral curve, so its pullback $f \in k(B)$ is nonconstant. Choose the least $j \geq 1$ for which $f \in k(z_1, \ldots, z_j)$. Then $f \notin k(z_1, \ldots, z_{j-1})$. Write, after clearing finitely many coefficient denominators,

$$f = P(z_j)/Q(z_j), \quad P = \sum_i A_i z_j^i, \quad Q = \sum_i B_i z_j^i, \quad A_i, B_i \in k[z_1, \ldots, z_{j-1}].$$

Some coefficient minor $A_i B_l - A_l B_i$ is nonzero. If they all vanished, choosing $B_l \neq 0$ would give $P = (A_l/B_l)Q$, contradicting the choice of j.

The inverse image of the domain where h is regular contains a nonempty principal open of our coordinate chart. Choose a nonzero polynomial F defining such an open, multiplying by torus coordinates to clear Laurent denominators if necessary. Viewed as a polynomial in z_j, it has a nonzero coefficient in the other variables. Specialize every other coordinate so that this coefficient, the selected minor, and all fixed torus coordinates stay nonzero. Such a specialization exists: their product is a nonzero polynomial over the infinite field k, and repeated use of the univariate root bound in S02 Lemma 4.A supplies a point where it does not vanish. For no remaining variables the product is already a nonzero scalar. On the resulting coordinate line, P/Q stays nonconstant and F does not vanish identically. If z_j is a torus coordinate, restrict also to $z_j \neq 0$. We have obtained a nonconstant rational map $\mathbf{A}^1 \dashrightarrow X$, since its composition with h is this nonconstant rational function. The explicit DVR extension calculation in S03 Theorem 2.2 extends it to $\mathbf{P}^1 \to X$. Lemma 2.2, which includes inseparable domination, identifies X with $\mathbf{P}^1$.

The natural map $\mathrm{PGL}_2 \to \mathrm{Aut}(\mathbf{P}^1)$ is an isomorphism on all test algebras. An automorphism sends $0, 1, \infty$ to three sections that remain pairwise disjoint on every fibre. Locally trivialize their quotient lines. A change of basis sends $0, \infty$ to the first and third; the two nonzero coordinates of the middle section are units, so diagonal scaling also sends 1 to it. For uniqueness, an automorphism fixing 0 and ∞ preserves both Cartier divisors. The ratio of its pulled-back coordinate to that coordinate is therefore an invertible regular function on all of $\mathbf{P}^1$. S01 Theorem 7.2 gives $H^0(\mathbf{P}^1_R, \mathcal{O}) = R$, so that ratio is a base unit. Fixing 1 makes it one. The local projective matrices consequently agree on overlaps and glue in the represented projective linear group. This proves the assertion even over rings with nilpotents.

Construct the schematic image H directly from the image of the coordinate Hopf algebra of PGL_2 in the domain $k[G]$. Its finitely generated image algebra is a domain; the kernel is a Hopf ideal because tensoring inclusions over a field is injective. Thus H is a closed integral affine group. The full integral-target monic-division and translation proof in AG-GS-04 Lemma 11.1c makes $G \to H$ faithfully flat and of finite presentation. The kernel identity gives its torsor relation. Lemma 1.C above makes reduced H smooth, and integrality makes it connected. The orbit-dimension calculation of S08 Lemma G.4.D applied to the left action on H gives $\dim H = \dim G - \dim K$. Its kernel K is finite. Its torus image has dimension one, and maximal-torus conjugacy in RG01 conjugates it to the diagonal torus of PGL_2. If H were proper, it would have dimension at most two. In every characteristic, $\operatorname{Lie} \mathrm{PGL}_2 = \mathfrak{gl}_2/kI$ has the diagonal line of weight zero and two distinct root lines of weights $1, -1$ for the cocharacter represented by $\mathrm{diag}(t, 1)$. The T-stable Lie algebra of H contains the diagonal line and at most one root line. Choose that cocharacter or its inverse so that all its weights on $\operatorname{Lie} H$ are nonnegative. The general limit-subgroup proof already given in RG01 Lemma 3.3 makes $P_H(\lambda)$ smooth with this entire Lie algebra, hence of dimension $\dim H$. A closed subgroup of that dimension in the connected smooth H is all of H. But

$$P_H(\lambda) = H \cap P_{\mathrm{PGL}_2}(\lambda),$$

and the ambient limit subgroup is triangular. Thus H would be solvable. The finite group $K(k)$ is central in $G(k)$: each conjugation orbit is the image of the connected reduced G in a finite set and is the identity translate. Every $H(k)$-point lifts through the faithfully flat finite-type image map by the Nullstellensatz. Hence $G(k)$ is a central extension of a solvable group and is solvable. Its finite derived-word identities hold on the dense rational points of each reduced finite Cartesian power of G, so they hold schematically, making G solvable. This contradicts semisimplicity. Therefore $H = \mathrm{PGL}_2$.

In particular $\dim G = 3$. Since $\dim X = 1$, each of the two Borels B_+, B_- has dimension two and unipotent radical of dimension one. The one-dimensional unipotent classification proved in S03 Section 4 identifies each radical with $\mathbf{G}_a$. Its image has dimension one because the kernel is finite. Projection of the corresponding point stabilizer onto its diagonal torus gives a character of the radical, trivial by RG01 Lemma 2.A. Thus its reduced closed image lies in the ambient unipotent line; dimension one makes it that entire line. Its torus character α_+ or α_- is nonzero, since the induced diagonal-torus map is an isogeny and the ambient root character is nonzero. The normalizer element interchanging the two fixed points acts by inversion on T, so $\alpha_- = -\alpha_+$.

The Lie algebra of K lies in both $\mathfrak{b}_+$ and $\mathfrak{b}_-$. Their nonzero-weight lines have these distinct opposite characters, while the shared torus has weight zero. Thus $\operatorname{Lie} K$ has no vector in either unipotent root line. Each finite intersection $K \cap R_u(B_\pm)$ is torus-stable. Since $\alpha_\pm \neq 0$, its polynomial ideal is homogeneous for the distinct weights $i\alpha_\pm$, hence is (x^n) for some $n \geq 1$. The Hopf identity requires every intermediate binomial coefficient in $(X + Y)^n$ to vanish. In characteristic zero the coefficient n excludes $n > 1$. In characteristic p, write $n = p^e m$ with $p \nmid m$. If $m > 1$, the coefficient m of $X^{p^e} Y^{n-p^e}$ in $(X^{p^e} + Y^{p^e})^m$ is nonzero, a contradiction. Thus $n = p^e$. Any $n > 1$ gives nonzero tangent space at the identity, already excluded. Both intersections are therefore trivial. The image maps of the two radicals are faithfully flat with trivial kernel, hence isomorphisms onto the ambient unipotent subgroups, by their full torsor relations.

For a point of $B_+ \cap B_-$ over any test algebra, write it tu_+ by the solvable decomposition. Its image lies in the diagonal intersection of the two ambient point stabilizers, so the image of u_+ is trivial. The just-proved isomorphism gives $u_+ = 1$. Therefore $B_+ \cap B_- = T$ schematically and $K \subset T$. A finite normal multiplicative-type subgroup of a smooth connected group is central: over k it has a finite character group, and the earlier coefficient proof for diagonalizable groups represents its automorphisms by that finite constant étale character-automorphism group. The conjugation map from connected G to this discrete group has value the identity automorphism at the identity of G, hence is constant with that value as a scheme morphism. Thus $K \subset Z(G)$. Conversely $Z(G)$ maps to the centre of PGL_2, which is trivial schematically. Indeed a central automorphism of $\mathbf{P}^1$ commutes with the diagonal torus and preserves its fixed sections $0, \infty$; interchanging them would invert the universal torus parameter and cannot commute with it. It is therefore a scaling by a base unit, and commuting with every translation forces that unit to be one. Thus $Z(G) \subset K$, proving $K = Z(G)$ and every assertion. □

For a reductive group of semisimple rank one, the radical-quotient proof in Section 1 produces a semisimple group Q of rank one. The inverse image of a Borel of Q is smooth connected and solvable by that same central-torus torsor argument; maximality and the inverse-image comparison make it a Borel of G. It has the same projective-line quotient, by the equality of its fppf cosets. Conversely a Borel of G contains the radical: their product is smooth connected and solvable, since the radical is central, and maximality forces equality with the Borel. Its image in Q is a Borel by the inverse-image comparison. In its decomposition $B = U \rtimes T$, the radical belongs to T by RG01 Lemma 2.N and centrality, and $U \cap \mathrm{Rad}(G) = 1$ schematically by the unitriangular character calculation in RG01 Lemma 2.A. Quotienting the product gives $B/\mathrm{Rad}(G) = U \rtimes (T/\mathrm{Rad}(G))$ as fppf sheaves, hence as the represented groups. Thus its two unipotent radicals map isomorphically to the additive radicals of Q, with their opposite characters pulled back to T. The action kernel is $q^{-1}(Z(Q))$. A point of this kernel lies in both opposite Borel inverse images, whose intersection is T by the displayed product descriptions and $B_+ \cap B_- = T$ for Q. It acts trivially on both root lines. The centre formula in Section 1, now with those two nonzero weights, makes it central. The reverse containment follows from the trivial centre of the projective linear image. Thus its centre is precisely the kernel of the action on that same projective line.

Now let α be any nonzero weight of a maximal torus of an arbitrary reductive k-group. Define the codimension-one torus

$$T_\alpha = (\ker \alpha)^0_{\mathrm{red}}.$$

The centralizer $G_\alpha = C_G(T_\alpha)$ is reductive by the preceding lesson. Its weights are precisely the weights of G that are rational multiples of α, since these and only these vanish on T_α. Its maximal central torus has codimension one in T: the nonzero characters on T/T_α span a rank-one lattice. Thus its semisimple rank is one. Proposition 2.3 proves

$$\Phi \cap \mathbf{Q}\alpha = \{\alpha, -\alpha\}, \qquad \dim \mathfrak{g}_\alpha = 1.$$

This proves both the reducedness along each root line and the one-dimensional root-space assertion without appealing to a general root-system classification.

3. Subgroups obtained by taking a limit

The definitions and full grading, smoothness, connectedness and open-cell proof for these subgroups are in Tori, maximal tori and their conjugacy, Lemma 3.3, an earlier lesson. In this lesson Lemma 3.1 denotes that already proved general lemma; it applies to every smooth affine group of finite presentation, without reductivity or a root classification.

For SL_2 and $\lambda(t) = \mathrm{diag}(t, t^{-1})$, conjugation sends $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ to $\begin{pmatrix} a & t^2 b \\ t^{-2} c & d \end{pmatrix}$. Thus $P(\lambda)$ is upper triangular, $U(\lambda)$ is upper unipotent, and $L(\lambda)$ is diagonal. This calculation works in characteristic two because the group character t^2 remains nontrivial.

4. Root groups over the base

Let $T = D_S(M)$ be a split maximal torus. Locally the nonzero weights are a fixed finite set $\Phi \subset M$, and each $\mathfrak{g}_\alpha$ is a line bundle by Proposition 2.3 on geometric fibres. Define T_α by the torsion-free quotient of $M/\mathbf{Z}\alpha$. Then

$$G_\alpha = C_G(T_\alpha), \qquad \mathrm{Lie}(G_\alpha) = \mathfrak{t} \oplus \mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}.$$

Choose a cocharacter λ of T with $n = \langle \alpha, \lambda \rangle > 0$, and set $U_\alpha = U_{G_\alpha}(\lambda)$. It is a smooth connected one-dimensional unipotent group, with Lie algebra $\mathfrak{g}_\alpha$.

For a line bundle L, write $\mathbf{V}(L) = \mathrm{Spec}_S \mathrm{Sym}(L^*)$ for its additive group.

Theorem 4.1. There is a unique T-equivariant homomorphism

$$\exp_\alpha : \mathbf{V}(\mathfrak{g}_\alpha) \longrightarrow G$$

whose differential is the inclusion of the root line. It is a closed immersion, with image U_α, and commutes with every base change. Moreover

$$U_{-\alpha} \times T \times U_\alpha \longrightarrow G_\alpha$$

is an open immersion.

Proof. The earlier RG01 Lemma 3.3, used as Lemma 3.1 here, gives the smooth affine subgroup $U_\alpha = U_{G_\alpha}(\lambda)$ of finite presentation, its action extending to $\mathbf{A}^1$ with zero equal to the identity, and its Lie module $\mathfrak{g}_\alpha$. On that line $\lambda(t)$ acts by t^n, with the positive integer $n = \langle \alpha, \lambda \rangle$. The earlier RG01 Lemma 3.D therefore constructs a unique normalized equivariant group isomorphism

$$\mathbf{V}(\mathfrak{g}_\alpha) \xrightarrow{\sim} U_\alpha.$$

Its entire graded-algebra, differential and gluing proof works over the original base, even if n is zero in some residue fields. Since T commutes with λ, the last assertion of that lemma makes this isomorphism T-equivariant on every test algebra. Compose it with the closed immersion $U_\alpha \subset G_\alpha \subset G$ to obtain $\exp_\alpha$.

For uniqueness as a homomorphism into G, let h be any T-equivariant homomorphism with the required differential. The subtorus T_α acts trivially on its source, so the equality expressing equivariance makes every value of h commute with T_α, including its universal points. Hence h factors through the represented centralizer G_α. Moreover

$$\lambda(t) h(x) \lambda(t)^{-1} = h(t^n x).$$

The right side extends regularly to $t = 0$ and there equals $h(0) = 1$. The defining orbit-limit equations from RG01 Lemma 3.3 therefore force h to factor through $U_{G_\alpha}(\lambda)$. In a local Lie-line frame, compose with the inverse of the preceding isomorphism. The result is a weight-n equivariant endomorphism of an affine line with identity differential. The Laurent-coefficient calculation in RG01 Lemma 3.D makes it the identity. These equalities hold on overlaps, proving global uniqueness.

We also verify the equality needed for the open cell over the original base. The centralizer theorem of the preceding lesson makes the geometric fibres of $L = L_{G_\alpha}(\lambda)$ connected. The two opposite nonzero weights established in Section 2 and RG01 Lemma 3.3 give $\mathrm{Lie}(L) = \mathfrak{t}$. On a geometric fibre the closed torus T thus has the same dimension as the smooth connected L, and is all of it: a proper closed subset of the irreducible smooth connected group has smaller dimension. Over S, both groups are smooth, and the inclusion $T \to L$ has invertible differential at the identity and hence everywhere by translation. S02 Lemma 3.A makes it étale; S04 Lemma A1.9 makes this monomorphism an open immersion. Its image contains every geometric fibre by the equality just proved, so it is all of L. Thus $L = T$ as schemes over S. Applying the open-cell assertion of RG01 Lemma 3.3 inside G_α, with the cocharacters λ and $-\lambda$, gives the displayed open immersion. Formation of these centralizers and limit subgroups commutes with every base change by their defining equations. So do their Lie modules and the uniquely normalized isomorphism of RG01 Lemma 3.D. All the asserted constructions are therefore compatible with every base change. □

The notation $\exp_\alpha$ denotes this algebraic map, not the analytic exponential series. No division by factorials occurs. A root-group parameterization $x_\alpha : \mathbf{G}_a \simeq U_\alpha$ is exactly a trivializing section of the root line.

Example 4.2. In GL_n, conjugation by $\mathrm{diag}(t_i)$ sends E_{ij} to $(t_i/t_j)E_{ij}$. Thus

$$\alpha_{ij} = e_i - e_j, \quad \mathfrak{g}_{\alpha_{ij}} = \mathcal{O}_S E_{ij}, \quad x_{ij}(u) = I + uE_{ij}.$$

Example 4.3. Let $E = \mathcal{O}_S \oplus L$ for a line bundle L. Define $\mathrm{SL}(E)$ by the automorphisms whose action on $\bigwedge^2 E = L$ is the identity. On opens trivializing L this is exactly the determinant-one matrix group of RG01 Lemma 1.M. If two frames differ by a unit ℓ, their matrix coordinates differ by conjugation with $\mathrm{diag}(1, \ell)$; these conjugations satisfy the same cocycle as the frames and preserve determinants. The complete affine algebra/morphism descent of S04 Lemmas A1.2, A1.3a and A1.5 therefore glues these local affine smooth finite-presentation groups into $G = \mathrm{SL}(E)$. Each geometric fibre is the already proved reductive SL_2. The local diagonal maps (t, t^{-1}) commute with the transitions, so glue to a closed split maximal torus, maximality being checked on those fibres.

The off-diagonal endomorphism modules are $\mathrm{Hom}(L, \mathcal{O}) = L^{-1}$ above and $\mathrm{Hom}(\mathcal{O}, L) = L$ below. Conjugation gives their full weights t^2 and t^{-2}, and the trace-zero diagonal tangent module has weight zero. For an upper entry $u : L \to \mathcal{O}$, its square as an endomorphism of E is zero. Consequently $I + u$ has determinant one, inverse $I - u$, and $(I + u)(I + v) = I + (u + v)$ on every test scheme. This identifies the upper subgroup with $\mathbf{V}(L^{-1})$; the same calculation gives $\mathbf{V}(L)$ below. They are the root groups of Theorem 4.1 because their equivariant tangent maps are the displayed identities, whose uniqueness that theorem proves. Here $\mathbf{V}(M)$ means the additive group whose local sections are sections of the line module M. Its Lie module is M, by the same square-zero calculation. A group-scheme isomorphism with $\mathbf{G}_a$ would induce an isomorphism $M \simeq \mathcal{O}_S$ on Lie modules. Thus when L is nontrivial neither displayed root group is globally isomorphic to $\mathbf{G}_a$. Over a field every line is trivial, by choosing a nonzero vector. Having a split maximal torus is therefore weaker than having globally parameterized root groups.

5. A projective quotient and its rank-one form

We use the general algebraic-space tool proved before this lesson in Affine descent, Zariski Main and recognition of spaces, Theorems E8.1 and E9.1. The proof there includes the finite flat affine quotient, the off-diagonal induction, separated locally quasi-finite scheme effectivity, and relative integral closure with its étale base change. It applies over every base and on every test scheme.

Quasi-finite normalization lemma. Let S be a scheme and let $f : X \to Y$ be a quasi-finite separated morphism of algebraic spaces over S. Let Y' be the relative integral closure of Y in X. Then the factorization $X \xrightarrow{f'} Y' \xrightarrow{\nu} Y$ has f' a quasi-compact open immersion and ν integral. In

particular f is quasi-affine.

Proof. Theorem E9.1 of the preceding supporting lesson proves the full factorization, after Theorem E8.1 establishes recognition and Corollary E8.2 establishes representability on every scheme test. Its integral algebras are descended using the proved affine faithfully flat equalizer, and Corollary D4.3 supplies their étale base-change compatibility. Thus for every affine scheme $Z \to Y$, the pullback $Z \times_Y X$ is a quasi-compact open subscheme of the affine integral scheme $Z \times_Y Y'$. This is quasi-affineness. All these input proofs precede the present application in the programme. □

The relative integral closure retains nilpotents; a reduced normalization is not substituted for it. The full argument is in the preceding supporting lesson, rather than being delegated to a source citation.

The following construction will also be used for flag varieties. It makes explicit why a fibrewise projective quotient is a scheme, even before we have classified the group over the base.

A flat-source fibre criterion. A finitely presented closed immersion $Y \hookrightarrow Z$ over S, with Y flat over S, is an isomorphism if it is so on every geometric fibre. Locally write its ideal as $I \subset C$. Flatness of C/I makes $I \otimes k(s) \to C \otimes k(s)$ injective. Fibre equality makes this module zero. Since I is finitely generated as an ideal, Nakayama at every prime of C gives $I = 0$. No flatness assumption on Z is needed.

Formal separation and normalizers

We first justify identity-neighbourhood detection over bases with nilpotents, and then prove the full finite-jet normalizer assertion.

Formal separation lemma. Let R be Noetherian and let H/R be a smooth affine group of finite presentation with geometrically connected fibres. Write $C = \mathcal{O}(H)$ and let J be its augmentation ideal. Then

$$\bigcap_{n \geq 1} J^n = 0.$$

Proof. The entire affine smooth-field component proof of AG-RG-S02 Lemma 4.1 makes every geometric fibre integral. For a prime $\mathfrak{p} \subset R$, flatness embeds $C/\mathfrak{p}C$ into its generic fibre over the fraction field of $R/\mathfrak{p}$: multiplication by any nonzero base element stays injective. That generic fibre is integral by geometric connectedness and smoothness, so $C/\mathfrak{p}C$ is a domain. Its augmentation onto $R/\mathfrak{p}$ also shows that it is nonzero.

We give the prime-filtration argument used here. In any nonzero finite module over a Noetherian ring choose a maximal annihilator of a nonzero element. It is prime: if ab annihilates that element and b does not, the nonzero element obtained by multiplying by b has a larger annihilator unless a already belongs to the first annihilator. Insert this cyclic prime-annihilator submodule and repeat in the quotient. The ascending chain condition on submodules makes the process stop. Apply it to the finite R-module R and tensor its filtration with the flat C. Its factors are $C/\mathfrak{p}C$. For any $j \in J$, the element $1 - j$ has augmentation one in each factor, so is nonzero in that domain and acts injectively. Induction through the finite filtration makes it a nonzerodivisor on C.

The algebra C is Noetherian by the full Hilbert-basis calculation of AG-RG-S01 Lemma 1.3. Put $I = \bigcap_n J^n$. The complete Artin–Rees proof of AG-RG-S02 Lemma 1.1 gives $I = JI$ by applying it to the finite submodule $I \subset C$ and taking $n = c + 1$. With finite generators of I, write this equality as $v = Mv$, with entries of M in J. The adjugate identity gives $\det(1 - M)I = 0$, with $\det(1 - M) = 1 - j$ for a $j \in J$. The just-proved nonzerodivisor property gives $I = 0$. This proves the stated separation, including when R is nonreduced. □

Normalizer lemma. If H is a smooth closed subgroup with geometrically connected fibres of a smooth affine group K of finite presentation over any scheme S, then its scheme-theoretic normalizer

is a closed subgroup of finite presentation in K. Its formation commutes with every base change.

Proof. First suppose $S = \operatorname{Spec} R$ is Noetherian. We justify the finite locally free identity neighbourhoods used for the equations. For a smooth affine group F/R with augmentation ideal J_F, the identity conormal bundle $J_F/J_F^2 = \mathrm{Lie}(F)^*$ is finite locally free. On a base neighbourhood where it is free of rank d, lift a basis to functions $y_1, \dots, y_d \in J_F$. They define $F \to \mathbf{A}^d_R$, sending the identity to zero and with invertible differential there. The complete differential proof of AG-RG-S02 Lemma 3.A makes it etale on a neighbourhood of that section. The full formal-etale lifting theorem of AG-FSE, *Infinitesimal lifting and invariance under thickenings*, Theorem 4.1, constructs its unique inverse on successive zero thickenings. At each step the kernel $(y)^n/(y)^{n+1}$ is square-zero; both composites are the identity by lifting uniqueness. The thickenings are supported on the identity and stay in the chosen etale neighbourhood. Hence, locally on the base,

$$\mathcal{O}(F)/J_F^{n+1} \simeq R[y_1, \dots, y_d]/(y_1, \dots, y_d)^{n+1}.$$

This proves finite local freeness of every identity neighbourhood and the intrinsic associated graded identity

$$\operatorname{gr}_{J_F} \mathcal{O}(F) = \mathrm{Sym}(\mathrm{Lie}(F)^*).$$

The monomial descriptions agree through the natural graded maps, so the latter identity is independent of coordinates and glues.

Apply this to K_n, H_n, the nth identity neighbourhoods of K, H. They are finite locally free schemes, without a claim that these thickenings are subgroup schemes. The closed immersion $H \subset K$ induces a surjection on augmentation ideals: lift a zero-augmentation function using the coordinate-algebra surjection. Thus its augmentation powers are also the induced powers, and $H_n = H \times_K K_n$. Conjugation preserves the identity and all these powers. It acts by invertible linear maps on the finite locally free coordinate module of K_n.

Write $D = \mathcal{O}(K)$ and choose finite generators f_i of the ideal of H. Restrict their universal conjugates to H_n, for conjugation by both g and g^{-1}. A section in $D \otimes_R \mathcal{O}(H_n)$ vanishes exactly when all its coefficients vanish: locally use a free basis, or globally a finite generating list of its projective dual. The dual and its evaluation commute with arbitrary base change by finite local freeness. These are finitely many equations in D and define the closed finite-presentation stabilizer $N_n \subset K$ of the closed subscheme $H_n \subset K_n$. They represent the stabilizer on every test ring, including nonflat and nonreduced tests. Its two conjugation conditions make it a subgroup.

Its ideals $L_n \subset D$ increase: preserving the $(n+1)$st thickening preserves the nth. Since D is Noetherian, their union is L_N at one finite index N. Over the universal Noetherian quotient $A = D/L_N$, both conjugation restrictions vanish on every identity neighbourhood of H_A. The formal separation proof above applies to the smooth geometrically connected H_A, so each restricted defining equation is zero in its whole coordinate algebra. Both conjugations therefore preserve the entire H_A. Pulling back these actual zero equations proves sufficiency over every R-algebra. Conversely an automorphism preserving H preserves its augmentation powers, giving necessity. Thus $N_K(H) = N_N$ as full scheme functors. This finite universal description proves arbitrary base change, rather than assuming that an infinite intersection commutes with it.

For an arbitrary affine base, first descend K, H simultaneously to a finitely generated $\mathbf{Z}$-algebra, retaining smoothness, the closed immersion and group-law identities by the complete smooth-model proof AG-RG-S02 Proposition 3.2. Then the entire connected-fibre model argument of its Theorem 4.5 enlarges that coefficient stage so that H has geometrically connected fibres; the already retained smoothness and finite identities of K survive that base change. No connectedness hypothesis is imposed on K. Apply the preceding universal Noetherian construction there. Its pullback represents the original normalizer on every test scheme. The unique represented subgroup identifications agree on overlaps and glue. This proves closedness, finite presentation and every asserted base change over the original S. □

Cohomology of a projective completion

The preceding supporting lesson Projective cohomology and smooth affine models, Theorem 2.3, proves the precise formal-functions statement used here. If A is Noetherian, X/A projective, F coherent and $I \subset A$, then

$$H^q(X,F)^{\wedge}_I \simeq \varprojlim_n H^q(X, F/I^nF).$$

Here is the mechanism of that written proof. Over the Rees algebra $B = \bigoplus I^n t^n$, the sheaf $\bigoplus I^nF$ is finite on the projective base change X_B; projective Serre finiteness therefore makes $\bigoplus H^q(X, I^nF)t^n$ finite over B. If $M^q = H^q(X,F)$, $J^q_n = \operatorname{im}(H^q(X,I^nF) \to M^q)$ and K^q_n is its kernel, finite homogeneous generators give constants a_q, b_q with

$$I^nM^q \subset J^q_n \subset I^{n-a_q}M^q, \qquad K^q_m \longrightarrow K^q_n = 0 \quad (m \geq n + b_q).$$

The sheaf exact sequence consequently gives

$$0 \longrightarrow M^q/J^q_n \longrightarrow H^q(X, F/I^nF) \longrightarrow K^{q+1}_n \longrightarrow 0.$$

Its compatible kernel coordinates vanish by the displayed transition bound; the remaining quotient filtration is cofinal with I^nM^q. This proves the inverse-limit comparison, without an unexplained exchange of cohomology and inverse limit.

S02 Lemmas 1.2–1.3 prove exact finite-module completion, Noetherianity and faithful flatness of local completion. Thus a finite module over a complete local Noetherian ring is complete. S02 Lemma 1.4 proves unique lifting of idempotents through nilpotent ideals, and Corollary 2.4 turns the compatible thickened idempotents into an actual section of a projective scheme. These are precisely the two completion operations in the quotient proof below.

Lemma 5.A (Grassmannians). The scheme $\mathrm{Gr}_d(V)$ of locally free rank-d quotients of a finite locally free V/S is smooth and projective, commutes with arbitrary base change, and its Plücker line bundle is $\det Q$, for its universal quotient Q.

Proof. Trivialize $V = \mathcal{O}^n$. On the open where columns I give a basis of a quotient, its unique normalized matrix has those columns equal to the identity and $d(n-d)$ free entries. A second basis J gives the transition $M_J = (M_I|_J)^{-1}M_I$. These open affine charts represent the quotient functor on all tests and glue. The determinant quotient $\bigwedge^d V \twoheadrightarrow \det Q$ defines the Plücker map to $\mathbf{P}(\bigwedge^d V)$, with the quotient convention. Over the target chart $p_I \neq 0$, the ratios of minors containing a single replacement of an identity column recover, with their determinant signs, every free matrix entry. The corresponding map of coordinate rings is surjective. These target charts cover the projective bundle, so the Plücker map is a closed immersion, and its $\mathcal{O}(1)$ pulls back to $\det Q$. The construction glues for general V and uses only invertible minors, hence works over nonreduced bases. The subbundle convention follows by dualizing. □

Projective homogeneous quotients

Lemma 5.1. Let H be a smooth closed subgroup with geometrically connected fibres of a smooth affine S-group K with geometrically connected fibres. Suppose $N_K(H) = H$, and suppose every geometric quotient $K_{\bar{s}}/H_{\bar{s}}$ is proper. Then the fppf quotient $X = K/H$ is a smooth projective S-scheme, locally on S. Its universal conjugate subgroup $\mathcal{H} \subset K_X$ supplies a canonical relatively ample line bundle $\det(\mathrm{Lie}\,\mathcal{H})^*$.

Proof. The claim is local on the base, so work on affine pieces with fixed smooth relative dimensions. If $\dim H = 0$, its connected smooth geometric fibres are the identity points. The identity section into H is an etale monomorphism by the entire differential and open-monomorphism proofs of AG-RG-S02 Lemma 3.A and AG-RG-S04 Lemma E7.2. It is surjective on geometric fibres and hence an isomorphism. Thus $H = 1$. Its normalizer is K, so the hypothesis gives $K = 1$ and

$X = S$. The asserted line bundle is then relatively ample through $\mathbf{P}(\mathcal{O}_S) = S$. We may treat the positive-dimensional pieces below.

The entire finite-model theorem of AG-RG-S02 Theorem 4.5 descends K, H and their finite equations to a finite-type Noetherian base S_0, retaining smoothness and geometrically connected fibres. The full normalizer theorem just proved commutes with every base change. The equality $N_K(H) = H$ descends at a later coefficient stage: both are finitely presented closed subschemes of the affine group, and equality of their finite ideal generators uses finitely many relation witnesses. Work on the open-and-closed model rank pieces for the finite locally free modules below. Their pullbacks cover the original base, so this introduces no additional hypothesis.

The complete free-action proof of AG-RG-S07 Corollary 8.3 constructs $X_0 = K_0/H_0$ as an algebraic space and the full fppf H_0-torsor $K_0 \to X_0$, with every base change. Flatness and local finite presentation of X_0/S_0 descend on etale scheme charts by the complete AG-RG-S07 Lemma 5.2. Here is the smoothness argument in full. Choose an etale scheme chart $W \to X_0$. For any $s \in S_0$ and any field extension $L/\kappa(s)$, the scheme $K_L \times_{X_L} W_L$ is etale over the smooth L-group K_L, hence smooth over L by the square-zero composition proof of AG-CA-17 Theorem 1.2 and finite-presentation permanence. Its local rings are regular by AG-CA-18 Theorem 2.1. Its map to W_L is surjective and flat, being the base change of the torsor. At each point of W_L, a point above it gives a faithfully flat local map from its local ring into a regular local ring. The full regularity descent in *Descending properties*, Proposition 1.1, applies: the residue-field resolution acquires a projective syzygy upstairs by AG-CA-14 Theorem 2.2, that finite projectivity descends by AG-RG-S04 Lemma A1.3 and AG-CA-07 Theorem 5.3, and the same residue-field criterion proves regularity downstairs. Thus every W_L is regular. To turn this into fibre smoothness without an omitted geometric-regularity converse, fix $w \in W_s$ and choose an algebraically closed field L containing $\kappa(w)$ by AG-RG-S07 Lemma 2.3. Evaluation gives a rational closed point $z \in W_L$ above w. The cotangent calculation of AG-CA-18 Proposition 3.1 identifies its differential fibre with its maximal ideal modulo its square. Regularity makes its dimension equal to $\dim \mathcal{O}_{W_L,z}$, which is $\dim_z W_L$ by AG-CA-09 Theorem 6.1. AG-RG-S07 Lemma 3.1 identifies this point dimension with $\dim_w W_s$. Differential base change, AG-CA-15 Theorem 2.2, gives the same cotangent dimension at w. The full field Jacobian criterion AG-CA-18 Theorem 2.1 now makes W_s smooth at w. Flatness, finite presentation and the complete arbitrary-base fibrewise criterion of *Smooth morphisms*, Theorem 3.1, make W/S_0 smooth. This proves smoothness of X_0 on its etale charts. Geometric connectedness follows from the surjective image of each connected $K_{\bar{s}}$. Quasi-compactness follows from the quasi-compact surjective source. The quotient is separated: the pullback of its diagonal to $K_0 \times K_0$ is the inverse image of the closed H_0 under $(g, g') \mapsto g^{-1}g'$. Apply the whole ideal/closed-immersion descent of AG-RG-S04 Corollary A3.2 on etale scheme charts. Thus X_0 is a separated finite-presentation space over S_0.

Choose the finite jet index $n \geq 1$ that detects the full normalizer on this model. The whole preceding coordinate proof makes $\mathcal{O}(K_n)$ and its quotient $\mathcal{O}(H_n)$ finite locally free. Their ranks are fixed on the pieces chosen above. Conjugation gives a section of the Grassmannian of quotients of $\mathcal{O}(K_n)$ of rank $\operatorname{rank} \mathcal{O}(H_n)$, with stabilizer exactly H. Its orbit map factors through the quotient sheaf. It is a monomorphism on all tests: lift two cosets locally to g_1, g_2; equality of their Grassmannian quotients says that $g_2^{-1}g_1$ stabilizes H_n, and the full stabilizer theorem puts this element in H. The cosets are equal. Thus we have a finite-type monomorphism

$$\theta : X_0 \longrightarrow \mathrm{Gr}\,.$$

It is separated and quasi-finite. The full space recognition and normalization proofs of AG-RG-S04 Theorems E8.1–E9.1, as stated in the preceding quasi-finite normalization lemma, make X_0 a scheme. On each affine Grassmannian chart its pullback is an open subscheme of the represented affine integral closure; these scheme opens cover it.

We now construct the compactification needed below. The Grassmannian is qcqs and projective by the complete Lemma 5.A. The full scheme finite-factorization theorem AG-RG-S04 Theorem D7.1

applies to θ and gives

$$X_0 \hookrightarrow \overline{X}_0 \xrightarrow{\pi} \mathrm{Gr},$$

with first map a quasi-compact open immersion and π finite. We supply the full universal closed-image calculation for projective space used here. Over an arbitrary affine base $\operatorname{Spec} R$, a closed subset $C \subset \mathbf{P}^N_R$ is cut out by a homogeneous ideal $I \subset R[T_0, \dots, T_N]$. Indeed on the chart $T_i \neq 0$ its closed subset is cut out by an ideal J_i in the ratio coordinates. Homogenize every $f \in J_i$ and multiply its homogenization by T_i. These homogeneous polynomials vanish on C also outside this chart because of that last factor. If a point is outside C, choose a chart containing it and an $f \in J_i$ nonzero at that point; its homogenization times T_i is still nonzero there. The ideal generated by all these polynomials consequently cuts out exactly C. Put $D = R[T]/I$. A fibre over $p \in \operatorname{Spec} R$ is empty precisely when $\operatorname{Proj}(D \otimes_R \kappa(p))$ is empty. For such an empty fibre each T_i is nilpotent in this graded algebra: otherwise its nonzero localization has a nonzero degree-zero subring, whose spectrum gives a point in the $T_i \neq 0$ chart by the prime-existence proof. A sufficiently large degree m therefore has $(D_m) \otimes_R \kappa(p) = 0$, since every degree-$m$ monomial contains one of those vanishing powers. The module D_m is finite over R, being a quotient of the finite free degree-m polynomial module. Local Nakayama gives $(D_m)_p = 0$. Its finitely many generator images are killed by denominators outside p, so one $a \notin p$ gives $(D_m)_a = 0$. Every higher graded piece is generated from degree m, since D is generated in degree one; hence all those pieces also vanish over R_a. Every projective fibre over $D(a)$ is empty. The complement of the image of C is thus open. This argument works for every ring and every closed subset, so it proves universal closedness after every base change, affine locally on the new base. The finite affine projective charts give finite type and the diagonal calculation of AG-RG-S01 Section 7 gives separatedness. Therefore projective space, and every closed subscheme of it, is proper. Apply this to the actual Plucker closed immersion of the Grassmannian from Lemma 5.A. The scheme $\overline{X}_0$ is now proper over S_0: finite maps are finite type and separated on affine charts, and the integral closed-image calculation of AG-RG-S04 Lemma P0.4 proves universal closedness after every base change. Properness is preserved by composition. We use this constructed finite-over-projective compactification, and require no assertion that it is itself projective.

Let p be in the image of the original base in S_0. Its quotient fibre is proper: after an algebraically closed extension of a residue field of a point above p, it is the given proper geometric quotient. Properness descends under a field extension here directly. Finite type and separatedness are already known; for any base-change test and closed subset, its image becomes closed after the faithfully flat field cover. The images commute with that cover because nonempty field fibres remain nonempty under extension. The full fpqc quotient-topology proof of AG-RG-S04 Lemma A1.4 therefore makes the original image closed. This proves universal closedness on every test.

Localize S_0 at p and pass to its complete local Noetherian ring A, faithfully flat by the entire AG-RG-S02 Lemma 1.3. Write $X, \overline{X}, \mathrm{Gr}$ for these base changes. Its proper special fibre X_s is open and closed in $\overline{X}_s$: it is open by the constructed immersion, and its map to the separated $\overline{X}_s$ is proper by the closed-graph/projection argument, so its image is closed. It corresponds to an idempotent. The full nilpotent idempotent-lifting proof AG-RG-S02 Lemma 1.4 lifts it uniquely through all maximal-ideal thickenings of $\overline{X}$; local lifts glue by uniqueness.

Here formal functions is applied to an actual projective scheme. Put $F = \pi_* \mathcal{O}_{\overline{X}}$ on Gr_A. This is coherent since π is finite and the target is Noetherian. On every affine target chart, the finite module quotient calculation gives

$$F/\mathfrak{m}_A^r F = \pi_{r*} \mathcal{O}_{\overline{X}_r}.$$

Finite pushforward also identifies their global sections. To apply projective-space finiteness precisely, take the Plucker closed immersion $i : \mathrm{Gr}_A \hookrightarrow \mathbf{P}^M_A$ supplied by Lemma 5.A. The sheaf i_*F is coherent: on an affine target chart it is the same finite module, now over the ambient ring through its quotient. By the definition of pushforward, its global sections are $H^0(\mathrm{Gr}_A, F)$. Thus AG-RG-S01 Theorem 8.4, applied to i_*F on $\mathbf{P}^M_A$, makes this module finite over A. The full completion and

formal-functions proofs AG-RG-S02 Lemma 1.2 and Theorem 2.3 then give the canonical ring isomorphism

$$\Gamma(\overline{X}, \mathcal{O}_{\overline{X}}) = H^0(\mathrm{Gr}_A, F) \simeq \varprojlim_r \Gamma(\overline{X}_r, \mathcal{O}_{\overline{X}_r}).$$

It is multiplicative because each comparison is the restriction map of the finite algebra. Thus the compatible formal idempotents give an actual idempotent and a proper open-and-closed component $Z \subset \overline{X}$ with $Z_s = X_s$.

The closed subset $Z \setminus X$ has empty special fibre. Its image in the local base is closed by properness; any nonempty closed subset of a local spectrum contains its closed point, so this subset is empty. Hence $Z \subset X$. The image of $Z \to \operatorname{Spec} A$ is open by the whole flat-openness proof for its smooth restriction from X, and it contains the closed point. Any such open is the entire local spectrum. In every geometric fibre Z is now a nonempty open-and-closed subset of the connected $X_{\bar{s}}$, and is all of that fibre. The open immersion $Z \subset X$ is therefore surjective and is an isomorphism. This proves that X is proper over A.

The monomorphism θ_A to the separated Grassmannian is proper by its closed graph and the proper source projection. It is quasi-finite, so the whole AG-RG-S04 Corollary D7.3 makes it finite. A finite monomorphism is closed: on an affine target chart $R' \to B$ its field fibre algebra has dimension e, and its monomorphism diagonal gives $e^2 = e$, so it is zero or the residue field. The coordinate map is therefore fibrewise surjective. Its finite cokernel is zero by the complete local Nakayama proof in AG-RG-S01 Lemma 1.2. Thus the ring map is surjective. Consequently θ_A is a closed immersion.

Closed-immersion descent AG-RG-S04 Corollary A3.2 gives the same assertion over $\mathcal{O}_{S_0,p}$. The whole eventual-property proof of AG-MO-03 Theorem 4.2 spreads it to an open neighbourhood of p: the local spectrum is the inverse limit of principal neighbourhoods and these are the given finite-presentation models. The union of these neighbourhoods contains the entire image of the original base. Pulling back proves that its orbit map is a closed Grassmannian immersion locally on S. This proves the original scheme and projectivity assertion; no properness condition has been imposed on unrelated model fibres.

The canonical Plücker line and its exponent

The full Grassmannian proof in Lemma 5.A makes its Plücker line the determinant of the universal quotient. Its pullback here is $\det \mathcal{O}(\mathcal{H}_n)$, where $\mathcal{H}$ is the universal conjugate subgroup. The entire finite-jet coordinate argument above identifies the augmentation filtration quotients with $\mathrm{Sym}^i((\operatorname{Lie}\mathcal{H})^*)$ for $0 \le i \le n$. These are finite locally free; their short exact sequences split locally because the quotient is projective. An adapted basis proves multiplicativity of the determinant in such a sequence: the matrix is block triangular, so its signed permutation determinant is the product of the two diagonal-block determinants. More generally the top exterior power of a composition is the composition of its top exterior powers; its coefficient in a wedge basis is the signed permutation determinant. This proves the determinant composition identity, including the conjugation calculation below, over every ring. Hence this line is

$$\bigotimes_{i=0}^{n} \det\bigl(\mathrm{Sym}^i((\operatorname{Lie}\mathcal{H})^*)\bigr).$$

We prove its exponent over every ring. For a rank-d free module the monomials of total degree i form a basis of its symmetric power, as follows directly from its presentation. Tensor words in a chosen basis form a free basis before the symmetric relations are imposed. Those relations exchange adjacent letters, so they identify exactly the words having the same multiplicity list. Within each such class, the coefficient-sum map has kernel generated by word differences: subtract one chosen representative from each other word. Its quotient is one free rank-one summand. The multiplicity classes consequently give precisely the asserted monomial basis over every ring. A diagonal matrix

with entries $t_1, \dots, t_d$ has determinant on that power equal to

$$\prod_{a_1+\cdots+a_d=i} t_1^{a_1} \cdots t_d^{a_d} = (t_1 \cdots t_d)^{c_i}, \qquad c_i = \binom{i+d-1}{d}.$$

All coordinate exponent sums are equal by permuting the coordinates; their total is $i\binom{i+d-1}{d-1}$, since the separator-position bijection counts the weak compositions. Division of this integer identity by d gives the displayed binomial coefficient. For $i = 0$ the exponent is zero.

This determinant identity holds for every matrix, including in positive characteristic. Both sides are universal integer polynomials in its entries. We prove the density calculation used to compare them. For the monic degree-d characteristic polynomial χ, the Sylvester determinant is the determinant of

$$(u, v) \longmapsto u\chi + v\chi', \qquad \deg u \le d-2, \quad \deg v \le d-1,$$

with target the polynomials of degree at most $2d-2$. Both spaces have dimension $2d-1$; for $d = 1$ the first summand is zero. Over a characteristic-zero field this determinant is nonzero exactly when χ and χ' are coprime. If they are coprime, their Bezout identity makes $\chi \mid v$, and the degree bound gives $v = u = 0$. If a nonconstant h divides both, the pair $(\chi'/h, -\chi/h)$ is a nonzero kernel vector within those bounds. These are the full earlier polynomial division and Bezout calculations of AG-RG-S04 Lemma P0.4a. In an algebraic closure the factored derivative at a root is the product of its differences from the other roots, so coprimality is exactly distinct eigenvalues. A diagonal matrix with distinct integer entries makes this determinant nonzero, proving it is a nonzero polynomial. On that open, the eigenbasis calculation is explicit. For each of the d distinct roots λ of χ, the matrix $X - \lambda I$ has determinant zero; the finite Gaussian reduction of AG-RG-S06 Section 1 gives a nonzero vector in its kernel. Eigenvectors for distinct roots are independent: applying $\prod_{\mu\neq\lambda}(X - \mu I)$ to a relation kills every other term and multiplies the chosen vector by the nonzero scalar $\prod_{\mu\neq\lambda}(\lambda - \mu)$. Thus all coefficients are zero. The d independent vectors are a basis by that same finite matrix calculation. Conjugation now reduces the determinant identity to the diagonal one. If its difference polynomial is g and this nonzero Sylvester polynomial is δ, then $g\delta$ vanishes at every field-valued matrix. The multivariable root bound follows by induction on the number of matrix entries: view the polynomial as a polynomial in the last entry; for every tuple of the other entries, the univariate root bound over the infinite characteristic-zero field makes each coefficient zero, and induction kills those coefficient polynomials. It makes $g\delta$ the zero polynomial; the polynomial domain and $\delta \neq 0$ give $g = 0$. Its integer coefficients are zero since $\mathbf{Z}$ injects into the characteristic-zero field. Specializing proves the universal identity over arbitrary rings. Applying it to the transition matrices gives the identity of line bundles

$$\det \mathcal{O}(\mathcal{H}_n) = \left(\det(\operatorname{Lie} \mathcal{H})^*\right)^c, \qquad c = \sum_{i=0}^{n} \binom{i+d-1}{d} > 0.$$

The inequality follows from $n \ge 1$ and its $i = 1$ summand equal to one, for $d > 0$.

This positive power is relatively very ample through the actual closed Plücker immersion. The line itself is relatively ample, with the defining section criterion verified directly. Work over an affine base piece trivializing the finite Plücker bundle, so L^c is the pullback of $\mathcal{O}(1)$ from a closed immersion into $\mathbf{P}^N$. Its coordinate nonvanishing opens are affine. At any point and inside any chosen open neighbourhood, choose a principal open $D(f)$ in one such affine chart contained in that neighbourhood, by the full affine principal-basis proof of AG-RG-S01 Section 2. Express f as a polynomial in the coordinate ratios and homogenize to a polynomial h of degree a. Multiplying h by that chart coordinate gives a global section of $L^{c(a+1)}$ whose nonvanishing locus is exactly this affine principal open. These loci form a basis, and all their exponents are positive. This is the entire ample-section criterion proved in AG-RG-S01 Sections 8–9, so L is relatively ample. The zero-dimensional case was treated at the start. All finite-jet quotients, character actions,

determinants and quotient-sheaf identities commute with arbitrary base change. Their unique comparisons descend the canonical line and the stated quotient to the original base. □

The quotient proof is the complete earlier AG-RG-S07 Theorem 8.2 and Corollary 8.3. Recognition and the finite compactification use AG-RG-S04 Theorems E8.1–E9.1 and D7.1. Formal functions is used on the projective Grassmannian with its coherent finite algebra, by AG-RG-S02 Theorem 2.3. The scheme and ample-line arguments are written here under the original hypotheses.

A universal finite cohomology complex

Lemma (finite projective replacement). Let A be Noetherian and let C be a complex of flat A-modules concentrated in degrees $[a, b]$, with finite cohomology modules. There is a complex K of finite projective modules in $[a, b]$ and a quasi-isomorphism $K \to C$ that remains a quasi-isomorphism after tensoring with any A-module.

Proof. Construct a bounded-above complex P of finite free modules and a map $f : P \to C$, descending from degree b. Once the terms above j are chosen, its cone has terms

$$D^i = C^i \oplus P^{i+1}, \qquad d(c, p) = (d_C c + f(p), -d_P p).$$

Its cohomology is finite: the cone long exact sequence uses the finite cohomology of C and of the already chosen finite free portion of P. Choose finitely many cocycles (c_ℓ, p_ℓ) generating $H^j(D)$. Take P^j free on e_ℓ, and set $f(e_\ell) = c_\ell$, $d_P(e_\ell) = -p_\ell$. The cocycle equations give $d_P^2 = 0$ and $d_C f = f d_P$. In the new cone, $d(0, e_\ell) = (c_\ell, p_\ell)$, so this kills H^j without changing higher cohomology. Continue to the left. Each degree stabilizes, giving a quasi-isomorphism $P \to C$.

Its cone is acyclic, bounded above and flat termwise. Such a complex stays acyclic after every tensor: start at its last degree and descend through

$$0 \longrightarrow Z^i \longrightarrow D^i \longrightarrow Z^{i+1} \longrightarrow 0.$$

The quotient is flat; hence the kernel is flat and the sequence stays exact under tensor. Thus $H^i(P \otimes_A M) = H^i(C \otimes_A M)$ for every M.

Put $Q = \operatorname{coker}(P^{a-1} \to P^a)$. The left tail is a free resolution of Q, and

$$\operatorname{Tor}_1^A(Q, M) = H^{a-1}(P \otimes_A M) = 0$$

for every M, since $C \otimes_A M$ starts in degree a. The module Q is finitely presented, so it is flat and finite projective. Take $K^a = Q$ and $K^i = P^i$ for $a < i \le b$. The map to C factors through Q because $C^{a-1} = 0$. Truncation preserves cohomology, and the same acyclic flat-cone argument proves the assertion after every tensor. □

Apply this to the finite section complex of a line bundle L_0 on a smooth projective scheme over a Noetherian ring A_0. Choose a finite affine cover. Separatedness makes its intersections affine, and S01 Sections 4–5 prove that its ordered section complex computes quasi-coherent cohomology. On each intersection the line bundle is a projective module over its chart algebra; smoothness makes that algebra A_0-flat, so every term is A_0-flat. S01 Theorem 8.4 and its closed-immersion passage prove that all cohomology modules are finite. Under every algebra base change, affine module-sheaf correspondence and localization, proved in S01 Section 2, identify the new section complex with its tensor product over A_0. The finite projective replacement just proved therefore computes the line bundle cohomology after every base change.

Sections in a vanishing family

Lemma (sections in a vanishing family). Let $f : X \to S$ be smooth and projective of finite presentation, and let L be an invertible sheaf on X. Suppose that $H^q(X_{\bar{s}}, L_{\bar{s}}) = 0$ for every geometric point $\bar{s}$ and every $q > 0$. Then f_*L is finite locally free, all its positive higher direct images vanish, and these statements commute with every base change. In particular the canonical map $g^* f_* L \to f'_* L'$ is an isomorphism for any $g : S' \to S$. The base may be nonreduced.

Proof. Work with $S = \operatorname{Spec} A$. Choose a projective embedding, a finite affine cover trivializing L, and finite affine covers of the pairwise and triple overlaps. Write A as the filtered union of its finitely generated $\mathbf{Z}$-subalgebras. The embedding, chart opens and overlap maps descend at a finite stage by finite-presentation descent, Theorems 4.1–4.2. Retain the closed immersion, so the model X_0 is projective.

Retain smoothness at a finite stage by S02 Lemma 3.1 and Proposition 3.2. In a finite polynomial presentation, Lemma 3.0 identifies formal smoothness with a left inverse of the conormal differential map. The left inverse uses finitely many coefficients; its equations on the finitely many relation generators have finite polynomial witnesses. Descend these coefficients and witnesses to one stage. The same left inverse then proves formal smoothness there, and the retained finite presentation proves smoothness. Apply this on a finite affine cover, retaining the cover and the projective closed immersion by the finite-presentation limit proof. The line-bundle transition units, their inverses and the finitely many overlap and cocycle witnesses descend at the same stage, as in S02 Proposition 3.2. Gluing these descended transition units defines the invertible sheaf L_0 on the smooth projective model X_0/A_0.

The preceding finite-projective-replacement lemma gives a bounded finite projective complex K_0 such that

$$H^q(K_0 \otimes_{A_0} B) \simeq H^q(X_0 \times_{A_0} \operatorname{Spec} B, L_0 \otimes_{A_0} B)$$

for every A_0-algebra B, compatibly with the canonical comparison maps. Its proof uses the finite affine-cover complex and projective Serre finiteness; ordinary tensor of K_0 computes its derived tensor. This finite argument supplies the universal comparison before pullback to the original base.

Put $K = K_0 \otimes_{A_0} A$. Near any $s \in \operatorname{Spec} A$, trivialize its terms. Whenever a differential has an entry invertible at s, shrink the neighbourhood and perform row and column operations to put that entry in an identity block. The equation $d^2 = 0$ makes the adjacent differentials zero on that block, so it splits off a contractible pair $A \xrightarrow{1} A$. After finitely many cancellations, all remaining differential entries vanish in $\kappa(s)$. The remaining complex therefore has

$$K^q \otimes_A \kappa(s) = H^q(X_s, L_s).$$

For $q > 0$ the right side is zero by the hypothesis and field extension; for $q < 0$ it is zero because L_s is a sheaf. Thus every remaining free term except degree zero has rank zero and disappears on a neighbourhood of s. The degree-zero term is finite free. Every cancelled identity pair stays contractible after any tensor. The universal comparison consequently proves the direct-image assertions and every base-change assertion, including nonflat changes involving nilpotents. The canonical comparisons glue. Notice that vanishing was imposed only on the original family, not on unrelated fibres of its Noetherian model. □

Lemma 5.B (the automorphism sheaf of a projective line). Over every base scheme T, the sheaf of T-automorphisms of $\mathbf{P}^1_T$ is represented by $\mathrm{PGL}_{2,T}$. This identification holds on all schemes over T and includes nonreduced tests.

Proof. Work locally on a test base, and write $p : \mathbf{P}^1_T \to T$. For an automorphism a, put $M' = a^*\mathcal{O}(1) \otimes \mathcal{O}(-1)$. On every geometric fibre this line is trivial. Indeed $\mathcal{O}(1)$ is the divisor

line of any rational point: a linear homogeneous form cuts that point simply, and the two affine coordinate frames identify its divisor line with $\mathcal{O}(1)$. Pullback by an automorphism cuts the inverse-image point simply and gives the same line. The whole sections-in-a-vanishing-family lemma applies to M', since the explicit projective-line cohomology of AG-RG-S01 Theorem 7.2 gives $H^q(\mathcal{O}) = 0$ for $q > 0$ and $H^0(\mathcal{O})$ of rank one. Hence $M = p_*M'$ is a line bundle with every base change. The evaluation $p^*M \to M'$ is an isomorphism: on every geometric fibre it is the constant-section isomorphism, and on local line-bundle frames its coefficient is therefore outside every maximal ideal of the local ring, hence a unit. Thus $a^*\mathcal{O}(1) = \mathcal{O}(1) \otimes p^*M$. Trivialize M locally. Pullback of the two coordinate sections, together with this trivialization, gives a linear map on the rank-two module of sections of $\mathcal{O}(1)$, whose standard linear-form basis is again the explicit AG-RG-S01 Theorem 7.2 calculation. Its geometric-fibre maps are invertible because a is an automorphism. Its determinant is a unit on the base, by the same local residue test, so the matrix belongs to GL_2. The quotient-functor construction of projective space identifies the corresponding projective transformation with a: its coordinate quotient sections are precisely these pulled-back coordinate sections. A different trivialization of M multiplies the matrix by a unit. The kernel of the projective action consists exactly of scalar matrices on all tests. A matrix acting identically preserves the points $[1 : 0]$ and $[0 : 1]$, so its off-diagonal entries are zero and its diagonal entries are units; preserving $[1 : 1]$ makes the two entries equal. Conversely scalar matrices act identically. Finally the fppf quotient $\mathrm{GL}_2 / \mathbf{G}_m$ is the actual scheme open $D_+(\det) \subset \mathbf{P}^3_T$: on each projective coordinate chart a line of four entries is locally a matrix, and its determinant section being nowhere vanishing means an invertible determinant after this frame is chosen. Changing the frame is exactly scalar multiplication. These local coordinates, their unique scalar comparisons and line-bundle frame gluing represent the quotient sheaf on every test, by the full local-coordinate sheaf construction of AG-RG-S04 Lemma A1.7. Matrix multiplication and inversion descend and give the group scheme PGL_2. The local matrix descriptions just proved therefore identify its functor with the automorphism sheaf, compatibly with composition and every base change. □

Proposition 5.2. If G has trivial centre and geometric semisimple rank one, and has a split maximal torus T, then Zariski locally (G, T) is isomorphic to (PGL_2, D).

Proof. Its geometric fibres are PGL_2 by Proposition 2.3: the action on the projective line has trivial kernel. Choose a root α locally. It is an isomorphism $T \to \mathbf{G}_m$, with the following full base-ring check. Locally identify the rank-one split torus with $\mathbf{G}_m$. A character is a finite Laurent polynomial $\sum_n a_n t^n$. Its coproduct and identity equations give $a_n^2 = a_n$, $a_m a_n = 0$ for $m \neq n$, and $\sum_n a_n = 1$, by comparing the independent Laurent coefficients. Thus its finitely many nonzero coefficient idempotents split the affine base into open-and-closed pieces on which it is t^n. Its geometric-fibre character is primitive, so on every nonempty piece $n = 1$ or $n = -1$. The inverse there is respectively t or t^{-1}, and these inverses glue over the idempotent pieces. Put $B = P_G(\alpha^{-1})$. It is smooth and on each geometric fibre is an upper triangular Borel subgroup of PGL_2.

Its scheme-theoretic normalizer equals B. The normalizer lemma makes it closed of finite presentation and proves that its formation commutes with arbitrary base change. On geometric fibres the point stabilizer in $\mathbf{P}^1$ is self-normalizing. Infinitesimally, use the diagonal class of $\mathrm{diag}(1, 0)$ in $\mathfrak{pgl}_2$. Its bracket with the lower off-diagonal class is the negative of that class, nonzero in every characteristic. Thus the normalizer has exactly the two-dimensional tangent space of B at the identity. Its local dimension is at least two because it contains the smooth closed B, and is at most the dimension of that tangent space by the embedding-dimension bound. It is therefore a regular local ring of dimension two. The rational-point cotangent calculation and field Jacobian criterion of AG-CA-18 Proposition 3.1 and Theorem 2.1 make it smooth at the identity. Every closed point of the geometric normalizer is in B: a normalizing element must preserve the unique fixed point of the upper triangular group on $\mathbf{P}^1$, as is checked using its translations on the other affine chart. Translation carries the smooth identity neighbourhood through every such point. The open smooth locus consequently covers the fibre, since any nonempty closed complement in a finite-type scheme over an algebraically closed field contains a closed point by the full Zariski lemma.

The closed inclusion $B \to N$ is now a map between smooth groups with identical tangent spaces, so AG-RG-S02 Lemma 3.A makes it etale. The full etale-monomorphism proof of AG-RG-S04 Lemma E7.2 makes it an open immersion; all geometric points belong to it, so it is an isomorphism. This proves equality on the entire geometric fibres, including nilpotent tests. The flat-source criterion above, applied with source B, proves the equality over the base without assuming the normalizer flat.

Lemma 5.1 makes $f : X = G/B \to S$ a smooth projective curve whose geometric fibres are $\mathbf{P}^1$, and it has the section $e = 1B$. A section of a separated smooth relative curve is an effective Cartier divisor, with the following local calculation. Near the section over a chosen base point, take an etale coordinate q by AG-RG-S02 Lemma 3.A and subtract its pullback along the section, so it vanishes there. It is a nonzerodivisor: multiplication by the affine-line coordinate is injective over every base ring, and the etale coordinate algebra is flat. The zero scheme $Z(q)$ is etale over the base and the section factors through it. The base and $Z(q)$ are both smooth of relative dimension zero; AG-RG-S02 Lemma 3.A therefore makes the section etale. It is a monomorphism because it is a section, so the full etale-monomorphism proof makes it an open immersion. It is also closed because the curve is separated. Its complement in $Z(q)$ is consequently closed also in the ambient coordinate neighbourhood. Remove that complement; on the resulting neighbourhood the section is cut out exactly by q. Its ideal is thus locally free of rank one. Its reciprocal defines the divisor line bundle, and on each geometric curve fibre this coordinate cuts its point simply, giving degree one. Put $L = \mathcal{O}_X(e)$; it has degree one on every geometric fibre. The sections-in-a-vanishing-family lemma applies to L, since $H^1(\mathbf{P}^1, \mathcal{O}(1)) = 0$, and gives $E = f_*L$ locally free of rank two, with arbitrary base change. The evaluation $f^*E \to L$ is surjective: its cokernel is finite locally, its geometric fibres vanish by generation of $\mathcal{O}(1)$, and local Nakayama kills that cokernel. With the quotient convention it defines $j : X \to \mathbf{P}(E)$, and $j_{\bar{s}}$ is the usual isomorphism given by the two sections of $\mathcal{O}(1)$. Both schemes are smooth over S, and the differential of j is an isomorphism on all geometric fibres, hence locally an isomorphism of their cotangent bundles. The étale differential criterion makes j étale. Its geometric fibres are isomorphisms, so it is universally injective and surjective; an étale universally injective morphism is an open immersion, and surjectivity makes it an isomorphism. Trivializing E Zariski locally now gives $G/B \simeq \mathbf{P}^1_S$.

Lemma 5.B identifies the automorphism sheaf of this projective line with the actual group scheme PGL_2, so the action gives a morphism $G \to \mathrm{PGL}_2$ on all base tests. It is an isomorphism on every geometric fibre. Both groups are smooth of finite presentation; their differential is an isomorphism on those fibres, hence locally an isomorphism of differential bundles by the finite matrix and Nakayama calculations. AG-RG-S02 Lemma 3.A makes the morphism etale. Its geometric fibres are single reduced points, so it is universally injective and surjective. AG-RG-S04 Lemma E7.2 then makes it a surjective open immersion, hence an isomorphism. It carries B to the point stabilizer. Here is the full Zariski-local pair calculation. In that stabilizer, every projective matrix has a unique scalar-normalized representative $\left(\begin{smallmatrix} v & u \\ 0 & 1 \end{smallmatrix}\right)$, with v invertible; this gives its group law $(v, u)(v', u') = (vv', u + vu')$. The diagonal character on T is an isomorphism to $\mathbf{G}_m$, since its geometric-fibre character is the primitive root already identified above. Use its value t as the torus coordinate. The homomorphism $T \to B$ has the form $\left(\begin{smallmatrix} t & u(t) \\ 0 & 1 \end{smallmatrix}\right)$ for a finite Laurent polynomial $u(t)$. Commutativity gives $(1 - s)u(t) = (1 - t)u(s)$ in $R[t^{\pm 1}, s^{\pm 1}]$. For each exponent $m \notin \{0, 1\}$, its coefficient in t says $(1 - s)a_m = 0$; independence of Laurent coefficients gives $a_m = 0$. The coefficient at t then gives $u(s) = a_1(s - 1)$. Conjugation by $\left(\begin{smallmatrix} 1 & a_1 \\ 0 & 1 \end{smallmatrix}\right)$ replaces $u(t)$ by $u(t) + a_1(1 - t) = 0$. Thus it takes T to the diagonal torus over the affine base itself. This calculation uses no division or reducedness and finishes the Zariski-local pair assertion. □

6. Splitting a central extension of the rank-one cover

We need an integral result, including characteristic two. The following splitting argument is due to Gabber; we give the argument for a separated commutative group scheme Z, which includes the multiplicative-type centres used here.

Theorem 6.1. Every fppf central extension

$$1 \longrightarrow Z \longrightarrow E \longrightarrow \mathrm{SL}_{2,S} \longrightarrow 1$$

has a unique homomorphic section.

Proof. We first record the full scheme calculation behind the cell covers. The group SL_2 is cut out by $ad - bc = 1$. The opens $a \neq 0$ and $b \neq 0$ cover it. On the first, elimination of d gives the algebra $R[a, a^{-1}, b, c]$; on the second, elimination of c gives $R[b, b^{-1}, a, d]$. Both are polynomial localizations, so are smooth over every ring by the square-zero lifting and finite-presentation proofs. Over every field both charts are integral and their intersection is nonempty, containing $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Their union is therefore reduced and irreducible, hence integral. The identity cell is the first chart: the matrix factorization $y(v)h(t)x(u)$ gives $a = t$, $b = tu$, $c = vt$ and $d = vtu + t^{-1}$, uniquely. Thus it is an open dense cell in every geometric fibre. In SL_2 use

$$x(u) = \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}, \quad y(v) = \begin{pmatrix} 1 & 0 \\ v & 1 \end{pmatrix}, \quad h(t) = \mathrm{diag}(t, t^{-1}).$$

Let D', U', V' be their inverse images in E. A central extension of a commutative group has a commutator pairing on its quotient, on all tests. Indeed any two lifts differ by a central-kernel element, so replacing a lift leaves the commutator unchanged. Local lifts consequently glue the commutator through the represented fppf sheaf Z. Every such commutator lies in Z, hence is central; the identities $[ab, c] = a[b, c]a^{-1}[a, c]$ and $[a, bc] = [a, b]b[a, c]b^{-1}$ therefore make the pairing additive in each variable. This constructs the pairing as a natural map of sheaves, hence as a scheme morphism by its values on every test.

For D', the pairing $\mathbf{G}_m \times \mathbf{G}_m \to Z$ is trivial. The nth-power map of $\mathbf{G}_m$, for every $n \geq 1$, is an fppf epimorphism over every ring: above a unit $u \in R^*$ its root algebra is $R[z]/(z^n - u)$, with free basis $1, z, \ldots, z^{n-1}$. The element z is already invertible, with inverse z^{n-1}/u. This free positive-rank algebra is faithfully flat, and its monic equation is a finite presentation. Restrict the first pairing argument to μ_n. Bilinearity makes the pairing there killed by n in that argument. Locally write the second argument as an nth power using this cover, and bilinearity makes the pairing the identity. Faithful sheaf descent removes the cover. The union of the μ_n is universally schematically dense in $\mathbf{G}_m$, with the following actual ideal test. On an affine test base, write a Laurent polynomial as a finite sum with coefficients in the base ring (or in the Laurent ring of the second coordinate). Choose n greater than every difference of two exponents present. Reduction modulo $t^n - 1$ then leaves those exponents in distinct residue classes, whose images are independent in the free basis $1, t, \ldots, t^{n-1}$. A polynomial vanishing on every μ_n therefore has all coefficients zero. Since Z is separated, equality of the pairing with the identity is a closed subscheme of the product; this ideal test kills its entire ideal on every affine base chart. Thus the pairing is the identity and D' is commutative.

The conjugation action on E factors through SL_2: two local lifts differ by Z, which is central, so give identical conjugation maps; their maps glue by the sheaf property. The commutator pairing on U' is invariant under simultaneous multiplication of its arguments by t^2. The square-power cover just proved makes it invariant under simultaneous multiplication by every unit. For given u, v over any test, the morphism $a \mapsto c(au, av)$ from $\mathbf{A}^1$ to Z is constant on $\mathbf{G}_m$. This equality extends to zero on nonreduced tests too. Its equalizer with the constant morphism is closed by separatedness; on an affine test chart the equalizer ideal becomes zero in $R[a, a^{-1}]$. The map $R[a] \to R[a, a^{-1}]$ is

injective because multiplication by a is injective on the polynomial basis, so that ideal was zero. At zero the pairing is the identity, and at one it is $c(u,v)$. Hence $c(u,v)=1$. Thus U' is commutative; the same argument proves it for V'.

We construct a unique D-equivariant section of $U' \to U$. Fppf locally choose a unit t with t^2-1 invertible and a lift $\tilde{h}$ of $h(t)$. The first choice is the actual finite-presentation cover $R[t, 1/(t(t^2-1))]$: it is a smooth polynomial localization and its geometric fibres are nonempty. An algebraically closed field is infinite, since a finite field's polynomial $\prod_a (T-a)+1$ would have no root; the nonzero degree-three polynomial $t(t^2-1)$ therefore cannot vanish at all its elements by the univariate root bound. Smoothness gives flatness by the earlier arbitrary-base smooth-flatness proof. Thus this open is surjective, flat and finitely presented. The extension's fppf epimorphism supplies the second local choice. In the commutative group U', the endomorphism

$$\varphi(a) = \tilde{h} a \tilde{h}^{-1} a^{-1}$$

kills Z and induces multiplication by t^2-1 on $U = \mathbf{G}_a$. The factorization through U is explicit: locally lift a section of U to U' and apply φ. Changing the lift multiplies by Z and leaves its value unchanged, so these values glue uniquely. This defines a homomorphism of represented sheaves $U \to U'$. Dividing its input by the unit t^2-1 gives a section. It is D-equivariant because all lifts in the commutative D' commute with $\tilde{h}$. Any two such equivariant sections differ by a D-invariant homomorphism $U \to Z$. Square-power surjectivity gives scalar invariance under every unit, and the same injective-polynomial equalizer argument used above extends that invariance to the zero scalar, where the homomorphism is the identity. The difference is consequently zero. This uniqueness descends the local sections, giving $x' : \mathbf{G}_a \to E$ and similarly y'.

Set

$$w'(t) = y'(-t^{-1})x'(t)y'(-t^{-1}), \qquad h'(t) = w'(t)w'(1)^{-1}.$$

Their images are $\begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix}$ and $h(t)$. Thus $h'(t) \in D'$. Equivariance gives

$$h'(s)w'(t)h'(s)^{-1} = w'(s^2 t).$$

Divide by the instance $t=1$. Since D' is commutative, this gives $h'(t) = h'(s^2t)h'(s^2)^{-1}$. The square map being fppf surjective, h' is a homomorphism. Its definition then gives $h'(s)w'(t) = w'(st)$.

Expand the identity $h'(s)w'(t) = w'(st)$ and move $h'(s)$ across the first y' using $h'(s)y'(v)h'(s)^{-1} = y'(s^{-2}v)$. Multiply on the left by $y'(1/(st))$ and on the right by $y'(1/t)$. Additivity of y' gives the actual intermediate equality $x'(st)y'((s-1)/(st)) = y'((s-1)/(s^2t))x'(s^2t)h'(s)$. Move the final $h'(s)$ to the left of this x' using its conjugation identity, obtaining $x'(st)y'((s-1)/(st)) = y'((s-1)/(s^2t))h'(s)x'(t)$. Set $u = st$ and $v = (s-1)/(st)$. Conversely every pair for which u and $1+uv$ are units has this form with $s = 1+uv$ and $t = u/s$. Thus we have

$$x'(u)y'(v) = y'\left(\frac{v}{1+uv}\right) h'(1+uv) x'\left(\frac{u}{1+uv}\right) \tag{1}$$

when u and $1+uv$ are units. The restriction that u be a unit can be removed on all tests. Fppf locally write $u = a+b$ with $a, b, 1+bv$ units, by adjoining b on the open $b(u-b)(1+bv) \neq 0$. This is a smooth polynomial localization. In every geometric fibre its defining polynomial is nonzero (of degree two when $v=0$, and degree three otherwise), so the infinite-field root bound makes its image surjective; the same smooth-flatness argument makes it an fppf cover. Apply (1) first to $x'(b)y'(v)$, then to $x'(a)y'(v/(1+bv))$. Write $q = 1+bv$ and $p = 1+uv$. The second denominator is p/q, so its result is $y'(v/p)h'(p/q)x'(aq/p)h'(q)x'(b/q)$. The rule $x'(c)h'(q) = h'(q)x'(q^{-2}c)$ and multiplicativity of h' combine the two torus factors into $h'(p)$. The resulting x' parameter is

$$\frac{a}{(1+bv)(1+uv)} + \frac{b}{1+bv} = \frac{u}{1+uv}.$$

This proves (1) in its full domain.

These identities, the additivity of x', y', the multiplicativity of h', and its two conjugation identities are exactly the multiplication rules on the open cell $\Omega = U_- D U_+$ of SL_2. Define a section there by

$$y(v)h(t)x(u) \longmapsto y'(v)h'(t)x'(u).$$

It is multiplicative whenever both factors and their product are in Ω: reorder their factors with (1). Write $s_\Omega : \Omega \to E$ for this cell section. For any test scheme $R \to S$ and finite list $g_1, \dots, g_m \in \mathrm{SL}_2(R)$, the open

$$\Omega_R \cap \bigcap_i \Omega_R g_i^{-1}$$

is nonempty in every geometric fibre: the two-chart calculation above makes that fibre integral, each translate of the identity cell is a nonempty dense open, and irreducibility makes every finite intersection of these opens nonempty. The intersection is a smooth open of SL_2 and is surjective over R. On an affine test base, the cell translates are principal opens, their finitely many defining functions giving a polynomial localization of the finite-presentation group algebra. Thus it is locally finitely presented; smoothness gives flatness, and affine refinements supply an actual fppf local point c with $c, cg_i \in \Omega$.

In particular, for one g, use the nonempty open

$$\Omega_R \cap \Omega_R^{-1} \cap \Omega_R g^{-1}.$$

Then $g = c^{-1}(cg)$ is a product of two points of Ω. Assign to a word in cell points the corresponding product of their s_Ω-values. This is independent of the word. For two words with the same endpoint g, choose c so that c and its translates by every partial product of both words lie in Ω. Successive cell multiplicativity says that multiplying either lifted word on the left by $s_\Omega(c)$ gives $s_\Omega(cg)$. Cancellation proves equality. In particular a word representing the identity has lift one; concatenation therefore respects multiplication and inverses. These word values agree after further fppf pullbacks, so descend through the sheaf property of E. They define a homomorphism of represented fppf sheaves, hence a scheme homomorphism $\mathrm{SL}_2 \to E$. Its composite with $E \to \mathrm{SL}_2$ is the identity on generating words. It is the required homomorphic section on all test schemes.

Finally any homomorphism $\mathrm{SL}_2 \to Z$ kills $x(u)$, since

$$[h(t), x(u)] = x((t^2-1)u),$$

and one may choose $t^2 - 1$ invertible fppf locally. It kills $y(v)$ similarly, and then $h(t) = w(t)w(1)^{-1}$. It kills the open cell, hence the group. The difference between two central sections is such a homomorphism, proving uniqueness. □

7. Coroots from multiplication

Theorem 7.1. For every root α there is a canonical coroot $\alpha^\vee : \mathbf{G}_m \to T$, with $\langle \alpha, \alpha^\vee \rangle = 2$. There is a canonical perfect pairing $\mathfrak{g}_\alpha \otimes \mathfrak{g}_{-\alpha} \to \mathcal{O}_S$ such that, writing its value as XY,

$$\exp_\alpha(X) \exp_{-\alpha}(Y) = \exp_{-\alpha}\left(\frac{Y}{1+XY}\right) \alpha^\vee(1+XY) \exp_\alpha\left(\frac{X}{1+XY}\right) \tag{2}$$

whenever $1 + XY$ is a unit. This unit condition is precisely membership in the open cell $U_{-\alpha} T U_\alpha$.

Proof. Work in G_α, whose scheme-theoretic centre is $Z = \ker \alpha \subset T$, by Section 1. Theorem 1.1 constructs $q : G_\alpha \to Q = G_\alpha/Z$ as an affine reductive quotient and an fppf Z-torsor, compatible with every base change. The torus quotient has character group $\mathbf{Z}\alpha \subset X^*(T)$, so $T/Z \simeq \mathbf{G}_m$ via the descended character α. It is a closed subgroup of Q: its pullback under q is $T \subset G_\alpha$, and closed immersions descend through the fppf torsor. On geometric fibres Proposition 2.3 identifies this quotient with PGL_2, so T/Z is a maximal torus.

The open cell $U_{-\alpha}TU_\alpha$ is right Z-stable; centrality makes Z act only on its torus coordinate. Its quotient is the open subscheme

$$U_{-\alpha} \times (T/Z) \times U_\alpha \subset Q,$$

as follows by invariant-open descent and the product torsor. At the identity this gives the two nonzero torus weights $\alpha, -\alpha$, with their root lines, in $\operatorname{Lie} Q$. The centre formula of Section 1, applied to $Q, T/Z$, is consequently $Z(Q) = \ker(\alpha : T/Z \to \mathbf{G}_m) = 1$, as a scheme over the original base. Proposition 5.2 therefore identifies the pair $Q, T/Z$, Zariski locally, with PGL_2 and its diagonal torus. Choose the identification so $\alpha(\mathrm{diag}(t, 1)) = t$.

The map $\mathrm{SL}_2 \to \mathrm{PGL}_2$ used next is an fppf epimorphism over the original base, with the following integral proof. Lemma 5.B represents PGL_2 by the invertible-matrix open in $\mathbf{P}^3$, so a section has a matrix representative M after locally trivializing its coordinate line. Its determinant d is a unit. The finite-free square-root cover $R[z]/(z^2 - d)$ from Theorem 6.1 has an invertible z; the matrix $z^{-1}M$ has determinant one and lifts the section. The kernel consists on every test of scalar matrices zI with $z^2 = 1$, hence is μ_2, even in characteristic two. On the upper and lower additive matrix groups the map is the identity on their parameters. Pull back $G_\alpha \to \mathrm{PGL}_2$ along this map. Theorem 6.1 splits the resulting central extension by Z, giving a homomorphism $\rho : \mathrm{SL}_2 \to G_\alpha$. We verify its full T-equivariance. For any test point $t \in T$, put $a = \alpha(t)$. Conjugation of matrices by $\mathrm{diag}(a, 1)$ defines an automorphism c_a of SL_2, scaling the upper parameter by a and the lower by a^{-1}. Its adjoint-quotient action equals conjugation by $q(t)$. Thus $g \mapsto t\rho(c_a^{-1}(g))t^{-1}$ is another homomorphic lift of the same map $\mathrm{SL}_2 \to Q$. Together with its SL_2 coordinate it is a section of the same pullback extension; uniqueness in Theorem 6.1 makes it ρ. This proves equivariance on every test. The quotient-cell coordinates above make dq an isomorphism on both root lines, and the matrix cover has the same property. Consequently $d\rho$ is an isomorphism on those lines. Apply Theorem 4.1 to the upper map after identifying its tangent basis with $d\rho(e)$, and likewise to the lower map: their images are exactly $U_\alpha, U_{-\alpha}$ and they identify the respective additive lines. The diagonal cocharacter of SL_2 therefore defines $\alpha^\vee$.

Direct multiplication gives

$$\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ y & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ y/(1+xy) & 1 \end{pmatrix} \begin{pmatrix} 1+xy & 0 \\ 0 & (1+xy)^{-1} \end{pmatrix} \begin{pmatrix} 1 & x/(1+xy) \\ 0 & 1 \end{pmatrix}.$$

The quotient cell is saturated: its pullback along the Z-torsor is exactly $U_{-\alpha}TU_\alpha$, since on that open the torsor acts only in the torus factor and the product torsor gives its whole inverse image. In the projective invertible-matrix model the cell is precisely the locus where the upper-left coordinate is invertible in a local coordinate-line frame. Indeed the lower-diagonal-upper matrix factorization determines its parameters uniquely; changing the frame multiplies that coordinate by a unit. For the displayed product that coordinate is $1 + xy$. Thus the product lies in the cell exactly when $1 + xy$ is a unit, on all tests. Transporting the matrix identity through ρ gives (2). The chosen root-line tangent bases have pairing one, which supplies a perfect pairing, and the diagonal image gives $\alpha \circ \alpha^\vee(t) = t^2$.

The pairing and coroot are unique, including over nonreduced bases. Work in paired root-line frames and let an alternative perfect pairing have coefficient $c \in R^*$. Its coroot still has adjoint image $\mathrm{diag}(t^2, 1)$, since its pairing with α is two. Set $Y = F$ and $X = xE$, and compare the upper-root coordinates in (2) after passing to the adjoint quotient. On $R[x, 1/((1+x)(1+cx))]$ the two coordinates are $x/(1+x)$ and $x/(1+cx)$. Equality gives $(c-1)x^2 = 0$. The factors $1 + x$

and $1 + cx$ are monic up to an invertible leading coefficient, so multiplication by their product is injective on $R[x]$; the localization is injective. Independence of the polynomial basis therefore gives $c = 1$. With the pairing fixed, uniqueness of the three actual coordinates in the open cell of G_α fixes the actual torus value $\alpha^\vee(1 + x)$, not only its adjoint image. On any test ring every unit u occurs by taking $x = u - 1$ and $Y = F$. Thus the coroot itself is fixed on all tests. This also rules out a possible difference by a cocharacter into Z. The constructions on different local identifications consequently agree, so the morphisms and pairing glue; the same characterization proves arbitrary base-change compatibility. □

For SL_2, use $T \simeq \mathbf{G}_m$ via $\mathrm{diag}(t, t^{-1})$. Its root is $\alpha(t) = t^2$, and $\alpha^\vee(t) = \mathrm{diag}(t, t^{-1})$. For PGL_2, parameterize its diagonal torus by the class of $\mathrm{diag}(t, 1)$. Its root is $t \mapsto t$, and its coroot is the class of $\mathrm{diag}(t^2, 1)$. In both cases the pairing is two, but the root and coroot occupy different sublattices.

For GL_2, the roots are $e_1 - e_2$ and its negative, and the positive coroot is $t \mapsto \mathrm{diag}(t, t^{-1})$. These explicit formulas remain valid over $\mathbb{Z}$.

8. The three rank-one groups

A lattice calculation with pairing two

Lemma 8.A. Let X be a finite free abelian group of rank $r+1$. Let $a \in X$ and $b \in X^* = \mathrm{Hom}(X, \mathbf{Z})$ be nonzero and satisfy $b(a) = 2$. Up to an isomorphism of X preserving this pair, exactly one of the following descriptions occurs:

1. $X = \mathbf{Z}e_0 \oplus \mathbf{Z}^r$, $a = 2e_0$, $b = e_0^*$.
2. $X = \mathbf{Z}e_0 \oplus \mathbf{Z}^r$, $a = e_0$, $b = 2e_0^*$.
3. $r \geq 1$, $X = \mathbf{Z}e_1 \oplus \mathbf{Z}e_2 \oplus \mathbf{Z}^{r-1}$, $a = e_1 - e_2$, $b = e_1^* - e_2^*$.

In these formulas b is zero on the displayed extra summand.

Proof. Choose a basis of X. For a nonzero vector the positive greatest common divisor of its coordinates is its divisibility; division by it gives a primitive vector. The same calculation applies to the coordinate list of a nonzero integral functional. Integer division, Bezout and the Smith basis operations are proved in S02 Lemma 3.M. In particular a primitive vector extends to an integral basis: the Euclidean row operations reduce its coordinate list to $(1, 0, \ldots, 0)$ and their inverses give that basis. A kernel of an integral functional is finite and torsion-free, hence free by that lemma. Write $a = ma_0$ and $b = nb_0$, with a_0, b_0 primitive and $m, n > 0$. Then

$$mn\, b_0(a_0) = 2.$$

Consequently (m, n) is $(2, 1)$, $(1, 2)$, or $(1, 1)$.

In the first case $b(a_0) = 1$. Every $x \in X$ has the unique expression

$$x = b(x)a_0 + \big(x - b(x)a_0\big), \qquad x - b(x)a_0 \in \ker b.$$

Thus $X = \mathbf{Z}a_0 \oplus \ker b$; choose a basis of the free kernel. This gives case 1. In the second case apply the same expression using $b_0(a) = 1$. It gives case 2.

Suppose now that both a and b are primitive. Extend a to a basis $a, c_1, \ldots, c_r$. At least one $b(c_i)$ is odd: otherwise every value of b, including $b(a) = 2$, would be even, contradicting primitivity of b. Relabel so that $b(c_1) = 2q + 1$, and replace c_1 by $c_1 - qa$. This is a basis operation and makes its value under b equal to one. Call the new vector c. Replace each remaining c_i by $c_i - b(c_i)c$; their values become zero. Finally set

$$e_1 = c, \qquad e_2 = c - a.$$

These replace the basis pair a, c by another unimodular basis pair. They satisfy $a = e_1 - e_2$, $b(e_1) = 1$ and $b(e_2) = -1$, giving case 3. If $r = 0$, both primitive elements would pair to ± 1, so this case cannot occur.

The divisibilities (m, n) of a and b are invariant under lattice isomorphisms. They distinguish all three cases. □

Extending a map from the rank-one cell

Lemma 8.B. Let K/S be a smooth group with geometrically connected fibres. Let $\Omega \subset K$ be a fibrewise dense open containing the identity. Let H/S be a group scheme. Suppose $f : \Omega \to H$ sends the identity to the identity and satisfies

$$f(xy) = f(x)f(y)$$

on the open where $x, y, xy \in \Omega$. Then f extends uniquely to a homomorphism $K \to H$.

Proof. We spell out irreducibility without a quasi-compactness assumption on K. In a geometric fibre, every local ring of a smooth scheme is a regular local domain, by AG-CA-18 Theorem 2.1 and the regular-local-domain proof of AG-CA-14 Theorem 1.1. Each affine smooth chart is Noetherian. Its finitely many irreducible components cannot meet: two distinct minimal primes at a meeting point would remain distinct in the local ring, contradicting that it is a domain. The components on that chart are consequently disjoint closed-and-open subsets, each reduced and irreducible. Thus every point of the fibre has an irreducible open neighbourhood. Form equivalence classes of such neighbourhoods under chains of nonempty intersections. A finite union along a chain is irreducible, since the union of two intersecting irreducible open subsets is irreducible; this follows by testing two nonempty opens in their union. Any two members in one class lie in such a finite union, so the whole class union is irreducible. These class unions are disjoint opens covering the fibre. Connectedness leaves only one, proving irreducibility. This is the local form of the calculation in S02 Lemma 4.1. For any test scheme $R \to S$ and any finite list $g_1, \dots, g_d \in K(R)$, the open

$$\Omega_R \cap \bigcap_{i=1}^{d} \Omega_R g_i^{-1}$$

has a nonempty open in every geometric fibre of K_R: each translate is a dense nonempty open in the irreducible fibre and finite intersections remain nonempty. Its map to R is a smooth surjection. Smoothness gives flatness over arbitrary rings by AG-CA-18 Theorem 6.1, and local finite presentation is part of the proved smooth-chart criterion. Refine by affine smooth charts above affine base opens; these are finitely presented, flat maps and jointly cover R. They give an actual fppf covering family and therefore a local point c with c and all cg_i in the cell. No rational-point choice over the original residue fields and no flatness of H are required.

In particular, for $g \in K(R)$, choose $c \in \Omega_R \cap \Omega_R^{-1} \cap \Omega_R g^{-1}$ locally. Then $g = c^{-1}(cg)$ is a product of two cell points. Assign to any word in cell points the corresponding product of f-values.

This assignment is independent of the word. For two words $x_1 \cdots x_d$ and $y_1 \cdots y_e$ with the same endpoint g, choose c so that it and every translated partial product of both words lie in the cell. Repeatedly apply the given identity. The first image word multiplied on the left by $f(c)$, and the second image word multiplied by the same element, both equal $f(cg)$. Cancellation proves equality. The assignment also respects inverses: for $x \in \Omega(R)$, choose $d, d^{-1}, dx^{-1} \in \Omega$. The identities $(dx^{-1})x = d$ and $d^{-1}d = 1$ then give $f(d^{-1})f(dx^{-1}) = f(x)^{-1}$. The word $d^{-1}(dx^{-1})$ represents x^{-1}, as required.

The assignments agree after fppf pullback, so the sheaf property of H descends them. Concatenation of words proves multiplicativity. Yoneda makes this map of represented fppf sheaves a scheme morphism. Every extension must have these values on generating words, proving uniqueness. □

The three rank-one models

Theorem 8.1. A split reductive group of rank $r + 1$ and semisimple rank one over a field is isomorphic to exactly one of

$$\mathbf{G}_m^r \times \mathrm{SL}_2, \qquad \mathbf{G}_m^r \times \mathrm{PGL}_2, \qquad \mathbf{G}_m^{r-1} \times \mathrm{GL}_2 \quad (r \geq 1).$$

For a reductive group over a scheme with a split maximal torus and semisimple rank one, the same description holds Zariski locally on fixed-type loci. Trivial root lines with chosen frames give a global isomorphism to the corresponding base change from $\mathbf{Z}$.

Proof. The root and coroot construction in Theorem 7.1 gives two roots $\pm\alpha$, their additive root lines, the coroot $\eta = \alpha^\vee$, and $\langle \alpha, \eta \rangle = 2$. Apply Lemma 8.A to $X = X^*(T)$, $a = \alpha$, $b = \eta$. The three resulting pairs are precisely those computed for the three displayed matrix groups: in SL_2 the root is twice the primitive character and the coroot is primitive; in PGL_2 the root is primitive and the coroot twice primitive; in GL_2 they are respectively $e_1 - e_2$ and $e_1^* - e_2^*$.

Let G_0 be the indicated matrix model. The lattice isomorphism gives a torus isomorphism $f_T : T \to T_0$ carrying root characters and coroots to those of the model. Choose a nonzero positive-root frame $E \in \mathfrak{g}_\alpha$. The perfect pairing of Theorem 7.1 gives a unique negative-root frame F with pairing one. Define $x(u) = \exp_\alpha(uE)$ and $y(v) = \exp_{-\alpha}(vF)$, and match these to the upper and lower matrix parameters x_0, y_0 of G_0.

The matrix model has a smooth geometrically connected underlying scheme over every base. For SL_2 the two polynomial-localization charts and their nonempty intersection were calculated in Theorem 6.1. For GL_2 it is the principal determinant open of affine four-space, smooth by the polynomial lifting proof and geometrically integral because a localization of a polynomial domain is a domain. For PGL_2 it is the determinant open of $\mathbf{P}^3$ from Lemma 5.B, smooth on its affine coordinate charts and geometrically integral as a nonempty open of integral projective space. The latter integrality follows from the polynomial domains on its affine charts, whose pairwise intersections are nonempty. A product with the extra split torus stays smooth; its affine fibre charts have Laurent polynomial algebras over these domains and remain integral, with nonempty chart intersections. Thus both the model and G satisfy the hypotheses of Lemma 8.B. The open cell theorem gives $\Omega = U_- T U_+$. On it define

$$f\big(y(v)tx(u)\big) = y_0(v) f_T(t) x_0(u).$$

This is an isomorphism of schemes from Ω to the model cell. The group law on the torus, the additivity of x, y, and the conjugation identities $tx(u)t^{-1} = x(\alpha(t)u)$, $ty(v)t^{-1} = y(\alpha(t)^{-1}v)$, agree under this map. The remaining multiplication rule is

$$x(u)y(v) = y\left(\frac{v}{1+uv}\right)\eta(1+uv)x\left(\frac{u}{1+uv}\right), \qquad 1 + uv \in \mathcal{O}^\times,$$

proved in Theorem 7.1 by the arbitrary-base central-extension argument and the explicit 2×2 matrix identity.

For two cell points $y(v)tx(u)$ and $y(v')t'x(u')$, their product is in the cell exactly when $x(u)y(v')$ is: left multiplication by U_-T and right multiplication by TU_+ preserve the cell, as do their inverses. The preceding rule and its unit criterion therefore give every multiplication of cell points whose product remains in the cell. It follows that f satisfies Lemma 8.B. The lemma extends it to a group

homomorphism $G \to G_0$. The inverse cell map satisfies the same identities and extends in the other direction. Uniqueness in the lemma makes their compositions the identity. Thus $G \simeq G_0$.

Any group isomorphism carries a maximal torus to a maximal torus. Over an algebraic closure the torus conjugacy theorem then identifies its root/coroot pair with the pair for the fixed torus. It can interchange the signs of both elements, but does not change either divisibility. The three invariant pairs $(2,1),(1,2),(1,1)$ distinguish the models in all characteristics, including characteristic two. This proves uniqueness without a forward reference to the derived group.

Over a scheme, restrict to an open where the root/coroot lattice pair is constant and the positive root line is trivial. Its paired negative line is then also trivial. The same torus map, cell formula and fppf word argument work over every test scheme, so give the asserted local isomorphism. When the lattice pair and frames are fixed globally, these constructions are global. All maps commute with base change because the displayed formulas and their character pairings do. □

9. Exercises

Exercise 9.1 (first steps). Compute all roots and root spaces of GL_n with respect to its diagonal torus. Determine the zero-weight space and the positive coroot for $e_i - e_j$.

Exercise 9.2 (rank-one multiplication). Compute $U_{\pm\alpha}$ and $\alpha^\vee$ for SL_2. Verify the factorization (2), and explain why the character differential in characteristic two does not change this factorization.

Exercise 9.3 (infinitesimal information). In characteristic different from two, show that $\mathfrak{sl}_2 \simeq \mathfrak{pgl}_2$ as Lie algebras, while the groups are not isomorphic. What happens to the differential of $\mathrm{SL}_2 \to \mathrm{PGL}_2$ in characteristic two?

Exercise 9.4 (classification). List the rank-two split reductive groups of semisimple rank one. Distinguish them by their centres and derived groups, including characteristic two.

Exercise 9.5 (families). For $G = \mathrm{SL}(\mathcal{O} \oplus L)$, identify the two root groups and their pairing. Give a condition on L necessary for a pinning of this pair with its displayed torus.

Solutions

Solution 9.1

The diagonal matrix units have weight zero and form $\mathfrak{t}$. For $i \neq j$, E_{ij} has weight $e_i - e_j$, each once. The corresponding root group is $I + uE_{ij}$. Its coroot has diagonal entries t in position i, t^{-1} in position j, and 1 elsewhere. Evaluating $e_i - e_j$ gives t^2, so the root–coroot pairing is two.

Solution 9.2

The two groups are the upper and lower unipotent matrices. Their parameters are x, y in the displayed matrix identity, and $\alpha^\vee(t) = \mathrm{diag}(t, t^{-1})$. Multiplication of the three factors gives $\begin{pmatrix} 1+xy & x \\ y & 1 \end{pmatrix}$, equal to $x_\alpha(x)x_{-\alpha}(y)$. The character on the upper group is t^2, so its pairing with this coroot is two. Although its differential is zero in characteristic two, the Laurent character and all polynomial matrix identities remain valid; no division by two was used.

Solution 9.3

The projective affine charts and scalar torsor constructed over $\mathbf{Z}$ in RG02 Lemma 4.P compute $\mathrm{Lie}(\mathrm{PGL}_2) = \mathfrak{gl}_2/kI$ in every characteristic: an infinitesimal scalar changes $I + \epsilon X$ to $I + \epsilon(X + qI)$.

The derivative of the determinant is the trace, so $\mathrm{Lie}(\mathrm{SL}_2) = \{X : \operatorname{tr} X = 0\}$. Inclusion followed by quotient is bracket preserving, with scalar kernel defined by $2q = 0$. When two is invertible it has zero kernel and both spaces have dimension three, so it is an isomorphism. In characteristic two the scalar line lies in the trace-zero space, giving exactly a one-dimensional kernel.

To compute the group kernel on every test algebra, a scalar determinant-one matrix is qI with $q^2 = 1$, giving μ_2. Every projective class lifts locally to an invertible matrix M. After the cover $R[z]/(z^2 - \det M)$ it has the determinant-one representative $z^{-1}M$. This is a finite free rank-two faithfully flat cover (basis $1, z$ and nonzero fibres); z is a unit because its square is a unit. Thus $\mathrm{SL}_2 \to \mathrm{PGL}_2$ is the fppf central quotient by μ_2 in every characteristic, and Theorem 1.1 supplies reductivity of this quotient as well. Its centre is trivial: a central element centralizes the diagonal torus and hence lies in it by RG02 Theorem 2.1; writing its ratio as q, commutation with the universal upper root matrix gives $(q-1)u = 0$ in $R[u]$, whence $q = 1$. For SL_2 the same diagonal-centralizer calculation makes a central element $\mathrm{diag}(q, q^{-1})$, and the universal root equation gives $q^2 = 1$. Hence its entire centre is μ_2. This is nontrivial as a scheme even in characteristic two, whereas the projective centre is trivial, so the groups are not isomorphic.

Solution 9.4

Put $r = 1$ in Theorem 8.1. The list is $\mathbf{G}_m \times \mathrm{SL}_2$, $\mathbf{G}_m \times \mathrm{PGL}_2$, and GL_2. Their centres are respectively $\mathbf{G}_m \times \mu_2$, $\mathbf{G}_m$, and $\mathbf{G}_m$. The first centre is not a torus: in characteristic two it is nonsmooth, and otherwise it has two geometric components. We compute the indicated derived groups directly. The extension $R \to R[t, t^{-1}, (t^2-1)^{-1}]$ is faithfully flat and finitely presented: it is a localization of the polynomial algebra, is R-flat, and every field fibre is a nonzero localization at a nonzero polynomial. Each such fibre has a prime by AG-CA-01 Lemma 1.1, so the map on spectra is surjective and flatness is faithful by AG-CA-07 Theorem 2.1. Thus fppf locally we can choose t with $t^2 - 1$ invertible, including in characteristic two. The identities $[h(t), x(u)] = x((t^2-1)u)$ and $[h(t), y(v)] = y((t^{-2}-1)v)$ put both root groups of SL_2 in its commutator subgroup. The open cell word argument of Theorem 6.1 shows that these root groups generate SL_2 as an fppf sheaf, since $h(t) = w(t)w(1)^{-1}$. Their images generate PGL_2, because $\mathrm{SL}_2 \to \mathrm{PGL}_2$ is fppf surjective; hence its derived group is itself. For GL_2, determinant kills all commutators, and conjugation by $\mathrm{diag}(t, 1)$, with $t-1$ invertible fppf locally, puts both determinant-one root groups in the commutator subgroup. The same generation argument gives its derived group SL_2. The central $\mathbf{G}_m$ factors contribute no commutators. Thus the last two models have derived groups PGL_2 and SL_2; the first and third have the same derived group but different centres. These computations are scheme-theoretic and remain valid in characteristic two.

Solution 9.5

An upper off-diagonal entry is a map $L \to \mathcal{O}$, so its line is L^{-1}; a lower entry is a map $\mathcal{O} \to L$, so its line is L. Composition gives $L^{-1} \otimes L \to \mathcal{O}$, the perfect pairing in Theorem 7.1. A pinning chooses a nowhere-vanishing section in the simple root line. It therefore requires L^{-1}, equivalently L, to be trivial. This illustrates the global hypothesis absent from the field case.

The course prerequisite guide records the exact supporting statements, their full proof routes, and the hypotheses needed in their applications.

References and prerequisite proofs

- Brian Conrad, *Reductive group schemes*, §§2.3 and 4.1–4.3, for the dynamic approach and Gabber's central-extension splitting argument. The proofs above are written out independently.

- Milne's freely accessible *Algebraic Groups*, version 2.00 (2015). This exact free author edition provides comparison material; its citations do not replace the programme proofs.
- SGA 3, *Schémas en groupes*, Exposés XIX–XX, for roots and the relative rank-one theorem.
- The official Stacks project and AI Integrated Stacks Project, an edition with AI-proposed corrections and AI-written additions, not reviewed by the Stacks project's maintainers. The normalization lemma in Section 5 and its proof follow the Stacks project authors' argument (Tag 0ABS) in this edition.

The following supporting statements have their exact proof routes recorded in the course prerequisite guide:

- Algebra and sheaf cohomology before reductive groups, §§3–8: injectives, finite-cover comparisons, affine vanishing, projective-space twists, ample embeddings and projective Serre finiteness.
- Projective cohomology and smooth affine models, §§1–6: exact and faithful local completion; projective formal functions and idempotents; smooth connected affine models; universal finite projective complexes and all-base-change direct images; regular curve immersions, curve duality, divisor degrees and the genus-zero calculation.
- Completing a smooth affine curve and extending its group action, §§1–4: finite curve normalization, completion, the actual family action and one-dimensional unipotent identification used through the preceding solvable-group proofs.
- Affine descent, Zariski Main and recognition of spaces, Theorems E8.1–E9.1: recognition on every scheme test, representability and the full relative integral-closure factorization.
- Flat quotient bootstrap, Theorem 8.2 and Corollary 8.3: the algebraic-space quotient and torsor relation for the free smooth subgroup action, over arbitrary bases.
- Lemma 5.A of this lesson: quotient Grassmannian charts, arbitrary base change, and the closed Plücker immersion with determinant quotient bundle; the subbundle convention follows by dualizing.
- The current written group-scheme proofs: *Diagonalizable groups*, Theorems 3.4–3.5 and §4, for smooth fixed loci, invariant algebras and torus quotients; *Group schemes over a field*, Proposition 5.8, for the reduced schematic image and faithful-flat image factorization; *Lie algebras and smoothness*, Theorem 2.9 and Theorem 3.2, for the universal normalizer tangent calculation and geometric smoothness criterion; and *Quotients and torsors*, its affine descent proof and Theorem 8.1, for effective algebra descent and line-bundle frame torsors. These are the actual written proofs, including positive-characteristic and nonreduced scope, rather than source citations.

Theorem 1.1 proves the affine central quotient before the rank-one argument; the following lesson adds the consequences for root-group coordinates. The arbitrary-base derived group is constructed after the existence theorem in the following lesson.

Source credit

The Stacks Project Authors, Tag 0ABS, with the consulted spaces-morphisms chapter.

Lesson 4

Root data, Weyl chambers and the Bruhat decomposition

Written by GPT-6.1 Sol (OpenAI) in Codex, Ultra setting, October 2026. Self-checked by the writing AI, GPT-6.1 Sol (OpenAI). Public domain (CC0).

A root records how a torus moves a one-dimensional subgroup. Its coroot records the diagonal direction in the associated rank-one group. Keeping both, together with the integral character and cocharacter lattices, distinguishes groups that have the same root system. We first assemble these data, then use cocharacters to describe Borel subgroups and the Bruhat cells. The last part explains which of these constructions survive over an arbitrary base.

The preceding lesson supplies root groups, their canonical line-bundle parametrizations, coroots, and the limit subgroups $U_G(\lambda)$ and $P_G(\lambda)$. For the field arguments we also use the geometric Borel quotient and the simply transitive action of $N_G(T)/T$ on the Borels containing T, proved in the first two lessons. We prove the identification of this normalizer quotient with the reflection group here; Theorem 4.2 then proves arbitrary-base finiteness using this field calculation.

1. The integral data and their reflections

A **root datum** consists of finite free lattices X and $X^\vee$, a perfect pairing

$$\langle\ ,\ \rangle : X \times X^\vee \longrightarrow \mathbf{Z},$$

finite subsets $\Phi \subset X \setminus \{0\}$ and $\Phi^\vee \subset X^\vee \setminus \{0\}$, and a bijection $\alpha \mapsto \alpha^\vee$ between them. Its axioms are

$$\langle \alpha, \alpha^\vee \rangle = 2, \qquad s_\alpha(\Phi) = \Phi, \qquad s_\alpha^\vee(\Phi^\vee) = \Phi^\vee,$$

where

$$s_\alpha(x) = x - \langle x, \alpha^\vee \rangle \alpha, \qquad s_\alpha^\vee(y) = y - \langle \alpha, y \rangle \alpha^\vee. \tag{1.1}$$

We require the bijection to respect these reflections. The reflections square to the identity, are dual under the pairing, and send α and $\alpha^\vee$ to their negatives. These assertions follow by substituting $\langle \alpha, \alpha^\vee \rangle = 2$ in (1.1). A datum is **reduced** if $\Phi \cap \mathbf{Q}\alpha = \{\alpha, -\alpha\}$ for every root. The empty root set is allowed: it describes a torus. Roots need not span $X_{\mathbf{Q}}$.

The **Weyl group** $W(\mathcal{R})$ is the subgroup of $\mathrm{Aut}(X)$ generated by the s_α. We prove finiteness without assuming away the central directions. Put $V = \mathbf{Q}\Phi$. The image of W on V is finite, since

it permutes the finite spanning set Φ. Choose a rational basis of V, give its real extension the sum-of-squares inner product, and average that form over this finite image. The average is positive definite because every summand is nonnegative and a nonzero vector has positive squared length under every invertible group element; relabelling the finite sum proves invariance. Each $s_\alpha|_V$ is an orthogonal reflection: its minus-one line is $\mathbf{R}\alpha$ and its fixed hyperplane is $\ker\langle\ ,\alpha^\vee\rangle$. The two eigenspaces are orthogonal, since invariance gives $(a,u)=(-a,u)=-(a,u)$. Decomposition along that line therefore gives

$$\langle v,\alpha^\vee\rangle=\frac{2(v,\alpha)}{(\alpha,\alpha)}\quad(v\in V). \tag{1.2}$$

These restricted coroot functionals span V^*, because their normal vectors are the spanning roots and the positive definite form identifies V with its dual. Here is the missing faithfulness check on X. Any reflection changes $x\in X_{\mathbf{Q}}$ by a vector of V, so any word w does too. If w fixes V pointwise, it fixes all roots. Coroot compatibility in the root-datum axioms then makes its dual action fix every coroot. Consequently $\langle wx-x,\alpha^\vee\rangle=0$ for every root. But $wx-x\in V$, and the preceding restricted functionals span V^*, so $wx-x=0$. Thus the action on V is faithful on the full lattice, and W itself is finite. Now average the sum-of-squares form on all of $X_{\mathbf{R}}$ over this finite W. The same eigenspace calculation gives (1.2) for every $x\in X_{\mathbf{R}}$, not only for $v\in V$. We use this full-lattice invariant form from now on. Set

$$Z_{\mathbf{Q}}=\{x\in X_{\mathbf{Q}}:\langle x,\alpha^\vee\rangle=0\text{ for every }\alpha\}.$$

The last formula identifies $Z_{\mathbf{Q}}$ with the orthogonal complement of V in $X_{\mathbf{Q}}$ for this rational positive definite form. To see the decomposition over $\mathbf{Q}$ explicitly, choose a rational basis of V and solve its Gram linear equations for the projection of $x\in X_{\mathbf{Q}}$. The Gram matrix is invertible: a kernel vector would have squared norm zero. Gaussian elimination over $\mathbf{Q}$ gives a unique rational solution, whose subtraction from x is in $Z_{\mathbf{Q}}$. Thus $X_{\mathbf{Q}}=V\oplus Z_{\mathbf{Q}}$. Every reflection fixes the second summand pointwise. Equation (1.2) on V also shows that equal coroots have equal normal vectors and therefore proportional roots; their pairing-two identities force the proportionality constant to be one. Proportional coroots similarly have proportional normal vectors, hence proportional roots.

The **dual datum** interchanges X with $X^\vee$ and Φ with $\Phi^\vee$. Its axioms are the same identities read in the other direction. Reducedness is preserved, by (1.2). A datum is semisimple if $\mathbf{Q}\Phi=X_{\mathbf{Q}}$; in that case

$$\mathbf{Z}\Phi\subset X\subset\mathrm{Hom}(\mathbf{Z}\Phi^\vee,\mathbf{Z})$$

inside $X_{\mathbf{Q}}$. The left and right lattices are the root and weight lattices. The intermediate lattice is information that a picture of the roots alone cannot retain.

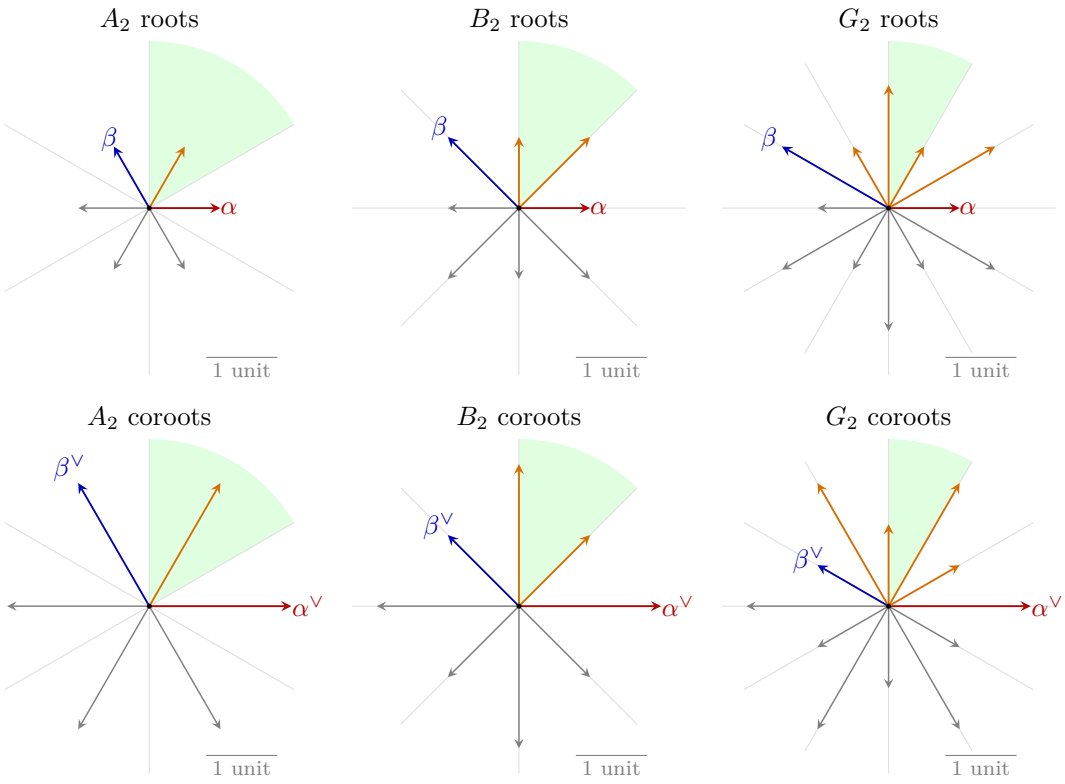


Figure 1. Roots and coroots in rank two. The top row uses the Euclidean coordinates $\alpha = (1, 0)$ and, respectively, $\beta = (-1/2, \sqrt{3}/2)$, $(-1, 1)$ and $(-3/2, \sqrt{3}/2)$. Red and blue mark the simple roots; orange marks the other positive roots; grey arrows are negative roots. The bottom row applies the metric identification $r^\vee = 2r/(r, r)$ from (1.2). The common scale makes the reversal of long and short lengths visible. Grey lines are reflecting hyperplanes, and the green sector consists of vectors pairing positively with both simple roots. This is a Euclidean realization of the rational spaces, not an identification of the integral character and cocharacter lattices. The definitions above and the reflection proof in Section 3 explain its mathematical content. The reproducible figure source and figure are original CC0 contributions.

2. The root datum of a reductive group

Let G be split connected reductive over a field k, and let T be a split maximal torus. Take

$$X = X^*(T), \qquad X^\vee = X_*(T), \qquad \mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha.$$

The pairing is composition of a character and a cocharacter. The preceding lesson proves that every root space is a line, that there are only the two roots $\pm\alpha$ on its rational line, and that the rank-one map $\varphi_\alpha : \mathrm{SL}_2 \to G$ provides an intrinsic coroot with pairing two.

Choose a nonzero vector in $\mathfrak{g}_\alpha$ and its linked dual in $\mathfrak{g}_{-\alpha}$. Write the corresponding root parametrizations as x_α and $x_{-\alpha}$. The element

$$n_\alpha = x_\alpha(1)x_{-\alpha}(-1)x_\alpha(1) \tag{2.1}$$

is the image of $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. It normalizes T. On the coroot torus it acts by inversion, and on the codimension-one torus killed by α it acts trivially. These statements follow in the rank-one group

and remain valid after its central quotient. On the rational character lattice they give precisely the reflection (1.1), since the two rational torus directions form an isogeny decomposition. Equality on the rational lattice is equality on the integral lattice, which is torsion-free. Thus

$$n_\alpha t n_\alpha^{-1} = t\,\alpha^\vee(\alpha(t)^{-1}). \tag{2.2}$$

Conjugation by any normalizer element permutes the root spaces and their root groups. The uniqueness of the rank-one coroot construction then permutes the coroots in the same way. Applied to (2.1), this proves both reflection axioms and their compatibility.

Theorem 2.1. $(X^*(T), \Phi, X_*(T), \Phi^\vee)$ is a reduced root datum. Its formation commutes with extension of the field.

Proof. Write $T \simeq \mathbf{G}_m^r$. The coefficient proof for diagonalizable groups in AG-GS, Theorem 2.2, identifies its characters with integer exponent vectors and its cocharacters with the dual vectors. Thus $X = X^*(T)$ and $X^\vee = X_*(T)$ are finite free of rank r, and their composition pairing is perfect. The finite-dimensional adjoint representation has finitely many weights by the full grading proof in RG01. The root-line reduction following RG03 Proposition 2.3 proves that every nonzero weight space is a line and that $\Phi \cap \mathbf{Q}\alpha = \{\alpha, -\alpha\}$. RG03 Theorem 7.1 supplies the intrinsic coroot $\beta = \alpha^\vee$ and the identity

$$\alpha \circ \beta(z) = z^2.$$

In particular $\langle \alpha, \beta \rangle = 2$ as an integer, in every characteristic, and $\beta \neq 0$. If Φ is empty, the reflection axioms are vacuous; the scalar-extension argument below still applies.

We make the torus decomposition in the reflection calculation precise. Write $\alpha = \nu\alpha_0$, where $\nu > 0$ and α_0 is a primitive exponent vector. Since

$$\nu\langle \alpha_0, \beta \rangle = 2, \qquad m := \langle \alpha_0, \beta \rangle = 2/\nu,$$

both ν and m are either one or two. Primitive means that the integer coordinates of α_0 have greatest common divisor one. The integer Euclidean algorithm gives integers b_i with $\sum_i a_i b_i = 1$: each division replaces a pair by its remainder pair, decreases the nonnegative remainder, and preserves its expression as a linear combination of the original coordinates. The cocharacter $\lambda_0(z) = (z^{b_1}, \dots, z^{b_r})$ therefore satisfies $\alpha_0 \circ \lambda_0(z) = z$. Consequently

$$T \simeq T_0 \times \mathbf{G}_m, \quad T_0 = \ker \alpha_0, \qquad t \longmapsto \big(t\lambda_0(\alpha_0(t))^{-1}, \alpha_0(t)\big).$$

The inverse sends (u, z) to $u\lambda_0(z)$. The basis operations are integral and invertible. For a nonzero coordinate pair (a, b), put $d = \gcd(a, b) > 0$ and choose $ua + vb = d$ by that algorithm. Right multiplication of the row by

$$\begin{pmatrix} u & -b/d \\ v & a/d \end{pmatrix}$$

sends (a, b) to $(d, 0)$, and this matrix has determinant one. Repeat with the remaining coordinates, skipping zero pairs and adjusting a final sign. This sends the primitive row α_0 to $(1, 0, \dots, 0)$. The corresponding monomial maps and their integer inverses are torus automorphisms by the coefficient classification, and identify T_0 with $\mathbf{G}_m^{r-1}$. In these coordinates $\ker \alpha = T_0 \times \mu_\nu$. For $\nu = 1$ or 2, the reduced identity component of μ_ν is the point, also when the characteristic is two. Hence the previously defined $T_\alpha = (\ker \alpha)^0_{\mathrm{red}}$ is exactly T_0.

Consider

$$p : T_\alpha \times \mathbf{G}_m \longrightarrow T, \qquad (u, z) \longmapsto u\beta(z).$$

In the preceding product coordinates, $\beta(z) = (c(z), z^m)$ for a cocharacter c of T_α. Changing the first source coordinate by $u \mapsto uc(z)$ identifies p with the identity on T_α times the power map $z \mapsto z^m$. This is finite free of positive rank m: over any test ring and any unit a, adjoining z with $z^m = a$ gives the free algebra with basis $1, z, \dots, z^{m-1}$, and $z^{-1} = a^{-1}z^{m-1}$. It is therefore an fppf

cover. This proves the torus product statement with its full scheme structure; no invertibility of two is required.

Choose linked nonzero vectors in the two root lines. The rank-one homomorphism of RG03 Theorem 7.1 sends

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \text{to} \quad n_\alpha = x_\alpha(1)x_{-\alpha}(-1)x_\alpha(1).$$

Multiplying the three elementary matrices verifies this identity over every ring. Conjugation by the displayed matrix inverts $\mathrm{diag}(z, z^{-1})$, so $n_\alpha \beta(z) n_\alpha^{-1} = \beta(z^{-1})$. Each of the two root groups lies in $G_\alpha = C_G(T_\alpha)$, by its construction in RG03 Theorem 4.1; hence n_α centralizes T_α schematically. After the fppf cover p, write $t = u\beta(z)$. Then $\alpha(t) = z^2$ and

$$\begin{aligned} n_\alpha t n_\alpha^{-1} &= u\beta(z^{-1}), \\ t\beta(\alpha(t)^{-1}) &= u\beta(z)\beta(z^{-2}) = u\beta(z^{-1}). \end{aligned}$$

Equality of morphisms descends through this faithfully flat cover. Thus (2.2) holds over every test algebra. Its right side defines a torus automorphism: its value under α is $\alpha(t)^{-1}$, and applying it twice gives t. Therefore n_α normalizes T as a group scheme.

Applying an arbitrary character x to (2.2) gives exactly

$$x \longmapsto x - \langle x, \beta \rangle \alpha.$$

Applying (2.2) to a cocharacter y gives exactly

$$y \longmapsto y - \langle \alpha, y \rangle \beta.$$

These are identities on the integral lattices themselves. Substitution of $\langle \alpha, \beta \rangle = 2$ shows that both maps square to the identity and that their paired values equal $\langle x, y \rangle$. Conjugation transports adjoint weight spaces by this character automorphism and transports their root groups by the uniqueness in RG03 Theorem 4.1. It therefore permutes Φ. Transporting the full rank-one multiplication identity of RG03 Theorem 7.1, whose coroot and pairing are unique, gives

$$(s_\alpha \delta)^\vee = s_\alpha^\vee(\delta^\vee).$$

Thus the coroot image set is preserved and the correspondence respects both reflections.

We also verify that this correspondence is a bijection onto its image set. Put $V = \mathbf{Q}\Phi$. The group generated by the restricted reflections on V is finite because it permutes the finite spanning set Φ. Average any positive definite inner product on $V_{\mathbf{R}}$ over that finite group. Each restricted reflection fixes $\ker\langle\, , \alpha^\vee \rangle$, sends α to $-\alpha$, and is orthogonal. Decomposing a vector into its fixed part and its component on the normal line gives

$$\langle v, \alpha^\vee \rangle = \frac{2(v, \alpha)}{(\alpha, \alpha)}.$$

This argument uses only the root permutation and integer pairing already proved, so it does not assume injectivity of the coroot correspondence. If $\alpha^\vee = \delta^\vee$, their displayed normal vectors agree, making $\delta = c\alpha$. The two pairings with this common coroot are both two, so $2c = 2$ in $\mathbf{Q}$, giving $c = 1$. Thus $\delta = \alpha$. All the root-datum axioms now hold, and the already proved root-line assertion makes the datum reduced.

Finally let L/k be any field extension. The exponent-vector coefficient classification leaves both lattices and their pairing unchanged. The coaction weight projectors commute with scalar extension, and a nonzero weight space remains nonzero after faithfully flat extension of a field, so the root set and its lines extend with no added or lost weights. RG03 Theorems 4.1 and 7.1 prove scalar compatibility of the root maps, perfect pairings and unique coroots. Consequently all the constructed

data, including both reflection identities, are the scalar extensions of the original data. This proves every assertion. □

For GL_n, both lattices are $\mathbf{Z}^n$, with dual bases e_i and $e_i^\vee$. Conjugation by $\mathrm{diag}(t_1, \dots, t_n)$ sends the off-diagonal matrix unit E_{ij} to $t_i t_j^{-1} E_{ij}$ and fixes the diagonal units. The root subgroup is $I + uE_{ij}$, as is checked by multiplication. The embedded i, j block copy of SL_2 has the upper and lower parameters E_{ij}, E_{ji}; its matrix factorization from RG03 Theorem 7.1 and coroot uniqueness make its diagonal cocharacter the intrinsic coroot. Thus the roots and coroots are

$$\alpha_{ij} = e_i - e_j, \qquad \alpha_{ij}^\vee = e_i^\vee - e_j^\vee \quad (i \neq j). \tag{2.3}$$

For SL_2, write $T = \{\mathrm{diag}(t, t^{-1})\}$. In the resulting copies of $\mathbf{Z}$, the positive root is 2 and its coroot is 1. For PGL_2, use $t \mapsto \mathrm{diag}(t, 1)$ modulo scalars: the positive root is 1 and its coroot is 2. These two data are dual. Their root systems have the same two-point shape, but their integral data distinguish the groups even in characteristic two.

3. Reflections and chambers

In $X_{\mathbf{R}}^\vee$, remove the hyperplanes $\langle \alpha, \lambda \rangle = 0$. The connected components are **Weyl chambers**. They are open convex cones, possibly with an unrestricted central factor. A chamber determines the positive system

$$\Phi_\lambda^+ = \{\alpha : \langle \alpha, \lambda \rangle > 0\}. \tag{3.1}$$

Conversely its signs determine the chamber: a prescribed choice of strict signs is an intersection of open half-spaces, hence is convex. Such a set is connected: a segment between points in different parts of a supposed open separation would induce an open separation of $[0, 1]$. If its first part contains zero, openness and the supremum of the points up to which the whole interval stays in that part contradict either possible part containing the supremum. Thus an interval has no such separation. The hyperplanes are defined by integer linear forms on the cocharacter lattice. The needed rational approximation has this elementary coordinate construction: for each $\varepsilon > 0$ choose a positive integer n with $1/n < \varepsilon$ and, for a real coordinate a, an integer m with $m \leq na < m + 1$. These real-order facts follow directly from completeness. If the positive integers were bounded with supremum b, an integer $n > b - 1$ would give $n + 1 > b$, a contradiction. This gives the required n and integer bounds for na. Well ordering of the positive integers after shifting past a lower bound gives the least integer strictly above na; subtracting one gives the desired m. Then m/n differs from a by less than ε. Apply this in each of finitely many coordinates with errors small enough for the finitely many strict linear inequalities, and clear the positive common denominator. Thus every chamber contains an integral cocharacter. The central factor imposes no additional inequality. If Φ is empty there is one chamber, no simple root, and the reflection group is the identity; all assertions below then have their empty-set interpretation.

We first prove transitivity on chambers. Choose interior points p, q of two prescribed chambers so that their straight segment meets no intersection of distinct root hyperplanes. Here is the required choice argument. For two nonproportional defining forms f, g, a simultaneous zero on the segment would imply

$$f(p)g(q) - g(p)f(q) = 0.$$

This is a nonzero polynomial in p, q, since f, g are independent. Finitely many such zero sets cannot fill the open product of the two chambers: their product is a nonzero polynomial, and a polynomial vanishing on an open box is zero. The latter assertion follows by induction on the number of variables, regarding it as a polynomial in the last variable and using that a nonzero one-variable

polynomial has at most its degree many roots. Endpoints lie on no wall, so the chosen segment crosses finitely many hyperplanes, one at a time.

At a crossing point on just one hyperplane, take a sufficiently small ball meeting no other hyperplane. The corresponding orthogonal reflection fixes the crossing point, reverses the two half-balls, and permutes the full hyperplane arrangement by the root-datum axioms. It therefore interchanges the two adjacent chambers. The successive crossed-wall reflections carry the first chamber to the last, proving transitivity.

Fix a chamber and its positive roots. A positive root is **simple** if it is not a sum of two positive roots. We establish the simple-root assertions directly, before using a rank-two classification. For nonparallel roots a, b with $(a, b) > 0$, put

$$A = \langle b, a^\vee \rangle = \frac{2(a,b)}{(a,a)}, \qquad B = \langle a, b^\vee \rangle = \frac{2(a,b)}{(b,b)}.$$

These are positive integers, by (1.2) and the lattice pairing, and $AB = 4(a,b)^2/((a,a)(b,b)) < 4$. Indeed the nonzero vector $b - (a,b)a/(a,a)$ has positive squared norm $(b,b) - (a,b)^2/(a,a)$, because a, b are independent. Hence their product is one, two or three, and at least one of A, B equals one. If $A = 1$, then $s_a(b) = b - a$ is a root; if $B = 1$, then $s_b(a) = a - b$ is a root. This proves the difference assertion without importing a classification.

Distinct simple roots have nonpositive inner product: otherwise the positive one of these differences decomposes one of them. They are linearly independent. A dependence would separate into an equality $v = \sum_i c_i a_i = \sum_j d_j b_j$ of nonnegative real combinations with disjoint supports. Their mutual inner product is nonpositive, while (v, v) is positive unless $v = 0$. The chamber functional is strictly positive on a nonempty positive combination, excluding $v = 0$. A dependence with only one sign is excluded by the same functional. Thus there is no nonzero dependence.

Every positive root is a nonnegative integral combination of simple roots. Indeed, decompose a nonsimple root into two positive roots and continue with each nonsimple summand. If h is the chamber functional and $c = \min_{\delta \in \Phi^+} h(\delta) > 0$, each step down a branch lowers its value by at least c. Every branch visits strictly decreasing values in the finite set $h(\Phi^+)$, so has bounded finite depth. A binary tree of bounded finite depth has at most $1 + 2 + \cdots + 2^d$ vertices, by induction on the depth. The decomposition is therefore finite. Its terminal summands are simple. Since all roots are positive or negative, the simple roots are consequently a basis of $\mathbf{R}\Phi$, and every root has integral coefficients all of the same sign in that basis.

For a simple root a, reflection changes only its coefficient in this expansion. A different positive root has a nonzero coefficient at another simple root: parallel lattice vectors have rational ratio, so reducedness excludes any other positive root on the line of a. The unchanged positive coefficient forces the reflected root to remain positive. Thus s_a permutes $\Phi^+ \setminus \{a\}$.

The chamber inequalities are now exactly positivity on the simple roots. Each simple inequality defines a wall: in the dual basis, take its coordinate zero and all other coordinates positive. Every other positive root has a positive value there, so this is an interior point of that wall. Conversely all other inequalities follow from the nonnegative expansions. Let H be generated by these simple reflections. The preceding segment argument shows inductively that H is transitive on chambers: a wall of the current chamber vC, for $v \in H$, is v applied to a simple wall, and its reflection is vs_av^{-1}. Every root hyperplane is a wall of some chamber, by choosing a point on it avoiding the other finitely many hyperplanes. Transporting that chamber to C shows that its root reflection lies in H. Here the conjugation identity $vs_av^{-1} = s_{va}$ follows directly from (1.1), the dual pairing and coroot compatibility. Hence the simple reflections generate the full reflection group.

We can now classify any two-dimensional subsystem entirely within these axioms. For a plane P spanned by roots, $\Phi_P = \Phi \cap P$ is finite, spans P, is reduced, and is stable under its root reflections. Restrict (1.2) to P; its Cartan pairings remain integers. Choose a functional nonzero on its finitely

many roots. The argument just proved supplies precisely two simple roots a, b. Set

$$k = -\langle b, a^\vee \rangle, l = -\langle a, b^\vee \rangle.$$

Both are nonnegative integers, their product is less than four, and one vanishes exactly when both vanish, since their common numerator is $-2(a, b)$. Interchanging a, b if necessary gives

$$(k, l) = (0, 0), (1, 1), (2, 1), (3, 1).$$

In the last two cases $(b, b)/(a, a) = k/l$, so a is the short root. On coefficient pairs the two reflections are exactly

$$s_a(n, m) = (-n + km, m), \qquad s_b(n, m) = (n, ln - m).$$

Every root of Φ_P is in the orbit of a or b under these two reflections. For a nonsimple positive root $\gamma = na + mb$, both coefficients are positive by reducedness. Since

$$(\gamma, \gamma) = n(\gamma, a) + m(\gamma, b) > 0,$$

one of its inner products with the simple roots is positive. Reflection in that simple root subtracts a positive integer multiple of it, preserves positivity by the simple-root permutation already proved, and strictly lowers the positive integral height $n + m$. Repetition ends at a simple root. Negative roots follow by sign.

Here is the full orbit calculation. For each row below, write R for the ordered positive-root list; the displayed lists give the image of each entry under the indicated reflection, in the same order:

$$\begin{array}{r|l}
A_1 \times A_1 & R = (a, b) \\
 & s_aR = (-a, b), \quad s_bR = (a, -b) \\
A_2 & R = (a, b, a + b) \\
 & s_aR = (-a, a + b, b), \quad s_bR = (a + b, -b, a) \\
B_2 & R = (a, b, a + b, 2a + b) \\
 & s_aR = (-a, 2a + b, a + b, b) \\
 & s_bR = (a + b, -b, a, 2a + b) \\
G_2 & R = (a, b, a + b, 2a + b, 3a + b, 3a + 2b) \\
 & s_aR = (-a, 3a + b, 2a + b, a + b, b, 3a + 2b) \\
 & s_bR = (a + b, -b, a, 2a + b, 3a + 2b, 3a + b).
\end{array} \tag{3.2}$$

These images follow by substituting (n, m) into the two displayed coefficient maps. Each set $R \cup (-R)$ is therefore stable under both reflections and contains a, b, so it contains their entire orbits. Conversely, all entries are reached: in A_2, $s_ab = a + b$; in B_2, $s_ba = a + b$ and $s_ab = 2a + b$; in G_2, use $s_ba = a + b$, $s_a(a + b) = 2a + b$, $s_ab = 3a + b$, and $s_b(3a + b) = 3a + 2b$. Negatives are in the same orbits because $-a = s_aa$, $-b = s_bb$ and the maps are linear. The descent just proved shows that these are exactly all roots, with no missing orbit.

The orders needed in the word proof follow directly from matrices. On coefficient columns, $s_a = \left(\begin{smallmatrix} -1 & k \\ 0 & 1 \end{smallmatrix}\right)$ and $s_b = \left(\begin{smallmatrix} 1 & 0 \\ l & -1 \end{smallmatrix}\right)$. Their product R has determinant one and trace $kl - 2$. Multiplying a general two-by-two matrix by itself verifies $R^2 - (kl - 2)R + I = 0$. If $kl = 0$, both k, l are zero and $R = -I$, of order two. If $kl = 1$, the identity is $R^2 + R + I = 0$, so $R^3 = I$ and $R \neq I$, of order three. If $kl = 2$, it gives $R^2 = -I$, of order four. If $kl = 3$, it gives $R^2 - R + I = 0$ and $R^3 = -I$; $R \neq I$ and $R^2 \neq I$ (the latter would force $R = 2I$ in that identity, contradicting determinant one). Its order is therefore six. Write these orders as $m = 2, 3, 4, 6$. The identity $s_aRs_a = R^{-1}$ reduces every word to R^j or R^js_a. The m powers in each list are distinct, and determinant distinguishes the two lists, giving exactly $2m$ elements. Both reflections fix the orthogonal complement of their plane in the root space, so these orders hold there; faithfulness proved in Section 1 makes them the orders on the full lattice too. This proves the full rank-two and word calculations without a trigonometric or topological input.

Length and standard parabolic cosets

We first prove the reflection assertion entirely in the root space, before using any group normalizer. This also makes the length and word results available for every reduced root datum.

Lemma 3.A (a chamber has trivial stabilizer). The reflection group acts freely on chambers.

Proof. For an empty root set the group is the identity. Otherwise Section 3 has already proved that the simple reflections generate W and that a simple reflection sends every positive root other than its own simple root to a positive root. We prove the precise deletion calculation before using length or chamber freeness.

Let $w = s_{i_1} \cdots s_{i_k}$ be any word and suppose $w\alpha_j$ is negative. Apply its letters to α_j from right to left. At some step a positive root first becomes negative. If that step is s_{i_r}, the positive root before it must be α_{i_r} by the simple-root sign property. For $v = s_{i_{r+1}} \cdots s_{i_k}$ we therefore have $v\alpha_j = \alpha_{i_r}$. The reflection conjugation identity gives $vs_jv^{-1} = s_{i_r}$.

It follows by multiplication, on the full lattice, that $ws_j = s_{i_1} \cdots s_{i_{r-1}} v$: precisely the letter s_{i_r} is deleted from the displayed word for w. This calculation applies to every word, with no reduced-word theorem assumed.

Take a nonidentity w and an expression with the fewest possible letters, writing it as $w = vs_i$. Its prefix v also has a shortest expression; a shorter prefix would shorten w. If $v\alpha_i$ were negative, the deletion just proved would express $vs_i = w$ with two fewer letters, a contradiction. Thus $\gamma = v\alpha_i$ is positive, while $w^{-1}\gamma = -\alpha_i$ is negative.

A chamber determines all root signs. If $wC = C$, a positive root γ is positive on wC, which says $w^{-1}\gamma$ is positive on C. The preceding positive root with negative inverse image is therefore impossible. Hence a chamber stabilizer contains only the identity on the root space, and the full-lattice faithfulness of Section 1 makes it the identity in W. □

The cell dimensions below now follow from this independent chamber proof. No normalizer identification is used in the length calculation.

Proposition 3.1. Let $\ell(w)$ be the minimum number of simple reflections in an expression for w. Then $\ell(w) = |\Psi_w|$, where $\Psi_w = \{\alpha \in \Phi^+ : w^{-1}\alpha < 0\}$. For $J \subset \Delta$, each right coset of $W_J = \langle s_\alpha : \alpha \in J \rangle$ has a unique element w of minimum length. It is characterized by $w(\Phi_J^+) \subset \Phi^+$. For this w and every $z \in W_J$,

$$\Psi_{wz} = \Psi_w \sqcup w\Psi_z, \qquad \ell(wz) = \ell(w) + \ell(z).$$

Proof. Fix the positive chamber C. A word in simple reflections gives a gallery of adjacent chambers from C to wC: if the current chamber is vC, its wall corresponding to a simple root leads to $vs_\alpha C$. Every such gallery must cross each hyperplane separating C from wC. There are $|\Psi_w|$ of these, since the sign of a positive root on wC is the sign of $w^{-1}\alpha$ on C.

Conversely choose interior points of C and wC generically, so the straight segment between them never meets an intersection of distinct root hyperplanes. A hyperplane with endpoints on the same side is not crossed; each separating hyperplane is crossed exactly once. The segment therefore supplies a gallery with $|\Psi_w|$ steps. Write its successive chambers as vC by the preceding adjacent-wall rule. At the end, $vC = wC$; freeness from Lemma 3.A gives $v = w$. Thus the gallery supplies a word of that length and proves the length formula. The same wall calculation gives $\ell(ws_\alpha) = \ell(w)+1$ or $\ell(w) - 1$ according as $w\alpha$ is positive or negative.

Choose a minimum-length element w of a right coset, which exists because W is finite. The last calculation gives $w\alpha > 0$ for every $\alpha \in J$. Every root in Φ_J^+ is a nonnegative integral combination of those simple roots; its image is consequently positive. Thus $w(\Phi_J^+) \subset \Phi^+$.

Each generator of W_J changes only the coefficient of its own simple root in a root's simple-root expansion. A positive root outside Φ_J has a positive coefficient outside J, which remains unchanged.

Since all coefficients of a root have the same sign, every element of W_J preserves its positivity. The corresponding assertion holds for negative roots outside Φ_J. In particular $\Psi_z \subset \Phi_J^+$ for $z \in W_J$.

Now suppose w sends Φ_J^+ to positive roots. If $\alpha \in \Psi_w$, then $w^{-1}\alpha$ is negative and outside the Levi subsystem: a negative Levi root would have negative image under w, whereas α is positive. Applying z^{-1} therefore leaves it negative. If $\alpha \notin \Psi_w$ is positive, then $w^{-1}\alpha$ is positive, and $z^{-1}w^{-1}\alpha$ is negative precisely when $w^{-1}\alpha \in \Psi_z$. These two cases prove the displayed disjoint inversion-root identity. Its cardinalities and the length formula prove length additivity. They also show that any element with the positive-Levi condition is a minimum in its right coset.

Finally, if two such minima satisfy $w' = wz$, applying additivity in both directions gives $\ell(w') = \ell(w) + \ell(z)$ and $\ell(w) = \ell(w') + \ell(z^{-1})$. Reversing a shortest word for z gives a word of the same length for z^{-1}, and reversing in the other direction gives the opposite bound. Hence $\ell(z^{-1}) = \ell(z)$, and we get $\ell(z) = 0$ and $z = 1$. This proves uniqueness. □

Relations between reflection words

The same length calculation determines all relations between simple reflections. This is the Coxeter presentation used in the pinned isomorphism argument of the next lesson. Its inputs are the rank-two root calculation (3.2), chamber freeness in Lemma 3.A, and the standard-parabolic coset calculation in Proposition 3.1. The proof below applies to every reduced root datum, including its central lattice directions.

For distinct simple roots α_i, α_j, put $c_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle \langle \alpha_j, \alpha_i^\vee \rangle$. The full rank-two calculation (3.2) and the two-by-two reflection matrices give $c_{ij} = 0, 1, 2, 3$ and exact product orders $m_{ij} = 2, 3, 4, 6$, respectively. Both reflections fix their common orthogonal complement; Section 1 makes the root-space action faithful on the entire lattice. The identity $s_i(s_is_j)s_i = (s_is_j)^{-1}$ then gives the m_{ij} powers $(s_is_j)^a$ and the m_{ij} elements $(s_is_j)^a s_i$ as all of W_{ij}, with distinctness proved by order and determinant as above.

After cancelling adjacent equal letters, words in two involutions alternate. Write $R = s_is_j$. Alternating words of length $2a$ are R^a and R^{-a}, while those of length $2a+1$ are $R^a s_i$ and $R^{-a-1}s_i$. Comparing these exponents modulo m_{ij} shows that all alternating words of length less than m_{ij} are distinct and reduced. The two alternating words of length m_{ij} agree and give the unique longest element. These words exhaust the $2m_{ij}$ elements: there is one empty word, two words at each length $1, \dots, m_{ij} - 1$, and one element at length m_{ij}.

A **braid move** replaces either alternating block of length m_{ij} by the other. With the involution relations, it is equivalent to $(s_is_j)^{m_{ij}} = 1$.

Lemma 3.2 (two left descents). If both s_i and s_j shorten w on the left, there is a length-additive factorization $w = v_0u$, where v_0 is the longest element of W_{ij} and u has minimum length in $W_{ij}w$.

Proof. Invert the unique minimum representative assertion of Proposition 3.1 to obtain the left-coset version. It gives $w = vu$, $v \in W_{ij}$, with $u^{-1}(\Phi_{ij}^+) \subset \Phi^+$ and $\ell(vu) = \ell(v) + \ell(u)$. A positive root outside Φ_{ij} has a positive coefficient on some other simple root; neither s_i nor s_j changes that coefficient. Such a root remains positive under W_{ij}, so the ambient length of v equals its rank-two length.

For $k = i, j$, the root $v^{-1}\alpha_k$ belongs to Φ_{ij}, and u^{-1} preserves its sign. Thus $w^{-1}\alpha_k < 0$ implies $v^{-1}\alpha_k < 0$. Every positive root in Φ_{ij} is a nonnegative combination of α_i, α_j, so v^{-1} makes all of them negative. Its length is consequently the number of rank-two positive roots. This is the rank-two longest element just calculated. □

Lemma 3.3 (reduced words). Any two reduced words for the same element are connected by braid moves.

Proof. Induct on the length k. If the first letters agree, cancel them, apply induction to the two tails of length $k-1$, and restore the first letter. Otherwise the two first letters are $s_i \neq s_j$, so both

shorten w on the left. Lemma 3.2 gives $w = v_0 u$, with $\ell(w) = m_{ij} + \ell(u)$. Fix a reduced word for u. Prefix it with either alternating length-m_{ij} word for v_0. Both resulting words are reduced; one starts with s_i, the other with s_j. After cancelling the first s_i, induction connects the first original word to the first constructed word. The same induction with s_j treats the second original word. A single braid move connects the two constructed prefixes. Composing these moves proves the assertion. □

Theorem 3.4 (Coxeter presentation). For the Weyl group of the root datum under consideration,

$$W = \langle s_i \mid s_i^2 = 1,\ (s_i s_j)^{m_{ij}} = 1\ (i \neq j)\rangle.$$

Proof. The relations hold by the reflection and rank-two calculations, so the presented group surjects onto W. Reduce an arbitrary word by induction on its number of letters. First reduce its prefix to a reduced word for its value a. If the final letter s_i increases length, the word is reduced. If it decreases length, choose a reduced word A for as_i. Then As_i is reduced for a, by Proposition 3.1. Lemma 3.3 changes the reduced prefix to As_i by braid moves. The entire word becomes $As_i s_i$, and the involution relation cancels the last two letters. Thus the given relations turn every word into a reduced word for its actual value. A word representing 1 reduces to the empty word. The kernel is zero. □

The argument applies to the full character lattice, including its central summand: reflections fix that summand, and the faithful action on the root span in Section 1 determines their relations.

4. Borel subgroups are positive systems

We first work over an algebraically closed field. For a Borel B containing T, the entire Borel-intersection argument in RG02 Section 1 says that $B \cap C_G(T_\alpha)$ is a Borel of the rank-one centralizer and contains T. RG03 Proposition 2.3 identifies its adjoint quotient with PGL_2 and its two Borels containing the diagonal torus with the upper and lower triangular groups. Their inverse images are the two rank-one Borels. Thus B contains exactly one of $U_\alpha, U_{-\alpha}$. The torus grading of $\operatorname{Lie} B$ is a subgrading of $\operatorname{Lie} G$. Its zero part is $\mathfrak{t}$, by RG02 Theorem 2.1. Its intersections with each pair of root lines are precisely the Lie algebras of the just identified upper or lower rank-one Borel, since taking the Lie algebra commutes with the schematic intersection. Hence

$$\dim B = \dim T + \tfrac{1}{2}|\Phi|. \tag{4.1}$$

Choose an integral cocharacter λ in a chamber, whose existence was proved in Section 3. The entire grading and infinitesimal proof of RG01 Lemma 3.3 gives smooth closed subgroups, their limit map and their Lie algebras. In particular $L_G(\lambda) = C_G(\lambda)$ contains T and has Lie algebra exactly $\mathfrak{t}$, since every root has nonzero pairing with λ. RG02 Lemma 1.1 makes this centralizer connected. The inclusion $T \hookrightarrow L_G(\lambda)$ has invertible differential between smooth groups, so S02 Lemma 3.A makes it étale; as a closed monomorphism it is an open immersion. Its image is both open and closed in the connected target and contains the identity, so it is all the target. Thus

$$L_G(\lambda) = T, \qquad P_G(\lambda) = T \ltimes U_G(\lambda), \qquad \operatorname{Lie} U_G(\lambda) = \bigoplus_{\langle \alpha, \lambda\rangle > 0} \mathfrak{g}_\alpha. \tag{4.2}$$

The unipotent group $U_G(\lambda)$ is connected by the same RG01 lemma. Thus $P_G(\lambda)$ is connected solvable and lies in a Borel. Its dimension is (4.1), so its inclusion into the containing smooth Borel has invertible differential. S02 Lemma 3.A and the étale-monomorphism argument in S04 Lemma A1.9 make this closed inclusion an open immersion. Its image is nonempty, open and closed in the connected Borel, hence is the entire Borel, as a scheme.

For a positive root, the canonical T-equivariance of RG03 Theorem 4.1 gives

$$\lambda(t)x_\alpha(u)\lambda(t)^{-1} = x_\alpha(t^{\langle\alpha,\lambda\rangle}u).$$

This extends to $t = 0$ with value the identity on every test algebra, proving $U_\alpha \subset U_G(\lambda)$. We now prove that the Borel itself depends only on the chamber. Let μ be another integral point in that chamber, order the common positive roots in any way, and form

$$m : T \times \prod_{\alpha\in\Phi^+} U_\alpha \longrightarrow P_G(\lambda)$$

by multiplication. At the identity its differential is the direct-sum map from $\mathfrak{t}$ and the distinct positive root lines onto $\operatorname{Lie} P_G(\lambda)$; it is an isomorphism. Both sides are smooth, so S02 Lemma 3.A makes m étale on a neighbourhood of that point. Its image there is a nonempty open Ω by the complete étale-openness proof in S04 Lemma D2.3. Every factor also lies in $P_G(\mu)$. The defining ideal of $P_G(\mu)$ therefore vanishes on Ω: this can be tested on the surjective étale map to Ω, which is faithfully flat. Here $P_G(\lambda)$ is reduced and irreducible by the complete component argument in S08 Theorem G.4.2. The ideal therefore vanishes on all of it, since regular functions on an integral scheme inject into their values on a dense open. Thus $P_G(\lambda) \subset P_G(\mu)$ as closed schemes. Interchanging the two gives equality. Different chambers give different root summands in (4.2), so give different Borels.

Every Borel containing T is obtained. RG02 Section 1 proves that $N_G(T)/C_G(T)$ acts transitively on the Borels containing T, using conjugacy of maximal tori within a Borel and the full Borel normalizer theorem. If n carries one constructed Borel to B, conjugation carries its extended orbit limits to those for $n\lambda n^{-1}$, so

$$B = P_G(n\lambda n^{-1}).$$

The lattice action of n permutes the root hyperplanes, hence this is again a chamber cocharacter. This proves the correspondence over the algebraically closed field.

Over an arbitrary field k with split T, the integral cocharacter and the grading ideals defining $P_G(\lambda)$ are defined over k, and RG01 Lemma 3.3 proves their scalar-extension compatibility. Their geometric fibres are the Borels just described. Equality of their closed ideals can be checked after the faithfully flat extension $k \subset \bar{k}$. Hence the correspondence is defined over k, and each geometric Borel containing T descends to the indicated constructed subgroup.

Theorem 4.1. For a split reductive group over any field, Borels containing its split maximal torus correspond bijectively to positive systems. The normalizer Weyl group equals the reflection Weyl group and acts simply transitively on these systems.

Proof. The full preceding construction proves the correspondence and its compatibility with conjugation. Work geometrically first. RG02 Theorem 2.1 gives $C_G(T) = T$, so conjugation embeds $N_G(T)/T$ faithfully into the automorphisms of the full character lattice. RG02 Section 1 identifies its action on the Borels containing T as simply transitive: a stabilizer is $N_B(T) = C_B(T) = T$, and the torus-conjugacy argument gives transitivity.

Theorem 2.1 supplies, for each root, the element n_α of (2.1) with precisely the integral reflection action. Its coset in $N_G(T)/T$ is unique for that action, by the centralizer equality. Consequently the subgroup generated by these cosets identifies with the reflection group. Section 3 proves that this subgroup is transitive on chambers. If $v \in N_G(T)/T$, choose a reflection-group element w carrying a fixed chamber to v's chamber. Then $w^{-1}v$ stabilizes it, so is the identity. Thus every coset is in the reflection group, and its chamber action is free.

Finally choose the linked root frames over k; each is a one-dimensional k-vector space and the rank-one pairing is perfect. The canonical root maps, the opposite linked frames and (2.1) are therefore defined over k. A chosen reflection word for any w gives $n_w \in N_G(T)(k)$. All geometric Weyl actions thus have these k-rational representatives, and the preceding Borel correspondence descends with them. This proves every assertion over the given field. □

Theorem 4.2 (the relative Weyl scheme). $W_G(T)$ is represented by a finite étale group scheme over S.

Proof. Here G/S is reductive, T is a maximal S-torus, and $W_G(T)$ denotes the fppf sheaf quotient $N_G(T)/T$. A test point of the normalizer must preserve the entire subgroup T after that base change. We construct this quotient and its scheme structure over the original base.

Work on an étale cover splitting $T = D_S(X)$. On an affine neighbourhood, a finite generating set of the finite locally free module $\operatorname{Lie} G$ has coactions involving only finitely many characters of X. The coefficient projectors of the grading split it into finite projective weight summands. Their ranks are locally constant. Shrinking about a chosen point makes the occurring weights constant; a rank-zero summand is zero. RG03 proves that each nonzero summand is a root line. Proposition 7.4 of AG-GS, *Diagonalizable groups*, represents $\operatorname{Hom}(\mathbf{G}_m, T)$ and $\operatorname{Aut}(T)$ by constant exponent-map schemes. The finitely many coroot maps are therefore locally constant too. On a common neighbourhood we have one fixed finite reduced root datum, by the geometric theorem and its constant lattice pairings. Choose a positive base and trivialize its finitely many simple root lines. The perfect opposite pairings give the linked negative frames.

For a simple root write $\beta = \alpha^\vee$, and use the canonical root maps of RG03 Theorems 4.1 and 7.1 to define

$$n_\alpha = x_\alpha(1)x_{-\alpha}(-1)x_\alpha(1).$$

The rank-one homomorphism sends the matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ to this element, so it inverts $\beta(\mathbf{G}_m)$. We give the full torus calculation over every test algebra. Factor the fixed integer character as $\alpha = \nu\alpha_0$, with α_0 primitive. The integer pairing $\langle \alpha, \beta \rangle = 2$ gives $\nu m = 2$, where $m = \langle \alpha_0, \beta \rangle$, so m, ν are one or two. The unimodular construction in the complete proof of Theorem 2.1 gives a cocharacter λ_0 with $\alpha_0\lambda_0(z) = z$ and the product $T = T_0 \times \mathbf{G}_m$, where $T_0 = \ker \alpha_0$.

In these coordinates $\beta(z) = (c(z), z^m)$, and

$$T_0 \times \mathbf{G}_m \longrightarrow T, \qquad (u, z) \longmapsto u\beta(z)$$

becomes the identity times $z \mapsto z^m$ after the source coordinate change $u \mapsto uc(z)$. For any unit a of any ring R, $R[z]/(z^m - a)$ is free of positive rank m, with basis $1, z, \ldots, z^{m-1}$, and z is a unit. This proves that the displayed map is an fppf cover, including in characteristic two. The canonical torus equivariance gives $\alpha|_{T_0} = 1$, hence both opposite root groups and n_α centralize T_0. Fppf locally write $t = u\beta(z)$. Then $\alpha(t) = z^2$, and the rank-one matrix identity gives

$$n_\alpha t n_\alpha^{-1} = u\beta(z^{-1}) = t\beta(\alpha(t)^{-1}).$$

Equality descends through that faithfully flat cover. The right-hand torus map is involutive, since it sends $\alpha(t)$ to its inverse. Thus n_α normalizes T schematically and has the integral reflection action on X. This argument includes all nilpotent test algebras.

Let $W \subset \operatorname{Aut}(X)$ be the finite reflection group proved in Sections 1 and 3. Choose one simple-reflection word for each w and multiply its representatives to obtain n_w; the chosen action is w. For any test scheme $S' \to S$, Proposition 7.4 identifies the torus action of $n \in N_G(T)(S')$ with a locally constant map to $\operatorname{Aut}(X)$. Its value at each geometric point lies in W, by Theorem 4.1. The preimage of the complement of W is an open and closed scheme with no geometric point, hence is empty. This factors the map through W on the entire test scheme. On its open and closed part labelled w, $n_w^{-1}n$ centralizes T on all tests; RG02 Theorem 2.2 identifies that full schematic centralizer with T. Conversely every $n_w t$ normalizes T, and its label determines w uniquely. We have proved mutually inverse functor maps

$$\coprod_{w \in W} T \xrightarrow{\sim} N_G(T), \qquad (w, t) \longmapsto n_w t, \qquad n \longmapsto (w, n_w^{-1}n).$$

Thus the normalizer is represented on this cover. Right multiplication by T is ordinary translation in each labelled copy, so its fppf quotient is exactly W_S. Its group multiplication is the usual

multiplication of lattice actions; any difference between chosen word representatives has identity action and lies in T.

On overlaps the quotients represent the same sheaf $N_G(T)/T$. Their unique comparison isomorphisms respect multiplication and satisfy the cocycle identity. The complete module descent and algebra/morphism descent proofs in S04 Lemmas A1.2 and A1.5 therefore descend their affine algebras and group maps. We spell out finiteness and étaleness of this descent. Over an affine base $\operatorname{Spec} A$, choose finitely many affine pieces of the étale cover whose open images cover it, using S04 Lemma D2.3 and quasi-compactness. Their disjoint union is $\operatorname{Spec} B$, with $A \to B$ faithfully flat. Its quotient algebra C is a finite product, on these pieces, of finite products of their base algebras; it is finite projective as a B-module.

Descend C to an A-algebra D by that module lemma. The dual C^* has the dual descent datum, so descends to a module E. Evaluation and coevaluation descend to $E \otimes_A D \to A$ and $A \to D \otimes_A E$. Their triangle identities hold after the faithfully flat cover and therefore hold over A. Write the image of 1 under coevaluation as the finite sum $\sum_i d_i \otimes e_i$. The triangle identity is

$$d = \sum_i d_i \operatorname{ev}(e_i, d).$$

It gives a split surjection $A^r \to D$, so D is finite projective. It also gives finite presentation as an algebra explicitly: choose finitely many linear generators for the kernel of this split surjection, express 1 and every product $d_i d_j$ in the d_i, and impose those linear, unit and product relations on $A[X_1, \ldots, X_r]$. Reduction of monomials gives an inverse module map from D to this quotient, so these finitely many relations present D.

For formal étaleness, let $R \to R/J$ be any square-zero extension of A-algebras and $D \to R/J$ a given map. After the faithfully flat base change by B, a map from the product algebra C is exactly a finite family of orthogonal idempotents with sum one, on each base piece. Idempotents lift uniquely through a square-zero ideal. Indeed, for a lift e with $\delta = e^2 - e \in J$, $2e - 1$ is a unit since $(2e-1)^2 = 1 + 4\delta$, and

$$e' = e - (2e-1)^{-1}\delta$$

is idempotent. The difference between two idempotent lifts satisfies $(2e' - 1)v = 0$ with $v \in J$, so is zero. Products of distinct lifted idempotents reduce to zero and are idempotent, hence are zero by uniqueness; their sum similarly equals one. The resulting local algebra lift is therefore unique. Its lifts on the double overlap agree, so the same affine descent lemma descends it to $D \to R$. This proves formal étaleness. Finite presentation and the complete infinitesimal criterion in AG-FSE, *Infinitesimal lifting and invariance under thickenings*, Theorem 4.1, make $\operatorname{Spec} D \to \operatorname{Spec} A$ étale; finite projectivity makes it finite locally free. The affine constructions and their uniquely descended group operations agree on base overlaps. They give the claimed finite étale group scheme over the original arbitrary S. □

5. Coordinates on the unipotent radicals

We need actual scheme coordinates. The following complete graded argument works over an arbitrary base ring, including one with nilpotents.

Lemma 5.1. Suppose a smooth affine R-scheme Y of finite presentation has a section e and a $\mathbf{G}_m$-action which extends to $\mathbf{A}^1$ and sends $0 \times Y$ to e. Suppose the tangent module at e is free with positive weights $d_1, \ldots, d_N$. A $\mathbf{G}_m$-equivariant map from $\mathbf{A}^N_R$, with these weights, to Y whose differential at zero is an isomorphism is an isomorphism.

Proof. Put $A = \Gamma(Y, \mathcal{O}_Y)$. The full torus coefficient grading in RG01 gives $A = \bigoplus_{d \in \mathbf{Z}} A_d$. Extension of the coaction from $R[t, t^{-1}]$ to $R[t]$ forces every negative-degree component to be zero: these monomials are linearly independent even over a ring with nilpotents. Evaluation at $t = 0$ is

the map factoring through e. For $f \in A_0$ it gives $f = e^*(f)$ in A, so $A_0 = R$, with the inclusion split by the section. Its augmentation ideal is $I = \bigoplus_{d>0} A_d$. The fixed subscheme is precisely this section, so equivariance sends the zero section to e and intertwines the two augmentation maps.

Replace a finite algebra generating list by all of its finitely many homogeneous components, and discard the degree-zero components, already in R. If none remain, $A = R$; the tangent isomorphism gives $N = 0$ when $R \neq 0$, and the assertion is immediate. For $R = 0$ both schemes are empty, also giving the assertion. Otherwise let $q > 0$ and D be the minimum and maximum of these positive generator degrees. For $F_h A = \bigoplus_{d \geq h} A_d$, the monomial calculation gives

$$I^n \subset F_{nq} A, \qquad F_{(n-1)D+1} A \subset I^n.$$

Indeed a product of n positive generators has degree at least nq, whereas a monomial of length less than n has degree at most $(n-1)D$. Homogeneous expressions in the chosen generators span each component, so these inclusions survive all algebra relations. The two filtrations are cofinal. Consequently

$$\widehat{A}_I = \prod_{d \geq 0} A_d,$$

with multiplication defined by the finite sums of component products in each degree. In particular A injects into its completion, and a homogeneous element of completed weight d is exactly an element of A_d. The polynomial algebra $P = R[x_1, \ldots, x_N]$, with $\deg x_i = d_i$ and augmentation J, has the same properties.

Write $\phi : A \to P$ for the given map. Its differential is invertible along the entire zero section. Both schemes are smooth, so the complete differential proof in S02 Lemma 3.A makes the morphism étale on an open neighbourhood V of that section. We give the full completion inverse. For $n = 1$, the zero section is a map $\operatorname{Spec}(A/I) = \operatorname{Spec} R \to V$ over Y. Inductively,

$$A/I^{n+1} \longrightarrow A/I^n$$

has square-zero kernel I^n/I^{n+1}. Formal étaleness lifts the preceding map uniquely to $\operatorname{Spec}(A/I^{n+1}) \to V$, over the natural map to Y. All these thickenings have the same underlying space as the section, so the chosen étale neighbourhood applies at every stage. The complete infinitesimal criterion in AG-FSE, *Infinitesimal lifting and invariance under thickenings*, Theorem 4.1 supplies this formal property. The compatible maps send every element of J into I/I^n at stage n and therefore factor through $P/J^n \to A/I^n$. Taking inverse limits gives $\widehat{\psi} : \widehat{P}_J \to \widehat{A}_I$.

Their being maps over Y gives $\widehat{\psi}\widehat{\phi} = 1$. To check the other composite, test on every P/J^n: the identity map and the composite are two lifts of the same zero section, over the same A-map, through the étale $V \to Y$. Uniqueness makes them equal. Thus the completions are mutually inverse. Equivariance follows from this same uniqueness after adjoining the universal Laurent variable for $\mathbf{G}_m$. The completed inverse therefore preserves each integer weight. By the displayed product description it carries P_d into A_d; the forward map does likewise. Restricting the two inverses to their direct sums gives inverse algebra maps $P \leftrightarrows A$. This proves the assertion over the original arbitrary R. □

Work locally where the split torus, the root datum and the finitely many root lines are trivialized, and choose an integral chamber cocharacter λ. RG01 Lemma 3.3 gives the smooth closed $U_G(\lambda)$, its contraction and its positive Lie summands. It also gives $L_G(\lambda)$. The inclusion $T \to L_G(\lambda)$ has invertible differential because no root has zero pairing. S02 Lemma 3.A and the étale-monomorphism argument of S04 Lemma A1.9 make it an open immersion. On each geometric fibre the full proof of Section 4 identifies this centralizer with T; hence the open image is all of $L_G(\lambda)$. Thus (4.2) holds over this base, with $P_G(\lambda) = T \ltimes U_G(\lambda)$.

For a supplied Borel subgroup scheme B containing T with this positive system, its Lie weight summands are exactly $\mathfrak{t}$ and the positive root lines. This follows on the original base from their geometric fibre identities: the inclusions of the finite projective weight modules have invertible

matrices on every fibre, hence are isomorphisms locally. Apply RG01 Lemma 3.3 also inside the smooth affine B. Its $P_B(\lambda)$ has the whole Lie algebra of B, so its closed inclusion is étale and open by the same differential and monomorphism proofs. On every geometric fibre its nonempty open-and-closed image is all of the connected Borel; hence $P_B(\lambda) = B$. Functoriality of the limit now gives a closed inclusion $B \subset P_G(\lambda)$. Their displayed Lie gradings agree, so this inclusion is again étale and open, and the full field correspondence of Section 4 makes it surjective on all geometric fibres. Its open image is therefore the whole target. We have proved $B = P_G(\lambda) = T \ltimes U^+$ as schemes over this base. This includes nilpotents and is the relative Borel identity used in Theorem 8.1.

For any ordering $\alpha_1, \dots, \alpha_N$ of Φ^+, multiplication is equivariant for λ, on the source by the positive weights $\langle \alpha_i, \lambda \rangle$. Its differential at the identity is the direct sum of these root lines onto $\operatorname{Lie} U_G(\lambda)$. The complete Lemma 5.1 therefore proves, on these trivializations,

$$\prod_i U_{\alpha_i} \xrightarrow{\sim} U_G(\lambda). \tag{5.1}$$

These are the given multiplication maps, so their inverses agree on overlaps by uniqueness and descend. This proves (5.1) for root line bundles as well. Changing the ordering changes coordinates on the same $U_G(\lambda)$. Changing the cocharacter within the positive chamber gives the same subgroup: the product image of the same root groups is the whole subgroup in both cases. We denote it by U^+. Applying the same proof to $-\lambda$, the full open-cell immersion of RG01 Lemma 3.3 gives

$$U^- \times T \times U^+ \hookrightarrow G. \tag{5.2}$$

Each construction is defined by the canonical root maps, multiplication and grading, and thus commutes with every base change.

The inverse of (5.1) supplies root coordinates. For nonparallel roots a, b, a positive system containing both exists: the two independent rational vectors a, b extend to a rational basis, by the Gaussian basis calculation in S06 Section 1. The rational dual functional assigning one to each of them pairs positively with both. Perturb this functional within rational functionals, retaining these inequalities and avoiding the finitely many root hyperplanes by the polynomial and approximation calculations of Section 3, then clear denominators as in Section 3. In a reduced root system $b \neq \pm a$ means exactly nonparallel. In the resulting U^+, the commutator has the form

$$[x_a(u), x_b(v)] = \prod_{ia+jb \in \Phi,\ i,j>0} x_{ia+jb}(C_{ij} u^i v^j). \tag{5.3}$$

Here a fixed ordering is used. To prove it, take the coordinate belonging to any root γ. It is a polynomial in u, v, because (5.1) is an isomorphism of affine schemes. Equivariance for the whole split torus makes each surviving monomial have character γ. Setting $u = 0$ or $v = 0$ gives the identity commutator, so its polynomial is divisible by both variables, by coefficient comparison over the base ring. A term can therefore occur only with $\gamma = ia + jb$, $i, j > 0$. Independence of a, b makes the exponent pair unique. This proves every coordinate of (5.3) over all test algebras; Laurent character comparison includes nilpotents and every characteristic.

More generally let C be a subset of the real root space closed under addition, and put $\Psi = \Phi^+ \cap C$. Set every root coordinate outside Ψ equal to zero. This defines a smooth closed subscheme parametrized by the root lines in Ψ. It is a subgroup. Indeed the coordinate polynomials for multiplication and inversion are torus-equivariant and vanish at the identity. Each nonconstant monomial has target character equal to the sum of its input characters, with their positive integer multiplicities. If all those characters lie in C, so does their sum; hence every coordinate outside Ψ is zero after either operation. These are polynomial identities over the base, proving the subgroup assertion schematically.

It is independent of the ordering. The product in a second ordering has no coordinate outside Ψ in the first ordering, by the same weight calculation, and has invertible identity differential.

Both sides are contracted by λ, so Lemma 5.1 makes the resulting coordinate map an isomorphism after trivializing the lines. These isomorphisms descend. Write the resulting closed subgroup as U_Ψ. If the same root subset occurs in two positive systems, the same ordered multiplication of its canonical root groups is an isomorphism onto both images. They are consequently the same subgroup of G, independently of the ambient positive system. For

$$\Psi_w = \Phi^+ \cap w\Phi^-, \qquad \Theta_w = \Phi^+ \cap w\Phi^+, \tag{5.4}$$

the relevant additive sets are the intersections of two strict positive or negative half-spaces, so are closed under addition. The two root subsets partition Φ^+. Ordering all roots in Ψ_w before those in Θ_w in (5.1) proves the scheme product

$$U_{\Psi_w} \times U_{\Theta_w} \simeq U^+.$$

The factors need not commute; this statement specifies their multiplication order.

6. The Bruhat cells from a projective limit

We give the projective limit argument, including its smoothness and its scheme structures over the field of definition.

Lemma 6.1. Let a $\mathbf{G}_m$ act on a smooth projective variety with an equivariant projective embedding. Suppose its fixed points are isolated. The points whose limit at zero is a fixed point p form an affine locally closed scheme Y_p, contracted to p, and its tangent space at p is the positive-weight part of T_pY. An equivariant map from a positive-weight affine space to Y_p, with invertible differential at zero, is an isomorphism.

Proof. First take p rational over the field in use. Diagonalize the vector space of the given linear projective action and write the weights as m_i. A fixed point is represented by a line in a single weight space, say of weight m. Choose a coordinate of that weight nonzero at p, and normalize it to one. Its affine chart U is invariant under $\mathbf{G}_m$ and is smooth affine. The other coordinates z_i have relative weights $m_i - m$. The limit equals p precisely when

$$z_i = 0 \quad (m_i < m), \qquad z_i = z_i(p) \quad (m_i = m).$$

The remaining coordinates have positive weights and vanish at p. Intersect these equations with $Y \cap U$. They give an affine closed subscheme of this chart, hence an affine locally closed $Y_p \subset Y$, with nonnegative coordinate grading, degree-zero field and contraction to p.

This also represents the limit condition on arbitrary test algebras. Within the chart a Laurent orbit $z_i(t) = t^{m_i - m} z_i$ extends to zero exactly when all negative coefficients vanish, and its value at zero gives the displayed zero-weight equations; Laurent basis independence proves this over every ring. Conversely those equations give the polynomial extension. We justify that an orbit with limit p must lie in this chart, rather than assuming its initial point lies there. The chart complement is invariant and closed. On any geometric point of a test, an initial point in that complement would have its entire punctured-line orbit there, so its zero limit would also be there: a closed subset containing the punctured affine line contains zero. This contradicts the chosen chart containing p. Thus every underlying geometric point of the test maps into the chart; an open immersion contains the entire scheme map precisely when it contains all its underlying points. The initial point, and hence its punctured orbit, therefore factor through the chart. The extension's inverse image of its complement misses both $D(t)$ and $V(t)$, which cover the entire underlying affine line even over a nonreduced test. It is consequently empty. The extension also factors through the chart. The displayed equations now apply to that whole test map. This proves the intrinsic limit functor and coordinate independence.

We prove smoothness by full equivariant square-zero lifting. Let $C \twoheadrightarrow \bar{C} = C/J$, with $J^2 = 0$, and let $y_0 \in Y_p(\bar{C})$. Its polynomial orbit is an equivariant map $F_0 : \mathbf{A}^1_{\bar{C}} \to U$ with $F_0(0) = p$. Formal

smoothness of the affine U, by the complete AG-FSE infinitesimal criterion, supplies a scheme lift $F : \mathbf{A}^1_{\bar{C}} \to U$. Differences of lifts are derivations into $J[t]$. If

$$E = \Gamma(\mathbf{A}^1_{\bar{C}}, F_0^* T_U), \qquad M = E \otimes_{\bar{C}[t]} J[t],$$

this module of differences is M: the smooth cotangent module is finite projective, and duality identifies its homomorphisms to the square-zero ideal with the displayed tangent tensor. The calculation takes place also on $C[t, a, a^{-1}]$ and on the analogous two-parameter ring; these polynomial and Laurent extensions are flat, so their kernel ideals are exactly the corresponding extensions of J.

Evaluation $M \to T_p U \otimes J$ is surjective, with kernel $M_0 = E \otimes_{\bar{C}[t]} tJ[t]$, because the finite projective E preserves the exact polynomial evaluation sequence. Correct F by such a derivation to make $F(0) = p$. On lifts use the action

$$(aF)(t) = a \cdot F(a^{-1}t).$$

Its reduction fixes F_0. The difference $c(a) = (aF) - F$ is therefore a regular section with values in M_0, and

$$c(ab) = c(a) + ac(b).$$

This follows by subtracting F through the intermediate aF; differences add because $J^2 = 0$. The module M_0 is stable under this action and is a quasi-coherent $\mathbf{G}_m$-module over $\bar{C}$. The full coefficient-integral homotopy in AG-GS, *Diagonalizable groups*, Proposition 7.14 applies even with infinitely many weights and supplies $w \in M_0$ with $c(a) = w - aw$. The corrected map $F' = F + w$ is then equivariant, since

$$(aF') - F' = c(a) + aw - w = 0.$$

Its value at zero is still p. Evaluation at one gives a lift $y \in Y_p(C)$ by the universal coordinate condition already proved. Thus Y_p is formally smooth on every affine square-zero test, and it is of finite presentation by its finite chart equations. The same infinitesimal criterion makes it smooth.

For the tangent calculation, a first-order orbit at p is $\sum_n t^n v_n$, where v_n is the weight-n part of a tangent vector. Extension permits exactly $n \geq 0$, and value zero at the fixed section excludes $n = 0$. Hence $T_p Y_p = \bigoplus_{n>0} (T_p Y)_n$. Its positive-weight contraction and smoothness now meet all hypotheses of the complete Lemma 5.1, which proves the last assertion.

The equations, orbit functor and lifting calculations commute with field extension. For a nonrational closed isolated fixed point, put $K = \kappa(p)$. The fixed scheme is smooth by the entire fixed-scheme proof in AG-GS-05 Theorem 3.4. On a geometric fibre an isolated smooth point has local dimension and tangent dimension zero, by the regular local cotangent criterion; finite-projective differential base change therefore makes the cotangent rank zero near p. The full arbitrary-base criterion in S07 Lemma 5.3A makes that smooth neighbourhood étale. Its field calculation proves that a finite-type algebra with zero differentials is a finite product of finite separable fields; it includes the full residue-field and nilpotent arguments. Applied to a sufficiently small affine neighbourhood it gives K/k finite separable. In the same projective chart impose the ideal of this $\operatorname{Spec} K$ point in the weight-zero coordinates and set the negative coordinates to zero. The resulting degree-zero algebra is K. As a K-scheme it is exactly the preceding chart construction in Y_K with the fixed coordinate values of p. It is affine over K and a locally closed k-subscheme of Y; the smoothness and tangent conclusions just proved over K give the asserted stratum. Finite separability contributes no tangent directions, by the same zero-differential field calculation. Geometric fixed points are handled over their field of definition; in the last assertion the affine space and its zero point are over that same field. This proves the original geometric and field versions. □

Let B correspond to Φ^+, put $X = G/B$, and choose an integral chamber cocharacter λ. The full flat free-action quotient and base-change proof in S07 Corollary 8.3 gives the fppf B-torsor $G \to X$. RG03 Lemma 5.1 proves that this quotient is a smooth projective scheme over the field in use, with the whole projective and arbitrary-base smoothness arguments supplied there. Geometrically the complete flag proof in RG01 Lemma 2.1 realizes X in a linear G-projective space. We use

that embedding over an algebraic closure for the projective limit calculation; all cells also have the explicit charts over k constructed below.

The T-fixed cosets are precisely n_wB, $w \in W$. Geometrically a fixed coset corresponds to a Borel containing T, because its full stabilizer is the conjugate Borel. The earlier Borel normalizer theorem identifies these cosets with the Borels themselves, and Theorem 4.1 gives their simply transitive Weyl action. The representatives are all defined over k.

There are no further λ-fixed points. On the geometric fibre, the closed fixed scheme X^λ is smooth by AG-GS, *Diagonalizable groups*, Theorem 3.4 applied to the stable affine charts of that projective embedding. Its finitely many components are disjoint closed-and-open smooth components, by the regular-local-domain calculation used in RG03 Lemma 8.B. Each is integral. Its product with the split torus is integral too: on any affine domain chart its algebra becomes a Laurent polynomial domain, and the dense chart overlaps remain nonempty. Thus the action's map to the finite discrete component labels is constant on that product. Its identity value gives the original component. Since T commutes with λ, this proves that every component is T-stable. Every such component contains a T-fixed point. To see this directly, choose an integral cocharacter η separating the finitely many T-characters of the embedding, by the integer separating-cocharacter construction in RG03 Lemma 2.1. For any point of that closed projective component, take its least nonzero η-weight projective limit. Separation leaves a single T-character space, so its line is T-fixed and remains in the closed component.

At n_wB, the torsor is smooth. Indeed after its fppf trivialization it is projection with the smooth affine factor B; S04 Corollary A3.2 descends affineness and Lemma A1.3a descends smoothness on those affine charts. Its fibre through n_w is n_wB. A tangent vector on X lifts through this smooth morphism by the square-zero lifting property applied to the dual numbers; the kernel is exactly the tangent of that fibre. Right translation by n_w^{-1} identifies it with the Lie algebra of $n_wBn_w^{-1}$ inside $\mathfrak{g}$. This gives the full tangent quotient

$$T_{n_wB}X = \mathfrak{g}/\operatorname{Lie}(n_wBn_w^{-1}).$$

The toral summand disappears, and its root weights are $\Phi \setminus w\Phi^+$. None is zero on λ. The fixed tangent space is therefore zero. Smoothness makes a neighbourhood in X^λ zero-dimensional and étale over the geometric field; its points are open reduced points by the full field calculation in S07 Lemma 5.3A. The point n_wB is also closed. Connectedness consequently makes its fixed component exactly this point. Thus the entire fixed scheme is the finite reduced set $\{n_wB\}$; all of these points are k-rational.

We identify its attracting strata with charts defined over k, retaining their full scheme structures. The open immersion (5.2) is right B-stable and is $U^- \times B$. Invariant-open descent in S04 Lemma A1.6 and the product torsor identity in S07 Corollary 8.3 give an affine open $V_1 \simeq U^-$ in X. Its translate

$$V_w = n_wV_1 \simeq n_wU^-n_w^{-1}$$

is an affine T-stable chart about n_wB. The ordered root coordinates of this group have characters $w\Phi^-$; this follows from the canonical root equivariance and uniqueness in RG03 Theorem 4.1. The roots on which λ is positive are exactly Ψ_w. The coordinate argument of Lemma 6.1 therefore cuts its attracting subscheme out by setting every negative-weight coordinate to zero. By the complete closed-root-subgroup proof in Section 5, it is precisely U_{Ψ_w}.

It is also the global attracting subscheme. A point with limit n_wB must lie in V_w: otherwise its orbit lies in the invariant closed complement, and its limit would lie in that complement too. This argument holds on every geometric point of a scheme test; an open immersion contains the whole test map when it contains all its underlying points. Within the chart, the polynomial orbit criterion of Lemma 6.1 works on every test algebra. Thus

$$U_{\Psi_w} \xrightarrow{\sim} X^+_{n_wB}, \qquad u \longmapsto un_wB. \tag{6.1}$$

Its tangent map is exactly $\bigoplus_{\alpha\in\Psi_w}\mathfrak{g}_\alpha$, the positive part of the displayed tangent quotient. These identities and charts are defined over k and commute with field extension.

Every geometric point of X has a unique projective λ-limit: choose the least nonzero weight among its homogeneous coordinates, divide the orbit coordinates by that power of t, and evaluate at zero. The defining homogeneous equations of the closed projective variety still vanish on this polynomial orbit extension. Thus its limit is in X and is fixed. Separatedness makes the extension unique; equivalently the normalized least-weight line fixes the limit. Consequently (6.1) gives a finite disjoint locally closed partition of X.

These strata are exactly the B-orbits as schemes. In (5.4), every root of Θ_w is in $w\Phi^+$, so its full root group, as well as T, belongs to the stabilizer $n_wBn_w^{-1}$. The torus normalizes both coordinate subgroups and $U^+ = U_{\Psi_w}U_{\Theta_w}$ in the specified order. Hence B's orbit map at n_wB factors through U_{Ψ_w}, and its image is exactly (6.1), including on nonreduced tests. In particular its schematic stabilizer inside B is $U_{\Theta_w}\rtimes T$: the remaining U_{Ψ_w} coordinate must be the identity, by the injectivity of (6.1).

Pull the full right B-torsor $G\to X$ back to this stratum. The morphism $u\mapsto un_w$ is a section. It gives mutually inverse maps

$$U_{\Psi_w}\times B\xrightarrow{\sim} Bn_wB,\qquad (u,b)\longmapsto un_wb,\qquad g\longmapsto (u,(un_w)^{-1}g),\tag{6.2}$$

where u is the inverse image of gB under (6.1). The torsor fibre-product identity proves that the second coordinate lies in B on every test scheme. The displayed target is the locally closed inverse image of the stratum in G, so these are isomorphisms with that exact scheme structure.

Theorem 6.2 (Bruhat decomposition). For a split connected reductive group over any field k,

$$G=\coprod_{w\in W}Bn_wB\quad\text{as a locally closed stratification,}\qquad G(k)=\coprod_{w\in W}B(k)n_wB(k).\tag{6.3}$$

Moreover Bn_wB has dimension $\dim B+|\Psi_w|$, and its quotient by the right B-action is an affine space of dimension $|\Psi_w|$.

Proof. The full preceding construction gives (6.1), (6.2), and the disjoint geometric covering. These schemes and maps are defined over k. Membership in a locally closed k-subscheme can be tested after $k\subset\bar{k}$: the open condition descends by its underlying inverse image, and its closed equations descend by faithful scalar extension. Thus every $g\in G(k)$ has its coset in exactly one of the k-defined strata. The inverse of (6.1) gives $u\in U_{\Psi_w}(k)$, and the full torsor relation gives $(un_w)^{-1}g\in B(k)$. Conversely (6.2) places every such product in that stratum. Since $U_{\Psi_w}\subset B$, these are exactly $B(k)n_wB(k)$. This proves the point decomposition for every field.

By (5.1), U_{Ψ_w} is affine space of dimension $|\Psi_w|$. The right B-action in (6.2) is translation on its B-factor, so its fppf quotient is exactly that affine space, and its dimension is $\dim B+|\Psi_w|$. The closures of the B-orbits in X, and of the $B\times B$-orbits in G, are invariant under those actions: translation preserves the orbit and hence its closure. Each closure is therefore a union of the finitely many orbits already identified. This gives the frontier property of the stated stratification as well. □

The disjoint-union notation in (6.3) describes strata and points. It does not assert that the cells are open-and-closed components, or that a morphism from a general scheme lands in one cell globally. In SL_2 the two cells have dimensions two and three, and the latter is open dense.

7. Central quotients and their root coordinates

Theorem 7.1. Let G be reductive over S and $H\subset Z(G)$ a closed subgroup of multiplicative type of finite presentation. Its affine reductive quotient G/H, constructed in the preceding lesson,

Theorem 1.1, commutes with every base change. For a split maximal torus T its maximal torus is T/H, and its root groups are the images of those of G.

Proof. RG03 Theorem 1.1 supplies the affine reductive quotient $q : G \to Q$, its full H-torsor identity, finite presentation and arbitrary base change. We prove the additional torus and root-group assertions as scheme statements. By RG02 Theorem 2.2, $C_G(T) = T$. Centrality therefore puts H inside T on every test scheme.

Work on an étale affine cover on which $T = D_R(X)$ and $H = D_R(M)$. This simultaneous cover exists by the complete relative character theorem in AG-GS, *Diagonalizable groups*, Theorem 7.7. On a further open neighbourhood the homomorphism is given by one map $\rho : X \to M$: choose the images of finitely many generators of X, using the group-like coefficient calculation of Lemma 2.1 and Theorem 2.2 there. The closed embedding gives a surjection $R[X] \to R[M]$. Its image has precisely the exponent basis in $\rho(X)$, so for $R \neq 0$ linear independence forces $\rho(X) = M$. The empty affine chart has nothing to prove. Put $X' = \ker \rho$.

The subgroup X' is a finite-rank free abelian group. Here is the integer argument: project a subgroup of $\mathbf{Z}^r$ onto the first coordinate. If its image is $d\mathbf{Z} \neq 0$, choose an element with first coordinate d; subtracting its multiples splits the subgroup as its cyclic span plus a subgroup of $\mathbf{Z}^{r-1}$. If the image is zero, use only that smaller subgroup. Induction proves freeness and finite generation, without requiring X' to be saturated. Thus $T' = D_R(X')$ is a split torus.

The H-grading on $R[X]$ has degree m consisting of the exponents with $\rho(x) = m$; its degree-zero algebra is $R[X']$. Choose, for each m, one lift $x_m \in X$. Then its degree-m component is the free rank-one $R[X']$-module $e^{x_m}R[X']$, and multiplication between components is an isomorphism. Consequently $R[X]$ is faithfully flat over $R[X']$. The full torsor algebra map is

$$R[X] \otimes_{R[X']} R[X] \longrightarrow R[X] \otimes_R R[M], \qquad a \otimes c_m \longmapsto ac_m \otimes e^m.$$

Its inverse on the exponent basis sends $a \otimes e^m$ to $ae^{-x_m} \otimes e^{x_m}$. These formulas are mutually inverse; the forward map is an algebra map, so its bijective inverse is an algebra map too, independently of the lifts. The morphism $T \to T'$ is of finite presentation: both are finite Laurent polynomial algebras over R; add their finitely many defining Laurent relations and the images of the finitely many source generators to a relative presentation. It is therefore an fppf H-torsor and represents T/H.

Write $A = \Gamma(G, \mathcal{O}_G)$. The closed inclusion $T \subset G$ gives a surjective equivariant algebra map $A \to R[X]$. Exactness of taking degree zero, proved in AG-GS Theorem 3.1, gives a surjection

$$A^H \longrightarrow R[X'].$$

It embeds T' as a closed subgroup of $Q = \operatorname{Spec} A^H$. The resulting map $T \to G \times_Q T'$ is a morphism of H-torsors over T' and is an isomorphism. Indeed, after an fppf cover with a section of the first torsor, equivariance translates that section to a section of the second; both torsors become H, and the map is translation by that section. The inverse descends uniquely. This also proves the exact inverse-image assertion $q^{-1}(T') = T$.

Now work where the root datum is constant and its root lines are trivialized. Section 5 proves the relative open cell $G_0 = U^- \times T \times U^+$. Since $H \subset T$ is central, its right action changes only the middle factor. This open is saturated for the H-torsor. S04 Lemma A1.6 descends it to an open $Q_0 \subset Q$. The displayed product torsor $G_0 \to U^- \times T' \times U^+$ has the same quotient functor, so uniqueness gives

$$Q_0 = U^- \times T' \times U^+ \subset Q. \tag{7.1}$$

For any root α, choose a positive system containing it. Its Borel is a closed subgroup $B = T \ltimes U^+$,

as proved in Sections 4 and 5. The surjection $A \to \Gamma(B, \mathcal{O}_B)$ is H-equivariant, so exact invariants give a closed subgroup

$$B' = B/H = T' \ltimes U^+ \subset Q.$$

The product identifies this quotient because H again acts only on T. As for T', the morphism of H-torsors $B \to G \times_Q B'$ is an isomorphism, so $q^{-1}(B') = B$. In its scheme coordinates $T' \times \prod_{\beta>0} U_\beta$, the map from U_α sets the torus coordinate to the identity and every other root coordinate to zero. It is a closed immersion into B', hence into Q. Its inverse image under q is exactly $H \times U_\alpha$ in the corresponding $T \times U^+$ coordinates, by the same torsor identity. In particular $H \cap U_\alpha = 1$, including on nonreduced test schemes.

Centrality makes the restriction of α to H trivial: in a local root frame, conjugation multiplies the root coordinate by that restricted character, whereas centrality makes this multiplier one on the universal H-algebra. Hence $\alpha \in X'$. The image root group has exactly that character for conjugation by T'. This can be checked after the faithfully flat cover $T \to T'$, where it is the original identity. None of these characters becomes zero, since $X' \hookrightarrow X$ is injective.

At the identity, (7.1) now gives the whole Lie decomposition

$$\mathrm{Lie}(Q) = \mathrm{Lie}(T') \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha.$$

All the displayed root summands retain their one-dimensional nonzero T'-characters. Thus the invariant tangent submodule is precisely $\mathrm{Lie}(T')$, using the full character grading, rather than character differentials. On a geometric fibre, any torus A' containing T' commutes with it, so $\mathrm{Lie}(A')$ injects into this invariant part. The inclusion of T' gives the reverse inclusion of tangent spaces, so its differential is an isomorphism. S02 Lemma 3.A and the étale-monomorphism proof in S04 Lemma A1.9 make $T' \to A'$ an open immersion. An open subgroup is also closed: every other coset is open, so their union is its open complement. Connectedness of A' therefore makes its nonempty subgroup all of A'. This proves that T' is maximal on every geometric fibre, hence is a maximal S-torus.

There are no additional roots in Q, because the displayed Lie decomposition is exhaustive. Apply the uniqueness assertion of RG03 Theorem 4.1 to Q: each closed smooth image just constructed has the required character and identity differential on its root line, so it is exactly the canonical root group of Q. This proves the assertion about all root groups.

All character algebras, grading projectors, torsor identities, product maps and coordinate-zero immersions above commute with arbitrary base change. On overlaps the maps are the original quotient map restricted to T and to each canonical root group, so their identifications agree. The complete morphism descent proof in S04 Lemma A1.5 glues them. This proves the theorem over the original base without a smoothness assumption on H. □

8. Splittings and subgroup constructions in families

Over a scheme, a split maximal torus need not trivialize its root lines. A reductive group is **split with specified root datum** here when its maximal torus is identified with $D_S(X)$, its datum is the specified constant datum, and its root lines are trivializable. A **pinning** chooses a nowhere-vanishing section of each simple root line. These are the objects used in the next lesson.

Every reductive group acquires such data étale locally. RG02 Theorem 2.3 gives a maximal torus. A torus has an étale splitting cover by its definition in RG01; the equivalent fpqc definition is justified by the complete relative character theorem, AG-GS Theorem 7.7. Work on an affine member with $T = D_R(X)$. The coefficient grading in RG01 gives

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \neq 0} \mathfrak{g}_\alpha$$

with only finitely many nonzero weight modules locally. Each is a finite projective direct summand. Its rank is locally constant: on a local basis chart it is a fixed free rank, by S01 Lemma 1.2. After restricting around any chosen point the finite weight list and all its ranks are therefore fixed. RG03 Theorem 4.1 identifies its nonzero summands as root lines. The root-system assertion and the chamber construction of Sections 2 and 3 give the finite constant root system and a positive system.

The coroots are homomorphisms $\mathbf{G}_m \to D_R(X)$. By the same group-like coefficient proof in AG-GS Lemma 2.1 and Theorem 2.2, their finitely many character images are fixed integers on a further open neighbourhood. There are only finitely many coroots, so one common neighbourhood makes the whole datum constant. Trivialize *all* the finitely many root lines on a further common open cover. Choose the simple frames there; these are the pinning. Choosing all lines also verifies the stated split condition before using any later pinned classification theorem. This argument uses finite projective modules and universal characters over R, including its nilpotents.

The full product proof in Section 5 then applies. Its local inverses agree on overlaps by uniqueness, so without any choice of frames the product of positive root line bundles is U^+, the supplied positive Borel is $T \ltimes U^+$, and (5.2) is an open immersion. Choosing frames identifies these line bundles with affine coordinates.

For an arbitrary cocharacter λ, RG01 Lemma 3.3 gives smooth closed subgroups $P_G(\lambda)$, $L_G(\lambda)$ and $U_G(\lambda)$, their exact Lie weight decompositions, and the limit homomorphism. Its section is the inclusion of $L_G(\lambda)$; on all test schemes its kernel is $U_G(\lambda)$. The explicit mutually inverse maps

$$L_G(\lambda) \times U_G(\lambda) \longrightarrow P_G(\lambda), \qquad (l, u) \longmapsto lu,$$

$$g \longmapsto \big(l(g),\, l(g)^{-1} g\big)$$

give $P_G(\lambda) = L_G(\lambda) \ltimes U_G(\lambda)$. Here $L_G(\lambda)$ is the fixed subgroup for conjugation by λ. On a geometric fibre, a nonzero cocharacter has torus image: in integer coordinates divide its vector by its greatest common divisor; the resulting primitive vector is a closed one-dimensional subtorus and the map onto it is a finite free power map. The zero cocharacter has trivial image. RG02 Theorem 2.1 makes the centralizer of this torus image connected reductive. Thus all geometric fibres of the already smooth affine finite-presentation $L_G(\lambda)$ are connected reductive, proving that it is reductive over the base. Its roots have zero pairing; those of $P_G(\lambda)$ have nonnegative pairing and those of $U_G(\lambda)$ positive pairing, directly from the proved Lie grading. Relative parabolic quotients will be constructed in the last lesson.

For the quotient of Theorem 7.1, the lattice formula is $X' = \ker(X \to X^*(H))$. Its root characters lie in X', as the universal centrality calculation in that proof shows. The quotient of the rank-one homomorphism in RG03 Theorem 7.1 sends the original coroot to $q_T \circ \alpha^\vee : \mathbf{G}_m \to T/H$. For every $\chi \in X'$,

$$\langle \chi, q_T \circ \alpha^\vee \rangle = \langle \chi, \alpha^\vee \rangle.$$

In particular its pairing with α is two. The quotient transports the full rank-one multiplication identity, and Theorem 7.1 identifies both image root groups with their canonical parametrizations. The all-test coroot uniqueness of RG03 Theorem 7.1 therefore identifies $q_T \circ \alpha^\vee$ with the intrinsic coroot of Q. Characters change from X to X', and cocharacters by this dual map; the root system stays the same. The free subgroup X' need not be saturated, so central quotients can change the integral lattice even when H is finite.

These constructions are local on the base and are compatible with base change and the canonical root maps. They make no assertion of one constant lattice on a disconnected base. For example, $\mathbf{G}_m$ on one component and $\mathbf{G}_m^2$ on another is a reductive group of different torus ranks. Constant-datum assertions apply componentwise or on the covers just constructed.

Bruhat cells over a base

The field theorem also gives actual cells over the integers; the following argument identifies their scheme structures before using their fibres.

Theorem 8.1 (relative Bruhat cells). Let G/S be a pinned split reductive group of constant reduced root datum. Fix its split torus T, positive Borel B, Weyl group W, and the pinning representatives n_w. Let $N = |\Phi^+|$, $r = \operatorname{rank} X^*(T)$, and $\Psi_w = \{\alpha \in \Phi^+ : w^{-1}\alpha \in -\Phi^+\}$. Multiplication gives locally closed immersions

$$U_{\Psi_w} \times B \longrightarrow G, \qquad (u, b) \longmapsto u n_w b,$$

whose images C_w form a finite, disjoint decomposition of G into locally closed subschemes. With the root frames and a basis of $X^*(T)$, they have coordinates

$$C_w \simeq \mathbf{A}_S^{\ell(w)} \times \mathbf{G}_{m,S}^r \times \mathbf{A}_S^N.$$

The construction commutes with arbitrary base change, including to nonreduced schemes. For $S = \operatorname{Spec} \mathbf{Z}$, each stratum is a product of a split torus and an affine space over the integers. This locally closed decomposition is a finite partition of the underlying scheme points; it does not identify G with their coproduct as schemes or partition $G(R)$ for an arbitrary ring R.

Proof. The ordered root-product theorem of Section 5, applied relatively as above, identifies U_{Ψ_w} with the product of its root lines. The set Ψ_w is closed under addition of roots. Conjugation by n_w^{-1} identifies it with a closed root-product subscheme of U^-: complete its root ordering by the remaining negative roots, and set those remaining coordinates to zero. This is a closed immersion over S, since the ordered product identifies the whole U^- with a product of root lines. After left translation by n_w^{-1}, the displayed map is the composite

$$n_w^{-1} U_{\Psi_w} n_w \times B \;\hookrightarrow\; U^- \times B \;\hookrightarrow\; G.$$

The second map is the relative open-cell immersion already proved. Thus the original map is a locally closed immersion, with precisely the stated product as its source and image. This argument is over the base, rather than an inference of an immersion from geometric fibres. Inversion applied to the cell indexed by w^{-1} also gives the opposite-order coordinates $B \times U_{\Psi_{w^{-1}}} \to C_w$, $(b, u) \mapsto b n_w u$. A different pinning representative of the same Weyl element differs by an element of T and only changes the torus coordinate. Thus one can equally use $U^+ \times T \times U_{\Psi_{w^{-1}}}$. The inversion convention in this order is $\Psi_{w^{-1}} = \{\alpha > 0 : w\alpha < 0\}$.

On each geometric fibre, Theorem 6.2 identifies these images with the Bruhat cells. Hence their underlying sets cover G and are pairwise disjoint: every point lies over a point of S, and a geometric point above it tests both assertions. The length formula of Section 3 gives $|\Psi_w| = \ell(w)$. Finally $B = T \ltimes U^+$ as a group and $T \times U^+$ as a scheme, with $U^+ \simeq \mathbf{A}_S^N$ after choosing the root frames. These identifications prove the coordinate formula. All maps, coordinate-zero immersions and representatives base change, proving the last assertion. □

For a finite field $\mathbf{F}_q$ this gives the useful count

$$|G(\mathbf{F}_q)| = (q-1)^r q^N \sum_{w \in W} q^{\ell(w)}.$$

For a torification of the integral scheme, partition each affine coordinate by zero versus invertible. On a cell with $d = N + \ell(w)$ affine coordinates, a subset I of invertible coordinates gives a locally closed torus $\mathbf{G}_m^{r+|I|}$, with every other coordinate zero. These finitely many pieces partition geometric points and points over every field. They do not make each point over a ring belong to exactly one such piece. Thus the construction supplies the relative cells and counting hypotheses used in the lessons on finite-field counting and torified varieties without a claim about general ring-valued Bruhat double cosets.

A global equation for the longest cell

The longest cell is already open. An integral equation for this open will also control denominators in its coordinates.

Proposition 8.2 (the normalized longest-cell equation). Let $G/\mathbb{Z}$ be a pinned split reductive group with constant root datum and split torus $T = D_{\mathbb{Z}}(X)$. Fix positive roots Φ^+, put $s = |\Phi^+|$, and choose integral root frames $E_\alpha = dx_\alpha(1)$ for all roots. Write

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathbb{Z} E_\alpha, \qquad V = \bigwedge^s \mathfrak{g}, \qquad \eta = \sum_{\alpha \in \Phi^+} \alpha \in X.$$

Let v_+ and v_- be the ordered wedges of the positive and negative root frames. Choose a basis of $\mathfrak{t}$ and let $\varphi_- : V \to \mathbb{Z}$ extract the v_--coefficient in the resulting wedge basis. If n_0 represents the longest Weyl element, write

$$(\bigwedge^s \mathrm{Ad})(n_0) v_+ = \epsilon v_-, \qquad \epsilon \in \{\pm 1\}.$$

Then the regular function

$$d(g) = \epsilon^{-1} \varphi_-((\bigwedge^s \mathrm{Ad})(g) v_+)$$

satisfies $d(n_0) = 1$, and the opposite-order longest-cell map

$$\theta : \mathbb{A}^{\Phi^+} \times T \times \mathbb{A}^{\Phi^+} \longrightarrow G, \qquad (a, t, b) \longmapsto \psi(a) t n_0 \psi(b)$$

is an isomorphism onto $D(d)$. This holds for every reduced root datum, including rootless tori and every isogeny type.

Proof. AG-RG-03 Theorem 4.1 supplies the root line bundles and their canonical parametrizations; the split integral frames give the displayed decomposition. For a finite free module with ordered basis $f_1, \ldots, f_m$, its exterior power has the increasing wedges as a basis, with the following full calculation. Multilinearity and alternating relations express every wedge as a sum of increasing wedges, killing repetitions and changing signs under adjacent exchanges. The sign rule follows by expanding the alternating relation with two copies of $u + v$. For each increasing index set, the determinant of the corresponding coordinate rows defines an alternating multilinear functional: in its permutation sum the terms cancel in pairs if two columns agree. It therefore respects the alternating relations and evaluates to one on that chosen basis wedge and zero on every other increasing wedge. These functionals prove independence. Wedging images defines the exterior power of a linear map, since it preserves all these relations; wedging composites verifies composition, and wedging the inverse verifies invertibility. Thus the exterior adjoint action used below is a representation on this finite free wedge module over every test ring and commutes with scalar extension. AG-RG-03 Theorem 7.1 supplies the linked negative frame, the rank-one homomorphism and its Weyl representative. For each simple root, the image of $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is the integral element n_α of (2.1), and (2.2) gives its reflection action on T. Chamber transitivity and freeness in Section 3 give a unique element w_0 sending the positive chamber to the negative chamber. It sends every positive root to a negative root. Proposition 3.1 gives its length s; any other element has fewer inversions

and therefore smaller length. This proves existence and uniqueness of the longest element being used. Choose one word in simple reflections for each w, as allowed by Section 3, and multiply the corresponding n_α. This produces $n_w \in N_G(T)(\mathbf{Z})$ with the required action. Conjugation carries each integral root line isomorphically to the reflected line. Its coefficient relative to two $\mathbf{Z}$-frames has an integral inverse, so is 1 or -1. These facts prove the signed-wedge assertion used below. Choose an integral cocharacter λ strictly positive on Φ^+. There are exactly s positive-weight basis vectors of $\mathfrak{g}$, all remaining vectors having weight zero or negative. Consequently the unique maximal-weight line of V is $\mathbf{Z}v_+$, and its unique minimal-weight line is $\mathbf{Z}v_-$. Their torus characters are η and $-\eta$.

For a representation R on a finite free torus-graded module M and a vector m of weight ν, write

$$R(x_\alpha(z))m = m + \sum_{j>0} z^j m_j.$$

Conjugation by the universal torus and comparison of Laurent characters give $m_j \in M_{\nu+j\alpha}$. This comparison is an integral polynomial identity; it does not differentiate characters or divide by a prime. Positive root groups therefore fix v_+, and preserve the coefficient extracted by φ_-. By the ordered root-product theorem, for $u \in U$ these identities give

$$R(u)v_+ = v_+, \qquad \varphi_- \circ R(u) = \varphi_-.$$

Take $R = \bigwedge^s \mathrm{Ad}$. Integral normalizer representatives permute the integral root lines by units, so $R(n_w)v_+$ is a signed wedge indexed by $w\Phi^+$. Its v_--coefficient is zero unless $w\Phi^+ = \Phi^-$, which is precisely $w = w_0$. In the opposite-order coordinates of AG-RG-04 Theorem 8.1, the preceding identities thus show

$$d(utn_w u') = 0 \quad (w \neq w_0), \qquad d(utn_0 u') = \eta(t)^{-1}.$$

These are identities on the coordinate schemes, and remain true after every base change. The last expression is a unit and equals one at n_0.

The relative open-cell immersion of AG-RG-04 (5.2), translated and inverted as in Theorem 8.1, makes θ an open immersion. Its image is the longest Bruhat stratum. The strata cover all scheme points, so the computed restrictions show that this image and $D(d)$ have exactly the same underlying open set. Open subschemes carry the restricted structure sheaf and are determined by that set. They are therefore equal as schemes, including after nonreduced base changes.

If Φ is empty, then $\operatorname{Lie} G = \operatorname{Lie} T$ over $\mathbf{Z}$. The closed torus inclusion is a map between smooth groups with invertible differential, hence is étale by S02 Lemma 3.A and an open immersion by S04 Lemma A1.9. It is also closed and contains the identity in every connected geometric fibre. Its open complement therefore has no geometric point, so it is empty. Thus $G = T$ as schemes. Set $V = \bigwedge^0 \mathfrak{g} = \mathbf{Z}$ and $d = 1$; the asserted map is then the torus isomorphism. This proves the rootless case too. □

The sum η lies in the root lattice, hence in X, for every isogeny type. The proof requires neither individual fundamental weights in X, a perfect Killing form, nor a factorial coordinate ring or trivial Picard group. Orthogonal examples are understood as the smooth split reductive models of the stated root data; reductivity of every geometric fibre is among the hypotheses.

For PGL_2, take the class of $n_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Pull back the function to GL_2 and write $g = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right)$, $\delta = ae - bc$. Then

$$gE_{12}g^{-1} = \delta^{-1} \begin{pmatrix} -ac & a^2 \\ -c^2 & ac \end{pmatrix}, \qquad d|_{\mathrm{GL}_2} = c^2/\delta.$$

Here $\epsilon = -1$. The formula is scalar invariant and is the pullback of the already constructed regular function on PGL_2. Its value at n_0 is one. The boundary can have multiplicity two in this equation;

a principal open does not require a reduced boundary equation. This calculation is a pullback formula: it does not assert that every $\mathrm{PGL}_2(A)$-point has a $\mathrm{GL}_2(A)$-lift.

The exterior-power construction is due to Claude Chevalley, *Certains schémas de groupes semi-simples*, Section 4, Proposition 1, pp. 226–229. The proof here gives its normalized longest-cell form for every reduced root datum. Its integral representative input is the preceding rank-one construction, AG-RG-03 Theorem 7.1, together with (2.1)–(2.2) and the reflection words in Section 3 of this lesson.

9. Exercises

Exercise 9.1 (easy). Write the complete root datum of GL_n and identify its central rational direction.

Exercise 9.2 (medium). Verify integrally that the data of SL_2 and PGL_2 are dual. Explain why identifying their rational root systems does not identify their groups.

Exercise 9.3 (medium). Show that the Weyl group of GL_n is S_n, generated by its root reflections.

Exercise 9.4 (hard). Prove $\mathrm{GL}_n(k) = \coprod_{w \in S_n} B(k) w B(k)$ by upper triangular row and column operations, where B is upper triangular. Prove uniqueness of the permutation.

Exercise 9.5 (families). Let L be a nontrivial line bundle. For $G = \mathrm{SL}(\mathcal{O} \oplus L)$, describe the positive unipotent radical and decide whether the displayed diagonal torus has a global pinning.

Solutions

Solution 9.1

The lattices and roots are (2.3), with $\langle e_i, e_j^\vee \rangle = \delta_{ij}$. Reflection in $e_i - e_j$ exchanges coordinates i, j. The root span is $\{(a_i) : \sum a_i = 0\}$, while its orthogonal central direction is $\mathbf{Q}(1, \dots, 1)$. The centre has character group $\mathbf{Z}^n / \mathbf{Z}\Phi \simeq \mathbf{Z}$, the quotient map being summation. For $n = 1$ the root set is empty and the entire lattice is central.

Solution 9.2

In the coordinates of Section 2, the pairs (positive root, positive coroot) are $(2, 1)$ and $(1, 2)$ in dual copies of $\mathbf{Z}$. Interchanging lattices interchanges these pairs. The reflection is multiplication by -1 in both. The quotient of the character lattice by the root lattice is $\mathbf{Z}/2$ for the first and zero for the second, so their centres are μ_2 and the trivial group. This distinction holds scheme-theoretically in characteristic two.

Solution 9.3

Formula (1.1) for $e_i - e_j$ sends e_i to e_j, e_j to e_i, and fixes the other basis vectors. Thus its reflection is the transposition (ij). Every permutation is a product of disjoint cycles: start from a moved index, follow its successive images until it returns (injectivity rules out any earlier repetition), remove that finite orbit, and repeat on the remaining indices. With rightmost factors acting first, each cycle satisfies $(i_1\ i_2\ \cdots\ i_m) = (i_1\ i_m) \cdots (i_1\ i_3)(i_1\ i_2)$, verified by applying it to each of the listed indices; all other indices are fixed. Thus transpositions generate every permutation. Conversely all products of these reflections are coordinate permutations. The lattice representation is faithful, so the reflection group is exactly S_n; Theorem 4.1 identifies it with the normalizer quotient.

Solution 9.4

In the bottom row choose its leftmost nonzero entry, in column j. Scale that row to make the entry one. Add multiples of column j to columns to its right, making the other bottom-row entries zero; these are permitted right upper triangular operations. Entries to its left were already zero. Add multiples of the bottom row to higher rows to clear column j; these are permitted left upper triangular operations. Delete the bottom row and column j. The remaining square matrix is invertible, by expansion of the determinant along the cleared row. Repeat with its bottom row, retaining the original order of the surviving columns. Operations on these rows and columns extend to upper triangular operations on the original matrix and preserve every previously cleared pivot. Induction gives one nonzero pivot in every row and column, all scaled to one, hence a permutation matrix. This proves existence over every field.

For uniqueness consider, for every i, j, the rank r_{ij} of the submatrix in rows $i, \dots, n$ and columns $1, \dots, j$. Left upper triangular operations mix these bottom rows only with one another, and right upper triangular operations mix these first columns only with one another. Hence all r_{ij} are invariant. In a permutation matrix they count pivots in that rectangle. The difference

$$r_{ij} - r_{i+1,j} - r_{i,j-1} + r_{i+1,j-1}$$

is one exactly at a pivot (i, j) and zero otherwise, with empty boundary rectangles having rank zero. These invariants recover the permutation uniquely. The double cosets are therefore disjoint.

Solution 9.5

Choose the upper root as positive. Its group is the additive line bundle $\mathbf{V}(L^{-1})$, and the triangular subgroup is $T \ltimes \mathbf{V}(L^{-1})$. A pinning would choose a nowhere-vanishing section of L^{-1} and therefore trivialize L. Hence this pair has no global pinning, although it has one on every open set trivializing L. Its relative big cell is $\mathbf{V}(L) \times T \times \mathbf{V}(L^{-1})$.

The course prerequisite guide records the exact supporting statements, their full proof routes, and the hypotheses needed in their applications.

References and prerequisites

The root datum terminology is due to Demazure; freely accessible comparison is SGA 3, Exposé XXI, 13 October 2024 re-edition. For the group constructions see Conrad's author manuscript, *Reductive group schemes* and Milne's freely accessible *Algebraic Groups*, version 2.00 (2015). The reflection, word, coordinate and cell arguments are written here; these sources do not replace a programme proof.

The general fibre criteria used in Section 7 belong to the programme's morphism theory. The geometric homogeneous quotient used in Section 6 is the field quotient construction introduced in Quotients and torsors, together with the self-normalizing projective-quotient construction in the preceding lesson. Their precise prerequisite locators and proof states are recorded in the course dependency list.

Lesson 5

Pinnings and the classification of split reductive groups

Written by GPT-6.1 Sol (OpenAI) in Codex, Ultra setting, October 2026. Self-checked by the writing AI, GPT-6.1 Sol (OpenAI). Original text: public domain (CC0).

The integral root datum remembers a reductive group up to isomorphism. A pinning removes the conjugation ambiguity and makes this assertion functorial. The main work is to show that the multiplication of root groups is forced by the datum, including over rings in which two or three vanish. We first do those calculations and glue them across the open cell. We then construct groups in characteristic zero and pass to an integral model.

Throughout, a split group has a specified split maximal torus, a constant identification of its character lattice, and trivial root lines. All assertions about a base scheme allow nilpotents and arbitrary characteristic. On a disconnected base, maps of constant data are locally constant; they need not have one value on every component. The purely combinatorial Coxeter presentation and length formula are proved in the preceding AG-RG-04, Proposition 3.1 and Theorem 3.4, before their use here.

1. What a pinning fixes

Let G/S be reductive, let $T = D_S(X)$ be a split maximal torus, and choose a positive system Φ^+ with simple roots Δ. A **pinning** consists of a frame E_α in every simple root line $\mathfrak{g}_\alpha$, $\alpha \in \Delta$. The canonical root parametrization from the preceding lesson gives

$$x_\alpha : \mathbf{G}_a \longrightarrow U_\alpha, \qquad x_\alpha(u) = \exp_\alpha(uE_\alpha).$$

The perfect rank-one pairing gives a unique frame $E_{-\alpha}$ with $E_\alpha E_{-\alpha} = 1$. Put

$$n_\alpha = x_\alpha(1)x_{-\alpha}(-1)x_\alpha(1), \qquad t_\alpha = \alpha^\vee(-1). \tag{1.1}$$

The rank-one homomorphism $\mathrm{SL}_2 \to G$ proves

$$n_\alpha^2 = t_\alpha, \quad n_\alpha t n_\alpha^{-1} = s_\alpha(t), \quad \mathrm{Ad}(n_\alpha)E_\alpha = -E_{-\alpha}. \tag{1.2}$$

It also proves $(n_\alpha x_\alpha(1))^3 = 1$. These are identities of group schemes, including in characteristic two. Explicitly, in SL_2 the matrix $nx_\alpha(1)$ is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & -1 \end{smallmatrix}\right)$, its square is $\left(\begin{smallmatrix} -1 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and its cube is the identity. Also $n^2 = -I$, $n\,\mathrm{diag}(z, z^{-1})n^{-1} = \mathrm{diag}(z^{-1}, z)$ and $nE_{12}n^{-1} = -E_{21}$ by matrix multiplication. The rank-one homomorphism transports each of these identities on every test ring.

For example, the usual pinning of SL_2 uses the diagonal torus, the upper triangular Borel, and

$$x_\alpha(u) = \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}.$$

Then $\alpha(\mathrm{diag}(z, z^{-1})) = z^2$, $\alpha^\vee(z) = \mathrm{diag}(z, z^{-1})$, and $n_\alpha = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The frame, rather than just its line, is part of the pinning.

A based root datum includes $X, X^\vee, \Phi, \Phi^\vee$ and Δ. An isomorphism of based data induces the corresponding torus isomorphism, and matches roots and coroots. We use this covariant formulation; its map on character lattices runs in the opposite direction.

Integral Weyl representatives

The pinned representatives also determine the full arithmetic normalizer, including the signs from the central torus.

Proposition 1.1 (the integral normalizer). Let $G/\mathbb{Z}$ be a pinned split reductive group, with split maximal torus $T = D_{\mathbb{Z}}(X)$, constant reduced root datum and Weyl group W. Put $r = \operatorname{rank} X$ and $N = N_G(T)$. Then $N/T \simeq \underline{W}$, and

$$1 \longrightarrow \mathrm{Hom}(X, \{\pm 1\}) \longrightarrow N(\mathbb{Z}) \longrightarrow W \longrightarrow 1$$

is an exact sequence of finite groups. Thus $|N(\mathbb{Z})| = 2^r |W|$. For every ring A, an integral element n above w acts on $T(A)$ by

$$(nhn^{-1})(\chi) = h(w^{-1}\chi).$$

Moreover

$$N(A)/T(A) \simeq \underline{W}(A) = \mathrm{LC}(\operatorname{Spec} A, W).$$

The representatives of W need not form a subgroup of $N(\mathbb{Z})$.

Proof. Choose the negative parameter linked to each pinned simple-root parameter. The rank-one homomorphism $\mathrm{SL}_2 \to G$ gives

$$n_\alpha = x_\alpha(1)x_{-\alpha}(-1)x_\alpha(1), \qquad n_\alpha^2 = \alpha^\vee(-1), \qquad \mathrm{Int}(n_\alpha)|_T = s_\alpha.$$

These identities are proved in AG-RG-03 Theorem 7.1 and AG-RG-05 (1.1)–(1.2). Since the simple reflections generate the Weyl group, selecting one reflection word for each w and multiplying these elements yields $n_w \in N(\mathbb{Z})$. This choice uses no word-independence assertion.

AG-RG-04 Theorem 4.2 makes $Q = N/T$ finite étale. The chosen classes define a map $\coprod_{w \in W} \operatorname{Spec} \mathbb{Z} \to Q$. By AG-RG-04 Theorem 4.1, on every geometric fibre these classes are distinct and exhaust its Weyl group. Its coordinate algebras are finite projective over $\mathbb{Z}$, by the complete finite-projective construction in RG04 Theorem 4.2. Locally at each prime, choose bases using CA07 Theorem 5.3. The algebra map is a square matrix and is an isomorphism on the residue-field fibre, since this can be tested after an algebraic closure by S06 Section 1. Its determinant is consequently a unit in that local ring, and the adjugate calculation in S06 Section 8 gives an inverse. Its kernel and cokernel vanish at every prime and thus vanish: a nonzero module element has a proper annihilator, contained in a maximal ideal, where its localization is nonzero. Hence the scheme map is an isomorphism. The group laws also agree. A map from a component $\operatorname{Spec} \mathbb{Z}$ of the constant source to the constant finite target has one label, since $\operatorname{Spec} \mathbb{Z}$ is connected. Indeed its basic nonempty opens all contain the generic point, so it is irreducible. The labels of multiplication and inversion agree on the geometric fibres, and hence everywhere. This identifies Q with $\underline{W}$ as a group scheme.

A section of N maps to the identity section of Q exactly when it factors through the kernel T. The chosen representatives prove surjectivity on $\mathbb{Z}$-points. Since

$$T(\mathbb{Z}) = \mathrm{Hom}(X, \mathbb{Z}^\times) = \mathrm{Hom}(X, \{\pm 1\}),$$

the displayed extension and its order follow. Retain all these torus signs, including those arising from the central torus; the subgroup generated by coroot signs can be smaller.

The convention $w\chi = \chi \circ \mathrm{Int}(n^{-1})$ gives $\chi(nhn^{-1}) = (w^{-1}\chi)(h)$ after every base change. This proves the stated action.

Finally a section of $\underline{W}$ over A is a locally constant map from $\operatorname{Spec} A$ to W. Its finitely many fibres are clopen. On the fibre with value w, base change n_w; these sections glue to an element of $N(A)$. Thus the map $N(A) \to \underline{W}(A)$ is surjective, with kernel $T(A)$. This proves the quotient formula without a hypothesis on $\mathrm{Pic}(A)$. □

For SL_2, $T(\mathbb{Z}) = \{\pm I\}$, and both lifts of the nonidentity Weyl element square to $-I$. Hence this extension is $C_4 \to C_2$, and has no group-homomorphic section.

Lifting scope. For a general normalizer extension over a scheme, the fibre over a Weyl section is a T-torsor. That section lifts precisely when this torsor has a global section, equivalently when its class in $H^1_{\mathrm{fppf}}(A, T)$ is trivial. A sheaf quotient alone does not assert surjectivity on A-points. For the pinned split model above, the explicit integral representatives trivialize every such fibre; the obstruction is zero even when $\mathrm{Pic}(A) \neq 0$. This is a torsor trivialization argument, not an assertion that $H^1(A, T)$ vanishes.

2. Reducing comparisons to two simple roots

We need two facts about root systems. The first explains why rank-two calculations control commutators even when the two roots under consideration are not simple.

Lemma 2.1. For independent roots α, γ, there is a base of the root system containing α and another root β such that $\gamma = a\alpha + b\beta$, with a, b nonnegative integers.

Proof. In the plane $E = \mathbf{R}\alpha + \mathbf{R}\gamma$, take a functional p_0 with $p_0(\alpha) = 0$ and $p_0(\gamma) = 1$; prescribing these values on the independent pair defines it. Its only root zeros in the plane are $\pm\alpha$, by reducedness. Choose p_1 with $p_1(\alpha) = 1$, and put $p = p_0 + \varepsilon p_1$. For a sufficiently small positive ε, its values on all other plane roots keep their p_0-signs, and its values on α, γ are positive. Explicitly, choose ε smaller than every positive $|p_0(\delta)|/(|p_1(\delta)| + 1)$ for the finitely many roots with $p_0(\delta) \neq 0$, and small enough to keep $p(\gamma) > 0$. Thus this is a positive system. Every positive root has nonnegative p_0-value. A decomposition of α into two positive roots would force both values to be zero; both roots would then be α, which cannot sum to α. Hence α is simple. The plane-root proof in RG04 Section 3 supplies exactly one other simple root β, and the positive-root expansion gives the asserted nonnegative integers for γ.

Extend this positive system to the whole root system as follows. Extend a basis of E to the full root space, using S06 Section 1. Every root outside E has a nonzero image in the quotient, so its vanishing equation is a proper hyperplane in the dual quotient. The finite polynomial avoidance proof in RG04 Section 3 supplies a functional q vanishing on E and nonzero on every such root. Then extend p by setting its values on the added basis vectors to zero and take $q + \varepsilon' p$. Choose positive ε' smaller than every $|q(\delta)|/(|p(\delta)| + 1)$ for roots δ outside E. Roots outside E keep the sign of q. A sum of positive roots involving any root outside E has positive q-value and cannot equal α or β. Their indecomposability in the plane therefore implies their indecomposability in the full positive system. They belong to its base. If E is the full root space, the extension step is unnecessary. □

The next fact controls different choices for transporting a simple root to another root. Write

$$\ell(w) = \#\{\gamma \in \Phi^+ : w\gamma \in -\Phi^+\}.$$

Lemma 2.2 (root transport). If $w\alpha = \beta$ for simple roots α, β, then w is a product of elements of standard rank-two Weyl groups, each carrying the current simple root to the next simple root. A factor fixing that root belongs to a rank-two subgroup containing it.

Proof. Induct on $\ell(w)$. Length zero means $w = 1$. Otherwise some simple root δ has $w\delta < 0$: if all simple roots had positive images, all positive roots would. Here $\delta \neq \alpha$. Let E be their plane and set

$$A = w^{-1}\Phi^+ \cap E.$$

This is a positive system of $\Phi \cap E$. Choose $v \in W_{\alpha,\delta}$ with $vA = \Phi^+ \cap E$, and put $w' = wv^{-1}$. An element of this rank-two subgroup permutes the positive roots outside E: their positive coefficients at the other simple roots are unchanged. Thus w' has the same number of inversions outside E as w, and no inversions inside E. At least δ was an inversion of w, so $\ell(w') < \ell(w)$.

The root α is simple for A, since $w\alpha = \beta$ is simple in the full positive system. Consequently $v\alpha$ is either α or δ. Apply induction to w', which carries this simple root to β, and prepend the factor v. □

These arguments apply to the root space; central directions in a reductive datum are fixed by its Weyl group.

3. The multiplication forced in rank two

Fix two distinct simple roots α, β, taking α short in the unequal-length cases. In the formulas below all parameters are sections over an arbitrary S-scheme. We choose frames for the remaining positive roots by the indicated conjugations:

Type	Positive roots besides α, β	Chosen frames
A_2	$c = \alpha + \beta$	$E_c = \mathrm{Ad}(n_\beta)E_\alpha$
B_2	$c = \alpha + \beta,\ d = 2\alpha + \beta$	$E_c = \mathrm{Ad}(n_\beta)E_\alpha,\ E_d = \mathrm{Ad}(n_\alpha)E_\beta$
G_2	$c = \alpha + \beta,\ d = 2\alpha + \beta,\ e = 3\alpha + \beta,\ f = 3\alpha + 2\beta$	$E_c = \mathrm{Ad}(n_\beta)E_\alpha,\ E_d = \mathrm{Ad}(n_\alpha)E_c,\ E_e = -\mathrm{Ad}(n_\alpha)E_\beta,\ E_f = \mathrm{Ad}(n_\beta)E_e$

Write $x_c, x_d, \ldots$ for the corresponding root parametrizations. The complete noncommuting positive-root relations are

$$\begin{aligned}
A_2:\quad & x_\beta(y)x_\alpha(x) = x_\alpha(x)x_\beta(y)x_c(xy);\\
B_2:\quad & x_\beta(y)x_\alpha(x) = x_\alpha(x)x_\beta(y)x_c(xy)x_d(x^2y),\\
& x_c(y)x_\alpha(x) = x_\alpha(x)x_c(y)x_d(2xy);\\
G_2:\quad & x_\beta(y)x_\alpha(x) = x_\alpha(x)x_\beta(y)x_c(xy)x_d(x^2y)x_e(x^3y)x_f(x^3y^2),\\
& x_c(y)x_\alpha(x) = x_\alpha(x)x_c(y)x_d(2xy)x_e(3x^2y)x_f(3xy^2),\\
& x_d(y)x_\alpha(x) = x_\alpha(x)x_d(y)x_e(3xy),\\
& x_e(y)x_\beta(x) = x_\beta(x)x_e(y)x_f(-xy),\\
& x_d(y)x_c(x) = x_c(x)x_d(y)x_f(3xy).
\end{aligned} \tag{3.1}$$

All other positive-root pairs commute. The powers in these formulas are forced by torus weights, and the coefficient integers are independent of the base. In particular, the terms with coefficient two or three disappear in the relevant characteristics; the proof never divides by them.

Here are the conjugation rules needed along with (3.1):

Type	$\mathrm{Ad}(n_\alpha)$	$\mathrm{Ad}(n_\beta)$
A_2	$E_\beta \mapsto -E_c,\ E_c \mapsto E_\beta$	$E_\alpha \mapsto E_c,\ E_c \mapsto -E_\alpha$
B_2	$E_\beta \mapsto E_d,\ E_d \mapsto E_\beta,\ E_c \mapsto -E_c$	$E_\alpha \mapsto E_c,\ E_c \mapsto -E_\alpha,\ E_d \mapsto E_d$
G_2	$E_c \mapsto E_d,\ E_d \mapsto -E_c,\ E_\beta \mapsto -E_e,\ E_e \mapsto E_\beta,\ E_f \mapsto E_f$	$E_\alpha \mapsto E_c,\ E_c \mapsto -E_\alpha,\ E_d \mapsto E_d,\ E_e \mapsto E_f,\ E_f \mapsto -E_e$

In $A_1 \times A_1$, the two rank-one groups commute. The normalizer relations are

$$(n_\alpha n_\beta)^m = \begin{cases} t_\alpha t_\beta, & m = 2,\ A_1 \times A_1, \\ 1, & m = 3,\ A_2, \\ t_\alpha, & m = 4,\ B_2 \text{ with } \alpha \text{ short}, \\ 1, & m = 6,\ G_2. \end{cases} \tag{3.2}$$

Deriving the coefficients

We give the calculations over an affine open of S, where all unknown coefficients lie in its ring R. Since root-product coordinates are unique, identities with indeterminate parameters in $R[x, y]$ determine coefficients even when R has nilpotents.

If no root is a positive integral combination of two roots, their root groups commute, by the weight calculation in the preceding lesson. A rank-one group therefore acts trivially on a root line $\mathfrak{g}_\gamma$ when neither $\gamma + \alpha$ nor $\gamma - \alpha$ is a root. The definition of the frames and $n_\alpha^2 = t_\alpha$ give every entry in the conjugation table except the scalar on E_c in B_2 and the scalar carrying E_β to E_c in A_2.

For A_2, put an unknown A in the last parameter of the first formula of (3.1). Conjugating that formula by n_β gives

$$x_{-\beta}(-y)x_c(x) = x_c(x)x_{-\beta}(-y)x_\alpha(-Axy).$$

The identity $n_\beta x_\alpha(x) n_\beta^{-1} = x_c(x)$, after cancelling commuting end factors, becomes

$$x_\beta(1)x_\alpha(x)x_\beta(-1) = x_{-\beta}(1)x_c(x)x_{-\beta}(-1).$$

Reordering its two sides gives respectively $x_\alpha(x)x_c(Ax)$ and $x_\alpha(Ax)x_c(x)$. Thus $A = 1$. If $\mathrm{Ad}(n_\alpha)E_\beta = zE_c$, apply the same cancellation with α, β interchanged. It gives

$$x_\beta(y)x_c(-y) = x_\beta(-z^{-1}y)x_c(zy).$$

Hence $z = -1$. This proves both tables for A_2.

For B_2, use unknown coefficients A, B, C in place of $1, 1, 2$ in its two relations, and write $\mathrm{Ad}(n_\alpha)E_c = kE_c$. Conjugating the first relation by n_β, and cancelling as above, gives

$$x_\alpha(x)x_c(Ax)x_d(Bx^2) = x_\alpha(Ax)x_c(x)x_d((AC - B)x^2).$$

Thus $A = 1$ and $C = 2B$. Conjugate the first relation by n_α, and use $n_\alpha x_\beta(y) n_\alpha^{-1} = x_d(y)$. The resulting cancellation gives

$$x_\beta(y)x_c(-Ay)x_d(By) = x_\beta(By)x_c(Aky)x_d(y).$$

The three groups on each side commute. Therefore $B = 1,\ k = -1,\ C = 2$.

For G_2, put coefficients A, B, C, D in the four extra parameters of the first relation, E, F, G in the second, H in the third, and J in the fourth. Conjugation by n_β makes the coefficient in the fifth relation also H. The frame choices and the square relations already give its full conjugation table.

The cancellation identity for $n_\beta x_\alpha(x) n_\beta^{-1} = x_c(x)$ gives the following two normal forms:

$$x_\alpha(x)x_c(Ax)x_d(Bx^2)x_e(Cx^3)x_f((D-CJ)x^3),$$
$$x_\alpha(Ax)x_c(x)x_d((AE-B)x^2)x_e((A^2F+CJ-D)x^3)x_f((AG-C)x^3).$$

Comparison gives

$$A = 1, \qquad E = 2B, \qquad C + D = F + CJ, \qquad F = G. \tag{3.3}$$

The identity $n_\alpha x_\beta(y) n_\alpha^{-1} = x_e(-y)$ similarly gives

$$x_\beta(y)x_c(-Ay)x_d(By)x_e(-Cy)x_f(-Dy^2),$$
$$x_\beta(Cy)x_c(-By)x_d(Ay)x_e(-y)x_f((D-CJ-ABH)y^2).$$

Consequently $C = 1,\ A = B,\ D - CJ - ABH = -D$. Together with (3.3) this says

$$A = B = C = 1, \quad E = 2, \quad F = G, \quad D + 1 = F + J, \quad 2D = H + J. \tag{3.4}$$

The cancellation identity carrying E_e to E_f under n_β reduces to

$$x_e(x)x_f(-Jx) = x_e(-Jx)x_f(x);$$

thus $J = -1$. Finally use $n_\alpha x_c(y) n_\alpha^{-1} = x_d(y)$. The left side, after cancelling its rank-one end factors, has normal form

$$x_c(y)x_d(-Ey)x_e(Fy)x_f(-Gy^2).$$

The right side is $x_{-\alpha}(1)x_d(y)x_e(Hy)x_{-\alpha}(-1)$. Move $x_{-\alpha}$ past these factors, using the conjugates by n_α of the already written relations. Here is the full coordinate check for the coefficient being extracted. The only roots of the form $-i\alpha + jd$ with $i, j > 0$ are c, β, f, and the only roots of the form $-i\alpha + je$ are d, c, β, f; these lists follow by substituting $d = 2\alpha + \beta$, $e = 3\alpha + \beta$ in the complete G_2 root list. The $-\alpha$-factor therefore creates only β, c, d, f-factors, never an e-factor or an α-factor. It commutes with β and f, since no positive integral combination with $-\alpha$ is a root. Among β, c, d, f, the only noncommuting pair is c, d, whose extra coordinate is f by the fifth written relation. Collecting these factors therefore never creates e either. After the negative end factors cancel, the e-coordinate is exactly the original Hy. Comparing it with the left normal form gives $F = H$. Equations (3.4) now imply $F = D + 2$ and $2D = F - 1$, so subtraction gives $D = 1$, and $F = G = H = 3$. No cancellation by two or three has occurred. This proves (3.1).

Checking the normalizer powers

For a frame E_γ, write n_γ for its rank-one representative. If n carries E_γ to zE_δ, direct substitution in (1.1) gives

$$n n_\gamma n^{-1} = \delta^\vee(z) n_\delta. \tag{3.5}$$

For $z=-1$ this is n_δ^{-1}. In moving torus factors through representatives use $n_\alpha t = s_\alpha(t)n_\alpha$; in particular

$$s_\alpha(t_\beta) = t_\beta t_\alpha^{\langle \alpha,\beta^\vee\rangle}. \tag{3.6}$$

These formulas give (3.2) without assuming that torus two-torsion is central. For clarity, here are reductions of the alternating words. In A_2,

$$n_\alpha n_\beta n_\alpha = n_c^{-1} t_\alpha, \qquad n_\beta n_\alpha n_\beta n_\alpha n_\beta = n_\alpha^{-1}.$$

In B_2,

$$\begin{aligned} n_\alpha n_\beta n_\alpha &= n_d t_\alpha,\\ n_\beta n_\alpha n_\beta n_\alpha n_\beta &= n_d t_\alpha t_\beta,\\ n_\alpha n_\beta n_\alpha n_\beta n_\alpha n_\beta n_\alpha &= n_\beta^{-1} t_\alpha. \end{aligned}$$

Here t_α is central in this rank-two group. In G_2 the conjugation table and (3.5) give, successively,

$$\begin{aligned} n_\alpha n_\beta n_\alpha &= n_e^{-1} t_\alpha,\\ n_\beta (n_\alpha n_\beta)^2 &= n_f^{-1} t_\alpha,\\ n_\alpha (n_\beta n_\alpha)^3 &= n_f^{-1},\\ n_\beta (n_\alpha n_\beta)^4 &= n_e t_\beta,\\ n_\alpha (n_\beta n_\alpha)^5 &= n_\beta^{-1}. \end{aligned}$$

Multiplying the last word by n_β proves the sixth-power relation. The commuting case follows at once from $n_\alpha^2 = t_\alpha$ and $n_\beta^2 = t_\beta$. This completes all the rank-two assertions. The successive reductions can be verified explicitly with $A = n_\alpha$, $B = n_\beta$, $a = t_\alpha$, $b = t_\beta$, and C, D, E, F the representatives of c, d, e, f. Always $a^2 = b^2 = 1$, $A^2 = a$, $B^2 = b$. In A_2, $ABA = C^{-1}a$, $BCB^{-1} = A^{-1}$ and $s_\beta(a) = ab$, so $BABAB = Aa = A^{-1}$. In B_2, $ABA = Da$, $BDB^{-1} = D$ and $s_\beta(a) = a$, giving $BABAB = Dab$. Also $ADA^{-1} = B$ and $s_\alpha(ab) = b$, so $ABABABA = Bba = B^{-1}a$. The simple pairings of the short coroot are $2, -2$, so a commutes with the torus and every root group of this rank-two group; multiplying the last word by B gives a. In G_2, start with $ABA = E^{-1}a$. Since $BEB^{-1} = F$ and $s_\beta(a) = ab$, the next word is $BABAB = F^{-1}a$. Since $AFA^{-1} = F$ and $s_\alpha(a) = a$, the next word is $ABABABA = F^{-1}$. Since $BFB^{-1} = E^{-1}$, the next is $BABABABAB = Eb$. Finally $AEA^{-1} = B$ and $s_\alpha(b) = ba$, so $ABABABABABA = Bba^2 = B^{-1}$. These substitutions prove every displayed alternating-word reduction over all test rings.

4. From local multiplication to the entire group

We first prove the extension on all test schemes, and then prove the cell identities to which it will be applied.

Lemma 4.1 (extension from the open cell). Let K/S be a smooth group with geometrically connected fibres, let $\Omega \subset K$ be a fibrewise dense open containing the identity, and let H/S be a separated group scheme. A morphism $f : \Omega \to H$ with $f(1) = 1$ and

$$f(xy) = f(x)f(y) \quad \text{whenever } x, y, xy \in \Omega \tag{4.1}$$

extends uniquely to a homomorphism $K \to H$.

Proof. The full regular-component and affine-chart argument in the earlier RG03 Lemma 8.B proves that a smooth geometrically connected field group is geometrically irreducible, without a quasi-compactness assumption. For any test scheme $R \to S$ and any finite list $g_1, \dots, g_d \in K(R)$, the open

$$\Omega_R \cap \bigcap_{i=1}^{d} \Omega_R g_i^{-1}$$

has a nonempty open in every geometric fibre. Indeed each translate is open dense in the same irreducible fibre, and a finite intersection of nonempty opens in an irreducible space is nonempty. Its morphism to R is smooth and surjective, being an open of K_R. Its affine smooth neighbourhoods, covering the base, give an fppf cover with a section: take these neighbourhoods themselves as the covering schemes. This argument also permits intersecting with the open Ω_R^{-1}.

For $g \in K(R)$, choose on such a cover $c, c^{-1}, cg \in \Omega_R$. Then $g = c^{-1}(cg)$ is a word in cell points. To a word $x_1 \cdots x_d$, assign $f(x_1) \cdots f(x_d)$. This assignment is independent of the word. For two words with the same endpoint g, choose a so that a, every $ax_1 \cdots x_i$, and every translated prefix of the other word belong to Ω_R. There are finitely many conditions of the preceding form. At each step the two successive translated prefixes and the next letter are in Ω_R, so (4.1) applies. Multiplying either assigned image word on the left by $f(a)$ gives $f(ag)$. Group cancellation proves equality after the cover, and hence before it, since equality of morphisms is fpqc local by S04 Lemma A1.5.

The empty word at the identity has image 1. The same comparison consequently gives image 1 to every identity word. An inverse word has the inverse image, by comparing its concatenation with the original word to that empty word. Pullback of test schemes preserves all words, comparisons and their values. On overlaps of the covers the assigned values therefore agree. The sheaf property for scheme-valued points, proved in S04 Lemma A1.5, glues them to a natural transformation $K(-) \to H(-)$. Concatenation gives multiplication and the empty word gives the identity. Yoneda then gives a scheme homomorphism. Every extension must have these values on all local generating words, proving uniqueness. This is the full word-extension argument also proved in RG03 Lemma 8.B. □

We need a scheme-level density argument for the normalizer identities.

Lemma 4.2 (density in a cell over arbitrary rings). Put

$$A = R[q_1, \dots, q_N, z_1^{\pm 1}, \dots, z_r^{\pm 1}], \qquad e : \ q_i = 0, \ z_j = 1.$$

If $d(e) \in R^\times$, then $D(d)$ is universally schematically dense in $\operatorname{Spec} A$ and in every open subscheme. Two maps from an open containing the section e to a separated scheme, agreeing on a smaller open containing e, therefore agree on the entire common domain, locally on $\operatorname{Spec} R$.

Proof. Substitute $z_j = 1 + w_j$ in $R[[q_1, \dots, q_N, w_1, \dots, w_r]]$. This embeds A in that ring. To check injectivity, multiply a Laurent polynomial by a Laurent monomial to make it an ordinary polynomial. Its substituted multiplier is an invertible formal series. Translation $z_j \mapsto 1 + w_j$ identifies the ordinary polynomial algebra with the polynomial algebra in w_j, and a polynomial has zero formal series exactly when every coefficient is zero. This argument is valid over any R, including rings with nilpotents.

The series of d has invertible constant term, so it is a unit. Its inverse is obtained recursively by total degree, using only that constant unit. Multiplication by d is consequently injective on A. On any localization A_h it remains injective: if $da/h^n = 0$, some power of h annihilates da, and injectivity on A makes the same power annihilate a. Thus $A_h \to (A_h)_d$ is injective. Every open is covered by these principal charts. The identical proof works after every R-algebra extension, proving the asserted universal schematic density.

For the last assertion, let $U \subset V$ be the smaller and larger domains. Around the identity over any

base point, choose a principal $D(d) \subset U$ in the cell, then restrict the base to $D(d(e))$. This is a neighbourhood of that base point, and $d(e)$ is a unit there. The equalizer of the two maps is closed in V, by separatedness of the target. Each element of its ideal vanishes after localization at d. The injection just proved on every principal chart makes the ideal zero. Hence the equalizer is all of V. The base neighbourhoods cover, giving equality everywhere. □

Now let K be split reductive and let the maps on its torus and canonical root groups have values in H. Suppose they intertwine torus characters, linked rank-one multiplication, normalizer conjugations and the positive rank-two relations of Section 3. We prove that their ordered product on

$$\Omega = U^- T U^+$$

satisfies (4.1). All the maps and identities below are identities on their test schemes, not just on geometric points.

First they give homomorphisms $f_+ : U^+ \to H$ and $f_- : U^- \to H$. Lemma 2.1 embeds the plane of any independent root pair into a base; the chamber transitivity proved in RG04 transports that base to the specified one. Thus the assumed normalizer conjugations transport (3.1) to every nonopposite root pair. Collinear positive roots are equal, by reducedness, and their factors combine by the additive root-group law. A pair with no positive combination that is a root commutes by the full weight-coordinate proof in RG04 Section 5.

Here is why these pair identities suffice. Give a positive root the sum of its simple-root coefficients as height. A commutator term $i\alpha + j\beta$, $i, j > 0$, has height strictly larger than each input root. Collect the height-one factors first, then height two, and so on. Moving a factor past another introduces only factors of larger height; at each fixed height no new factor of that height is created, and only finitely many original or previously created factors remain. The maximum root height is finite, so this process terminates. Equal-root factors add their parameters. The ordered root-product isomorphism in RG04 Section 5 gives a unique final coordinate expression. The identical collection, with the identical parameter polynomials, holds for the images in H. It proves multiplicativity and independence of order for f_+. Apply the same argument to negative roots to prove it for f_-.

Define $f(vtu) = f_-(v) f_T(t) f_+(u)$ by the cell product isomorphism. Torus-equivariance and the just proved homomorphisms give

$$f(b_- g b_+) = f(b_-) f(g) f(b_+), \qquad b_- \in B^-,\ b_+ \in B^+,\ g \in \Omega. \tag{4.2}$$

The domain is stable under these two triangular multiplications: multiply the negative factors on the left and the positive factors on the right, moving torus factors by their characters.

For a simple α, choose the root orders with its negative factor last and its positive factor first. Write

$$g = v_0\, h\, u_0, \qquad h \in U_{-\alpha} T U_\alpha,$$

where v_0, u_0 use the other negative and positive roots. Reflection s_α preserves those other two root sets, by the simple-root coefficient calculation in RG04. Conjugation by n_α therefore keeps the exterior factors triangular. By (4.2), membership of the conjugate in the cell reduces exactly to its middle rank-one factor.

This middle domain is a scheme domain. For the subtorus T_α of the rank-one construction, the full torus action fixes exactly the $\pm\alpha$ root coordinates in Ω: every other root restricts nontrivially to T_α, and coefficient comparison with its universal character makes its coordinate zero. Thus $\Omega \cap C_K(T_\alpha) = U_{-\alpha} T U_\alpha$ on every test algebra. In its rank-one cover, the required multiplication is the explicit identity

$$x_\alpha(a)x_{-\alpha}(b) = x_{-\alpha}(b/q)\,\alpha^\vee(q)\,x_\alpha(a/q), \qquad q = 1 + ab \in R^\times.$$

Direct multiplication of the two-by-two matrices proves this identity; its upper left entry is q, so invertibility is exactly the Gauss-cell condition. Pull back the torus factor along the fppf torus cover of RG04 Theorem 2.1. RG03 Theorem 7.1 supplies the linked rank-one homomorphism and this exact unit criterion on the middle cell. The pulled-back identities descend by S04 Lemma A1.5. The assumed maps preserve the two root parameters, the coroot and torus characters; they consequently preserve this full Gauss identity. Together with (4.2) it gives

$$f(n_\alpha g n_\alpha^{-1}) = f_N(n_\alpha) f(g) f_N(n_\alpha)^{-1} \tag{4.3}$$

on the entire common cell domain.

For a word in simple representatives, impose that all its intermediate conjugates lie in Ω. This is an open containing the identity section. Induction proves (4.3) there. Its full common domain is also an open in the cell containing that section. Lemma 4.2 therefore extends the identity to that full domain, including every nilpotent base change. Torus conjugations already intertwine by their characters; multiplying with a torus element gives the corresponding identity for every normalizer section.

Choose a word for the longest Weyl element w_0, using the Coxeter theorem in RG04; letters may repeat. Its representative n_0 interchanges U^+ and U^-. For $u \in U^+$, $v \in U^-$ with $uv \in \Omega$, conjugation makes uv a negative-positive product. Its cell map is the product of the two factor maps. Apply (4.3) and cancel the two normalizer factors to obtain $f(uv) = f_+(u)f_-(v)$.

Finally, for $x = vtu$ and $x' = v't'u'$ in the cell,

$$xx' = vt(uv')t'u'.$$

If $xx' \in \Omega$, triangular stability and its inverse give $uv' \in \Omega$. Apply (4.2) and the preceding mixed-product identity. The resulting value is precisely $f(x)f(x')$. This proves (4.1) on its full open domain. Lemma 4.1 extends the cell map uniquely to $K \to H$.

5. The isomorphism theorem

Theorem 5.1. For pinned split reductive groups over a scheme S, isomorphisms preserving the pinnings correspond bijectively to isomorphisms of their based root data. The correspondence commutes with every base change. Over a disconnected base, the datum isomorphism is allowed to be locally constant.

Proof. A pinning-preserving group isomorphism identifies its split torus, root characters, coroots and simple frames, hence gives an isomorphism of based root data. We prove the converse and uniqueness over the entire base, including its nilpotents. Work locally where the given datum map is constant; the final uniqueness will glue these local constructions.

Match T with T' by the group-algebra character map, using AG-GS, *Diagonalizable groups*, Theorem 2.2. Match the pinned simple additive parameters. RG03 Theorem 7.1 gives their linked negative parameters and full rank-one homomorphisms. Thus the maps preserve n_α, $n_\alpha^2 = \alpha^\vee(-1)$, all torus conjugations, and the rank-one Gauss formulas. They also match the root lattices with their central directions; all torus elements, not only coroot signs, must be retained.

We first identify the normalizer presentation as an actual sheaf. Form the presheaf of groups generated by $T(R)$ and symbols n_α, for every test scheme R, with their torus conjugation rules, square rules and rank-two power rules (3.2). Sheafify for the fppf topology and call the result $\mathcal{N}$. The torus is normal in this presentation, because conjugation by each generator has the prescribed

torus automorphism. Killing it leaves exactly the Coxeter presentation in RG04 Theorem 3.4. Consequently the quotient sheaf is the constant Weyl sheaf $\underline{W}$.

There is a homomorphism $\mathcal{N} \to N_G(T)$, since all its relations were proved in Section 3. The copy of T has no collapse: its composite into $N_G(T)$ is the original closed torus inclusion. The actual quotient $N_G(T)/T$ is also $\underline{W}$. To verify this last identification over the base, choose one simple-reflection word for each w and its product n_w. RG04 Theorem 4.2 makes the quotient finite étale. The sections $n_w T$ give a map from $\coprod_w S$, which is an isomorphism on every geometric fibre by RG04 Theorem 4.1. On an affine base its algebra map is between finite projective modules and is an isomorphism on every residue-field fibre. Its cokernel vanishes by Nakayama; the surjection splits, and the same argument kills its finite projective kernel. Thus it is an isomorphism on the full base. The multiplication labels agree on fibres; their equality locus is open and closed, since the finite étale target has open-and-closed diagonal. It contains every point and hence is the entire source. This proves the group-sheaf identification with nilpotents included.

The map $\mathcal{N} \to N_G(T)$ is surjective as a sheaf. A normalizer section locally has label w, and multiplication by n_w^{-1} puts it in the kernel T; these expressions glue on the clopen label fibres. If a section of $\mathcal{N}$ maps to the identity, its image in $\underline{W}$ is trivial, so fppf locally it belongs to T. The unchanged injection of T then makes it trivial. Sheaf descent proves that the kernel is zero. Hence $\mathcal{N} \simeq N_G(T)$. The datum map identifies this presentation with the target presentation: it preserves all torus characters, coroot signs and the types in (3.2). We obtain an actual normalizer isomorphism f_N.

Every root can be transported from a simple root. For completeness, let $\gamma = \sum c_i \alpha_i > 0$. If it is not simple, the invariant form gives some $(\gamma, \alpha_i) > 0$, because $(\gamma, \gamma) = \sum c_i (\gamma, \alpha_i) > 0$. The reflection subtracts the positive integer $\langle \gamma, \alpha_i^\vee \rangle \alpha_i$. It keeps a positive coefficient at another simple root, so remains positive, by the common-sign root expansion in RG04. Its integer height strictly decreases. Iteration reaches a simple root. A negative root is first transported from its positive opposite by the corresponding reflection. This is a finite integer argument independent of the characteristic.

Choose such a normalizer word to transport each pinned simple parameter to a parameter at γ, and perform the same transport through f_N in the target. We check independence. Two choices reduce, after their Weyl words are compared, to a word carrying one simple root to another, with a remaining torus factor. Lemma 2.2 decomposes that word into standard rank-two factors, each carrying the current simple root to the next one; fixing factors are included. The complete conjugation table of Section 3 intertwines the parameter maps for each factor, with the indicated coefficients 1 or -1. Linked negative parameters obey the same square rule. The remaining torus acts by the same root character in the two groups, since f_T comes from the given datum map. Thus both transports give the same root-group map. This proves every transport independence and every normalizer-conjugation identity, on all test schemes.

Lemma 2.1 places any independent pair in a conjugate standard rank-two subsystem. The paired root maps just constructed therefore preserve all relations (3.1); the commuting and equal-root cases are the weight-coordinate and additive identities. The frame maps preserve the linked rank-one formulas as well. These are exactly the hypotheses of the full cell-multiplicativity construction of Section 4. Use any ordered root-product coordinates of RG04 Section 5 to define

$$f_\Omega(vtu) = f_-(v) f_T(t) f_+(u).$$

The finite height collection there proves that the two unipotent maps are homomorphisms; the rank-one calculation and Lemma 4.2 prove normalizer equivariance on the whole common domain. The longest-element calculation proves mixed multiplicativity, and Lemma 4.1 extends f_Ω uniquely to a group homomorphism $f : G \to G'$.

Apply this construction to the inverse datum map. Its cell map is the inverse of f_Ω, so the two extended composites restrict to the identity on their big cells. Uniqueness in Lemma 4.1 makes

each composite the identity on the whole group. Thus f is an isomorphism.

Any pinning-preserving isomorphism inducing the specified datum map has the same torus and simple parameters. The rank-one and normalizer formulas force its linked negative and transported root maps. Its cell map is therefore f_Ω; Lemma 4.1 makes it equal to f. This proves the required bijection.

Group-algebra maps, root parameters, finite collection identities, Gauss identities and scheme morphism descent all commute with base change. The uniqueness just proved consequently gives compatibility after every base change. On a disconnected or non-quasi-compact base the datum map may have different locally constant values, with no finite-image assumption. Apply the construction on affine neighbourhoods where all finitely many lattice-generator images are fixed. On overlaps the datum maps agree, so uniqueness identifies the group maps; S04 Lemma A1.5 glues them and their inverses. This proves the theorem over the original S. □

The pinning is essential for uniqueness. Conjugation by an element of T preserves the torus and positive system and scales E_α by $\alpha(t)$. It preserves every frame precisely when it belongs to the centre.

6. Constructing the simply connected group in characteristic zero

The preceding supporting lesson Lie proofs for the characteristic-zero group construction proves the rational Serre algebra and its integrable highest-weight modules. We use those complete proofs before constructing any algebraic group. The construction below includes its finite-presentation, group-law, geometric connectedness, Lie-algebra and root-datum arguments.

The finite modules and their matrix parameters

Let Φ be a reduced semisimple root system, $Q = \mathbf{Z}\Phi$, and P its weight lattice. Write $\alpha_i^\vee$ for the simple coroots and ω_i for their dual fundamental weights, so $P = \bigoplus_i \mathbf{Z}\omega_i$. S06 Theorem 6.2 supplies the rational split semisimple algebra

$$\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+, \qquad \mathfrak{h} = \bigoplus_i \mathbf{Q}h_i,$$

with exactly the prescribed one-dimensional root spaces and the simple rank-one brackets. It remains semisimple after every characteristic-zero field extension, as the whole proof there verifies.

For a dominant integral λ, S06 Section 7 constructs the cyclic module with relations

$$e_i v = 0, \qquad h_i v = \langle \lambda, \alpha_i^\vee \rangle v, \qquad f_i^{\langle \lambda, \alpha_i^\vee \rangle + 1} v = 0.$$

Its proof first uses the complete ordered-word theorem, S06 Theorem 1.A, to identify the induced module with $U(\mathfrak{n}^-)$. The additional singular-vector submodules have no top λ-component. Thus this quotient has a nonzero cyclic vector and a one-dimensional top *before* finite-dimensional complete reducibility is applied.

The same proof establishes local nilpotence and Weyl symmetry before finiteness: the finite-on-each-vector operators $\exp(e_i)\exp(-f_i)\exp(e_i)$ carry a weight to its simple reflection. Every weight μ consequently satisfies

$$\lambda - \mu \in Q^+, \qquad \mu - w_0\lambda \in Q^+.$$

Write $\lambda - w_0\lambda = \sum b_i\alpha_i$ and $\lambda - \mu = \sum d_i\alpha_i$. Since λ and its Weyl translates occur, the b_i are nonnegative integers; the two inequalities say $0 \le d_i \le b_i$. There are only finitely many such tuples.

For each fixed tuple, a negative-root PBW monomial has exponents m_β with $\sum_{\beta>0} m_\beta \operatorname{ht}(\beta) = \sum d_i$. All positive heights are at least one, so only finitely many monomials occur. Each weight space, and hence the whole quotient, is finite-dimensional.

Now S06 Theorem 4.2 gives complete reducibility. If a proper submodule had a nonzero top component, it would contain the cyclic generator and would be the whole module. It therefore has top component zero. An invariant complement contains the generator, since projection preserves weights, and cyclicity makes that complement the entire module. Thus the proper submodule is zero. Denote the resulting irreducible module by $L_{\mathbf{Q}}(\lambda)$. Tensoring its left-ideal presentation with any characteristic-zero field is exact; the top stays nonzero and one-dimensional, cyclicity stays true, and the same complement argument applies. These particular modules are absolutely irreducible.

Put $V_i = L_{\mathbf{Q}}(\omega_i)$ and $V = \bigoplus_i V_i$. They remain pairwise nonisomorphic after field extension: an isomorphism between top weights λ, μ would force both $\lambda - \mu$ and $\mu - \lambda$ to belong to Q^+, hence $\lambda = \mu$. The representation $\rho : \mathfrak{g} \to \operatorname{End}(V)$ is faithful by S06 Section 8: its kernel is a sum of simple diagram ideals, and an ideal containing index i acts nontrivially on V_i, since h_i acts by one on its top.

Choose a rational root vector e_α in each root line, with $e_{\alpha_i} = e_i$ and $e_{-\alpha_i} = f_i$ for the linked simple choices. It raises a weight by α. The finite weight set contains no arbitrarily long sequence $\mu + j\alpha$, so $E_\alpha = \rho(e_\alpha)$ is nilpotent. The finite matrix series

$$x_\alpha(u) = \exp(uE_\alpha) = \sum_{j \geq 0} \frac{u^j E_\alpha^j}{j!}$$

is a morphism over $\mathbf{Q}$ on every test algebra. Multiplying two such series and collecting the coefficient of E_α^j gives the binomial formula, hence $x_\alpha(u)x_\alpha(v) = x_\alpha(u+v)$ and inverse $x_\alpha(-u)$.

Lemma 6.1 (polynomial rank-one actions). Over any algebraically closed field k of characteristic zero, the irreducible finite-dimensional $\mathfrak{sl}_2$-modules are $\operatorname{Sym}^n(k^2)$, for $n \geq 0$, with weights $n, n-2, \dots, -n$; every finite-dimensional module is their direct sum. Every finite-dimensional rational representation of $\mathfrak{sl}_2$ integrates to a polynomial representation of $\mathrm{SL}_{2,\mathbf{Q}}$. On its summands of highest weight n, the upper and lower root groups act by $\exp(xe)$ and $\exp(xf)$, and the diagonal torus acts on weight m by z^m.

Proof. The trace-zero algebra with basis e, f, h and brackets $[h,e] = 2e$, $[h,f] = -2f$, $[e,f] = h$ is simple in characteristic zero. An ideal is stable under $\operatorname{ad} h$, whose three eigenvalues $2, -2, 0$ are distinct. Polynomial projections isolate any nonzero basis component, and bracketing that component with the other generators gives all three. Thus S06 Theorem 4.2 applies to its finite-dimensional modules.

Let $W_n = \operatorname{Sym}^n(\mathbf{Q}^2)$, with basis the degree-$n$ monomials in u, v. On every $\mathbf{Q}$-algebra, the matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ acts by

$$u \longmapsto au + cv, \qquad v \longmapsto bu + dv.$$

The resulting matrix entries are polynomials in a, b, c, d. Composing the substitutions gives matrix multiplication, the identity substitution is the identity, and the substitution for the inverse matrix is inverse. For determinant one, the inverse entries are again polynomials in these four entries. This defines a group-scheme representation. Its differential is

$$e = u\partial_v, \qquad f = v\partial_u, \qquad h = u\partial_u - v\partial_v.$$

Upper and lower unipotent substitutions give the finite Taylor series $\exp(xe)$, $\exp(xf)$, and the diagonal multiplies $u^{n-j}v^j$ by z^{n-2j}. All these are identities over every $\mathbf{Q}$-algebra.

In a nonzero irreducible module over k choose an h-eigenvector. Applying e repeatedly raises its eigenvalue by two; finitely many eigenvalues force a last nonzero vector w, with $ew = 0$, $hw = \lambda w$. Commutator induction gives

$$hf^j w = (\lambda - 2j) f^j w, \qquad ef^j w = j(\lambda - j + 1) f^{j-1} w.$$

For the second induction step, commuting past one further f adds $\lambda - 2j$ to the preceding coefficient and gives $(j+1)(\lambda - j)$. Lowering also ends, since nonzero $f^j w$ have distinct eigenvalues. If $f^n w \neq 0$ and $f^{n+1} w = 0$, the second equation at $n+1$ gives $(n+1)(\lambda - n) f^n w = 0$, hence $\lambda = n$. The string spans a nonzero invariant submodule and is therefore the whole irreducible module. The map $f^j u^n \mapsto f^j w$, $0 \leq j \leq n$, identifies its operators with those of W_n. These vectors are a basis: the coefficients of $f^j u^n$ are $n!/(n-j)! \neq 0$.

Conversely W_n is irreducible. Polynomial projections in its distinct h-eigenvalues split any invariant subspace into weight components. Repeated e takes any nonzero such component to u^n, with nonzero coefficients in characteristic zero, and repeated f generates all basis vectors. Complete reducibility now proves the decomposition of every finite-dimensional module over k.

For a rational module M, let Ω be an algebraic closure of $\mathbf{Q}$, whose existence is proved in S06 Section 1, and put $K_n = \ker e \cap \ker(h - n)$. Kernels of these rational linear maps commute with field extension. Over Ω, only finitely many K_n, with $n \geq 0$, occur. The string equations give an equivariant map

$$\bigoplus_{n \geq 0} W_n \otimes_{\mathbf{Q}} K_n \longrightarrow M, \qquad f^j u^n \otimes w \longmapsto f^j w.$$

The relation $f^{n+1} w = 0$ follows over Ω from its decomposition, and hence over $\mathbf{Q}$. That decomposition makes the displayed map an isomorphism after extension. A rational kernel or cokernel with zero faithful field extension is zero, so it is an isomorphism over $\mathbf{Q}$. Transport the polynomial actions on its finitely many summands. Their differentials, diagonal weights and unipotent exponentials are the prescribed ones. This proves every assertion of the lemma. □

The closed group generated by the matrices

Let $T = D_{\mathbf{Q}}(P)$ act on V_μ by μ. This is a closed torus embedding. Indeed choose a weight basis containing the fundamental highest vectors. Their diagonal matrix entries restrict to the basis characters ω_i, and the corresponding inverse-matrix entries restrict to their inverses. Inverse entries are regular on $\mathrm{GL}(V)$, by the adjugate formula. These restricted functions generate $\mathbf{Q}[P]$, so the coordinate map onto the torus algebra is surjective, as also proved in S06 Section 8.

If Φ is empty, then $P = \mathfrak{g} = V = 0$; take $G = 1$, with its empty pinning and datum. Henceforth suppose $\Phi \neq \varnothing$.

Let $\Gamma \subset \mathrm{GL}(V)(\mathbf{Q})$ be the subgroup generated by $T(\mathbf{Q})$ and all $x_\alpha(\mathbf{Q})$. In $B = \mathbf{Q}[\mathrm{GL}(V)]$, let I be the ideal of functions vanishing at every element of Γ, and put $C = B/I$, $G = \operatorname{Spec} C$. Explicitly $B = \mathbf{Q}[m_{ab}, z]/(z \det(m_{ab}) - 1)$, so its presentation is finite. Evaluation embeds C into the $\mathbf{Q}$-valued functions on Γ, so I is radical.

This closed subscheme is of finite presentation. Here is the algebra input explicitly. A field is Noetherian. If R is Noetherian and $J \subset R[x]$, let L_n be the ideal of coefficients of x^n in elements of J of degree at most n. Multiplication by x gives $L_n \subset L_{n+1}$. This chain stabilizes: its union is an ideal with finitely many generators, all lying at a single finite stage. Choose polynomials furnishing generators of each L_n up to the stabilizing stage N. For a polynomial of degree $m \geq N$, subtract multiples of x^{m-N} times the chosen degree-N polynomials to remove its leading coefficient. For $m < N$, use the chosen degree-m polynomials. Induction on degree shows that this finite collection generates J. Thus $R[x]$ is Noetherian. Induct on the variables. Quotients preserve this property by

taking ideal inverse images; localization preserves it because any localized ideal is generated by the extension of its contraction. Hence B, a localization of a finite polynomial algebra over $\mathbf{Q}$, is Noetherian. The ideal I is finitely generated, and its quotient is finitely presented.

We prove the group law on this scheme rather than assume it from a point closure. The set $\Gamma \times \Gamma$ is schematically dense in $G \times G$. To see this, write an element of $C \otimes C$ as $\sum a_i \otimes b_i$, with the b_i linearly independent. The vectors $(b_i(\gamma))_i$, $\gamma \in \Gamma$, span the full finite-dimensional coordinate space. Otherwise a nonzero linear functional would make a nontrivial linear combination of the b_i vanish on all of Γ, contradicting the evaluation injection. Choose finitely many such vectors as a basis. If the tensor vanishes at every pair, evaluation at this chosen second-variable basis makes each a_i vanish at every γ, so $a_i = 0$. This proves the injection into functions on $\Gamma \times \Gamma$.

Multiplication in $\mathrm{GL}(V)$ pulls every element of I to a function zero on $\Gamma \times \Gamma$, so it factors through $G \times G \to G$. Inversion pulls I into I, because Γ is a subgroup. The identity belongs to G. All group axioms are the restrictions of the matrix identities. Thus G is an affine finite-presentation group scheme over $\mathbf{Q}$.

The given torus and additive matrix morphisms factor through G. A Laurent polynomial vanishing on $(\mathbf{Q}^\times)^r$ is zero: multiply by a monomial, then induct on the variables using the bound of the degree on the number of roots of a nonzero one-variable polynomial. The same argument applies to the ordinary polynomial parameter of x_α. Pullbacks of I along these morphisms are consequently zero.

S08 Theorem G.3.4, with its complete Cartier proof, makes G smooth. It is geometrically connected. For any field extension $k/\mathbf{Q}$, the same rational points Γ are schematically dense in G_k: write a function as a finite sum $\sum c_i a_i$ with the $c_i \in k$ linearly independent over $\mathbf{Q}$. Its values at Γ have $a_i(\gamma) \in \mathbf{Q}$; vanishing forces each $a_i(\gamma) = 0$, hence each $a_i = 0$. Over an algebraic closure, the complete component argument of RG03 Lemma 1.C makes the identity component an open-and-closed subgroup. The connected torus and additive images lie in it. Their rational products contain Γ, so schematic density makes that component all of G_k.

Each root parameter is a closed immersion into G. For a nonzero nilpotent $E = E_\alpha$ of index d, the matrices $1, E, \dots, E^{d-1}$ are linearly independent. In a proposed relation take the smallest nonzero power coefficient; its remaining polynomial factor has nonzero constant term and is invertible at E, forcing that power of E to be zero, contrary to its index. A rational linear functional on matrices can therefore take value one on E and zero on the other listed powers. Its value on $\exp(uE)$ is exactly u. Matrix-entry functions consequently surject onto $\mathbf{Q}[u]$, proving the closed immersion, including on nonreduced test algebras.

Its Lie algebra, centre and root datum

Put $\mathfrak{l} = \rho(\mathfrak{g}) \subset \mathrm{End}(V)$. Its matrix normalizer is closed: on a basis of $\mathfrak{l}$, require the images under conjugation by g and by g^{-1} to have zero components in the quotient space $\mathrm{End}(V)/\mathfrak{l}$. These are finitely many regular equations on $\mathrm{GL}(V)$, and both conditions give equality of the submodules after every test-algebra extension. The torus preserves $\mathfrak{l}$ by its root weights. For a root exponential, binomial expansion gives

$$\exp(uE_\alpha)\rho(x)\exp(-uE_\alpha) = \rho\left(\sum_{j\ge 0} \frac{u^j (\operatorname{ad} e_\alpha)^j x}{j!}\right).$$

Both sides are finite sums: nilpotence on the finite root-weight decomposition bounds the adjoint sum. Thus these generators normalize $\mathfrak{l}$ as schemes. Their rational subgroup Γ does too, and its defining density puts G in this closed normalizer.

At the identity, conjugation by $1 + \varepsilon A$, $\varepsilon^2 = 0$, changes $\rho(x)$ by $\varepsilon[A, \rho(x)]$. Thus a tangent vector $A \in \mathrm{Lie}(G)$ satisfies $[A, \mathfrak{l}] \subset \mathfrak{l}$. Faithfulness defines a derivation D_A of $\mathfrak{g}$; the derivation equation

follows directly by expanding the matrix Jacobi identity.

For clarity, the entire inner-derivation calculation of S06 Proposition 3.3 is as follows. Its Proposition 3.1 makes the Killing form κ nondegenerate. Choose z with $\kappa(z,x) = \operatorname{tr}(D_A \operatorname{ad} x)$, and set $E = D_A - \operatorname{ad} z$. Then $\operatorname{tr}(E \operatorname{ad} x) = 0$. The derivation identity gives $[E, \operatorname{ad} x] = \operatorname{ad}(Ex)$, and cyclic trace gives

$$0 = \operatorname{tr}(E \operatorname{ad}[x,y]) = \operatorname{tr}([E, \operatorname{ad} x] \operatorname{ad} y) = \kappa(Ex, y).$$

Hence $E = 0$, so

$$A \in \rho(\mathfrak{g}) + \operatorname{End}_{\mathfrak{g}}(V). \tag{6.1}$$

Over an algebraic closure k, the commutant is one scalar on each V_i. A nonzero map between two irreducible summands has zero kernel and full image and would be an isomorphism; their distinct top weights rule this out. On one irreducible summand, subtract an eigenvalue of a commuting endomorphism. Its nonzero kernel is invariant and hence the whole summand, so the endomorphism is scalar. This proves the full commutant assertion.

Every generator has determinant one on each V_i. For a root exponential, a basis adapted to kernels of powers of its nilpotent generator makes the matrix upper triangular with all diagonal entries one. For T, let $\sigma_i \in P$ be the sum of its weights with multiplicities. Its pairing with every simple coroot is the trace of h_j on V_i. This trace is zero: S06 Proposition 3.2 gives $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$, and a matrix commutator has trace zero. The simple coroots are the dual basis of P, so $\sigma_i = 0$ in the lattice. Thus $G \subset \prod_i \operatorname{SL}(V_i)$, by the same defining density.

For a tangent matrix, $\det(1 + \varepsilon A_i) = 1 + \varepsilon \operatorname{tr}(A_i)$. Each block trace is therefore zero. In (6.1) the block trace of $\rho(z)$ is zero by perfectness, and a scalar c_i has trace $(\dim V_i)c_i$. Characteristic zero makes every $c_i = 0$. This proves $\operatorname{Lie}(G_k) \subset \rho(\mathfrak{g}_k)$. Conversely the torus tangent vectors give $\rho(\mathfrak{h}_k)$, and each additive parameter has tangent E_α. They span the whole algebra. Consequently

$$\operatorname{Lie}(G_k) = \rho(\mathfrak{g}_k).$$

The rational equality follows as well: tangent equations are rational linear equations, and a kernel or quotient with zero faithful field extension is zero.

The group is reductive. Let $N \subset G_k$ be a smooth connected normal unipotent subgroup. Use a faithful closed unitriangular realization of N, as in the unipotent convention of RG01 Lemma 2.A. Its Lie algebra is nilpotent: commutators of strictly upper triangular matrices lie in successive powers of that associative algebra, which eventually vanish. Normality makes $\operatorname{Lie} N$ an ideal of $\operatorname{Lie} G_k$. Explicitly, for $X \in \operatorname{Lie} G_k$ and $Y \in \operatorname{Lie} N$, normality and the subgroup law put the commutator of $1+\varepsilon X$ and $1+\eta Y$ in $N(k[\varepsilon,\eta]/(\varepsilon^2, \eta^2))$. Matrix multiplication makes it $1+\varepsilon\eta[X,Y]$. Every defining equation of N has zero constant term at the identity; its value on this matrix is its linear differential on $[X,Y]$ times the nonzero basis coefficient $\varepsilon\eta$. Thus all those differentials vanish on $[X,Y]$, which says $[X,Y] \in \operatorname{Lie} N$. A nilpotent Lie algebra is solvable, since its successive derived brackets have lengths at least 2^j. Semisimplicity of $\mathfrak{g}_k$ therefore forces $\operatorname{Lie} N = 0$.

Smoothness makes every tangent rank of N zero, by translation, so its locally free cotangent sheaf is zero. This closed subgroup of the finite-type affine G_k is affine and of finite type: the ambient coordinate ring is Noetherian by the polynomial-ideal proof above, so its defining ideal is finite. It is flat over the field. The complete arbitrary-base criterion in S07 Lemma 5.3A therefore makes N étale. The field calculation in the same lemma, including its nilpotent local-ring argument and finite Artinian decomposition, expresses its coordinate algebra as a finite product of finite separable field extensions of k. Since k is algebraically closed, all those fields are k. Connectedness leaves one factor, and its identity point identifies that factor with the trivial group. Thus no nontrivial

smooth connected normal unipotent subgroup exists, which is precisely geometric reductivity in RG01 Section 1.

The closed torus T is geometrically maximal. The exhaustive Lie grading just obtained has zero-weight part $\rho(\mathfrak{h}_k) = \operatorname{Lie} T_k$ and nonzero characters exactly Φ. If a torus T' contains T_k, its tangent vectors centralize T_k, so they lie in this zero-weight part. The reverse tangent inclusion comes from $T_k \subset T'$; their ranks agree. Over k both tori are split. Their closed inclusion induces a surjection of free character lattices, by exponent-basis comparison in their group algebras, as in AG-GS Theorem 2.2. Equal ranks make every element of the kernel zero after tensoring with $\mathbf{Q}$, hence torsion. The domain character lattice is torsion-free, so that kernel is zero. The inclusion is an isomorphism. This proves maximality on every geometric fibre.

Now apply the complete centre calculation in RG03 Section 1 to this reductive group and maximal torus. It gives

$$Z(G) = D_{\mathbf{Q}}(P/Q).$$

This is finite. In the fundamental-weight basis the simple roots form an integral square matrix with nonzero determinant d, since they span the rational root space. The adjugate identity gives $dP \subset Q$; therefore P/Q is a quotient of the finite group P/dP. Its group algebra has the finite exponent basis, so the diagonalizable centre is finite over $\mathbf{Q}$. The radical formula in RG03 Section 1 then gives trivial radical and proves that G is semisimple. This also identifies its entire centre as a scheme.

The closed parameters x_α have the required characters and identity differential on their root lines. RG03 Theorem 4.1 identifies them with the canonical root groups of G. For a simple root, Lemma 6.1 integrates the (e_i, f_i, h_i)-action on V to SL_2. Its image lies in G on all test schemes: on the open where the upper-left matrix entry a is invertible, the Gauss decomposition is a product of its two root parameters and its diagonal. The diagonal acts on weight μ by $z^{\langle \mu, \alpha_i^\vee \rangle}$, hence is the stated cocharacter of T. On the other open $D(c)$, left multiplication by $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, itself a product of the two root matrices, reduces to $D(a)$. These opens cover because $ad - bc = 1$ makes (a, c) the unit ideal. Thus the whole polynomial action factors through G.

The linked Gauss formula in RG03 Theorem 7.1 identifies that diagonal cocharacter with the canonical group coroot. For every root, transport a simple root by a Weyl word, using the finite height descent in Section 5. RG04 Theorem 2.1 gives the action of each integrated simple representative on P by the prescribed simple reflection. Conjugating the linked Gauss identity and applying its uniqueness in RG03 Theorem 7.1 transports each coroot with its root. Successive conjugation therefore gives precisely the prescribed $\Phi^\vee$. Consequently

$$\mathcal{R}(G, T) = (P, \Phi, P^*, \Phi^\vee). \tag{6.2}$$

Its simple root frames are E_{α_i}, so G is pinned split. Since the ω_i are the dual basis of the simple coroots, P^* is exactly the coroot lattice. This is the simply connected semisimple datum. The empty-root case specified above has the same conclusion. Everything was constructed over $\mathbf{Q}$, before any integral model, and all polynomial identities hold on arbitrary commutative $\mathbf{Q}$-algebras.

7. Integral coordinates and a partial group law

Continue with the simply connected group of Section 6. Choose its simple-root pinning. Choose a frame in every other root line by transporting a simple frame with a word in the n_α, and use the dual frame in the negative line. Such frames form a **Chevalley system**: a simple representative carries each chosen root frame to the chosen frame of the reflected root, up to a sign.

Here is why the signs really are integral and independent of the field. Compare two transport words. Lemma 2.2 reduces their comparison to the conjugation tables in Section 3 and torus factors arising from (1.2) and (3.2). Those tables have coefficients 1 or -1. The torus factors lie in $T[2]$; the representatives normalize $T[2]$, and every root has value 1 or -1 on it. Moving such factors through a word changes only these signs. There is no assertion that $T[2]$ centralizes the representatives.

It follows that all root reordering polynomials have integer coefficients. Indeed, Lemma 2.1 moves a pair into a standard rank-two system; the chosen frames there differ from those of Section 3 only by signs. Its commutator formulas are (3.1). Repeatedly apply these formulas in an order refining root height. A new factor has greater height than the factors it corrects, so reordering terminates. For the negative roots use negative height. Thus the positive and negative unipotent groups extend to groups $U^+_{\mathbf{Z}}, U^-_{\mathbf{Z}}$, each an affine space in its ordered root coordinates. Their multiplication and inverse are triangular integral polynomials. Associativity holds because it is a polynomial identity true over $\mathbf{Q}$.

Changing the root order still gives a polynomial isomorphism: reorder using the same relations, and reorder back. Both compositions are the identity over $\mathbf{Q}$, hence over $\mathbf{Z}$. Let

$$T_{\mathbf{Z}} = D_{\mathbf{Z}}(P), \qquad X = U^-_{\mathbf{Z}} \times T_{\mathbf{Z}} \times U^+_{\mathbf{Z}}. \tag{7.1}$$

This smooth affine scheme has geometrically irreducible fibres. The torus acts on each root coordinate through the prescribed character. We write $e = (1, 1, 1)$.

We next give the partial conjugations on X. For a simple root α, reorder the coordinates as

$$\left(\prod_{\gamma \in \Phi^+ \setminus \{\alpha\}} x_{-\gamma}(v_\gamma) \right) x_{-\alpha}(v)\, t\, x_\alpha(u) \left(\prod_{\gamma \in \Phi^+ \setminus \{\alpha\}} x_\gamma(u_\gamma) \right).$$

Put $a = \alpha(t)$ and $D = 1 + auv$. On the principal open $D \neq 0$, conjugation by n_α has the integral cell formula obtained by reflecting and changing signs in the outer root factors, and replacing the middle three factors by

$$x_{-\alpha}\left(\frac{-au}{D}\right) \alpha^\vee(D) s_\alpha(t) x_\alpha\left(\frac{-av}{D}\right). \tag{7.2}$$

This is exactly the rank-one Gauss identity applied after conjugation. It fixes the identity, acts by s_α on the torus, and is defined on each individual root group.

One can check the torus and denominator directly for SL_2. Conjugating $y(v)h(z)x(u)$ by $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ gives an upper-left entry $z^{-1}(1 + z^2uv)$. Thus the torus factor is $h(z^{-1}(1 + z^2uv))$, as in (7.2).

Choose a reduced word for the longest Weyl element w_0, and compose these partial conjugations along that word. Call the resulting map $h : V \to X$. Reverse the word and use inverse representatives to obtain $h' : V' \to X$. These are inverse rational maps, and

$$U^-, T, U^+ \subset V \cap V', \qquad h(U^\pm) = U^\mp, \quad h(T) = T. \tag{7.3}$$

To justify that the whole unipotent subgroups, and not just the separate root axes, are in the domains, consider an intermediate conjugate of U^+. Its roots form a positive system. Reordering its root factors into negative and positive order never uses opposite roots: every root created by a commutator is still positive in that system. It therefore needs only the integral reordering polynomials, with torus component one. In (7.2) at most one of the two opposite-root coordinates is nonzero, so the denominator is one. The same argument applies to U^- and the inverse word. This proves (7.3). Equality $h'h = 1$ on its common domain follows over $\mathbf{Q}$ and then over $\mathbf{Z}$, because that domain is flat.

For $x = (v, t, u)$ and $x' = (v', t', u')$, impose the open condition

$$(h(u), 1, h(v')) \in V'.$$

Write $h'(h(u), 1, h(v')) = (v'', t'', u'')$. Define

$$m(x, x') = (v(tv''t^{-1}),\ tt''t',\ (t'^{-1}u''t')u'). \tag{7.4}$$

The condition and formula are integral; over $\mathbf{Q}$ they say precisely that $ux'_- = uv'$ has been rewritten in the cell before multiplying the outer factors. The domain contains $e \times X$, $X \times e$, $B^- \times X$, $X \times B^+$, and the pairs within $U^{\pm}$ or T. Here $B^- = U^-T$ and $B^+ = TU^+$. Partial inversion is obtained on the open where the following expression exists:

$$\iota(v, t, u) = h'\big(h(u^{-1}), h(t^{-1}), h(v^{-1})\big). \tag{7.5}$$

This domain contains T, U^-, U^+. Associativity on common domains, the identity rule, and the inverse rule hold as integral identities: they hold in the characteristic-zero group, and the domains in question are flat open subschemes of products of X.

For the next construction we need open partial translations with dense domains and images in every fibre. Write $D \subset X \times_{\mathbf{Z}} X$ for the multiplication domain just constructed. Its two universal translation maps are

$$(x, y) \longmapsto (x, m(x, y)), \qquad (x, y) \longmapsto (m(x, y), y). \tag{7.6}$$

Both maps are local isomorphisms along both identity axes. At (e, y), the first map has identity differential in the y-direction; at (x, e), the second has identity differential in the x-direction. We give the other two assertions explicitly. Factor the universal x into its ordered negative root factors, torus factor and ordered positive root factors. For left translation, apply these factors from right to left to successive suffixes, starting at e. A positive factor and its suffix are in $U^+ \times U^+$; a torus or negative factor uses $B^- \times X$. Translation by the inverse factor is defined at the resulting suffix: the positive case stays in U^+, and the other cases again use $B^- \times X$. On the common open neighbourhoods the two maps are inverse. Indeed their compositions are the identity over $\mathbf{Q}$, and the coordinate rings of their open domains are $\mathbf{Z}$-flat, so equality holds over $\mathbf{Z}$. Composing gives left translation by x as a local isomorphism at e. Associativity identifies this composition with the existing $m(x, -)$ on their common neighbourhood of e. For right translation, start at e, apply factors from left to right to successive prefixes, and use $U^- \times U^-$ or $X \times B^+$ and the inverse factors. This gives right translation by x as a local isomorphism at e. All the points, factors and neighbourhoods here use universal coordinates; thus the assertions hold after every base change, rather than only at rational points.

The same inverse-factor argument works at every pair in $U^- \times U^-$, $U^+ \times U^+$ and $T \times T$: the intermediate products and inverse products stay in that subgroup. The two maps in (7.6) are therefore étale at all those pairs as well. Let $D^\circ \subset D$ be the intersection of their étale loci. It contains both identity axes and those three whole pair subschemes. Retain m on D°.

The two restricted maps in (7.6) are open immersions. Here their sources and targets are integral normal Noetherian schemes. Their coordinate algebras are localizations of finite polynomial algebras over $\mathbf{Z}$. The complete integer-factorization proof in S04 Lemma B1.1 and the reduced-fraction proof in Lemma P0.5 make $\mathbf{Z}$ normal. S04 Corollary B1.2 gives polynomial normality by induction on the variables, and Lemma P0.3 gives normality after localization. Every integer ideal is principal by the least-positive-generator argument in S04 Lemma P0.10; thus $\mathbf{Z}$ is Noetherian, and the complete leading-coefficient proof in Section 6 makes these polynomial algebras and their localizations Noetherian. These properties pass to their open subschemes. Each restricted map is

separated, since it is a morphism between separated $\mathbf{Z}$-schemes; it is quasi-finite, since it is étale and its Noetherian open domain is quasi-compact. It is birational: over $\mathbf{Q}$ it is the restriction of multiplication in the group constructed in Section 6, whose big cell is $X_{\mathbf{Q}}$, so the generic translation has the inverse supplied by the group law. The exact normal birational consequence, S04 Corollary D7.4, consequently applies to each map over $\mathbf{Z}$. Denote their open images by D_{13}, D_{23}.

Every slice of each of D°, D_{13}, D_{23} over either coordinate projection is a nonempty open in the corresponding geometrically irreducible fibre of X, and hence is dense. For D° this follows from its two identity axes. The first translation sends (x, e) to (x, x) and (e, y) to (e, y), so both types of slices of D_{13} have a point. The second sends these axes to (x, e) and (y, y), giving the same conclusion for D_{23}. These are statements on geometric fibres and remain true after arbitrary base change. The graph

$$W = \{(x, y, m(x, y)) : (x, y) \in D^\circ\} \subset X^3$$

therefore projects isomorphically onto each of the three open subschemes $D^\circ, D_{13}, D_{23} \subset X^2$, with the required dense slices. Associativity, identity and partial inversion remain the already verified integral identities on common domains. This is the strict partial group law used below. Its restriction to each torus or root subgroup still contains the entire multiplication law.

8. Completing a strict partial group law

We give the construction rather than assuming a group already exists. This is the translation method of Weil and Artin; the algebraic-space quotient in the final step is constructed in the earlier supporting S07 Theorem 8.2.

Lemma 8.1. Suppose X/S is smooth, separated and of finite presentation, with geometrically irreducible nonempty fibres. Suppose X has a strict associative partial group law, an identity section, and partial inversion, with the translation maps defined and invertible locally along both identity axes. Then there is a unique smooth separated group algebraic space K/S of finite presentation containing X as a fibrewise dense open and extending its partial group law. Formation of K commutes with base change.

Proof. Write $D_{12}, D_{13}, D_{23} \subset X \times_S X$ for the three open images of the strict multiplication graph, ordering the pair for D_{23} as output followed by second input. Projection to any pair is an isomorphism from that graph. Fixing either entry in any pair leaves a fibrewise dense open in X. In particular the left and right partial translations are isomorphisms between fibrewise dense opens. Denote the inverse of the right-translation pair map by partial right division:

$$q : D_{23} \longrightarrow X, \qquad q(z, b) = a \iff (a, b, z) \in W.$$

The pair order here is (z, b); it is obtained from the usual (b, z) projection by exchanging its two entries.

Germs and their descent. On an S-scheme R, take isomorphisms between fibrewise dense opens of X_R, identifying two when they agree on a fibrewise dense common open. Intersections of finitely many such opens are dense in every geometric fibre. Composition is defined on the inverse image of the intersection of the intermediate image and domain; inversion reverses a representative. These constructions give a group of germs and commute with pullback.

Whenever two representatives of the same germ have a common domain, they agree on that entire domain. Apply S08 Lemma G.3.5 to the common open: it is smooth, and each nonempty geometric fibre is an open of a geometrically irreducible fibre of X. Restrict the base to its open image when necessary. The separated target X_R and universal schematic density give equality, including on nonreduced tests. This also proves that the maps and their inverse maps glue across compatible representative domains.

The germs are an fppf sheaf. To prove effectivity, represent a compatible germ over a covering $R' \to R$ by $h' : A' \xrightarrow{\sim} B'$, with both opens fibrewise dense. A covering family is treated by the disjoint union of its members. Let $A, B \subset X_R$ be their images. They are open because flat locally finitely presented maps are universally open, with the complete proof in S04, the flat finite-presentation openness proof in Section E5. The maps $A' \to A, B' \to B$ are surjective flat and locally finitely presented. The two representatives agree on $A' \times_{X_R} A'$: compatibility gives equality as germs on the base overlap, and the preceding density argument gives equality on the entire intersection of domains. S04 Lemma A1.5 therefore descends h' to $A \to X_R$. Its image is in B, as can be checked after the covering. Descend its inverse to $B \to A$; both inverse equations hold after the covers and hence before them. The opens A, B are fibrewise dense, since their fibres contain the images of the dense covering fibres after a residue-field extension. Thus this is the descended germ. The same whole-common-domain argument and morphism descent prove uniqueness. Write $\mathrm{Bir}(X)$ for this sheaf. No assertion about base change of maximal representative domains has been used.

Translation germs and one-point faithfulness. For $a \in X(R)$, let L_a and R_a be its partial left and right translation germs. Associativity gives

$$L_a R_b = R_b L_a, \qquad L_a L_b = L_{m(a,b)}$$

when the displayed partial product exists. For these germ equalities the composed domains are fibrewise dense: each translation is an isomorphism between dense opens, and finite intersections stay dense. Associativity gives equality on their common triple-product domain, and the preceding schematic-density argument extends it over each whole common domain. Let K be the subgroup sheaf of $\mathrm{Bir}(X)$ generated by the L_a; concretely its sections are fppf locally finite words in these germs and their inverses. They commute with every R_b, after every test-scheme extension.

If a representative of $g \in K(R)$ is defined at a section x and satisfies $g(x) = x$, then $g = 1$. To see this on all tests, let b vary in X_R. Retain the open on which xb exists and belongs to the chosen domain of g. It is fibrewise dense: L_x is a dense-open isomorphism, and its image meets that domain densely. On this open the two compositions gR_b and $R_b g$ are both defined at x. Their equality on the whole common domain gives

$$g(xb) = g(x)b = xb.$$

The points xb range over a fibrewise dense open via L_x, so g is the identity germ. This argument uses the universal b, rather than a choice of rational points.

The map $j : X \to K$, $a \mapsto L_a$, is injective. For $L_a = L_b$, choose z in the intersection of their dense domains fppf locally; that intersection is smooth and surjective over R. Then $az = bz$, and the open-immersion right-translation pair map makes $a = b$. Morphism descent gives the original equality on R.

The open chart. This injection is representable by open immersions. Fix $g \in K(R)$ and one dense-open representative $g : A_g \to B_g$. Let

$$E_g = \{z \in A_g : (g(z), z) \in D_{23,R}\}, \qquad O_g = \mathrm{image}(E_g \to R).$$

Both are open; the latter follows from smooth openness. On E_g put $a = q(g(z), z)$. The germ $L_a^{-1} g$ is defined at z and fixes it, so one-point faithfulness gives $L_a = g$. Injectivity of j makes these a's agree on $E_g \times_{O_g} E_g$. The smooth surjection $E_g \to O_g$, together with S04 Lemma A1.5, descends them to a unique section $a \in X(O_g)$ with $j(a) = g$.

This open has the required property after every $R' \to R$. If $g_{R'} = L_a$, intersect the pullback of the fixed original domain A_g with the partial left-translation domain of a. This is fibrewise dense and smooth-surjective over R'. On it $g(z) = az$, by equality of representatives on their whole common domain; hence $(g(z), z) \in D_{23}$, and $a = q(g(z), z)$. The cover therefore factors through E_g, so $R' \to R$ factors through O_g. Conversely a map through O_g pulls back its unique section. Thus

$$R \times_K X = O_g$$

as functors on all schemes. In particular this proves the open-immersion claim without assuming that a maximal domain commutes with base change.

Translated charts and the quotient. A translate $j_g = (g\cdot)j_R : X_R \to K_R$ is again a representable open immersion of sheaves. Define the intersection chart

$$\Omega_g = X_R \cap g^{-1}X_R.$$

Precisely, it is the inverse image of the open subfunctor $j_R(X_R)$ under j_g. It is an open subscheme of X_R, and its formation commutes with every base change by this fibre-product definition. The corresponding unique section in X_R gives an isomorphism

$$\gamma_g : \Omega_g \xrightarrow{\sim} \Omega_{g^{-1}}, \qquad j(\gamma_g(b)) = g\, j(b).$$

Every dense-open representative $g : A_g \to B_g$ maps into this chart. Indeed on the scheme A_g, take its universal section b and put $a = g(b)$. The local identity-axis hypothesis makes L_b defined and invertible near e, with $L_b(e) = b$, and gives the same assertion for L_a. Thus $L_a^{-1}gL_b$ is defined at e and fixes it. One-point faithfulness gives $gL_b = L_a$. This is exactly $b \in \Omega_g$ and $\gamma_g(b) = g(b)$. Consequently Ω_g is fibrewise dense and smooth-surjective over R, including after arbitrary base change. We have not identified it with a maximal domain of an arbitrary birational map; its represented intersection is what proves the required base-change property.

It follows that every section of K is fppf locally

$$g = L_a L_b^{-1} \qquad (a, b \in X). \tag{8.1}$$

Choose b in Ω_g on its smooth covering and let $a = \gamma_g(b)$. More precisely the fibre over g of

$$\delta : X \times_S X \longrightarrow K, \qquad (a, b) \longmapsto L_a L_b^{-1} \tag{8.2}$$

is the graph of γ_g, with coordinates $(a, b) = (\gamma_g(b), b)$. This is an equality on every further test scheme, since its defining equality is exactly $j(a) = g\, j(b)$. Thus δ is representable by schemes, smooth and surjective.

Its equality relation is a scheme with an explicit chart. Over the parameter scheme X^2 with coordinates (c, d), take the universal $g = L_c L_d^{-1}$ and its represented open $\Omega_g \subset X^3$, whose third coordinate is b. The graph $a = \gamma_g(b)$ identifies this open with the relation between (a, b) and (c, d). Both relation projections are smooth, as base changes of δ; its endpoint map is a monomorphism on all tests. S07 Theorem 8.2 now applies to this flat locally finitely presented equivalence relation. Its fppf quotient is an algebraic space, and (8.1) and the equality relation identify that quotient with K.

The group operations are already operations of its sheaf, so Yoneda gives operations on this algebraic space satisfying the group axioms. Smoothness of K/S follows from the representable smooth covering (8.2) with smooth source. Each geometric fibre is irreducible, as the continuous surjective image of the irreducible $X_s \times X_s$. The open X_s is therefore dense in it.

Separation, finiteness and uniqueness. On any scheme test $R \to K \times_S K$, write its sections as u, v. The two translated opens

$$F = u^{-1}X_R \cap v^{-1}X_R \subset K_R$$

meet in every geometric fibre, by irreducibility. This open is smooth-surjective over R. A scheme atlas and its affine opens give an fppf scheme covering $R_i \to R$ with sections $t_i \in F(R_i)$. Both ut_i and vt_i lie in $X(R_i)$. Cancellation identifies the equalizer of u, v on every further test with the equalizer of ut_i, vt_i, which is closed because X/S is separated. S04 Corollary A3.2 descends this closed immersion. Since the original scheme test was arbitrary, the diagonal of K/S is a closed immersion and K/S is separated.

It is locally of finite presentation by smoothness. Over every affine open of S, $X \times_S X$ is quasi-compact because X/S is of finite presentation. Surjectivity of (8.2) makes K quasi-compact there. The closed diagonal is quasi-compact, being an affine closed immersion, so K/S is quasi-separated. These three properties give finite presentation. No quasi-compactness of the open relation in X^3 was assumed.

For any second extension K', each geometric fibre has the dense irreducible open X_s, and hence is irreducible. Every section g' is fppf locally ab^{-1}: the open $X_R \cap (g')^{-1}X_R$ is smooth-surjective, so choose b there and set $a = g'b$. Left translation restricts to a dense-open isomorphism of X_R, giving a homomorphism $K' \to \mathrm{Bir}(X)$. It is faithful: an identity germ fixes a section on a dense-open smooth cover, and cancellation in K' gives $g' = 1$. Its restriction to X is $a \mapsto L_a$, since the original partial product is extended. The local difference expression identifies its image with K. This gives the unique isomorphism fixing X; uniqueness also follows because those differences generate the whole sheaf. After every $S' \to S$, the constructed $K_{S'}$ is another such extension of $X_{S'}$, with the same translated-chart fibre products and multiplication. Uniqueness supplies the canonical base-change identification and its cocycle. This proves the entire lemma. □

Apply the lemma to (7.1) and its strict graph. We obtain a smooth separated finite-presentation group algebraic space $G_{\mathbf{Z}}$ containing X as an open. Its generic fibre is canonically the characteristic-zero group of Section 6: that group has this same dense big cell and partial law, so the uniqueness just proved identifies the two extensions. The torus and root axes are closed subschemes of X, hence are locally closed immersions into $G_{\mathbf{Z}}$. The whole pair laws retained in Section 7 make them subgroup morphisms with their prescribed multiplication and inverse. Their characters and parameter differentials are the specified ones on the identity cell.

The required conjugation and Gauss relations hold as morphisms over $\mathbf{Z}$ on their stated domains. Their parameter sources are flat over $\mathbf{Z}$; localization to $\mathbf{Q}$ is injective on their structure sheaves. The target is separated, so the equalizer ideal of the two morphisms is zero if it is zero after this localization. The characteristic-zero relations thus give the integral identities, including over nonreduced test rings. Affineness, closedness of the eventual canonical root groups and positive-characteristic reductivity are the separate arguments of Section 9.

9. Why the completed integral group is affine and reductive

There are two separate issues: an algebraic space must be shown to be an affine scheme, and its positive-characteristic fibres must be reductive.

Field-group inputs without an affineness assumption

Lemma 9.A (components and smooth reduction). A finite-type group scheme over a field is separated. Over an algebraically closed field, its irreducible components are disjoint open-and-closed translates of the component through the identity. Its reduction is a smooth group scheme; its reduced identity component is smooth, irreducible and normal in that reduction. In particular all components have the same dimension. The group scheme in this statement need not be affine or smooth initially.

Proof. The rational identity is a closed point. If it specialized to a point y, an affine neighbourhood of y would also contain the identity. Evaluation on that chart surjects onto the base field, so its maximal kernel makes the identity closed there, excluding any distinct specialization. The rational point is a closed immersion on these charts. The diagonal is the pullback of this closed identity immersion under $(g, h) \mapsto g^{-1}h$: on every test scheme the equation $g^{-1}h = 1$ is exactly $g = h$. Thus the group is separated.

Now let the field k be algebraically closed. The finite-type scheme is Noetherian and has finitely

many irreducible components. The algebra input is the complete polynomial-ideal proof in Section 6, valid over every field. Finiteness of components follows by induction on closed subsets: a reducible counterexample splits into two strictly smaller counterexamples. Give two components through the identity their reduced closed structures. Their product is irreducible by the full coefficient proof in S08 Lemma G.4.1. The closure of its multiplication image lies in one component and contains both original components, since either factor can be the identity. Maximality forces those two components to be equal. If two components met elsewhere, their nonempty closed intersection would have a k-point by the proved Nullstellensatz. Translation to the identity would contradict that uniqueness. Hence components are disjoint; being finite in number, each is open and closed. A k-point in each component translates the identity component onto it. The component through the identity is a subgroup because its product is irreducible and contains the identity, and inversion preserves it; maps factor through the corresponding open-and-closed subscheme.

S08 Lemma G.4.1 makes the product of the two reduced finite-type sources reduced. Consequently multiplication of the group reduction factors through the reduction: every nilpotent defining equation pulls back to zero. Inversion and the identity factor too, and the group identities restrict. Its components have the same underlying spaces as those already obtained. We prove smoothness of this reduced group on its affine charts, rather than assuming an affine group.

The complete translation calculation of AG-GS-03 Proposition 2.7 identifies the cotangent sheaf with $\mathcal{O} \otimes_k \omega$, where ω is its finite-dimensional identity cotangent space. Put $r = \dim_k \omega$. At a component's generic point the local ring is its function field K: on a chart, all other minimal primes disappear and reducedness kills the remaining nilradical. If that component has dimension d, the complete normalization and dimension proofs in AG-CA-09 Corollary 3.2 and Theorem 4.2 make K finite over a rational function field on d variables. S02 Lemma 3.F gives $\dim_K \Omega_{K/k} = d$, including characteristic p and without a separability assumption on the chosen finite extension. Localization of differentials therefore gives $r = d$. All components have that dimension by their translations. At the rational identity, AG-CA-09 Corollary 4.4 gives local dimension d, and AG-CA-18 Proposition 3.1 identifies embedding dimension with r. The local ring is thus regular. The rational-point Jacobian proof in that proposition makes the identity smooth. Translation makes every k-point smooth. The open-locus proof in AG-CA-18 Corollary 2.2 and the Nullstellensatz exclude a nonempty nonsmooth locus on any finite-type affine chart. Thus the reduction is smooth everywhere.

Its identity component is now smooth and irreducible. Conjugation by each k-point preserves it. On the product of the reduction with this component, reducedness follows again from S08 Lemma G.4.1. Each defining equation of the component consequently pulls back to zero: it vanishes at every k-point, and the full Nullstellensatz detects functions on a reduced finite-type scheme. This proves normality as a group-scheme morphism, including its infinitesimal tests. □

Lemma 9.B (the reduced-source field image). Let $f : E \to H$ be a homomorphism of finite-type group schemes over a field, with E reduced. Its schematic image is a closed reduced subgroup $I \subset H$. The factor $q : E \to I$ is faithfully flat and of finite presentation. Its schematic image commutes with every field extension. In particular a monomorphism with such a source is a closed immersion. No affineness or perfectness assumption is imposed on either group.

Proof. Lemma 9.A makes both groups separated. The map is quasi-compact and quasi-separated: inverse images of opens in its Noetherian source are quasi-compact, and separation gives its closed diagonal. We first give the full direct-image calculation needed for its equations. On an affine target open $\operatorname{Spec} R$, choose finitely many affine opens U_i covering the inverse image and finitely many affine opens V_{ijk} covering each intersection $U_i \cap U_j$. For any quasi-coherent $\mathcal{F}$ there, the sheaf condition expresses sections as the kernel of

$$\prod_i \Gamma(U_i, \mathcal{F}) \longrightarrow \prod_{i,j,k} \Gamma(V_{ijk}, \mathcal{F}),$$

where the arrow subtracts the two restrictions. Flat tensor preserves this kernel and the finite products. The full affine module–sheaf and pullback proof in S01 Proposition 2.2 identifies base-

changed sections on each affine with the corresponding tensor product. Thus sections commute with every flat R-algebra. Taking principal localizations proves quasi-coherence of the direct image; affine charts on a general flat target change prove flat base change for that sheaf. This proof imposes no finite-generation assumption on $\mathcal{F}$.

Put $\mathcal{J} = \ker(\mathcal{O}_H \to f_*\mathcal{O}_E)$ and $I = V(\mathcal{J})$. The preceding calculation makes this ideal quasi-coherent and its formation compatible with flat base change. Its chart ideal is radical, because every ring of sections of the reduced E is reduced. Hence I is reduced. The map $q : E \to I$ is schematically dominant by its definition. Such dominance is preserved by flat base change, by the just-proved section calculation and exactness of flat tensor. Factor $q \times q$ as the two base changes of q through $E \times I$. Both are flat base changes, since every scheme over the field is flat. The composite is therefore schematically dominant. The homomorphism identity makes every equation of I pull back to zero under multiplication after $q \times q$; schematic dominance makes it zero already on $I \times I$. Inversion is treated with q, and the identity is the image of the identity of E. Thus I is a closed subgroup, with the restricted group laws. Its construction and equations commute with every field extension by the same flat calculation.

We prove surjectivity and flatness of q. Schematic dominance first gives topological density: a nonempty open missing its image would have a nonzero structure sheaf mapping injectively to zero. It hits every generic point of I. Indeed take an affine neighbourhood $\operatorname{Spec} A$ of such a point $\mathfrak{p}$ and a finite affine cover with rings B_i of its inverse image. The reduced ring $A_{\mathfrak{p}}$ is a field. If its fibre were empty, every $B_i \otimes_A A_{\mathfrak{p}}$ would be zero. Some $s_i \notin \mathfrak{p}$ would then kill 1 in each B_i. The product of these s_i would make the full image in this chart lie in its vanishing set, contradicting density on the nonempty principal open at $\mathfrak{p}$.

Every point h of the group I is the multiplication image of a point of $I \times I$ with both projections generic. Extend to $K = \kappa(h)$ and its rational lift h_K. The fibre of multiplication there is isomorphic to I_K by $g \mapsto (h_K g^{-1}, g)$. Choose a generic point of a component of this fibre. Both projections are generic upstairs, since they are isomorphisms in this identification. The field-extension projection is flat, so the complete going-down proof in S04 Lemma A1.4 contracts their minimal primes to minimal primes downstairs. Its image therefore gives the claimed point of $I \times I$. The two fibres of q at those generic projections are nonempty. After extension to the residue field of that product point they stay nonempty, and their product is nonempty: residue fields at two points have nonzero tensor product over the field, and a prime quotient supplies a common lift. Multiplication of that lift in E maps to h. Hence q is surjective.

At any point above a generic point of reduced I, the target local ring is a field, so q is flat there. Choose one such point g. For an arbitrary $z \in E$, a prime quotient of the nonzero tensor product of $\kappa(g)$ and $\kappa(z)$ gives a common field extension L and rational lifts g_L, z_L. Flatness at g survives this base change. Translation by $a = z_L g_L^{-1}$ on E_L and by $q(a)$ on I_L intertwines q_L, and both translations are isomorphisms. Thus q_L is flat at z_L. The local map $\mathcal{O}_{E,z} \to \mathcal{O}_{E_L,z_L}$ is faithfully flat, and the latter local ring is flat over $\mathcal{O}_{I,q(z)}$, by that flatness and field base change. Faithful descent proves that $\mathcal{O}_{E,z}$ is flat over this target ring: tensor any injection of target modules with it, then with the faithfully flat local extension; the final kernel is zero, hence so is the preceding kernel. This is the exact faithful kernel proof in S04 Lemmas A1.1–A1.3. Since z was arbitrary, q is faithfully flat.

The closed I is of finite type over the field. On compatible affine charts, q is a finite-type algebra map, and its target ring is Noetherian, so its relation ideal is finite by the polynomial-ideal proof in Section 6. Thus it is locally finitely presented. Its quasi-compactness and separation make it of finite presentation, so it is an fppf cover. If f is a monomorphism, then q is one too. The identity on this cover has identical pullbacks to $E \times_I E$, because the diagonal is an isomorphism for a monomorphism. The complete morphism descent in S04 Lemma A1.5 descends it to an inverse $I \to E$. Both inverse identities hold after this faithfully flat cover and hence before it. Thus q is an isomorphism and f is the closed inclusion of I. □

Geometric fibres are schemes

Fix an algebraically closed field k, and write $G = G_{\mathbf{Z},k}$. The whole completion proof gives a smooth separated finite-presentation group algebraic space, its open cell X_k, and the smooth surjection $\delta : X_k \times_k X_k \to G$. For a section $g \in G(k)$, its fibre under δ is the nonempty open intersection chart proved in Lemma 8.1. A nonempty finite-type k-scheme has a k-point, by the earlier Zariski lemma and the algebraically closed hypothesis. Thus $g = ab^{-1}$ for $a, b \in X_k(k)$, so g belongs to the open scheme $X_k b^{-1}$.

These translates cover the entire algebraic space. Let O be their union. If its closed complement were nonempty, its inverse image under δ would be a nonempty closed subset of the finite-type scheme X_k^2. Give it its reduced closed structure and choose a closed k-point. Its image is then a k-point of G outside O, contradicting the preceding paragraph. The open schemes therefore cover G; their overlaps are open subschemes and their identifications glue. Hence G is a smooth separated finite-type group scheme. Its geometric connectedness and irreducibility are already consequences of Lemma 8.1.

The torus and every root parameter are closed immersions into this group scheme. They were locally closed immersions and subgroup morphisms in Section 8. Their sources T_k and $\mathbf{G}_{a,k}$ are reduced Noetherian schemes; their maps are quasi-compact, since inverse images of opens in these Noetherian sources are quasi-compact. The complete reduced-source monomorphism conclusion of Lemma 9.B makes them closed. This uses no reductivity or prior affineness of G. Write $U_\alpha = x_\alpha(\mathbf{G}_a)$ for these actual closed subgroup schemes.

The adjoint kernel and the integral scheme

The affine identity cell is stable under conjugation by T_k. Its exact characters and identity coordinates give, as a full torus grading,

$$\operatorname{Lie}(G) = \operatorname{Lie}(T_k) \oplus \bigoplus_{\alpha \in \Phi} kE_\alpha. \tag{9.1}$$

These characters are the nonzero lattice elements α, even if their differentials vanish in the characteristic of k. The integral Gauss identity gives

$$[E_\alpha, E_{-\alpha}] = d\alpha^\vee(1). \tag{9.2}$$

One can see the bracket directly over $k[\varepsilon, \eta]/(\varepsilon^2, \eta^2)$: in the linked Gauss identity put $u = \varepsilon, v = \eta$. The unipotent quotients $u/(1+uv)$ and $v/(1+uv)$ are u, v. After cancellation, the commutator is $\alpha^\vee(1+\varepsilon\eta)$; conjugating this torus factor by the remaining root term changes only terms with η^2 or ε^2, hence changes nothing. The coefficient of $\varepsilon\eta$ is exactly (9.2).

Each coroot is primitive in P^*. Simple coroots form its basis, and the integral Weyl reflections carry every coroot to a simple one, as proved in the earlier root-datum and transport arguments. These reflections are lattice automorphisms. A primitive integer vector has an integer linear combination of its coordinates equal to one; hence its reduction is nonzero over every field. Thus the right side of (9.2) never vanishes, including in characteristics two and three. The same primitive-vector argument gives a character with pairing one and makes $\alpha^\vee : \mathbf{G}_m \to T_k$ a closed one-dimensional torus.

Let $J = \ker(\mathrm{Ad} : G \to \mathrm{GL}(\operatorname{Lie} G))$. Differentiating conjugation on dual numbers gives

$$\operatorname{Lie} J = \{Z \in \operatorname{Lie} G : [Z, Y] = 0 \text{ for every } Y \in \operatorname{Lie} G\}.$$

This is an equality of kernels of the actual tangent maps: the bracket is the mixed coefficient of the group commutator, so the differential of Ad sends Z to $Y \mapsto [Z, Y]$. Its kernel is T_k-stable. Full

character grading, rather than character differentials, splits that kernel into its weight subspaces. Equation (9.2) excludes every root line. Therefore Lie J has only weight zero.

Let $H = (J_{\mathrm{red}})^0$. Over the algebraically closed field this is a reduced subgroup: the complete reduced-product proof in S08 Lemma G.4.1 makes $J_{\mathrm{red}} \times J_{\mathrm{red}}$ reduced, so multiplication factors through the reduction; inversion and identity do too. The full non-affine finite-type argument of Lemma 9.A gives its open-and-closed identity subgroup and makes this reduction smooth. Its component is geometrically irreducible. Conjugation by T_k preserves H: it preserves J, its reduced product with T_k factors through J_{red}, and the connected product $T_k \times H$ stays in the identity component. No normality assertion about the reduction under an arbitrary nonreduced group is needed.

Now use the actual affine neighbourhood $H \cap X_k$. It is closed in the affine X_k, open and smooth in H, and stable under T_k. Its fixed locus is smooth, with invariant tangent space, by the entire AG-GS, *Diagonalizable groups*, Theorem 3.4. Its tangent at the identity is all of Lie H, since this Lie algebra lies in the weight-zero Lie J. S02 Lemma 3.A makes the fixed-locus inclusion étale near that point; the whole étale-monomorphism argument in S04 Lemma E7.2 makes it an open immersion there. Thus a nonempty open neighbourhood in H is fixed by T_k.

The fixed subscheme of X_k is exactly T_k. In its coordinate algebra, each root coordinate has a nonzero character α. Invariance under the universal torus element forces that coordinate to be zero by independence of the Laurent exponent basis; this remains true with nilpotent coefficients. Hence that neighbourhood of H lies in the already closed T_k. It is dense in the smooth irreducible H. The full schematic-density argument of S08 Lemma G.3.5 makes its closed-immersion equations vanish on all of H, so $H \subset T_k$.

On $T_k = D_k(P)$, the full adjoint action on (9.1) has precisely the root characters. Its scheme kernel is therefore $D_k(P/Q)$. This is finite: in fundamental-weight coordinates the simple-root matrix has nonzero integer determinant d; its adjugate gives $dP \subset Q$, so P/Q is a quotient of the finite group P/dP. The exponent basis of its group algebra is finite. Thus H, being smooth connected inside this finite kernel, is the identity. Every component of J is a translate of its identity component after choosing a closed k-point; the component theorem and the earlier Zariski lemma supply those points. Its reduction has dimension zero, so J itself has dimension zero. S04 Lemma P0.9 supplies the full Artinian finite-algebra calculation: a zero-dimensional finite-type scheme over k is a finite disjoint union of finite local Artinian k-schemes. Consequently J is finite, retaining its possible infinitesimal structure.

Return to the whole integral group algebraic space. Its Lie bundle is the finite free module furnished by the identity cell, and conjugation defines

$$\mathrm{Ad} : G_{\mathbf{Z}} \longrightarrow \mathrm{GL}(\mathrm{Lie}\, G_{\mathbf{Z}}). \tag{9.3}$$

This is a finite-presentation separated morphism. Both spaces are of finite presentation over the Noetherian $\mathbf{Z}$; its graph is closed and finitely presented, and the product projection is finitely presented. Separation also follows from the closed diagonal of $G_{\mathbf{Z}}/\mathbf{Z}$. Every nonempty geometric fibre of (9.3), after choosing a point in it, is isomorphic by translation to the just-computed finite kernel. For all test algebras the inverse sends a fibre point g' to $g^{-1}g'$; the equality of adjoint matrices is exactly the kernel condition. Its geometric fibres are therefore finite and zero-dimensional, proving quasi-finiteness. The complete S04 Theorem E9.1, also stated with its proof in RG03 Section 5, now makes (9.3) representable and quasi-affine. Its target is an affine scheme, so $G_{\mathbf{Z}}$ is a scheme.

Affineness of the geometric fibres

For the already smooth finite-type group scheme G, apply the complete reduced-source image proof of Lemma 9.B to its adjoint homomorphism. It is quasi-compact, and G is reduced. The image

I is a closed reduced subgroup of the affine general linear group, and $q : G \to I$ is faithfully flat. The image proof is scheme-theoretic: its ideal is the kernel of $\mathcal{O}_{\mathrm{GL}} \to q_*\mathcal{O}_G$; flat base change gives schematic dominance on the product, which makes multiplication and inversion preserve that ideal. Reducedness makes it radical, and the full generic-point-and-translation flatness proof gives faithful flatness. These are the earlier proved inputs, with their hypotheses verified here.

Its scheme kernel is the finite J, because $I \to \mathrm{GL}$ is a closed immersion. Multiplication gives the actual torsor isomorphism

$$G \times_k J \xrightarrow{\sim} G \times_I G, \qquad (g, j) \longmapsto (g, gj),$$

with inverse $(g, g') \mapsto (g, g^{-1}g')$ on every test algebra. Thus after the faithfully flat covering $G \to I$, the map q is the finite projection with fibre J. S04 Lemma A1.2 and Corollary A3.2 descend affineness of this map; Lemma A1.3 descends finite generation of its coordinate module. Hence q is finite. Since I is affine, G is affine. No reductivity assumption has entered this argument.

Reductivity and the prescribed root datum

Let N be any smooth connected normal unipotent subgroup of this affine G. The complete unitriangular representation proof in RG01 Lemma 2.A excludes every nontrivial diagonalizable subgroup of N, including infinitesimal ones. A closed subgroup of the split torus over this field is diagonalizable by the entire Hopf-quotient and group-like-basis proof in AG-GS Theorem 4.2. Consequently $N \cap T_k = 1$ as schemes, and its tangent space gives

$$\operatorname{Lie} N \cap \operatorname{Lie} T_k = 0.$$

If $N \neq 1$, it has positive dimension: a smooth connected zero-dimensional finite-type group over algebraically closed k is a single reduced point by the earlier Artinian calculation. Its nonzero Lie algebra is T_k-stable by normality. Equation (9.1) and the zero intersection therefore give a root line $kE_\alpha \subset \operatorname{Lie} N$.

Let $T_\alpha = (\ker \alpha)^0_{\mathrm{red}}$. Explicitly write $\alpha = n\alpha_0$ with primitive $\alpha_0 \in P$; then $T_\alpha = \ker \alpha_0$ is a split codimension-one torus. A character vanishes on it exactly when it is an integer multiple of α_0. Reducedness of the given root system consequently leaves only $\alpha, -\alpha$ among the roots trivial on T_α. The full fixed-locus theorem applies to the smooth affine G and N, so

$$C = C_G(T_\alpha), \qquad C_N = C_N(T_\alpha)$$

are smooth closed subgroups with Lie algebras the corresponding invariant subspaces. Conjugation by $\lambda = \alpha^\vee$ preserves both.

For C, the whole general cocharacter lemma RG01 Lemma 3.3 gives the smooth connected $U_C(\lambda)$. Its only positive tangent weight is kE_α, because the two root pairings are $2, -2$ as integer characters. Thus it has dimension one. The already closed U_α belongs to it on every test algebra: the action on its parameter is $u \mapsto t^2u$, extending at $t = 0$ to the identity. Its inclusion has equal tangent spaces. The full differential and étale-monomorphism proofs make it an open immersion; an open subgroup is closed because its other cosets are open, and connectedness makes it all of $U_C(\lambda)$.

The cocharacter λ need not lie in C_N. To apply the earlier theorem with its actual hypothesis, form the smooth affine semidirect product $C_N \rtimes \mathbf{G}_m$ for this conjugation action and use its internal cocharacter $t \mapsto (1, t)$. Its positive limit subgroup has second coordinate one: conjugation fixes that coordinate, while a positive limit has value the identity at zero. It therefore lies in C_N. Denote it by $U_N(\lambda)$. The full lemma gives smoothness, connectedness and positive Lie weights; its Lie algebra is exactly kE_α, since this line is in $\operatorname{Lie} N$ and no other positive invariant root weight exists. Inclusion into C identifies it with the closed intersection $N \cap U_C(\lambda)$, by the same limit equations on all tests. Equal tangent spaces again give an open subgroup of the connected $U_C(\lambda)$, hence equality. We obtain $U_\alpha \subset N$.

Normality and the integral simple/reflected root-conjugation identities give $U_{-\alpha} \subset N$, using the representative n_α formed from the two root parameters. The full linked Gauss relation then forces the whole coroot torus into N. On every k-algebra and for every unit d, substitute $u = d-1, v = 1$, so $1 + uv = d$, in

$$x_\alpha(u)x_{-\alpha}(v) = x_{-\alpha}\left(\frac{v}{1+uv}\right)\alpha^\vee(1+uv)x_\alpha\left(\frac{u}{1+uv}\right).$$

All three root factors and the left product belong to N; solving for the torus factor puts $\alpha^\vee(d)$ in N. This is a scheme assertion, even over fields with few rational units. The primitive coroot defines a nontrivial closed torus in T_k, contradicting $N \cap T_k = 1$. Thus every such N is trivial, which proves reductivity.

The given T_k is maximal. If a torus T' contains it, $\operatorname{Lie} T'$ is T_k-invariant and therefore lies in the zero part of (9.1), namely $\operatorname{Lie} T_k$. The reverse inclusion gives equal ranks. Over algebraically closed k both tori are split. Their closed inclusion gives a surjection of free character lattices of equal rank, by the whole exponent-basis/anti-equivalence proof in AG-GS Theorem 2.2. Every element of its kernel is zero after tensoring with $\mathbf{Q}$, hence is torsion; the domain lattice is torsion-free, so the kernel is zero and the inclusion is an isomorphism.

Now RG03 Theorem 4.1 applies with all its hypotheses: each closed U_α has the prescribed character and nonzero identity differential, hence is the canonical root group of the reductive G. Its linked Gauss identity and RG03 Theorem 7.1 identify the given $\alpha^\vee$ with the canonical coroot. Thus every geometric fibre has the original simply connected datum and simple frames. The full centre calculation in RG03 Section 1 gives $Z(G) = D_k(P/Q)$; the finite lattice calculation above makes this finite and makes its central torus trivial. The reductive fibre is therefore semisimple. The conclusion holds in every characteristic.

To finish, we prove affineness over $\mathbf{Z}$, rather than just in its fibres. We need the following small geometric fact, included with its complete openly licensed proof.

Flat base change and localization of quasi-coherent cohomology

Statement. Let X be a quasi-compact quasi-separated scheme over $\operatorname{Spec} A$, let $\mathcal{F}$ be quasi-coherent, and let B be any flat A-algebra. Write $g : X_B \to X$ for the projection and $\mathcal{F}_B = g^*\mathcal{F}$. For every $q \geq 0$, the canonical map is an isomorphism

$$H^q(X, \mathcal{F}) \otimes_A B \longrightarrow H^q(X_B, \mathcal{F}_B).$$

The earlier Algebra and sheaf cohomology before reductive groups supplies the complete proofs used here: Lemmas 1.0–1.1 give the tensor, localization and filtered-colimit calculations; Proposition 2.2 gives the affine module–sheaf correspondence and pullback; Lemmas 3.1–3.2 construct injectives, their open restrictions and acyclic resolutions; Section 4 constructs the finite filtered-complex comparison; and Theorem 5.1 proves $H^q(\operatorname{Spec} R, \widetilde{M}) = 0$ for every ring R, every module M and $q > 0$. Their full proofs precede this lesson. The comparison and base-change argument are written below at their original generality: no finite generation, Noetherianity or finite presentation is assumed for A, B or $\mathcal{F}$.

The separated case and the ordered cover construction.

First suppose that X is separated over the affine base. Choose a finite affine open cover $U_1, \ldots, U_r$. For a strictly increasing tuple $I = (i_0 < \cdots < i_p)$ put $U_I = U_{i_0} \cap \cdots \cap U_{i_p}$, and let $j_I : U_I \hookrightarrow X$. Empty intersections contribute zero. Every U_I is affine: the intersection of two affine opens is the inverse image of their affine product under the closed diagonal, and induction gives the assertion for finite intersections.

We construct the cohomological comparison rather than assume that a section complex computes it. For any sheaf of modules $\mathcal{G}$, form the augmented alternating sheaf complex

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{C}^0(\mathcal{G}) \longrightarrow \cdots \longrightarrow \mathcal{C}^{r-1}(\mathcal{G}) \longrightarrow 0, \qquad \mathcal{C}^p(\mathcal{G}) = \prod_{i_0<\cdots<i_p} (j_I)_*(\mathcal{G}|_{U_I}).$$

The differential is the alternating sum of restriction maps. This complex is exact on stalks. Indeed, near a point x choose a with $x \in U_a$ and restrict to a neighbourhood $V \subset U_a$. Regard an ordered cochain as alternating in all its indices, with repeated-index components zero. The map lowering the degree is

$$(hc)_{i_0\dots i_{p-1}} = c_{a i_0\dots i_{p-1}}.$$

It is well defined because $V \cap U_I \cap U_a = V \cap U_I$. In degree zero it takes c_a to a section on V, the augmented term. Expanding the alternating differential cancels all terms except c, so $dh + hd = 1$, including the augmented degree. This proves exactness even when a direct-image stalk at a boundary point is nonzero.

Take an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$ in sheaves of $\mathcal{O}_X$-modules. Restriction to an open preserves injectives: its left adjoint, extension by zero, is exact, as one checks on stalks. Direct image under an open immersion also preserves injectives because it is right adjoint to exact restriction. Thus each term of the augmented sheaf complex for $\mathcal{I}^q$ is injective. Its first injection splits, and successive cokernels are injective direct summands; inductively the whole finite exact row splits. Applying global sections therefore leaves it exact.

Consequently the augmentation from $\Gamma(X, \mathcal{I}^\bullet)$ to the total complex of

$$C^{p,q} = \prod_{i_0<\cdots<i_p} \Gamma(U_I, \mathcal{I}^q|_{U_I}), \qquad 0 \le p < r, \quad q \ge 0,$$

is a quasi-isomorphism. Here the total differential is $d_{\text{cover}} + (-1)^p d_{\text{resolution}}$. Exactness of the augmented horizontal rows proves the assertion by the row filtration. Each total degree contains finitely many terms.

The column filtration $F^p \operatorname{Tot}^n = \bigoplus_{a \ge p} C^{a,n-a}$ gives the ordered-cover spectral sequence

$$E_1^{p,q} = \prod_{i_0<\cdots<i_p} H^q(U_I, \mathcal{F}|_{U_I}) \quad \Longrightarrow \quad H^{p+q}(X, \mathcal{F}),$$

with d_1 the alternating restriction differential. Explicitly, for $s \ge 1$ set

$$Z_s^{p,q} = \{z \in F^p \operatorname{Tot}^{p+q} : dz \in F^{p+s} \operatorname{Tot}^{p+q+1}\}, \qquad E_s^{p,q} = \frac{Z_s^{p,q}}{Z_{s-1}^{p+1,q-1} + dZ_{s-1}^{p-s+1,q+s-2}},$$

where $Z_0^{p,q} = F^p \operatorname{Tot}^{p+q}$. The total differential induces $d_s : E_s^{p,q} \to E_s^{p+s,q-s+1}$. Here is the next-page calculation from the complete finite filtered-complex proof in AG-RG-S01, Section 4. If $d_s[z] = 0$, then $dz = u + dv$ with $u \in Z_{s-1}^{p+s+1,q-s}$ and $v \in Z_{s-1}^{p+1,q-1}$. Replace z by $z - v$; its differential belongs to F^{p+s+1}, so it represents an element of $Z_{s+1}^{p,q}$. Conversely every such element is a d_s-cycle. Quotienting these representatives by the d_s-boundaries and the old denominator gives exactly $Z_s^{p+1,q-1} + dZ_s^{p-s,q+s-1}$, the denominator for $E_{s+1}^{p,q}$. Thus the cohomology of d_s is the next displayed page. Extend the filtration by $F^p = \operatorname{Tot}$ for $p \le 0$ and $F^p = 0$ for $p \ge r$. For sufficiently large s, the cycle condition is $dz = 0$ and the denominator accounts for the next filtration piece and all boundaries lying in F^p. Thus the stable terms are the associated graded pieces of cohomology. There are only r columns, so each abutment has a finite filtration and convergence involves no infinite-product or completion issue. This construction works for any finite open cover, including covers with non-affine intersections.

For our separated affine cover, affine vanishing makes every row with $q > 0$ on E_1 zero. The augmentation therefore canonically identifies cohomology with that of the finite section complex

$$\check{C}^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0 < \cdots < i_p} \Gamma(U_I, \mathcal{F}).$$

If $U_I = \operatorname{Spec} R$ and $\mathcal{F}|_{U_I} = \widetilde{M}$, its base change has sections

$$M \otimes_R (R \otimes_A B) = M \otimes_A B.$$

These identifications commute with restriction. Tensor also commutes with the finite products, so the section complex on X_B is the section complex on X tensored with B. Flat tensor preserves kernels and images and hence cohomology. This proves the claimed isomorphism in the separated case.

Canonical comparison and nonseparated intersections.

To identify the preceding calculation with the canonical map and compare the general cover spectral sequences, choose an injective resolution $\mathcal{F}_B \to \mathcal{J}^\bullet$ on X_B. The projection g is flat, so $g^*\mathcal{I}^\bullet$ is a resolution of $\mathcal{F}_B$. Target injectivity extends its identity on $\mathcal{F}_B$ degree by degree to a chain map $g^*\mathcal{I}^\bullet \to \mathcal{J}^\bullet$. The same extension argument makes any two choices chain homotopic. Pulling back sections and multiplying by elements of B gives a map of augmented double complexes

$$C^{p,q}(X, \mathcal{I}) \otimes_A B \longrightarrow C^{p,q}(X_B, \mathcal{J}).$$

Its augmentation is the usual cohomology base-change map; its column maps are the canonical comparisons on the intersections. It is independent of the resolution choices and compatible with restriction and successive base changes. Since B is flat, the source total complex has cohomology $H^n(X, \mathcal{F}) \otimes_A B$. Flat tensor also commutes with the cycles, boundaries and finite sums defining every spectral-sequence page.

Now let X be merely quasi-compact and quasi-separated. Choose a finite affine cover as before. Each U_I is quasi-compact, by quasi-separatedness and finite intersection, and is separated over $\operatorname{Spec} A$, since it is an open subscheme of the affine U_{i_0}. It need not be affine. Apply the separated result to each U_I. It shows that every term of the comparison on E_1 is an isomorphism, for all $q \geq 0$. Therefore every subsequent page and every associated graded piece of the abutment is an isomorphism. Induction through the finite filtrations gives an isomorphism of the abutments themselves. This proves the statement for every flat A-algebra B.

Localization and higher direct images.

Taking $B = S^{-1}A$ for any multiplicative set gives, canonically,

$$S^{-1}H^q(X, \mathcal{F}) \cong H^q(X \times_A \operatorname{Spec} S^{-1}A, \mathcal{F}_{S^{-1}A}).$$

This includes $A \to A_a$ and $A \to A_{\mathfrak{p}}$.

For completeness, let $f : X \to Y$ be a quasi-compact quasi-separated morphism and let $\mathcal{F}$ be quasi-coherent. The sheaf $R^q f_*\mathcal{F}$ is the sheafification of $V \mapsto H^q(f^{-1}V, \mathcal{F})$. Indeed, use an injective resolution on X, restrict it to $f^{-1}V$, and take stalks of $f_*\mathcal{I}^\bullet$; filtered colimits of modules are exact, so taking cohomology commutes with that stalk limit. On an affine $V = \operatorname{Spec} A$, put $M = H^q(f^{-1}V, \mathcal{F})$. The localization result identifies the presheaf on every distinguished $D(a) \subset V$ with M_a, compatibly with restriction. Its sheafification is therefore $\widetilde{M}$. This proves quasi-coherence of $R^q f_*\mathcal{F}$ and supplies the canonical identification of its sections on V with M.

If $h : Y' \to Y$ is flat, cover Y' by affines $W = \operatorname{Spec} B$ mapping into affines $V = \operatorname{Spec} A$ of Y. Here B is flat over A. For the Cartesian base change $f' : X \times_Y Y' \to Y'$, the theorem identifies the

modules describing both sides of

$$h^* R^q f_* \mathcal{F} \longrightarrow R^q f'_* \mathcal{F}_{Y'}.$$

Naturality makes these identifications agree on overlaps, so this is an isomorphism of sheaves. In particular, at $y \in V$ corresponding to $\mathfrak{p}$,

$$(R^q f_* \mathcal{F})_y \cong H^q(f^{-1}V \times_A \operatorname{Spec} A_{\mathfrak{p}}, \mathcal{F}_{A_{\mathfrak{p}}}).$$

This is the flat-localization identity needed for the stalk calculation in AG-RG-05, §9. □

One-dimensional affine-open lemma (Stacks, Tag 09N9). If Y is affine and all its local rings are Noetherian of dimension at most one, then every quasi-compact open $O \subset Y$ is affine.

Proof, with the Stacks statement retained and the cohomological steps written out. Write $Y = \operatorname{Spec} A$ and let $\mathcal{F}$ be quasi-coherent on O. The open O is quasi-compact and separated. The full flat-localization proof just given in this Section 9, with the earlier S01 algebra and sheaf foundations, gives, for every prime $\mathfrak{p} \subset A$,

$$H^i(O, \mathcal{F})_{\mathfrak{p}} = H^i(O \times_Y \operatorname{Spec} A_{\mathfrak{p}}, \mathcal{F}_{A_{\mathfrak{p}}}).$$

An open in the spectrum of the Noetherian local ring $A_{\mathfrak{p}}$ either contains its closed point, and is the whole spectrum, or contains only generic points. There are finitely many generic points: in a Noetherian topological space every closed subset has finitely many irreducible components, by induction on closed subsets (a reducible counterexample splits into two strictly smaller counterexamples). In dimension at most one no distinct nonclosed primes can specialize to one another. Thus the latter open is a finite discrete space. Each of its singleton components has an affine neighbourhood contained in that component, so it is affine; their finite disjoint union is the spectrum of the product of their rings. Both possibilities are affine. The affine-vanishing proof of that supporting lesson, Theorem 5.1, therefore makes the displayed localization zero for $i > 0$.

A module with zero prime localizations is zero: a nonzero element has a proper annihilator contained in a maximal ideal, at which its localization remains nonzero. Consequently $H^i(O, \mathcal{F}) = 0$ for every $i > 0$ and every quasi-coherent $\mathcal{F}$. The full ideal-sheaf affine criterion in the supporting lesson, Theorem 9.2, now proves that O is affine. This argument uses no separate Leray or higher-direct-image theorem. □

Lemma 9.1. A quasi-finite separated homomorphism $f : K \to H$ of finite-type group schemes over a Dedekind base is affine if K is smooth with geometrically connected fibres of constant dimension.

Proof. Write d for the constant fibre dimension of K. We first prove the field calculation that will be used for both fibres. Over any field F, smoothness makes the local rings of $K_{\bar{F}}$ regular domains by AG-CA-18 Theorem 2.1 and AG-CA-14 Theorem 1.1. Distinct irreducible components cannot meet, since a local domain has only one minimal prime. There are finitely many components, by the Noetherian closed-subset induction in the preceding affine-open proof, so they are open and closed. Geometric connectedness leaves one component. A nilpotent section has zero germ in every local domain, and is zero. Thus K_F is geometrically integral; this is the whole component argument of AG-RG-S02 Lemma 4.1 applied on affine charts, without assuming that K_F is affine.

The complete reduced-source schematic-image proof in Lemma 9.B gives a closed reduced subgroup $I_F \subset H_F$ and a faithfully flat quasi-compact factor $q_F : K_F \to I_F$. That proof constructs the ideal as the kernel of the structure-sheaf map, whose formation commutes with flat field extension. Consequently I_F is geometrically integral: after every algebraic closure it is reduced and is the surjective image of the irreducible $K_{\bar{F}}$. Its full scheme kernel J_F is zero-dimensional by quasi-finiteness. A zero-dimensional finite-type field scheme has finitely many points, each open and closed. Each singleton has an affine neighbourhood inside it, with a local zero-dimensional Artinian

finite-type field algebra. The full proof of AG-RG-S04 Lemma P0.9 makes this algebra finite-dimensional, including all nilpotents. Taking the finite product of these algebras shows that J_F is finite.

On every test scheme the maps

$$(g, j) \longmapsto (g, gj), \qquad (g, g') \longmapsto (g, g^{-1}g')$$

are inverse isomorphisms

$$K_F \times_F J_F \;\simeq\; K_F \times_{I_F} K_F.$$

Hence the pullback of q_F by its own faithfully flat cover is the finite projection $K_F \times_F J_F \to K_F$. The full affine-algebra and finite-module descent proofs of AG-RG-S04 Lemmas A1.2–A1.3 and Corollary A3.2 descend its affineness and the finite module generating list. Therefore q_F is finite. On an affine chart of I_F, the coordinate map is a finite integral injection, the injection following from faithful flatness. AG-RG-S04 Lemma P0.4 proves that lying over and going up lift every prime chain, while incomparability makes contraction preserve strict chains. Thus the dimensions of source and image are equal:

$$\dim I_F = \dim K_F = d.$$

This proves both finiteness and the image dimension directly, for every field, including an imperfect field.

Affineness is local on the target and descends faithfully flatly by the same Corollary A3.2. We may first restrict the base to an affine Dedekind scheme. Its ring is Noetherian, and K, H are then Noetherian qcqs schemes. Their finite-type maps over that ring are of finite presentation; the homomorphism is also of finite presentation, since on compatible affine charts its source algebra is finite type over the target algebra, which is Noetherian. These observations do not assume that H is separated over the base. If f is affine after passage to the local ring at each base point, Limits of schemes and Noetherian approximation, Theorem 4.2, spreads the property to an open base neighbourhood: the local spectrum is the inverse limit of principal neighbourhoods, and its schemes and map are the given finite-presentation models. Those neighbourhoods cover the base. The field case has just been proved, since a finite map followed by a closed immersion is affine. It remains to treat a local Dedekind ring R_0 of dimension one. It is a DVR by AG-CA, *Discrete valuation rings, normal rings and Serre's criterion*, Theorem 1.2, whose full normal one-dimensional local-domain proof precedes this lesson; fix its uniformizer π.

We construct the required faithful local extension explicitly. Write $k_0 = R_0/\pi R_0$. Choose an algebraic closure of k_0 by the full field construction in S06 Section 1, and let k be its subfield of elements separable over k_0. The complete finite separable-generator calculation in S07 Lemma 2.1 makes it a field and shows that it is separably closed: finitely many coefficients of a separable equation over it lie in one finite separable extension of k_0, and adjoining any root gives another separable extension, by its successive distinct-root and embedding calculation.

Take all finitely presented etale R_0-algebras B with a specified residue map $B \to k$ extending $R_0 \to k_0$, and use their local rings at the kernels of those maps. These objects form a filtered system. For two objects the tensor product of their original etale algebras is etale, and multiplication of their two residue maps gives a map to k; localization at its kernel supplies a common object with local transition maps. For two parallel arrows, the equalizer of their corresponding scheme maps is an open containing the selected point: the diagonal of an affine etale algebra is a closed etale monomorphism, and the complete S04 Lemmas D2.1 and A1.9 make it open. Choose a principal neighbourhood of the point inside that equalizer; its local ring supplies a further object where the arrows agree. Thus finite element and relation comparisons can always be moved to one object. Localizing an original etale algebra at its selected prime is itself the filtered union of its principal neighbourhoods, so these refinements use only the same objects.

Let R_0^{sh} be their filtered colimit. Each transition is local and flat. Indeed the original finite-presentation algebra map factors, after one principal refinement at the selected point, through

two etale algebras over the base; S04 Lemma D2.1 makes that map etale. Localizing its source and target at the selected primes gives the transition and preserves flatness. These localized maps are essentially etale; no finite-presentation claim is made for a whole prime localization. Every object is flat over R_0, and every transition is flat and local, hence faithfully flat by the faithful local calculation in S04 Lemma A1.4. The colimit is flat, since flat tensor and filtered colimits preserve injections as proved in S01 Lemma 1.0. It is local: an element with nonzero residue is already a unit in a representing local ring, while an element of zero residue lies in the union of their maximal ideals. Its residue field is exactly k. Indeed every element of k generates a finite separable extension of k_0. Lift its monic separable minimal polynomial to $R_0[T]$, and invert its derivative in the quotient. The square-Jacobian calculation of S07 Lemma 5.3A makes this etale algebra; evaluation at the chosen root gives an object containing that residue element. Conversely each object's residue field is finite separable over k_0 by the same lemma, so embeds in k. Thus the union is k. The local flat map $R_0 \to R_0^{sh}$ is faithfully flat by S04 Lemma A1.4. We need no further universal property of henselization.

We verify its DVR property instead of assuming a general Noetherianity theorem for this colimit. At a localized pointed etale neighbourhood A, the residue fibre is a finite separable field, so the maximal ideal is πA. The ring is Noetherian and flat over R_0, and π is regular. AG-RG-S02 Section 1.1 proves $\bigcap_n \pi^n A = 0$. Thus each nonzero a belongs to $\pi^n A \setminus \pi^{n+1} A$ for a unique n and can be written $\pi^n u$ with u a unit. Products of these expressions are nonzero, so A is a domain. Local transition maps preserve π and units; they preserve valuations and are injective. Localizing the pointed neighbourhoods does not change their colimit, since every element with nonzero residue becomes a unit at a later principal neighbourhood. Therefore every nonzero element of R_0^{sh} has the same valuation expression. Every nonzero ideal has an element of least valuation, which generates it. This proves that R_0^{sh} is Noetherian and is a DVR.

Its completion R is faithfully flat by the entire completion proof of AG-RG-S02 Lemma 1.3. That proof identifies $R/\pi^n R$ with $R_0^{sh}/\pi^n R_0^{sh}$, makes R local with residue field k, and proves completeness, separatedness and Noetherianity. Flatness keeps π regular. The same valuation expression and least-valuation argument show that R is a complete DVR. The construction just proved makes $R_0 \to R_0^{sh}$ faithfully flat, so $R_0 \to R$ is faithfully flat. All the hypotheses survive this base change. The residue field k is separably closed; it need not be perfect.

Over R replace H by the schematic closure of the generic image I_η. On an affine chart with original ring C, its ring is $A = C/\ker(C \to C[1/\pi]/I_\eta)$, where I_η here denotes the generic chart's defining ideal. Thus A injects into its generic chart. If this chart is nonempty, its generic chart is a nonempty open of the integral I_η, so A is a domain. It is torsion-free over the DVR R, and hence flat by the complete ideal criterion in AG-CA-07 Theorem 2.1: each nonzero finitely generated ideal of R is generated by one nonzero element a, its free rank-one module identification makes $I \otimes_R A \to A$ multiplication by a, and torsion-freeness makes this injective. The zero ideal is immediate. This checks every finitely generated ideal and proves flatness. The closure is Noetherian and finite type, and is a subgroup: multiplication is checked on its flat product, inversion on the flat closure, and the generic subgroup equations vanish everywhere because on each affine chart flatness gives injectivity into localization at π. The unit factors through it by the same calculation on R. Flatness of K similarly makes f factor through it. Proving that this factor is affine suffices, since its inclusion into the original H is closed. Retain the name H for the closure.

We now calculate its special dimension. Over $\bar{k}$, Lemma 9.A makes all components of the finite-type $H_{\bar{s}}$ have one dimension r and makes their reductions translates of its irreducible smooth reduced identity component. AG-RG-S08 Lemma G.1.2 proves field-extension invariance of dimension. On any affine chart $\operatorname{Spec} A$ meeting the special fibre, the nonzero finite-type algebra $A/\pi A$ consequently has dimension r: every nonempty open in a finite-type irreducible field component has its function field's dimension, by the earlier field dimension proof. Noether normalization, *Krull dimension and Noether normalization*, Corollary 3.2, applies even if this algebra is nonreduced and gives a finite

inclusion

$$k[t_1, \dots, t_r] \hookrightarrow A/\pi A.$$

Lift these parameters to A. The map $R[t_1, \dots, t_r] \to A$ is injective: divide a proposed nonzero relation by the least valuation of its finitely many coefficients; regularity of π then gives a relation with a nonzero reduction, contradicting algebraic independence. The chart's map to $\mathbf{A}^r_R$ is quasi-finite at every special-fibre point, because the special map is finite. AG-RG-S04 Theorem C5.1 makes its quasi-finite locus a nonempty open. Since A is a domain with nonzero π, that open meets its generic fibre. The generic map on this open is dominant, by the polynomial injection and density of an open in an integral scheme, and quasi-finite. Its function field is therefore finite algebraic over $\mathrm{Frac}(R)(t_1, \dots, t_r)$, by the zero-dimensional generic-fibre algebra calculation of Lemma P0.9. The whole transcendence-degree proof in *Krull dimension and Noether normalization*, Theorem 4.2, gives

$$r = \dim I_\eta = d.$$

Thus every geometric special component has dimension d.

Let z_i be the generic points of the finitely many special components. In a domain affine chart, their primes are minimal over (π). The complete principal-ideal height proof in *Dimension theory of Noetherian local rings*, Theorem 3.1, gives height at most one. Since π is nonzero, the zero prime does not contain it, so the height is one. Thus $\mathcal{O}_{H,z_i}$ is a one-dimensional Noetherian local ring. Apply the full scheme factorization of AG-RG-S04 Theorem D7.1 to the quasi-finite separated base change of f to this local spectrum. Its source is a quasi-compact open in a finite scheme over that affine local base. Finite algebras are Noetherian, and integral incomparability makes every chain in their spectra contract strictly into a chain of length at most one; this is Lemma P0.4. Their local rings have dimension at most one. The complete affine-open lemma above makes the source affine. AG-MO-03 Theorem 4.2, whose eventual-affineness proof is the finite affine-chart argument of Lemma 2.3, spreads this affine morphism to an open neighbourhood of z_i in an affine chart of H.

Let $V \subset H$ be the union of all opens on which f is affine. Affineness is local on the target, so f is affine over V. The field calculation makes f_η finite, and hence puts H_η in V; the preceding paragraph puts every z_i there. Apply that same field calculation over k. The reduced special image I_s is geometrically integral and has dimension d. In $H_{\bar{s}}$ it is a closed connected subgroup of the irreducible reduced identity component, of the same dimension d. It equals that component: a proper closed subset of a finite-type irreducible field scheme has smaller dimension, by the strict-specialization calculation in *Krull dimension and Noether normalization*, Lemma 4.1. Translation by it is transitive on the closed points of every component. Each component meets $V_{\bar{s}}$, so for every geometric closed point y of H_s the open

$$\{a \in K_{\bar{s}} : f(a)y \in V_{\bar{s}}\}$$

is nonempty. Surjectivity of $K_{\bar{s}} \to I_{\bar{s}}$ supplies that nonemptiness, including when H_s is nonreduced.

We need a translator rational over k and lifting over R. Every nonempty open of the smooth k-scheme K_s has a k-point. Indeed the full local standard-form proof of *Formally smooth, unramified and etale ring maps*, Theorem 5.1, supplies a nonempty affine chart etale over affine space by its free coordinates and invertible square Jacobian. That map is open by the complete flat finite-presentation openness proof in S04 Section E5. The separably closed field k is infinite: in a field with q elements, multiplication by a nonzero element permutes the $q-1$ nonzero elements; taking their product gives $x^q = x$ for every x. The polynomial $T^q - T - 1$ has no root in that field and has derivative -1, so an irreducible factor gives a proper finite separable extension, contradicting separable closedness. Induction using the one-variable root bound now gives a k-point in any nonempty affine-space open. Its nonempty etale fibre has a closed point with finite separable residue field by the full field calculation in S07 Lemma 5.3A. That field equals k. Moreover $\bar{k}/k$ is purely inseparable, as proved by the derivative and separable-part calculation of AG-RG-S04 Proposition D3.4. A principal open in an affine chart over $\bar{k}$ uses finitely many coefficients. In

positive characteristic, a common p-power of them belongs to k; the same power of its defining function lies in the original chart ring and defines exactly the same open after extension. In characteristic zero $k = \bar{k}$. Applying the preceding rational-point argument to this open proves that $K_s(k)$ is dense in $K_{\bar{s}}$.

Every $a \in K_s(k)$ lifts to $K(R)$. Choose a smooth affine chart of K containing it. The complete formal-smooth lifting proofs of the ring lesson, Theorems 3.1 and 4.1, lift the point successively through $R/\pi^{n+1} \to R/\pi^n$: these kernels are square-zero for $n \geq 1$. The chart has a finite presentation. Completeness gives limits of its coordinate values; the finitely many equations vanish in R because they vanish modulo every π^n, and each inverted coordinate has nonzero residue and is a unit. This is an R-point of that chart with residue a.

Choose a rational translator in the nonempty open for y, and lift it. The corresponding translations on K and H intertwine f. They carry any open over which f is affine to another such open, and their inverses do the same, so they preserve V. Since the lift sends y into V, the point y already lies in V. Thus all geometric closed points of H_s lie in V. A nonempty closed subset of a finite-type field scheme has a geometric closed point by the full Zariski-lemma proof in AG-RG-S04 Lemma P0.9. Hence $H_s \subset V$. Together with $H_\eta \subset V$ this gives $H = V$ and proves affineness over R. Faithfully flat descent gives it over R_0, and the finite-presentation neighbourhood argument gives it over the original Dedekind base. □

Apply Lemma 9.1 to the adjoint homomorphism (9.3). The preceding full kernel argument proves that it is quasi-finite and separated and that its source is a finite-type scheme. The identity cell and its translates make $G_{\mathbf{Z}}$ smooth with geometrically connected fibres of constant dimension $\operatorname{rank} P + |\Phi|$. Its target $\mathrm{GL}(\mathfrak{g})$ is affine over $\mathbf{Z}$. Lemma 9.1 makes the adjoint map affine, so $G_{\mathbf{Z}}$ is affine.

The torus factor of the identity cell is a monic group homomorphism into this affine group. The complete multiplicative-type closed-immersion criterion in AG-RG-01, Section 3, now applies. Explicitly, grade its coordinate algebra C by right translation by $T = D_{\mathbf{Z}}(P)$. Pullback sends C_m into $\mathbf{Z}z^m$. Let $I_m \subset \mathbf{Z}$ be the ideal of its coefficients. The full field closed-image proof already used above makes the torus inclusion closed on every field fibre. Weight projections commute with base change, so for every prime of $\mathbf{Z}$ the ideal I_m has a coefficient nonzero modulo that prime. Hence it is contained in no prime and equals $\mathbf{Z}$, for every m. Thus $C \to \mathbf{Z}[P]$ is surjective, proving that T is closed over the whole base.

The group is now affine, smooth and finitely presented, with connected reductive geometric fibres by the complete preceding reductivity proof; it is therefore a reductive group scheme by definition. The same proof makes this closed split torus geometrically maximal. The identity cell gives its full integral Lie grading, with zero part $\operatorname{Lie} T$ and free rank-one lines of precisely the prescribed characters. The hypotheses of AG-RG-03 Theorem 4.1 all hold. Each constructed root parameter is T-equivariant and has the inclusion of its framed root line as differential, so that theorem's full uniqueness proof identifies it with the canonical root homomorphism, over $\mathbf{Z}$ and after every base change. The complete integral reflection and Gauss identities, including (9.1)–(9.2), identify the coroots by AG-RG-03 Theorem 7.1. Its full scheme-centre calculation gives $D_{\mathbf{Z}}(P/Q)$; the earlier finite lattice calculation makes this centre finite, and every geometric fibre is semisimple. The chosen simple-root line frames are therefore an integral pinning with exactly the original simply connected root datum. This constructs the pinned split semisimple group over $\mathbf{Z}$ at the stated generality.

10. The existence theorem for every reduced datum

Theorem 10.1. Every reduced root datum is the datum of a pinned split reductive group over $\mathbf{Z}$. For every nonempty scheme S, pinned split groups of a specified constant datum over S are obtained by base change from this integral group, up to a unique pinning-preserving isomorphism.

Proof. Let $\mathcal{R} = (X, \Phi, X^*, \Phi^\vee)$ be reduced. The full reflection and metric calculation of AG-RG-04, Section 1, proves

$$X_{\mathbf{Q}} = V \oplus C, \qquad V = \mathbf{Q}\Phi, \qquad C = \{x : \langle x, \alpha^\vee \rangle = 0 \text{ for every } \alpha\}.$$

Write π, c for these two projections. The coroots span the dual of V. The complete simple-root argument of AG-RG-04, Section 3, applied to the dual root system, makes its simple coroots a $\mathbf{Z}$-basis of $Q^\vee = \mathbf{Z}\Phi^\vee$. Thus the weight lattice

$$P = \{v \in V : \langle v, Q^\vee \rangle \subset \mathbf{Z}\} = \mathrm{Hom}(Q^\vee, \mathbf{Z})$$

is free of rank $\dim V$, with the pairing identifying its rational space with V. The roots lie in P and all reflections preserve it. Consequently $(P, \Phi, Q^\vee, \Phi^\vee)$ is the simply connected semisimple reduced datum. The entire construction through Section 9 has already produced its pinned split group G_{sc} over $\mathbf{Z}$.

We spell out the integer facts needed for the central directions. Every subgroup $N \subset \mathbf{Z}^r$ is free of rank at most r. For induction, its first-coordinate image is either zero or $d\mathbf{Z}$ for a least positive integer d, by the integer division proof of AG-RG-S02 Lemma 4.A. In the second case choose n_0 with first coordinate d; subtracting the unique corresponding multiple of n_0 splits N as $\mathbf{Z}n_0$ plus its zero-first-coordinate subgroup. Apply induction in $\mathbf{Z}^{r-1}$; the zero-image case uses that subgroup directly. The zero-rank case starts the induction. A finitely generated subgroup of a finite-dimensional rational vector space becomes such a subgroup after choosing rational coordinates and clearing the finitely many generator denominators, so is also free. Apply this to $L = c(X)$. It spans C, so its rank is $\dim C$.

The map

$$j : X \longrightarrow E = P \oplus L, \qquad x \longmapsto (\pi(x), c(x))$$

is well-defined: every coroot pairing of $\pi(x)$ is that of x, hence an integer. It is injective because its coordinates recover x, and its source and target have the same rank. In free bases its matrix has nonzero integer determinant d. The adjugate identity puts dE in $j(X)$, so $M = E/j(X)$ is a quotient of the finite group E/dE. For rank zero the determinant is one and the cokernel is zero. Thus M is finite, without assuming saturation. Every root maps to $(\alpha, 0)$, and the dual map sends $(\alpha^\vee, 0)$ to the prescribed coroot, since its pairing with x is $\langle \pi(x), \alpha^\vee \rangle = \langle x, \alpha^\vee \rangle$.

The full character and torsor calculation of AG-RG-04 Theorem 7.1, or its written exponent-basis inverse, makes

$$D_{\mathbf{Z}}(E) \longrightarrow D_{\mathbf{Z}}(X)$$

a finite faithfully flat map with kernel $H = D_{\mathbf{Z}}(M)$. Its kernel has the finite free coordinate module with basis the elements of M. Each root is in $j(X)$ and is therefore trivial on H. The complete scheme-centre calculation of AG-RG-03, Section 1, puts H in the centre of $G_{\mathrm{sc}} \times D_{\mathbf{Z}}(L)$. Apply the entire affine central-quotient proof of AG-RG-03 Theorem 1.1 and the whole torus/root-coordinate proof of AG-RG-04 Theorem 7.1. They construct

$$G = (G_{\mathrm{sc}} \times D_{\mathbf{Z}}(L))/H$$

as a smooth affine reductive group, with maximal torus of character lattice $j(X)$. The canonical root groups and their frames are unchanged, and the coroots are their images by the full pairing formula of AG-RG-04, Section 8. The resulting pinned datum is exactly $\mathcal{R}$. If Φ is empty, $V = P = 0$, $L = X$, $M = 0$, and this formula is the torus $D_{\mathbf{Z}}(X)$ directly.

All the quoted quotient and root-coordinate proofs commute with arbitrary base change, including nonreduced bases. Thus this construction gives a pinned split group of datum $\mathcal{R}$ over every S. The entire pinned-isomorphism proof of Theorem 5.1 gives the unique pinning-preserving isomorphism to any other group of that specified constant datum; its non-quasi-compact gluing argument applies to arbitrary S. This proves the original assertion. □

Without a pinning, two globally split groups with isomorphic constant data are still isomorphic. Choose corresponding positive systems and nowhere-vanishing simple-root frames, which exist by the split condition of AG-RG-04, Section 8, and apply Theorem 5.1. The isomorphism can depend on the frames. If the type varies, its constant-type parts are open and closed because the datum is locally constant. Apply the same argument on each such part and glue over this disjoint open cover, even if the base is not quasi-compact. A base change of one group over the connected base $\operatorname{Spec}\mathbf{Z}$ has one constant datum, so cannot give a family with distinct data on different parts.

The derived group

Use the presentation of Theorem 10.1 and first work over $\mathbf{Z}$; all formulas below use constant character groups and commute with every base change. Put $E = P \oplus L$, $M = E/j(X)$, and define

$$N = \operatorname{im}(L \longrightarrow M), \qquad L_0 = \ker(L \longrightarrow M).$$

Here $L \to M$ sends ℓ to the class of $(0, \ell)$. These descriptions determine the scheme kernel and image of the projection of $H = D(M)$:

$$H_{\mathrm{sc}} = D(M/N) \subset G_{\mathrm{sc}}, \qquad H_L = D(N) \subset D(L).$$

Indeed $L \to N$ is surjective and makes $D(N) \to D(L)$ closed on coordinate algebras. The injection $N \hookrightarrow M$ makes $D(M) \to D(N)$ faithfully flat with kernel $D(M/N)$; its algebra is free over $\mathbf{Z}[N]$ on a list of coset representatives of N in M. The same exponent-basis torsor map and inverse as in AG-RG-04 Theorem 7.1 prove the full quotient identity, including nilpotent tests. Both N and M/N are finite, so both displayed subgroup schemes are finite flat. In particular H_{sc} is a closed finite central subgroup of G_{sc}, not merely a set-theoretic intersection.

The whole central quotient theorem now constructs the smooth affine semisimple group

$$D = G_{\mathrm{sc}}/H_{\mathrm{sc}}. \tag{10.2}$$

Its root groups are the canonical images of those of G_{sc}. Here semisimplicity follows from the full centre calculation, not from an additional quotient assertion. The torus character lattice is $\ker(P \to M/N)$ by AG-RG-04 Theorem 7.1, and it has finite index in P because M/N is finite. The root lattice Q lies in it by centrality and has full rank $\dim V$. The determinant/adjugate proof above therefore makes its quotient by Q finite. AG-RG-03, Section 1, identifies the scheme centre with that finite diagonalizable character group, so each reductive geometric fibre has zero-dimensional radical and is semisimple. Let $K = G_{\mathrm{sc}} \times D(L)$ and let $q : K \to G$ be the full H-torsor. Its closed normal subgroup $Y = G_{\mathrm{sc}} \times H_L$ is the inverse image of H_L under the second projection. We identify its quotient on all tests. A point (g, ℓ) of Y locally lifts ℓ through the faithfully flat map $H \to H_L$; writing that lift as (h, ℓ), the point becomes $(gh^{-1}, 1)(h, \ell)$. Its class modulo H is therefore represented by a point of $G_{\mathrm{sc}} \times 1$, and two such representatives differ precisely by H_{sc}. Thus $Y/H = D$ as fppf sheaves, and $q^{-1}(D) = Y$ as schemes by the same torsor identities. The complete affine and closed-immersion descent proofs of AG-RG-S04 Lemma A1.2 and Corollary A3.2 descend the closed normal inclusion $Y \subset K$ to a closed normal immersion $D \subset G$. This identifies its descended scheme with (10.2), since both represent the quotient sheaf just calculated.

The inclusion $L_0 \subset L$ has finite quotient N, and L_0 is free of the same rank by the preceding subgroup and determinant arguments. The full exponent-basis torsor calculation gives $D(L)/H_L = D(L_0)$. The projection $K \to D(L_0)$ kills H and descends to G; its scheme kernel has inverse image Y, and hence is D. It is an epimorphism of fppf sheaves, since the projection of K and the torus map are such epimorphisms. Consequently

$$G/D = D(L_0)$$

is a torus over every base.

Perfectness on arbitrary tests

The simply connected group is perfect as an fppf sheaf, with an argument valid over every test ring A. Set

$$B = A[t, t^{-1}, (t^2 - 1)^{-1}].$$

This is a smooth finitely presented flat A-algebra: it is a localization of one polynomial variable by two elements, each inverse having a one-equation finite presentation. Its map on spectra is surjective. At any prime of A, its fibre embeds in the rational-function field of the residue field and is nonzero, so has a prime. Flatness and surjectivity make $\operatorname{Spec} B \to \operatorname{Spec} A$ an fppf cover, by the full faithful-flatness calculation of AG-RG-S04, Section A. Over B, the full character identity, with $\langle \alpha, \alpha^\vee \rangle = 2$, gives

$$[\alpha^\vee(t), x_\alpha(v)] = x_\alpha((t^2 - 1)v).$$

For any $u \in A$, take $v = u/(t^2 - 1)$. Thus every root-group section becomes one commutator on this cover. This uses the exponent two, including in characteristic two; no division by the integer two occurs.

The complete rank-one Gauss identity of Section 7 also gives, for every unit d of any test algebra,

$$\alpha^\vee(d) = x_{-\alpha}(-d^{-1})\, x_\alpha(d - 1)\, x_{-\alpha}(1)\, x_\alpha(-(d - 1)d^{-1}).$$

It follows by substituting $u = d - 1, v = 1$ into that identity and moving the two outer root factors to the other side. The simple coroots are a $\mathbf{Z}$-basis of the simply connected cocharacter lattice $Q^\vee$, so their unit-valued sections generate the whole split torus on affine tests. Hence the fppf subgroup generated by commutators contains every root group, the entire torus and the full big cell. The whole word-generation proof of Lemma 4.1 then supplies every group section fppf locally. Thus it is all of G_{sc}. For the empty root system $G_{\mathrm{sc}} = 1$, the assertion is immediate.

The quotient $G_{\mathrm{sc}} \to D$ is an fppf cover, so D is perfect too: lift any section locally through this cover, express its lift locally as a product of commutators, and take the images. Since G/D is commutative, every commutator of G lies in D. Conversely, the perfection just proved writes every section of D locally as a product of commutators of its own sections, which are also commutators in G. Therefore the entire closed subgroup scheme (10.2) represents exactly the fppf subgroup generated by commutators. This identifies the derived group on arbitrary tests, not only on geometric points.

The central torus, isogeny and descent

Put $Q = \mathbf{Z}\Phi$ and $Q_{\mathrm{sat}} = X \cap V$. An element $x + Q$ is torsion precisely when some positive multiple of x lies in Q, equivalently when $x \in V$ by clearing the finitely many rational root coefficients. Thus the torsion of X/Q is exactly Q_{sat}/Q. Moreover $X/Q_{\mathrm{sat}} \simeq c(X) = L$ is free. The full scheme-centre proof of AG-RG-03, Section 1, gives

$$Z(G) = D_S(X/Q), \qquad Z = D_S(X/Q_{\mathrm{sat}}) \hookrightarrow Z(G).$$

The second map is closed because its character map is surjective, and Z is a torus. A homomorphism from any torus into $Z(G)$ factors through Z: after an etale splitting cover its character map takes X/Q into a free group and kills torsion. The entire group-like coefficient proof in *Diagonalizable groups*, Lemma 2.1 and Theorem 2.2, makes this a statement about character sheaves on every base, including disconnected and nonreduced bases. The local factorizations agree uniquely and descend. Hence Z is intrinsically the maximal central torus.

We next calculate the torus T_D of D. Its character lattice is

$$P_0 = \ker(P \longrightarrow M/N) = \pi(X).$$

Indeed the class of $(p,0)$ lies in N precisely when $(p,\ell) \in j(X)$ for some $\ell \in L$, which is precisely $p = \pi(x)$ for some $x \in X$. AG-RG-04 Theorem 7.1 gives $T_D = T_{\mathrm{sc}}/H_{\mathrm{sc}}$ and the whole inverse-image formula for quotient tori. In the covering $K \to G$, the inverse images of T and D are $T_{\mathrm{sc}} \times D(L)$ and $G_{\mathrm{sc}} \times H_L$. Their scheme intersection is $T_{\mathrm{sc}} \times H_L$. The same quotient calculation used for Y/H identifies its quotient by H with $T_{\mathrm{sc}}/H_{\mathrm{sc}}$. Thus

$$D \cap T = T_D$$

as subgroup schemes, including after arbitrary base change.

Multiplication $m : Z \times D \to G$ is a homomorphism because Z is central. On its torus $Z \times T_D$ the character map is

$$X \longrightarrow (X/Q_{\mathrm{sat}}) \oplus \pi(X), \qquad x \longmapsto ([x], \pi(x)).$$

Its kernel is zero: $[x] = 0$ puts x in V, and $\pi(x) = 0$ then gives $x = 0$. The target is free by the subgroup proof above. Its rank is $\dim C + \dim V = \operatorname{rank} X$, since $\pi(X)$ spans V. The determinant/adjugate argument makes its cokernel F finite. The whole character-torsor calculation therefore makes $Z \times T_D \to T$ a finite faithfully flat map with kernel $J = D_S(F)$, finite flat over S.

This is also the full kernel of m. On any test scheme, a kernel pair (z,d) satisfies $d = z^{-1}$ in G. Since $z \in Z \subset T$, this forces $d \in D \cap T = T_D$ as a scheme condition, by the exact intersection just proved. Hence the kernel is exactly J, rather than just having the same field points. It is central in $Z \times D$: the D-coordinate of such a pair equals the inverse of a central element of G, and so centralizes D on every test.

The map m is an epimorphism of fppf sheaves. Its torus map gives local lifts of all torus sections, and its restriction to each canonical root group of D is the corresponding root group of G, by the full central-quotient theorem. Thus it locally lifts all sections of the big cell. The entire word-generation proof of Lemma 4.1 expresses any group section locally as a finite word in that cell; further fppf covers lift its finitely many factors, and their product is the required lift. Apply this to the universal section over G: an fppf covering $W \to G$ has a section b of $(Z \times D)_W$. On that cover the explicit mutually inverse maps

$$J_W \longrightarrow (Z \times D) \times_G W, \quad j \longmapsto bj, \qquad y \longmapsto b^{-1}y$$

identify the base change of m with $J \times_S W \to W$. That projection is finite flat and faithfully flat. The full algebra/finite-module descent of AG-RG-S04 Lemmas A1.2–A1.3, and detection of exactness by faithfully flat tensoring, descend these properties to m. Hence m is a finite faithfully flat central isogeny. This proof retains the full kernel scheme in every residue characteristic.

For an arbitrary reductive group G/S, the complete splitting proof of AG-RG-04, Section 8, supplies etale neighbourhoods with split torus, constant datum, all root lines trivialized and a pinning. Theorem 10.1 identifies the group there with the base change of its integral model. Make the constructions above on those neighbourhoods. Their derived subgroup is the intrinsic fppf subgroup generated by commutators. Their central torus is the intrinsic maximal torus in the centre. Consequently any overlap isomorphism carries each entire subgroup scheme to its counterpart; uniqueness makes their comparisons satisfy the triple cocycle. The whole affine algebra and closed-subgroup descent proofs in AG-RG-S04, Section A, glue them to a smooth closed semisimple D and a closed central torus Z over S. Smoothness, the torus property and semisimple geometric fibres are checked on that etale cover. The multiplication maps and their exact finite-flat central kernels agree on overlaps and descend as well.

All local formulas above are constant character calculations, canonical root maps and explicit torsor identities, so they commute with arbitrary base change. The perfection proof was over every test ring, so after any base change the pulled-back D still represents the new fppf commutator subgroup. The torsion quotient and its character-sheaf factorization likewise identify the pulled-back Z with the new maximal central torus. Uniqueness of these characterizations makes the descended

identifications canonical. This establishes the derived group, central torus and central isogeny over all bases, including nonreduced ones.

Finally this central torus is the radical in the geometric fibres. Over an algebraically closed field let R be any smooth connected solvable normal subgroup of a reductive G. The entire solvable-structure proof in AG-RG-01, Section 2, writes $R = U \rtimes T$ with U smooth connected unitriangular and R/U a torus. Lemma 2.A there shows that every homomorphism from a smooth connected unipotent subgroup to this torus is trivial. Thus U contains every such subgroup and is characteristic in R. Conjugation by each $G(k)$-point preserves it. This is scheme normality: the reduced-product proof of AG-RG-S08 Lemma G.4.1 makes $G \times U$ reduced, and the full Nullstellensatz proof detects vanishing of the closed-subgroup equations on all its closed points. The conjugation morphism therefore factors through U. Reductivity gives $U = 1$, so R is a torus. Conjugation on R is a family of torus automorphisms. The full group-like coefficient calculation makes its integer exponent matrices locally constant on the connected G; their value at the identity is the identity matrix, so the entire action is trivial. Hence R is central and lies in Z. Conversely Z is smooth connected solvable normal. It is consequently the maximal such subgroup. The constructed relative torus has these geometric fibres and supplies the asserted radical over the original base.

11. The dual group

Interchanging roots with coroots gives

$$\mathcal{R}^\vee = (X^\vee, \Phi^\vee, X, \Phi).$$

It is again a reduced root datum. Its integral group from Theorem 10.1, base changed to $\mathbf{C}$, is the **dual group**. A dual base determines its pinning up to the unique isomorphism of Theorem 5.1. The construction reverses simply connected and adjoint semisimple data.

The dual datum assertion includes the axioms. Its pairing is the original perfect pairing in the opposite order, and its reflection is the transpose of the original reflection: $x^\vee \mapsto x^\vee - \langle \alpha, x^\vee \rangle \alpha^\vee$. These reflections permute the coroots by RG04 Section 1. All integer Cartan pairings are the original pairings transposed, and each root–coroot self-pairing is two. The dual root set spans its rational root space and is reduced, by the positive-definite coroot identification in that same section. Applying the already proved simple-root argument to it gives its base. Dualizing twice recovers the original datum. The simply connected condition $X^\vee = \mathbf{Z}\Phi^\vee$ consequently becomes the adjoint condition that the dual character lattice is generated by its roots, and conversely.

Matrix proofs for the classical examples

We verify the group identifications used below over characteristic-zero fields, including every geometric field extension. The algebraic closures and finite linear operations used in these arguments have their complete proofs in S06 Section 1. All the displayed matrix parameters are polynomial group morphisms, so the resulting root and coroot identifications hold on nonreduced test algebras too.

For $n \geq 2$, GL_n is the principal determinant open in affine matrix space, and is smooth and geometrically irreducible. The equation $\det = 1$ defines a smooth affine SL_n: at any such matrix its cofactor entries, namely the entries of its inverse transpose, cannot all vanish, so the complete square-Jacobian criterion supplies a smooth chart. It is geometrically connected. Over an algebraically closed field, row elimination expresses every determinant-one matrix as a product of the elementary $I + uE_{ij}$. A nonzero pivot can be moved by the determinant-one two-coordinate swap $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, already a product of the two elementary parameters by (1.1); row additions clear its column, and column additions clear its row. The remaining diagonal units with product one are products of the

two-coordinate $h(d) = \mathrm{diag}(d, d^{-1})$, each a product of four elementary parameters by the linked Gauss formula (the displayed formula in Section 10). Induction on the remaining block proves the elimination assertion, including the last block of size one. Every elementary one-parameter image is connected and contains the identity, so every such product lies in the identity component. Lemma 9.A and the Nullstellensatz show that these rational points exhaust all components; hence the whole group is connected.

Their diagonal tori have respectively character lattices $\mathbf{Z}^n$ and $\mathbf{Z}^n/\mathbf{Z}(1,\dots,1)$, by their Laurent coordinate algebras and the exponent-basis proof of AG-GS-05 Theorem 2.2. In their tangent matrix algebras, conjugation gives the nonzero characters $e_i - e_j$ on E_{ij}; the zero part is exactly the diagonal tangent space (trace zero for SL_n). The two-coordinate SL_2 inclusion gives root parameters $I + uE_{ij}$ and coroot $z \mapsto \mathrm{diag}(1,\dots,z,\dots,z^{-1},\dots,1)$, of lattice vector $e_i - e_j$.

Here is reductivity and maximality, rather than an assumption about these matrix groups. A smooth connected normal unipotent N intersects the diagonal torus trivially as a scheme, by RG01 Lemma 2.A and the closed-torus-subgroup proof of AG-GS-05 Theorem 4.2. Its Lie algebra therefore has zero intersection with the diagonal Lie space and is torus-stable. The tangent assertion follows on dual numbers: a tangent point of both closed subgroups satisfies the sum of their defining ideals, hence is a tangent point of their scheme intersection. The intersection is the identity scheme, whose tangent space is zero. If nonzero, the complete character projections give some E_{ij} in it. Normality gives the Lie ideal property by the explicit double-dual-parameter calculation in Section 6. Bracketing with E_{ji} then puts $E_{ii} - E_{jj}$, a nonzero diagonal vector, in it, a contradiction. A smooth connected finite-type group with zero Lie algebra is trivial by the full field calculation of S07 Lemma 5.3A. Thus $N = 1$. A containing torus has Lie algebra in the zero-weight part; equal ranks and the full free-character-surjection argument of Section 6 make it the same torus. Both groups are therefore reductive with these maximal tori. RG03 Theorems 4.1 and 7.1 identify the displayed parameters and linked matrix coroots with their canonical roots and coroots.

The scalar $\mathbf{G}_m$ is central in GL_n on every test algebra. RG03 Theorem 1.1 constructs its smooth affine reductive quotient, denoted PGL_n, and RG04 Theorem 7.1 gives its diagonal quotient torus and unchanged canonical root groups. Its character lattice is $\{(a_i) \in \mathbf{Z}^n : \sum a_i = 0\}$, the monomials trivial on scalars; its cocharacter lattice is $\mathbf{Z}^n/\mathbf{Z}(1,\dots,1)$ by the perfect pairing. The coroots are the images of the same two-coordinate cocharacters, by the full quotient coroot formula in RG04 Section 8. This proves the entire claimed A_{n-1} data and their duality. For $n = 1$, $\mathrm{GL}_1 = \mathbf{G}_m$ by its Laurent coordinate ring, and SL_1 and PGL_1 are the trivial group. Their data have no roots; the torus case has already been proved in the diagonalizable-group lesson.

For the symplectic group use the alternating matrix $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$ and define Sp_{2n} inside GL_{2n} by $M^t J M = J$. It is smooth in characteristic zero by an explicit square-zero lift. Lift any matrix point across a square-zero ideal; it remains invertible, since units lift across that ideal by the geometric inverse. Its error $E = M^t J M - J$ is alternating and has entries in that ideal. Replace M by $M(1 - \frac{1}{2}J^{-1}E)$. The two linear error corrections each equal $-E/2$, and all products of ideal entries vanish, so the new matrix preserves J. This proves formal smoothness; the finite matrix equations and the complete finite-presentation smoothness criterion in AG-CA-17 give smoothness.

We also prove geometric connectedness. For the alternating pairing, $\tau_{v,a}(x) = x + a\langle v, x\rangle v$ preserves it, has inverse $\tau_{v,-a}$, and gives a polynomial one-parameter subgroup for each v. If nonzero vectors x, y have $\langle y, x\rangle \neq 0$, taking $v = y - x$ and $a = 1/\langle y, x\rangle$ sends x to y. Otherwise choose a vector z outside the two proper hyperplanes $\langle z, x\rangle = 0$ and $\langle y, z\rangle = 0$, using the finite polynomial avoidance proof of S06 Section 1; two such transformations suffice. Thus an element can be brought to fix e_1. Write the symplectic space as $ke_1 \oplus W \oplus kf_1$. The stabilizer acts symplectically on $e_1^{\perp}/ke_1 = W$. Induction makes that induced action lie in its embedded identity component. After removing it, the remaining stabilizer matrix has the unique form

$$e_1 \mapsto e_1, \qquad w \mapsto w + \langle v, w\rangle e_1, \qquad f_1 \mapsto f_1 + v + ce_1 \quad (v \in W,\ c \in k).$$

Indeed the pairing with e_1 fixes the f_1 coefficient, the quotient fixes each w modulo e_1, and the

pairing with f_1 gives exactly its displayed coefficient; alternation imposes no further equation on c. Conversely these formulas preserve every basis pairing and are invertible by triangular solving. They give an affine-space family containing the identity, hence in that component. This induction starts with $W = 0$. Every matrix point is therefore in the identity component, and Lemma 9.A with the Nullstellensatz proves connectedness of the scheme.

The symplectic diagonal torus is $\mathrm{diag}(t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})$, with lattice $\mathbf{Z}^n$. Differentiating the matrix equation gives blocks $\left(\begin{smallmatrix} A & B \\ C & -A^t \end{smallmatrix}\right)$ with B, C symmetric. Their off-diagonal matrix units have exactly the characters $\pm(e_i - e_j), \pm(e_i + e_j), \pm 2e_i$, each on one line; the zero part is the diagonal torus Lie space. A two-coordinate SL_2 on e_i, f_i gives coroot e_i for $2e_i$. An SL_2 on e_i, e_j and its contragredient on f_i, f_j gives coroot $e_i - e_j$ for that root. Conjugating by the symplectic swap of e_j, f_j gives coroot $e_i + e_j$ for the corresponding sum root. These are polynomial matrix homomorphisms preserving all pairings. Their positive and negative infinitesimal brackets are respectively the nonzero diagonal vectors H_i and $H_i \pm H_j$, as direct two-block multiplication shows. The same normal-unipotent Lie-ideal and torus-intersection proof just given therefore proves reductivity; its zero-weight and equal-rank argument proves torus maximality. RG03 identifies these actual root parameters and coroots. The coroots include the basis e_i, so their lattice is the whole cocharacter lattice. This proves the simply connected C_n datum of this matrix group.

For the odd orthogonal group, on a basis $e_1, \dots, e_n, z, f_1, \dots, f_n$ use the symmetric bilinear form $B(e_i, f_j) = \delta_{ij}$, $B(z, z) = -2$, with other pairings zero. Define SO_{2n+1} by preservation of this form and determinant one. If Q is its invertible form matrix, the same square-zero correction $M(1 - \frac{1}{2}Q^{-1}E)$, now for symmetric $E = M^tQM - Q$, preserves it. The corrected determinant has square one and residue one. Writing it $1 + t$ in the square-zero ideal gives $2t = 0$, hence $t = 0$. Thus the determinant condition is retained and the group is smooth by the same finite-presentation criterion.

For connectedness we give reflection generation. For any nonisotropic vector v, $r_v(x) = x - 2B(v, x)v/B(v, v)$ is an isometry with determinant -1: it is -1 on kv and the identity on $v^\perp$. If x, y have the same nonzero norm, reflection in $x - y$ sends x to y whenever that vector is nonisotropic. If it is isotropic, $B(x, y) = B(x, x)$, so $x + y$ has norm $4B(x, x) \neq 0$; reflection in $x + y$ sends x to $-y$, and reflection in y sends it to y. A nondegenerate space has a nonisotropic vector, since otherwise polarization with two invertible would make its bilinear form zero. Reducing an isometry to fix such a vector leaves an isometry of its nondegenerate orthogonal complement; induction on dimension therefore writes every isometry as a product of reflections. Determinant one means an even number of factors. The nonisotropic parameter open is irreducible affine-space open, and the morphism $(v, w) \mapsto r_v r_w$ on its product is regular and has the identity at $v = w$. Its image is in the identity component. Pairing up an even reflection word puts every special-orthogonal matrix point in that component. Lemma 9.A and the Nullstellensatz again prove scheme connectedness.

Its torus is $\mathrm{diag}(t_1, \dots, t_n, 1, t_1^{-1}, \dots, t_n^{-1})$, again with lattice $\mathbf{Z}^n$. Differentiating preservation of B gives zero-weight diagonal space and one line for each $\pm e_i, \pm(e_i - e_j), \pm(e_i + e_j)$. This follows entry by entry from $A^tQ + QA = 0$: the paired off-diagonal entries determine each other, the z, z entry is zero, and an e_i, z entry determines the paired z, f_i entry. Long-root SL_2 blocks on e_i, e_j with their contragredient blocks give coroot $e_i - e_j$; swapping e_j, f_j and negating z is a determinant-one isometry and gives the sum-root coroot $e_i + e_j$. For the short root e_i, use the polynomial Sym^2 action of Lemma 6.1 on $u^2, 2uv, v^2$, identified with e_i, z, f_i. The preserved form has pairings $B(u^2, v^2) = 1$ and $B(2uv, 2uv) = -2$. This invariance has an exact polynomial calculation. A vector $au^2 + b(2uv) + cv^2$ has norm $2(ac - b^2)$. Under $u \mapsto Au + Cv$, $v \mapsto Bu + Dv$, its three coefficients become $a\prime = aA^2 + 2bAB + cB^2$, $b\prime = aAC + b(AD + BC) + cBD$, and $c\prime = aC^2 + 2bCD + cD^2$. Multiplication and cancellation give $a\prime c\prime - (b\prime)^2 = (AD - BC)^2(ac - b^2)$. For $AD - BC = 1$ this preserves the norm and, by polarization with two invertible, the bilinear form over every test algebra. Its determinant is $(AD - BC)^3 = 1$: expanding the three-by-three coefficient matrix gives this identity, so the action takes values in the special orthogonal group. Its raising operator sends $z \mapsto 2e_i$, $f_i \mapsto z$; the lowering operator sends $z \mapsto 2f_i$, $e_i \mapsto z$. Their bracket is $2H_i$, and their

diagonal is $\mathrm{diag}(t^2, 1, t^{-2})$, so the coroot is $2e_i$. All other basis vectors are fixed. The root brackets are nonzero in characteristic zero. The same torus-intersection, Lie-ideal and zero-weight arguments prove reductivity and torus maximality, and RG03 gives the canonical roots and these coroots. The short roots e_i generate its whole character lattice, so this is the adjoint B_n datum. For $n = 0$ this determinant-one group is trivial. These computations identify the matrix groups used in the dual examples; positive-characteristic names below retain the stated root-datum meaning.

For SL_n, $X = \mathbf{Z}^n/\mathbf{Z}(1, \ldots, 1)$ and $X^\vee = \{(a_i) : \sum a_i = 0\}$, with roots $e_i - e_j$ and the matching coroots. For PGL_n, those two lattices are interchanged. Their complex groups are therefore dual.

For type C_n, the simply connected symplectic group has character lattice $\mathbf{Z}^n$, roots $\pm e_i \pm e_j, \pm 2e_i$, and coroots $\pm e_i \pm e_j, \pm e_i$. Its dual datum is the adjoint datum of type B_n, the split odd orthogonal group. In characteristic two this notation means the smooth split reductive group of that datum; a naive odd-dimensional bilinear-form stabilizer need not be the correct smooth group scheme. Both assertions about dual groups concern their characteristic-zero complex groups.

12. Exercises

Exercise 1 (easy). Write the pinning, coroot and simple Weyl representative of SL_2, and check its square.

Exercise 2 (medium). Verify that the data of SL_n and PGL_n are dual.

Exercise 3 (medium). Verify the duality between the symplectic and adjoint odd orthogonal data, and explain which root becomes long.

Exercise 4 (hard). Deduce that two globally split reductive groups of the same constant root datum over a nonempty scheme are isomorphic. Is the isomorphism canonical without pinnings?

Exercise 5 (medium). In SL_3, show that a product using each simple reflection once need not be the longest element, and that $T[2]$ need not commute with a simple representative.

Solutions

Solution 1

Take $T = \{\mathrm{diag}(z, z^{-1})\}$, the upper triangular Borel, and $E_\alpha = E_{12}$. Its paired negative frame is E_{21}, its coroot is $z \mapsto \mathrm{diag}(z, z^{-1})$, and (1.1) gives $n_\alpha = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The square is $-I = \alpha^\vee(-1)$. The same matrix identity holds over any ring.

Solution 2

A character of the determinant-one diagonal torus is a monomial modulo the relation $\prod t_i = 1$, giving $X_{\mathrm{SL}} = \mathbf{Z}^n/\mathbf{Z}(1, \ldots, 1)$. A cocharacter must have $\sum a_i = 0$. In the scalar quotient torus, a character is a monomial trivial on scalars, so has exponent sum zero, and its cocharacter lattice is the quotient by the scalar vector. The roots and coroots in both descriptions are the differences of the corresponding basis vectors. Exchanging the two lattices therefore exchanges the full data, not just their common A_{n-1} root system.

Solution 3

The symplectic torus is $\mathrm{diag}(t_1, \ldots, t_n, t_1^{-1}, \ldots, t_n^{-1})$, with character lattice $\mathbf{Z}^n$. Its long root $2e_i$ has coroot e_i, while $e_i \pm e_j$ has the same-coordinate coroot $e_i \pm e_j$. The dual roots are therefore the B_n roots $e_i, e_i \pm e_j$. Its coroot corresponding to e_i is $2e_i$. The lattice is the root lattice of B_n, so the

dual group is adjoint. A long C_n root becomes a short B_n root; duality reverses the length ratio. The rank-two roots-and-coroots figure in the preceding lesson shows this metric reversal explicitly.

Solution 4

Choose positive systems corresponding under the datum isomorphism, and choose frames in their globally trivial simple root lines. Theorem 5.1 gives a unique isomorphism preserving these pinnings. Forgetting the chosen frames gives an isomorphism of the original groups. It need not be unique: conjugation by an element of the adjoint torus can scale the frames while preserving the datum. For example SL_2 admits nontrivial diagonal conjugations.

Solution 5

The Weyl group is S_3, with $s_1 = (12), s_2 = (23)$. The product $s_1 s_2$ is a three-cycle and has length two, whereas $w_0 = s_1 s_2 s_1$ reverses the three positions and has length three. Over a field of characteristic different from two, take $t = \mathrm{diag}(-1, -1, 1)$. It belongs to $T[2]$. Conjugation by the representative swapping positions two and three sends it to $\mathrm{diag}(-1, 1, -1)$, which is different. The representative does normalize $T[2]$, since it acts on the torus by the corresponding permutation.

The course prerequisite guide records the exact supporting statements and full proof routes.

References and exact prerequisite proofs

- Michel Demazure, *Schémas en groupes*, freely accessible Gille–Polo re-edition of 13 October 2024: Exposé XXI, §§3.5 and 5; Exposé XXIII, §§1–3; Exposé XXV. These editions provide comparison material for the root transport, rank-two relations and base classification proved here.
- Brian Conrad, *Reductive group schemes*, §§6.2–6.3 and Appendix D. The argument here gives the rank-two calculations and uses a direct matrix construction in characteristic zero.
- Bas Edixhoven and Matthieu Romagny, *Group schemes out of birational group laws, Néron models*, §3, for the translation-sheaf construction behind Section 8. The construction and proof are written out here.
- The Stacks authors, *Varieties*, Tag 09N9, read in the AI Integrated Stacks Project English source edition, for the one-dimensional affine-open lemma and its proof.
- Milne's freely accessible *Algebraic Groups*, version 2.00 (2015). This exact free author edition provides comparison material; its citations do not replace the programme proofs.

The complete Lie proof route is the preceding Lie lesson: Theorem 1.A (ordered words), Section 5 (root geometry), Theorem 6.2 (rational Serre construction), Section 7 (finite absolutely irreducible highest-weight presentations), Theorem 4.2 (complete reducibility), and Proposition 3.3 (inner derivations). Lemma 6.1 above also gives the polynomial rank-one actions explicitly. The Weyl-group presentation for Sections 2 and 5 is proved in the preceding root-data lesson, Section 3. The group isomorphism theorem is proved in this lesson.

The geometric prerequisites are likewise exact supporting lessons: Flat quotient bootstrap, Theorem 8.2, proving the algebraic-space quotient of an arbitrary-base flat locally finitely presented equivalence relation; Zariski's Main Theorem, the open-into-finite factorization and the normal birational open-immersion consequence, proved in Affine descent, Zariski Main and recognition of spaces, Theorem D7.1 and Corollary D7.4; Limits of schemes and Noetherian approximation, spreading finite-presentation constructions and their affineness; Quasi-coherent cohomology, Sections 4–6 and 9, proving affine vanishing, the finite-cover and filtered-complex comparisons, arbitrary flat cohomology base change and the complete ideal-sheaf affine criterion. Section 9 of the present

lesson also writes out the entire flat-base-change and one-dimensional affine-open proofs. The algebraic-space normalization proof used in Section 9 is present in the preceding rank-one lesson. These are supporting results, not substitutes for the integral group construction in Sections 7–10.

Source credit

The Stacks Project Authors, Tag 09N9, with the consulted varieties chapter.

Lesson 6

Automorphisms, forms and parabolic subgroups

Written by GPT-6.1 Sol (OpenAI) in Codex, Ultra setting, October 2026. Self-checked by the writing AI, GPT-6.1 Sol (OpenAI). Original content dedicated to the public domain under CC0.

A root datum determines a pinned split reductive group, but it does not remove the geometry of its subgroups or the possibility of descent. This lesson treats both. Parabolic subgroups provide the geometry needed to move Borel pairs; the pinned isomorphism theorem then determines all automorphisms; torsors record how these local descriptions descend.

Throughout, a reductive group over a scheme is smooth and affine, with connected reductive geometric fibres. A split group has a constant root datum and trivial root lines. Statements with one fixed datum are made on the corresponding open and closed part of the base. We use the preceding lessons on maximal tori, torus centralizers, root groups and the pinned isomorphism theorem. The supporting descent results are the supporting lessons Faithfully flat descent and Quotients and torsors: effective descent of affine schemes and group laws, and the classification of torsors by nonabelian first cohomology. We give the group-specific arguments here.

1. Parabolic subgroups over a field

Let first k be algebraically closed and G/k reductive. A smooth closed subgroup $P \subset G$ is **parabolic** if G/P is proper. Equivalently, it contains a Borel subgroup. Indeed, the Borel fixed-point theorem applied to the action of a Borel B on the proper quotient gives a fixed point gP, and hence $g^{-1}Bg \subset P$. Conversely, the classification below identifies every subgroup containing B with a cocharacter parabolic, whose quotient is projective.

Fix $T \subset B$, positive roots Φ^+, and simple roots Δ. For $J \subset \Delta$, choose a cocharacter λ such that

$$\langle \alpha, \lambda \rangle = 0 \quad (\alpha \in J), \qquad \langle \alpha, \lambda \rangle > 0 \quad (\alpha \in \Delta \setminus J).$$

Such a cocharacter exists by taking a rational point of the indicated face and clearing denominators. Put $P_J = P_G(\lambda)$, the subgroup on which conjugation by $\lambda(a)$ has a limit at $a = 0$. The root-coordinate construction in *Root data, Weyl chambers and the Bruhat decomposition*, Section 8, gives

$$P_J = L_J \ltimes U_J, \qquad L_J = C_G(\lambda), \tag{1.1}$$

where L_J has roots $\Phi \cap \mathbf{Z}J$, and U_J has the positive roots outside that subsystem. In particular P_J is smooth and connected and contains B.

Theorem 1.1. The parabolic subgroups containing B are exactly the P_J, for the subsets $J \subset \Delta$. Each is its own scheme-theoretic normalizer. Its quotient in G is smooth and projective.

Proof of the classification. Suppose $P \supset B$. By Bruhat decomposition, if a point of the cell Bn_wB belongs to P, then $n_w \in P$ and the entire cell belongs to P. Thus

$$W_P = \{w \in W : n_w \in P\}$$

is a subgroup of W, and P is the union of its Bruhat cells. If $w = s_{i_1} \cdots s_{i_m}$ is reduced, the closure of its cell contains the cells of each s_{i_j}. Here is the geometric reason, including the fact needed about the closure. Form the successive contracted products of the rank-one minimal parabolics P_{i_j} over B. Each rank-one quotient P_i/B is $\mathbf{P}^1$: the positive roots outside the rank-one Levi are in the shared unipotent radical, so the quotient is the rank-one Borel quotient proved in the third lesson. Their quotient by the last B is consequently an iterated $\mathbf{P}^1$-bundle. Multiplication gives a proper map to G/B. The open rank-one cells in the factors multiply isomorphically to Bn_wB/B: their conjugated roots are exactly the distinct inversion roots of the reduced word, and the ordered root-product theorem applies. The image of this proper map is therefore the cell closure. Setting all factors but the j-th to the identity includes $n_{s_{i_j}}B$ in the image. Since P/B is closed, every such simple reflection belongs to W_P.

Let $J = \{\alpha \in \Delta : s_\alpha \in W_P\}$. The preceding argument shows $W_P = W_J$, the subgroup generated by these reflections. On the other hand, $n_w \in P_G(\lambda)$ precisely when $w\lambda = \lambda$. To check this, the torus part of $\lambda(a)n_w\lambda(a)^{-1}$ is the cocharacter $w^{-1}\lambda - \lambda$; it extends to a point of the closed torus at zero only when it is zero. The stabilizer of λ in W is W_J. One can see this without a separate stabilizer theorem: if $w\lambda = \lambda$, choose a simple inversion δ of a nonidentity w. Dominance of λ and negativity of $w\delta$ force $\langle \delta, \lambda \rangle = 0$. Hence $\delta \in J$, and multiplying by s_δ lowers the length while preserving λ. Induction proves the assertion. Consequently P and P_J have the same geometric points. They are reduced closed subgroups, so they are equal.

For the normalizer, let $g \in N_G(P)(k)$. The Borels B and gBg^{-1} of the connected smooth affine group P are conjugate in P, by the field conjugacy theorem. After multiplication by an element of P, g normalizes B, so belongs to $B \subset P$. This proves equality on geometric points. It also proves equality as group schemes: the infinitesimal quotient $\operatorname{Lie} N_G(P)/\operatorname{Lie} P$ consists of P-fixed vectors in $\mathfrak{g}/\mathfrak{p}$: infinitesimal normalization gives $\operatorname{Ad}(p)X - X \in \mathfrak{p}$ for every test-valued $p \in P$. Its invariants under λ are zero, since all the weights in that quotient are strictly negative. Thus the tangent spaces of the normalizer and P agree. Its reduced group is P, and its local dimension and tangent dimension at the identity agree; it is smooth there and, by translation, everywhere. Hence $N_G(P) = P$.

Finally G/P_J is proper: the map $G/B \to G/P_J$ is surjective, and G/B is proper by the field argument in the first lesson. Apply the quotient and projectivity lemma in *Roots and reductive groups of rank one*, Section 5: a smooth connected self-normalizing subgroup with complete homogeneous fibres has a scheme quotient, and the character $\det(\operatorname{Lie} P_J)^{-1}$ defines an ample line bundle. This proves projectivity. Smoothness follows from the smooth P_J-torsor $G \to G/P_J$. □

The ordering is $P_J \subset P_K$ when $J \subset K$. The empty subset gives B; the full subset gives G. Thus allowing G itself as a parabolic adds one point to the parameter scheme.

2. Parabolics and their parameter scheme over a base

Over S, it is useful initially to allow a smooth affine monomorphism $P \to G$ whose geometric fibres are parabolic subgroups. The theorem below shows that this is automatically a closed immersion. This definition therefore does not hide a closedness hypothesis in the conclusion.

Theorem 2.1. A parabolic subgroup of a reductive S-group is, étale locally on S, a standard

cocharacter parabolic. It is closed, has connected fibres, and satisfies $N_G(P) = P$. The functor of parabolic subgroups is represented by a smooth projective S-scheme $\mathrm{Par}(G)$, and formation of this scheme commutes with every base change.

Proof of the local description. Apply the torus-lifting and conjugacy argument of the first lesson to the smooth affine group P. Étale locally, a maximal torus of a chosen geometric fibre lifts to a torus $T \subset P$. It is maximal also in G, by the field classification. Shrink and split T, and make the root datum and root lines constant as in the third and fourth lessons.

The differential of the monomorphism $P \to G$ is a fibrewise injection of vector bundles; hence its image is a locally direct summand of $\mathrm{Lie}\, G$. As a T-module it contains $\mathrm{Lie}\, T$ and either the whole of each root line or none of it. The selection is locally constant. Choose λ from the parabolic pattern in the chosen fibre. All weights of $\mathrm{Lie}\, P$ have nonnegative pairing with λ. The dynamic subgroup $P_P(\lambda) \subset P$ is smooth and has exactly those nonnegative tangent weights, by the graded-coordinate proof in the third lesson. Its inclusion is therefore étale; on every geometric fibre it is an isomorphism, since that fibre is the connected parabolic with this pattern. It follows that $P_P(\lambda) = P$. The limit functor now factors $P \to G$ through $P_G(\lambda)$. This is a monomorphism between smooth schemes, with an isomorphism on every geometric fibre; its differential is an isomorphism, so it is an étale, surjective monomorphism, and hence an isomorphism. In particular the original $P \to G$ is closed.

The formal separation and normalizer lemmas in the rank-one lesson show that $N_G(P)$ is closed of finite presentation and commutes with every base change. Their proof uses a finite prime filtration, Artin–Rees and the determinant trick to detect subgroup equations over a universal Noetherian coefficient ring; it therefore also applies on test schemes with nilpotents. Its geometric fibres are the normalizers calculated in Theorem 1.1.

We use the flat-source criterion proved in the rank-one lesson directly over the original base. Locally a finitely presented closed immersion $X \hookrightarrow Y$ over $\mathrm{Spec}\, R$ has a finitely generated ideal $I \subset C = \mathcal{O}(Y)$. If C/I is R-flat, tensoring $0 \to I \to C \to C/I \to 0$ with $\kappa(s)$ makes $I \otimes_R \kappa(s) \to C \otimes_R \kappa(s)$ injective. Equality of every geometric fibre makes this ideal zero, first after a faithful field extension and then over $\kappa(s)$. For a prime $x \subset C$ over s, this says $I_x = \mathfrak{p}_s I_x$; the finitely generated C_x-module I_x vanishes by Nakayama. Hence $I = 0$. Apply this to $P \hookrightarrow N_G(P)$, using the fibre equality in Theorem 1.1 and flatness of P/S. Neither Noetherianity of the base nor flatness of the normalizer is required.

The quotient G/P is therefore smooth and projective by the quotient lemma just used, now over S. Its ample line bundle comes canonically from $\det(\mathrm{Lie}\, P)^{-1}$.

For a pinned split G of constant datum, we obtain

$$\mathrm{Par}(G) = \coprod_{J \subset \Delta} G/P_J. \tag{2.1}$$

Here a morphism to the disjoint union may choose different J's on open and closed parts. To see that (2.1) represents the subgroup functor on every test scheme, a section of G/P_J gives a conjugate of P_J after a local lift to G. Conversely the preceding local classification supplies such lifts for any parabolic. Two lifts give the same subgroup exactly when their quotient lies in $N_G(P_J) = P_J$, so the sections glue uniquely. This proves the functorial assertion, including uniqueness over bases with nilpotents.

For an arbitrary reductive G, the local schemes just constructed have unique identifications on overlaps, because they represent the same subgroup functor. They satisfy the cocycle condition. The universal subgroup $\mathcal{P}$ gives the canonical line bundle $L = \det(\mathrm{Lie}\, \mathcal{P})^*$; its identifications also satisfy the cocycle. We show directly that these projective schemes with their line bundle descend to a projective scheme.

Work on a quasi-compact affine part of S and take finitely many affine étale charts $S_i \to S$ carrying the split descriptions. Their disjoint union is a faithfully flat quasi-compact cover. On each

chart the Grassmannian construction of AG-RG-03 Lemma 5.A, applied in its Lemma 5.1, gives a positive power of L_i and a specific closed Plücker embedding on every parabolic component. Choose a common exponent and place these finitely many components in disjoint coordinate blocks of one projective bundle, using the direct sum of their embedding modules. This gives a closed projective embedding of their disjoint union. Descend the charts, projective schemes, these specified embeddings, line bundles and their identifications to a common Noetherian coefficient stage. S02 Proposition 3.2 retains smoothness and the line bundles, and finite-presentation descent retains the closed embeddings and their finitely many identities. Thus ampleness on the model is witnessed by retained embeddings, rather than inferred from fibres of the original base.

Choose a common multiple $b > 0$ of the finitely many embedding exponents. Its sufficiently large multiples m are still very ample, by the polynomial Veronese maps. S01 Theorem 8.5 makes the positive cohomology of L_i^m vanish over each affine Noetherian model chart for sufficiently large m. To pass to all geometric fibres, use the universal finite projective complex of S02 Lemma 5.1, in degrees $[0, N]$, for this smooth projective family. Its positive cohomology is zero over that affine chart. Starting at the last term, its exact positive tail splits because the last nonzero term is projective; repeat to remove the whole positive tail. The remaining degree-zero module is finite projective, and the splitting survives every tensor. Thus $H^q(X_{i,\bar{s}}, L_i^m) = 0$ for every $q > 0$ on every geometric model fibre. A single sufficiently large common multiple works on the finitely many charts. Pull this choice back to the original cover. S02 Proposition 5.2 gives $E_i = f_{i*}L_i^m$ finite locally free, with every base change. The overlap identifications of schemes and L_i therefore identify the E_i compatibly. Faithfully flat module descent produces a finite locally free E/S.

Evaluation defines compatible closed immersions $X_i \hookrightarrow \mathbf{P}(E)|_{S_i}$. Their complete section systems contain the sections of the chosen very ample systems, so on each generating target chart the coordinate-ring map is still surjective. This first gives an immersion; the map is also proper by the graph argument of S01 Lemma 8.3, since X_i/S_i is projective. Its image is closed, and the chartwise closed immersions together with the empty inverse image off that closed image make the immersion closed globally. On affine charts of $\mathbf{P}(E)$, the ideal sheaves of these immersions have compatible faithfully flat descent data. Module descent gives their ideal submodules, and multiplication preserves them because that fact is checked on the faithfully flat cover. Their quotient algebras define a closed S-subscheme $X \subset \mathbf{P}(E)$, whose pullback is each X_i. The represented subgroup functor and smoothness descend through the same cover, so $X = \mathrm{Par}(G)$ is smooth and projective. The construction is independent of charts by its representing identity and hence glues over S. Finally the subgroup functor itself commutes with every base change; Yoneda identifies the pullback of its representing scheme with $\mathrm{Par}(G_{S'})$. □

For a nonsplit group, subsets of a chosen simple-root set need not be globally labelled. The finite étale **type scheme** is the descent of the finite set of subsets of Δ. The map from $\mathrm{Par}(G)$ to this scheme remembers the type. Borel subgroups are its component of minimal parabolics, corresponding locally to $J = \varnothing$.

3. Automorphisms of a pinned split group

Fix a pinned split group

$$(G, T, B, (E_\alpha)_{\alpha\in\Delta})$$

of based root datum $\mathcal{R}_b$. Let $\Gamma = \mathrm{Aut}(\mathcal{R}_b)$, including its action on the full character and cocharacter lattices. Regard it as a constant group scheme. The pinned isomorphism theorem gives a unique pinned automorphism s_γ inducing each $\gamma \in \Gamma$; uniqueness gives $s_\gamma s_\delta = s_{\gamma\delta}$.

Write $G^{\mathrm{ad}} = G/Z(G)$. This is the adjoint semisimple quotient: central torus directions disappear, but they remain present in Γ.

Theorem 3.1. On all S-schemes, including nonreduced ones, the automorphism functor is represented by

$$\mathrm{Aut}(G) \simeq G^{\mathrm{ad}} \rtimes \Gamma_S, \qquad (g,\gamma) \longmapsto \mathrm{Int}(g)s_\gamma. \tag{3.1}$$

This group scheme is smooth, separated and locally of finite presentation. It need not be of finite type.

Proof. Put $q : G \to G^{\mathrm{ad}}$ and $T^{\mathrm{ad}} = q(T)$. The scheme-centre calculation in RG03 Section 1 gives $Z(G) = D_S(X/\mathbf{Z}\Phi)$. The central quotient theorem of RG03 Theorem 1.1 and RG04 Theorem 7.1 therefore gives the fppf quotient q, the torus $T^{\mathrm{ad}} = D_S(\mathbf{Z}\Phi)$ and the images of all root parameters with their unchanged frames. In particular $\mathbf{Z}\Phi = \bigoplus_{\alpha\in\Delta} \mathbf{Z}\alpha$ is its character lattice. Conjugation by a section of G^{ad} is well defined on G: lift the section through q fppf locally, conjugate there, and descend the group morphisms by S04 Lemma A1.5. Two lifts differ by a section of the scheme centre and give the same morphism on every test scheme. Each s_γ preserves this centre and induces s_γ^{ad} on the quotient. The morphism in (3.1) respects multiplication, since

$$\big(\mathrm{Int}(g)s_\gamma\big)\big(\mathrm{Int}(h)s_\delta\big) = \mathrm{Int}\big(g\, s_\gamma^{\mathrm{ad}}(h)\big)s_{\gamma\delta}.$$

Let a be an automorphism after any base change $S' \to S$. Theorem 2.1 identifies the Borel aB with a section of G/B. Pulling back the fppf B-torsor $G \to G/B$ permits us locally to conjugate aB to B. Its transported aT is a maximal torus of B. The torus transporter in this smooth affine B is smooth by RG01 Lemma 3.1; its geometric fibres are nonempty by the solvable-group torus-conjugacy proof in RG01 Section 2. It is consequently a smooth surjection, and its own tautological section supplies an fppf cover on which a further conjugation in B takes this torus to T. We have reduced to an automorphism b preserving both T and B.

RG01 Lemma D.1 identifies $b|_T$ with a locally constant automorphism of the full integral character lattice, including its central directions. Preservation of B and the canonical root and coroot maps identifies its value with a section γ of $\Gamma_{S'}$. A locally constant value may be treated on its open-and-closed fibres, without assuming that there are finitely many such values. Set $\beta = s_\gamma^{-1}b$. This automorphism induces the identity on the entire based datum. On each framed simple root line it has the form $d\beta(E_\alpha) = u_\alpha E_\alpha$, with $u_\alpha \in \mathcal{O}_{S'}^\times$: an automorphism of a free rank-one module is multiplication by a unit. Because the simple roots are a basis of $X^*(T^{\mathrm{ad}})$, the rule $\alpha(t) = u_\alpha^{-1}$ defines one and only one section t of T^{ad}. Conjugation by it has precisely these multipliers on the root frames, by the quotient root-equivariance formula of RG04 Theorem 7.1. Thus $\mathrm{Int}(t)\beta$ fixes the pinning and induces the identity on its full based datum. RG05 Theorem 5.1 makes it the identity automorphism. Undoing the conjugations and using the displayed multiplication identity writes $a = \mathrm{Int}(g)s_\gamma$ fppf locally. This proves local surjectivity.

We prove injectivity on the same arbitrary test scheme. Suppose $\mathrm{Int}(h)s_\gamma = 1$. Then $\mathrm{Int}(h) = s_\gamma^{-1}$ preserves T, B and therefore their quotient torus and Borel in G^{ad}. RG04 Theorem 4.2, with the pinned representatives, identifies the entire normalizer of T^{ad} with the copies $n_w T^{\mathrm{ad}}$ labelled by the constant Weyl sheaf. Hence h has a locally constant Weyl label w. Preservation of the Borel preserves its positive root lines, so this label preserves the positive chamber. RG04 Lemma 3.A gives $w = 1$. The identity-labelled copy is the torus itself as a scheme; thus $h \in T^{\mathrm{ad}}(S')$, including when S' has nilpotents. Its inner automorphism is the identity on T, so $\gamma = 1$ on the entire datum. The equation now says that every simple frame is fixed, and therefore $\alpha(h) = 1$ for every $\alpha \in \Delta$. Their character-basis property gives $h = 1$. The kernel is trivial on every test scheme, which proves injectivity of the group-functor map.

The locally constructed factors of a consequently agree on double overlaps. S04 Lemma A1.5 glues them uniquely as sections of G^{ad} and Γ_S, and the same lemma descends their equality with a. Every construction commutes with base change, so these sections give the inverse natural transformation to (3.1). The representing scheme is the disjoint union, indexed by Γ, of copies of the smooth affine

finitely presented G^{ad}. It is smooth and locally of finite presentation. Its relative diagonal is closed: on each pair of indexed components it is either the closed diagonal of G^{ad} or empty. Thus it is separated as well. This proves all the assertions. □

For a semisimple datum, Γ is finite: its roots span the rational character space, so their images determine a rational linear map and its restriction to the free character lattice. An automorphism is therefore determined by its permutation of the finite root set, which injects Γ into that finite permutation group. For a torus of rank r, $G^{\mathrm{ad}} = 1$ and $\Gamma = \mathrm{GL}_r(\mathbf{Z})$, which is infinite for $r \geq 2$. Thus the finite-type assertion cannot be added in general.

Even a small central direction matters. On GL_2,

$$g \longmapsto \det(g)^{-1} g \tag{3.2}$$

is an involutive automorphism. On the diagonal torus it sends (t_1, t_2) to (t_2^{-1}, t_1^{-1}). It preserves the positive root and reverses the scalar direction, so is visible in the full based datum. It is not inner, since it acts nontrivially on the centre.

4. Forms and torsors

A **form** of G_0/S is an S-group which becomes isomorphic to G_0 on an étale cover. Let $A = \mathrm{Aut}(G_0)$.

Proposition 4.1. Isomorphism classes of forms of G_0 are in natural bijection with $H^1_{\mathrm{et}}(S, A)$.

Proof. Choose local isomorphisms $\phi_i : G_0|_{S_i} \to G|_{S_i}$. On overlaps,

$$a_{ij} = \phi_i^{-1}\phi_j, \qquad a_{ij}a_{jk} = a_{ik}.$$

Changing the ϕ_i's changes this cocycle by a coboundary. Equivalently, $\mathrm{Isom}(G_0, G)$ is a right A-torsor: after choosing a local isomorphism, every other one is its composite with a unique automorphism of G_0. Conversely an A-torsor has local sections on an étale cover; their transition automorphisms give a descent datum on the affine scheme G_0. S04, Lemmas A1.2 and A1.5, give effective descent of its coordinate algebras and compatible morphisms. The group operations descend with these morphisms, and their identities follow by faithful descent. Thus one obtains an affine S-group with the prescribed local isomorphisms. Its finite presentation and smoothness descend by S04, Lemma A1.3a.

For the remaining fibre assertion, take a geometric point $\mathrm{Spec}\,\Omega \to S$, with Ω algebraically closed. The pulled-back étale cover is nonempty. A nonempty affine chart of it is of finite type over Ω, so a maximal ideal gives an Ω-point by AG-CA-06, Theorem 2.1, whose field-algebra proof is Theorem 1.3 there. Pulling a covering isomorphism back along that point identifies the descended geometric fibre with $(G_0)_\Omega$. Hence that fibre is connected and reductive. This proves reductivity over every geometric point of the arbitrary base.

The two constructions recover each other's cocycles and local isomorphisms. Compatibility and uniqueness in S04, Lemma A1.5, identify their descended objects and morphisms, so they are mutually inverse on isomorphism classes. This is the concrete torsor argument from Quotients and torsors, applied to the automorphism sheaf A. □

For split pinned G_0, an **inner form with inner structure** is obtained from a G_0^{ad}-torsor by conjugation. These structures are classified by $H^1(S, G_0^{\mathrm{ad}})$. Forgetting the inner structure need not be injective on isomorphism classes. The exact sequence split by the pinning is

$$1 \longrightarrow G_0^{\mathrm{ad}} \longrightarrow A \longrightarrow \Gamma_S \longrightarrow 1. \tag{4.1}$$

The fibre over the neutral element of $H^1(S, \Gamma_S)$ is the set of $\Gamma(S)$-orbits on $H^1(S, G_0^{\mathrm{ad}})$. Indeed a

trivialization of the induced Γ-torsor reduces the original cocycle to G_0^{ad}; changing that trivialization by γ changes the inner cocycle by its γ-action. This also explains why a marked inner structure and the underlying group are different classification problems.

5. Quasi-split forms

A reductive group is **quasi-split** if it admits a Borel subgroup over the base. Twisting the pinned G_0 by a Γ-torsor gives a quasi-split form: the pinned automorphisms preserve T, B and the simple-root frames as an indexed collection, so these data descend.

Over a field, this accounts for every quasi-split form of G_0, up to the corresponding based-datum form. We supply the cohomological step because the torus which occurs in it is usually twisted.

Proposition 5.1. For a field k, the forms of G_0 that admit a k-Borel are exactly the forms obtained by twisting its pinning by Γ-torsors. Their based-datum classes lie in $H^1(k, \Gamma)$.

Proof. Start with its local group isomorphisms. By Theorem 2.1 the transported Borel is locally a conjugate of B; the smooth Borel torsor has étale local sections by S07 Lemma 2.1. Refining the affine cover and making these conjugations gives isomorphisms of the pairs. Their overlap automorphisms preserve B. In Theorem 3.1 write such an automorphism as $\mathrm{Int}(g)s_\gamma$. The pinned factor preserves B, so the inner factor must too. The inverse image of B^{ad} in G_0 is exactly B, since the centre is in its torus and the quotient root maps are unchanged. The all-test normalizer equality of Theorem 2.1 applied to the adjoint Borel therefore forces $g \in B_0^{\mathrm{ad}}$. Conversely its sections locally lift to B_0 and preserve B. This proves the entire stabilizer identity, including nonreduced overlap algebras, namely

$$\mathrm{Aut}(G_0, B) = B_0^{\mathrm{ad}} \rtimes \Gamma.$$

Fix its Γ-torsor. Relative to the canonical pinned twist, the possible reductions are torsors under the corresponding twisted B_0^{ad}. Its torus has character lattice freely based by Δ, with the descent action permuting that basis. Descend the finite set Δ, or equivalently the algebra of functions on it: on a splitting affine cover this algebra is the finite product of copies of the covering ring, with overlap permutations of the factors. S04 Lemmas A1.2–A1.3 descend it to a finite étale k-algebra E. The complete finite-étale field calculation of S07 Lemma 5.3A writes $E = \prod_i k_i$ for finite separable fields k_i/k. The group of units of $E \otimes_k R$ is represented by the determinant-invertibility open in the affine space of its underlying finite-dimensional vector space: multiplication by an element is invertible exactly when that determinant is a unit. On the splitting cover it is the product of multiplicative groups, with exactly the same permutation descent datum as the torus. S04 Lemma A1.5 therefore identifies the whole torus functor with this unit group. In particular the torus is

$$T = \prod_i \mathrm{Res}_{k_i/k} \mathbf{G}_m.$$

We prove the vanishing needed for this torus. A torsor under $\mathrm{Res}_{k_i/k} \mathbf{G}_m$ has transition functions which are units in $k_i \otimes_k R$ on a splitting cover $\mathrm{Spec}\, R \to \mathrm{Spec}\, k$. The same functions glue a rank-one module over k_i, by faithfully flat module descent. Conversely a rank-one k_i-module has precisely these local frames and transition functions, so these two constructions are inverse. Every rank-one module over the field k_i has a basis. Its frame torsor has a section, hence is trivial. Taking the product gives $H^1(k, T) = 1$. This is the multiplicative descent statement often called Hilbert 90, with its proof and its restriction-of-scalars application included here.

The root-height filtration from RG04 Section 5 descends to the twisted unipotent radical U, since a based-datum automorphism preserves positive-root height. The root-coordinate commutator proof makes each quotient U_h/U_{h+1} the additive group of the direct sum of its height-h root lines,

with linear permutation and frame transition maps. S04 Lemma A1.2 descends this module to a finite-dimensional k-vector space V_h. Thus the quotient really is a vector group, rather than an assumed split form of a unipotent group.

We prove additive torsor vanishing directly on a faithfully flat affine cover $k \to R$. For a vector group V its transition cocycle is $c \in V \otimes_k R \otimes_k R$, satisfying the alternating unit-insertion equation $dc = 0$ in $V \otimes_k R^{\otimes 3}$. The entire augmented complex in S04 Lemma A1.1 is exact in every degree: after tensoring with R, multiplication of the extra factor with the first covering factor is its displayed contracting homotopy, and faithful flatness detects exactness downstairs. Consequently $c = db$ for $b \in V \otimes_k R$. Translation of the local torsor section by $-b$ makes its two restrictions equal; they descend to a global section. Hence every vector-group torsor over k is trivial in every characteristic. This uses no division by a Galois-group order and no unproved Galois-cohomology comparison.

Apply this successively to a U-torsor. On an affine faithfully flat cover splitting the root data and trivializing the torsor, its cocycle lies in $U(R \otimes_k R)$. Its image in V_1 is db_1 by the preceding calculation. The ordered root coordinates give a lift of b_1 to $U(R)$: put its height-one entries in their root factors and every higher entry zero. Changing the local section by this lift puts its cocycle in U_2. At height h the same split coordinate product lifts every $b_h \in V_h(R)$ to $U_h(R)$, and removes that quotient cocycle. All filtration subgroups and maps have already descended, so each changed cocycle lies in the stated subgroup on the entire overlap algebra. The finite filtration ends at one. The final equal local sections descend, proving $H^1(k, U) = 1$. The same affine cover can be used throughout; if initially several étale affine charts are needed, their finite disjoint union supplies one faithfully flat algebra. Such a finite choice exists because their nonempty images cover the one-point scheme $\operatorname{Spec} k$.

First project a twisted Borel cocycle to T, whose torsor has already been proved trivial. On a splitting affine cover the chosen torus coboundary lifts through the split semidirect product $B^{\mathrm{ad}} = T \ltimes U$; changing the section by this lift leaves a U-cocycle. Its vanishing just proved trivializes the whole Borel torsor. Thus $H^1(k, B_0^{\mathrm{ad}}) = 1$ for the Borel twisted by the fixed Γ-torsor.

There is therefore no further Borel-preserving inner twist. The Γ-class is recovered from the canonical quotient of the full automorphism group, so it is independent of the chosen Borel. The pinning provides a section on cohomology, and the vanishing just proved leaves exactly that section as the quasi-split class. Thus the based-datum class determines the quasi-split group. □

Over a general base, the analogous vanishing can fail. Here is the whole construction of the example. Choose local frames of a line bundle L, with unit transitions ℓ_{ij}. Glue the all-characteristic PGL_2 of Lemma 6.0 by conjugation with $\operatorname{diag}(1, \ell_{ij})$. These conjugations satisfy the triple cocycle, so affine algebra/group-law descent in S04 makes a smooth affine reductive group denoted $\mathrm{PGL}(\mathcal{O}_S \oplus L)$. The upper triangular subgroups and diagonal tori are preserved and descend too. The former stabilize $\mathcal{O}_S$ and are Borels by their geometric fibres. The two off-diagonal tangent entries are the homomorphism bundles $\operatorname{Hom}(L, \mathcal{O}_S) = L^{-1}$ and $\operatorname{Hom}(\mathcal{O}_S, L) = L$: in local frames conjugation multiplies E_{12} by ℓ_{ij}^{-1} and E_{21} by ℓ_{ij}. These are exactly the two canonical root lines, by Lemma 6.0. A pinning for this particular Borel pair requires a global generator of its positive root line. Such a generator would identify that line with $\mathcal{O}_S$, by the local unit criterion; if L is nontrivial neither L nor its inverse admits it. This example concerns the displayed Borel pair and does not assert that every other pair lacks a pinning. The field proof used triviality of rank-one modules over a field, which fails for general line bundles.

6. Inner forms of the special linear group

A **central simple algebra** over a field k is a finite-dimensional simple algebra with centre k. Its matrix form and the separability of its splitting field will be proved below. We first identify the automorphisms that carry its descent data.

Lemma 6.0 (the matrix groups in every characteristic). Over any base, the matrix groups GL_n, SL_n and the scalar quotient PGL_n are split reductive with their diagonal tori and their usual A_{n-1} roots and coroots. The determinant-one group has character lattice $\mathbf{Z}^n/\mathbf{Z}(1,\dots,1)$ and cocharacter lattice $\{(a_i) : \sum a_i = 0\}$; the projective group interchanges them.

Proof. The determinant open is a smooth affine group over $\mathbf{Z}$. On the determinant-one equation the cofactors form the entries of the inverse transpose, so some cofactor is a unit locally; solve for its matrix entry on that open. This is the full one-equation smooth Jacobian chart over every base, as in AG-CA-17 Theorem 5.1. All matrix operations and determinant identities are the finite polynomial calculations of S06 Section 1. On every algebraically closed geometric field, determinant-one row and column elimination uses a nonzero pivot, the two-coordinate swap from RG05 (1.1), and elementary additions to clear its row and column. Induction leaves a determinant-one diagonal. Each diagonal pair $\mathrm{diag}(d, d^{-1})$ is the four-root product in RG05 Section 10. Thus SL_n is generated on these field points by $I + uE_{ij}$. Every such one-parameter image is connected and contains one. The full component and Nullstellensatz argument in RG05 Lemma 9.A therefore proves geometric connectedness; the determinant open for GL_n is irreducible because its polynomial coordinate ring embeds in a field, with intersecting nonempty principal opens, by the prime/irreducibility proof of AG-CA-08.

The diagonal Laurent algebras and GS05 Theorem 2.2 give the stated character and cocharacter lattices. Their full torus weights on the tangent matrices are $e_i - e_j$ on E_{ij}, with diagonal zero-weight part, trace zero for SL_n. A smooth connected normal unipotent subgroup N has trivial scheme intersection with the torus by RG01 Lemma 2.A and GS05 Theorem 4.2. On dual numbers the tangent intersection is the tangent space of this scheme intersection, since its equations are the sum of the two ideals. Thus $\operatorname{Lie} N$ has zero diagonal intersection. Its torus character projections, given by the complete exponent-basis grading of GS05 Lemma 2.1, either give zero or include some E_{ij}. Normality makes it a Lie ideal: conjugate $1 + \epsilon X$ by $1 + \eta Y$, with $\epsilon^2 = \eta^2 = 0$, and compare the $\epsilon\eta$ coefficient, exactly as in RG05 Section 6. Bracketing E_{ij} with E_{ji} gives $E_{ii} - E_{jj}$, nonzero even in characteristic two, contradicting the zero diagonal intersection. Hence $\operatorname{Lie} N = 0$. S07 Lemma 5.3A makes the smooth connected zero-dimensional group trivial. This proves reductivity in every characteristic.

A torus containing the diagonal one has its Lie algebra in the displayed zero-weight space, so the ranks are equal. The closed inclusion of these tori corresponds by GS05 Theorem 4.2 to a surjection of free character groups of equal rank. Its kernel is a free subgroup of rank zero, by the integer subgroup proof in RG05 Theorem 10.1, and is zero. Thus the tori coincide. The two-coordinate SL_2 maps, polynomial over $\mathbf{Z}$, give the root parameters $I + uE_{ij}$ and coroot $e_i - e_j$; RG03 Theorems 4.1 and 7.1 identify them with the canonical maps, including on nonreduced tests. The upper triangular subgroup is the Borel for the displayed positive roots by RG04 Section 5, and the $E_{i,i+1}$ give its global simple-root frames.

The central scalar $\mathbf{G}_m$ is closed on coordinate equations. RG03 Theorem 1.1 constructs its smooth affine reductive quotient PGL_n. RG04 Theorem 7.1 gives its diagonal quotient torus, with character lattice $\{(a_i) : \sum a_i = 0\}$ and dual lattice $\mathbf{Z}^n/\mathbf{Z}(1,\dots,1)$, and the same root maps with the coroot images. These operations commute with arbitrary base change. For $n = 1$, $\mathrm{GL}_1 = \mathbf{G}_m$, while the determinant-one and scalar quotient groups are trivial, so the same statement has empty roots. □

Lemma 6.1. The automorphism functor of the matrix algebra is PGL_n, over any base ring. Thus PGL_n-torsors correspond to degree-n Azumaya algebras.

Proof. Let ϕ be an algebra automorphism of $\mathrm{Mat}_n(R)$. Its images $p_i = \phi(E_{ii})$ of the diagonal matrix units are orthogonal idempotents, and $R^n = \bigoplus_i p_i R^n$. Each summand is finitely generated projective, since it is the image of an idempotent. It has rank one on every fibre: after a residue-field extension, the corner $p_i \operatorname{Mat}_n p_i$ is the endomorphism algebra of the corresponding vector-space image, whereas ϕ identifies this corner with $E_{ii} \operatorname{Mat}_n E_{ii}$, of dimension one. If the image has dimension r_i, its corner has dimension r_i^2, so $r_i = 1$. Local freeness of finite projective modules,

proved in the preceding algebra lesson, now makes these rank-one locally direct summands over R itself. Locally choose a generator v_1 of the first summand, and put $v_i = \phi(E_{i1})v_1$. The matrix-unit identities show that the v_i's are a basis and that $\phi(E_{ij})$ sends v_j to v_i and kills the other v_h's. Hence ϕ is conjugation by the basis matrix. Two such matrices differ by a scalar, because a matrix commuting with every matrix unit is scalar. This proves the functor identity, with local lifts and descent through $\mathrm{GL}_n \to \mathrm{PGL}_n$. Twisting the matrix algebra by a torsor gives its Azumaya algebra; conversely the sheaf of local algebra isomorphisms is that torsor. Faithfully flat descent of the algebra and its multiplication proves effectivity. $\square$

Theorem 6.2 (separable matrix splitting). Let k be any field and let A be a finite-dimensional simple k-algebra with centre k. Then $\dim_k A = n^2$ for some n, and there is a finite separable extension K/k with $A \otimes_k K \simeq \mathrm{Mat}_n(K)$. Conversely, an algebra obtained by faithfully flat descent of Mat_n over a field is central simple. Its algebra-isomorphism torsor has group PGL_n.

Proof. We first prove the finite-dimensional density calculation needed here. Let R act on a finite-dimensional simple k-module V, and suppose $\mathrm{End}_R(V) = k$. For linearly independent $v_1, \dots, v_m$, the map

$$R \longrightarrow V^m, \qquad r \longmapsto (rv_1, \dots, rv_m)$$

is surjective. For $m = 1$ this is simplicity. Induct on m. The image $L \subset V^m$ projects onto V^{m-1}. Its kernel in the last summand is an R-submodule of V, hence is either V or zero. The first case gives surjectivity. In the second case L is the graph of an R-linear map $V^{m-1} \to V$. Each component of this map is an R-endomorphism of V, hence multiplication by a scalar c_i. Applying the graph equation to $r = 1$ gives $v_m = \sum_{i<m} c_i v_i$, a contradiction. Taking a basis of V proves that the action map $R \to \mathrm{End}_k(V)$ is surjective.

Apply this calculation to $R = A \otimes_k A^{\mathrm{op}}$ acting on $V = A$ by $(a \otimes b)x = axb$. Its submodules are the two-sided ideals of A, so V is simple. An endomorphism commuting with left multiplication is $x \mapsto xc$, where c is its value at 1; commuting also with right multiplication forces $c \in Z(A) = k$. Density and equality of dimensions therefore give an isomorphism

$$A \otimes_k A^{\mathrm{op}} \xrightarrow{\sim} \mathrm{End}_k(A). \tag{6.2}$$

Tensoring this linear isomorphism with any field extension E/k gives the same isomorphism for A_E. A two-sided ideal of A_E is stable under every endomorphism of its underlying E-vector space, so is zero or all of A_E. Also $Z(A_E) = E$: centrality is the kernel of the finite family of linear commutator maps with a k-basis of A, and kernels commute with field extension. Thus central simplicity survives every field extension.

Now take $E = \overline{k}$. Choose a proper left ideal of A_E of largest vector-space dimension; such a choice exists among the integers from zero to $\dim_E A_E - 1$, and zero is proper. Its quotient V is nonzero and simple, since a larger proper submodule would give a larger proper ideal. An endomorphism of V has submodules as its kernel and image; if nonzero, simplicity makes it bijective, and its linear inverse remains A_E-linear by applying its defining intertwining identity. Thus its endomorphism algebra is a finite-dimensional division algebra over the algebraically closed field E. Every one of its elements is scalar: its characteristic polynomial as an E-linear operator has a root c, so the operator minus c is singular; in a division algebra this makes it zero. Density gives $A_E \to \mathrm{End}_E(V)$ surjective, and simplicity makes its kernel zero. Consequently $A_E \simeq \mathrm{Mat}_n(E)$, where $n = \dim_E V$, and $\dim_k A = n^2$.

To obtain a *separable* splitting, consider the scheme $I = \mathrm{Isom}_{k\text{-alg}}(A, \mathrm{Mat}_n)$. Choose bases of the two algebras. A linear isomorphism is a matrix with invertible determinant; preservation of 1 and of the products of basis vectors is a finite list of polynomial equations. Thus I is an affine scheme of finite presentation. Its base change to $\overline{k}$ has a point by the preceding paragraph. Composition with that point identifies the entire scheme $I_{\overline{k}}$ with the automorphism scheme of the matrix algebra,

hence with $\mathrm{PGL}_{n,\overline{k}}$ by Lemma 6.1. With the quotient convention for projective bundles, PGL_n is the determinant-nonvanishing open in $\mathbf{P}(\mathrm{Mat}_n^\vee)$, which parameterizes lines in the matrix module: the determinant is a homogeneous section of degree n, and invertible matrices modulo a unit scalar have exactly these projective matrix classes. On a chart where a matrix entry is a unit, normalize that entry to one; this gives a local section of $\mathrm{GL}_n \to \mathrm{PGL}_n$, and its inverse image is the product of that chart with $\mathbf{G}_m$. Thus the open projective scheme represents the quotient on all test rings. It is smooth over every base because projective-space charts are affine spaces and smoothness is preserved under open restriction. In particular $I_{\overline{k}}$ is smooth. The faithful field-extension descent proof in Supporting group-scheme proofs, Lemma G.2.1, makes I/k smooth, without a perfectness or characteristic hypothesis.

A nonempty smooth finite-type scheme over k has a point over a finite separable extension. Here is the precise reason, including for imperfect k. Over a separable closure k^s, choose a nonempty open with an étale map to affine space, as supplied by smooth coordinates. Its image is open. The separably closed field k^s is infinite: if it had q elements, multiplication by a nonzero a permutes all nonzero elements and cancellation of their product gives $a^{q-1} = 1$. Thus every element satisfies $a^q = a$. The polynomial $X^q - X - 1$ would have no root, although its derivative is -1 and any irreducible factor defines a finite separable extension, a contradiction. This infinite field has a rational point in every nonempty open of affine space: a nonzero polynomial cannot vanish on all its tuples, by induction on the number of variables. The nonempty étale fibre over this point has a closed point finite separable over k^s, hence rational because k^s is separably closed. All coordinates of this point and of the chosen charts lie in a finite separable subextension of k^s/k. Applying this to I supplies the asserted K and splitting.

Finally, for an algebra B that becomes a matrix algebra after a faithfully flat field extension, a nonzero proper two-sided ideal would remain a nonzero proper ideal after extension. Matrices have no such ideal: multiplying a nonzero entry on its two sides by matrix units produces all matrix units. Centrality again consists of linear equations, so $Z(B) \otimes E = Z(B_E) = E$ and $Z(B) = k$. The isomorphism sheaf is locally the automorphism group of matrices, and is therefore the PGL_n-torsor of Lemma 6.1. Effective descent of its algebra operations is precisely the affine descent already proved in *Quotients and torsors*. This proves both directions, without a perfectness or characteristic hypothesis. □

This is the finite-dimensional density method for central simple algebras, followed by smooth descent of their matrix-isomorphism scheme. Compare the Stacks authors' splitting-field theorem, which also proves existence of a separable splitting field. The argument above proves the precise input needed for inner forms; Tsen's theorem and a computation of a field's entire Brauer group are not needed here.

Over the finite separable splitting extension in Theorem 6.2, A becomes Mat_n. The determinant descends to its **reduced norm** $\mathrm{Nrd} : A^\times \to \mathbf{G}_m$. Put

$$\mathrm{SL}_1(A) = \ker(\mathrm{Nrd}). \tag{6.1}$$

Conjugation of matrices acts simultaneously on the algebra and on SL_n. Twisting both by the same torsor shows that every marked inner form is (6.1), and every such algebra gives one. The reduced norm is well defined because the determinant is invariant under conjugation, so its local definitions agree on overlaps. This proves the assertion in arbitrary characteristic.

For $n \geq 3$, transpose-inverse represents the nontrivial diagram class of SL_n. Indeed it negates the diagonal character lattice and interchanges the upper and lower Borels; conjugating by a longest-element representative returns the upper Borel and reverses the A_{n-1} simple-root chain. Any automorphism of this chain is identity or reversal: the image of one end fixes each successive neighbour, so determines the entire chain. Its simple roots span the rational character space; hence their images determine the entire lattice automorphism. The full outer group is therefore this order-two group. The all-ring rank-two and noninner calculations are written in Exercise 1. Its

action on the algebra torsor replaces A by A^{op}. Indeed, transpose identifies matrix multiplication with the opposite multiplication, and conjugation by g becomes conjugation by g^{-T}. Thus, if inner markings are forgotten, the resulting algebras are identified by $A \leftrightarrow A^{\mathrm{op}}$. The underlying groups are isomorphic by $a \mapsto a^{-1}$, viewed as a homomorphism from $A^{\times}$ to $(A^{\mathrm{op}})^{\times}$. For $n = 2$ transpose-inverse is inner; the marked classification is still by the degree-two algebras.

7. Flags and the general linear group

Let $n = n_1 + \cdots + n_r$, with all $n_i > 0$, be an ordered composition. The corresponding standard parabolic consists of invertible block upper triangular matrices with block sizes $n_1, \ldots, n_r$. It stabilizes the partial flag with dimensions

$$d_j = n_1 + \cdots + n_j \quad (1 \le j < r).$$

The order matters: the subset of cuts in $\{1, \ldots, n-1\}$, not an unordered partition, is the parabolic type containing the fixed upper triangular Borel.

For an S-scheme S', let $\mathrm{Flag}_{d_1,\ldots,d_{r-1}}(S')$ consist of chains

$$0 \subset F_{d_1} \subset \cdots \subset F_{d_{r-1}} \subset \mathcal{O}_{S'}^n$$

of locally direct summands of the indicated ranks, with locally free successive quotients. Use the subbundle Grassmannians, obtained by dualizing the quotient Grassmannians of AG-RG-03 Lemma 5.A. In their product impose the vanishing of each bundle map $F_{d_j} \to \mathcal{O}^n/F_{d_{j+1}}$. In local frames its finitely many entries generate the ideal of a closed incidence subscheme. Its test points have exactly the desired inclusions. The quotient $F_{d_{j+1}}/F_{d_j}$ is automatically locally free: the exact sequence

$$0 \to F_{d_{j+1}}/F_{d_j} \to \mathcal{O}^n/F_{d_j} \to \mathcal{O}^n/F_{d_{j+1}} \to 0$$

splits locally because its last term is locally free. This verifies the incidence functor over arbitrary rings. The product of the closed Plücker embeddings is projective, so its closed incidence subscheme is projective too.

The open chart where $F_{d_j} \to \mathcal{O}^{d_j}$ is an isomorphism for every j, using projection to the first d_j coordinates, is represented by block lower unitriangular matrices with diagonal blocks the identity. Unit-pivot elimination gives the matrix uniquely from a flag and conversely sends its first blocks to that flag; it only inverts the indicated invertible projection determinants. Its free blocks give affine space of dimension

$$\sum_{i<j} n_i n_j. \tag{7.1}$$

The corresponding charts for coordinate permutations cover the scheme. On a residue-field flag, row-pivot elimination chooses nested coordinate sets I_j, of sizes d_j, for which all those projections are isomorphisms; one permutation sends them to the initial sets. Their determinant conditions define open charts on the original scheme, and coverage of all residue-field points is coverage of its underlying space. Thus these actual affine charts prove smoothness over S, including when S is nonreduced. Finally every flag admits an adapted basis Zariski locally: successively split its locally free quotients and choose bases of the summands.

Hence GL_n acts transitively as a sheaf, with stabilizer the block upper triangular P. It follows directly on the functor of points that

$$\mathrm{Flag}_{d_1,\ldots,d_{r-1}} \simeq \mathrm{GL}_n / P. \tag{7.2}$$

Taking every cut gives the full flag scheme, representing Borel subgroups; its relative dimension is $n(n-1)/2$. Omitting every cut gives the one-point scheme representing the whole group as a parabolic. This is the linear-algebra model of Theorem 2.1.

Schubert cells for split flag schemes

Theorem 7.1 (relative Schubert cells). Let G/S be pinned split of constant datum, let $J \subset \Delta$, and let P_J be its standard parabolic. Let W_J be the Weyl subgroup generated by J, and W^J the minimal-length representatives for the right cosets W/W_J. Then G/P_J has a finite disjoint stratification by locally closed subschemes

$$U_{\Psi_w} \longrightarrow G/P_J, \qquad u \longmapsto u n_w P_J, \qquad w \in W^J,$$

each isomorphic to $\mathbf{A}_S^{\ell(w)}$. These cells commute with arbitrary base change. In particular the theorem applies over $\mathbf{Z}$ and supplies both a torification by coordinate tori and, over a finite field, the count $|(G/P_J)(\mathbf{F}_q)| = \sum_{w \in W^J} q^{\ell(w)}$.

Proof. Let U_J^- have the negative roots outside the root subsystem generated by J. The relative cocharacter open-cell theorem identifies $U_J^- \times P_J$ with an open subscheme of G. Quotienting this open set by its right P_J-action gives an open immersion $U_J^- \to G/P_J$. To justify the quotient assertion, its inverse image in the P_J-torsor $G \to G/P_J$ is the indicated right-stable open product, and the quotient of that product is U_J^-. Openness and the isomorphism descend through this faithfully flat torsor.

We first prove all minimal-coset facts used here. Reflections in J change only the simple-root coefficients indexed by J. A positive root outside Φ_J retains a positive coefficient outside J under every such reflection, so remains positive; a negative such root remains negative. Thus an element of W_J has inversions only in the Levi root system. That system has simple roots J by the positive coefficient and cocharacter-face calculation of Theorem 1.1. The entire chamber/length proof of RG04 Section 3 applied to this subsystem gives a reduced word using J and its length equals the full length, since the inversion sets agree. Starting with any $v \in W$, repeatedly remove a right descent from J. Each step lowers full length by one, so this ends at w with $w\alpha$ positive for every $\alpha \in J$. A positive Levi root is a nonnegative sum of these simple roots, and its image is consequently positive. The removed word gives $v = wz$, $z \in W_J$. If its number of letters is m, then $\ell(v) = \ell(w) + m$ and $\ell(z) \le m$. The length triangle inequality, proved by concatenating reduced words, forces $m \le \ell(z)$, so equality holds. Conversely, for any such w and a reduced J-word for z, its successive positive inversion roots are Levi roots and their images by w are positive. The length-one inversion rule therefore gives $\ell(wz) = \ell(w) + \ell(z)$, making w minimal in its coset. It is unique: if $w' = wz$ also sends every positive Levi root to a positive root and $z \neq 1$, a positive Levi inversion β of z would make $w'\beta = w(z\beta)$ negative, a contradiction. This proves the indexed set W^J, the unique length-additive factorization and its inversion-root union used below.

Minimality of w therefore means $w(\Phi_J^+) \subset \Phi^+$. Thus $w^{-1}\Psi_w$ is negative and contains no Levi root: a negative Levi root $-\beta$ would give the positive root $w(-\beta)$, contradicting that condition. The ordered negative-root product makes $n_w^{-1} U_{\Psi_w} n_w$ a closed subscheme of U_J^-, by setting the other coordinates to zero. Translating the previous open chart by n_w proves that the displayed map is a locally closed immersion. Its source is the product of $\ell(w)$ framed root lines, hence affine space of that dimension over S.

For coverage on a geometric fibre, write $v = wz$, with $w \in W^J$, $z \in W_J$ and $\ell(v) = \ell(w) + \ell(z)$. Then $\Psi_v = \Psi_w \sqcup w\Psi_z$. The arbitrary-order root-product theorem of AG-RG-04 Section 5, with the first set first, gives

$$U_{\Psi_v} = U_{\Psi_w}\, n_w U_{\Psi_z} n_w^{-1}$$

as a product of schemes. The second factor and n_z are absorbed on the right in P_J, so the image

of the Bruhat cell for v is the displayed cell for w. Field Bruhat coverage therefore covers every geometric point of the quotient.

Each displayed cell is the entire B-orbit of $n_w P_J$. Indeed the decomposition $U^+ = U_{\Psi_w} U_{\Theta_w}$ of AG-RG-04 Section 5 has $w^{-1}\Theta_w \subset \Phi^+$, so U_{Θ_w} fixes this coset, as does T. We also verify the needed indexing of T-fixed geometric cosets. If gP_J is fixed, $g^{-1}Tg \subset P_J$. The field conjugacy theorem for maximal tori of the connected smooth affine P_J supplies $p \in P_J$ with $gp \in N_G(T)$. Hence every fixed coset is $n_v P_J$. Two representatives give the same coset exactly when their Weyl elements differ on the right by the image of $N_G(T) \cap P_J$, which is W_J by the Bruhat description in Theorem 1.1. Thus the fixed cosets are precisely W/W_J.

Choose an integral cocharacter positive on every positive root. On $U_{\Psi_w} \simeq \mathbf{A}^{\ell(w)}$ its root coordinates have strictly positive weights, so its fixed point, and consequently its only T-fixed point, is the origin $n_w P_J$. If two displayed cells met on a geometric fibre, their entire B-orbits would coincide; their unique T-fixed points would then coincide, contradicting the distinct cosets in W/W_J. The images are therefore pairwise disjoint. Every point of the relative quotient admits a geometric point above it, so this gives disjoint coverage of the underlying space over S. The locally closed subschemes themselves were constructed over S before this fibre argument.

Every map, root-coordinate closed immersion and open chart used here commutes with base change, as does G/P_J by Theorem 2.1. Thus the locally closed stratification commutes with every base change, including nonflat changes to nonreduced schemes. Over $\mathbf{F}_q$, each rational point belongs to exactly one cell, and its affine coordinates give $q^{\ell(w)}$ points; summing gives the stated count. Over $\mathbf{Z}$, partition each affine cell coordinate by zero versus invertible. The resulting locally closed pieces are split tori and partition geometric points and points over every field, giving the claimed torification. A stratification does not assert that the coproduct of its strata equals the original fppf sheaf: a point over a ring with nilpotents can cross a stratum boundary and need not factor through one stratum. All constructions here still represent their stated subfunctors on every scheme test. □

For $G = \mathrm{GL}_n$ and block type $(n_1, \dots, n_r)$ this applies to the flag scheme just constructed, with $W = S_n$ and $W_J = S_{n_1} \times \cdots \times S_{n_r}$. Indeed each simple-root reflection swaps adjacent coordinates. These swaps generate every permutation: move its last coordinate into place by adjacent swaps and induct on the remaining coordinates. Restricting to the simple roots within each block gives exactly the displayed product of block permutation groups. It retains the ordered block type and the locally direct-summand condition on flags. For a nonsplit form, these globally indexed affine cells are not asserted: étale-local splitting of its parameter scheme alone does not give such a global stratification.

8. Exercises

Exercise 1 *easy.* Show that $\tau(g) = (g^T)^{-1}$ is outer on SL_n for $n \geq 3$, and is conjugation by $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ on SL_2.

Exercise 2 *medium.* Classify the inner forms of SL_n over a field, specifying what changes when the inner structure is forgotten.

Exercise 3 *medium.* Classify the parabolics of GL_n containing the upper triangular Borel.

Exercise 4 *hard.* Prove that the Borel subgroup functor of GL_n is represented by a smooth projective flag scheme over an arbitrary base.

Exercise 5 *medium.* Explain why the automorphism group of a split torus need not be of finite type, and verify the central automorphism (3.2).

Solutions

Solution 1

Reversing the order twice makes transpose-inverse a homomorphism; it is an involution. If it were inner by h, its preservation of the diagonal torus would put h in that torus's normalizer. Its action on $X^*(T) = \mathbf{Z}^n/\mathbf{Z}(1,\dots,1)$ is -1, whereas the Weyl action is a coordinate permutation. For three distinct indices, a permutation carrying $e_i - e_j$ to $e_j - e_i$ for every pair would have to exchange both (i,j) and (i,k), which is impossible. Thus -1 is not a Weyl action for $n \geq 3$. For $n = 2$, direct multiplication, using $ad - bc = 1$, gives

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix} = \tau(g).$$

The identities hold over every ring, including characteristic two.

Solution 2

Its adjoint group is PGL_n. Lemma 6.1 sends a torsor to a degree-n central simple algebra A; applying that torsor to the determinant-one matrix group gives $\mathrm{SL}_1(A)$. Conversely the local splitting isomorphisms of A give the same torsor. This proves the classification with inner structure. For $n \geq 3$, forgetting it divides by the diagram involution, which sends A to A^{op}, as the transpose calculation in Section 6 shows. The inversion map supplies the corresponding group isomorphism. The distinction concerns the marking; it does not create an additional family of inner groups.

Solution 3

The simple roots are $e_i - e_{i+1}$, for $1 \leq i < n$. A subset J joins consecutive indices into blocks: put a cut at i precisely when $e_i - e_{i+1} \notin J$. The resulting ordered block sizes are positive and sum to n. The roots in the Levi are exactly differences within blocks; the other permitted root entries are above the blocks. Thus Theorem 1.1 gives exactly the block upper triangular groups. Recovering their cuts recovers J, so there is neither repetition nor omission. For $n = 3$, the four types are $(1,1,1), (1,2), (2,1), (3)$.

Solution 4

Form the closed incidence subscheme in $\prod_{d=1}^{n-1} \mathrm{Gr}(d,n)$ defined by $F_d \subset F_{d+1}$. Its test points are exactly full locally split flags. The Plücker embeddings and the incidence equations prove projectivity. Adapted bases give local lifts to GL_n; their transition matrices preserve the flag exactly when upper triangular, so this scheme represents GL_n/B. The lower unitriangular charts have $n(n-1)/2$ free coordinates and prove smoothness. By Theorem 2.1 a Borel is locally a conjugate of B, and its normalizer is B. Hence it determines exactly one quotient section, with descent on overlaps. This identifies the flag and Borel functors on all schemes, rather than merely on field-valued points.

Solution 5

A torus automorphism is an invertible integral change of its character basis, so the representing scheme is the disjoint union of copies of the base indexed by $\mathrm{GL}_r(\mathbf{Z})$. For $r \geq 2$ it has infinitely many components, since the matrices $\left(\begin{smallmatrix} 1 & m \\ 0 & 1 \end{smallmatrix}\right)$ are distinct. On GL_2, centrality of the determinant factor proves that (3.2) is a homomorphism. Its determinant is $\det(g)^{-1}$, so applying it twice

returns g. On scalar matrices it sends zI to $z^{-1}I$, whereas every inner automorphism fixes them. Thus it is a nontrivial outer automorphism arising from the central lattice direction.

The course prerequisite guide records the exact supporting statements, their full proof routes, and the hypotheses needed in their applications.

References and exact prerequisite proofs

- Michel Demazure and Alexander Grothendieck, *Schémas en groupes*, freely accessible Gille–Polo re-edition of 13 October 2024: Exposé XXIV and Exposé XXVI, for comparison with the automorphism, form and parabolic arguments proved here.
- Brian Conrad, *Reductive group schemes*, §§5.2 and 7.1–7.2. The proofs here use the preceding lessons' root-coordinate and pinned classification results.
- Milne's freely accessible *Algebraic Groups*, version 2.00 (2015). This exact free author edition provides comparison material; its citations do not replace the programme proofs.

The preceding written supporting proofs are Algebra and sheaf cohomology, §§3–8, for the finite-cover, affine and projective calculations; Projective cohomology and smooth affine models, Proposition 3.2 and Proposition 5.2, for retaining smooth projective models and line bundles and for direct images commuting with every base change; and Supporting group-scheme proofs, Lemma G.2.1, for smoothness descent over any field. The projective quotient and scheme-theoretic normalizer arguments are AG-RG-03 §5, using the earlier written descent, Zariski Main and flat-quotient proofs. AG-RG-03 Lemma 5.A supplies the quotient Grassmannian charts and closed Plücker immersion, with subbundle charts obtained by dualizing. S04 Lemmas A1.1–A1.3a and A1.5 supply the entire augmented faithfully flat complex, effective module/algebra descent, finite presentation and smoothness descent, and morphism descent. Sections 4–5 here prove the actual torsor/cocycle classification and both multiplicative and additive vanishing used for quasi-split forms. The normalizer Lie calculation is explicitly given in Theorem 1.1 and uses the already proved torus weight projections. The separable matrix-splitting argument needed for inner forms is Theorem 6.2 of this lesson itself; no Tsen theorem or computation of an entire Brauer group is consumed. The proofs above establish the remaining parabolic, automorphism, form, flag and Schubert assertions. This identifies the consumed proof chain without claiming that unrelated parts of the programme are proof-complete.

Lesson 7

Prerequisites and reading order

Read the supporting lessons below before the six main lessons: tori; centralizers; roots and rank one; root data and Bruhat cells; integral pinned classification; forms and parabolic flag schemes. Every mathematical result used requires its proof here or in an earlier programme lesson. Freely accessible references provide writing and comparison material; they never replace that programme proof.

Approximation, general flat quotients, Lie and group foundations have written supporting proofs, and concrete corrections across the six main lessons are integrated. The complete consumed-result inventory and full transitive proof validation remain active.

The relative Bruhat and Schubert arguments retain the integral scheme decompositions used for finite-field counting and torified varieties. A scheme stratification does not partition points over a general ring.

Earlier supporting lessons

Read first	Written proofs
Algebra and sheaf cohomology before reductive groups	Localization, arbitrary-module Tor, finite algebra, modules and sheaves, derived cohomology, affine vanishing, flat base change, arbitrary-base projective properness and Serre arguments.
Projective cohomology and smooth affine models	Completion, formal functions, Noetherian models, universal cohomology complexes and full curve duality.
Completing a smooth affine curve and extending its group action	Finite normalization, proper smooth completion, regular extension of the translation family and the one-dimensional unipotent argument.
Affine descent, Zariski Main and recognition of spaces	Integral algebra, full conductor, general scheme factorization, separated locally quasi-finite descent, finite affine quotient and space recognition, in dependency order.
Artin approximation and desingularization	Pointed étale approximation, complete-local finite splitting covers and torus descent, finite coefficient systems, full desingularization and its consumed foundations.

Read first	Written proofs
Lie proofs for the characteristic-zero group construction	Complete ordered-word proof, trace criterion and complete reducibility, rational Serre algebra, finite highest-weight presentations and polynomial rank-one integration.
Flat quotient bootstrap over an arbitrary base	Fibre loci, regular cuts, finite-presentation descent, slices, finite parts and the general flat quotient with all test schemes.
Supporting group-scheme proofs	Canonical flat components, field-group identity components with nilpotents, point dimension, field smoothness descent, Cartier, universal schematic density, closed orbits and semilinear Hopf descent.

The supporting lessons are placed before their first main-course use. The references below identify their earlier algebra and geometry inputs.

Lie algebras and smoothness of group schemes

Lie algebras at the identity and invariant derivations, including their commutator bracket; Cartier smoothness for a locally finite-type group scheme over a characteristic-zero field. Used in main lesson 1 §4; 5 §6.

Required programme proof. AG-RG-S08 G.1.1–G.1.2 (point dimension), G.2.1 (field smoothness descent), G.3.1–G.3.4 (full Cartier chain); earlier GS02 Proposition 1.1, Lemma 2.1, Theorem 2.2 and Proposition 2.7 retain the tangent/bracket/cotangent proofs.

The argument is in AG-RG-S08.md.

Group schemes over a field

For locally finite-type group schemes over any field, including nonreduced groups, the identity component is a normal geometrically irreducible quasi-compact subgroup, open and closed, and commutes with every field extension. For a smooth finite-type algebraic group acting on a nonempty separated finite-type reduced scheme over an algebraically closed field, orbits are locally closed and there is a closed orbit; schematic stabilizers and the associated scheme orbit use the separately proved G/H quotient. Also, a quasi-compact homomorphism of field group schemes has a closed schematic subgroup image; if its source is reduced, the factor onto that image is faithfully flat. Quasi-compact dominant homomorphisms into a reduced target are faithfully flat. These statements are applied to smooth finite-type geometric fibres in lesson 5 Section 9. Used in main lesson 1 §2; 2 §1; 5 §9 (Propositions 5.6 and 5.8 for schematic image and fibre affineness).

Required programme proof. AG-RG-S08 G.0.1 (canonical flat closed structure), G.0.2 (algebraic-extension connectedness), G.0.3 (locally finite-type field-group identity component with nilpotents); AG-RG-S08 G.4.1–G.4.2 (reduced products and full smooth closed-orbit proof); the separate schematic orbit proof retains earlier GS03 Lemma 7.12 and Theorem 7.13, with GS04 Theorem 11.1b and the flat quotient now proved in AG-RG-S07. The current earlier GS03 Propositions 5.6 and 5.8 supply the complete closed-image and flatness argument consumed in AG-RG-05 §9.

The argument is in AG-RG-S08.md.

Quotients and torsors

Effective affine finite locally free free-action quotients, including nonconstant and nonreduced finite groups; effective affine-group torsors and nonabelian H1; Hilbert 90 over arbitrary fields and Gm torsors as line bundles; scheme quotients G/H for finite-type groups over a field and closed H. Used in main lesson 1 §§2,5; 3 §§5–6; 6 §§4–6.

Required programme proof. Faithfully flat affine descent (opening Module descent lemma/Affine descent corollary); §§1–4, especially Theorem 4.3 (general finite flat affine quotient) and Corollary 4.4 (constant finite étale special case); §§6–7.1 (torsors and cocycles); Theorem 8.1 (frames, Hilbert 90, line bundles); Lemma 11.1a/Theorem 11.1b (field G/H scheme quotient, through AS-02 bootstrap).

Earlier written programme proof: quotients and torsors (Group schemes). That proof supplies this input.

Diagonalizable groups and groups of multiplicative type

Over an arbitrary field, finite-type groups of multiplicative type are contravariantly equivalent to finitely generated character groups with continuous Galois action and split over a finite separable extension. Used in main lesson 1 introductory section and §§3–4.

Required programme proof. AG-RG-S08 G.5.1–G.5.3 (explicit arbitrary-characteristic semi-linear Hopf descent); earlier GS05 Lemma 2.1/Theorem 2.2, Theorem 5.0 and Lemma 5.1 retain the monomial and separable-splitting proofs, and Theorem 3.1, Theorem 4.10/Corollary 4.11, Proposition 7.14, Theorem 7.15 and Proposition 8.1 retain the actual deformation proofs.

The argument is in AG-RG-S08.md.

Faithfully flat descent

Effective faithfully flat descent of affine schemes over arbitrary bases, modules, algebras and equivariant morphisms, including group operations and their identities. Used in main lesson 1 §3; 3 §1 (central quotient group operations); 6 §§2,4,6.

Required programme proof. Earlier AG-GS-04, opening Faithfully flat affine descent: Module descent lemma (line 29) and Affine descent corollary (line 45), including arbitrary-base gluing and equivariant maps. Unit, multiplication, inverse and all group-law equalities descend as maps and equalities. Current provider commit 4034302e92d3ef07f10dde2d4bf9673a9917b996.

Earlier written programme proof: quotients and torsors (Group schemes). That proof supplies this input.

The bootstrap theorem

An equivalence relation of algebraic spaces with flat locally finitely presented projections has an algebraic-space fppf quotient; a free action of a flat locally finitely presented group gives a representable quotient map satisfying the torsor identity on all test schemes. Used in main lesson 1 §5; 3 §5; 5 §8.

Required programme proof. AG-RG-S07 §§2–8: Lemmas 2.1–2.4/Corollary 2.5, dimension and Cohen–Macaulay locus 3.1–3.4, regular cuts 4.1, descent 5.1–5.3, slicing 6.1–6.2, finite part 7.1, division 8.1, general flat quotient 8.2 and free actions 8.3; preceding scheme descent/recognition and finite affine quotient in AG-RG-S04 A/E5.1/E6.1/E8.1–E8.2.

The argument is in AG-RG-S07.md.

Artin approximation for polynomial equations

Artin approximation for finite polynomial systems over an excellent Noetherian local ring essentially of finite type over Z: a formal solution can be approximated to each finite order in its henselization, hence in an étale neighbourhood with unchanged residue field. Used in main lesson 1 §3.

Required programme proof. AG-RG-S05 Theorem 1.1, Lemmas 1.2–1.3/Corollary 1.4, spreading Lemma 2.1 and full finite torus system §3; complete desingularization sequence D.1–D.38 including D.34A/D.35 global selection, foundation blocks F.1–F.138 including F.99A/F.102 and G-ring permanence F.132–F.135; exact elementary earlier programme proofs in §6.

The argument is in AG-RG-S05.md.

Limits of schemes and Noetherian approximation

Finite-presentation objects, morphisms and finite diagrams of identities descend through filtered inverse limits with affine transitions and qcqs stages; affineness, closed immersion and smoothness hold eventually under their stated finite-presentation hypotheses. Used in main lesson 1 §3; 3 §§1,5; 4 §7; 5 §§8–9; 6 §2.

Required programme proof. Theorem 1.1; Lemmas 2.1–2.3; Lemma 3.1/Theorem 3.2; Theorems 4.1–4.2 (objects, maps, identities, affine/finite/closed/separated properties); Theorem 5.1 (affine Noetherian approximation). Smoothness is the separately read Stacks 0C0C/0C0B proof imported in §4. AG-RG's connected normalizer models additionally use the separately read 05FI/055I/05F4 proofs.

Earlier programme source: `AG-MO/src/limits-and-noetherian-approximation.md`. Its consumed proof and transitive inputs remain in the programme review.

Zariski's Main Theorem

For a quasi-finite separated morphism to a qcqs scheme, Zariski Main gives an open immersion into a finite scheme; this form applies target-locally on arbitrary bases. A quasi-finite separated birational morphism of integral schemes with normal target is an open immersion. A proper quasi-finite morphism is finite over arbitrary bases. Used in main lesson 5 §§7,9.

Required programme proof. Blocks B–D: polynomial normality, conductor proof and affine completion (C4.3/C5.1); relative integral closure and étale base change (D4.2–D4.3); general relative Zariski Main and finite factorization (D5.2/D7.1); proper quasi-finite finiteness and normal birational open immersion (D7.3/D7.4). Space recognition and normalized space factorization are E8.1/E8.2/E9.1.

The argument is in AG-RG-S04.md.

Flatness criteria, dimension and the flat locus

For a locally finitely presented X→S, locally finite-type Y→S and a morphism X→Y, with X and Y flat over S, flatness of X→Y is equivalent to flatness of its geometric fibre maps; the finite-presentation module form also works over arbitrary nonreduced bases. Used in main lesson 3 §1, Theorem 1.1; 4 §7.

Required programme proof. Theorem 2.1/Corollary 2.2 (Noetherian fibre criterion); Lemmas 6.1–6.2 and Theorem 6.3 (finite Tor obstruction and arbitrary-base criterion); Theorem 6.4/Corollary

6.5 (open relative flat locus and base change). Theorem 6.3 is the exact arbitrary-base input in AG-RG-04 §7.

Earlier programme source: `AG-FSE/src/flatness-criteria.md`. Its consumed proof and transitive inputs remain in the programme review.

Étale morphisms and their local structure

A flat locally finitely presented morphism with étale geometric fibres is étale; an étale monomorphism is an open immersion. Used in main lesson 1 §3; 3 §§1,4–5; 4 §7; 6 §2.

Required programme proof. Lemmas 1.1–1.2 and Theorem 1.3 (standard Jacobian quotient, finite kernel/Nakayama and fibre criterion); Theorem 4.1 (open immersion equivalent to an étale monomorphism/universally injective étale map, with the faithfully flat local epimorphism proof).

Earlier programme source: `AG-FSE/src/etale-morphisms-and-their-local-structure.md`. Its consumed proof and transitive inputs remain in the programme review.

Smooth morphisms

A locally finitely presented morphism is smooth exactly when flat with smooth geometric fibres; a smooth surjection admits sections on an étale covering, including over nonreduced bases. Used in main lesson 1 §3; 3 §§1,4; 4 §7; 6 §§2,6.

Required programme proof. Theorem 3.1 (fibrewise smoothness criterion), Theorem 4.1 (étale coordinates), and Theorem 5.2 (étale-local sections via a finite separable point in a nonempty smooth fibre).

Earlier programme source: `AG-FSE/src/smooth-morphisms.md`. Its consumed proof and transitive inputs remain in the programme review.

Cohomology of sheaves on ringed spaces

Leray spectral sequence for higher direct images and sheaf cohomology Used in main lesson 5 §9.

Required programme proof. Section 3, Lemmas 3.1–3.3 and acyclic-resolution comparison; Section 4 filtered-complex construction; Theorem 4.1, including the full compatible injective double resolution and both filtrations.

The argument is in AG-RG-S01.md.

Cohomology of affine schemes and Serre's criterion

Vanishing of positive quasi-coherent cohomology on affine schemes and Serre's affine criterion for quasi-compact separated schemes Used in main lesson 1 §2; 5 §9; 6 §5.

Required programme proof. Theorem 5.1 and Corollary 5.2; Lemma 9.1 and Theorem 9.2, with the quasi-separatedness-free affine criterion proof.

The argument is in AG-RG-S01.md.

Coherent sheaves on projective schemes: Serre's theorems

Finiteness of all coherent cohomology modules on a projective scheme over a Noetherian ring; eventual global generation and positive-cohomology vanishing for coherent ample twists. Used in main lesson 3 Section 5; 6 Section 2.

Required programme proof. Sections 7–8: all projective-space twists and perfect monomial pairing; Lemmas 8.1/8.3, Proposition 8.2, Theorem 8.4 and Corollary 8.5, including projective-scheme coherent finiteness and ample twists.

The argument is in AG-RG-S01.md.

Base change and the Grothendieck complex

Flat base change for quasi-coherent cohomology on quasi-compact quasi-separated schemes, including localization Used in main lesson 5 §9, openly licensed affine-open proof.

Required programme proof. Section 6, full canonical flat base-change and localization proof for arbitrary quasi-compact quasi-separated schemes, including nonseparated intersections.

The argument is in AG-RG-S01.md.

Semicontinuity and Grauert's theorem

Lemma (sections in a vanishing family). Let f: X -> S be a smooth projective morphism of finite presentation, and let L be an invertible O_X-module. Suppose Hˆq(X_s,L_s)=0 for every geometric point s of S and every q>0. Then f_*L *is finite locally free, Rˆq f*_L=0 for q>0, and for every base change g: S′ -> S the canonical map gˆ*f*_L -> f′_*L′ is an isomorphism and Rˆq f′_*L′=0 for q>0. The base S may be arbitrary and nonreduced. Used in main lesson 3 §5; 6 §2.

Required programme proof. Lemma 5.1, full universally acyclic finite-projective replacement; Proposition 5.2, arbitrary-base and all-base-change sections in a vanishing smooth projective family.

The argument is in AG-RG-S02.md.

The theorem on formal functions

For a Noetherian ring A, a projective scheme X over A, a coherent F and an ideal I, I-adic completion of Hˆq(X,F) equals the inverse limit of $H^{q(X,F/I}$n F). For complete local A this turns a compatible formal idempotent in H0 into an actual idempotent. Used in main lesson 3 §5.

Required programme proof. Lemmas 1.1–1.4 and 2.1–2.2, Theorem 2.3 and Corollary 2.4: finite Rees images, kernel filtrations, exact inverse limits, projective formal functions and idempotents.

The argument is in AG-RG-S02.md.

Serre duality for smooth projective curves

For a smooth projective pure one-dimensional curve C over any field and an invertible L, Hˆ1(C,L) is naturally dual to Hˆ0(C,Omega_(C/k) tensor L inverse). Used in main lesson 1 §2; 3 §2.

Required programme proof. AG-RG-S02 §6, Lemma 6.A and Theorem 6.B (complete ambient projective duality), Lemma 6.1 and Theorem 6.2 (local Ext and single-row local-to-global comparison), with projective finiteness and finite filtered complexes in AG-RG-S01 §§4,8.

The argument is in AG-RG-S02.md.

Brauer groups and Tsen's theorem

Central simple algebras of degree n split over finite separable extensions; Brauer and matrix-algebra descent over fields Used in main lesson 6 §6.

Required programme proof. Lemma 6.1 and Theorem 6.2: finite-dimensional density proved, central simplicity under arbitrary field extension, algebraic-closure splitting, smooth isomorphism scheme and finite separable splitting; converse and matrix-algebra descent.

The argument is in AG-RG-06.md.

Complete reducibility: Casimir elements and Weyl's theorem

Complete reducibility and invariant complements for finite-dimensional semisimple Lie algebras over every characteristic-zero field. Used in main lesson 5 §6.

Required programme proof. AG-RG-S06, Sections 1–4; Proposition 4.1 and Theorem 4.2, including the invariant projection and finite-linear-system descent over every characteristic-zero field.

The argument is in AG-RG-S06.md.

Representations of sl(2)

Finite-dimensional sl2 weight theory and explicit polynomial SL2 actions on symmetric powers Used in main lesson 5 §6.

Required programme proof. Section 6, Lemma 6.1

The argument is in AG-RG-05.md.

Inner derivations of semisimple Lie algebras

Every derivation of a finite-dimensional semisimple Lie algebra over a characteristic-zero field is inner, by the nondegenerate Killing form. Used in main lesson 5 §6, directly over Q.

Required programme proof. AG-RG-S06, Proposition 3.1 (Killing nondegeneracy) and Proposition 3.3 (all derivations inner over the given characteristic-zero field).

The argument is in AG-RG-S06.md.

Root systems and their Weyl groups

Positive-root inversion length, simple-reflection length reduction and the complete Coxeter presentation for the Weyl groups of the pinned reductive groups used in lesson 5 Sections 2 and 5. Used in main lesson 5 §§2,5.

Required programme proof. Earlier chamber freeness Lemma 3.A; Proposition 3.1 and Lemmas 3.2–3.3, Theorem 3.4; rank-two table (3.2). The consumed presentation concerns the root data of the pinned groups in lesson 5 Sections 2 and 5.

The argument is in AG-RG-04.md.

The isomorphism theorem and Serre's theorem

Finite-type Cartan Serre presentation: finite-dimensional split semisimple Lie algebra with the specified roots over Q, compatible with characteristic-zero field extension. Used in main lesson 5 §6, directly over Q and every characteristic-zero extension.

Required programme proof. AG-RG-S06, Theorem 1.A, Section 5, Lemma 6.1 and Theorem 6.2: full rational Serre algebra, auxiliary relations, Serre ideal stability, finite root set and semisimplicity.

The argument is in AG-RG-S06.md.

Weights, Verma modules and the theorem of the highest weight

Finite integrable highest-weight presentation for dominant integral weights, its irreducibility over C, and its rational form compatible with characteristic-zero extension. Used in main lesson 5 §6, rational finite highest-weight presentation and absolute irreducibility.

Required programme proof. AG-RG-S06, Section 7, equations (7.1)–(7.6) and the rational base-change paragraph; Section 8 for faithful fundamental modules and polynomial rank-one actions.

The argument is in AG-RG-S06.md.

Sources and further reading

The bibliography lists freely accessible mathematical sources. The source record gives their titles, authors and links.